



U.S. SECURITIES AND EXCHANGE COMMISSION

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UNITED STATES SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-K

ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF
THE SECURITIES EXCHANGE ACT OF 1934

Commission File Number: 001-14965

For the fiscal year ended December 31, 2024

The Goldman Sachs Group, Inc.

(Exact name of registrant as specified in its charter)

Delaware

(State or other jurisdiction of
incorporation or organization)

200 West Street, New York, NY
(Address of principal executive offices)

13-4019460

(I.R.S. Employer
Identification No.)

10282
(Zip Code)

(212) 902-1000

(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Exchange on which registered
Common stock, par value \$0.01 per share	GS	NYSE
Depository Shares, Each Representing 1/1,000th Interest in a Share of Floating Rate Non-Cumulative Preferred Stock, Series A	GS PRA	NYSE
Depository Shares, Each Representing 1/1,000th Interest in a Share of Floating Rate Non-Cumulative Preferred Stock, Series C	GS PRC	NYSE
Depository Shares, Each Representing 1/1,000th Interest in a Share of Floating Rate Non-Cumulative Preferred Stock, Series D	GS PRD	NYSE
5.793% Fixed-to-Floating Rate Normal Automatic Preferred Enhanced Capital Securities of Goldman Sachs Capital II	GS/43PE	NYSE
Floating Rate Normal Automatic Preferred Enhanced Capital Securities of Goldman Sachs Capital III	GS/43PF	NYSE
Medium-Term Notes, Series F, Callable Fixed and Floating Rate Notes due March 2031 of GS Finance Corp.	GS/31B	NYSE
Medium-Term Notes, Series F, Callable Fixed and Floating Rate Notes due May 2031 of GS Finance Corp.	GS/31X	NYSE

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. ☐ Yes ☒ No

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. ☐ Yes ☒ No

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. ☒ Yes ☐ No

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). ☒ Yes ☐ No

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to § 240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). ☐ Yes ☒ No

As of June 30, 2024, the aggregate market value of the common stock of the registrant held by non-affiliates of the registrant was approximately \$42.3 billion.

As of February 7, 2025, there were 312,039,033 shares of the registrant's common stock outstanding.

Documents incorporated by reference: Portions of The Goldman Sachs Group, Inc.'s Proxy Statement for its 2025 Annual Meeting of Shareholders are incorporated by reference in the Annual Report on Form 10-K in response to Part III, Items 10, 11, 12, 13 and 14.

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PART I

Item 1. Business

Introduction

Goldman Sachs is a leading global financial institution that delivers a broad range of financial services to a large and diversified client base that includes corporations, financial institutions, governments and individuals. Our purpose is to advance sustainable economic growth and financial opportunity. Our goal, reflected in our One Goldman Sachs initiative, is to deliver the full range of our services and expertise to support our clients in a more accessible, comprehensive and efficient manner, across businesses and product areas.

When we use the terms “Goldman Sachs,” “we,” “us,” “our” and “the firm,” we mean The Goldman Sachs Group, Inc. (Group Inc. or parent company), a Delaware corporation, and its consolidated subsidiaries. When we use the term “our subsidiaries,” we mean the consolidated subsidiaries of Group Inc. References to “this Form 10-K” are to our Annual Report on Form 10-K for the year ended December 31, 2024. All references to 2024, 2023 and 2022 refer to our years ended, or the dates, as the context requires, December 31, 2024, December 31, 2023 and December 31, 2022, respectively.

Group Inc. is a bank holding company (BHC) and a financial holding company (FHC) regulated by the Board of Governors of the Federal Reserve System (FRB). Our U.S. depository institution subsidiary, Goldman Sachs Bank USA (GS Bank USA), is a New York State-chartered bank.

Our Business Segments

We manage and report our activities in three business segments: Global Banking & Markets, Asset & Wealth Management and Platform Solutions. Global Banking & Markets generates revenues from investment banking fees, including advisory, and equity and debt underwriting fees, Fixed Income, Currency and Commodities (FICC) intermediation and financing activities and Equities intermediation and financing activities, as well as relationship lending and acquisition financing (and related hedges) and investing activities related to our Global Banking & Markets activities. Asset & Wealth Management generates revenues from management and other fees, incentive fees, private banking and lending, equity investments and debt investments. Platform Solutions generates revenues from consumer platforms and transaction banking and other.

The chart below presents our three business segments and their revenue sources.



Global Banking & Markets

Global Banking & Markets serves public and private sector clients and we seek to develop and maintain long-term relationships with a diverse global group of institutional clients, including corporations, governments, states and municipalities. Our goal is to deliver to our institutional clients all of our resources in a seamless fashion, with our advisory and underwriting activities serving as the main initial point of contact. We make markets and facilitate client transactions in fixed income, currency, commodity and equity products and offer market expertise on a global basis. In addition, we make markets in, and clear client transactions on, major stock, options and futures exchanges worldwide. Our clients include companies that raise capital and funding to grow and strengthen their businesses, and engage in mergers and acquisitions, divestitures, corporate defense, restructurings and spin-offs, as well as companies that are professional market participants, who buy and sell financial products and manage risk, and investment entities whose ultimate clients include individual investors investing for their retirement, buying insurance or saving surplus cash.

As a market maker, we provide prices to clients globally across thousands of products in all major asset classes and markets. At times, we take the other side of transactions ourselves if a buyer or seller is not readily available, and at other times we connect our clients to other parties who want to transact. Our willingness to make markets, commit capital and take risk in a broad range of products is crucial to our client relationships. Market makers provide liquidity and play a critical role in price discovery, which contributes to the overall efficiency of the capital markets. In connection with our market-making activities, we maintain (i) market-making positions, typically for a short period of time, in response to, or in anticipation of, client demand, and (ii) positions to actively manage our risk exposures that arise from these market-making activities (collectively, inventory).

We execute a high volume of transactions for our clients in large, highly liquid markets (such as markets for U.S. Treasury securities, stocks and certain agency mortgage pass-through securities). We also execute transactions for our clients in less liquid markets (such as mid-cap corporate bonds, emerging market currencies and certain non-agency mortgage-backed securities) for spreads and fees that are generally somewhat larger than those charged in more liquid markets. Additionally, we structure and execute transactions involving customized or tailor-made products that address our clients' risk exposures, investment objectives or other complex needs, as well as derivative transactions related to client advisory and underwriting activities.

Through our global sales force, we maintain relationships with our clients, receiving orders and distributing investment research, trading ideas, market information and analysis. Much of this connectivity between us and our clients is maintained on technology platforms, including *Marquee*, and operates globally where markets are open for trading. *Marquee* provides institutional investors with market intelligence, risk analytics, proprietary datasets and trade execution across multiple asset classes.

Our businesses are supported by our Global Investment Research business, which, as of December 2024, provided fundamental research on approximately 3,000 companies worldwide and on approximately 50 national economies, as well as on industries, currencies and commodities.

Our activities are organized by asset class and include both "cash" and "derivative" instruments. "Cash" refers to trading the underlying instrument (such as a stock, bond or barrel of oil). "Derivative" refers to instruments that derive their value from underlying asset prices, indices, reference rates and other inputs, or a combination of these factors (such as an option, which is the right or obligation to buy or sell a certain bond, stock or other asset on a specified date in the future at a certain price, or an interest rate swap, which is the agreement to convert a fixed rate of interest into a floating rate or vice versa).

Global Banking & Markets generates revenues from the following:

Investment banking fees. We provide advisory and underwriting services and help companies raise capital to strengthen and grow their businesses.

Investment banking fees includes the following:

- **Advisory.** We have been a leader for many years in providing advisory services, including strategic advisory assignments with respect to mergers and acquisitions, divestitures, corporate defense activities, restructurings and spin-offs. In particular, we help clients execute large, complex transactions for which we provide multiple services, including cross-border structuring expertise. We also assist our clients in managing their asset and liability exposures and their capital.
- **Underwriting.** We help companies raise capital to fund their businesses. As a financial intermediary, our job is to match the capital of our investing clients, who aim to grow the savings of millions of people, with the needs of our public and private sector clients, who need financing to generate growth, create jobs and deliver products and services. Our underwriting activities include public offerings and private placements in both local and cross-border transactions of a wide range of securities and other financial instruments, including acquisition financing. Underwriting consists of the following:

Equity underwriting. We underwrite common stock, preferred stock, convertible securities and exchangeable securities. We regularly receive mandates for large, complex transactions and have held a leading position in worldwide public common stock offerings and worldwide initial public offerings for many years.

Debt underwriting. We originate and underwrite various types of debt instruments, including investment-grade and high-yield debt, bank and bridge loans, including in connection with acquisition financing, and emerging- and growth-market debt, which may be issued by, among others, corporate, sovereign, municipal and agency issuers. In addition, we underwrite and originate structured securities, which include mortgage-related securities and other asset-backed securities.

FICC. FICC generates revenues from intermediation and financing activities.

- **FICC intermediation.** Includes client execution activities related to making markets in both cash and derivative instruments, as detailed below.

Interest Rate Products. Government bonds (including inflation-linked securities) across maturities, other government-backed securities, and interest rate swaps, options and other derivatives.

Credit Products. Investment-grade and high-yield corporate securities, credit derivatives, exchange-traded funds (ETFs), bank and bridge loans, municipal securities, distressed debt and trade claims.

Mortgages. Commercial mortgage-related securities, loans and derivatives, residential mortgage-related securities, loans and derivatives (including U.S. government agency-issued collateralized mortgage obligations and other securities and loans), and other asset-backed securities, loans and derivatives.

Currencies. Currency options, spot/forwards and other derivatives on G-10 currencies and emerging-market products.

Commodities. Commodity derivatives and, to a lesser extent, physical commodities, involving crude oil and petroleum products, natural gas, agricultural, base, precious and other metals, electricity, including renewable power, environmental products and other commodity products.

- **FICC financing.** Includes (i) secured lending to our clients through structured credit and asset-backed lending, including warehouse loans backed by mortgages (including residential and commercial mortgage loans), corporate loans and consumer loans (including auto loans and private student loans), (ii) financing through securities purchased under agreements to resell (resale agreements) and (iii) commodity financing to clients through structured transactions.

Equities. Equities generates revenues from intermediation and financing activities.

- **Equities intermediation.** We make markets in equity securities and equity-related products, including ETFs, convertible securities, options, futures and over-the-counter (OTC) derivative instruments. As a principal, we facilitate client transactions by providing liquidity to our clients, including by transacting in large blocks of stocks or derivatives, requiring the commitment of our capital.

We also structure and make markets in derivatives on indices, industry sectors, financial measures and individual company stocks. We develop strategies and provide information about portfolio hedging and restructuring and asset allocation transactions for our clients. We also work with our clients to create specially tailored instruments to enable sophisticated investors to establish or liquidate investment positions or undertake hedging strategies. We are one of the leading participants in the trading and development of equity derivative instruments.

Our exchange-based market-making activities include making markets in stocks and ETFs, futures and options on major exchanges worldwide.

In addition, we generate commissions and fees from executing and clearing institutional client transactions on major stock, options and futures exchanges worldwide, as well as OTC transactions. We provide our clients with access to a broad spectrum of equity execution services, including electronic “low-touch” access and more complex “high-touch” execution through both traditional and electronic platforms.

- **Equities financing.** Includes prime financing, which provides financing to our clients for their securities trading activities through margin loans that are generally collateralized by securities or cash. Prime financing also includes services which involve lending securities to cover institutional clients’ short sales and borrowing securities to cover our short sales and to make deliveries into the market. We are also an active participant in broker-to-broker securities lending and third-party agency lending activities. In addition, we execute swap transactions to provide our clients with exposure to securities and indices. Financing activities also include portfolio financing, which clients can utilize to manage their investment portfolios, and other equity financing activities, including securities-based loans to individuals.

Other. We lend to corporate clients, including through relationship lending and acquisition financing. The hedges related to this lending and financing activity are also reported as part of Other. Other also includes equity and debt investing activities related to our Global Banking & Markets activities.

Asset & Wealth Management

Asset & Wealth Management provides investment services to help clients preserve and grow their financial assets and achieve their financial goals. We provide these services to our clients, both institutional and individuals, including investors who primarily access our products through a network of third-party distributors around the world.

We manage client assets across a broad range of investment strategies and asset classes, including equity, fixed income and alternative investments. Alternative investments primarily includes hedge funds, credit funds, private equity, real estate, currencies, commodities and asset allocation strategies. Our investment offerings include those managed on a fiduciary basis by our portfolio managers, as well as those managed by third-party managers. We offer our investment solutions in a variety of structures, including separately managed accounts, mutual funds, private partnerships and other commingled vehicles.

We also provide customized investment advisory solutions designed to address our clients' investment needs. These solutions begin with identifying clients' objectives and continue through portfolio construction, ongoing asset allocation and risk management and investment realization. We draw from a variety of third-party managers, as well as our proprietary offerings, to implement solutions for clients.

We also provide tailored wealth advisory services, primarily to ultra-high-net worth clients. We operate globally, serving individuals, families, family offices, and foundations and endowments. Our relationships are established directly or introduced through companies that sponsor financial wellness or financial planning programs for their employees, as well as through corporate referrals.

We offer personalized financial planning to individuals and also provide customized investment advisory solutions, and offer structuring and execution capabilities in securities and derivative products across all major global markets. In addition, we offer clients a full range of private banking services, including a variety of deposit alternatives and loans that our clients use to finance investments in both financial and nonfinancial assets, bridge cash flow timing gaps or provide liquidity and flexibility for other needs. We also raise deposits from consumers through *Marcus by Goldman Sachs* (Marcus).

We invest alongside our clients that invest in investment funds that we raise or manage. We also have investments in alternative assets across a range of asset classes. Our investing activities, which are typically longer-term, include investments in corporate equity, credit, real estate and infrastructure assets. See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Results of Operations — Asset & Wealth Management" in Part II, Item 7 of this Form 10-K for information about our targets to reduce our historical principal investments.

Asset & Wealth Management generates revenues from the following:

- **Management and other fees.** We receive fees related to managing assets for institutional and individual clients, providing investing and wealth advisory solutions, providing financial planning and counseling services, and executing brokerage transactions for wealth management clients. The vast majority of revenues in management and other fees consists of asset-based fees on client assets that we manage. The fees that we charge vary by asset class, client channel and the types of services provided, and are affected by investment performance, as well as asset inflows and redemptions.
- **Incentive fees.** In certain circumstances, we also receive incentive fees based on a percentage of a fund's or a separately managed account's return, or when the return exceeds a specified benchmark or other performance targets. Such fees include overrides, which consist of the increased share of the income and gains derived primarily from our private equity and credit funds when the return on a fund's investments over the life of the fund exceeds certain threshold returns.
- **Private banking and lending.** Our private banking and lending activities include issuing loans to our wealth management clients. Such loans are generally secured by commercial and residential real estate, securities or other assets. We also raise deposits from wealth management clients, including through Marcus. Private banking and lending revenues include net interest income allocated to deposits and net interest income earned on loans to individual clients.
- **Equity investments.** Includes investing activities related to our asset management activities primarily related to public and private equity investments in corporate, real estate and infrastructure assets. We also make investments through consolidated investment entities, substantially all of which are engaged in real estate investment activities. In addition, we make investments in connection with our activities to satisfy requirements under the Community Reinvestment Act (CRA), primarily through our Urban Investment Group.
- **Debt investments.** Includes lending activities related to our asset management activities, including investing in corporate debt, lending to middle-market clients, and providing financing for real estate and other assets. These activities include investments in mezzanine debt, senior debt and distressed debt securities.

Platform Solutions

Platform Solutions includes our consumer platforms and transaction banking and other.

Platform Solutions generates revenues from the following:

Consumer platforms. Our Consumer platforms business issues credit cards, and raises deposits from Apple Card customers. Consumer platforms revenues primarily includes net interest income earned on credit card lending activities. During 2024, we entered into an agreement to transition the General Motors (GM) credit card program to another issuer. The transition is expected to be completed in the third quarter of 2025.

Transaction banking and other. We provide transaction banking and other services, such as deposit-taking, payment solutions and other cash management services, for corporate and institutional clients. Transaction banking revenues include net interest income attributed to transaction banking deposits.

During 2023 and 2024, we narrowed our focus with respect to consumer-related activities by taking several actions. See “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Regulatory and Other Matters — Other Matters — Narrowing our Focus on Consumer-Related Activities” for further information.

Business Continuity and Information Security

Business continuity and information security, including cybersecurity, are high priorities for us. Their importance has been highlighted by (i) numerous highly publicized events in recent years, including cyber attacks against financial institutions, governmental agencies, large consumer-based companies, software and information technology service providers and other organizations, some of which have resulted in the unauthorized access to or disclosure of personal information and other sensitive or confidential information, the theft and destruction of corporate information, requests for ransom payments, and disruptions to organizations’ operations, (ii) extreme weather events and (iii) the COVID-19 pandemic. See “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Cybersecurity Risk Management” in Part II, Item 7 of this Form 10-K for further information about cybersecurity.

Our Business Continuity & Technology Resilience Program has been developed to provide reasonable assurance of business continuity in the event of disruptions at our critical facilities or of our systems, and to comply with regulatory requirements, including those of FINRA. Because we are a BHC, our Business Continuity & Technology Resilience Program is also subject to review by the FRB. The key elements of the program are crisis management, business continuity, technology resilience, business recovery, assurance and verification, and process improvement. In the area of information security, we have developed and implemented a framework of principles, policies and technology designed to protect the information provided to us by our clients and our own information from cyber attacks and other misappropriation, corruption or loss. Safeguards are designed to maintain the confidentiality, integrity and availability of information.

Human Capital Management

Our people are our greatest asset. We believe that a major strength and principal reason for our success is the quality, dedication, determination and collaboration of our people, which enables us to serve our clients, generate long-term value for our shareholders and contribute to the broader community.

Our Workforce

Our goal is to attract, retain, and promote an exceptionally skilled workforce. We invest heavily in developing and supporting our people throughout their careers, and we strive to maintain a work environment that fosters professionalism, excellence, high standards of business ethics, teamwork and cooperation among our employees worldwide.

We believe that the diversity of our workforce, including diversity of perspectives, enhances our performance-based culture and is critical to our commercial success. We previously set forth five-year aspirational hiring and representation goals which expire in 2025, and we are proud of the progress we have made. We remain focused on the importance of attracting and retaining diverse exceptional talent. In that connection, we will continue to develop programs consistent with our fundamental commitment to inclusive merit-based promotion and in compliance with the law.

Talent Development and Retention

We seek to help our people achieve their full potential by investing in them and supporting a culture of continuous development. Our goals are to maximize individual capabilities, increase commercial effectiveness and innovation, reinforce our culture, expand professional opportunities, and help our people contribute positively to their communities. As of December 2024, the average tenure of the members of our Management Committee was approximately 24 years, and that of all of our employees was approximately 6 years. In addition, more than 40% of our partners were campus hires.

Instilling our culture in all employees is a continuous process, in which training plays an important part. We offer our employees the opportunity to participate in ongoing educational offerings and periodic seminars facilitated by our Learning & Engagement team. To accelerate their integration into the firm and our culture, new hires have the opportunity to receive training as soon as they start working via orientation programs that emphasize culture and networking, and nearly all employees participate in at least one training event each year. For our more senior employees, we provide guidance and training on how to manage people and projects effectively, exhibit strong leadership and exemplify our culture. We are also focused on developing a high performing, diverse leadership pipeline and career planning for our next generation of leaders. We maintain a variety of programs aimed at employees' professional growth and leadership development. For example, we are focused on ensuring that vice presidents have the necessary coaching, sponsorship and advocacy to support their career trajectories and strengthen their leadership platforms. Many other career development initiatives are aimed at fostering talent at the analyst and associate levels. We also work closely with our leadership teams to promote diversity and inclusion. Our global and regional Inclusion Networks and Interest Forums are open to all of our professionals to promote and advance these goals.

Enhancing our people's experience of internal mobility is a key focus, as we believe that this will inspire employees, help retain top talent and create diverse experiences to build future leaders.

Another important part of instilling our culture is our employee performance review process. Employees are reviewed by supervisors, co-workers and employees whom they supervise in a 360-degree review process that is integral to our team approach and includes an evaluation of an employee's performance. Our approach to evaluating employee performance centers on providing robust, timely and actionable feedback that facilitates professional development. We have directed our managers, as leaders at the firm, to take an active coaching role with their teams. We also engage in "The Three Conversations at GS" through which managers establish goals with their team members at the start of the year, check in mid-year on progress and then close out the year with a conversation on performance against goals.

We believe that our people value opportunities to contribute to their communities and that these opportunities enhance their job satisfaction. We also believe that being able to volunteer together with colleagues and support community organizations through completing local service projects strengthens our people's bond with us. Community TeamWorks, our signature volunteering initiative, enables our people to participate in high-impact, team-based volunteer opportunities, including projects coordinated with hundreds of nonprofit partner organizations worldwide. During 2024, our people volunteered approximately 103,000 hours of service globally through Community TeamWorks, with approximately 19,000 employees partnering with 660 nonprofit organizations on approximately 1,400 community projects.

Wellness

We recognize that for our people to be successful in the workplace they need support in their personal, as well as their professional, lives and that is why our wellness framework is designed to promote health and fitness, resilience, and work-life balance. We provide a number of policies for our employees that support taking time away from the office when needed, including a minimum of 20 weeks of parental leave and up to four weeks of family care leave in order to assist with the care of family members with a serious health condition, death of an immediate family member or miscarriage, in addition to bereavement leave. We allow managing directors to take time off without a fixed vacation day entitlement, and have also set a minimum annual expected vacation usage of 15 days for all employees. For longer-tenured employees, we offer an unpaid sabbatical leave.

We also continue to advance our resilience programs, offering our people a range of counseling, coaching, medical advisory and personal wellness services. We have introduced and globally scaled the internationally recognized Mental Health First Aid certification to our people. As of December 2024, over 1,300 employees were certified across the firm, surpassing our goal to train 1,000 Mental Health First Aiders by the end of 2024. We have evolved and strengthened virtual offerings to enhance access to support, with the aim of maintaining the physical and mental well-being of our people, and enhancing their effectiveness and productivity. We also launched a manager mental health training to help support leaders globally, achieving a completion rate of approximately 85%.

We understand the crucial role caregiving plays in the lives of our employees. To help enable employees to better balance their roles at work and their responsibilities at home, we offer a variety of family-centered benefits, including adoption and surrogacy stipends, as well as adult or childcare options to help our people navigate caregiving across various life stages. Employees also have access to our family care coordination services for comprehensive support and coaching across the caregiving continuum—from their transition to new parenthood to care for adult family members.

In addition, to support the financial wellness of our employees, we offer a variety of resources that help them manage their personal financial health and decision-making, including financial education information sessions, live and on-demand webinars, articles and interactive digital tools.

Global Reach and Strategic Locations

As a firm with a global client base, we take a strategic approach to attracting, developing and managing a global workforce. Our clients are located worldwide and we are an active participant in financial markets around the world. As of December 2024, we had headcount of 46,500, offices in over 40 countries, and 50% of our headcount was based in the Americas, 20% in Europe, Middle East and Africa (EMEA) and 30% in Asia. Our employees come from over 190 countries and speak more than 170 languages as of December 2024.

In addition to maintaining offices in major financial centers around the world, we have established key strategic locations, including in Bengaluru, Salt Lake City, Dallas, Singapore, Warsaw, Birmingham and Hyderabad. We continue to evaluate the expanded use of strategic locations, including cities in which we do not currently have a presence.

As of December 2024, 43% of our employees were working in strategic locations. We believe our investment in these strategic locations enables us to build centers of excellence around specific capabilities that support our business initiatives.

Sustainability

We have a long-standing commitment to sustainability. Our two priorities in this area are helping clients across industries decarbonize their businesses to support their transitions to a low-carbon economy (Climate Transition) and to advance solutions that expand access, increase affordability, and drive outcomes to support sustainable economic growth (Inclusive Growth). Our strategy is to advance these two priorities through our work with our clients, and with strategic partners whose strengths and areas of focus complement our own, as well as through our supply chain.

Our Sustainable Finance Group serves as the centralized group that drives our sustainability strategy and related efforts across our firm, including the commercial approach alongside our businesses, to advance Climate Transition and Inclusive Growth. Our sustainable finance-related efforts continue to evolve. For example, within Global Banking & Markets, we established the Sustainable Banking Group, a group focused on delivering analysis, advice, and capital solutions for clients focused on their sustainability objectives. Within Asset & Wealth Management, we provide clients sustainability-related capabilities, across public and private markets and open architecture, proprietary and third-party products, portfolio strategy and implementation, and specialist sustainability teams and strategies.

Our activities relating to sustainability present both financial and nonfinancial risks, and we have processes for managing these risks, similar to the other risks we face. We have integrated oversight of climate-related risks into our risk management governance structure, from senior management to our Board and its committees, including the Risk and Public Responsibilities committees. The Risk Committee of the Board oversees firmwide financial and nonfinancial risks, which include climate risk, and, as part of its oversight, receives updates on our risk management approach to climate risk. The Public Responsibilities Committee of the Board assists the Board in its oversight of our firmwide sustainability strategy and sustainability issues affecting us, including with respect to climate change. As part of its oversight, the Public Responsibilities Committee receives periodic updates on our sustainability strategy and disclosures, and also periodically reviews our governance and related policies and processes for climate and other sustainability-related matters. We have also implemented the Environmental & Social Due Diligence Guidelines to guide our overall risk-based due diligence approach when evaluating relevant transactions for environmental and social risks and impacts.

Relevant employees also receive training with respect to environmental and social risks, including for sectors and industries that we believe have higher potential for these risks. See “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Other Risk Management — Climate-Related and Environmental Risk Management” in Part II, Item 7 of this Form 10-K for further information about our climate-related and environmental risk management.

As a leading financial institution, we acknowledge the importance of Climate Transition and Inclusive Growth for our business. We have completed sustainability debt issuances, which align with our sustainable finance framework and fund a range of on-balance sheet sustainable finance activity. We believe we can advance sustainability by partnering with our clients across our businesses, including by developing new sustainability-linked financing solutions, offering strategic advice, or co-investing alongside our clients in energy companies. We have announced a target to deploy \$750 billion in sustainable financing, investing and advisory activity by the beginning of 2030. As of December 2024, we achieved over 80% of that goal, with the majority dedicated to Climate Transition.

With respect to Climate Transition, we have announced our goal to align our financing activities with a net-zero-by-2050 pathway. We have an initial set of 2030 targets for our energy, power and auto manufacturing portfolios, three sectors where we see an opportunity to proactively engage our clients and investors, deploy capital required for transition, and invest in new commercial solutions to facilitate decarbonization in the real economy. Carbon neutrality is also a priority for the operation of our firm. We have been carbon neutral in our operations and business travel since 2015 through a combination of emissions reduction efforts and the procurement of Energy Attribute Certificates and third-party-verified carbon offsets. We have since expanded our operational carbon commitment to include our supply chain, prioritizing emissions reductions.

In addition to Climate Transition, our approach to sustainability also centers on Inclusive Growth where we seek to help drive solutions that expand access, increase affordability, and support outcomes to advance sustainable economic growth. Commercial solutions that seek to support Inclusive Growth include, among others, those of our Urban Investment Group and our Sustainable Investing Group. We also seek to support Inclusive Growth through our corporate engagement programs, such as *10,000 Small Businesses*, *10,000 Women* and *One Million Black Women*. These efforts are further strengthened by strategic partnerships that we have established in areas where we have identified gaps or believe we are able to drive even greater impact through collaboration. We believe our ability to achieve our sustainability objectives is critically dependent on the strengths and talents of our people. See “Business — Human Capital Management” for information about our human capital management programs and policies.

Competition

The financial services industry and all of our businesses are intensely competitive, and we expect them to remain so. Our competitors provide investment banking, market-making and asset management services, private banking and lending, commercial lending, credit cards, transaction banking, deposit-taking and other banking products and services, and make investments in securities, commodities, derivatives, real estate, loans and other financial assets. Our competitors include brokers and dealers, investment banking firms, commercial banks, credit card issuers, insurance companies, investment advisers, mutual funds, hedge funds, private equity funds, private credit funds, merchant banks and financial technology and other internet-based companies. Some of our competitors operate globally and others regionally, and we compete based on a number of factors, including transaction execution, client experience, products and services, innovation, reputation and price.

We have faced, and expect to continue to face, pressure to retain market share by committing capital to businesses or transactions on terms that offer returns that may not be commensurate with their risks. In particular, corporate clients seek such commitments (such as agreements to participate in their loan facilities) from financial services firms in connection with investment banking and other assignments.

Consolidation and convergence have significantly increased the capital base and geographic reach of some of our competitors and have also hastened the globalization of the securities and other financial services markets. As a result, we have had to commit capital to support our international operations and to execute large global transactions. To capitalize on some of our most significant opportunities, we will have to compete successfully with financial institutions that are larger and have more capital and that may have a stronger local presence and longer operating history outside the U.S.

We also compete with smaller institutions that offer more targeted services, such as independent advisory firms. Some clients may perceive these firms to be less susceptible to potential conflicts of interest than we are, and, as described below, our ability to effectively compete with them could be affected by regulations and limitations on activities that apply to us but may not apply to them.

A number of our businesses are subject to intense price competition. Efforts by our competitors to gain market share have resulted in pricing pressure in our investment banking, market-making, consumer, wealth management and asset management businesses. For example, the increasing volume of trades executed electronically, through the internet and through alternative trading systems, has increased the pressure on trading commissions, in that commissions for electronic trading are generally lower than those for non-electronic trading. It appears that this trend toward low-commission trading will continue. Price competition has also led to compression in the difference between the price at which a market participant is willing to sell an instrument and the price at which another market participant is willing to buy it (i.e., bid/offer spread), which has affected our market-making businesses. The increasing prevalence of passive investment strategies that typically have lower fees than other strategies we offer has affected the competitive and pricing dynamics for our asset management products and services. In addition, we believe that we will continue to experience competitive pressures in these and other areas in the future as some of our competitors seek to obtain market share by further reducing prices, and as we enter into or expand our presence in markets that rely more heavily on electronic trading and execution. We and other banks also compete for deposits on the basis of the rates we offer. Higher short-term interest rates in recent years have resulted in and may continue to result in more intense competition in deposit pricing, as well as competition from non-deposit financial products.

We also compete on the basis of the types of financial products and client experiences that we and our competitors offer. In some circumstances, our competitors may offer financial products that we do not offer and that our clients may prefer, including cryptocurrencies and other digital assets that we cannot or may choose not to provide. Our competitors may also develop technology platforms that provide a better client experience.

The provisions of the U.S. Dodd-Frank Wall Street Reform and Consumer Protection Act (Dodd-Frank Act), the requirements promulgated by the Basel Committee on Banking Supervision (Basel Committee) and other financial regulations could affect our competitive position to the extent that limitations on activities, increased fees and compliance costs or other regulatory requirements do not apply, or do not apply equally, to all of our competitors or are not implemented uniformly across different jurisdictions. For example, the provisions of the Dodd-Frank Act that prohibit proprietary trading and restrict investments in certain hedge and private equity funds differentiate between U.S.-based and non-U.S.-based banking organizations and give non-U.S.-based banking organizations greater flexibility to trade outside of the U.S. and to form and invest in funds outside the U.S.

Likewise, the obligations with respect to derivative transactions under Title VII of the Dodd-Frank Act depend, in part, on the location of the counterparties to the transaction. The impact of regulatory developments on our competitive position has depended and will continue to depend to a large extent on the manner in which the required rulemaking and regulatory guidance evolve, the extent of international convergence, and the development of market practice and structures under the evolving regulatory regimes, as described further in “Regulation” below.

We also face intense competition in attracting and retaining qualified employees. Our ability to continue to compete effectively has depended and will continue to depend upon our ability to attract new employees, retain and motivate our existing employees and to continue to compensate employees competitively amid intense public and regulatory scrutiny on the compensation practices of large financial institutions, including in jurisdictions such as New York State where we are required to publish certain compensation information as part of the employee hiring process. Our pay practices and those of certain of our competitors are subject to review by, and the standards of, the FRB and other regulators inside and outside the U.S., including the Prudential Regulation Authority (PRA) and the Financial Conduct Authority (FCA) in the U.K. We also compete for employees with institutions whose pay practices are not subject to regulatory oversight. See “Regulation — Compensation Practices” and “Risk Factors — Competition — Our businesses would be adversely affected if we are unable to hire and retain qualified employees” in Part I, Item 1A of this Form 10-K for further information about such regulation.

Regulation

As a participant in the global financial services industry, we are subject to extensive regulation and supervision worldwide. The regulatory regimes applicable to our operations have been, and continue to be, subject to significant changes.

New regulations have been adopted or are being considered by regulators and policy makers worldwide, as described below. The impacts of any changes to the regulations affecting our businesses, including as a result of the proposals described below, are uncertain and will not be known until such changes are finalized and market practices and structures develop under the revised regulations.

Group Inc. is a BHC under the U.S. Bank Holding Company Act of 1956 (BHC Act) and an FHC under amendments to the BHC Act effected by the U.S. Gramm-Leach-Bliley Act of 1999 (GLB Act), and is subject to supervision and examination by the FRB, which is our primary regulator.

Under the system of “functional regulation” established under the GLB Act, the primary regulators of our U.S. non-bank subsidiaries directly regulate the activities of those subsidiaries, with the FRB exercising a supervisory role. Such “functionally regulated” subsidiaries include broker-dealers and security-based swap dealers registered with the SEC, such as our principal U.S. broker-dealer, entities registered with or regulated by the CFTC with respect to futures-related and swaps-related activities and investment advisers registered with the SEC with respect to their investment advisory activities.

Our principal subsidiaries operating in the U.S. include GS Bank USA, Goldman Sachs & Co., LLC (GS&Co.), J. Aron & Company LLC (J. Aron) and Goldman Sachs Asset Management, L.P.

GS Bank USA is our principal U.S. bank subsidiary and is supervised and regulated by the FRB, the FDIC, the New York State Department of Financial Services (NYDFS) and the Consumer Financial Protection Bureau (CFPB). GS Bank USA also has a London branch, which is regulated by the FCA and PRA. We conduct a number of our activities partially or entirely through GS Bank USA and its subsidiaries, including: corporate loans (including leveraged lending); securities-based and collateralized loans; credit card loans; residential mortgages; transaction banking; deposit-taking; interest rate, credit, currency and other derivatives; and agency lending.

GS&Co. is our principal U.S. broker-dealer and is registered as a broker-dealer, a security-based swap dealer, a municipal advisor and an investment adviser with the SEC and as a broker-dealer in all 50 states and the District of Columbia. U.S. self-regulatory organizations, such as FINRA and the NYSE, have adopted rules that apply to broker-dealers, such as GS&Co.

Our principal subsidiaries operating in Europe include: Goldman Sachs International (GSI), Goldman Sachs International Bank (GSIB), Goldman Sachs Asset Management International (GSAMI), Goldman Sachs Bank Europe SE (GSBE), Goldman Sachs Asset Management B.V., and Goldman Sachs Paris Inc. et Cie (GSPIC).

Our E.U. subsidiaries are subject to various E.U. regulations, as well as national laws, including those implementing European directives. GSBE is directly supervised by the European Central Bank (ECB) and additionally by BaFin and Deutsche Bundesbank in the context of the E.U. Single Supervisory Mechanism. GSBE engages in certain activities primarily in the E.U., including underwriting and market making in debt and equity securities and derivatives, investment, asset and wealth management services, deposit-taking, lending (including securities lending), and financial advisory services. GSBE is also registered with the CFTC as a swap dealer and with the SEC as a security-based swap dealer and as a primary dealer for government bonds issued by E.U. sovereigns. Like our other foreign bank subsidiaries, GSBE is subject to limits on the nature and scope of its activities under the FRB's Regulation K, including limits on its underwriting and market making in equity securities based on GSBE's and/or GS Bank USA's capital.

GSPIC is an investment firm under the French Prudential Supervision and Resolution Authority and the French Financial Markets Authority. GSPIC's activities include certain activities that GSBE is prevented from undertaking. GSPIC is subject to the E.U. Investment Firm Regulation, the prudential regime for E.U. investment firms.

GSI is a U.K. broker-dealer and a designated investment firm, and GSIB is a U.K. bank. Both GSI and GSIB are regulated by the PRA and the FCA. As a designated investment firm, GSI is subject to prudential requirements similar to those applicable to banks, including capital and liquidity requirements. GSI provides broker-dealer services in and from the U.K. and is registered with the CFTC as a swap dealer and with the SEC as a security-based swap dealer. GSIB engages in lending (including securities lending) and deposit-taking activities (including by taking retail deposits) and is a primary dealer for U.K. government bonds. GSI and GSIB maintain branches outside of the U.K. and are subject to the laws and regulations of the jurisdictions where they are located.

Our principal subsidiary operating in Asia is Goldman Sachs Japan Co., Ltd. (GSJCL). GSJCL is our regulated Japanese broker-dealer subsidiary and is regulated by Japan's Financial Services Agency, the Tokyo Stock Exchange, the Bank of Japan and the Ministry of Finance, among others.

Banking Supervision and Regulation

The Basel Committee is the primary global standard setter for prudential bank regulation. However, the Basel Committee's standards do not become effective in a jurisdiction until the relevant regulators have adopted rules to implement its standards. The implications of Basel Committee standards and related regulations for our businesses depend to a large extent on their implementation by the relevant regulators globally, and the market practices and structures that develop.

Capital and Liquidity Requirements. We and GS Bank USA are subject to risk-based regulatory capital and leverage requirements that are calculated in accordance with the regulations of the FRB (Capital Framework). The Capital Framework is largely based on the Basel Committee's framework for strengthening the regulation, supervision and risk management of banks (Basel III) and also implements certain provisions of the Dodd-Frank Act. Under the U.S. federal bank regulatory agencies' tailoring framework, we and GS Bank USA are subject to "Category I" standards because we have been designated as a global systemically important bank (G-SIB). Accordingly, we and GS Bank USA are "Advanced approach" banking organizations. Under the Capital Framework, we and GS Bank USA must meet specific regulatory capital requirements that involve quantitative measures of assets, liabilities and certain off-balance sheet items. The sufficiency of our capital levels is also subject to qualitative judgments by regulators. We and GS Bank USA are also subject to liquidity requirements established by the U.S. federal bank regulatory agencies.

GSBE is subject to capital and liquidity requirements prescribed in the E.U. Capital Requirements Regulation, as amended (CRR), and the E.U. Capital Requirements Directive, as amended (CRD), which are largely based on Basel III.

GSI and GSIB are subject to the U.K. capital and liquidity frameworks prescribed in the PRA Rulebook and the U.K. Capital Requirements Regulation, which are also largely based on Basel III and are generally aligned with the E.U. capital and liquidity frameworks.

Risk-Based Capital Ratios. As Advanced approach banking organizations, we and GS Bank USA calculate risk-based capital ratios in accordance with both the Standardized and Advanced Capital Rules. Both the Standardized and Advanced Capital Rules include minimum risk-based capital requirements and additional capital conservation buffer requirements that must be satisfied solely with Common Equity Tier 1 (CET1) capital. Failure to satisfy a buffer requirement in full would result in constraints on capital distributions and discretionary executive compensation. The severity of the constraints would depend on the amount of the shortfall and the organization's "eligible retained income," defined as the greater of (i) net income for the four preceding quarters, net of distributions and associated tax effects not reflected in net income; and (ii) the average of net income over the preceding four quarters. For Group Inc., the capital conservation buffer requirements consist of a 2.5% buffer (under the Advanced Capital Rules), a stress capital buffer (SCB) (under the Standardized Capital Rules), and both a countercyclical buffer and the G-SIB surcharge (under both Capital Rules). For GS Bank USA, the capital conservation buffer requirements consist of a 2.5% buffer and the countercyclical capital buffer.

In 2023, the FRB issued a proposal to implement a revised G-SIB assessment methodology and to revise certain systemic indicators to be based on daily or monthly average values during each year, instead of year-end values.

The SCB is based on the results of the Federal Reserve's supervisory stress tests and our planned common stock dividends and can change significantly over time based on the results of the annual supervisory stress tests. See "Stress Tests and Capital Planning" below. The countercyclical capital buffer is designed to counteract systemic vulnerabilities and currently applies only to banking organizations subject to Category I, II or III standards, including us and GS Bank USA. Several other national supervisors also require countercyclical capital buffers. The G-SIB surcharge and countercyclical capital buffer applicable to us may change in the future, including due to additional guidance from our regulators and/or positional changes. As a result, the minimum capital ratios to which we are subject are likely to change over time.

The capital requirements applicable to GSBE, GSI and GSIB include both minimum requirements and buffers. See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Capital Management and Regulatory Capital" in Part II, Item 7 of this Form 10-K and Note 20 to the consolidated financial statements in Part II, Item 8 of this Form 10-K for information about our capital ratios and those of GS Bank USA, GSBE, GSI and GSIB.

The Basel Committee standards include guidelines for calculating incremental capital ratio requirements for banking institutions that are systemically significant from a domestic but not global perspective (D-SIBs). Depending on how these guidelines are implemented by national regulators, they may apply to certain subsidiaries of G-SIBs. These guidelines are in addition to the framework for G-SIBs, but are more principles-based. The U.S. federal bank regulatory agencies have not designated any D-SIBs. The CRD and CRR provide that institutions that are systemically important at the E.U. or member state level, known as other systemically important institutions (O-SIIs), may be subject to additional capital ratio requirements, according to their degree of systemic importance (O-SII buffers). BaFin has identified GSBE as an O-SII in Germany and set an O-SII buffer.

In the U.K., the PRA has identified Goldman Sachs Group UK Limited (GSG UK), the parent company of GSI and GSIB, as an O-SII but has not applied an O-SII buffer.

The Basel Committee has finalized revisions to the Basel III Capital Requirements (Basel III Revisions), and in 2023, the U.S. federal bank regulatory agencies proposed a rule implementing the Basel III Revisions, including the Fundamental Review of the Trading Book (FRTB). The FRTB, among other things, revises the standardized and internal model-based approaches used to calculate market risk requirements and clarifies the scope of positions subject to market risk capital requirements.

The U.S. proposal has a three-year transition period for the calculation of Expanded Risk-Based approach risk-weighted assets (RWAs). The proposal includes the replacement of the Advanced approach with an Expanded Risk-Based approach, which eliminates the use of internal models to calculate RWAs for credit and operational risk. The proposal incorporates the application of the SCB requirements in the Expanded Risk-Based approach. The credit risk component of the Expanded Risk-Based approach would include new risk weights for many counterparty and exposure types, a revised collateral haircut approach for certain collateralized transactions and additional restrictions for recognizing collateral in certain securities financing transactions. Under the proposed rules, the RWAs for operational risk would be calculated primarily based on revenues and historical losses. In addition, the proposal introduces the FRTB, which would replace the market risk rule for both the Standardized and Expanded Risk-Based approaches and introduce a new credit valuation adjustment (CVA) risk RWA calculation for the Expanded Risk-Based approach. The FRB has indicated that it expects to work with the other U.S. federal bank regulatory agencies in 2025 on a revised proposal.

In 2024, the E.U. adopted rules to implement the Basel III Revisions through amendments to the CRR and the CRD, referred to as CRR III and CRD VI. The amendments include the FRTB rules, revised rules for credit risk capital, a new standardized approach for operational risk and CVA risk capital and a floor on internally modeled capital requirements at a percentage of the capital requirements under the standardized approach, commonly known as the “output floor.” Substantial parts of these rules became effective in January 2025, though certain provisions applied beginning in July 2024. The FRTB rules are currently expected to apply from January 2026.

The PRA issued near final rules with a proposed effective date of January 1, 2027, implementing Basel III Revisions, including new rules covering the FRTB, credit risk, counterparty credit risk, CVA risk and operational risk for the U.K. Under the PRA near final rules, our U.K. subsidiaries are not expected to be subject to a floor on internally modeled capital requirements.

The Basel Committee has published an updated securitization framework and a revised G-SIB assessment methodology. The U.S. federal bank regulatory agencies’ July 2023 proposal would implement the updated securitization framework. The updated securitization framework has been implemented in the E.U. and U.K.

The Basel Committee has also published a final standard on the prudential treatment of cryptoasset exposures. The Basel Committee contemplates that national regulators will have incorporated the standard into local capital requirements by January 1, 2026. U.S. federal bank regulatory agencies and E.U. and U.K. authorities have not yet proposed rules implementing the standards.

Leverage Ratios. Under the Capital Framework, we and GS Bank USA are subject to Tier 1 leverage ratios and supplementary leverage ratios (SLRs) established by the FRB. As a G-SIB, the SLR requirements applicable to us include both a minimum requirement and a buffer requirement, which operates in the same manner as the risk-based buffer requirements described above.

GSBE and certain of our U.K. entities are also subject to requirements relating to leverage ratios, which are generally based on the Basel Committee leverage ratio standards.

See “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Capital Management and Regulatory Capital” in Part II, Item 7 of this Form 10-K and Note 20 to the consolidated financial statements in Part II, Item 8 of this Form 10-K for information about our and GS Bank USA’s Tier 1 leverage ratios and SLRs, and GSI’s leverage ratio.

Liquidity Ratios. The Basel Committee’s framework for liquidity risk measurement, standards and monitoring requires banking organizations to measure their liquidity against two specific liquidity tests: the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR).

The LCR rule issued by the U.S. federal bank regulatory agencies and applicable to both us and GS Bank USA is generally consistent with the Basel Committee’s framework and is designed to ensure that a banking organization maintains an adequate level of unencumbered, high-quality liquid assets equal to or greater than the expected net cash outflows under an acute short-term liquidity stress scenario. We and GS Bank USA are required to maintain a minimum LCR of 100%.

GSBE is subject to the LCR rule approved by the European Parliament and Council, and GSI and GSIB are subject to the U.K. regulatory authorities’ LCR rules, which are generally consistent with the Basel Committee’s framework.

The NSFR is designed to promote medium- and long-term stable funding of the assets and off-balance sheet activities of banking organizations over a one-year time horizon. The Basel Committee’s NSFR framework requires banking organizations to maintain a minimum NSFR of 100%. We and GS Bank USA are subject to the U.S. NSFR rule.

The CRR implements the NSFR for certain E.U. financial institutions, including GSBE. The NSFR requirement implemented in the U.K. is applicable to both GSI and GSIB.

The FRB’s enhanced prudential standards require BHCs with \$100 billion or more in total consolidated assets to comply with enhanced liquidity and overall risk management standards, which include maintaining a level of highly liquid assets based on projected funding needs for 30 days, and increased involvement by boards of directors in liquidity and overall risk management. Although the liquidity requirement under these rules has some similarities to the LCR, it is a separate requirement. GSBE also has its own liquidity planning process, which incorporates internally designed stress tests and those required under German regulatory requirements and the ECB Guide to Internal Liquidity Adequacy Assessment Process (ILAAP). GSI and GSIB have their own liquidity planning processes, which incorporate internally designed stress tests developed in accordance with the guidelines of the PRA’s ILAAP.

See “Available Information” below and “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Overview and Structure of Risk Management” and “— Liquidity Risk Management — Liquidity Regulatory Framework” in Part II, Item 7 of this Form 10-K for information about the LCR and NSFR, as well as our risk management practices and liquidity.

Stress Tests and Capital Planning. The FRB's Comprehensive Capital Analysis and Review (CCAR) is designed to ensure that large BHCs, including us, have sufficient capital to permit continued operations during times of economic and financial stress. As required by the FRB, we perform an annual capital stress test and incorporate the results into an annual capital plan, which we submit to the FRB for review. See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Capital Management and Regulatory Capital — Capital Management — Capital Planning and Stress Testing Process" in Part II, Item 7 of this Form 10-K for further information about our annual capital plan. As described in "Available Information" below, summary results of the annual stress test are published on our website.

As part of the CCAR process, the FRB evaluates our plan to make capital distributions across a range of macroeconomic and company-specific assumptions, based on our and the FRB's own stress tests. In December 2024 and February 2025, the FRB indicated that it intends to propose significant changes to the stress test framework in 2025 and, for the 2025 stress test, take steps to reduce the volatility of results and to begin to improve model transparency. Also in December 2024, trade groups representing major U.S. banks, including us, filed a lawsuit against the FRB concerning certain inadequacies with its stress testing process.

Under the FRB's rule applicable to BHCs with \$100 billion or more in total consolidated assets, including us, the SCB applies to the Standardized approach capital requirements. The SCB reflects stressed losses estimated under the supervisory severely adverse scenario of the CCAR stress tests, as calculated by the FRB, and includes four quarters of planned common stock dividends. The SCB, which is subject to a 2.5% floor, is generally effective on October 1 of each year and remains in effect until October 1 of the following year, unless it is reset in connection with the resubmission of a capital plan. See "Available Information" below and "Management's Discussion and Analysis of Financial Condition and Results of Operations — Capital Management and Regulatory Capital" in Part II, Item 7 of this Form 10-K for information about our SCB requirement.

The SCB rule requires a BHC to receive the FRB's approval for any dividend, stock repurchase or other capital distribution, other than a capital distribution on a newly issued capital instrument, if the BHC is required to resubmit its capital plan, which may occur if the BHC determines there has been or will be a "material change" in its risk profile, financial condition or corporate structure since the plan was last submitted, or if the FRB directs the BHC to revise and resubmit its capital plan.

U.S. depository institutions with total consolidated assets of \$250 billion or more that are subsidiaries of U.S. G-SIBs, such as GS Bank USA, are required to submit annual company-run stress test results to the FRB. GSBE also has its own capital and stress testing process, which incorporates internally designed stress tests and those required under German regulatory requirements and the ECB Guide to Internal Capital Adequacy Assessment Process (ICAAP). In addition, GSI and GSIB have their own capital planning and stress testing processes, which incorporate internally designed stress tests developed in accordance with the PRA's ICAAP guidelines.

Limitations on the Payment of Dividends. U.S. federal and state laws impose limitations on the payment of dividends by U.S. depository institutions, such as GS Bank USA. In general, the amount of dividends that may be paid by GS Bank USA is limited to the lesser of the amounts calculated under a recent earnings test and an undivided profits test. Under the recent earnings test, a dividend may not be paid if the total of all dividends declared by the entity in any calendar year is in excess of the current year's net income combined with the retained net income of the two preceding years, unless the entity obtains regulatory approval. Under the undivided profits test, a dividend may not be paid in excess of the entity's undivided profits (generally, accumulated net profits that have not been paid out as dividends or transferred to surplus), unless the entity receives regulatory and stockholder approval.

The applicable U.S. banking regulators have authority to prohibit or limit the payment of dividends if, in the banking regulator's opinion, payment of a dividend would constitute an unsafe or unsound practice in light of the financial condition of the banking organization.

Source of Strength. The Dodd-Frank Act requires BHCs to act as a source of strength to their U.S. bank subsidiaries and to commit capital and financial resources to support those subsidiaries. This support may be required by the FRB at times when BHCs might otherwise determine not to provide it. Capital loans by a BHC to a U.S. subsidiary bank are subordinate in right of payment to deposits and to certain other indebtedness of the subsidiary bank. In addition, if a BHC commits to a U.S. federal bank regulatory agency that it will maintain the capital of its bank subsidiary, whether in response to the FRB's invoking its source-of-strength authority or in response to other regulatory measures, that commitment will be assumed by the bankruptcy trustee for the BHC and the bank will be entitled to priority payment in respect of that commitment, ahead of other creditors of the BHC.

Transactions Between Affiliates. Transactions between GS Bank USA or its subsidiaries, including GSBE, and Group Inc. or its other subsidiaries and affiliates are subject to restrictions under the Federal Reserve Act and regulations issued by the FRB. These laws and regulations generally limit the types and amounts of transactions (such as loans and other credit extensions, including credit exposure arising from resale agreements, securities borrowing and derivative transactions, from GS Bank USA or its subsidiaries to Group Inc. or its other subsidiaries and affiliates and purchases of assets by GS Bank USA or its subsidiaries from Group Inc. or its other subsidiaries and affiliates) that may take place and generally require those transactions, to the extent permitted, to be on market terms or better to GS Bank USA or its subsidiaries. These laws and regulations generally do not apply to transactions between GS Bank USA and its subsidiaries. Similarly, German regulatory requirements provide that certain transactions between GSBE and GS Bank USA or its other affiliates, including Group Inc., must be on market terms and are subject to special internal approval requirements. PRA rules also provide requirements for transactions between GSI and GSIB and their respective affiliates.

Resolution and Recovery Plans. We are required by the FRB and the FDIC to submit a periodic plan for our rapid and orderly resolution in the event of material financial distress or failure (resolution plan). If these regulators jointly determine that an institution has failed to remediate identified deficiencies in its resolution plan or that its resolution plan, after any permitted resubmission, is not credible or would not facilitate an orderly resolution under the U.S. Bankruptcy Code, they may jointly impose more stringent capital, leverage or liquidity requirements or restrictions on growth, activities or operations, or may jointly order the institution to divest assets or operations, in order to facilitate orderly resolution in the event of failure. The FRB and FDIC require U.S. G-SIBs to submit resolution plans every two years (alternating between submissions of full plans and targeted plans that include only select information). We submitted our 2023 resolution plan, which was a full submission, in June 2023. In June 2024, the FRB and the FDIC provided feedback on our 2023 resolution plan and identified one shortcoming, other areas for additional focus to address resolution readiness and additional information required to be included in our 2025 resolution plan. In August 2024, we submitted a description of our key actions to address the shortcoming. Our next required submission is a targeted submission by July 1, 2025. See “Risk Factors — Legal and Regulatory — The application of Group Inc.’s proposed resolution strategy could result in greater losses for Group Inc.’s security holders” in Part I, Item 1A of this Form 10-K and “Available Information” in Part I, Item 1 of this Form 10-K for further information about our resolution plan.

We are also required by the FRB to submit, on a periodic basis, a global recovery plan that outlines the steps that we could take to reduce risk, maintain sufficient liquidity and conserve capital in times of prolonged stress. Certain of our subsidiaries are also subject to similar recovery plan requirements in their local jurisdictions.

GS Bank USA is required to provide a resolution plan to the FDIC that must, among other things, demonstrate that it is adequately protected from risks arising from our other entities. GS Bank USA’s most recent resolution plan was submitted in December 2023. In June 2024, the FDIC adopted revisions to its rule that requires the submission of resolution plans by insured depository institutions (IDIs) with \$50 billion or more in total assets. These revisions modify the requirements regarding the content and timing of resolution submissions, as well as interim supplements to those submissions provided to the FDIC. IDIs with \$100 billion or more in total assets, including GS Bank USA, are required to submit full resolution plans biennially. GS Bank USA’s next required full submission is due by July 1, 2026.

The U.S. federal bank regulatory agencies have adopted rules imposing restrictions on qualified financial contracts (QFCs) entered into by G-SIBs. The rules are intended to facilitate the orderly resolution of a failed G-SIB by limiting the ability of the G-SIB to enter into a QFC unless (i) the counterparty waives certain default rights in such contract arising upon the entry of the G-SIB or one of its affiliates into resolution, (ii) the contract does not contain enumerated prohibitions on the transfer of such contract and/or any related credit enhancement, and (iii) the counterparty agrees that the contract will be subject to the special resolution regimes set forth in the Dodd-Frank Act orderly liquidation authority (OLA) and the Federal Deposit Insurance Act (FDIA), described below. GS Bank USA has achieved compliance by adhering to the International Swaps and Derivatives Association Universal Resolution Stay Protocol (ISDA Universal Protocol) and International Swaps and Derivatives Association 2018 U.S. Resolution Stay Protocol (U.S. ISDA Protocol) described below.

Certain of our other subsidiaries also adhere to these protocols. The ISDA Universal Protocol imposes a stay on certain cross-default and early termination rights within standard ISDA derivative contracts and securities financing transactions between adhering parties in the event that one of them is subject to resolution in its home jurisdiction, including a resolution under OLA or the FDIA in the U.S. The U.S. ISDA Protocol, which was based on the ISDA Universal Protocol, was created to allow market participants to comply with the final QFC rules adopted by the federal bank regulatory agencies.

The E.U. Bank Recovery and Resolution Directive (BRRD), as amended by the BRRD II, establishes a framework for the recovery and resolution of financial institutions in the E.U., such as GSBE. The BRRD provides national supervisory authorities with tools and powers to pre-emptively address potential financial crises in order to promote financial stability and minimize taxpayers' exposure to losses. The BRRD requires E.U. member states to grant certain resolution powers to national and, where relevant, E.U. resolution authorities, including the power to impose a temporary stay, and to recapitalize a failing entity by writing down its unsecured debt or converting its unsecured debt into equity. Financial institutions in the E.U. must provide that contracts governed by non-E.U. law recognize those temporary stay and bail-in powers unless doing so would be impracticable. GSBE is under the direct authority of the Single Resolution Board for resolution planning. E.U. law requires financial institutions in the E.U., including subsidiaries of non-E.U. groups, to submit recovery plans and to assist the relevant resolution authority in constructing resolution plans for the E.U. entities. GSBE's primary regulator with respect to recovery planning is the ECB, and it is also regulated by BaFin and Deutsche Bundesbank.

The U.K. Special Resolution Regime confers substantially the same powers on the Bank of England, as the U.K. resolution authority, and substantially the same requirements on U.K. financial institutions. Further, certain U.K. financial institutions, including GSI and GSIB, are required to meet the Bank of England's expectations contained in the U.K. Resolution Assessment Framework, including with respect to loss absorbency, contractual stays, operational continuity and funding in resolution. They are also required by the PRA to submit solvent wind-down plans on how they could be wound down in a stressed environment. The PRA is also the regulatory authority in the U.K. that supervises recovery planning, and GSI and GSIB are each required to submit recovery plans to the PRA.

Total Loss-Absorbing Capacity (TLAC). The FRB's TLAC rule, among other things, establishes minimum TLAC requirements and establishes minimum requirements for "eligible long-term debt" (i.e., debt that is unsecured, has a maturity of at least one year from issuance and satisfies certain additional criteria). In 2023, the FRB proposed to introduce a minimum denomination requirement for eligible long-term debt, among other changes.

The rule also prohibits a BHC that has been designated as a U.S. G-SIB from (i) guaranteeing subsidiaries' liabilities that are subject to early termination provisions if the BHC enters into an insolvency or receivership proceeding, subject to an exception for guarantees permitted by rules of the U.S. federal bank regulatory agencies imposing restrictions on QFCs; (ii) incurring liabilities guaranteed by subsidiaries; (iii) issuing short-term debt to third parties; or (iv) entering into derivatives and certain other financial contracts with external counterparties.

Additionally, the rule caps, at 5% of the value of the parent company's eligible TLAC, the amount of unsecured non-contingent third-party liabilities that are not eligible long-term debt that could rank equally with or junior to eligible long-term debt.

The CRR, the BRRD and U.K. financial services regime also impose minimum TLAC requirements on G-SIBs. For example, the CRR requires E.U. subsidiaries of a non-E.U. G-SIB that exceed the threshold of 5% of the G-SIB's RWAs, operating income or leverage exposure, such as GSBE, to meet requirements for the issuance of instruments to qualifying affiliates (internal TLAC) in order to be able to transfer losses or otherwise recapitalize those subsidiaries. Under the U.K. financial services regime, GSG UK exceeds the applicable thresholds and therefore, it is subject to internal TLAC requirements.

The CRD requires a non-E.U. group with more than €40 billion of assets in the E.U., such as us, to have an E.U. intermediate holding company (E.U. IHC) if it has, as in our case, two or more of certain types of E.U. financial institution subsidiaries, including broker-dealers and banks. The ECB granted GSBE and GSPIC an exemption to operate under two E.U. IHCs. The CRR requires E.U. IHCs to satisfy capital and liquidity requirements, a minimum requirement for own funds and eligible liabilities (MREL), and certain other prudential requirements at a consolidated level. The U.K. has not implemented a similar requirement to establish an IHC; however, the PRA requires that certain U.K. financial holding companies or a designated U.K. group entity be responsible for the U.K. group's regulatory compliance. We have designated GSI for that responsibility.

The BRRD II and the U.K. resolution regime subject institutions to a MREL, which is generally consistent with the Financial Stability Board's (FSB's) TLAC standard. The Single Resolution Board imposes internal MREL requirements applicable to GSBE. GSI is required to maintain a minimum level of internal MREL and provide the Bank of England the right to exercise bail-in triggers over certain intercompany regulatory capital and senior debt instruments issued by GSI. These triggers enable the Bank of England to write down such instruments or convert such instruments to equity. The triggers can be exercised by the Bank of England if it determines that GSI has reached the point of non-viability and the FRB and the FDIC have not objected to the bail-in or if Group Inc. enters bankruptcy or similar proceedings.

Insolvency of a BHC or IDI. The Dodd-Frank Act created a resolution regime, OLA, for BHCs and their affiliates that are systemically important. Under OLA, the FDIC may be appointed as receiver for the systemically important institution and its failed non-bank subsidiaries if, upon the recommendation of applicable regulators, the U.S. Secretary of the Treasury determines, among other things, that the institution is in default or in danger of default, that the institution's failure would have serious adverse effects on the U.S. financial system and that resolution under OLA would avoid or mitigate those effects.

If the FDIC is appointed as receiver under OLA, then the powers of the receiver, and the rights and obligations of creditors and other parties who have dealt with the institution, would be determined under OLA, and not under the bankruptcy or insolvency law that would otherwise apply. The powers of the receiver under OLA are generally based on the powers of the FDIC as receiver for depository institutions under the FDIA, described below.

Substantial differences in the rights of creditors exist between OLA and the U.S. Bankruptcy Code, including the right of the FDIC under OLA to disregard the strict priority of creditor claims in some circumstances, the use of an administrative claims procedure to determine creditors' claims (as opposed to the judicial procedure utilized in bankruptcy proceedings), and the right of the FDIC to transfer claims to a "bridge" entity. In addition, OLA limits the ability of creditors to enforce certain contractual cross-defaults against affiliates of the institution in receivership. The FDIC has issued a notice that it would likely resolve a failed FHC by transferring its assets to a "bridge" holding company under its "single point of entry" or "SPOE" strategy pursuant to OLA.

Under the FDIA, if the FDIC is appointed as conservator or receiver for an IDI such as GS Bank USA, upon its insolvency or in certain other events, the FDIC has broad powers, including the power:

- To transfer any of the IDI's assets and liabilities to a new obligor, including a newly formed "bridge" bank, without the approval of the depository institution's creditors;
- To enforce the IDI's contracts pursuant to their terms without regard to any provisions triggered by the appointment of the FDIC in that capacity; or
- To repudiate or disaffirm any contract or lease to which the IDI is a party, the performance of which is determined by the FDIC to be burdensome and the repudiation or disaffirmance of which is determined by the FDIC to promote the orderly administration of the IDI.

In addition, the claims of holders of domestic deposit liabilities and certain claims for administrative expenses against an IDI would be afforded a priority over other general unsecured claims, including deposits at non-U.S. branches and claims of debtholders of the IDI, in the "liquidation or other resolution" of such an institution by any receiver. As a result, whether or not the FDIC ever sought to repudiate any debt obligations of GS Bank USA, the debtholders (other than depositors at U.S. branches) would be treated differently from, and could receive, if anything, substantially less than, the depositors at U.S. branches of GS Bank USA.

Deposit Insurance. Deposits at GS Bank USA have the benefit of FDIC insurance up to the applicable limits. The FDIC's Deposit Insurance Fund is funded by assessments on IDIs. GS Bank USA's assessment (subject to adjustment by the FDIC) is currently based on its average total consolidated assets less its average tangible equity during the assessment period, its supervisory ratings and specified forward-looking financial measures used to calculate the assessment rate. In addition, the FDIC must recover, by special assessment, losses to the FDIC deposit insurance fund as a result of the FDIC's use of the systemic risk exception to the least cost resolution test under the FDIA. See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Results of Operations — Operating Expenses" in Part II, Item 7 of this Form 10-K for information about the estimated impact of the FDIC special assessment fee. The deposits of GSBE are covered by the German statutory deposit protection program to the extent provided by law. In addition, GSBE has elected to participate in the German voluntary deposit protection program which provides further insurance for certain eligible deposits beyond the coverage of the German statutory deposit program. Eligible deposits at GSIB and the London branch of GS Bank USA are covered by the U.K. Financial Services Compensation Scheme up to the applicable limits.

Prompt Corrective Action. The U.S. Federal Deposit Insurance Corporation Improvement Act of 1991 (FDICIA) requires the U.S. federal bank regulatory agencies to take “prompt corrective action” in respect of depository institutions that do not meet specified capital requirements. FDICIA establishes five capital categories for FDIC-insured banks, such as GS Bank USA: well-capitalized, adequately capitalized, undercapitalized, significantly undercapitalized and critically undercapitalized.

An institution may be downgraded to, or deemed to be in, a capital category that is lower than is indicated by its capital ratios if it is determined to be in an unsafe or unsound condition or if it receives an unsatisfactory examination rating with respect to certain matters. FDICIA imposes progressively more restrictive constraints on operations, management and capital distributions, as the capital category of an institution declines. Failure to meet the capital requirements could also require a depository institution to raise capital. Ultimately, critically undercapitalized institutions are subject to the appointment of a receiver or conservator, as described in “Insolvency of an IDI or a BHC” above.

The prompt corrective action regulations do not apply to BHCs. However, the FRB is authorized to take appropriate action at the BHC level, based upon the undercapitalized status of the BHC’s depository institution subsidiaries. In certain instances, relating to an undercapitalized depository institution subsidiary, the BHC would be required to guarantee the performance of the undercapitalized subsidiary’s capital restoration plan and might be liable for civil money damages for failure to fulfill its commitments on that guarantee. Furthermore, in the event of the bankruptcy of the BHC, the guarantee would take priority over the BHC’s general unsecured creditors, as described in “Source of Strength” above.

Volcker Rule and Other Restrictions on Activities. As a BHC, we are subject to limitations on the types of business activities in which we may engage.

Volcker Rule. The Volcker Rule prohibits “proprietary trading,” but permits activities such as underwriting, market making and risk-mitigation hedging, requires an extensive compliance program and includes additional reporting and record-keeping requirements.

In addition, the Volcker Rule limits the sponsorship of, and investment in, “covered funds” (as defined in the rule) by banking entities, including us. It also limits certain types of transactions between us and our sponsored and advised funds, similar to the limitations on transactions between depository institutions and their affiliates. Covered funds include our private equity funds, certain of our credit and real estate funds, our hedge funds and certain other investment structures. The limitation on investments in covered funds requires us to limit our investment in each such fund to 3% or less of the fund’s net asset value, and to limit our aggregate investment in all such funds to 3% or less of our Tier 1 capital.

Other Restrictions. FHCs generally can engage in a broader range of financial and related activities than are otherwise permissible for BHCs as long as they continue to meet the eligibility requirements for FHCs. The broader range of permissible activities for FHCs includes underwriting, dealing and making markets in securities and making investments in non-FHCs (merchant banking activities). In addition, certain FHCs, including us, are permitted to engage in certain commodities activities in the U.S. that may otherwise be impermissible for BHCs, so long as the assets held pursuant to these activities do not equal 5% or more of their consolidated assets.

The FRB, however, has the authority to limit an FHC’s ability to conduct activities that would otherwise be permissible, and will likely do so if the FHC does not satisfactorily meet certain requirements of the FRB. For example, if an FHC or any of its U.S. depository institution subsidiaries ceases to maintain its status as well-capitalized or well-managed, the FRB may impose corrective capital and/or managerial requirements, as well as additional limitations or conditions. If the deficiencies persist, the FHC may be required to divest its U.S. depository institution subsidiaries or to cease engaging in activities other than the business of banking and certain closely related activities.

In addition, we are required to obtain prior FRB approval before certain acquisitions and before engaging in certain banking and other financial activities both within and outside the U.S.

U.S. G-SIBs, like us, are also required to comply with a rule regarding single counterparty credit limits, which imposes more stringent requirements for credit exposures to major financial institutions.

The New York State banking law imposes lending limits (which take into account credit exposure from derivative transactions) and other requirements that have in the past impacted and could in the future impact the manner and scope of GS Bank USA’s activities.

The U.S. federal bank regulatory agencies have issued guidance that focuses on transaction structures and risk management frameworks and that outlines high-level principles for safe-and-sound leveraged lending, including underwriting standards, valuation and stress testing. This guidance has, among other things, limited the percentage amount of debt that can be included in certain transactions.

As a German credit institution, GSBE is subject to Volcker Rule-type prohibitions under German banking law and regulations because its financial assets exceed certain thresholds. Prohibited activities include (i) proprietary trading, (ii) high-frequency trading at a German trading venue, and (iii) lending and guarantee businesses with German hedge funds, German funds of hedge funds or any non-German substantially leveraged alternative investment funds, unless an exclusion or an exemption applies.

As part of its implementation of the Basel III Revisions, the E.U. introduced new restrictions on the provision of certain “core” banking services (i.e., deposit-taking, lending and the provision of commitments and guarantees) cross-border into the E.U. Contracts in place before June 2026 will be subject to a “grandfathering” provision and new core banking services will need to be executed out of our subsidiaries established in the E.U., including GSBE.

U.K. banks that have over £35 billion of core retail deposits are required to separate their retail banking services from their investment and international banking activities, commonly known as “ring-fencing.” GSIB is not currently subject to the ring-fencing requirement.

CRA. In 2023, GS Bank USA ceased to be assessed as a “wholesale bank” for CRA and New York Community Reinvestment Act (NYCRA) compliance purposes. GS Bank USA instead adopted a strategic plan that was approved by the FRB and NYDFS. The 2023 strategic plan will be in effect through 2028. While the plan is in effect, its terms will not be impacted by the revised federal CRA regulations, jointly published by the FDIC, FRB, and OCC in 2023. The revised federal CRA regulations tailor CRA evaluations to bank size and type, with many of the changes applying only to banks with over \$2 billion in assets and several applying only to banks with over \$10 billion in assets, including GS Bank USA. A court has issued a preliminary injunction enjoining the U.S. federal bank regulatory agencies from enforcing the revised regulations pending resolution of the lawsuit challenging the regulations.

The CRA does not establish specific lending requirements or programs for financial institutions nor does it limit an institution’s discretion to develop the types of products and services that it believes are best suited to its particular community, but depository institutions may only receive CRA credit for certain types of lending and for lending, investments and services that support community development, as defined in the CRA regulations. The CRA and its regulations require each appropriate federal bank regulatory agency, in connection with its examination of a depository institution, to assess such institution’s record of meeting the credit needs of the communities served by that institution, including the needs of low- and moderate-income borrowers and neighborhoods, and to make such assessment available to the public.

The assessment also is part of the FRB’s consideration of applications to acquire, merge or consolidate with another banking institution or its holding company, to assume deposits of or acquire assets from another depository institution, to establish a new domestic branch office that will raise deposits, or to relocate an office. In the case of a BHC applying for approval to acquire a bank or another BHC, the FRB will assess the records of performance under the CRA of the IDIs involved in the transaction, and such records may be the basis for denying the application.

If GS Bank USA fails to maintain at least a “satisfactory” rating under the CRA, we would be subject to restrictions on certain new activities and acquisitions.

We are also subject to provisions of the New York Banking Law that impose continuing and affirmative obligations upon New York State-chartered banks, such as GS Bank USA, to serve the credit needs of its local community (NYCRA). Such obligations are substantially similar to those imposed by the CRA. The NYCRA requires the NYDFS to make a periodic written assessment of an institution’s compliance with the NYCRA, and to make such assessment available to the public. The NYCRA also requires the NYDFS to consider the NYCRA rating when reviewing an application to engage in certain transactions, including mergers, asset purchases and the establishment of domestic branch offices, and provides that such assessment may serve as a basis for the denial of any such application.

Broker-Dealer and Securities Regulation

Our broker-dealer subsidiaries, including GS&Co., are subject to regulations that cover all aspects of the securities business, including sales methods, trade practices, the use and safekeeping of clients' funds and securities, capital structure, record-keeping, the financing of clients' purchases, and the conduct of directors, officers and employees. In the U.S., the SEC is the federal agency responsible for the administration of the federal securities laws.

U.S. state securities and other U.S. regulators also have regulatory or oversight authority over GS&Co. For a description of net capital requirements applicable to GS&Co., see "Management's Discussion and Analysis of Financial Condition and Results of Operations — Capital Management and Regulatory Capital — Subsidiary Capital Requirements — U.S. Regulated Broker-Dealer Subsidiaries" in Part II, Item 7 of this Form 10-K.

The SEC requires lenders of securities to provide the material terms of securities lending transactions to FINRA and for FINRA to make certain terms publicly available. Reporting under this requirement will begin in January 2026.

The SEC requires broker-dealers to act in the best interest of their retail customers. SEC rules require broker-dealers to provide a standardized, short-form disclosure highlighting services offered, applicable standards of conduct, fees and costs, the differences between brokerage and advisory services, and any conflicts of interest. In addition, several states have adopted or proposed adopting uniform fiduciary duty standards applicable to broker-dealers.

The SEC has proposed four rules to reform the U.S. equity market structure, two of which have been adopted. In 2024, the SEC adopted a rule, effective December 2025, to revise and expand reporting and disclosure requirements relating to execution quality. In 2024, the SEC also adopted a rule to update the minimum pricing increments, with variable price increments based on the trading characteristics of stocks. In December 2024, the SEC stayed the implementation of this rule, which would apply starting in November 2025, pending the outcome of litigation challenging the rule.

The SEC, FINRA and regulators in various non-U.S. jurisdictions have imposed both conduct-based and disclosure-based requirements with respect to research reports and research analysts and may impose additional regulations.

The SEC prohibits participants involved in the creation of asset-backed securities, including any underwriter, placement agent, initial purchaser or sponsor of an asset-backed security (or any affiliate or subsidiary), from engaging in any transaction that involves or results in a material conflict of interest between the securitization participant and an investor in an asset-backed security, including reducing its exposure to the asset-backed securities, subject to certain exceptions.

The SEC requires that SEC-registered clearing agencies set up policies and procedures that would, among other things, require many market participants to clear cash and repurchase transactions involving U.S. Treasury securities through such a clearing agency by December 2025 for cash transactions and by June 2026 for repurchase transactions.

GS&Co. and other U.S. subsidiaries are also subject to rules adopted by U.S. federal agencies pursuant to the Dodd-Frank Act that require any person who organizes or initiates certain asset-backed securities transactions to retain a portion (generally, at least five percent) of any credit risk that the person conveys to a third party. For certain securitization transactions, retention by third-party purchasers may satisfy this requirement.

In Europe, we provide broker-dealer services, including through GSBE, GSPIC and GSI, that are subject to oversight by European and national regulators. These services are regulated in accordance with E.U., U.K. and other national laws and regulations. These laws require, among other things, compliance with certain capital adequacy and liquidity standards, customer protection requirements and market conduct and trade reporting rules. Certain of our European subsidiaries are also regulated by the securities, derivatives and commodities exchanges of which they are members.

In the E.U. and the U.K., the Markets in Financial Instruments Directives (MiFID II) and the Markets in Financial Instruments Regulations (MiFIR) (as amended from time to time, including the proposed amendments to MiFID II and MiFIR), have established trading venue categories for the purposes of discharging the obligation to trade OTC derivatives on a trading platform, established enhanced pre- and post-trade transparency covering a wide range of financial instruments, placed volume caps on non-transparent liquidity trading for equities trading venues, limited the use of broker-dealer equities crossing networks and created a regime for systematic internalizers in certain financial instruments (which are investment firms that execute transactions outside a trading venue). Additional control requirements apply to algorithmic trading, high frequency trading and direct electronic access. Commodities trading firms are required to calculate their positions and adhere to specific position limits. MiFID II and MiFIR also require transaction reporting, transparency on costs and charges to clients for portfolio management and investment advice services, restrictions on the way investment managers can pay for the receipt of investment research, rules limiting the payment and receipt of soft commissions and other forms of inducements, and rules addressing bundling for broker-dealers between execution and other major services. Certain of our non-U.S. subsidiaries, including GSBE, GSI and GSIB, are subject to E.U. and U.K. regulation applicable to securitization activities, which will require them to conduct upfront due diligence and ongoing monitoring in connection with their investment in securitization positions and may impose ongoing risk retention and transparency requirements where they are acting as a sponsor, original lender or originator in respect of any E.U. or U.K. securitizations.

GSJCL, our regulated Japanese broker-dealer, is subject to capital requirements imposed by Japan's Financial Services Agency. GSJCL is also regulated by the Tokyo Stock Exchange, the Bank of Japan and the Ministry of Finance, among others.

The Securities and Futures Commission in Hong Kong, the China Securities Regulatory Commission, the Reserve Bank of India, the Securities and Exchange Board of India, the Australian Securities and Investments Commission, the Australian Securities Exchange, the Monetary Authority of Singapore, the Korean Financial Supervisory Service and the Central Bank of Brazil, among others, regulate various of our subsidiaries and also have capital standards and other requirements comparable to the rules of the U.S. regulators.

Our exchange-based market-making activities are subject to extensive regulation by a number of securities exchanges. As a market maker on exchanges, we are required to maintain orderly markets in the securities to which we are assigned.

Swaps, Derivatives and Commodities Regulation

The commodity futures, commodity options and swaps industry in the U.S. is subject to regulation under the U.S. Commodity Exchange Act (CEA). The CFTC is the U.S. federal agency charged with the administration of the CEA. In addition, the SEC is the U.S. federal agency charged with the regulation of security-based swaps. The rules and regulations of various self-regulatory organizations, such as the Chicago Mercantile Exchange, other futures exchanges and the National Futures Association (NFA), also govern commodity futures, commodity options and swaps activities.

The terms "swaps" and "security-based swaps" include a wide variety of derivative instruments in addition to those conventionally referred to as swaps (including certain forward contracts and options), and relate to a wide variety of underlying assets or obligations, including currencies, commodities, interest or other monetary rates, yields, indices, securities, credit events, loans and other financial obligations.

CFTC rules require registration of swap dealers, mandatory clearing and execution of interest rate and credit default swaps and real-time public reporting and adherence to business conduct standards for all in-scope swaps. A number of these requirements, particularly those regarding recordkeeping and reporting, also apply to transactions that do not involve a registered swap dealer. GS&Co. and other subsidiaries, including GS Bank USA, GSBE, GSI and J. Aron, are registered with the CFTC as swap dealers. The CFTC has rules establishing capital requirements for swap dealers that are not subject to the capital rules of a prudential regulator, such as the FRB. The CFTC also has financial reporting requirements for covered swap entities and capital rules for CFTC-registered futures commission merchants that provide explicit capital requirements for proprietary positions in swaps and security-based swaps that are not cleared by a clearing organization. Certain of our registered swap dealers, including J. Aron, are subject to the CFTC's capital requirements.

Our affiliates registered as swap dealers are subject to the margin rules issued by the CFTC (in the case of our non-bank swap dealers) and the FRB (in the case of GS Bank USA and GSBE). Inter-affiliate transactions under the CFTC and FRB margin rules are generally exempt from initial margin requirements.

Our affiliates registered as swap dealers are also subject to NFA regulation, including requirements pertaining to cybersecurity and supervision, and the NFA examines them for compliance with these requirements as well as compliance with CFTC rules.

SEC rules govern the registration and regulation of security-based swap dealers. Security-based swaps are defined as swaps on single securities, single loans or narrow-based baskets or indices of securities. The SEC has adopted a number of rules for security-based swap dealers, including (i) capital, margin and segregation requirements; (ii) record-keeping, financial reporting and notification requirements; (iii) business conduct standards; (iv) regulatory and public trade reporting; and (v) the application of risk mitigation techniques to uncleared portfolios of security-based swaps. GS&Co., GS Bank USA, GSI, GSBE and Goldman Sachs Financial Markets, L.P. (GSFM) are registered with the SEC as security-based swap dealers and subject to the SEC's regulations regarding security-based swaps. The SEC has proposed additional regulations regarding security-based swaps that would, among other things, require public reporting of large positions in security-based swaps.

GS Bank USA and GSBE are also subject to the FRB's swaps margin rules. These rules require the exchange of initial and variation margin in connection with transactions in swaps and security-based swaps that are not cleared through a registered or exempt clearinghouse. GS Bank USA and GSBE are required to post and collect margin in connection with transactions with swap dealers, security-based swap dealers, major swap participants and major security-based swap participants, or financial end users.

The CFTC and the SEC have adopted rules relating to cross-border regulation of swaps and security-based swaps, and business conduct and registration requirements. The CFTC and the SEC have entered into agreements with certain non-U.S. regulators regarding the cross-border regulation of derivatives and the mutual recognition of cross-border execution facilities and clearinghouses, and have approved substituted compliance with certain non-U.S. regulations related to certain business conduct requirements and margin rules, among other requirements. The U.S. prudential regulators have not yet made a determination with respect to substituted compliance for transactions subject to non-U.S. margin rules.

Similar types of regulation have been proposed or adopted in jurisdictions outside the U.S., including in the E.U. and Japan. Under the European Market Infrastructure Regulation (EMIR), for example, the E.U. and the U.K. have established regulatory requirements relating to portfolio reconciliation and reporting, clearing certain OTC derivatives and margining for uncleared derivatives activities. In addition, under the European Markets in Financial Instruments Directive and Regulation, transactions in certain types of derivatives are required to be executed on regulated platforms or exchanges.

The CFTC has adopted rules that limit the size of positions in physical commodity derivatives that can be held by any entity, or any group of affiliates or other parties trading under common ownership or control. The CFTC position limits apply to futures on physical commodities and options on such futures, apply to both physically and cash settled positions and to swaps that are economically equivalent to such futures and options. The position limit rules initially impose limits in the spot month only (i.e., during the delivery period for the physical commodities, which is typically a period of several days).

J. Aron is authorized by the U.S. Federal Energy Regulatory Commission (FERC) to sell wholesale physical power at market-based rates. As a FERC-authorized power marketer, J. Aron is subject to regulation under the U.S. Federal Power Act and FERC regulations and to the oversight of FERC. As a result of our investing activities, Group Inc. is also an "exempt holding company" under the U.S. Public Utility Holding Company Act of 2005 and applicable FERC rules.

In addition, as a result of our power-related and commodities activities, we are subject to energy, environmental and other governmental laws and regulations, as described in "Risk Factors — Legal and Regulatory — Our commodities activities, particularly our physical commodities activities, subject us to extensive regulation and involve certain potential risks, including environmental, reputational and other risks that may expose us to significant liabilities and costs" in Part I, Item 1A of this Form 10-K.

GS&Co. is registered with the CFTC as a futures commission merchant, and several of our subsidiaries, including GS&Co., are registered with the CFTC and act as commodity pool operators and commodity trading advisors. GSFM is registered with the SEC as an OTC derivatives dealer.

Asset Management and Wealth Management Regulation

Our asset management and wealth management businesses are subject to extensive oversight by regulators around the world relating to, among other things, the fair treatment of clients, safeguarding of client assets, offerings of funds, marketing activities, transactions among affiliates and our management of client funds.

The federal securities laws impose fiduciary duties on investment advisers, including GS&Co., Goldman Sachs Asset Management, L.P. and our other U.S. registered investment adviser subsidiaries, and SEC rules prescribe mandatory disclosures.

The SEC requires certain institutional investment managers that meet or exceed certain specified reporting thresholds to report on a monthly basis specific short position data and short activity data for equity securities. Reporting under this rule was required beginning in January 2025.

In 2024, the SEC issued guidance addressing the rules governing liquidity risk management of open-end management investment companies, such as mutual funds, and adopted amendments to the reporting requirements for certain registered investment companies, requiring such investment companies to, among other things, file reports about their portfolios and each of their portfolio holdings on a monthly basis within 30 days of the end of each month (compared to 60 days of the end of the fiscal quarter under the previous rule) and, for open-end management investment companies such as mutual funds, to identify and provide certain information about the service providers used to fulfill the rules governing liquidity risk management. Compliance with these amendments is required by November 17, 2025. Timely compliance with these accelerated or new reporting requirements will require us to enhance systems and disclosure controls and procedures.

Certain of our European subsidiaries, including GSBE in the E.U. and GSAMI in the U.K., are subject to MiFID II and/or related regulations (including the U.K. legislation making such regulations part of U.K. law), which govern the approval, organizational, marketing and reporting requirements of E.U. or U.K.-based investment managers and the ability of investment fund managers located outside the E.U. or the U.K. to access those markets. Goldman Sachs Asset Management BV is subject to similar requirements as a management company licensed under the E.U. Undertakings for Collective Investment in Transferable Securities (UCITS) Directive and the E.U. Alternative Investment Fund Managers (AIFM) Directive with additional authorizations for certain activities regulated under MiFID II. Our asset management business in the E.U. and the U.K. significantly depends on our ability to delegate parts of our activities to other affiliates.

GSAMI is also subject to the prudential regime for U.K. investment firms, the Investment Firms Prudential Regime, which governs the prudential requirements for U.K. investment firms prudentially regulated by the FCA.

Consumer Regulation

Our U.S. consumer-oriented activities are subject to supervision and regulation by the CFPB with respect to federal consumer protection laws, including laws relating to fair lending and the prohibition of unfair, deceptive or abusive acts or practices in connection with the offer, sale or provision of consumer financial products and services. Our consumer-oriented activities are also subject to various state and local consumer protection laws, rules and regulations, which, among other things, impose obligations relating to marketing, origination, servicing and collections activities in our consumer businesses. In addition, our U.K. consumer deposit-taking activities are subject to U.K. consumer protection laws and regulations.

Compensation Practices

Our compensation practices are subject to oversight by the FRB and, with respect to some of our subsidiaries and employees, by other regulatory bodies worldwide.

The FSB has released standards for implementation by local regulators that are designed to encourage sound compensation practices at banks and other financial companies. The U.S. federal bank regulatory agencies have also provided guidance designed to ensure that incentive compensation arrangements at banking organizations take into account risk and are consistent with safe and sound practices. The guidance sets forth the following three key principles with respect to incentive compensation arrangements: (i) the arrangements should provide employees with incentives that appropriately balance risk and financial results in a manner that does not encourage employees to expose their organizations to imprudent risk; (ii) the arrangements should be compatible with effective controls and risk management; and (iii) the arrangements should be supported by strong corporate governance. The guidance provides that supervisory findings with respect to incentive compensation will be incorporated, as appropriate, into the organization's supervisory ratings, which can affect its ability to make acquisitions or perform other actions. The guidance also notes that enforcement actions may be taken against a banking organization if its incentive compensation arrangements or related risk management, control or governance processes pose a risk to the organization's safety and soundness.

The Dodd-Frank Act requires U.S. financial regulators, including the FRB and SEC, to adopt rules on incentive-based payment arrangements at specified regulated entities having at least \$1 billion in total assets. The U.S. financial regulators proposed revised rules in 2016, which have not been finalized. In accordance with an SEC rule, securities exchanges have adopted rules mandating, in the case of a restatement, the recovery or "clawback" of excess incentive-based compensation paid to current or former executive officers and requiring listed issuers to disclose any recovery analysis where recovery is triggered by a restatement.

The NYDFS' guidance emphasizes that any incentive compensation arrangements tied to employee performance indicators at banking institutions regulated by the NYDFS, including GS Bank USA, must be subject to effective risk management, oversight and control.

In the E.U., certain provisions in the CRR and CRD are designed to meet the FSB's compensation standards. These provisions limit the ratio of variable to fixed compensation of all employees at GSBE and of certain employees at our other operating subsidiaries in the E.U., including those employees identified as having a material impact on the risk profile of regulated entities. CRR II and CRD V amended certain aspects of these rules, including, by increasing minimum variable compensation deferral periods.

The E.U. and the U.K. have each also introduced investment firm regimes, including rules regulating compensation for certain persons providing services to certain investment funds.

Anti-Money Laundering and Anti-Bribery Rules and Regulations

The U.S. Bank Secrecy Act, as amended (BSA), including by the USA PATRIOT Act of 2001 and the Anti-Money Laundering Act of 2020 (AMLA), contains anti-money laundering and financial transparency laws and authorizes or mandates the promulgation of various regulations applicable to financial institutions, including standards for verifying client identification at account opening, and obligations to monitor client transactions and report suspicious activities. Through these and other provisions, the BSA seeks, among other things, to promote the identification of parties that may be involved in terrorism, money laundering or other suspicious activities.

The AMLA was intended to comprehensively reform and modernize U.S. anti-money laundering laws. Among other things, the AMLA codifies a risk-based approach to anti-money laundering compliance for financial institutions; requires the U.S. Department of the Treasury to periodically promulgate priorities for anti-money laundering and countering the financing of terrorism; requires the development of standards by the U.S. Department of the Treasury for testing technology and internal processes for BSA compliance; expands enforcement- and investigation-related authority, including a significant expansion in the available sanctions for certain BSA violations; and expands BSA whistleblower incentives and protections. Certain statutory provisions in the AMLA require rulemakings beyond those that have already been finalized, reports and other measures. The impact of the AMLA will depend on, among other things, these additional rulemakings and implementation guidance. The Financial Crimes Enforcement Network (FinCEN), a bureau of the U.S. Department of Treasury, has issued the priorities for anti-money laundering and countering the financing of terrorism, as required under the AMLA. The priorities include: corruption, cybercrime, terrorist financing, fraud, transnational crime, drug trafficking organization activity, human trafficking and proliferation financing.

The Corporate Transparency Act (CTA) was enacted in 2021 as part of the AMLA and designed to establish beneficial ownership information reporting requirements for certain types of entities, and in 2024, FinCEN's rule implementing the reporting requirements of the CTA became effective. Under the rule, domestic companies formed before 2024 would be required to file their initial report by January 1, 2025 and domestic companies formed in or after 2024 would be required to file their initial report within, respectively, 90 or 30 days of their formation, in each case subject to certain exemptions. On February 19, 2025, FinCEN announced that the earliest initial reporting deadline is March 21, 2025.

In 2024, FinCEN and the SEC proposed a rule that would require certain investment advisers to implement reasonable procedures to identify and verify the identities of their customers. In 2024, FinCEN proposed to amend the anti-money laundering/countering the financing of terrorism (AML/CFT) program requirements for all financial institutions subject to the BSA that have AML/CFT program obligations, including us. The proposal would, among other things, require a financial institution's risk assessment process to identify, evaluate and document the financial institution's money laundering, terrorist financing and other illicit activity risks, and update such risk assessments on a periodic basis. In 2024, the U.S. federal bank regulatory agencies proposed amendments to their respective BSA program rules to align those rules with the FinCEN proposal. FinCEN has also adopted a rule, effective on January 1, 2026, that includes certain investment advisers, such as us, in the definition of "financial institutions" under FinCEN's rules implementing the BSA and, among other things, prescribes minimum standards for AML/CFT programs to be established by such investment advisers and requires them to report suspicious activity to FinCEN.

We are subject to other laws and regulations worldwide relating to anti-money laundering and financial transparency, including the E.U. Anti-Money Laundering Directives. In addition, we are subject to the U.S. Foreign Corrupt Practices Act (FCPA), the U.K. Bribery Act and other laws and regulations worldwide regarding corrupt and illegal payments, or providing anything of value, for the benefit of government officials and others. The scope of the types of payments or other benefits covered by these laws is very broad. These laws and regulations include requirements relating to the identification of clients, monitoring for and reporting suspicious transactions, monitoring direct and indirect payments to politically exposed persons, providing information to regulatory authorities and law enforcement agencies, and sharing information with other financial institutions.

Privacy and Cybersecurity Regulation

Our businesses are subject to numerous laws and regulations relating to the privacy of information regarding clients, employees and others. These include, but are not limited to, the GLB Act, the California Consumer Privacy Act of 2018, as amended by the California Privacy Rights Act of 2020 (CCPA), the E.U.'s General Data Protection Regulation (GDPR), the U.K.'s Data Protection Act 2018 and U.K. GDPR, the Swiss Federal Data Protection Act, the Japanese Personal Information Protection Act, the Personal Information Protection Law of the People's Republic of China, the Australian Privacy Act 1988, and India's Digital Personal Data Protection Act. Generally, privacy laws impose obligations with regard to the collection, use and disclosure of personal information and require public disclosure of privacy practices. Some privacy laws offer individuals certain rights about how their personal information is processed, provide for significant penalties for non-compliance, and, under certain circumstances, impose requirements for transfers of personal data across national borders.

In 2023, the SEC proposed to amend Regulation Systems Compliance and Integrity (SCI). The proposed amendments to Regulation SCI would, among other things, expand the types of entities covered by the regulation, require additional policies and procedures to address cybersecurity risks, and require disclosure of additional types of cybersecurity events to the SEC.

In 2024, the SEC amended Regulation S-P that implements the GLB Act. The amendments to Regulation S-P require broker-dealers, investment companies and investment advisers registered with the SEC to adopt written policies and procedures for incident response programs to address unauthorized access to or use of customer information. The amended Regulation S-P requires covered entities to notify within 30 days individuals affected by an incident involving sensitive customer information and provide them with details about the incident and other information intended to help affected individuals respond appropriately. Larger covered entities, such as GS&Co., will have until December 2025, and smaller covered entities will have until June 2026, to comply with the amended Regulation S-P. In 2024, the California Privacy Protection Agency proposed regulations under the CCPA relating to cybersecurity audits, risk assessments, and automated decision-making technology.

Our businesses are also subject to laws and regulations governing cybersecurity and related risks, and which require regulatory disclosures, and, in some instances, individual disclosures, of certain security incidents. These include, but are not limited to, the NYDFS Cybersecurity Requirements for Financial Services Companies. The NYDFS also requires financial institutions regulated by the NYDFS, including GS Bank USA, to, among other things, (i) establish and maintain a cybersecurity program designed to ensure the confidentiality, integrity and availability of their information systems; (ii) implement and maintain a written cybersecurity policy setting forth policies and procedures for the protection of their information systems and nonpublic information; and (iii) designate a Chief Information Security Officer. In 2023, the NYDFS adopted amendments to its cybersecurity regulations that will impose heightened or additional requirements with respect to cybersecurity incident notifications, risk management and governance. Most of these amendments became effective in 2024 although certain amendments have a longer transition period and become effective in 2025.

In 2023, the E.U. Digital Operational Resilience Act (DORA) became effective and applies from January 2025. DORA requires E.U. financial entities, such as GSBE, to have a comprehensive governance and control framework for the management of information and communications technology risk. In addition, in 2024, the E.U. Artificial Intelligence Act (E.U. AI Act) became effective. Certain provisions of the E.U. AI Act apply from February 2025, with other provisions applying between August 2025 and August 2027. The E.U. AI Act establishes rules for placing on the market, putting into service, and using artificial intelligence systems in the E.U.

In 2024, the CFPB adopted a rule regarding personal financial data rights that requires financial institutions that offer consumer deposit accounts and issue credit cards, such as GS Bank USA, to provide consumers electronic access to at least 24 months of transaction data and certain account information and are prohibited from imposing any fees or charges for maintaining or providing access to such data. The rule also imposes data accuracy, retention and other obligations, and data use limitations and other obligations on entities obtaining access to such personal financial data, such as GS&Co. The current compliance deadline for large financial institutions, including GS Bank USA, is April 1, 2026. However, this timeline is potentially subject to change as a result of CFPB action or currently pending litigation.

See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Cybersecurity Risk Management" in Part II, Item 7 of this Form 10-K for further information about our cybersecurity risk management, strategy and governance.

Environmental, Social and Governance (ESG)

Policymakers, lawmakers and regulators in the U.S. and other jurisdictions have recently increased their focus on ESG-related risk oversight, disclosure, and practices at financial institutions and other companies.

In 2023, the federal bank regulatory agencies jointly issued principles for climate-related financial risk management for large financial institutions, which apply to regulated financial institutions with more than \$100 billion in total consolidated assets, including us. The principles are intended to support efforts by large financial institutions to focus on key aspects of climate-related financial risk management and consist of six general principles: (1) governance; (2) policies, procedures, and limits; (3) strategic planning; (4) risk management; (5) data, risk measurement, and reporting; and (6) scenario analysis. In 2023, the SEC adopted amendments to Rule 35d-1 (Names Rule) under the Investment Company Act of 1940. The previous Names Rule generally required a fund with a name suggesting a focus in a particular type of investment, or in investments in a particular industry or geographic region, to adopt a policy to invest at least 80% of the value of its assets in the type of investment, or in investments in the industry, country or geographic region, suggested by its name. The amendments expand such 80% investment policy to apply to any fund name with terms suggesting that the fund focuses in investments that have, or investments whose issuers have, particular characteristics, including names that suggest the fund incorporates ESG factors in its investment decisions. In 2022, the SEC proposed a rule that would require enhanced disclosures by certain investment advisers and investment companies about their ESG investment practices.

In 2024, the SEC adopted final rules requiring registrants to provide certain climate-related disclosures, including Scope 1 and Scope 2 greenhouse gas emissions to the extent they are material. These rules require certain disclosures related to severe weather events and other natural conditions in the notes to audited financial statements. These disclosures are required to be phased-in over multiple years beginning with fiscal year 2025 for large accelerated filers like us. However, the SEC has stayed the implementation of these rules, pending the outcome of litigation challenging the rules.

Several states in which we operate have enacted or proposed statutes, regulations or guidance addressing climate change and other ESG issues, including climate disclosure laws and climate-related financial risk management guidance. For example, in 2023, the NYDFS issued guidance on climate-related financial risk management applicable to NYDFS-regulated banking and mortgage organizations, including GS Bank USA. The guidance addresses material financial risks related to climate change faced by these organizations in the context of risk assessment, risk management, and risk appetite setting.

Certain states have also enacted, or are considering enacting, “fair access” statutes that generally prohibit financial institutions from denying or canceling services on the basis of factors, such as political opinions, religious beliefs, “social credit scores,” or any factor that is not quantitative, impartial, and risk-based.

Certain of our entities are expected to be subject, in varying degrees, to sustainability-related laws being implemented across certain jurisdictions, including by E.U. member states. E.U. rules include directives, such as the Corporate Sustainability Reporting Directive (CSRD) and the Corporate Sustainability Due Diligence Directive (CSDDD), both of which would, if implemented in current form, significantly expand the scope of ESG disclosure requirements applicable to us. There remains uncertainty around timing, scope and impacts to us, as numerous member states have not yet transposed some of these directives into national law, and the European Commission stated its intention to revisit these directives, among potentially others, with a desire to pursue an Omnibus package, which could impact us. Our regulated banking subsidiaries in the E.U. are also subject to supervisory expectations and potential enforcement actions for, among others, the management of climate-related financial risks and related disclosure.

The CRR and the E.U. member states’ legislation implementing CRD require large institutions with securities traded on a regulated market of a member state to make qualitative and quantitative disclosures relating to environmental, social and governance risks on a semi-annual basis. GSBE is expected to become subject to this requirement in January 2026.

In 2021, the FCA introduced mandatory Taskforce on Climate-related Financial Disclosures (TCFD)-aligned disclosure requirements for certain FCA-regulated firms, including GSI and GSAMI. These entities are also subject to climate-related financial disclosures required under the U.K. Companies Act. In addition, during 2024, new FCA rules on sustainability requirements and investment labels became effective. We continue to assess the impact of other ESG-related regulatory frameworks that will, or are proposed to, in the future apply to our FCA- and/or PRA-regulated subsidiaries. Our PRA-regulated banking subsidiaries are also subject to the PRA’s supervisory expectations for the management of climate-related financial risks, including with respect to governance, risk management, scenario analysis and disclosure.

Information about our Executive Officers

Set forth below are the name, age, present title, principal occupation and certain biographical information for the executive officers who have been appointed by, and serve at the pleasure of, Group Inc.'s Board.

Denis P. Coleman III, 51

Mr. Coleman has been Chief Financial Officer since January 2022. He had previously served as Deputy Chief Financial Officer from September 2021 and, prior to that, Co-Head of the Global Financing Group from June 2018 to September 2021. From 2016 to June 2018, he was Head of the EMEA Financing Group, and from 2009 to 2016 he was Head of EMEA Credit Finance in London.

Sheara J. Fredman, 49

Ms. Fredman has been Controller and Chief Accounting Officer since November 2019. She had previously served as Head of Regulatory Controllers from September 2017 and, prior to that, she had served as Global Product Controller.

Alex Golten, 49

Mr. Golten has been Chief Risk Officer since January 2025. He had previously served as Head of Finance Risk from July 2024 to December 2024, as Head of Enterprise Risk from June 2022 to July 2024 and as Chief Market Risk Officer from November 2020 to January 2025. Prior to that, he was Chief Credit Risk Officer from January 2018 to April 2021.

Carey Halio, 51

Ms. Halio has been Global Treasurer since May 2024. She had previously served as Chief Strategy Officer from October 2022 to June 2024, and as Global Head of Investor Relations from May 2021 to April 2024. Prior to that, she was Chief Executive Officer of GS Bank USA from October 2018 to May 2021, Deputy Treasurer from September 2019 to May 2021 and Chief Financial Officer of GS Bank USA from June 2014 to September 2018.

John F.W. Rogers, 68

Mr. Rogers has been an Executive Vice President since April 2011 and Secretary to the Board since December 2001. He also served as Chief of Staff from December 2001 to September 2023.

Kathryn H. Ruemmler, 53

Ms. Ruemmler has been the Chief Legal Officer, General Counsel and Secretary since March 2021, and was previously Global Head of Regulatory Affairs from April 2020. From June 2014 to April 2020, Ms. Ruemmler was a Litigation Partner at Latham & Watkins LLP, a global law firm, where she was Global Chair of the White Collar Defense and Investigations practice.

David Solomon, 63

Mr. Solomon has been Chairman of the Board since January 2019 and Chief Executive Officer and a director since October 2018. He had previously served as President and Chief or Co-Chief Operating Officer from January 2017 and Co-Head of the Investment Banking Division from July 2006 to December 2016.

John E. Waldron, 55

Mr. Waldron has been President and Chief Operating Officer since October 2018. He had previously served as Co-Head of the Investment Banking Division from December 2014. Prior to that he was Global Head of Investment Banking Services/Client Coverage for the Investment Banking Division and had oversight of the Investment Banking Services Leadership Group, and from 2007 to 2009 was Global Co-Head of the Financial Sponsors Group.

Available Information

Our internet address is www.goldmansachs.com and the investor relations section of our website is located at www.goldmansachs.com/investor-relations, where we make available, free of charge, our annual reports on Form 10-K, quarterly reports on Form 10-Q and current reports on Form 8-K and amendments to those reports filed or furnished pursuant to Section 13(a) or 15(d) of the Exchange Act, as well as proxy statements, as soon as reasonably practicable after we electronically file such material with, or furnish it to, the SEC. Also posted on our website, and available in print upon request of any shareholder to our Investor Relations Department (Investor Relations), are our certificate of incorporation and by-laws, charters for our Audit, Risk, Compensation, Corporate Governance and Nominating, and Public Responsibilities Committees, our Policy Regarding Director Independence Determinations, our Policy on Reporting of Concerns Regarding Accounting and Other Matters, our Corporate Governance Guidelines and our Code of Business Conduct and Ethics governing our directors, officers and employees. Within the time period required by the SEC, we will post on our website any amendment to the Code of Business Conduct and Ethics and any waiver applicable to any executive officer, director or senior financial officer.

Our website also includes information about (i) purchases and sales of our equity securities by our executive officers and directors; (ii) disclosure relating to certain non-GAAP financial measures (as defined in the SEC's Regulation G) that we may make public orally, telephonically, by webcast, by broadcast or by other means; (iii) our U.S. Dodd-Frank Wall Street Reform and Consumer Protection Act Stress Tests results; (iv) the public portion of our and GS Bank USA's resolution plan submissions; (v) our Pillar 3 disclosure; (vi) our average daily LCR; (vii) our average daily NSFR; (viii) our People Strategy Report; (ix) our Sustainability Report; and (x) our TCFD Report.

Investor Relations can be contacted at The Goldman Sachs Group, Inc., 200 West Street, 29th Floor, New York, New York 10282, Attn: Investor Relations, telephone: 212-902-0300, e-mail: gs-investor-relations@gs.com. We use the following, as well as other social media channels, to disclose public information to investors, the media and others:

- Our website (www.goldmansachs.com);
- Our X, formerly known as Twitter, account (x.com/GoldmanSachs); and
- Our Instagram account ([instagram.com/GoldmanSachs](https://www.instagram.com/GoldmanSachs)).

Our officers may use similar social media channels to disclose public information. It is possible that certain information we or our officers post on our website and on social media could be deemed material, and we encourage investors, the media and others interested in Goldman Sachs to review the business and financial information we or our officers post on our website and on the social media channels identified above. The information on our website and those social media channels is not incorporated by reference into this Form 10-K.

Forward-Looking Statements

We have included in this Form 10-K, and our management may make, statements that constitute "forward-looking statements" within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. Forward-looking statements are not historical facts or statements of current conditions, but instead represent only our beliefs regarding future events, many of which, by their nature, are inherently uncertain and outside our control.

By identifying these statements for you in this manner, we are alerting you to the possibility that our actual results, financial condition, liquidity and capital actions may differ, possibly materially, from the anticipated results, financial condition, liquidity and capital actions in these forward-looking statements. Important factors that could cause our results, financial condition, liquidity and capital actions to differ from those in these statements include, among others, those described below and in "Risk Factors" in Part I, Item 1A of this Form 10-K.

These statements may relate to, among other things, (i) our future plans and results, including our target return on average common shareholders' equity (ROE), return on average tangible common shareholders' equity (ROTE), efficiency ratio, CET1 capital ratio and firmwide total credit alternative assets, and how they can be achieved, (ii) trends in or growth opportunities for our businesses, including the timing, costs, profitability, benefits and other aspects of business and strategic initiatives and their impact on our efficiency ratio, as well as the opportunities and challenges presented by artificial intelligence (AI), (iii) our level of future compensation expense, (iv) our Investment banking fees backlog and future advisory and capital markets results, (v) our expected interest income and interest expense, (vi) our expense savings and strategic locations initiatives, (vii) expenses we may incur, including future litigation expense, (viii) the projected growth of our deposits and other funding, asset liability management and funding strategies and related interest expense savings, (ix) our business initiatives, (x) our planned 2025 benchmark debt issuances, (xi) the amount, composition and location of global core liquid assets (GCLA) we expect to hold, (xii) our credit exposures, (xiii) our expected provision for credit losses, (xiv) the adequacy of our allowance for credit losses, (xv) the narrowing of our consumer business, (xvi) the objectives and effectiveness of our business continuity planning (BCP), information security program, risk management and liquidity policies, (xvii) our resolution plan and its implications for stakeholders, (xviii) the design and effectiveness of our resolution capital and liquidity models and triggers and alerts framework, (xix) the results of stress tests, the effect of changes to regulations, and our future status, activities or reporting under banking and financial regulation, (xx) our expected tax rate, (xxi) the future state of our liquidity and regulatory capital ratios, and our prospective capital distributions (including dividends and repurchases), (xxii) our expected SCB and G-SIB surcharge, (xxiii) legal proceedings, governmental investigations or other contingencies, (xxiv) the asset recovery guarantee and our remediation activities related to our 1Malaysia Development Berhad (1MDB) settlements, (xxv) the effectiveness of our management of our human capital, (xxvi) our sustainability and carbon neutrality targets and goals, (xxvii) future inflation, (xxviii) the impact of Russia's invasion of Ukraine and related sanctions and other developments on our business, results and financial position, (xxix) our ability to sell, and the terms of any proposed sales of, Asset & Wealth Management historical principal investments and our ability to transition the GM credit card program to another issuer, (xxx) the impact of the conflicts in the Middle East, (xxxi) our ability to manage our commercial real estate exposures, (xxxii) the profitability of Platform Solutions and (xxxiii) the effectiveness of our cybersecurity risk management process.

Statements about our target ROE, ROTE, efficiency ratio and expense savings, and how they can be achieved, are based on our current expectations regarding our business prospects and are subject to the risk that we may be unable to achieve our targets due to, among other things, changes in our business mix and inability to grow our businesses and execute our strategy.

Statements about our target ROE, ROTE and CET1 capital ratio, and how they can be achieved, are based on our current expectations regarding the capital requirements applicable to us and are subject to the risk that our actual capital requirements may be higher than currently anticipated because of, among other factors, changes in the regulatory capital requirements applicable to us resulting from changes in regulations, including as a result of any revisions to the U.S. bank regulatory capital rules, or the interpretation or application of existing regulations or changes in the nature and composition of our activities.

Statements about our total credit alternative assets targets are based on our current expectations regarding our fundraising prospects and are subject to the risk that actual inflows may be lower than expected due to, among other factors, competition from other asset managers, changes in investment preferences and changes in economic or market conditions.

Statements about the timing, costs, profitability, benefits and other aspects of business and expense savings initiatives, the level and composition of more durable revenues and increases in market share and the narrowing of our consumer business are based on our current expectations regarding our ability to implement these initiatives, and actual results may differ, possibly materially, from our current expectations due to, among other things, a delay in the timing of these initiatives, increased competition and an inability to reduce expenses and grow businesses with more durable revenues or to exit certain consumer businesses.

Statements about the level of future compensation expense, including as a percentage of both operating expenses and net revenues, net of provision for credit losses, and our efficiency ratio are subject to the risks that the compensation and other costs to operate our businesses may be greater than currently expected.

Statements about our Investment banking fees backlog and future advisory and capital market results are subject to the risk that advisory and capital market activity may not increase as the firm expects or that such transactions may be modified or may not be completed at all, and related net revenues may not be realized or may be materially less than expected. Important factors that could have such a result include, for underwriting transactions, a decline or weakness in general economic conditions, an outbreak or worsening of hostilities, including those in Ukraine and the Middle East, continuing volatility in the securities markets or an adverse development with respect to the issuer of the securities and, for advisory transactions, a decline in the securities markets, an inability to obtain adequate financing, an adverse development with respect to a party to the transaction or a failure to obtain a required regulatory approval.

Statements about the projected growth of our deposits and other funding, asset liability management and funding strategies and related interest expense savings, and our platform solutions business, are subject to the risk that actual growth, savings and profitability may differ, possibly materially, from that currently anticipated due to, among other things, changes in interest rates and competition from other similar products.

Statements about planned 2025 benchmark debt issuances and the amount, composition and location of GCLA we expect to hold are subject to the risk that actual issuances and GCLA levels may differ, possibly materially, from that currently expected due to changes in market conditions, business opportunities or our funding and projected liquidity needs.

Statements about our expected provision for credit losses are subject to the risk that actual credit losses may differ and our expectations may change, possibly materially, from that currently anticipated due to, among other things, changes to the composition of our loan portfolio and changes in the economic environment in future periods and our forecasts of future economic conditions, as well as changes in our models, policies and other management judgments.

Statements about our future effective income tax rate are subject to the risk that it may differ from the anticipated rate indicated in such statements, possibly materially, due to, among other things, changes in the tax rates applicable to us, changes in our earnings mix, our profitability and entities in which we generate profits, the assumptions we have made in forecasting our expected tax rate, the interpretation or application of existing tax statutes and regulations, as well as any corporate tax legislation that may be enacted or any guidance that may be issued by the U.S. Internal Revenue Service or in the other jurisdictions in which we operate (including Global Anti-Base Erosion (Pillar II) guidance).

Statements about the future state of our liquidity and regulatory capital ratios (including our SCB and G-SIB surcharge), and our prospective capital distributions (including dividends and repurchases), are subject to the risk that our actual liquidity, regulatory capital ratios and capital distributions may differ, possibly materially, from what is currently expected due to, among other things, the need to use capital to support clients, increased regulatory requirements resulting from changes in regulations or the interpretation or application of existing regulations, results of applicable supervisory stress tests, changes to the composition of our balance sheet and our results of operations. Statements about the estimated impact of proposed, but not finalized, capital rules are subject to change as the proposed rules may change, the final rules may differ from the proposed rules and our balance sheet composition will change. As a consequence, we may underestimate the actual impact of the final rules.

Statements about the risk exposure related to the asset recovery guarantee provided to the Government of Malaysia are subject to the risk that we may be unsuccessful in our arbitration against the Government of Malaysia. Statements about the progress or the status of remediation activities relating to 1MDB are based on our expectations regarding our current remediation plans. Accordingly, our ability to complete the remediation activities may change, possibly materially, from what is currently expected.

Statements about our objectives in management of our human capital are based on our current expectations and are subject to the risk that we may not achieve these objectives.

Statements about our sustainability and carbon neutrality, net-zero or other sustainability-related targets and goals are based on our current expectations and are subject to the risk that we may not achieve these targets and goals due to, among other things, global socio-demographic and economic trends, energy prices, lack of technological innovations, climate-related conditions and weather events, legislative and regulatory changes, consumer behavior and demand, and other unforeseen events or conditions.

Statements about future inflation are subject to the risk that actual inflation may differ, possibly materially, due to, among other things, changes in economic growth, unemployment or consumer demand.

Statements about the impact of Russia's invasion of Ukraine and related sanctions, the impact of the conflicts in the Middle East and other developments on our business, results and financial position are subject to the risks that hostilities may escalate and expand, that sanctions may increase and that the actual impact may differ, possibly materially, from what is currently expected.

Statements about the proposed sales of Asset & Wealth Management historical principal investments are subject to the risks that buyers may not bid on these assets or bid at levels, or with terms, that are unacceptable to us, and that the performance of these activities may deteriorate as a result of the proposed sales, and statements about our ability to transition the GM credit card program to another issuer are subject to the risk that the transaction may not close on the anticipated timeline or at all, including due to a failure to obtain requisite regulatory approval.

Statements about the effectiveness of our cybersecurity risk management process are subject to the risk that measures we have implemented to safeguard our systems (and third parties that we interface with) may not be sufficient to prevent a successful cybersecurity attack or a material security breach that results in the disclosure of confidential information or otherwise disrupts our operations.

Item 1A. Risk Factors

We face a variety of risks that are substantial and inherent in our businesses.

The following is a summary of some of the more important factors that could affect our businesses:

Market

- Our businesses have been and may in the future be adversely affected by conditions in the global financial markets and broader economic conditions.
- Our businesses have been and may in the future be adversely affected by declining asset values, particularly where we have net "long" positions, receive fees based on the value of assets managed, or receive or post collateral.
- Our market-making activities have been and may in the future be affected by changes in the levels of market volatility.
- Our investment banking, client intermediation, asset management and wealth management businesses have been adversely affected and may in the future be adversely affected by market uncertainty or lack of confidence among investors and CEOs due to declines in economic activity and other unfavorable economic, geopolitical or market conditions.
- Our asset management and wealth management businesses have been and may in the future be adversely affected by the poor investment performance of our investment products or a client preference for products other than those which we offer or for products that generate lower fees.
- Inflation has had, and could continue to have, a negative effect on our business, results of operations and financial condition.

Liquidity

- Our liquidity, profitability and businesses may be adversely affected by an inability to access the debt capital markets or to sell assets.
- Our businesses have been and may in the future be adversely affected by disruptions or lack of liquidity in the credit markets, including reduced access to credit and higher costs of obtaining credit.
- Reductions in our credit ratings or an increase in our credit spreads may adversely affect our liquidity and cost of funding.
- Group Inc. is a holding company and its liquidity depends on payments and loans from its subsidiaries, many of which are subject to legal, regulatory and other restrictions on providing funds or assets to Group Inc.

Credit

- Our businesses, profitability and liquidity may be adversely affected by deterioration in the credit quality of or defaults by third parties.
- Concentration of risk increases the potential for significant losses in our market-making, underwriting, investing and financing activities.
- Derivative transactions and delayed documentation or settlements expose us to credit risk, unexpected risks and potential losses.

Operational

- A failure in our or third-party operational systems or human error, malfeasance or other misconduct, could impair our liquidity, disrupt our businesses, result in the disclosure of confidential information, damage our reputation and cause losses.
- A failure or disruption in our infrastructure, or in the operational systems or infrastructure of third parties, could impair our liquidity, disrupt our businesses, damage our reputation and cause losses.
- The development and use of AI present risks and challenges that may adversely impact our business.
- A failure to protect our computer systems, networks and information, and our clients' information, against cyber attacks and similar threats could impair our ability to conduct our businesses, result in the disclosure, theft or destruction of confidential information, damage our reputation and cause losses.
- We have in the past incurred and may in the future incur losses as a result of ineffective risk management processes and strategies.

Legal and Regulatory

- Our businesses and those of our clients are subject to extensive and pervasive regulation around the world.
- A failure to appropriately identify and address potential conflicts of interest has in the past and may in the future adversely affect our businesses.
- We may be adversely affected by increased governmental and regulatory scrutiny or negative publicity.
- Substantial civil or criminal liability or significant regulatory action against us has in the past had and may in the future have material adverse financial effects and significant reputational consequences, which in turn could seriously harm our business prospects.
- In conducting our businesses around the world, we are subject to political, legal, regulatory, tax and other risks that are inherent in operating in many countries.
- The application of regulatory strategies and requirements in the U.S. and in non-U.S. jurisdictions to facilitate the orderly resolution of large financial institutions could create greater risk of loss for Group Inc.'s security holders.

- The application of Group Inc.'s proposed resolution strategy could result in greater losses for Group Inc.'s security holders.

- Our commodities activities, particularly our physical commodities activities, subject us to extensive regulation and involve certain potential risks, including environmental, reputational and other risks that may expose us to significant liabilities and costs.

Competition

- Our results have been and may in the future be adversely affected by the composition of our client base.
- The financial services industry is highly competitive.
- The growth of electronic trading and the introduction of new products and technologies, including trading and distributed ledger technologies, such as cryptocurrencies, and AI technologies, has increased competition.
- Our businesses would be adversely affected if we are unable to hire and retain qualified employees.

Market Developments and General Business Environment

- Our businesses, financial condition, liquidity and results of operations have been and may in the future be adversely affected by unforeseen or catastrophic events, including pandemics, terrorist attacks, wars, extreme weather events or other natural disasters.
- Climate change could disrupt our businesses and adversely affect client activity levels and the creditworthiness of our clients and counterparties, and our actual or perceived action or inaction relating to climate change could result in damage to our reputation.
- Our business, financial condition, liquidity and results of operations have been adversely affected by disruptions in the global economy caused by conflicts, and related sanctions and other developments.
- Certain of our businesses and our funding instruments may be adversely affected by changes in reference rates, currencies, indexes, baskets or ETFs to which products we offer or funding that we raise are linked.
- Our business, financial condition, liquidity and results of operations may be adversely affected by disruptions in the global economy caused by escalating tensions between the U.S. and China.
- We face enhanced risks as we operate in new locations and transact with a broader array of clients and counterparties.
- We may not be able to fully realize the expected benefits or synergies from acquisitions or other business initiatives in the time frames we expect, or at all.

The following are detailed descriptions of our Risk Factors summarized above:

Market

Our businesses have been and may in the future be adversely affected by conditions in the global financial markets and broader economic conditions.

Many of our businesses, by their nature, do not produce predictable earnings, and all of our businesses are materially affected by conditions in the global financial markets and economic conditions generally, both directly and through their impact on client activity levels and creditworthiness. These conditions can change suddenly and negatively.

Our financial performance is highly dependent on the environment in which our businesses operate. A favorable business environment is generally characterized by, among other factors, high global gross domestic product growth, regulatory and market conditions that result in transparent, liquid and efficient capital markets, low inflation, business, consumer and investor confidence, stable geopolitical conditions and strong business earnings.

Unfavorable or uncertain economic and market conditions can be caused by: low levels of or declines in economic growth, business activity or investor, business or consumer confidence; concerns over a potential recession; changes in consumer spending or borrowing patterns; pandemics; limitations on the availability or increases in the cost of credit and capital; illiquid markets; increases in inflation, interest rates, exchange rate or basic commodity price volatility or default rates; high levels of inflation or stagflation; concerns about U.S. and other sovereign defaults; uncertainty concerning fiscal or monetary policy, government shutdowns, debt ceilings or funding; the extent of and uncertainty about potential changes in tax rates and regulatory changes; limitations on international trade and travel; changes in immigration policies; laws and regulations that limit trading in, or the issuance of, securities of issuers outside their domestic markets; outbreaks or worsening of domestic or international tensions or hostilities, terrorism, nuclear proliferation, cybersecurity threats or attacks and other forms of disruption to or curtailment of global communication, energy transmission or transportation networks or other geopolitical instability or uncertainty; corporate, political or other scandals that reduce investor confidence in capital markets; extreme weather events or other natural disasters; or a combination of these or other factors.

The financial services industry and the securities and other financial markets have been materially and adversely affected in the past by significant declines in the values of nearly all asset classes, by a severe lack of liquidity and by high levels of borrower defaults. In addition, concerns about actual or potential increases in interest rates, inflation and other borrowing costs, a public health emergency, sovereign debt risk and its impact on the relevant sovereign banking system, and limitations on international trade, have, at times, negatively impacted the levels of client activity.

General uncertainty about economic, political and market activities, and the scope, timing and impact of regulatory reform, as well as weak consumer, investor and CEO confidence resulting in large part from such uncertainty, has in the past negatively impacted client activity, which has in the past adversely affected and could in the future adversely affect many of our businesses. Periods of low volatility and periods of high volatility combined with a lack of liquidity have at times had an unfavorable impact on our market-making businesses.

Changes, or proposed changes, to U.S. international trade and investment policies, particularly with important trading partners, have in recent years negatively impacted financial markets. Continued or escalating tensions may result in further actions taken by the U.S. or other countries that could disrupt international trade and investment and adversely affect financial markets. Those actions could include, among others, the implementation of or increase in sanctions, tariffs or foreign exchange measures, the large-scale sale of U.S. Treasury securities or other restrictions on cross-border trade, investment, or transfer of information or technology. Such developments have in the past affected and could in the future adversely affect our or our clients' businesses.

Financial institution returns may be negatively impacted by increased funding costs due in part to the lack of perceived government support of such institutions in the event of future financial crises relative to financial institutions in countries in which governmental support is maintained. In addition, liquidity in the financial markets has in the past been, and could in the future be, negatively impacted as market participants and market practices and structures adjust to evolving regulatory frameworks.

In 2023, the U.S. federal government suspended the federal debt limit until 2025. If Congress does not raise the debt ceiling, the U.S. could default on its obligations, including Treasury securities that play an integral role in financial markets. A default by the U.S. could result in unprecedented market volatility and illiquidity, heightened operational risks relating to the clearance and settlement of transactions, margin and other disputes with clients and counterparties, an adverse impact to investors including money market funds that invest in U.S. Treasuries, downgrades in the U.S. credit rating, further increases in interest rates and borrowing costs and a recession in the U.S. or other economies. Continued uncertainty relating to the debt ceiling could result in downgrades of the U.S. credit rating, which could adversely affect market conditions, lead to margin disputes, increases in interest rates and borrowing costs and necessitate significant operational changes among market participants, including us. A downgrade of the U.S. federal government's credit rating could also materially and adversely affect the market for repurchase agreements, securities borrowing and lending, and other financings typically collateralized by U.S. Treasury or agency obligations. Further, the fair value, liquidity and credit ratings of securities issued by, or other obligations of, agencies of the U.S. government or related to the U.S. government or its agencies, as well as municipal bonds could be similarly adversely affected. An increasing frequency of government shutdowns, or near shutdowns, in the U.S. could also lead to uncertainty as to the continued funding of the U.S. government, which could, in turn, adversely affect the credit ratings of the U.S. and the market for U.S. Treasury or agency obligations.

In 2024, numerous elections were held globally, including the recent U.S. presidential election. The outcomes of the elections are expected to result in changes in policy, which could also have adverse effects on us or the business environment in which we operate more generally. For example, the new U.S. presidential administration has imposed or increased tariffs, including on imports from China, and proposed imposing or increasing tariffs on U.S. trading partners, which could adversely affect markets, the business environment and some of our businesses.

Our businesses have been and may in the future be adversely affected by declining asset values, particularly where we have net "long" positions, receive fees based on the value of assets managed, or receive or post collateral.

Many of our businesses have net "long" positions in debt securities, loans, derivatives, mortgages, equities (including private equity and real estate) and most other asset classes. These include positions we take when we act as a principal to facilitate our clients' activities, including our exchange-based market-making activities, or commit large amounts of capital to maintain positions in interest rate and credit products, as well as through our currencies, commodities, equities and mortgage-related activities. In addition, we invest in similar asset classes. Substantially all of our investing and market-making positions and a portion of our loans are marked-to-market on a daily or other periodic basis and declines in asset values directly and promptly impact our earnings, unless we have effectively "hedged" our exposures to those declines.

In certain circumstances, it may not be possible or economic to hedge our exposures and, to the extent that we do so, the hedge may be ineffective or may greatly reduce our ability to profit from increases in the values of the assets. This is particularly the case for credit products, including leveraged loans, and private equities or other securities that are not freely tradable or lack established and liquid trading markets. Sudden declines and significant volatility in the prices of assets have in the past substantially curtailed or eliminated, and may in the future substantially curtail or eliminate, the trading markets for certain assets, which may make it difficult to sell, hedge or value such assets. We may incur losses from time to time as trading markets deteriorate or cease to function, including with respect to loan commitments we have made or securities offerings we have underwritten. The inability to sell or effectively hedge assets reduces our ability to limit losses in such positions and the difficulty in valuing assets has in the past negatively affected, and may in the future negatively affect, our capital, liquidity or leverage ratios, our funding costs and our ability to deploy capital.

In our exchange-based market-making activities, we are obligated by stock exchange rules to maintain an orderly market, including by purchasing securities in a declining market. In markets where asset values are declining and in volatile markets, this results in losses and an increased need for liquidity.

We receive asset-based management fees based on the value of our clients' portfolios or investment in funds managed by us and, in some cases, we also receive incentive fees based on increases in the value of such investments. Declines in asset values would ordinarily reduce the value of our clients' portfolios or fund assets, which in turn would typically reduce the fees we earn for managing such assets.

We post collateral to support our obligations and receive collateral that supports the obligations of our clients and counterparties. When the value of the assets posted as collateral or the credit ratings of the party posting collateral decline, the party posting the collateral may need to provide additional collateral or, if possible, reduce its trading position. An example of such a situation is a “margin call” in connection with a brokerage account. Therefore, declines in the value of asset classes used as collateral mean that either the cost of funding positions is increased or the size of positions is decreased. If we are the party providing collateral, this can increase our costs and reduce our profitability and if we are the party receiving collateral, this can also reduce our profitability by reducing the level of business done with our clients and counterparties.

In addition, volatile or less liquid markets increase the difficulty of valuing assets, which can lead to costly and time-consuming disputes over asset values and the level of required collateral, as well as increased credit risk to the recipient of the collateral due to delays in receiving adequate collateral. In cases where we foreclose on collateral, sudden declines in the value or liquidity of the collateral have in the past resulted in, and may in the future result in, significant losses to us, especially where there is a single type of collateral supporting the obligation. In addition, we have been and may in the future be subject to claims that the foreclosure was not permitted under the legal documents, was conducted in an improper manner, including in violation of law, or caused a client or counterparty to incur significant losses or go out of business.

Our market-making activities have been and may in the future be affected by changes in the levels of market volatility.

Certain of our market-making activities depend on market volatility to provide trading and arbitrage opportunities to our clients, and decreases in volatility have reduced and may in the future reduce these opportunities and the level of client activity associated with them and have adversely affected and may in the future adversely affect the results of these activities. While increased volatility can increase trading volumes and spreads, it also increases risk as measured by Value-at-Risk (VaR) and increases risks in connection with our market-making activities and can cause us to reduce our inventory. Limiting the size of our market-making positions can adversely affect our profitability. In periods when volatility is increasing, but asset values are declining significantly, it may not be possible to sell assets at all or it may only be possible to do so at steep discounts. In those circumstances, we have been and may in the future be forced to either take on additional risk or to realize losses in order to decrease our VaR. In addition, increases in volatility increase the level of our RWAs, which increases the amount of capital that we are required to hold, and this can reduce our profitability and reduce our ability to distribute capital to our shareholders. For example, in August 2024, market volatility increased significantly, which adversely affected activity levels, increased our market RWAs and adversely impacted our results on some days.

Our investment banking, client intermediation, asset management and wealth management businesses have been adversely affected and may in the future be adversely affected by market uncertainty or lack of confidence among investors and CEOs due to declines in economic activity and other unfavorable economic, geopolitical or market conditions.

Our investment banking business has been and may in the future be adversely affected by market conditions. Poor economic conditions and other uncertain geopolitical conditions may adversely affect and have in the past adversely affected investor and CEO confidence, resulting in significant industry-wide declines in the size and number of underwritings and of advisory transactions, which would likely have, and have in the past had, an adverse effect on our revenues and our profit margins. In particular, because a significant portion of our investment banking revenues is derived from our participation in large transactions, a decline in the number of large transactions has in the past and would in the future adversely affect our investment banking business. Similarly, in recent years, cross-border initial public offerings and other securities offerings have accounted for a significant proportion of new issuance activity. Legislative, regulatory or other changes that limit trading in, or the issuance of, securities outside the issuers’ domestic markets, that result in or could result in the delisting or removal of securities from exchanges or indices, have in the past adversely affected and would in the future adversely affect our underwriting and client intermediation businesses. Furthermore, changes, or proposed changes, to international trade and investment policies of the U.S. and other countries could negatively affect market activity levels and our revenues.

In certain circumstances, market uncertainty or general declines in market or economic activity may adversely affect our client intermediation businesses by decreasing levels of overall activity or by decreasing volatility.

Market uncertainty, volatility and adverse economic conditions, as well as declines in asset values, may cause our clients to transfer their assets out of our funds or other products or their brokerage accounts and result in reduced net revenues, principally in our asset management and wealth management businesses. Even if clients do not withdraw their funds, they may invest them in products that generate less fee income.

Our asset management and wealth management businesses have been and may in the future be adversely affected by the poor investment performance of our investment products or a client preference for products other than those which we offer or for products that generate lower fees.

Poor investment returns in our asset management and wealth management businesses, due to either general market conditions or underperformance (relative to our competitors or to benchmarks) by funds or accounts that we manage or investment products that we design or sell, affect our ability to retain existing assets and to attract new clients or additional assets from existing clients. This could affect the management and incentive fees that we earn on assets under supervision (AUS) or the commissions and net spreads that we earn for selling other investment products. To the extent that our clients choose to invest in products that we do not currently offer, we will suffer outflows and a loss of management fees. Further, if, due to changes in investor sentiment or the relative performance of certain asset classes or otherwise, clients continue to invest in products that generate lower fees (e.g., passively managed or fixed income products), our average effective management fee will decline further and our asset management and wealth management businesses could be adversely affected.

Inflation has had, and could continue to have, a negative effect on our business, results of operations and financial condition.

Inflationary pressures in recent years have affected economies, financial markets and market participants worldwide. Inflationary pressures in recent years have increased certain of our operating expenses, and have adversely affected consumer sentiment and CEO confidence. Central bank responses to inflationary pressures in recent years have also resulted in higher market interest rates, which, in turn, have contributed to lower activity levels across financial markets, in particular for debt underwriting transactions and mortgage originations, and resulted in lower values for certain financial assets which have adversely affected our equity and debt investments. Higher interest rates increase our borrowing costs and have in recent years required us to increase interest paid on our deposits. If inflationary pressures increase, our expenses may increase; we may be unable to achieve our efficiency ratio target; activity levels for certain of our businesses, in particular debt underwriting and mortgages, may decline; our interest expense could increase faster than our interest income, reducing our net interest income and net interest margin; certain of our investments could incur losses or generally low levels of returns; AUS could decline, or the composition of our AUS could shift to lower fee products, reducing management and other fees; economies worldwide could experience recessions; and we could continue to operate in a generally unfavorable economic and market environment.

Liquidity

Our liquidity, profitability and businesses may be adversely affected by an inability to access the debt capital markets or to sell assets.

Liquidity is essential to our businesses. It is of critical importance to us, as most of the failures of financial institutions have occurred in large part due to insufficient liquidity. Our liquidity may be impaired by an inability to access secured and/or unsecured debt markets, an inability to raise or retain deposits, an inability to access funds from our subsidiaries or otherwise allocate liquidity optimally, an inability to sell assets or redeem our investments, lack of timely settlement of transactions, unusual deposit outflows, or other unforeseen outflows of cash or collateral. This situation may arise due to circumstances that we may be unable to control, such as a general market or economic disruption or an operational problem that affects third parties or us, or even by the perception among market participants that we, or other market participants, are experiencing greater liquidity risk.

We employ structured products to benefit our clients and hedge our own risks. The financial instruments that we hold and the contracts to which we are a party are often complex, and these complex structured products often do not have readily available markets to access in times of liquidity stress. Our investing and financing activities may lead to situations where the holdings from these activities represent a significant portion of specific markets, which could restrict liquidity for our positions.

Further, our ability to sell assets may be impaired if there is not generally a liquid market for such assets, as well as in circumstances where other market participants are seeking to sell similar otherwise generally liquid assets at the same time, as is likely to occur in a liquidity or other market crisis or in response to changes to rules or regulations. For example, in 2021, an investment management firm with large positions with several financial institutions defaulted, resulting in rapidly declining prices in the securities underlying those positions. In addition, clearinghouses, exchanges and other financial institutions with which we interact may exercise set-off rights or the right to require additional collateral, including in difficult market conditions, which could further impair our liquidity.

Numerous regulations impose stringent liquidity requirements on large financial institutions, including us. These regulations require us to hold large amounts of highly liquid assets and reduce our flexibility to source and deploy funding. In addition, our need to manage our operations in light of certain regulatory requirements when applicable thresholds are met has in the past limited and may in the future limit our ability to raise deposits in GSIB or other funding, which could adversely affect our liquidity or ability to respond efficiently to liquidity stress.

Our businesses have been and may in the future be adversely affected by disruptions or lack of liquidity in the credit markets, including reduced access to credit and higher costs of obtaining credit.

Widening credit spreads, as well as significant declines in the availability of credit, have in the past adversely affected our ability to borrow on a secured and unsecured basis and may do so in the future. We fund ourselves on an unsecured basis by primarily issuing long-term debt and commercial paper, by raising deposits at our bank subsidiaries, by issuing hybrid financial instruments and by obtaining loans or lines of credit from commercial or other banking entities. We seek to finance many of our assets on a secured basis. Any disruptions in the credit markets may make it harder and more expensive to obtain funding for our businesses. If our available funding is limited or we are forced to fund our operations at a higher cost, these conditions may require us to curtail our business activities and increase our cost of funding, both of which could reduce our profitability, particularly in our businesses that involve investing, lending and market making.

Our clients engaging in mergers, acquisitions and other types of strategic transactions often rely on access to the secured and unsecured credit markets to finance their transactions. A lack of available credit or an increased cost of credit can adversely affect the size, volume and timing of our clients' mergers and acquisitions transactions, particularly large transactions, and adversely affect our advisory and underwriting businesses.

Our credit businesses have been and may in the future be negatively affected by a lack of liquidity in credit markets. A lack of liquidity reduces price transparency, increases price volatility and decreases transaction volumes and size, all of which can increase transaction risk or decrease the profitability of these businesses.

Reductions in our credit ratings or an increase in our credit spreads may adversely affect our liquidity and cost of funding.

Our credit ratings are important to our liquidity. A reduction in our credit ratings could adversely affect our liquidity and competitive position, increase our borrowing costs, limit our access to the capital markets or trigger our obligations under certain provisions in some of our trading and collateralized financing contracts. Under these provisions, counterparties could be permitted to terminate contracts with us or require us to post additional collateral. Termination of our trading and collateralized financing contracts could cause us to sustain losses and impair our liquidity by requiring us to find other sources of financing or to make significant cash payments or securities movements.

As of December 2024, our counterparties could have called for additional collateral or termination payments related to our net derivative liabilities under bilateral agreements in an aggregate amount of \$315 million in the event of a one-notch downgrade of our credit ratings and \$1.20 billion in the event of a two-notch downgrade of our credit ratings. A downgrade by any one rating agency, depending on the agency's relative ratings of us at the time of the downgrade, may have an impact which is comparable to the impact of a downgrade by all rating agencies. For further information about our credit ratings, see "Management's Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Liquidity Risk Management — Credit Ratings" in Part II, Item 7 of this Form 10-K.

Our cost of obtaining long-term unsecured funding is directly related to our credit spreads (the amount in excess of the interest rate of benchmark securities that we need to pay). Increases in our credit spreads can significantly increase our cost of this funding. Changes in credit spreads are continuous, market-driven, and subject at times to unpredictable and highly volatile movements. Our credit spreads are also influenced by market perceptions of our creditworthiness and movements in the costs to purchasers of credit default swaps referenced to our long-term debt. The market for credit default swaps has proven to be extremely volatile and at times has lacked a high degree of transparency or liquidity.

Group Inc. is a holding company and its liquidity depends on payments and loans from its subsidiaries, many of which are subject to legal, regulatory and other restrictions on providing funds or assets to Group Inc.

Group Inc. is a holding company and, therefore, depends on dividends, distributions, loans and other payments from its subsidiaries to fund share repurchases and dividend payments and to fund payments on its obligations, including debt obligations. Many of our subsidiaries, including our broker-dealer and bank subsidiaries, are subject to laws that restrict dividend payments or authorize regulatory bodies to block or reduce the flow of funds from those subsidiaries to Group Inc.

In addition, our broker-dealer and bank entities and their subsidiaries are subject to restrictions on their ability to lend or transact with affiliates and to minimum regulatory capital and other requirements, as well as restrictions on their ability to use funds deposited with them in brokerage or bank accounts to fund their businesses. Additional restrictions on related-party transactions, increased capital and liquidity requirements and additional limitations on the use of funds on deposit in bank or brokerage accounts, as well as lower earnings, can reduce the amount of funds available to meet the obligations of Group Inc., including under the FRB's source of strength requirement, and even require Group Inc. to provide additional funding to such subsidiaries. Restrictions or regulatory action of that kind could impede access to funds that Group Inc. needs to make payments on its obligations, including debt obligations, or dividend payments. In addition, Group Inc.'s right to participate in a distribution of assets upon a subsidiary's liquidation or reorganization is subject to the prior claims of the subsidiary's creditors.

There has been a trend towards increased regulation and supervision of our branches and subsidiaries by the governments and regulators in the countries in which those branches and subsidiaries are located or do business. Concerns about protecting clients and creditors of branches and subsidiaries of financial institutions that are located outside of the country in which such branches or subsidiaries are located or do business have caused or may cause a number of governments and regulators to take additional steps to "ring fence" or require internal total loss-absorbing capacity (which may also be subject to "bail-in" powers, as described below) at those branches and subsidiaries in order to protect clients and creditors of those branches and subsidiaries in the event of financial difficulties involving those branches and subsidiaries. The result has been and may continue to be additional limitations on our ability to efficiently move capital and liquidity among our affiliated entities, or to Group Inc., including in times of stress, thereby increasing the overall level of capital and liquidity required by us on a consolidated basis.

Furthermore, Group Inc. has guaranteed the payment obligations of certain of its subsidiaries, including GS&Co. and GS Bank USA, subject to certain exceptions. In addition, Group Inc. guarantees many of the obligations of its other consolidated subsidiaries on a transaction-by-transaction basis, as negotiated with counterparties. These guarantees may require Group Inc. to provide substantial funds or assets to its subsidiaries or their creditors or counterparties at a time when Group Inc. is in need of liquidity to fund its own obligations.

The requirements for us and certain of our subsidiaries to develop and submit recovery and resolution plans to regulators, and the incorporation of feedback received from regulators, may require us to increase capital or liquidity levels or issue additional long-term debt at Group Inc. or particular subsidiaries or otherwise incur additional or duplicative operational or other costs at multiple entities, and may reduce our ability to provide Group Inc. guarantees of the obligations of our subsidiaries or raise debt at Group Inc. Resolution planning may also impair our ability to structure our intercompany and external activities in a manner that we may otherwise deem most operationally efficient. Furthermore, arrangements to facilitate our resolution planning may cause us to be subject to additional taxes. Any such limitations or requirements would be in addition to the legal and regulatory restrictions described above on our ability to engage in capital actions or make intercompany dividends or payments.

See "Business — Regulation" in Part I, Item 1 of this Form 10-K for further information about regulatory restrictions.

Credit

Our businesses, profitability and liquidity may be adversely affected by deterioration in the credit quality of or defaults by third parties.

We are exposed to the risk that third parties that owe us money, securities or other assets will not perform their obligations. These parties may default on their obligations to us due to bankruptcy, lack of liquidity, operational failure or other reasons. A failure of a significant market participant, or even concerns about a default by such an institution, has in the past and could in the future lead to significant liquidity problems, losses or defaults by other institutions, which in turn could adversely affect us. We are also exposed to the risk of a special assessment, including under the FDIA or OLA in the event of the failure of a bank or non-bank financial institution, which have in the past, and may in the future, adversely affect our results of operations.

We are also subject to the risk that our rights against third parties may not be enforceable in all circumstances. In addition, deterioration in the credit quality of third parties whose securities or obligations we hold, including a deterioration in the value of collateral posted by third parties to secure their obligations to us, including under derivative contracts and loan agreements, could result in losses and/or adversely affect our ability to rehypothecate or otherwise use those securities or obligations for liquidity purposes.

A significant downgrade in the credit ratings of our counterparties could also have a negative impact on our results. While in many cases we are permitted to require additional collateral from counterparties that experience financial difficulty, disputes may arise as to the amount of collateral we are entitled to receive and the value of pledged assets. The termination of contracts and the foreclosure on collateral may subject us to claims for the improper exercise of our rights. Default rates, downgrades and disputes with counterparties as to the valuation of collateral typically increase significantly in times of market stress, increased volatility and illiquidity.

As part of our clearing and prime financing activities, we finance our clients' positions, and we could be held responsible for the defaults or misconduct of our clients. Default risk may arise from events or circumstances that are difficult to detect or foresee.

Concentration of risk increases the potential for significant losses in our market-making, underwriting, investing and financing activities.

Concentration of risk increases the potential for significant losses in our market-making, underwriting, investing and financing activities. The number and size of these transactions has affected and may in the future affect our results of operations in a given period. Moreover, because of concentrated risk, we may suffer losses even when economic and market conditions are generally favorable for our competitors. Disruptions in the credit markets can make it difficult to hedge these credit exposures effectively or economically. In addition, we extend large commitments as part of our credit origination activities. Disruptions in the credit markets have in the past substantially curtailed or eliminated, and may in the future substantially curtail or eliminate, the trading markets for loans we originate. These disruptions have in the past made, and may in the future make, it difficult for us to sell or value such assets, which have in the past resulted, and may in the future result, in losses for us.

Rules adopted under the Dodd-Frank Act, and similar rules adopted in other jurisdictions, require issuers of certain asset-backed securities and any person who organizes and initiates certain asset-backed securities transactions to retain economic exposure to the asset, which has affected the cost of and structures used in connection with these securitization activities. Our inability to reduce our credit risk by selling, syndicating or securitizing these positions, including during periods of market stress, has in the past negatively affected and may in the future negatively affect our results of operations due to a decrease in the fair value of the positions, including due to the insolvency or bankruptcy of borrowers, as well as the loss of revenues associated with selling such securities or loans.

In the ordinary course of business, we are at times subject to a concentration of credit risk to a particular counterparty, borrower, issuer (including sovereign issuers) or geographic area or group of related countries, such as the E.U., and a failure or downgrade of, or default by, such entity could negatively impact our businesses, perhaps materially, and the systems by which we set limits and monitor the level of our credit exposure to individual entities, industries, countries and regions may not function as we have anticipated. Regulatory reform, including the Dodd-Frank Act, has led to increased centralization of trading activity through particular clearinghouses, agent banks or exchanges, which has significantly increased our concentration of risk with respect to these entities. While our activities expose us to many different industries, counterparties and countries, we routinely execute a high volume of transactions with counterparties engaged in financial services activities, including brokers and dealers, commercial banks, clearinghouses, exchanges and investment funds. This has resulted in significant credit concentration with respect to these counterparties.

Derivative transactions and delayed documentation or settlements expose us to credit risk, unexpected risks and potential losses.

We are party to a large number of derivative transactions, including credit derivatives. Many of these derivative instruments are individually negotiated and non-standardized, which can make exiting, transferring or settling positions difficult. Many credit derivatives require that we deliver to the counterparty the underlying security, loan or other obligation in order to receive payment. In a number of cases, we do not hold the underlying security, loan or other obligation and may not be able to obtain the underlying security, loan or other obligation. This could cause us to forfeit the payments due to us under these contracts or result in settlement delays with the attendant credit and operational risk, as well as increased costs to us.

Derivative transactions also involve the risk that documentation has not been properly executed, that executed agreements may not be enforceable against the counterparty, or that obligations under such agreements may not be able to be “netted” against other obligations with such counterparty. In addition, counterparties may claim that such transactions were not appropriate or authorized.

As a signatory to the ISDA Universal Protocol or U.S. ISDA Protocol (ISDA Protocols) and being subject to the FRB’s and FDIC’s rules on QFCs and similar rules in other jurisdictions, we may not be able to exercise remedies against counterparties and, as this regime has not yet been tested, we may suffer risks or losses that we would not have expected to suffer if we could immediately close out transactions upon a termination event. The ISDA Protocols and these rules and regulations extend to repurchase agreements and other instruments that are not derivative contracts.

Derivative contracts and other transactions, including secondary bank loan purchases and sales, entered into with third parties are not always confirmed by the counterparties or settled on a timely basis. While the transaction remains unconfirmed or during any delay in settlement, we are subject to heightened credit and operational risk and in the event of a default may find it more difficult to enforce our rights.

In addition, as new complex derivative products are created, covering a wider array of underlying credit and other instruments, disputes about the terms of the underlying contracts could arise, which could impair our ability to effectively manage our risk exposures from these products and subject us to increased costs. The provisions of the Dodd-Frank Act requiring central clearing of credit derivatives and other OTC derivatives, or a market shift toward standardized derivatives, could reduce the risk associated with these transactions, but under certain circumstances could also limit our ability to develop derivatives that best suit the needs of our clients and to hedge our own risks, and could adversely affect our profitability. In addition, these provisions have increased our credit exposure to central clearing platforms.

Operational

A failure in our or third-party operational systems or human error, malfeasance or other misconduct, could impair our liquidity, disrupt our businesses, result in the disclosure of confidential information, damage our reputation and cause losses.

Our businesses are highly dependent on our ability to process and monitor, on a daily basis, a very large number of transactions, many of which are highly complex and occur at high volumes and frequencies, across numerous and diverse markets in many currencies. These transactions, as well as the information technology services we provide to clients, often must adhere to client-specific guidelines, as well as legal and regulatory standards.

Many rules and regulations worldwide govern our obligations to execute transactions and report transactions and other information to regulators, exchanges and investors. Compliance with these legal and reporting requirements can be challenging, and we have been and may in the future be subject to regulatory fines and penalties for failing to follow these rules or to report timely, accurate and complete information in accordance with these rules.

As the volume, speed, frequency and complexity of transactions, especially electronic transactions (as well as the requirements to report such transactions on a real-time basis to clients, regulators and exchanges) increase, developing and maintaining our operational systems and infrastructure has become more challenging, and the risk of systems or human error by us or our third-party service providers in connection with such transactions has increased, as have the potential consequences of errors due to the speed and volume of transactions involved and the potential difficulty associated with discovering errors quickly enough to limit the resulting consequences. For example, the transition to a T+1 settlement timeframe in the U.S. in 2024 has subjected us to, and will continue to subject us to, increased operational risks with respect to reporting and timely settlement of transactions. These risks are exacerbated in times of increased volatility. As with other similarly situated institutions, we utilize credit underwriting models in connection with our businesses, including our consumer-oriented activities. Allegations or publicity, whether or not accurate, that our underwriting decisions do not treat consumers or clients fairly, or comply with the applicable law or regulation, have in the past resulted and may in the future result in negative publicity, reputational damage and governmental and regulatory scrutiny, investigations and enforcement actions.

Our financial, accounting, data processing or other operational systems and facilities have in the past not operated properly in certain respects and may in the future not operate properly or become disabled as a result of events that are wholly or partially beyond our control, such as a spike in transaction volume or an operational disruption at a third-party service provider, adversely affecting our ability to process these transactions or provide these services. We must continuously update our systems to support our operations and growth and to respond to changes in regulations and markets, and invest heavily in systemic controls and training to pursue our objective of ensuring that such transactions do not violate applicable rules and regulations or, due to errors in processing such transactions, adversely affect markets, our clients and counterparties or us. Enhancements and updates to systems, as well as the requisite training, including in connection with the integration of new businesses, entail significant costs and create risks associated with implementing new systems and integrating them with existing ones.

The use of computing devices, phones and other mobile devices is critical to the work done by our employees and the operation of our systems and businesses and those of our clients and our third-party service providers and vendors. Their importance has continued to increase, for both our regular operations and business continuity plans. Computers and computer networks are subject to various risks, including, among others, cyber attacks, inherent technological defects, system disruptions and failures and human error. For example, fundamental security flaws in computer chips found in many types of these computing devices and phones have been reported in the past and may occur in the future, and in July 2024 there was a widely publicized information technology outage as a result of a faulty update to a cybersecurity software product that affected many businesses worldwide. The use of personal devices by our employees or by our vendors for work-related activities also presents risks related to potential violations of record retention and other requirements. Cloud technologies are also critical to the operation of our systems and platforms and our reliance on cloud technologies is growing. Service disruptions have resulted, and may result in the future, in delays in accessing, or the loss of, data that is important to our businesses and may hinder our clients' access to our platforms. There have been a number of widely publicized cases of outages in connection with access to cloud computing providers. Addressing these and similar issues could be costly and affect the performance of these businesses and systems. Applying fixes can introduce operational risks, and, despite the fixes, there may still be residual security risks.

Notwithstanding the proliferation of technology and technology-based risk and control systems, our businesses ultimately rely on people as our greatest resource, and, from time to time, they have in the past and may in the future make mistakes or engage in violations of applicable policies, laws, rules or procedures that are not always caught immediately by our technological processes or by our controls and other procedures, which are intended to prevent and detect such errors or violations. These have in the past and may in the future include calculation errors, mistakes in addressing emails, errors in software or model development or implementation, or simple errors in judgment, as well as intentional efforts to ignore or circumvent applicable policies, laws, rules or procedures. Human errors, malfeasance and other misconduct, including the intentional misuse of client information in connection with insider trading or for other purposes, even if promptly discovered and remediated, has in the past resulted and may in the future result in reputational damage and losses and liabilities for us.

The majority of the employees in our primary locations, including the New York metropolitan area, London, Bengaluru, Hyderabad, Hong Kong, Tokyo, Salt Lake City, Dallas, Singapore, Warsaw and Birmingham, work in close proximity to one another. Our headquarters is located in the New York metropolitan area, and we have our largest employee concentration occupying two principal office buildings near the Hudson River waterfront. They are subject to potential catastrophic events, including, but not limited to, terrorist attacks, extreme weather, or other hostile events that could negatively affect our business. Notwithstanding our efforts to maintain business continuity, business disruptions impacting our offices and employees could lead to our employees' inability to occupy the offices, communicate with or travel to other office locations or work remotely. As a result, our ability to service and interact with clients may be adversely impacted, due to our failure or inability to successfully implement business contingency plans.

A failure or disruption in our infrastructure, or in the operational systems or infrastructure of third parties, could impair our liquidity, disrupt our businesses, damage our reputation and cause losses.

We face the risk of operational failure or significant operational delay, termination or capacity constraints of any of the clearing agents, exchanges, clearinghouses or other financial intermediaries we use to facilitate our securities and derivatives transactions, and as our interconnectivity with our clients grows, we increasingly face the risk of operational failure or significant operational delay with respect to our clients' systems.

There has been significant consolidation among clearing agents, exchanges and clearinghouses and an increasing number of derivative transactions are cleared on exchanges, which has increased our exposure to operational failure or significant operational delay, termination or capacity constraints of the particular financial intermediaries that we use and could affect our ability to find adequate and cost-effective alternatives in the event of any such failure, delay, termination or constraint. Industry consolidation, whether among market participants or financial intermediaries, increases the risk of operational failure or significant operational delay as disparate complex systems need to be integrated, often on an accelerated basis.

The interconnectivity of multiple financial institutions with agent banks, exchanges and clearinghouses, and the increased centrality of these entities, increases the risk that an operational failure at one institution or entity may cause an industry-wide operational failure that could materially impact our ability to conduct business. Interconnectivity of financial institutions with other companies through, among other things, application programming interfaces or APIs presents similar risks. Any such failure, termination or constraint could adversely affect our ability to effect transactions, service our clients, manage our exposure to risk or expand our businesses or result in financial loss or liability to our clients, impairment of our liquidity, disruption of our businesses, regulatory intervention or reputational damage.

Despite our resiliency plans and facilities, our ability to conduct business may be adversely impacted by a disruption in the infrastructure that supports our businesses and the communities where we are located. This may include a disruption involving electrical, satellite, undersea cable or other communications, internet, transportation or other facilities used by us, our employees or third parties with which we conduct business, including cloud service providers. These disruptions may occur as a result of events that affect only our buildings or systems or those of third parties, or as a result of events with a broader impact globally, regionally or in the cities where those buildings or systems are located, including, but not limited to, natural disasters, war, civil unrest, terrorism, economic or political developments, pandemics and weather events.

In addition, although we seek to diversify our third-party vendors to increase our resiliency, we are exposed to risks if our vendors operate in the same area and are also exposed to the risk that a disruption or other information technology event at a common service provider to our vendors could impede their ability to provide products or services to us. We may not be able to effectively monitor or mitigate operational risks relating to our vendors' use of common service providers.

Additionally, although the prevalence and scope of applications of distributed ledger technology, cryptocurrency and similar technologies is growing, the technology is nascent and may be vulnerable to cyber attacks or have other inherent weaknesses. We are exposed to risks, and may become exposed to additional risks, related to distributed ledger technology, including through our facilitation of clients' activities involving financial products that use distributed ledger technology, such as blockchain, cryptocurrencies or other digital assets, our investments in companies that seek to develop platforms based on distributed ledger technology, the use of distributed ledger technology by third-party vendors, clients, counterparties, clearinghouses and other financial intermediaries, and the receipt of cryptocurrencies or other digital assets as collateral. Market volatility of financial products using distributed ledger technology may increase these risks.

The development and use of AI present risks and challenges that may adversely impact our business.

We or our third-party vendors, clients or counterparties have in the past developed or incorporated, and may in the future develop or incorporate, AI technology in certain business processes, services or products. The development and use of AI present a number of risks and challenges to our business. The legal and regulatory environment relating to AI is uncertain and rapidly evolving, both in the U.S. and internationally, and includes regulation targeted specifically at AI as well as provisions in intellectual property, privacy, consumer protection, employment and other laws applicable to the use of AI. These evolving laws and regulations could require changes in our implementation of AI technology and increase our compliance costs and the risk of non-compliance. AI models, particularly generative AI models, may produce output or take action that is incorrect, that result in the release of private, confidential or proprietary information, that reflect biases included in the data on which they are trained, infringe on the intellectual property rights of others, or that is otherwise harmful. In addition, the complexity of many AI models makes it challenging to understand why they are generating particular outputs. This limited transparency increases the challenges associated with assessing the proper operation of AI models, understanding and monitoring the capabilities of the AI models, reducing erroneous output, eliminating bias and complying with regulations that require documentation or explanation of the basis on which decisions are made. Further, we may rely on AI models developed by third parties, and, to that extent, would be dependent in part on the manner in which those third parties develop and train their models, including risks arising from the inclusion of any unauthorized material in the training data for their models, and the effectiveness of the steps these third parties have taken to limit the risks associated with the output of their models, matters over which we may have limited visibility. Additionally, we are exposed to risks related to the use of AI technologies by third-party vendors, clients, counterparties, clearinghouses and other financial intermediaries. Any of these risks could expose us to liability or adverse legal or regulatory consequences and harm our reputation and the public perception of our business or the effectiveness of our security measures.

In addition to our use of AI technologies, we are exposed to risks arising from the use of AI technologies by bad actors to commit fraud and misappropriate funds and to facilitate cyberattacks. Generative AI, if used to perpetrate fraud or launch cyberattacks, could result in losses, liquidity outflows or other adverse effects at a particular financial institution or exchange.

A failure to protect our computer systems, networks and information, and our clients' information, against cyber attacks and similar threats could impair our ability to conduct our businesses, result in the disclosure, theft or destruction of confidential information, damage our reputation and cause losses.

Our operations rely on the secure processing, storage and transmission of confidential and other information in our computer systems and networks and those of our vendors. There have been a number of highly publicized cases involving financial services companies, consumer-based companies, software and information technology service providers, governmental agencies and other organizations reporting the unauthorized access or disclosure of client, customer or other confidential information in recent years, as well as cyber attacks involving the dissemination, theft and destruction of corporate information or other assets, as a result of inadequate procedures or the failure to follow procedures by employees or contractors or as a result of actions by third parties, including actions by foreign governments. There have also been a number of highly publicized cases where hackers have requested "ransom" payments in exchange for not disclosing customer information or for restoring access to information or systems.

We are regularly the target of attempted cyber attacks, including denial-of-service attacks, and must continuously monitor and develop our systems to protect the integrity and functionality of our technology infrastructure and access to and the security of our data. We have faced a high volume of cyber attacks as we expand our mobile- and other internet-based products and services, as well as our usage of mobile and cloud technologies, and as we provide these services to individual consumers. Further, the use of AI by cybercriminals may increase the frequency and severity of cybersecurity attacks against us or our third-party vendors and clients. The use of employee-owned devices presents additional risks of cyber attacks, as do hybrid work arrangements. In addition, due to our interconnectivity with third-party vendors (and their respective service providers), agent banks, exchanges, clearinghouses and other financial institutions, we could be adversely impacted if any of them is subject to a successful cyber attack or other information security event. These impacts could include the loss of access to information or services from the third party subject to the cyber attack or other information security event or could result in unauthorized access to or disclosure of client, customer or other confidential information, which could, in turn, interrupt certain of our businesses or adversely affect our results of operations and reputation.

Despite our efforts to ensure the integrity of our systems and information, we may not be able to anticipate, detect or implement effective preventive measures against all cyber threats, including because the techniques used are increasingly sophisticated, change frequently and are often not recognized until launched. Cyber attacks can originate from a variety of sources, including third parties who are affiliated with or sponsored by foreign governments or are involved with organized crime or terrorist organizations. Third parties may also attempt to place individuals in our offices or induce employees, clients or other users of our systems to disclose sensitive information or provide access to our data or that of our clients, and these types of risks may be difficult to detect or prevent.

Although we take protective measures proactively and endeavor to modify them as circumstances warrant, our computer systems, software and networks may be vulnerable to unauthorized access, misuse, computer viruses or other malicious code, cyber attacks on our vendors and other events that could have a security impact. Risks relating to cyber attacks on our vendors have been increasing given the greater frequency and severity in recent years of supply chain attacks affecting software and information technology service providers. Due to the complexity and interconnectedness of our systems, the process of enhancing our protective measures can itself create a risk of systems disruptions and security issues. In addition, protective measures that we employ to compartmentalize our data may reduce our visibility into, and adversely affect our ability to respond to, cyber threats and issues with our systems.

If one or more of these types of events occur, it potentially could jeopardize our, our clients', our counterparties' or third parties' confidential and other information processed, stored in, or transmitted through our computer systems and networks, or otherwise cause interruptions or malfunctions in our operations or those of our clients, counterparties or third parties, which could impact their ability to transact with us or otherwise result in legal or regulatory action, significant losses or reputational damage. In addition, such an event could persist for an extended period of time before being properly detected or escalated, and, following detection or escalation, it could take considerable time for us to obtain full and reliable information about the extent, amount and type of information compromised. During the course of an investigation, we may not know the full impact of the event and how to remediate it, and actions, decisions and mistakes that are taken or made may further increase the negative effects of the event on our business, results of operations and reputation. Moreover, regulations require us to disclose information on a timely basis about material cybersecurity incidents, including those that may not have been resolved or fully investigated at the time of disclosure.

We have expended, and expect to continue to expend, significant resources on an ongoing basis to modify our protective measures and to investigate and remediate vulnerabilities or other exposures, but these measures may be ineffective and we may be subject to legal or regulatory action, as well as financial losses that are either not insured against or not fully covered through any insurance maintained by us. Regulatory agencies have become increasingly focused on cybersecurity incidents.

Our clients' confidential information may also be at risk from the compromise of clients' accounts, including as a result of a data security breach at an unrelated company. Losses due to unauthorized account activity could harm our reputation and may have adverse effects on our business, financial condition and results of operations.

The increased use of mobile and cloud technologies heightens these and other operational risks, as do hybrid work arrangements. Certain aspects of the security of these technologies are unpredictable or beyond our control, and the failure by mobile technology and cloud service providers to adequately safeguard their systems and prevent cyber attacks could disrupt our operations and result in misappropriation, corruption or loss of confidential and other information. In addition, there is a risk that encryption and other protective measures, despite their sophistication, may be defeated, particularly to the extent that new computing technologies, such as quantum computing, vastly increase the speed and computing power available.

We routinely transmit and receive personal, confidential and proprietary information by email and other electronic means. We have discussed and worked with clients, vendors, service providers, counterparties and other third parties to develop secure transmission capabilities and protect against cyber attacks, but we do not have, and may be unable to put in place, secure capabilities with all of our clients, vendors, service providers, counterparties and other third parties and we may not be able to ensure that these third parties have appropriate controls in place to protect the confidentiality of the information. An interception, misuse or mishandling of personal, confidential or proprietary information being sent to or received from a client, vendor, service provider, counterparty or other third party could result in legal liability, regulatory action and reputational harm.

We have in the past incurred and may in the future incur losses as a result of ineffective risk management processes and strategies.

We seek to monitor and control our risk exposure through a risk and control framework encompassing a variety of separate but complementary financial, credit, operational, compliance and legal systems, internal controls, management review processes and other mechanisms. Our risk management process seeks to balance our ability to profit from market-making, investing or lending positions, and underwriting activities, with our exposure to potential losses. While we employ a broad and diversified set of risk monitoring and risk mitigation techniques, those techniques and the judgments that accompany their application cannot anticipate every economic and financial outcome or the specifics and timing of such outcomes. Thus, in the course of our activities, we have incurred and may in the future incur losses. Market conditions in recent years have involved unprecedented dislocations and highlight the limitations inherent in using historical data to manage risk.

The models that we use to assess and control our risk exposures reflect assumptions about the degrees of correlation or lack thereof among prices of various asset classes or other market indicators. In times of market stress or other unforeseen circumstances, previously uncorrelated indicators may become correlated, or conversely previously correlated indicators may move in different directions. These types of market movements have at times limited the effectiveness of our hedging strategies and have caused us to incur significant losses, and they may do so in the future. These changes in correlation have been and may in the future be exacerbated where other market participants are using risk or trading models with assumptions or algorithms that are similar to ours. In these and other cases, it may be difficult to reduce our risk positions due to the activity of other market participants or widespread market dislocations, including circumstances where asset values are declining significantly or no market exists for certain assets.

In addition, the use of models in connection with risk management and numerous other critical activities presents risks that the models may be ineffective, either because of poor design, ineffective testing, or improper or flawed inputs, as well as unpermitted access to the models resulting in unapproved or malicious changes to the model or its inputs.

To the extent that we have positions through our market-making or origination activities or we make investments directly through our investing activities, including private equity or private credit, that do not have an established liquid trading market or are otherwise subject to restrictions on sale or hedging, we may not be able to reduce our positions and therefore reduce our risk associated with those positions. In addition, to the extent permitted by applicable law and regulation, we invest our own capital in private equity, credit, real estate and hedge funds that we manage and limitations on our ability to withdraw some or all of our investments in these funds, whether for legal, reputational or other reasons, may make it more difficult for us to control the risk exposures relating to these investments.

Prudent risk management, as well as regulatory restrictions, may cause us to limit our exposure to counterparties, geographic areas or markets, which may limit our business opportunities and increase the cost of our funding or hedging activities.

Our consumer offerings present us with different risks, and we have needed and continue to need to expand and adapt our risk monitoring and mitigation activities to account for these business activities. A failure to adequately assess and control such risk exposures has in the past resulted and could in the future result in losses to us.

For further information about our risk management policies and procedures, see “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Risk Management” in Part II, Item 7 of this Form 10-K.

Legal and Regulatory

Our businesses and those of our clients are subject to extensive and pervasive regulation around the world.

As a participant in the financial services industry and a systemically important financial institution, we are subject to extensive regulation in jurisdictions around the world. We face the risk of significant intervention by law enforcement, regulatory and taxing authorities, as well as private litigation, in all jurisdictions in which we conduct our businesses. In many cases, our activities have been and may continue to be subject to overlapping and divergent regulation in different jurisdictions. Among other things, as a result of law enforcement authorities, regulators or private parties challenging our compliance with existing laws and regulations, we or our employees have been, and could be, fined, criminally charged or sanctioned; prohibited from engaging in some of our business activities; subjected to limitations or conditions on our business activities, including higher capital requirements; or subjected to new or substantially higher taxes or other governmental charges in connection with the conduct of our businesses or with respect to our employees. These limitations or conditions may limit our business activities and negatively impact our profitability.

In addition to the impact on the scope and profitability of our business activities, day-to-day compliance with existing laws and regulations has involved and will continue to involve significant amounts of time, including that of our senior leaders and that of a large number of dedicated compliance and other reporting and operational personnel, in connection with which we expect to continue to add personnel, all of which may negatively impact our profitability.

Our revenues and profitability and those of our competitors have been and will continue to be impacted by requirements relating to capital, leverage, liquidity and long-term funding levels, requirements related to resolution and recovery planning, derivatives clearing and margin rules and levels of regulatory oversight, as well as limitations on which and, if permitted, how certain business activities may be carried out by financial institutions. The laws, regulations and accounting standards, that apply to our businesses are often complex and, in many cases, we must make interpretive decisions regarding the application of those laws, regulations and accounting standards to our business activities. Changes in interpretations, whether in response to regulatory guidance, industry conventions, our own reassessments or otherwise, could adversely affect our businesses, results of operations or ability to satisfy applicable regulatory requirements, such as capital or liquidity requirements.

If there are new laws or regulations or changes in the interpretation or enforcement of existing laws or regulations applicable to our businesses or those of our clients, including capital, liquidity, leverage, long-term debt, total loss-absorbing capacity and margin requirements, restrictions on leveraged lending or other business practices, reporting requirements, requirements relating to recovery and resolution planning, tax burdens and compensation restrictions, that are imposed on a limited subset of financial institutions (whether based on size, method of funding, activities, geography or other criteria), compliance with these new laws or regulations, or changes in the enforcement of existing laws or regulations, could adversely affect our ability to compete effectively with other institutions that are not affected in the same way. In addition, regulation imposed on financial institutions or market participants generally, such as taxes on stock transfers, share repurchases and other financial transactions, could adversely impact levels of market activity more broadly, and thus impact our businesses. Changes to laws or regulations, such as tax laws, could also have a disproportionate impact on us, based on the way those laws or regulations are applied to financial services and financial firms or due to our corporate structure or where or how we provide these services. Recent political developments have added additional uncertainty with respect to new laws and regulations or changes in the interpretations or enforcement of existing laws and regulations.

These developments could impact our profitability in the affected jurisdictions, or even make it uneconomic for us to continue to conduct all or certain of our businesses in those jurisdictions, or could cause us to incur significant costs associated with changing our business practices, restructuring our businesses, moving all or certain of our businesses and our employees to other locations or complying with applicable capital requirements, including reducing dividends or share repurchases, liquidating assets or raising capital in a manner that adversely increases our funding costs or otherwise adversely affects our shareholders and creditors.

U.S. and non-U.S. regulatory developments, in particular the Dodd-Frank Act and Basel III, have significantly altered the regulatory framework within which we operate and have adversely affected and may in the future adversely affect our profitability. Among the aspects of the Dodd-Frank Act that have affected or may in the future affect our businesses are: increased capital, liquidity and reporting requirements; limitations on activities in which we may engage; increased regulation of and restrictions on OTC derivatives markets and transactions; limitations on incentive compensation; limitations on affiliate transactions; requirements to reorganize or limit activities in connection with recovery and resolution planning; increased deposit insurance assessments; and increased standards of care for broker-dealers and investment advisers in dealing with clients. The implementation of higher capital requirements, more stringent requirements relating to liquidity, long-term debt and total loss-absorbing capacity and the prohibition on proprietary trading and the sponsorship of, or investment in, covered funds by the Volcker Rule may continue to adversely affect our profitability and competitive position, particularly where these requirements do not apply equally to our competitors. We may become subject to higher and more stringent capital and other regulatory requirements as a result of the implementation of the Basel Committee's finalization of the post-crisis regulatory capital reforms as well as future Basel Committee standards. Substantial parts of the E.U. rules implementing the Basel III Revisions became effective on January 1, 2025; the E.U. has delayed implementation of the FRTB rules to January 1, 2026; the U.K. issued near final rules implementing the Basel III revisions with a proposed effective date of January 1, 2027; and the FRB has indicated that it expects to work with the other U.S. federal bank regulatory agencies in 2025 on a revised proposal. See "Business — Regulation — Banking Supervision and Regulation — Risk-Based Capital Ratios" in Part I, Item 1 of this Form 10-K for further information about proposed and adopted regulatory requirements.

As described in "Business — Regulation — Banking Supervision and Regulation — Risk-Based Capital Ratios" in Part I, Item 1 of this Form 10-K, the SCB has replaced the capital conservation buffer under the Standardized Capital Rules and resulted in higher and more volatile Standardized capital ratio requirements. Failure to comply with these requirements could limit our ability to, among other things, repurchase shares, pay dividends and make certain discretionary compensation payments. In addition, if we are required to resubmit our capital plan, we generally may not make capital distributions, such as share repurchases or dividends, without the prior approval of the FRB. Dividends and repurchases are also subject to oversight by the FRB, which can result in limitations. Limitations on our ability to make capital distributions could, among other things, prevent us from returning capital to our shareholders and impact our return on equity. Additionally, as a G-SIB, we are subject to the G-SIB surcharge. Our G-SIB surcharge is updated annually based on financial data from the prior year. Expansion of our businesses, growth in our balance sheet and increased reliance on short-term wholesale funding have resulted in increases and in the future may result in further increases in our G-SIB surcharge and a corresponding increase in our capital requirements. Effective on January 1, 2026, our G-SIB surcharge is expected to increase from 3.0% to 3.5%. The July 2023 proposal from the FRB would introduce additional granularity in the surcharge buckets and increase the amount of financial data used in the calculation of the G-SIB surcharge based on averages over the year, as opposed to period-end values, which could increase our G-SIB surcharge.

We are also subject to laws and regulations, such as the GDPR and the California Consumer Privacy Act, relating to the privacy of the information of clients, employees or others, and any failure to comply with these laws and regulations could expose us to liability and/or reputational damage. As new privacy-related laws and regulations are implemented, the time and resources needed for us to comply with such laws and regulations, as well as our potential liability for non-compliance and reporting obligations in the case of data breaches, may significantly increase.

In addition, our businesses are increasingly subject to laws and regulations relating to surveillance, encryption and data on-shoring in the jurisdictions in which we operate. Compliance with these laws and regulations may require us to change our policies, procedures and technology for information security, which could, among other things, make us more vulnerable to cyber attacks and misappropriation, corruption or loss of information or technology.

Our consumer-oriented deposit-taking and credit card businesses subject us to numerous additional regulations in the jurisdictions in which these businesses operate. Not only are these regulations extensive, but they involve types of regulations and supervision, as well as regulatory compliance risks, that have not historically applied to us. The level of regulatory scrutiny and the scope of regulations affecting financial interactions with consumers is often much greater than that associated with doing business with institutions and high-net-worth individuals. Complying with these regulations is time-consuming, costly and presents new and increased risks.

GS Bank USA is assessed pursuant to a strategic plan for CRA compliance purposes. Any failure to comply with CRA requirements could negatively impact GS Bank USA's CRA ratings, cause reputational harm and result in limits on our ability to make future acquisitions or engage in certain new activities.

Increasingly, regulators and courts have sought to hold financial institutions liable for the misconduct of their clients where they have determined that the financial institution should have detected that the client was engaged in wrongdoing, even though the financial institution had no direct knowledge of the activities engaged in by its client. Regulators and courts have also increasingly found liability as a "control person" for activities of entities in which financial institutions or funds controlled by financial institutions have an investment, but which they do not actively manage. In addition, regulators and courts continue to seek to establish "fiduciary" obligations to counterparties to which no such duty had been thought to exist. To the extent that such efforts are successful, the cost of, and liabilities associated with, engaging in brokerage, clearing, market-making, prime financing, investing and other similar activities could increase significantly. To the extent that we have fiduciary obligations in connection with acting as a financial adviser or investment adviser or in other roles for individual, institutional, sovereign or investment fund clients, any breach, or even an alleged breach, of such obligations could have materially negative legal, regulatory and reputational consequences.

For information about the extensive regulation to which our businesses are subject, see "Business — Regulation" in Part I, Item 1 of this Form 10-K.

A failure to appropriately identify and address potential conflicts of interest has in the past and may in the future adversely affect our businesses.

Due to the broad scope of our businesses and our client base, we regularly address potential conflicts of interest, including situations where our services to a particular client or our own investments or other interests conflict, or are perceived to conflict, with the interests of that client or another client, as well as situations where one or more of our businesses have access to material non-public information that may not be shared with our other businesses and situations where we may be a creditor of an entity with which we also have an advisory or other relationship.

In addition, our status as a BHC subjects us to heightened regulation and increased regulatory scrutiny by the FRB with respect to transactions between GS Bank USA and its subsidiaries and entities that are or could be viewed as affiliates of ours and, under the Volcker Rule, transactions between us and covered funds.

We have extensive procedures and controls that are designed to identify and address conflicts of interest, including those designed to prevent the improper sharing of information among our businesses. However, appropriately identifying and dealing with conflicts of interest is complex and difficult, and our reputation, which is one of our most important assets, could be damaged and the willingness of clients to enter into transactions with us may be adversely affected if we fail, or appear to fail, to identify, disclose and deal appropriately with conflicts of interest. In addition, potential or perceived conflicts have in the past and may in the future give rise to litigation, government investigations or enforcement actions. Additionally, our One Goldman Sachs initiative, as well as the alignment of our businesses, aim to increase collaboration among our businesses, which may increase the potential for actual or perceived conflicts of interest and improper information sharing.

We may be adversely affected by increased governmental and regulatory scrutiny or negative publicity.

Governmental scrutiny from regulators, legislative bodies and law enforcement agencies with respect to matters relating to compensation, our business practices, our past actions and other matters remains at high levels. Political and public sentiment regarding financial institutions has in the past resulted and may in the future result in a significant amount of adverse press coverage, as well as adverse statements or charges by regulators or other government officials. Press coverage and other public statements that assert some form of wrongdoing (including, in some cases, press coverage and public statements that do not directly involve us) often result in some type of investigation by regulators, legislators and law enforcement officials or in lawsuits.

Responding to these investigations and lawsuits, regardless of the ultimate outcome of the proceeding, is time-consuming and expensive and can divert the time and effort of our senior management from our business. Penalties and fines sought by regulatory authorities have increased substantially and certain regulators have been more likely in recent years to commence enforcement actions or to support legislation targeted at the financial services industry. Governmental authorities may also be more likely to pursue criminal or other actions, including seeking admissions of wrongdoing or guilty pleas, in connection with the resolution of an inquiry or investigation to the extent a company is viewed as having previously engaged in criminal, regulatory or other misconduct. Adverse publicity, governmental scrutiny and legal and enforcement proceedings can also have a negative impact on our reputation and on the morale and performance of our employees, which could adversely affect our businesses and results of operations. Further, we are subject to regulatory settlements, orders and feedback that require significant remediation activities and enhancements to existing controls, systems and procedures, which has required and will require us to commit significant resources, including hiring, as well as testing the operation and effectiveness of new controls, policies and procedures. The failure to complete these remediation activities in a timely manner could lead to higher operating expenses, reputational damage and other negative consequences.

The financial services industry generally and our businesses in particular have been subject to negative publicity. Our reputation and businesses may be adversely affected by negative publicity or information regarding our businesses, personnel, corporate engagement programs and other initiatives, whether or not accurate or true, that may be posted on social media or other internet forums or published by news organizations. Postings on these types of forums may also adversely impact risk positions of our clients and other parties that owe us money, securities or other assets and increase the chance that they will not perform their obligations to us or reduce the revenues we receive from their use of our services. The speed and pervasiveness with which information can be disseminated through these channels, in particular social media, may magnify risks relating to negative publicity.

In 2023, the rapid dissemination of negative information through social media, in part, is believed to have led to the collapse of Silicon Valley Bank (SVB). SVB suffered a level of deposit withdrawals within a time period not previously experienced by a financial institution. We could also be subject to rapid deposit withdrawals or other outflows as a result of negative social media posts or other negative publicity.

Substantial civil or criminal liability or significant regulatory action against us has in the past had and may in the future have material adverse financial effects and significant reputational consequences, which in turn could seriously harm our business prospects.

We face significant legal risks in our businesses, and the volume of claims and amount of damages and penalties claimed in litigation and regulatory proceedings against financial institutions remain high. See Notes 18 and 27 to the consolidated financial statements in Part II, Item 8 of this Form 10-K for information about certain of our legal and regulatory proceedings and investigations. We have seen legal claims by consumers and clients increase in a market downturn and employment-related claims increase following periods in which we have reduced our headcount. Additionally, governmental entities have been plaintiffs and are parties in certain of our legal proceedings, and we may face future civil or criminal actions or claims by the same or other governmental entities, as well as follow-on civil litigation that is often commenced after regulatory settlements.

Significant settlements by large financial institutions, including, in some cases, us, with governmental entities have become common. The trend of large settlements with governmental entities may adversely affect the outcomes for other financial institutions, including, in some cases, us, in similar actions, especially where governmental officials have announced that the large settlements will be used as the basis or a template for other settlements. The uncertain regulatory enforcement environment makes it difficult to estimate probable losses, which can lead to substantial disparities between legal reserves and subsequent actual settlements or penalties.

Claims of collusion or anti-competitive conduct have become more common. Financial institutions (including us) have been subject to civil cases and investigatory demands relating to alleged bid-rigging, group boycotts or other anti-competitive practices. Antitrust laws generally provide for joint and several liability and treble damages. These claims have resulted in significant settlements and fines for us in the past and may do so in the future.

We are subject to laws and regulations worldwide, including the FCPA and the U.K. Bribery Act, relating to corrupt and illegal payments to, and hiring practices with regard to, government officials and others. Violation of these or similar laws and regulations have in the past resulted in and could in the future result in significant monetary penalties. Such violations could also result in severe restrictions on our activities and damage to our reputation.

Certain law enforcement authorities have recently required admissions of wrongdoing, and, in some cases, criminal pleas, as part of the resolutions of matters brought against financial institutions or their employees. See for example, “1MDB-Related Matters” in Note 27 to the consolidated financial statements in Part II, Item 8 of this Form 10-K. Any such resolution of a criminal matter involving us or our employees could lead to increased exposure to civil litigation, could adversely affect our reputation, could result in penalties or limitations on our ability to conduct our activities generally or in certain circumstances and could have other negative effects. Further, as a result of this type of settlement, we are no longer a “well-known seasoned issuer,” which places limitations on the manner in which we can market our securities.

In conducting our businesses around the world, we are subject to political, legal, regulatory, tax and other risks that are inherent in operating in many countries.

In conducting our businesses and supporting our global operations, we are subject to risks of possible nationalization, expropriation, price controls, capital controls, exchange controls, communications and other content restrictions, and other restrictive governmental actions. For example, sanctions have been imposed by the U.S. and the E.U. on certain individuals and companies in Russia and Venezuela. In many countries, the laws and regulations applicable to the securities and financial services industries and many of the transactions in which we are involved are uncertain and evolving, and it may be difficult for us to determine the exact requirements of local laws in every market. We have been, in some cases, subject to divergent and conflicting laws and regulations across markets, and we are increasingly subject to the risk that the jurisdictions in which we operate have implemented or may implement laws and regulations that directly conflict with those of another jurisdiction. Any determination by local regulators that we have not acted in compliance with the application of local laws in a particular market or our failure to develop effective working relationships with local regulators could have a significant and negative effect not only on our businesses in that market, but also on our reputation generally. Further, in some jurisdictions a failure, or alleged failure, to comply with laws and regulations has subjected and may in the future subject us and our personnel not only to civil actions, but also criminal actions and other sanctions. We are also subject to the enhanced risk that transactions we structure might not be legally enforceable in all cases.

While business and other practices throughout the world differ, our principal entities are subject in their operations worldwide to rules and regulations relating to corrupt and illegal payments, hiring practices and money laundering, as well as laws relating to doing business with certain individuals, groups and countries, such as the FCPA, the BSA and the U.K. Bribery Act. While we have invested and continue to invest significant resources in training and in compliance monitoring, the geographical diversity of our operations, employees, clients and consumers, as well as the vendors and other third parties that we deal with, greatly increases the risk that we may be found in violation of such rules or regulations and such violations have in the past and could in the future subject us to significant penalties or adversely affect our reputation. See for example, “1MDB-Related Matters” in Note 27 to the consolidated financial statements in Part II, Item 8 of this Form 10-K.

In addition, there have been a number of highly publicized cases around the world, involving actual or alleged fraud or other misconduct by employees in the financial services industry, and we have had and may in the future have employee misconduct. This misconduct has included and may also in the future include intentional efforts to ignore or circumvent applicable policies, rules or procedures or misappropriation of funds and the theft of proprietary information, including proprietary software. It is not always possible to deter or prevent employee misconduct and the precautions we take to prevent and detect this activity have not been and may not be effective in all cases, as reflected by the settlements relating to 1MDB.

The application of regulatory strategies and requirements in the U.S. and in non-U.S. jurisdictions to facilitate the orderly resolution of large financial institutions could create greater risk of loss for Group Inc.'s security holders.

As described in "Business — Regulation — Banking Supervision and Regulation — Insolvency of an IDI or a BHC," if the FDIC is appointed as receiver under OLA, the rights of Group Inc.'s creditors would be determined under OLA, and substantial differences exist in the rights of creditors between OLA and the U.S. Bankruptcy Code, including the right of the FDIC under OLA to disregard the strict priority of creditor claims in some circumstances, which could have a material adverse effect on our debtholders.

The FDIC has announced that a single point of entry strategy may be a desirable strategy under OLA to resolve a large financial institution in a manner that would, among other things, impose losses on shareholders, debtholders and other creditors of the top-tier BHC (in our case, Group Inc.), while the BHC's subsidiaries may continue to operate. It is possible that the application of the single point of entry strategy under OLA, in which Group Inc. would be the only entity to enter resolution proceedings (and its material broker-dealer, bank and other operating entities would not enter resolution proceedings), would result in greater losses to Group Inc.'s security holders (including holders of our fixed rate, floating rate and indexed debt securities), than the losses that would result from the application of a bankruptcy proceeding or a different resolution strategy, such as a multiple point of entry resolution strategy for Group Inc. and certain of its material subsidiaries.

Assuming Group Inc. entered resolution proceedings and support from Group Inc. or other resources available to its subsidiaries was sufficient to enable the subsidiaries to remain solvent, losses at the subsidiary level would be transferred to Group Inc. and ultimately borne by Group Inc.'s security holders, third-party creditors of Group Inc.'s subsidiaries would receive full recoveries on their claims, and Group Inc.'s security holders (including our shareholders, debtholders and other unsecured creditors) could face significant and possibly complete losses. In that case, Group Inc.'s security holders would face losses while the third-party creditors of Group Inc.'s subsidiaries would incur no losses because the subsidiaries would continue to operate and would not enter resolution or bankruptcy proceedings. In addition, holders of Group Inc.'s eligible long-term debt and holders of Group Inc.'s other debt securities could face losses ahead of its other similarly situated creditors in a resolution under OLA if the FDIC exercised its right, described above, to disregard the priority of creditor claims.

OLA also provides the FDIC with authority to cause creditors and shareholders of the financial company in receivership to bear losses before taxpayers are exposed to such losses, and amounts owed to the U.S. government would generally receive a statutory payment priority over the claims of private creditors, including senior creditors.

In addition, under OLA, claims of creditors (including debtholders) could be satisfied through the issuance of equity or other securities in a bridge entity to which Group Inc.'s assets are transferred. If such a securities-for-claims exchange were implemented, there can be no assurance that the value of the securities of the bridge entity would be sufficient to repay or satisfy all or any part of the creditor claims for which the securities were exchanged. While the FDIC has issued regulations to implement OLA, not all aspects of how the FDIC might exercise this authority are known and additional rulemaking is possible.

In addition, certain jurisdictions, including the U.K. and the E.U., have implemented resolution regimes to provide resolution authorities with the ability to recapitalize a failing entity by writing down its unsecured debt or converting its unsecured debt into equity. Such “bail-in” powers are intended to enable the recapitalization of a failing institution by allocating losses to its shareholders and unsecured debtholders. For example, the Bank of England requires a certain amount of intercompany funding that we provide to our material U.K. subsidiaries to contain a contractual trigger to expressly permit the Bank of England to exercise such “bail-in” powers in certain circumstances. If the intercompany funding we provide to our subsidiaries is “bailed in,” Group Inc.’s claims on its subsidiaries would be subordinated to the claims of the subsidiaries’ third-party creditors or written down. U.S. regulators are considering and non-U.S. authorities have adopted requirements that certain subsidiaries of large financial institutions maintain minimum amounts of total loss-absorbing capacity that would pass losses up from the subsidiaries to the top-tier BHC and, ultimately, to security holders of the top-tier BHC in the event of failure.

The application of Group Inc.’s proposed resolution strategy could result in greater losses for Group Inc.’s security holders.

In our resolution plan, Group Inc. would be resolved under the U.S. Bankruptcy Code. The strategy described in our resolution plan is a variant of the single point of entry strategy: Group Inc. and Goldman Sachs Funding LLC (Funding IHC), a wholly-owned, direct subsidiary of Group Inc., would recapitalize and provide liquidity to certain major subsidiaries, including through the forgiveness of intercompany indebtedness, the extension of the maturities of intercompany indebtedness and the extension of additional intercompany loans. If this strategy were successful, creditors of some or all of Group Inc.’s major subsidiaries would receive full recoveries on their claims, while Group Inc.’s security holders could face significant and possibly complete losses.

To facilitate the execution of our resolution plan, we formed Funding IHC. In exchange for an unsecured subordinated funding note and equity interest, Group Inc. transferred certain intercompany receivables and substantially all of its GCLA to Funding IHC, and agreed to transfer additional GCLA above prescribed thresholds.

We also put in place a Capital and Liquidity Support Agreement (CLSA) among Group Inc., Funding IHC and our major subsidiaries. Under the CLSA, Funding IHC has provided Group Inc. with a committed line of credit that allows Group Inc. to draw sufficient funds to meet its cash needs during the ordinary course of business. In addition, if our financial resources deteriorate so severely that resolution may be imminent, (i) the committed line of credit will automatically terminate and the unsecured subordinated funding note will automatically be forgiven, (ii) all intercompany receivables owed by the major subsidiaries to Group Inc. will be transferred to Funding IHC or their maturities will be extended to five years, (iii) Group Inc. will be obligated to transfer substantially all of its remaining intercompany receivables and GCLA (other than an amount to fund anticipated bankruptcy expenses) to Funding IHC, and (iv) Funding IHC will be obligated to provide capital and liquidity support to the major subsidiaries. Group Inc.’s and Funding IHC’s obligations under the CLSA are secured pursuant to a related security agreement. Such actions would materially and adversely affect Group Inc.’s liquidity. As a result, during a period of severe stress, Group Inc. might commence bankruptcy proceedings at an earlier time than it otherwise would if the CLSA and related security agreement had not been implemented.

If Group Inc.’s proposed resolution strategy were successful, Group Inc.’s security holders could face losses while the third-party creditors of Group Inc.’s major subsidiaries would incur no losses because those subsidiaries would continue to operate and not enter resolution or bankruptcy proceedings. As part of the strategy, Group Inc. could also seek to elevate the priority of its guarantee obligations relating to its major subsidiaries’ derivative contracts or transfer them to another entity so that cross-default and early termination rights would be stayed under the ISDA Protocols, as applicable, which would result in holders of Group Inc.’s eligible long-term debt and holders of Group Inc.’s other debt securities incurring losses ahead of the beneficiaries of those guarantee obligations. It is also possible that holders of Group Inc.’s eligible long-term debt and other debt securities could incur losses ahead of other similarly situated creditors of Group Inc.’s major subsidiaries.

If Group Inc.’s proposed resolution strategy were not successful, Group Inc.’s financial condition would be adversely impacted and Group Inc.’s security holders, including debtholders, may as a consequence be in a worse position than if the strategy had not been implemented. In all cases, any payments to debtholders are dependent on our ability to make such payments and are therefore subject to our credit risk.

As a result of our recovery and resolution planning processes, including incorporating feedback from our regulators, we may incur increased operational, funding or other costs and face limitations on our ability to structure our internal organization or engage in internal or external activities in a manner that we may otherwise deem most operationally efficient.

Our commodities activities, particularly our physical commodities activities, subject us to extensive regulation and involve certain potential risks, including environmental, reputational and other risks that may expose us to significant liabilities and costs.

As part of our commodities business, we purchase and sell certain physical commodities, arrange for their storage and transport, and engage in market making of commodities. The commodities involved in these activities may include crude oil, refined oil products, natural gas, liquefied natural gas, electric power, agricultural products, base, precious and other metals, minerals (including unenriched uranium), emission credits, coal, freight and related products and indices.

We make investments in and finance entities that engage in the production, storage and transportation of numerous commodities, including many of the commodities referenced above.

These activities subject us and/or the entities in which we invest to extensive and evolving federal, state and local energy, environmental, antitrust and other governmental laws and regulations worldwide, including environmental laws and regulations relating to, among others, air quality, water quality, waste management, transportation of hazardous substances, natural resources, site remediation and health and safety. Additionally, rising climate change concerns have led to additional laws and regulations, regulatory scrutiny and disclosure obligations that have increased and could further increase the operating costs and could adversely affect the profitability of certain of our investments and activities.

There may be substantial costs in complying with current or future laws and regulations relating to our commodities-related activities and investments. Compliance with these laws and regulations requires significant commitments of capital toward environmental and operational monitoring, development of appropriate operational and supervisory procedures and processes, payment of emission fees and carbon or other taxes, and application for, and holding of, permits and licenses.

Commodities involved in our intermediation activities and investments are also subject to the risk of unforeseen or catastrophic events, which are likely to be outside of our control, including those arising from the breakdown or failure of third-party or service providers' transport vessels, storage facilities or other equipment or processes or other mechanical malfunctions, fires, leaks, spills or release of hazardous substances, performance below expected levels of output or efficiency, terrorist attacks, extreme weather events or other natural disasters or other hostile or catastrophic events. In addition, we rely on third-party suppliers or service providers to perform their contractual obligations and any failure on their part, including the failure to supply or to safely transport or store commodities, could expose us to costs or losses. Also, while we seek to insure against potential risks, we do not have insurance to cover some of these risks and the insurance that we have may be inadequate to cover our losses.

The occurrence of any of such events may prevent us from performing under our agreements with clients, may impair our operations or financial results and may result in litigation, regulatory action, negative publicity or other reputational harm.

We have made changes to and may also be required to divest or discontinue certain of these activities for regulatory or legal reasons or due to the transition to a less carbon-dependent economy in response to climate change.

Competition

Our results have been and may in the future be adversely affected by the composition of our client base.

Our client base is not the same as that of our major competitors. Our businesses may have a higher or lower percentage of clients in certain industries or markets than some or all of our competitors. Therefore, unfavorable industry developments or market conditions affecting certain industries or markets have resulted in the past and may result in the future in our businesses underperforming relative to similar businesses of a competitor if our businesses have a higher concentration of clients in such industries or markets. For example, our market-making businesses have a higher percentage of clients with actively managed assets than some of our competitors and these clients have in the past been and may in the future be disproportionately affected by low volatility.

Correspondingly, favorable or simply less adverse developments or market conditions involving industries or markets in a business where we have a lower concentration of clients in that industry or market have also resulted in the past and may result in the future in our underperforming relative to a similar business of a competitor that has a higher concentration of clients in that industry or market. For example, we have a smaller corporate client base in our market-making businesses than some of our peers and therefore those competitors may benefit more from increased activity by corporate clients. Similarly, we have not historically engaged in retail equities intermediation to the same extent as other financial institutions, which has in the past affected and could in the future adversely affect our market share in equities execution.

The financial services industry is highly competitive.

The financial services industry and all of our businesses are intensely competitive, and we expect them to remain so. We compete on the basis of a number of factors, including transaction execution, our products and services, innovation, reputation, creditworthiness and price. There has been substantial consolidation and convergence among companies in the financial services industry. This has hastened the globalization of the securities and other financial services markets. As a result, we have had to commit capital to support our international operations and to execute large global transactions. As we have expanded into new business areas and new geographic regions, we have faced competitors with more experience and more established relationships with clients, regulators and industry participants in the relevant market, which could adversely affect our ability to expand our businesses.

Governments and regulators have adopted regulations, imposed taxes, adopted compensation restrictions or otherwise put forward various proposals that have impacted or may impact our ability to conduct certain of our businesses in a cost-effective manner or at all in certain or all jurisdictions, including proposals relating to restrictions on the type of activities in which financial institutions are permitted to engage. These or other similar rules, many of which do not apply to all our U.S. or non-U.S. competitors, could impact our ability to compete effectively.

Pricing and other competitive pressures in our businesses have continued to increase, particularly in situations where some of our competitors may seek to increase market share by reducing prices. For example, in connection with investment banking and other assignments, in response to competitive pressure we have experienced, we have extended and priced credit at levels that, in some cases, have not fully compensated us for the risks we undertook.

The financial services industry is highly interrelated in that a significant volume of transactions occur among a limited number of members of that industry. Many transactions are syndicated to other financial institutions, and financial institutions are often counterparties in transactions. This has led to claims by other market participants and regulators that such institutions have colluded in order to manipulate markets or market prices, including allegations that antitrust laws have been violated. While we have extensive procedures and controls that are designed to identify and prevent such activities, they may not be effective. Allegations of such activities, particularly by regulators, can have a negative reputational impact and can subject us to large fines and settlements, and potentially significant penalties, including treble damages.

The growth of electronic trading and the introduction of new products and technologies, including trading and distributed ledger technologies, such as cryptocurrencies, and AI technologies, has increased competition.

Technology is fundamental to our business and our industry. The growth of electronic trading and the introduction of new technologies is changing our businesses and presenting us with new challenges. Securities, futures and options transactions are increasingly occurring electronically, both on our own systems and through other alternative trading systems, and it appears that the trend toward alternative trading systems will continue. Some of these alternative trading systems compete with us, particularly our exchange-based market-making activities, and we may experience continued competitive pressures in these and other areas. In addition, the increased use by our clients of low-cost electronic trading systems and direct electronic access to trading markets has caused and could continue to cause a reduction in commissions and spreads. As our clients increasingly use our systems to trade directly in the markets, we may incur liabilities as a result of their use of our order routing and execution infrastructure.

We have invested significant resources into the development of electronic trading systems and expect to continue to do so, but there is no assurance that the revenues generated by these systems will yield an adequate return, particularly given the generally lower commissions arising from electronic trades.

In addition, the emergence, adoption and evolution of new technologies, including distributed ledgers, such as digital assets and blockchain, and AI technologies, have required us to invest resources to adapt our existing products and services, and we expect to continue to make such investments, which could be material. The adoption and evolution of such new technologies may also increase our compliance and regulatory costs. Further, technologies, such as those based on distributed ledgers, that do not require intermediation could also significantly disrupt payments processing and other financial services. Regulatory limitations on our involvement in products and platforms involving digital assets and distributed ledger technologies may not apply equally or, in some cases, at all to certain of our competitors. We may not be as timely or successful in developing or integrating, or even able to develop or integrate, new products and technologies, such as those built on distributed ledgers or AI technologies, into our existing products and services, adapting to changes in client preferences or achieving market acceptance of our products and services. For example, our competitors may be more timely or successful in developing or integrating AI technologies to increase their productivity and reduce their costs or to provide better transaction execution or improved products and services to clients. Any of the foregoing could affect our ability to attract or retain clients, cause us to lose market share or result in service disruptions and in turn reduce our revenues or otherwise adversely affect us.

Our businesses would be adversely affected if we are unable to hire and retain qualified employees.

Our performance is largely dependent on the talents and efforts of highly skilled people; therefore, our continued ability to compete effectively in our businesses, to manage our businesses effectively and to expand into new businesses and geographic areas depends on our ability to attract and retain a talented workforce. Factors that affect our ability to attract and retain such employees include the level and composition of our compensation and benefits, our reputation as a successful business with a culture of fairly hiring, training and promoting qualified employees and government policies, including immigration policy. As a significant portion of the compensation that we pay to our employees is in the form of year-end discretionary compensation, a significant portion of which is in the form of deferred equity-related awards, declines in our profitability, or in the outlook for our future profitability, as well as regulatory limitations on compensation levels and terms, can negatively impact our ability to hire and retain highly qualified employees.

Competition from within the financial services industry and from businesses outside the financial services industry, including the technology industry, for qualified employees has often been intense. We have experienced increased competition in hiring and retaining employees to address the demands of our consumer-oriented businesses and our technology initiatives. This is also the case in emerging and growth markets, where we are often competing for qualified employees with entities that have a significantly greater presence or more extensive experience in the region.

Laws or regulations in jurisdictions in which our operations are located that affect taxes on our employees' income or the amount or composition of compensation, or that require us to disclose our or our competitors' compensation practices, may also adversely affect our ability to hire and retain qualified employees in those jurisdictions.

As described further in "Business — Regulation — Compensation Practices" in Part I, Item 1 of this Form 10-K, our compensation practices are subject to review by, and the standards of, the FRB. As a large global financial and banking institution, we are subject to limitations on compensation practices (which may or may not affect the companies with which we compete for talent) by the FRB, the PRA, the FCA, the FDIC and other regulators worldwide. These limitations have shaped our compensation practices, which has, in some cases, adversely affected our ability to attract and retain talented employees, in particular in relation to companies not subject to these limitations, and future legislation or regulation may have similar adverse effects.

Our operating expenses and efficiency ratio depend, in part, on our overall headcount and the proportion of our employees located in strategic locations. Our future human capital resource requirements and the benefits provided by strategic locations are uncertain, and we may not realize the benefits we anticipate.

Market Developments and General Business Environment

Our businesses, financial condition, liquidity and results of operations have been and may in the future be adversely affected by unforeseen or catastrophic events, including pandemics, terrorist attacks, wars, extreme weather events or other natural disasters.

The occurrence of unforeseen or catastrophic events, including pandemics or other widespread health emergencies (or concerns over the possibility of such an emergency), terrorist attacks, wars, extreme weather events, solar events or other natural disasters, could adversely affect our business, financial condition, liquidity and results of operations. These events could have such effects through economic or financial market disruptions or challenging economic or market conditions more generally, the deterioration of our creditworthiness or that of our counterparties, changes in consumer sentiment and consumer borrowing, spending and savings patterns, liquidity stress, or operational difficulties (such as travel limitations and limitations on occupancy in our offices) that impair our ability to manage our businesses.

Climate change could disrupt our businesses and adversely affect client activity levels and the creditworthiness of our clients and counterparties, and our actual or perceived action or inaction relating to climate change could result in damage to our reputation.

Climate change may cause or be a contributing factor to extreme weather events that disrupt operations at one or more of our primary locations, which may negatively affect our ability to service and interact with our clients, adversely affect the value of our investments, including our real estate investments, and reduce the availability or increase the cost of insurance. Climate change and the transition to a less carbon-dependent economy may also have a negative impact on the operations or financial condition of our clients and counterparties, which may decrease revenues from those clients and counterparties and increase the credit risk associated with loans and other credit exposures to those clients and counterparties. In addition, climate change may impact the broader economy.

We are also exposed to risks resulting from changes in public policy, laws and regulations, or market and public perceptions and preferences in connection with the transition to a less carbon-dependent economy. These changes could adversely affect our business, results of operations and reputation. In addition, due to divergent stakeholder views regarding climate change, we are at increased risk that any actual or perceived action, or lack thereof, by us in connection with the transition to a less carbon-dependent economy will be perceived negatively by some stakeholders and adversely affect our business and reputation. If our response to climate change is subject to criticism, our business, reputation and efforts to recruit and retain employees may suffer.

New laws, regulations or guidance relating to climate change, as well as the perspectives of government officials, regulators, shareholders, employees and other stakeholders regarding climate change, may affect whether and on what terms and conditions we engage in certain activities or offer certain products. Federal and state, and non-U.S. banking regulators and supervisory authorities, shareholders and other stakeholders have increasingly viewed financial institutions as playing an important role in helping to address risks related to climate change, both directly and with respect to their clients, which may result in financial institutions coming under increased requirements and expectations regarding the disclosure and management of their climate risks and related lending, investment and advisory activities. For example, in 2023 we participated in a pilot climate scenario analysis exercise conducted by the FRB, the results of which were released in 2024. We are also subject to interagency guidance jointly issued by the FRB, FDIC, and OCC in 2023 regarding principles for climate-related financial risk management for large financial institutions. In addition, in 2023, the NYDFS issued guidance on the management of material financial risks from climate change, which applies to New York State-regulated banking and mortgage institutions, including GS Bank USA. Additionally, certain states in which we operate have enacted or proposed laws or regulations regarding ESG matters, including laws or regulations that require climate-related disclosures, that address the consideration of ESG factors in state investments or contracting, and that require financial institutions to broadly provide customers with access to financial services, many of which may have broad, extraterritorial application.

In the E.U., certain of our entities are expected to be subject in varying degrees to sustainability-related laws being implemented, including directives, such as the CSRD and the CSDDD, which would significantly expand the scope of ESG disclosure requirements applicable to us and impose stringent due diligence requirements with respect to adverse human rights and environmental impacts in our upstream business partners' operations, as well as require us to put into effect a climate transition plan.

These, as well as any additional or heightened laws, regulations, guidance and expectations, have in the past subjected and may in the future, subject us to different and potentially conflicting requirements and expectations in the various jurisdictions in which we operate, and have in the past resulted in and could in the future result in increased regulatory, compliance or other costs or higher capital requirements. The risks associated with, and the perspective of government officials, regulators, shareholders, employees and other stakeholders regarding, climate change are continuing to evolve rapidly, which can make it difficult to assess the ultimate impact on us of climate change-related risks and uncertainties, but we expect that climate change-related risks will increase over time.

Our business, financial condition, liquidity and results of operations have been adversely affected by disruptions in the global economy caused by conflicts, and related sanctions and other developments.

The conflict between Russia and Ukraine has negatively affected the global economy. Governments around the world have responded to Russia's invasion by imposing economic sanctions and export controls on certain industry sectors, including price caps on Russian oil, and on Russian businesses and persons. Compliance with economic sanctions and restrictions imposed by governments has increased our costs and otherwise adversely affected our business and may continue to do so. Russia has responded with its own restrictions against investors and countries outside Russia and has proposed additional measures aimed at non-Russian owned businesses. Businesses in the U.S. and globally have experienced shortages in materials and increased costs for transportation, energy, and raw materials due in part to the negative effects of the conflict on the global economy.

The conflicts in the Middle East could also affect and harm our business and increase market uncertainty. The impact of these conflicts on our business and operations is uncertain and therefore cannot be predicted.

The escalation or continuation of these conflicts or other hostilities could result in, among other things, an increased risk of cyber attacks, an increased frequency and volume of failures to settle securities transactions, supply chain disruptions, higher inflation, lower consumer demand and increased volatility in commodity, currency and other financial markets. The extent and duration of the conflicts, sanctions and resulting market disruptions are impossible to predict, and the consequences for our business could be significant. If international political instability and geopolitical tensions continue or increase in any region in which we do business, our business and results of operations could be harmed. See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Credit Risk Management — Selected Exposures — Country Exposures" for further information about our credit exposure to Russia and Ukraine.

Certain of our businesses and our funding instruments may be adversely affected by changes in reference rates, currencies, indexes, baskets or ETFs to which products we offer or funding that we raise are linked.

Many of the products that we own or that we offer, such as structured notes, warrants, swaps or security-based swaps, pay interest or determine the principal amount to be paid at maturity or in the event of default by reference to rates or by reference to an index, currency, basket, ETF or other financial metric (the underlier). In the event that the composition of the underlier is significantly changed, by reference to rules governing such underlier or otherwise, the underlier ceases to exist (for example, in the event that a country withdraws from the Euro or links its currency to or delinks its currency from another currency or benchmark, an index or ETF sponsor materially alters the composition of an index or ETF, or stocks in a basket are delisted or become impermissible to be included in the index or ETF), the underlier ceases to be recognized as an acceptable market benchmark or there are legal or regulatory constraints on linking a financial instrument to the underlier, we may experience adverse effects.

Our business, financial condition, liquidity and results of operations may be adversely affected by disruptions in the global economy caused by escalating tensions between the U.S. and China.

Continued or escalating tensions between the U.S. and China have resulted in and may result in additional changes to U.S. international trade and investment policies, which could disrupt international trade and investment, adversely affect financial markets, including market activity levels, and adversely impact our revenues. Continued or escalating tensions may also lead to the U.S., China or other countries taking other actions, which could include the implementation of sanctions, tariffs or foreign exchange measures, the large-scale sale of U.S. Treasury securities or restrictions on cross-border trade, investment or transfer of information or technology. Any such developments could adversely affect our or our clients' businesses, as well as our financial condition, liquidity and results of operations, possibly materially.

A conflict, or concerns about a potential conflict, involving China and Taiwan, the U.S. or other countries could negatively impact financial markets and our or our clients' businesses. Trade restrictions by the U.S. or other countries in response to a conflict or potential conflict involving China, including financial and economic sanctions and export controls against certain organizations or individuals, or actions taken by China in response to trade restrictions, could negatively impact our or our clients' ability to conduct business in certain countries or with certain counterparties and could negatively impact regional and global financial markets and economic conditions. Any of the foregoing could adversely affect our business, financial condition, liquidity and results of operations, possibly materially.

We face enhanced risks as we operate in new locations and transact with a broader array of clients and counterparties.

Our businesses, have in the past, and may in the future, bring us into contact, directly or indirectly, with individuals and entities that are not within our traditional client and counterparty base, expose us to new asset classes and new markets, and present us with integration challenges. For example, we continue to transact business and invest in new regions, including a wide range of emerging and growth markets, and we expect this trend to continue. Various emerging and growth market countries have experienced severe economic and financial disruptions, including significant devaluations of their currencies, defaults or threatened defaults on sovereign debt, capital and currency exchange controls, and low or negative growth rates in their economies. The possible effects of any of these conditions include an adverse impact on our businesses and increased volatility in financial markets generally.

In our consumer-oriented activities, we have faced and continue to face compliance, legal and regulatory risk, increased reputational risk and increased operational risk due to, among other things, higher transaction volumes and significantly increased retention and transmission of consumer and client information. We are also subject to additional legal requirements, including with respect to suitability and consumer protection (for example, Regulation Best Interest, fair lending laws and regulations and privacy laws and regulations). Further, identity fraud may increase and credit reporting practices may change in a manner that makes it more difficult for financial institutions, such as us, to evaluate the creditworthiness of consumers.

In connection with our transaction banking activities, we face compliance, legal and regulatory risk, including with respect to know-your-customer, anti-money laundering and reporting requirements and prohibitions on transfers of property belonging to countries, entities and individuals subject to sanctions by U.S. or other governmental authorities. We are making significant enhancements to existing controls, systems and procedures to manage these risks.

New business initiatives expose us to new and enhanced risks, including risks associated with dealing with governmental entities, reputational concerns arising from dealing with different types of clients, business partners, counterparties and investors, greater regulatory scrutiny of these activities, increased credit-related, market, sovereign and operational risks, risks arising from accidents or acts of terrorism, and reputational concerns with the manner in which certain assets are being operated or held or in which we interact with these clients, business partners, counterparties and investors. Legal, regulatory and reputational risks may also exist in connection with activities and transactions involving new products or markets where there is regulatory uncertainty or where there are different or conflicting regulations depending on the regulator or the jurisdiction involved, particularly where transactions in such products may involve multiple jurisdictions.

We have developed and pursued new business and strategic initiatives, including acquisitions, and may continue to do so. If and to the extent we are unable to successfully execute those initiatives, we may incur unanticipated costs and losses, and face other adverse consequences, such as negative reputational effects. In addition, the actual effects of pursuing those initiatives may differ, possibly materially, from the benefits that we expect to realize from them, such as generating additional revenues, achieving expense savings, reducing operational risk exposures or using capital and funding more efficiently. Engaging in new activities exposes us to a variety of risks, including that we may be unable to successfully develop new, competitive, efficient and effective systems and processes, and hire and retain the necessary personnel. Due to our lack of historical experience with unsecured consumer lending, our loan loss assumptions may prove to be incorrect and we may incur losses significantly above those which we originally anticipated in entering the business.

In recent years, we have invested, and may continue to invest, more in businesses that we expect will generate a higher level of more consistent revenues. Such investments and acquisitions may not be successful or have returns similar to our other businesses.

We may not be able to fully realize the expected benefits or synergies from acquisitions or other business initiatives in the time frames we expect, or at all.

We have engaged in selective acquisitions and may continue to do so in the future and these acquisitions may, individually or in the aggregate, be material to us. Any future acquisitions could involve the issuance of common stock and/or the payment of cash as consideration. The success of our acquisitions will depend, in part, on our ability to integrate the acquired businesses and realize anticipated synergies, cost savings and growth opportunities. For example, in 2024, we sold GreenSky Holdings, LLC, which we had previously acquired, and in connection with the disposition we incurred a write-down of intangible assets and goodwill. Also in 2024, we entered into an agreement to transition the GM credit card program to another issuer and have incurred a write-down of intangible assets in connection with that transaction. In any future acquisitions, we may face numerous risks and uncertainties in combining and integrating the relevant businesses and systems, including the need to combine or separate accounting and data processing systems and management controls and to integrate relationships with clients, counterparties, regulators and others in connection with acquisitions. Integration of acquired businesses is time-consuming and could disrupt our ongoing businesses, produce unforeseen regulatory or operating difficulties, cause us to incur incremental expenses or require incremental financial, management and other resources. It is also possible that an acquisition, once announced, may not close due to the failure to satisfy applicable closing conditions, such as the receipt of necessary shareholder or regulatory approvals.

There is no assurance that any of our acquisitions will be successfully integrated or yield all of the expected benefits and synergies in the time frames that we expect, or at all. If we are not able to integrate our acquisitions successfully, our results of operations, financial condition and cash flows could be adversely affected.

Item 1B. Unresolved Staff Comments

There are no material unresolved written comments that were received from the SEC staff 180 days or more before the end of our fiscal year relating to our periodic or current reports under the Exchange Act.

Item 1C. Cybersecurity

See “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Risk Management — Cybersecurity Risk Management” in Part II, Item 7 of this Form 10-K for further information about cybersecurity.

Item 2. Properties

In the U.S. and elsewhere in the Americas, we have offices consisting of approximately 6.3 million square feet of leased and owned space. Our principal executive offices are located at 200 West Street, New York, New York and consist of approximately 2.1 million square feet. The building is located on a parcel leased from Battery Park City Authority pursuant to a ground lease. Under the lease, Battery Park City Authority holds title to all improvements, including the office building, subject to our right of exclusive possession and use until June 2069, the expiration date of the lease. Under the terms of the ground lease, we made a lump sum ground rent payment in June 2007 of \$161 million for rent through the term of the lease.

In Europe, the Middle East and Africa, we have offices consisting of approximately 1.8 million square feet of leased and owned space. Our European headquarters is located in London at Plumtree Court, consisting of approximately 826,000 square feet under a lease which can be terminated in 2039.

In Asia, Australia and New Zealand, we have offices consisting of approximately 2.9 million square feet, including our offices in India, and regional headquarters in Tokyo and Hong Kong. In India, we have offices with approximately 1.9 million square feet, the majority of which have leases that will expire starting in 2028.

In the preceding paragraphs, square footage figures are provided only for properties that are used in the operation of our businesses. We regularly evaluate our space capacity in relation to current and projected headcount. We may incur exit costs in the future if we (i) reduce our space capacity or (ii) commit to, or occupy, new properties in locations in which we operate and dispose of existing space that had been held for potential growth. These costs may be material to our operating results in a given period.

Item 3. Legal Proceedings

We are involved in a number of judicial, regulatory and arbitration proceedings concerning matters arising in connection with the conduct of our businesses. Many of these proceedings are in early stages, and many of these cases seek an indeterminate amount of damages. We have estimated the upper end of the range of reasonably possible aggregate loss for matters where we have been able to estimate a range and we believe, based on currently available information, that the results of matters where we have not been able to estimate a range of reasonably possible loss, in the aggregate, will not have a material adverse effect on our financial condition, but may be material to our operating results in a given period. Given the range of litigation and investigations presently under way, our litigation expenses may remain high. See “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Use of Estimates” in Part II, Item 7 of this Form 10-K. See Notes 18 and 27 to the consolidated financial statements in Part II, Item 8 of this Form 10-K for information about our reasonably possible aggregate loss estimate and judicial, regulatory and legal proceedings.

Item 4. Mine Safety Disclosures

Not applicable.

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

The principal market on which our common stock is traded is the NYSE under the symbol "GS." Information relating to the performance of our common stock from December 31, 2019 through December 31, 2024 is set forth in "Supplemental Financial Information — Common Stock Performance" in Part II, Item 8 of this Form 10-K. As of February 7, 2025, there were 5,251 holders of record of our common stock.

The table below presents purchases made by or on behalf of Group Inc. or any "affiliated purchaser" (as defined in Rule 10b-18(a)(3) under the Exchange Act) of our common stock during the fourth quarter of 2024.

	Total Shares Purchased	Average Price Paid Per Share	Total Shares Purchased as Part of a Publicly Announced Program	Dollar Value of Remaining Authorized Repurchases (\$ in millions)
October	1,146,242	\$ 523.45	1,146,242	\$ 17,604
November	1,717,264	\$ 582.31	1,717,264	\$ 16,604
December	668,161	\$ 598.52	668,161	\$ 16,204
Total	3,531,667		3,531,667	

In 2023, our Board approved a share repurchase program authorizing repurchases of up to \$30 billion of our common stock. This program replaced our previous share repurchase program and has no set expiration or termination date. The share repurchases are effected primarily through regular open-market purchases (which may include repurchase plans designed to comply with Rule 10b5-1 and accelerated share repurchases), the amounts and timing of which are determined primarily by our current and projected capital position, and capital deployment opportunities, but which may also be influenced by the evolution of current and future regulatory capital requirements, general market conditions and the prevailing price and trading volumes of our common stock. See "Management's Discussion and Analysis of Financial Condition and Results of Operations — Capital Management and Regulatory Capital — Capital Management — Share Repurchase Program" in Part II, Item 7 of this Form 10-K for further information.

Information relating to compensation plans under which our equity securities are authorized for issuance is presented in Part III, Item 12 of this Form 10-K.

Management's Discussion and Analysis

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Introduction

The Goldman Sachs Group, Inc. (Group Inc. or parent company), a Delaware corporation, together with its consolidated subsidiaries, is a leading global financial institution that delivers a broad range of financial services to a large and diversified client base that includes corporations, financial institutions, governments and individuals. Founded in 1869, we are headquartered in New York and maintain offices in all major financial centers around the world. We manage and report our activities in three business segments: Global Banking & Markets, Asset & Wealth Management and Platform Solutions. See "Results of Operations" for further information about our business segments.

When we use the terms "we," "us" and "our," we mean Group Inc. and its consolidated subsidiaries. When we use the term "our subsidiaries," we mean the consolidated subsidiaries of Group Inc. References to "this Form 10-K" are to our Annual Report on Form 10-K for the year ended December 31, 2024. All references to "the consolidated financial statements" or "Supplemental Financial Information" are to Part II, Item 8 of this Form 10-K. All references to 2024, 2023 and 2022 refer to our years ended, or the dates, as the context requires, December 31, 2024, December 31, 2023 and December 31, 2022, respectively. Any reference to a future year refers to a year ending on December 31 of that year. Certain reclassifications have been made to previously reported amounts to conform to the current presentation.

Group Inc. is a bank holding company and a financial holding company regulated by the Board of Governors of the Federal Reserve System (FRB).

In this discussion and analysis of our financial condition and results of operations, we have included information that constitutes "forward-looking statements" within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. Forward-looking statements are not historical facts or statements of current conditions, but instead represent only our beliefs regarding future events, many of which, by their nature, are inherently uncertain and outside our control.

By identifying these statements for you in this manner, we are alerting you to the possibility that our actual results, financial condition, liquidity and capital actions may differ, possibly materially, from the anticipated results, financial condition, liquidity and capital actions in these forward-looking statements. Important factors that could cause our results, financial condition, liquidity and capital actions to differ from those in these statements include, among others, those described in "Risk Factors" in Part I, Item 1A of this Form 10-K and "Forward-Looking Statements" in Part I, Item 1 of this Form 10-K.

Management's Discussion and Analysis

These statements may relate to, among other things, (i) our future plans and results, including our target return on average common shareholders' equity (ROE), return on average tangible common shareholders' equity (ROTE), efficiency ratio, Common Equity Tier 1 (CET1) capital ratio and firmwide total credit alternative assets, and how they can be achieved, (ii) trends in or growth opportunities for our businesses, including the timing, costs, profitability, benefits and other aspects of business and strategic initiatives and their impact on our efficiency ratio, as well as the opportunities and challenges presented by artificial intelligence (AI), (iii) our level of future compensation expense, (iv) our Investment banking fees backlog and future advisory and capital markets results, (v) our expected interest income and interest expense, (vi) our expense savings and strategic locations initiatives, (vii) expenses we may incur, including future litigation expense, (viii) the projected growth of our deposits and other funding, asset liability management and funding strategies and related interest expense savings, (ix) our business initiatives, (x) our planned 2025 benchmark debt issuances, (xi) the amount, composition and location of global core liquid assets (GCLA) we expect to hold, (xii) our credit exposures, (xiii) our expected provision for credit losses, (xiv) the adequacy of our allowance for credit losses, (xv) the narrowing of our consumer business, (xvi) the objectives and effectiveness of our business continuity planning (BCP), information security program, risk management and liquidity policies, (xvii) our resolution plan and its implications for stakeholders, (xviii) the design and effectiveness of our resolution capital and liquidity models and triggers and alerts framework, (xix) the results of stress tests, the effect of changes to regulations, and our future status, activities or reporting under banking and financial regulation, (xx) our expected tax rate, (xxi) the future state of our liquidity and regulatory capital ratios, and our prospective capital distributions (including dividends and repurchases), (xxii) our expected stress capital buffer (SCB) and global systemically important bank (G-SIB) surcharge, (xxiii) legal proceedings, governmental investigations or other contingencies, (xxiv) the asset recovery guarantee and our remediation activities related to our 1Malaysia Development Berhad (1MDB) settlements, (xxv) the effectiveness of our management of our human capital, (xxvi) our sustainability and carbon neutrality targets and goals, (xxvii) future inflation, (xxviii) the impact of Russia's invasion of Ukraine and related sanctions and other developments on our business, results and financial position, (xxix) our ability to sell, and the terms of any proposed sales of, Asset & Wealth Management historical principal investments, and our ability to transition the General Motors (GM) credit card program, (xxx) the impact of the conflicts in the Middle East, (xxxi) our ability to manage our commercial real estate exposures, (xxxii) the profitability of Platform Solutions and (xxxiii) the effectiveness of our cybersecurity risk management process.

Executive Overview

We generated net earnings of \$14.28 billion for 2024, compared with \$8.52 billion for 2023. Diluted earnings per common share (EPS) was \$40.54 for 2024, compared with \$22.87 for 2023. ROE was 12.7% for 2024, compared with 7.5% for 2023. Book value per common share was \$336.77 as of December 2024, 7.4% higher compared with December 2023.

Net revenues were \$53.51 billion for 2024, 16% higher than 2023, primarily reflecting higher net revenues in Global Banking & Markets and Asset & Wealth Management. The increase in net revenues in Global Banking & Markets primarily reflected higher net revenues in Equities, significantly higher Investment banking fees and higher net revenues in Fixed Income, Currency and Commodities (FICC). The increase in net revenues in Asset & Wealth Management primarily reflected significantly higher net revenues in Equity investments and higher Management and other fees. Net revenues in Platform Solutions were slightly higher.

Provision for credit losses was \$1.35 billion for 2024, compared with \$1.03 billion for 2023. Provisions for 2024 reflected net provisions related to the credit card portfolio (primarily driven by net charge-offs). Provisions for 2023 reflected net provisions related to both the credit card portfolio (primarily driven by net charge-offs) and wholesale loans (primarily driven by impairments), partially offset by reserve reductions related to the transfer of the GreenSky loan portfolio to held for sale and the sale of substantially all of the *Marcus by Goldman Sachs* (Marcus) loan portfolio.

Operating expenses were \$33.77 billion for 2024, 2% lower than 2023, reflecting decreases driven by significantly lower expenses, including impairments, related to commercial real estate in consolidated investment entities (CIEs) and other significant expenses recognized in the prior year, including the write-down of identifiable intangible assets related to GreenSky Holdings, LLC (GreenSky), an impairment of goodwill related to Consumer platforms and the FDIC special assessment fee. These decreases were partially offset by higher compensation and benefits expenses (reflecting improved operating performance) and higher transaction based expenses. Our efficiency ratio (total operating expenses divided by total net revenues) was 63.1% for 2024, compared with 74.6% for 2023.

During 2024, we returned a total of \$11.80 billion of capital to common shareholders, including \$8.00 billion of common share repurchases and \$3.80 billion of common stock dividends. As of December 2024, our CET1 capital ratio was 15.0% under the Standardized Capital Rules and 15.3% under the Advanced Capital Rules. See Note 20 to the consolidated financial statements for further information about our capital ratios.

Management's Discussion and Analysis

Business Environment

In 2024, the global economy grew, but was impacted throughout the year by broad macroeconomic and geopolitical concerns. Concerns regarding inflation and ongoing geopolitical stresses, including tensions with China and the conflicts in Ukraine and the Middle East, remained elevated. Despite these concerns, the economy in the U.S. has remained resilient and equity markets have reacted favorably to the outcomes of national elections. Additionally, markets were focused on policy interest rate cuts by several central banks, including the first rate cut by U.S. Federal Reserve since it began increasing the rate in 2022.

There remains uncertainty and concerns about geopolitical risks, central bank policy and inflation. See "Results of Operations — Segment Assets and Operating Results — Segment Operating Results" for further information about the operating environment for each of our business segments.

Critical Accounting Policies

Fair Value

Fair Value Hierarchy. Trading assets and liabilities, certain investments and loans, and certain other financial assets and liabilities, are included in our consolidated balance sheets at fair value (i.e., marked-to-market), with related gains or losses generally recognized in our consolidated statements of earnings. The use of fair value to measure financial instruments is fundamental to our risk management practices and is our most critical accounting policy.

The fair value of a financial instrument is the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. We measure certain financial assets and liabilities as a portfolio (i.e., based on its net exposure to market and/or credit risks). In determining fair value, the hierarchy under U.S. generally accepted accounting principles (U.S. GAAP) gives (i) the highest priority to unadjusted quoted prices in active markets for identical, unrestricted assets or liabilities (level 1 inputs), (ii) the next priority to inputs other than level 1 inputs that are observable, either directly or indirectly (level 2 inputs), and (iii) the lowest priority to inputs that cannot be observed in market activity (level 3 inputs). In evaluating the significance of a valuation input, we consider, among other factors, a portfolio's net risk exposure to that input. Assets and liabilities are classified in their entirety based on the lowest level of input that is significant to their fair value measurement.

The fair values for substantially all of our financial assets and liabilities are based on observable prices and inputs and are classified in levels 1 and 2 of the fair value hierarchy. Certain level 2 and level 3 financial assets and liabilities may require appropriate valuation adjustments that a market participant would require to arrive at fair value for factors, such as counterparty and our credit quality, funding risk, transfer restrictions, liquidity and bid/offer spreads.

Instruments classified in level 3 of the fair value hierarchy are those which require one or more significant inputs that are not observable. Level 3 financial assets represented 1.2% as of December 2024 and 1.5% as of December 2023 of our total assets. See Notes 4 and 5 to the consolidated financial statements for further information about level 3 financial assets, including changes in level 3 financial assets and related fair value measurements. Absent evidence to the contrary, instruments classified in level 3 of the fair value hierarchy are initially valued at transaction price, which is considered to be the best initial estimate of fair value. Subsequent to the transaction date, we use other methodologies to determine fair value, which vary based on the type of instrument. Estimating the fair value of level 3 financial instruments requires judgments to be made. These judgments include:

- Determining the appropriate valuation methodology and/or model for each type of level 3 financial instrument;
- Determining model inputs based on an evaluation of all relevant empirical market data, including prices evidenced by market transactions, interest rates, credit spreads, volatilities and correlations; and
- Determining appropriate valuation adjustments, including those related to illiquidity or counterparty credit quality.

Regardless of the methodology, valuation inputs and assumptions are only changed when corroborated by substantive evidence.

Management's Discussion and Analysis

Controls Over Valuation of Financial Instruments. Market makers and investment professionals in our revenue-producing units are responsible for pricing our financial instruments. Our control infrastructure is independent of the revenue-producing units and is fundamental to ensuring that all of our financial instruments are appropriately valued at market-clearing levels. In the event that there is a difference of opinion in situations where estimating the fair value of financial instruments requires judgment (e.g., calibration to market comparables or trade comparison, as described below), the final valuation decision is made by senior managers in our independent price verification function within Controllers. This independent price verification is critical to ensuring that our financial instruments are properly valued.

Price Verification. All financial instruments at fair value classified in levels 1, 2 and 3 of the fair value hierarchy are subject to our independent price verification process. The objective of price verification is to have an informed and independent opinion with regard to the valuation of financial instruments under review. Instruments that have one or more significant inputs which cannot be corroborated by external market data are classified in level 3 of the fair value hierarchy. Price verification strategies utilized by our independent price verification function within Controllers include:

- **Trade Comparison.** Analysis of trade data (both internal and external, where available) is used to determine the most relevant pricing inputs and valuations.
- **External Price Comparison.** Valuations and prices are compared to pricing data obtained from third parties (e.g., brokers or dealers, S&P Global Services, Bloomberg, ICE Data Services, Pricing Direct, TRACE). Data obtained from various sources is compared to ensure consistency and validity. When broker or dealer quotations or third-party pricing vendors are used for valuation or price verification, greater priority is generally given to executable quotations.
- **Calibration to Market Comparables.** Market-based transactions are used to corroborate the valuation of positions with similar characteristics, risks and components.
- **Relative Value Analyses.** Market-based transactions are analyzed to determine the similarity, measured in terms of risk, liquidity and return, of one instrument relative to another or, for a given instrument, of one maturity relative to another.
- **Collateral Analyses.** Margin calls on derivatives are analyzed to determine implied values, which are used to corroborate our valuations.
- **Execution of Trades.** Where appropriate, market-making desks are instructed to execute trades in order to provide evidence of market-clearing levels.

- **Backtesting.** Valuations are corroborated by comparison to values realized upon sales.

See Note 4 to the consolidated financial statements for further information about fair value measurements.

Review of Net Revenues. We seek to ensure adherence to our pricing policy through a combination of daily procedures, including the explanation and attribution of net revenues based on the underlying factors. Through this process, we independently validate net revenues, identify and resolve potential fair value or trade booking issues on a timely basis and seek to ensure that risks are being properly categorized and quantified.

Review of Valuation Models. Our independent model risk management group (Model Risk), consisting of quantitative professionals who are separate from model developers, performs an independent model review and validation process of our valuation models. New or changed models are reviewed and approved prior to implementation. Models are reviewed annually to assess the impact of any changes in the product or market and any market developments in pricing theories. See "Risk Management — Model Risk Management" for further information about the review and validation of our valuation models.

Allowance for Credit Losses

We estimate and record an allowance for credit losses related to our loans held for investment that are accounted for at amortized cost. To determine the allowance for credit losses, we classify our loans accounted for at amortized cost into wholesale and consumer portfolios. These portfolios represent the level at which we have developed and documented our methodology to determine the allowance for credit losses. The allowance for credit losses is measured on a collective basis for loans that exhibit similar risk characteristics using a modeled approach and on an asset-specific basis for loans that do not share similar risk characteristics.

Management's Discussion and Analysis

The allowance for credit losses takes into account the weighted average of a range of forecasts of future economic conditions over the expected life of the loans and lending commitments. The expected life of each loan or lending commitment is determined based on the contractual term adjusted for extension options or demand features, or is modeled in the case of revolving credit card loans. The forecasts include baseline, favorable and adverse economic scenarios over a three-year period. For loans with expected lives beyond three years, the model reverts to historical loss information based on a non-linear modeled approach. We apply judgment in weighting individual scenarios each quarter based on a variety of factors, including our internally derived economic outlook, market consensus, recent macroeconomic conditions and industry trends. The forecasted economic scenarios consider a number of risk factors relevant to the wholesale and consumer portfolios. Risk factors for wholesale loans include internal credit ratings, industry default and loss data, expected life, macroeconomic indicators (e.g., unemployment rates and GDP), the borrower's capacity to meet its financial obligations, the borrower's country of risk and industry, loan seniority and collateral type. In addition, for loans backed by real estate, risk factors include the loan-to-value ratio, debt service ratio and home price index. The allowance for loan losses for wholesale loans that do not share similar risk characteristics, such as nonaccrual loans, is calculated using the present value of expected future cash flows discounted at the loan's effective rate, the observable market price of the loan, or, in the case of collateral dependent loans, the fair value of the collateral less estimated costs to sell, if applicable. Risk factors for installment and credit card loans include Fair Isaac Corporation (FICO) credit scores, delinquency status, loan vintage and macroeconomic indicators.

The allowance for credit losses also includes qualitative components which allow management to reflect the uncertain nature of economic forecasting, capture uncertainty regarding model inputs, and account for model imprecision and concentration risk. The qualitative factors considered by management include, among others, changes and trends in loan portfolios, uncertainties associated with the macroeconomic and geopolitical environments, credit concentrations, changes in volume and severity of past due and criticized loans, idiosyncratic events and deterioration within an industry or region.

Our estimate of credit losses entails judgment about collectability at the reporting dates, and there are uncertainties inherent in those judgments. The allowance for credit losses is subject to a governance process that involves senior management within Risk and Controllers. Personnel within Risk are responsible for forecasting the economic variables that underlie the economic scenarios that are used in the modeling of expected credit losses. While we use the best information available to determine this estimate, future adjustments to the allowance may be necessary based on, among other things, changes in the economic environment or variances between actual results and the original assumptions used. Loans are charged off against the allowance for loan losses when deemed to be uncollectible.

We also record an allowance for credit losses on lending commitments which are held for investment that are accounted for at amortized cost. Such allowance is determined using the same methodology as the allowance for loan losses, while also taking into consideration the probability of drawdowns or funding, and whether such commitments are cancellable by us.

To estimate the potential impact of an adverse macroeconomic environment on our allowance for credit losses, we, among other things, compared the expected credit losses under the weighted average forecast used in the calculation of allowance for credit losses as of December 2024 (which was weighted towards the baseline and adverse economic scenarios) to the expected credit losses under a 100% weighted adverse economic scenario. The adverse economic scenario of the forecast model reflects a global recession in the first quarter of 2025 through the first quarter of 2026, resulting in an economic contraction and rising unemployment rates. A 100% weighting to the adverse economic scenario would have resulted in an approximate \$0.9 billion increase in our allowance for credit losses as of December 2024. This hypothetical increase does not take into consideration any potential adjustments to qualitative reserves. The forecasts of macroeconomic conditions are inherently uncertain and do not take into account any other offsetting or correlated effects. The actual credit loss in an adverse macroeconomic environment may differ significantly from this estimate. See Note 9 to the consolidated financial statements for further information about the allowance for credit losses.

Use of Estimates

U.S. GAAP requires us to make certain estimates and assumptions. In addition to the estimates we make in connection with fair value measurements and the allowance for credit losses on loans and lending commitments held for investment and accounted for at amortized cost, the use of estimates and assumptions is also important in determining the accounting for goodwill and identifiable intangible assets, provisions for losses that may arise from litigation and regulatory proceedings (including governmental investigations), and accounting for income taxes.

Management's Discussion and Analysis

Goodwill is assessed for impairment annually in the fourth quarter or more frequently if events occur or circumstances change that indicate an impairment may exist. When assessing goodwill for impairment, first, a qualitative assessment can be made to determine whether it is more likely than not that the estimated fair value of a reporting unit is less than its carrying value. If the results of the qualitative assessment are not conclusive, a quantitative goodwill test is performed. Alternatively, a quantitative goodwill test can be performed without performing a qualitative assessment. Estimating the fair value of our reporting units requires judgment. Critical inputs to the fair value estimates include projected earnings, allocated equity, price-to-earnings multiples and price-to-book multiples. There is inherent uncertainty in the projected earnings. The carrying value of each reporting unit reflects an allocation of total shareholders' equity and represents the estimated amount of total shareholders' equity required to support the activities of the reporting unit under currently applicable regulatory capital requirements. During the third quarter of 2024, in connection with the planned sale of our seller financing loan portfolio, we performed a quantitative goodwill test and determined that the goodwill associated with Transaction banking and other was impaired, and accordingly, recorded a \$14 million impairment. In the fourth quarter of 2024, we performed our annual assessment of goodwill for impairment, for each of our reporting units with goodwill, by performing a qualitative assessment. As a result of the annual assessment, we determined that it was more likely than not that the estimated fair value of each reporting unit with goodwill exceeded its respective carrying value. Therefore, we determined that goodwill for each reporting unit was not impaired and that a quantitative goodwill test was not required. See Note 12 to the consolidated financial statements for further information about our annual assessment of goodwill for impairment. If we experience a prolonged or severe period of weakness in the business environment, financial markets, the performance of one or more of our reporting units or our common stock price, or additional increases in capital requirements, our goodwill could be impaired in the future.

Identifiable intangible assets are tested for impairment when events or changes in circumstances suggest that an asset's or asset group's carrying value may not be fully recoverable. Judgment is required to evaluate whether indications of potential impairment have occurred, and to test identifiable intangible assets for impairment, if required. An impairment is recognized if the estimated undiscounted cash flows relating to the asset or asset group is less than the corresponding carrying value. During 2024, in connection with the planned transition of the GM credit card program to another issuer, we classified the GM credit card program to held for sale and recognized a \$72 million write-down of identifiable intangible assets. See Note 12 to the consolidated financial statements for further information about identifiable intangible assets.

We also estimate and provide for potential losses that may arise out of litigation and regulatory proceedings to the extent that such losses are probable and can be reasonably estimated. In addition, we estimate the upper end of the range of reasonably possible aggregate loss in excess of the related reserves for litigation and regulatory proceedings where we believe the risk of loss is more than slight. See Notes 18 and 27 to the consolidated financial statements for information about certain judicial, litigation and regulatory proceedings. Significant judgment is required in making these estimates and our final liabilities may ultimately be materially different. Our total estimated liability in respect of litigation and regulatory proceedings is determined on a case-by-case basis and represents an estimate of probable losses after considering, among other factors, the progress of each case, proceeding or investigation, our experience and the experience of others in similar cases, proceedings or investigations, and the opinions and views of legal counsel.

In accounting for income taxes, we recognize tax positions in the financial statements only when it is more likely than not that the position will be sustained on examination by the relevant taxing authority based on the technical merits of the position. As of December 2024, our liability for unrecognized tax benefits was \$2.16 billion. We use estimates to recognize current and deferred income taxes in the U.S. federal, state and local and non-U.S. jurisdictions in which we operate. The income tax laws in these jurisdictions are complex and can be subject to different interpretations between taxpayers and taxing authorities. Disputes may arise over these interpretations and can be settled by audit, administrative appeals or judicial proceedings. We do not expect that the resolution of any such dispute will have a material impact on our financial condition, but it may be material to the operating results for a particular period, depending, in part, on the operating results for that period. Our interpretations are reevaluated quarterly based on guidance currently available, tax examination experience and the opinions of legal counsel, among other factors. We recognize deferred taxes based on the amount that will more likely than not be realized in the future based on enacted income tax laws. As of December 2024, we had \$11.01 billion of deferred tax assets with a related valuation allowance of \$2.06 billion. Our estimate for deferred taxes includes estimates for future taxable earnings, including the level and character of those earnings, and various tax planning strategies. See Note 24 to the consolidated financial statements for further information about income taxes.

Recent Accounting Developments

See Note 3 to the consolidated financial statements for information about Recent Accounting Developments.

Management's Discussion and Analysis

Results of Operations

The composition of our net revenues has varied over time as financial markets and the scope of our operations have changed. The composition of net revenues can also vary over the shorter term due to fluctuations in U.S. and global economic and market conditions. See "Risk Factors" in Part I, Item 1A of this Form 10-K for further information about the impact of economic and market conditions on our results of operations. For a discussion of our 2023 financial results compared with 2022, see Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations" in our Annual Report on Form 10-K for the year ended December 31, 2023.

Financial Overview

The table below presents an overview of our financial results and selected financial ratios.

\$ in millions, except per share amounts	Year Ended December		
	2024	2023	2022
Net revenues	\$ 53,512	\$ 46,254	\$ 47,365
Pre-tax earnings	\$ 18,397	\$ 10,739	\$ 13,486
Net earnings	\$ 14,276	\$ 8,516	\$ 11,261
Net earnings to common	\$ 13,525	\$ 7,907	\$ 10,764
Diluted EPS	\$ 40.54	\$ 22.87	\$ 30.06
ROE	12.7 %	7.5 %	10.2 %
ROTE	13.5 %	8.1 %	11.0 %
Net earnings to average assets	0.9 %	0.5 %	0.7 %
Return on shareholders' equity	12.0 %	7.3 %	9.7 %
Average equity to average assets	7.1 %	7.5 %	7.5 %
Dividend payout ratio	28.4 %	45.9 %	29.9 %

Our target (through-the-cycle) is to achieve ROE within a range of 14% to 16% and ROTE within a range of 15% to 17%.

In the table above:

- Net earnings to common represents net earnings applicable to common shareholders, which is calculated as net earnings less preferred stock dividends.
- ROE is calculated by dividing net earnings to common by average monthly common shareholders' equity.
- ROTE is calculated by dividing net earnings to common by average monthly tangible common shareholders' equity. Tangible common shareholders' equity is calculated as total shareholders' equity less preferred stock, goodwill and identifiable intangible assets. We believe that tangible common shareholders' equity is meaningful because it is a measure that we and investors use to assess capital adequacy and that ROTE is meaningful because it measures the performance of businesses consistently, whether they were acquired or developed internally. Tangible common shareholders' equity and ROTE are non-GAAP measures and may not be comparable to similar non-GAAP measures used by other companies.

The table below presents our average equity and the reconciliation of average common shareholders' equity to average tangible common shareholders' equity.

\$ in millions	Average for the Year Ended December		
	2024	2023	2022
Total shareholders' equity	\$ 119,204	\$ 116,699	\$ 115,990
Preferred stock	(12,430)	(10,895)	(10,703)
Common shareholders' equity	106,774	105,804	105,287
Goodwill	(5,895)	(6,147)	(5,726)
Identifiable intangible assets	(1,003)	(1,736)	(1,583)
Tangible common shareholders' equity	\$ 99,876	\$ 97,921	\$ 97,978

- Net earnings to average assets is calculated by dividing net earnings by average total assets.
- Return on shareholders' equity is calculated by dividing net earnings by average monthly shareholders' equity.
- Average equity to average assets is calculated by dividing average total shareholders' equity by average total assets.
- Dividend payout ratio is calculated by dividing dividends declared per common share by diluted EPS.

Net Revenues

The table below presents our net revenues by line item.

\$ in millions	Year Ended December		
	2024	2023	2022
Investment banking	\$ 7,738	\$ 6,218	\$ 7,360
Investment management	10,596	9,532	9,005
Commissions and fees	4,086	3,789	4,034
Market making	18,390	18,238	18,634
Other principal transactions	4,646	2,126	654
Total non-interest revenues	45,456	39,903	39,687
Interest income	81,397	68,515	29,024
Interest expense	73,341	62,164	21,346
Net interest income	8,056	6,351	7,678
Total net revenues	\$ 53,512	\$ 46,254	\$ 47,365

In the table above:

- Investment banking consists of revenues (excluding net interest) from financial advisory and underwriting assignments. These activities are included in Global Banking & Markets.
- Investment management consists of revenues (excluding net interest) from providing asset management and wealth advisory services across all major asset classes to a diverse set of clients. These activities are included in Asset & Wealth Management.
- Commissions and fees consists of revenues from executing and clearing client transactions on major stock, options and futures exchanges worldwide, as well as over-the-counter (OTC) transactions. Substantially all of these activities are included in Global Banking & Markets.

Management's Discussion and Analysis

- Market making consists of revenues (excluding net interest) from client execution activities related to making markets in interest rate products, credit products, mortgages, currencies, commodities and equity products. These activities are included in Global Banking & Markets.
- Other principal transactions consists of revenues (excluding net interest) from our equity investing activities, including revenues related to our consolidated investments (included in Asset & Wealth Management), and debt investing and lending activities (included across our three segments).

Operating Environment. During 2024, the operating environment was generally characterized by continued broad macroeconomic concerns, including concerns and uncertainty about inflation, ongoing geopolitical tensions, central bank policy and the potential outcome of national elections. In investment banking, industry-wide underwriting volumes increased compared with 2023, driven by strong levels of debt offerings and improved levels of equity offerings, while industry-wide completed mergers and acquisitions volumes remained below historical averages. In market making, activity levels were mixed, as activity levels in fixed income-related products decreased compared with the prior year, while activity levels in equity-related products increased. Global equity prices were generally higher compared with the end of 2023, and concerns about the commercial real estate market persisted. In the U.S., the rate of unemployment remained low and the pace of growth in consumer spending increased slightly compared with 2023.

If uncertainty and concerns about geopolitical tensions and the economic outlook remain elevated or grow, including those about central bank policy, inflation and the commercial real estate sector, it may lead to a decline in asset prices, a decline in market-making activity levels, a decline in industry-wide investment banking volumes, and net revenues and provision for credit losses would likely be negatively impacted. See "Segment Assets and Operating Results — Segment Operating Results" for information about the operating environment and material trends and uncertainties that may impact our results of operations.

2024 versus 2023

Net revenues in the consolidated statements of earnings were \$53.51 billion for 2024, 16% higher than 2023, primarily reflecting significantly higher other principal transactions revenues, net interest income and investment banking revenues and higher investment management revenues.

Non-Interest Revenues. Investment banking revenues in the consolidated statements of earnings were \$7.74 billion for 2024, 24% higher than 2023, primarily reflecting significantly higher revenues in debt underwriting, primarily driven by leveraged finance activity, and in equity underwriting, primarily driven by secondary and initial public offerings. In addition, revenues in advisory were higher, reflecting an increase in completed mergers and acquisitions transactions.

Investment management revenues in the consolidated statements of earnings were \$10.60 billion for 2024, 11% higher than 2023, primarily due to higher management and other fees, primarily reflecting the impact of higher average assets under supervision (AUS).

Commissions and fees in the consolidated statements of earnings were \$4.09 billion for 2024, 8% higher than 2023, due to higher commissions and fees in Equities, reflecting generally higher market volumes and increased transaction fees, partially offset by a loss related to the planned transition of the GM credit card program to another issuer.

Market making revenues in the consolidated statements of earnings were \$18.39 billion for 2024, essentially unchanged compared with 2023. Market making revenues from intermediation activities were slightly higher, primarily reflecting significantly higher revenues in equity cash products, currencies and mortgages, offset by significantly lower revenues in commodities and equity derivatives. Market making revenues from financing activities were essentially unchanged, reflecting significantly lower revenues in FICC financing, offset by significantly higher revenues from Equities financing.

Other principal transactions revenues in the consolidated statements of earnings were \$4.65 billion for 2024, 119% higher than 2023, primarily reflecting significantly higher net gains from investments in private equities, significantly higher net gains from derivatives related to our funding activities, the impact of the sale of the Marcus loan portfolio in 2023 (including net revenues of approximately \$(370) million related to the sale of substantially all of the portfolio) and significantly lower net losses on hedges related to our relationship lending portfolio.

Management's Discussion and Analysis

Net Interest Income. Net interest income in the consolidated statements of earnings was \$8.06 billion for 2024, 27% higher than 2023, reflecting an increase in interest income, partially offset by an increase in interest expense. The increase in interest income primarily related to trading assets and investments (each reflecting the impact of higher average balances and higher average interest rates), collateralized agreements and other interest-earning assets (each reflecting the impact of higher average interest rates), partially offset by a decrease in interest income related to deposits with banks (reflecting the impact of lower average balances). The increase in interest expense primarily related to collateralized financings and deposits (each reflecting the impact of higher average balances and higher average interest rates) and other interest-bearing liabilities (reflecting the impact of higher average interest rates). See "Supplemental Financial Information — Statistical Disclosures — Distribution of Assets, Liabilities and Shareholders' Equity" for further information about our sources of net interest income.

Provision for Credit Losses

Provision for credit losses consists of provision for credit losses on financial assets and commitments accounted for at amortized cost, including loans and lending commitments held for investment. See Note 9 to the consolidated financial statements for further information about the provision for credit losses on loans and lending commitments.

The table below presents our provision for credit losses.

\$ in millions	Year Ended December		
	2024	2023	2022
Provision for credit losses	\$ 1,348	\$ 1,028	\$ 2,715

2024 versus 2023. Provision for credit losses in the consolidated statements of earnings was \$1.35 billion for 2024, compared with \$1.03 billion for 2023. Provisions for 2024 reflected net provisions related to the credit card portfolio (primarily driven by net charge-offs). Provisions for 2023 reflected net provisions related to both the credit card portfolio (primarily driven by net charge-offs) and wholesale loans (primarily driven by impairments), partially offset by reserve reductions of \$637 million related to the transfer of the GreenSky loan portfolio to held for sale and \$442 million related to the sale of substantially all of the Marcus loan portfolio.

Operating Expenses

Our operating expenses are primarily influenced by compensation, headcount and levels of business activity. Compensation and benefits includes salaries, year-end discretionary compensation, amortization of equity awards and other items such as benefits. Discretionary compensation is significantly impacted by, among other factors, the level of net revenues, net of provision for credit losses, overall financial performance, prevailing labor markets, business mix, the structure of our share-based awards and the external environment.

The table below presents our operating expenses by line item and headcount.

\$ in millions	Year Ended December		
	2024	2023	2022
Compensation and benefits	\$ 16,706	\$ 15,499	\$ 15,148
Transaction based	6,724	5,698	5,312
Market development	646	629	812
Communications and technology	1,991	1,919	1,808
Depreciation and amortization	2,392	4,856	2,455
Occupancy	973	1,053	1,026
Professional fees	1,652	1,623	1,887
Other expenses	2,683	3,210	2,716
Total operating expenses	\$ 33,767	\$ 34,487	\$ 31,164
Headcount at period-end	46,500	45,300	48,500

2024 versus 2023. Operating expenses in the consolidated statements of earnings were \$33.77 billion for 2024, 2% lower than 2023. Our efficiency ratio was 63.1% for 2024, compared with 74.6% for 2023.

Operating expenses, compared with 2023, reflected decreases driven by significantly lower expenses, including impairments (\$1.46 billion recognized in 2023), related to commercial real estate in CIEs (largely in depreciation and amortization) and other significant expenses recognized in the prior year, including the write-down of identifiable intangible assets related to GreenSky of \$506 million and an impairment of goodwill related to Consumer platforms of \$504 million (both in depreciation and amortization), and the FDIC special assessment fee of \$529 million (in other expenses). These decreases were partially offset by higher compensation and benefits expenses (reflecting improved operating performance) and higher transaction based expenses. An incremental expense for the FDIC special assessment fee of \$71 million was recognized in 2024, as the FDIC notified banks subject to the special assessment fee of the updated estimated cost to the Deposit Insurance Fund resulting from the closures in 2023 of Silicon Valley Bank and Signature Bank. Net provisions for litigation and regulatory proceedings were \$166 million for 2024 compared with \$115 million for 2023.

As of December 2024, headcount increased 3% compared with December 2023, primarily due to increases in Asset & Wealth Management, Risk and Compliance, partially offset by the impact of the sale of GreenSky.

Provision for Taxes

The effective income tax rate for 2024 was 22.4%, up from the full year income tax rate of 20.7% for 2023, primarily due to a decrease in the impact of permanent tax benefits for 2024 compared with 2023, partially offset by changes in the geographic mix of earnings.

Management's Discussion and Analysis

The Organisation for Economic Co-operation and Development (OECD) Global Anti-Base Erosion Model Rules (Pillar II) aim to ensure that multinationals with revenues in excess of EUR 750 million pay a minimum effective corporate tax rate of 15% (minimum tax) in each jurisdiction in which they operate. The U.K. and other jurisdictions in which we operate have adopted certain portions of the OECD directive (Pillar II legislation) effective beginning in calendar year 2024. The Pillar II legislation did not have a material impact on our effective tax rate for 2024. We expect additional guidance or legislation to be issued by the OECD and various jurisdictions during 2025 which could impact any minimum tax we owe in future periods, possibly materially, and our effective tax rate could increase in 2025 and thereafter. This minimum tax, if any, will be recognized in the period in which it is incurred.

On August 26, 2024, the U.S. Tax Court issued a decision in *Varian Medical Systems, Inc. v. Commissioner* (Varian decision). The Varian decision reduced the U.S. tax on the deemed repatriation of unremitted foreign earnings of applicable non-U.S. subsidiaries in the transition year of the Tax Cuts and Jobs Act. We are monitoring the Varian decision and evaluating its impact, which could be a material income tax benefit, on the deemed repatriation tax we incurred for the 2018 tax year. No income tax benefit has been recognized in the provision for income taxes as a result of the Varian decision as of December 2024.

We expect our 2025 annual effective tax rate to be approximately 21%.

Segment Assets and Operating Results

Segment Assets. The table below presents assets by segment.

\$ in millions	As of December	
	2024	2023
Global Banking & Markets	\$ 1,420,142	\$ 1,381,247
Asset & Wealth Management	193,328	191,863
Platform Solutions	62,502	68,484
Total	\$ 1,675,972	\$ 1,641,594

The allocation process for segment assets is based on the activities of these segments. The allocation of assets includes allocation of GCLA (which consists of unencumbered, highly liquid securities and cash), which is included within cash and cash equivalents, collateralized agreements, trading assets and investments on our balance sheet. Due to the integrated nature of these segments, estimates and judgments are made in allocating these assets. See "Risk Management — Liquidity Risk Management" for further information about our GCLA.

Segment Operating Results. The table below presents our segment operating results.

\$ in millions	Year Ended December		
	2024	2023	2022
Global Banking & Markets			
Net revenues	\$ 34,943	\$ 29,996	\$ 32,487
Provision for credit losses	40	401	468
Compensation and benefits expenses	9,426	8,571	8,661
Other operating expenses	10,554	9,469	9,190
Total operating expenses	19,980	18,040	17,851
Pre-tax earnings	\$ 14,923	\$ 11,555	\$ 14,168
Net earnings to common	\$ 10,998	\$ 8,703	\$ 11,458
Average common equity	\$ 75,796	\$ 71,863	\$ 69,951
Return on average common equity	14.5 %	12.1 %	16.4 %
Asset & Wealth Management			
Net revenues	\$ 16,142	\$ 13,880	\$ 13,376
Provision for credit losses	(232)	(508)	519
Compensation and benefits expenses	6,595	6,144	5,927
Other operating expenses	5,230	6,885	5,623
Total operating expenses	11,825	13,029	11,550
Pre-tax earnings	\$ 4,549	\$ 1,359	\$ 1,307
Net earnings to common	\$ 3,386	\$ 952	\$ 979
Average common equity	\$ 26,405	\$ 30,078	\$ 31,762
Return on average common equity	12.8 %	3.2 %	3.1 %
Platform Solutions			
Net revenues	\$ 2,427	\$ 2,378	\$ 1,502
Provision for credit losses	1,540	1,135	1,728
Compensation and benefits expenses	685	784	560
Other operating expenses	1,277	2,634	1,203
Total operating expenses	1,962	3,418	1,763
Pre-tax earnings/(loss)	\$ (1,075)	\$ (2,175)	\$ (1,989)
Net earnings/(loss) to common	\$ (859)	\$ (1,748)	\$ (1,673)
Average common equity	\$ 4,573	\$ 3,863	\$ 3,574
Return on average common equity	(18.8)%	(45.2)%	(46.8)%
Total			
Net revenues	\$ 53,512	\$ 46,254	\$ 47,365
Provision for credit losses	1,348	1,028	2,715
Compensation and benefits expenses	16,706	15,499	15,148
Other operating expenses	17,061	18,988	16,016
Total operating expenses	33,767	34,487	31,164
Pre-tax earnings	\$ 18,397	\$ 10,739	\$ 13,486
Net earnings to common	\$ 13,525	\$ 7,907	\$ 10,764
Average common equity	\$ 106,774	\$ 105,804	\$ 105,287
Return on average common equity	12.7 %	7.5 %	10.2 %

Net revenues in our segments include allocations of interest income and expense based on the funding generated by, or the funding and liquidity requirements of, the respective segments. See Note 25 to the consolidated financial statements for further information about our business segments.

The allocation of common shareholders' equity and preferred stock dividends to each segment is based on the estimated amount of equity required to support the activities of the segment under relevant regulatory capital requirements. Net earnings for each segment is calculated by applying the firmwide tax rate to each segment's pre-tax earnings.

Management's Discussion and Analysis

Compensation and benefits expenses within our segments reflect, among other factors, our overall performance, as well as the performance of individual businesses. Consequently, pre-tax margins in one segment of our business may be significantly affected by the performance of our other business segments. A description of segment operating results follows.

Global Banking & Markets

Global Banking & Markets generates revenues from the following:

Investment banking fees. We provide advisory and underwriting services and help companies raise capital to strengthen and grow their businesses. Investment banking fees includes the following:

- **Advisory.** Includes strategic advisory assignments with respect to mergers and acquisitions, divestitures, corporate defense activities, restructurings and spin-offs.
- **Underwriting.** Includes public offerings and private placements in both local and cross-border transactions of a wide range of securities and other financial instruments, including acquisition financing.

FICC. FICC generates revenues from intermediation and financing activities.

- **FICC intermediation.** Includes client execution activities related to making markets in both cash and derivative instruments, as detailed below.

Interest Rate Products. Government bonds (including inflation-linked securities) across maturities, other government-backed securities, and interest rate swaps, options and other derivatives.

Credit Products. Investment-grade and high-yield corporate securities, credit derivatives, exchange-traded funds (ETFs), bank and bridge loans, municipal securities, distressed debt and trade claims.

Mortgages. Commercial mortgage-related securities, loans and derivatives, residential mortgage-related securities, loans and derivatives (including U.S. government agency-issued collateralized mortgage obligations and other securities and loans), and other asset-backed securities, loans and derivatives.

Currencies. Currency options, spot/forwards and other derivatives on G-10 currencies and emerging-market products.

Commodities. Commodity derivatives and, to a lesser extent, physical commodities, involving crude oil and petroleum products, natural gas, agricultural, base, precious and other metals, electricity, including renewable power, environmental products and other commodity products.

- **FICC financing.** Includes (i) secured lending to our clients through structured credit and asset-backed lending, including warehouse loans backed by mortgages (including residential and commercial mortgage loans), corporate loans and consumer loans (including auto loans and private student loans), (ii) financing through securities purchased under agreements to resell (resale agreements) and (iii) commodity financing to clients through structured transactions.

Equities. Equities generates revenues from intermediation and financing activities.

- **Equities intermediation.** We make markets in equity securities and equity-related products, including ETFs, convertible securities, options, futures and OTC derivative instruments. We also structure and make markets in derivatives on indices, industry sectors, financial measures and individual company stocks. Our exchange-based market-making activities include making markets in stocks and ETFs, futures and options on major exchanges worldwide. In addition, we generate commissions and fees from executing and clearing institutional client transactions on major stock, options and futures exchanges worldwide, as well as OTC transactions.

- **Equities financing.** Includes prime financing, which provides financing to our clients for their securities trading activities through margin loans that are generally collateralized by securities or cash. Prime financing also includes services which involve lending securities to cover institutional clients' short sales and borrowing securities to cover our short sales and to make deliveries into the market. We are also an active participant in broker-to-broker securities lending and third-party agency lending activities. In addition, we execute swap transactions to provide our clients with exposure to securities and indices. Financing activities also include portfolio financing, which clients can utilize to manage their investment portfolios, and other equity financing activities, including securities-based loans to individuals.

Market-Making Activities

As a market maker, we facilitate transactions in both liquid and less liquid markets, primarily for institutional clients, such as corporations, financial institutions, investment funds and governments, to assist clients in meeting their investment objectives and in managing their risks. In this role, we seek to earn the difference between the price at which a market participant is willing to sell an instrument to us and the price at which another market participant is willing to buy it from us, and vice versa (i.e., bid/offer spread). In addition, we maintain (i) market-making positions, typically for a short period of time, in response to, or in anticipation of, client demand, and (ii) positions to actively manage our risk exposures that arise from these market-making activities (collectively, inventory). Our inventory is recorded in trading assets (long positions) or trading liabilities (short positions) in our consolidated balance sheets.

Management's Discussion and Analysis

Our results are influenced by a combination of interconnected drivers, including (i) client activity levels and transactional bid/offer spreads (collectively, client activity), and (ii) changes in the fair value of our inventory and interest income and interest expense related to the holding, hedging and funding of our inventory (collectively, market-making inventory changes). Due to the integrated nature of our market-making activities, disaggregation of net revenues into client activity and market-making inventory changes is judgmental and has inherent complexities and limitations.

The amount and composition of our net revenues vary over time as these drivers are impacted by multiple interrelated factors affecting economic and market conditions, including volatility and liquidity in the market, changes in interest rates, currency exchange rates, credit spreads, equity prices and commodity prices, investor confidence, and other macroeconomic concerns and uncertainties.

In general, assuming all other market-making conditions remain constant, increases in client activity levels or bid/offer spreads tend to result in increases in net revenues, and decreases tend to have the opposite effect. However, changes in market-making conditions can materially impact client activity levels and bid/offer spreads, as well as the fair value of our inventory. For example, a decrease in liquidity in the market could have the impact of (i) increasing our bid/offer spread, (ii) decreasing investor confidence and thereby decreasing client activity levels, and (iii) widening of credit spreads on our inventory positions.

Other. We lend to corporate clients, including through relationship lending and acquisition financing. The hedges related to this lending and financing activity are also reported as part of Other. Other also includes equity and debt investing activities related to our Global Banking & Markets activities.

The table below presents our Global Banking & Markets assets.

\$ in millions	As of December	
	2024	2023
Cash and cash equivalents	\$ 129,687	\$ 168,857
Collateralized agreements	356,637	401,554
Customer and other receivables	113,646	117,633
Trading assets	508,379	435,275
Investments	161,381	122,350
Loans	130,670	117,464
Other assets	19,742	18,114
Total	\$ 1,420,142	\$ 1,381,247

The table below presents details about our Global Banking & Markets loans.

\$ in millions	As of December	
	2024	2023
Corporate	\$ 22,595	\$ 24,159
Real estate	37,705	34,813
Securities-based	4,279	3,758
Other collateralized	67,080	55,527
Installment	70	173
Other	128	475
Loans, gross	131,857	118,905
Allowance for loan losses	(1,187)	(1,441)
Total loans	\$ 130,670	\$ 117,464

Our average Global Banking & Markets gross loans were \$126.87 billion for 2024 and \$112.07 billion for 2023.

The table below presents our Global Banking & Markets operating results.

\$ in millions	Year Ended December		
	2024	2023	2022
Advisory	\$ 3,534	\$ 3,299	\$ 4,704
Equity underwriting	1,677	1,153	848
Debt underwriting	2,521	1,764	1,808
Investment banking fees	7,732	6,216	7,360
FICC intermediation	9,564	9,318	11,890
FICC financing	3,640	2,742	2,786
FICC	13,204	12,060	14,676
Equities intermediation	7,937	6,489	6,662
Equities financing	5,494	5,060	4,326
Equities	13,431	11,549	10,988
Other	576	171	(537)
Total net revenues	34,943	29,996	32,487
Provision for credit losses	40	401	468
Compensation and benefits expenses	9,426	8,571	8,661
Other operating expenses	10,554	9,469	9,190
Total operating expenses	19,980	18,040	17,851
Pre-tax earnings	14,923	11,555	14,168
Provision for taxes	3,343	2,392	2,338
Net earnings	11,580	9,163	11,830
Preferred stock dividends	582	460	372
Net earnings to common	\$ 10,998	\$ 8,703	\$ 11,458
Average common equity	\$ 75,796	\$ 71,863	\$ 69,951
Return on average common equity	14.5 %	12.1 %	16.4 %

Management's Discussion and Analysis

The table below presents our FICC and Equities net revenues by line item in the consolidated statements of earnings.

<i>\$ in millions</i>	FICC	Equities
Year Ended December 2024		
Market making	\$ 9,020	\$ 9,370
Commissions and fees	-	4,289
Other principal transactions	1,203	68
Net interest income	2,981	(296)
Total	\$ 13,204	\$ 13,431
Year Ended December 2023		
Market making	\$ 10,632	7,606
Commissions and fees	-	3,736
Other principal transactions	656	81
Net interest income	772	126
Total	\$ 12,060	\$ 11,549
Year Ended December 2022		
Market making	\$ 12,422	6,212
Commissions and fees	-	3,791
Other principal transactions	377	41
Net interest income	1,877	944
Total	\$ 14,676	\$ 10,988

In the table above:

- See "Net Revenues" for information about market making revenues, commissions and fees, other principal transactions revenues and net interest income. See Note 25 to the consolidated financial statements for net interest income by segment.
- The primary driver of net revenues for FICC intermediation for all periods was client activity.
- The increase in net interest income within FICC for 2024 compared with 2023 reflected an increase in interest-earning assets. Due to the nature of activities within FICC and Equities and the composition of their associated balance sheet, we assess the performance of these businesses based on total net revenues, as offsets can occur across revenue line items. For example, cash instruments that generate interest income are, in some cases, hedged or funded by derivatives for which changes in fair value are reflected in market making revenues. Also, certain activities produce market making revenues but incur interest expense related to the funding of the related inventory.

The table below presents our financial advisory and underwriting transaction volumes.

<i>\$ in billions</i>	Year Ended December		
	2024	2023	2022
Announced mergers and acquisitions	\$ 1,037	\$ 933	\$ 1,173
Completed mergers and acquisitions	\$ 901	\$ 1,012	\$ 1,356
Equity and equity-related offerings	\$ 57	\$ 43	\$ 33
Debt offerings	\$ 295	\$ 209	\$ 223

In the table above:

- Volumes are per Dealogic.

- Announced and completed mergers and acquisitions volumes are based on full credit to each of the advisors in a transaction. Equity and equity-related and debt offerings are based on full credit for single book managers and equal credit for joint book managers. Transaction volumes may not be indicative of net revenues in a given period. In addition, transaction volumes for prior periods may vary from amounts previously reported due to the subsequent withdrawal or a change in the value of a transaction.
- Equity and equity-related offerings includes Rule 144A and public common stock offerings, convertible offerings and rights offerings.
- Debt offerings includes non-convertible preferred stock, mortgage-backed securities, asset-backed securities and taxable municipal debt. It also includes publicly registered and Rule 144A issues and excludes leveraged loans.

In January 2025, we formed the Capital Solutions Group in Global Banking & Markets, which provides a more comprehensive suite of our financing, origination, structuring and risk management offerings across both public and private markets. This group includes the current capabilities of our financing group and expands its coverage to financial sponsors and alternative asset management firms. It also includes an alternatives origination group focused on sourcing, to provide seamless coverage to our private credit and private equity clients. We believe that a more integrated set of these capabilities will allow us to better serve our clients as these private markets continue to grow.

Operating Environment. During 2024, Global Banking & Markets operated in an environment generally characterized by continued broad macroeconomic concerns, including concerns and uncertainty about inflation, prolonged geopolitical stresses and central bank policy, and the potential outcomes of the national elections.

In investment banking, industry-wide debt underwriting volumes for the year increased significantly compared with the prior year, driven by strong levels of leveraged finance and investment-grade offerings. However, industry-wide equity underwriting volumes, despite improving year-over-year, and industry-wide completed mergers and acquisitions volumes remained below historical averages.

In interest rates, the yields on 10-year U.S. and U.K. government bonds increased during the year. In equities, the S&P 500 Index increased by 23% and the MSCI World Index increased by 16% compared with the end of 2023.

In the future, if market and economic conditions deteriorate, and market-making activity levels decline, industry-wide investment banking volumes decline, or credit spreads related to hedges on our relationship lending portfolio tighten, net revenues in Global Banking & Markets would likely be negatively impacted. In addition, if economic conditions deteriorate or if the creditworthiness of borrowers deteriorates, provision for credit losses would likely be negatively impacted.

Management's Discussion and Analysis

2024 versus 2023. Net revenues in Global Banking & Markets were \$34.94 billion for 2024, 16% higher than 2023.

Investment banking fees were \$7.73 billion, 24% higher than 2023, primarily reflecting significantly higher net revenues in Debt underwriting, primarily driven by leveraged finance activity, and in Equity underwriting, primarily driven by secondary and initial public offerings. In addition, net revenues in Advisory were higher, reflecting an increase in completed mergers and acquisitions transactions.

As of December 2024, our Investment banking fees backlog increased compared with the end of 2023, primarily reflecting higher estimated net revenues from potential advisory transactions.

Our backlog represents an estimate of our net revenues from future transactions where we believe that future revenue realization is more likely than not. We believe changes in our backlog may be a useful indicator of client activity levels which, over the long term, impact our net revenues. However, the time frame for completion and corresponding revenue recognition of transactions in our backlog varies based on the nature of the assignment, as certain transactions may remain in our backlog for longer periods of time. In addition, our backlog is subject to certain limitations, such as assumptions about the likelihood that individual client transactions will occur in the future. Transactions may be cancelled or modified, and transactions not included in the estimate may also occur.

Net revenues in FICC were \$13.20 billion, 9% higher than 2023, primarily reflecting significantly higher net revenues in FICC financing, primarily driven by mortgages and structured lending. Net revenues in FICC intermediation were slightly higher, driven by significantly higher net revenues in currencies, mortgages and credit products, largely offset by lower net revenues in interest rate products and significantly lower net revenues in commodities.

The increase in FICC intermediation net revenues reflected the impact of improved market-making conditions on our inventory, partially offset by lower client activity. The following provides information about our FICC intermediation net revenues by business, compared with results for 2023:

- Net revenues in currencies, mortgages and credit products reflected the impact of improved market-making conditions on our inventory.
- Net revenues in interest rate products and commodities primarily reflected lower client activity.

Net revenues in Equities were \$13.43 billion, 16% higher than 2023, reflecting significantly higher net revenues in Equities intermediation, primarily driven by derivatives, and higher net revenues in Equities financing, driven by prime financing.

Net revenues in Other were \$576 million for 2024, compared with \$171 million for 2023, with the increase primarily reflecting significantly lower net losses on hedges.

Provision for credit losses was \$40 million for 2024, compared with \$401 million for 2023. Provisions for 2023 primarily reflected net provisions related to the commercial real estate portfolio.

Operating expenses were \$19.98 billion for 2024, 11% higher than 2023, primarily due to significantly higher transaction based expenses and higher compensation and benefits expenses (reflecting improved operating performance). Pre-tax earnings were \$14.92 billion for 2024, 29% higher than 2023.

Asset & Wealth Management

Asset & Wealth Management provides investment services to help clients preserve and grow their financial assets and achieve their financial goals. We provide these services to our clients, both institutional and individuals, including investors who primarily access our products through a network of third-party distributors around the world.

We manage client assets across a broad range of investment strategies and asset classes, including equity, fixed income and alternative investments. We provide investment solutions, including those managed on a fiduciary basis by our portfolio managers, as well as those managed by third-party managers. We offer our investment solutions in a variety of structures, including separately managed accounts, mutual funds, private partnerships and other commingled vehicles.

We also provide tailored wealth advisory services, primarily to ultra-high-net worth clients. We operate globally, serving individuals, families, family offices, and foundations and endowments. Our relationships are established directly or introduced through companies that sponsor financial wellness or financial planning programs for their employees, as well as through corporate referrals.

We offer personalized financial planning to individuals and also provide customized investment advisory solutions, and offer structuring and execution capabilities in securities and derivative products across all major global markets. In addition, we offer clients a full range of private banking services, including a variety of deposit alternatives and loans that our clients use to finance investments in both financial and nonfinancial assets, bridge cash flow timing gaps or provide liquidity and flexibility for other needs. We also raise deposits from consumers through Marcus.

Management's Discussion and Analysis

We invest alongside our clients that invest in investment funds that we raise or manage. We also have investments in alternative assets across a range of asset classes. Our investing activities, which are typically longer-term, include investments in corporate equity, credit, real estate and infrastructure assets.

Asset & Wealth Management generates revenues from the following:

- **Management and other fees.** We receive fees related to managing assets for institutional and individual clients, providing investing and wealth advisory solutions, providing financial planning and counseling services, and executing brokerage transactions for wealth management clients. The vast majority of revenues in management and other fees consists of asset-based fees on client assets that we manage. For further information about assets under supervision, see "Assets Under Supervision" below. The fees that we charge vary by asset class, client channel and the types of services provided, and are affected by investment performance, as well as asset inflows and redemptions.
- **Incentive fees.** In certain circumstances, we also receive incentive fees based on a percentage of a fund's or a separately managed account's return, or when the return exceeds a specified benchmark or other performance targets. Such fees include overrides, which consist of the increased share of the income and gains derived primarily from our private equity and credit funds when the return on a fund's investments over the life of the fund exceeds certain threshold returns.
- **Private banking and lending.** Our private banking and lending activities include issuing loans to our wealth management clients. Such loans are generally secured by commercial and residential real estate, securities or other assets. We also raise deposits from wealth management clients, including through Marcus. Private banking and lending revenues include net interest income allocated to deposits and net interest income earned on loans to individual clients.
- **Equity investments.** Includes investing activities related to our asset management activities primarily related to public and private equity investments in corporate, real estate and infrastructure assets. We also make investments through CIEs, substantially all of which are engaged in real estate investment activities. In addition, we make investments in connection with our activities to satisfy requirements under the Community Reinvestment Act, primarily through our Urban Investment Group.
- **Debt investments.** Includes lending activities related to our asset management activities, including investing in corporate debt, lending to middle-market clients, and providing financing for real estate and other assets. These activities include investments in mezzanine debt, senior debt and distressed debt securities.

The table below presents our Asset & Wealth Management assets.

\$ in millions	As of December	
	2024	2023
Cash and cash equivalents	\$ 36,364	\$ 48,677
Collateralized agreements	12,126	14,020
Customer and other receivables	19,999	14,859
Trading assets	41,724	27,324
Investments	23,130	24,487
Loans	46,694	45,866
Other assets	13,291	16,630
Total	\$ 193,328	\$ 191,863

The table below presents details about our Asset & Wealth Management loans.

\$ in millions	As of December	
	2024	2023
Corporate	\$ 7,377	\$ 11,715
Real estate	18,053	16,603
Securities-based	12,198	10,863
Other collateralized	8,027	6,698
Other	1,951	1,121
Loans, gross	47,606	47,000
Allowance for loan losses	(912)	(1,134)
Total loans	\$ 46,694	\$ 45,866

In the table above, gross loans included \$38 billion of loans as of December 2024 and \$33 billion of loans as of December 2023 that were related to Private banking and lending.

The average Asset & Wealth Management gross loans were \$45.84 billion for 2024 and \$51.98 billion for 2023.

The table below presents our Asset & Wealth Management operating results.

\$ in millions	Year Ended December		
	2024	2023	2022
Management and other fees	\$ 10,425	\$ 9,486	\$ 8,781
Incentive fees	393	161	359
Private banking and lending	2,881	2,576	2,458
Equity investments	1,359	342	610
Debt investments	1,084	1,315	1,168
Total net revenues	16,142	13,880	13,376
Provision for credit losses	(232)	(508)	519
Compensation and benefits expenses	6,595	6,144	5,927
Other operating expenses	5,230	6,885	5,623
Total operating expenses	11,825	13,029	11,550
Pre-tax earnings	4,549	1,359	1,307
Provision for taxes	1,019	281	215
Net earnings	3,530	1,078	1,092
Preferred stock dividends	144	126	113
Net earnings to common	\$ 3,386	\$ 952	\$ 979
Average common equity	\$ 26,405	\$ 30,078	\$ 31,762
Return on average common equity	12.8 %	3.2 %	3.1 %

In the table above, Management and other fees included fees from alternatives of \$2.18 billion for 2024, \$2.13 billion for 2023 and \$1.85 billion for 2022.

In 2024, we surpassed our target to achieve annual firmwide management and other fees of more than \$10 billion, including more than \$2 billion from alternatives.

Management's Discussion and Analysis

In 2024, we achieved our target of pre-tax margins in the mid-twenties for Asset & Wealth Management. We also have a target to achieve ROE in the mid-teens within the medium term (three to five year time horizon from year-end 2022) for Asset & Wealth Management. The pre-tax margin for Asset & Wealth Management was 28% for 2024, including the positive impact of 4 percentage points from the results of historical principal investments.

The table below presents our Asset management and Wealth management net revenues by line item in Asset & Wealth Management.

<i>\$ in millions</i>	Asset management	Wealth management	Asset & Wealth Management
Year Ended December 2024			
Management and other fees	\$ 4,576	\$ 5,849	\$ 10,425
Incentive fees	393	-	393
Private banking and lending	-	2,881	2,881
Equity investments	1,357	2	1,359
Debt investments	1,084	-	1,084
Total	\$ 7,410	\$ 8,732	\$ 16,142
Year Ended December 2023			
Management and other fees	\$ 4,207	\$ 5,279	\$ 9,486
Incentive fees	161	-	161
Private banking and lending	-	2,576	2,576
Equity investments	(7)	349	342
Debt investments	1,315	-	1,315
Total	\$ 5,676	\$ 8,204	\$ 13,880
Year Ended December 2022			
Management and other fees	\$ 3,817	\$ 4,964	\$ 8,781
Incentive fees	359	-	359
Private banking and lending	-	2,458	2,458
Equity investments	610	-	610
Debt investments	1,168	-	1,168
Total	\$ 5,954	\$ 7,422	\$ 13,376

The table below presents our Equity investments net revenues by equity type and asset class.

<i>\$ in millions</i>	Year Ended December		
	2024	2023	2022
Equity Type			
Private equity	\$ 1,303	\$ 361	\$ 2,078
Public equity	56	(19)	(1,468)
Total	\$ 1,359	\$ 342	\$ 610
Asset Class			
Real estate	\$ 289	(181)	\$ 1,482
Corporate	1,070	523	(872)
Total	\$ 1,359	\$ 342	\$ 610

The table below presents details about our Debt investments net revenues.

<i>\$ in millions</i>	Year Ended December		
	2024	2023	2022
Fair value net gains/(losses)	\$ 154	(61)	(415)
Net interest income	930	1,376	1,583
Total	\$ 1,084	\$ 1,315	\$ 1,168

Operating Environment. During 2024, Asset & Wealth Management operated in an environment generally characterized by continued broad macroeconomic concerns, including persistent concerns about the commercial real estate market. However, global equity prices were generally higher compared with the end of 2023, positively affecting assets under supervision.

In the future, if market and economic conditions deteriorate, it may lead to a decline in asset prices, or investors transitioning to asset classes that typically generate lower fees or withdrawing their assets, and net revenues in Asset & Wealth Management would likely be negatively impacted.

2024 versus 2023. Net revenues in Asset & Wealth Management were \$16.14 billion for 2024, 16% higher than 2023, primarily reflecting significantly higher net revenues in Equity investments and higher Management and other fees. In addition, net revenues in Private banking and lending and Incentive fees were higher, while net revenues in Debt investments were lower.

The increase in Equity investments net revenues primarily reflected significantly higher net gains from investments in private equities (largely reflecting the impact of net losses in real estate investments in the prior year). The increase in Management and other fees primarily reflected the impact of higher average assets under supervision. The increase in Private banking and lending net revenues primarily reflected the impact of the sale of the Marcus loan portfolio in 2023 (including net revenues of approximately \$(370) million related to the sale of substantially all of the portfolio) and the impact of higher direct-to-consumer deposit balances. The increase in Incentive fees was driven by harvesting. The decrease in Debt investments net revenues reflected lower net interest income due to a reduction in the debt investments balance sheet, partially offset by net gains in the current year compared with net losses (particularly in real estate investments) in the prior year.

Provision for credit losses was a net benefit of \$232 million for 2024, compared with a net benefit of \$508 million for 2023. The net benefit for 2024 reflected a net benefit related to the wholesale portfolio (driven by paydowns). The net benefit for 2023 primarily reflected reserve reductions related to the sale of substantially all of the Marcus loan portfolio and lower balances in corporate loans, partially offset by impairments.

Operating expenses were \$11.83 billion for 2024, 9% lower than 2023, due to significantly lower expenses, including impairments, related to commercial real estate in CIEs, partially offset by higher compensation and benefits expenses (reflecting improved operating performance). Pre-tax earnings were \$4.55 billion for 2024, compared with \$1.36 billion for 2023.

Management's Discussion and Analysis

Assets Under Supervision. AUS includes our institutional clients' assets, assets sourced through third-party distributors and high-net-worth clients' assets where we earn a fee for managing assets on a discretionary basis. This includes net assets in our mutual funds, hedge funds, credit funds, private equity funds, real estate funds, and separately managed accounts for institutional and individual investors. AUS also includes client assets invested with third-party managers, private bank deposits and advisory relationships where we earn a fee for advisory and other services, but do not have investment discretion. AUS does not include the self-directed brokerage assets of our clients.

The table below presents information about our firmwide period-end AUS by asset class, client channel, region and vehicle.

\$ in billions	As of December		
	2024	2023	2022
Asset Class			
Alternative investments	\$ 336	\$ 295	\$ 263
Equity	772	658	563
Fixed income	1,184	1,122	1,010
Total long-term AUS	2,292	2,075	1,836
Liquidity products	845	737	711
Total AUS	\$ 3,137	\$ 2,812	\$ 2,547
Client Channel			
Institutional	\$ 1,078	\$ 1,033	\$ 905
Wealth management	929	798	712
Third-party distributed	1,130	981	930
Total AUS	\$ 3,137	\$ 2,812	\$ 2,547
Region			
Americas	\$ 2,235	\$ 1,951	\$ 1,806
EMEA	683	653	548
Asia	219	208	193
Total AUS	\$ 3,137	\$ 2,812	\$ 2,547
Vehicle			
Separate accounts	\$ 1,687	\$ 1,557	\$ 1,388
Public funds	1,004	901	862
Private funds and other	446	354	297
Total AUS	\$ 3,137	\$ 2,812	\$ 2,547

In the table above:

- Liquidity products includes money market funds and private bank deposits.
- EMEA represents Europe, Middle East and Africa.

Total wealth management client assets (consisting of AUS, brokerage assets and Marcus deposits) were approximately \$1.6 trillion as of December 2024.

The table below presents changes in our AUS.

\$ in billions	Year Ended December		
	2024	2023	2022
Beginning balance	\$ 2,812	\$ 2,547	\$ 2,470
Net inflows/(outflows):			
Alternative investments	38	25	19
Equity	15	(3)	13
Fixed income	53	52	18
Total long-term AUS net inflows/(outflows)	106	74	50
Liquidity products	108	27	16
Total AUS net inflows/(outflows)	214	101	66
Acquisitions/(dispositions)	-	(23)	316
Net market appreciation/(depreciation)	111	187	(305)
Ending balance	\$ 3,137	\$ 2,812	\$ 2,547

In the table above:

- During 2024, our AUS increased \$325 billion due to net inflows (across all asset classes) and net market appreciation (primarily in equity assets).
- During 2023, our AUS increased \$265 billion due to net market appreciation (primarily in equity and fixed income assets) and net inflows (driven by fixed income assets, liquidity products and alternative investments assets), partially offset by the impact of dispositions (related to the sale of Personal Financial Management (PFM)).
- During 2022, our AUS increased \$77 billion due to the impact of acquisitions (primarily related to the acquisition of NN Investment Partners) and net inflows (across all asset classes), partially offset by net market depreciation (primarily in fixed income and equity assets).

The table below presents information about our total AUS net inflows/(outflows) by client channel.

\$ in billions	Year Ended December		
	2024	2023	2022
Institutional	\$ 38	\$ 38	\$ 16
Wealth management	61	31	39
Third-party distributed	115	32	11
Total AUS net inflows/(outflows)	\$ 214	\$ 101	\$ 66

Management's Discussion and Analysis

The table below presents information about our average monthly firmwide AUS by asset class.

\$ in billions	Average for the Year Ended December		
	2024	2023	2022
Asset Class			
Alternative investments	\$ 314	\$ 269	\$ 253
Equity	731	610	581
Fixed income	1,164	1,050	992
Total long-term AUS	2,209	1,929	1,826
Liquidity products	751	749	693
Total AUS	\$ 2,960	\$ 2,678	\$ 2,519

In addition to our AUS, we have discretion over alternative investments where we currently do not earn management fees (non-fee-earning alternative assets).

We earn management fees on client assets that we manage and also receive incentive fees based on a percentage of a fund's or a separately managed account's return, or when the return exceeds a specified benchmark or other performance targets. These incentive fees are recognized when it is probable that a significant reversal of such fees will not occur. Our estimated unrecognized incentive fees were \$4.12 billion as of December 2024 and \$3.77 billion as of December 2023. Such amounts are based on the completion of the funds' financial statements, which is generally one quarter in arrears. These fees will be recognized, assuming no decline in fair value, if and when it is probable that a significant reversal of such fees will not occur, which is generally when such fees are no longer subject to fluctuations in the market value of the assets.

The table below presents our average effective management fee (which excludes non-asset-based fees) earned on our firmwide AUS by asset class.

Effective fees (bps)	Year Ended December		
	2024	2023	2022
Alternative investments	62	64	64
Equity	55	57	57
Fixed income	17	17	17
Liquidity products	15	15	14
Total average effective fee	31	31	31

The table below presents details about our monthly average AUS for alternative investments and the average effective management fee we earned on such assets.

\$ in billions	Direct strategies	Fund of funds	Total
Year Ended December 2024			
Average AUS			
Corporate equity	\$ 34	\$ 84	\$ 118
Credit	48	12	60
Real estate	13	14	27
Hedge funds and other	45	26	71
Funds and discretionary accounts	\$ 140	\$ 136	\$ 276
Advisory accounts			38
Total average AUS for alternative investments			\$ 314
Effective Fees (bps)			
Corporate equity	122	55	75
Credit	82	10	71
Real estate	88	32	58
Hedge funds and other	68	45	59
Funds and discretionary accounts	88	47	68
Advisory accounts			17
Total average effective fee			62
Year Ended December 2023			
Average AUS			
Corporate equity	\$ 29	\$ 70	\$ 99
Credit	44	2	46
Real estate	11	9	20
Hedge funds and other	42	22	64
Funds and discretionary accounts	\$ 126	\$ 103	\$ 229
Advisory accounts			40
Total average AUS for alternative investments			\$ 269
Effective Fees (bps)			
Corporate equity	125	61	80
Credit	80	37	78
Real estate	82	42	64
Hedge funds and other	67	53	62
Funds and discretionary accounts	86	57	73
Advisory accounts			16
Total average effective fee			64
Year Ended December 2022			
Average AUS			
Corporate equity	\$ 27	\$ 61	\$ 88
Credit	36	2	38
Real estate	10	8	18
Hedge funds and other	45	22	67
Funds and discretionary accounts	\$ 118	\$ 93	\$ 211
Advisory accounts			42
Total average AUS for alternative investments			\$ 253
Effective Fees (bps)			
Corporate equity	133	61	83
Credit	81	51	80
Real estate	87	50	70
Hedge funds and other	64	49	59
Funds and discretionary accounts	87	57	74
Advisory accounts			16
Total average effective fee			64

In the table above, direct strategies primarily includes our private equity, growth equity, private credit, liquid alternatives and real estate strategies. Fund of funds primarily includes our business which invests in leading private equity, hedge fund, real estate and credit third-party managers as a limited partner, secondary-market investor, co-investor or management company partner.

Management's Discussion and Analysis

The table below presents information about our period-end AUS for alternative investments, non-fee-earning alternative investments and total alternative investments.

<i>\$ in billions</i>	AUS	Non-fee-earning alternative assets	Total alternative assets
As of December 2024			
Corporate equity	\$ 131	\$ 73	\$ 204
Credit	62	79	141
Real estate	30	24	54
Hedge funds and other	76	3	79
Funds and discretionary accounts	299	179	478
Advisory accounts	37	2	39
Total alternative investments	\$ 336	\$ 181	\$ 517
As of December 2023			
Corporate equity	\$ 109	\$ 77	\$ 186
Credit	55	75	130
Real estate	22	32	54
Hedge funds and other	66	3	69
Funds and discretionary accounts	252	187	439
Advisory accounts	43	3	46
Total alternative investments	\$ 295	\$ 190	\$ 485
As of December 2022			
Corporate equity	\$ 94	\$ 76	\$ 170
Credit	44	73	117
Real estate	18	36	54
Hedge funds and other	65	2	67
Funds and discretionary accounts	221	187	408
Advisory accounts	42	-	42
Total alternative investments	\$ 263	\$ 187	\$ 450

In the table above:

- Corporate equity primarily includes private equity.
- Total alternative assets included uncalled capital that is available for future investing of \$61 billion as of December 2024 and \$58 billion as of December 2023.
- Non-fee-earning alternative assets primarily includes investments that we hold on our balance sheet, our unfunded commitments, unfunded commitments of our clients (where we do not charge fees on commitments), credit facilities collateralized by fund assets and employee funds. Our calculation of non-fee-earning alternative assets may not be comparable to similar calculations used by other companies.
- Non-fee-earning alternative assets primarily includes our direct investing strategies, including private equity, growth equity, private credit and real estate strategies.

Our target is to grow our total credit alternative assets to \$300 billion by the end of 2028.

The table below presents information about third-party commitments raised in our alternatives business from the beginning of 2020 through 2024.

<i>\$ in billions</i>	As of December 2024
Included in AUS	\$ 242
Included in non-fee-earning alternative assets	81
Third-party commitments raised	\$ 323

In the table above, commitments included in non-fee-earning alternative assets included approximately \$61 billion, which will begin to earn fees (and become AUS) if and when the commitments are drawn and assets are invested. In 2024, we raised \$72 billion in third-party commitments in our alternatives business, including \$28 billion in corporate equity, \$19 billion in credit, \$6 billion in real estate and \$19 billion in hedge funds and other. Since 2019, we have raised \$323 billion of third-party commitments in our alternatives business and expect fundraising in 2025 to be consistent with levels achieved in recent years.

The table below presents information about alternative investments in Asset & Wealth Management that we hold on our balance sheet by asset type.

<i>\$ in billions</i>	As of December	
	2024	2023
Loans	\$ 8.5	\$ 12.9
Debt securities	9.0	10.8
Equity securities	13.4	13.2
Other	5.6	9.3
Total	\$ 36.5	\$ 46.2

The table below presents further information about our alternative investments in Asset & Wealth Management that we hold on our balance sheet.

<i>\$ in billions</i>	As of December	
	2024	2023
Client co-invest	\$ 18.4	\$ 21.3
Firmwide initiatives	8.7	8.6
Historical principal investments:		
Loans	1.6	3.5
Debt securities	2.6	3.6
Equity securities	3.5	4.0
Other	1.7	5.2
Total historical principal investments	9.4	16.3
Total	\$ 36.5	\$ 46.2

In the table above:

- Client co-invest primarily includes our investments in funds that we raise and manage or where we have invested alongside our clients.
- Firmwide initiatives primarily includes our investments related to the Community Reinvestment Act and our corporate engagement programs, such as *One Million Black Women*.
- Historical principal investments includes our remaining balance sheet alternative investments portfolio that we plan to reduce. This portfolio was approximately \$30 billion as of December 2022 and we expect to sell down the vast majority of this portfolio by the end of 2026. The impact of historical principal investments to our pre-tax earnings was \$939 million for 2024. Attributed equity associated with historical principal investments was approximately \$4.0 billion as of December 2024.

Management's Discussion and Analysis

The table below presents the rollforward of our alternative investments categorized as historical principal investments for 2024.

<i>\$ in billions</i>	Historical principal investments
Beginning balance	\$ 16.3
Additions	0.7
Dispositions	(7.9)
Net markups/(markdowns)	0.3
Ending balance	\$ 9.4

In the table above, dispositions included approximately \$400 million of investments that were transferred out of historical principal investments into client co-invest.

Loans and Debt Securities. The table below presents the concentration of loans and debt securities within our alternative investments by accounting classification, region and industry.

<i>\$ in billions</i>	As of December	
	2024	2023
Loans	\$ 8.5	\$ 12.9
Debt securities	9.0	10.8
Total	\$ 17.5	\$ 23.7
Accounting Classification		
Debt securities at fair value	51 %	45 %
Loans at amortized cost	45 %	49 %
Loans at fair value	3 %	3 %
Loans held for sale	1 %	3 %
Total	100 %	100 %
Region		
Americas	54 %	52 %
EMEA	35 %	37 %
Asia	11 %	11 %
Total	100 %	100 %
Industry		
Consumer & Retail	11 %	11 %
Financial Institutions	9 %	6 %
Healthcare	12 %	15 %
Industrials	14 %	18 %
Natural Resources & Utilities	2 %	2 %
Real Estate	13 %	11 %
Technology, Media & Telecommunications	29 %	28 %
Other	10 %	9 %
Total	100 %	100 %

Equity Securities. The table below presents the concentration of equity securities within our alternative investments by region and industry.

<i>\$ in billions</i>	As of December	
	2024	2023
Equity securities	\$ 13.4	\$ 13.2
Region		
Americas	68 %	70 %
EMEA	17 %	15 %
Asia	15 %	15 %
Total	100 %	100 %
Industry		
Consumer & Retail	5 %	6 %
Financial Institutions	15 %	11 %
Healthcare	6 %	6 %
Industrials	7 %	10 %
Natural Resources & Utilities	14 %	13 %
Real Estate	27 %	30 %
Technology, Media & Telecommunications	24 %	22 %
Other	2 %	2 %
Total	100 %	100 %

In the table above:

- Equity securities included \$12.6 billion as of December 2024 and \$12.1 billion as of December 2023 of private equity positions, and \$0.8 billion as of December 2024 and \$1.1 billion as of December 2023 of public equity positions that converted from private equity upon the initial public offerings of the underlying companies.
- The concentrations for real estate equity securities as of December 2024 were 14% for multifamily (13% as of December 2023), 5% for mixed use (8% as of December 2023), 3% for industrials (3% as of December 2023), 2% for office (2% as of December 2023) and 3% for other real estate equity securities (4% as of December 2023).

The table below presents the concentration of equity securities within our alternative investments by vintage.

	Vintage
As of December 2024	
2017 or earlier	22 %
2018 - 2020	28 %
2021 - thereafter	50 %
Total	100 %
As of December 2023	
2016 or earlier	25 %
2017 - 2019	26 %
2020 - thereafter	49 %
Total	100 %

Management's Discussion and Analysis

Other. Other investments include tax credit investments (accounted for under the proportional amortization method of accounting) of \$3.2 billion as of December 2024 and \$3.4 billion as of December 2023. Additionally, other investments includes CIEs, which held assets (generally accounted for at historical cost less depreciation) of \$2.4 billion as of December 2024 and \$5.9 billion as of December 2023, and were funded with liabilities of approximately \$1.2 billion as of December 2024 and \$3.5 billion as of December 2023. Substantially all such liabilities were nonrecourse, thereby reducing our equity at risk.

The table below presents the concentration of CIE assets, net of financings, within our alternative investments by region and asset class.

<i>\$ in billions</i>	As of December	
	2024	2023
CIE assets, net of financings	\$ 1.2	\$ 2.4
Region		
Americas	72 %	61 %
EMEA	15 %	25 %
Asia	13 %	14 %
Total	100 %	100 %
Asset Class		
Hospitality	7 %	6 %
Industrials	23 %	16 %
Multifamily	15 %	13 %
Office	29 %	24 %
Retail	6 %	7 %
Senior Housing	4 %	15 %
Student Housing	1 %	7 %
Other	15 %	12 %
Total	100 %	100 %

The table below presents the concentration of CIE assets, net of financings, within our alternative investments by vintage.

	Vintage
As of December 2024	
2017 or earlier	29 %
2018 - 2020	37 %
2021 - thereafter	34 %
Total	100 %
As of December 2023	
2016 or earlier	12 %
2017 - 2019	57 %
2020 - thereafter	31 %
Total	100 %

Platform Solutions

Platform Solutions includes our consumer platforms and transaction banking and other.

Platform Solutions generates revenues from the following:

Consumer platforms. Our Consumer platforms business issues credit cards, and raises deposits from Apple Card customers. Consumer platforms revenues primarily includes net interest income earned on credit card lending activities. See "Regulatory and Other Matters — Other Matters — Narrowing our Focus on Consumer-Related Activities" for further information.

Transaction banking and other. We provide transaction banking and other services, such as deposit-taking, payment solutions and other cash management services, for corporate and institutional clients. Transaction banking revenues include net interest income attributed to transaction banking deposits. See "Regulatory and Other Matters — Other Matters — Narrowing our Focus on Consumer-Related Activities" for further information.

The table below presents our Platform Solutions assets.

<i>\$ in millions</i>	As of December	
	2024	2023
Cash and cash equivalents	\$ 16,041	\$ 24,043
Collateralized agreements	5,944	7,651
Customer and other receivables	72	3
Trading assets	20,452	14,911
Investments	3	2
Loans	18,836	20,028
Other assets	1,154	1,846
Total	\$ 62,502	\$ 68,484

The table below presents details about our Platform Solutions loans.

<i>\$ in millions</i>	As of December	
	2024	2023
Installment	\$ -	\$ 3,125
Credit cards	21,403	19,361
Other	-	17
Loans, gross	21,403	22,503
Allowance for loan losses	(2,567)	(2,475)
Total loans	\$ 18,836	\$ 20,028

The average Platform Solutions gross loans were \$20.48 billion for 2024 and \$21.48 billion for 2023.

Management's Discussion and Analysis

The table below presents our Platform Solutions operating results.

\$ in millions	Year Ended December		
	2024	2023	2022
Consumer platforms	\$ 2,147	\$ 2,072	\$ 1,176
Transaction banking and other	280	306	326
Total net revenues	2,427	2,378	1,502
Provision for credit losses	1,540	1,135	1,728
Compensation and benefits expenses	685	784	560
Other operating expenses	1,277	2,634	1,203
Total operating expenses	1,962	3,418	1,763
Pre-tax earnings/(loss)	(1,075)	(2,175)	(1,989)
Provision/(benefit) for taxes	(241)	(450)	(328)
Net earnings/(loss)	(834)	(1,725)	(1,661)
Preferred stock dividends	25	23	12
Net earnings/(loss) to common	\$ (859)	\$ (1,748)	\$ (1,673)
Average common equity	\$ 4,573	\$ 3,863	\$ 3,574
Return on average common equity	(18.8)%	(45.2)%	(46.8)%

Our target is to achieve pre-tax breakeven by the end of 2025 for Platform Solutions.

Operating Environment. The operating environment for Platform Solutions is mainly impacted by the economic environment in the U.S., which, during 2024, was generally characterized by concerns about inflation (although some measures had begun to improve), a continued low rate of unemployment and a slight increase in the pace of growth in consumer spending compared with 2023.

In the future, if economic conditions deteriorate, it may lead to a decrease in consumer spending or a deterioration in consumer credit, and net revenues and provision for credit losses in Platform Solutions would likely be negatively impacted.

2024 versus 2023. Net revenues in Platform Solutions were \$2.43 billion for 2024, 2% higher than 2023.

Notwithstanding our strategic decision to narrow the focus on consumer-related activities, Consumer platforms net revenues were slightly higher compared with 2023, reflecting higher average credit card balances and higher average deposit balances, largely offset by the impact of the planned transition of the GM credit card program to another issuer. Transaction banking and other net revenues were lower, primarily reflecting lower net revenues related to the seller financing loan portfolio that was sold during 2024. See "Regulatory and Other Matters — Other Matters — Narrowing our Focus on Consumer-Related Activities" for further information.

Provision for credit losses was \$1.54 billion for 2024, compared with \$1.14 billion for 2023. Provisions for 2024 reflected net provisions related to the credit card portfolio (primarily driven by net charge-offs). The net provision for 2023 reflected net provisions related to the credit card portfolio (primarily driven by net charge-offs), partially offset by a net release related to the GreenSky loan portfolio (including a reserve reduction related to the transfer of the portfolio to held for sale).

Operating expenses were \$1.96 billion for 2024, 43% lower than 2023, primarily due to the write-down of identifiable intangible assets related to GreenSky and an impairment of goodwill related to Consumer platforms in the prior year period. Pre-tax loss was \$1.08 billion for 2024, compared with a pre-tax loss of \$2.18 billion for 2023.

Geographic Data

See Note 25 to the consolidated financial statements for a summary of our total net revenues, pre-tax earnings and net earnings by geographic region.

Balance Sheet and Funding Sources

Balance Sheet Management

One of our risk management disciplines is our ability to manage the size and composition of our balance sheet. While our asset base changes due to client activity, market fluctuations and business opportunities, the size and composition of our balance sheet also reflects factors, including (i) our overall risk tolerance, (ii) the amount of capital we hold and (iii) our funding profile, among other factors. See "Capital Management and Regulatory Capital — Capital Management" for information about our capital management process.

Although our balance sheet fluctuates on a day-to-day basis, our total assets at quarter-end are generally not materially different from those occurring within our reporting periods.

In order to ensure appropriate risk management, we seek to maintain a sufficiently liquid balance sheet and have processes in place to dynamically manage our assets and liabilities, which include (i) balance sheet planning, (ii) setting balance sheet targets, (iii) monitoring of key metrics and (iv) scenario analyses.

Management's Discussion and Analysis

Balance Sheet Planning. We prepare a balance sheet plan that combines our projected total assets and composition of assets with our expected funding sources over a three-year time horizon. This plan is reviewed quarterly and may be adjusted in response to changing business needs or market conditions. The objectives of this planning process are:

- To develop our balance sheet projections, taking into account the general state of the financial markets and expected business activity levels, as well as regulatory requirements;
- To allow Corporate Treasury to evaluate balance sheet targets of our revenue-producing units and requests to change such targets in the context of our overall balance sheet constraints, including our liability profile and capital levels, and key metrics; and
- To inform the target amount, tenor and type of funding to raise, based on our projected assets and contractual maturities.

Corporate Treasury and Risk, along with our revenue-producing units, review current and prior period information and expectations for the year to prepare our balance sheet plan. The specific information reviewed includes asset and liability size and composition, target utilization, risk and performance measures, and capital usage.

Our consolidated balance sheet plan, including our balance sheets by business, funding projections and projected key metrics, is reviewed and approved by the Firmwide Asset Liability Committee. See "Risk Management — Overview and Structure of Risk Management" for an overview of our risk management structure.

Setting Balance Sheet Targets. We set balance sheet targets with the aim of ensuring that our consolidated balance sheet, as well as the balance sheets for our businesses remain within our risk appetite. The Firmwide Asset Liability Committee has the responsibility to review and approve balance sheet targets at least quarterly. Our balance sheet targets are set at levels which are close to actual operating levels, rather than at levels which reflect our maximum risk appetite, in order to ensure prompt escalation and discussion among our revenue-producing units, Corporate Treasury and Risk. Requests for changes in targets are evaluated after giving consideration to their impact on our key metrics. Compliance with targets is monitored by our revenue-producing units, Corporate Treasury and Risk.

Monitoring of Key Metrics. We monitor key balance sheet metrics both by business and on a consolidated basis, including asset and liability size and composition, target utilization and risk measures. We attribute assets to businesses and review and analyze movements resulting from new business activity, as well as market fluctuations.

Scenario Analyses. We conduct various scenario analyses, including as part of the Comprehensive Capital Analysis and Review (CCAR) and U.S. Dodd-Frank Wall Street Reform and Consumer Protection Act Stress Tests (DFAST), as well as our resolution and recovery planning. See "Capital Management and Regulatory Capital — Capital Management" for further information about these scenario analyses. These scenarios cover short- and long-term time horizons using various macroeconomic and firm-specific assumptions, based on a range of economic scenarios. We use these analyses to assist us in developing our longer-term balance sheet management strategy, including the level and composition of assets, funding and capital. Additionally, these analyses help us develop approaches for maintaining appropriate funding, liquidity and capital across a variety of situations, including a severely stressed environment.

Balance Sheet Analysis and Metrics

As of December 2024, total assets in our consolidated balance sheets were \$1.68 trillion, an increase of \$34.38 billion from December 2023, reflecting increases in trading assets of \$93.05 billion (primarily due to an increase in government obligations, reflecting the impact of our and our clients' activities), investments of \$37.68 billion (primarily due to an increase in U.S. government obligations accounted for as available-for-sale) and loans of \$12.84 billion (primarily reflecting our clients' activities), partially offset by decreases in cash and cash equivalents of \$59.49 billion (primarily reflecting our activity) and collateralized agreements of \$48.52 billion (primarily reflecting our activity). See "Liquidity Risk Management — Cash Flows" for further information about cash and cash equivalents.

As of December 2024, total liabilities in our consolidated balance sheets were \$1.55 trillion, an increase of \$29.29 billion from December 2023, primarily reflecting increases in collateralized financings of \$35.03 billion (reflecting the impact of our and our clients' activities), and deposits of \$4.60 billion (due to an increase in consumer deposits, partially offset by decreases in transaction banking deposits and other deposits), partially offset by decreases in customer and other payables of \$7.47 billion (primarily reflecting our clients' activities) and borrowings of \$5.48 billion (driven by net maturities).

Management's Discussion and Analysis

Our total securities sold under agreements to repurchase (repurchase agreements), accounted for as collateralized financings, were \$274.38 billion as of December 2024 and \$249.89 billion as of December 2023, which were 5% higher as of December 2024 and 21% higher as of December 2023 than the average daily amount of repurchase agreements over the respective quarters, and 9% higher as of December 2024 and 26% higher as of December 2023 than the average daily amount of repurchase agreements over the respective years. As of December 2024, the increase in our repurchase agreements relative to the average daily amount of repurchase agreements during the quarter and year resulted from lower levels of our and our clients' activities at the end of the period.

The level of our repurchase agreements fluctuates between and within periods, primarily due to providing clients with access to highly liquid collateral, such as certain government and agency obligations, through collateralized financing activities.

The table below presents information about our balance sheet and leverage ratios.

<i>\$ in millions</i>	As of December	
	2024	2023
Total assets	\$ 1,675,972	\$ 1,641,594
Unsecured long-term borrowings	\$ 242,634	\$ 241,877
Total shareholders' equity	\$ 121,996	\$ 116,905
Leverage ratio	13.7x	14.0x
Debt-to-equity ratio	2.0x	2.1x

In the table above:

- The leverage ratio equals total assets divided by total shareholders' equity and measures the proportion of equity and debt we use to finance assets. This ratio is different from the leverage ratios included in Note 20 to the consolidated financial statements.
- The debt-to-equity ratio equals unsecured long-term borrowings divided by total shareholders' equity.

The table below presents information about our shareholders' equity and book value per common share, including the reconciliation of common shareholders' equity to tangible common shareholders' equity.

<i>\$ in millions, except per share amounts</i>	As of December	
	2024	2023
Total shareholders' equity	\$ 121,996	\$ 116,905
Preferred stock	(13,253)	(11,203)
Common shareholders' equity	108,743	105,702
Goodwill	(5,853)	(5,916)
Identifiable intangible assets	(847)	(1,177)
Tangible common shareholders' equity	\$ 102,043	\$ 98,609
Book value per common share	\$ 336.77	\$ 313.56
Tangible book value per common share	\$ 316.02	\$ 292.52

In the table above:

- Tangible common shareholders' equity is calculated as total shareholders' equity less preferred stock, goodwill and identifiable intangible assets. We believe that tangible common shareholders' equity is meaningful because it is a measure that we and investors use to assess capital adequacy. Tangible common shareholders' equity is a non-GAAP measure and may not be comparable to similar non-GAAP measures used by other companies.
- Book value per common share and tangible book value per common share are based on common shares outstanding and restricted stock units granted to employees with no future service requirements and not subject to performance or market conditions (collectively, basic shares) of 322.9 million as of December 2024 and 337.1 million as of December 2023. We believe that tangible book value per common share (tangible common shareholders' equity divided by basic shares) is meaningful because it is a measure that we and investors use to assess capital adequacy. Tangible book value per common share is a non-GAAP measure and may not be comparable to similar non-GAAP measures used by other companies.

Management's Discussion and Analysis

Funding Sources

Our primary sources of funding are deposits, collateralized financings, unsecured short- and long-term borrowings, and shareholders' equity. We seek to maintain broad and diversified funding sources globally across products, programs, markets, currencies and creditors to avoid funding concentrations.

The table below presents information about our funding sources.

\$ in millions	As of December			
	2024		2023	
Deposits	\$ 433,013	35 %	\$ 428,417	36 %
Collateralized financings	358,590	29 %	323,564	27 %
Unsecured short-term borrowings	69,709	6 %	75,945	6 %
Unsecured long-term borrowings	242,634	20 %	241,877	21 %
Total shareholders' equity	121,996	10 %	116,905	10 %
Total	\$ 1,225,942	100 %	\$ 1,186,708	100 %

Our funding is primarily raised in U.S. dollar, Euro, British pound and Japanese yen. We generally distribute our funding products through our own sales force and third-party distributors to a large, diverse creditor base in a variety of markets in the Americas, Europe and Asia. We believe that our relationships with our creditors are critical to our liquidity. Our creditors include banks, governments, securities lenders, corporations, pension funds, insurance companies, mutual funds and individuals. We have imposed various internal guidelines to monitor creditor concentration across our funding programs.

Deposits. Our deposits provide us with a diversified source of funding and reduce our reliance on wholesale funding. We raise deposits, including savings, demand and time deposits, from consumers, private bank clients, through internal and third-party broker-dealers, transaction banking clients and other institutional clients. Substantially all of our deposits are raised through Goldman Sachs Bank USA (GS Bank USA), Goldman Sachs International Bank (GSIB) and Goldman Sachs Bank Europe SE (GSBE).

The table below presents the types and sources of deposits.

<i>\$ in millions</i>	Savings and Demand		Time		Total
As of December 2024					
Consumer	\$	126,694	\$	54,541	\$ 181,235
Private bank		90,013		6,489	96,502
Brokered certificates of deposit		-		41,014	41,014
Deposit sweep programs		30,927		-	30,927
Transaction banking		60,925		1,820	62,745
Other		1,776		18,814	20,590
Total	\$	310,335	\$	122,678	\$ 433,013
As of December 2023					
Consumer	\$	120,211	\$	36,903	\$ 157,114
Private bank		86,457		6,855	93,312
Brokered certificates of deposit		-		46,860	46,860
Deposit sweep programs		31,916		-	31,916
Transaction banking		68,177		3,643	71,820
Other		1,568		25,827	27,395
Total	\$	308,329	\$	120,088	\$ 428,417

In the table above:

- Savings and demand accounts consist of money market deposit accounts, negotiable order of withdrawal accounts and demand deposit accounts that have no stated maturity or expiration date.
- Time deposits had a weighted average maturity of approximately 0.6 years as of both December 2024 and December 2023.
- Consumer deposits consist of deposits from both Marcus and Apple Card customers.
- Deposit sweep programs include contractual agreements with U.S. broker-dealers who sweep client cash to FDIC-insured deposits.
- Transaction banking deposits consist of deposits that we raised through our cash management services business for corporate and other institutional clients.
- Other deposits are substantially all from institutional clients.
- Deposits insured by the FDIC were \$234.54 billion as of December 2024 and \$221.52 billion as of December 2023.
- Deposits insured by non-U.S. insurance programs were \$25.98 billion as of December 2024 and \$26.00 billion as of December 2023.

Management's Discussion and Analysis

See Note 13 to the consolidated financial statements for further information about our deposits, including a maturity profile of our time deposits.

Secured Funding. We fund a significant amount of inventory and a portion of investments on a secured basis. Secured funding includes collateralized financings in the consolidated balance sheets. See Note 11 to the consolidated financial statements for further information about our collateralized financings, including its maturity profile. We may also pledge our inventory and investments as collateral for securities borrowed under a securities lending agreement. We also use our own inventory and investments to cover transactions in which we or our clients have sold securities that have not yet been purchased. Secured funding is less sensitive to changes in our credit quality than unsecured funding, due to our posting of collateral to our lenders. Nonetheless, we analyze the refinancing risk of our secured funding activities, taking into account trade tenors, maturity profiles, counterparty concentrations, collateral eligibility and counterparty rollover probabilities. We seek to mitigate our refinancing risk by executing term trades with staggered maturities, diversifying counterparties, raising excess secured funding and pre-funding residual risk through our GCLA.

We seek to raise secured funding with a term appropriate for the liquidity of the assets that are being financed, and we seek longer maturities for secured funding collateralized by asset classes that may be harder to fund on a secured basis, especially during times of market stress. Our secured funding, excluding funding collateralized by liquid government and agency obligations, is primarily executed for tenors of one month or greater and is primarily executed through term repurchase agreements and securities loaned contracts.

Assets that may be harder to fund on a secured basis during times of market stress include certain financial instruments in the following categories: mortgage- and other asset-backed loans and securities, non-investment-grade corporate debt securities, equity securities and emerging market securities.

We also raise financing through other types of collateralized financings, such as secured loans and notes. GS Bank USA has access to funding from the Federal Home Loan Bank. Our outstanding borrowings from the Federal Home Loan Bank were \$5.04 billion as of December 2024 and we had no outstanding borrowings as of December 2023. Additionally, we have access to funding through the Federal Reserve discount window, but we do not rely on this funding in our liquidity planning and stress testing.

Unsecured Short-Term Borrowings. A significant portion of our unsecured short-term borrowings was originally long-term debt that is scheduled to mature within one year of the reporting date. We use unsecured short-term borrowings, including U.S. and non-U.S. hybrid financial instruments and commercial paper, to finance liquid assets and for other cash management purposes. In accordance with regulatory requirements, Group Inc. does not issue debt with an original maturity of less than one year, other than to its subsidiaries. See Note 14 to the consolidated financial statements for further information about our unsecured short-term borrowings.

Unsecured Long-Term Borrowings. Unsecured long-term borrowings, including structured notes, are raised through syndicated U.S. registered offerings, U.S. registered and Rule 144A medium-term note programs, offshore medium-term note offerings and other debt offerings. We issue in different tenors, currencies and products to maximize the diversification of our investor base.

The table below presents our quarterly unsecured long-term borrowings maturity profile.

<i>\$ in millions</i>	First Quarter	Second Quarter	Third Quarter	Fourth Quarter	Total
As of December 2024					
2026	\$ 10,016	\$ 6,947	\$ 8,179	\$ 11,788	\$ 36,930
2027	\$ 13,058	\$ 8,700	\$ 8,091	\$ 11,289	\$ 41,138
2028	\$ 11,495	\$ 6,191	\$ 4,542	\$ 6,976	\$ 29,204
2029	\$ 4,490	\$ 10,308	\$ 6,912	\$ 11,026	\$ 32,736
2030 - thereafter					\$ 102,626
Total					\$ 242,634

The weighted average maturity of our unsecured long-term borrowings as of December 2024 was approximately seven years. To mitigate refinancing risk, we seek to limit the principal amount of debt maturing over the course of any monthly, quarterly, semi-annual or annual time horizon. We enter into interest rate swaps to convert a portion of our unsecured long-term borrowings into floating-rate obligations to manage our exposure to interest rates. See Note 14 to the consolidated financial statements for further information about our unsecured long-term borrowings.

Shareholders' Equity. Shareholders' equity is a stable and perpetual source of funding. See Note 19 to the consolidated financial statements for further information about our shareholders' equity.

Management's Discussion and Analysis

Capital Management and Regulatory Capital

Capital adequacy is of critical importance to us. We have in place a comprehensive capital management policy that provides a framework, defines objectives and establishes guidelines to assist us in maintaining the appropriate level and composition of capital in both business-as-usual and stressed conditions.

Capital Management

We determine the appropriate amount and composition of our capital by considering multiple factors, including our current and future regulatory capital requirements, the results of our capital planning and stress testing process, the results of resolution capital models and other factors, such as rating agency guidelines, subsidiary capital requirements, the business environment and conditions in the financial markets.

We manage our capital requirements and the levels of our capital usage principally by setting targets on our balance sheet and risk-weighted assets (RWAs), in each case at both the firmwide and business levels.

We principally manage the level and composition of our capital through issuances and repurchases of our common stock.

We may issue, redeem or repurchase our preferred stock and subordinated debt or other forms of capital as business conditions warrant. Prior to such redemptions or repurchases, we must receive approval from the FRB. See Notes 14 and 19 to the consolidated financial statements for further information about our subordinated debt and preferred stock.

Capital Planning and Stress Testing Process. As part of capital planning, we project sources and uses of capital given a range of business environments, including stressed conditions. Our stress testing process is designed to identify and measure material risks associated with our business activities, including market risk, credit risk, operational risk and liquidity risk, as well as our ability to generate revenues.

Our capital planning process incorporates an internal capital adequacy assessment with the objective of ensuring that we are appropriately capitalized relative to the risks in our businesses. We incorporate stress scenarios into our capital planning process with a goal of holding sufficient capital to ensure we remain adequately capitalized after experiencing a severe stress event. Our assessment of capital adequacy is viewed in tandem with our assessment of liquidity adequacy and is integrated into our overall risk management structure, governance and policy framework.

Our stress tests incorporate our internally designed stress scenarios, including our internally developed severely adverse scenario, and those required by the FRB, and are designed to capture our specific vulnerabilities and risks. We provide further information about our stress test processes and a summary of the results on our website as described in "Business — Available Information" in Part I, Item 1 of this Form 10-K.

As required by the FRB's CCAR rules, we submit an annual capital plan for review by the FRB. The purpose of the FRB's review is to ensure that we have a robust, forward-looking capital planning process that accounts for our unique risks and that permits continued operation during times of economic and financial stress.

The FRB evaluates us based, in part, on whether we have the capital necessary to continue operating under the baseline and severely adverse scenarios provided by the FRB and those developed internally. This evaluation also takes into account our process for identifying risk, our controls and governance for capital planning, and our guidelines for making capital planning decisions. In addition, the FRB evaluates our plan to make capital distributions (i.e., dividend payments and repurchases or redemptions of stock, subordinated debt or other capital securities) and issue capital, across the range of macroeconomic scenarios and firm-specific assumptions. The FRB determines the SCB applicable to us based on its own annual stress test. The SCB under the Standardized approach is calculated as (i) the difference between our starting and minimum projected CET1 capital ratios under the supervisory severely adverse scenario and (ii) our planned common stock dividends for each of the fourth through seventh quarters of the planning horizon, expressed as a percentage of RWAs.

Based on our 2024 CCAR submission, the FRB increased our SCB from 5.5% to 6.2%, resulting in a Standardized CET1 capital ratio requirement of 13.7% for the period from October 1, 2024 through September 30, 2025. See "Share Repurchase Program" for further information about common stock repurchases and dividends and "Consolidated Regulatory Capital" for further information about the G-SIB surcharge. We published a summary of our annual DFAST results in June 2024. See "Business — Available Information" in Part I, Item 1 of this Form 10-K.

GS Bank USA is required to conduct stress tests on an annual basis and publish a summary of certain results. GS Bank USA published a summary of its annual DFAST results in June 2024. See "Business — Available Information" in Part I, Item 1 of this Form 10-K.

Management's Discussion and Analysis

Goldman Sachs International (GSI), GSIB and GSBE also have their own capital planning and stress testing processes, which incorporate internally designed stress tests developed in accordance with the guidelines of their respective regulators.

Contingency Capital Plan. As part of our comprehensive capital management policy, we maintain a contingency capital plan. Our contingency capital plan provides a framework for analyzing and responding to a perceived or actual capital deficiency, including, but not limited to, identification of drivers of a capital deficiency, as well as mitigants and potential actions. It outlines the appropriate communication procedures to follow during a crisis period, including internal dissemination of information, as well as timely communication with external stakeholders.

Capital Attribution. We assess the capital usage of each of our businesses based on our attributed equity framework. This framework considers many factors, including our internal assessment of risks as well as the regulatory capital requirements related to our business activities.

We review and make any necessary adjustments to our attributed equity in January each year, to reflect, among other things, our most recent stress test results and changes to our regulatory capital requirements. On January 1, 2024, our allocation of attributed equity changed (relative to the allocation as of December 2023) as follows: attributed equity increased by approximately \$1.6 billion for Platform Solutions, while attributed equity decreased by approximately \$1.2 billion for Asset & Wealth Management and approximately \$0.4 billion for Global Banking & Markets. On January 1, 2025, our allocation of attributed equity changed (relative to the allocation as of December 2024) as follows: attributed equity increased by approximately \$0.4 billion for Global Banking & Markets, while attributed equity decreased by approximately \$0.3 billion for Asset & Wealth Management and approximately \$0.1 billion for Platform Solutions. See "Results of Operations — Segment Assets and Operating Results — Segment Operating Results" for information about our average quarterly attributed equity by segment.

Share Repurchase Program. We use our share repurchase program to help maintain the appropriate level of common equity. On an annual basis, we submit a Board of Directors of Group Inc. (Board) approved capital plan to the Federal Reserve, which includes planned share repurchases for each quarter. The share repurchases are effected primarily through regular open-market purchases (which may include repurchase plans designed to comply with Rule 10b5-1 and accelerated share repurchases), the amounts and timing of which are determined primarily by our current and projected capital position, and capital deployment opportunities, but which may also be influenced by general market conditions and the prevailing price and trading volumes of our common stock.

In 2023, the Board approved a share repurchase program authorizing repurchases of up to \$30 billion of our common stock. The program has no set expiration or termination date. See "Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities" in Part II, Item 5 of this Form 10-K and Note 19 to the consolidated financial statements for further information about our share repurchase program, and see above for information about our capital planning and stress testing process.

During 2024, we returned a total of \$11.80 billion of capital to common shareholders, including \$8.00 billion of common share repurchases and \$3.80 billion of common stock dividends. Consistent with our capital management philosophy, we will continue prioritizing deployment of capital for our clients where returns are attractive and distribute any excess capital to shareholders through dividends and share repurchases, while targeting a 50 to 100 basis point buffer above our capital requirement.

We are subject to a one percent non-deductible federal excise tax (buyback tax) that is applicable to the fair market value of certain corporate share repurchases. The fair market value of share repurchases subject to the tax is reduced by the fair market value of any applicable stock issued during the calendar year, including stock issued to employees. The buyback tax did not have a material impact on our financial condition, results of operations or cash flows for 2024.

Resolution Capital Models. In connection with our resolution planning efforts, we have established a Resolution Capital Adequacy and Positioning framework, which is designed to ensure that our major subsidiaries (GS Bank USA, Goldman Sachs & Co. LLC (GS&Co.), GSI, GSIB, GSBE, Goldman Sachs Japan Co., Ltd. (GSJCL), Goldman Sachs Asset Management, L.P. and Goldman Sachs Asset Management International) have access to sufficient loss-absorbing capacity (in the form of equity, subordinated debt and unsecured senior debt) so that they are able to wind down following a Group Inc. bankruptcy filing in accordance with our preferred resolution strategy.

In addition, we have established a triggers and alerts framework, which is designed to provide the Board with information needed to make an informed decision on whether and when to commence bankruptcy proceedings for Group Inc.

Management's Discussion and Analysis

Rating Agency Guidelines

The credit rating agencies assign credit ratings to the obligations of Group Inc., which directly issues or guarantees substantially all of our senior unsecured debt obligations. GS&Co. and GSI have been assigned long- and short-term issuer ratings by certain credit rating agencies. GS Bank USA, GSIB and GSBE have also been assigned long- and short-term issuer ratings, as well as ratings on their long- and short-term bank deposits. In addition, credit rating agencies have assigned ratings to debt obligations of certain other subsidiaries of Group Inc.

The level and composition of our capital are among the many factors considered in determining our credit ratings. Each agency has its own definition of eligible capital and methodology for evaluating capital adequacy, and assessments are generally based on a combination of factors rather than a single calculation. See "Risk Management — Liquidity Risk Management — Credit Ratings" for further information about credit ratings of Group Inc., GS Bank USA, GSIB, GSBE, GS&Co. and GSI.

Consolidated Regulatory Capital

We are subject to consolidated regulatory capital requirements which are calculated in accordance with the regulations of the FRB (Capital Framework). Under the Capital Framework, we are an "Advanced approaches" banking organization and have been designated as a G-SIB. In managing our capital, we consider a number of different capital requirements, the most binding of which can vary over time.

The capital requirements calculated under the Capital Framework include the capital conservation buffer requirements, which are comprised of a 2.5% buffer (under the Advanced Capital Rules), the SCB (under the Standardized Capital Rules), a countercyclical capital buffer (under both Capital Rules) and the G-SIB surcharge (under both Capital Rules). Our G-SIB surcharge is 3.0% for both 2024 and 2025 and is expected to be 3.5% beginning in 2026. The G-SIB surcharge and countercyclical capital buffer in the future may differ due to additional guidance from our regulators and/or positional changes, and our SCB can change significantly from year to year based on the results of the annual supervisory stress tests. Our target is to maintain capital ratios equal to the regulatory requirements plus a buffer of 50 to 100 basis points.

See Note 20 to the consolidated financial statements for further information about our risk-based capital ratios and leverage ratios, and the Capital Framework.

Total Loss-Absorbing Capacity (TLAC)

We are also subject to the FRB's TLAC and related requirements. Failure to comply with the TLAC and related requirements would result in restrictions being imposed by the FRB and could limit our ability to repurchase shares, pay dividends and make certain discretionary compensation payments.

The table below presents TLAC and external long-term debt requirements.

	As of December	
	2024	2023
TLAC to RWAs	22.0 %	22.0 %
TLAC to leverage exposure	9.5 %	9.5 %
External long-term debt to RWAs	9.0 %	9.0 %
External long-term debt to leverage exposure	4.5 %	4.5 %

In the table above:

- The TLAC to RWAs requirement included (i) the 18% minimum, (ii) the 2.5% buffer, (iii) the countercyclical capital buffer, which the FRB has set to zero percent and (iv) the 1.5% G-SIB surcharge (Method 1).
- The TLAC to leverage exposure requirement includes (i) the 7.5% minimum and (ii) the 2.0% leverage exposure buffer.
- The external long-term debt to RWAs requirement includes (i) the 6% minimum and (ii) the 3.0% G-SIB surcharge (Method 2).
- The external long-term debt to total leverage exposure is the 4.5% minimum.

The table below presents information about our TLAC and external long-term debt ratios.

\$ in millions	For the Three Months Ended or as of December	
	2024	2023
TLAC	\$ 275,904	\$ 278,188
External long-term debt	\$ 150,682	\$ 154,300
RWAs	\$ 688,541	\$ 692,737
Leverage exposure	\$ 2,120,756	\$ 1,995,756
TLAC to RWAs	40.1 %	40.2 %
TLAC to leverage exposure	13.0 %	13.9 %
External long-term debt to RWAs	21.9 %	22.3 %
External long-term debt to leverage exposure	7.1 %	7.7 %

In the table above:

- TLAC includes common and preferred stock, and eligible long-term debt issued by Group Inc. Eligible long-term debt represents unsecured debt, which has a remaining maturity of at least one year and satisfies additional requirements.
- External long-term debt consists of eligible long-term debt subject to a haircut if it is due to be paid between one and two years.

Management's Discussion and Analysis

- In accordance with the TLAC rules, the higher of Standardized or Advanced RWAs are used in the calculation of TLAC and external long-term debt ratios and applicable requirements. RWAs represent Standardized RWAs as of both December 2024 and December 2023.
- Leverage exposure consists of average adjusted total assets and certain off-balance sheet exposures.

See "Business — Regulation" in Part I, Item 1 of this Form 10-K for further information about TLAC.

Subsidiary Capital Requirements

Many of our subsidiaries, including our bank and broker-dealer subsidiaries, are subject to separate regulation and capital requirements of the jurisdictions in which they operate.

Bank Subsidiaries. GS Bank USA is our primary U.S. banking subsidiary and GSIB and GSBE are our primary non-U.S. banking subsidiaries. These entities are subject to regulatory capital requirements. See Note 20 to the consolidated financial statements for further information about the regulatory capital requirements for GS Bank USA.

- **GSIB.** GSIB is our U.K. bank subsidiary regulated by the Prudential Regulation Authority (PRA) and the Financial Conduct Authority (FCA). GSIB is subject to the U.K. capital framework, which is largely based on the Basel Committee on Banking Supervision's (Basel Committee) capital framework for strengthening international capital standards (Basel III). The eligible retail deposits of GSIB are covered by the U.K. Financial Services Compensation Scheme to the extent provided by law.

The table below presents GSIB's risk-based capital requirements.

	As of December	
	2024	2023
Risk-based capital requirements		
CET1 capital ratio	11.9 %	10.1 %
Tier 1 capital ratio	14.7 %	12.4 %
Total capital ratio	18.4 %	15.4 %

The table below presents information about GSIB's risk-based capital ratios.

	As of December	
\$ in millions	2024	2023
Risk-based capital and risk-weighted assets		
CET1 capital	\$ 4,336	\$ 3,936
Tier 1 capital	\$ 4,336	\$ 3,936
Tier 2 capital	\$ 826	\$ 826
Total capital	\$ 5,162	\$ 4,762
RWAs	\$ 17,767	\$ 16,546
Risk-based capital ratios		
CET1 capital ratio	24.4 %	23.8 %
Tier 1 capital ratio	24.4 %	23.8 %
Total capital ratio	29.1 %	28.8 %

In the table above, the risk-based capital ratios as of December 2024 reflected profits that are still subject to annual audit by GSIB's external auditors and approval by GSIB's Board of Directors for inclusion in risk-based capital. These profits contributed 213 basis points to the CET1 capital ratio as of December 2024.

The table below presents GSIB's leverage ratio requirement and leverage ratio.

	As of December	
	2024	2023
Leverage ratio requirement	3.7 %	3.6 %
Leverage ratio	8.9 %	7.4 %

In the table above, the leverage ratio as of December 2024 reflected profits that are still subject to annual audit by GSIB's external auditors and approval by GSIB's Board of Directors for inclusion in risk-based capital. These profits contributed 87 basis points to the leverage ratio as of December 2024.

GSIB is subject to minimum reserve requirements at central banks in certain of the jurisdictions in which it operates. As of both December 2024 and December 2023, GSIB was in compliance with these requirements.

- **GSBE.** GSBE is our German bank subsidiary supervised by the European Central Bank, BaFin and Deutsche Bundesbank. GSBE is a non-U.S. banking subsidiary of GS Bank USA and is also subject to standalone regulatory capital requirements noted below. GSBE is subject to the capital requirements prescribed in the amended E.U. Capital Requirements Directive (CRD) and E.U. Capital Requirements Regulation (CRR), which are largely based on Basel III. The deposits of GSBE are covered by the German statutory deposit protection program to the extent provided by law. In addition, GSBE has elected to participate in the German voluntary deposit protection program which provides further insurance for certain eligible deposits beyond the coverage of the German statutory deposit program.

The table below presents GSBE's risk-based capital requirements.

	As of December	
	2024	2023
Risk-based capital requirements		
CET1 capital ratio	10.3 %	10.0 %
Tier 1 capital ratio	12.3 %	12.1 %
Total capital ratio	15.0 %	14.8 %

Management's Discussion and Analysis

The table below presents information about GSBE's risk-based capital ratios.

\$ in millions	As of December	
	2024	2023
Risk-based capital and risk-weighted assets		
CET1 capital	\$ 13,871	\$ 14,212
Tier 1 capital	\$ 13,871	\$ 14,212
Tier 2 capital	\$ 21	\$ 22
Total capital	\$ 13,892	\$ 14,234
RWAs	\$ 43,426	\$ 39,797
Risk-based capital ratios		
CET1 capital ratio	31.9 %	35.7 %
Tier 1 capital ratio	31.9 %	35.7 %
Total capital ratio	32.0 %	35.8 %

In the table above, the risk-based capital ratios as of December 2024 reflected profits that are still subject to annual audit by GSBE's external auditors and approval by GSBE's shareholder (GS Bank USA) for inclusion in risk-based capital. These profits contributed 151 basis points to the CET1 capital ratio as of December 2024.

The table below presents GSBE's leverage ratio requirement and leverage ratio.

	As of December	
	2024	2023
Leverage ratio requirement	3.0 %	3.0 %
Leverage ratio	9.8 %	11.4 %

In the table above, the leverage ratio as of December 2024 reflected profits that are still subject to annual audit by GSBE's external auditors and approval by GSBE's shareholder (GS Bank USA) for inclusion in risk-based capital. These profits contributed 54 basis points to the leverage ratio as of December 2024.

GSBE is subject to minimum reserve requirements at central banks in certain of the jurisdictions in which it operates. As of both December 2024 and December 2023, GSBE was in compliance with these requirements.

GSBE is a registered swap dealer with the CFTC and a registered security-based swap dealer with the SEC. As of both December 2024 and December 2023, GSBE was subject to and in compliance with applicable capital requirements for swap dealers and security-based swap dealers.

U.S. Regulated Broker-Dealer Subsidiaries. GS&Co., our primary U.S. regulated broker-dealer subsidiary, is also a registered futures commission merchant and a registered swap dealer with the CFTC, and a registered security-based swap dealer with the SEC, and therefore is subject to regulatory capital requirements imposed by the SEC, the Financial Industry Regulatory Authority, Inc., the CFTC, the Chicago Mercantile Exchange and the National Futures Association. Rule 15c3-1 of the SEC and Rules 1.17 and Part 23 Subpart E of the CFTC specify uniform minimum net capital requirements, as defined, for their registrants, and also effectively require that a significant part of the registrants' assets be kept in relatively liquid form. GS&Co. has elected to calculate its SEC minimum capital requirements in accordance with the "Alternative Net Capital Requirement" as permitted by Rule 15c3-1 of the SEC.

GS&Co. had regulatory net capital, as defined by Rule 15c3-1 of the SEC, of \$21.31 billion as of December 2024 and \$20.25 billion as of December 2023, which exceeded the greater of the minimum amounts required under Rule 15c3-1 of the SEC and Rules 1.17 and Part 23 Subpart E of the CFTC by \$15.87 billion as of December 2024 and \$15.07 billion as of December 2023. In addition to its alternative minimum net capital requirements, GS&Co. is also required to hold tentative net capital in excess of \$5 billion and net capital in excess of \$1 billion in accordance with Rule 15c3-1. GS&Co. is also required to notify the SEC in the event that its tentative net capital is less than \$6 billion. As of both December 2024 and December 2023, GS&Co. had tentative net capital and net capital in excess of both the minimum and the notification requirements.

Non-U.S. Regulated Broker-Dealer Subsidiaries. Our principal non-U.S. regulated broker-dealer subsidiaries include GSI and GSJCL.

GSI, our U.K. broker-dealer, is regulated by the PRA and the FCA. GSI is subject to the U.K. capital framework, which is largely based on Basel III.

The table below presents GSI's risk-based capital requirements.

	As of December	
	2024	2023
Risk-based capital requirements		
CET1 capital ratio	9.1 %	9.1 %
Tier 1 capital ratio	11.0 %	11.0 %
Total capital ratio	13.6 %	13.7 %

Management's Discussion and Analysis

The table below presents information about GSI's risk-based capital ratios.

\$ in millions	As of December	
	2024	2023
Risk-based capital and risk-weighted assets		
CET1 capital	\$ 32,697	\$ 32,403
Tier 1 capital	\$ 38,197	\$ 37,903
Tier 2 capital	\$ 6,874	\$ 6,877
Total capital	\$ 45,071	\$ 44,780
RWAs	\$ 265,944	\$ 257,956
Risk-based capital ratios		
CET1 capital ratio	12.3 %	12.6 %
Tier 1 capital ratio	14.4 %	14.7 %
Total capital ratio	16.9 %	17.4 %

In the table above, the risk-based capital ratios as of December 2024 reflected profits that are still subject to annual audit by GSI's external auditors and approval by GSI's Board of Directors for inclusion in risk-based capital. These profits contributed 14 basis points to the CET1 capital ratio as of December 2024.

The table below presents GSI's leverage ratio requirement and leverage ratio.

	As of December	
	2024	2023
Leverage ratio requirement	3.5 %	3.5 %
Leverage ratio	5.3 %	4.9 %

In the table above, the leverage ratio as of December 2024 reflected profits that are still subject to annual audit by GSI's external auditors and approval by GSI's Board of Directors for inclusion in risk-based capital. These profits contributed 3 basis points to the leverage ratio as of December 2024.

GSI is a registered swap dealer with the CFTC and a registered security-based swap dealer with the SEC. As of both December 2024 and December 2023, GSI was subject to and in compliance with applicable capital requirements for swap dealers and security-based swap dealers.

GSJCL, our Japanese broker-dealer, is regulated by Japan's Financial Services Agency. GSJCL and certain other non-U.S. subsidiaries are also subject to capital requirements promulgated by authorities of the countries in which they operate. As of both December 2024 and December 2023, these subsidiaries were in compliance with their local capital requirements.

Regulatory and Other Matters**Regulatory Matters**

Our businesses are subject to extensive regulation and supervision worldwide. Regulations have been adopted or are being considered by regulators and policy makers worldwide. Given that many of the new and proposed rules are highly complex, the full impact of regulatory reform will not be known until the rules are implemented and market practices develop under the final regulations.

See "Business — Regulation" in Part I, Item 1 of this Form 10-K for further information about the laws, rules and regulations and proposed laws, rules and regulations that apply to us and our operations.

Other Matters**Narrowing our Focus on Consumer-Related Activities.**

During 2023 and 2024, we narrowed our focus with respect to consumer-related activities by taking the following actions:

- We completed the sale of substantially all of the Marcus loan portfolio in 2023 (included within Asset & Wealth Management).
- We sold our PFM business in 2023 (included within Asset & Wealth Management).
- We sold the majority of the GreenSky loan portfolio in 2023 and, during 2024, completed the sale of GreenSky (included within Platform Solutions).
- During 2024, we entered into an agreement to transition the GM credit card program (included within Platform Solutions) to another issuer. The transition is expected to be completed in the third quarter of 2025.
- During 2024, we sold our seller financing loan portfolio (included within Platform Solutions). This portfolio consisted of loans that were extended to small- and medium-sized retailers.

We remain committed to supporting the products and servicing customers through the various transition arrangements for our consumer-related activities.

Management's Discussion and Analysis

The table below presents the impact to pre-tax earnings of the items that we sold or have announced the decision to sell (with respect to the narrowing of our focus on consumer-related activities).

\$ in millions	Year Ended December	
	2024	2023
Marcus loan portfolio	\$ -	\$ 233
PFM	-	276
GreenSky	(27)	(1,227)
GM credit card program	(557)	(65)
Seller financing loan portfolio	(84)	(28)
Total	\$ (668)	\$ (811)

In the table above, pre-tax earnings related to GreenSky, the GM credit card program and the seller financing loan portfolio were included within Platform Solutions and the pre-tax earnings related to the Marcus loan portfolio and PFM were included within Asset & Wealth Management.

We have the following remaining consumer-related activities within Platform Solutions:

- We issue credit cards to and raise deposits from Apple Card customers.
- We will continue to support existing GM customers and issue credit cards to new GM customers until the transition of the GM credit card program to another issuer is completed.

Future decisions we may make in connection with the narrowing of our focus on consumer-related activities could have a material impact on our results of operations in the period such decisions are made.

See "Results of Operations — Platform Solutions" for the drivers of changes in our net revenues for Consumer platforms.

Impact of Los Angeles County Wildfires. In January 2025, a series of wildfires started in Los Angeles County that spread throughout the region. We are in ongoing dialogue with key stakeholders to assess the health and safety conditions of our office locations and the well-being of our employees. We are monitoring the ongoing developments of the wildfires and the potential impact on the broader economy. As of the date of this filing, the wildfires did not have a material impact on our results of operations.

Off-Balance Sheet Arrangements

In the ordinary course of business, we enter into various types of off-balance sheet arrangements. Our involvement in these arrangements can take many different forms, including:

- Purchasing or retaining residual and other interests in special purpose entities, such as mortgage-backed and other asset-backed securitization vehicles;
- Holding senior and subordinated debt, interests in limited and general partnerships, and preferred and common stock in other nonconsolidated vehicles;
- Entering into interest rate, foreign currency, equity, commodity and credit derivatives, including total return swaps; and
- Providing guarantees, indemnifications, commitments, letters of credit and representations and warranties.

We enter into these arrangements for a variety of business purposes, including securitizations. The securitization vehicles that purchase mortgages, corporate bonds and other types of financial assets are critical to the functioning of several significant investor markets, including the mortgage-backed and other asset-backed securities markets, since they offer investors access to specific cash flows and risks created through the securitization process.

We also enter into these arrangements to underwrite client securitization transactions; provide secondary market liquidity; make investments in performing and nonperforming debt, distressed loans, power-related assets, equity securities, real estate and other assets; and provide investors with credit-linked and asset-repackaged notes.

The table below presents where information about our various off-balance sheet arrangements may be found in this Form 10-K. In addition, see Note 3 to the consolidated financial statements for information about our consolidation policies.

Off-Balance Sheet Arrangement	Disclosure in Form 10-K
Variable interests and other obligations, including contingent obligations, arising from variable interests in nonconsolidated variable interest entities	See Note 17 to the consolidated financial statements.
Guarantees, and lending and other commitments	See Note 18 to the consolidated financial statements.
Derivatives	See "Risk Management — Credit Risk Management — Credit Exposures — OTC Derivatives" and Notes 4, 5, 7 and 18 to the consolidated financial statements.

Management's Discussion and Analysis

Risk Management

Risks are inherent in our businesses and include liquidity, market, credit, operational, cybersecurity, model, legal, compliance, conduct, regulatory and reputational risks. For further information about our risk management processes, see "Overview and Structure of Risk Management," and for information about our areas of risk, see "Liquidity Risk Management," "Market Risk Management," "Credit Risk Management," "Operational Risk Management," "Cybersecurity Risk Management," "Model Risk Management" and "Other Risk Management," as well as "Risk Factors" in Part I, Item 1A of this Form 10-K.

Overview and Structure of Risk Management

Overview

Effective risk management is critical to our success. Accordingly, we have established an enterprise risk management framework that employs a comprehensive, integrated approach to risk management and is designed to enable comprehensive risk management processes through which we identify, assess, monitor and manage the risks we assume in conducting our activities. Our risk management structure is built around three core components: governance, processes and people.

Governance. Risk management governance starts with the Board, which both directly and through its committees, including its Risk Committee, oversees our approach to managing our risks through the enterprise risk management framework. The Board is also responsible for the annual review and approval of our risk appetite statement. The risk appetite statement describes the levels and types of risk we are willing to accept or to avoid in order to achieve our objectives included in our strategy and business plan, while remaining in compliance with regulatory requirements. The Board reviews our strategy and business plan and is ultimately responsible for overseeing and providing direction about our strategy and risk appetite.

The Board, including through its committees, receives regular briefings on firmwide risks, including liquidity risk, market risk, credit risk, operational risk, model risk and climate risk, from our chief risk officer, on cybersecurity threats and risks from our chief information security officer (CISO), on compliance risk and conduct risk from our chief compliance officer, on legal and regulatory enforcement matters from our chief legal officer, and on other matters impacting our reputation from the chair and/or vice-chairs of our Firmwide Reputational Risk Committee. The chief risk officer reports to our chief executive officer and to the Risk Committee of the Board. As part of the review of the firmwide risk portfolio, the chief risk officer regularly advises the Risk Committee of the Board of relevant risk metrics and material exposures, including risk limits and thresholds established in our risk appetite statement.

Enterprise Risk, which reports to our chief risk officer, is responsible for ensuring that our enterprise risk management framework provides the Board, our risk committees and senior management with a consistent and integrated approach to managing our various risks in a manner consistent with our risk appetite.

Our first line of defense consists of our revenue-producing units, Conflicts Resolution, Controllers, Engineering, Corporate Treasury and certain other corporate functions. The first line of defense is responsible for its risk-generating activities, as well as for the design and execution of controls to mitigate such risks.

Our Risk and Compliance functions are considered our second line of defense and provide independent assessment, oversight and challenge of the risks taken by our first line of defense, as well as lead and participate in firmwide risk committees.

Internal Audit is considered our third line of defense, and our director of Internal Audit reports to the Audit Committee of the Board and administratively to our chief executive officer. Internal Audit includes professionals with a broad range of audit and industry experience, including risk management expertise. Internal Audit is responsible for independently assessing and validating the effectiveness of key controls, including those within the risk management framework, and providing timely reporting to the Audit Committee of the Board, senior management and regulators.

The three lines of defense structure promotes the accountability of first line risk takers, provides a framework for effective challenge by the second line and empowers independent review from the third line.

Management's Discussion and Analysis

Processes. We maintain various processes that are critical components of our risk management framework, including (i) risk identification and assessment, (ii) risk appetite, limits, thresholds and alerts, (iii) control monitoring and testing, and (iv) risk reporting.

- **Risk Identification and Assessment.** We believe the identification and assessment of our risks is a critical step in providing our Board and senior management transparency and insight into the range and materiality of our risks. We have a comprehensive data collection process, including firmwide policies and procedures that require all employees to report and escalate risk events. Our approach for risk identification and assessment is comprehensive across all risk types, is dynamic and forward-looking to reflect and adapt to our changing risk profile and business environment, leverages subject matter expertise, and allows for prioritization of our most critical risks. We perform risk assessments periodically with the aim of ensuring that our material financial and nonfinancial risks are mitigated through controls to an acceptable tolerance level in accordance with our risk appetite. Our risk assessments include, among other things, the use of stress testing as well as an assessment of our internal control processes designed to mitigate such risks.

Firmwide stress testing is an important part of our risk management process. It allows us to quantify our exposure to tail risks, highlight potential loss concentrations, undertake risk/reward analysis, and assess and mitigate our risk positions. Firmwide stress tests are performed on a regular basis and are designed to ensure a comprehensive analysis of our vulnerabilities and idiosyncratic risks combining financial and nonfinancial risks, including, but not limited to, credit, market, liquidity and funding, operational and compliance, strategic, systemic and emerging risks into our stress scenarios. We also perform ad hoc stress tests in anticipation of market events or conditions. Stress tests are also used to assess capital adequacy as part of our capital planning and stress testing process. See “Capital Management and Regulatory Capital — Capital Management” for further information.

We maintain a daily discipline of marking substantially all of our inventory to current market levels. We carry our inventory at fair value, with changes in valuation reflected immediately in our risk management systems and in net revenues. We do so because we believe this discipline is one of the most effective tools for assessing and managing risk and that it provides transparent and realistic insight into our inventory exposures.

- **Risk Appetite, Limits, Thresholds and Alerts.** We apply risk limits, thresholds and alerts to control and monitor risk across transactions, products, businesses and markets. The Board, directly or indirectly through its Risk Committee, approves limits, thresholds and alerts included in our risk appetite statement at firmwide, business and product levels. In addition, the Firmwide Risk Appetite Committee, through delegated authority from the Firmwide Enterprise Risk Committee, is responsible for approving our risk limits, thresholds and alerts policy, subject to the overall limits directly or indirectly approved by the Board, and monitoring these limits.

The Firmwide Risk Appetite Committee is responsible for approving and monitoring limits at firmwide, business and product levels. Certain limits may be set at levels that will require periodic adjustment, rather than at levels that reflect our maximum risk appetite. This fosters an ongoing dialogue about risk among our first and second lines of defense, committees and senior management, as well as rapid escalation of risk-related matters. The Firmwide Risk Appetite Committee also authorizes Risk to set limits and thresholds to support monitoring and oversight at a more granular level. For example, Market Risk sets limits at certain product and desk levels, and Credit Risk sets limits for individual counterparties and their subsidiaries, industries and countries. Limits are reviewed regularly and amended on a permanent or temporary basis to reflect changes to our strategic business plan, as well as changing market conditions, business conditions or risk tolerance. Risks limits are monitored by the respective Risk functions.

Management's Discussion and Analysis

- **Control Monitoring and Testing.** We perform control monitoring and testing to measure the effectiveness of our key controls and to ensure that we are in compliance with policies, codes of conduct, control standards and regulatory requirements. Monitoring and testing is performed by dedicated teams within the first and second lines of defense. These teams establish procedures, develop risk-based annual plans, perform control testing and escalate identified issues.

Issues identified by the dedicated teams, as well as self-identified issues by our employees, are assessed for appropriate escalation and resolution. Where material or thematic issues exist, we develop a plan to remediate them, as appropriate, and monitor the remediation activities.

- **Risk Reporting.** Effective risk reporting depends on our ability to get the right information to the right people at the right time. Risk reporting is designed to be both forward- and backward-looking and consider detailed information on existing and emerging risk exposures. Risk reporting may include stress testing and scenario analysis, information about the risk profiles for financial and nonfinancial risks, utilization of risk limits and thresholds, details of new and emerging risks identified through our risk identification processes, details of issues, significant internal and external events, and information related to the effectiveness of our controls and remediation plans. As such, we focus on the rigor and effectiveness of our risk systems, with the objective of ensuring that our risk management technology systems provide us with complete, accurate and timely information. Our risk reporting process is designed to take into account information about both existing and emerging risks, thereby enabling our risk committees and senior management to perform their responsibilities with the appropriate level of insight into risk exposures.

We make extensive use of risk committees and councils that meet regularly and serve as an important means to facilitate and foster ongoing discussions to manage and mitigate risks.

We maintain strong and proactive communication about risk and we have a culture of collaboration in decision-making among our first and second lines of defense, committees and senior management. While our first line of defense is accountable and responsible for management of their risk, we dedicate extensive resources to our second line of defense in order to reinforce the importance of having effective oversight and challenge, and a strong culture of escalation and accountability across all functions.

People. Even the best technology serves only as a tool for helping to make informed decisions in real time about the risks we are taking. Ultimately, effective risk management requires our people to interpret our risk data on an ongoing and timely basis and adjust risk positions accordingly. The experience of our professionals, and their understanding of the nuances and limitations of each risk measure, guides us in assessing exposures and maintaining them within prudent levels.

We reinforce a culture of effective risk management, consistent with our risk appetite, in our training and development programs, as well as in the way we evaluate performance, and recognize and reward our people. Our training and development programs, including certain sessions led by our most senior leaders, are focused on the importance of risk management, client relationships and reputational excellence. As part of our performance review process, we assess reputational excellence, including how an employee exercises good risk management and reputational judgment, and adheres to our code of conduct and compliance policies. Our review and reward processes are designed to communicate and reinforce to our professionals the link between behavior and how people are recognized, the need to focus on our clients and our reputation, and the need to always act in accordance with our highest standards.

Management's Discussion and Analysis

Structure

Ultimate oversight of risk is the responsibility of our Board. The Board oversees risk both directly and through its committees, including its Risk Committee. We also have a series of committees that generally consist of senior managers, including from both our first and second lines of defense, with specific risk management mandates that have oversight or decision-making responsibilities for risk management activities. We have an established policy for these committees so that appropriate information barriers are in place. Our primary risk committees, most of which also have additional sub-committees, councils or working groups, are described below. In addition to these committees, we have other risk committees that provide oversight for different businesses, activities, products, regions and entities. All of our committees have responsibility for considering the impact on our reputation of the transactions and activities that they oversee.

Membership of our risk committees is reviewed regularly and updated to reflect changes in the responsibilities of the committee members. Accordingly, the length of time that members serve on the respective committees varies as determined by the committee chairs and based on the responsibilities of the members.

The chart below presents an overview of our risk management governance structure.



Management Committee. The Management Committee oversees our global activities. It provides this oversight directly and through delegated authority. This committee consists of our most senior leaders, and is chaired by our chief executive officer. Most members of the Management Committee are also members of other committees. The following are the committees that are principally involved in firmwide risk management.

Firmwide Enterprise Risk Committee. The Firmwide Enterprise Risk Committee is responsible for overseeing all of our financial and nonfinancial risks. As part of such oversight, the committee is responsible for the ongoing review, approval and monitoring of our enterprise risk management framework, as well as our risk limits, and thresholds and alerts policy, through delegated authority to the Firmwide Risk Appetite Committee. The Firmwide Enterprise Risk Committee also reviews new significant strategic business initiatives to determine whether they are consistent with our risk appetite and risk management capabilities. Additionally, the Firmwide Enterprise Risk Committee performs enhanced reviews of significant risk events, the top residual and emerging risks, and the overall risk and control environment in each of our business units in order to propose uplifts, identify elements that are common to all business units and analyze the consolidated residual risks that we face. This committee, which reports to the Management Committee, is co-chaired by our president and chief operating officer and our chief risk officer, who are appointed as chairs by our chief executive officer, and the vice-chair is our chief financial officer, who is appointed as vice-chair by the chairs of the Firmwide Enterprise Risk Committee. The Firmwide Enterprise Risk Committee also periodically provides updates to, and receives guidance from, the Risk Committee of the Board. The following are the primary committees that report to the Firmwide Enterprise Risk Committee:

- **Firmwide New Activity Committee.** The Firmwide New Activity Committee is responsible for reviewing new activities and, upon referral by the Firmwide Enterprise Risk Committee, significant strategic business initiatives. Additionally, the Firmwide New Activity Committee may review previously approved activities that are significant and/or that have changed in complexity and/or structure or present different reputational and suitability concerns over time to consider whether these activities remain appropriate. This committee is chaired by our controller and chief accounting officer, who is appointed as chair by the chairs of the Firmwide Enterprise Risk Committee.
- **Firmwide Technology Risk Committee.** The Firmwide Technology Risk Committee is responsible for reviewing matters related to the design, development, deployment and use of technology. This committee oversees cybersecurity matters, as well as technology risk management frameworks and methodologies, and monitors their effectiveness. This committee is co-chaired by our CISO and our chief technology officer, who are appointed as chairs by the chairs of the Firmwide Enterprise Risk Committee. To assist the Firmwide Technology Risk Committee in carrying out its mandate, the Firmwide Artificial Intelligence Risk and Controls Committee, which oversees risks associated with the use of AI, reports to the Firmwide Technology Risk Committee.

Management's Discussion and Analysis

- **Firmwide Compliance and Operational Risk Committee.**

The Firmwide Compliance and Operational Risk Committee is responsible for overseeing compliance and operational risk. This committee is co-chaired by our chief administrative officer for EMEA, our head of Operational Risk, and our chief compliance officer, who are appointed as chairs by the chairs of the Firmwide Enterprise Risk Committee.

- **Firmwide Risk Appetite Committee.** The Firmwide Risk Appetite Committee (through delegated authority from the Firmwide Enterprise Risk Committee) is responsible for the ongoing approval and monitoring of risk frameworks, policies and parameters related to our risk management processes, as well as limits, thresholds and alerts, at firmwide, business and product levels. In addition, this committee is responsible for overseeing our financial and model risks and reviews the results of stress tests and scenario analyses. To assist the Firmwide Risk Appetite Committee in carrying out its mandate, a number of other risk committees with dedicated oversight for stress testing, model risks, Volcker Rule compliance, as well as our investments or other capital commitments that may give rise to financial risk, report into the Firmwide Risk Appetite Committee. This committee is chaired by our chief risk officer, who is appointed as chair by the chairs of the Firmwide Enterprise Risk Committee. The Firmwide Capital Committee and Firmwide Commitments Committee report to the Firmwide Risk Appetite Committee.

- **Firmwide Reputational Risk Committee.** The Firmwide Reputational Risk Committee is responsible for assessing reputational risks arising from opportunities that have been identified as having potential heightened reputational risk, including transactions identified pursuant to the criteria established by the Firmwide Reputational Risk Committee and as determined by committee leadership. This committee is also responsible for overseeing client-related business standards and addressing client-related reputational risk. This committee is chaired by our president and chief operating officer, who is appointed as chair by our chief executive officer, and the vice-chairs are our chief legal officer and the head of Conflicts Resolution, who are appointed as vice-chairs by the chair of the Firmwide Reputational Risk Committee. This committee periodically provides updates to, and receives guidance from, the Public Responsibilities Committee of the Board. The Firmwide Suitability Committee reports to the Firmwide Reputational Risk Committee.

- **Firmwide Data Governance Committee.** The Firmwide Data Governance Committee is responsible for overseeing the firmwide data governance framework, and its implementation, to help ensure that data governance and data quality are appropriate. This committee is co-chaired by our chief information officer and our chief risk officer, who are appointed as chairs by the chairs of the Firmwide Enterprise Risk Committee.

Firmwide Asset Liability Committee. The Firmwide Asset Liability Committee reviews and approves the strategic direction for our financial resources, including capital, liquidity, funding and balance sheet. This committee has oversight responsibility for asset liability management, including interest rate and currency risk, funds transfer pricing, capital allocation and incentives, and credit ratings. This committee makes recommendations as to any adjustments to asset liability management and financial resource allocation in light of current events, risks, exposures, and regulatory requirements and approves related policies. This committee is co-chaired by our chief financial officer and our global treasurer, who are appointed as chairs by our chief executive officer, and reports to the Management Committee.

Liquidity Risk Management

Overview

Liquidity risk is the risk that we will be unable to fund ourselves or meet our liquidity needs in the event of firm-specific, broader industry or market liquidity stress events. We have in place a comprehensive and conservative set of liquidity and funding policies. Our principal objective is to be able to fund ourselves and to enable our core businesses to continue to serve clients and generate revenues, even under adverse circumstances.

Corporate Treasury is responsible for our liquidity, including developing and executing our liquidity and funding strategy.

Liquidity Risk, which is part of our second line of defense and reports to our chief risk officer, has primary responsibility for assessing, monitoring and managing our liquidity risk by providing independent firmwide oversight and challenge across our global businesses. Liquidity Risk is also responsible for the establishment of stress testing and limits frameworks.

Management's Discussion and Analysis

Liquidity Risk Management Principles

We manage liquidity risk according to three principles: (i) hold sufficient excess liquidity in the form of GCLA to cover outflows during a stressed period, (ii) maintain appropriate Asset-Liability Management and (iii) maintain a viable Contingency Funding Plan.

GCLA. GCLA is liquidity that we maintain to meet a broad range of potential cash outflows and collateral needs in a stressed environment. A primary liquidity principle is to pre-fund our estimated potential cash and collateral needs during a liquidity crisis and hold this liquidity in the form of unencumbered, highly liquid securities and cash. We believe that the securities held in our GCLA would be readily convertible to cash in a matter of days, through liquidation, by entering into repurchase agreements or from maturities of resale agreements, and that this cash would allow us to meet immediate obligations without needing to sell other assets or depend on additional funding from credit-sensitive markets.

Our GCLA reflects the following principles:

- The first days or weeks of a liquidity crisis are the most critical to a company's survival;
- Focus must be maintained on all potential cash and collateral outflows, not just disruptions to financing flows. Our businesses are diverse, and our liquidity needs are determined by many factors, including market movements, collateral requirements and client commitments, all of which can change dramatically in a difficult funding environment;
- During a liquidity crisis, credit-sensitive funding, including unsecured debt, certain deposits and some types of secured financing agreements, may be unavailable, and the terms (e.g., interest rates, collateral provisions and tenor) or availability of other types of secured financing may change and certain deposits may be withdrawn; and
- As a result of our policy to pre-fund liquidity that we estimate may be needed in a crisis, we hold more unencumbered securities and have larger funding balances than our businesses would otherwise require. We believe that our liquidity is stronger with greater balances of highly liquid unencumbered securities, even though it increases our total assets and our funding costs.

We maintain our GCLA across Group Inc., Goldman Sachs Funding LLC (Funding IHC) and Group Inc.'s major broker-dealer and bank subsidiaries, asset types and clearing agents with the goal of providing us with sufficient operating liquidity to ensure timely settlement in all major markets, even in a difficult funding environment. In addition to the GCLA, we maintain cash balances and securities in several of our other entities, primarily for use in specific currencies, entities or jurisdictions where we do not have immediate access to parent company liquidity.

Asset-Liability Management. Our liquidity risk management policies are designed to ensure we have a sufficient amount of financing, even when funding markets experience persistent stress. We manage the maturities and diversity of our funding across markets, products and counterparties, and seek to maintain a diversified funding profile with an appropriate tenor, taking into consideration the characteristics and liquidity profile of our assets.

Our approach to asset-liability management includes:

- Conservatively managing the overall characteristics of our funding book, with a focus on maintaining long-term, diversified sources of funding in excess of our current requirements. See "Balance Sheet and Funding Sources — Funding Sources" for further information;
- Actively managing and monitoring our asset base, with particular focus on the liquidity, holding period and ability to fund assets on a secured basis. We assess our funding requirements and our ability to liquidate assets in a stressed environment while appropriately managing risk. This enables us to determine the most appropriate funding products and tenors. See "Balance Sheet and Funding Sources — Balance Sheet Management" for further information about our balance sheet management process and "— Funding Sources — Secured Funding" for further information about asset classes that may be harder to fund on a secured basis; and
- Raising secured and unsecured financing that has a long tenor relative to the liquidity profile of our assets. This reduces the risk that our liabilities will come due in advance of our ability to generate liquidity from the sale of our assets. Because we maintain a highly liquid balance sheet, the holding period of certain of our assets may be materially shorter than their contractual maturity dates.

Our goal is to ensure that we maintain sufficient liquidity to fund our assets and meet our contractual and contingent obligations in normal times, as well as during periods of market stress. Through our dynamic balance sheet management process, we use actual and projected asset balances to determine secured and unsecured funding requirements. Funding plans are reviewed and approved by the Firmwide Asset Liability Committee. In addition, Risk and the Firmwide Asset Liability Committee review our total unsecured long-term borrowings and total shareholders' equity to help ensure that we maintain a level of long-term funding that is sufficient to meet our long-term financing requirements. In a liquidity crisis, we would begin by liquidating and monetizing our GCLA before selling other assets. However, we recognize that orderly asset sales may be prudent or necessary in a severe or persistent liquidity crisis.

Management's Discussion and Analysis

Subsidiary Funding Policies

The majority of our unsecured borrowings is raised by Group Inc., which provides the necessary funds to Funding IHC and other subsidiaries, some of which are regulated, to meet their asset financing, liquidity and capital requirements. In addition, Group Inc. provides its regulated subsidiaries with the necessary capital to meet their regulatory requirements. The benefits of this approach to subsidiary funding are enhanced control and greater flexibility to meet the funding requirements of our subsidiaries. Funding is also raised at the subsidiary level through a variety of products, including deposits, secured funding and unsecured borrowings.

Our intercompany funding policies assume that a subsidiary's funds or securities are not freely available to its parent, Funding IHC or other subsidiaries unless (i) legally provided for and (ii) there are no additional regulatory, tax or other restrictions. In particular, many of our subsidiaries are subject to laws that authorize regulatory bodies to block or reduce the flow of funds from those subsidiaries to Group Inc. or Funding IHC. Regulatory action of that kind could impede access to funds that Group Inc. needs to make payments on its obligations. Accordingly, we assume that the capital provided to our regulated subsidiaries is not available to Group Inc. or other subsidiaries and any other financing provided to our regulated subsidiaries is not available to Group Inc. or Funding IHC until the maturity of such financing.

Group Inc. has provided substantial amounts of equity and subordinated indebtedness, directly or indirectly, to its regulated subsidiaries. For example, as of December 2024, Group Inc. had \$38.69 billion of equity and subordinated indebtedness invested in GS&Co., its principal U.S. registered broker-dealer; \$47.21 billion invested in GSI, a regulated U.K. broker-dealer; \$2.08 billion invested in GSJCL, a regulated Japanese broker-dealer; \$62.81 billion invested in GS Bank USA, a regulated New York State-chartered bank; and \$5.30 billion invested in GSIB, a regulated U.K. bank. Group Inc. also provides financing, directly or indirectly, in the form of: \$131.82 billion of unsubordinated loans (including secured loans of \$59.97 billion) and \$32.92 billion of collateral and cash deposits to these entities as of December 2024. In addition, as of December 2024, Group Inc. had significant amounts of capital invested in and loans to its other regulated subsidiaries.

Contingency Funding Plan. We maintain a contingency funding plan to provide a framework for analyzing and responding to a liquidity crisis situation or periods of market stress. Our contingency funding plan outlines a list of potential risk factors, key reports and metrics that are reviewed on an ongoing basis to assist in assessing the severity of, and managing through, a liquidity crisis and/or market dislocation. The contingency funding plan also describes in detail our potential responses if our assessments indicate that we have entered a liquidity crisis, which include pre-funding for what we estimate will be our potential cash and collateral needs, as well as utilizing secondary sources of liquidity. Mitigants and action items to address specific risks which may arise are also described and assigned to individuals responsible for execution.

The contingency funding plan identifies key groups of individuals and their responsibilities, which include fostering effective coordination, control and distribution of information, implementing liquidity maintenance activities and managing internal and external communication, all of which are critical in the management of a crisis or period of market stress.

Stress Tests

In order to determine the appropriate size of our GCLA, we model liquidity outflows over a range of scenarios and time horizons. One of our primary internal liquidity risk models, referred to as the Modeled Liquidity Outflow, quantifies our liquidity risks over a 30-day stress scenario. We also consider other factors, including, but not limited to, an assessment of our potential intraday liquidity needs through an additional internal liquidity risk model, referred to as the Intraday Liquidity Model, the results of our long-term stress testing models, our resolution liquidity models and other applicable regulatory requirements and a qualitative assessment of our condition, as well as the financial markets. The results of the Modeled Liquidity Outflow, the Intraday Liquidity Model, the long-term stress testing models and the resolution liquidity models are reported to senior management on a regular basis. We also perform firmwide stress tests. See "Overview and Structure of Risk Management" for information about firmwide stress tests.

Management's Discussion and Analysis

Modeled Liquidity Outflow. Our Modeled Liquidity Outflow is based on conducting multiple scenarios that include combinations of market-wide and firm-specific stress. These scenarios are characterized by the following qualitative elements:

- Severely challenged market environments, which include low consumer and corporate confidence, financial and political instability, and adverse changes in market values, including potential declines in equity markets and widening of credit spreads; and
- A firm-specific crisis potentially triggered by material losses, reputational damage (including, as a result of, the dissemination of negative information through social media), litigation and/or a ratings downgrade.

The following are key modeling elements of our Modeled Liquidity Outflow:

- Liquidity needs over a 30-day scenario;
- A two-notch downgrade of our long-term senior unsecured credit ratings;
- Changing conditions in funding markets, which limit our access to unsecured and secured funding;
- No support from additional government funding facilities. Although we have access to various central bank funding programs, we do not assume reliance on additional sources of funding in a liquidity crisis; and
- A combination of contractual outflows and contingent outflows arising from both our on- and off-balance sheet arrangements. Contractual outflows include, among other things, upcoming maturities of unsecured debt, term deposits and secured funding. Contingent outflows include, among other things, the withdrawal of customer credit balances in our prime brokerage business, increase in variation margin requirements due to adverse changes in the value of our exchange-traded and OTC-cleared derivatives, draws on unfunded commitments and withdrawals of deposits that have no contractual maturity. See notes to the consolidated financial statements for further information about contractual outflows, including Note 11 for collateralized financings, Note 13 for deposits, Note 14 for unsecured long-term borrowings and Note 15 for operating lease payments, and "Off-Balance Sheet Arrangements" for further information about our various types of off-balance sheet arrangements.

Intraday Liquidity Model. Our Intraday Liquidity Model measures our intraday liquidity needs in a scenario where access to sources of intraday liquidity may become constrained. The intraday liquidity model considers a variety of factors, including historical settlement activity.

Long-Term Stress Testing. We utilize longer-term stress tests to take a forward view on our liquidity position through prolonged stress periods in which we experience a severe liquidity stress and recover in an environment that continues to be challenging. We are focused on ensuring conservative asset-liability management to prepare for a prolonged period of potential stress, seeking to maintain a diversified funding profile with an appropriate tenor, taking into consideration the characteristics and liquidity profile of our assets.

Resolution Liquidity Models. In connection with our resolution planning efforts, we have established our Resolution Liquidity Adequacy and Positioning framework, which estimates liquidity needs of our major subsidiaries in a stressed environment. The liquidity needs are measured using our Modeled Liquidity Outflow assumptions and include certain additional inter-affiliate exposures. We have also established our Resolution Liquidity Execution Need framework, which measures the liquidity needs of our major subsidiaries to stabilize and wind down following a Group Inc. bankruptcy filing in accordance with our preferred resolution strategy.

In addition, we have established a triggers and alerts framework, which is designed to provide the Board with information needed to make an informed decision on whether and when to commence bankruptcy proceedings for Group Inc.

Limits

We use liquidity risk limits at various levels and across liquidity risk types to manage the size of our liquidity exposures. Limits are measured relative to acceptable levels of risk given our liquidity risk tolerance. See "Overview and Structure of Risk Management" for information about the limit approval process.

Limits are monitored by Corporate Treasury and Liquidity Risk. Liquidity Risk is responsible for identifying and escalating to senior management and/or the appropriate risk committee, on a timely basis, instances where limits have been exceeded.

GCLA and Unencumbered Metrics

GCLA. Based on the results of our internal liquidity risk models, described above, as well as our consideration of other factors, including, but not limited to, a qualitative assessment of our condition, as well as the financial markets, we believe our liquidity position as of both December 2024 and December 2023 was appropriate. We strictly limit our GCLA to a narrowly defined list of securities and cash because they are highly liquid, even in a difficult funding environment. We do not include other potential sources of excess liquidity in our GCLA, such as less liquid unencumbered securities or committed credit facilities.

Management's Discussion and Analysis

The table below presents information about our GCLA.

\$ in millions	Average for the			
	Three Months Ended December		Year Ended December	
	2024	2023	2024	2023
Denomination				
U.S. dollar	\$ 311,839	\$ 282,414	\$ 300,535	\$ 282,307
Non-U.S. dollar	110,252	131,176	128,645	124,691
Total	\$ 422,091	\$ 413,590	\$ 429,180	\$ 406,998
Asset Class				
Overnight cash deposits	\$ 143,563	\$ 204,929	\$ 184,370	\$ 231,066
U.S. government obligations	177,721	150,806	163,983	133,000
U.S. agency obligations	46,979	22,895	32,102	16,387
Non-U.S. government obligations	53,828	34,960	48,725	26,545
Total	\$ 422,091	\$ 413,590	\$ 429,180	\$ 406,998
Entity Type				
Group Inc. and Funding IHC	\$ 60,609	\$ 65,952	\$ 65,965	\$ 66,803
Major broker-dealer subsidiaries	113,996	117,818	117,204	114,824
Major bank subsidiaries	247,486	229,820	246,011	225,371
Total	\$ 422,091	\$ 413,590	\$ 429,180	\$ 406,998

In the table above:

- The U.S. dollar-denominated GCLA consists of (i) unencumbered U.S. government and agency obligations (including highly liquid U.S. agency mortgage-backed obligations), all of which are eligible as collateral in Federal Reserve open market operations and (ii) certain overnight U.S. dollar cash deposits.
- The non-U.S. dollar-denominated GCLA consists of non-U.S. government obligations (only unencumbered German, French, Japanese and U.K. government obligations) and certain overnight cash deposits in highly liquid currencies.

We maintain our GCLA to enable us to meet current and potential liquidity requirements of our parent company, Group Inc., and its subsidiaries. Our Modeled Liquidity Outflow and Intraday Liquidity Model incorporate a requirement for Group Inc., as well as a standalone requirement for each of our major broker-dealer and bank subsidiaries. Funding IHC is required to provide the necessary liquidity to Group Inc. during the ordinary course of business, and is also obligated to provide capital and liquidity support to major subsidiaries in the event of our material financial distress or failure. Liquidity held directly in each of our major broker-dealer and bank subsidiaries is intended for use only by that subsidiary to meet its liquidity requirements and is assumed not to be available to Group Inc. or Funding IHC unless (i) legally provided for and (ii) there are no additional regulatory, tax or other restrictions. In addition, the Modeled Liquidity Outflow and Intraday Liquidity Model also incorporate a broader assessment of standalone liquidity requirements for other subsidiaries and we hold a portion of our GCLA directly at Group Inc. or Funding IHC to support such requirements.

Other Unencumbered Assets. In addition to our GCLA, we have a significant amount of other unencumbered cash and financial instruments, including other government obligations, high-grade money market securities, corporate obligations, marginable equities, loans and cash deposits not included in our GCLA. The fair value of our unencumbered assets averaged \$295.49 billion for the three months ended December 2024, \$286.51 billion for the three months ended December 2023, \$292.22 billion for the year ended December 2024 and \$281.95 billion for the year ended December 2023. We do not consider these assets liquid enough to be eligible for our GCLA.

Liquidity Regulatory Framework

We are subject to a minimum Liquidity Coverage Ratio (LCR) under the LCR rule approved by the U.S. federal bank regulatory agencies. The LCR rule requires organizations to maintain an adequate ratio of eligible high-quality liquid assets (HQLA) to expected net cash outflows under an acute, short-term liquidity stress scenario. Eligible HQLA excludes HQLA held by subsidiaries that is in excess of their minimum requirement and is subject to transfer restrictions. We are required to maintain a minimum LCR of 100%. We expect that fluctuations in client activity, business mix and the market environment will impact our LCR.

The table below presents information about our average daily LCR.

\$ in millions	Average for the Three Months Ended		
	December	September	December
	2024	2024	2023
Total HQLA	\$ 407,348	\$ 434,256	\$ 401,721
Eligible HQLA	\$ 352,494	\$ 369,119	\$ 326,181
Net cash outflows	\$ 279,368	\$ 277,825	\$ 255,106
LCR	126 %	133 %	128 %

In the table above, our average quarterly LCR represents the average of our daily LCRs during the quarter.

We are also subject to a minimum Net Stable Funding Ratio (NSFR) under the NSFR rule approved by the U.S. federal bank regulatory agencies. The NSFR rule requires large U.S. banking organizations to maintain available stable funding (ASF) above their required stable funding (RSF) over a one-year time horizon. Total ASF excludes ASF held by subsidiaries that is in excess of their minimum requirement and is subject to transfer restrictions. We are required to maintain a minimum NSFR of 100%. We expect that fluctuations in client activity, business mix and the market environment will impact our NSFR.

The table below presents information about our average daily NSFR.

\$ in millions	Average for the Three Months Ended		
	December	September	December
	2024	2024	2023
Total ASF	\$ 692,474	\$ 673,860	\$ 628,734
Total RSF	\$ 595,352	\$ 577,525	\$ 542,089
NSFR	116 %	117 %	116 %

In the table above, our average quarterly NSFR represents the average of our daily NSFRs during the quarter.

Management's Discussion and Analysis

The following provides information about our subsidiary liquidity regulatory requirements:

- **GS Bank USA.** GS Bank USA is subject to a minimum LCR of 100% under the LCR rule approved by the U.S. federal bank regulatory agencies. As of December 2024, GS Bank USA's LCR exceeded the minimum requirement. The NSFR requirement described above also applies to GS Bank USA. As of December 2024, GS Bank USA's NSFR exceeded the minimum requirement.
- **GSI and GSIB.** GSI and GSIB are subject to a minimum LCR of 100% under the LCR rule approved by the U.K. regulatory authorities. GSI's and GSIB's average monthly LCR for the trailing twelve-month period ended December 2024 exceeded the minimum requirement. GSI and GSIB are subject to the applicable NSFR requirement in the U.K. As of December 2024, both GSI's and GSIB's NSFR exceeded the minimum requirement.
- **GSBE.** GSBE is subject to a minimum LCR of 100% under the LCR rule approved by the European Parliament and Council. GSBE's average monthly LCR for the trailing twelve-month period ended December 2024 exceeded the minimum requirement. GSBE is subject to the applicable NSFR requirement in the E.U. As of December 2024, GSBE's NSFR exceeded the minimum requirement.
- **Other Subsidiaries.** We monitor local regulatory liquidity requirements of our subsidiaries to ensure compliance. For many of our subsidiaries, these requirements either have changed or are likely to change in the future due to the implementation of the Basel Committee's framework for liquidity risk measurement, standards and monitoring, as well as other regulatory developments.

The implementation of these rules and any amendments adopted by the regulatory authorities could impact our liquidity and funding requirements and practices in the future.

Credit Ratings

We rely on the short- and long-term debt capital markets to fund a significant portion of our day-to-day operations, and the cost and availability of debt financing is influenced by our credit ratings. Credit ratings are also important when we are competing in certain markets, such as OTC derivatives, and when we seek to engage in longer-term transactions. See "Risk Factors" in Part I, Item 1A of this Form 10-K for information about the risks associated with a reduction in our credit ratings.

The table below presents the unsecured credit ratings and outlook of Group Inc.

	As of December 2024				
	DBRS	Fitch	Moody's	R&I	S&P
Short-term debt	R-1 (middle)	F1	P-1	a-1	A-2
Long-term debt	A (high)	A	A2	A	BBB+
Subordinated debt	A	BBB+	Baa2	A-	BBB
Trust preferred	A	BBB-	Baa3	N/A	BB+
Preferred stock	BBB (high)	BBB-	Ba1	N/A	BB+
Ratings outlook	Stable	Stable	Stable	Stable	Stable

In the table above:

- The ratings and outlook are by DBRS, Inc. (DBRS), Fitch, Inc. (Fitch), Moody's Investors Service (Moody's), Rating and Investment Information, Inc. (R&I), and Standard & Poor's Ratings Services (S&P).
- The ratings for trust preferred relate to the guaranteed preferred beneficial interests issued by Goldman Sachs Capital I.
- The DBRS, Fitch, Moody's and S&P ratings for preferred stock include the APEX issued by Goldman Sachs Capital II and Goldman Sachs Capital III.

The table below presents the unsecured credit ratings and outlook of GS Bank USA, GSIB, GSBE, GS&Co. and GSI.

	As of December 2024		
	Fitch	Moody's	S&P
GS Bank USA			
Short-term debt	F1	P-1	A-1
Long-term debt	A+	A1	A+
Short-term bank deposits	F1+	P-1	N/A
Long-term bank deposits	AA-	A1	N/A
Ratings outlook	Stable	Stable	Stable
GSIB			
Short-term debt	F1	P-1	A-1
Long-term debt	A+	A1	A+
Short-term bank deposits	F1	P-1	N/A
Long-term bank deposits	A+	A1	N/A
Ratings outlook	Stable	Stable	Stable
GSBE			
Short-term debt	F1	P-1	A-1
Long-term debt	A+	A1	A+
Short-term bank deposits	N/A	P-1	N/A
Long-term bank deposits	N/A	A1	N/A
Ratings outlook	Stable	Stable	Stable
GS&Co.			
Short-term debt	F1	N/A	A-1
Long-term debt	A+	N/A	A+
Ratings outlook	Stable	N/A	Stable
GSI			
Short-term debt	F1	P-1	A-1
Long-term debt	A+	A1	A+
Ratings outlook	Stable	Stable	Stable

Management's Discussion and Analysis

We believe our credit ratings are primarily based on the credit rating agencies' assessment of:

- Our liquidity, market, credit and operational risk management practices;
- Our level and variability of earnings;
- Our capital base;
- Our franchise, reputation and management;
- Our corporate governance; and
- The external operating and economic environment, including, in some cases, the assumed level of government support or other systemic considerations, such as potential resolution.

Certain of our derivatives have been transacted under bilateral agreements with counterparties who may require us to post collateral or terminate the transactions based on changes in our credit ratings. We manage our GCLA to ensure we would, among other potential requirements, be able to make the additional collateral or termination payments that may be required in the event of a two-notch reduction in our long-term credit ratings, as well as collateral that has not been called by counterparties, but is available to them. See Note 7 to the consolidated financial statements for further information about derivatives with credit-related contingent features and the additional collateral or termination payments related to our net derivative liabilities under bilateral agreements that could have been called by counterparties in the event of a one- or two-notch downgrade in our credit ratings.

Cash Flows

As a global financial institution, our cash flows are complex and bear little relation to our net earnings and net assets. Consequently, we believe that traditional cash flow analysis is less meaningful in evaluating our liquidity position than the liquidity and asset-liability management policies described above. Cash flow analysis may, however, be helpful in highlighting certain macro trends and strategic initiatives in our businesses.

Year Ended December 2024. Our cash and cash equivalents decreased by \$59.49 billion to \$182.09 billion at the end of 2024, primarily due to net cash used for investing activities and operating activities, partially offset by financing activities. The net cash used for investing activities primarily reflected net purchases of U.S. government obligations accounted for as available-for-sale securities and an increase in net lending activities (reflecting increases in other collateralized loans). The net cash used for operating activities primarily reflected cash outflows from trading assets, partially offset by cash inflows from collateralized transactions (reflecting both an increase in collateralized financings and a decrease in collateralized agreements). The net cash provided by financing activities primarily reflected cash inflows from other secured financings and deposits (reflecting increases in consumer deposits, partially offset by decreases in transaction banking deposits and other deposits), partially offset by common stock repurchases and net repayments of unsecured long-term borrowings.

Year Ended December 2023. Our cash and cash equivalents decreased by \$248 million to \$241.58 billion at the end of 2023, due to net cash used for investing and operating activities, partially offset by net cash provided by financing activities and the effect of exchange rate changes on cash and cash equivalents. The net cash used for investing activities primarily reflected net purchases of U.S. government obligations accounted for as held-to-maturity securities. The net cash used for operating activities primarily reflected cash outflows from trading assets and customer and other receivables and payables, net (reflecting a decrease in customer and other payables, partially offset by a decrease in customer and other receivables), partially offset by cash inflows from collateralized transactions (reflecting an increase in collateralized financings, partially offset by an increase in collateralized agreements), net earnings and trading liabilities. The net cash provided by financing activities primarily reflected cash inflows from deposits (reflecting increases in consumer deposits, brokered certificates of deposit and other deposits, partially offset by decreases in deposits sweep program balances and private bank deposits), partially offset by net repayments of unsecured long-term borrowings. The increase in cash and cash equivalents as a result of changes in foreign exchange rates was due to the U.S. dollar weakening during 2023.

For an analysis of cash flows for the year ended December 2022, see Part II, Item 7 "Management's Discussion and Analysis of Financial Condition and Results of Operations" in our Annual Report on Form 10-K for the year ended December 31, 2023.

Management's Discussion and Analysis

Market Risk Management

Overview

Market risk is the risk of an adverse impact to our earnings due to changes in market conditions. Our assets and liabilities that give rise to market risk primarily include positions held for market making for our clients and for our investing and financing activities, and these positions change based on client demands and our investment opportunities. We employ a variety of risk measures, each described in the respective sections below, to monitor market risk. Categories of market risk include the following:

- Interest rate risk: results from exposures to changes in the level, slope and curvature of yield curves, the volatilities of interest rates, prepayment speeds and credit spreads;
- Equity price risk: results from exposures to changes in prices and volatilities of individual equities, baskets of equities and equity indices;
- Currency rate risk: results from exposures to changes in spot prices, forward prices and volatilities of currency rates; and
- Commodity price risk: results from exposures to changes in spot prices, forward prices and volatilities of commodities, such as crude oil, petroleum products, natural gas, electricity, and precious and base metals.

Market Risk, which is part of our second line of defense and reports to our chief risk officer, has primary responsibility for assessing, monitoring and managing our market risk by providing independent firmwide oversight and challenge across our global businesses.

Managers in revenue-producing units, Corporate Treasury and Market Risk discuss market information, positions and estimated loss scenarios on an ongoing basis. Managers in revenue-producing units and Corporate Treasury are accountable for managing risk within prescribed limits. These managers have in-depth knowledge of their positions, markets and the instruments available to hedge their exposures.

Market Risk Management Process

Our process for managing market risk includes the critical components of our risk management framework described in the "Overview and Structure of Risk Management," as well as the following:

- Monitoring compliance with established market risk limits and reporting our exposures;
- Diversifying exposures;
- Controlling position sizes; and
- Evaluating mitigants, such as economic hedges in related securities or derivatives.

Our market risk management systems enable us to perform an independent calculation of Value-at-Risk (VaR), Earnings-at-Risk (EaR) and other stress measures, capture risk measures at individual position levels, attribute risk measures to individual risk factors of each position, report many different views of the risk measures (e.g., by desk, business, product type or entity) and produce ad hoc analyses in a timely manner.

Risk Measures

We produce risk measures and monitor them against established market risk limits. These measures reflect an extensive range of scenarios and the results are aggregated at product, business and firmwide levels.

We use a variety of risk measures to estimate the size of potential losses for small, moderate and more extreme market moves over both short- and long-term time horizons. Our primary risk measures are VaR, EaR and other stress tests.

Our risk reports detail key risks, drivers and changes for each desk and business, and are distributed daily to senior management of both our revenue-producing units and Risk.

Value-at-Risk. VaR is the potential loss in value due to adverse market movements over a defined time horizon with a specified confidence level. For assets and liabilities included in VaR, see "Financial Statement Linkages to Market Risk Measures." We typically employ a one-day time horizon with a 95% confidence level. We use a single VaR model, which captures risks, including those related to interest rates, equity prices, currency rates and commodity prices. As such, VaR facilitates comparison across portfolios of different risk characteristics. VaR also captures the diversification of aggregated risk at the firmwide level.

Management's Discussion and Analysis

We are aware of the inherent limitations to VaR and therefore use a variety of risk measures in our market risk management process. Inherent limitations to VaR include:

- VaR does not estimate potential losses over longer time horizons where moves may be extreme;
- VaR does not take account of the relative liquidity of different risk positions; and
- Previous moves in market risk factors may not produce accurate predictions of all future market moves.

To comprehensively capture our exposures and relevant risks in our VaR calculation, we use historical simulations with full valuation of market factors at the position level by simultaneously shocking the relevant market factors for that position. These market factors include spot prices, credit spreads, funding spreads, yield curves, volatility and correlation, and are updated periodically based on changes in the composition of positions, as well as variations in market conditions. We sample from five years of historical data to generate the scenarios for our VaR calculation. The historical data is weighted so that the relative importance of the data reduces over time. This gives greater importance to more recent observations and reflects current asset volatilities, which improves the accuracy of our estimates of potential loss. As a result, even if our positions included in VaR were unchanged, our VaR would increase with increasing market volatility and vice versa.

Given its reliance on historical data, VaR is most effective in estimating risk exposures in markets in which there are no sudden fundamental changes or shifts in market conditions.

Our VaR measure does not include:

- Positions that are not accounted for at fair value, such as held-to-maturity securities and loans, deposits and unsecured borrowings that are accounted for at amortized cost;
- Available-for-sale securities for which the related unrealized fair value gains and losses are included in accumulated other comprehensive income/(loss);
- Positions that are best measured and monitored using sensitivity measures; and
- The impact of changes in counterparty and our own credit spreads on derivatives, as well as changes in our own credit spreads on financial liabilities for which the fair value option was elected.

We perform daily backtesting of our VaR model (i.e., comparing daily net revenues for positions included in VaR to the VaR measure calculated as of the prior business day) at the firmwide level and for each of our businesses and major regulated subsidiaries.

Earnings-at-Risk. We manage our interest rate risk using the EaR metric. EaR measures the estimated impact of changes in interest rates to our net revenues and preferred stock dividends over a defined time horizon. EaR complements the VaR metric, which measures the impact of interest rate changes that have an immediate impact on the fair values of our assets and liabilities (i.e., mark-to-market changes). Our exposure to interest rate risk occurs due to a variety of factors, including, but not limited to:

- Differences in maturity or repricing dates of assets, liabilities, preferred stock and certain off-balance sheet instruments.
- Differences in the amounts of assets, liabilities, preferred stock and certain off-balance sheet instruments with the same maturity or repricing dates.
- Certain interest rate sensitive fees.

Corporate Treasury manages the aggregated interest rate risk from all businesses using both cash and derivatives instruments, including available-for-sale and held-to-maturity securities and interest rate derivatives. We measure EaR over a one-year time horizon following a 100- and 200-basis point instantaneous parallel shock in both short- and long-term interest rates. This sensitivity is calculated relative to a baseline market scenario, which takes into consideration, among other things, the market's expectation of forward rates, as well as our expectation of future business activity. These scenarios include contractual elements of assets, liabilities, preferred stock, and certain off-balance sheet instruments, such as rates of interest, principal repayment schedules, maturity and reset dates, and any interest rate ceilings or floors, as well as assumptions with respect to our balance sheet size and composition, prepayment behavior and deposit repricing. Deposit repricing is captured by evaluating the change in deposit rate paid relative to the change in market rates (deposit beta) and we calibrate the deposit betas used in our models by using a number of factors, including observed historical behavior, future expectations, funding needs and the competitive landscape. We continuously monitor the performance of our key assumptions against observed behavior and regularly review their sensitivity on our risk metrics.

We manage EaR with a goal to reduce potential volatility resulting from changes in interest rates so it remains within our EaR risk appetite. Our EaR scenario is regularly evaluated and updated, if necessary, to reflect changes in our business plans, market conditions and other macroeconomic factors. While management uses the best information available to estimate EaR, actual results may differ materially as a result of, among other things, changes in the economic environment or assumptions used in the process. We also measure the sensitivity of the economic value of our equity (EVE) to changes in interest rates. Compared to EaR, EVE provides a longer-term measurement of the interest rate risk exposure, primarily on non-trading assets and liabilities, by capturing the net impact of changes in interest rates to the present value of their cash flows.

Management's Discussion and Analysis

Corporate Treasury is responsible for our aggregated interest rate risk, including assessing and monitoring EaR and EVE sensitivity, and interest rate risk stress tests and assumptions.

Risk, which is part of our second line of defense and reports to our chief risk officer, has primary responsibility for assessing, monitoring and managing our interest rate risk (including EaR and EVE sensitivity) by providing independent firmwide oversight and challenge across our global businesses.

Stress Testing. Stress testing is a method of determining the effect of various hypothetical stress scenarios. We use stress tests to examine risks of specific portfolios, as well as the potential impact of our significant risk exposures. We use a variety of stress testing techniques to calculate the potential loss from a wide range of market moves on our portfolios, including firmwide stress tests, sensitivity analysis and scenario analysis. The results of our various stress tests are analyzed together for risk management purposes. See "Overview and Structure of Risk Management" for information about firmwide stress tests.

Sensitivity analysis is used to quantify the impact of a market move in a single risk factor across all positions (e.g., equity prices or credit spreads) using a variety of defined market shocks, ranging from those that could be expected over a one-day time horizon up to those that could take many months to occur. We also use sensitivity analysis to quantify the impact of the default of any single entity, which captures the risk of large or concentrated exposures.

Scenario analysis is used to quantify the impact of a specified event, including how the event impacts multiple risk factors simultaneously. For example, for sovereign stress testing we calculate potential direct exposure associated with our sovereign positions, as well as the corresponding debt, equity and currency exposures associated with our non-sovereign positions that may be impacted by the sovereign distress. When conducting scenario analysis, we often consider a number of possible outcomes for each scenario, ranging from moderate to severely adverse market impacts. In addition, these stress tests are constructed using both historical events and forward-looking hypothetical scenarios.

Unlike VaR measures, which have an implied probability because they are calculated at a specified confidence level, there may not be an implied probability that our stress testing scenarios will occur. Instead, stress testing is used to model both moderate and more extreme moves in underlying market factors. When estimating potential loss, we generally assume that our positions cannot be reduced or hedged (although experience demonstrates that we are generally able to do so).

Limits

We use market risk limits at various levels to manage the size of our market exposures. These limits are set based on VaR, EaR and on a range of stress tests relevant to our exposures. See "Overview and Structure of Risk Management" for information about the limit approval process.

Limits are monitored by Corporate Treasury and Risk. Risk is responsible for identifying and escalating to senior management and/or the appropriate risk committee, on a timely basis, instances where limits have been exceeded (e.g., due to positional changes or changes in market conditions, such as increased volatilities or changes in correlations). Such instances are remediated by a reduction in the positions we hold and/or a temporary or permanent increase to the limit, if warranted.

Metrics

We analyze VaR at the firmwide level and a variety of more detailed levels, including by risk category, business and region. Diversification effect in the tables below represents the difference between total VaR and the sum of the VaRs for the four risk categories. This effect arises because the four market risk categories are not perfectly correlated. Substantially all positions in VaR are included within Global Banking & Markets.

The table below presents our average daily VaR.

\$ in millions	Year Ended December	
	2024	2023
Categories		
Interest rates	\$ 81	\$ 96
Equity prices	37	29
Currency rates	26	24
Commodity prices	19	19
Diversification effect	(71)	(69)
Total	\$ 92	\$ 99

Our average daily VaR decreased to \$92 million in 2024 from \$99 million in 2023, due to lower levels of volatility, partially offset by increased exposures. The total decrease was primarily driven by a decrease in the interest rates category, partially offset by an increase in the equity prices category.

Management's Discussion and Analysis

The table below presents our period-end VaR.

\$ in millions	As of December	
	2024	2023
Categories		
Interest rates	\$ 84	\$ 93
Equity prices	41	25
Currency rates	38	15
Commodity prices	14	14
Diversification effect	(86)	(54)
Total	\$ 91	\$ 93

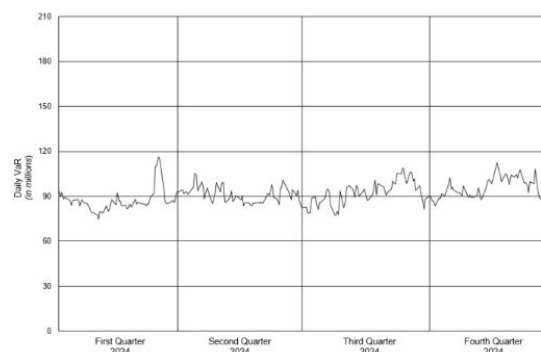
Our period-end VaR decreased to \$91 million as of December 2024 from \$93 million as of December 2023, due to lower levels of volatility, partially offset by increased exposures. The total decrease was driven by an increase in the diversification effect and a decrease in the interest rates category, partially offset by an increase in the currency rates and equity prices categories.

During 2024, there was a permanent increase to the firmwide VaR risk limit due to higher levels of volatility and increased exposures. The firmwide VaR risk limit was not exceeded during this period. During 2023, the firmwide VaR risk limit was not exceeded and there were no permanent changes to the firmwide VaR risk limit. However, the firmwide VaR risk limit was temporarily changed on four occasions as a result of changes in the market environment in the first half of 2023.

The table below presents our high and low VaR.

\$ in millions	Year Ended December			
	2024		2023	
	High	Low	High	Low
Categories				
Interest rates	\$ 121	\$ 57	\$ 148	\$ 70
Equity prices	\$ 65	\$ 25	\$ 49	\$ 22
Currency rates	\$ 47	\$ 10	\$ 47	\$ 9
Commodity prices	\$ 31	\$ 12	\$ 32	\$ 11
Firmwide				
VaR	\$ 116	\$ 75	\$ 142	\$ 79

The chart below presents our daily VaR for 2024.



The table below presents, by number of business days, the frequency distribution of our daily net revenues for positions included in VaR.

\$ in millions	Year Ended December	
	2024	2023
>\$100	62	52
\$75 - \$100	39	40
\$50 - \$75	57	52
\$25 - \$50	46	47
\$0 - \$25	31	22
\$(25) - \$0	12	30
\$(50) - \$(25)	3	4
\$(75) - \$(50)	-	2
\$(100) - \$(75)	-	-
<\$(100)	2	1
Total	252	250

Daily net revenues for positions included in VaR are compared with VaR calculated as of the end of the prior business day. Net losses incurred on a single day for such positions exceeded our 95% one-day VaR (i.e., a VaR exception) on two occasions during 2024 and on one occasion during 2023.

During periods in which we have significantly more positive net revenue days than net revenue loss days, we expect to have fewer VaR exceptions because, under normal conditions, our business model generally produces positive net revenues. In periods in which our franchise revenues are adversely affected, we generally have more loss days, resulting in more VaR exceptions. The daily net revenues for positions included in VaR used to determine VaR exceptions reflect the impact of any intraday activity, including bid/offer net revenues, which are more likely than not to be positive by their nature.

Management's Discussion and Analysis

Sensitivity Measures

Certain portfolios and individual positions are not included in VaR because VaR is not the most appropriate risk measure. Other sensitivity measures we use to analyze market risk are described below.

10% Sensitivity Measures. The table below presents our market risk by asset category for positions accounted for at fair value or accounted for at the lower of cost or fair value, that are not included in VaR.

\$ in millions	As of December	
	2024	2023
Equity	\$ 1,567	\$ 1,562
Debt	1,904	2,446
Total	\$ 3,471	\$ 4,008

In the table above:

- The market risk of these positions is determined by estimating the potential reduction in net revenues of a 10% decline in the value of the underlying positions.
- Equity positions relate to private and public equity securities, which primarily include investments in corporate, real estate and infrastructure assets. Substantially all such equity positions are included within Asset & Wealth Management.
- Debt positions include mezzanine and senior debt, and corporate and real estate loans, substantially all of which are included within Asset & Wealth Management. Debt positions also included approximately \$1.8 billion as of December 2024 and approximately \$2.0 billion as of December 2023 of GM co-branded credit card loans, and approximately \$3.0 billion as of December 2023 of GreenSky loans that were classified as held for sale. These held for sale loans were included within Platform Solutions.
- Funded equity and debt positions are included in our consolidated balance sheets in investments and loans, and the related hedges are included in our consolidated balance sheets in derivatives. See Note 8 to the consolidated financial statements for further information about investments, Note 9 to the consolidated financial statements for further information about loans and Note 7 to the consolidated financial statements for further information about derivatives.
- These measures do not reflect the diversification effect across asset categories or across other market risk measures.

Credit and Funding Spread Sensitivity on Derivatives and Financial Liabilities.

VaR excludes the impact of changes in counterparty credit spreads, our own credit spreads and unsecured funding spreads on derivatives, as well as changes in our own credit spreads (debt valuation adjustment) on financial liabilities for which the fair value option was elected. The estimated sensitivity to a one basis point increase in credit spreads (counterparty and our own) and unsecured funding spreads on derivatives (including hedges) was a loss of \$2 million as of both December 2024 and December 2023. In addition, the estimated sensitivity to a one basis point increase in our own credit spreads on financial liabilities for which the fair value option was elected was a gain of \$43 million as of December 2024 and \$42 million as of December 2023. However, the actual net impact of a change in our own credit spreads is also affected by the liquidity, duration and convexity (as the sensitivity is not linear to changes in yields) of those financial liabilities for which the fair value option was elected, as well as the relative performance of any hedges undertaken.

Earnings-at-Risk. The table below presents the impact of a parallel shift in rates on our net revenues and preferred stock dividends over the next 12 months relative to the baseline scenario.

\$ in millions	As of December	
	2024	2023
+100 basis points parallel shift in rates	\$ 140	\$ 225
-100 basis points parallel shift in rates	\$ (270)	\$ (232)
+200 basis points parallel shift in rates	\$ 196	\$ 445
-200 basis points parallel shift in rates	\$ (525)	\$ (475)

In the table above, the EaR metric utilized various assumptions, including, among other things, balance sheet size and composition, prepayment behavior and deposit repricing, all of which have inherent uncertainties. The EaR metric does not represent a forecast of our net revenues and preferred stock dividends. We expect our EaR to be more sensitive to short-term interest rates than long-term rates.

Other Market Risk Considerations

We make investments in securities that are accounted for as available-for-sale, held-to-maturity or under the equity method which are included in investments in the consolidated balance sheets. See Note 8 to the consolidated financial statements for further information.

Direct investments in real estate are accounted for at cost less accumulated depreciation. See Note 12 to the consolidated financial statements for further information about other assets.

Management's Discussion and Analysis

Financial Statement Linkages to Market Risk Measures

We employ a variety of risk measures, each described in the respective sections above, to monitor market risk across the consolidated balance sheets and consolidated statements of earnings. The related gains and losses on these positions are included in market making, other principal transactions, interest income and interest expense in the consolidated statements of earnings, and debt valuation adjustment and unrealized gains/(losses) on available-for-sale securities in the consolidated statements of comprehensive income.

The table below presents certain assets and liabilities accounted for at fair value or accounted for at the lower of cost or fair value in our consolidated balance sheets and the market risk measures used to assess those assets and liabilities.

Assets or Liabilities	Market Risk Measures
Collateralized agreements and financings	VaR
Customer and other receivables	10% Sensitivity Measures
Trading assets and liabilities	VaR Credit Spread Sensitivity 10% Sensitivity Measures
Investments	VaR 10% Sensitivity Measures
Loans	VaR 10% Sensitivity Measures
Other assets and liabilities	VaR
Deposits	VaR Credit Spread Sensitivity
Unsecured borrowings	VaR Credit Spread Sensitivity

In addition to the above, we measure the interest rate risk for all positions within our consolidated balance sheets using the EaR metric.

Credit Risk Management

Overview

Credit risk represents the potential for loss due to the default or deterioration in credit quality of a counterparty (e.g., an OTC derivatives counterparty or a borrower) or an issuer of securities or other instruments we hold. Our exposure to credit risk comes mostly from client transactions in OTC derivatives and loans and lending commitments. Credit risk also comes from cash placed with banks, securities financing transactions (i.e., resale and repurchase agreements and securities borrowing and lending activities) and customer and other receivables.

Credit Risk, which is part of our second line of defense and reports to our chief risk officer, has primary responsibility for assessing, monitoring and managing our credit risk by providing independent firmwide oversight and challenge across our global businesses. In addition, we hold other positions that give rise to credit risk (e.g., bonds and secondary bank loans). These credit risks are captured as a component of market risk measures, which are monitored and managed by Market Risk. We also enter into derivatives to manage market risk exposures. Such derivatives also give rise to credit risk, which is monitored and managed by Credit Risk.

Credit Risk Management Process

Our process for managing credit risk includes the critical components of our risk management framework described in the "Overview and Structure of Risk Management," as well as the following:

- Monitoring compliance with established credit risk limits and reporting our credit exposures and credit concentrations;
- Establishing or approving underwriting standards;
- Assessing the likelihood that a counterparty will default on its payment obligations;
- Measuring our current and potential credit exposure and losses resulting from a counterparty default;
- Using credit risk mitigants, including collateral and hedging; and
- Maximizing recovery through active workout and restructuring of claims.

We also perform credit analyses, which incorporate initial and ongoing evaluations of the capacity and willingness of a counterparty to meet its financial obligations. For substantially all of our credit exposures, the core of our process is an annual counterparty credit evaluation or more frequently if deemed necessary as a result of events or changes in circumstances. We determine an internal credit rating for the counterparty by considering the results of the credit evaluations and assumptions with respect to the nature of and outlook for the counterparty's industry and the economic environment. For collateralized loans, we also take into consideration collateral received or other credit support arrangements when determining an internal credit rating. Senior personnel, with expertise in specific industries, inspect and approve credit reviews and internal credit ratings.

Our risk assessment process may also include, where applicable, reviewing certain key metrics, including, but not limited to, delinquency status, collateral value, FICO credit scores and other risk factors.

Management's Discussion and Analysis

Our credit risk management systems capture credit exposure to individual counterparties and on an aggregate basis to counterparties and their subsidiaries. These systems also provide management with comprehensive information about our aggregate credit risk by product, internal credit rating, industry, country and region.

Risk Measures

We measure our credit risk based on the potential loss in the event of non-payment by a counterparty using current and potential exposure. For derivatives and securities financing transactions, current exposure represents the amount presently owed to us after taking into account applicable netting and collateral arrangements, while potential exposure represents our estimate of the future exposure that could arise over the life of a transaction based on market movements within a specified confidence level. Potential exposure also takes into account netting and collateral arrangements. For loans and lending commitments, the primary measure is a function of the notional amount of the position.

Stress Tests

We conduct regular stress tests to calculate the credit exposures, including potential concentrations that would result from applying shocks to counterparty credit ratings or credit risk factors (e.g., currency rates, interest rates, equity prices). These shocks cover a wide range of moderate and more extreme market movements, including shocks to multiple risk factors, consistent with the occurrence of a severe market or economic event. In the case of sovereign default, we estimate the direct impact of the default on our sovereign credit exposures, changes to our credit exposures arising from potential market moves in response to the default, and the impact of credit market deterioration on corporate borrowers and counterparties that may result from the sovereign default. Unlike potential exposure, which is calculated within a specified confidence level, stress testing does not generally assume a probability of these events occurring. We also perform firmwide stress tests. See "Overview and Structure of Risk Management" for information about firmwide stress tests.

To supplement these regular stress tests, as described above, we also conduct tailored stress tests on an ad hoc basis in response to specific events that we deem significant. We also utilize these stress tests to estimate the indirect impact of certain hypothetical events on our country exposures, such as the impact of credit market deterioration on corporate borrowers and counterparties along with the shocks to the risk factors described above. The parameters of these shocks vary based on the scenario reflected in each stress test. We review estimated losses produced by the stress tests in order to understand their magnitude, highlight potential loss concentrations, and assess and seek to mitigate our exposures, where necessary.

Limits

We use credit risk limits at various levels, as well as underwriting standards to manage the size and nature of our credit exposures. Limits for industries and countries are based on our risk appetite and are designed to allow for regular monitoring, review, escalation and management of credit risk concentrations. See "Overview and Structure of Risk Management" for information about the limit approval process.

Credit Risk is responsible for monitoring these limits, and identifying and escalating to senior management and/or the appropriate risk committee, on a timely basis, instances where limits have been exceeded.

Risk Mitigants

To reduce our credit exposures on derivatives and securities financing transactions, we may enter into netting agreements with counterparties that permit us to offset receivables and payables with such counterparties. We may also reduce credit risk with counterparties by entering into agreements that enable us to obtain collateral from them on an upfront or contingent basis and/or to terminate transactions if the counterparty's credit rating falls below a specified level. We monitor the fair value of the collateral to ensure that our credit exposures are appropriately collateralized. We seek to minimize exposures where there is a significant positive correlation between the creditworthiness of our counterparties and the market value of collateral we receive.

For loans and lending commitments, depending on the credit quality of the borrower and other characteristics of the transaction, we employ a variety of potential risk mitigants. Risk mitigants include collateral provisions, guarantees, covenants, structural seniority of the bank loan claims and, for certain lending commitments, provisions in the legal documentation that allow us to adjust loan amounts, pricing, structure and other terms as market conditions change. The type and structure of risk mitigants employed can significantly influence the degree of credit risk involved in a loan or lending commitment.

When we do not have sufficient visibility into a counterparty's financial strength or when we believe a counterparty requires support from its parent, we may obtain third-party guarantees of the counterparty's obligations. We may also seek to mitigate our credit risk using credit derivatives or participation agreements.

Management's Discussion and Analysis

Credit Exposures

As of December 2024, our aggregate credit exposure decreased slightly compared with December 2023, primarily reflecting a decrease in cash deposits with central banks, partially offset by an increase in loans and lending commitments. The percentage of our credit exposures arising from non-investment-grade counterparties (based on our internally determined public rating agency equivalents) increased compared with December 2023, primarily reflecting a decrease in investment-grade credit exposure related to cash deposits with central banks. Our credit exposures are described further below.

Cash and Cash Equivalents. Our credit exposure on cash and cash equivalents arises from our unrestricted cash, and includes both interest-bearing and non-interest-bearing deposits. We seek to mitigate the risk of credit loss, by placing substantially all of our deposits with highly rated banks and central banks.

The table below presents our credit exposure from unrestricted cash and cash equivalents, and the concentration by industry, region and internally determined public rating agency equivalents.

\$ in millions	As of December	
	2024	2023
Cash and Cash Equivalents	\$167,253	\$224,493
Industry		
Financial Institutions	10 %	9 %
Sovereign	90 %	91 %
Total	100 %	100 %
Region		
Americas	67 %	50 %
EMEA	24 %	34 %
Asia	9 %	16 %
Total	100 %	100 %
Credit Quality (Credit Rating Equivalent)		
AAA	79 %	65 %
AA	6 %	15 %
A	14 %	20 %
BBB	1 %	-
Total	100 %	100 %

The table above excludes cash segregated for regulatory and other purposes of \$14.84 billion as of December 2024 and \$17.08 billion as of December 2023.

OTC Derivatives. Our credit exposure on OTC derivatives arises primarily from our market-making activities. As a market maker, we enter into derivative transactions to provide liquidity to clients and to facilitate the transfer and hedging of their risks. We also enter into derivatives to manage market risk exposures. We manage our credit exposure on OTC derivatives using the credit risk process, measures, limits and risk mitigants described above.

We generally enter into OTC derivatives transactions under bilateral collateral arrangements that require the daily exchange of collateral. As credit risk is an essential component of fair value, we include a credit valuation adjustment (CVA) in the fair value of derivatives to reflect counterparty credit risk, as described in Note 7 to the consolidated financial statements. CVA is a function of the present value of expected exposure, the probability of counterparty default and the assumed recovery upon default.

The table below presents our net credit exposure from OTC derivatives and the concentration by industry and region.

\$ in millions	As of December	
	2024	2023
OTC derivative assets	\$41,655	\$42,950
Collateral (not netted under U.S. GAAP)	(15,821)	(14,420)
Net credit exposure	\$25,834	\$28,530
Industry		
Consumer & Retail	3 %	3 %
Diversified Industrials	10 %	11 %
Financial Institutions	19 %	21 %
Funds	28 %	20 %
Healthcare	1 %	2 %
Municipalities & Nonprofit	2 %	4 %
Natural Resources & Utilities	16 %	17 %
Sovereign	11 %	14 %
Technology, Media & Telecommunications	8 %	6 %
Other (including Special Purpose Vehicles)	2 %	2 %
Total	100 %	100 %
Region		
Americas	43 %	48 %
EMEA	49 %	45 %
Asia	8 %	7 %
Total	100 %	100 %

Our credit exposure (before any potential recoveries) to OTC derivative counterparties that defaulted during 2024 remained low, representing less than 2% of our total credit exposure from OTC derivatives.

In the table above:

- OTC derivative assets, included in the consolidated balance sheets, are reported on a net-by-counterparty basis (i.e., the net receivable for a given counterparty) when a legal right of setoff exists under an enforceable netting agreement (counterparty netting) and are accounted for at fair value, net of cash collateral received under enforceable credit support agreements (cash collateral netting).
- Collateral represents cash collateral and the fair value of securities collateral, primarily U.S. and non-U.S. government and agency obligations, received under credit support agreements, that we consider when determining credit risk, but such collateral is not eligible for netting under U.S. GAAP.

Management's Discussion and Analysis

The table below presents the distribution of our net credit exposure from OTC derivatives by tenor.

<i>\$ in millions</i>	Investment-Grade	Non-Investment-Grade / Unrated	Total
As of December 2024			
Less than 1 year	\$ 24,256	\$ 5,247	\$ 29,503
1 - 5 years	16,762	5,240	22,002
Greater than 5 years	49,709	2,622	52,331
Total	90,727	13,109	103,836
Netting	(72,077)	(5,925)	(78,002)
Net credit exposure	\$ 18,650	\$ 7,184	\$ 25,834
As of December 2023			
Less than 1 year	\$ 19,314	\$ 7,700	\$ 27,014
1 - 5 years	19,673	6,331	26,004
Greater than 5 years	51,944	3,999	55,943
Total	90,931	18,030	108,961
Netting	(72,412)	(8,019)	(80,431)
Net credit exposure	\$ 18,519	\$ 10,011	\$ 28,530

In the table above:

- Tenor is based on remaining contractual maturity for substantially all OTC derivative assets.
- Netting includes counterparty netting across tenor categories and collateral that we consider when determining credit risk (including collateral that is not eligible for netting under U.S. GAAP). Counterparty netting within the same tenor category is included within such tenor category.

The tables below present the distribution of our net credit exposure from OTC derivatives by tenor and internally determined public rating agency equivalents.

	Investment-Grade					
<i>\$ in millions</i>	AAA	AA	A	BBB	Total	
As of December 2024						
Less than 1 year	\$ 781	\$ 5,243	\$ 11,397	\$ 6,835	\$ 24,256	
1 - 5 years	855	4,301	6,689	4,917	16,762	
Greater than 5 years	2,431	13,970	17,824	15,484	49,709	
Total	4,067	23,514	35,910	27,236	90,727	
Netting	(1,753)	(20,812)	(30,083)	(19,429)	(72,077)	
Net credit exposure	\$ 2,314	\$ 2,702	\$ 5,827	\$ 7,807	\$ 18,650	
As of December 2023						
Less than 1 year	\$ 583	\$ 4,383	\$ 7,718	\$ 6,630	\$ 19,314	
1 - 5 years	1,226	4,850	6,755	6,842	19,673	
Greater than 5 years	5,963	13,417	15,507	17,057	51,944	
Total	7,772	22,650	29,980	30,529	90,931	
Netting	(5,308)	(18,364)	(25,470)	(23,270)	(72,412)	
Net credit exposure	\$ 2,464	\$ 4,286	\$ 4,510	\$ 7,259	\$ 18,519	

	Non-Investment-Grade / Unrated		
<i>\$ in millions</i>	≤ BB	Unrated	Total
As of December 2024			
Less than 1 year	\$ 5,020	\$ 227	\$ 5,247
1 - 5 years	5,201	39	5,240
Greater than 5 years	2,578	44	2,622
Total	12,799	310	13,109
Netting	(5,888)	(37)	(5,925)
Net credit exposure	\$ 6,911	\$ 273	\$ 7,184
As of December 2023			
Less than 1 year	\$ 7,274	\$ 426	\$ 7,700
1 - 5 years	6,244	87	6,331
Greater than 5 years	3,887	112	3,999
Total	17,405	625	18,030
Netting	(7,975)	(44)	(8,019)
Net credit exposure	\$ 9,430	\$ 581	\$ 10,011

Lending Activities. We manage our lending activities using the credit risk process, measures, limits and risk mitigants described above. Other lending positions, including secondary trading positions, are risk-managed as a component of market risk.

The table below presents our loans and lending commitments.

<i>\$ in millions</i>	Loans	Lending Commitments	Total
As of December 2024			
Corporate	\$ 29,972	\$ 162,529	\$ 192,501
Commercial real estate	29,789	5,016	34,805
Residential real estate	25,969	1,848	27,817
Securities-based	16,477	1,542	18,019
Other collateralized	75,107	33,536	108,643
Consumer:			
Installment	70	-	70
Credit cards	21,403	78,099	99,502
Other	2,079	872	2,951
Total	\$ 200,866	\$ 283,442	\$ 484,308
Allowance for loan losses	\$ (4,666)	\$ (674)	\$ (5,340)
As of December 2023			
Corporate	\$ 35,874	\$ 144,463	\$ 180,337
Commercial real estate	26,028	3,440	29,468
Residential real estate	25,388	1,471	26,859
Securities-based	14,621	691	15,312
Other collateralized	62,225	23,731	85,956
Consumer:			
Installment	3,298	2,250	5,548
Credit cards	19,361	70,824	90,185
Other	1,613	888	2,501
Total	\$ 188,408	\$ 247,758	\$ 436,166
Allowance for loan losses	\$ (5,050)	\$ (620)	\$ (5,670)

In the table above, lending commitments excluded \$5.69 billion as of December 2024 and \$5.81 billion as of December 2023 related to issued letters of credit which are classified as guarantees in our consolidated financial statements. See Note 18 to the consolidated financial statements for further information about guarantees.

See Note 9 to the consolidated financial statements for information about net charge-offs on wholesale and consumer loans, as well as past due and nonaccrual loans accounted for at amortized cost.

Management's Discussion and Analysis

Corporate. Corporate loans and lending commitments include term loans, revolving lines of credit, letter of credit facilities and bridge loans, and are principally used for operating and general corporate purposes, or in connection with acquisitions. Corporate loans are secured (typically by a senior lien on the assets of the borrower) or unsecured, depending on the loan purpose, the risk profile of the borrower and other factors.

The table below presents our credit exposure from corporate loans and lending commitments, and the concentration by industry, region, internally determined public rating agency equivalents and other credit metrics.

\$ in millions	Loans	Lending Commitments	Total
As of December 2024			
Corporate	\$29,972	\$162,529	\$192,501
Industry			
Consumer & Retail	9 %	13 %	12 %
Diversified Industrials	16 %	20 %	20 %
Financial Institutions	9 %	9 %	9 %
Funds	5 %	3 %	3 %
Healthcare	9 %	11 %	11 %
Natural Resources & Utilities	9 %	16 %	15 %
Real Estate	14 %	5 %	6 %
Technology, Media & Telecommunications	24 %	22 %	22 %
Other (including Special Purpose Vehicles)	5 %	1 %	2 %
Total	100 %	100 %	100 %
Region			
Americas	66 %	76 %	75 %
EMEA	26 %	22 %	22 %
Asia	8 %	2 %	3 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
AAA	-	1 %	1 %
AA	1 %	4 %	4 %
A	6 %	17 %	16 %
BBB	22 %	41 %	37 %
BB or lower	71 %	37 %	42 %
Total	100 %	100 %	100 %
As of December 2023			
Corporate	\$35,874	\$144,463	\$180,337
Industry			
Consumer & Retail	11 %	13 %	12 %
Diversified Industrials	17 %	20 %	20 %
Financial Institutions	8 %	9 %	9 %
Funds	4 %	3 %	3 %
Healthcare	9 %	11 %	10 %
Natural Resources & Utilities	8 %	18 %	16 %
Real Estate	13 %	5 %	7 %
Technology, Media & Telecommunications	25 %	20 %	21 %
Other (including Special Purpose Vehicles)	5 %	1 %	2 %
Total	100 %	100 %	100 %
Region			
Americas	63 %	77 %	74 %
EMEA	29 %	22 %	23 %
Asia	8 %	1 %	3 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
AAA	-	1 %	1 %
AA	1 %	5 %	4 %
A	5 %	20 %	17 %
BBB	20 %	41 %	37 %
BB or lower	74 %	33 %	41 %
Total	100 %	100 %	100 %

Commercial Real Estate. Commercial real estate includes originated loans and lending commitments that are directly or indirectly secured by hotels, retail stores, multifamily housing complexes and commercial and industrial properties. Commercial real estate also includes loans and lending commitments extended to clients who warehouse assets that are directly or indirectly backed by commercial real estate. In addition, commercial real estate includes loans purchased by us.

The table below presents our credit exposure from commercial real estate loans and lending commitments, and the concentration by region, internally determined public rating agency equivalents and other credit metrics.

\$ in millions	Loans	Lending Commitments	Total
As of December 2024			
Commercial Real Estate	\$29,789	\$5,016	\$34,805
Region			
Americas	78 %	83 %	78 %
EMEA	18 %	16 %	18 %
Asia	4 %	1 %	4 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	61 %	61 %	61 %
Non-investment-grade	39 %	38 %	39 %
Unrated	-	1 %	-
Total	100 %	100 %	100 %
As of December 2023			
Commercial Real Estate	\$26,028	\$3,440	\$29,468
Region			
Americas	80 %	74 %	79 %
EMEA	17 %	25 %	18 %
Asia	3 %	1 %	3 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	47 %	46 %	47 %
Non-investment-grade	52 %	54 %	52 %
Unrated	1 %	-	1 %
Total	100 %	100 %	100 %

In the table above, the concentration of loans and lending commitments by asset class as of December 2024 was 50% for warehouse and other indirect, 11% for multifamily, 7% for industrials, 5% for hospitality, 4% for office, 3% for mixed use and 20% for other asset classes. The concentration of loans and lending commitments by asset class as of December 2023 was 42% for warehouse and other indirect, 13% for multifamily, 12% for industrials, 7% for office, 7% for hospitality, 7% for mixed use and 12% for other asset classes.

Management's Discussion and Analysis

In addition, we also have credit exposure to commercial real estate loans held for securitization of \$568 million as of December 2024 and \$119 million as of December 2023. Such loans are included in trading assets in our consolidated balance sheets.

Residential Real Estate. Residential real estate loans and lending commitments are primarily extended to wealth management clients and to clients who warehouse assets that are directly or indirectly secured by residential real estate. In addition, residential real estate includes loans purchased by us.

The table below presents our credit exposure from residential real estate loans and lending commitments, and the concentration by region, internally determined public rating agency equivalents and other credit metrics.

<i>\$ in millions</i>	Loans	Lending Commitments	Total
As of December 2024			
Residential Real Estate	\$25,969	\$1,848	\$27,817
Region			
Americas	94 %	99 %	94 %
EMEA	5 %	-	5 %
Asia	1 %	1 %	1 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	39 %	38 %	39 %
Non-investment-grade	13 %	36 %	15 %
Other metrics	48 %	24 %	46 %
Unrated	-	2 %	-
Total	100 %	100 %	100 %
As of December 2023			
Residential Real Estate	\$25,388	\$1,471	\$26,859
Region			
Americas	95 %	93 %	95 %
EMEA	4 %	7 %	4 %
Asia	1 %	-	1 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	42 %	56 %	43 %
Non-investment-grade	13 %	25 %	13 %
Other metrics	45 %	16 %	43 %
Unrated	-	3 %	1 %
Total	100 %	100 %	100 %

In the table above:

- Credit exposure included loans and lending commitments of \$14.35 billion as of December 2024 and \$14.45 billion as of December 2023 which are extended to clients who warehouse assets that are directly or indirectly secured by residential real estate.
- Substantially all residential real estate loans included in the other metrics category consists of loans extended to wealth management clients. As of both December 2024 and December 2023, substantially all of such loans had a loan-to-value ratio of less than 80% and were performing in accordance with the contractual terms. Additionally, as of both December 2024 and December 2023, the vast majority of such loans had a FICO credit score of greater than 740.

In addition, we also have credit exposure to residential real estate loans held for securitization of \$10.18 billion as of December 2024 and \$7.65 billion as of December 2023. Such loans are included in trading assets in our consolidated balance sheets.

Management's Discussion and Analysis

Securities-Based. Securities-based includes loans and lending commitments that are secured by stocks, bonds, mutual funds, and exchange-traded funds. These loans and commitments are primarily extended to our wealth management clients and used for purposes other than purchasing, carrying or trading margin stocks. Securities-based loans require borrowers to post additional collateral on a daily basis (daily margin requirement) based on changes in the underlying collateral's fair value.

The table below presents our credit exposure from securities-based loans and lending commitments, and the concentration by region, internally determined public rating agency equivalents and other credit metrics.

<i>\$ in millions</i>	Loans	Lending Commitments	Total
As of December 2024			
Securities-based	\$16,477	\$1,542	\$18,019
Region			
Americas	76 %	50 %	73 %
EMEA	24 %	50 %	27 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	77 %	63 %	76 %
Non-investment-grade	2 %	-	2 %
Other metrics	21 %	37 %	22 %
Total	100 %	100 %	100 %
As of December 2023			
Securities-based	\$14,621	\$691	\$15,312
Region			
Americas	79 %	98 %	80 %
EMEA	20 %	2 %	19 %
Asia	1 %	-	1 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	75 %	25 %	73 %
Non-investment-grade	4 %	2 %	4 %
Other metrics	21 %	73 %	23 %
Total	100 %	100 %	100 %

In the table above, the vast majority of securities-based loans included in the other metrics category had a loan-to-value ratio of less than 80% and were performing in accordance with the contractual terms as of both December 2024 and December 2023.

Other Collateralized. Other collateralized includes loans and lending commitments that are backed by specific collateral (other than securities-based loans where there is a daily margin requirement and real estate loans). Such loans and lending commitments are extended to clients who warehouse assets that are directly or indirectly secured by corporate loans, consumer loans and other assets. Other collateralized also includes loans and lending commitments to investment funds (managed by third parties) that are collateralized by capital commitments of the funds' investors or assets held by the fund, as well as other secured loans and lending commitments extended to our wealth management and corporate clients.

The table below presents our credit exposure from other collateralized loans and lending commitments, and the concentration by region, internally determined public rating agency equivalents and other credit metrics.

<i>\$ in millions</i>	Loans	Lending Commitments	Total
As of December 2024			
Other Collateralized	\$75,107	\$33,536	\$108,643
Region			
Americas	86 %	89 %	87 %
EMEA	12 %	10 %	12 %
Asia	2 %	1 %	1 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	85 %	83 %	84 %
Non-investment-grade	14 %	16 %	15 %
Other metrics	1 %	-	-
Unrated	-	1 %	1 %
Total	100 %	100 %	100 %
As of December 2023			
Other Collateralized	\$62,225	\$23,731	\$85,956
Region			
Americas	89 %	94 %	90 %
EMEA	10 %	5 %	9 %
Asia	1 %	1 %	1 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	78 %	80 %	79 %
Non-investment-grade	21 %	18 %	20 %
Unrated	1 %	2 %	1 %
Total	100 %	100 %	100 %

In the table above, credit exposure included loans and lending commitments extended to clients who warehouse assets of \$31.67 billion as of December 2024 and \$21.78 billion as of December 2023.

Management's Discussion and Analysis

Credit Cards and Installment Loans. We provide credit card loans (pursuant to revolving lines of credit) to consumers in the Americas. The credit card lines are cancellable by us and therefore do not result in credit exposure. We also have installment loans to consumers in the Americas but have ceased originating such loans.

The tables below present our credit exposure from credit card funded loans and originated installment loans, and the concentration by the five most concentrated U.S. states.

<i>\$ in millions</i>	Credit Cards
As of December 2024	
Loans, gross	\$21,403
California	17 %
Texas	9 %
Florida	9 %
New York	8 %
Illinois	4 %
Other	53 %
Total	100 %

As of December 2023	
Loans, gross	\$19,361
California	17 %
Texas	9 %
Florida	8 %
New York	8 %
Illinois	4 %
Other	54 %
Total	100 %

<i>\$ in millions</i>	Installment
As of December 2024	
Loans, gross	\$70
New Jersey	44 %
Minnesota	20 %
California	4 %
Texas	3 %
New York	3 %
Other	26 %
Total	100 %

As of December 2023	
Loans, gross	\$3,298
California	8 %
Texas	8 %
Florida	7 %
New York	5 %
New Jersey	5 %
Other	67 %
Total	100 %

In addition, we had credit exposure of \$2.25 billion as of December 2023 related to our commitments to extend unsecured installment loans to consumers.

See Note 9 to the consolidated financial statements for further information about the credit quality indicators of credit card and installment loans.

Other. Other includes unsecured loans extended to wealth management clients and unsecured consumer and credit card loans purchased by us.

The table below presents our credit exposure from other loans and lending commitments, and the concentration by region, internally determined public rating agency equivalents and other credit metrics.

<i>\$ in millions</i>	Loans	Lending Commitments	Total
As of December 2024			
Other	\$2,079	\$872	\$2,951
Region			
Americas	96 %	99 %	97 %
EMEA	4 %	1 %	3 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	87 %	90 %	87 %
Non-investment-grade	8 %	10 %	9 %
Other metrics	5 %	-	4 %
Total	100 %	100 %	100 %
As of December 2023			
Other	\$1,613	\$888	\$2,501
Region			
Americas	97 %	100 %	98 %
EMEA	3 %	-	2 %
Total	100 %	100 %	100 %
Credit Quality (Credit Rating Equivalent)			
Investment-grade	61 %	87 %	70 %
Non-investment-grade	9 %	13 %	11 %
Other metrics	30 %	-	19 %
Total	100 %	100 %	100 %

In the table above, other metrics primarily includes consumer and credit card loans purchased by us. Our risk assessment process for such loans includes reviewing certain key metrics, such as expected cash flows, delinquency status and other risk factors.

In addition, we also have credit exposure to other loans held for securitization of \$1.22 billion as of both December 2024 and December 2023. Such loans are included in trading assets in our consolidated balance sheets.

Credit Hedges. We seek to mitigate the credit risk associated with our lending activities by obtaining credit protection on certain loans and lending commitments through credit default swaps, both single-name and index-based contracts.

Management's Discussion and Analysis

Securities Financing Transactions. We enter into securities financing transactions in order to, among other things, facilitate client activities, invest excess cash, acquire securities to cover short positions and finance certain activities. We bear credit risk related to resale agreements and securities borrowed only to the extent that cash advanced or the value of securities pledged or delivered to the counterparty exceeds the value of the collateral received. We also have credit exposure on repurchase agreements and securities loaned to the extent that the value of securities pledged or delivered to the counterparty for these transactions exceeds the amount of cash or collateral received. Securities collateral for these transactions primarily includes U.S. and non-U.S. government and agency obligations.

The table below presents our credit exposure from securities financing transactions and the concentration by industry, region and internally determined public rating agency equivalents.

\$ in millions	As of December	
	2024	2023
Securities Financing Transactions	\$39,299	\$40,201
Industry		
Financial Institutions	39 %	30 %
Funds	27 %	33 %
Municipalities & Nonprofit	10 %	7 %
Sovereign	24 %	29 %
Other (including Special Purpose Vehicles)	-	1 %
Total	100 %	100 %
Region		
Americas	55 %	45 %
EMEA	31 %	38 %
Asia	14 %	17 %
Total	100 %	100 %
Credit Quality (Credit Rating Equivalent)		
AAA	18 %	14 %
AA	22 %	31 %
A	42 %	38 %
BBB	9 %	7 %
BB or lower	9 %	10 %
Total	100 %	100 %

The table above reflects both netting agreements and collateral that we consider when determining credit risk.

Other Credit Exposures. We are exposed to credit risk from our receivables from brokers, dealers and clearing organizations and customers and counterparties. Receivables from brokers, dealers and clearing organizations primarily consist of initial margin placed with clearing organizations and receivables related to sales of securities which have traded, but not yet settled. These receivables generally have minimal credit risk due to the low probability of clearing organization default and the short-term nature of receivables related to securities settlements. Receivables from customers and counterparties generally consist of collateralized receivables related to customer securities transactions and generally have minimal credit risk due to both the value of the collateral received and the short-term nature of these receivables.

The table below presents our other credit exposures and the concentration by industry, region and internally determined public rating agency equivalents.

\$ in millions	As of December	
	2024	2023
Other Credit Exposures	\$48,013	\$50,820
Industry		
Financial Institutions	77 %	80 %
Funds	13 %	13 %
Other (including Special Purpose Vehicles)	10 %	7 %
Total	100 %	100 %
Region		
Americas	44 %	35 %
EMEA	41 %	54 %
Asia	15 %	11 %
Total	100 %	100 %
Credit Quality (Credit Rating Equivalent)		
AAA	4 %	2 %
AA	49 %	57 %
A	24 %	26 %
BBB	8 %	6 %
BB or lower	14 %	8 %
Unrated	1 %	1 %
Total	100 %	100 %

The table above reflects collateral that we consider when determining credit risk.

Management's Discussion and Analysis

Selected Exposures

We have credit and market exposures, as described below, that have had heightened focus given recent events and broad market concerns. Credit exposure represents the potential for loss due to the default or deterioration in credit quality of a counterparty or borrower. Market exposure represents the potential for loss in value of our long and short positions due to changes in market prices.

Country Exposures. The Russian invasion of Ukraine has negatively affected the global economy and increased macroeconomic uncertainty. Our total credit exposure to Ukrainian counterparties or borrowers was not material as of December 2024. Our total market exposure to Ukrainian issuers as of December 2024 was \$126 million, primarily to sovereign issuers. Such exposure consisted of \$134 million related to debt and \$(8) million related to credit derivatives. Our credit exposure to Russian counterparties or borrowers and our market exposure to Russian issuers were not material as of December 2024. See "Risk Factors" in Part I, Item 1A of this Form 10-K for further information about our risks related to Russia's invasion of Ukraine.

In addition, economic and/or political uncertainties in Ethiopia, Lebanon and Venezuela have led to concerns about their financial stability. Our credit exposure to counterparties or borrowers and our market exposure to issuers relating to each of these countries was not material as of December 2024.

We have a comprehensive framework to monitor, measure and assess our country exposures and to determine our risk appetite. We determine the country of risk by the location of the counterparty, issuer's assets, where they generate revenue, the country in which they are headquartered, the jurisdiction where a claim against them could be enforced, and/or the government whose policies affect their ability to repay their obligations. We monitor our credit exposure to a specific country both at the individual counterparty level, as well as at the aggregate country level. See "Stress Tests" for information about stress tests that are designed to estimate the direct and indirect impact of events involving the above countries.

Operational Risk Management

Overview

Operational risk is the risk of an adverse outcome resulting from inadequate or failed internal processes, people, systems or from external events. Our exposure to operational risk arises from routine processing errors, as well as extraordinary incidents, such as major systems failures or legal and regulatory matters, that could occur for us or our third-party vendors.

Potential types of loss events related to internal and external operational risk include:

- Execution, delivery and process management;
- Business disruption and system failures;
- Employment practices and workplace safety;
- Clients, products and business practices;
- Third-party risk, including vendor risk;
- Damage to physical assets;
- Internal fraud; and
- External fraud.

Operational Risk, which is part of our second line of defense and reports to our chief risk officer, has primary responsibility for developing and implementing a formalized framework for assessing, monitoring and managing operational risk to support firmwide oversight and challenge of our global businesses, with the goal of maintaining our exposure to operational risk at levels that are within our risk appetite.

Operational Risk Management Process

Our process for managing operational risk includes the critical components of our risk management framework described in the "Overview and Structure of Risk Management," including a comprehensive data collection process, as well as firmwide policies and procedures, for operational risk events.

We combine top-down and bottom-up approaches to manage and measure operational risk. From a top-down perspective, our senior management assesses firmwide and business-level operational risk profiles. From a bottom-up perspective, our first and second lines of defense are responsible for risk identification and risk management on a day-to-day basis, including escalating operational risks and risk events to senior management.

Management's Discussion and Analysis

We seek to maintain a comprehensive control framework designed to provide a well-controlled environment to minimize operational risks. The Firmwide Compliance and Operational Risk Committee is responsible for overseeing compliance and operational risk for our business.

Our operational risk management framework is designed to comply with the operational risk measurement rules under the Capital Framework and has evolved based on the changing needs of our businesses and regulatory guidance.

We have established policies that require all employees and consultants to report and escalate operational risk events. When operational risk events are identified, our policies require that the events be documented and analyzed to determine whether changes are required in our systems and/or processes to further mitigate the risk of future events.

We use operational risk management applications to capture, analyze, aggregate and report operational risk event data and key metrics. One of our key risk identification and control assessment tools is an operational risk and control self-assessment process, which is performed by our managers. This process consists of the identification and rating of operational risks, on a forward-looking basis, and the related controls. The results from this process are analyzed to evaluate operational risk exposures and identify businesses, activities or products with heightened levels of operational risk.

Risk Measurement

We measure our operational risk exposure using both statistical modeling and scenario analyses, which involve qualitative and quantitative assessments of internal and external operational risk event data and internal control factors for each of our businesses. Operational risk measurement also incorporates an assessment of business environment factors, including:

- Evaluations of the complexity of our business activities;
- The degree of automation in our processes;
- New activity information;
- The legal and regulatory environment; and
- Changes in the markets for our products and services, including the diversity and sophistication of our customers and counterparties.

The results from these scenario analyses are used to monitor changes in operational risk and to determine business lines that may have heightened exposure to operational risk. We also perform firmwide stress tests. See "Overview and Structure of Risk Management" for information about firmwide stress tests.

Types of Operational Risks

Increased reliance on technology and third-party relationships has resulted in increased operational risks, such as third-party risk, business resilience risk and cybersecurity risk. See "Cybersecurity Risk Management" for information about our cybersecurity risk management process. We manage third-party and business resilience risks as follows:

Third-Party Risk. Third-party risk, including vendor risk, is the risk of an adverse impact due to reliance on third parties performing services or activities on our behalf. These risks may include legal, regulatory, information security, cybersecurity, reputational, operational or other risks inherent in engaging a third party. We identify, manage and report key third-party risks and conduct due diligence across multiple risk domains, including information security and cybersecurity, resilience and additional supply chain dependencies. We evaluate whether vendors design, implement, and maintain information security controls consistent with our security policies and standards. Vendors that access and process our information on their infrastructure external to our network are required to undergo an initial risk assessment, resulting in the assignment of a vendor inherent risk rating that is determined based on a number of factors, including the type of data stored and processed by a particular vendor. Subsequently, we conduct re-certifications at a depth and frequency that is commensurate with each vendor's inherent risk rating as a component of our risk-based approach to vendor oversight. Vendors are required to agree to standard contractual provisions before receiving sensitive information from us. These provisions have specific information security control requirements, which apply to vendors that store, access, transmit or otherwise process sensitive information on our behalf. The Third-Party Risk Program monitors, reviews and reassesses third-party risks on an ongoing basis. See "Risk Factors" in Part I, Item 1A of this Form 10-K for further information about third-party risk.

Business Resilience Risk. Business resilience risk is the risk of disruption to our critical processes. We monitor threats and assess risks and seek to ensure our state of readiness in the event of a significant operational disruption to the normal operations of our critical functions or their dependencies, such as critical facilities, systems, third parties, data and/or personnel. Our resilience framework defines the fundamental principles for BCP and crisis management to ensure that critical functions can continue to operate in the event of a disruption. We seek to maintain a business continuity program that is comprehensive, consistent on a firmwide basis, and up-to-date, incorporating new information, including resilience capabilities. Our resilience assurance program encompasses testing of response and recovery strategies on a regular basis with the objective of minimizing and preventing significant operational disruptions. See "Business — Business Continuity and Information Security" in Part I, Item 1 of this Form 10-K for further information about business continuity.

Management's Discussion and Analysis

Cybersecurity Risk Management

Overview

Cybersecurity risk is the risk of compromising the confidentiality, integrity or availability of our data and systems, leading to an adverse impact to us, our reputation, our clients and/or the broader financial system. We seek to minimize the occurrence and impact of unauthorized access, disruption or use of information and/or information systems. We deploy and operate preventive and detective controls and processes to mitigate emerging and evolving information security and cybersecurity threats, including monitoring our network for known vulnerabilities and signs of unauthorized attempts to access our data and systems. There is increased information risk through diversification of our data across external service providers, including use of a variety of cloud-provided or -hosted services and applications. In addition, new AI technologies may increase the frequency and severity of cybersecurity attacks. See "Risk Factors" in Part I, Item 1A of this Form 10-K for further information about information and cybersecurity risk.

Cybersecurity Risk Management Process

Our cybersecurity risk management processes are integrated into our overall risk management processes described in the "Overview and Structure of Risk Management." We have established an Information Security and Cybersecurity Program (the Cybersecurity Program), administered by Technology Risk within Engineering, and overseen by our CISO. This program is designed to identify, assess, document and mitigate threats, govern, establish and evaluate compliance with information security mandates, adopt and apply our security control framework, and prevent, detect and respond to security incidents. The Cybersecurity Program is periodically reviewed and modified to respond to changing threats and conditions. A dedicated Operational Risk team, which reports to the chief risk officer, provides oversight and challenge of the Cybersecurity Program, independent of Technology Risk, and assesses the operating effectiveness of the program against industry standard frameworks and Board risk appetite-approved operational risk limits and thresholds.

Our process for managing cybersecurity risk includes the critical components of our risk management framework described in the "Overview and Structure of Risk Management," as well as the following:

- Training and education, to enable our people to recognize information and cybersecurity threats and respond accordingly;
- Identity and access management, including entitlement management and production access;
- Application and software security, including software change management, open source software, and backup and restoration;
- Infrastructure security, including monitoring our network for known vulnerabilities and signs of unauthorized attempts to access our data and systems;
- Mobile security, including mobile applications;
- Data security, including cryptography and encryption, database security, data erasure and media disposal;
- Cloud computing, including governance and security of cloud applications, and software-as-a-service data onboarding;
- Technology operations, including change management, incident management, capacity and resilience; and
- Third-party risk management, including vendor management and governance, and cybersecurity and business resiliency on vendor assessments.

In conjunction with third-party vendors and consultants, we perform risk assessments to gauge the performance of the Cybersecurity Program, to estimate our risk profile and to assess compliance with relevant regulatory requirements. We perform periodic assessments of control efficacy through our internal risk and control self-assessment process, as well as a variety of external technical assessments, including external penetration tests and "red team" engagements where third parties test our defenses. The results of these risk assessments, together with control performance findings, are used to establish priorities, allocate resources, and identify and improve controls. We use third parties, such as outside forensics firms, to augment our cyber incident response capabilities. We have a vendor management program that documents a risk-based framework for managing third-party vendor relationships. Information security risk management is built into our vendor management process, which covers vendor selection, onboarding, performance monitoring and risk management. See "Third-Party Risk" for further information about vendor risk.

Management's Discussion and Analysis

During 2024, we did not identify any cybersecurity threats that have materially affected or are reasonably likely to materially affect our business strategy, results of operations or financial condition. Technology Risk monitors cybersecurity threats and risks from information security and cybersecurity matters on an ongoing basis, and allocates resources and directs operations in a manner designed to mitigate those risks. For example, in response to the proliferation of AI-enabled fraud and ransomware attacks that continue to be reported globally, we have emphasized phishing and cybersecurity training for our employees and allocated additional resources for business continuity. However, despite these efforts, we cannot eliminate all cybersecurity risks or provide assurances that we have not had occurrences of undetected cybersecurity incidents.

Governance

The Board, both directly and through its committees, including its Risk Committee and Technology Risk Subcommittee, oversees our risk management policies and practices, including cybersecurity risks, and information security and cybersecurity matters. Our chief risk officer, chief information officer and chief technology officer, among others, periodically brief the Board on operational and technology risks, including cybersecurity risks, relevant to us. The Board also receives regular briefings from our CISO on a range of cybersecurity-related topics, including the status of our Cybersecurity Program, emerging cybersecurity threats, mitigation strategies and related regulatory engagements. In addition, these are topics on which various directors maintain an ongoing dialogue with our CISO, chief information officer and chief technology officer.

Our CISO is responsible for managing and implementing the Cybersecurity Program and reports directly to our chief information officer. Our CISO oversees our Technology Risk team, which assesses and manages material risks from cybersecurity threats, sets firmwide control requirements, assesses adherence to controls, and oversees incident detection and response.

In addition, we have a series of committees and steering groups that oversee the implementation of our cybersecurity risk management strategy and framework. These committees and steering groups are informed about cybersecurity incidents and risks by designated members of Technology Risk, who periodically report to these committees and steering groups about the Cybersecurity Program, including the efforts of the Technology Risk teams to prevent, detect, mitigate and remediate incidents and threats. These committees and steering groups enable formal escalation and reporting of risks, and our CISO and other members of Technology Risk provide regular briefings to senior management.

The Firmwide Technology Risk Committee is responsible for reviewing matters related to the design, development, deployment and use of technology. This committee oversees cybersecurity matters, as well as technology risk management frameworks and methodologies, and monitors their effectiveness. This committee is co-chaired by our CISO and our chief technology officer, and reports to the Firmwide Enterprise Risk Committee. To assist the Firmwide Technology Risk Committee in carrying out its mandate, the Firmwide Artificial Intelligence Risk and Controls Committee, which oversees risks associated with the use of AI, reports to the Firmwide Technology Risk Committee. See "Overview and Structure of Risk Management" for further information about this committee.

The Digital Risk Office Steering Group oversees Engineering risk decisions, monitors control performance and reviews approaches to comply with current and emerging regulation applicable to Engineering. This steering group is co-chaired by our CISO, chief technology officer and chief digital risk officer, and reports to the Firmwide Technology Risk Committee.

Our CISO, senior management within Technology Risk and Operational Risk, as well as management personnel overseeing the Cybersecurity Program, all have substantial relevant expertise in the areas of information security and cybersecurity risk management.

Management's Discussion and Analysis

Model Risk Management

Overview

Model risk is the potential for adverse consequences from decisions made based on model outputs that may be incorrect or used inappropriately. We rely on quantitative models across our business activities primarily to value certain financial assets and liabilities, to monitor and manage our risk, and to measure and monitor our regulatory capital.

Model Risk, which is part of our second line of defense, is independent of our model developers, model owners and model users, and reports to our chief risk officer, has primary responsibility for assessing, monitoring and managing our model risk by providing firmwide oversight and challenge across our global businesses.

Our model risk management framework is managed through a governance structure and risk management controls, which encompass standards designed to ensure we maintain a comprehensive model inventory, including risk assessment and classification, sound model development practices, independent review and model-specific usage controls. The Firmwide Model Risk Control Committee oversees our model risk management framework.

Model Review and Validation Process

Model Risk consists of quantitative professionals who perform an independent review, validation and approval of our models. This review includes an analysis of the model documentation, independent testing, an assessment of the appropriateness of the methodology used, and verification of compliance with model development and implementation standards.

We regularly refine and enhance our models to reflect changes in market or economic conditions and our business mix. All models are reviewed on an annual basis, and new models or significant changes to existing models and their assumptions are approved prior to implementation.

The model validation process incorporates a review of models and trade and risk parameters across a broad range of scenarios (including extreme conditions) in order to critically evaluate and verify:

- The model's conceptual soundness, including the reasonableness of model assumptions, and suitability for intended use;
- The testing strategy utilized by the model developers to ensure that the models function as intended;
- The suitability of the calculation techniques incorporated in the model;
- The model's accuracy in reflecting the characteristics of the related product and its significant risks;
- The model's consistency with models for similar products; and
- The model's sensitivity to input parameters and assumptions.

See "Critical Accounting Policies — Fair Value — Review of Valuation Models," "Liquidity Risk Management," "Market Risk Management," "Credit Risk Management" and "Operational Risk Management" for further information about our use of models within these areas.

Management's Discussion and Analysis**Other Risk Management**

In addition to the areas of risks discussed above, we also manage other risks, including capital, climate, compliance, conflicts and reputational. These areas of risks are discussed below.

Capital Risk Management

Capital risk is the risk that our capital is insufficient to support our business activities under normal and stressed market conditions or we face capital reductions or RWA increases, including from new or revised rules or changes in interpretations of existing rules, and are therefore unable to meet our internal capital targets or external regulatory capital requirements. Capital adequacy is of critical importance to us. Accordingly, we have in place a comprehensive capital management policy that provides a framework, defines objectives and establishes guidelines to maintain an appropriate level and composition of capital in both business-as-usual and stressed conditions. Our capital management framework is designed to provide us with the information needed to identify and comprehensively manage risk, and develop and apply projected stress scenarios that capture idiosyncratic vulnerabilities with a goal of holding sufficient capital to remain adequately capitalized even after experiencing a severe stress event. See "Capital Management and Regulatory Capital" for further information about our capital management process.

We have established a comprehensive governance structure to manage and oversee our day-to-day capital management activities and to ensure compliance with capital rules and related policies. Our capital management activities are overseen by the Board and its committees. The Board is responsible for approving our annual capital plan and the Risk Committee of the Board approves our capital management policy, which details the risk committees and members of senior management who are responsible for the ongoing monitoring of our capital adequacy and evaluation of current and future regulatory capital requirements, the review of the results of our capital planning and stress tests processes, and the results of our capital models. In addition, our risk committees and senior management are responsible for the review of our contingency capital plan, key capital adequacy metrics, including regulatory capital ratios, and capital plan metrics, such as the payout ratio, as well as monitoring capital targets and potential breaches of capital requirements.

Our process for managing capital risk also includes independent oversight by Risk that assesses our capital management framework, regulatory capital policies and related interpretations and escalates certain interpretations to senior management and/or the appropriate risk committee. This oversight includes, among other things, independent review and challenge of our capital ratio targets, planned capital actions and regulatory capital calculations; analysis of the related documentation; independent testing; and an assessment of the appropriateness of the calculations and their alignment with the relevant regulatory capital rules.

Climate-Related and Environmental Risk Management

We categorize climate-related and environmental risks into physical risk and transition risk. Physical risk is the risk that asset values may decline or operations may be disrupted as a result of changes in the climate, while transition risk is the risk that asset values may decline because of changes in climate policies or changes in the underlying economy due to decarbonization.

Climate-related and environmental risks manifest in different ways across our businesses. We have continued to make significant enhancements to our climate risk management framework, including steps to further integrate climate risk into our broader risk management processes. We have integrated oversight of climate-related risks into our risk management governance structure, from senior management to our Board and its committees, including the Risk and Public Responsibilities committees. The Risk Committee of the Board oversees firmwide financial and nonfinancial risks, which include climate risk, and, as part of its oversight, receives updates on our risk management approach to climate risk, including our approaches towards scenario analysis and integration into existing risk management processes. The Public Responsibilities Committee of the Board assists the Board in its oversight of our firmwide sustainability strategy and sustainability issues affecting us, including with respect to climate change. As part of its oversight, the Public Responsibilities Committee receives periodic updates on our sustainability strategy and disclosures, and also periodically reviews our governance and related policies and processes for climate and other sustainability-related matters. Senior management within Risk, in coordination with senior management in our revenue-producing units, is responsible for the development of the climate-related and environmental risk program. The objective of this program is to integrate climate-related and environmental risks into existing risk disciplines and business considerations, such as the integration of climate risk into our credit evaluation and underwriting processes for select industries.

Management's Discussion and Analysis

See "Business — Sustainability" in Part I, Item 1 and "Risk Factors" in Part I, Item 1A of this Form 10-K for information about our sustainability initiatives, including in relation to climate transition.

Compliance Risk Management

Compliance risk is the risk of legal or regulatory sanctions, material financial loss or damage to our reputation arising from our failure to comply with the requirements of applicable laws, rules and regulations, and our internal policies and procedures. Compliance risk is inherent in all activities through which we conduct our businesses. Our Compliance Risk Management Program, administered by Compliance, assesses our compliance, regulatory and reputational risk; monitors for compliance with new or amended laws, rules and regulations; designs and implements controls, policies, procedures and training; conducts independent testing; investigates, surveils and monitors for compliance risks and breaches; and leads our responses to regulatory examinations, audits and inquiries. We monitor and review business practices to assess whether they meet or exceed minimum regulatory and legal standards in all markets and jurisdictions in which we conduct business.

Conflicts Management

Conflicts of interest and our approach to dealing with them are fundamental to our client relationships, our reputation and our long-term success. The term "conflict of interest" does not have a universally accepted meaning, and conflicts can arise in many forms within a business or between businesses. The responsibility for identifying potential conflicts, as well as complying with our policies and procedures, is shared by all of our employees.

We have a multilayered approach to resolving conflicts and addressing reputational risk. Our senior management oversees policies related to conflicts resolution and, in conjunction with Conflicts Resolution, Legal and Compliance, and internal committees, formulates policies, standards and principles, and assists in making judgments regarding the appropriate resolution of particular conflicts. Resolving potential conflicts necessarily depends on the facts and circumstances of a particular situation and the application of experienced and informed judgment.

As a general matter, Conflicts Resolution reviews financing and advisory assignments in Global Banking & Markets and certain of our investing, lending and other activities. In addition, we have various transaction oversight committees that also review new underwritings, loans, investments and structured products. These groups and committees work with internal and external counsel and Compliance to evaluate and address any actual or potential conflicts. The head of Conflicts Resolution reports to our chief legal officer, who reports to our chief executive officer.

We regularly assess our policies and procedures that address conflicts of interest in an effort to conduct our business in accordance with the highest ethical standards and in compliance with all applicable laws, rules and regulations.

Reputational Risk Management

Reputational risk is the potential risk that negative publicity regarding our business practices, whether true or not, will cause a decline in our customer base, costly litigation or revenue reductions. Our reputation is critical to effectively serving our clients and fostering and maintaining long-term client relationships, and it is integral to how we are viewed by our key stakeholders.

In evaluating business opportunities, reputational risk is one of the most significant components we consider. We evaluate the ethics, suitability and transparency of transactions undertaken by us. Our employees are responsible for considering the reputational impacts that our business activities may have.

We have implemented a comprehensive program designed to monitor reputational risk. The Firmwide Reputational Risk Committee, which reports into the Firmwide Enterprise Risk Committee, is responsible for assessing reputational risks arising from business opportunities that have been identified as having potential heightened reputational risk. This committee is also responsible for overseeing client-related business standards and addressing client-related reputational risk and considers, among other things, the potential effects any business opportunities, products, transactions, new activities, acquisitions, dispositions or investments could have on our reputation.

For further information about our risk management processes, see "Overview and Structure of Risk Management" and "Risk Factors" in Part I, Item 1A of this Form 10-K.

Item 7A. Quantitative and Qualitative Disclosures About Market Risk

Quantitative and qualitative disclosures about market risk are set forth in “Management’s Discussion and Analysis of Financial Condition and Results of Operations — Risk Management” in Part II, Item 7 of this Form 10-K.

Item 8. Financial Statements and Supplementary Data

Management’s Report on Internal Control over Financial Reporting

Management of The Goldman Sachs Group, Inc., together with its consolidated subsidiaries (the firm), is responsible for establishing and maintaining adequate internal control over financial reporting. The firm’s internal control over financial reporting is a process designed under the supervision of the firm’s principal executive and principal financial officers to provide reasonable assurance regarding the reliability of financial reporting and the preparation of the firm’s financial statements for external reporting purposes in accordance with U.S. generally accepted accounting principles.

As of December 31, 2024, management conducted an assessment of the firm’s internal control over financial reporting based on the framework established in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO). Based on this assessment, management has determined that the firm’s internal control over financial reporting as of December 31, 2024 was effective.

Our internal control over financial reporting includes policies and procedures that pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect transactions and dispositions of assets; provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with U.S. generally accepted accounting principles, and that receipts and expenditures are being made only in accordance with authorizations of management and the directors of the firm; and provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of the firm’s assets that could have a material effect on our financial statements.

The firm’s internal control over financial reporting as of December 31, 2024 has been audited by PricewaterhouseCoopers LLP (PCAOB ID 238), an independent registered public accounting firm, as stated in its report appearing below, which expresses an unqualified opinion on the effectiveness of the firm’s internal control over financial reporting as of December 31, 2024.

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Shareholders of The Goldman Sachs Group, Inc.

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of The Goldman Sachs Group, Inc. and its subsidiaries (the Company) as of December 31, 2024 and 2023, and the related consolidated statements of earnings, of comprehensive income, of changes in shareholders' equity and of cash flows for each of the three years in the period ended December 31, 2024, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2024 and 2023, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2024 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2024, based on criteria established in Internal Control – Integrated Framework (2013) issued by the COSO.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Report of Independent Registered Public Accounting Firm

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matters communicated below are matters arising from the current period audit of the consolidated financial statements that were communicated or required to be communicated to the audit committee and that (i) relate to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matters below, providing separate opinions on the critical audit matters or on the accounts or disclosures to which they relate.

Valuation of Certain Level 3 Financial Instruments

As described in Notes 4 and 5 to the consolidated financial statements, as of December 31, 2024, the Company carries financial instruments at fair value, which includes \$20.4 billion of financial assets and \$25.7 billion of financial liabilities classified in level 3 of the fair value hierarchy (a portion of which relates to certain equity securities and hybrid financial instruments), as one or more inputs to the financial instrument's valuation technique are significant and unobservable. Significant unobservable inputs used by management to value certain of the level 3 financial instruments included market multiples, discount rates, and capitalization rates for equity securities; and correlation and volatility for hybrid financial instruments.

The principal considerations for our determination that performing procedures relating to the valuation of certain level 3 financial instruments is a critical audit matter are (i) the significant judgment by management when developing the fair value estimate of certain level 3 financial instruments, (ii) a high degree of auditor judgment, subjectivity, and effort in performing procedures and evaluating audit evidence related to the significant unobservable inputs related to market multiples, discount rates, and capitalization rates for equity securities; and correlation for hybrid financial instruments, and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the valuation of financial instruments, including controls over the methods and aforementioned significant unobservable inputs used in the valuation of certain level 3 financial instruments. These procedures also included, among others, testing the completeness and accuracy of data used by management, as well as (i) testing management's process for developing the fair value estimate of certain level 3 financial instruments, (ii) evaluating the appropriateness of the techniques used by management, (iii) evaluating the reasonableness of the aforementioned significant unobservable inputs, and either (iv) the use of professionals with specialized skill and knowledge to assist in evaluating (a) the appropriateness of the techniques used by management and (b) the reasonableness of the aforementioned significant unobservable inputs, or (v) the use of professionals with specialized skill and knowledge to assist in evaluating the reasonableness of management's estimate by (a) evaluating the appropriateness of the techniques used by management and (b) developing an independent estimate of fair value for a sample of certain level 3 securities using independently determined assumptions, and comparing the independent range of prices to management's estimate.

Report of Independent Registered Public Accounting Firm

Allowance for Credit Losses - Wholesale Loan Portfolio

As described in Note 9 to the consolidated financial statements, the allowance for credit losses was \$4.7 billion as of December 31, 2024, of which \$2.1 billion relates to the wholesale loan portfolio. The allowance for credit losses for wholesale loans that exhibit similar risk characteristics is measured using a modeled approach. These models determine the probability of default and loss given default based on various risk factors, including internal credit ratings, industry default and loss data, expected life and macroeconomic indicators (the most significant to the forecast model being the forecasted U.S. unemployment and U.S. GDP growth rates), the borrower's capacity to meet its financial obligations, the borrower's country of risk and industry, loan seniority and collateral type. The allowance for credit losses also includes qualitative components which allow management to reflect the uncertain nature of economic forecasting, capture uncertainty regarding model inputs, and account for model imprecision and concentration risk.

The principal considerations for our determination that performing procedures relating to the allowance for credit losses for the wholesale loan portfolio is a critical audit matter are (i) the significant judgment by management when developing the allowance for credit losses for the wholesale loan portfolio using a modeled approach, (ii) a high degree of auditor judgment and effort in performing procedures and evaluating audit evidence related to management's internal credit ratings, forecasted U.S. unemployment and U.S. GDP growth rates, and the qualitative component to reflect the uncertain nature of economic forecasting, and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to the allowance for credit losses for the wholesale loan portfolio using a modeled approach, including controls over the determination of management's internal credit ratings, forecasted U.S. unemployment and U.S. GDP growth rates, and the qualitative component to reflect the uncertain nature of economic forecasting. These procedures also included, among others, (i) testing management's process for developing the allowance for credit losses for the wholesale loan portfolio using a modeled approach (ii) testing the completeness and accuracy of data used in the estimate, and (iii) the involvement of professionals with specialized skill and knowledge to assist in evaluating (a) the appropriateness of the forecast model and methodology used by management (b) the reasonableness of management's internal credit ratings, forecasted U.S. unemployment and U.S. GDP growth rates, and (c) the reasonableness of the qualitative component to reflect the uncertain nature of economic forecasting.

/s/ PricewaterhouseCoopers LLP

New York, New York
February 26, 2025

We have served as the Company's auditor since 1922.

Consolidated Statements of Earnings

	Year Ended December		
	2024	2023	2022
<i>in millions, except per share amounts</i>			
Revenues			
Investment banking	\$ 7,738	\$ 6,218	\$ 7,360
Investment management	10,596	9,532	9,005
Commissions and fees	4,086	3,789	4,034
Market making	18,390	18,238	18,634
Other principal transactions	4,646	2,126	654
Total non-interest revenues	45,456	39,903	39,687
Interest income	81,397	68,515	29,024
Interest expense	73,341	62,164	21,346
Net interest income	8,056	6,351	7,678
Total net revenues	53,512	46,254	47,365
Provision for credit losses	1,348	1,028	2,715
Operating expenses			
Compensation and benefits	16,706	15,499	15,148
Transaction based	6,724	5,698	5,312
Market development	646	629	812
Communications and technology	1,991	1,919	1,808
Depreciation and amortization	2,392	4,856	2,455
Occupancy	973	1,053	1,026
Professional fees	1,652	1,623	1,887
Other expenses	2,683	3,210	2,716
Total operating expenses	33,767	34,487	31,164
Pre-tax earnings	18,397	10,739	13,486
Provision for taxes	4,121	2,223	2,225
Net earnings	14,276	8,516	11,261
Preferred stock dividends	751	609	497
Net earnings applicable to common shareholders	\$ 13,525	\$ 7,907	\$ 10,764
Earnings per common share			
Basic	\$ 41.07	\$ 23.05	\$ 30.42
Diluted	\$ 40.54	\$ 22.87	\$ 30.06
Average common shares			
Basic	328.1	340.8	352.1
Diluted	333.6	345.8	358.1

Consolidated Statements of Comprehensive Income

	Year Ended December		
	2024	2023	2022
<i>\$ in millions</i>			
Net earnings	\$ 14,276	\$ 8,516	\$ 11,261
Other comprehensive income/(loss) adjustments, net of tax:			
Currency translation	32	(62)	(47)
Debt valuation adjustment	(263)	(1,015)	1,403
Pension and postretirement liabilities	47	(76)	(172)
Available-for-sale securities	401	1,245	(2,126)
Cash flow hedges	(1)	-	-
Other comprehensive income/(loss)	216	92	(942)
Comprehensive income	\$ 14,492	\$ 8,608	\$ 10,319

The accompanying notes are an integral part of these consolidated financial statements.

Consolidated Balance Sheets

\$ in millions	As of December	
	2024	2023
Assets		
Cash and cash equivalents	\$ 182,092	\$ 241,577
Collateralized agreements:		
Securities purchased under agreements to resell (includes \$179,793 and \$223,543 at fair value)	180,062	223,805
Securities borrowed (includes \$46,902 and \$44,930 at fair value)	194,645	199,420
Customer and other receivables (includes \$23 and \$23 at fair value)	133,717	132,495
Trading assets (at fair value and includes \$148,417 and \$110,567 pledged as collateral)	570,555	477,510
Investments:		
Available-for-sale securities (at fair value; amortized cost of \$80,777 and \$51,001)	79,458	49,141
Held-to-maturity securities	78,713	70,310
Other investments (includes \$25,284 and \$26,626 at fair value)	26,343	27,388
Loans (net of allowance of \$4,666 and \$5,050, and includes \$5,460 and \$6,506 at fair value)	196,200	183,358
Other assets (includes \$194 and \$366 at fair value)	34,187	36,590
Total assets	\$ 1,675,972	\$ 1,641,594
Liabilities and shareholders' equity		
Deposits (includes \$44,855 and \$29,460 at fair value)	\$ 433,013	\$ 428,417
Collateralized financings:		
Securities sold under agreements to repurchase (at fair value)	274,380	249,887
Securities loaned (includes \$10,246 and \$8,934 at fair value)	56,060	60,483
Other secured financings (includes \$27,985 and \$12,554 at fair value)	28,150	13,194
Customer and other payables	223,255	230,728
Trading liabilities (at fair value)	202,555	200,355
Unsecured short-term borrowings (includes \$50,367 and \$46,127 at fair value)	69,709	75,945
Unsecured long-term borrowings (includes \$89,189 and \$86,410 at fair value)	242,634	241,877
Other liabilities (includes \$84 and \$266 at fair value)	24,220	23,803
Total liabilities	1,553,976	1,524,689
Commitments, contingencies and guarantees		
Shareholders' equity		
Preferred stock; aggregate liquidation preference of \$13,253 and \$11,203	13,253	11,203
Common stock; 927,499,667 and 922,895,030 shares issued, and 310,653,708 and 323,376,354 shares outstanding	9	9
Share-based awards	5,148	5,121
Nonvoting common stock; no shares issued and outstanding	-	-
Additional paid-in capital	61,376	60,247
Retained earnings	153,412	143,688
Accumulated other comprehensive loss	(2,702)	(2,918)
Stock held in treasury, at cost; 616,845,961 and 599,518,678 shares	(108,500)	(100,445)
Total shareholders' equity	121,996	116,905
Total liabilities and shareholders' equity	\$ 1,675,972	\$ 1,641,594

The accompanying notes are an integral part of these consolidated financial statements.

Consolidated Statements of Changes in Shareholders' Equity

\$ in millions	Year Ended December		
	2024	2023	2022
Preferred stock			
Beginning balance	\$ 11,203	\$ 10,703	\$ 10,703
Issued	4,250	1,500	-
Redeemed	(2,200)	(1,000)	-
Ending balance	13,253	11,203	10,703
Common stock			
Beginning balance	9	9	9
Issued	-	-	-
Ending balance	9	9	9
Share-based awards			
Beginning balance	5,121	5,696	4,211
Issuance and amortization of share-based awards	2,741	2,098	4,110
Delivery of common stock underlying share-based awards	(2,483)	(2,504)	(2,468)
Forfeiture of share-based awards	(231)	(169)	(157)
Ending balance	5,148	5,121	5,696
Additional paid-in capital			
Beginning balance	60,247	59,050	56,396
Delivery of common stock underlying share-based awards	2,476	2,549	2,516
Cancellation of share-based awards in satisfaction of withholding tax requirements	(1,331)	(1,345)	(1,591)
Preferred stock issuance costs	10	5	-
Issuance of common stock in connection with acquisition	-	-	1,730
Other	(26)	(12)	(1)
Ending balance	61,376	60,247	59,050
Retained earnings			
Beginning balance	143,688	139,372	131,811
Net earnings	14,276	8,516	11,261
Dividends and dividend equivalents declared on common stock and share-based awards	(3,801)	(3,591)	(3,203)
Dividends declared on preferred stock	(717)	(599)	(497)
Preferred stock redemption premium	(34)	(10)	-
Ending balance	153,412	143,688	139,372
Accumulated other comprehensive income/(loss)			
Beginning balance	(2,918)	(3,010)	(2,068)
Other comprehensive income/(loss)	216	92	(942)
Ending balance	(2,702)	(2,918)	(3,010)
Stock held in treasury, at cost			
Beginning balance	(100,445)	(94,631)	(91,136)
Repurchased	(8,000)	(5,796)	(3,500)
Reissued	33	29	20
Other	(88)	(47)	(15)
Ending balance	(108,500)	(100,445)	(94,631)
Total shareholders' equity	\$ 121,996	\$ 116,905	\$ 117,189

Consolidated Statements of Cash Flows

\$ in millions	Year Ended December		
	2024	2023	2022
Cash flows from operating activities			
Net earnings	\$ 14,276	\$ 8,516	\$ 11,261
Adjustments to reconcile net earnings to net cash provided by/(used for) operating activities:			
Depreciation and amortization	2,392	4,856	2,455
Deferred income taxes	(800)	(1,360)	(2,412)
Share-based compensation	2,663	2,085	4,083
Provision for credit losses	1,348	1,028	2,715
Changes in operating assets and liabilities:			
Customer and other receivables and payables, net	(8,367)	(28,219)	35,014
Collateralized transactions (excluding other secured financings), net	68,582	160,227	(100,996)
Trading assets	(86,991)	(163,807)	45,278
Trading liabilities	(212)	5,751	8,062
Loans held for sale, net	537	1,635	3,161
Other, net	(6,640)	(3,299)	87
Net cash provided by/(used for) operating activities	(13,212)	(12,587)	8,708
Cash flows from investing activities			
Purchase of property, leasehold improvements and equipment	(2,091)	(2,316)	(3,748)
Proceeds from sales of property, leasehold improvements and equipment	1,613	3,278	2,706
Net cash received from/(used for) business dispositions or acquisitions	3,622	487	(2,115)
Available-for-sale securities:			
Purchases	(60,852)	(9,185)	(3,753)
Proceeds from sales	19,005	3,161	2
Proceeds from paydowns and maturities	12,809	8,130	25
Held-to-maturity securities:			
Purchases	(23,018)	(26,238)	(50,099)
Proceeds from paydowns and maturities	15,549	8,604	3,671
Other investments:			
Purchases	(8,226)	(4,833)	(6,684)
Proceeds from sales, paydowns and maturities	9,010	6,953	9,263
Loans (excluding loans held for sale), net	(17,045)	(5,353)	(25,228)
Net cash used for investing activities	(49,624)	(17,312)	(75,960)
Cash flows from financing activities			
Unsecured short-term borrowings, net	934	2,050	321
Other secured financings (short-term), net	13,707	673	(2,283)
Proceeds from issuance of other secured financings (long-term)	5,243	3,047	1,800
Repayment of other secured financings (long-term), including the current portion	(2,036)	(3,570)	(3,407)
Proceeds from issuance of unsecured long-term borrowings	67,252	47,153	84,522
Repayment of unsecured long-term borrowings, including the current portion	(74,519)	(54,066)	(42,806)
Derivative contracts with a financing element, net	2,390	3,280	1,797
Deposits, net	5,894	39,723	28,074
Preferred stock redemption	(2,200)	(1,000)	-
Common stock repurchased	(8,000)	(5,796)	(3,500)
Settlement of share-based awards in satisfaction of withholding tax requirements	(1,331)	(1,345)	(1,595)
Dividends and dividend equivalents paid on common stock, preferred stock and share-based awards	(4,497)	(4,189)	(3,682)
Proceeds from issuance of preferred stock, net of issuance costs	4,239	1,496	-
Other financing, net	247	344	361
Net cash provided by financing activities	7,323	27,800	59,602
Effect of exchange rate changes on cash and cash equivalents	(3,972)	1,851	(11,561)
Net decrease in cash and cash equivalents	(59,485)	(248)	(19,211)
Cash and cash equivalents, beginning balance	241,577	241,825	261,036
Cash and cash equivalents, ending balance	\$ 182,092	\$ 241,577	\$ 241,825
Supplemental disclosures:			
Cash payments for interest, net of capitalized interest	\$ 72,623	\$ 60,026	\$ 19,022
Cash payments for income taxes, net	\$ 3,673	\$ 2,389	\$ 4,555

See Notes 9, 12 and 16 for information about non-cash activities.

Notes to Consolidated Financial Statements**Note 1.****Description of Business**

The Goldman Sachs Group, Inc. (Group Inc. or parent company), a Delaware corporation, together with its consolidated subsidiaries (collectively, the firm), is a leading global financial institution that delivers a broad range of financial services to a large and diversified client base that includes corporations, financial institutions, governments and individuals. Founded in 1869, the firm is headquartered in New York and maintains offices in all major financial centers around the world.

The firm manages and reports its activities in the following three business segments:

Global Banking & Markets

The firm provides a broad range of services to a diverse group of corporations, financial institutions, investment funds and governments. Services include strategic advisory assignments with respect to mergers and acquisitions, divestitures, corporate defense activities, restructurings and spin-offs, and equity and debt underwriting of public offerings and private placements. The firm facilitates client transactions and makes markets in fixed income, equity, currency and commodity products. In addition, the firm makes markets in and clears institutional client transactions on major stock, options and futures exchanges worldwide and provides prime financing (including securities lending, margin lending and swaps), portfolio financing and other types of equity financing (including securities-based loans to individuals). The firm also provides lending to corporate clients, including through relationship lending and acquisition financing, and secured lending, through structured credit and asset-backed lending. In addition, the firm provides commodity financing to clients through structured transactions and also provides financing through securities purchased under agreements to resell (resale agreements). The firm also makes equity and debt investments related to Global Banking & Markets activities.

Asset & Wealth Management

The firm manages assets and offers investment products across all major asset classes to a diverse set of clients, both institutional and individuals, including through a network of third-party distributors around the world. The firm also provides investing and wealth advisory solutions, including financial planning and counseling, and executing brokerage transactions for wealth management clients. The firm issues loans to wealth management clients and raises deposits through its consumer banking digital platform, *Marcus by Goldman Sachs* (Marcus), and through its private bank. The firm makes equity investments, which include investing activities related to public and private equity investments in corporate, real estate and infrastructure assets, as well as investments through consolidated investment entities (CIEs), substantially all of which are engaged in real estate investment activities. The firm also invests in debt instruments and engages in lending activities to middle-market clients, and provides financing for real estate and other assets.

Platform Solutions

The firm issues credit cards through partnership arrangements, raises deposits from Apple Card customers and provides transaction banking and other services, such as deposit-taking, payment solutions and other cash management services, for corporate and institutional clients. See Note 9 for information about the General Motors (GM) credit card program and the firm's seller financing loans.

Notes to Consolidated Financial Statements**Note 2.****Basis of Presentation**

These consolidated financial statements are prepared in accordance with accounting principles generally accepted in the United States (U.S. GAAP) and include the accounts of Group Inc. and all other entities in which the firm has a controlling financial interest. Intercompany transactions and balances have been eliminated.

All references to 2024, 2023 and 2022 refer to the firm's years ended, or the dates, as the context requires, December 31, 2024, December 31, 2023 and December 31, 2022, respectively. Any reference to a future year refers to a year ending on December 31 of that year. Certain reclassifications have been made to previously reported amounts to conform to the current presentation.

Note 3.**Significant Accounting Policies**

The firm's significant accounting policies include when and how to measure the fair value of assets and liabilities, measuring the allowance for credit losses on loans and lending commitments accounted for at amortized cost, and when to consolidate an entity. See Note 4 for policies on fair value measurements, Note 9 for policies on the allowance for credit losses, and below and Note 17 for policies on consolidation accounting. All other significant accounting policies are either described below or included in the following footnotes:

Fair Value Measurements	Note 4
Fair Value Hierarchy	Note 5
Trading Assets and Liabilities	Note 6
Derivatives and Hedging Activities	Note 7
Investments	Note 8
Loans	Note 9
Fair Value Option	Note 10
Collateralized Agreements and Financings	Note 11
Other Assets	Note 12
Deposits	Note 13
Unsecured Borrowings	Note 14
Other Liabilities	Note 15
Securitization Activities	Note 16
Variable Interest Entities	Note 17
Commitments, Contingencies and Guarantees	Note 18
Shareholders' Equity	Note 19
Regulation and Capital Adequacy	Note 20
Earnings Per Common Share	Note 21
Transactions with Affiliated Funds	Note 22
Interest Income and Interest Expense	Note 23
Income Taxes	Note 24
Business Segments	Note 25
Credit Concentrations	Note 26
Legal Proceedings	Note 27
Employee Benefit Plans	Note 28
Employee Incentive Plans	Note 29
Parent Company	Note 30

Notes to Consolidated Financial Statements**Consolidation**

The firm consolidates entities in which the firm has a controlling financial interest. The firm determines whether it has a controlling financial interest in an entity by first evaluating whether the entity is a voting interest entity or a variable interest entity (VIE).

Voting Interest Entities. Voting interest entities are entities in which (i) the total equity investment at risk is sufficient to enable the entity to finance its activities independently and (ii) the equity holders have the power to direct the activities of the entity that most significantly impact its economic performance, the obligation to absorb the losses of the entity and the right to receive the residual returns of the entity. The usual condition for a controlling financial interest in a voting interest entity is ownership of a majority voting interest. If the firm has a controlling majority voting interest in a voting interest entity, the entity is consolidated.

Variable Interest Entities. A VIE is an entity that lacks one or more of the characteristics of a voting interest entity. The firm has a controlling financial interest in a VIE when the firm has a variable interest or interests that provide it with (i) the power to direct the activities of the VIE that most significantly impact the VIE's economic performance and (ii) the obligation to absorb losses of the VIE or the right to receive benefits from the VIE that could potentially be significant to the VIE. See Note 17 for further information about VIEs.

Equity-Method Investments. When the firm does not have a controlling financial interest in an entity but can exert significant influence over the entity's operating and financial policies, the investment is generally accounted for at fair value by electing the fair value option available under U.S. GAAP. Significant influence generally exists when the firm owns 20% to 50% of the entity's common stock or in-substance common stock.

In certain cases, the firm applies the equity method of accounting to new investments that are strategic in nature or closely related to the firm's principal business activities, when the firm has a significant degree of involvement in the cash flows or operations of the investee or when cost-benefit considerations are less significant. See Note 8 for further information about equity-method investments.

Investment Funds. The firm has formed investment funds with third-party investors. These funds are typically organized as limited partnerships or limited liability companies for which the firm acts as general partner or manager. Generally, the firm does not hold a majority of the economic interests in these funds. These funds are usually voting interest entities and generally are not consolidated because third-party investors typically have rights to terminate the funds or to remove the firm as general partner or manager. Investments in these funds are generally measured at net asset value (NAV) and are included in investments. See Notes 8, 18 and 22 for further information about investments in funds.

Use of Estimates

Preparation of these consolidated financial statements requires management to make certain estimates and assumptions, the most important of which relate to fair value measurements, the allowance for credit losses on loans and lending commitments accounted for at amortized cost, accounting for goodwill and identifiable intangible assets, provisions for losses that may arise from litigation and regulatory proceedings (including governmental investigations), and accounting for income taxes. These estimates and assumptions are based on the best available information, but actual results could be materially different.

Revenue Recognition

Financial Assets and Liabilities at Fair Value. Trading assets and liabilities and certain investments are carried at fair value either under the fair value option or in accordance with other U.S. GAAP. In addition, the firm has elected to account for certain of its loans and other financial assets and liabilities at fair value by electing the fair value option. The fair value of a financial instrument is the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. Financial assets are marked to bid prices and financial liabilities are marked to offer prices. Fair value measurements do not include transaction costs. Fair value gains or losses are generally included in market making or other principal transactions. See Note 4 for further information about fair value measurements.

Notes to Consolidated Financial Statements

Revenue from Contracts with Clients. The firm recognizes revenue earned from contracts with clients for services, such as investment banking, investment management, and execution and clearing (contracts with clients), when the performance obligations related to the underlying transaction are completed.

Revenues from contracts with clients represent approximately 45% of total non-interest revenues for both 2024 and 2023 (including approximately 85% of investment banking revenues, approximately 95% of investment management revenues and all commissions and fees), and approximately 50% of total non-interest revenues for 2022 (including approximately 85% of investment banking revenues, approximately 95% of investment management revenues and all commissions and fees). See Note 25 for information about net revenues by business segment.

Investment Banking

Advisory. Fees from financial advisory assignments are recognized in revenues when the services related to the underlying transaction are completed under the terms of the assignment. Non-refundable deposits and milestone payments in connection with financial advisory assignments are recognized in revenues upon completion of the underlying transaction or when the assignment is otherwise concluded.

Expenses associated with financial advisory assignments are recognized when incurred and are included in transaction based expenses. Client reimbursements for such expenses are included in investment banking revenues.

Underwriting. Fees from underwriting assignments are recognized in revenues upon completion of the underlying transaction based on the terms of the assignment.

Expenses associated with underwriting assignments are generally deferred until the related revenue is recognized or the assignment is otherwise concluded. Such expenses are included in transaction based expenses for completed assignments.

Investment Management

The firm earns management fees and incentive fees for investment management services, which are included in investment management revenues. The firm makes payments to brokers and advisors related to the placement of the firm's investment funds (distribution fees), which are included in transaction based expenses.

Management Fees. Management fees for mutual funds are calculated as a percentage of daily NAV and are received monthly. Management fees for hedge funds are calculated as a percentage of month-end NAV and are generally received quarterly. Management fees for separately managed accounts are calculated as a percentage of either the daily or monthly NAV and are received quarterly. Management fees for private equity funds are calculated as a percentage of monthly invested capital or committed capital and are generally received quarterly, semi-annually or annually, depending on the fund. Management fees are recognized over time in the period the services are provided.

Distribution fees paid by the firm are calculated based on either a percentage of the management fee, the investment fund's NAV or the committed capital. Such fees are included in transaction based expenses.

Incentive Fees. Incentive fees are calculated as a percentage of a fund's or separately managed account's return, or excess return above a specified benchmark or other performance target. Incentive fees are generally based on investment performance over a twelve-month period or over the life of a fund. Fees that are based on performance over a twelve-month period are subject to adjustment prior to the end of the measurement period. For fees that are based on investment performance over the life of the fund, future investment underperformance may require fees previously distributed to the firm to be returned to the fund.

Incentive fees earned from a fund or separately managed account are recognized when it is probable that a significant reversal of such fees will not occur, which is generally when such fees are no longer subject to fluctuations in the market value of investments held by the fund or separately managed account. Therefore, incentive fees recognized during the period may relate to performance obligations satisfied in previous periods.

Notes to Consolidated Financial Statements**Commissions and Fees**

The firm earns substantially all commissions and fees from executing and clearing client transactions on stock, options and futures markets, as well as over-the-counter (OTC) transactions. Commissions and fees are recognized on the day the trade is executed. The firm also provides third-party research services to clients in connection with certain soft-dollar arrangements. Third-party research costs incurred by the firm in connection with such arrangements are presented net within commissions and fees.

Remaining Performance Obligations

Remaining performance obligations are services that the firm has committed to perform in the future in connection with its contracts with clients. The firm's remaining performance obligations are generally related to its financial advisory assignments and certain investment management activities. Revenues associated with remaining performance obligations relating to financial advisory assignments cannot be determined until the outcome of the transaction. For the firm's investment management activities, where fees are calculated based on the NAV of the fund or separately managed account, future revenues associated with such remaining performance obligations cannot be determined as such fees are subject to fluctuations in the market value of investments held by the fund or separately managed account.

The firm is able to determine the future revenues associated with management fees calculated based on committed capital. As of December 2024, substantially all future net revenues associated with such remaining performance obligations will be recognized through 2033. Annual revenues associated with such performance obligations average less than \$300 million through 2033.

Transfers of Financial Assets

Transfers of financial assets are accounted for as sales when the firm has relinquished control over the assets transferred. For transfers of financial assets accounted for as sales, any gains or losses are recognized in net revenues. Assets or liabilities that arise from the firm's continuing involvement with transferred financial assets are initially recognized at fair value. For transfers of financial assets that are not accounted for as sales, the assets are generally included in trading assets and the transfer is accounted for as a collateralized financing, with the related interest expense recognized over the life of the transaction. See Note 11 for further information about transfers of financial assets accounted for as collateralized financings and Note 16 for further information about transfers of financial assets accounted for as sales.

Cash and Cash Equivalents

The firm defines cash equivalents as highly liquid overnight deposits held in the ordinary course of business. Cash and cash equivalents included cash and due from banks of \$5.36 billion as of December 2024 and \$7.93 billion as of December 2023. Cash and cash equivalents also included interest-bearing deposits with banks of \$176.73 billion as of December 2024 and \$233.65 billion as of December 2023.

The firm segregates cash for regulatory and other purposes related to client activity. Cash and cash equivalents segregated for regulatory and other purposes were \$14.84 billion as of December 2024 and \$17.08 billion as of December 2023. In addition, the firm segregates securities for regulatory and other purposes related to client activity. See Note 11 for further information about segregated securities.

Customer and Other Receivables

Customer and other receivables included receivables from customers and counterparties of \$92.88 billion as of December 2024 and \$90.16 billion as of December 2023, and receivables from brokers, dealers and clearing organizations of \$40.84 billion as of December 2024 and \$42.33 billion as of December 2023. Such receivables primarily consist of customer margin loans, collateral posted in connection with certain derivative transactions, and receivables resulting from unsettled transactions.

Substantially all of these receivables are accounted for at amortized cost net of any allowance for credit losses, which generally approximates fair value. As these receivables are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these receivables been included in the firm's fair value hierarchy, substantially all would have been classified in level 2 as of both December 2024 and December 2023. See Note 10 for further information about customer and other receivables accounted for at fair value under the fair value option. Interest on customer and other receivables is recognized over the life of the transaction and included in interest income.

Customer and other receivables includes receivables from contracts with clients and contract assets. Contract assets represent the firm's right to receive consideration for services provided in connection with its contracts with clients for which collection is conditional and not merely subject to the passage of time. The firm's receivables from contracts with clients were \$4.08 billion as of December 2024 and \$3.59 billion as of December 2023. As of both December 2024 and December 2023, contract assets were not material.

Notes to Consolidated Financial Statements**Customer and Other Payables**

Customer and other payables included payables to customers and counterparties of \$217.15 billion as of December 2024 and \$220.71 billion as of December 2023, and payables to brokers, dealers and clearing organizations of \$6.11 billion as of December 2024 and \$10.02 billion as of December 2023. Such payables primarily consist of customer credit balances related to the firm's prime brokerage activities. Customer and other payables are accounted for at cost plus accrued interest, which generally approximates fair value. As these payables are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these payables been included in the firm's fair value hierarchy, substantially all would have been classified in level 2 as of both December 2024 and December 2023. Interest on customer and other payables is recognized over the life of the transaction and included in interest expense.

Offsetting Assets and Liabilities

To reduce credit exposures on derivatives and securities financing transactions, the firm may enter into master netting agreements or similar arrangements (collectively, netting agreements) with counterparties that permit it to offset receivables and payables with such counterparties. A netting agreement is a contract with a counterparty that permits net settlement of multiple transactions with that counterparty, including upon the exercise of termination rights by a non-defaulting party. Upon exercise of such termination rights, all transactions governed by the netting agreement are terminated and a net settlement amount is calculated. In addition, the firm receives and posts cash and securities collateral with respect to its derivatives and securities financing transactions, subject to the terms of the related credit support agreements or similar arrangements (collectively, credit support agreements). An enforceable credit support agreement grants the non-defaulting party exercising termination rights the right to liquidate the collateral and apply the proceeds to any amounts owed. In order to assess enforceability of the firm's right of setoff under netting and credit support agreements, the firm evaluates various factors, including applicable bankruptcy laws, local statutes and regulatory provisions in the jurisdiction of the parties to the agreement.

Derivatives are reported on a net-by-counterparty basis (i.e., the net payable or receivable for derivative assets and liabilities for a given counterparty) in the consolidated balance sheets when a legal right of setoff exists under an enforceable netting agreement. Resale agreements and securities sold under agreements to repurchase (repurchase agreements) and securities borrowed and loaned transactions with the same settlement date are presented on a net-by-counterparty basis in the consolidated balance sheets when such transactions meet certain settlement criteria and are subject to netting agreements.

In the consolidated balance sheets, derivatives are reported net of cash collateral received and posted under enforceable credit support agreements, when transacted under an enforceable netting agreement. In the consolidated balance sheets, resale and repurchase agreements, and securities borrowed and loaned, are not reported net of the related cash and securities received or posted as collateral. See Note 11 for further information about collateral received and pledged, including rights to deliver or repledge collateral. See Notes 7 and 11 for further information about offsetting assets and liabilities.

Foreign Currency Translation

Assets and liabilities denominated in non-U.S. currencies are translated at rates of exchange prevailing on the date of the consolidated balance sheets and revenues and expenses are translated at average rates of exchange for the period. Foreign currency remeasurement gains or losses on transactions in nonfunctional currencies are recognized in earnings. Gains or losses on translation of the financial statements of a non-U.S. operation, when the functional currency is other than the U.S. dollar, are included, net of hedges and taxes, in the consolidated statements of comprehensive income.

Notes to Consolidated Financial Statements**Recent Accounting Developments**

Troubled Debt Restructurings and Vintage Disclosures (ASC 326). In March 2022, the FASB issued ASU No. 2022-02, "Financial Instruments — Credit Losses (Topic 326) — Troubled Debt Restructurings and Vintage Disclosures." This ASU eliminates the recognition and measurement guidance for troubled debt restructurings and requires enhanced disclosures about loan modifications for borrowers experiencing financial difficulty. This ASU also requires enhanced disclosure for loans that have been charged off. This ASU became effective for the firm in January 2023 under a prospective approach. Adoption of this ASU did not have a material impact on the firm's consolidated financial statements.

Fair Value Measurement of Equity Securities Subject to Contractual Sale Restrictions (ASC 820). In June 2022, the FASB issued ASU No. 2022-03, "Fair Value Measurement of Equity Securities Subject to Contractual Sale Restrictions." This ASU clarifies that a contractual restriction on the sale of an equity security should not be considered in measuring its fair value. In addition, the ASU requires specific disclosures related to equity securities that are subject to contractual sale restrictions. This ASU became effective for the firm in January 2024 under a prospective approach. Adoption of this ASU did not have a material impact on the firm's consolidated financial statements.

Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method (ASC 323). In March 2023, the FASB issued ASU No. 2023-02, "Investments — Equity Method and Joint Ventures (Topic 323) — Accounting for Investments in Tax Credit Structures Using the Proportional Amortization Method." This ASU expands the proportional amortization method election currently associated with low-income housing tax credits to other qualifying tax credits and requires incremental disclosures for programs in which the proportional amortization method is elected. This ASU became effective for the firm in January 2024 under a modified retrospective approach. Adoption of this ASU did not have a material impact on the firm's consolidated financial statements.

Improvements to Reportable Segment Disclosures (ASC 280). In November 2023, the FASB issued ASU No. 2023-07, "Improvements to Reportable Segment Disclosures." This ASU requires enhanced disclosures primarily about significant segment expenses that are regularly provided to the chief operating decision maker. This ASU became effective for the firm for annual periods beginning in January 2024, and is effective for interim periods beginning in January 2025 under a retrospective approach. Since this ASU only requires additional disclosures, adoption of this ASU did not have an impact on the firm's financial condition, results of operations or cash flows. See Note 25 for further information.

Improvements to Income Tax Disclosures (ASC 740). In December 2023, the FASB issued ASU No. 2023-09, "Improvements to Income Tax Disclosures." This ASU requires incremental disclosures primarily related to the reconciliation of the statutory income tax rate to the effective income tax rate, as well as income taxes paid. This ASU is effective for the firm for annual periods beginning in January 2025 under a prospective approach with the option to apply it retrospectively. Since this ASU only requires additional disclosures, adoption of this ASU will not have an impact on the firm's financial condition, results of operations or cash flows.

Disaggregation of Income Statement Expenses (ASC 220). In November 2024, the FASB issued ASU No. 2024-03, "Disaggregation of Income Statement Expenses." This ASU requires additional disaggregation of certain expenses within the footnotes to the financial statements. This ASU is effective for the firm for annual periods beginning in January 2027, and interim periods beginning in January 2028 under a prospective approach. Early adoption and retrospective application is permitted. Since this ASU only requires additional disclosures, adoption of this ASU will not have an impact on the firm's financial condition, results of operations or cash flows.

Notes to Consolidated Financial Statements**Note 4.****Fair Value Measurements**

The fair value of a financial instrument is the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. Financial assets are marked to bid prices and financial liabilities are marked to offer prices. Fair value measurements do not include transaction costs. The firm measures certain financial assets and liabilities as a portfolio (i.e., based on its net exposure to market and/or credit risks).

The best evidence of fair value is a quoted price in an active market. If quoted prices in active markets are not available, fair value is determined by reference to prices for similar instruments, quoted prices or recent transactions in less active markets, or internally developed models that primarily use market-based or independently sourced inputs, including, but not limited to, interest rates, volatilities, equity or debt prices, foreign exchange rates, commodity prices, credit spreads and funding spreads (i.e., the spread or difference between the interest rate at which a borrower could finance a given financial instrument relative to a benchmark interest rate).

U.S. GAAP has a three-level hierarchy for disclosure of fair value measurements. This hierarchy prioritizes inputs to the valuation techniques used to measure fair value, giving the highest priority to level 1 inputs and the lowest priority to level 3 inputs. A financial instrument's level in this hierarchy is based on the lowest level of input that is significant to its fair value measurement. In evaluating the significance of a valuation input, the firm considers, among other factors, a portfolio's net risk exposure to that input. The fair value hierarchy is as follows:

Level 1. Inputs are unadjusted quoted prices in active markets to which the firm had access at the measurement date for identical, unrestricted assets or liabilities.

Level 2. Inputs to valuation techniques are observable, either directly or indirectly.

Level 3. One or more inputs to valuation techniques are significant and unobservable.

The fair values for substantially all of the firm's financial assets and liabilities are based on observable prices and inputs and are classified in levels 1 and 2 of the fair value hierarchy. Certain level 2 and level 3 financial assets and liabilities may require valuation adjustments that a market participant would require to arrive at fair value for factors, such as counterparty and the firm's credit quality, funding risk, transfer restrictions, liquidity and bid/offer spreads. Valuation adjustments are generally based on market evidence.

The table below presents financial assets and liabilities carried at fair value.

\$ in millions	As of December	
	2024	2023
Total level 1 financial assets	\$ 436,298	\$ 332,549
Total level 2 financial assets	497,514	519,130
Total level 3 financial assets	20,358	25,100
Investments in funds at NAV	2,547	3,000
Counterparty and cash collateral netting	(49,048)	(51,134)
Total financial assets at fair value	\$ 907,669	\$ 828,645
Total assets	\$ 1,675,972	\$ 1,641,594
Total level 3 financial assets divided by:		
Total assets	1.2 %	1.5 %
Total financial assets at fair value	2.2 %	3.0 %
Total level 1 financial liabilities	\$ 100,350	\$ 125,715
Total level 2 financial liabilities	611,340	523,709
Total level 3 financial liabilities	25,721	28,704
Counterparty and cash collateral netting	(37,750)	(44,135)
Total financial liabilities at fair value	\$ 699,661	\$ 633,993
Total liabilities	\$ 1,553,976	\$ 1,524,689
Total level 3 financial liabilities divided by:		
Total liabilities	1.7 %	1.9 %
Total financial liabilities at fair value	3.7 %	4.5 %

In the table above:

- Counterparty netting among positions classified in the same level is included in that level.
- Counterparty and cash collateral netting represents the impact on derivatives of netting across levels.

The table below presents a summary of level 3 financial assets.

\$ in millions	As of December	
	2024	2023
Trading assets:		
Trading cash instruments	\$ 1,213	\$ 1,791
Derivatives	4,126	5,161
Investments	14,142	17,138
Loans	683	823
Other assets	194	187
Total	\$ 20,358	\$ 25,100

Level 3 financial assets as of December 2024 decreased compared with December 2023, primarily reflecting a decrease in level 3 investments. See Note 5 for further information about level 3 financial assets (including information about unrealized gains and losses related to level 3 financial assets and transfers into and out of level 3).

Notes to Consolidated Financial Statements

The valuation techniques and nature of significant inputs used to determine the fair value of the firm's financial instruments are described below. See Note 5 for further information about significant unobservable inputs used to value level 3 financial instruments.

Valuation Techniques and Significant Inputs for Trading Cash Instruments, Investments and Loans

Level 1. Level 1 instruments include U.S. government obligations, most non-U.S. government obligations, certain agency obligations, certain corporate debt instruments, certain money market instruments and actively traded listed equities. These instruments are valued using quoted prices for identical unrestricted instruments in active markets. The firm defines active markets for equity instruments based on the average daily trading volume both in absolute terms and relative to the market capitalization for the instrument. The firm defines active markets for debt instruments based on both the average daily trading volume and the number of days with trading activity.

Level 2. Level 2 instruments include certain non-U.S. government obligations, most agency obligations, most mortgage-backed loans and securities, most corporate debt instruments, most state and municipal obligations, most money market instruments, most other debt obligations, restricted or less liquid listed equities, certain private equities, commodities and certain lending commitments.

Valuations of level 2 instruments can be verified to quoted prices, recent trading activity for identical or similar instruments, broker or dealer quotations or alternative pricing sources with reasonable levels of price transparency. Consideration is given to the nature of the quotations (e.g., indicative or executable) and the relationship of recent market activity to the prices provided from alternative pricing sources.

Valuation adjustments are typically made to level 2 instruments (i) if the instrument is subject to transfer restrictions and/or (ii) for other premiums and liquidity discounts that a market participant would require to arrive at fair value. Valuation adjustments are generally based on market evidence.

Level 3. Level 3 instruments have one or more significant valuation inputs that are not observable. Absent evidence to the contrary, level 3 instruments are initially valued at transaction price, which is considered to be the best initial estimate of fair value. Subsequently, the firm uses other methodologies to determine fair value, which vary based on the type of instrument. Valuation inputs and assumptions are changed when corroborated by substantive observable evidence, including values realized on sales.

Valuation techniques of level 3 instruments vary by instrument, but are generally based on discounted cash flow techniques. The valuation techniques and the nature of significant inputs used to determine the fair values of each type of level 3 instrument are described below:

Loans and Securities Backed by Commercial Real Estate

Loans and securities backed by commercial real estate are directly or indirectly collateralized by a single property or a portfolio of properties, and may include tranches of varying levels of subordination. Significant inputs are generally determined based on relative value analyses and include:

- Market yields implied by transactions of similar or related assets and/or current levels and changes in market indices, such as the CMBX (an index that tracks the performance of commercial mortgage bonds);
- Transaction prices in both the underlying collateral and instruments with the same or similar underlying collateral;
- A measure of expected future cash flows in a default scenario (recovery rates) implied by the value of the underlying collateral, which is mainly driven by current performance of the underlying collateral and capitalization rates. Recovery rates are expressed as a percentage of notional or face value of the instrument and reflect the benefit of credit enhancements on certain instruments; and
- Timing of expected future cash flows (duration) which, in certain cases, may incorporate the impact of any loan forbearances and other unobservable inputs (e.g., prepayment speeds).

Loans and Securities Backed by Residential Real Estate

Loans and securities backed by residential real estate are directly or indirectly collateralized by portfolios of residential real estate and may include tranches of varying levels of subordination. Significant inputs are generally determined based on relative value analyses, which incorporate comparisons to instruments with similar collateral and risk profiles. Significant inputs include:

- Market yields implied by transactions of similar or related assets;
- Transaction prices in both the underlying collateral and instruments with the same or similar underlying collateral; and
- Duration, driven by underlying loan prepayment speeds and residential property liquidation timelines.

Notes to Consolidated Financial Statements**Corporate Debt Instruments**

Corporate debt instruments includes corporate loans, debt securities and convertible debentures. Significant inputs for corporate debt instruments are generally determined based on relative value analyses, which incorporate comparisons both to prices of credit default swaps that reference the same or similar underlying instrument or entity and to other debt instruments for the same or similar issuer for which observable prices or broker quotations are available. Significant inputs include:

- Market yields implied by transactions of similar or related assets and/or current levels and trends of market indices, such as the CDX (an index that tracks the performance of corporate credit);
- Current performance and recovery assumptions and, where the firm uses credit default swaps to value the related instrument, the cost of borrowing the underlying reference obligation;
- Duration; and
- Market and transaction multiples for corporate debt instruments with convertibility or participation options.

Equity Securities

Equity securities consists of private equities. Recent third-party completed or pending transactions (e.g., merger proposals, debt restructurings, tender offers) are considered the best evidence for any change in fair value. When these are not available, the following valuation methodologies are used, as appropriate:

- Industry multiples (primarily EBITDA and revenue multiples) and public comparables;
- Transactions in similar instruments;
- Discounted cash flow techniques; and
- Third-party appraisals.

The firm also considers changes in the outlook for the relevant industry and financial performance of the issuer as compared to projected performance. Significant inputs include:

- Market and transaction multiples;
- Discount rates and capitalization rates; and
- For equity securities with debt-like features, market yields implied by transactions of similar or related assets, current performance and recovery assumptions, and duration.

Other Trading Cash Instruments, Investments and Loans

The significant inputs to the valuation of other trading cash instruments, investments and loans are generally determined based on relative value analyses, which incorporate comparisons both to prices of credit default swaps that reference the same or similar underlying instrument or entity and to other debt instruments for the same issuer for which observable prices or broker quotations are available. Significant inputs include:

- Market yields implied by transactions of similar or related assets and/or current levels and trends of market indices;
- Current performance and recovery assumptions and, where the firm uses credit default swaps to value the related instrument, the cost of borrowing the underlying reference obligation; and
- Duration.

Valuation Techniques and Significant Inputs for Derivatives

The firm's level 2 and level 3 derivatives are valued using derivative pricing models (e.g., discounted cash flow models, correlation models and models that incorporate option pricing methodologies, such as Monte Carlo simulations). Price transparency of derivatives can generally be characterized by product type, as described below.

- **Interest Rate.** In general, the key inputs used to value interest rate derivatives are transparent, even for most long-dated contracts. Interest rate swaps and options denominated in the currencies of leading industrialized nations are characterized by high trading volumes and tight bid/offer spreads. Interest rate derivatives that reference indices, such as an inflation index, or the shape of the yield curve (e.g., 10-year swap rate vs. 2-year swap rate) are more complex, but the key inputs are generally observable.
- **Credit.** Price transparency for credit default swaps, including both single names and baskets of credits, varies by market and underlying reference entity or obligation. Credit default swaps that reference indices, large corporates and major sovereigns generally exhibit the most price transparency. For credit default swaps with other underliers, price transparency varies based on credit rating, the cost of borrowing the underlying reference obligations, and the availability of the underlying reference obligations for delivery upon the default of the issuer. Credit default swaps that reference loans, asset-backed securities and emerging market debt instruments tend to have less price transparency than those that reference corporate bonds. In addition, more complex credit derivatives, such as those sensitive to the correlation between two or more underlying reference obligations, generally have less price transparency.

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- **Currency.** Prices for currency derivatives based on the exchange rates of leading industrialized nations, including those with longer tenors, are generally transparent. The primary difference between the price transparency of developed and emerging market currency derivatives is that emerging markets tend to be only observable for contracts with shorter tenors.
- **Commodity.** Commodity derivatives include transactions referenced to energy (e.g., oil, natural gas and electricity), metals (e.g., precious and base) and soft commodities (e.g., agricultural). Price transparency varies based on the underlying commodity, delivery location, tenor and product quality (e.g., diesel fuel compared to unleaded gasoline). In general, price transparency for commodity derivatives is greater for contracts with shorter tenors and contracts that are more closely aligned with major and/or benchmark commodity indices.
- **Equity.** Price transparency for equity derivatives varies by market and underlier. Options on indices and the common stock of corporates included in major equity indices exhibit the most price transparency. Equity derivatives generally have observable market prices, except for contracts with long tenors or reference prices that differ significantly from current market prices. More complex equity derivatives, such as those sensitive to the correlation between two or more individual stocks, generally have less price transparency.

Liquidity is essential to the observability of all product types. If transaction volumes decline, previously transparent prices and other inputs may become unobservable. Conversely, even highly structured products may at times have trading volumes large enough to provide observability of prices and other inputs.

Level 1. Level 1 derivatives include short-term contracts for future delivery of securities when the underlying security is a level 1 instrument, and exchange-traded derivatives if they are actively traded and are valued at their quoted market price.

Level 2. Level 2 derivatives include OTC derivatives for which all significant valuation inputs are corroborated by market evidence and exchange-traded derivatives that are not actively traded and/or that are valued using models that calibrate to market-clearing levels of OTC derivatives.

The selection of a particular model to value a derivative depends on the contractual terms of and specific risks inherent in the instrument, as well as the availability of pricing information in the market. For derivatives that trade in liquid markets, model selection does not involve significant management judgment because outputs of models can be calibrated to market-clearing levels.

Valuation models require a variety of inputs, such as contractual terms, market prices, yield curves, discount rates (including those derived from interest rates on collateral received and posted as specified in credit support agreements for collateralized derivatives), credit curves, measures of volatility, prepayment rates, loss severity rates and correlations of such inputs. Significant inputs to the valuations of level 2 derivatives can be verified to market transactions, broker or dealer quotations or other alternative pricing sources with reasonable levels of price transparency. Consideration is given to the nature of the quotations (e.g., indicative or executable) and the relationship of recent market activity to the prices provided from alternative pricing sources.

Level 3. Level 3 derivatives are valued using models which utilize observable level 1 and/or level 2 inputs, as well as unobservable level 3 inputs. The significant unobservable inputs used to value the firm's level 3 derivatives are described below.

- For level 3 interest rate and currency derivatives, significant unobservable inputs include correlations of certain currencies and interest rates (e.g., the correlation between Euro inflation and Euro interest rates) and specific interest rate and currency volatilities.
- For level 3 credit derivatives, significant unobservable inputs include illiquid credit spreads and upfront credit points, which are unique to specific reference obligations and reference entities, and recovery rates.
- For level 3 commodity derivatives, significant unobservable inputs include volatilities for options with strike prices that differ significantly from current market prices and prices or spreads for certain products for which the product quality or physical location of the commodity is not aligned with benchmark indices.
- For level 3 equity derivatives, significant unobservable inputs generally include equity volatility inputs for options that are long-dated and/or have strike prices that differ significantly from current market prices. In addition, the valuation of certain structured trades requires the use of level 3 correlation inputs, such as the correlation of the price performance of two or more individual stocks or the correlation of the price performance for a basket of stocks to another asset class, such as commodities.

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Subsequent to the initial valuation of a level 3 derivative, the firm updates the level 1 and level 2 inputs to reflect observable market changes and any resulting gains and losses are classified in level 3. Level 3 inputs are changed when corroborated by evidence, such as similar market transactions, third-party pricing services and/or broker or dealer quotations or other empirical market data. In circumstances where the firm cannot verify the model value by reference to market transactions, it is possible that a different valuation model could produce a materially different estimate of fair value. See Note 5 for further information about significant unobservable inputs used in the valuation of level 3 derivatives.

Valuation Adjustments. Valuation adjustments are integral to determining the fair value of derivative portfolios and are used to adjust the mid-market valuations produced by derivative pricing models to the exit price valuation. These adjustments incorporate bid/offer spreads, the cost of liquidity, and credit and funding valuation adjustments, which account for the credit and funding risk inherent in the uncollateralized portion of derivative portfolios. The firm also makes funding valuation adjustments to collateralized derivatives where the terms of the agreement do not permit the firm to deliver or repledge collateral received. Market-based inputs are generally used when calibrating valuation adjustments to market-clearing levels.

In addition, for derivatives that include significant unobservable inputs, the firm makes model or exit price adjustments to account for the valuation uncertainty present in the transaction.

Valuation Techniques and Significant Inputs for Other Financial Assets and Liabilities at Fair Value

In addition to trading cash instruments, derivatives, and certain investments and loans, the firm accounts for certain of its other financial assets and liabilities at fair value under the fair value option. Such instruments include repurchase agreements and substantially all resale agreements; certain securities borrowed and loaned transactions; certain customer and other receivables, including certain margin loans; certain time deposits, including structured certificates of deposit, which are hybrid financial instruments; substantially all other secured financings, including structured financing arrangements and transfers of assets accounted for as financings; certain unsecured short- and long-term borrowings, substantially all of which are hybrid financial instruments; and certain other assets and liabilities. These instruments are generally valued based on discounted cash flow techniques, which incorporate inputs with reasonable levels of price transparency, and are generally classified in level 2 because the inputs are observable. Valuation adjustments may be made for liquidity and for counterparty and the firm's credit quality. The significant inputs used to value the firm's other financial assets and liabilities are described below.

Resale and Repurchase Agreements and Securities Borrowed and Loaned. The significant inputs to the valuation of resale and repurchase agreements and securities borrowed and loaned are funding spreads, the amount and timing of expected future cash flows and interest rates.

Customer and Other Receivables. The significant inputs to the valuation of receivables are interest rates, the amount and timing of expected future cash flows and funding spreads.

Deposits. The significant inputs to the valuation of time deposits are interest rates and the amount and timing of future cash flows. The inputs used to value the embedded derivative component of hybrid financial instruments are consistent with the inputs used to value the firm's other derivative instruments described above. See Note 7 for further information about derivatives and Note 13 for further information about deposits.

Other Secured Financings. The significant inputs to the valuation of other secured financings are the amount and timing of expected future cash flows, interest rates, funding spreads and the fair value of the collateral delivered by the firm (determined using the amount and timing of expected future cash flows, market prices, market yields and recovery assumptions). See Note 11 for further information about other secured financings.

Unsecured Short- and Long-Term Borrowings. The significant inputs to the valuation of unsecured short- and long-term borrowings are the amount and timing of expected future cash flows, interest rates, the credit spreads of the firm and commodity prices for prepaid commodity transactions. The inputs used to value the embedded derivative component of hybrid financial instruments are consistent with the inputs used to value the firm's other derivative instruments described above. See Note 7 for further information about derivatives and Note 14 for further information about borrowings.

Other Assets and Liabilities. The significant inputs to the valuation of other assets and liabilities are the amount and timing of expected future cash flows, interest rates, market yields, volatility and correlation inputs. The inputs used to value the embedded derivative component of hybrid financial instruments are consistent with the inputs used to value the firm's other derivative instruments described above. See Note 7 for further information about derivatives.

Notes to Consolidated Financial Statements**Note 5.****Fair Value Hierarchy**

Financial assets and liabilities at fair value includes trading cash instruments, derivatives, and certain investments, loans and other financial assets and liabilities at fair value.

Trading Cash Instruments

Fair Value by Level. The table below presents trading cash instruments by level within the fair value hierarchy.

<i>\$ in millions</i>	Level 1	Level 2	Level 3	Total
As of December 2024				
Assets				
Government and agency obligations:				
U.S.	\$ 169,121	\$ 66,958	\$ -	\$ 236,079
Non-U.S.	44,427	25,071	41	69,539
Loans and securities backed by:				
Commercial real estate	-	1,450	38	1,488
Residential real estate	-	11,364	57	11,421
Corporate debt instruments	172	46,739	728	47,639
State and municipal obligations	-	529	-	529
Other debt obligations	1	2,236	95	2,332
Equity securities	141,821	1,143	242	143,206
Commodities	-	10,971	12	10,983
Total	\$ 355,542	\$ 166,461	\$ 1,213	\$ 523,216

Liabilities

Government and agency obligations:				
U.S.	\$ (21,181)	\$ (52)	\$ -	\$ (21,233)
Non-U.S.	(37,466)	(3,283)	(3)	(40,752)
Loans and securities backed by:				
Commercial real estate	-	(32)	(1)	(33)
Residential real estate	-	(20)	-	(20)
Corporate debt instruments	(75)	(23,755)	(52)	(23,882)
Other debt obligations	-	(72)	-	(72)
Equity securities	(41,528)	(12)	(19)	(41,559)
Commodities	-	(24)	-	(24)
Total	\$ (100,250)	\$ (27,250)	\$ (75)	\$ (127,575)

As of December 2023**Assets**

Government and agency obligations:				
U.S.	\$ 85,190	\$ 58,862	\$ -	\$ 144,052
Non-U.S.	61,981	25,702	91	87,774
Loans and securities backed by:				
Commercial real estate	-	916	45	961
Residential real estate	-	8,940	99	9,039
Corporate debt instruments	177	37,883	1,415	39,475
State and municipal obligations	-	371	-	371
Other debt obligations	80	2,086	37	2,203
Equity securities	135,032	1,739	103	136,874
Commodities	-	5,640	1	5,641
Total	\$ 282,460	\$ 142,139	\$ 1,791	\$ 426,390

Liabilities

Government and agency obligations:				
U.S.	\$ (26,400)	\$ (32)	\$ -	\$ (26,432)
Non-U.S.	(50,825)	(2,343)	-	(53,168)
Loans and securities backed by:				
Commercial real estate	-	(27)	-	(27)
Residential real estate	-	(5)	-	(5)
Corporate debt instruments	(124)	(15,317)	(70)	(15,511)
Equity securities	(48,347)	(37)	(8)	(48,392)
Commodities	-	(66)	-	(66)
Total	\$ (125,696)	\$ (17,827)	\$ (78)	\$ (143,601)

Trading cash instruments consists of instruments held in connection with the firm's market-making or risk management activities. These instruments are carried at fair value and the related fair value gains and losses are recognized in the consolidated statements of earnings.

In the table above:

- Assets are shown as positive amounts and liabilities are shown as negative amounts.
- Corporate debt instruments includes corporate loans, debt securities, convertible debentures, prepaid commodity transactions and transfers of assets accounted for as secured loans rather than purchases.
- Other debt obligations includes other asset-backed securities and money market instruments.
- Equity securities includes public equities and exchange-traded funds.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of trading cash instruments.

Significant Unobservable Inputs. The table below presents the amount of level 3 trading cash instrument assets, and ranges and weighted averages of significant unobservable inputs used to value such trading cash instrument assets.

<i>\$ in millions</i>	As of December 2024		As of December 2023	
	Amount or Range	Weighted Average	Amount or Range	Weighted Average
Loans and securities backed by real estate				
Level 3 assets	\$ 95		\$ 144	
Yield	7.2% to 24.3%	11.5 %	3.8% to 26.1%	12.8 %
Recovery rate	23.3% to 69.2%	50.9 %	35.5% to 76.0%	44.6 %
Duration (years)	1.9 to 12.1	3.7	0.3 to 15.3	5.2
Corporate debt instruments				
Level 3 assets	\$ 728		\$ 1,415	
Yield	3.0% to 35.9%	13.0 %	2.8% to 40.0%	9.3 %
Recovery rate	6.8% to 69.0%	53.6 %	7.3% to 65.0%	39.4 %
Duration (years)	1.6 to 3.3	2.3	0.9 to 11.3	3.4
Other				
Level 3 assets	\$ 390		\$ 232	
Yield	4.7% to 37.9%	15.2 %	3.6% to 31.3%	14.6 %
Multiples	N/A	N/A	0.7x to 4.5x	3.9x
Duration (years)	1.5 to 3.5	2.4	2.3 to 6.4	4.1

Notes to Consolidated Financial Statements

In the table above:

- Other includes government and agency obligations, other debt obligations, equity securities and commodities.
- Ranges represent the significant unobservable inputs that were used in the valuation of each type of trading cash instrument.
- Weighted averages are calculated by weighting each input by the relative fair value of the trading cash instruments.
- The ranges and weighted averages of these inputs are not representative of the appropriate inputs to use when calculating the fair value of any one trading cash instrument. For example, the highest recovery rate for corporate debt instruments is appropriate for valuing a specific corporate debt instrument, but may not be appropriate for valuing any other corporate debt instrument. Accordingly, the ranges of inputs do not represent uncertainty in, or possible ranges of, fair value measurements of level 3 trading cash instruments.
- Increases in yield or duration used in the valuation of level 3 trading cash instruments would have resulted in a lower fair value measurement, while increases in recovery rate or multiples would have resulted in a higher fair value measurement as of both December 2024 and December 2023. Due to the distinctive nature of each level 3 trading cash instrument, the interrelationship of inputs is not necessarily uniform within each product type.
- Trading cash instruments are valued using discounted cash flows.
- The significant unobservable inputs for multiples related to other trading cash instruments as of December 2024 did not have a range (and there was no weighted average) as it related to a single position. Therefore, such unobservable inputs are not included in the table above.

Level 3 Rollforward. The table below presents a summary of the changes in fair value for level 3 trading cash instruments.

\$ in millions	Year Ended December	
	2024	2023
Assets		
Beginning balance	\$ 1,791	\$ 1,734
Net realized gains/(losses)	161	154
Net unrealized gains/(losses)	7	(32)
Purchases	512	825
Sales	(618)	(515)
Settlements	(382)	(286)
Transfers into level 3	222	167
Transfers out of level 3	(480)	(256)
Ending balance	\$ 1,213	\$ 1,791
Liabilities		
Beginning balance	\$ (78)	\$ (64)
Net realized gains/(losses)	4	3
Net unrealized gains/(losses)	(25)	(66)
Purchases	57	90
Sales	(43)	(77)
Settlements	1	1
Transfers into level 3	(9)	(3)
Transfers out of level 3	18	38
Ending balance	\$ (75)	\$ (78)

In the table above:

- Changes in fair value are presented for all trading cash instruments that are classified in level 3 as of the end of the period.
- Net unrealized gains/(losses) relates to trading cash instruments that were still held at period-end.
- Transfers between levels of the fair value hierarchy are reported at the beginning of the reporting period in which they occur. If a trading cash instrument was transferred to level 3 during a reporting period, its entire gain or loss for the period is classified in level 3.
- For level 3 trading cash instrument assets, increases are shown as positive amounts, while decreases are shown as negative amounts. For level 3 trading cash instrument liabilities, increases are shown as negative amounts, while decreases are shown as positive amounts.
- Level 3 trading cash instruments are frequently economically hedged with level 1 and level 2 trading cash instruments and/or level 1, level 2 or level 3 derivatives. Accordingly, gains or losses that are classified in level 3 can be partially offset by gains or losses attributable to level 1 or level 2 trading cash instruments and/or level 1, level 2 or level 3 derivatives. As a result, gains or losses included in the level 3 rollforward below do not necessarily represent the overall impact on the firm's results of operations, liquidity or capital resources.

Notes to Consolidated Financial Statements

The table below presents information, by product type, for assets included in the summary table above.

\$ in millions	Year Ended December	
	2024	2023
Loans and securities backed by real estate		
Beginning balance	\$ 144	\$ 154
Net realized gains/(losses)	18	10
Net unrealized gains/(losses)	(11)	5
Purchases	37	28
Sales	(33)	(57)
Settlements	(28)	(16)
Transfers into level 3	6	31
Transfers out of level 3	(38)	(11)
Ending balance	\$ 95	\$ 144
Corporate debt instruments		
Beginning balance	\$ 1,415	\$ 1,238
Net realized gains/(losses)	87	56
Net unrealized gains/(losses)	2	(26)
Purchases	253	656
Sales	(389)	(277)
Settlements	(327)	(201)
Transfers into level 3	116	98
Transfers out of level 3	(429)	(129)
Ending balance	\$ 728	\$ 1,415
Other		
Beginning balance	\$ 232	\$ 342
Net realized gains/(losses)	56	88
Net unrealized gains/(losses)	16	(11)
Purchases	222	141
Sales	(196)	(181)
Settlements	(27)	(69)
Transfers into level 3	100	38
Transfers out of level 3	(13)	(116)
Ending balance	\$ 390	\$ 232

In the table above, other includes government and agency obligations, other debt obligations, equity securities and commodities.

Level 3 Rollforward Commentary for the Year Ended December 2024. The net realized and unrealized gains on level 3 trading cash instrument assets of \$168 million (reflecting \$161 million of net realized gains and \$7 million of net unrealized gains) for 2024 included gains of \$100 million reported in market making and \$68 million reported in interest income.

The drivers of net unrealized gains on level 3 trading cash instrument assets for 2024 were not material.

Transfers into level 3 trading cash instrument assets during 2024 primarily reflected transfers of certain corporate debt instruments and certain other debt obligations (included in other cash instruments) from level 2 (in each case, principally due to reduced price transparency as a result of a lack of market evidence, including fewer market transactions in these instruments).

Transfers out of level 3 trading cash instrument assets during 2024 primarily reflected transfers of certain corporate debt instruments to level 2 (principally due to increased price transparency as a result of market evidence, including market transactions in these instruments).

Level 3 Rollforward Commentary for the Year Ended December 2023. The net realized and unrealized gains on level 3 trading cash instrument assets of \$122 million (reflecting \$154 million of net realized gains and \$32 million of net unrealized losses) for 2023 included gains of \$60 million reported in market making and \$62 million reported in interest income.

The drivers of net unrealized losses on level 3 trading cash instrument assets for 2023 were not material.

The drivers of transfers into level 3 trading cash instrument assets during 2023 were not material.

Transfers out of level 3 trading cash instrument assets during 2023 primarily reflected transfers of certain corporate debt instruments and other debt obligations (included in other cash instruments) to level 2 (in each case, principally due to increased price transparency as a result of market evidence, including market transactions in these instruments).

Notes to Consolidated Financial Statements**Derivatives**

Fair Value by Level. The table below presents derivatives on a gross basis by level and product type, as well as the impact of netting.

<i>\$ in millions</i>	Level 1	Level 2	Level 3	Total
As of December 2024				
Assets				
Interest rates	\$ 102	\$ 156,054	\$ 580	\$ 156,736
Credit	-	9,249	2,188	11,437
Currencies	-	114,500	174	114,674
Commodities	-	12,134	1,192	13,326
Equities	136	79,301	742	80,179
Gross fair value	238	371,238	4,876	376,352
Counterparty netting in levels	-	(279,215)	(750)	(279,965)
Subtotal	\$ 238	\$ 92,023	\$ 4,126	\$ 96,387
Cross-level counterparty netting				(947)
Cash collateral netting				(48,101)
Net fair value			\$	47,339
Liabilities				
Interest rates	\$ (92)	\$ (124,235)	\$ (692)	\$ (125,019)
Credit	-	(9,060)	(970)	(10,030)
Currencies	-	(114,488)	(127)	(114,615)
Commodities	-	(14,111)	(414)	(14,525)
Equities	(8)	(126,650)	(1,848)	(128,506)
Gross fair value	(100)	(388,544)	(4,051)	(392,695)
Counterparty netting in levels	-	279,215	750	279,965
Subtotal	\$ (100)	\$ (109,329)	\$ (3,301)	\$ (112,730)
Cross-level counterparty netting				947
Cash collateral netting				36,803
Net fair value			\$	(74,980)
As of December 2023				
Assets				
Interest rates	\$ 15	\$ 241,850	\$ 758	\$ 242,623
Credit	-	9,964	2,861	12,825
Currencies	-	89,694	210	89,904
Commodities	-	15,393	1,449	16,842
Equities	2	59,220	816	60,038
Gross fair value	17	416,121	6,094	422,232
Counterparty netting in levels	-	(319,045)	(933)	(319,978)
Subtotal	\$ 17	\$ 97,076	\$ 5,161	\$ 102,254
Cross-level counterparty netting				(1,411)
Cash collateral netting				(49,723)
Net fair value			\$	51,120
Liabilities				
Interest rates	\$ (14)	\$ (213,861)	\$ (1,197)	\$ (215,072)
Credit	-	(8,923)	(1,211)	(10,134)
Currencies	-	(97,436)	(168)	(97,604)
Commodities	-	(17,122)	(821)	(17,943)
Equities	(5)	(78,222)	(1,887)	(80,114)
Gross fair value	(19)	(415,564)	(5,284)	(420,867)
Counterparty netting in levels	-	319,045	933	319,978
Subtotal	\$ (19)	\$ (96,519)	\$ (4,351)	\$ (100,889)
Cross-level counterparty netting				1,411
Cash collateral netting				42,724
Net fair value			\$	(56,754)

In the table above:

- Gross fair values exclude the effects of both counterparty netting and collateral netting, and therefore are not representative of the firm's exposure.
- Counterparty netting is reflected in each level to the extent that receivable and payable balances are netted within the same level and is included in counterparty netting in levels. Where the counterparty netting is across levels, the netting is included in cross-level counterparty netting.
- Assets are shown as positive amounts and liabilities are shown as negative amounts.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of derivatives.

Significant Unobservable Inputs. The table below presents the amount of level 3 derivative assets (liabilities), and ranges, averages and medians of significant unobservable inputs used to value such derivatives.

<i>\$ in millions, except inputs</i>	As of December 2024		As of December 2023	
	Amount or Range	Average/ Median	Amount or Range	Average/ Median
Interest rates, net	\$ (112)		\$ (439)	
Correlation	(10)% to 95%	60%/72%	(10)% to 75%	60%/66%
Volatility (bps)	31 to 101	63/59	31 to 101	56/49
Credit, net	\$ 1,218		\$ 1,650	
Credit spreads (bps)	8 to 1,328	134/91	3 to 1,750	130/85
Upfront credit points	(10) to 100	24/14	0 to 100	26/15
Recovery rates	20% to 70%	46%/50%	20% to 70%	43%/40%
Currencies, net	\$ 47		\$ 42	
Correlation	20% to 68%	34%/23%	20% to 90%	41%/43%
Volatility	17% to 17%	17%/17%	15% to 16%	16%/16%
Commodities, net	\$ 778		\$ 628	
Volatility	21% to 120%	37%/33%	23% to 98%	42%/39%
Natural gas spread	\$(2.82) to \$3.76	\$(0.14)/\$(0.16)	\$(1.39) to \$3.06	\$(0.32)/\$(0.35)
Oil spread	\$(6.42) to \$22.10	\$1.77/\$(3.80)	\$(5.39) to \$31.69	\$15.39/\$19.35
Electricity price	\$1.89 to \$587.75	\$52.18/\$32.68	\$2.72 to \$1,088.00	\$48.15/\$35.16
Equities, net	\$ (1,106)		\$ (1,071)	
Correlation	(75)% to 100%	56%/52%	(70)% to 100%	65%/71%
Volatility	2% to 101%	15%/12%	1% to 106%	14%/13%

In the table above:

- Assets are shown as positive amounts and liabilities are shown as negative amounts.
- Ranges represent the significant unobservable inputs that were used in the valuation of each type of derivative.

Notes to Consolidated Financial Statements

- **Averages** represent the arithmetic average of the inputs and are not weighted by the relative fair value or notional amount of the respective financial instruments. An average greater than the median indicates that the majority of inputs are below the average. For example, the difference between the average and the median for credit spreads indicates that the majority of the inputs fall in the lower end of the range.
- The ranges, averages and medians of these inputs are not representative of the appropriate inputs to use when calculating the fair value of any one derivative. For example, the highest correlation for interest rate derivatives is appropriate for valuing a specific interest rate derivative but may not be appropriate for valuing any other interest rate derivative. Accordingly, the ranges of inputs do not represent uncertainty in, or possible ranges of, fair value measurements of level 3 derivatives.
- Interest rates, currencies and equities derivatives are valued using option pricing models, credit derivatives are valued using option pricing, correlation and discounted cash flow models, and commodities derivatives are valued using option pricing and discounted cash flow models.
- The fair value of any one instrument may be determined using multiple valuation techniques. For example, option pricing models and discounted cash flow models are typically used together to determine fair value. Therefore, the level 3 balance encompasses both of these techniques.
- Correlation within currencies and equities includes cross-product type correlation.
- Natural gas spread represents the spread per million British thermal units of natural gas.
- Oil spread represents the spread per barrel of oil and refined products.
- Electricity price represents the price per megawatt hour of electricity.

Range of Significant Unobservable Inputs. The following provides information about the ranges of significant unobservable inputs used to value the firm's level 3 derivative instruments:

- **Correlation.** Ranges for correlation cover a variety of underliers both within one product type (e.g., equity index and equity single stock names) and across product types (e.g., correlation of an interest rate and a currency), as well as across regions. Generally, cross-product type correlation inputs are used to value more complex instruments and are lower than correlation inputs on assets within the same derivative product type.
 - **Volatility.** Ranges for volatility cover numerous underliers across a variety of markets, maturities and strike prices. For example, volatility of equity indices is generally lower than volatility of single stocks.
 - **Credit spreads, upfront credit points and recovery rates.** The ranges for credit spreads, upfront credit points and recovery rates cover a variety of underliers (index and single names), regions, sectors, maturities and credit qualities (high-yield and investment-grade). The broad range of this population gives rise to the width of the ranges of significant unobservable inputs.
 - **Commodity prices and spreads.** The ranges for commodity prices and spreads cover variability in products, maturities and delivery locations.
- Sensitivity of Fair Value Measurement to Changes in Significant Unobservable Inputs.** The following is a description of the directional sensitivity of the firm's level 3 fair value measurements to changes in significant unobservable inputs, in isolation, as of each period-end:
- **Correlation.** In general, for contracts where the holder benefits from the convergence of the underlying asset or index prices (e.g., interest rates, credit spreads, foreign exchange rates, inflation rates and equity prices), an increase in correlation results in a higher fair value measurement.
 - **Volatility.** In general, for purchased options, an increase in volatility results in a higher fair value measurement.
 - **Credit spreads, upfront credit points and recovery rates.** In general, the fair value of purchased credit protection increases as credit spreads or upfront credit points increase or recovery rates decrease. Credit spreads, upfront credit points and recovery rates are strongly related to distinctive risk factors of the underlying reference obligations, which include reference entity-specific factors, such as leverage, volatility and industry, market-based risk factors, such as borrowing costs or liquidity of the underlying reference obligation, and macroeconomic conditions.
 - **Commodity prices and spreads.** In general, for contracts where the holder is receiving a commodity, an increase in the spread (price difference from a benchmark index due to differences in quality or delivery location) or price results in a higher fair value measurement.
- Due to the distinctive nature of each of the firm's level 3 derivatives, the interrelationship of inputs is not necessarily uniform within each product type.

Notes to Consolidated Financial Statements

Level 3 Rollforward. The table below presents a summary of the changes in fair value for level 3 derivatives.

\$ in millions	Year Ended December	
	2024	2023
Total level 3 derivatives, net		
Beginning balance	\$ 810	\$ 1,521
Net realized gains/(losses)	(129)	65
Net unrealized gains/(losses)	25	(327)
Purchases	480	366
Sales	(885)	(1,420)
Settlements	573	466
Transfers into level 3	(25)	(90)
Transfers out of level 3	(24)	229
Ending balance	\$ 825	\$ 810

In the table above:

- Changes in fair value are presented for all derivative assets and liabilities that are classified in level 3 as of the end of the period.
- Net unrealized gains/(losses) relates to instruments that were still held at period-end.
- Transfers between levels of the fair value hierarchy are reported at the beginning of the reporting period in which they occur. If a derivative was transferred into level 3 during a reporting period, its entire gain or loss for the period is classified in level 3.
- Positive amounts for transfers into level 3 and negative amounts for transfers out of level 3 represent net transfers of derivative assets. Negative amounts for transfers into level 3 and positive amounts for transfers out of level 3 represent net transfers of derivative liabilities.
- A derivative with level 1 and/or level 2 inputs is classified in level 3 in its entirety if it has at least one significant level 3 input.
- If there is one significant level 3 input, the entire gain or loss from adjusting only observable inputs (i.e., level 1 and level 2 inputs) is classified in level 3.
- Gains or losses that have been classified in level 3 resulting from changes in level 1 or level 2 inputs are frequently offset by gains or losses attributable to level 1 or level 2 derivatives and/or level 1, level 2 and level 3 trading cash instruments. As a result, gains/(losses) included in the level 3 rollforward below do not necessarily represent the overall impact on the firm's results of operations, liquidity or capital resources.

The table below presents information, by product type, for derivatives included in the summary table above.

\$ in millions	Year Ended December	
	2024	2023
Interest rates, net		
Beginning balance	\$ (439)	\$ (459)
Net realized gains/(losses)	21	9
Net unrealized gains/(losses)	63	(78)
Purchases	4	85
Sales	(29)	(408)
Settlements	290	423
Transfers into level 3	(36)	(81)
Transfers out of level 3	14	70
Ending balance	\$ (112)	\$ (439)
Credit, net		
Beginning balance	\$ 1,650	\$ 1,460
Net realized gains/(losses)	94	(58)
Net unrealized gains/(losses)	9	274
Purchases	85	89
Sales	(74)	(50)
Settlements	(354)	(138)
Transfers into level 3	79	(20)
Transfers out of level 3	(271)	93
Ending balance	\$ 1,218	\$ 1,650
Currencies, net		
Beginning balance	\$ 42	\$ 162
Net realized gains/(losses)	(27)	80
Net unrealized gains/(losses)	1	(182)
Purchases	44	2
Sales	(9)	(1)
Settlements	(45)	(56)
Transfers into level 3	3	4
Transfers out of level 3	38	33
Ending balance	\$ 47	\$ 42
Commodities, net		
Beginning balance	\$ 628	\$ 919
Net realized gains/(losses)	(340)	(113)
Net unrealized gains/(losses)	91	(373)
Purchases	192	4
Sales	(27)	(17)
Settlements	57	68
Transfers into level 3	14	122
Transfers out of level 3	163	18
Ending balance	\$ 778	\$ 628
Equities, net		
Beginning balance	\$ (1,071)	\$ (561)
Net realized gains/(losses)	123	147
Net unrealized gains/(losses)	(139)	32
Purchases	155	186
Sales	(746)	(944)
Settlements	625	169
Transfers into level 3	(85)	(115)
Transfers out of level 3	32	15
Ending balance	\$ (1,106)	\$ (1,071)

Notes to Consolidated Financial Statements

Level 3 Rollforward Commentary for the Year Ended December 2024. The net realized and unrealized losses on level 3 derivatives of \$104 million (reflecting \$129 million of net realized losses and \$25 million of net unrealized gains) for 2024 included gains/(losses) of \$(115) million reported in market making and \$11 million reported in other principal transactions.

The net unrealized gains on level 3 derivatives for 2024 primarily reflected gains on certain commodity derivatives (principally due to the impact of changes in commodity prices) and gains on certain interest rate derivatives (principally due to an increase in interest rates), partially offset by losses on certain equity derivatives (principally due to the impact of changes in equity prices).

The drivers of transfers into level 3 derivatives during 2024 were not material.

Transfers out of level 3 derivatives during 2024 reflected transfers of certain credit derivative assets to level 2 (principally due to increased transparency of certain credit spread inputs used to value these instruments), partially offset by transfers of certain commodity derivative liabilities to level 2 (principally due to increased transparency of certain commodity price inputs used to value these instruments).

Level 3 Rollforward Commentary for the Year Ended 2023. The net realized and unrealized losses on level 3 derivatives of \$262 million (reflecting \$65 million of net realized gains and \$327 million of net unrealized losses) for 2023 included losses of \$251 million reported in market making and \$11 million reported in other principal transactions.

The net unrealized losses on level 3 derivatives for 2023 primarily reflected losses on certain commodity derivatives (principally due to the impact of changes in commodity prices), losses on certain currency derivatives (principally due to the impact of a decrease in interest rates), partially offset by gains on certain credit derivatives (principally due to the impact of changes in foreign exchange rates and a decrease in interest rates).

Transfers into level 3 derivatives during 2023 primarily reflected transfers of certain equity derivative liabilities (principally due to reduced transparency of certain unobservable volatility inputs used to value these instruments) and transfers of certain interest rate derivative liabilities from level 2 (principally due to certain unobservable volatility inputs becoming significant to the valuation of these instruments), partially offset by transfers of certain commodity derivative assets from level 2 (principally due to certain unobservable volatility inputs becoming significant to the valuation of these instruments).

The drivers of transfers out of level 3 derivatives during 2023 were not material.

Investments

Fair Value by Level. The table below presents investments accounted for at fair value by level within the fair value hierarchy.

<i>\$ in millions</i>	Level 1	Level 2	Level 3	Total
As of December 2024				
Government and agency obligations:				
U.S.	\$ 75,410	\$ -	\$ -	\$ 75,410
Non-U.S.	4,048	62	-	4,110
Corporate debt securities	137	2,629	4,510	7,276
Securities backed by real estate	-	7	562	569
Money market instruments	327	1,453	-	1,780
Other debt obligations	22	53	328	403
Equity securities	574	3,331	8,742	12,647
Subtotal	\$ 80,518	\$ 7,535	\$ 14,142	\$ 102,195
Investments in funds at NAV				2,547
Total investments				\$ 104,742
As of December 2023				
Government and agency obligations:				
U.S.	\$ 46,731	\$ -	\$ -	\$ 46,731
Non-U.S.	2,399	144	-	2,543
Corporate debt securities	160	2,299	6,533	8,992
Securities backed by real estate	-	2	687	689
Money market instruments	52	999	-	1,051
Other debt obligations	9	14	244	267
Equity securities	721	2,099	9,674	12,494
Subtotal	\$ 50,072	\$ 5,557	\$ 17,138	\$ 72,767
Investments in funds at NAV				3,000
Total investments				\$ 75,767

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of investments.

Notes to Consolidated Financial Statements

Significant Unobservable Inputs. The table below presents the amount of level 3 investments, and ranges and weighted averages of significant unobservable inputs used to value such investments.

	As of December 2024		As of December 2023	
	Amount or Range	Weighted Average	Amount or Range	Weighted Average
<i>\$ in millions</i>				
Corporate debt securities				
Level 3 assets	\$ 4,510		\$ 6,533	
Yield	5.0% to 28.8%	12.2 %	6.0% to 31.0%	12.1 %
Recovery rate	5.8% to 41.0%	25.2 %	7.3% to 41.2%	27.6 %
Duration (years)	0.3 to 9.0	3.6	0.4 to 5.3	3.0
Multiples	1.1x to 34.2x	6.5x	0.9x to 53.3x	7.7x
Securities backed by real estate				
Level 3 assets	\$ 562		\$ 687	
Yield	9.5% to 16.0%	13.6 %	7.4% to 18.8%	14.1 %
Duration (years)	1.1 to 2.8	2.8	0.4 to 4.1	3.9
Other debt obligations				
Level 3 assets	\$ 328		\$ 244	
Yield	7.0% to 8.7%	7.7 %	7.6% to 8.8%	8.2 %
Equity securities				
Level 3 assets	\$ 8,742		\$ 9,674	
Multiples	0.4x to 34.2x	8.6x	0.5x to 25.2x	8.3x
Discount rate/yield	6.0% to 27.9%	13.3 %	6.0% to 38.5%	12.3 %
Capitalization rate	4.4% to 9.1%	5.4 %	4.5% to 8.0%	5.3 %

In the table above:

- Ranges represent the significant unobservable inputs that were used in the valuation of each type of investment.
- Weighted averages are calculated by weighting each input by the relative fair value of the investment.
- The ranges and weighted averages of these inputs are not representative of the appropriate inputs to use when calculating the fair value of any one investment. For example, the highest multiple for private equity securities is appropriate for valuing a specific private equity security but may not be appropriate for valuing any other private equity security. Accordingly, the ranges of inputs do not represent uncertainty in, or possible ranges of, fair value measurements of level 3 investments.
- Increases in yield, discount rate, capitalization rate or duration used in the valuation of level 3 investments would have resulted in a lower fair value measurement, while increases in recovery rate or multiples would have resulted in a higher fair value measurement as of both December 2024 and December 2023. Due to the distinctive nature of each level 3 investment, the interrelationship of inputs is not necessarily uniform within each product type.

- Corporate debt securities, securities backed by real estate and other debt obligations are valued using discounted cash flows, and equity securities are valued using market comparables and discounted cash flows.

- The fair value of any one instrument may be determined using multiple valuation techniques. For example, market comparables and discounted cash flows may be used together to determine fair value. Therefore, the level 3 balance encompasses both of these techniques.

Level 3 Rollforward. The table below presents a summary of the changes in fair value for level 3 investments.

<i>\$ in millions</i>	Year Ended December	
	2024	2023
Beginning balance	\$ 17,138	\$ 16,942
Net realized gains/(losses)	342	463
Net unrealized gains/(losses)	(287)	(722)
Purchases	1,395	1,651
Sales	(916)	(1,186)
Settlements	(2,322)	(1,762)
Transfers into level 3	953	2,918
Transfers out of level 3	(2,161)	(1,166)
Ending balance	\$ 14,142	\$ 17,138

In the table above:

- Changes in fair value are presented for all investments that are classified in level 3 as of the end of the period.
- Net unrealized gains/(losses) relates to investments that were still held at period-end.
- Transfers between levels of the fair value hierarchy are reported at the beginning of the reporting period in which they occur. If an investment was transferred to level 3 during a reporting period, its entire gain or loss for the period is classified in level 3.

Notes to Consolidated Financial Statements

The table below presents information, by product type, for investments included in the summary table above.

\$ in millions	Year Ended December	
	2024	2023
Corporate debt securities		
Beginning balance	\$ 6,533	\$ 7,003
Net realized gains/(losses)	179	360
Net unrealized gains/(losses)	(169)	1
Purchases	600	590
Sales	(306)	(458)
Settlements	(1,672)	(1,110)
Transfers into level 3	462	755
Transfers out of level 3	(1,117)	(608)
Ending balance	\$ 4,510	\$ 6,533
Securities backed by real estate		
Beginning balance	\$ 687	\$ 827
Net realized gains/(losses)	52	12
Net unrealized gains/(losses)	(82)	(166)
Purchases	53	58
Sales	(33)	(148)
Settlements	(114)	(62)
Transfers into level 3	1	171
Transfers out of level 3	(2)	(5)
Ending balance	\$ 562	\$ 687
Other debt obligations		
Beginning balance	\$ 244	\$ 256
Net realized gains/(losses)	5	4
Purchases	141	2
Settlements	(31)	(18)
Transfers out of level 3	(31)	—
Ending balance	\$ 328	\$ 244
Equity securities		
Beginning balance	\$ 9,674	\$ 8,856
Net realized gains/(losses)	106	87
Net unrealized gains/(losses)	(36)	(557)
Purchases	601	1,001
Sales	(577)	(580)
Settlements	(505)	(572)
Transfers into level 3	490	1,992
Transfers out of level 3	(1,011)	(553)
Ending balance	\$ 8,742	\$ 9,674

Level 3 Rollforward Commentary for the Year Ended December 2024. The net realized and unrealized gains on level 3 investments of \$55 million (reflecting \$342 million of net realized gains and \$287 million of net unrealized losses) for 2024 included gains/(losses) of \$(224) million reported in other principal transactions and \$279 million reported in interest income.

The net unrealized losses on level 3 investments for 2024 primarily reflected losses on certain corporate debt securities (principally due to corporate performance and company-specific events).

Transfers into level 3 investments during 2024 primarily reflected transfers of certain equity securities from level 2 (principally due to reduced price transparency as a result of a lack of market evidence, including fewer market transactions in these instruments) and transfers of certain corporate debt securities from level 2 (principally due to certain unobservable yield inputs becoming significant to the valuation of these instruments).

Transfers out of level 3 investments during 2024 primarily reflected transfers of certain corporate debt securities to level 2 (principally due to certain unobservable yield inputs no longer being significant to the valuation of these instruments and increased price transparency as a result of market evidence, including market transactions in these instruments) and transfers of certain equity securities to level 2 (principally due to increased price transparency as a result of market evidence, including market transactions in these instruments).

Level 3 Rollforward Commentary for the Year Ended December 2023. The net realized and unrealized losses on level 3 investments of \$259 million (reflecting \$463 million of net realized gains and \$722 million of net unrealized losses) for 2023 included gains/(losses) of \$(820) million reported in other principal transactions and \$561 million reported in interest income.

The net unrealized losses on level 3 investments for 2023 primarily reflected losses on certain equity securities and securities backed by real estate (in each case, principally due to ongoing weakness in the commercial real estate market).

Transfers into level 3 investments during 2023 primarily reflected transfers of certain equity securities and corporate debt securities from level 2 (in each case, principally due to reduced price transparency as a result of a lack of market evidence, including fewer market transactions in these instruments, and certain unobservable yield inputs becoming significant to the valuation of these instruments).

Transfers out of level 3 investments during 2023 primarily reflected transfers of certain corporate debt securities to level 2 (principally due to increased price transparency as a result of market evidence, including market transactions in these instruments, and certain unobservable yield inputs becoming less significant to the valuation of these instruments), and transfers of certain equity securities to level 2 (principally due to increased price transparency as a result of market evidence, including market transactions in these instruments).

Notes to Consolidated Financial Statements**Loans**

Fair Value by Level. The table below presents loans held for investment accounted for at fair value under the fair value option by level within the fair value hierarchy.

<i>\$ in millions</i>	Level 1	Level 2	Level 3	Total
As of December 2024				
Loan Type				
Corporate	\$ -	\$ 64	\$ 403	\$ 467
Real estate:				
Commercial	-	349	75	424
Residential	-	3,684	42	3,726
Other collateralized	-	648	135	783
Other	-	32	28	60
Total	\$ -	\$ 4,777	\$ 683	\$ 5,460
As of December 2023				
Loan Type				
Corporate	\$ -	\$ 415	\$ 344	\$ 759
Real estate:				
Commercial	-	360	203	563
Residential	-	4,087	58	4,145
Other collateralized	-	775	136	911
Other	-	46	82	128
Total	\$ -	\$ 5,683	\$ 823	\$ 6,506

The gains/(losses) as a result of changes in the fair value of loans held for investment for which the fair value option was elected were not material for both 2024 and 2023. These gains/(losses) were included in other principal transactions.

Significant Unobservable Inputs. The table below presents the amount of level 3 loans, and ranges and weighted averages of significant unobservable inputs used to value such loans.

<i>\$ in millions</i>	As of December 2024		As of December 2023	
	Amount or Range	Weighted Average	Amount or Range	Weighted Average
Corporate				
Level 3 assets	\$ 403		\$ 344	
Yield	11.6% to 22.4%	17.5 %	8.0% to 17.1%	10.5 %
Recovery rate	37.2% to 95.6%	72.9 %	2.0% to 95.0%	74.0 %
Duration (years)	0.6 to 9.3	6.0	0.7 to 2.3	1.7
Real estate				
Level 3 assets	\$ 117		\$ 261	
Yield	6.1% to 10.9%	6.7 %	5.0% to 21.4%	18.1 %
Recovery rate	3.3% to 99.2%	73.7 %	5.3% to 99.2%	66.0 %
Duration (years)	0.2 to 3.6	0.6	0.5 to 6.2	1.6
Other collateralized				
Level 3 assets	\$ 135		\$ 136	
Yield	6.2% to 6.8%	6.3 %	5.6% to 8.7%	6.1 %
Other				
Level 3 assets	\$ 28		\$ 82	
Yield	N/A	N/A	7.3% to 13.5%	9.6 %
Duration (years)	N/A	N/A	3.6 to 5.2	4.2

In the table above:

- Ranges represent the significant unobservable inputs that were used in the valuation of each type of loan.
- Weighted averages are calculated by weighting each input by the relative fair value of the loan.
- The ranges and weighted averages of these inputs are not representative of the appropriate inputs to use when calculating the fair value of any one loan. For example, the highest yield for corporate loans is appropriate for valuing a specific corporate loan but may not be appropriate for valuing any other corporate loan. Accordingly, the ranges of inputs do not represent uncertainty in, or possible ranges of, fair value measurements of level 3 loans.
- Increases in yield or duration used in the valuation of level 3 loans would have resulted in a lower fair value measurement, while increases in recovery rate would have resulted in a higher fair value measurement as of both December 2024 and December 2023. Due to the distinctive nature of each level 3 loan, the interrelationship of inputs is not necessarily uniform within each product type.
- Loans are valued using discounted cash flows.
- The significant unobservable inputs for yield and duration (related to other loans) as of December 2024 did not have a range (and there was no weighted average) as each pertained to a single position. Therefore, such unobservable inputs are not included in the table above.

Notes to Consolidated Financial Statements

Level 3 Rollforward. The table below presents a summary of the changes in fair value for level 3 loans.

\$ in millions	Year Ended December	
	2024	2023
Beginning balance	\$ 823	\$ 1,837
Net realized gains/(losses)	33	26
Net unrealized gains/(losses)	(41)	(169)
Purchases	135	53
Sales	(61)	(540)
Settlements	(286)	(318)
Transfers into level 3	124	2
Transfers out of level 3	(44)	(68)
Ending balance	\$ 683	\$ 823

In the table above:

- Changes in fair value are presented for loans that are classified in level 3 as of the end of the period.
- Net unrealized gains/(losses) relates to loans that were still held at period-end.
- Purchases includes originations and secondary purchases.
- Transfers between levels of the fair value hierarchy are reported at the beginning of the reporting period in which they occur. If a loan was transferred to level 3 during a reporting period, its entire gain or loss for the period is classified in level 3.

The table below presents information, by loan type, for loans included in the summary table above.

\$ in millions	Year Ended December	
	2024	2023
Corporate		
Beginning balance	\$ 344	\$ 637
Net realized gains/(losses)	12	10
Net unrealized gains/(losses)	(26)	(124)
Purchases	127	10
Sales	(5)	(47)
Settlements	(160)	(113)
Transfers into level 3	111	2
Transfers out of level 3	-	(31)
Ending balance	\$ 403	\$ 344
Real estate		
Beginning balance	\$ 261	\$ 785
Net realized gains/(losses)	9	11
Net unrealized gains/(losses)	(10)	(42)
Purchases	2	7
Sales	(52)	(272)
Settlements	(93)	(192)
Transfers out of level 3	-	(36)
Ending balance	\$ 117	\$ 261
Other collateralized		
Beginning balance	\$ 136	\$ 140
Net realized gains/(losses)	4	1
Net unrealized gains/(losses)	1	(6)
Purchases	6	5
Sales	(3)	(2)
Settlements	(22)	(2)
Transfers into level 3	13	-
Ending balance	\$ 135	\$ 136
Other		
Beginning balance	\$ 82	\$ 275
Net realized gains/(losses)	8	4
Net unrealized gains/(losses)	(6)	3
Purchases	-	31
Sales	(1)	(219)
Settlements	(11)	(11)
Transfers out of level 3	(44)	(1)
Ending balance	\$ 28	\$ 82

Notes to Consolidated Financial Statements

Level 3 Rollforward Commentary for the Year Ended December 2024. The net realized and unrealized losses on level 3 loans of \$8 million (reflecting \$33 million of net realized gains and \$41 million of net unrealized losses) for 2024 included gains/(losses) of \$(13) million reported in other principal transactions and \$5 million reported in interest income.

The drivers of the net unrealized losses on level 3 loans for 2024 were not material.

Transfers into level 3 loans during 2024 primarily reflected transfers of certain corporate loans from level 2 (principally due to reduced price transparency as a result of a lack of market evidence, including fewer market transactions in these instruments).

The drivers of the transfers out of level 3 loans during 2024 were not material.

Level 3 Rollforward Commentary for the Year Ended December 2023. The net realized and unrealized losses on level 3 loans of \$143 million (reflecting \$26 million of net realized gains and \$169 million of net unrealized losses) for 2023 included gains/(losses) of \$(152) million reported in other principal transactions and \$9 million reported in interest income.

The net unrealized losses on level 3 loans for 2023 primarily reflected losses on corporate loans (principally due to company-specific events).

The drivers of both transfers into and transfers out of level 3 loans during 2023 were not material.

Other Financial Assets and Liabilities

Fair Value by Level. The table below presents, by level within the fair value hierarchy, other financial assets and liabilities at fair value, substantially all of which are accounted for at fair value under the fair value option.

<i>\$ in millions</i>	Level 1	Level 2	Level 3	Total
As of December 2024				
Assets				
Resale agreements	\$ -	\$ 179,793	\$ -	\$ 179,793
Securities borrowed	-	46,902	-	46,902
Customer and other receivables	-	23	-	23
Other assets	-	-	194	194
Total	\$ -	\$ 226,718	\$ 194	\$ 226,912
Liabilities				
Deposits	\$ -	\$ (41,810)	\$ (3,045)	\$ (44,855)
Repurchase agreements	-	(274,380)	-	(274,380)
Securities loaned	-	(10,246)	-	(10,246)
Other secured financings	-	(27,434)	(551)	(27,985)
Unsecured borrowings:				
Short-term	-	(45,073)	(5,294)	(50,367)
Long-term	-	(75,810)	(13,379)	(89,189)
Other liabilities	-	(8)	(76)	(84)
Total	\$ -	\$ (474,761)	\$ (22,345)	\$ (497,106)
As of December 2023				
Assets				
Resale agreements	\$ -	\$ 223,543	\$ -	\$ 223,543
Securities borrowed	-	44,930	-	44,930
Customer and other receivables	-	23	-	23
Other assets	-	179	187	366
Total	\$ -	\$ 268,675	\$ 187	\$ 268,862
Liabilities				
Deposits	\$ -	\$ (26,723)	\$ (2,737)	\$ (29,460)
Repurchase agreements	-	(249,887)	-	(249,887)
Securities loaned	-	(8,934)	-	(8,934)
Other secured financings	-	(10,532)	(2,022)	(12,554)
Unsecured borrowings:				
Short-term	-	(40,538)	(5,589)	(46,127)
Long-term	-	(72,562)	(13,848)	(86,410)
Other liabilities	-	(187)	(79)	(266)
Total	\$ -	\$ (409,363)	\$ (24,275)	\$ (433,638)

In the table above, assets are shown as positive amounts and liabilities are shown as negative amounts.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of other financial assets and liabilities.

Notes to Consolidated Financial Statements

Significant Unobservable Inputs. See below for information about the significant unobservable inputs used to value level 3 other financial assets and liabilities at fair value as of both December 2024 and December 2023.

Other Secured Financings. The ranges and weighted averages of significant unobservable inputs used to value level 3 other secured financings are presented below. These ranges and weighted averages exclude unobservable inputs that are only relevant to a single instrument, and therefore are not meaningful.

As of December 2024:

- Yield: 3.8% to 12.3% (weighted average: 8.9%)
- Duration: 2.0 to 5.4 years (weighted average: 2.9 years)

As of December 2023:

- Yield: 6.7% to 11.3% (weighted average: 8.5%)
- Duration: 0.1 to 4.5 years (weighted average: 0.9 years)

Generally, increases in yield or duration, in isolation, would have resulted in a lower fair value measurement as of period-end. Due to the distinctive nature of each of level 3 other secured financings, the interrelationship of inputs is not necessarily uniform across such financings. See Note 11 for further information about other secured financings.

Deposits, Unsecured Borrowings and Other Assets and Liabilities. Substantially all of the firm's deposits, unsecured short- and long-term borrowings, and other assets and liabilities that are classified in level 3 are hybrid financial instruments. As the significant unobservable inputs used to value hybrid financial instruments primarily relate to the embedded derivative component of these deposits, unsecured borrowings and other assets and liabilities, these unobservable inputs are incorporated in the firm's derivative disclosures. See Note 12 for further information about other assets, Note 13 for further information about deposits, Note 14 for further information about unsecured borrowings and Note 15 for further information about other liabilities.

Level 3 Rollforward. The table below presents a summary of the changes in fair value for level 3 other financial assets and liabilities accounted for at fair value.

\$ in millions	Year Ended December	
	2024	2023
Assets		
Beginning balance	\$ 187	\$ 74
Net realized gains/(losses)	-	(2)
Net unrealized gains/(losses)	18	95
Purchases	-	20
Sales	(11)	-
Ending balance	\$ 194	\$ 187
Liabilities		
Beginning balance	\$ (24,275)	\$ (18,826)
Net realized gains/(losses)	(210)	(212)
Net unrealized gains/(losses)	715	(1,667)
Issuances	(9,768)	(8,153)
Settlements	11,040	8,298
Transfers into level 3	(1,835)	(4,542)
Transfers out of level 3	1,988	827
Ending balance	\$ (22,345)	\$ (24,275)

In the table above:

- Changes in fair value are presented for all other financial assets and liabilities that are classified in level 3 as of the end of the period.
- Net unrealized gains/(losses) relates to other financial assets and liabilities that were still held at period-end.
- Transfers between levels of the fair value hierarchy are reported at the beginning of the reporting period in which they occur. If a financial instrument was transferred to level 3 during a reporting period, its entire gain or loss for the period is classified in level 3.
- For level 3 other financial assets, increases are shown as positive amounts, while decreases are shown as negative amounts. For level 3 other financial liabilities, increases are shown as negative amounts, while decreases are shown as positive amounts.
- Level 3 other financial assets and liabilities are frequently economically hedged with trading assets and liabilities. Accordingly, gains or losses that are classified in level 3 can be partially offset by gains or losses attributable to level 1, 2 or 3 trading assets and liabilities. As a result, gains or losses included in the level 3 rollforward below do not necessarily represent the overall impact on the firm's results of operations, liquidity or capital resources.

Notes to Consolidated Financial Statements

The table below presents information, by the consolidated balance sheet line items, for other financial liabilities included in the summary table above.

\$ in millions	Year Ended December	
	2024	2023
Deposits		
Beginning balance	\$ (2,737)	\$ (2,743)
Net realized gains/(losses)	-	(2)
Net unrealized gains/(losses)	(96)	(140)
Issuances	(1,086)	(506)
Settlements	928	773
Transfers into level 3	(173)	(153)
Transfers out of level 3	119	34
Ending balance	\$ (3,045)	\$ (2,737)
Other secured financings		
Beginning balance	\$ (2,022)	\$ (1,842)
Net realized gains/(losses)	(1)	(19)
Net unrealized gains/(losses)	12	(50)
Issuances	(51)	(657)
Settlements	1,615	1,479
Transfers into level 3	(104)	(941)
Transfers out of level 3	-	8
Ending balance	\$ (551)	\$ (2,022)
Unsecured short-term borrowings		
Beginning balance	\$ (5,589)	\$ (4,090)
Net realized gains/(losses)	(6)	(36)
Net unrealized gains/(losses)	9	(563)
Issuances	(4,883)	(4,315)
Settlements	4,700	3,418
Transfers into level 3	(101)	(281)
Transfers out of level 3	576	278
Ending balance	\$ (5,294)	\$ (5,589)
Unsecured long-term borrowings		
Beginning balance	\$ (13,848)	\$ (10,066)
Net realized gains/(losses)	(203)	(155)
Net unrealized gains/(losses)	787	(914)
Issuances	(3,748)	(2,675)
Settlements	3,797	2,622
Transfers into level 3	(1,457)	(3,167)
Transfers out of level 3	1,293	507
Ending balance	\$ (13,379)	\$ (13,848)
Other liabilities		
Beginning balance	\$ (79)	\$ (85)
Net unrealized gains/(losses)	3	-
Settlements	-	6
Ending balance	\$ (76)	\$ (79)

Level 3 Rollforward Commentary for the Year Ended December 2024. The net realized and unrealized gains on level 3 other financial liabilities of \$505 million (reflecting \$210 million of net realized losses and \$715 million of net unrealized gains) for 2024 included gains/(losses) of \$491 million reported in market making, \$(42) million reported in other principal transactions and \$(1) million reported in interest expense in the consolidated statements of earnings, and \$57 million reported in debt valuation adjustment in the consolidated statements of comprehensive income.

The net unrealized gains on level 3 other financial liabilities for 2024 primarily reflected gains on certain hybrid financial instruments included in unsecured long-term borrowings (principally due to the impact of increases in interest rates and changes in foreign exchange rates), partially offset by losses on certain hybrid financial instruments included in deposits (principally due to an increase in equity prices).

Transfers into level 3 other financial liabilities during 2024 primarily reflected transfers of certain hybrid financial instruments included in unsecured long-term borrowings from level 2 (principally due to reduced transparency of certain credit spreads and volatility inputs used to value these instruments).

Transfers out of level 3 other financial liabilities during 2024 primarily reflected transfers of certain hybrid financial instruments included in long- and short-term borrowings to level 2 (principally due to increased transparency of certain volatility inputs used to value these instruments).

Level 3 Rollforward Commentary for the Year Ended December 2023. The net realized and unrealized losses on level 3 other financial liabilities of \$1.88 billion (reflecting \$212 million of net realized losses and \$1.67 billion of net unrealized losses) for 2023 included losses of \$1.41 billion reported in market making, \$23 million reported in other principal transactions and \$22 million reported in interest expense in the consolidated statements of earnings, and \$427 million reported in debt valuation adjustment in the consolidated statements of comprehensive income.

The net unrealized losses on level 3 other financial liabilities for 2023 primarily reflected losses on certain hybrid financial instruments included in unsecured long- and short-term borrowings (principally due to an increase in global equity prices).

Transfers into level 3 other financial liabilities during 2023 primarily reflected transfers of certain hybrid financial instruments included in unsecured long-term borrowings from level 2 (principally due to reduced price transparency of certain credit spread and volatility inputs used to value these instruments) and transfers of certain other secured financings from level 2 (principally due to reduced price transparency of certain yield and duration inputs used to value these instruments).

Transfers out of level 3 other financial liabilities during 2023 primarily reflected transfers of certain hybrid financial instruments included in unsecured long- and short-term borrowings to level 2 (principally due to increased price transparency of certain volatility inputs used to value these instruments).

Notes to Consolidated Financial Statements**Note 6.****Trading Assets and Liabilities**

Trading assets and liabilities include trading cash instruments and derivatives held in connection with the firm's market-making or risk management activities. These assets and liabilities are carried at fair value either under the fair value option or in accordance with other U.S. GAAP, and the related fair value gains and losses are generally recognized in the consolidated statements of earnings.

The table below presents a summary of trading assets and liabilities.

<i>\$ in millions</i>	Trading Assets	Trading Liabilities
As of December 2024		
Trading cash instruments	\$ 523,216	\$ 127,575
Derivatives	47,339	74,980
Total	\$ 570,555	\$ 202,555
As of December 2023		
Trading cash instruments	\$ 426,390	\$ 143,601
Derivatives	51,120	56,754
Total	\$ 477,510	\$ 200,355

See Note 5 for further information about trading cash instruments and Note 7 for further information about derivatives.

Gains and Losses from Market Making

The table below presents market making revenues by major product type.

<i>\$ in millions</i>	Year Ended December		
	2024	2023	2022
Interest rates	\$ 715	\$ 4,437	\$ (4,890)
Credit	2,467	1,141	1,095
Currencies	6,292	2,827	11,662
Equities	7,632	7,938	7,734
Commodities	1,284	1,895	3,033
Total	\$ 18,390	\$ 18,238	\$ 18,634

In the table above:

- Gains/(losses) include both realized and unrealized gains and losses. Gains/(losses) exclude related interest income and interest expense. See Note 23 for further information about interest income and interest expense.
- Gains/(losses) included in market making are primarily related to the firm's trading assets and liabilities, including both derivative and non-derivative financial instruments.
- Gains/(losses) are not representative of the manner in which the firm manages its business activities because many of the firm's market-making and client facilitation strategies utilize financial instruments across various product types. Accordingly, gains or losses in one product type frequently offset gains or losses in other product types. For example, most of the firm's longer-term derivatives across product types are sensitive to changes in interest rates and may be economically hedged with interest rate swaps. Similarly, a significant portion of the firm's trading cash instruments and derivatives across product types has exposure to foreign currencies and may be economically hedged with foreign currency contracts.

Notes to Consolidated Financial Statements**Note 7.****Derivatives and Hedging Activities****Derivative Activities**

Derivatives are instruments that derive their value from underlying asset prices, indices, reference rates and other inputs, or a combination of these factors. Derivatives may be traded on an exchange (exchange-traded) or they may be privately negotiated contracts, which are usually referred to as OTC derivatives. Certain of the firm's OTC derivatives are cleared and settled through central clearing counterparties (OTC-cleared), while others are bilateral contracts between two counterparties (bilateral OTC).

Market Making. As a market maker, the firm enters into derivative transactions to provide liquidity to clients and to facilitate the transfer and hedging of their risks. In this role, the firm typically acts as principal and is required to commit capital to provide execution, and maintains market-making positions in response to, or in anticipation of, client demand.

Risk Management. The firm also enters into derivatives to actively manage risk exposures that arise from its market-making and investing and financing activities. The firm's holdings and exposures are hedged, in many cases, on either a portfolio or risk-specific basis, as opposed to an instrument-by-instrument basis. The offsetting impact of this economic hedging is reflected in the same business segment as the related revenues. In addition, the firm may enter into derivatives designated as hedges under U.S. GAAP. These derivatives are used to manage interest rate exposure of certain fixed-rate unsecured borrowings and deposits and certain U.S. and non-U.S. government securities classified as available-for-sale, foreign exchange risk of certain available-for-sale securities, the net investment in certain non-U.S. operations and the exposure to the variability of the forecasted cash flows associated with certain floating-rate assets.

The firm enters into various types of derivatives, including:

- **Futures and Forwards.** Contracts that commit counterparties to purchase or sell financial instruments, commodities or currencies in the future.
- **Swaps.** Contracts that require counterparties to exchange cash flows, such as currency or interest payment streams. The amounts exchanged are based on the specific terms of the contract with reference to specified rates, financial instruments, commodities, currencies or indices.
- **Options.** Contracts in which the option purchaser has the right, but not the obligation, to purchase from or sell to the option writer financial instruments, commodities or currencies within a defined time period for a specified price.

Derivatives are reported on a net-by-counterparty basis (i.e., the net payable or receivable for derivative assets and liabilities for a given counterparty) when a legal right of setoff exists under an enforceable netting agreement (counterparty netting). Derivatives are accounted for at fair value, net of cash collateral received or posted under enforceable credit support agreements (cash collateral netting). Derivative assets are included in trading assets and derivative liabilities are included in trading liabilities. Realized and unrealized gains and losses on derivatives not designated as hedges are included in market making (for derivatives included in Fixed Income, Currency and Commodities (FICC) and Equities within Global Banking & Markets), and other principal transactions (for derivatives included in Investment banking fees and Other within Global Banking & Markets, as well as derivatives in Asset & Wealth Management) in the consolidated statements of earnings. For both 2024 and 2023, substantially all of the firm's derivatives were included in Global Banking & Markets.

Notes to Consolidated Financial Statements

The tables below present the gross fair value and the notional amounts of derivative contracts by major product type, the amounts of counterparty and cash collateral netting in the consolidated balance sheets, as well as cash and securities collateral posted and received under enforceable credit support agreements that do not meet the criteria for netting under U.S. GAAP.

\$ in millions	Fair Value as of December			
	2024		2023	
	Derivative Assets	Derivative Liabilities	Derivative Assets	Derivative Liabilities
Not accounted for as hedges				
Exchange-traded	\$ 2,706	\$ 1,068	\$ 3,401	\$ 1,129
OTC-cleared	1,746	2,428	67,815	64,490
Bilateral OTC	152,083	121,515	171,109	149,444
Total interest rates	156,535	125,011	242,325	215,063
OTC-cleared	1,787	1,893	1,271	1,533
Bilateral OTC	9,650	8,137	11,554	8,601
Total credit	11,437	10,030	12,825	10,134
Exchange-traded	142	6	708	15
OTC-cleared	1,325	993	1,033	1,632
Bilateral OTC	112,838	113,608	88,158	95,742
Total currencies	114,305	114,607	89,899	97,389
Exchange-traded	5,563	5,917	5,468	5,998
OTC-cleared	558	650	635	711
Bilateral OTC	7,205	7,958	10,739	11,234
Total commodities	13,326	14,525	16,842	17,943
Exchange-traded	55,049	83,475	31,315	39,247
OTC-cleared	189	131	122	171
Bilateral OTC	24,941	44,900	28,601	40,696
Total equities	80,179	128,506	60,038	80,114
Subtotal	375,782	392,679	421,929	420,643
Accounted for as hedges				
Bilateral OTC	201	8	298	9
Total interest rates	201	8	298	9
OTC-cleared	114	3	-	7
Bilateral OTC	255	5	5	208
Total currencies	369	8	5	215
Subtotal	570	16	303	224
Total gross fair value	\$ 376,352	\$ 392,695	\$ 422,232	\$ 420,867
Offset in the consolidated balance sheets				
Exchange-traded	\$ (57,776)	\$ (57,776)	\$ (32,722)	\$ (32,722)
OTC-cleared	(4,867)	(4,867)	(67,272)	(67,272)
Bilateral OTC	(218,269)	(218,269)	(221,395)	(221,395)
Counterparty netting	(280,912)	(280,912)	(321,389)	(321,389)
OTC-cleared	(412)	(105)	(1,335)	(486)
Bilateral OTC	(47,689)	(36,698)	(48,388)	(42,238)
Cash collateral netting	(48,101)	(36,803)	(49,723)	(42,724)
Total amounts offset	\$ (329,013)	\$ (317,715)	\$ (371,112)	\$ (364,113)
Included in the consolidated balance sheets				
Exchange-traded	\$ 5,684	\$ 32,690	\$ 8,170	\$ 13,667
OTC-cleared	440	1,126	2,269	786
Bilateral OTC	41,215	41,164	40,681	42,301
Total	\$ 47,339	\$ 74,980	\$ 51,120	\$ 56,754
Not offset in the consolidated balance sheets				
Cash collateral	\$ (600)	\$ (1,271)	\$ (877)	\$ (2,732)
Securities collateral	(14,938)	(8,731)	(13,425)	(6,516)
Total	\$ 31,801	\$ 64,978	\$ 36,818	\$ 47,506

\$ in millions	Notional Amounts as of December	
	2024	2023
Not accounted for as hedges		
Exchange-traded	\$ 2,332,117	\$ 3,854,689
OTC-cleared	12,571,690	16,007,915
Bilateral OTC	10,569,501	12,390,595
Total interest rates	25,473,308	32,253,199
Exchange-traded	322	299
OTC-cleared	660,181	498,720
Bilateral OTC	619,068	619,975
Total credit	1,279,571	1,118,994
Exchange-traded	9,264	11,586
OTC-cleared	429,858	268,293
Bilateral OTC	6,031,944	6,363,700
Total currencies	6,471,066	6,643,579
Exchange-traded	324,159	306,787
OTC-cleared	3,087	3,323
Bilateral OTC	183,174	199,270
Total commodities	510,420	509,380
Exchange-traded	1,868,855	1,564,341
OTC-cleared	1,475	1,487
Bilateral OTC	1,256,905	1,204,140
Total equities	3,127,235	2,769,968
Subtotal	36,861,600	43,295,120
Accounted for as hedges		
OTC-cleared	246,765	241,160
Bilateral OTC	3,588	2,914
Total interest rates	250,353	244,074
OTC-cleared	5,041	1,227
Bilateral OTC	10,328	9,130
Total currencies	15,369	10,357
Subtotal	265,722	254,431
Total notional amounts	\$ 37,127,322	\$ 43,549,551

In the tables above:

- Gross fair values exclude the effects of both counterparty netting and collateral, and therefore are not representative of the firm's exposure.
- Amounts presented for collateral not offset in the consolidated balance sheets consists of collateral received or posted in connection with OTC-cleared and bilateral OTC derivatives under enforceable credit support agreements that do not meet the criteria for netting under U.S. GAAP. In addition to collateral presented in the table above, the firm also posts or receives collateral in connection with its transactions with certain exchanges in accordance with the exchanges' margin requirements. Such collateral may be calculated based on the firm's total exposure to the respective exchange across all product types, including both derivative and non-derivative instruments. See Note 11 for further information about collateral received and pledged.
- Total gross fair value of derivatives included derivative assets of \$8.55 billion as of December 2024 and \$8.98 billion as of December 2023, and derivative liabilities of \$10.84 billion as of December 2024 and \$16.03 billion as of December 2023, which are not subject to an enforceable netting agreement or are subject to a netting agreement that the firm has not yet determined to be enforceable. The collateral received or posted in connection with such derivative agreements has not been netted.
- Notional amounts, which represent the sum of gross long and short derivative contracts, provide an indication of the volume of the firm's derivative activity and do not represent anticipated losses.

Notes to Consolidated Financial Statements

- During 2024, as permitted under the rules of a clearing organization in Europe, Middle East and Africa (EMEA), the firm elected to settle its transactions with this clearing organization on a daily basis. The impact of reflecting transactions with this clearing organization as settled would have been a reduction in gross derivative assets of \$64.19 billion and a reduction in gross derivative liabilities of \$62.86 billion as of December 2023, and a corresponding decrease in counterparty and cash collateral netting, with no impact to the consolidated balance sheets.

OTC Derivatives

The table below presents OTC derivative assets and liabilities by tenor and major product type.

<i>\$ in millions</i>	Less than 1 Year	1 - 5 Years	Greater than 5 Years	Total
As of December 2024				
Assets				
Interest rates	\$ 5,880	\$ 9,552	\$ 46,625	\$ 62,057
Credit	1,664	2,576	1,605	5,845
Currencies	17,249	8,280	5,387	30,916
Commodities	1,908	1,882	1,026	4,816
Equities	5,483	2,395	1,959	9,837
Counterparty netting in tenors	(2,681)	(2,683)	(4,271)	(9,635)
Subtotal	\$ 29,503	\$ 22,002	\$ 52,331	\$ 103,836
Cross-tenor counterparty netting				(14,080)
Cash collateral netting				(48,101)
Total OTC derivative assets				\$ 41,655
Liabilities				
Interest rates	\$ 5,074	\$ 10,858	\$ 16,049	\$ 31,981
Credit	999	2,474	963	4,436
Currencies	12,931	9,645	8,417	30,993
Commodities	2,012	2,448	1,201	5,661
Equities	11,435	15,113	3,189	29,737
Counterparty netting in tenors	(2,681)	(2,683)	(4,271)	(9,635)
Subtotal	\$ 29,770	\$ 37,855	\$ 25,548	\$ 93,173
Cross-tenor counterparty netting				(14,080)
Cash collateral netting				(36,803)
Total OTC derivative liabilities				\$ 42,290
As of December 2023				
Assets				
Interest rates	\$ 9,511	\$ 12,178	\$ 49,045	\$ 70,734
Credit	1,814	3,283	1,961	7,058
Currencies	9,117	7,579	5,479	22,175
Commodities	2,993	2,574	1,451	7,018
Equities	6,625	3,155	1,655	11,435
Counterparty netting in tenors	(3,046)	(2,765)	(3,648)	(9,459)
Subtotal	\$ 27,014	\$ 26,004	\$ 55,943	\$ 108,961
Cross-tenor counterparty netting				(16,288)
Cash collateral netting				(49,723)
Total OTC derivative assets				\$ 42,950
Liabilities				
Interest rates	\$ 11,952	\$ 15,972	\$ 17,540	\$ 45,464
Credit	792	2,508	1,067	4,367
Currencies	15,335	7,934	7,299	30,568
Commodities	2,526	3,643	1,419	7,588
Equities	10,183	10,048	3,340	23,571
Counterparty netting in tenors	(3,046)	(2,765)	(3,648)	(9,459)
Subtotal	\$ 37,742	\$ 37,340	\$ 27,017	\$ 102,099
Cross-tenor counterparty netting				(16,288)
Cash collateral netting				(42,724)
Total OTC derivative liabilities				\$ 43,087

In the table above:

- Tenor is based on the remaining contractual maturity for substantially all OTC derivative assets and liabilities.
- Counterparty netting within the same product type and tenor category is included within such product type and tenor category.
- Counterparty netting across product types within the same tenor category is included in counterparty netting in tenors. Where the counterparty netting is across tenor categories, the netting is included in cross-tenor counterparty netting.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of derivatives, and Note 5 for further information about derivatives within the fair value hierarchy.

Credit Derivatives

The firm enters into a broad array of credit derivatives to facilitate client transactions and to manage the credit risk associated with market-making and investing and financing activities. Credit derivatives are actively managed based on the firm's net risk position. Credit derivatives are generally individually negotiated contracts and can have various settlement and payment conventions. Credit events include failure to pay, bankruptcy, acceleration of indebtedness, restructuring, repudiation and dissolution of the reference entity.

The firm enters into the following types of credit derivatives:

- Credit Default Swaps.** Single-name credit default swaps protect the buyer against the loss of principal on one or more bonds, loans or mortgages (reference obligations) in the event the issuer of the reference obligations suffers a credit event. The buyer of protection pays an initial or periodic premium to the seller and receives protection for the period of the contract. If there is no credit event, as defined in the contract, the seller of protection makes no payments to the buyer. If a credit event occurs, the seller of protection is required to make a payment to the buyer, calculated according to the terms of the contract.
- Credit Options.** In a credit option, the option writer assumes the obligation to purchase or sell a reference obligation at a specified price or credit spread. The option purchaser buys the right, but does not assume the obligation, to sell the reference obligation to, or purchase it from, the option writer. The payments on credit options depend either on a particular credit spread or the price of the reference obligation.

Notes to Consolidated Financial Statements

• **Credit Indices, Baskets and Tranches.** Credit derivatives may reference a basket of single-name credit default swaps or a broad-based index. If a credit event occurs in one of the underlying reference obligations, the protection seller pays the protection buyer. The payment is typically a pro-rata portion of the transaction's total notional amount based on the underlying defaulted reference obligation. In certain transactions, the credit risk of a basket or index is separated into various portions (tranches), each having different levels of subordination. The most junior tranches cover initial defaults and once losses exceed the notional amount of these junior tranches, any excess loss is covered by the next most senior tranche.

• **Total Return Swaps.** A total return swap transfers the risks relating to economic performance of a reference obligation from the protection buyer to the protection seller. Typically, the protection buyer receives a floating rate of interest and protection against any reduction in fair value of the reference obligation, and the protection seller receives the cash flows associated with the reference obligation, plus any increase in the fair value of the reference obligation.

The firm economically hedges its exposure to written credit derivatives primarily by entering into offsetting purchased credit derivatives with identical underliers. Substantially all of the firm's purchased credit derivative transactions are with financial institutions and are subject to stringent collateral thresholds. In addition, upon the occurrence of a specified trigger event, the firm may take possession of the reference obligations underlying a particular written credit derivative, and consequently may, upon liquidation of the reference obligations, recover amounts on the underlying reference obligations in the event of default.

The table below presents information about credit derivatives.

	Credit Spread on Underlier (basis points)					
				Greater than 1,000		
<i>\$ in millions</i>	0 - 250	251 - 500	501 - 1,000		Total	
As of December 2024						
Maximum Payout/Notional Amount of Written Credit Derivatives by Tenor						
Less than 1 year	\$ 137,145	\$ 23,875	\$ 717	\$ 3,041	\$ 164,778	
1 - 5 years	378,743	15,349	4,242	6,194	404,528	
Greater than 5 years	29,196	2,457	471	64	32,188	
Total	\$ 545,084	\$ 41,681	\$ 5,430	\$ 9,299	\$ 601,494	
Maximum Payout/Notional Amount of Purchased Credit Derivatives						
Offsetting	\$ 463,919	\$ 20,691	\$ 4,781	\$ 8,264	\$ 497,655	
Other	165,662	11,721	1,879	1,160	180,422	
Total	\$ 629,581	\$ 32,412	\$ 6,660	\$ 9,424	\$ 678,077	
Fair Value of Written Credit Derivatives						
Asset	\$ 6,429	\$ 664	\$ 31	\$ 59	\$ 7,183	
Liability	985	314	307	1,138	2,744	
Net asset/(liability)	\$ 5,444	\$ 350	\$ (276)	\$ (1,079)	\$ 4,439	
As of December 2023						
Maximum Payout/Notional Amount of Written Credit Derivatives by Tenor						
Less than 1 year	\$ 126,667	\$ 12,594	\$ 892	\$ 3,611	\$ 143,764	
1 - 5 years	324,577	11,371	5,613	5,802	347,363	
Greater than 5 years	30,406	1,316	671	249	32,642	
Total	\$ 481,650	\$ 25,281	\$ 7,176	\$ 9,662	\$ 523,769	
Maximum Payout/Notional Amount of Purchased Credit Derivatives						
Offsetting	\$ 396,984	\$ 11,857	\$ 6,241	\$ 8,246	\$ 423,328	
Other	155,468	12,862	1,948	1,619	171,897	
Total	\$ 552,452	\$ 24,719	\$ 8,189	\$ 9,865	\$ 595,225	
Fair Value of Written Credit Derivatives						
Asset	\$ 11,147	\$ 654	\$ 221	\$ 165	\$ 12,187	
Liability	1,723	47	201	1,034	3,005	
Net asset/(liability)	\$ 9,424	\$ 607	\$ 20	\$ (869)	\$ 9,182	

In the table above:

- Fair values exclude the effects of both netting of receivable balances with payable balances under enforceable netting agreements, and netting of cash received or posted under enforceable credit support agreements, and therefore are not representative of the firm's credit exposure.
- Tenor is based on the remaining contractual maturity for substantially all written credit derivatives.
- The credit spread on the underlier, together with the tenor of the contract, are indicators of payment/performance risk. The firm is less likely to pay or otherwise be required to perform where the credit spread and the tenor are lower.
- Offsetting purchased credit derivatives represent the notional amount of purchased credit derivatives that economically hedge written credit derivatives with identical underliers.
- Other purchased credit derivatives represent the notional amount of all other purchased credit derivatives not included in offsetting.
- Written and purchased credit derivatives primarily consist of credit default swaps.

Notes to Consolidated Financial Statements**Impact of Credit and Funding Spreads on Derivatives**

The firm realizes gains or losses on its derivative contracts. These gains or losses include credit valuation adjustments (CVAs) relating to uncollateralized derivative assets and liabilities, which represent the gains or losses (including hedges) attributable to the impact of changes in credit exposure, counterparty credit spreads, liability funding spreads (which include the firm's own credit), probability of default and assumed recovery. These gains or losses also include funding valuation adjustments (FVA) relating to uncollateralized derivative assets, which represent the gains or losses (including hedges) attributable to the impact of changes in expected funding exposures and funding spreads.

The table below presents information about CVA and FVA.

\$ in millions	Year Ended December		
	2024	2023	2022
CVA, net of hedges	\$ 187	\$ (139)	\$ 320
FVA, net of hedges	142	131	(193)
Total	\$ 329	\$ (8)	\$ 127

Bifurcated Embedded Derivatives

The table below presents the fair value and the notional amount of derivatives that have been bifurcated from their related borrowings.

\$ in millions	As of December	
	2024	2023
Fair value of assets	\$ 467	\$ 450
Fair value of liabilities	(175)	(307)
Net asset/(liability)	\$ 292	\$ 143
Notional amount	\$ 8,106	\$ 8,082

In the table above, derivatives that have been bifurcated from their related borrowings are recorded at fair value and primarily consist of interest rate, equity and commodity products. These derivatives are included in unsecured short- and long-term borrowings, as well as other secured financings, with the related borrowings.

Derivatives with Credit-Related Contingent Features

Certain of the firm's derivatives have been transacted under bilateral agreements with counterparties who may require the firm to post collateral or terminate the transactions based on changes in the firm's credit ratings. The firm assesses the impact of these bilateral agreements by determining the collateral or termination payments that would occur assuming a downgrade by all rating agencies. A downgrade by any one rating agency, depending on the agency's relative ratings of the firm at the time of the downgrade, may have an impact which is comparable to the impact of a downgrade by all rating agencies.

The table below presents information about net derivative liabilities under bilateral agreements (excluding collateral posted), the fair value of collateral posted and additional collateral or termination payments that could have been called by counterparties in the event of a one- or two-notch downgrade in the firm's credit ratings.

\$ in millions	As of December	
	2024	2023
Net derivative liabilities under bilateral agreements	\$ 31,575	\$ 30,021
Collateral posted	\$ 20,262	\$ 20,758
Additional collateral or termination payments:		
One-notch downgrade	\$ 315	\$ 271
Two-notch downgrade	\$ 1,200	\$ 1,584

Hedge Accounting

The firm applies hedge accounting for (i) interest rate swaps used to manage the interest rate exposure of certain fixed-rate unsecured long- and short-term borrowings, certain fixed-rate certificates of deposit and certain U.S. and non-U.S. government securities classified as available-for-sale, (ii) foreign currency forward contracts used to manage the foreign exchange risk of certain securities classified as available-for-sale, (iii) foreign currency forward contracts and foreign currency-denominated debt used to manage foreign exchange risk on the firm's net investment in certain non-U.S. operations and (iv) interest rate swaps used to manage the variability of the forecasted cash flows associated with certain floating-rate assets.

To qualify for hedge accounting, the hedging instrument must be highly effective at reducing the risk from the exposure being hedged. Additionally, the firm must formally document the hedging relationship at inception and assess the hedging relationship at least on a quarterly basis to ensure the hedging instrument continues to be highly effective over the life of the hedging relationship.

Fair Value Hedges

The firm designates interest rate swaps as fair value hedges of certain fixed-rate unsecured long- and short-term debt and fixed-rate certificates of deposit and of certain U.S. and non-U.S. government securities classified as available-for-sale. These interest rate swaps hedge changes in fair value attributable to the designated benchmark interest rate (e.g., Secured Overnight Financing Rate (SOFR), Overnight Index Swap Rate or Sterling Overnight Index Average), effectively converting a substantial portion of these fixed-rate financial instruments into floating-rate financial instruments.

The firm applies a statistical method that utilizes regression analysis when assessing the effectiveness of these hedging relationships in achieving offsetting changes in the fair values of the hedging instrument and the risk being hedged (i.e., interest rate risk). An interest rate swap is considered highly effective in offsetting changes in fair value attributable to changes in the hedged risk when the regression analysis results in a coefficient of determination of 80% or greater and a slope between 80% and 125%.

Notes to Consolidated Financial Statements

For qualifying interest rate fair value hedges, gains or losses on derivatives are included in interest income/expense. The change in fair value of the hedged items attributable to the risk being hedged is reported as an adjustment to its carrying value (hedging adjustment) and is also included in interest income/expense. When a derivative is no longer designated as a hedge, any remaining difference between the carrying value and par value of the hedged item is amortized in interest income/expense over the remaining life of the hedged item using the effective interest method. See Note 23 for further information about interest income and interest expense.

The table below presents the gains/(losses) from interest rate derivatives accounted for as hedges and the related hedged items.

\$ in millions	Year Ended December		
	2024	2023	2022
Investments			
Interest rate hedges	\$ 63	\$ (109)	\$ 366
Hedged investments	(80)	111	(350)
Gains/(losses)	\$ (17)	\$ 2	\$ 16
Borrowings and deposits			
Interest rate hedges	\$ (581)	\$ 3,859	\$ (22,183)
Hedged borrowings and deposits	181	(4,344)	21,662
Gains/(losses)	\$ (400)	\$ (485)	\$ (521)

The table below presents the carrying value of investments, deposits and unsecured borrowings that are designated in an interest rate hedging relationship and the related cumulative hedging adjustment (increase/(decrease)) from current and prior hedging relationships included in such carrying values.

\$ in millions	Carrying Value		Cumulative Hedging Adjustment
	2024	2023	2022
As of December 2024			
Assets			
Investments	\$ 34,755	\$ (279)	
Liabilities			
Deposits	\$ 1,840	\$ (52)	
Unsecured short-term borrowings	\$ 14,720	\$ (113)	
Unsecured long-term borrowings	\$ 130,161	\$ (10,757)	
As of December 2023			
Assets			
Investments	\$ 16,523	\$ (104)	
Liabilities			
Deposits	\$ 3,435	\$ (123)	
Unsecured short-term borrowings	\$ 14,449	\$ (94)	
Unsecured long-term borrowings	\$ 134,992	\$ (10,810)	

In the table above:

- Cumulative hedging adjustment included \$(5.81) billion as of December 2024 and \$(5.63) billion as of December 2023 of hedging adjustments from prior hedging relationships that were de-designated and substantially all were related to unsecured long-term borrowings.
- The amortized cost of investments was \$35.29 billion as of December 2024 and \$17.33 billion as of December 2023.

In addition, cumulative hedging adjustments for items no longer designated in a hedging relationship were not material as of both December 2024 and December 2023.

The firm designates certain foreign currency forward contracts as fair value hedges of the foreign exchange risk of non-U.S. government securities classified as available-for-sale. See Note 8 for information about the amortized cost and fair value of such securities. The effectiveness of such hedges is assessed based on changes in spot rates. The gains/(losses) on the hedges (relating to both spot and forward points) and the foreign exchange gains/(losses) on the related available-for-sale securities are included in market making. The net gains/(losses) on hedges and the related hedged available-for-sale securities were not material for 2024, were \$(2) million (reflecting a loss of \$127 million related to hedges and a gain of \$125 million on the related hedged available-for-sale securities) for 2023 and were \$(30) million (reflecting a gain of \$266 million related to hedges and a loss of \$296 million on the related hedged available-for-sale securities) for 2022.

Net Investment Hedges

The firm seeks to reduce the impact of fluctuations in foreign exchange rates on its net investments in certain non-U.S. operations through the use of foreign currency forward contracts and foreign currency-denominated debt. For foreign currency forward contracts designated as hedges, the effectiveness of the hedge is assessed based on the overall changes in the fair value of the forward contracts (i.e., based on changes in forward rates). For foreign currency-denominated debt designated as a hedge, the effectiveness of the hedge is assessed based on changes in spot rates. For qualifying net investment hedges, all gains or losses on the hedging instruments are included in currency translation.

The table below presents the gains/(losses) from net investment hedging.

\$ in millions	Year Ended December		
	2024	2023	2022
Hedges:			
Foreign currency forward contracts	\$ 1,064	\$ (276)	\$ 1,713
Foreign currency-denominated debt	\$ 1,633	\$ (550)	\$ (269)

Gains or losses on individual net investments in non-U.S. operations are reclassified from accumulated other comprehensive income/(loss) to earnings when such net investments are sold or substantially liquidated. The gross and net gains/(losses) reclassified to earnings from accumulated other comprehensive income/(loss) were not material for 2024, were \$(49) million (reflecting a gain of \$90 million related to hedges and a loss of \$139 million on the related net investments in non-U.S. operations) for 2023 and were not material for 2022.

The firm had designated \$22.10 billion as of December 2024 and \$27.52 billion as of December 2023 of foreign currency-denominated debt, included in unsecured long- and short-term borrowings, as hedges of net investments in non-U.S. subsidiaries.

Notes to Consolidated Financial Statements**Cash Flow Hedges**

During the fourth quarter of 2024, the firm designated certain interest rate swaps as cash flow hedges. These interest rate swaps hedge the firm's exposure to the variability of the forecasted cash flows due to changes in the contractually specified interest rates associated with certain floating-rate assets.

The firm applies a statistical method that utilizes regression analysis when assessing hedge effectiveness. A cash flow hedge is considered highly effective in offsetting the variability of the forecasted cash flows attributable to the hedged risk when the regression analysis results in a coefficient of determination of 80% or greater and a slope between 80% and 125%.

For qualifying cash flow hedges, the gains or losses on derivatives are included in "Cash flow hedges" within the consolidated statements of comprehensive income. Such gains or losses are reclassified to interest income/expense within the consolidated statements of earnings in the same period that the forecasted hedged cash flows impact earnings.

The gains/(losses) included within other comprehensive income/(loss) and the gains/(losses) reclassified to earnings from accumulated other comprehensive income/(loss) related to cash flow hedges were not material for 2024 and are not expected to be material for 2025. The maximum length of time over which the forecasted cash flows are hedged is approximately one year.

Note 8.**Investments**

Investments includes debt securities classified as available-for-sale and held-to-maturity that are generally held in connection with the firm's asset-liability management activities. In addition, investments includes equity securities and debt instruments that are accounted for at fair value and equity securities that are accounted for under the equity method that are generally held by the firm in connection with its long-term investing activities.

The table below presents information about investments.

<i>\$ in millions</i>	As of December	
	2024	2023
Available-for-sale securities, at fair value	\$ 79,458	\$ 49,141
Held-to-maturity securities	78,713	70,310
Equity securities, at fair value	13,832	13,747
Debt instruments, at fair value	11,452	12,879
Equity-method investments	1,059	762
Total other investments	26,343	27,388
Total investments	\$ 184,514	\$ 146,839

Beginning in the fourth quarter of 2024, as the balances have increased during the year, investments are further disaggregated between available-for-sale securities, held-to-maturity securities and other investments in the consolidated balance sheets and the related cash flows are disaggregated in the consolidated statements of cash flows. Previously, the disaggregation of investments was provided in this footnote. Prior period disclosures have been conformed to the current presentation.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of investments, and Note 5 for information about investments within the fair value hierarchy.

Notes to Consolidated Financial Statements**Available-for-Sale Securities, at Fair Value**

Available-for-sale securities are accounted for at fair value, and the related unrealized fair value gains and losses are included in accumulated other comprehensive income/(loss) unless designated in a fair value hedging relationship. See Note 7 for information about available-for-sale securities that are designated in a hedging relationship.

The table below presents information about available-for-sale securities by type and tenor.

<i>\$ in millions</i>	Amortized Cost	Fair Value	Weighted Average Yield
As of December 2024			
Less than 1 year	\$ 21,176	\$ 21,011	2.62 %
1 year to 5 years	48,564	47,931	3.64 %
5 years to 10 years	6,620	6,468	3.93 %
Total U.S. government obligations	76,360	75,410	3.38 %
1 year to 5 years	4,224	3,893	1.73 %
5 years to 10 years	193	155	0.72 %
Total non-U.S. government obligations	4,417	4,048	1.69 %
Total available-for-sale securities	\$ 80,777	\$ 79,458	3.29 %
As of December 2023			
Less than 1 year	\$ 20,027	\$ 19,687	0.45 %
1 year to 5 years	27,592	26,500	1.83 %
5 years to 10 years	586	544	2.05 %
Total U.S. government obligations	48,205	46,731	1.25 %
Less than 1 year	11	11	0.01 %
1 year to 5 years	1,635	1,420	0.10 %
5 years to 10 years	1,150	979	0.84 %
Total non-U.S. government obligations	2,796	2,410	0.40 %
Total available-for-sale securities	\$ 51,001	\$ 49,141	1.21 %

In the table above:

- The weighted average yield is presented on a pre-tax basis and computed using the effective interest rate of each security at the end of the period, weighted based on the fair value of each security. The effective interest rate considers the contractual coupon, the amortization of premiums and accretion of discounts, and excludes the effect of related hedges.

- As of December 2024, the gross unrealized gains included in accumulated other comprehensive income/(loss) were not material and the gross unrealized losses included in accumulated other comprehensive income/(loss) were \$1.38 billion. Of such losses, \$983 million related to securities (fair value of \$21.10 billion, primarily consisting of U.S. government obligations) that were in a continuous unrealized loss position for 12 months or longer and \$399 million related to securities (fair value of \$32.71 billion, substantially all consisting of U.S. government obligations) that were in a continuous unrealized loss position for less than 12 months.

As of December 2023, the gross unrealized gains included in accumulated other comprehensive income/(loss) were not material and the gross unrealized losses included in accumulated other comprehensive income/(loss) were \$1.89 billion, and primarily related to U.S. government obligations in a continuous unrealized loss position for 12 months or longer.

Net unrealized gains included in other comprehensive income/(loss) were \$541 million (\$401 million, net of tax) for 2024 and \$1.65 billion (\$1.25 billion, net of tax) for 2023.

- Substantially all available-for-sale securities were classified in level 1 of the fair value hierarchy.
- If the fair value of available-for-sale securities is less than amortized cost, such securities are considered impaired. If the firm has the intent to sell the debt security, or if it is more likely than not that the firm will be required to sell the debt security before recovery of its amortized cost, the difference between the amortized cost (net of allowance, if any) and the fair value of the securities is recognized as an impairment loss in earnings. The firm did not record any such impairment losses during either 2024 or 2023. Impaired available-for-sale debt securities that the firm has the intent and ability to hold are reviewed to determine if an allowance for credit losses should be recorded. The firm considers various factors in such determination, including market conditions, changes in issuer credit ratings and severity of the unrealized losses. The firm did not record any provision for credit losses on such securities during either 2024 or 2023.

The gross realized gains and gross realized losses relating to the sales of available-for-sale securities were not material for each of 2024, 2023 and 2022. The specific identification method is used to determine realized gains on available-for-sale securities.

Notes to Consolidated Financial Statements**Held-to-Maturity Securities**

Held-to-maturity securities are accounted for at amortized cost.

The table below presents information about held-to-maturity securities by type and tenor.

<i>\$ in millions</i>	Amortized Cost	Fair Value	Weighted Average Yield
As of December 2024			
Less than 1 year	\$ 15,449	\$ 15,409	3.46 %
1 year to 5 years	42,420	41,939	3.66 %
Total government obligations	57,869	57,348	3.61 %
Greater than 10 years	20,637	20,482	5.42 %
Total U.S. agency obligations	20,637	20,482	5.42 %
1 year to 5 years	2	2	7.50 %
Greater than 10 years	205	206	5.25 %
Total securities backed by real estate	207	208	5.29 %
Total held-to-maturity securities	\$ 78,713	\$ 78,038	4.09 %
As of December 2023			
Less than 1 year	\$ 13,475	\$ 13,382	2.90 %
1 year to 5 years	54,789	54,352	3.58 %
5 years to 10 years	1,848	1,861	3.94 %
Total government obligations	70,112	69,595	3.46 %
1 year to 5 years	3	2	7.92 %
Greater than 10 years	195	195	5.98 %
Total securities backed by real estate	198	197	6.02 %
Total held-to-maturity securities	\$ 70,310	\$ 69,792	3.47 %

In the table above:

- Substantially all of the government obligations consist of U.S. government obligations.
- U.S. agency obligations consist of U.S. agency issued mortgage-backed securities.
- Substantially all of the securities backed by real estate consist of securities backed by residential real estate.
- As these securities are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these securities been included in the firm's fair value hierarchy, government obligations would have been classified in level 1, U.S. agency obligations would have been classified in level 2 and securities backed by real estate would have been primarily classified in level 2 of the fair value hierarchy.
- The weighted average yield is presented on a pre-tax basis and computed using the effective interest rate of each security at the end of the period, weighted based on the amortized cost of each security. The effective interest rate considers the contractual coupon and the amortization of premiums and accretion of discounts.

- The gross unrealized gains were \$121 million as of December 2024 and \$383 million as of December 2023. The gross unrealized losses were \$796 million as of December 2024 and \$901 million as of December 2023.
- Held-to-maturity securities are reviewed to determine if an allowance for credit losses should be recorded in the consolidated statements of earnings. The firm considers various factors in such determination, including market conditions, changes in issuer credit ratings, historical credit losses and sovereign guarantees. Provision for credit losses on such securities was not material during either 2024 or 2023.

Equity Securities and Debt Instruments, at Fair Value

Equity securities and debt instruments, at fair value are accounted for at fair value either under the fair value option or in accordance with other U.S. GAAP, and the related fair value gains and losses are recognized in the consolidated statements of earnings.

Equity Securities, at Fair Value. Equity securities, at fair value consists of the firm's public and private equity investments in corporate and real estate entities.

The table below presents information about equity securities, at fair value.

<i>\$ in millions</i>	As of December	
	2024	2023
Equity securities, at fair value	\$ 13,832	\$ 13,747
Equity Type		
Public equity	6 %	9 %
Private equity	94 %	91 %
Total	100 %	100 %
Asset Class		
Corporate	75 %	73 %
Real estate	25 %	27 %
Total	100 %	100 %

In the table above:

- Equity securities, at fair value included investments accounted for at fair value under the fair value option where the firm would otherwise apply the equity method of accounting of \$5.04 billion as of December 2024 and \$5.18 billion as of December 2023. Gains/(losses) recognized as a result of changes in the fair value of equity securities for which the fair value option was elected were \$(172) million for 2024 and \$(638) million for 2023. These gains/(losses) are included in other principal transactions.
- Equity securities, at fair value includes investments in private equity, real estate and hedge funds that are measured at NAV.
- Equity securities subject to contractual sale restrictions were not material as of both December 2024 and December 2023.

Notes to Consolidated Financial Statements

Debt Instruments, at Fair Value. Debt instruments, at fair value primarily includes mezzanine, senior and distressed debt.

The table below presents information about debt instruments, at fair value.

\$ in millions	As of December	
	2024	2023
Corporate debt securities	\$ 7,276	\$ 8,992
Securities backed by real estate	569	689
Money market instruments	1,780	1,051
Other	1,827	2,147
Total	\$ 11,452	\$ 12,879

In the table above, money market instruments primarily consist of time deposits and other primarily includes investments in credit funds that are measured at NAV.

Investments in Funds at Net Asset Value Per Share.

Equity securities and debt instruments, at fair value include investments in funds that are measured at NAV of the investment fund. The firm uses NAV to measure the fair value of fund investments when (i) the fund investment does not have a readily determinable fair value and (ii) the NAV of the investment fund is calculated in a manner consistent with the measurement principles of investment company accounting, including measurement of the investments at fair value.

Substantially all of the firm's investments in funds at NAV consist of investments in firm-sponsored private equity, credit, real estate and hedge funds where the firm co-invests with third-party investors.

Private equity funds primarily invest in a broad range of industries worldwide, including leveraged buyouts, recapitalizations, growth investments and distressed investments. Credit funds generally invest in loans and other fixed income instruments and are focused on providing private high-yield capital for leveraged and management buyout transactions, recapitalizations, financings, refinancings, acquisitions and restructurings for private equity firms, private family companies and corporate issuers. Real estate funds invest globally, primarily in real estate companies, loan portfolios, debt recapitalizations and property. Substantially all private equity, credit and real estate funds are closed-end funds in which the firm's investments are generally not eligible for redemption. Distributions will be received from these funds as the underlying assets are liquidated or distributed, the timing of which is uncertain.

The firm also invests in hedge funds, primarily multi-disciplinary hedge funds that employ a fundamental bottom-up investment approach across various asset classes and strategies. The firm's investments in hedge funds primarily include interests where the underlying assets are illiquid in nature, and proceeds from redemptions will not be received until the underlying assets are liquidated or distributed, the timing of which is uncertain.

The table below presents the fair value of investments in funds at NAV and the related unfunded commitments.

<i>\$ in millions</i>	Fair Value of Investments		Unfunded Commitments
As of December 2024			
Private equity funds	\$	881	\$ 432
Credit funds		1,281	364
Hedge funds		31	-
Real estate funds		354	159
Total	\$	2,547	\$ 955
As of December 2023			
Private equity funds	\$	875	\$ 484
Credit funds		1,733	248
Hedge funds		46	-
Real estate funds		346	65
Total	\$	3,000	\$ 797

Notes to Consolidated Financial Statements**Note 9.****Loans**

Loans includes (i) loans held for investment that are accounted for at amortized cost net of allowance for loan losses or at fair value under the fair value option and (ii) loans held for sale that are accounted for at the lower of cost or fair value. Interest on loans is recognized over the life of the loan and is recorded on an accrual basis.

The table below presents information about loans.

<i>\$ in millions</i>	Amortized Cost	Fair Value	Held For Sale	Total
As of December 2024				
Loan Type				
Corporate	\$ 28,689	\$ 467	\$ 816	\$ 29,972
Commercial real estate	28,899	424	466	29,789
Residential real estate	22,243	3,726	-	25,969
Securities-based	16,477	-	-	16,477
Other collateralized	74,008	783	316	75,107
Consumer:				
Installment	-	-	70	70
Credit cards	19,615	-	1,788	21,403
Other	1,950	60	69	2,079
Total loans, gross	191,881	5,460	3,525	200,866
Allowance for loan losses	(4,666)	-	-	(4,666)
Total loans	\$ 187,215	\$ 5,460	\$ 3,525	\$ 196,200
As of December 2023				
Loan Type				
Corporate	\$ 33,866	\$ 759	\$ 1,249	\$ 35,874
Commercial real estate	25,025	563	440	26,028
Residential real estate	21,243	4,145	-	25,388
Securities-based	14,621	-	-	14,621
Other collateralized	61,105	911	209	62,225
Consumer:				
Installment	250	-	3,048	3,298
Credit cards	17,432	-	1,929	19,361
Other	1,333	128	152	1,613
Total loans, gross	174,875	6,506	7,027	188,408
Allowance for loan losses	(5,050)	-	-	(5,050)
Total loans	\$ 169,825	\$ 6,506	\$ 7,027	\$ 183,358

In the table above:

- Loans held for investment that are accounted for at amortized cost include net deferred fees and costs, and unamortized premiums and discounts, which are amortized over the life of the loan. These amounts were less than 1% of loans accounted for at amortized cost as of both December 2024 and December 2023.
- Substantially all loans had floating interest rates as of both December 2024 and December 2023.
- During 2024, the firm sold the seller financing loan portfolio (included in installment loans). The net carrying value of such loans at the time of the sale was not material.
- During 2023, the firm sold \$3.24 billion of the Marcus loan portfolio (included in installment loans).
- During 2023, the firm sold approximately \$4.0 billion of the GreenSky loan portfolio (included in installment loans) and during 2024, sold the remaining GreenSky loan portfolio of \$3.69 billion.
- During 2023, the firm transferred approximately \$2.0 billion of the GM co-branded credit card portfolio to held for sale. During 2024, we entered into an agreement to transition the GM credit card program to another issuer. The transition is expected to be completed in the third quarter of 2025.
- During 2023, the firm purchased a portfolio of approximately \$15.0 billion of private equity capital call credit facilities (including approximately \$9.0 billion of funded loans) from the FDIC's auction of Signature Bank's loans.

Notes to Consolidated Financial Statements

The following is a description of the loan types in the table above:

- **Corporate.** Corporate loans includes term loans, revolving lines of credit, letter of credit facilities and bridge loans, and are principally used for operating and general corporate purposes, or in connection with acquisitions. Corporate loans are secured (typically by a senior lien on the assets of the borrower) or unsecured, depending on the loan purpose, the risk profile of the borrower and other factors.
- **Commercial Real Estate.** Commercial real estate loans includes originated loans that are directly or indirectly secured by hotels, retail stores, multifamily housing complexes and commercial and industrial properties. Commercial real estate loans also includes loans extended to clients who warehouse assets that are directly or indirectly backed by commercial real estate. In addition, commercial real estate includes loans purchased by the firm.
- **Residential Real Estate.** Residential real estate loans primarily includes loans extended to wealth management clients and to clients who warehouse assets that are directly or indirectly secured by residential real estate. In addition, residential real estate includes loans purchased by the firm.
- **Securities-Based.** Securities-based loans includes loans that are secured by stocks, bonds, mutual funds, and exchange-traded funds. These loans are primarily extended to the firm's wealth management clients and used for purposes other than purchasing, carrying or trading margin stocks. Securities-based loans require borrowers to post additional collateral on a daily basis (daily margin requirement) based on changes in the underlying collateral's fair value.

- **Other Collateralized.** Other collateralized loans includes loans that are backed by specific collateral (other than securities-based loans where there is a daily margin requirement and real estate loans). Such loans are extended to clients who warehouse assets that are directly or indirectly secured by corporate loans, consumer loans and other assets. Other collateralized loans also includes loans to investment funds (managed by third parties) that are collateralized by capital commitments of the funds' investors or assets held by the fund, as well as other secured loans extended to the firm's wealth management and corporate clients.
- **Installment.** Installment loans are unsecured loans that were originated by the firm.
- **Credit Cards.** Credit card loans are loans made pursuant to revolving lines of credit issued to consumers by the firm.
- **Other.** Other loans primarily includes unsecured loans extended to wealth management clients and unsecured consumer loans purchased by the firm.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of loans, and Note 5 for information about loans within the fair value hierarchy.

Credit Quality

Risk Assessment. The firm's risk assessment process includes evaluating the credit quality of its loans by Risk. For corporate loans and a majority of securities-based, real estate, other collateralized and other loans, the firm performs credit analyses which incorporate initial and ongoing evaluations of the capacity and willingness of a borrower to meet its financial obligations. These credit evaluations are performed on an annual basis or more frequently if deemed necessary as a result of events or changes in circumstances. The firm determines an internal credit rating for the borrower by considering the results of the credit evaluations and assumptions with respect to the nature of and outlook for the borrower's industry and the economic environment. For collateralized loans, the firm also takes into consideration collateral received or other credit support arrangements when determining an internal credit rating. For consumer loans and for loans that are not assigned an internal credit rating, including U.S. residential mortgage loans extended to wealth management clients, the firm reviews certain key metrics, including, but not limited to, the Fair Isaac Corporation (FICO) credit scores, loan to value ratios, delinquency status, collateral value and other risk factors.

Notes to Consolidated Financial Statements

The table below presents gross loans by an internally determined public rating agency equivalent or other credit metrics and the concentration of secured and unsecured loans.

<i>\$ in millions</i>	Investment- Grade	Non-Investment- Grade	Other Metrics/Unrated	Total
As of December 2024				
Accounting Method				
Amortized cost	\$ 113,986	\$ 45,595	\$ 32,300	\$ 191,881
Fair value	505	856	4,099	5,460
Held for sale	869	745	1,911	3,525
Total	\$ 115,360	\$ 47,196	\$ 38,310	\$ 200,866
Loan Type				
Corporate	\$ 8,601	\$ 21,370	\$ 1	\$ 29,972
Real estate:				
Commercial	18,175	11,514	100	29,789
Residential	10,227	3,375	12,367	25,969
Securities-based	12,662	320	3,495	16,477
Other collateralized	63,896	10,442	769	75,107
Consumer:				
Installment	-	-	70	70
Credit cards	-	-	21,403	21,403
Other	1,799	175	105	2,079
Total	\$ 115,360	\$ 47,196	\$ 38,310	\$ 200,866
Secured	93 %	90 %	43 %	83 %
Unsecured	7 %	10 %	57 %	17 %
Total	100 %	100 %	100 %	100 %
As of December 2023				
Accounting Method				
Amortized cost	\$ 91,324	\$ 54,200	\$ 29,351	\$ 174,875
Fair value	1,212	1,213	4,081	6,506
Held for sale	255	1,628	5,144	7,027
Total	\$ 92,791	\$ 57,041	\$ 38,576	\$ 188,408
Loan Type				
Corporate	\$ 9,408	\$ 26,328	\$ 138	\$ 35,874
Real estate:				
Commercial	12,097	13,574	357	26,028
Residential	10,771	3,217	11,400	25,388
Securities-based	10,991	561	3,069	14,621
Other collateralized	48,536	13,207	482	62,225
Consumer:				
Installment	-	-	3,298	3,298
Credit cards	-	-	19,361	19,361
Other	988	154	471	1,613
Total	\$ 92,791	\$ 57,041	\$ 38,576	\$ 188,408
Secured	91 %	92 %	40 %	81 %
Unsecured	9 %	8 %	60 %	19 %
Total	100 %	100 %	100 %	100 %

In the table above:

- Substantially all residential real estate loans included in the other metrics/unrated category consists of loans extended to wealth management clients. As of both December 2024 and December 2023, substantially all of such loans had a loan-to-value ratio of less than 80% and were performing in accordance with the contractual terms. Additionally, as of both December 2024 and December 2023, the vast majority of such loans had a FICO credit score of greater than 740.
- The vast majority of securities-based loans included in the other metrics/unrated category had a loan-to-value ratio of less than 80% and were performing in accordance with the contractual terms as of both December 2024 and December 2023.
- For installment and credit card loans included in the other metrics/unrated category, the evaluation of credit quality incorporates the borrower's FICO credit score. FICO credit scores are periodically refreshed by the firm to assess the updated creditworthiness of the borrower. See "Vintage" below for information about installment and credit card loans by FICO credit scores.

The firm also assigns a regulatory risk rating to its loans based on the definitions provided by the U.S. federal bank regulatory agencies. Total loans included 93% of loans as of December 2024 and 92% of loans as of December 2023 that were rated pass/non-criticized.

Notes to Consolidated Financial Statements

Vintage. The tables below present gross loans accounted for at amortized cost (excluding installment and credit card loans) by an internally determined public rating agency equivalent or other credit metrics and origination year for term loans.

As of December 2024					
<i>\$ in millions</i>	Investment-Grade	Non-Investment-Grade	Other Metrics/Unrated	Total	
2024	\$ 1,447	\$ 2,545	\$ -	\$	3,992
2023	1,522	1,446	-		2,968
2022	727	2,084	1		2,812
2021	215	2,244	-		2,459
2020	102	1,287	-		1,389
2019 or earlier	376	1,910	-		2,286
Revolving	4,001	8,696	-		12,697
Revolving converted to term	-	86	-		86
Corporate	8,390	20,298	1		28,689
2024	2,988	1,669	27		4,684
2023	1,079	1,252	-		2,331
2022	1,018	1,664	-		2,682
2021	624	1,901	-		2,525
2020	273	766	-		1,039
2019 or earlier	972	738	18		1,728
Revolving	10,355	2,944	5		13,304
Revolving converted to term	201	405	-		606
Commercial real estate	17,510	11,339	50		28,899
2024	713	584	1,746		3,043
2023	224	9	1,414		1,647
2022	87	46	2,537		2,670
2021	21	122	2,598		2,741
2020	-	6	41		47
2019 or earlier	-	19	306		325
Revolving	9,182	2,588	-		11,770
Residential real estate	10,227	3,374	8,642		22,243
2024	1,528	78	16		1,622
2023	35	-	-		35
2022	5	-	-		5
2019 or earlier	-	22	-		22
Revolving	11,094	220	3,479		14,793
Securities-based	12,662	320	3,495		16,477
2024	5,033	2,009	151		7,193
2023	3,816	1,279	150		5,245
2022	910	144	42		1,096
2021	546	739	72		1,357
2020	854	566	26		1,446
2019 or earlier	196	45	25		266
Revolving	51,373	5,211	15		56,599
Revolving converted to term	710	96	-		806
Other collateralized	63,438	10,089	481		74,008
2024	257	73	-		330
2023	113	10	-		123
2022	36	6	-		42
2021	16	-	16		32
2020	-	2	-		2
Revolving	1,337	84	-		1,421
Other	1,759	175	16		1,950
Total	\$ 113,986	\$ 45,595	\$ 12,685	\$	172,266
Percentage of total	66 %	27 %	7 %		100 %

As of December 2023					
<i>\$ in millions</i>	Investment-Grade	Non-Investment-Grade	Other Metrics/Unrated	Total	
2023	\$ 2,475	\$ 1,912	\$ 16	\$	4,403
2022	1,223	3,284	-		4,507
2021	848	4,045	-		4,893
2020	306	2,098	-		2,404
2019	45	1,909	-		1,954
2018 or earlier	371	2,102	-		2,473
Revolving	3,857	9,355	20		13,232
Corporate	9,125	24,705	36		33,866
2023	553	1,547	38		2,138
2022	1,251	2,838	-		4,089
2021	1,134	2,661	-		3,795
2020	271	1,234	-		1,505
2019	430	631	-		1,061
2018 or earlier	832	744	-		1,576
Revolving	7,129	3,192	309		10,630
Revolving converted to term	231	-	-		231
Commercial real estate	11,831	12,847	347		25,025
2023	619	54	1,627		2,300
2022	108	41	2,687		2,836
2021	22	249	2,724		2,995
2020	3	23	81		107
2019	6	-	89		95
2018 or earlier	-	20	254		274
Revolving	9,813	2,823	-		12,636
Residential real estate	10,571	3,210	7,462		21,243
2023	8	-	-		8
2022	5	-	-		5
2018 or earlier	-	303	-		303
Revolving	10,978	258	3,069		14,305
Securities-based	10,991	561	3,069		14,621
2023	5,412	2,767	245		8,424
2022	1,940	293	69		2,302
2021	1,883	845	102		2,830
2020	1,256	469	32		1,757
2019	177	74	9		260
2018 or earlier	436	66	21		523
Revolving	35,605	8,242	1		43,848
Revolving converted to term	1,161	-	-		1,161
Other collateralized	47,870	12,756	479		61,105
2023	60	21	-		81
2022	67	9	-		76
2021	6	8	51		65
2020	-	3	218		221
2019	-	-	4		4
2018 or earlier	-	-	3		3
Revolving	803	80	-		883
Other	936	121	276		1,333
Total	\$ 91,324	\$ 54,200	\$ 11,669	\$	157,193
Percentage of total	58 %	35 %	7 %		100 %

Notes to Consolidated Financial Statements

The table below presents gross installment loans accounted for at amortized cost by refreshed FICO credit scores and origination year and gross credit card loans by refreshed FICO credit scores.

<i>\$ in millions</i>	Greater than or equal to 660		Less than 660		Total
As of December 2024					
Credit cards	\$	13,090	\$	6,525	\$ 19,615
Percentage		67 %		33 %	100 %
As of December 2023					
2023	\$	79	\$	10	\$ 89
2022		132		18	150
2021 or earlier		11		-	11
Installment		222		28	250
Credit cards		11,119		6,313	17,432
Total	\$	11,341	\$	6,341	\$ 17,682
Percentage of total:					
Installment		89 %		11 %	100 %
Credit cards		64 %		36 %	100 %
Total		64 %		36 %	100 %

In the table above, credit card loans consist of revolving lines of credit.

Credit Concentrations. The table below presents the concentration of gross loans by region.

<i>\$ in millions</i>	Carrying Value	Americas	EMEA	Asia	Total
As of December 2024					
Corporate	\$ 29,972	66 %	26 %	8 %	100 %
Commercial real estate	29,789	78 %	18 %	4 %	100 %
Residential real estate	25,969	94 %	5 %	1 %	100 %
Securities-based	16,477	76 %	24 %	-	100 %
Other collateralized	75,107	86 %	12 %	2 %	100 %
Consumer:					
Installment	70	100 %	-	-	100 %
Credit cards	21,403	100 %	-	-	100 %
Other	2,079	96 %	4 %	-	100 %
Total	\$ 200,866	83 %	14 %	3 %	100 %
As of December 2023					
Corporate	\$ 35,874	63 %	29 %	8 %	100 %
Commercial real estate	26,028	80 %	17 %	3 %	100 %
Residential real estate	25,388	95 %	4 %	1 %	100 %
Securities-based	14,621	79 %	20 %	1 %	100 %
Other collateralized	62,225	89 %	10 %	1 %	100 %
Consumer:					
Installment	3,298	100 %	-	-	100 %
Credit cards	19,361	100 %	-	-	100 %
Other	1,613	97 %	3 %	-	100 %
Total	\$ 188,408	84 %	13 %	3 %	100 %

In the table above:

- The top five industry concentrations for corporate loans as of December 2024 were 24% for technology, media & telecommunications, 16% for diversified industrials, 14% for real estate, 9% for financial institutions and 9% for healthcare.
- The top five industry concentrations for corporate loans as of December 2023 were 25% for technology, media & telecommunications, 17% for diversified industrials, 13% for real estate, 11% for consumer & retail and 9% for healthcare.

Nonaccrual, Past Due and Modified Loans. Loans accounted for at amortized cost (other than credit card loans) are placed on nonaccrual status when it is probable that the firm will not collect all principal and interest due under the contractual terms, regardless of the delinquency status or if a loan is past due for 90 days or more, unless the loan is both well collateralized and in the process of collection. At that time, all accrued but uncollected interest is reversed against interest income and interest subsequently collected is recognized on a cash basis to the extent the loan balance is deemed collectible. Otherwise, all cash received is used to reduce the outstanding loan balance. A loan is considered past due when a principal or interest payment has not been made according to its contractual terms. Credit card loans are not placed on nonaccrual status and accrue interest until the loan is paid in full or is charged off.

The table below presents information about past due loans.

<i>\$ in millions</i>	30-89 days	90 days or more	Total
As of December 2024			
Corporate	\$ -	\$ 15	\$ 15
Commercial real estate	186	286	472
Residential real estate	3	18	21
Securities-based	6	-	6
Other collateralized	-	5	5
Consumer:			
Credit cards	417	456	873
Total	\$ 612	\$ 780	\$ 1,392
Total divided by gross loans at amortized cost			
			0.7 %
As of December 2023			
Corporate	\$ 45	\$ 73	\$ 118
Commercial real estate	137	352	489
Residential real estate	12	4	16
Securities-based	2	-	2
Other collateralized	9	7	16
Consumer:			
Installment	6	7	13
Credit cards	463	486	949
Other	7	11	18
Total	\$ 681	\$ 940	\$ 1,621
Total divided by gross loans at amortized cost			
			0.9 %

Notes to Consolidated Financial Statements

The table below presents information about nonaccrual loans.

\$ in millions	As of December	
	2024	2023
Corporate	\$ 1,977	\$ 1,779
Commercial real estate	800	1,466
Residential real estate	104	19
Securities-based	2	-
Other collateralized	757	860
Other	12	17
Total	\$ 3,652	\$ 4,141
Total divided by gross loans at amortized cost	1.9 %	2.4 %

In the table above:

- Nonaccrual loans included \$322 million as of December 2024 and \$600 million as of December 2023 of loans that were 30 days or more past due.
- Loans that were 90 days or more past due and still accruing were not material as of both December 2024 and December 2023.
- Allowance for loan losses as a percentage of total nonaccrual loans was 127.8% as of December 2024 and 122.0% as of December 2023.
- Commercial real estate, residential real estate, securities-based and other collateralized loans are collateral dependent loans and the repayment of such loans is generally expected to be provided by the operation or sale of the underlying collateral. The allowance for credit losses for such nonaccrual loans is determined by considering the fair value of the collateral less estimated cost to sell, if applicable. See Note 4 for further information about fair value measurements.

The firm may modify the terms of a loan agreement for a borrower experiencing financial difficulty. Such modifications may include, among other things, forbearance of interest or principal, payment extensions or interest rate reductions.

The table below presents the carrying value of loans, as of both December 2024 and December 2023, that were modified during either 2024 or 2023.

\$ in millions	Year Ended December	
	2024	2023
Modified loans	\$ 1,208	\$ 846

In the table above:

- Loan modifications during both 2024 and 2023 were primarily in the form of term extensions. These extensions increased the weighted average term by 19 months for loans modified during 2024 and by 16 months for loans modified during 2023.
- Substantially all of the modified loans were related to corporate loans, commercial real estate loans and credit cards. Modified loans represented approximately 2% of corporate loans (at amortized cost), and approximately 1% of both commercial real estate loans (at amortized cost) and credit card loans (at amortized cost).
- Lending commitments related to modified loans were \$156 million as of December 2024 and were not material as of December 2023.
- During 2024, loans that defaulted after being modified were not material. During 2023, the firm charged off approximately \$100 million of loans that had defaulted after being modified. Substantially all of the remaining modified loans were performing in accordance with the modified contractual terms as of both December 2024 and December 2023.

Notes to Consolidated Financial Statements**Allowance for Credit Losses**

The firm's allowance for credit losses consists of the allowance for losses on loans and lending commitments accounted for at amortized cost. Loans and lending commitments accounted for at fair value or accounted for at the lower of cost or fair value are not subject to an allowance for credit losses.

To determine the allowance for credit losses, the firm classifies its loans and lending commitments accounted for at amortized cost into wholesale and consumer portfolios. These portfolios represent the level at which the firm has developed and documented its methodology to determine the allowance for credit losses. The allowance for credit losses is measured on a collective basis for loans that exhibit similar risk characteristics using a modeled approach and on an asset-specific basis for loans that do not share similar risk characteristics.

The allowance for credit losses takes into account the weighted average of a range of forecasts of future economic conditions over the expected life of the loans and lending commitments. The expected life of each loan or lending commitment is determined based on the contractual term adjusted for extension options or demand features, or is modeled in the case of revolving credit card loans. The forecasts include baseline, favorable and adverse economic scenarios over a three-year period. For loans with expected lives beyond three years, the model reverts to historical loss information based on a non-linear modeled approach. The forecasted economic scenarios consider a number of risk factors relevant to the wholesale and consumer portfolios described below. The firm applies judgment in weighing individual scenarios each quarter based on a variety of factors, including the firm's internally derived economic outlook, market consensus, recent macroeconomic conditions and industry trends.

The allowance for credit losses also includes qualitative components which allow management to reflect the uncertain nature of economic forecasting, capture uncertainty regarding model inputs, and account for model imprecision and concentration risk. The qualitative factors considered by management include, among others, changes and trends in loan portfolios, uncertainties associated with the macroeconomic and geopolitical environments, credit concentrations, changes in volume and severity of past due and criticized loans, idiosyncratic events and deterioration within an industry or region.

Management's estimate of credit losses entails judgment about the expected life of the loan and loan collectability at the reporting dates, and there are uncertainties inherent in those judgments. The allowance for credit losses is subject to a governance process that involves senior management within Risk and Controllers. Personnel within Risk are responsible for forecasting the economic variables that underlie the economic scenarios that are used in the modeling of expected credit losses. While management uses the best information available to determine this estimate, future adjustments to the allowance may be necessary based on, among other things, changes in the economic environment or variances between actual results and the original assumptions used.

The table below presents gross loans and lending commitments accounted for at amortized cost by portfolio.

\$ in millions	As of December			
	2024		2023	
	Loans	Lending Commitments	Loans	Lending Commitments
Wholesale				
Corporate	\$ 28,689	\$ 156,562	\$ 33,866	\$ 141,976
Commercial real estate	28,899	4,969	25,025	3,379
Residential real estate	22,243	1,742	21,243	1,431
Securities-based	16,477	1,542	14,621	691
Other collateralized	74,008	33,136	61,105	23,020
Other	1,950	872	1,333	888
Consumer				
Installment	-	-	250	1
Credit cards	19,615	63,781	17,432	56,479
Total	\$ 191,881	\$ 262,604	\$ 174,875	\$ 227,865

In the table above, wholesale loans included \$3.65 billion as of December 2024 and \$4.14 billion as of December 2023 of nonaccrual loans for which the allowance for credit losses was measured on an asset-specific basis. The allowance for credit losses on these loans was \$735 million as of December 2024 and \$778 million as of December 2023. These loans included \$585 million as of December 2024 and \$625 million as of December 2023 of loans which did not require a reserve as the loan was deemed to be recoverable.

See Note 18 for further information about lending commitments.

Notes to Consolidated Financial Statements

The following is a description of the methodology used to calculate the allowance for credit losses:

Wholesale. The allowance for credit losses for wholesale loans and lending commitments that exhibit similar risk characteristics is measured using a modeled approach. These models determine the probability of default and loss given default based on various risk factors, including internal credit ratings, industry default and loss data, expected life, macroeconomic indicators, the borrower's capacity to meet its financial obligations, the borrower's country of risk and industry, loan seniority and collateral type. For lending commitments, the methodology also considers the probability of drawdowns or funding. In addition, for loans backed by real estate, risk factors include the loan-to-value ratio, debt service ratio and home price index. The most significant inputs to the forecast model for wholesale loans and lending commitments include unemployment rates, GDP, credit spreads, commercial and industrial delinquency rates, short- and long-term interest rates, and oil prices.

The allowance for loan losses for wholesale loans that do not share similar risk characteristics, such as nonaccrual loans, is calculated using the present value of expected future cash flows discounted at the loan's effective rate, the observable market price of the loan, or, in the case of collateral dependent loans, the fair value of the collateral less estimated costs to sell, if applicable. Wholesale loans are charged off against the allowance for loan losses when deemed to be uncollectible.

Consumer. The allowance for credit losses for consumer loans that exhibit similar risk characteristics is calculated using a modeled approach which classifies consumer loans into pools based on borrower-related and exposure-related characteristics that differentiate a pool's risk characteristics from other pools. The factors considered in determining a pool are generally consistent with the risk characteristics used for internal credit risk measurement and management and include key metrics, such as FICO credit scores, delinquency status, loan vintage and macroeconomic indicators. The most significant inputs to the forecast model for consumer loans include unemployment rates and delinquency rates. The expected life of revolving credit card loans is determined by modeling expected future draws and the timing and amount of repayments allocated to the funded balance. The firm does not recognize an allowance for credit losses on credit card lending commitments as they are cancellable by the firm.

Credit card loans are charged off when they are 180 days past due. Installment loans were charged off when they were 120 days past due.

Allowance for Credit Losses Rollforward

The table below presents information about the allowance for credit losses.

<i>\$ in millions</i>	Wholesale	Consumer	Total
Year Ended December 2024			
Allowance for loan losses			
Beginning balance	\$ 2,576	\$ 2,474	\$ 5,050
Charge-offs	(169)	(1,471)	(1,640)
Recoveries	121	104	225
Net (charge-offs)/recoveries	(48)	(1,367)	(1,415)
Provision	(292)	1,540	1,248
Other	(137)	(80)	(217)
Ending balance	\$ 2,099	\$ 2,567	\$ 4,666
Allowance ratio	1.2 %	13.1 %	2.4 %
Net charge-off ratio	-	7.6 %	0.8 %
Allowance for losses on lending commitments			
Beginning balance	\$ 620	\$ -	\$ 620
Provision	56	-	56
Other	(2)	-	(2)
Ending balance	\$ 674	\$ -	\$ 674
Year Ended December 2023			
Allowance for loan losses			
Beginning balance	\$ 2,562	\$ 2,981	\$ 5,543
Charge-offs	(455)	(1,246)	(1,701)
Recoveries	55	98	153
Net (charge-offs)/recoveries	(400)	(1,148)	(1,548)
Provision	540	641	1,181
Other	(126)	-	(126)
Ending balance	\$ 2,576	\$ 2,474	\$ 5,050
Allowance ratio	1.6 %	14.0 %	2.9 %
Net charge-off ratio	0.3 %	5.5 %	0.9 %
Allowance for losses on lending commitments			
Beginning balance	\$ 711	\$ 63	\$ 774
Provision	(90)	(63)	(153)
Other	(1)	-	(1)
Ending balance	\$ 620	\$ -	\$ 620

In the table above:

- Other primarily represented the reduction to the allowance related to loans transferred to held for sale.
- The allowance ratio is calculated by dividing the allowance for loan losses by gross loans accounted for at amortized cost.
- The net charge-off ratio is calculated by dividing net (charge-offs)/recoveries by average gross loans accounted for at amortized cost.

Notes to Consolidated Financial Statements**Forecast Model Inputs as of December 2024**

When modeling expected credit losses, the firm employs a weighted, multi-scenario forecast, which includes baseline, adverse and favorable economic scenarios. As of December 2024, this multi-scenario forecast was weighted towards the baseline and adverse economic scenarios.

The table below presents the forecasted U.S. unemployment and U.S. GDP growth rates used in the baseline economic scenario of the forecast model.

As of December 2024	
U.S. unemployment rate	
Forecast for the quarter ended:	
June 2025	4.4 %
December 2025	4.4 %
June 2026	4.3 %
Growth in U.S. GDP	
Forecast for the year:	
2025	2.0 %
2026	1.7 %
2027	1.7 %

The adverse economic scenario of the forecast model reflects a global recession in the first quarter of 2025 through the first quarter of 2026, resulting in an economic contraction and rising unemployment rates. In this scenario, the U.S. unemployment rate peaks at approximately 7.4% during the first quarter of 2026 and the maximum decline in the quarterly U.S. GDP relative to the fourth quarter of 2024 is approximately 2.7%, which occurs during the fourth quarter of 2025.

In the table above:

- U.S. unemployment rate represents the rate forecasted as of the respective quarter-end.
- Growth in U.S. GDP represents the year-over-year growth rate forecasted for the respective years.
- While the U.S. unemployment and U.S. GDP growth rates are significant inputs to the forecast model, the model contemplates a variety of other inputs across a range of scenarios to provide a forecast of future economic conditions. Given the complex nature of the forecasting process, no single economic variable can be viewed in isolation and independently of other inputs.

Allowance for Credit Losses Commentary

Year Ended December 2024. The allowance for credit losses decreased by \$330 million during 2024, primarily reflecting a reserve release relating to the wholesale portfolio due to improved macroeconomic environment, partially offset by growth in the credit card portfolio.

Charge-offs for 2024 for wholesale loans (principally related to term loans originated in 2022 and 2021) were primarily related to corporate loans and charge-offs for consumer loans were primarily related to credit cards.

Year Ended December 2023. The allowance for credit losses decreased by \$647 million during 2023, reflecting a net release related to the GreenSky installment loan portfolio (including a reserve reduction of \$637 million related to the partial sale and transfer of the portfolio to held for sale), a reserve reduction of \$442 million associated with the sale of Marcus loans, a reserve reduction of \$160 million related to the transfer of the GM co-branded credit card portfolio to held for sale, a reserve release in the consumer portfolio based on actual repayment experience and lower balances in corporate loans due to sales and paydowns, partially offset by asset-specific provisions and ratings downgrades in the wholesale portfolio and seasoning of the credit card portfolio.

Charge-offs for 2023 for wholesale loans (principally related to term loans originated in 2021 and revolving loans) were primarily related to corporate loans and charge-offs for consumer loans were primarily related to credit cards.

Estimated Fair Value

The table below presents the estimated fair value of loans that are not accounted for at fair value and in what level of the fair value hierarchy they would have been classified if they had been included in the firm's fair value hierarchy.

\$ in millions	Carrying Value	Estimated Fair Value		
		Level 2	Level 3	Total
<u>As of December 2024</u>				
Amortized cost	\$ 187,215	\$ 99,790	\$ 89,540	\$ 189,330
Held for sale	\$ 3,525	\$ 2,928	\$ 600	\$ 3,528
<u>As of December 2023</u>				
Amortized cost	\$ 169,825	\$ 88,485	\$ 83,288	\$ 171,773
Held for sale	\$ 7,027	\$ 3,992	\$ 3,038	\$ 7,030

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of loans, and Note 5 for information about loans within the fair value hierarchy.

Notes to Consolidated Financial Statements**Note 10.****Fair Value Option****Other Financial Assets and Liabilities at Fair Value**

In addition to trading assets and liabilities, and certain investments and loans, the firm accounts for certain of its other financial assets and liabilities at fair value, substantially all under the fair value option. The primary reasons for electing the fair value option are to:

- Reflect economic events in earnings on a timely basis;
- Mitigate volatility in earnings from using different measurement attributes (e.g., transfers of financial assets accounted for as financings are recorded at fair value, whereas the related secured financing would be recorded on an accrual basis absent electing the fair value option); and
- Address simplification and cost-benefit considerations (e.g., accounting for hybrid financial instruments at fair value in their entirety versus bifurcation of embedded derivatives and hedge accounting for debt hosts).

Hybrid financial instruments are instruments that contain bifurcable embedded derivatives and do not require settlement by physical delivery of nonfinancial assets (e.g., physical commodities). Unless the firm has elected to account for the entire hybrid financial instrument at fair value under the fair value option, the embedded derivative is bifurcated from the associated host contract, the derivative is accounted for at fair value and the host contract is accounted for at amortized cost, adjusted for the effective portion of any fair value hedges.

Other financial assets and liabilities accounted for at fair value under the fair value option include:

- Repurchase agreements and substantially all resale agreements;
- Certain securities borrowed and loaned transactions;
- Certain customer and other receivables and certain other assets and liabilities;
- Certain time deposits (deposits with no stated maturity are not eligible for a fair value option election), including structured certificates of deposit, which are hybrid financial instruments;
- Substantially all other secured financings, including structured financing arrangements and transfers of assets accounted for as financings; and
- Certain unsecured short- and long-term borrowings, substantially all of which are hybrid financial instruments.

See Note 4 for an overview of the firm's fair value measurement policies, valuation techniques and significant inputs used to determine the fair value of other financial assets and liabilities, and Note 5 for information about other financial assets and liabilities within the fair value hierarchy.

Gains and Losses on Other Financial Assets and Liabilities Accounted for at Fair Value Under the Fair Value Option

The table below presents the gains and losses recognized in earnings as a result of the election to apply the fair value option to certain financial assets and liabilities.

\$ in millions	Year Ended December		
	2024	2023	2022
Unsecured short-term borrowings	\$ (2,749)	\$ (4,341)	\$ 4,055
Unsecured long-term borrowings	(3,295)	(4,937)	6,506
Other	(356)	(513)	1,072
Total	\$ (6,400)	\$ (9,791)	\$ 11,633

In the table above:

- Gains/(losses) were substantially all included in market making.
- Gains/(losses) exclude contractual interest, which is included in interest income and interest expense, for all instruments other than hybrid financial instruments. See Note 23 for further information about interest income and interest expense.
- Gains/(losses) included in unsecured short- and long-term borrowings were substantially all related to the embedded derivative component of hybrid financial instruments. These gains and losses would have been recognized under other U.S. GAAP even if the firm had not elected to account for the entire hybrid financial instrument at fair value.
- Gains/(losses) included in other were primarily related to resale and repurchase agreements, deposits and other secured financings.
- Other financial assets and liabilities at fair value are frequently economically hedged with trading assets and liabilities. Accordingly, gains or losses on such other financial assets and liabilities can be partially offset by gains or losses on trading assets and liabilities. As a result, gains or losses on other financial assets and liabilities do not necessarily represent the overall impact on the firm's results of operations, liquidity or capital resources.

Notes to Consolidated Financial Statements

Gains/(losses) on trading assets and liabilities accounted for at fair value under the fair value option are included in market making. See Note 6 for further information about gains/(losses) from market making. See Note 8 for information about gains/(losses) on equity securities and Note 9 for information about gains/(losses) on loans which are accounted for at fair value under the fair value option.

Long-Term Debt Instruments

The difference between the aggregate contractual principal amount and the related fair value of long-term other secured financings, for which the fair value option was elected was not material as of December 2024, and the aggregate contractual principal amount exceeded the fair value by \$147 million as of December 2023.

The aggregate contractual principal amount of unsecured long-term borrowings, for which the fair value option was elected, exceeded the related fair value by \$4.23 billion as of December 2024 and \$3.37 billion as of December 2023.

These debt instruments include both principal-protected and non-principal-protected long-term borrowings.

Debt Valuation Adjustment

The firm calculates the fair value of financial liabilities for which the fair value option is elected by discounting future cash flows at a rate which incorporates the firm's credit spreads.

The table below presents information about the net debt valuation adjustment (DVA) gains/(losses) on financial liabilities for which the fair value option was elected.

\$ in millions	Year Ended December		
	2024	2023	2022
Pre-tax DVA	\$ (351)	\$ (1,355)	\$ 1,882
After-tax DVA	\$ (263)	\$ (1,015)	\$ 1,403

In the table above:

- After-tax DVA is included in debt valuation adjustment in the consolidated statements of comprehensive income.
- The gains/(losses) reclassified to market making in the consolidated statements of earnings from accumulated other comprehensive income/(loss) upon extinguishment of such financial liabilities were not material for each of 2024, 2023 and 2022.

Loans and Lending Commitments

The table below presents the difference between the aggregate fair value and the aggregate contractual principal amount for loans (included in trading assets and loans in the consolidated balance sheets) for which the fair value option was elected.

\$ in millions	As of December	
	2024	2023
Performing loans		
Aggregate contractual principal in excess of fair value	\$ 965	\$ 1,893
Loans on nonaccrual status and/or more than 90 days past due		
Aggregate contractual principal in excess of fair value	\$ 2,402	\$ 2,305
Aggregate fair value	\$ 1,454	\$ 1,508

In the table above, the aggregate contractual principal amount of loans on nonaccrual status and/or more than 90 days past due (which excludes loans carried at zero fair value and considered uncollectible) exceeds the related fair value primarily because the firm regularly purchases loans, such as distressed loans, at values significantly below the contractual principal amounts.

The total contractual amount of unfunded lending commitments for which the fair value option was elected was \$568 million as of December 2024 and \$878 million as of December 2023, and the related fair value of these lending commitments was not material as of both December 2024 and December 2023. See Note 18 for further information about lending commitments.

Impact of Credit Spreads on Loans and Lending Commitments

The estimated net loss attributable to changes in instrument-specific credit spreads on loans and lending commitments for which the fair value option was elected was not material for 2024, \$125 million for 2023 and \$281 million for 2022. The firm generally calculates the fair value of loans and lending commitments for which the fair value option is elected by discounting future cash flows at a rate which incorporates the instrument-specific credit spreads. For floating-rate loans and lending commitments, substantially all changes in fair value are attributable to changes in instrument-specific credit spreads, whereas for fixed-rate loans and lending commitments, changes in fair value are also attributable to changes in interest rates.

Notes to Consolidated Financial Statements**Note 11.****Collateralized Agreements and Financings**

Collateralized agreements are resale agreements and securities borrowed. Collateralized financings are repurchase agreements, securities loaned and other secured financings. The firm enters into these transactions in order to, among other things, facilitate client activities, invest excess cash, acquire securities to cover short positions and finance certain firm activities.

Collateralized agreements and financings with the same settlement date are presented on a net-by-counterparty basis when such transactions meet certain settlement criteria and are subject to netting agreements. Interest on collateralized agreements, which is included in interest income, and collateralized financings, which is included in interest expense, is recognized over the life of the transaction. See Note 23 for further information about interest income and interest expense.

Resale and Repurchase Agreements

A resale agreement is a transaction in which the firm purchases financial instruments from a seller, typically in exchange for cash, and simultaneously enters into an agreement to resell the same or substantially the same financial instruments to the seller at a stated price plus accrued interest at a future date.

A repurchase agreement is a transaction in which the firm sells financial instruments to a buyer, typically in exchange for cash, and simultaneously enters into an agreement to repurchase the same or substantially the same financial instruments from the buyer at a stated price plus accrued interest at a future date.

Even though repurchase and resale agreements (including “repos- and reverses-to-maturity”) involve the legal transfer of ownership of financial instruments, they are accounted for as financing arrangements because they require the financial instruments to be repurchased or resold before or at the maturity of the agreement. The financial instruments purchased or sold in resale and repurchase agreements typically include U.S. government and agency obligations, and investment-grade sovereign obligations.

The firm receives financial instruments purchased under resale agreements and makes delivery of financial instruments sold under repurchase agreements. To mitigate credit exposure, the firm monitors the market value of these financial instruments on a daily basis, and delivers or obtains additional collateral due to changes in the market value of the financial instruments, as appropriate. For resale agreements, the firm typically requires collateral with a fair value approximately equal to the carrying value of the relevant assets in the consolidated balance sheets.

Repurchase agreements and substantially all resale agreements are recorded at fair value under the fair value option. See Note 5 for further information about repurchase and resale agreements.

Securities Borrowed and Loaned Transactions

In a securities borrowed transaction, the firm borrows securities from a counterparty in exchange for cash or securities. When the firm returns the securities, the counterparty returns the cash or securities. Interest is generally paid periodically over the life of the transaction.

In a securities loaned transaction, the firm lends securities to a counterparty in exchange for cash or securities. When the counterparty returns the securities, the firm returns the cash or securities posted as collateral. Interest is generally paid periodically over the life of the transaction.

In a transaction where the firm lends securities and receives securities that can be delivered or pledged as collateral, the firm recognizes the securities received within securities borrowed and the obligation to return those securities within securities loaned in the consolidated balance sheets.

The firm receives securities borrowed and makes delivery of securities loaned. To mitigate credit exposure, the firm monitors the market value of these securities on a daily basis, and delivers or obtains additional collateral due to changes in the market value of the securities, as appropriate. For securities borrowed transactions, the firm typically requires collateral with a fair value approximately equal to the carrying value of the securities borrowed transaction.

Securities borrowed and loaned within FICC financing are recorded at fair value under the fair value option. See Note 5 for further information about securities borrowed and loaned accounted for at fair value.

Substantially all of the securities borrowed and loaned within Equities financing are recorded based on the amount of cash collateral advanced or received plus accrued interest. The firm also reviews such securities borrowed to determine if an allowance for credit losses should be recorded by taking into consideration the fair value of collateral received. As these agreements generally can be terminated on demand, they exhibit little, if any, sensitivity to changes in interest rates. Therefore, the carrying value of such agreements approximates fair value. As these agreements are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these agreements been included in the firm's fair value hierarchy, they would have been classified in level 2 as of both December 2024 and December 2023.

Notes to Consolidated Financial Statements**Offsetting Arrangements**

The table below presents resale and repurchase agreements and securities borrowed and loaned transactions included in the consolidated balance sheets, as well as the amounts not offset in the consolidated balance sheets.

\$ in millions	Assets		Liabilities	
	Resale agreements	Securities borrowed	Repurchase agreements	Securities loaned
As of December 2024				
Included in the consolidated balance sheets				
Gross carrying value	\$ 313,924	\$ 205,259	\$ 408,242	\$ 66,674
Counterparty netting	(133,862)	(10,614)	(133,862)	(10,614)
Total	180,062	194,645	274,380	56,060
Amounts not offset	(176,390)	(187,474)	(270,150)	(55,910)
Total	\$ 3,672	\$ 7,171	\$ 4,230	\$ 150
As of December 2023				
Included in the consolidated balance sheets				
Gross carrying value	\$ 315,112	\$ 199,753	\$ 341,194	\$ 60,816
Counterparty netting	(91,307)	(333)	(91,307)	(333)
Total	223,805	199,420	249,887	60,483
Amounts not offset	(218,494)	(192,291)	(246,634)	(60,180)
Total	\$ 5,311	\$ 7,129	\$ 3,253	\$ 303

In the table above:

- Substantially all of the gross carrying values of these arrangements are subject to enforceable netting agreements.
- Where the firm has received or posted collateral under credit support agreements, but has not yet determined such agreements are enforceable, the related collateral has not been netted.
- Amounts not offset includes counterparty netting that does not meet the criteria for netting under U.S. GAAP and the fair value of collateral received or posted subject to enforceable credit support agreements.
- Resale agreements included in the consolidated balance sheets of \$179.79 billion as of December 2024 and \$223.54 billion as of December 2023 and all repurchase agreements included in the consolidated balance sheets are carried at fair value under the fair value option. See Note 5 for further information about resale agreements and repurchase agreements accounted for at fair value.
- Securities borrowed included in the consolidated balance sheets of \$46.90 billion as of December 2024 and \$44.93 billion as of December 2023, and securities loaned included in the consolidated balance sheets of \$10.25 billion as of December 2024 and \$8.93 billion as of December 2023 were at fair value under the fair value option. See Note 5 for further information about securities borrowed and securities loaned accounted for at fair value.

Gross Carrying Value of Repurchase Agreements and Securities Loaned

The table below presents the gross carrying value of repurchase agreements and securities loaned by class of collateral pledged.

\$ in millions	Repurchase agreements	Securities loaned
As of December 2024		
Money market instruments	\$ 61	\$ -
U.S. government and agency obligations	276,341	-
Non-U.S. government and agency obligations	95,812	461
Securities backed by commercial real estate	407	-
Securities backed by residential real estate	1,154	-
Corporate debt securities	11,521	376
State and municipal obligations	573	-
Other debt obligations	289	-
Equity securities	22,084	65,837
Total	\$ 408,242	\$ 66,674
As of December 2023		
Money market instruments	\$ 3	\$ -
U.S. government and agency obligations	228,718	216
Non-U.S. government and agency obligations	85,230	376
Securities backed by commercial real estate	135	-
Securities backed by residential real estate	641	-
Corporate debt securities	10,585	230
State and municipal obligations	57	-
Other debt obligations	144	-
Equity securities	15,681	59,994
Total	\$ 341,194	\$ 60,816

The table below presents the gross carrying value of repurchase agreements and securities loaned by maturity.

\$ in millions	As of December 2024	
	Repurchase agreements	Securities loaned
No stated maturity and overnight	\$ 189,068	\$ 40,189
2 - 30 days	115,949	1,638
31 - 90 days	41,361	1,634
91 days - 1 year	38,967	12,068
Greater than 1 year	22,897	11,145
Total	\$ 408,242	\$ 66,674

In the table above:

- Repurchase agreements and securities loaned that are repayable prior to maturity at the option of the firm are reflected at their contractual maturity dates.
- Repurchase agreements and securities loaned that are redeemable prior to maturity at the option of the holder are reflected at the earliest dates such options become exercisable.

Notes to Consolidated Financial Statements**Other Secured Financings**

In addition to repurchase agreements and securities loaned transactions, the firm funds certain assets through the use of other secured financings and pledges financial instruments and other assets as collateral in these transactions. These other secured financings include:

- Liabilities of CIEs and consolidated VIEs;
- Transfers of assets accounted for as financings rather than sales (e.g., pledged commodities, bank loans and mortgage whole loans); and
- Other structured financing arrangements.

Other secured financings included nonrecourse arrangements. Nonrecourse other secured financings were \$3.74 billion as of December 2024 and \$5.57 billion as of December 2023.

The firm has elected to apply the fair value option to substantially all other secured financings because the use of fair value eliminates non-economic volatility in earnings that would arise from using different measurement attributes. See Note 5 for further information about other secured financings that are accounted for at fair value.

Other secured financings that are not recorded at fair value are recorded based on the amount of cash received plus accrued interest, which generally approximates fair value. As these financings are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these financings been included in the firm's fair value hierarchy, substantially all would have been classified in level 3 as of both December 2024 and December 2023.

The table below presents information about other secured financings.

<i>\$ in millions</i>	U.S. Dollar	Non-U.S. Dollar	Total
As of December 2024			
Other secured financings			
Short-term	\$ 16,333	\$ 4,582	\$ 20,915
Long-term	1,377	5,858	7,235
Total other secured financings	\$ 17,710	\$ 10,440	\$ 28,150
Other secured financings collateralized by:			
Financial instruments	\$ 17,094	\$ 8,644	\$ 25,738
Other assets	\$ 616	\$ 1,796	\$ 2,412
As of December 2023			
Other secured financings			
Short-term	\$ 3,385	\$ 3,819	\$ 7,204
Long-term	2,144	3,846	5,990
Total other secured financings	\$ 5,529	\$ 7,665	\$ 13,194
Other secured financings collateralized by:			
Financial instruments	\$ 3,122	\$ 6,755	\$ 9,877
Other assets	\$ 2,407	\$ 910	\$ 3,317

In the table above:

- Short-term other secured financings includes financings maturing within one year of the financial statement date and financings that are redeemable within one year of the financial statement date at the option of the holder.
- Other secured financings included \$27.99 billion as of December 2024 and \$12.55 billion as of December 2023 of financings accounted for at fair value under the fair value option.
- Non-U.S. dollar-denominated short-term other secured financings had a weighted average interest rate of 0.47% as of December 2023. This rate includes the effect of hedging activities and excludes other secured financings held at fair value under the fair value option. As of December 2024, all of the non-U.S. dollar-denominated short-term other secured financings were held at fair value under the fair value option.
- U.S. dollar-denominated long-term other secured financings had a weighted average interest rate of 7.16% as of December 2024 and 3.44% as of December 2023. These rates include the effect of hedging activities and excludes other secured financings held at fair value under the fair value option.
- All U.S. dollar denominated short-term and non-U.S. dollar denominated long-term other secured financings were held at fair value under the fair value option.
- Total other secured financings included \$2.50 billion as of December 2024 and \$2.34 billion as of December 2023 related to transfers of financial assets accounted for as financings rather than sales. Such financings were collateralized by financial assets, primarily included in trading assets, of \$2.50 billion as of December 2024 and \$2.36 billion as of December 2023.
- Other secured financings collateralized by financial instruments included \$24.39 billion as of December 2024 and \$8.38 billion as of December 2023 of other secured financings collateralized by trading assets, investments and loans, and included \$1.35 billion as of December 2024 and \$1.49 billion as of December 2023 of other secured financings collateralized by financial instruments received as collateral and repledged.

Notes to Consolidated Financial Statements

The table below presents other secured financings by maturity.

\$ in millions	As of	
	December 2024	
Other secured financings (short-term)	\$	20,915
Other secured financings (long-term):		
2026		4,626
2027		475
2028		1,223
2029		184
2030 - thereafter		727
Total other secured financings (long-term)		7,235
Total other secured financings	\$	28,150

In the table above:

- Long-term other secured financings that are repayable prior to maturity at the option of the firm are reflected at their contractual maturity dates.
- Long-term other secured financings that are redeemable prior to maturity at the option of the holder are reflected at the earliest dates such options become exercisable.

Collateral Received and Pledged

The firm receives cash and securities (e.g., U.S. government and agency obligations, other sovereign and corporate obligations, as well as equity securities) as collateral, primarily in connection with resale agreements, securities borrowed, derivative transactions and customer margin loans. The firm obtains cash and securities as collateral on an upfront or contingent basis for derivative instruments and collateralized agreements to reduce its credit exposure to individual counterparties.

In many cases, the firm is permitted to deliver or repledge financial instruments received as collateral when entering into repurchase agreements and securities loaned transactions, primarily in connection with secured client financing activities. The firm is also permitted to deliver or repledge these financial instruments in connection with other secured financings, collateralized derivative transactions and firm or customer settlement requirements.

The firm also pledges certain trading assets in connection with repurchase agreements, securities loaned transactions and other secured financings, and other assets (substantially all real estate and cash) in connection with other secured financings to counterparties who may or may not have the right to deliver or repledge them.

The table below presents financial instruments at fair value received as collateral that were available to be delivered or repledged and were delivered or repledged.

\$ in millions	As of December	
	2024	2023
Collateral available to be delivered or repledged	\$ 1,038,740	\$ 1,002,891
Collateral that was delivered or repledged	\$ 912,863	\$ 862,988

The table below presents information about assets pledged.

\$ in millions	As of December	
	2024	2023
Pledged to counterparties that had the right to deliver or repledge		
Trading assets	\$ 148,417	\$ 110,567
Pledged to counterparties that did not have the right to deliver or repledge		
Trading assets	\$ 173,254	\$ 138,404
Investments	\$ 8,712	\$ 22,165
Loans	\$ 12,065	\$ 8,865
Other assets	\$ 1,590	\$ 3,924

The firm also segregates securities for regulatory and other purposes related to client activity. Such securities are segregated from trading assets and investments, as well as from securities received as collateral under resale agreements and securities borrowed transactions. Securities segregated by the firm were \$64.21 billion as of December 2024 and \$49.26 billion as of December 2023.

Note 12.**Other Assets**

The table below presents other assets by type.

\$ in millions	As of December	
	2024	2023
Property, leasehold improvements and equipment	\$ 8,024	\$ 11,244
Goodwill	5,853	5,916
Identifiable intangible assets	847	1,177
Operating lease right-of-use assets	1,967	2,171
Income tax-related assets	9,131	8,157
Miscellaneous receivables and other	8,365	7,925
Total	\$ 34,187	\$ 36,590

Property, Leasehold Improvements and Equipment

Property, leasehold improvements and equipment is net of accumulated depreciation and amortization of \$13.64 billion as of both December 2024 and December 2023. Property, leasehold improvements and equipment included \$6.57 billion as of December 2024 and \$6.65 billion as of December 2023 that the firm uses in connection with its operations, and \$52 million as of December 2024 and \$124 million as of December 2023 of foreclosed real estate. The remainder is held by investment entities, including VIEs, consolidated by the firm. Substantially all property and equipment is depreciated on a straight-line basis over the useful life of the asset. Leasehold improvements are amortized on a straight-line basis over the shorter of the useful life of the improvement or the term of the lease. Capitalized costs of software developed or obtained for internal use are amortized on a straight-line basis over three years.

Notes to Consolidated Financial Statements

The firm tests property, leasehold improvements and equipment for impairment when events or changes in circumstances suggest that an asset's or asset group's carrying value may not be fully recoverable. To the extent the carrying value of an asset or asset group exceeds the projected undiscounted cash flows expected to result from the use and eventual disposal of the asset or asset group, the firm determines the asset or asset group is impaired and records an impairment equal to the difference between the estimated fair value and the carrying value of the asset or asset group. In addition, the firm will recognize an impairment prior to the sale of an asset or asset group if the carrying value of the asset or asset group exceeds its estimated fair value. Any impairments recognized are included in depreciation and amortization. The firm had impairments of \$228 million during 2024 and \$314 million during 2022, substantially all related to commercial real estate included in CIEs within Asset & Wealth Management. During 2023, the firm had impairments of \$1.46 billion related to commercial real estate included in CIEs within Asset & Wealth Management and \$118 million related to capitalized software substantially all within Platform Solutions and Asset & Wealth Management.

Goodwill

Goodwill is the cost of acquired companies in excess of the fair value of net assets, including identifiable intangible assets, at the acquisition date.

The table below presents the carrying value of goodwill by reporting unit.

\$ in millions	As of December	
	2024	2023
Global Banking & Markets:		
Investment banking	\$ 267	\$ 267
FICC	269	269
Equities	2,647	2,647
Asset & Wealth Management:		
Asset management	1,361	1,410
Wealth management	1,309	1,309
Platform Solutions:		
Transaction banking and other	-	14
Total	\$ 5,853	\$ 5,916

Goodwill is assessed for impairment annually in the fourth quarter or more frequently if events occur or circumstances change that indicate an impairment may exist. When assessing goodwill for impairment, first, a qualitative assessment can be made to determine whether it is more likely than not that the estimated fair value of a reporting unit is less than its carrying value. If the results of the qualitative assessment are not conclusive, a quantitative goodwill test is performed. Alternatively, a quantitative goodwill test can be performed without performing a qualitative assessment.

The quantitative goodwill test compares the estimated fair value of each reporting unit with its carrying value (including goodwill and identifiable intangible assets). If the reporting unit's estimated fair value exceeds its carrying value, goodwill is not impaired. An impairment is recognized if the estimated fair value of a reporting unit is less than its carrying value and any such impairment is included in depreciation and amortization.

When performing a quantitative goodwill test, the estimated fair value of each reporting unit is based on valuation techniques the firm believes market participants would use to value these reporting units. Estimated fair values are generally derived from utilizing a relative value technique, which applies observable price-to-earnings multiples or price-to-book multiples of comparable competitors to the reporting units' net earnings or net book value, or a discounted cash flow valuation approach, for reporting units with businesses in early stages of development. The carrying value of each reporting unit reflects an allocation of total shareholders' equity and represents the estimated amount of total shareholders' equity required to support the activities of the reporting unit under currently applicable regulatory capital requirements.

During the third quarter of 2024, in connection with the planned sale of the firm's seller financing loan portfolio, the firm performed a quantitative goodwill test and determined that the goodwill associated with Transaction banking and other was impaired, and accordingly, recorded a \$14 million impairment.

In the fourth quarter of 2024, the firm performed its annual assessment of goodwill for impairment, for each of its reporting units with goodwill, by performing a qualitative assessment. Multiple factors were assessed with respect to each of these reporting units to determine whether it was more likely than not that the estimated fair value of each of those reporting units was less than its carrying value.

The firm considered the following factors in the qualitative annual assessment:

- **Performance Indicators.** During 2024, the firm's net revenues, efficiency ratio (total operating expenses divided by total net revenues), diluted earnings per common share (EPS), return on average common shareholders' equity and book value per common share all improved from 2023 and from 2022 (when a quantitative test was last performed). Within the reporting units with goodwill, there continued to be solid fundamentals underlying our businesses, where the firm continued to maintain industry leadership positions and execute on strategic goals.
- **Macroeconomic Indicators.** Despite broad macroeconomic and geopolitical concerns, the global economy continued to grow in 2024.
- **Firm and Industry Events.** There were no events, entity-specific or otherwise, that would have had a significant negative impact on the valuation of the firm's reporting units with goodwill.
- **Fair Value Indicators.** Fair value indicators for both the firm and its peers have generally improved since the annual assessment performed in 2023 and from 2022 (when a quantitative test was last performed).

Notes to Consolidated Financial Statements

As a result of the annual assessment, the firm determined that it was more likely than not that the estimated fair value of each reporting unit with goodwill exceeded its respective carrying value. Therefore, the firm determined that goodwill for each reporting unit was not impaired and that a quantitative goodwill test was not required.

Identifiable Intangible Assets

The table below presents identifiable intangible assets by type.

\$ in millions	As of December	
	2024	2023
Customer lists		
Gross carrying value	\$ 2,187	\$ 2,339
Accumulated amortization	(1,358)	(1,292)
Net carrying value	829	1,047
Other		
Gross carrying value	82	866
Accumulated amortization	(64)	(736)
Net carrying value	18	130
Total gross carrying value	2,269	3,205
Total accumulated amortization	(1,422)	(2,028)
Total net carrying value	\$ 847	\$ 1,177

In the table above:

- The decrease in the net carrying value of identifiable intangible assets from December 2023 to December 2024 reflected a \$110 million reduction due to the sale of GreenSky Holdings, LLC (GreenSky) and a \$72 million write-down in connection with the classification of the GM credit card program (included within Platform Solutions) as held for sale in 2024.
- Substantially all of the firm's identifiable intangible assets have finite useful lives and are amortized over their estimated useful lives generally using the straight-line method.

The tables below present information about the amortization of identifiable intangible assets.

\$ in millions	Year Ended December		
	2024	2023	2022
Amortization	\$ 176	\$ 681	\$ 174

In the table above, amortization for 2024 included the write-down related to the GM credit card program noted above. Amortization for 2023 included a \$506 million write-down related to GreenSky.

	As of	
<i>\$ in millions</i>	December 2024	
Estimated future amortization		
2025	\$	79
2026	\$	73
2027	\$	73
2028	\$	72
2029	\$	72

The firm tests identifiable intangible assets for impairment when events or changes in circumstances suggest that an asset's or asset group's carrying value may not be fully recoverable. To the extent the carrying value of an asset or asset group exceeds the projected undiscounted cash flows expected to result from the use and eventual disposal of the asset or asset group, the firm determines the asset or asset group is impaired and records an impairment equal to the difference between the estimated fair value and the carrying value of the asset or asset group. In addition, the firm will recognize an impairment prior to the sale of an asset or asset group if the carrying value of the asset or asset group exceeds its estimated fair value. Other than as noted above, there were no material impairments or write-downs during each of 2024, 2023 and 2022.

Operating Lease Right-of-Use Assets

The firm enters into operating leases for real estate, office equipment and other assets, substantially all of which are used in connection with its operations. For leases longer than one year, the firm recognizes a right-of-use asset representing the right to use the underlying asset for the lease term, and a lease liability representing the liability to make payments. The lease term is generally determined based on the contractual maturity of the lease. For leases where the firm has the option to terminate or extend the lease, an assessment of the likelihood of exercising the option is incorporated into the determination of the lease term. Such assessment is initially performed at the inception of the lease and is updated if events occur that impact the original assessment.

An operating lease right-of-use asset is initially determined based on the operating lease liability, adjusted for initial direct costs, lease incentives and amounts paid at or prior to lease commencement. This amount is then amortized over the lease term. Right-of-use assets and operating lease liabilities recognized (in non-cash transactions for leases entered into or assumed) by the firm were \$167 million for 2024, \$333 million for 2023 and \$256 million for 2022. See Note 15 for information about operating lease liabilities.

For leases where the firm will derive no economic benefit from leased space that it has vacated or where the firm has shortened the term of a lease when space is no longer needed, the firm will record an impairment or accelerated amortization of right-of-use assets. There were no material impairments or accelerated amortizations during each of 2024, 2023 and 2022.

Notes to Consolidated Financial Statements**Miscellaneous Receivables and Other**

Miscellaneous receivables and other included:

- Investments in qualified affordable housing and renewable energy projects of \$3.46 billion as of December 2024 and \$3.39 billion as of December 2023. The firm receives tax credits for such investments. See Note 17 for further information about these investments.
- Assets classified as held for sale were \$517 million as of December 2024 and \$518 million as of December 2023. See below for further information.

Assets Held for Sale. During 2024, in connection with the planned transition of the GM credit card program to another issuer, the firm classified the GM credit card program (within Platform Solutions) as held for sale. The firm had previously classified the GM co-branded credit card loans as held for sale in 2023. As of December 2024, the assets related to the GM credit card program consisted of the GM co-branded credit card portfolio of \$1.8 billion (included in loans). See Note 9 for further information about loans classified as held for sale.

Assets held for sale also included \$517 million as of December 2024 and \$327 million as of December 2023 of assets related to certain of the firm's consolidated investments within Asset & Wealth Management. Substantially all of these assets consisted of property and equipment and were included in miscellaneous receivables and other within other assets. In addition, as of December 2023, GreenSky (within Platform Solutions) was classified as held for sale. Assets related to GreenSky were approximately \$3.4 billion and consisted of loans of approximately \$3.0 billion (included in loans), segregated cash of approximately \$110 million (included in cash and cash equivalents), identifiable intangible assets of approximately \$110 million (included in identifiable intangible assets within other assets) and other assets of approximately \$190 million (included in miscellaneous receivables and other within other assets). During 2024, the firm completed the sale of GreenSky. See Note 9 for further information about loans classified as held for sale, above for further information about identifiable intangible assets, and Note 15 for information about liabilities classified as held for sale.

Note 13.**Deposits**

The table below presents information about deposits.

\$ in millions	As of December	
	2024	2023
U.S. offices	\$ 341,711	\$ 333,116
Non-U.S. offices	91,302	95,301
Total	\$ 433,013	\$ 428,417

In the table above:

- Deposits include savings, demand and time deposits.
- All U.S. deposits were held at Goldman Sachs Bank USA (GS Bank USA). Substantially all non-U.S. deposits were held at Goldman Sachs International Bank (GSIB) and Goldman Sachs Bank Europe SE (GSBE).
- Substantially all deposits are interest-bearing.

The table below presents maturities of time deposits held in U.S. and non-U.S. offices.

\$ in millions	As of December 2024		
	U.S.	Non-U.S.	Total
2025	\$ 86,122	\$ 23,167	\$ 109,289
2026	6,325	543	6,868
2027	2,192	215	2,407
2028	1,031	194	1,225
2029	1,382	192	1,574
2030 - thereafter	1,265	50	1,315
Total	\$ 98,317	\$ 24,361	\$ 122,678

In the table above:

- The aggregate amount of time deposits in denominations that met or exceeded the applicable insurance limits, or were otherwise not covered by insurance, were \$19.00 billion in U.S. deposits and \$23.92 billion in non-U.S. deposits.
- Time deposits included \$44.86 billion as of December 2024 and \$29.46 billion as of December 2023 of deposits accounted for at fair value under the fair value option. See Note 10 for further information about deposits accounted for at fair value.

The firm's savings and demand deposits are recorded based on the amount of cash received plus accrued interest, which approximates fair value. In addition, the firm designates certain derivatives as fair value hedges to convert a portion of its time deposits not accounted for at fair value from fixed-rate obligations into floating-rate obligations. The carrying value of time deposits not accounted for at fair value approximated fair value as of both December 2024 and December 2023. As these savings and demand deposits and time deposits are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these deposits been included in the firm's fair value hierarchy, they would have been classified in level 2 as of both December 2024 and December 2023.

Notes to Consolidated Financial Statements**Note 14.****Unsecured Borrowings**

The table below presents information about unsecured borrowings.

\$ in millions	As of December	
	2024	2023
Unsecured short-term borrowings	\$ 69,709	\$ 75,945
Unsecured long-term borrowings	242,634	241,877
Total	\$ 312,343	\$ 317,822

Unsecured Short-Term Borrowings

Unsecured short-term borrowings includes the portion of unsecured long-term borrowings maturing within one year of the financial statement date and unsecured long-term borrowings that are redeemable within one year of the financial statement date at the option of the holder.

The firm accounts for certain hybrid financial instruments at fair value under the fair value option. See Note 10 for further information about unsecured short-term borrowings that are accounted for at fair value. In addition, the firm designates certain derivatives as fair value hedges to convert a portion of its unsecured short-term borrowings not accounted for at fair value from fixed-rate obligations into floating-rate obligations. The carrying value of unsecured short-term borrowings that are not recorded at fair value generally approximates fair value due to the short-term nature of the obligations. As these unsecured short-term borrowings are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these borrowings been included in the firm's fair value hierarchy, substantially all would have been classified in level 2 as of both December 2024 and December 2023.

The table below presents information about unsecured short-term borrowings.

\$ in millions	As of December	
	2024	2023
Current portion of unsecured long-term borrowings	\$ 38,521	\$ 49,361
Hybrid financial instruments	29,130	23,073
Commercial paper	-	1,213
Other unsecured short-term borrowings	2,058	2,298
Total unsecured short-term borrowings	\$ 69,709	\$ 75,945
Weighted average interest rate	5.87 %	3.64 %

In the table above, the weighted average interest rates for these borrowings include the effect of hedging activities and exclude unsecured short-term borrowings accounted for at fair value under the fair value option. See Note 7 for further information about hedging activities.

Unsecured Long-Term Borrowings

The table below presents information about unsecured long-term borrowings.

<i>\$ in millions</i>	U.S. Dollar		Non-U.S. Dollar	Total
As of December 2024				
Fixed-rate obligations:				
Group Inc.	\$	116,077	\$ 24,613	\$ 140,690
Subsidiaries		7,034	3,708	10,742
Floating-rate obligations:				
Group Inc.		18,017	8,816	26,833
Subsidiaries		42,713	21,656	64,369
Total	\$	183,841	\$ 58,793	\$ 242,634
As of December 2023				
Fixed-rate obligations:				
Group Inc.	\$	112,821	\$ 31,023	\$ 143,844
Subsidiaries		1,992	3,739	5,731
Floating-rate obligations:				
Group Inc.		18,936	12,555	31,491
Subsidiaries		38,117	22,694	60,811
Total	\$	171,866	\$ 70,011	\$ 241,877

In the table above:

- Unsecured long-term borrowings consists principally of senior borrowings, which have maturities extending through 2061.
- Unsecured long-term borrowings included \$89.19 billion as of December 2024 and \$86.41 billion as of December 2023 of borrowings accounted for at fair value under the fair value option. The carrying value of unsecured long-term borrowings for which the firm did not elect the fair value option was \$153.44 billion as of December 2024 and \$155.47 billion as of December 2023. The estimated fair value of such unsecured long-term borrowings was \$156.31 billion as of December 2024 and \$157.75 billion as of December 2023. As these borrowings are not accounted for at fair value, they are not included in the firm's fair value hierarchy in Notes 4 and 5. Had these borrowings been included in the firm's fair value hierarchy, substantially all would have been classified in level 2 as of both December 2024 and December 2023.
- Floating-rate obligations includes equity-linked, credit-linked and indexed instruments. Floating interest rates are generally based on SOFR and Euro Interbank Offered Rate.

Notes to Consolidated Financial Statements

- U.S. dollar-denominated debt had interest rates ranging from 0.86% to 6.75% (with a weighted average rate of 4.10%) as of December 2024 and 0.86% to 6.75% (with a weighted average rate of 3.73%) as of December 2023. These rates exclude unsecured long-term borrowings accounted for at fair value under the fair value option.
- Non-U.S. dollar-denominated debt had interest rates ranging from 0.25% to 7.25% (with a weighted average rate of 2.04%) as of December 2024 and 0.25% to 7.25% (with a weighted average rate of 2.11%) as of December 2023. These rates exclude unsecured long-term borrowings accounted for at fair value under the fair value option.

The table below presents unsecured long-term borrowings by maturity.

\$ in millions	As of December 2024		
	Group Inc.	Subsidiaries	Total
2026	\$ 22,884	\$ 14,046	\$ 36,930
2027	22,389	18,749	41,138
2028	20,728	8,476	29,204
2029	21,205	11,531	32,736
2030 - thereafter	80,317	22,309	102,626
Total	\$ 167,523	\$ 75,111	\$ 242,634

In the table above:

- Unsecured long-term borrowings maturing within one year of the financial statement date and unsecured long-term borrowings that are redeemable within one year of the financial statement date at the option of the holder are excluded as they are included in unsecured short-term borrowings.
- Unsecured long-term borrowings that are repayable prior to maturity at the option of the firm are reflected at their contractual maturity dates.
- Unsecured long-term borrowings that are redeemable prior to maturity at the option of the holder are reflected at the earliest dates such options become exercisable.
- Unsecured long-term borrowings included \$(10.74) billion of adjustments to the carrying value of certain unsecured long-term borrowings resulting from the application of hedge accounting by year of maturity as follows: \$(250) million in 2026, \$(837) million in 2027, \$(867) million in 2028, \$(874) million in 2029 and \$(7.91) billion in 2030 and thereafter.

The firm designates certain derivatives as fair value hedges to convert a portion of fixed-rate unsecured long-term borrowings not accounted for at fair value into floating-rate obligations. See Note 7 for further information about hedging activities.

The table below presents unsecured long-term borrowings, after giving effect to such hedging activities.

\$ in millions	Group Inc.	Subsidiaries	Total
As of December 2024			
Fixed-rate obligations	\$ 19,883	\$ 5,783	\$ 25,666
Floating-rate obligations	147,640	69,328	216,968
Total	\$ 167,523	\$ 75,111	\$ 242,634
As of December 2023			
Fixed-rate obligations	\$ 15,460	\$ 4,912	\$ 20,372
Floating-rate obligations	159,875	61,630	221,505
Total	\$ 175,335	\$ 66,542	\$ 241,877

In the table above, the aggregate amounts of unsecured long-term borrowings had weighted average interest rates of 5.72% (4.39% related to fixed-rate obligations and 5.82% related to floating-rate obligations) as of December 2024 and 6.13% (3.44% related to fixed-rate obligations and 6.27% related to floating-rate obligations) as of December 2023. These rates exclude unsecured long-term borrowings accounted for at fair value under the fair value option.

Subordinated Borrowings

Unsecured long-term borrowings includes subordinated debt and junior subordinated debt. Subordinated debt that matures within one year is included in unsecured short-term borrowings. Junior subordinated debt is junior in right of payment to other subordinated borrowings, which are junior to senior borrowings. Subordinated debt had maturities ranging from 2025 to 2045 as of both December 2024 and December 2023.

The table below presents information about subordinated borrowings.

\$ in millions	Par Amount	Carrying Value	Rate
As of December 2024			
Subordinated debt	\$ 12,131	\$ 11,217	6.89 %
Junior subordinated debt	968	1,004	5.88 %
Total	\$ 13,099	\$ 12,221	6.82 %
As of December 2023			
Subordinated debt	\$ 12,215	\$ 11,898	7.79 %
Junior subordinated debt	968	1,053	6.30 %
Total	\$ 13,183	\$ 12,951	7.68 %

In the table above:

- The par amount of subordinated debt issued by Group Inc. was \$12.13 billion as of December 2024 and \$12.22 billion as of December 2023, and the carrying value of subordinated debt issued by Group Inc. was \$11.22 billion as of December 2024 and \$11.90 billion as of December 2023.
- The rate is the weighted average interest rate for these borrowings (excluding borrowings accounted for at fair value under the fair value option), including the effect of fair value hedges used to convert fixed-rate obligations into floating-rate obligations. See Note 7 for further information about hedging activities.

Notes to Consolidated Financial Statements**Junior Subordinated Debt**

In 2004, Group Inc. issued \$2.84 billion of junior subordinated debt to Goldman Sachs Capital I, a Delaware statutory trust. Goldman Sachs Capital I issued \$2.75 billion of guaranteed preferred beneficial interests (Trust Preferred securities) to third parties and \$85 million of common beneficial interests to Group Inc. As of both December 2024 and December 2023, the outstanding par amount of junior subordinated debt held by Goldman Sachs Capital I was \$968 million and the outstanding par amount of Trust Preferred securities and common beneficial interests issued by Goldman Sachs Capital I was \$939 million and \$29 million, respectively. Goldman Sachs Capital I is a wholly-owned finance subsidiary of the firm for regulatory and legal purposes but is not consolidated for accounting purposes.

The firm pays interest semi-annually on the junior subordinated debt at an annual rate of 6.345% and the debt matures on February 15, 2034. The coupon rate and the payment dates applicable to the beneficial interests are the same as the interest rate and payment dates for the junior subordinated debt. The firm has the right, from time to time, to defer payment of interest on the junior subordinated debt, and therefore cause payment on Goldman Sachs Capital I's preferred beneficial interests to be deferred, in each case up to ten consecutive semi-annual periods. During any such deferral period, the firm will not be permitted to, among other things, pay dividends on or make certain repurchases of its common stock. Goldman Sachs Capital I is not permitted to pay any distributions on the common beneficial interests held by Group Inc. unless all dividends payable on the preferred beneficial interests have been paid in full.

Note 15.**Other Liabilities**

The table below presents other liabilities by type.

\$ in millions	As of December	
	2024	2023
Compensation and benefits	\$ 8,770	\$ 7,804
Income tax-related liabilities	3,544	2,947
Operating lease liabilities	2,062	2,232
Noncontrolling interests	401	363
Employee interests in consolidated funds	16	19
Accrued expenses and other	9,427	10,438
Total	\$ 24,220	\$ 23,803

Operating Lease Liabilities

For leases longer than one year, the firm recognizes a right-of-use asset representing the right to use the underlying asset for the lease term, and a lease liability representing the liability to make payments. See Note 12 for information about operating lease right-of-use assets.

The table below presents information about operating lease liabilities.

\$ in millions	Operating lease liabilities
As of December 2024	
2025	\$ 295
2026	315
2027	278
2028	252
2029	225
2030 - thereafter	1,298
Total undiscounted lease payments	2,663
Imputed interest	(601)
Total operating lease liabilities	\$ 2,062
Weighted average remaining lease term	12 years
Weighted average discount rate	4.25 %
As of December 2023	
2024	\$ 325
2025	325
2026	288
2027	256
2028	231
2029 - thereafter	1,462
Total undiscounted lease payments	2,887
Imputed interest	(655)
Total operating lease liabilities	\$ 2,232
Weighted average remaining lease term	12 years
Weighted average discount rate	4.13 %

In the table above, the weighted average discount rate represents the firm's incremental borrowing rate as of the date of adoption of ASU No. 2016-02, "Leases (Topic 842)," for operating leases existing on the date of adoption and as of the lease inception date for leases entered into subsequent to the adoption of this ASU.

Operating lease costs were \$473 million for 2024, \$484 million for 2023 and \$462 million for 2022. Variable lease costs, which are included in operating lease costs, were not material for each of 2024, 2023 and 2022. Total occupancy expenses for space held in excess of the firm's current requirements were not material for each of 2024, 2023 and 2022.

Lease payments relating to operating lease arrangements that were signed but had not yet commenced were \$1.10 billion as of December 2024.

Notes to Consolidated Financial Statements**Accrued Expenses and Other**

Accrued expenses and other included:

- Liabilities classified as held for sale were not material as of December 2024 and were \$257 million as of December 2023, substantially all of which related to GreenSky within Platform Solutions and consisted primarily of customer and other payables. See Note 12 for further information about assets held for sale.
- Contract liabilities, which represent consideration received by the firm in connection with its contracts with clients prior to providing the service, were not material as of both December 2024 and December 2023.
- Accrued unfunded commitments related to investments in qualified affordable housing projects were \$2.15 billion as of December 2024 and \$2.26 billion as of December 2023. See Note 17 for further information about these investments.

Note 16.**Securitization Activities**

The firm securitizes residential and commercial mortgages, corporate bonds, loans and other types of financial assets by selling these assets to securitization vehicles (e.g., trusts, corporate entities and limited liability companies) or through a resecuritization. The firm acts as underwriter of the beneficial interests that are sold to investors. The firm's residential mortgage securitizations are primarily in connection with government agency securitizations.

The firm accounts for a securitization as a sale when it has relinquished control over the transferred financial assets. Prior to securitization, the firm generally accounts for assets pending transfer at fair value and therefore does not typically recognize significant gains or losses upon the transfer of assets. Net revenues from underwriting activities are recognized in connection with the sales of the underlying beneficial interests to investors.

The firm generally receives cash in exchange for the transferred assets but may also have continuing involvement with the transferred financial assets, including ownership of beneficial interests in securitized financial assets, primarily in the form of debt instruments. The firm may also purchase senior or subordinated securities issued by securitization vehicles (which are typically VIEs) in connection with secondary market-making activities.

The primary risks included in beneficial interests and other interests from the firm's continuing involvement with securitization vehicles are the performance of the underlying collateral, the position of the firm's investment in the capital structure of the securitization vehicle and the market yield for the security. Interests accounted for at fair value are primarily classified in level 2 of the fair value hierarchy. Interests not accounted for at fair value are carried at amounts that approximate fair value. See Note 4 for further information about fair value measurements.

The table below presents the amount of financial assets securitized and the cash flows received on retained interests in securitization entities in which the firm had continuing involvement as of the end of the period.

\$ in millions	Year Ended December		
	2024	2023	2022
Residential mortgages	\$ 33,210	\$ 20,276	\$ 26,717
Commercial mortgages	7,238	4,446	13,935
Other financial assets	2,782	5,951	3,617
Total financial assets securitized	\$ 43,230	\$ 30,673	\$ 44,269
Retained interests cash flows	\$ 722	\$ 493	\$ 551

The firm securitized assets of \$364 million during 2024, \$369 million during 2023 and \$792 million during 2022, in a non-cash exchange for loans and investments.

The table below presents information about nonconsolidated securitization entities to which the firm sold assets and had continuing involvement as of the end of the period.

\$ in millions	Outstanding		
	Principal Amount	Retained Interests	Purchased Interests
As of December 2024			
U.S. government agency-issued CMOs	\$ 34,049	\$ 3,053	\$ -
Other residential mortgage-backed	33,069	1,357	14
Other commercial mortgage-backed	59,562	945	57
Corporate debt and other asset-backed	12,059	493	5
Total	\$ 138,739	\$ 5,848	\$ 76
As of December 2023			
U.S. government agency-issued CMOs	\$ 31,140	\$ 2,260	\$ -
Other residential mortgage-backed	28,767	1,162	78
Other commercial mortgage-backed	61,648	1,192	61
Corporate debt and other asset-backed	12,501	685	56
Total	\$ 134,056	\$ 5,299	\$ 195

Notes to Consolidated Financial Statements

In the table above:

- CMOs represents collateralized mortgage obligations.
- The outstanding principal amount is presented for the purpose of providing information about the size of the securitization entities and is not representative of the firm's risk of loss.
- The firm's risk of loss from retained or purchased interests is limited to the carrying value of these interests.
- Purchased interests represent senior and subordinated interests, purchased in connection with secondary market-making activities, in securitization entities in which the firm also holds retained interests.
- Substantially all of the total outstanding principal amount and total retained interests relate to securitizations during 2019 and thereafter.
- The fair value of retained interests was \$5.78 billion as of December 2024 and \$5.26 billion as of December 2023.

In addition to the interests in the table above, the firm had other continuing involvement in the form of derivative transactions and commitments with certain nonconsolidated VIEs. The carrying value of these derivatives and commitments was a net asset of \$847 million as of December 2024 and \$120 million as of December 2023, and the notional amount of these derivatives and commitments was \$2.78 billion as of December 2024 and \$1.95 billion as of December 2023. The notional amounts of these derivatives and commitments are included in maximum exposure to loss in the nonconsolidated VIE table in Note 17. Additionally, the firm provided seller financing of \$2.12 billion in connection with the sale of \$4.44 billion of loans (the vast majority of which were related to GreenSky) during 2024 and of approximately \$2.7 billion in connection with the sale of \$3.24 billion of Marcus loans during 2023. The principal and interest repayments received from these financings were \$2.85 billion for 2024 and were \$1.0 billion for 2023. Seller financing related to the GreenSky loans sale was fully repaid as of December 2024. The total outstanding principal amount of these seller financings were \$1.32 billion as of December 2024 and \$1.81 billion as of December 2023.

The table below presents information about the weighted average key economic assumptions used in measuring the fair value of mortgage-backed retained interests.

\$ in millions	As of December	
	2024	2023
Fair value of retained interests	\$ 5,292	\$ 4,590
Weighted average life (years)	6.1	5.7
Constant prepayment rate	10.0%	12.2%
Impact of 10% adverse change	\$ (57)	\$ (50)
Impact of 20% adverse change	\$ (110)	\$ (94)
Discount rate	7.8%	7.6%
Impact of 10% adverse change	\$ (136)	\$ (117)
Impact of 20% adverse change	\$ (281)	\$ (226)

In the table above:

- Amounts do not reflect the benefit of other financial instruments that are held to mitigate risks inherent in these retained interests.
- Changes in fair value based on an adverse variation in assumptions generally cannot be extrapolated because the relationship of the change in assumptions to the change in fair value is not usually linear.
- The impact of a change in a particular assumption is calculated independently of changes in any other assumption. In practice, simultaneous changes in assumptions might magnify or counteract the sensitivities disclosed above.
- The constant prepayment rate is included only for positions for which it is a key assumption in the determination of fair value.
- The discount rate for retained interests that relate to U.S. government agency-issued CMOs does not include any credit loss. Expected credit loss assumptions are reflected in the discount rate for the remainder of retained interests.

The firm has other retained interests not reflected in the table above with a fair value of \$491 million and a weighted average life of 5.0 years as of December 2024, and a fair value of \$674 million and a weighted average life of 5.0 years as of December 2023. Due to the nature and fair value of certain of these retained interests, the weighted average assumptions for constant prepayment and discount rates and the related sensitivity to adverse changes are not meaningful as of both December 2024 and December 2023. The firm's maximum exposure to adverse changes in the value of these interests is the carrying value of \$493 million as of December 2024 and \$685 million as of December 2023.

Notes to Consolidated Financial Statements**Note 17.****Variable Interest Entities**

A variable interest in a VIE is an investment (e.g., debt or equity) or other interest (e.g., derivatives or loans and lending commitments) that will absorb portions of the VIE's expected losses and/or receive portions of the VIE's expected residual returns.

The firm's variable interests in VIEs include senior and subordinated debt; loans and lending commitments; limited and general partnership interests; preferred and common equity; derivatives that may include foreign currency, equity and/or credit risk; guarantees; and certain of the fees the firm receives from investment funds. Certain interest rate, foreign currency and credit derivatives the firm enters into with VIEs are not variable interests because they create, rather than absorb, risk.

VIEs generally finance the purchase of assets by issuing debt and equity securities that are either collateralized by or indexed to the assets held by the VIE. The debt and equity securities issued by a VIE may include tranches of varying levels of subordination. The firm's involvement with VIEs includes securitization of financial assets, as described in Note 16, and investments in and loans to other types of VIEs, as described below. See Note 3 for the firm's consolidation policies, including the definition of a VIE.

VIE Consolidation Analysis

The enterprise with a controlling financial interest in a VIE is known as the primary beneficiary and consolidates the VIE. The firm determines whether it is the primary beneficiary of a VIE by performing an analysis that principally considers:

- Which variable interest holder has the power to direct the activities of the VIE that most significantly impact the VIE's economic performance;
- Which variable interest holder has the obligation to absorb losses or the right to receive benefits from the VIE that could potentially be significant to the VIE;
- The VIE's purpose and design, including the risks the VIE was designed to create and pass through to its variable interest holders;
- The VIE's capital structure;
- The terms between the VIE and its variable interest holders and other parties involved with the VIE; and
- Related-party relationships.

The firm reassesses its evaluation of whether an entity is a VIE when certain reconsideration events occur. The firm reassesses its determination of whether it is the primary beneficiary of a VIE on an ongoing basis based on current facts and circumstances.

VIE Activities

The firm is principally involved with VIEs through the following business activities:

Mortgage-Backed VIEs. The firm sells residential and commercial mortgage loans and securities to mortgage-backed VIEs and may retain beneficial interests in the assets sold to these VIEs. The firm purchases and sells beneficial interests issued by mortgage-backed VIEs in connection with market-making activities. In addition, the firm may enter into derivatives with certain of these VIEs, primarily interest rate swaps, which are typically not variable interests. The firm generally enters into derivatives with other counterparties to mitigate its risk.

Tax Credit and Other Investing VIEs. The firm makes equity investments in VIEs that invest in qualified affordable housing and renewable energy projects designed to generate a return through the realization of tax credits and related tax benefits. The firm also purchases equity and debt securities issued by and makes loans to VIEs that hold real estate, performing and nonperforming debt, distressed loans and equity securities. In addition, the firm makes equity investments in certain investment fund VIEs it manages and is entitled to receive fees from these VIEs. The firm generally does not sell assets to, or enter into derivatives with, these VIEs.

Corporate Debt and Other Asset-Backed VIEs. The firm structures VIEs that issue notes to clients, purchases and sells beneficial interests issued by corporate debt and other asset-backed VIEs in connection with market-making activities, and makes loans to VIEs that warehouse corporate debt. Certain of these VIEs synthetically create the exposure for the beneficial interests they issue by entering into credit derivatives with the firm, rather than purchasing the underlying assets. In addition, the firm may enter into derivatives, such as total return swaps, with certain corporate debt and other asset-backed VIEs, under which the firm pays the VIE a return due to the beneficial interest holders and receives the return on the collateral owned by the VIE. The collateral owned by these VIEs is primarily other asset-backed loans and securities. The firm may be removed as the total return swap counterparty and may enter into derivatives with other counterparties to mitigate its risk related to these swaps. The firm may sell assets to the corporate debt and other asset-backed VIEs it structures.

Notes to Consolidated Financial Statements

Principal-Protected Note VIEs. The firm structures VIEs that issue principal-protected notes to clients. These VIEs own portfolios of assets, principally with exposure to hedge funds. Substantially all of the principal protection on the notes issued by these VIEs is provided by the asset portfolio rebalancing that is required under the terms of the notes. The firm enters into total return swaps with these VIEs under which the firm pays the VIE the return due to the principal-protected note holders and receives the return on the assets owned by the VIE. The firm may enter into derivatives with other counterparties to mitigate its risk. The firm also obtains funding through these VIEs.

Nonconsolidated VIEs

The table below presents a summary of the nonconsolidated VIEs in which the firm holds variable interests.

\$ in millions	As of December	
	2024	2023
Total nonconsolidated VIEs		
Assets in VIEs	\$ 206,500	\$ 193,934
Carrying value of variable interests — assets	\$ 16,710	\$ 15,478
Carrying value of variable interests — liabilities	\$ 2,754	\$ 2,750
Maximum exposure to loss:		
Retained interests	\$ 5,848	\$ 5,299
Purchased interests	767	902
Commitments and guarantees	5,034	4,159
Derivatives	8,974	8,636
Debt and equity	6,878	6,927
Total	\$ 27,501	\$ 25,923

In the table above:

- The nature of the firm's variable interests is described in the rows under maximum exposure to loss.
- The firm's exposure to the obligations of VIEs is generally limited to its interests in these entities. In certain instances, the firm provides guarantees, including derivative guarantees, to VIEs or holders of variable interests in VIEs.
- The maximum exposure to loss excludes the benefit of offsetting financial instruments that are held to mitigate the risks associated with these variable interests.
- The maximum exposure to loss from retained interests, purchased interests, and debt and equity is the carrying value of these interests.
- The maximum exposure to loss from commitments and guarantees, and derivatives is the notional amount, which does not represent anticipated losses and has not been reduced by unrealized losses. As a result, the maximum exposure to loss exceeds liabilities recorded for commitments and guarantees, and derivatives.

The table below presents information, by principal business activity, for nonconsolidated VIEs included in the summary table above.

\$ in millions	As of December	
	2024	2023
Mortgage-backed		
Assets in VIEs	\$ 128,378	\$ 123,108
Carrying value of variable interests — assets	\$ 5,618	\$ 4,867
Maximum exposure to loss:		
Retained interests	\$ 5,355	\$ 4,614
Purchased interests	263	253
Commitments and guarantees	-	35
Derivatives	1	2
Total	\$ 5,619	\$ 4,904
Tax credit and other investing		
Assets in VIEs	\$ 55,311	\$ 47,638
Carrying value of variable interests — assets	\$ 6,978	\$ 6,716
Carrying value of variable interests — liabilities	\$ 2,315	\$ 2,220
Maximum exposure to loss:		
Commitments and guarantees	\$ 4,138	\$ 3,893
Debt and equity	4,831	4,824
Total	\$ 8,969	\$ 8,717
Corporate debt and other asset-backed		
Assets in VIEs	\$ 22,811	\$ 23,188
Carrying value of variable interests — assets	\$ 4,114	\$ 3,895
Carrying value of variable interests — liabilities	\$ 439	\$ 530
Maximum exposure to loss:		
Retained interests	\$ 493	\$ 685
Purchased interests	504	649
Commitments and guarantees	896	231
Derivatives	8,973	8,634
Debt and equity	2,047	2,103
Total	\$ 12,913	\$ 12,302

Beginning in the fourth quarter of 2024, investment fund VIEs are included in tax credit and other investing in the table above. Prior to the fourth quarter of 2024, such VIEs were disclosed separately in investments in funds. The carrying value of the firm's variable interests in such VIEs was \$9 million as of December 2024 and \$91 million as of December 2023. Prior period amounts have been conformed to the current presentation.

As of both December 2024 and December 2023, the carrying values of the firm's variable interests in nonconsolidated VIEs are included in the consolidated balance sheets as follows:

- Mortgage-backed: Assets primarily included in trading assets and loans.
- Tax credit and other investing: Assets primarily included in investments and other assets, and liabilities included in trading liabilities and other liabilities.
- Corporate debt and other asset-backed: Assets included in loans and trading assets, and liabilities included in trading liabilities.

Notes to Consolidated Financial Statements**Tax Credit VIEs**

The firm makes equity investments in nonconsolidated tax credit VIEs that invest in qualified affordable housing and renewable energy projects. These VIEs are generally organized as limited partnerships or similar entities and a third-party is typically the general partner or the managing member. The firm invests in the entity as a limited partner and receives income tax credits and other income tax benefits for such investments. In connection with the adoption of ASU No. 2023-02, as of January 1, 2024, the firm elected the proportional amortization method for qualified affordable housing and renewable energy projects that receive production tax credits. The investments that meet the criteria for the proportional amortization method of accounting are amortized in proportion to the income tax credits and other income tax benefits received on such investments. The amortization of investments and the related income tax credits and other income tax benefits are recorded as a component of the provision for taxes, and are included in other operating activities in the consolidated statements of cash flows.

The table below presents information about investments (included in miscellaneous receivables and other within other assets in the consolidated balance sheets) in qualified affordable housing and renewable energy projects that met the criteria of the proportional amortization method of accounting.

\$ in millions	As of December	
	2024	2023
Carrying value of investments	\$ 3,456	\$ 3,394

In the table above, investments included \$2.15 billion as of December 2024 and \$2.26 billion as of December 2023 of accrued unfunded commitments. As of December 2024, a majority of such accrued unfunded commitments were expected to be funded by year-end 2026.

The table below presents information about the amortization and income tax credits and other income tax benefits related to investments in qualified affordable housing and renewable energy projects that met the criteria of the proportional amortization method of accounting.

\$ in millions	Year Ended December		
	2024	2023	2022
Amortization	\$ 381	\$ 301	\$ 127
Tax credits and other benefits	\$ 452	\$ 370	\$ 151

Investments in qualified affordable housing projects that did not meet the criteria for the proportional amortization method of accounting were not material as of both December 2024 and December 2023.

The firm's existing investments in renewable energy projects that receive production tax credits were not eligible for transition to the proportional amortization method of accounting upon adoption of ASU No. 2023-02. Such investments were \$1.39 billion as of December 2024 and \$1.40 billion as of December 2023, were included in investments in the consolidated balance sheets and were accounted for at fair value under the fair value option.

Consolidated VIEs

The table below presents a summary of the carrying value and balance sheet classification of assets and liabilities in consolidated VIEs.

\$ in millions	As of December	
	2024	2023
Total consolidated VIEs		
<i>Assets</i>		
Cash and cash equivalents	\$ 172	\$ 439
Customer and other receivables	325	347
Trading assets	95	95
Investments	170	80
Loans	9	267
Other assets	69	248
Total	\$ 840	\$ 1,476
<i>Liabilities</i>		
Other secured financings	\$ 661	\$ 850
Customer and other payables	7	2
Unsecured short-term borrowings	5	14
Unsecured long-term borrowings	15	17
Other liabilities	165	91
Total	\$ 853	\$ 974

In the table above:

- Assets and liabilities are presented net of intercompany eliminations and exclude the benefit of offsetting financial instruments that are held to mitigate the risks associated with the firm's variable interests.
- VIEs in which the firm holds a majority voting interest are excluded if (i) the VIE meets the definition of a business and (ii) the VIE's assets can be used for purposes other than the settlement of its obligations.
- Substantially all assets can only be used to settle obligations of the VIE.

Notes to Consolidated Financial Statements

The table below presents information, by principal business activity, for consolidated VIEs included in the summary table above.

\$ in millions	As of December	
	2024	2023
Real estate, credit-related and other investing		
<i>Assets</i>		
Cash and cash equivalents	\$ 17	\$ 417
Trading assets	18	28
Investments	170	80
Loans	9	267
Other assets	69	248
Total	\$ 283	\$ 1,040
<i>Liabilities</i>		
Other secured financings	\$ 4	\$ 143
Customer and other payables	7	2
Other liabilities	165	91
Total	\$ 176	\$ 236
Corporate debt and other asset-backed		
<i>Assets</i>		
Cash and cash equivalents	\$ 155	\$ 22
Total	\$ 155	\$ 22
<i>Liabilities</i>		
Other secured financings	\$ 334	\$ 374
Total	\$ 334	\$ 374
Principal-protected notes		
<i>Assets</i>		
Customer and other receivables	\$ 325	\$ 347
Trading assets	77	67
Total	\$ 402	\$ 414
<i>Liabilities</i>		
Other secured financings	\$ 323	\$ 333
Unsecured short-term borrowings	5	14
Unsecured long-term borrowings	15	17
Total	\$ 343	\$ 364

In the table above, creditors and beneficial interest holders of real estate, credit-related and other investing VIEs do not have recourse to the general credit of the firm.

Note 18.**Commitments, Contingencies and Guarantees****Commitments**

The table below presents commitments by type.

\$ in millions	As of December	
	2024	2023
Commitment Type		
Commercial lending:		
Investment-grade	\$ 124,001	\$ 111,202
Non-investment-grade	67,755	54,298
Warehouse financing	13,587	9,184
Consumer	78,099	73,074
Total lending	283,442	247,758
Risk participations	5,014	8,167
Collateralized agreement	95,282	100,503
Collateralized financing	49,333	84,276
Investment	5,832	4,592
Other	8,223	8,258
Total commitments	\$ 447,126	\$ 453,554

The table below presents commitments by expiration.

\$ in millions	As of December 2024			
	2025	2026 - 2027	2028 - 2029	2030 - Thereafter
	2025	2027	2029	
Commitment Type				
Commercial lending:				
Investment-grade	\$ 19,773	\$ 41,607	\$ 59,574	\$ 3,047
Non-investment-grade	5,752	24,247	28,326	9,430
Warehouse financing	1,296	7,972	3,774	545
Consumer	78,099	-	-	-
Total lending	104,920	73,826	91,674	13,022
Risk participations	583	2,202	2,203	26
Collateralized agreement	94,438	844	-	-
Collateralized financing	49,333	-	-	-
Investment	1,868	754	366	2,844
Other	7,796	244	40	143
Total commitments	\$ 258,938	\$ 77,870	\$ 94,283	\$ 16,035

Lending Commitments

The firm's commercial and warehouse financing lending commitments are agreements to lend with fixed termination dates and depend on the satisfaction of all contractual conditions to borrowing. These commitments are presented net of amounts syndicated to third parties. The total commitment amount does not necessarily reflect actual future cash flows because the firm may syndicate portions of these commitments. In addition, commitments can expire unused or be reduced or cancelled at the counterparty's request. The firm also provides credit to consumers by issuing credit card lines. The firm also provided credit to consumers through commitments to extend unsecured installment loans and beginning in the third quarter of 2024, ceased providing such commitments.

Notes to Consolidated Financial Statements

The table below presents information about lending commitments.

\$ in millions	As of December	
	2024	2023
Held for investment	\$ 262,604	\$ 227,865
Held for sale	20,258	19,129
At fair value	580	764
Total	\$ 283,442	\$ 247,758

In the table above:

- Held for investment lending commitments are accounted for at amortized cost. The carrying value of lending commitments was a liability of \$929 million (including allowance for credit losses of \$674 million) as of December 2024 and \$845 million (including allowance for credit losses of \$620 million) as of December 2023. The estimated fair value of such lending commitments was a liability of \$4.84 billion as of December 2024 and \$5.29 billion as of December 2023. Had these lending commitments been carried at fair value and included in the fair value hierarchy, \$2.44 billion as of December 2024 and \$3.10 billion as of December 2023 would have been classified in level 2, and \$2.40 billion as of December 2024 and \$2.19 billion as of December 2023 would have been classified in level 3.
- Held for sale lending commitments are accounted for at the lower of cost or fair value. The carrying value of lending commitments held for sale was a liability of \$43 million as of December 2024 and \$70 million as of December 2023. The estimated fair value of such lending commitments approximates the carrying value. Had these lending commitments been included in the fair value hierarchy, they would have been primarily classified in level 3 as of both December 2024 and December 2023.
- Gains or losses related to lending commitments at fair value, if any, are generally recorded net of any fees in other principal transactions.

Commercial Lending. The firm's commercial lending commitments were primarily extended to investment-grade corporate borrowers. Such commitments primarily included \$148.78 billion as of December 2024 and \$137.11 billion as of December 2023, related to relationship lending activities (principally used for operating and general corporate purposes), and \$11.26 billion as of December 2024 and \$4.21 billion as of December 2023, related to other investment banking activities (generally extended for contingent acquisition financing and are often intended to be short-term in nature, as borrowers often seek to replace them with other funding sources). The firm also extends lending commitments in connection with other types of corporate lending, commercial real estate financing and other collateralized lending. See Note 9 for further information about funded loans.

To mitigate the credit risk associated with the firm's commercial lending activities, the firm obtains credit protection on certain loans and lending commitments through credit default swaps, both single-name and index-based contracts.

Warehouse Financing. The firm provides financing to clients who warehouse financial assets. These arrangements are collateralized by the warehoused assets, primarily consisting of residential real estate, consumer and corporate loans.

Consumer. The firm's consumer lending commitments includes:

- Credit card lines issued by the firm to consumers were \$78.10 billion as of December 2024 and \$70.82 billion as of December 2023. Such credit card lines included \$14.32 billion as of December 2024 and \$14.35 billion as of December 2023 of commitments classified as held for sale in connection with the planned sale of the GM co-branded credit card portfolio. These credit card lines are cancellable by the firm.
- Commitments to provide unsecured installment loans to consumers were \$2.25 billion as of December 2023 and such commitments were classified as held for sale in connection with the sale of GreenSky. During 2024, the firm completed the sale of GreenSky.

Risk Participations

The firm also risk participates certain of its commercial lending commitments to other financial institutions. In the event of a risk participant's default, the firm will be responsible to fund the borrower.

Collateralized Agreement Commitments/ Collateralized Financing Commitments

Collateralized agreement commitments includes forward starting resale and securities borrowing agreements, and collateralized financing commitments includes forward starting repurchase and secured lending agreements that settle at a future date. Collateralized agreement commitments also includes transactions where the firm has entered into commitments to provide contingent financing to its clients and counterparties through resale agreements. The firm's funding of these commitments depends on the satisfaction of all contractual conditions to the resale agreement and these commitments can expire unused.

Investment Commitments

Investment commitments includes commitments to invest in private equity, real estate and other assets directly and through funds that the firm raises and manages. Investment commitments included \$1.10 billion as of December 2024 and \$963 million as of December 2023, related to commitments to invest in funds managed by the firm. If these commitments are called, they would be funded at market value on the date of investment.

Notes to Consolidated Financial Statements**Contingencies**

Legal Proceedings. See Note 27 for information about legal proceedings.

Guarantees

The table below presents derivatives that meet the definition of a guarantee, securities lending and clearing guarantees and certain other financial guarantees.

<i>\$ in millions</i>	Derivatives	Securities lending and clearing	Other financial guarantees
As of December 2024			
Carrying Value of Net Liability	\$ 3,535	\$ -	\$ 425
Maximum Payout/Notional Amount by Period of Expiration			
2025	\$ 181,940	\$ 58,056	\$ 2,523
2026 - 2027	78,419	-	3,086
2028 - 2029	17,074	-	1,843
2030 - thereafter	31,819	-	505
Total	\$ 309,252	\$ 58,056	\$ 7,957
As of December 2023			
Carrying Value of Net Liability	\$ 5,240	\$ -	\$ 430
Maximum Payout/Notional Amount by Period of Expiration			
2024	\$ 177,895	\$ 28,787	\$ 2,325
2025 - 2026	98,843	-	3,108
2027 - 2028	19,282	-	2,109
2029 - thereafter	29,030	-	231
Total	\$ 325,050	\$ 28,787	\$ 7,773

In the table above:

- The maximum payout is based on the notional amount of the contract and does not represent anticipated losses.
- Amounts exclude certain commitments to issue standby letters of credit that are included in lending commitments. See the tables in "Commitments" above for a summary of the firm's commitments.
- The carrying value for derivatives included derivative assets of \$464 million as of December 2024 and \$359 million as of December 2023, and derivative liabilities of \$4.00 billion as of December 2024 and \$5.60 billion as of December 2023.

Derivative Guarantees. The firm enters into various derivatives that meet the definition of a guarantee under U.S. GAAP, including written equity and commodity put options, written currency contracts and interest rate caps, floors and swaptions. These derivatives are risk managed together with derivatives that do not meet the definition of a guarantee, and therefore the amounts in the table above do not reflect the firm's overall risk related to derivative activities. Disclosures about derivatives are not required if they may be cash settled and the firm has no basis to conclude it is probable that the counterparties held the underlying instruments at the inception of the contract. The firm has concluded that these conditions have been met for certain large, internationally active commercial and investment bank counterparties, central clearing counterparties, hedge funds and certain other counterparties. Accordingly, the firm has not included such contracts in the table above. See Note 7 for information about credit derivatives that meet the definition of a guarantee, which are not included in the table above.

Derivatives are accounted for at fair value and therefore the carrying value is considered the best indication of payment/performance risk for individual contracts. However, the carrying values in the table above exclude the effect of counterparty and cash collateral netting.

Securities Lending and Clearing Guarantees. Securities lending and clearing guarantees include the indemnifications and guarantees that the firm provides in its capacity as an agency lender and in its capacity as a sponsoring member of the Fixed Income Clearing Corporation.

As an agency lender, the firm indemnifies most of its securities lending customers against losses incurred in the event that borrowers do not return securities and the collateral held is insufficient to cover the market value of the securities borrowed. The maximum payout of such indemnifications was \$10.62 billion as of December 2024 and \$14.19 billion as of December 2023. Collateral held by the lenders in connection with securities lending indemnifications was \$11.02 billion as of December 2024 and \$14.63 billion as of December 2023. Because the contractual nature of these arrangements requires the firm to obtain collateral with a market value that exceeds the value of the securities lent to the borrower, there is minimal performance risk associated with these indemnifications.

Notes to Consolidated Financial Statements

As a sponsoring member of the Government Securities Division of the Fixed Income Clearing Corporation, the firm guarantees the performance of its sponsored member clients to the Fixed Income Clearing Corporation in connection with certain resale and repurchase agreements. To minimize potential losses on such guarantees, the firm obtains a security interest in the collateral that the sponsored client placed with the Fixed Income Clearing Corporation. Therefore, the risk of loss on such guarantees is minimal. The maximum payout on this guarantee was \$47.44 billion as of December 2024 and \$14.60 billion as of December 2023. The related collateral held was \$46.96 billion as of December 2024 and \$14.69 billion as of December 2023.

Other Financial Guarantees. In the ordinary course of business, the firm provides other financial guarantees of the obligations of third parties (e.g., standby letters of credit and other guarantees to enable clients to complete transactions and fund-related guarantees). These guarantees represent obligations to make payments to beneficiaries if the guaranteed party fails to fulfill its obligation under a contractual arrangement with that beneficiary. Other financial guarantees also include a guarantee that the firm has provided to the Government of Malaysia that it will receive, by August 2025, at least \$1.4 billion in assets and proceeds from assets seized by governmental authorities around the world related to 1Malaysia Development Berhad, a sovereign wealth fund in Malaysia (1MDB). In connection with this guarantee, the firm agreed to make a one-time interim payment of \$250 million towards the \$1.4 billion if the Government of Malaysia did not receive at least \$500 million in assets and proceeds by August 2022. The firm does not believe that any interim payment is required. Any amounts paid by the firm would, in any event, be subject to reimbursement in the event the assets and proceeds received by the Government of Malaysia through August 18, 2028 exceed \$1.4 billion.

On October 11, 2023, the firm filed a demand for arbitration alleging that the Government of Malaysia had, as of August 2022, recovered assets and proceeds well in excess of \$500 million; it had recovered substantial additional assets and proceeds that should be credited against the guarantee; and it had not used all reasonable efforts to recover other assets and proceeds that could be credited against the guarantee. On November 8, 2023, the Government of Malaysia filed a response to the firm's demand for arbitration and on June 10, 2024, filed an application for a partial award to immediately enforce the interim payment (plus interest). On July 24, 2024, the arbitral tribunal rejected that application. Final determinations on all remaining issues, including any subsequent enforcement of the interim payment, are to be made at a final hearing. The arbitral process is ongoing. See Note 27 for further information about matters related to 1MDB.

Guarantees of Securities Issued by Trusts. The firm has established trusts, including Goldman Sachs Capital I, Goldman Sachs Capital II and Goldman Sachs Capital III (the Trusts), and other entities, for the limited purpose of issuing securities to third parties, lending the proceeds to the firm and entering into contractual arrangements with the firm and third parties related to this purpose. The firm does not consolidate these entities. See Notes 14 and 19 for further information about the transactions involving the Trusts.

The firm effectively provides for the full and unconditional guarantee of the securities issued by these entities. Timely payment by the firm of amounts due to these entities under the guarantee, borrowing, preferred stock and related contractual arrangements will be sufficient to cover payments due on the securities issued by these entities. No subsidiary of Group Inc. guarantees the securities of the Trusts.

Management believes that it is unlikely that any circumstances will occur, such as nonperformance on the part of paying agents or other service providers, that would make it necessary for the firm to make payments related to these entities other than those required under the terms of the guarantee, borrowing, preferred stock and related contractual arrangements and in connection with certain expenses incurred by these entities.

Indemnities and Guarantees of Service Providers. In the ordinary course of business, the firm indemnifies and guarantees certain service providers, such as clearing and custody agents, trustees and administrators, against specified potential losses in connection with their acting as an agent of, or providing services to, the firm or its affiliates.

The firm may also be liable to some clients or other parties for losses arising from its custodial role or caused by acts or omissions of third-party service providers, including sub-custodians and third-party brokers. In certain cases, the firm has the right to seek indemnification from these third-party service providers for certain relevant losses incurred by the firm. In addition, the firm is a member of payment, clearing and settlement networks, as well as securities exchanges around the world that may require the firm to meet the obligations of such networks and exchanges in the event of member defaults and other loss scenarios.

In connection with the firm's prime brokerage and clearing businesses, the firm agrees to clear and settle transactions entered into by clients with other brokerage firms. The firm's obligations in respect of such transactions are secured by the assets in the client's account and proceeds received from the transactions cleared and settled by the firm on behalf of the client. In connection with joint venture investments, the firm may issue loan guarantees under which it may be liable in the event of fraud, misappropriation, environmental liabilities and other matters involving the borrower.

Notes to Consolidated Financial Statements

The firm is unable to develop an estimate of the maximum payout under these guarantees and indemnifications as this depends upon the occurrence of future events, including an assessment of claims that have not yet occurred. However, management believes that it is unlikely the firm will have to make any material payments under these arrangements, and no material liabilities related to these guarantees and indemnifications have been recognized in the consolidated balance sheets as of both December 2024 and December 2023.

Other Representations, Warranties and Indemnifications.

The firm provides representations and warranties to counterparties in connection with a variety of commercial transactions and occasionally indemnifies them against potential losses caused by the breach of those representations and warranties. The firm may also provide indemnifications protecting against changes in or adverse application of certain U.S. tax laws in connection with ordinary-course transactions, such as securities issuances, borrowings or derivatives.

In addition, the firm may provide indemnifications to some counterparties to protect them in the event additional taxes are owed or payments are withheld, due either to a change in or an adverse application of certain non-U.S. tax laws. These indemnifications, as well as indemnifications provided by the firm on other contractual or other obligations, generally are standard contractual terms and are entered into in the ordinary course of business. Generally, there are no stated or notional amounts included in these indemnifications, and the contingencies triggering the obligation to indemnify are not expected to occur. Future changes in tax laws and how such laws would apply to these indemnifications cannot be determined. Therefore, the firm is unable to develop an estimate of the maximum payout under these guarantees and indemnifications. However, management believes that it is unlikely the firm will have to make any material payments under these arrangements, and no material liabilities related to these arrangements have been recognized in the consolidated balance sheets as of both December 2024 and December 2023.

Guarantees of Subsidiaries. Group Inc. is the entity that fully and unconditionally guarantees the securities issued by GS Finance Corp., a wholly-owned finance subsidiary of the firm. Group Inc. has guaranteed the payment obligations of Goldman Sachs & Co. LLC (GS&Co.), GS Bank USA and Goldman Sachs Paris Inc. et Cie, subject to certain exceptions. Group Inc. also guarantees many of the obligations of its other consolidated subsidiaries on a transaction-by-transaction basis, as negotiated with counterparties. In addition, Group Inc. has provided guarantees to Goldman Sachs International (GSI) and GSBE related to agreements that each entity has entered into with certain of its counterparties. Given the obligations of the consolidated subsidiaries are recognized in the consolidated balance sheets or reflected as commitments, Group Inc.'s liabilities as guarantor are not separately disclosed.

Note 19.**Shareholders' Equity****Common Equity**

As of both December 2024 and December 2023, the firm had 4.00 billion authorized shares of common stock and 200 million authorized shares of nonvoting common stock, each with a par value of \$0.01 per share.

The firm's share repurchase program is intended to help maintain the appropriate level of common equity. The share repurchase program is effected primarily through regular open-market purchases (which may include repurchase plans designed to comply with Rule 10b5-1 and accelerated share repurchases), the amounts and timing of which are determined primarily by the firm's current and projected capital position, and capital deployment opportunities, but which may also be influenced by the evolution of current and future regulatory capital requirements, general market conditions and the prevailing price and trading volumes of the firm's common stock.

The table below presents information about common stock repurchases.

<i>in millions, except per share amounts</i>	Year Ended December		
	2024	2023	2022
Common share repurchases	17.5	16.8	10.1
Average cost per share	\$ 457.82	\$ 345.87	\$ 346.07
Total cost of common share repurchases	\$ 8,000	\$ 5,796	\$ 3,500

Pursuant to the terms of certain share-based awards, employees may remit shares to the firm or the firm may cancel share-based awards to satisfy statutory employee tax withholding requirements. Under these plans, 1,197 shares in 2024, 868 shares in 2023 and 11,644 shares in 2022 were remitted with a total value of \$0.5 million in 2024, \$0.3 million in 2023 and \$4 million in 2022, and the firm cancelled 3.4 million share-based awards in 2024, 3.9 million in 2023 and 4.6 million in 2022, with a total value of \$1.33 billion in 2024, \$1.35 billion in 2023 and \$1.59 billion in 2022. The cash settlement of share-based awards is included in other in additional paid-in capital in the consolidated statements of shareholders' equity and in other financing, net in the consolidated statements of cash flows. The amount of cash used to settle share-based awards was not material for each of 2024, 2023 and 2022.

The table below presents common stock dividends declared.

	Year Ended December		
	2024	2023	2022
Dividends declared per common share	\$ 11.50	\$ 10.50	\$ 9.00

On January 14, 2025, the Board of Directors of Group Inc. declared a dividend of \$3.00 per common share to be paid on March 28, 2025 to common shareholders of record on February 28, 2025.

Notes to Consolidated Financial Statements**Preferred Equity**

The tables below present information about the perpetual preferred stock issued and outstanding as of December 2024.

Series	Shares Authorized	Shares Issued	Shares Outstanding	Depository Shares Per Share
A	50,000	30,000	29,999	1,000
C	25,000	8,000	8,000	1,000
D	60,000	54,000	53,999	1,000
E	17,500	7,667	7,667	N.A.
F	5,000	1,615	1,615	N.A.
O	26,000	26,000	26,000	25
Q	20,000	20,000	20,000	25
R	24,000	24,000	24,000	25
S	14,000	14,000	14,000	25
T	27,000	27,000	27,000	25
U	30,000	30,000	30,000	25
V	30,000	30,000	30,000	25
W	60,000	60,000	60,000	25
X	90,000	90,000	90,000	25
Y	80,000	80,000	80,000	25
Total	558,500	502,282	502,280	

Series	Earliest Redemption Date	Liquidation Preference	Redemption Value (\$ in millions)
A	Currently redeemable	\$ 25,000	750
C	Currently redeemable	\$ 25,000	200
D	Currently redeemable	\$ 25,000	1,350
E	Currently redeemable	\$ 100,000	767
F	Currently redeemable	\$ 100,000	161
O	November 10, 2026	\$ 25,000	650
Q	Currently redeemable	\$ 25,000	500
R	February 10, 2025	\$ 25,000	600
S	February 10, 2025	\$ 25,000	350
T	May 10, 2026	\$ 25,000	675
U	August 10, 2026	\$ 25,000	750
V	November 10, 2026	\$ 25,000	750
W	February 10, 2029	\$ 25,000	1,500
X	May 10, 2029	\$ 25,000	2,250
Y	November 10, 2034	\$ 25,000	2,000
Total		\$	13,253

In the tables above:

- All shares have a par value of \$0.01 per share and, where applicable, each share is represented by the specified number of depository shares.
- The earliest redemption date represents the date on which each share of non-cumulative preferred stock is redeemable at the firm's option.
- Prior to redeeming preferred stock, the firm must receive approval from the Board of Governors of the Federal Reserve System (FRB).
- In April 2024, the firm issued 90,000 shares of Series X 7.50% Fixed-Rate Reset Non-Cumulative Preferred Stock (Series X Preferred Stock).
- In September 2024, the firm issued 80,000 shares of Series Y 6.125% Fixed-Rate Reset Non-Cumulative Preferred Stock (Series Y Preferred Stock).

- The redemption price per share for Series A through F and Series Q through Y Preferred Stock is the liquidation preference plus declared and unpaid dividends. The redemption price per share for Series O Preferred Stock is the liquidation preference plus accrued and unpaid dividends.

- All series of preferred stock are pari passu and have a preference over the firm's common stock on liquidation.

- The firm's ability to declare or pay dividends on, or purchase, redeem or otherwise acquire, its common stock is subject to certain restrictions in the event that the firm fails to pay or set aside full dividends on the preferred stock for the latest completed dividend period.

- Series E and Series F Preferred Stock are held by Goldman Sachs Capital II and Goldman Sachs Capital III, respectively. These trusts are Delaware statutory trusts sponsored by the firm and wholly-owned finance subsidiaries of the firm for regulatory and legal purposes but are not consolidated for accounting purposes.

In 2024, the firm redeemed all outstanding shares of its (i) Series K 6.375% Fixed-to-Floating Rate Non-Cumulative Preferred Stock (Series K Preferred Stock) with a redemption value of \$700 million (\$25,000 per share), plus accrued and unpaid dividends and (ii) Series P 5.00% Fixed-to-Floating Rate Non-Cumulative Preferred Stock (Series P Preferred Stock) with a redemption value of \$1.50 billion (\$25,000 per share), plus accrued and unpaid dividends. The difference between the redemption value and net carrying value at the time of these redemptions was \$34 million, which was recorded as an addition to preferred stock dividends in 2024.

In 2023, the firm redeemed all outstanding shares of its Series J 5.50% Fixed-to-Floating Rate Non-Cumulative Preferred Stock (Series J Preferred Stock) with a redemption value of \$1 billion (\$25,000 per share), plus accrued and unpaid dividends. The difference between the redemption value and net carrying value was \$10 million, which was recorded as an addition to preferred stock dividends in 2023.

In January 2025, the firm issued 76,000 shares of Series Z 6.85% Fixed-Rate Reset Non-Cumulative Preferred Stock (Series Z Preferred Stock). Each share of Series Z Preferred Stock issued and outstanding has a liquidation preference of \$25,000 per share, is represented by 25 depository shares and is redeemable at the firm's option beginning February 10, 2030 at a redemption price equal to \$25,000 per share plus declared and unpaid dividends. Dividends on Series Z Preferred Stock, if declared, are payable semi-annually at (i) 6.85% per annum from the issuance date to, but excluding, February 10, 2030 and, thereafter, (ii) 2.461% per annum plus the five-year treasury rate.

Notes to Consolidated Financial Statements

The preferred stock issuance costs in the consolidated statements of shareholders' equity reflects reclassifications of issuance costs to retained earnings on redemptions, net of issuance costs relating to new issuances.

The table below presents the dividend rates of perpetual preferred stock as of December 2024.

Series	Per Annum Dividend Rate
A	3 month term SOFR + 1.01161%, with floor of 3.75%, payable quarterly
C	3 month term SOFR + 1.01161%, with floor of 4.00%, payable quarterly
D	3 month term SOFR + 0.93161%, with floor of 4.00%, payable quarterly
E	3 month term SOFR + 1.02911%, with floor of 4.00%, payable quarterly
F	3 month term SOFR + 1.03161%, with floor of 4.00%, payable quarterly
O	5.30%, payable semi-annually, from issuance date to, but excluding, November 10, 2026; 3 month term SOFR + 4.09561%, payable quarterly, thereafter
Q	5 year treasury rate + 3.623%, payable semi-annually
R	4.95%, payable semi-annually, from issuance date to, but excluding, February 10, 2025; 5 year treasury rate + 3.224%, payable semi-annually, thereafter
S	4.40%, payable semi-annually, from issuance date to, but excluding, February 10, 2025; 5 year treasury rate + 2.85%, payable semi-annually thereafter
T	3.80%, payable semi-annually, from issuance date to, but excluding, May 10, 2026; 5 year treasury rate + 2.969%, payable semi-annually, thereafter
U	3.65%, payable semi-annually, from issuance date to, but excluding, August 10, 2026; 5 year treasury rate + 2.915%, payable semi-annually, thereafter
V	4.125%, payable semi-annually, from issuance date to, but excluding, November 10, 2026; 5 year treasury rate + 2.949%, payable semi-annually, thereafter
W	7.50%, payable semi-annually, from issuance date to, but excluding, February 10, 2029; 5 year treasury rate + 3.156%, payable semi-annually, thereafter
X	7.50%, payable semi-annually, from issuance date to, but excluding, May 10, 2029; 5 year treasury rate + 2.809%, payable semi-annually, thereafter
Y	6.125%, payable semi-annually, from issuance date to, but excluding, November 10, 2034; 10 year treasury rate + 2.40%, payable semi-annually, thereafter

In the table above, dividends on each series of preferred stock are payable in arrears for the periods specified.

The table below presents preferred stock dividends declared.

Series	2024		2023		2022	
	per share	\$ in millions	per share	\$ in millions	per share	\$ in millions
Year Ended December						
A	\$ 1,606.63	\$ 48	\$ 1,484.27	\$ 44	\$ 950.51	\$ 28
C	\$ 1,606.63	13	\$ 1,484.27	12	\$ 1,013.90	8
D	\$ 1,586.18	86	\$ 1,463.99	79	\$ 1,013.90	55
E	\$ 6,423.37	50	\$ 6,074.57	46	\$ 4,055.55	31
F	\$ 6,425.91	10	\$ 6,077.09	10	\$ 4,055.55	6
J	\$ -	-	\$ 1,261.02	51	\$ 1,375.00	55
K	\$ 796.88	20	\$ 1,593.76	44	\$ 1,593.76	45
O	\$ 1,325.00	34	\$ 1,325.00	34	\$ 1,325.00	34
P	\$ 1,623.09	97	\$ 2,022.64	121	\$ 1,250.00	75
Q	\$ 1,375.00	28	\$ 1,375.00	28	\$ 1,375.00	28
R	\$ 1,237.50	30	\$ 1,237.50	30	\$ 1,237.50	30
S	\$ 1,100.00	15	\$ 1,100.00	16	\$ 1,100.00	16
T	\$ 950.00	26	\$ 950.00	26	\$ 950.00	26
U	\$ 912.50	27	\$ 912.50	27	\$ 942.92	28
V	\$ 1,031.25	31	\$ 1,031.25	31	\$ 1,062.76	32
W	\$ 1,833.33	110	\$ -	-	\$ -	-
X	\$ 1,026.04	92	\$ -	-	\$ -	-
Total	\$ 717		\$ 599		\$ 497	

On January 7, 2025, Group Inc. declared dividends of \$345.49 per share of Series A Preferred Stock, \$345.49 per share of Series C Preferred Stock, \$340.49 per share of Series D Preferred Stock, \$922.38 per share of Series Q Preferred Stock, \$618.75 per share of Series R Preferred Stock, \$550.00 per share of Series S Preferred Stock, \$456.25 per share of Series U Preferred Stock and \$937.50 per share of Series W Preferred Stock that were paid on February 10, 2025 to preferred shareholders of record on January 26, 2025. In addition, the firm declared dividends of \$1,397.48 per share of Series E Preferred Stock and \$1,398.11 per share of Series F Preferred Stock that will be paid on March 3, 2025 to preferred shareholders of record on February 16, 2025.

Accumulated Other Comprehensive Income/(Loss)

The table below presents changes in accumulated other comprehensive income/(loss), net of tax, by type.

\$ in millions	Beginning balance	Other comprehensive income/(loss) adjustments, net of tax	Ending balance
Year Ended December 2024			
Currency translation	\$ (847)	\$ 32	\$ (815)
Debt valuation adjustment	(123)	(263)	(386)
Pension and postretirement liabilities	(575)	47	(528)
Available-for-sale securities	(1,373)	401	(972)
Cash flow hedges	-	(1)	(1)
Total	\$ (2,918)	\$ 216	\$ (2,702)
Year Ended December 2023			
Currency translation	\$ (785)	(62)	(847)
Debt valuation adjustment	892	(1,015)	(123)
Pension and postretirement liabilities	(499)	(76)	(575)
Available-for-sale securities	(2,618)	1,245	(1,373)
Cash flow hedges	-	-	-
Total	\$ (3,010)	\$ 92	\$ (2,918)
Year Ended December 2022			
Currency translation	\$ (738)	(47)	(785)
Debt valuation adjustment	(511)	1,403	892
Pension and postretirement liabilities	(327)	(172)	(499)
Available-for-sale securities	(492)	(2,126)	(2,618)
Cash flow hedges	-	-	-
Total	\$ (2,068)	\$ (942)	\$ (3,010)

Notes to Consolidated Financial Statements**Note 20.****Regulation and Capital Adequacy**

The FRB is the primary regulator of Group Inc., a bank holding company under the U.S. Bank Holding Company Act of 1956 and a financial holding company under amendments to this Act. The firm is subject to consolidated regulatory capital requirements which are calculated in accordance with the regulations of the FRB (Capital Framework).

The capital requirements are expressed as risk-based capital and leverage ratios that compare measures of regulatory capital to risk-weighted assets (RWAs), average assets and off-balance sheet exposures. Failure to comply with these capital requirements would result in restrictions being imposed by the firm's regulators and could limit the firm's ability to repurchase shares, pay dividends and make certain discretionary compensation payments. The firm's capital levels are also subject to qualitative judgments by the regulators about components of capital, risk weightings and other factors. Furthermore, certain of the firm's subsidiaries are subject to separate regulations and capital requirements.

Capital Framework

The regulations under the Capital Framework are largely based on the Basel Committee on Banking Supervision's (Basel Committee) capital framework for strengthening international capital standards (Basel III) and also implement certain provisions of the U.S. Dodd-Frank Wall Street Reform and Consumer Protection Act. Under the Capital Framework, the firm is an "Advanced approaches" banking organization and has been designated as a global systemically important bank (G-SIB).

The Capital Framework includes the minimum risk-based capital and the capital conservation buffer requirements. The buffer must consist entirely of capital that qualifies as Common Equity Tier 1 (CET1) capital.

The firm calculates its CET1 capital, Tier 1 capital and Total capital ratios in accordance with both the Standardized and Advanced Capital Rules. Each of the ratios calculated under the Standardized and Advanced Capital Rules must meet its respective capital requirements.

Under the Capital Framework, the firm is also subject to leverage requirements which consist of a minimum Tier 1 leverage ratio and a minimum supplementary leverage ratio (SLR), as well as the SLR buffer.

Consolidated Regulatory Capital Requirements

Risk-Based Capital Ratios. The table below presents the risk-based capital requirements.

	Standardized	Advanced
As of December 2024		
CET1 capital ratio	13.7 %	10.0 %
Tier 1 capital ratio	15.2 %	11.5 %
Total capital ratio	17.2 %	13.5 %
As of December 2023		
CET1 capital ratio	13.0 %	10.0 %
Tier 1 capital ratio	14.5 %	11.5 %
Total capital ratio	16.5 %	13.5 %

In the table above:

- As of both December 2024 and December 2023, under both the Standardized and Advanced Capital Rules, the CET1 capital ratio requirement includes a minimum of 4.5%, the Tier 1 capital ratio requirement includes a minimum of 6.0% and the Total capital ratio requirement includes a minimum of 8.0%. These requirements also include the capital conservation buffer requirements, consisting of the G-SIB surcharge (Method 2) of 3.0% and the countercyclical capital buffer, which the FRB has set to zero percent. In addition, the capital conservation buffer requirements include the stress capital buffer (SCB) of 6.2% as of December 2024 and 5.5% as of December 2023 under the Standardized Capital Rules and a buffer of 2.5% as of both December 2024 and December 2023 under the Advanced Capital Rules.
- The G-SIB surcharge is updated annually based on financial data from the prior year and is generally applicable for the following year. The G-SIB surcharge is calculated using two methodologies, the higher of which is reflected in the firm's risk-based capital requirements. The first calculation (Method 1) is based on the Basel Committee's methodology which, among other factors, relies upon measures of the size, activity and complexity of each G-SIB. The second calculation (Method 2) uses similar inputs but includes a measure of reliance on short-term wholesale funding.

Notes to Consolidated Financial Statements

The table below presents information about risk-based capital ratios.

<i>\$ in millions</i>	Standardized		Advanced	
As of December 2024				
CET1 capital	\$	103,065	\$	103,065
Tier 1 capital	\$	115,647	\$	115,647
Tier 2 capital	\$	14,125	\$	10,164
Total capital	\$	129,772	\$	125,811
RWAs	\$	688,541	\$	674,812
CET1 capital ratio		15.0 %		15.3 %
Tier 1 capital ratio		16.8 %		17.1 %
Total capital ratio		18.8 %		18.6 %
As of December 2023				
CET1 capital	\$	99,442	\$	99,442
Tier 1 capital	\$	110,288	\$	110,288
Tier 2 capital	\$	14,874	\$	10,684
Total capital	\$	125,162	\$	120,972
RWAs	\$	692,737	\$	665,348
CET1 capital ratio		14.4 %		14.9 %
Tier 1 capital ratio		15.9 %		16.6 %
Total capital ratio		18.1 %		18.2 %

Leverage Ratios. The table below presents the leverage requirements.

	As of December	
	2024	2023
Tier 1 leverage ratio	4.0 %	4.0 %
SLR	5.0 %	5.0 %

In the table above, the SLR requirement of 5% includes a minimum of 3% and a 2% buffer applicable to G-SIBs.

The table below presents information about leverage ratios.

<i>\$ in millions</i>	For the Three Months Ended or as of December	
	2024	2023
Tier 1 capital	\$ 115,647	\$ 110,288
Average total assets	\$ 1,699,419	\$ 1,579,237
Deductions from Tier 1 capital	(6,919)	(7,167)
Average adjusted total assets	1,692,500	1,572,070
Off-balance sheet and other exposures	428,256	423,686
Total leverage exposure	\$ 2,120,756	\$ 1,995,756
Tier 1 leverage ratio	6.8 %	7.0 %
SLR	5.5 %	5.5 %

In the table above:

- Average total assets represents the average daily assets for the quarter adjusted for the impact of Current Expected Credit Losses (CECL) transition.
- Off-balance sheet and other exposures primarily includes the monthly average of off-balance sheet exposures, consisting of derivatives, securities financing transactions, commitments and guarantees.
- Tier 1 leverage ratio is calculated as Tier 1 capital divided by average adjusted total assets.
- SLR is calculated as Tier 1 capital divided by total leverage exposure.

Risk-Based Capital. The table below presents information about risk-based capital.

<i>\$ in millions</i>	As of December	
	2024	2023
Common shareholders' equity	\$ 108,743	\$ 105,702
Impact of CECL transition	276	553
Deduction for goodwill	(5,159)	(5,224)
Deduction for identifiable intangible assets	(638)	(950)
Other adjustments	(157)	(639)
CET1 capital	103,065	99,442
Preferred stock	13,253	11,203
Deduction for investments in covered funds	(669)	(354)
Other adjustments	(2)	(3)
Tier 1 capital	\$ 115,647	\$ 110,288
Standardized Tier 2 and Total capital		
Tier 1 capital	\$ 115,647	\$ 110,288
Qualifying subordinated debt	9,124	9,886
Allowance for credit losses	5,011	5,012
Other adjustments	(10)	(24)
Standardized Tier 2 capital	14,125	14,874
Standardized Total capital	\$ 129,772	\$ 125,162
Advanced Tier 2 and Total capital		
Tier 1 capital	\$ 115,647	\$ 110,288
Standardized Tier 2 capital	14,125	14,874
Allowance for credit losses	(5,011)	(5,012)
Other adjustments	1,050	822
Advanced Tier 2 capital	10,164	10,684
Advanced Total capital	\$ 125,811	\$ 120,972

In the table above:

- Beginning in January 2022, the firm started to phase in the estimated reduction to regulatory capital as a result of adopting the CECL model. The total amount of reduction to be phased in from January 1, 2022 through January 1, 2025 (at 25% per year) was \$1.11 billion, of which \$829 million had been phased in as of December 2024. The total amount to be phased in includes the impact of adopting CECL as of January 1, 2020, as well as 25% of the increase in the allowance for credit losses from January 1, 2020 through December 31, 2021. The impact of CECL transition reflects the remaining amount of reduction to be phased in as of both December 2024 and December 2023.
- Deduction for goodwill was net of deferred tax liabilities of \$694 million as of December 2024 and \$692 million as of December 2023.
- Deduction for identifiable intangible assets was net of deferred tax liabilities of \$209 million as of December 2024 and \$227 million as of December 2023.
- Deduction for investments in covered funds represents the firm's aggregate investments in applicable covered funds as defined in the Volcker Rule.
- Other adjustments within CET1 capital and Tier 1 capital primarily include CVAs on derivative liabilities, the overfunded portion of the firm's defined benefit pension plan obligation net of associated deferred tax liabilities, disallowed deferred tax assets, debt valuation adjustments and other required credit risk-based deductions. Other adjustments within Advanced Tier 2 capital include eligible credit reserves.

Notes to Consolidated Financial Statements

- Qualifying subordinated debt is subordinated debt issued by Group Inc. with an original maturity of five years or greater. The outstanding amount of subordinated debt qualifying for Tier 2 capital is reduced upon reaching a remaining maturity of five years. See Note 14 for further information about the firm's subordinated debt.

The table below presents changes in CET1 capital, Tier 1 capital and Tier 2 capital.

<i>\$ in millions</i>	Standardized	Advanced
Year Ended December 2024		
CET1 capital		
Beginning balance	\$ 99,442	\$ 99,442
Change in:		
Common shareholders' equity	3,041	3,041
Impact of CECL transition	(277)	(277)
Deduction for goodwill	65	65
Deduction for identifiable intangible assets	312	312
Other adjustments	482	482
Ending balance	\$ 103,065	\$ 103,065
Tier 1 capital		
Beginning balance	\$ 110,288	\$ 110,288
Change in:		
CET1 capital	3,623	3,623
Preferred stock	2,050	2,050
Deduction for investments in covered funds	(315)	(315)
Other adjustments	1	1
Ending balance	115,647	115,647
Tier 2 capital		
Beginning balance	14,874	10,684
Change in:		
Qualifying subordinated debt	(762)	(762)
Allowance for credit losses	(1)	-
Other adjustments	14	242
Ending balance	14,125	10,164
Total capital	\$ 129,772	\$ 125,811

RWAs. RWAs are calculated in accordance with both the Standardized and Advanced Capital Rules.

Credit Risk

Credit RWAs are calculated based on measures of exposure, which are then risk weighted under the Standardized and Advanced Capital Rules:

- The Standardized Capital Rules apply prescribed risk-weights, which depend largely on the type of counterparty. The exposure measures for derivatives and securities financing transactions are based on specific formulas which take certain factors into consideration.
- Under the Advanced Capital Rules, the firm computes risk-weights for wholesale and retail credit exposures in accordance with the Advanced Internal Ratings-Based approach. The exposure measures for derivatives and securities financing transactions are computed utilizing internal models.
- For both Standardized and Advanced credit RWAs, the risk-weights for securitizations and equities are based on specific required formulaic approaches.

Market Risk

RWAs for market risk in accordance with the Standardized and Advanced Capital Rules are generally consistent. Market RWAs are calculated based on measures of exposure which include the following:

- Value-at-Risk (VaR) is the potential loss in value of trading assets and liabilities, as well as certain investments, loans, and other financial assets and liabilities accounted for at fair value, due to adverse market movements over a defined time horizon with a specified confidence level.

For both risk management purposes and regulatory capital calculations, the firm uses a single VaR model which captures risks, including those related to interest rates, equity prices, currency rates and commodity prices. However, VaR used for risk management purposes differs from VaR used for regulatory capital requirements (regulatory VaR) due to differences in time horizons, confidence levels and the scope of positions on which VaR is calculated. For risk management purposes, a 95% one-day VaR is used, whereas for regulatory capital requirements, a 99% 10-day VaR is used to determine Market RWAs and a 99% one-day VaR is used to determine regulatory VaR exceptions. In addition, the daily net revenues used to determine risk management VaR exceptions (i.e., comparing the daily net revenues to the VaR measure calculated as of the end of the prior business day) include intraday activity, whereas the Capital Framework requires that intraday activity be excluded from daily net revenues when calculating regulatory VaR exceptions. Intraday activity includes bid/offer net revenues, which are more likely than not to be positive by their nature. As a result, there may be differences in the number of VaR exceptions and the amount of daily net revenues calculated for regulatory VaR compared to the amounts calculated for risk management VaR.

The firm's positional losses observed on a single day exceeded its 99% one-day regulatory VaR on two occasions during 2024 and on one occasion during 2023. There was no change in the firm's VaR multiplier used to calculate Market RWAs;

- Stressed VaR is the potential loss in value of trading assets and liabilities, as well as certain investments, loans, and other financial assets and liabilities accounted for at fair value, during a period of significant market stress;
- Incremental risk is the potential loss in value of non-securitized positions due to the default or credit migration of issuers of financial instruments over a one-year time horizon;

Notes to Consolidated Financial Statements

- Comprehensive risk is the potential loss in value, due to price risk and defaults, within the firm's credit correlation positions; and
- Specific risk is the risk of loss on a position that could result from factors other than broad market movements, including event risk, default risk and idiosyncratic risk. The standardized measurement method is used to determine specific risk RWAs, by applying supervisory defined risk-weighting factors after applicable netting is performed.

Operational Risk

Operational RWAs are only required to be included under the Advanced Capital Rules. The firm utilizes an internal risk-based model to quantify Operational RWAs.

The table below presents information about RWAs.

<i>\$ in millions</i>	Standardized	Advanced
As of December 2024		
Credit RWAs		
Derivatives	\$ 146,368	\$ 99,766
Commitments, guarantees and loans	247,140	199,816
Securities financing transactions	97,174	22,846
Equity investments	30,018	31,457
Other	71,013	97,129
Total Credit RWAs	591,713	451,014
Market RWAs		
Regulatory VaR	19,995	19,995
Stressed VaR	48,249	48,249
Incremental risk	7,054	7,054
Comprehensive risk	2,057	2,057
Specific risk	19,473	19,473
Total Market RWAs	96,828	96,828
Total Operational RWAs	-	126,970
Total RWAs	\$ 688,541	\$ 674,812
As of December 2023		
Credit RWAs		
Derivatives	\$ 146,357	\$ 96,322
Commitments, guarantees and loans	243,094	194,236
Securities financing transactions	103,704	23,637
Equity investments	34,223	36,920
Other	76,481	96,755
Total Credit RWAs	603,859	447,870
Market RWAs		
Regulatory VaR	16,457	16,457
Stressed VaR	48,496	48,496
Incremental risk	5,032	5,032
Comprehensive risk	2,718	2,718
Specific risk	16,175	16,175
Total Market RWAs	88,878	88,878
Total Operational RWAs	-	128,600
Total RWAs	\$ 692,737	\$ 665,348

In the table above:

- Securities financing transactions represents resale and repurchase agreements and securities borrowed and loaned transactions.
- Other includes receivables, certain debt securities, cash and cash equivalents, and other assets.

The table below presents changes in RWAs.

<i>\$ in millions</i>	Standardized	Advanced
Year Ended December 2024		
RWAs		
Beginning balance	\$ 692,737	\$ 665,348
Credit RWAs		
Change in:		
Derivatives	11	3,444
Commitments, guarantees and loans	4,046	5,580
Securities financing transactions	(6,530)	(791)
Equity investments	(4,205)	(5,463)
Other	(5,468)	374
Change in Credit RWAs	(12,146)	3,144
Market RWAs		
Change in:		
Regulatory VaR	3,538	3,538
Stressed VaR	(247)	(247)
Incremental risk	2,022	2,022
Comprehensive risk	(661)	(661)
Specific risk	3,298	3,298
Change in Market RWAs	7,950	7,950
Change in Operational RWAs	-	(1,630)
Ending balance	\$ 688,541	\$ 674,812

RWAs Rollforward Commentary

Year Ended December 2024. Standardized Credit RWAs as of December 2024 decreased by \$12.15 billion compared with December 2023, primarily reflecting a decrease in securities financing transactions (principally due to reduced funding exposures), a decrease in other credit RWAs (principally due to decreases in other assets), and a decrease in equity investments (principally due to reduced exposures). These decreases were partially offset by an increase in commitments, guarantees and loans (principally due to increased lending exposures). Standardized Market RWAs as of December 2024 increased by \$7.95 billion compared with December 2023, primarily reflecting an increase in regulatory VaR (principally due to increased exposures) and an increase in specific risk (principally due to increased exposures to debt and equity instruments held for market-making purposes).

Advanced Credit RWAs as of December 2024 increased by \$3.14 billion compared with December 2023, primarily reflecting an increase in commitments, guarantees and loans (principally due to increased lending exposures) and an increase in derivatives (principally due to increased counterparty credit risk). These increases were partially offset by a decrease in equity investments (principally due to reduced exposures). Advanced Market RWAs as of December 2024 increased by \$7.95 billion compared with December 2023, primarily reflecting an increase in regulatory VaR (principally due to increased exposures) and an increase in specific risk (principally due to increased exposures to debt and equity instruments held for market-making purposes).

Notes to Consolidated Financial Statements**GS Bank USA**

GS Bank USA is the firm's primary U.S. bank subsidiary. GS Bank USA is a New York State-chartered bank and a member of the Federal Reserve System, is supervised and regulated by the FRB, the FDIC, the New York State Department of Financial Services (NYDFS) and the Consumer Financial Protection Bureau (CFPB), and is subject to regulatory capital requirements that are calculated under the Capital Framework. GS Bank USA is an "Advanced approaches" banking organization under the Capital Framework. The deposits of GS Bank USA are insured by the FDIC to the extent provided by law.

The Capital Framework includes the minimum risk-based capital and the capital conservation buffer requirements (consisting of a 2.5% buffer and the countercyclical capital buffer). The buffer must consist entirely of capital that qualifies as CET1 capital. In addition, the Capital Framework includes the leverage ratio requirement. GS Bank USA is required to calculate the CET1 capital, Tier 1 capital and Total capital ratios in accordance with both the Standardized and Advanced Capital Rules. The lower of each risk-based capital ratio under the Standardized and Advanced Capital Rules is the ratio against which GS Bank USA's compliance with its risk-based capital requirements is assessed. In addition, under the regulatory framework for prompt corrective action applicable to GS Bank USA, in order to meet the quantitative requirements for a "well-capitalized" depository institution, GS Bank USA must also meet the "well-capitalized" requirements in the table below. GS Bank USA's capital levels and prompt corrective action classification are also subject to qualitative judgments by the regulators about components of capital, risk weightings and other factors. Failure to comply with the capital requirements, including a breach of the buffers described below, would result in restrictions being imposed by the regulators.

The table below presents GS Bank USA's risk-based capital, leverage and "well-capitalized" requirements.

	As of December			
	2024	2023	2024	2023
			"Well-capitalized"	
	Requirements		Requirements	
Risk-based capital requirements				
CET1 capital ratio	7.0 %	7.0 %	6.5 %	6.5 %
Tier 1 capital ratio	8.5 %	8.5 %	8.0 %	8.0 %
Total capital ratio	10.5 %	10.5 %	10.0 %	10.0 %
Leverage requirements				
Tier 1 leverage ratio	4.0 %	4.0 %	5.0 %	5.0 %
SLR	3.0 %	3.0 %	6.0 %	6.0 %

In the table above:

- The CET1 capital ratio requirement includes a minimum of 4.5%, the Tier 1 capital ratio requirement includes a minimum of 6.0% and the Total capital ratio requirement includes a minimum of 8.0%. These requirements also include the capital conservation buffer requirements consisting of a 2.5% buffer and the countercyclical capital buffer, which the FRB has set to zero percent.
- The "well-capitalized" requirements are the binding requirements for leverage ratios.

The table below presents information about GS Bank USA's risk-based capital ratios.

<i>\$ in millions</i>	Standardized		Advanced	
As of December 2024				
CET1 capital	\$	62,022	\$	62,022
Tier 1 capital	\$	62,022	\$	62,022
Tier 2 capital	\$	4,209	\$	955
Total capital	\$	66,231	\$	62,977
RWAs	\$	386,922	\$	284,624
CET1 capital ratio		16.0 %		21.8 %
Tier 1 capital ratio		16.0 %		21.8 %
Total capital ratio		17.1 %		22.1 %
As of December 2023				
CET1 capital	\$	53,781	\$	53,781
Tier 1 capital	\$	53,781	\$	53,781
Tier 2 capital	\$	6,314	\$	2,951
Total capital	\$	60,095	\$	56,732
RWAs	\$	380,774	\$	288,938
CET1 capital ratio		14.1 %		18.6 %
Tier 1 capital ratio		14.1 %		18.6 %
Total capital ratio		15.8 %		19.6 %

In the table above:

- The lower of the Standardized or Advanced ratio is the ratio against which GS Bank USA's compliance with the capital requirements is assessed under the risk-based Capital Rules, and therefore, the Standardized ratios applied to GS Bank USA as of both December 2024 and December 2023.
- Beginning in January 2022, GS Bank USA started to phase in the estimated reduction to regulatory capital as a result of adopting the CECL model at 25% per year through January 2025. The total amount to be phased in includes the impact of adopting CECL as of January 1, 2020, as well as 25% of the increase in the allowance for credit losses from January 1, 2020 through December 31, 2021.
- The Standardized and Advanced risk-based capital ratios increased from December 2023 to December 2024, primarily reflecting an increase in capital, principally due to net earnings, and a decrease in Market RWAs, partially offset by an increase in Credit RWAs.

Notes to Consolidated Financial Statements

The table below presents information about GS Bank USA's leverage ratios.

\$ in millions	For the Three Months Ended or as of December	
	2024	2023
Tier 1 capital	\$ 62,022	\$ 53,781
Average adjusted total assets	\$ 565,513	\$ 523,546
Total leverage exposure	\$ 775,170	\$ 722,465
Tier 1 leverage ratio	11.0 %	10.3 %
SLR	8.0 %	7.4 %

In the table above:

- Average adjusted total assets represents the average daily assets for the quarter adjusted for deductions from Tier 1 capital and the impact of CECL transition.
- Tier 1 leverage ratio is calculated as Tier 1 capital divided by average adjusted total assets.
- SLR is calculated as Tier 1 capital divided by total leverage exposure.

The FRB requires that GS Bank USA maintain cash reserves with the Federal Reserve. As of both December 2024 and December 2023, the reserve requirement ratio was zero percent. See Note 26 for further information about cash deposits held by the firm at the Federal Reserve.

GS Bank USA is a registered swap dealer with the CFTC and a registered security-based swap dealer with the SEC. As of both December 2024 and December 2023, GS Bank USA was subject to and in compliance with applicable capital requirements for swap dealers and security-based swap dealers.

Restrictions on Payments

Group Inc. may be limited in its ability to access capital held at certain subsidiaries as a result of regulatory, tax or other constraints. These limitations include provisions of applicable law and regulations and other regulatory restrictions that limit the ability of those subsidiaries to declare and pay dividends without prior regulatory approval. For example, the amount of dividends that may be paid by GS Bank USA are limited to the lesser of the amounts calculated under a recent earnings test and an undivided profits test.

In addition, subsidiaries not subject to separate regulatory capital requirements may hold capital to satisfy local tax and legal guidelines, rating agency requirements (for entities with assigned credit ratings) or internal policies, including policies concerning the minimum amount of capital a subsidiary should hold based on its underlying level of risk.

Group Inc.'s equity investment in subsidiaries was \$140.79 billion as of December 2024 and \$133.75 billion as of December 2023, of which Group Inc. was required to maintain \$98.48 billion as of December 2024 and \$95.80 billion as of December 2023, of minimum equity capital in its regulated subsidiaries in order to satisfy the regulatory requirements of such subsidiaries.

Group Inc.'s capital invested in certain non-U.S. dollar functional currency subsidiaries is exposed to foreign exchange risk, substantially all of which is managed through a combination of non-U.S. dollar-denominated debt and derivatives. See Note 7 for information about the firm's net investment hedges used to hedge this risk.

Notes to Consolidated Financial Statements**Note 21.****Earnings Per Common Share**

Basic EPS is calculated by dividing net earnings to common by the weighted average number of common shares outstanding and restricted stock units (RSUs) for which the delivery of the underlying common stock is not subject to satisfaction of future service, performance or market conditions (collectively, basic shares). Diluted EPS includes the determinants of basic EPS and, in addition, reflects the dilutive effect of the common stock deliverable for RSUs for which the delivery of the underlying common stock is subject to satisfaction of future service, performance or market conditions.

The table below presents information about basic and diluted EPS.

	Year Ended December		
<i>in millions, except per share amounts</i>	2024	2023	2022
Net earnings to common	\$ 13,525	\$ 7,907	\$ 10,764
Weighted average basic shares	328.1	340.8	352.1
Effect of dilutive RSUs	5.5	5.0	6.0
Weighted average diluted shares	333.6	345.8	358.1
Basic EPS	\$ 41.07	\$ 23.05	\$ 30.42
Diluted EPS	\$ 40.54	\$ 22.87	\$ 30.06

In the table above:

- Net earnings to common represents net earnings applicable to common shareholders, which is calculated as net earnings less preferred stock dividends.
- Unvested share-based awards that have non-forfeitable rights to dividends or dividend equivalents are treated as a separate class of securities under the two-class method. Distributed earnings allocated to these securities reduce net earnings to common to calculate EPS under this method. The impact of applying this methodology was a reduction in basic EPS of \$0.15 for each of 2024, 2023 and 2022.
- Diluted EPS does not include antidilutive RSUs, including those that are subject to market or performance conditions, of 0.1 million for 2024, 0.4 million for 2023 and 0.5 million for 2022.

Note 22.**Transactions with Affiliated Funds**

The firm has formed nonconsolidated investment funds with third-party investors. As the firm generally acts as the investment manager for these funds, it is entitled to receive management fees and, in certain cases, advisory fees or incentive fees from these funds. Additionally, the firm invests alongside its clients in certain funds.

The tables below present information about affiliated funds.

	Year Ended December		
<i>\$ in millions</i>	2024	2023	2022
Fees earned from funds	\$ 5,399	\$ 4,726	\$ 4,553

	As of December	
<i>\$ in millions</i>	2024	2023
Fees receivable from funds	\$ 1,589	\$ 1,536
Aggregate carrying value of interests in funds	\$ 4,079	\$ 4,042

In the ordinary course of business, the firm may choose to provide voluntary financial support to funds, although any such support is not expected to be material to the results of operations of the firm. The firm has waived or deferred collection of management fees and has deferred reimbursement of expenses, and in the future may waive or defer collection of management fees, from select funds. The impact of these waivers and deferrals was not material to the firm's results of operations for both 2024 and 2023. Management fees waived were \$123 million for 2022. Except as noted above, the firm did not provide any additional financial support to its affiliated funds during each of 2024, 2023 and 2022.

In addition, in the ordinary course of business, the firm may also engage in other activities with its affiliated funds, including, among others, securities lending, trade execution, market-making, custody, and acquisition and bridge financing. See Note 18 for information about the firm's investment commitments related to these funds.

Notes to Consolidated Financial Statements**Note 23.****Interest Income and Interest Expense**

Interest is recorded over the life of the instrument on an accrual basis based on contractual interest rates.

The table below presents sources of interest income and interest expense.

\$ in millions	Year Ended December		
	2024	2023	2022
Deposits with banks	\$ 9,282	\$ 10,949	\$ 3,233
Collateralized agreements	19,895	16,405	4,468
Trading assets	14,316	8,460	5,087
Investments	5,998	3,856	2,199
Loans	16,162	14,905	9,059
Other interest	15,744	13,940	4,978
Total interest income	81,397	68,515	29,024
Deposits	20,282	17,010	5,823
Collateralized financings	17,362	12,705	2,808
Trading liabilities	2,911	2,453	1,923
Short-term borrowings	2,112	1,322	541
Long-term borrowings	11,010	11,084	5,716
Other interest	19,664	17,590	4,535
Total interest expense	73,341	62,164	21,346
Net interest income	\$ 8,056	\$ 6,351	\$ 7,678

In the table above:

- Collateralized agreements includes rebates paid and interest income on securities borrowed.
- Loans excludes interest on loans held for sale that are accounted for at the lower of cost or fair value. Such interest is included within other interest.
- Other interest income includes interest income on customer debit balances, other interest-earning assets and loans held for sale that are accounted for at the lower of cost or fair value.
- Collateralized financings consists of repurchase agreements and securities loaned.
- Short- and long-term borrowings include both secured and unsecured borrowings.
- Other interest expense includes rebates received on other interest-bearing liabilities and interest expense on customer credit balances.

Note 24.**Income Taxes****Provision for Income Taxes**

Income taxes are provided for using the asset and liability method under which deferred tax assets and liabilities are recognized for temporary differences between the financial reporting and tax bases of assets and liabilities. The firm reports interest expense related to income tax matters in provision for taxes and income tax penalties in other expenses.

The table below presents information about the provision for taxes.

\$ in millions	Year Ended December		
	2024	2023	2022
Current taxes			
U.S. federal	\$ 2,000	\$ 1,230	\$ 2,356
State and local	659	389	623
Non-U.S.	2,262	1,964	1,658
Total current tax expense	4,921	3,583	4,637
Deferred taxes			
U.S. federal	(869)	(954)	(2,079)
State and local	(140)	(356)	(436)
Non-U.S.	209	(50)	103
Total deferred tax (benefit)/expense	(800)	(1,360)	(2,412)
Provision for taxes	\$ 4,121	\$ 2,223	\$ 2,225

The table below presents a reconciliation of the U.S. federal statutory income tax rate to the effective income tax rate.

	Year Ended December		
	2024	2023	2022
U.S. federal statutory income tax rate	21.0 %	21.0 %	21.0 %
State and local taxes, net of U.S. federal benefit	2.4	0.6	1.3
Settlement of employee share-based awards	(1.2)	(1.8)	(2.4)
Non-U.S. operations	(1.4)	(2.4)	(3.4)
GILTI	2.8	4.4	1.8
Tax credits	(0.8)	(1.6)	(0.9)
Tax-exempt income, including dividends	(0.6)	(1.0)	(2.2)
Non-deductible legal expenses	-	0.2	0.8
Other	0.2	1.3	0.5
Effective income tax rate	22.4 %	20.7 %	16.5 %

In the table above:

- The firm recognizes income tax expense associated with Global Intangible Low Taxed Income (GILTI) in the period in which it is incurred.
- Beginning in the fourth quarter of 2023, GILTI is presented as a separate reconciliation item. Previously, GILTI was included in non-U.S. operations. Prior period amounts have been conformed to the current presentation.

Notes to Consolidated Financial Statements**Deferred Income Taxes**

Deferred income taxes reflect the net tax effects of temporary differences between the financial reporting and tax bases of assets and liabilities. These temporary differences result in taxable or deductible amounts in future years and are measured using the tax rates and laws that will be in effect when such differences are expected to reverse. Valuation allowances are established to reduce deferred tax assets to the amount that more likely than not will be realized and primarily relate to the ability to utilize losses and tax credits in various tax jurisdictions. Tax assets are included in other assets and tax liabilities are included in other liabilities.

The table below presents information about deferred tax assets and liabilities, excluding the impact of netting within tax jurisdictions.

\$ in millions	As of December	
	2024	2023
Deferred tax assets		
Compensation and benefits	\$ 1,888	\$ 1,773
ASC 740 asset related to unrecognized tax benefits	366	331
Non-U.S. operations	1,507	1,278
Unrealized losses	2,786	2,100
Net operating losses	786	929
Occupancy-related	85	98
Other comprehensive income/(loss)-related	714	1,198
Tax credits carryforward	461	236
Operating lease liabilities	581	636
Allowance for credit losses	1,412	1,450
Other, net	421	165
Subtotal	11,007	10,194
Valuation allowance	(2,064)	(1,978)
Total deferred tax assets	\$ 8,943	\$ 8,216
Deferred tax liabilities		
Depreciation and amortization	\$ 982	\$ 752
Operating lease right-of-use assets	512	574
Total deferred tax liabilities	\$ 1,494	\$ 1,326

The firm has recorded deferred tax assets of \$ 786 million as of December 2024 and \$929 million as of December 2023, in connection with U.S. federal, state and local and foreign net operating loss carryforwards. The firm also recorded a valuation allowance of \$328 million as of December 2024 and \$396 million as of December 2023, related to these net operating loss carryforwards.

As of December 2024, the U.S. federal net operating loss carryforward was \$1.3 billion, the state and local net operating loss carryforward was \$3.3 billion, and the foreign net operating loss carryforward was \$1.5 billion. If not utilized, the U.S. federal, the state and local, and foreign net operating loss carryforwards will begin to expire in 2025. If these carryforwards expire, they will not have a material impact on the firm's results of operations. As of December 2024, the firm has recorded deferred tax assets of \$405 million in connection with foreign tax credit carryforwards and a related valuation allowance of \$289 million. As of December 2024, the firm has recorded deferred tax assets of \$45 million in connection with general business credit carryforwards and \$11 million in connection with state and local tax credit carryforwards. If not utilized, the foreign tax credit carryforward will begin to expire in 2033, the general business credit carryforward will begin to expire in 2025 and the state and local tax credit carryforward will begin to expire in 2026.

As of both December 2024 and December 2023, the firm had no U.S. federal capital loss carryforwards and no related net deferred income tax assets. As of December 2024, the firm had deferred tax assets of \$7 million in connection with state and local capital loss carryforwards and a valuation allowance of \$1 million related to these capital loss carryforwards. As of December 2024, the firm had deferred tax assets of \$365 million in connection with foreign capital loss carryforwards and a valuation allowance of \$346 million related to these capital loss carryforwards.

The valuation allowance increased by \$86 million during 2024 and \$409 million during 2023. The increases in both 2024 and 2023 were primarily due to an increase in deferred tax assets from which the firm does not expect to realize any benefit.

The firm permanently reinvested eligible earnings of certain foreign subsidiaries. As of both December 2024 and December 2023, all U.S. taxes were accrued on these subsidiaries' distributable earnings.

Notes to Consolidated Financial Statements**Unrecognized Tax Benefits**

The firm recognizes tax positions in the consolidated financial statements only when it is more likely than not that the position will be sustained on examination by the relevant taxing authority based on the technical merits of the position. A position that meets this standard is measured at the largest amount of benefit that will more likely than not be realized on settlement. A liability is established for differences between positions taken in a tax return and amounts recognized in the consolidated financial statements.

The accrued liability for interest expense related to income tax matters and income tax penalties was \$475 million as of December 2024 and \$320 million as of December 2023. The firm recognized interest expense and income tax penalties of \$113 million for 2024, \$90 million for 2023 and \$59 million for 2022. It is reasonably possible that unrecognized tax benefits could change significantly during the twelve months subsequent to December 2024 due to potential audit settlements. However, at this time it is not possible to estimate any potential change.

The table below presents the changes in the liability for unrecognized tax benefits, which is included in other liabilities.

\$ in millions	Year Ended or as of December		
	2024	2023	2022
Beginning balance	\$ 1,726	\$ 1,533	\$ 1,446
Increases based on current year tax positions	381	143	190
Increases based on prior years' tax positions	87	164	10
Decreases based on prior years' tax positions	(23)	(92)	(32)
Decreases related to settlements	(9)	(20)	(76)
Exchange rate fluctuations	—	(2)	(5)
Ending balance	\$ 2,162	\$ 1,726	\$ 1,533

In the table above, the liability for unrecognized tax benefits included \$1.8 billion as of December 2024, \$1.4 billion as of December 2023 and \$1.2 billion as of December 2022 of unrecognized tax benefits which, if recognized, would reduce the annual effective tax rate. The remaining unrecognized tax benefits in the table above would not affect the annual tax rate, as such benefits have jurisdictional offsets or relate to temporary differences.

Regulatory Tax Examinations

The firm is subject to examination by the U.S. Internal Revenue Service (IRS) and other taxing authorities in jurisdictions where the firm has significant business operations, such as the United Kingdom, Japan, Hong Kong and various states, such as New York. The tax years under examination vary by jurisdiction. The firm does not expect completion of these audits to have a material impact on the firm's financial condition, but it may be material to operating results for a particular period, depending, in part, on the operating results for that period.

The table below presents the earliest tax years that remain subject to examination by major jurisdiction.

Jurisdiction	As of December 2024
U.S. Federal	2011
New York State and City	2015
United Kingdom	2017
Japan	2018
Hong Kong	2018

The firm has been accepted into the Compliance Assurance Process (CAP) program by the IRS for each of the tax years from 2013 through 2025. This program allows the firm to work with the IRS to identify and resolve potential U.S. Federal tax issues before the filing of tax returns. All issues addressed through the CAP program for the 2011 through 2018 tax years have been resolved and completion is pending final review by the Joint Committee on Taxation. All issues for the 2019 through 2022 tax years have been resolved and will be effectively settled pending administrative completion by the IRS. Final completion of tax years 2011 through 2022 will not have a material impact on the effective tax rate. The 2023 tax year remains subject to post-filing review. New York State and City examinations of tax years 2015 through 2018 commenced during 2021.

All years, including and subsequent to the years in the table above, remain open to examination by the taxing authorities. The firm believes that the liability for unrecognized tax benefits it has established is adequate in relation to the potential for additional assessments.

Notes to Consolidated Financial Statements**Note 25.****Business Segments**

The firm manages and reports its activities in three business segments: Global Banking & Markets, Asset & Wealth Management and Platform Solutions. These business segments are determined and organized based on products and services provided, and the types of customers and counterparties served. See Note 1 for a description of the firm's business segments.

The firm's chief operating decision maker (CODM) is its president and chief operating officer. The CODM makes operating decisions, assesses the performance of, and allocates resources to, the firm's operating segments principally based on the total net revenues of the segments, revenues net of provision for credit losses, total operating expenses, pre-tax earnings, net earnings applicable to common shareholders and the return on average common equity to assess the performance of the segments. The CODM evaluates segment operating performance against the firm's targets and industry metrics and considers the current and future business and operating environment.

The accounting policies used to prepare the operating results and other metrics for the segments are consistent with those described in Note 3. The following provides a description of the primary components of the firm's segment results disclosed in the table below.

- The firm fully allocates its revenues, expenses, assets and shareholders' equity to the firm's three business segments.
- Revenues and expenses directly associated with each segment are included in determining pre-tax earnings for the respective segment.
- Net revenues in the firm's segments include allocations of interest income and expense based on the funding generated by, or the funding and liquidity requirements of the respective segments. Net interest is included in segment net revenues as it is consistent with how management assesses segment performance.
- Expenses not directly associated with specific segments are allocated among the business segments based on an estimate of support provided to each segment.
- Compensation and benefits expenses in the firm's segments reflect, among other factors, the overall performance of the firm, as well as the performance of individual businesses. Consequently, pre-tax margins in one segment of the firm's business may be significantly affected by the performance of the firm's other business segments.
- Certain assets (including allocations of global core liquid assets and cash, and secured client financing), not directly associated with specific segments are generally allocated among the business segments based on the funding and liquidity requirements of the segments.
- Common shareholders' equity and preferred stock dividends are allocated to each segment based on the estimated amount of equity required to support the activities of the segment under relevant regulatory capital requirements.
- Net earnings for each segment is calculated by applying the firmwide tax rate to each segment's pre-tax earnings.
- Management believes that this allocation provides a reasonable representation of each segment's contribution to consolidated net earnings to common, return on average common equity and total assets. Due to the integrated nature of these segments, estimates and judgments are made in allocating these assets, revenues and expenses. Transactions between segments are based on specific criteria or approximate third-party rates.

Notes to Consolidated Financial Statements**Segment Results**

The table below presents a summary of the firm's segment results.

\$ in millions	Year Ended December		
	2024	2023	2022
Global Banking & Markets			
Non-interest revenues	\$ 32,538	\$ 29,254	\$ 30,042
Net interest income	2,405	742	2,445
Total net revenues	34,943	29,996	32,487
Provision for credit losses	40	401	468
Compensation and benefits expenses	9,426	8,571	8,661
Other operating expenses	10,554	9,469	9,190
Total operating expenses	19,980	18,040	17,851
Pre-tax earnings	\$ 14,923	\$ 11,555	\$ 14,168
Net earnings	\$ 11,580	\$ 9,163	\$ 11,830
Net earnings to common	\$ 10,998	\$ 8,703	\$ 11,458
Average common equity	\$ 75,796	\$ 71,863	\$ 69,951
Return on average common equity	14.5 %	12.1 %	16.4 %
Asset & Wealth Management			
Non-interest revenues	\$ 13,346	\$ 10,790	\$ 9,843
Net interest income	2,796	3,090	3,533
Total net revenues	16,142	13,880	13,376
Provision for credit losses	(232)	(508)	519
Compensation and benefits expenses	6,595	6,144	5,927
Other operating expenses	5,230	6,885	5,623
Total operating expenses	11,825	13,029	11,550
Pre-tax earnings	\$ 4,549	\$ 1,359	\$ 1,307
Net earnings	\$ 3,530	\$ 1,078	\$ 1,092
Net earnings to common	\$ 3,386	\$ 952	\$ 979
Average common equity	\$ 26,405	\$ 30,078	\$ 31,762
Return on average common equity	12.8 %	3.2 %	3.1 %
Platform Solutions			
Non-interest revenues	\$ (428)	\$ (141)	\$ (198)
Net interest income	2,855	2,519	1,700
Total net revenues	2,427	2,378	1,502
Provision for credit losses	1,540	1,135	1,728
Compensation and benefits expenses	685	784	560
Other operating expenses	1,277	2,634	1,203
Total operating expenses	1,962	3,418	1,763
Pre-tax earnings/(loss)	\$ (1,075)	\$ (2,175)	\$ (1,989)
Net earnings/(loss)	\$ (834)	\$ (1,725)	\$ (1,661)
Net earnings/(loss) to common	\$ (859)	\$ (1,748)	\$ (1,673)
Average common equity	\$ 4,573	\$ 3,863	\$ 3,574
Return on average common equity	(18.8)%	(45.2)%	(46.8)%
Total			
Non-interest revenues	\$ 45,456	\$ 39,903	\$ 39,687
Net interest income	8,056	6,351	7,678
Total net revenues	53,512	46,254	47,365
Provision for credit losses	1,348	1,028	2,715
Compensation and benefits expenses	16,706	15,499	15,148
Other operating expenses	17,061	18,988	16,016
Total operating expenses	33,767	34,487	31,164
Pre-tax earnings	\$ 18,397	\$ 10,739	\$ 13,486
Net earnings	\$ 14,276	\$ 8,516	\$ 11,261
Net earnings to common	\$ 13,525	\$ 7,907	\$ 10,764
Average common equity	\$ 106,774	\$ 105,804	\$ 105,287
Return on average common equity	12.7 %	7.5 %	10.2 %

In the table above:

- Other operating expenses for Global Banking & Markets primarily included transaction based, communications and technology, and depreciation and amortization expenses.
- Other operating expenses for Asset & Wealth Management primarily included transaction based expenses, depreciation and amortization expenses, and professional fees.
- Other operating expenses for Platform Solutions primarily included professional fees, depreciation and amortization expenses, and communications and technology expenses.

The table below presents depreciation and amortization expenses by segment.

\$ in millions	Year Ended December		
	2024	2023	2022
Global Banking & Markets	\$ 1,119	\$ 1,109	\$ 1,033
Asset & Wealth Management	1,017	2,425	1,212
Platform Solutions	256	1,322	210
Total	\$ 2,392	\$ 4,856	\$ 2,455

In the table above:

- Asset & Wealth Management included impairments related to commercial real estate in CIEs of \$1.46 billion for 2023.
- Platform Solutions included a write-down related to GreenSky of \$506 million for 2023, and an impairment of goodwill related to Consumer platforms of \$504 million for 2023.

Segment Assets

The table below presents assets by segment.

\$ in millions	As of December	
	2024	2023
Global Banking & Markets	\$ 1,420,142	\$ 1,381,247
Asset & Wealth Management	193,328	191,863
Platform Solutions	62,502	68,484
Total	\$ 1,675,972	\$ 1,641,594

Notes to Consolidated Financial Statements**Geographic Information**

Due to the highly integrated nature of international financial markets, the firm manages its businesses based on the profitability of the enterprise as a whole. The methodology for allocating profitability to geographic regions is dependent on estimates and management judgment because a significant portion of the firm's activities require cross-border coordination in order to facilitate the needs of the firm's clients. Geographic results are generally allocated as follows:

- Global Banking & Markets: Investment banking fees and Other: location of the client and investment banking team; FICC intermediation and Equities intermediation: location of the market-making desk; FICC financing and Equities financing: location of the desk.
- Asset & Wealth Management (excluding direct-to-consumer business, Equity investments and Debt investments): location of the sales team; Direct-to-consumer business: location of the client; Equity investments and Debt investments: location of the investment or investment professional.
- Platform Solutions: location of the client.

The table below presents total net revenues, pre-tax earnings and net earnings by geographic region.

<i>\$ in millions</i>	2024		2023		2022	
Year Ended December						
Americas	\$ 34,448	64 %	\$ 29,335	64 %	\$ 28,669	61 %
EMEA	12,250	23 %	11,744	25 %	12,860	27 %
Asia	6,814	13 %	5,175	11 %	5,836	12 %
Total net revenues	\$ 53,512	100 %	\$ 46,254	100 %	\$ 47,365	100 %
Americas	\$ 12,106	66 %	\$ 6,038	56 %	\$ 7,016	52 %
EMEA	4,418	24 %	4,033	38 %	5,260	39 %
Asia	1,873	10 %	668	6 %	1,210	9 %
Total pre-tax earnings	\$ 18,397	100 %	\$ 10,739	100 %	\$ 13,486	100 %
Americas	\$ 9,354	66 %	\$ 4,849	57 %	\$ 6,067	54 %
EMEA	3,470	24 %	3,137	37 %	4,164	37 %
Asia	1,452	10 %	530	6 %	1,030	9 %
Total net earnings	\$ 14,276	100 %	\$ 8,516	100 %	\$ 11,261	100 %

In the table above:

- Americas pre-tax earnings for 2023 were impacted by impairments related to commercial real estate in CIEs, the write-down related to GreenSky, an impairment of goodwill related to Consumer platforms and the FDIC special assessment fee.
- Substantially all of the amounts in the Americas were attributable to the U.S.
- Asia includes Australia and New Zealand.

Note 26.**Credit Concentrations**

The firm's concentrations of credit risk arise from its market-making, client facilitation, investing, underwriting, lending and collateralized transactions, and cash management activities, and may be impacted by changes in economic, industry or political factors. These activities expose the firm to many different industries and counterparties, and may also subject the firm to a concentration of credit risk to a particular central bank, counterparty, borrower or issuer, including sovereign issuers, or to a particular clearinghouse or exchange. The firm seeks to mitigate credit risk by actively monitoring exposures and obtaining collateral from counterparties as deemed appropriate.

The firm measures and monitors its credit exposure based on amounts owed to the firm after taking into account risk mitigants that the firm considers when determining credit risk. Such risk mitigants include netting and collateral arrangements and economic hedges, such as credit derivatives, futures and forward contracts. Netting and collateral agreements permit the firm to offset receivables and payables with such counterparties and/or enable the firm to obtain collateral on an upfront or contingent basis.

The table below presents the credit concentrations included in trading cash instruments and investments.

<i>\$ in millions</i>	As of December	
	2024	2023
U.S. government and agency obligations	\$ 389,148	\$ 260,531
Percentage of total assets	23.2 %	15.9 %
Non-U.S. government and agency obligations	\$ 74,496	\$ 90,681
Percentage of total assets	4.4 %	5.5 %

In addition, the firm had \$151.84 billion as of December 2024 and \$206.07 billion as of December 2023 of cash deposits held at central banks (included in cash and cash equivalents), of which \$105.78 billion as of December 2024 and \$105.66 billion as of December 2023 was held at the Federal Reserve.

As of both December 2024 and December 2023, the firm did not have credit exposure to any other counterparty that exceeded 2% of total assets.

Collateral obtained by the firm related to derivative assets is principally cash and is held by the firm or a third-party custodian. Collateral obtained by the firm related to resale agreements and securities borrowed transactions is primarily U.S. government and agency obligations, and non-U.S. government and agency obligations. See Note 11 for further information about collateralized agreements and financings.

Notes to Consolidated Financial Statements

The table below presents U.S. government and agency obligations, and non-U.S. government and agency obligations that collateralize resale agreements and securities borrowed transactions.

\$ in millions	As of December	
	2024	2023
U.S. government and agency obligations	\$ 129,942	\$ 154,056
Non-U.S. government and agency obligations	\$ 76,932	\$ 92,833

In the table above:

- Non-U.S. government and agency obligations primarily consists of securities issued by the governments of the U.K., Japan, Germany and France.
- Given that the firm's primary credit exposure on such transactions is to the counterparty to the transaction, the firm would be exposed to the collateral issuer only in the event of counterparty default.

Note 27.**Legal Proceedings**

The firm is involved in a number of judicial, regulatory and arbitration proceedings (including those described below) concerning matters arising in connection with the conduct of the firm's businesses. Many of these proceedings are in early stages, and many of these cases seek an indeterminate amount of damages.

Under ASC 450, an event is "reasonably possible" if "the chance of the future event or events occurring is more than remote but less than likely" and an event is "remote" if "the chance of the future event or events occurring is slight." Thus, references to the upper end of the range of reasonably possible loss for cases in which the firm is able to estimate a range of reasonably possible loss mean the upper end of the range of loss for cases for which the firm believes the risk of loss is more than slight.

With respect to matters described below for which management has been able to estimate a range of reasonably possible loss where (i) actual or potential plaintiffs have claimed an amount of money damages, (ii) the firm is being, or threatened to be, sued by purchasers in a securities offering and is not being indemnified by a party that the firm believes will pay the full amount of any judgment, or (iii) the purchasers are demanding that the firm repurchase securities, management has estimated the upper end of the range of reasonably possible loss based on (a) in the case of (i), the amount of money damages claimed, (b) in the case of (ii), the difference between the initial sales price of the securities that the firm sold in such offering and the estimated lowest subsequent price of such securities prior to the action being commenced and (c) in the case of (iii), the price that purchasers paid for the securities less the estimated value, if any, as of December 2024 of the relevant securities, in each of cases (i), (ii) and (iii), taking into account any other factors believed to be relevant to the particular matter or matters of that type. As of the date hereof, the firm has estimated the upper end of the range of reasonably possible aggregate loss for such matters and for any other matters described below where management has been able to estimate a range of reasonably possible aggregate loss to be approximately \$1.7 billion in excess of the aggregate reserves for such matters.

Management is generally unable to estimate a range of reasonably possible loss for matters other than those included in the estimate above, including where (i) actual or potential plaintiffs have not claimed an amount of money damages, except in those instances where management can otherwise determine an appropriate amount, (ii) matters are in early stages, (iii) matters relate to regulatory investigations or reviews, except in those instances where management can otherwise determine an appropriate amount, (iv) there is uncertainty as to the likelihood of a class being certified or the ultimate size of the class, (v) there is uncertainty as to the outcome of pending appeals or motions, (vi) there are significant factual issues to be resolved, and/or (vii) there are novel legal issues presented. For example, the firm's potential liabilities with respect to the investigations and reviews described below in "Regulatory Investigations and Reviews and Related Litigation" generally are not included in management's estimate of reasonably possible loss. However, management does not believe, based on currently available information, that the outcomes of such other matters will have a material adverse effect on the firm's financial condition, though the outcomes could be material to the firm's operating results for any particular period, depending, in part, upon the operating results for such period.

Notes to Consolidated Financial Statements**1MDB-Related Matters**

Between 2012 and 2013, subsidiaries of the firm acted as arrangers or purchasers of approximately \$6.5 billion of debt securities of 1MDB.

On November 1, 2018, the U.S. Department of Justice (DOJ) unsealed a criminal information and guilty plea by Tim Leissner, a former participating managing director of the firm, and an indictment against Ng Chong Hwa, a former managing director of the firm. On August 28, 2018, Leissner was adjudicated guilty by the U.S. District Court for the Eastern District of New York of conspiring to launder money and to violate the U.S. Foreign Corrupt Practices Act's (FCPA) anti-bribery and internal accounting controls provisions. Ng was charged with conspiring to launder money and to violate the FCPA's anti-bribery and internal accounting controls provisions, and on April 8, 2022, Ng was found guilty on all counts following a trial.

On August 18, 2020, the firm announced that it entered into a settlement agreement with the Government of Malaysia to resolve the criminal and regulatory proceedings in Malaysia involving the firm, which includes a guarantee that the Government of Malaysia receives at least \$1.4 billion in assets and proceeds from assets seized by governmental authorities around the world related to 1MDB. See Note 18 for further information about this guarantee, including related arbitration proceedings.

On October 22, 2020, the firm announced that it reached settlements of governmental and regulatory investigations relating to 1MDB with the DOJ, the SEC, the FRB, the NYDFS, the Financial Conduct Authority, the Prudential Regulation Authority, the Singapore Attorney General's Chambers, the Singapore Commercial Affairs Department, the Monetary Authority of Singapore and the Hong Kong Securities and Futures Commission. Group Inc. entered into a three-year deferred prosecution agreement with the DOJ, in which a charge against the firm, one count of conspiracy to violate the FCPA, was filed and was later dismissed on May 6, 2024 in accordance with the agreement. In addition, GS Malaysia pleaded guilty to one count of conspiracy to violate the FCPA, and was sentenced on June 9, 2021. In May 2021, the U.S. Department of Labor granted the firm a five-year exemption to maintain its status as a qualified professional asset manager (QPAM).

On December 20, 2018, a putative securities class action lawsuit was filed in the U.S. District Court for the Southern District of New York against Group Inc. and certain former officers of the firm alleging violations of the anti-fraud provisions of the Exchange Act with respect to Group Inc.'s disclosures and public statements concerning 1MDB and seeking unspecified damages. The plaintiff filed the second amended complaint on October 28, 2019. On June 28, 2021, the court dismissed the claims against one of the individual defendants but denied the defendants' motion to dismiss with respect to the firm and the remaining individual defendants. On August 4, 2023, the plaintiff filed a third amended complaint. On September 29, 2023, the plaintiff moved for class certification. On April 5, 2024, the Magistrate Judge recommended that the plaintiff's motion for class certification be granted in part and denied in part. On May 3, 2024, the defendants filed objections to the Magistrate Judge's report and recommendation with the district court.

Mortgage-Related Matters

Complaints were filed in the U.S. District Court for the Southern District of New York on July 25, 2019 and May 29, 2020 against Goldman Sachs Mortgage Company and GS Mortgage Securities Corp. by U.S. Bank National Association, as trustee for two residential mortgage-backed securitization trusts that issued \$1.7 billion of securities. The complaints generally allege that mortgage loans in the trusts failed to conform to applicable representations and warranties and seek specific performance or, alternatively, compensatory damages and other relief. On November 23, 2020, the court granted in part and denied in part defendants' motion to dismiss the complaint in the first action and denied defendants' motion to dismiss the complaint in the second action. On January 14, 2021, amended complaints were filed in both actions.

Currencies-Related Litigation

GS&Co. is among the defendants named in a putative class action filed in the U.S. District Court for the Southern District of New York on August 4, 2021. The amended complaint, filed on January 6, 2022, generally asserts claims under federal antitrust law and state common law in connection with an alleged conspiracy among the defendants to manipulate auctions for foreign exchange transactions on an electronic trading platform, as well as claims under the Racketeer Influenced and Corrupt Organizations Act. The complaint seeks declaratory and injunctive relief, as well as unspecified amounts of treble and other damages. On May 18, 2023, the court dismissed certain state common law claims, but denied dismissal of the remaining claims. On July 7, 2023, the plaintiffs filed a second amended complaint.

Notes to Consolidated Financial Statements**Banco Espirito Santo S.A. and Oak Finance**

In December 2014, September 2015 and December 2015, the Bank of Portugal (BoP) rendered decisions to reverse an earlier transfer to Novo Banco of an \$835 million facility agreement (the Facility), structured by GSI, between Oak Finance Luxembourg S.A. (Oak Finance), a special purpose vehicle formed in connection with the Facility, and Banco Espirito Santo S.A. (BES) prior to the failure of BES. In response, GSI and, with respect to the BoP's December 2015 decision, GSIB commenced actions beginning in February 2015 against Novo Banco S.A. (Novo Banco) in the English Commercial Court and the BoP in the Portuguese Administrative Court. In July 2018, the English Supreme Court found that the English courts will not have jurisdiction over GSI's action unless and until the Portuguese Administrative Court finds against BoP in GSI's parallel action. In July 2018, the Liquidation Committee for BES issued a decision seeking to claw back from GSI \$54 million paid to GSI and \$50 million allegedly paid to Oak Finance in connection with the Facility, alleging that GSI acted in bad faith in extending the Facility, including because GSI allegedly knew that BES was at risk of imminent failure. In October 2018, GSI commenced an action in the Lisbon Commercial Court challenging the Liquidation Committee's decision and has since also issued a claim against the Portuguese State seeking compensation for losses of approximately \$222 million related to the failure of BES, together with a contingent claim for the \$104 million sought by the Liquidation Committee. On April 11, 2023, GSI commenced administrative proceedings against the BoP, seeking the nullification of the BoP's September 2015 and December 2015 decisions on new grounds.

Financial Advisory Services

Group Inc. and certain of its affiliates are from time to time parties to various civil litigation and arbitration proceedings and other disputes with clients and third parties relating to the firm's financial advisory activities. These claims generally seek, among other things, compensatory damages and, in some cases, punitive damages, and in certain cases allege that the firm did not appropriately disclose or deal with conflicts of interest.

Archegos-Related Matters

GS&Co. is among the underwriters named as defendants in a putative securities class action filed on August 13, 2021 in New York Supreme Court, County of New York, relating to ViacomCBS Inc.'s (ViacomCBS) March 2021 public offerings of \$1.7 billion of common stock and \$1.0 billion of preferred stock. In addition to the underwriters, the defendants include ViacomCBS and certain of its officers and directors. GS&Co. underwrote 646,154 shares of common stock representing an aggregate offering price of approximately \$55 million and 323,077 shares of preferred stock representing an aggregate offering price of approximately \$32 million. The complaint asserts claims under the federal securities laws and alleges that the offering documents contained material misstatements and omissions, including, among other things, that the offering documents failed to disclose that Archegos Capital Management, LP (Archegos) had substantial exposure to ViacomCBS, including through total return swaps to which certain of the underwriters (the trading underwriters), including GS&Co., were allegedly counterparties, and that such underwriters failed to disclose their exposure to Archegos. On December 21, 2021, the plaintiffs filed a corrected amended complaint. The complaint seeks rescission and compensatory damages in unspecified amounts. On February 6, 2023, the trial court dismissed the claims against ViacomCBS and the individual defendants, but denied the defendants' motions to dismiss with respect to GS&Co. and the other underwriter defendants. On January 4, 2024, the trial court granted the plaintiffs' motion for class certification, and on February 14, 2024, the underwriter defendants appealed. On April 4, 2024, the Appellate Division for the First Department affirmed the trial court's dismissal of the claims against ViacomCBS and the individual defendants, reversed the trial court's failure to dismiss the claims against the non-trading underwriter defendants, and affirmed the trial court's denial of the motion to dismiss claims against the trading underwriter defendants, including GS&Co.

Group Inc. is also a defendant in putative securities class actions filed beginning in October 2021 and consolidated in the U.S. District Court for the Southern District of New York. The complaints allege that Group Inc., along with another financial institution, sold shares in Baidu Inc. (Baidu), Discovery Inc. (Discovery), GSX Techedu Inc. (Gaotu), iQIYI Inc. (iQIYI), Tencent Music Entertainment Group (Tencent), ViacomCBS, and Vipshop Holdings Ltd. (Vipshop) based on material nonpublic information regarding the liquidation of Archegos' position in Baidu, Discovery, Gaotu, iQIYI, Tencent, ViacomCBS and Vipshop, respectively. The complaints generally assert violations of Sections 10(b), 20A and 20(a) of the Exchange Act and seek unspecified damages. In May 2023, the plaintiffs in the class actions filed second amended complaints, and on March 28, 2024, the court granted the defendants' motion to dismiss the second amended complaints with prejudice. On April 26, 2024, the plaintiffs appealed to the U.S. Court of Appeals for the Second Circuit.

Notes to Consolidated Financial Statements**Silicon Valley Bank Matters**

GS&Co. is among the underwriters named as defendants in a putative securities class action filed on April 7, 2023 and consolidated in the U.S. District Court for the Northern District of California and an individual action filed on January 25, 2024 in the same court relating to SVB Financial Group's (SVBFG) January 2021 public offerings of \$500 million principal amount of senior notes and \$750 million of depositary shares representing interests in preferred stock, March 2021 public offering of approximately \$1.2 billion of common stock, May 2021 public offerings of \$1.0 billion of depositary shares representing interests in preferred stock and \$500 million principal amount of senior notes, August 2021 public offering of approximately \$1.3 billion of common stock, and April 2022 public offering of \$800 million aggregate principal amount of senior notes, among other public offerings of securities. In addition to the underwriters, the defendants include certain of SVBFG's officers and directors and its auditor. GS&Co. underwrote an aggregate of 831,250 depositary shares representing an aggregate offering price of approximately \$831 million, an aggregate of 3,266,108 shares of common stock representing an aggregate offering price of approximately \$1.8 billion and senior notes representing an aggregate price to the public of approximately \$727 million. The complaints generally assert claims under the federal securities laws and allege that the offering documents contained material misstatements and omissions. The complaints seek compensatory damages in unspecified amounts. On March 17, 2023, SVBFG filed for Chapter 11 bankruptcy in the U.S. Bankruptcy Court for the Southern District of New York. On January 16, 2024, the plaintiffs filed a consolidated amended complaint in the putative class action, and on April 3, 2024, the defendants moved to dismiss the consolidated amended complaint.

The firm is also cooperating with and providing information to various governmental bodies in connection with their investigations and inquiries regarding SVBFG and its affiliates (collectively SVB), including the firm's business with SVB in or around March 2023, when SVB engaged the firm to assist with a proposed capital raise and SVB sold the firm a portfolio of securities.

Underwriting Litigation

Firm affiliates are among the defendants in a number of proceedings in connection with securities offerings. In these proceedings, including those described below, the plaintiffs assert class action or individual claims under federal and state securities laws and, in some cases, other applicable laws, allege that the offering documents for the securities that they purchased contained material misstatements and omissions, and generally seek compensatory and rescissory damages in unspecified amounts, as well as rescission. Certain of these proceedings involve additional allegations.

Uber Technologies, Inc. GS&Co. is among the underwriters named as defendants in several putative securities class actions filed beginning in September 2019 in California Superior Court, County of San Francisco and the U.S. District Court for the Northern District of California, relating to Uber Technologies, Inc.'s (Uber) \$8.1 billion May 2019 initial public offering. In addition to the underwriters, the defendants include Uber and certain of its officers and directors. GS&Co. underwrote 35,864,408 shares of common stock representing an aggregate offering price of approximately \$1.6 billion. On November 16, 2020, the court in the state court action granted defendants' motion to dismiss the consolidated amended complaint filed on February 11, 2020, and on December 16, 2020, plaintiffs appealed. On August 7, 2020, defendants' motion to dismiss the district court action was denied. On September 25, 2020, the plaintiffs in the district court action moved for class certification. On December 5, 2020, the plaintiffs in the state court action filed a complaint in the district court, which was consolidated with the existing district court action on January 25, 2021. On May 14, 2021, the plaintiffs filed a second amended complaint in the district court, purporting to add the plaintiffs from the state court action as additional class representatives. On October 1, 2021, defendants' motion to dismiss the additional class representatives from the second amended complaint was denied, and on July 26, 2022, the district court granted the plaintiffs' motion for class certification. On February 27, 2023, the U.S. Court of Appeals for the Ninth Circuit denied the defendants' petition seeking interlocutory review of the district court's grant of class certification. On December 4, 2024, the court in the federal court action approved a settlement, which does not require a contribution from GS&Co.

Notes to Consolidated Financial Statements

Array Technologies, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on May 14, 2021 in the U.S. District Court for the Southern District of New York relating to Array Technologies, Inc.'s (Array) \$1.2 billion October 2020 initial public offering of common stock, \$1.3 billion December 2020 offering of common stock and \$993 million March 2021 offering of common stock. In addition to the underwriters, the defendants include Array and certain of its officers and directors. GS&Co. underwrote an aggregate of 31,912,213 shares of common stock in the three offerings representing an aggregate offering price of approximately \$877 million. On December 7, 2021, the plaintiffs filed an amended consolidated complaint, and on May 19, 2023, the court granted the defendants' motion to dismiss the amended consolidated complaint. On July 5, 2023, the court denied the plaintiffs' request for leave to amend the amended consolidated complaint and dismissed the case with prejudice. On August 4, 2023, plaintiffs appealed to the U.S. Court of Appeals for the Second Circuit.

ContextLogic Inc. GS&Co. is among the underwriters named as defendants in putative securities class actions filed beginning on May 17, 2021 and consolidated in the U.S. District Court for the Northern District of California, relating to ContextLogic Inc.'s (ContextLogic) \$1.1 billion December 2020 initial public offering of common stock. In addition to the underwriters, the defendants include ContextLogic and certain of its officers and directors. GS&Co. underwrote 16,169,000 shares of common stock representing an aggregate offering price of approximately \$388 million. On July 15, 2022, the plaintiffs filed a consolidated amended complaint, and on March 10, 2023, the court granted the defendants' motion to dismiss the consolidated amended complaint with leave to amend. On April 10, 2023, the plaintiffs filed a second consolidated amended complaint, and on December 22, 2023, the court granted in part and denied in part the defendants' motion to dismiss the second consolidated amended complaint with leave to amend. On February 15, 2024, the plaintiffs filed a third consolidated amended complaint, and on August 22, 2024, the court granted the defendants' motion to dismiss the third consolidated amended complaint without leave to amend. On September 19, 2024, the plaintiffs filed a motion to alter the court's judgment.

DiDi Global Inc. Goldman Sachs (Asia) L.L.C. (GS Asia) is among the underwriters named as defendants in putative securities class actions filed beginning on July 6, 2021 in the U.S. District Courts for the Southern District of New York and the Central District of California and New York Supreme Court, County of New York, relating to DiDi Global Inc.'s (DiDi) \$4.4 billion June 2021 initial public offering of American Depositary Shares (ADS). In addition to the underwriters, the defendants include DiDi and certain of its officers and directors. GS Asia underwrote 104,554,000 ADS representing an aggregate offering price of approximately \$1.5 billion. On September 22, 2021, plaintiffs in the California action voluntarily dismissed their claims without prejudice. On May 5, 2022, plaintiffs in the consolidated federal action filed a second consolidated amended complaint, which includes allegations of violations of Sections 10(b) and 20A of the Exchange Act against the underwriter defendants. On March 14, 2024, the court denied the defendants' motions to dismiss the second consolidated amended complaint. On January 6, 2025, the plaintiffs moved for class certification.

Vroom Inc. GS&Co. is among the underwriters named as defendants in an amended complaint for a putative securities class action filed on October 4, 2021 in the U.S. District Court for the Southern District of New York relating to Vroom Inc.'s (Vroom) approximately \$589 million September 2020 public offering of common stock. In addition to the underwriters, the defendants include Vroom and certain of its officers and directors. GS&Co. underwrote 3,886,819 shares of common stock representing an aggregate offering price of approximately \$212 million. On December 20, 2021, the defendants served a motion to dismiss the consolidated complaint. On November 13, 2024, Vroom filed a Chapter 11 bankruptcy petition in the U.S. Bankruptcy Court for the Southern District of Texas.

Notes to Consolidated Financial Statements

Zymergen Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on August 4, 2021 in the U.S. District Court for the Northern District of California relating to Zymergen Inc.'s (Zymergen) \$575 million April 2021 initial public offering of common stock. In addition to the underwriters, the defendants include Zymergen, certain of its officers and directors and certain of its shareholders. GS&Co. underwrote 5,750,345 shares of common stock representing an aggregate offering price of approximately \$178 million. On February 24, 2022, the plaintiffs filed an amended complaint, and on November 29, 2022, the court granted in part and denied in part the defendants' motion to dismiss the amended complaint, denying dismissal of the claims for violations of Section 11 of the Securities Act. On August 11, 2023, the court granted the plaintiffs' motion for class certification. On October 3, 2023, Zymergen and three affiliates filed Chapter 11 bankruptcy petitions in the U.S. Bankruptcy Court for the District of Delaware. On March 4, 2024, the plaintiffs filed a second amended complaint.

Sea Limited. GS Asia is among the underwriters named as defendants in putative securities class actions filed on February 11, 2022 and June 17, 2022, respectively, in New York Supreme Court, County of New York, relating to Sea Limited's approximately \$4.0 billion September 2021 public offering of ADS and approximately \$2.9 billion September 2021 public offering of convertible senior notes, respectively. In addition to the underwriters, the defendants include Sea Limited, certain of its officers and directors and certain of its shareholders. GS Asia underwrote 8,222,500 ADS representing an aggregate offering price of approximately \$2.6 billion and convertible senior notes representing an aggregate offering price of approximately \$1.9 billion. On August 3, 2022, the actions were consolidated, and on August 9, 2022, the plaintiffs filed a consolidated amended complaint. The defendants had previously moved to dismiss the action on July 15, 2022, with the parties stipulating that the motion would apply to the consolidated amended complaint. On May 15, 2023, the court granted the defendants' motion to dismiss the consolidated amended complaint with prejudice. On May 28, 2024, the Appellate Division for the First Department reversed the trial court's dismissal of the consolidated amended complaint, and on June 27, 2024, the defendants moved to reargue or alternatively, for leave to appeal the reversal. On November 11, 2024, the parties reached a settlement in principle, subject to final documentation and court approval, to resolve this action, which does not require a contribution from GS Asia.

Rivian Automotive Inc. GS&Co. is among the underwriters named as defendants in putative securities class actions filed on March 7, 2022 and February 28, 2023 in the U.S. District Court for the Central District of California and in the Superior Court of the State of California, County of Orange, respectively, relating to Rivian Automotive Inc.'s (Rivian) approximately \$13.7 billion November 2021 initial public offering. In addition to the underwriters, the defendants include Rivian and certain of its officers and directors. GS&Co. underwrote 44,733,050 shares of common stock representing an aggregate offering price of approximately \$3.5 billion. On March 2, 2023, the plaintiffs in the federal court action filed an amended consolidated complaint, and on July 3, 2023, the court denied the defendants' motion to dismiss the amended consolidated complaint. On June 30, 2023, the court in the state court action granted the defendants' motion to dismiss the complaint, and on September 1, 2023, the plaintiffs appealed. On July 17, 2024, the court in the federal court action granted the plaintiffs' motion for class certification.

Natera Inc. GS&Co. is among the underwriters named as defendants in putative securities class actions in New York Supreme Court, County of New York and the U.S. District Court for the Western District of Texas filed on March 10, 2022 and October 7, 2022, respectively, relating to Natera Inc.'s (Natera) approximately \$585 million July 2021 public offering of common stock. In addition to the underwriters, the defendants include Natera and certain of its officers and directors. GS&Co. underwrote 1,449,000 shares of common stock representing an aggregate offering price of approximately \$164 million. On July 15, 2022, the parties in the state court action filed a stipulation and proposed order approving the discontinuance of the action without prejudice. On September 11, 2023, the federal court granted in part and denied in part the defendants' motion to dismiss. On June 4, 2024, the plaintiffs in the federal court action moved for class certification, and on January 28, 2025, the Magistrate Judge recommended that the plaintiffs' motion for class certification be granted. On February 21, 2025, the underwriter defendants filed objections to the Magistrate Judge's report and recommendation with the district court.

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Robinhood Markets, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on December 17, 2021 in the U.S. District Court for the Northern District of California relating to Robinhood Markets, Inc.'s (Robinhood) approximately \$2.2 billion July 2021 initial public offering. In addition to the underwriters, the defendants include Robinhood and certain of its officers and directors. GS&Co. underwrote 18,039,706 shares of common stock representing an aggregate offering price of approximately \$686 million. On February 10, 2023, the court granted the defendants' motion to dismiss the complaint with leave to amend, and on March 13, 2023, the plaintiffs filed a second amended complaint. On January 24, 2024, the court granted the defendants' motion to dismiss the second amended complaint without leave to amend. On February 21, 2024, the plaintiffs appealed to the U.S. Court of Appeals for the Ninth Circuit.

ON24, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on November 3, 2021 in the U.S. District Court for the Northern District of California relating to ON24, Inc.'s (ON24) approximately \$492 million February 2021 initial public offering of common stock. In addition to the underwriters, the defendants include ON24 and certain of its officers and directors, including a director who was a Managing Director of GS&Co. at the time of the initial public offering. GS&Co. underwrote 3,616,785 shares of common stock representing an aggregate offering price of approximately \$181 million. On March 18, 2022, the plaintiffs filed a consolidated complaint, and on July 7, 2023, the court granted the defendants' motion to dismiss the consolidated complaint with leave to amend. On September 1, 2023, the plaintiffs filed an amended consolidated complaint, and on March 5, 2024, the court granted the defendants' motion to dismiss the amended consolidated complaint with prejudice. On April 4, 2024, the plaintiffs appealed to the U.S. Court of Appeals for the Ninth Circuit.

Oscar Health, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on May 12, 2022 in the U.S. District Court for the Southern District of New York relating to Oscar Health, Inc.'s (Oscar Health) approximately \$1.4 billion March 2021 initial public offering. In addition to the underwriters, the defendants include Oscar Health and certain of its officers and directors. GS&Co. underwrote 12,760,633 shares of common stock representing an aggregate offering price of approximately \$498 million. On December 5, 2022, the plaintiffs filed an amended complaint. On April 4, 2023, the defendants moved to dismiss the amended complaint.

Oak Street Health, Inc. GS&Co. is among the underwriters named as defendants in an amended complaint for a putative securities class action filed on May 25, 2022 in the U.S. District Court for the Northern District of Illinois relating to Oak Street Health, Inc.'s (Oak Street) \$377 million August 2020 initial public offering, \$298 million December 2020 secondary equity offering, \$691 million February 2021 secondary equity offering and \$747 million May 2021 secondary equity offering. In addition to the underwriters, the defendants include Oak Street, certain of its officers and directors and certain of its shareholders. GS&Co. underwrote 4,157,103 shares of common stock in the August 2020 initial public offering representing an aggregate offering price of approximately \$87 million, 1,503,944 shares of common stock in the December 2020 secondary equity offering representing an aggregate offering price of approximately \$69 million, 3,083,098 shares of common stock in the February 2021 secondary equity offering representing an aggregate offering price of approximately \$173 million and 3,013,065 shares of common stock in the May 2021 secondary equity offering representing an aggregate offering price of approximately \$187 million. On February 10, 2023, the court granted in part and denied in part the defendants' motion to dismiss, dismissing the claim alleging a violation of Section 12(a)(2) of the Securities Act and, with respect to the May 2021 secondary equity offering only, the claim alleging a violation of Section 11 of the Securities Act, but declining to dismiss the remaining claims. On December 15, 2023, the plaintiffs moved for class certification. On December 12, 2024, the court approved a settlement, which does not require a contribution from GS&Co.

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Bright Health Group, Inc. GS&Co. is among the underwriters named as defendants in an amended complaint for a putative securities class action filed on June 24, 2022 in the U.S. District Court for the Eastern District of New York relating to Bright Health Group, Inc.'s (Bright Health) approximately \$924 million June 2021 initial public offering of common stock. In addition to the underwriters, the defendants include Bright Health and certain of its officers and directors. GS&Co. underwrote 11,297,000 shares of common stock representing an aggregate offering price of approximately \$203 million. On September 30, 2024, the court granted the defendants' motion to dismiss the amended complaint, and on November 27, 2024, the plaintiffs appealed to the U.S. Court of Appeals for the Second Circuit.

MINISO Group Holding Limited. GS Asia is among the underwriters named as defendants in a putative securities class action filed on August 17, 2022 in the U.S. District Court for the Central District of California and transferred to the U.S. District Court for the Southern District of New York on November 18, 2022 relating to MINISO Group Holding Limited's (MINISO) approximately \$656 million October 2020 initial public offering of ADS. In addition to the underwriters, the defendants include MINISO and certain of its officers and directors. GS Asia underwrote 16,408,093 ADS representing an aggregate offering price of approximately \$328 million. On April 24, 2023, the plaintiffs filed a second amended complaint, and on February 23, 2024, the court granted the defendants' motion to dismiss the second amended complaint with leave to amend.

Coupang, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on August 26, 2022 in the U.S. District Court for the Southern District of New York relating to Coupang, Inc.'s (Coupang) approximately \$4.6 billion March 2021 initial public offering of common stock. In addition to the underwriters, the defendants include Coupang and certain of its officers and directors. GS&Co. underwrote 42,900,000 shares of common stock representing an aggregate offering price of approximately \$1.5 billion. On May 24, 2023, the plaintiffs filed an amended complaint, and on July 28, 2023, the defendants moved to dismiss the amended complaint.

Yatsen Holding Limited. GS Asia is among the underwriters named as defendants in a putative securities class action filed on September 23, 2022 in the U.S. District Court for the Southern District of New York relating to Yatsen Holding Limited's (Yatsen) approximately \$617 million November 2020 initial public offering of ADS. In addition to the underwriters, the defendants include Yatsen and certain of its officers and directors. GS Asia underwrote 22,912,500 ADS representing an aggregate offering price of approximately \$241 million. On October 4, 2023, the plaintiffs filed an amended complaint, and on July 22, 2024, the court granted the defendants' motion to dismiss the amended complaint.

Rent the Runway, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on November 14, 2022 in the U.S. District Court for the Eastern District of New York relating to Rent the Runway, Inc.'s (Rent the Runway) \$357 million October 2021 initial public offering of common stock. In addition to the underwriters, the defendants include Rent the Runway and certain of its officers and directors. GS&Co. underwrote 5,254,304 shares of common stock representing an aggregate offering price of approximately \$110 million. On September 5, 2023, the plaintiffs filed an amended complaint, and on September 25, 2024, the court granted in part and denied in part the defendants' motion to dismiss the amended complaint. On October 9, 2024, the defendants filed a motion for reconsideration of the court's order.

Opendoor Technologies Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on November 22, 2022 in the U.S. District Court for the District of Arizona relating to, among other things, Opendoor Technologies Inc.'s (Opendoor) approximately \$886 million February 2021 public offering of common stock. In addition to the underwriters, the defendants include Opendoor and certain of its officers and directors. GS&Co. underwrote 10,173,401 shares of common stock representing an aggregate offering price of approximately \$275 million. On April 17, 2023, the plaintiffs filed a consolidated amended complaint, and on February 28, 2024, the court granted the defendants' motion to dismiss the consolidated amended complaint with leave to amend. On May 14, 2024, the court granted the plaintiffs' motion for reconsideration and vacated the dismissal of certain of the plaintiffs' claims, and on September 9, 2024, the court denied the defendants' motion for certification of an interlocutory appeal as to the plaintiffs' surviving claims.

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FIGS, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on December 8, 2022 in the U.S. District Court for the Central District of California relating to FIGS, Inc.'s (FIGS) approximately \$668 million May 2021 initial public offering and approximately \$413 million September 2021 secondary equity offering. In addition to the underwriters, the defendants include FIGS, certain of its officers and directors and certain of its shareholders. GS&Co. underwrote 9,545,073 shares of common stock in the May 2021 initial public offering representing an aggregate offering price of approximately \$210 million and 3,179,047 shares of common stock in the September 2021 secondary equity offering representing an aggregate offering price of approximately \$128 million. On April 10, 2023, the plaintiffs filed a consolidated complaint, and on January 17, 2024, the court granted the defendants' motions to dismiss the consolidated complaint with leave to amend. On March 19, 2024, the plaintiffs filed a first amended complaint, and on January 10, 2025, the court granted in part and denied in part the defendants' motions to dismiss the first amended complaint with leave to amend, resulting in the dismissal of all claims against the underwriter defendants, including GS&Co. On February 10, 2025, the plaintiffs appealed to the U.S. Court of Appeals for the Ninth Circuit.

Silvergate Capital Corporation. GS&Co. is among the underwriters and sales agents named as defendants in a putative securities class action filed on January 19, 2023 in the U.S. District Court for the Southern District of California, as amended on May 11, 2023, relating to Silvergate Capital Corporation's (Silvergate) approximately \$288 million January 2021 public offering of common stock, approximately \$300 million "at-the-market" offering of common stock conducted from March through May 2021, approximately \$200 million July 2021 public offering of depositary shares representing interests in preferred stock, and approximately \$552 million December 2021 public offering of common stock. In addition to the underwriters and sales agents, the defendants include Silvergate and certain of its officers and directors. GS&Co. underwrote 1,711,313 shares of common stock in the January 2021 public offering of common stock representing an aggregate offering price of approximately \$108 million, acted as a sales agent with respect to up to a \$300 million aggregate offering price of shares of common stock in the March through May 2021 "at-the-market" offering, underwrote 1,600,000 depositary shares in the July 2021 public offering representing an aggregate offering price of approximately \$40 million, and underwrote 1,375,397 shares of common stock in the December 2021 public offering of common stock representing an aggregate offering price of approximately \$199 million. On July 10, 2023, the defendants moved to dismiss the consolidated amended complaint. On September 17, 2024, Silvergate and two affiliates filed Chapter 11 bankruptcy petitions in the U.S. Bankruptcy Court for the District of Delaware.

F45 Training Holdings Inc. GS&Co. is among the underwriters named as defendants in an amended complaint for a putative securities class action filed on May 19, 2023 in the U.S. District Court for the Western District of Texas relating to F45 Training Holdings Inc.'s (F45) approximately \$350 million July 2021 initial public offering of common stock. In addition to the underwriters, the defendants include F45, certain of its officers and directors and certain of its shareholders. GS&Co. acted as a qualified independent underwriter for the offering and underwrote 8,303,744 shares of common stock representing an aggregate offering price of approximately \$133 million. On August 7, 2023, the defendants filed a motion to dismiss the amended complaint. On January 25, 2024, the plaintiffs filed a second amended complaint, and on March 11, 2024, the defendants moved to dismiss the second amended complaint.

Olaplex Holdings, Inc. GS&Co. is among the underwriters named as defendants in a putative securities class action filed on April 28, 2023 in the U.S. District Court for the Central District of California relating to Olaplex Holdings, Inc.'s (Olaplex) approximately \$1.8 billion September 2021 initial public offering of common stock. In addition to the underwriters, the defendants include Olaplex, certain of its officers and directors and selling shareholders. GS&Co. underwrote 19,419,420 shares of common stock representing an aggregate offering price of approximately \$408 million. On June 22, 2023, the plaintiffs filed a revised consolidated complaint, and on February 7, 2025, the court granted in part and denied in part the defendants' motions to dismiss the revised consolidated complaint, resulting in the dismissal of all claims against the underwriter defendants, including GS&Co.

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agilon health, inc. GS&Co. is among the underwriters named as defendants in putative securities class actions filed beginning on March 19, 2024 and consolidated in the U.S. District Court for the Western District of Texas, relating to agilon health, inc.'s (agilon) approximately \$1.2 billion April 2021 initial public offering, approximately \$587 million September 2021 secondary equity offering and approximately \$1.8 billion May 2023 secondary equity offering. In addition to the underwriters, the defendants include agilon, certain of its officers and directors and certain of its shareholders. GS&Co. underwrote 10,631,949 shares of common stock in the April 2021 initial public offering representing an aggregate offering price of approximately \$245 million, 3,759,588 shares of common stock in the September 2021 secondary equity offering representing an aggregate offering price of approximately \$113 million and 26,879,772 shares of common stock in the May 2023 secondary equity offering, of which 2,731,638 shares were purchased by agilon, representing an aggregate offering price of approximately \$519 million sold to third parties. On September 6, 2024, the plaintiffs filed a consolidated complaint, and on November 8, 2024, the defendants moved to dismiss the consolidated complaint.

Investment Management Services

Group Inc. and certain of its affiliates are parties to various civil litigation and arbitration proceedings and other disputes with clients relating to losses allegedly sustained as a result of the firm's investment management services. These claims generally seek, among other things, restitution or other compensatory damages and, in some cases, punitive damages.

Variable Rate Demand Obligations Antitrust Litigation

Group Inc. and GS&Co. were among the defendants named in a putative class action relating to variable rate demand obligations (VRDOs), filed beginning in February 2019 under separate complaints and consolidated in the U.S. District Court for the Southern District of New York. The consolidated amended complaint, filed on May 31, 2019, generally asserts claims under federal antitrust law and state common law in connection with an alleged conspiracy among the defendants to manipulate the market for VRDOs. The complaint seeks declaratory and injunctive relief, as well as unspecified amounts of compensatory, treble and other damages. Group Inc. was voluntarily dismissed from the putative class action on June 3, 2019. On November 2, 2020, the court granted in part and denied in part the defendants' motion to dismiss, dismissing the state common law claims against GS&Co., but denying dismissal of the federal antitrust law claims.

GS&Co. is also among the defendants named in a related putative class action filed on June 2, 2021 in the U.S. District Court for the Southern District of New York. The complaint alleges the same conspiracy in the market for VRDOs as that alleged in the consolidated amended complaint filed on May 31, 2019, and asserts federal antitrust law, state law and state common law claims against the defendants. The complaint seeks declaratory and injunctive relief, as well as unspecified amounts of compensatory, treble and other damages. On August 6, 2021, plaintiffs in the May 31, 2019 action filed an amended complaint consolidating the June 2, 2021 action with the May 31, 2019 action. On September 14, 2021, defendants filed a joint partial motion to dismiss the August 6, 2021 amended consolidated complaint. On June 28, 2022, the court granted in part and denied in part the defendants' motion to dismiss, dismissing the state breach of fiduciary duty claim against GS&Co., but declining to dismiss any portion of the federal antitrust law claims. On September 21, 2023, the court granted the plaintiffs' motion for class certification. On February 5, 2024, the U.S. Court of Appeals for the Second Circuit granted the defendants' petition seeking interlocutory review of the district court's grant of class certification. On February 15, 2024, the district court granted the defendants' request to stay the proceedings pending their appeal of the district court's grant of class certification.

Notes to Consolidated Financial Statements**Interest Rate Swap Antitrust Litigation**

Group Inc., GS&Co., GSI, GS Bank USA and Goldman Sachs Financial Markets, L.P. are among the defendants named in a putative antitrust class action relating to the trading of interest rate swaps, filed in November 2015 and consolidated in the U.S. District Court for the Southern District of New York. The same Goldman Sachs entities are also among the defendants named in two antitrust actions relating to the trading of interest rate swaps, commenced in April 2016 and June 2018, respectively, in the U.S. District Court for the Southern District of New York by three operators of swap execution facilities and certain of their affiliates. These actions have been consolidated for pretrial proceedings. The complaints generally assert claims under federal antitrust law and state common law in connection with an alleged conspiracy among the defendants to preclude exchange trading of interest rate swaps. The complaints in the individual actions also assert claims under state antitrust law. The complaints seek declaratory and injunctive relief, as well as treble damages in an unspecified amount. Defendants moved to dismiss the class and the first individual action and the district court dismissed the state common law claims asserted by the plaintiffs in the first individual action and otherwise limited the state common law claim in the putative class action and the antitrust claims in both actions to the period from 2013 to 2016. On November 20, 2018, the court granted in part and denied in part the defendants' motion to dismiss the second individual action, dismissing the state common law claims for unjust enrichment and tortious interference, but denying dismissal of the federal and state antitrust claims. On March 13, 2019, the court denied the plaintiffs' motion in the putative class action to amend their complaint to add allegations related to conduct from 2008 to 2012, but granted the motion to add limited allegations from 2013 to 2016, which the plaintiffs added in a fourth consolidated amended complaint filed on March 22, 2019. On December 15, 2023, the court denied the plaintiffs' motion for class certification, and on December 28, 2023, the plaintiffs filed a petition with the U.S. Court of Appeals for the Second Circuit seeking interlocutory review of the district court's denial of class certification. On July 11, 2024, the court preliminarily approved a settlement among the plaintiffs and certain defendants, including the firm, to resolve the class action. The firm has paid the full amount of its proposed contribution to the settlement into an escrow account. The individual actions remain pending.

Commodities-Related Litigation

GSI is among the defendants named in putative class actions relating to trading in platinum and palladium, filed beginning on November 25, 2014 and most recently amended on May 15, 2017, in the U.S. District Court for the Southern District of New York. The amended complaint generally alleges that the defendants violated federal antitrust laws and the Commodity Exchange Act in connection with an alleged conspiracy to manipulate a benchmark for physical platinum and palladium prices and seek declaratory and injunctive relief, as well as treble damages in an unspecified amount. On March 29, 2020, the court granted the defendants' motions to dismiss and for reconsideration, resulting in the dismissal of all claims, and on February 27, 2023, the U.S. Court of Appeals for the Second Circuit reversed the district court's dismissal of certain plaintiffs' antitrust claims and vacated the district court's dismissal of the plaintiffs' Commodity Exchange Act claim. On April 12, 2023, the defendants' petition for rehearing or rehearing en banc with the U.S. Court of Appeals for the Second Circuit was denied. On July 21, 2023, the defendants filed a motion for judgment on the pleadings. On January 17, 2025, the court approved a settlement to resolve this action. The firm has paid the full amount of its contribution to the settlement.

Corporate Bonds Antitrust Litigation

Group Inc. and GS&Co. are among the dealers named as defendants in a putative class action relating to the secondary market for odd-lot corporate bonds, filed on April 21, 2020 in the U.S. District Court for the Southern District of New York. The amended consolidated complaint, filed on October 29, 2020, asserts claims under federal antitrust law in connection with alleged anti-competitive conduct by the defendants in the secondary market for odd-lots of corporate bonds, and seeks declaratory and injunctive relief, as well as unspecified monetary damages, including treble and punitive damages and restitution. On October 25, 2021, the court granted defendants' motion to dismiss with prejudice. On November 10, 2022, the district court denied the plaintiffs' motion for an indicative ruling that the judgment should be vacated because the wife of the district judge owned stock in one of the defendants and the district judge did not recuse himself. On July 2, 2024, the U.S. Court of Appeals for the Second Circuit vacated the district court's dismissal and remanded the case for further proceedings. On September 3, 2024, the plaintiffs filed a second amended complaint, and on October 18, 2024, the defendants moved to dismiss the second amended complaint.

Notes to Consolidated Financial Statements**Credit Default Swap Antitrust Litigation**

Group Inc., GS&Co. and GSI were among the defendants named in a putative antitrust class action relating to the settlement of credit default swaps, filed on June 30, 2021 in the U.S. District Court for the District of New Mexico. The complaint generally asserts claims under federal antitrust law and the Commodity Exchange Act in connection with an alleged conspiracy among the defendants to manipulate the benchmark price used to value credit default swaps for settlement. The complaint also asserts a claim for unjust enrichment under state common law. The complaint seeks declaratory and injunctive relief, as well as unspecified amounts of treble and other damages. On November 15, 2021, the defendants filed a motion to dismiss the complaint. On February 4, 2022, the plaintiffs filed an amended complaint and voluntarily dismissed Group Inc. from the action. On June 5, 2023, the court dismissed the claims against certain foreign defendants for lack of personal jurisdiction but denied the defendants' motion to dismiss with respect to GS&Co., GSI and the remaining defendants. On January 24, 2024, the court granted the defendants' motion to stay the proceedings pending the resolution of the motion filed by the defendants on November 3, 2023 in the U.S. District Court for the Southern District of New York to enforce a 2015 settlement and release among the parties. On January 26, 2024, the U.S. District Court for the Southern District of New York granted the defendants' motion to enforce the settlement and release and enjoined the plaintiffs from pursuing any claims against the defendants in the New Mexico action for any alleged violation of law based on conduct before June 30, 2014, and on February 23, 2024, the plaintiffs appealed to the U.S. Court of Appeals for the Second Circuit.

Consumer Investigation and Review

The firm has been cooperating with the CFPB and other governmental bodies relating to investigations and/or inquiries concerning GS Bank USA's credit card account management practices and is providing information regarding the application of refunds, crediting of nonconforming payments, billing error resolution, advertisements, reporting to credit bureaus, and any other consumer-related information requested by them, and GS Bank USA has entered into a consent order, without admitting or denying the findings, to resolve the CFPB's investigation. The consent order requires a \$45 million penalty, approximately \$20 million in restitution (to be offset by restitution GS Bank USA has already provided to consumers), and certain non-monetary remedial measures. The firm has paid the full amount of the settlement.

Regulatory Investigations and Reviews and Related Litigation

Group Inc. and certain of its affiliates are subject to a number of other investigations and reviews by, and, in some cases, have received subpoenas and requests for documents and information from, various governmental and regulatory bodies and self-regulatory organizations and litigation and shareholder requests relating to various matters relating to the firm's businesses and operations, including:

- The securities offering process and underwriting practices;
- The firm's investment management and financial advisory services;
- Conflicts of interest;
- Research practices, including research independence and interactions between research analysts and other firm personnel, including investment banking personnel, as well as third parties;
- Transactions involving government-related financings and other matters, municipal securities, including wall-cross procedures and conflict of interest disclosure with respect to state and municipal clients, the trading and structuring of municipal derivative instruments in connection with municipal offerings, political contribution rules, municipal advisory services and the possible impact of credit default swap transactions on municipal issuers;
- Consumer lending, as well as residential mortgage lending, servicing and securitization, and compliance with related consumer laws;

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- The offering, auction, sales, trading and clearance of corporate and government securities, currencies, commodities and other financial products and related sales and other communications and activities, as well as the firm's supervision and controls relating to such activities, including compliance with applicable short sale rules, algorithmic, high-frequency and quantitative trading, the firm's U.S. alternative trading system (dark pool), futures trading, options trading, when-issued trading, transaction and regulatory reporting, technology systems and controls, communications recordkeeping and recording, securities lending practices, prime brokerage activities, trading and clearance of credit derivative instruments and interest rate swaps, commodities activities and metals storage, private placement practices, allocations of and trading in securities, and trading activities and communications in connection with the establishment of benchmark rates, such as currency rates;
- Compliance with the FCPA;
- The firm's hiring and compensation practices;
- The firm's system of risk management and controls; and
- Insider trading, the potential misuse and dissemination of material nonpublic information regarding corporate and governmental developments and the effectiveness of the firm's insider trading controls and information barriers.

The firm is cooperating with all such governmental and regulatory investigations and reviews.

Note 28.**Employee Benefit Plans**

The firm sponsors various pension plans and certain other postretirement benefit plans, primarily healthcare and life insurance. The firm also provides certain benefits to former or inactive employees prior to retirement.

Defined Benefit Pension Plans and Postretirement Plans

Employees of certain non-U.S. subsidiaries participate in various defined benefit pension plans. These plans generally provide benefits based on years of credited service and a percentage of eligible compensation. The firm maintains a defined benefit pension plan for certain U.K. employees. As of April 2008, the U.K. defined benefit plan was closed to new participants and frozen for existing participants as of March 31, 2016. The non-U.S. plans do not have a material impact on the firm's consolidated results of operations.

The firm also maintains a defined benefit pension plan for substantially all U.S. employees hired prior to November 1, 2003. As of November 2004, this plan was closed to new participants and frozen for existing participants. In addition, the firm maintains unfunded postretirement benefit plans that provide medical and life insurance for eligible retirees and their dependents covered under these programs. These plans do not have a material impact on the firm's consolidated results of operations.

The firm recognizes the funded status of its defined benefit pension and postretirement plans, measured as the difference between the fair value of the plan assets and the benefit obligation, in the consolidated balance sheets. As of December 2024, other assets included \$73 million (related to overfunded pension plans) and other liabilities included \$344 million related to these plans. As of December 2023, other assets included \$93 million (related to overfunded pension plans) and other liabilities included \$449 million related to these plans.

Defined Contribution Plans

The firm contributes to employer-sponsored U.S. and non-U.S. defined contribution plans. The firm's contribution to these plans was \$382 million for 2024, \$377 million for 2023 and \$378 million for 2022.

Notes to Consolidated Financial Statements**Note 29.****Employee Incentive Plans**

The cost of employee services received in exchange for a share-based award is generally measured based on the grant-date fair value of the award. Share-based awards that do not require future service (i.e., vested awards, including awards granted to retirement-eligible employees) are expensed immediately. Share-based awards that require future service are amortized over the relevant service period. Forfeitures are recorded when they occur.

Cash dividend equivalents paid on RSUs are generally charged to retained earnings. If RSUs that require future service are forfeited, the related dividend equivalents originally charged to retained earnings are reclassified to compensation expense in the period in which forfeiture occurs.

The firm generally issues new shares of common stock upon delivery of share-based awards. In limited cases, as outlined in the applicable award agreements, the firm may cash settle share-based awards accounted for as equity instruments. For these awards, additional paid-in capital is adjusted to the extent of the difference between the value of the award at the time of cash settlement and the grant-date value of the award. The tax effect related to the settlement of share-based awards and payments of dividend equivalents is recorded in income tax benefit or expense.

Stock Incentive Plan

The firm sponsors a stock incentive plan, The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (2021 SIP), which provides for grants of RSUs, restricted stock, dividend equivalent rights, incentive stock options, nonqualified stock options, stock appreciation rights, and other share-based awards, each of which may be subject to terms and conditions, including performance or market conditions. On April 29, 2021, shareholders approved the 2021 SIP. The 2021 SIP is a successor to several predecessor stock incentive plans, the first of which was adopted on April 30, 1999, and each of which was approved by the firm's shareholders.

As of December 2024, 55.7 million shares were available to be delivered pursuant to awards granted under the 2021 SIP. If any shares of common stock underlying awards granted under the 2021 SIP or awards granted under predecessor stock incentive plans are not delivered because such awards are forfeited, terminated or canceled, or if shares of common stock underlying such awards are surrendered or withheld to satisfy any obligation of the grantee (including taxes), those shares will become available to be delivered pursuant to awards granted under the 2021 SIP. Shares available to be delivered under the 2021 SIP also are subject to adjustment for certain events or changes in corporate structure as provided under the 2021 SIP. The 2021 SIP is scheduled to terminate on the date of the 2025 Annual Meeting of Shareholders.

Restricted Stock Units

The firm grants RSUs (including RSUs subject to performance or market conditions) to employees, which are generally valued based on the closing price of the underlying shares on the date of grant, after taking into account a liquidity discount for any applicable post-vesting and delivery transfer restrictions. The value of equity awards also considers the impact of material non-public information, if any, that the firm expects to make available shortly following grant. RSUs not subject to performance or market conditions generally vest and underlying shares of common stock are delivered (net of required withholding tax) over a three-year period as outlined in the applicable award agreements. Award agreements generally provide that vesting is accelerated in certain circumstances, such as on retirement, death, disability and, in certain cases, conflicted employment. Delivery of the underlying shares of common stock is conditioned on the grantees satisfying certain vesting and other requirements outlined in the award agreements.

RSUs that are subject to performance or market conditions generally are settled after the end of a three- to five-year period. For awards that are subject to performance or market conditions, generally the final award is adjusted from zero up to 150% of the original grant based on the extent to which those conditions are satisfied. Dividend equivalents that accrue on these awards are paid when the awards settle.

Notes to Consolidated Financial Statements

The table below presents the 2024 activity related to stock settled RSUs.

	Restricted Stock Units Outstanding		Weighted Average Grant-Date Fair Value of Restricted Stock Units Outstanding	
	Future Service Required	No Future Service Required	Future Service Required	No Future Service Required
Beginning balance	4,379,141	15,306,921	\$ 310.31	\$ 304.37
Granted	2,795,500	4,302,501	\$ 381.27	\$ 354.89
Forfeited	(585,760)	(418,815)	\$ 291.64	\$ 321.77
Delivered	-	(8,502,057)	-	\$ 301.08
Vested	(3,103,014)	3,103,014	\$ 339.86	\$ 339.86
Ending balance	3,485,867	13,791,564	\$ 344.05	\$ 329.61

In the table above:

- The weighted average grant-date fair value of RSUs granted was \$365.28 during 2024, \$329.23 during 2023 and \$316.98 during 2022. The grant-date fair value of these RSUs included an average liquidity discount of 3.9% during 2024, 4.5% during 2023 and 6.0% during 2022, to reflect post-vesting and delivery transfer restrictions, generally of 1 year for each of 2024, 2023 and 2022. In addition, delivered RSUs include RSUs that have been settled in cash.
- The aggregate fair value of awards that vested was \$3.15 billion during 2024, \$2.47 billion during 2023 and \$3.91 billion during 2022.
- The ending balance included restricted stock subject to future service requirements of 4,579 shares as of December 2024 and 347,240 shares as of December 2023.
- The ending balance included RSUs subject to future service requirements and performance or market conditions of 466,731 RSUs as of December 2024 and 617,655 RSUs as of December 2023, and the maximum amount of such RSUs that may be earned was 700,097 RSUs as of December 2024 and 913,551 RSUs as of December 2023.
- The ending balance also included RSUs not subject to future service requirements but subject to performance conditions of 1,587,795 RSUs as of December 2024 and 1,620,470 RSUs as of December 2023, and the maximum amount of such RSUs that may be earned was 2,381,693 RSUs as of December 2024 and 2,430,705 RSUs as of December 2023.

In relation to 2024 year-end, during the first quarter of 2025, the firm granted to its employees 5.0 million RSUs (of which 1.5 million RSUs require future service as a condition for delivery of the related shares of common stock). These RSUs are subject to additional conditions as outlined in the award agreements. Shares underlying these RSUs, net of required withholding tax, generally are delivered over a three-year period and are generally subject to a one-year post-vesting and delivery transfer restriction. These awards are not included in the table above.

As of December 2024, there was \$630 million of total unrecognized compensation cost related to non-vested share-based awards. This cost is expected to be recognized over a weighted average period of 1.70 years. In addition, there is unrecognized compensation cost related to share-based awards subject to performance conditions. The maximum payout related to these awards is \$299 million. This cost is expected to be recognized over a weighted average period of 0.65 years.

The table below presents the share-based compensation and the related excess tax benefit.

\$ in millions	Year Ended December		
	2024	2023	2022
Share-based compensation	\$ 2,772	\$ 2,098	\$ 4,107
Excess net tax benefit for share-based awards	\$ 213	\$ 198	\$ 324

In the table above, excess net tax benefit for share-based awards includes the net tax benefit on dividend equivalents paid on RSUs and the delivery of common stock underlying share-based awards.

Overrides

The firm shares a portion of its overrides related to investment management services with approximately 900 employees, including with the firm's executive officers. The fair value of these overrides is recognized as compensation expense over the vesting period. Such expense was \$376 million for 2024, \$407 million for 2023 and \$493 million for 2022.

Notes to Consolidated Financial Statements**Note 30.****Parent Company****Group Inc. - Condensed Statements of Earnings**

\$ in millions	Year Ended December		
	2024	2023	2022
Revenues			
Dividends from subsidiaries and other affiliates:			
Bank	\$ 62	\$ 58	\$ 101
Nonbank	9,021	11,499	6,243
Other revenues	(1,214)	(1,965)	(3,590)
Total non-interest revenues	7,869	9,592	2,754
Interest income	20,533	18,839	8,367
Interest expense	23,527	21,479	9,428
Net interest loss	(2,994)	(2,640)	(1,061)
Total net revenues	4,875	6,952	1,693
Operating expenses			
Compensation and benefits	676	287	328
Other expenses	693	219	685
Total operating expenses	1,369	506	1,013
Pre-tax earnings	3,506	6,446	680
Benefit for taxes	(1,617)	(1,070)	(1,398)
Undistributed earnings of subsidiaries and other affiliates	9,153	1,000	9,183
Net earnings	14,276	8,516	11,261
Preferred stock dividends	751	609	497
Net earnings applicable to common shareholders	\$ 13,525	\$ 7,907	\$ 10,764

Supplemental Disclosures:

In the condensed statements of earnings above, revenues and expenses included the following with subsidiaries and other affiliates:

- Dividends from bank subsidiaries included cash dividends of \$62 million for 2024, \$58 million for 2023 and \$97 million for 2022.
- Dividends from nonbank subsidiaries and other affiliates included cash dividends of \$9.02 billion for 2024, \$11.49 billion for 2023 and \$6.14 billion for 2022.
- Other revenues included \$(1.72) billion for 2024, \$(892) million for 2023 and \$(3.34) billion for 2022.
- Interest income included \$17.65 billion for 2024, \$16.82 billion for 2023 and \$7.47 billion for 2022.
- Interest expense included \$11.91 billion for 2024, \$9.94 billion for 2023 and \$3.80 billion for 2022.
- Other expenses included \$104 million for 2024, \$105 million for 2023 and \$116 million for 2022.

Group Inc.'s other comprehensive income/(loss) was \$216 million for 2024, \$92 million for 2023 and \$(942) million for 2022.

Group Inc. - Condensed Balance Sheets

\$ in millions	As of December	
	2024	2023
Assets		
Cash and cash equivalents:		
With third-party banks	\$ 19	\$ 35
With subsidiary bank	2	7
Loans to and receivables from subsidiaries:		
Bank	5,738	1,833
Nonbank (\$15,494 and \$4,813 at fair value)	282,580	286,688
Investments in subsidiaries and other affiliates:		
Bank	63,427	55,164
Nonbank	77,362	78,591
Trading assets (at fair value)	438	3,197
Investments (\$35,205 and \$22,443 at fair value)	80,697	79,300
Other assets	8,300	7,408
Total assets	\$ 518,563	\$ 512,223
Liabilities and shareholders' equity		
Repurchase agreements with subsidiaries (at fair value)	\$ 78,145	\$ 78,776
Secured borrowings with subsidiaries	28,151	19,233
Payables to subsidiaries	1,803	568
Trading liabilities (at fair value)	1,107	898
Unsecured short-term borrowings:		
With third parties (\$4,583 and \$4,721 at fair value)	22,409	31,833
With subsidiaries	3,526	5,710
Unsecured long-term borrowings:		
With third parties (\$29,051 and \$28,966 at fair value)	167,523	175,335
With subsidiaries	89,883	79,316
Other liabilities	4,020	3,649
Total liabilities	396,567	395,318
Commitments, contingencies and guarantees		
Shareholders' equity		
Preferred stock	13,253	11,203
Common stock	9	9
Share-based awards	5,148	5,121
Additional paid-in capital	61,376	60,247
Retained earnings	153,412	143,688
Accumulated other comprehensive loss	(2,702)	(2,918)
Stock held in treasury, at cost	(108,500)	(100,445)
Total shareholders' equity	121,996	116,905
Total liabilities and shareholders' equity	\$ 518,563	\$ 512,223

Supplemental Disclosures:

Goldman Sachs Funding LLC, a wholly-owned, direct subsidiary of Group Inc., has provided Group Inc. with a committed line of credit that allows Group Inc. to draw sufficient funds to meet its cash needs in the ordinary course of business.

Trading assets included derivative contracts with subsidiaries of \$261 million as of December 2024 and \$155 million as of December 2023.

Trading liabilities included derivative contracts with subsidiaries of \$1.11 billion as of December 2024 and \$898 million as of December 2023.

As of December 2024, unsecured long-term borrowings with subsidiaries by maturity date are \$87.66 billion in 2026, \$160 million in 2027, \$224 million in 2028, \$243 million in 2029 and \$1.60 billion in 2030-thereafter.

Notes to Consolidated Financial Statements**Group Inc. - Condensed Statements of Cash Flows**

\$ in millions	Year Ended December		
	2024	2023	2022
Cash flows from operating activities			
Net earnings	\$ 14,276	\$ 8,516	\$ 11,261
Adjustments to reconcile net earnings to net cash provided by operating activities:			
Undistributed earnings of subsidiaries and other affiliates	(9,153)	(1,000)	(9,183)
Depreciation and amortization	25	13	9
Deferred income taxes	(844)	(380)	(1,523)
Share-based compensation	361	(11)	378
Changes in operating assets and liabilities:			
Collateralized transactions (excluding secured borrowings, net)	(631)	11,937	66,839
Trading assets	3,685	7,620	(23,451)
Trading liabilities	209	(1,646)	1,428
Other, net	5,090	(221)	5,933
Net cash provided by operating activities	13,018	24,828	51,691
Cash flows from investing activities			
Purchase of property, leasehold improvements and equipment	(55)	(48)	(64)
Repayments/(issuances) of short-term loans to subsidiaries, net	9,578	3,145	2,210
Issuance of term loans to subsidiaries	(22,275)	(25,473)	(1,859)
Repayments of term loans by subsidiaries	12,626	921	2,311
Purchase of investments	(30,473)	(25,904)	(47,247)
Sales/paydowns of investments	30,239	17,801	3,162
Capital distributions from/(contributions to) subsidiaries, net	127	1,205	(5,665)
Net cash used for investing activities	(233)	(28,353)	(47,152)
Cash flows from financing activities			
Secured borrowings with subsidiary, net	8,518	3,810	(36,389)
Unsecured short-term borrowings, net:			
With third parties	(54)	87	13
With subsidiaries	8,152	19,314	27,803
Issuance of unsecured long-term borrowings	77,389	127,728	78,803
Repayment of unsecured long-term borrowings	(94,943)	(136,618)	(65,960)
Preferred stock redemption	(2,200)	(1,000)	-
Common stock repurchased	(8,000)	(5,796)	(3,500)
Settlement of share-based awards in satisfaction of withholding tax requirements	(1,331)	(1,345)	(1,595)
Dividends and dividend equivalents paid on stock and share-based awards	(4,497)	(4,189)	(3,682)
Issuance of preferred stock, net of costs	4,239	1,496	-
Other financing, net	(79)	(1)	-
Net cash provided by/(used for) financing activities	(12,806)	3,486	(4,507)
Net increase/(decrease) in cash and cash equivalents	(21)	(39)	32
Cash and cash equivalents, beginning balance	42	81	49
Cash and cash equivalents, ending balance	\$ 21	\$ 42	\$ 81

Supplemental Disclosures:

Cash payments for interest, net of capitalized interest, were \$22.43 billion for 2024, \$20.53 billion for 2023 and \$8.54 billion for 2022, and included \$11.60 billion for 2024, \$9.40 billion for 2023 and \$3.55 billion for 2022 of payments to subsidiaries.

Cash payments for income taxes, net, were \$1.29 billion for 2024, \$671 million for 2023 and \$2.59 billion for 2022.

There were no material non-cash activities during the year ended December 2024.

Non-cash activities during the year ended December 2023:

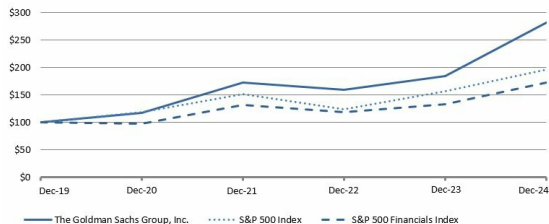
- Group Inc. exchanged \$1.42 billion of equity investment in its wholly-owned subsidiaries for loans.

Non-cash activities during the year ended December 2022:

- Group Inc. issued \$1.75 billion of equity in connection with the acquisition of GreenSky. Upon closing of the transaction, GreenSky became a wholly-owned subsidiary of GS Bank USA.

Common Stock Performance

The graph and table below compare the performance of an investment in the firm's common stock from December 31, 2019 (the last trading day before the firm's 2020 fiscal year) through December 31, 2024, with the S&P 500 Index (S&P 500) and the S&P 500 Financials Index (S&P 500 Financials).



	As of December					
	2019	2020	2021	2022	2023	2024
Group Inc.	\$ 100.00	\$ 117.48	\$ 173.39	\$ 159.76	\$ 185.18	\$ 281.53
S&P 500	\$ 100.00	\$ 118.39	\$ 152.34	\$ 124.72	\$ 157.48	\$ 196.85
S&P 500 Financials	\$ 100.00	\$ 98.24	\$ 132.50	\$ 118.49	\$ 132.83	\$ 173.34

The graph and table above assume \$100 was invested on December 31, 2019 in each of the firm's common stock, the S&P 500 and the S&P 500 Financials, and the dividends were reinvested without payment of any commissions. The performance shown represents past performance and should not be considered an indication of future performance.

Statistical Disclosures

Distribution of Assets, Liabilities and Shareholders' Equity

The tables below present information about average balances, interest and average interest rates.

\$ in millions	Average Balance for the Year Ended December		
	2024	2023	2022
Assets			
U.S.	\$ 110,144	\$ 129,718	\$ 151,152
Non-U.S.	96,783	127,250	107,843
Deposits with banks	206,927	256,968	258,995
U.S.	242,469	214,772	241,968
Non-U.S.	148,594	158,347	169,621
Collateralized agreements	391,063	373,119	411,589
U.S.	288,793	205,013	165,331
Non-U.S.	196,305	146,655	123,332
Trading assets	485,098	351,668	288,663
U.S.	151,248	123,828	97,221
Non-U.S.	14,625	15,003	14,696
Investments	165,873	138,831	111,917
U.S.	168,198	156,349	144,781
Non-U.S.	16,514	19,112	22,067
Loans	184,712	175,461	166,848
U.S.	81,030	85,373	95,513
Non-U.S.	57,549	55,043	64,301
Other interest-earning assets	138,579	140,416	159,814
Interest-earning assets	1,572,252	1,436,463	1,397,826
Cash and due from banks	5,681	6,372	7,715
Other non-interest-earning assets	100,915	109,042	137,418
Assets	\$ 1,678,848	\$ 1,551,877	\$ 1,542,959
Liabilities			
U.S.	\$ 332,750	\$ 307,686	\$ 302,678
Non-U.S.	96,184	82,321	74,662
Interest-bearing deposits	428,934	390,007	377,340
U.S.	202,997	154,341	107,008
Non-U.S.	113,693	93,697	83,783
Collateralized financings	316,690	248,038	190,791
U.S.	58,696	62,254	80,950
Non-U.S.	79,147	76,057	83,657
Trading liabilities	137,843	138,311	164,607
U.S.	52,398	47,878	34,322
Non-U.S.	35,746	27,166	28,675
Short-term borrowings	88,144	75,044	62,997
U.S.	193,173	197,442	221,598
Non-U.S.	54,435	45,611	37,656
Long-term borrowings	247,608	243,053	259,254
U.S.	145,407	149,883	166,200
Non-U.S.	88,223	94,915	98,130
Other interest-bearing liabilities	233,630	244,798	264,330
Interest-bearing liabilities	1,452,849	1,339,251	1,319,319
Non-interest-bearing deposits	5,029	4,733	4,811
Other non-interest-bearing liabilities	101,766	91,194	102,839
Liabilities	1,559,644	1,435,178	1,426,969
Shareholders' equity			
Preferred stock	12,430	10,895	10,703
Common stock	106,774	105,804	105,287
Shareholders' equity	119,204	116,699	115,990
Liabilities and shareholders' equity	\$ 1,678,848	\$ 1,551,877	\$ 1,542,959
Percentage attributable to non-U.S. operations			
Interest-earning assets	33.73 %	36.30 %	35.90 %
Interest-bearing liabilities	32.17 %	31.34 %	30.82 %

Supplemental Financial Information

\$ in millions	Interest for the Year Ended December		
	2024	2023	2022
Assets			
U.S.	\$ 5,861	\$ 7,074	\$ 2,793
Non-U.S.	3,421	3,875	440
Deposits with banks	9,282	10,949	3,233
U.S.	13,947	11,301	3,463
Non-U.S.	5,948	5,104	1,005
Collateralized agreements	19,895	16,405	4,468
U.S.	9,459	5,717	3,362
Non-U.S.	4,857	2,743	1,725
Trading assets	14,316	8,460	5,087
U.S.	5,333	3,055	1,656
Non-U.S.	665	801	543
Investments	5,998	3,856	2,199
U.S.	14,814	13,332	7,967
Non-U.S.	1,348	1,573	1,092
Loans	16,162	14,905	9,059
U.S.	9,431	8,266	3,236
Non-U.S.	6,313	5,674	1,742
Other interest-earning assets	15,744	13,940	4,978
Interest-earning assets	\$ 81,397	\$ 68,515	\$ 29,024
Liabilities			
U.S.	\$ 15,800	\$ 13,658	\$ 4,959
Non-U.S.	4,482	3,352	864
Interest-bearing deposits	20,282	17,010	5,823
U.S.	11,953	8,750	2,027
Non-U.S.	5,409	3,955	781
Collateralized financings	17,362	12,705	2,808
U.S.	1,319	969	872
Non-U.S.	1,592	1,484	1,051
Trading liabilities	2,911	2,453	1,923
U.S.	1,745	1,200	408
Non-U.S.	367	122	133
Short-term borrowings	2,112	1,322	541
U.S.	10,728	10,838	5,570
Non-U.S.	282	246	146
Long-term borrowings	11,010	11,084	5,716
U.S.	11,870	11,228	2,356
Non-U.S.	7,794	6,362	2,179
Other interest-bearing liabilities	19,664	17,590	4,535
Interest-bearing liabilities	\$ 73,341	\$ 62,164	\$ 21,346
Net interest income			
U.S.	\$ 5,430	\$ 2,102	\$ 6,285
Non-U.S.	2,626	4,249	1,393
Net interest income	\$ 8,056	\$ 6,351	\$ 7,678

	Average Rate for the Year Ended December		
	2024	2023	2022
Assets			
U.S.	5.32 %	5.45 %	1.85 %
Non-U.S.	3.53 %	3.05 %	0.41 %
Deposits with banks	4.49 %	4.26 %	1.25 %
U.S.	5.75 %	5.26 %	1.43 %
Non-U.S.	4.00 %	3.22 %	0.59 %
Collateralized agreements	5.09 %	4.40 %	1.09 %
U.S.	3.28 %	2.79 %	2.03 %
Non-U.S.	2.47 %	1.87 %	1.40 %
Trading assets	2.95 %	2.41 %	1.76 %
U.S.	3.53 %	2.47 %	1.70 %
Non-U.S.	4.55 %	5.34 %	3.69 %
Investments	3.62 %	2.78 %	1.96 %
U.S.	8.81 %	8.53 %	5.50 %
Non-U.S.	8.16 %	8.23 %	4.95 %
Loans	8.75 %	8.49 %	5.43 %
U.S.	11.64 %	9.68 %	3.39 %
Non-U.S.	10.97 %	10.31 %	2.71 %
Other interest-earning assets	11.36 %	9.93 %	3.11 %
Interest-earning assets	5.18 %	4.77 %	2.08 %
Liabilities			
U.S.	4.75 %	4.44 %	1.64 %
Non-U.S.	4.66 %	4.07 %	1.16 %
Interest-bearing deposits	4.73 %	4.36 %	1.54 %
U.S.	5.89 %	5.67 %	1.89 %
Non-U.S.	4.76 %	4.22 %	0.93 %
Collateralized financings	5.48 %	5.12 %	1.47 %
U.S.	2.25 %	1.56 %	1.08 %
Non-U.S.	2.01 %	1.95 %	1.26 %
Trading liabilities	2.11 %	1.77 %	1.17 %
U.S.	3.33 %	2.51 %	1.19 %
Non-U.S.	1.03 %	0.45 %	0.46 %
Short-term borrowings	2.40 %	1.76 %	0.86 %
U.S.	5.55 %	5.49 %	2.51 %
Non-U.S.	0.52 %	0.54 %	0.39 %
Long-term borrowings	4.45 %	4.56 %	2.20 %
U.S.	8.16 %	7.49 %	1.42 %
Non-U.S.	8.83 %	6.70 %	2.22 %
Other interest-bearing liabilities	8.42 %	7.19 %	1.72 %
Interest-bearing liabilities	5.05 %	4.64 %	1.62 %
Interest rate spread	0.13 %	0.13 %	0.46 %
U.S.	0.52 %	0.23 %	0.70 %
Non-U.S.	0.50 %	0.81 %	0.28 %
Net yield on interest-earning assets	0.51 %	0.44 %	0.55 %

In the tables above:

- Assets, liabilities and interest are classified as U.S. and non-U.S. based on the location of the legal entity in which the assets and liabilities are held.
- Derivative instruments and commodities are included in other non-interest-earning assets and other non-interest-bearing liabilities.
- Average collateralized agreements included \$188.06 billion of resale agreements and \$203.00 billion of securities borrowed for 2024, \$179.35 billion of resale agreements and \$193.77 billion of securities borrowed for 2023, and \$216.73 billion of resale agreements and \$194.86 billion of securities borrowed for 2022.
- Other interest-earning assets primarily consists of certain receivables from customers and counterparties.

Supplemental Financial Information

- Collateralized financings included \$252.88 billion of repurchase agreements and \$63.81 billion of securities loaned for 2024, \$198.26 billion of repurchase agreements and \$49.78 billion of securities loaned for 2023, and \$150.23 billion of repurchase agreements and \$40.56 billion of securities loaned for 2022.
- Substantially all of the other interest-bearing liabilities consists of certain payables to customers and counterparties.
- Interest rates for borrowings include the effects of interest rate swaps accounted for as hedges.
- Loans exclude loans held for sale that are accounted for at the lower of cost or fair value. Such loans are included within other interest-earning assets.
- Short- and long-term borrowings include both secured and unsecured borrowings.

Changes in Net Interest Income, Volume and Rate Analysis

The tables below present the effect on net interest income of volume and rate changes. In this analysis, changes due to volume/rate variance have been allocated to volume.

Year Ended December 2024 versus December 2023			
\$ in millions	Increase (decrease) due to change in:		
	Volume	Rate	Net Change
Interest-earning assets			
U.S.	\$ (1,042)	\$ (171)	\$ (1,213)
Non-U.S.	(1,077)	623	(454)
Deposits with banks	(2,119)	452	(1,667)
U.S.	1,593	1,053	2,646
Non-U.S.	(390)	1,234	844
Collateralized agreements	1,203	2,287	3,490
U.S.	2,744	998	3,742
Non-U.S.	1,228	886	2,114
Trading assets	3,972	1,884	5,856
U.S.	967	1,311	2,278
Non-U.S.	(17)	(119)	(136)
Investments	950	1,192	2,142
U.S.	1,044	438	1,482
Non-U.S.	(212)	(13)	(225)
Loans	832	425	1,257
U.S.	(505)	1,670	1,165
Non-U.S.	275	364	639
Other interest-earning assets	(230)	2,034	1,804
Change in interest income	4,608	8,274	12,882
Interest-bearing liabilities			
U.S.	1,190	952	2,142
Non-U.S.	646	484	1,130
Interest-bearing deposits	1,836	1,436	3,272
U.S.	2,865	338	3,203
Non-U.S.	951	503	1,454
Collateralized financings	3,816	841	4,657
U.S.	(80)	430	350
Non-U.S.	62	46	108
Trading liabilities	(18)	476	458
U.S.	151	394	545
Non-U.S.	88	157	245
Short-term borrowings	239	551	790
U.S.	(237)	127	(110)
Non-U.S.	46	(10)	36
Long-term borrowings	(191)	117	(74)
U.S.	(365)	1,007	642
Non-U.S.	(591)	2,023	1,432
Other interest-bearing liabilities	(956)	3,030	2,074
Change in interest expense	4,726	6,451	11,177
Change in net interest income	\$ (118)	\$ 1,823	\$ 1,705

Year Ended December 2023 versus December 2022			
\$ in millions	Increase (decrease) due to change in:		
	Volume	Rate	Net Change
Interest-earning assets			
U.S.	\$ (1,169)	\$ 5,450	\$ 4,281
Non-U.S.	591	2,844	3,435
Deposits with banks	(578)	8,294	7,716
U.S.	(1,431)	9,269	7,838
Non-U.S.	(363)	4,462	4,099
Collateralized agreements	(1,794)	13,731	11,937
U.S.	1,107	1,248	2,355
Non-U.S.	436	582	1,018
Trading assets	1,543	1,830	3,373
U.S.	656	743	1,399
Non-U.S.	16	242	258
Investments	672	985	1,657
U.S.	986	4,379	5,365
Non-U.S.	(243)	724	481
Loans	743	5,103	5,846
U.S.	(982)	6,012	5,030
Non-U.S.	(954)	4,886	3,932
Other interest-earning assets	(1,936)	10,898	8,962
Change in interest income	(1,350)	40,841	39,491
Interest-bearing liabilities			
U.S.	222	8,477	8,699
Non-U.S.	312	2,176	2,488
Interest-bearing deposits	534	10,653	11,187
U.S.	2,683	4,040	6,723
Non-U.S.	418	2,756	3,174
Collateralized financings	3,101	6,796	9,897
U.S.	(291)	388	97
Non-U.S.	(148)	581	433
Trading liabilities	(439)	969	530
U.S.	340	452	792
Non-U.S.	(7)	(4)	(11)
Short-term borrowings	333	448	781
U.S.	(1,326)	6,594	5,268
Non-U.S.	43	57	100
Long-term borrowings	(1,283)	6,651	5,368
U.S.	(1,222)	10,094	8,872
Non-U.S.	(215)	4,398	4,183
Other interest-bearing liabilities	(1,437)	14,492	13,055
Change in interest expense	809	40,009	40,818
Change in net interest income	\$ (2,159)	\$ 832	\$ (1,327)

Supplemental Financial Information**Deposits**

The table below presents information about interest-bearing deposits.

\$ in millions	Year Ended December	
	2024	2023
Average balances		
U.S.		
Savings and demand	\$ 242,378	\$ 241,356
Time	90,372	66,330
Total U.S.	332,750	307,686
Non-U.S.		
Demand	64,678	57,506
Time	31,506	24,815
Total non-U.S.	96,184	82,321
Total	\$ 428,934	\$ 390,007
Average interest rates		
U.S.		
Savings and demand	4.72 %	4.57 %
Time	4.81 %	3.97 %
Total U.S.	4.75 %	4.44 %
Non-U.S.		
Demand	4.53 %	3.99 %
Time	4.92 %	4.26 %
Total non-U.S.	4.66 %	4.07 %
Total	4.73 %	4.36 %

In the table above, deposits are classified as U.S. and non-U.S. based on the location of the entity in which such deposits are held.

The amount of deposits in U.S. offices held by non-U.S. depositors was \$6.06 billion as of December 2024 and \$10.34 billion as of December 2023.

The amount of uninsured deposits in U.S. offices was \$107.17 billion as of December 2024 and \$111.60 billion as of December 2023. These amounts exclude cash collateral on derivatives that is considered by the FDIC when determining uninsured deposits. Such collateral is either netted against the derivative balances or included in payables to customer and counterparties in our consolidated balance sheets.

The amount of uninsured deposits in non-U.S. offices was \$65.32 billion as of December 2024 and \$69.30 billion as of December 2023.

The table below presents uninsured time deposits by maturity.

\$ in millions	As of December 2024	
	U.S.	Non-U.S.
3 months or less	\$ 4,999	\$ 11,167
3 to 6 months	3,553	9,449
6 to 12 months	2,760	1,788
Greater than 12 months	711	1,204
Total	\$ 12,023	\$ 23,608

In the table above:

- All U.S. time deposits were in accounts eligible for FDIC insurance and non-U.S. time deposits include deposits in accounts eligible for insurance in their local jurisdictions, as well as deposits in uninsured accounts.
- The insurance limit (for account holders who have both time and other deposits that, in aggregate, exceed the insurance limit) is allocated between time and other deposits based on regulatory methodologies defined by each local jurisdiction.

Loan Portfolio

The table below presents information about gross loans.

\$ in millions	As of December			
	2024		2023	
Corporate	\$ 29,972	15 %	\$ 35,874	19 %
Commercial real estate	29,789	15 %	26,028	14 %
Residential real estate	25,969	13 %	25,388	13 %
Securities-based	16,477	8 %	14,621	8 %
Other collateralized	75,107	37 %	62,225	33 %
Consumer:				
Installment	70	-	3,298	2 %
Credit cards	21,403	11 %	19,361	10 %
Other	2,079	1 %	1,613	1 %
Total	\$ 200,866	100 %	\$ 188,408	100 %

Supplemental Financial Information

Maturities and Interest Rates. The table below presents gross loans by tenor.

\$ in millions	As of December 2024					Total
	1 year or less	More than 1 year to 5 years	More than 5 years to 15 years	More than 15 years		
Corporate	\$ 2,134	\$ 25,357	\$ 2,480	\$ 1	\$ 29,972	
Commercial real estate	4,432	23,845	1,510	2	29,789	
Residential real estate	4,999	8,624	132	12,214	25,969	
Securities-based	14,778	1,683	16	-	16,477	
Other collateralized	25,806	47,133	1,598	570	75,107	
Consumer:						
Installment	-	70	-	-	70	
Credit cards	21,403	-	-	-	21,403	
Other	1,060	941	74	4	2,079	
Total	\$ 74,612	\$ 107,653	\$ 5,810	\$ 12,791	\$ 200,866	

The table below presents the gross loans by tenor and for loans with tenors greater than one year, the distributions of such loans between fixed and floating interest rates.

\$ in millions	As of December 2024				Total
	1 year or less	More than one year			
		Fixed-rate	Floating-rate		
Corporate	\$ 2,134	\$ 340	\$ 27,498	\$ 29,972	
Commercial real estate	4,432	706	24,651	29,789	
Residential real estate	4,999	11,923	9,047	25,969	
Securities-based	14,778	44	1,655	16,477	
Other collateralized	25,806	323	48,978	75,107	
Consumer:					
Installment	-	-	70	70	
Credit cards	21,403	-	-	21,403	
Other	1,060	36	983	2,079	
Total	\$ 74,612	\$ 13,372	\$ 112,882	\$ 200,866	

Allowance for Loan Losses

The table below presents information about the allowance for loan losses.

\$ in millions	As of December	
	2024	2023
Corporate	\$ 1,033	\$ 1,307
Commercial real estate	637	765
Residential real estate	144	129
Securities-based	1	-
Other collateralized	279	340
Other	5	35
Wholesale	2,099	2,576
Installment	-	23
Credit cards	2,567	2,451
Consumer	2,567	2,474
Total	\$ 4,666	\$ 5,050

The table below presents information about the net charge-off ratio for loans accounted for at amortized cost.

\$ in millions	Net charge-offs	Average balance	Net charge-off ratio
Year Ended December 2024			
Wholesale	\$ 48	\$ 164,688	-
Installment	13	155	8.4 %
Credit cards	1,354	17,730	7.6 %
Consumer	1,367	17,885	7.6 %
Total	\$ 1,415	\$ 182,573	0.8 %
Year Ended December 2023			
Wholesale	\$ 400	\$ 151,834	0.3 %
Installment	86	3,721	2.3 %
Credit cards	1,062	17,028	6.2 %
Consumer	1,148	20,749	5.5 %
Total	\$ 1,548	\$ 172,583	0.9 %

In the table above, the net charge-off ratio is calculated by dividing the net charge-offs by average gross loans accounted for at amortized cost. Net charge-offs for wholesale loans were not material for 2024 and were primarily related to corporate loans for 2023.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

There were no changes in or disagreements with accountants on accounting and financial disclosure during the last two years.

Item 9A. Controls and Procedures

As of the end of the period covered by this report, an evaluation was carried out by our management, with the participation of our Chief Executive Officer and Chief Financial Officer, of the effectiveness of our disclosure controls and procedures (as defined in Rule 13a-15(e) under the Exchange Act). Based on that evaluation, our Chief Executive Officer and Chief Financial Officer concluded that these disclosure controls and procedures were effective as of the end of the period covered by this report. In addition, no change in our internal control over financial reporting (as defined in Rule 13a-15(f) under the Exchange Act) occurred during the fourth quarter of our year ended December 31, 2024 that has materially affected, or is reasonably likely to materially affect, our internal control over financial reporting.

Management's Report on Internal Control over Financial Reporting and the Report of Independent Registered Public Accounting Firm are set forth in Part II, Item 8 of this Form 10-K.

Item 9B. Other Information

Rule 10b5-1 Trading Plans

During the quarter ended December 2024, no directors or executive officers entered into, modified or terminated, contracts, instructions or written plans for the sale or purchase of Group Inc.'s securities that were intended to satisfy the affirmative defense conditions of Rule 10b5-1 or that constituted non-Rule 10b5-1 trading arrangements (as defined in Item 408 of Regulation S-K).

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

Not applicable.

PART III

Item 10. Directors, Executive Officers and Corporate Governance

Information about our executive officers is included in "Business — Information about our Executive Officers" in Part I, Item 1 of this Form 10-K. Information about our directors, including our audit committee and audit committee financial experts and the procedures by which shareholders can recommend director nominees, and our executive officers will be in our definitive Proxy Statement for our 2025 Annual Meeting of Shareholders, which will be filed within 120 days of the end of 2024 (2025 Proxy Statement) and is incorporated in this Form 10-K by reference. Information about our Code of Business Conduct and Ethics, which applies to our senior financial officers, is included in "Business — Available Information" in Part I, Item 1 of this Form 10-K.

We have adopted an insider trading policy governing the purchase, sale and/or other disposition of our securities by our directors, officers and employees and other covered persons, as well as Group Inc. itself, that we believe is reasonably designed to promote compliance with insider trading laws, rules and regulations and New York Stock Exchange listing standards. A copy of our insider trading policy is filed as Exhibit 19.1 to this Annual Report on Form 10-K.

Item 11. Executive Compensation

Information relating to our executive officer and director compensation and the compensation committee of the Board will be in the 2025 Proxy Statement and is incorporated in this Form 10-K by reference.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

Information relating to security ownership of certain beneficial owners of our common stock and information relating to the security ownership of our management will be in the 2025 Proxy Statement and is incorporated in this Form 10-K by reference.

The table below presents information as of December 31, 2024 regarding securities to be issued pursuant to outstanding restricted stock units (RSUs) and securities remaining available for issuance under our equity compensation plans that were in effect during 2024.

Plan Category	Securities to be Issued Upon Exercise of Outstanding Options and Rights (a)	Weighted Average Exercise Price of Outstanding Options (b)	Securities Available For Future Issuance Under Equity Compensation Plans (c)
Equity compensation plans:			
Approved by security holders	18,300,116	N/A	55,703,255
Not approved by security holders	-	-	-
Total	18,300,116		55,703,255

In the table above:

- Securities to be Issued Upon Exercise of Outstanding Options and Rights includes 18,300,116 shares that may be issued pursuant to outstanding RSUs. These awards are subject to vesting and other conditions to the extent set forth in the respective award agreements, and the underlying shares will be delivered net of any required tax withholding. As of December 31, 2024, there were no outstanding options.
- Shares underlying RSUs are deliverable without the payment of any consideration, and therefore the weighted average exercise price is not applicable for these awards.
- Securities Available For Future Issuance Under Equity Compensation Plans represents shares remaining to be issued under our current stock incentive plan (SIP), excluding shares reflected in column (a). If any shares of common stock underlying awards granted under our current SIP or certain of our prior SIPs are not delivered due to forfeiture, termination or cancellation or are surrendered or withheld, those shares will again become available to be delivered under our current SIP. Shares available for grant are also subject to adjustment for certain changes in corporate structure as permitted under our current SIP.

Item 13. Certain Relationships and Related Transactions, and Director Independence

Information regarding certain relationships and related transactions and director independence will be in the 2025 Proxy Statement and is incorporated in this Form 10-K by reference.

Item 14. Principal Accountant Fees and Services

Information regarding principal accountant fees and services will be in the 2025 Proxy Statement and is incorporated in this Form 10-K by reference.

PART IV

Item 15. Exhibit and Financial Statement Schedules

(a) Documents filed as part of this Report:

1. Consolidated Financial Statements

The consolidated financial statements required to be filed in this Form 10-K are included in Part II, Item 8 hereof.

2. Exhibits

- 2 . 1 Plan of Incorporation (incorporated by reference to Exhibit 2.1 to the Registrant's Registration Statement on Form S-1 (No. 333-74449)).
- 3 . 1 Restated Certificate of Incorporation of The Goldman Sachs Group, Inc., amended as of February 12, 2025 (incorporated by reference to Exhibit 3.1 to the Registrant's Current Report on Form 8-K, filed on February 13, 2025).
- 3 . 2 Amended and Restated By-Laws of The Goldman Sachs Group, Inc., amended as of November 3, 2023 (incorporated by reference to Exhibit 3.2 to the Registrant's Quarterly Report on Form 10-Q for the quarter ended September 30, 2023).
- 4 . 1 Description of The Goldman Sachs Group, Inc.'s Securities registered pursuant to Section 12 of the Securities Exchange Act of 1934.
- 4 . 2 Indenture, dated as of May 19, 1999, between The Goldman Sachs Group, Inc. and The Bank of New York, as trustee (incorporated by reference to Exhibit 6 to the Registrant's Registration Statement on Form 8-A, filed on June 29, 1999).

- 4.3 Subordinated Debt Indenture, dated as of February 20, 2004, between The Goldman Sachs Group, Inc. and The Bank of New York, as trustee (incorporated by reference to Exhibit 4.2 to the Registrant's Annual Report on Form 10-K for the fiscal year ended November 28, 2003).
- 4.4 Senior Debt Indenture, dated as of December 4, 2007, among GS Finance Corp., as issuer, The Goldman Sachs Group, Inc., as guarantor, and The Bank of New York, as trustee (incorporated by reference to Exhibit 4.69 to the Registrant's Post-Effective Amendment No. 10 to Form S-3, filed on December 4, 2007).
- 4.5 Senior Debt Indenture, dated as of July 16, 2008, between The Goldman Sachs Group, Inc. and The Bank of New York Mellon, as trustee (incorporated by reference to Exhibit 4.82 to the Registrant's Post-Effective Amendment No. 11 to Form S-3 (No. 333-130074), filed on July 17, 2008).
- 4.6 Fourth Supplemental Indenture, dated as of December 31, 2016, between The Goldman Sachs Group, Inc. and The Bank of New York Mellon, as trustee, with respect to the Senior Debt Indenture, dated as of July 16, 2008 (incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K, filed on January 6, 2017).
- 4.7 Senior Debt Indenture, dated as of October 10, 2008, among GS Finance Corp., as issuer, The Goldman Sachs Group, Inc., as guarantor, and The Bank of New York Mellon, as trustee (incorporated by reference to Exhibit 4.70 to the Registrant's Registration Statement on Form S-3 (No. 333-154173), filed on October 10, 2008).
- 4.8 First Supplemental Indenture, dated as of February 20, 2015, among GS Finance Corp., as issuer, The Goldman Sachs Group, Inc., as guarantor, and The Bank of New York Mellon, as trustee, with respect to the Senior Debt Indenture, dated as of October 10, 2008 (incorporated by reference to Exhibit 4.7 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014).
- 4.9 Fourth Supplemental Indenture, dated as of August 21, 2018, among GS Finance Corp., as issuer, The Goldman Sachs Group, Inc., as guarantor, and The Bank of New York Mellon, as trustee, with respect to the Senior Debt Indenture, dated as of October 10, 2008 (incorporated by reference to Exhibit 4.1 to the Registrant's Quarterly Report on Form 10-Q for the period ended September 30, 2018).
- 4.10 Ninth Supplemental Subordinated Debt Indenture, dated as of May 20, 2015, between The Goldman Sachs Group, Inc. and The Bank of New York Mellon, as trustee, with respect to the Subordinated Debt Indenture, dated as of February 20, 2004 (incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K, filed on May 22, 2015).
- 4.11 Tenth Supplemental Subordinated Debt Indenture, dated as of July 7, 2017, between The Goldman Sachs Group, Inc. and The Bank of New York Mellon, as trustee, with respect to the Subordinated Debt Indenture, dated as of February 20, 2004 (incorporated by reference to Exhibit 4.89 to the Registrant's Registration Statement on Form S-3 (No. 333-219206), filed on July 10, 2017).
- 4.12 Seventh Supplemental Indenture, dated as of July 1, 2020, among GS Finance Corp., as issuer, The Goldman Sachs Group, Inc., as guarantor, and The Bank of New York Mellon, as trustee, with respect to the Senior Debt Indenture, dated as of October 10, 2008 (incorporated by reference to Exhibit 4.69 to the Registrant's Registration Statement on Form S-3 (No. 333-239610), filed on July 1, 2020).
- 4.13 Eighth Supplemental Indenture, dated as of October 14, 2020, among GS Finance Corp., as issuer, The Goldman Sachs Group, Inc., as guarantor, and The Bank of New York Mellon, as trustee, with respect to the Senior Debt Indenture, dated as of October 10, 2008 (incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K, filed on October 14, 2020).

Certain instruments defining the rights of holders of long-term debt securities of the Registrant and its subsidiaries are omitted pursuant to Item 601(b)(4)(iii) of Regulation S-K. The Registrant hereby undertakes to furnish to the SEC, upon request, copies of any such instruments.

- 10.1 Amended and Restated The Goldman Sachs Group, Inc. Clawback Policy, effective as of December 1, 2023 (incorporated by reference to Exhibit 10.1 to the Registrant's Quarterly Report on Form 10-Q for the period ended March 31, 2024).
- 10.2 The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (incorporated by reference to Annex C to the Registrant's Definitive Proxy Statement on Schedule 14A, filed on March 19, 2021). †
- 10.3 The Goldman Sachs Partner Compensation Plan (incorporated by reference to Exhibit 10.18 to the Registrant's Registration Statement on Form S-1 (No. 333-74449)). †
- 10.4 The Goldman Sachs Amended and Restated Restricted Partner Compensation Plan (incorporated by reference to Exhibit 10.1 to the Registrant's Quarterly Report on Form 10-Q for the period ended February 24, 2006). †
- 10.5 Form of Employment Agreement for Participating Managing Directors (incorporated by reference to Exhibit 10.19 to the Registrant's Registration Statement on Form S-1 (No. 333-75213)). †
- 10.6 Form of Agreement Relating to Noncompetition and Other Covenants (incorporated by reference to Exhibit 10.20 to the Registrant's Registration Statement on Form S-1 (No. 333-75213)). †
- 10.7 Amended and Restated Shareholders' Agreement, effective as of December 31, 2019, among The Goldman Sachs Group, Inc. and various parties (incorporated by reference to Exhibit 10.6 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2019).
- 10.8 Form of Amendment, dated November 27, 2004, to Agreement Relating to Noncompetition and Other Covenants, dated May 7, 1999 (incorporated by reference to Exhibit 10.32 to the Registrant's Annual Report on Form 10-K for the fiscal year ended November 26, 2004). †
- 10.9 Form of Non-Employee Director RSU Award Agreement (pre-2015) (incorporated by reference to Exhibit 10.21 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.10 Ground Lease, dated August 23, 2005, between Battery Park City Authority d/b/a/ Hugh L. Carey Battery Park City Authority, as Landlord, and Goldman Sachs Headquarters LLC, as Tenant (incorporated by reference to Exhibit 10.1 to the Registrant's Current Report on Form 8-K, filed on August 26, 2005).
- 10.11 General Guarantee Agreement, dated January 30, 2006, made by The Goldman Sachs Group, Inc. relating to certain obligations of Goldman Sachs & Co. LLC (incorporated by reference to Exhibit 10.45 to the Registrant's Annual Report on Form 10-K for the fiscal year ended November 25, 2005).
- 10.12 Goldman Sachs & Co. LLC Executive Life Insurance Policy and Certificate with Metropolitan Life Insurance Company for Participating Managing Directors (incorporated by reference to Exhibit 10.1 to the Registrant's Quarterly Report on Form 10-Q for the period ended August 25, 2006). †
- 10.13 Form of Goldman Sachs & Co. LLC Executive Life Insurance Policy with Pacific Life & Annuity Company for Participating Managing Directors, including policy specifications and form of restriction on Policy Owner's Rights (incorporated by reference to Exhibit 10.2 to the Registrant's Quarterly Report on Form 10-Q for the period ended August 25, 2006). †
- 10.14 Form of Second Amendment, dated November 25, 2006, to Agreement Relating to Noncompetition and Other Covenants, dated May 7, 1999, as amended effective November 27, 2004 (incorporated by reference to Exhibit 10.51 to the Registrant's Annual Report on Form 10-K for the fiscal year ended November 24, 2006). †
- 10.15 Description of PMD Retiree Medical Program (incorporated by reference to Exhibit 10.20 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2021). †
- 10.16 General Guarantee Agreement, dated December 1, 2008, made by The Goldman Sachs Group, Inc. relating to certain obligations of Goldman Sachs Bank USA (incorporated by reference to Exhibit 4.80 to the Registrant's Post-Effective Amendment No. 2 to Form S-3, filed on March 19, 2009).
- 10.17 Form of One-Time RSU Award Agreement (pre-2015) (incorporated by reference to Exhibit 10.32 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.18 Amendments to Certain Non-Employee Director Equity Award Agreements (incorporated by reference to Exhibit 10.69 to the Registrant's Annual Report on Form 10-K for the fiscal year ended November 28, 2008). †

- 10.19 Form of Year-End RSU Award Agreement (not fully vested) (pre-2015) (incorporated by reference to Exhibit 10.36 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.20 Form of Year-End RSU Award Agreement (fully vested) (pre-2015) (incorporated by reference to Exhibit 10.37 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.21 Form of Year-End RSU Award Agreement (Base and/or Supplemental) (pre-2015) (incorporated by reference to Exhibit 10.38 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.22 Form of Year-End Restricted Stock Award Agreement (fully vested) (pre-2015) (incorporated by reference to Exhibit 10.41 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2013). †
- 10.23 Form of Year-End Restricted Stock Award Agreement (Base and/or Supplemental) (pre-2015) (incorporated by reference to Exhibit 10.41 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.24 Form of Fixed Allowance RSU Award Agreement (pre-2015) (incorporated by reference to Exhibit 10.43 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2014). †
- 10.25 Form of Deed of Gift (incorporated by reference to Exhibit 10.1 to the Registrant's Quarterly Report on Form 10-Q for the period ended June 30, 2010). †
- 10.26 The Goldman Sachs Long-Term Performance Incentive Plan, dated December 17, 2010 (incorporated by reference to Exhibit 10.1 to the Registrant's Current Report on Form 8-K, filed on December 23, 2010). †
- 10.27 Form of Performance-Based Restricted Stock Unit Award Agreement (pre-2015) (incorporated by reference to Exhibit 10.2 to the Registrant's Current Report on Form 8-K, filed on December 23, 2010). †
- 10.28 Form of Performance-Based Option Award Agreement (incorporated by reference to Exhibit 10.3 to the Registrant's Current Report on Form 8-K, filed on December 23, 2010). †
- 10.29 Form of Performance-Based Cash Compensation Award Agreement (pre-2015) (incorporated by reference to Exhibit 10.4 to the Registrant's Current Report on Form 8-K, filed on December 23, 2010). †
- 10.30 Amended and Restated General Guarantee Agreement, dated November 21, 2011, made by The Goldman Sachs Group, Inc. relating to certain obligations of Goldman Sachs Bank USA (incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K, filed on November 21, 2011).
- 10.31 Form of Aircraft Time Sharing Agreement (incorporated by reference to Exhibit 10.61 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2011). †
- 10.32 Form of Non-Employee Director RSU Award Agreement. †
- 10.33 Form of Non-Employee Director RSU Award Agreement (Cash-Settled)
- 10.34 Form of One-Time/Year-End RSU Award Agreement. †
- 10.35 Form of One-Time/Year-End RSU Award Agreement (fully vested). †
- 10.36 Form of One-Time/Year-End RSU Award Agreement (Base (not fully vested) and/or Supplemental). †
- 10.37 Form of One-Time/Year-End Short-Term RSU Award Agreement. †
- 10.38 Form of Year-End Restricted Stock Award Agreement (incorporated by reference to Exhibit 10.46 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2020). †
- 10.39 Form of Year-End Restricted Stock Award Agreement (fully vested) (incorporated by reference to Exhibit 10.53 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2017). †
- 10.40 Form of Year-End Short-Term Restricted Stock Award Agreement (incorporated by reference to Exhibit 10.57 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2015). †
- 10.41 Form of Fixed Allowance Deferred Cash Award Agreement (incorporated by reference to Exhibit 10.59 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2015). †
- 10.42 Form of One-Time/Year-End Performance-Based Restricted Stock Unit Award Agreement (fully vested). †
- 10.43 Form of One-Time/Year-End Performance-Based Restricted Stock Unit Award Agreement (not fully vested). †

- 10.44 Form of Performance-Based Cash Compensation Award Agreement (incorporated by reference to Exhibit 10.61 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2015). †
- 10.45 Form of Signature Card for Equity Awards. †
- 10.46 Form of Amended and Restated Agreement of Limited Partnership for Participants in the Long Term Executive Carried Interest Incentive Program. †
- 10.47 Form of Subscription Agreement and Materials for Participants in the Long Term Executive Carried Interest Incentive Program. †
- 10.48 Amended and Restated General Guarantee Agreement, dated September 28, 2018, made by The Goldman Sachs Group, Inc. relating to certain obligations of Goldman Sachs Bank USA (incorporated by reference to Exhibit 4.1 to the Registrant's Current Report on Form 8-K, filed on September 28, 2018).
- 10.49 Amended and Restated General Guarantee Agreement, dated September 28, 2018, made by The Goldman Sachs Group, Inc. relating to certain obligations of Goldman Sachs & Co. LLC (incorporated by reference to Exhibit 99.1 to the Registrant's Current Report on Form 8-K, filed on September 28, 2018).
- 10.50 Lease, dated August 17, 2018, between Farrington Street Partners Limited and Farrington Street (Nominee) Limited, as Landlord, and Goldman Sachs International, as Tenant (incorporated by reference to Exhibit 10.3 to the Registrant's Quarterly Report on Form 10-Q for the period ended September 30, 2018).
- 9.1 The Goldman Sachs Group, Inc. Firmwide Insider Trading Policy.
- 21.1 List of significant subsidiaries of The Goldman Sachs Group, Inc.
- 22.1 Issuers of guaranteed securities (incorporated by reference to Exhibit 22.1 to the Registrant's Post-Effective Amendment No. 1 to Form S-3, filed on February 18, 2021).
- 23.1 Consent of Independent Registered Public Accounting Firm.
- 31.1 Rule 13a-14(a) Certifications.
- 32.1 Section 1350 Certifications (This information is furnished and not filed for purposes of Sections 11 and 12 of the Securities Act of 1933 and Section 18 of the Securities Exchange Act of 1934).
- 97.1 The Goldman Sachs Group, Inc. Policy for the Recovery of Erroneously Awarded Compensation, effective as of December 1, 2023 (incorporated by reference to Exhibit 97.1 to the Registrant's Annual Report on Form 10-K for the fiscal year ended December 31, 2023).
- 101 Pursuant to Rules 405 and 406 of Regulation S-T, the following information is formatted in iXBRL (Inline eXtensible Business Reporting Language): (i) the Consolidated Statements of Earnings for the years ended December 31, 2024, December 31, 2023 and December 31, 2022, (ii) the Consolidated Statements of Comprehensive Income for the years ended December 31, 2024, December 31, 2023 and December 31, 2022, (iii) the Consolidated Balance Sheets as of December 31, 2024 and December 31, 2023, (iv) the Consolidated Statements of Changes in Shareholders' Equity for the years ended December 31, 2024, December 31, 2023 and December 31, 2022, (v) the Consolidated Statements of Cash Flows for the years ended December 31, 2024, December 31, 2023 and December 31, 2022, (vi) the notes to the Consolidated Financial Statements and (vii) the cover page.
- 104 Cover Page Interactive Data File (formatted in iXBRL in Exhibit 101).

† This exhibit is a management contract or a compensatory plan or arrangement.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

THE GOLDMAN SACHS GROUP, INC.

By: /s/ Denis P. Coleman III
Name: Denis P. Coleman III
Title: Chief Financial Officer
Date: February 26, 2025

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

By: /s/ David Solomon
Name: David Solomon
Title: Director, Chairman and Chief Executive Officer (Principal Executive Officer)
Date: February 26, 2025

By: /s/ M. Michele Burns
Name: M. Michele Burns
Title: Director
Date: February 26, 2025

By: /s/ Mark A. Flaherty
Name: Mark A. Flaherty
Title: Director
Date: February 26, 2025

By: /s/ Kimberley D. Harris
Name: Kimberley D. Harris
Title: Director
Date: February 26, 2025

By: /s/ John B. Hess
Name: John B. Hess
Title: Director
Date: February 26, 2025

By: /s/ Kevin R. Johnson
Name: Kevin R. Johnson
Title: Director
Date: February 26, 2025

By: /s/ Ellen J. Kullman
Name: Ellen J. Kullman
Title: Director
Date: February 26, 2025

By: /s/ KC McClure
Name: KC McClure
Title: Director
Date: February 26, 2025

By: /s/ Lakshmi N. Mittal
Name: Lakshmi N. Mittal
Title: Director
Date: February 26, 2025

By: /s/ Thomas Montag
Name: Thomas Montag
Title: Director
Date: February 26, 2025

By: /s/ Peter Oppenheimer
Name: Peter Oppenheimer
Title: Director
Date: February 26, 2025

By: /s/ Jan E. Tighe
Name: Jan E. Tighe
Title: Director
Date: February 26, 2025

By: /s/ David A. Viniar
Name: David A. Viniar
Title: Director
Date: February 26, 2025

By: /s/ John E. Waldron
Name: John E. Waldron
Title: Director
Date: February 26, 2025

By: /s/ Denis P. Coleman III
Name: Denis P. Coleman III
Title: Chief Financial Officer
(Principal Financial Officer)
Date: February 26, 2025

By: /s/ Sheara J. Fredman
Name: Sheara J. Fredman
Title: Chief Accounting Officer
(Principal Accounting Officer)
Date: February 26, 2025

THE GOLDMAN SACHS GROUP, INC.

**DESCRIPTION OF SECURITIES REGISTERED PURSUANT TO SECTION 12 OF THE SECURITIES EXCHANGE ACT OF 1934
AS OF DECEMBER 31, 2024**

The following is a description of each class of securities of The Goldman Sachs Group, Inc. (the “Company”) that is registered under Section 12 of the Securities and Exchange Act of 1934, as amended, as of December 31, 2024.

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DESCRIPTION OF COMMON STOCK

The following is a brief description of the material terms of the Company's common stock. The following summary does not purport to be complete and is qualified in its entirety by reference to the pertinent sections of the Company's restated certificate of incorporation and the Company's amended and restated by-laws, which are exhibits to the Annual Report of which this exhibit is a part. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

Pursuant to the Company's restated certificate of incorporation, the Company's authorized capital stock consists of 4,350,000,000 shares, each with a par value of \$0.01 per share, of which 4,000,000,000 shares are designated as common stock and 200,000,000 shares are designated as nonvoting common stock. All outstanding shares of common stock are validly issued, fully paid and nonassessable, which means that its holders have paid their purchase price in full and that the Company may not ask them to surrender additional funds.

The Company's Shareholders' Agreement governs, among other things, the voting of shares of common stock owned by participating managing directors of the Company. Shares of common stock subject to the Company's Shareholders' Agreement are called "voting shares." Before any of the Company's shareholders vote, a separate, preliminary vote is held by the persons covered by the Company's Shareholders' Agreement. In the election of directors, all voting shares will be voted in favor of the election of the director nominees receiving the highest numbers of votes cast by the covered persons in the preliminary vote. For all other matters, all voting shares will be voted in accordance with the majority of the votes cast by the covered persons in the preliminary vote.

Common Stock

Each holder of common stock is entitled to one vote for each share owned of record on all matters submitted to a vote of shareholders. There are no cumulative voting rights. Accordingly, the holders of a plurality of the shares of common stock voting in a contested election of directors can elect all the directors if they choose to do so, subject to any voting rights of holders of preferred stock to elect directors. In an uncontested director election, a director must receive a majority of the votes cast for or against the director to be elected.

Subject to the preferential rights of any holders of any outstanding series of preferred stock, the holders of common stock, together with the holders of the nonvoting common stock, are entitled to such dividends and distributions, whether payable in cash or otherwise, as may be declared from time to time by the Company's board of directors from legally available funds. Subject to the preferential rights of holders of any outstanding series of preferred stock, upon the Company's liquidation, dissolution or winding up and after payment of all prior claims, the holders of common stock, with the shares of the common stock and the nonvoting common stock being considered as a single class for this purpose, will be entitled to receive *pro rata* all the Company's assets. Holders of common stock have no redemption or conversion rights or preemptive rights to purchase or subscribe for securities of the Company.

Nonvoting Common Stock

The nonvoting common stock has the same rights and privileges as, ranks equally and shares proportionately with, and is identical in all respects as to all matters to, the common stock, except that the nonvoting common stock has no voting rights other than those voting rights required by law.

Section 203 of the Delaware General Corporation Law

The Company is subject to the provisions of Section 203 of the Delaware General Corporation Law. In general, Section 203 prohibits a publicly held Delaware corporation from engaging in a "business combination" with an "interested stockholder" for a period of three years after the date of the transaction in which the person became

an interested stockholder, unless the business combination is approved in a prescribed manner. A “business combination” includes a merger, asset sale or a transaction resulting in a financial benefit to the interested stockholder. An “interested stockholder” is a person who, together with affiliates and associates, owns (or, in certain cases, within the preceding three years, did own) 15% or more of the corporation’s outstanding voting stock. Under Section 203, a business combination between the Company and an interested stockholder is prohibited unless it satisfies one of the following conditions:

- prior to the stockholder becoming an interested stockholder, the board of directors of the Company must have previously approved either the business combination or the transaction that resulted in the stockholder becoming an interested stockholder;
- on consummation of the transaction that resulted in the stockholder becoming an interested stockholder, the interested stockholder owned at least 85% of the voting stock of the Company outstanding at the time the transaction commenced, excluding, for purposes of determining the number of shares outstanding (but not the outstanding voting stock owned by the interested stockholder), those shares owned (i) by persons who are directors and also officers and (ii) employee stock plans in which employee participants do not have the right to determine confidentially whether shares held subject to the plan will be tendered in a tender or exchange offer; or
- at or subsequent to such time the business combination is approved by the board of directors and authorized at an annual or special meeting of stockholders, and not by written consent, by the affirmative vote of at least 66 2/3% of the outstanding voting stock which is not owned by the interested stockholder.

The Company’s board of directors has adopted a resolution providing that the shareholders’ agreement will not create an “interested stockholder.”

Certain Anti-Takeover Matters

The Company’s restated certificate of incorporation and amended and restated by-laws include a number of provisions that may have the effect of encouraging persons considering unsolicited tender offers or other unilateral takeover proposals to negotiate with the Company’s board of directors rather than pursue non-negotiated takeover attempts. These provisions include:

Constituency Provision

In accordance with the Company’s restated certificate of incorporation, a director of the Company may (but is not required to) in taking any action (including an action that may involve or relate to a change or potential change in control of the Company), consider, among other things, the effects that the Company’s actions may have on other interests or persons (including its employees, former partners of The Goldman Sachs Group, L.P. and the community) in addition to the Company’s shareholders.

Advance Notice Requirements

The Company’s amended and restated by-laws establish advance notice procedures with regard to shareholder proposals relating to the nomination of candidates for election as directors or new business to be brought before meetings of shareholders of the Company. These procedures provide that notice of such shareholder proposals must be timely given in writing to the Secretary of the Company prior to the meeting at which the action is to be taken. The time periods vary depending on the nature of the proposal. The notice must contain certain information specified in the amended and restated by-laws and must otherwise comply with the amended and restated by-laws.

Limitation on Ability of Shareholders to Call Special Meetings

The Company’s restated certificate of incorporation and amended and restated by-laws provide procedures pursuant to which holders of record of not less than 25% of the voting power of outstanding shares of the

Company's common stock may call a special meeting of shareholders. The Company's amended and restated by-laws impose certain procedural requirements on shareholders requesting such a meeting (including the provision of the same information required for shareholder proposals at annual meetings under the Company's advance notice by-law provisions described above), as well as qualifications designed to prevent duplicative and unnecessary meetings.

No Written Consent of Shareholders

The Company's restated certificate of incorporation requires all shareholder actions to be taken by a vote of the shareholders at an annual or special meeting, and does not permit the Company's shareholders to act by written consent without a meeting.

Blank Check Preferred Stock

The Company's restated certificate of incorporation provides for 150,000,000 authorized shares of preferred stock. The existence of authorized but unissued shares of preferred stock may enable the board of directors to render more difficult or to discourage an attempt to obtain control of the Company by means of a merger, tender offer, proxy contest or otherwise. For example, if in the due exercise of its fiduciary obligations, the board of directors were to determine that a takeover proposal is not in the best interests of the Company, the board of directors could cause shares of preferred stock to be issued without shareholder approval in one or more private offerings or other transactions that might dilute the voting or other rights of the proposed acquiror or insurgent shareholder or shareholder group. In this regard, the restated certificate of incorporation grants the Company's board of directors broad power to establish the rights and preferences of authorized and unissued shares of preferred stock. The issuance of shares of preferred stock could decrease the amount of earnings and assets available for distribution to holders of shares of common stock. The issuance may also adversely affect the rights and powers, including voting rights, of such holders and may have the effect of delaying, deterring or preventing a change in control of the Company.

Listing

The common stock of the Company is listed on the NYSE under the ticker symbol "GS."

DESCRIPTION OF THE DEPOSITARY SHARES, EACH REPRESENTING 1/1,000TH INTEREST IN A SHARE OF FLOATING RATE NON-CUMULATIVE PREFERRED STOCK, SERIES A

DESCRIPTION OF SERIES A PREFERRED STOCK

The depositary is the sole holder of the Company's Floating Rate Non-Cumulative Preferred Stock, Series A (the "Series A Preferred Stock"), and all references herein to the holders of the Series A Preferred Stock shall mean the depositary. However, the holders of depositary shares are entitled, through the depositary, to exercise the rights and preferences of the holders of the Series A Preferred Stock, as described below under "Description of Depositary Shares."

The following is a brief description of the material terms of the Series A Preferred Stock. The following summary of the terms and provisions of the Series A Preferred Stock does not purport to be complete and is qualified in its entirety by reference to the pertinent sections of the Company's restated certificate of incorporation, which is an exhibit to the Annual Report of which this exhibit is a part. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

General

The Company's authorized capital stock includes 150,000,000 shares of preferred stock, par value \$0.01 per share. The Series A Preferred Stock is part of a single series of the Company's authorized preferred stock. The Company may from time to time, without notice to or the consent of holders of the Series A Preferred Stock, issue additional shares of the Series A Preferred Stock, up to the maximum number of authorized but unissued shares.

Shares of the Series A Preferred Stock rank senior to the Company's common stock, equally with each other series of the Company's preferred stock outstanding as of December 31, 2020 and at least equally to each other series of preferred stock that the Company may issue (except for any senior series that may be issued with the requisite consent of the holders of the Series A Preferred Stock), with respect to the payment of dividends and distributions of assets upon liquidation, dissolution or winding up. In addition, the Company will generally be able to pay dividends and distributions upon liquidation, dissolution or winding up only out of lawfully available funds for such payment (*i.e.*, after taking account of all indebtedness and other non-equity claims). The Series A Preferred Stock is fully paid and nonassessable, which means that its holders have paid their purchase price in full and that the Company may not ask them to surrender additional funds. Holders of Series A Preferred Stock do not have preemptive or subscription rights to acquire more stock of the Company.

The Series A Preferred Stock is not convertible into, or exchangeable for, shares of any other class or series of stock or other securities of the Company, except under certain limited circumstances described below under "— Regulatory Changes Relating to Capital Adequacy." The Series A Preferred Stock has no stated maturity and is not subject to any sinking fund or other obligation of the Company to redeem or repurchase the Series A Preferred Stock. The Series A Preferred Stock represents non-withdrawable capital, is not a bank deposit and is not insured by the FDIC or any other governmental agency, nor is it the obligation of, or guaranteed by, a bank.

Dividends

Dividends on shares of the Series A Preferred Stock are not mandatory. Holders of Series A Preferred Stock are entitled to receive, when, as and if declared by the Company's board of directors or a duly authorized committee of the board, out of funds legally available for the payment of dividends under Delaware law, non-cumulative cash dividends from the original issue date, quarterly in arrears on the 10th day of February, May, August and November of each year (each, a "dividend payment date"). These dividends accrue, with respect to each dividend period, on the liquidation preference amount of \$25,000 per share (equivalent to \$25 per depositary share) at a rate per annum equal to the greater of (1) 0.75% above LIBOR (as described below) on

the related LIBOR determination date (as described below) or (2) 3.75%. In the event that the Company issues additional shares of Series A Preferred Stock after the original issue date, dividends on such shares may accrue from the original issue date or any other date the Company specifies at the time such additional shares are issued.

Dividends will be payable to holders of record of Series A Preferred Stock as they appear on the Company's books on the applicable record date, which shall be the 15th calendar day before that dividend payment date or such other record date fixed by the Company's board of directors (or a duly authorized committee of the board) that is not more than 60 nor less than 10 days prior to such dividend payment date (each, a "dividend record date"). These dividend record dates will apply regardless of whether a particular dividend record date is a business day. The corresponding record dates for the depositary shares are the same as the record dates for the Series A Preferred Stock.

A dividend period is the period from and including a dividend payment date to but excluding the next dividend payment date. Dividends payable on the Series A Preferred Stock are computed on the basis of a 360-day year and the actual number of days elapsed in the dividend period. If any date on which dividends would otherwise be payable is not a business day, then the dividend payment date will be the next succeeding business day unless such day falls in the next calendar month, in which case the dividend payment date will be the immediately preceding day that is a business day.

For any dividend period, LIBOR shall be determined by the calculation agent on the second London business day immediately preceding the first day of such dividend period in the following manner:

- LIBOR will be the offered rate per annum for three-month deposits in U.S. dollars, beginning on the first day of such period, as that rate appears on Moneyline Telerate Page 3750 as of 11:00 A.M., London time, on the second London business day immediately preceding the first day of such dividend period.
- If the rate described above does not appear on Moneyline Telerate page 3750, LIBOR will be determined on the basis of the rates, at approximately 11:00 A.M., London time, on the second London business day immediately preceding the first day of such dividend period, at which deposits of the following kind are offered to prime banks in the London interbank market by four major banks in that market selected by the calculation agent: three-month deposits in U.S. dollars, beginning on the first day of such dividend period, and in a Representative Amount. The calculation agent will request the principal London office of each of these banks to provide a quotation of its rate. If at least two quotations are provided, LIBOR for the second London business day immediately preceding the first day of such dividend period will be the arithmetic mean of the quotations.
- If fewer than two quotations are provided as described above, LIBOR for the second London business day immediately preceding the first day of such dividend period will be the arithmetic mean of the rates for loans of the following kind to leading European banks quoted, at approximately 11:00 A.M. New York City time on the second London business day immediately preceding the first day of such dividend period, by three major banks in New York City selected by the calculation agent: three-month loans of U.S. dollars, beginning on the first day of such dividend period, and in a Representative Amount.
- If fewer than three banks selected by the calculation agent are quoting as described above, LIBOR for the new dividend period will be LIBOR in effect for the prior dividend period.
- Under the Federal Reserve System's ("FRB") final rule and the Adjustable Interest Rate Act ("LIBOR Act"), on the first London business day following June 30, 2023, LIBOR was replaced with three-month term Secured Overnight Financing Rate ("SOFR") plus the statutorily prescribed tenor spread, with three-month term SOFR determined by the calculation agent on the second U.S. government securities business day immediately preceding the first day of the applicable dividend period.

The calculation agent's determination of any dividend rate, and its calculation of the amount of dividends for any dividend period, will be on file at the Company's principal offices, will be made available to any stockholder upon request and will be final and binding in the absence of manifest error.

This subsection uses several terms that have special meanings relevant to calculating LIBOR. Those terms have the following meanings:

The term "Representative Amount" means an amount that, in the calculation agent's judgment, is representative of a single transaction in the relevant market at the relevant time.

The term "Moneyline Telerate Page" means the display on Moneyline Telerate, Inc., or any successor service, on the page or pages specified herein or any replacement page or pages on that service.

The term "business day" means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is not a day on which banking institutions in New York City generally are authorized or obligated by law or executive order to close.

The term "London business day" means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is a day on which dealings in U.S. dollars are transacted in the London interbank market.

Dividends on shares of Series A Preferred Stock are not cumulative. Accordingly, if the board of directors of the Company, or a duly authorized committee of the board, does not declare a dividend on the Series A Preferred Stock payable in respect of any dividend period before the related dividend payment date, such dividend will not accrue and the Company will have no obligation to pay a dividend for that dividend period on the dividend payment date or at any future time, whether or not dividends on the Series A Preferred Stock are declared for any future dividend period.

So long as any share of Series A Preferred Stock remains outstanding, no dividend shall be paid or declared on the Company's common stock or any other shares of the Company's junior stock (as defined below) (other than a dividend payable solely in junior stock), and no common stock or other junior stock shall be purchased, redeemed or otherwise acquired for consideration by the Company, directly or indirectly (other than as a result of a reclassification of junior stock for or into other junior stock, or the exchange or conversion of one share of junior stock for or into another share of junior stock and other than through the use of the proceeds of a substantially contemporaneous sale of junior stock), during a dividend period, unless the full dividends for the latest completed dividend period on all outstanding shares of Series A Preferred Stock have been declared and paid (or declared and a sum sufficient for the payment thereof has been set aside). However, the foregoing provision shall not restrict the ability of Goldman Sachs & Co. LLC, or any of the Company's other affiliates, to engage in any market-making transactions in the Company's junior stock in the ordinary course of business.

As used in this description of the Series A Preferred Stock, "junior stock" means any class or series of stock of the Company that ranks junior to the Series A Preferred Stock either as to the payment of dividends or as to the distribution of assets upon any liquidation, dissolution or winding up of the Company. Junior stock includes the Company's common stock.

When dividends are not paid (or duly provided for) on any dividend payment date (or, in the case of parity stock, as defined below, having dividend payment dates different from the dividend payment dates pertaining to the Series A Preferred Stock, on a dividend payment date falling within the related dividend period for the Series A Preferred Stock) in full upon the Series A Preferred Stock and any shares of parity stock, all dividends declared upon the Series A Preferred Stock and all such equally ranking securities payable on such dividend payment date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Series A Preferred Stock, on a dividend payment date falling within the related dividend period for the Series A Preferred Stock) shall be declared *pro rata* so that the respective amounts of such dividends shall bear the same ratio to each other as all accrued but unpaid dividends per share on the Series A Preferred Stock and all parity stock payable on such dividend payment date (or, in the case of parity

stock having dividend payment dates different from the dividend payment dates pertaining to the Series A Preferred Stock, on a dividend payment date falling within the related dividend period for the Series A Preferred Stock) bear to each other.

As used in this description of the Series A Preferred Stock, “parity stock” means any other class or series of stock of the Company that ranks equally with the Series A Preferred Stock in the payment of dividends and in the distribution of assets on any liquidation, dissolution or winding up of the Company.

Subject to the foregoing, such dividends (payable in cash, stock or otherwise) as may be determined by the Company’s board of directors (or a duly authorized committee of the board) may be declared and paid on the Company’s common stock and any other stock ranking equally with or junior to the Series A Preferred Stock from time to time out of any funds legally available for such payment, and the shares of the Series A Preferred Stock shall not be entitled to participate in any such dividend.

Liquidation Rights

Upon any voluntary or involuntary liquidation, dissolution or winding up of the Company, holders of the Series A Preferred Stock are entitled to receive out of assets of the Company available for distribution to stockholders, after satisfaction of liabilities to creditors, if any, before any distribution of assets is made to holders of common stock or of any of the Company’s other shares of stock ranking junior as to such a distribution to the shares of Series A Preferred Stock, a liquidating distribution in the amount of \$25,000 per share (equivalent to \$25 per depositary share) plus declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of the Series A Preferred Stock will not be entitled to any other amounts from the Company after they have received their full liquidation preference.

In any such distribution, if the assets of the Company are not sufficient to pay the liquidation preferences in full to all holders of the Series A Preferred Stock and all holders of any other shares of the Company’s stock ranking equally as to such distribution with the Series A Preferred Stock, the amounts paid to the holders of Series A Preferred Stock and to the holders of all such other stock will be paid *pro rata* in accordance with the respective aggregate liquidation preferences of those holders. In any such distribution, the “liquidation preference” of any holder of preferred stock means the amount payable to such holder in such distribution, including any declared but unpaid dividends (and any unpaid, accrued cumulative dividends in the case of any holder of stock on which dividends accrue on a cumulative basis). If the liquidation preference has been paid in full to all holders of Series A Preferred Stock, the holders of the Company’s other stock shall be entitled to receive all remaining assets of the Company according to their respective rights and preferences.

For purposes of this description of the Series A Preferred Stock, the merger or consolidation of the Company with any other entity, including a merger or consolidation in which the holders of Series A Preferred Stock receive cash, securities or property for their shares, or the sale, lease or exchange of all or substantially all of the assets of the Company for cash, securities or other property shall not constitute a liquidation, dissolution or winding up of the Company.

Redemption

The Series A Preferred Stock is not subject to any mandatory redemption, sinking fund or other similar provisions. The Series A Preferred Stock is currently redeemable at the Company’s option, in whole or in part, upon not less than 30 nor more than 60 days’ notice, at a redemption price equal to \$25,000 per share (equivalent to \$25 per depositary share), plus declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of Series A Preferred Stock have no right to require the redemption or repurchase of the Series A Preferred Stock.

If shares of the Series A Preferred Stock are to be redeemed, the notice of redemption shall be given by first-class mail to the holders of record of the Series A Preferred Stock to be redeemed, mailed not less than 30 days nor more than 60 days prior to the date fixed for redemption thereof (*provided that*, if the depositary

shares representing the Series A Preferred Stock are held in book-entry form through The Depository Trust Company, or “DTC,” the Company may give such notice in any manner permitted by the DTC). Each notice of redemption will include a statement setting forth: (i) the redemption date, (ii) the number of shares of the Series A Preferred Stock to be redeemed and, if less than all the shares held by such holder are to be redeemed, the number of such shares to be redeemed from such holder, (iii) the redemption price and (iv) the place or places where holders may surrender certificates evidencing shares of Series A Preferred Stock for payment of the redemption price. If notice of redemption of any shares of Series A Preferred Stock has been given and if the funds necessary for such redemption have been set aside by the Company for the benefit of the holders of any shares of Series A Preferred Stock so called for redemption, then, from and after the redemption date, dividends will cease to accrue on such shares of Series A Preferred Stock, such shares of Series A Preferred Stock shall no longer be deemed outstanding and all rights of the holders of such shares will terminate, except the right to receive the redemption price.

In case of any redemption of only part of the shares of the Series A Preferred Stock at the time outstanding, the shares to be redeemed shall be selected either *pro rata* or in such other manner as the Company may determine to be fair and equitable.

See “Description of Depositary Shares” below for information about redemption of the depositary shares relating to the Company’s Series A Preferred Stock.

Regulatory Changes Relating to Capital Adequacy

The SEC had previously approved the application of the Company and Goldman Sachs & Co. LLC to be supervised by the SEC as a consolidated supervised entity (“CSE”) pursuant to the SEC’s rules at that time relating to CSEs (referred to as the “CSE Rules”). The Company treated the Series A Preferred Stock as allowable capital in accordance with the CSE Rules (such capital is referred to below as “Allowable Capital”).

If the regulatory capital requirements that apply to the Company change in the future, the Series A Preferred Stock may be converted, at the Company’s option and without consent of the holders, into a new series of preferred stock, subject to the limitations described below. The Company will be entitled to exercise this conversion right as follows.

If both of the following occur:

- the Company (by election or otherwise) becomes subject to any law, rule, regulation or guidance (together, “regulations”) relating to the Company’s capital adequacy, which regulation (i) modifies the existing requirements for treatment as Allowable Capital, (ii) provides for a type or level of capital characterized as “Tier 1” or its equivalent pursuant to regulations of any governmental body having jurisdiction over the Company (or any of the Company’s subsidiaries or consolidated affiliates) and implementing capital standards published by the Basel Committee on Banking Supervision, the SEC, the Board of Governors of the Federal Reserve System (the “Federal Reserve Board”) or any other United States national governmental body, or any other applicable regime based on capital standards published by the Basel Committee on Banking Supervision or its successor, or (iii) provides for a type of capital that in the Company’s judgment (after consultation with counsel of recognized standing) is substantially equivalent to such “Tier 1” capital (such capital described in either (ii) or (iii) is referred to below as “Tier 1 Capital Equivalent”), and
- the Company affirmatively elects to qualify the Series A Preferred Stock for such Allowable Capital or Tier 1 Capital Equivalent treatment without any sublimit or other quantitative restriction on the inclusion of the Series A Preferred Stock in Allowable Capital or Tier 1 Capital Equivalent (other than any limitation the Company elects to accept and any limitation requiring that common equity or a specified form of common equity constitute the dominant form of Allowable Capital or Tier 1 Capital Equivalent) under such regulations,

then, upon such affirmative election, the Series A Preferred Stock shall be convertible at the Company’s option into a new series of preferred stock having terms and provisions substantially identical to those of the Series A Preferred Stock, *except* that such new series may have such additional or modified rights, preferences,

privileges and voting powers, and such limitations and restrictions thereof, as are necessary, in the Company's judgment (after consultation with counsel of recognized standing), to comply with the Required Unrestricted Capital Provisions (defined below), *provided* that the Company will not cause any such conversion unless the Company determines that the rights, preferences, privileges and voting powers of such new series of preferred stock, taken as a whole, are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Series A Preferred Stock, taken as a whole. For example, the Company could agree to restrict its ability to pay dividends on or redeem the new series of preferred stock for a specified period or indefinitely, to the extent permitted by the terms and provisions of the new series of preferred stock, since such a restriction would be permitted in the Company's discretion under the terms and provisions of the Series A Preferred Stock.

The Company will provide notice to holders of the Series A Preferred Stock of any election to qualify the Series A Preferred Stock for Allowable Capital or Tier 1 Capital Equivalent treatment and of any determination to convert the Series A Preferred Stock into a new series of preferred stock, promptly upon the effectiveness of any such election or determination. A copy of any such notice and of the relevant regulations will be on file at the Company's principal offices and, upon request, will be made available to any stockholder.

As used above, the term "Required Unrestricted Capital Provisions" means the terms that are, in the Company's judgment (after consultation with counsel of recognized standing), required for preferred stock to be treated as Allowable Capital or Tier 1 Capital Equivalent, as applicable, without any sublimit or other quantitative restriction on the inclusion of such preferred stock in Allowable Capital or Tier 1 Capital Equivalent (other than any limitation the Company elects to accept and any limitation requiring that common equity or a specified form of common equity constitute the dominant form of Allowable Capital or Tier 1 Capital Equivalent) pursuant to applicable regulations.

Voting Rights

Except as provided below, the holders of the Series A Preferred Stock have no voting rights.

Whenever dividends on any shares of the Series A Preferred Stock shall have not been declared and paid for the equivalent of six or more dividend payments, whether or not for consecutive dividend periods (as used in this section, a "Nonpayment"), the holders of such shares, voting together as a class with holders of any and all other series of voting preferred stock (as defined below) then outstanding, will be entitled to vote for the election of a total of two additional members of the Company's board of directors (as used in this section, the "Preferred Stock Directors"), *provided* that the election of any such directors shall not cause the Company to violate the corporate governance requirement of the New York Stock Exchange (or any other exchange on which the Company's securities may be listed) that listed companies must have a majority of independent directors. In that event, the number of directors on the Company's board of directors shall automatically increase by two, and the new directors shall be elected at a special meeting called at the request of the holders of record of at least 20% of the Series A Preferred Stock or of any other series of voting preferred stock (unless such request is received less than 90 days before the date fixed for the next annual or special meeting of the stockholders, in which event such election shall be held at such next annual or special meeting of stockholders), and at each subsequent annual meeting. These voting rights will continue until dividends on the shares of the Series A Preferred Stock and any such series of voting preferred stock for at least four dividend periods, whether or not consecutive, following the Nonpayment shall have been fully paid (or declared and a sum sufficient for the payment of such dividends shall have been set aside for payment).

As used in this description of the Series A Preferred Stock, "voting preferred stock" means any other class or series of preferred stock of the Company ranking equally with the Series A Preferred Stock either as to dividends or the distribution of assets upon liquidation, dissolution or winding up and upon which like voting rights have been conferred and are exercisable. Whether a plurality, majority or other portion of the shares of Series A Preferred Stock and any other voting preferred stock have been voted in favor of any matter shall be determined by reference to the liquidation amounts of the shares voted.

If and when dividends for at least four dividend periods, whether or not consecutive, following a Nonpayment have been paid in full (or declared and a sum sufficient for such payment shall have been set aside), the holders of the Series A Preferred Stock shall be divested of the foregoing voting rights (subject to revesting in the event of each subsequent Nonpayment) and, if such voting rights for all other holders of voting preferred stock have terminated, the term of office of each Preferred Stock Director so elected shall terminate and the number of directors on the board of directors shall automatically decrease by two. In determining whether dividends have been paid for four dividend periods following a Nonpayment, the Company may take account of any dividend the Company elects to pay for such a dividend period after the regular dividend date for that period has passed. Any Preferred Stock Director may be removed at any time without cause by the holders of record of a majority of the outstanding shares of the Series A Preferred Stock when they have the voting rights described above (voting together as a class with all series of voting preferred stock then outstanding). So long as a Nonpayment shall continue, any vacancy in the office of a Preferred Stock Director (other than prior to the initial election after a Nonpayment) may be filled by the written consent of the Preferred Stock Director remaining in office, or if none remains in office, by a vote of the holders of record of a majority of the outstanding shares of Series A Preferred Stock when they have the voting rights described above (voting together as a class with all series of voting preferred stock then outstanding). The Preferred Stock Directors shall each be entitled to one vote per director on any matter.

So long as any shares of Series A Preferred Stock remain outstanding, the Company will not, without the affirmative vote or consent of the holders of at least two-thirds of the outstanding shares of the Series A Preferred Stock and all other series of voting preferred stock entitled to vote thereon, voting together as a single class, given in person or by proxy, either in writing or at a meeting:

- amend or alter the provisions of the Company's restated certificate of incorporation so as to authorize or create, or increase the authorized amount of, any class or series of stock ranking senior to the Series A Preferred Stock with respect to payment of dividends or the distribution of assets upon liquidation, dissolution or winding up of the Company;
- amend, alter or repeal the provisions of the Company's restated certificate of incorporation so as to materially and adversely affect the special rights, preferences, privileges and voting powers of the Series A Preferred Stock, taken as a whole; or
- consummate a binding share exchange or reclassification involving the Series A Preferred Stock or a merger or consolidation of the Company with another entity, unless in each case (i) the shares of Series A Preferred Stock remain outstanding or, in the case of any such merger or consolidation with respect to which the Company is not the surviving or resulting entity, are converted into or exchanged for preference securities of the surviving or resulting entity or its ultimate parent, and (ii) such shares remaining outstanding or such preference securities, as the case may be, have such rights, preferences, privileges and voting powers, taken as a whole, as are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Series A Preferred Stock, taken as a whole;

provided, however, that any increase in the amount of the authorized or issued Series A Preferred Stock or authorized preferred stock or the creation and issuance, or an increase in the authorized or issued amount, of other series of preferred stock ranking equally with and/or junior to the Series A Preferred Stock with respect to the payment of dividends (whether such dividends are cumulative or non-cumulative) and/or the distribution of assets upon liquidation, dissolution or winding up of the Company will not be deemed to adversely affect the special rights, preferences, privileges or voting powers of the Series A Preferred Stock. In addition, any conversion of the Series A Preferred Stock upon the occurrence of certain regulatory events, as discussed above under "— Regulatory Changes Relating to Capital Adequacy," will not be deemed to adversely affect the special rights, preferences, privileges or voting powers of the Series A Preferred Stock.

If an amendment, alteration, repeal, share exchange, reclassification, merger or consolidation described above would adversely affect one or more but not all series of voting preferred stock (including the Series A Preferred Stock for this purpose), then only the series affected and entitled to vote shall vote as a class in lieu of all such series of preferred stock.

Without the consent of the holders of the Series A Preferred Stock, so long as such action does not adversely affect the special rights, preferences, privileges and voting powers of the Series A Preferred Stock, taken as a whole, the Company may amend, alter, supplement or repeal any terms of the Series A Preferred Stock:

- to cure any ambiguity, or to cure, correct or supplement any provision contained in the certificate of designation for the Series A Preferred Stock that may be defective or inconsistent; or
- to make any provision with respect to matters or questions arising with respect to the Series A Preferred Stock that is not inconsistent with the provisions of the certificate of designations.

The foregoing voting provisions will not apply if, at or prior to the time when the act with respect to which such vote would otherwise be required shall be effected, all outstanding shares of Series A Preferred Stock shall have been redeemed or called for redemption upon proper notice and sufficient funds shall have been set aside by the Company for the benefit of the holders of the Series A Preferred Stock to effect such redemption.

DESCRIPTION OF DEPOSITARY SHARES

Please note that as used in this section, references to “holders” of depositary shares mean those who own depositary shares registered in their own names, on the books that the Company or the depositary maintain for this purpose, and not indirect holders who own beneficial interests in depositary shares registered in street name or issued in book-entry form through The Depository Trust Company.

General

The Company has issued fractional interests in shares of preferred stock in the form of depositary shares, each representing a 1/1,000th ownership interest in a share of Series A Preferred Stock and evidenced by a depositary receipt. The shares of Series A Preferred Stock represented by depositary shares are deposited under a deposit agreement among the Company, the depositary and the holders from time to time of the depositary receipts evidencing the depositary shares. Subject to the terms of the deposit agreement, each holder of a depositary share is entitled, through the depositary, in proportion to the applicable fraction of a share of Series A Preferred Stock represented by such depositary share, to all the rights and preferences of the Series A Preferred Stock represented thereby (including dividend, voting, redemption and liquidation rights).

Dividends and Other Distributions

The depositary will distribute any cash dividends or other cash distributions received in respect of the deposited Series A Preferred Stock to the record holders of depositary shares relating to the underlying Series A Preferred Stock in proportion to the number of depositary shares held by the holders. The depositary will distribute any property received by it other than cash to the record holders of depositary shares entitled to those distributions, unless it determines that the distribution cannot be made proportionally among those holders or that it is not feasible to make a distribution. In that event, the depositary may, with the Company’s approval, sell the property and distribute the net proceeds from the sale to the holders of the depositary shares in proportion to the number of depositary shares they hold.

Record dates for the payment of dividends and other matters relating to the depositary shares will be the same as the corresponding record dates for the Series A Preferred Stock.

The amounts distributed to holders of depositary shares will be reduced by any amounts required to be withheld by the depositary or by the Company on account of taxes or other governmental charges.

Redemption of Depositary Shares

If the Company redeems the Series A Preferred Stock represented by the depositary shares, the depositary shares will be redeemed from the proceeds received by the depositary resulting from the redemption of the

Series A Preferred Stock held by the depositary. The redemption price per depositary share will be equal to 1/1,000th of the redemption price per share payable with respect to the Series A Preferred Stock (or \$25 per depositary share). Whenever the Company redeems shares of Series A Preferred Stock held by the depositary, the depositary will redeem, as of the same redemption date, the number of depositary shares representing shares of Series A Preferred Stock so redeemed.

In case of any redemption of less than all of the outstanding depositary shares, the depositary shares to be redeemed will be selected by the depositary *pro rata* or in such other manner determined by the depositary to be equitable. In any such case, the Company will redeem depositary shares only in increments of 1,000 shares and any multiple thereof.

Voting the Series A Preferred Stock

When the depositary receives notice of any meeting at which the holders of the Series A Preferred Stock are entitled to vote, the depositary will mail the information contained in the notice to the record holders of the depositary shares relating to the Series A Preferred Stock. Each record holder of the depositary shares on the record date, which will be the same date as the record date for the Series A Preferred Stock, may instruct the depositary to vote the amount of the Series A Preferred Stock represented by the holder's depositary shares. To the extent possible, the depositary will vote the amount of the Series A Preferred Stock represented by depositary shares in accordance with the instructions it receives. The Company will agree to take all reasonable actions that the depositary determines are necessary to enable the depositary to vote as instructed. If the depositary does not receive specific instructions from the holders of any depositary shares representing the Series A Preferred Stock, it will vote all depositary shares of that series held by it proportionately with instructions received.

Listing

The depositary shares are listed on the New York Stock Exchange under the ticker symbol "GS PrA."

Form of Preferred Stock and Depositary Shares

The depositary shares are issued in book-entry form through The Depository Trust Company. The Series A Preferred Stock is issued in registered form to the depositary.

DESCRIPTION OF THE DEPOSITARY SHARES, EACH REPRESENTING 1/1,000TH INTEREST IN A SHARE OF FLOATING RATE NON-CUMULATIVE PREFERRED STOCK, SERIES C

DESCRIPTION OF SERIES C PREFERRED STOCK

The depositary is the sole holder of the Company's Floating Rate Non-Cumulative Preferred Stock, Series C (the "Series C Preferred Stock"), and all references herein to the holders of the Series C Preferred Stock shall mean the depositary. However, the holders of depositary shares are entitled, through the depositary, to exercise the rights and preferences of the holders of the Series C Preferred Stock, as described below under "Description of Depositary Shares."

The following is a brief description of the material terms of the Series C Preferred Stock. The following summary of the terms and provisions of the Series C Preferred Stock does not purport to be complete and is qualified in its entirety by reference to the pertinent sections of the Company's restated certificate of incorporation, which is an exhibit to the Annual Report of which this exhibit is a part. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

General

The Company's authorized capital stock includes 150,000,000 shares of preferred stock, par value \$0.01 per share. The Series C Preferred Stock is part of a single series of the Company's authorized preferred stock. The Company may from time to time, without notice to or the consent of holders of the Series C Preferred Stock, issue additional shares of the Series C Preferred Stock, up to the maximum number of authorized but unissued shares.

Shares of the Series C Preferred Stock rank senior to the Company's common stock, equally with each other series of the Company's preferred stock outstanding as of December 31, 2020 and at least equally with each other series of preferred stock that the Company may issue (except for any senior series that may be issued with the requisite consent of the holders of the Series C Preferred Stock), with respect to the payment of dividends and distributions of assets upon liquidation, dissolution or winding up. In addition, the Company will generally be able to pay dividends and distributions upon liquidation, dissolution or winding up only out of lawfully available funds for such payment (*i.e.*, after taking account of all indebtedness and other non-equity claims). The Series C Preferred Stock is fully paid and nonassessable, which means that its holders have paid their purchase price in full and that the Company may not ask them to surrender additional funds. Holders of Series C Preferred Stock do not have preemptive or subscription rights to acquire more stock of the Company.

The Series C Preferred Stock is not convertible into, or exchangeable for, shares of any other class or series of stock or other securities of the Company, except under certain limited circumstances described below under "— Regulatory Changes Relating to Capital Adequacy." The Series C Preferred Stock has no stated maturity and is not subject to any sinking fund or other obligation of the Company to redeem or repurchase the Series C Preferred Stock. The Series C Preferred Stock represents non-withdrawable capital, is not a bank deposit and is not insured by the FDIC or any other governmental agency, nor is it the obligation of, or guaranteed by, a bank.

Dividends

Dividends on shares of the Series C Preferred Stock are not mandatory. Holders of Series C Preferred Stock are entitled to receive, when, as and if declared by the Company's board of directors or a duly authorized committee of the board, out of funds legally available for the payment of dividends under Delaware law, non-cumulative cash dividends from the original issue date, quarterly in arrears on the 10th day of February, May, August, and November of each year (each, a "dividend payment date"). These dividends accrue, with respect to each dividend period, on the liquidation preference amount of \$25,000 per share (equivalent to \$25 per depositary share) at a rate per annum equal to the greater of (1) 0.75% above LIBOR (as described below) on

the related LIBOR determination date (as described below) or (2) 4.00%. In the event that the Company issues additional shares of Series C Preferred Stock after the original issue date, dividends on such shares may accrue from the original issue date or any other date the Company specifies at the time such additional shares are issued.

Dividends will be payable to holders of record of Series C Preferred Stock as they appear on the Company's books on the applicable record date, which shall be the 15th calendar day before that dividend payment date or such other record date fixed by the Company's board of directors (or a duly authorized committee of the board) that is not more than 60 nor less than 10 days prior to such dividend payment date (each, a "dividend record date"). These dividend record dates will apply regardless of whether a particular dividend record date is a business day. The corresponding record dates for the depositary shares are the same as the record dates for the Series C Preferred Stock.

A dividend period is the period from and including a dividend payment date to but excluding the next dividend payment date. Dividends payable on the Series C Preferred Stock are computed on the basis of a 360-day year and the actual number of days elapsed in the dividend period. If any date on which dividends would otherwise be payable is not a business day, then the dividend payment date will be the next succeeding business day unless such day falls in the next calendar month, in which case the dividend payment date will be the immediately preceding day that is a business day.

For any dividend period, LIBOR shall be determined by the calculation agent on the second London business day immediately preceding the first day of such dividend period in the following manner:

- LIBOR will be the offered rate per annum for three-month deposits in U.S. dollars, beginning on the first day of such period, as that rate appears on Moneyline Telerate Page 3750 as of 11:00 A.M., London time, on the second London business day immediately preceding the first day of such dividend period.
- If the rate described above does not appear on Moneyline Telerate page 3750, LIBOR will be determined on the basis of the rates, at approximately 11:00 A.M., London time, on the second London business day immediately preceding the first day of such dividend period, at which deposits of the following kind are offered to prime banks in the London interbank market by four major banks in that market selected by the calculation agent: three-month deposits in U.S. dollars, beginning on the first day of such dividend period, and in a Representative Amount. The calculation agent will request the principal London office of each of these banks to provide a quotation of its rate. If at least two quotations are provided, LIBOR for the second London business day immediately preceding the first day of such dividend period will be the arithmetic mean of the quotations.
- If fewer than two quotations are provided as described above, LIBOR for the second London business day immediately preceding the first day of such dividend period will be the arithmetic mean of the rates for loans of the following kind to leading European banks quoted, at approximately 11:00 A.M. New York City time on the second London business day immediately preceding the first day of such dividend period, by three major banks in New York City selected by the calculation agent: three-month loans of U.S. dollars, beginning on the first day of such dividend period, and in a Representative Amount.
- If fewer than three banks selected by the calculation agent are quoting as described above, LIBOR for the new dividend period will be LIBOR in effect for the prior dividend period.

Under the FRB's final rule and the LIBOR Act, on the first London business day following June 30, 2023, LIBOR was replaced with three-month term SOFR plus the statutorily prescribed tenor spread, with three-month term SOFR determined by the calculation agent on the second U.S. government securities business day immediately preceding the first day of the applicable dividend period.

The calculation agent's determination of any dividend rate, and its calculation of the amount of dividends for any dividend period, will be on file at the Company's principal offices, will be made available to any stockholder upon request and will be final and binding in the absence of manifest error.

This subsection uses several terms that have special meanings relevant to calculating LIBOR. Those terms have the following meanings:

The term “Representative Amount” means an amount that, in the calculation agent’s judgment, is representative of a single transaction in the relevant market at the relevant time.

The term “Moneyline Telerate Page” means the display on Moneyline Telerate, Inc., or any successor service, on the page or pages specified herein or any replacement page or pages on that service.

The term “business day” means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is not a day on which banking institutions in New York City generally are authorized or obligated by law or executive order to close.

The term “London business day” means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is a day on which dealings in U.S. dollars are transacted in the London interbank market.

Dividends on shares of Series C Preferred Stock are not cumulative. Accordingly, if the board of directors of the Company, or a duly authorized committee of the board, does not declare a dividend on the Series C Preferred Stock payable in respect of any dividend period before the related dividend payment date, such dividend will not accrue and the Company will have no obligation to pay a dividend for that dividend period on the dividend payment date or at any future time, whether or not dividends on the Series C Preferred Stock are declared for any future dividend period.

So long as any share of Series C Preferred Stock remains outstanding, no dividend shall be paid or declared on the Company’s common stock or any other shares of the Company’s junior stock (as defined below) (other than a dividend payable solely in junior stock), and no common stock or other junior stock shall be purchased, redeemed or otherwise acquired for consideration by the Company, directly or indirectly (other than as a result of a reclassification of junior stock for or into other junior stock, or the exchange or conversion of one share of junior stock for or into another share of junior stock and other than through the use of the proceeds of a substantially contemporaneous sale of junior stock), during a dividend period, unless the full dividends for the latest completed dividend period on all outstanding shares of Series C Preferred Stock have been declared and paid (or declared and a sum sufficient for the payment thereof has been set aside). However, the foregoing provision shall not restrict the ability of Goldman Sachs & Co. LLC, or any of the Company’s other affiliates, to engage in any market-making transactions in the Company’s junior stock in the ordinary course of business.

As used in this description of the Series C Preferred Stock, “junior stock” means any class or series of stock of the Company that ranks junior to the Series C Preferred Stock either as to the payment of dividends or as to the distribution of assets upon any liquidation, dissolution or winding up of the Company. Junior stock includes the Company’s common stock.

When dividends are not paid (or duly provided for) on any dividend payment date (or, in the case of parity stock, as defined below, having dividend payment dates different from the dividend payment dates pertaining to the Series C Preferred Stock, on a dividend payment date falling within the related dividend period for the Series C Preferred Stock) in full upon the Series C Preferred Stock and any shares of parity stock, all dividends declared upon the Series C Preferred Stock and all such equally ranking securities payable on such dividend payment date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Series C Preferred Stock, on a dividend payment date falling within the related dividend period for the Series C Preferred Stock) shall be declared *pro rata* so that the respective amounts of such dividends shall bear the same ratio to each other as all accrued but unpaid dividends per share on the Series C Preferred Stock and all parity stock payable on such dividend payment date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Series C Preferred Stock, on a dividend payment date falling within the related dividend period for the Series C Preferred Stock) bear to each other.

As used in this description of the Series C Preferred Stock, “parity stock” means any other class or series of stock of the Company that ranks equally with the Series C Preferred Stock in the payment of dividends and in the distribution of assets on any liquidation, dissolution or winding up of the Company.

Subject to the foregoing, such dividends (payable in cash, stock or otherwise) as may be determined by the Company’s board of directors (or a duly authorized committee of the board) may be declared and paid on the Company’s common stock and any other stock ranking equally with or junior to the Series C Preferred Stock from time to time out of any funds legally available for such payment, and the shares of the Series C Preferred Stock shall not be entitled to participate in any such dividend.

Liquidation Rights

Upon any voluntary or involuntary liquidation, dissolution or winding up of the Company, holders of the Series C Preferred Stock are entitled to receive out of assets of the Company available for distribution to stockholders, after satisfaction of liabilities to creditors, if any, before any distribution of assets is made to holders of common stock or of any of the Company’s other shares of stock ranking junior as to such a distribution to the shares of Series C Preferred Stock, a liquidating distribution in the amount of \$25,000 per share (equivalent to \$25 per depositary share) plus declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of the Series C Preferred Stock will not be entitled to any other amounts from the Company after they have received their full liquidation preference.

In any such distribution, if the assets of the Company are not sufficient to pay the liquidation preferences in full to all holders of the Series C Preferred Stock and all holders of any other shares of the Company’s stock ranking equally as to such distribution with the Series C Preferred Stock, the amounts paid to the holders of Series C Preferred Stock and to the holders of all such other stock will be paid *pro rata* in accordance with the respective aggregate liquidation preferences of those holders. In any such distribution, the “liquidation preference” of any holder of preferred stock means the amount payable to such holder in such distribution, including any declared but unpaid dividends (and any unpaid, accrued cumulative dividends in the case of any holder of stock on which dividends accrue on a cumulative basis). If the liquidation preference has been paid in full to all holders of the Company’s Series C Preferred Stock and any other shares of the Company’s stock ranking equally as to the liquidation distribution, the holders of the Company’s other stock shall be entitled to receive all remaining assets of the Company according to their respective rights and preferences.

For purposes of this description of the Series C Preferred Stock, the merger or consolidation of the Company with any other entity, including a merger or consolidation in which the holders of Series C Preferred Stock receive cash, securities or property for their shares, or the sale, lease or exchange of all or substantially all of the assets of the Company for cash, securities or other property shall not constitute a liquidation, dissolution or winding up of the Company.

Redemption

The Series C Preferred Stock is not subject to any mandatory redemption, sinking fund or other similar provisions. The Series C Preferred Stock is currently redeemable at the Company’s option, in whole or in part, upon not less than 30 nor more than 60 days’ notice, at a redemption price equal to \$25,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of Series C Preferred Stock have no right to require the redemption or repurchase of the Series C Preferred Stock.

If shares of the Series C Preferred Stock are to be redeemed, the notice of redemption shall be given by first class mail to the holders of record of the Series C Preferred Stock to be redeemed, mailed not less than 30 days nor more than 60 days prior to the date fixed for redemption thereof (*provided that*, if the depositary shares representing the Series C Preferred Stock are held in book-entry form through The Depository Trust Company, or “DTC,” the Company may give such notice in any manner permitted by the DTC). Each notice of redemption will include a statement setting forth: (i) the redemption date, (ii) the number of shares of the Series

C Preferred Stock to be redeemed and, if less than all the shares held by such holder are to be redeemed, the number of such shares to be redeemed from such holder, (iii) the redemption price and (iv) the place or places where holders may surrender certificates evidencing shares of Series C Preferred Stock for payment of the redemption price. If notice of redemption of any shares of Series C Preferred Stock has been given and if the funds necessary for such redemption have been set aside by the Company for the benefit of the holders of any shares of Series C Preferred Stock so called for redemption, then, from and after the redemption date, dividends will cease to accrue on such shares of Series C Preferred Stock, such shares of Series C Preferred Stock shall no longer be deemed outstanding and all rights of the holders of such shares will terminate, except the right to receive the redemption price.

In case of any redemption of only part of the shares of the Series C Preferred Stock at the time outstanding, the shares to be redeemed shall be selected either *pro rata* or in such other manner as the Company may determine to be fair and equitable.

See “Description of Depositary Shares” below for information about redemption of the depositary shares relating to the Company’s Series C Preferred Stock.

Regulatory Changes Relating to Capital Adequacy

The Company was previously regulated by the SEC as a consolidated supervised entity (“CSE”) pursuant to the SEC’s rules at that time relating to CSEs (referred to as the “CSE Rules”). The Company treated the Series C Preferred Stock as allowable capital in accordance with the CSE Rules (such capital is referred to below as “Allowable Capital”).

If the regulatory capital requirements that apply to the Company change in the future, the Series C Preferred Stock may be converted, at the Company’s option and without consent of the holders, into a new series of preferred stock, subject to the limitations described below. The Company will be entitled to exercise this conversion right as follows.

If both of the following occur:

- the Company (by election or otherwise) becomes subject to any law, rule, regulation or guidance (together, “regulations”) relating to the Company’s capital adequacy, which regulation (i) modifies the existing requirements for treatment as Allowable Capital, (ii) provides for a type or level of capital characterized as “Tier 1” or its equivalent pursuant to regulations of any governmental body having jurisdiction over the Company (or any of the Company’s subsidiaries or consolidated affiliates) and implementing capital standards published by the Basel Committee on Banking Supervision, the SEC, the Federal Reserve Board or any other United States national governmental body, or any other applicable regime based on capital standards published by the Basel Committee on Banking Supervision or its successor, or (iii) provides for a type of capital that in the Company’s judgment (after consultation with counsel of recognized standing) is substantially equivalent to such “Tier 1” capital (such capital described in either (ii) or (iii) is referred to below as “Tier 1 Capital Equivalent”), and
- the Company affirmatively elects to qualify the Series C Preferred Stock for such Allowable Capital or Tier 1 Capital Equivalent treatment without any sublimit or other quantitative restriction on the inclusion of the Series C Preferred Stock in Allowable Capital or Tier 1 Capital Equivalent (other than any limitation the Company elects to accept and any limitation requiring that common equity or a specified form of common equity constitute the dominant form of Allowable Capital or Tier 1 Capital Equivalent) under such regulations,

then, upon such affirmative election, the Series C Preferred Stock shall be convertible at the Company’s option into a new series of preferred stock having terms and provisions substantially identical to those of the Series C Preferred Stock, *except* that such new series may have such additional or modified rights, preferences, privileges and voting powers, and such limitations and restrictions thereof, as are necessary, in the Company’s judgment (after consultation with counsel of recognized standing), to comply with the Required Unrestricted Capital Provisions (defined below), *provided* that the Company will not cause any such conversion unless the

Company determines that the rights, preferences, privileges and voting powers of such new series of preferred stock, taken as a whole, are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Series C Preferred Stock, taken as a whole. For example, the Company could agree to restrict its ability to pay dividends on or redeem the new series of preferred stock for a specified period or indefinitely, to the extent permitted by the terms and provisions of the new series of preferred stock, since such a restriction would be permitted in the Company's discretion under the terms and provisions of the Series C Preferred Stock.

The Company will provide notice to holders of the Series C Preferred Stock of any election to qualify the Series C Preferred Stock for Allowable Capital or Tier 1 Capital Equivalent treatment and of any determination to convert the Series C Preferred Stock into a new series of preferred stock, promptly upon the effectiveness of any such election or determination. A copy of any such notice and of the relevant regulations will be on file at the Company's principal offices and, upon request, will be made available to any stockholder.

As used above, the term "Required Unrestricted Capital Provisions" means the terms that are, in the Company's judgment (after consultation with counsel of recognized standing), required for preferred stock to be treated as Allowable Capital or Tier 1 Capital Equivalent, as applicable, without any sublimit or other quantitative restriction on the inclusion of such preferred stock in Allowable Capital or Tier 1 Capital Equivalent (other than any limitation the Company elects to accept and any limitation requiring that common equity or a specified form of common equity constitute the dominant form of Allowable Capital or Tier 1 Capital Equivalent) pursuant to applicable regulations.

Voting Rights

Except as provided below, the holders of the Series C Preferred Stock have no voting rights.

Whenever dividends on any shares of the Series C Preferred Stock shall have not been declared and paid for the equivalent of six or more dividend payments, whether or not for consecutive dividend periods (as used in this section, a "Nonpayment"), the holders of such shares, voting together as a class with holders of any and all other series of voting preferred stock (as defined below) then outstanding, will be entitled to vote for the election of a total of two additional members of the Company's board of directors (as used in this section, the "Preferred Stock Directors"), *provided* that the election of any such directors shall not cause the Company to violate the corporate governance requirement of the New York Stock Exchange (or any other exchange on which the Company's securities may be listed) that listed companies must have a majority of independent directors and *provided further* that the Company's board of directors shall at no time include more than two Preferred Stock Directors. In that event, the number of directors on the Company's board of directors shall automatically increase by two, and the new directors shall be elected at a special meeting called at the request of the holders of record of at least 20% of the Series C Preferred Stock or of any other series of voting preferred stock (unless such request is received less than 90 days before the date fixed for the next annual or special meeting of the stockholders, in which event such election shall be held at such next annual or special meeting of stockholders), and at each subsequent annual meeting. These voting rights will continue until dividends on the shares of the Series C Preferred Stock and any such series of voting preferred stock for at least four dividend periods, whether or not consecutive, following the Nonpayment shall have been fully paid (or declared and a sum sufficient for the payment of such dividends shall have been set aside for payment).

As used in this description of the Series C Preferred Stock, "voting preferred stock" means any other class or series of preferred stock of the Company ranking equally with the Series C Preferred Stock either as to dividends or the distribution of assets upon liquidation, dissolution or winding up and upon which like voting rights have been conferred and are exercisable. Whether a plurality, majority or other portion of the shares of Series C Preferred Stock and any other voting preferred stock have been voted in favor of any matter shall be determined by reference to the liquidation amounts of the shares voted.

If and when dividends for at least four dividend periods, whether or not consecutive, following a Nonpayment have been paid in full (or declared and a sum sufficient for such payment shall have been set aside), the

holders of the Series C Preferred Stock shall be divested of the foregoing voting rights (subject to revesting in the event of each subsequent Nonpayment) and, if such voting rights for all other holders of voting preferred stock have terminated, the term of office of each Preferred Stock Director so elected shall terminate and the number of directors on the board of directors shall automatically decrease by two. In determining whether dividends have been paid for four dividend periods following a Nonpayment, the Company may take account of any dividend the Company elects to pay for such a dividend period after the regular dividend date for that period has passed. Any Preferred Stock Director may be removed at any time without cause by the holders of record of a majority of the outstanding shares of the Series C Preferred Stock when they have the voting rights described above (voting together as a class with all series of voting preferred stock then outstanding). So long as a Nonpayment shall continue, any vacancy in the office of a Preferred Stock Director (other than prior to the initial election after a Nonpayment) may be filled by the written consent of the Preferred Stock Director remaining in office, or if none remains in office, by a vote of the holders of record of a majority of the outstanding shares of Series C Preferred Stock and all voting preferred stock when they have the voting rights described above (voting together as a class). The Preferred Stock Directors shall each be entitled to one vote per director on any matter.

So long as any shares of Series C Preferred Stock remain outstanding, the Company will not, without the affirmative vote or consent of the holders of at least two-thirds of the outstanding shares of the Series C Preferred Stock and all other series of voting preferred stock entitled to vote thereon, voting together as a single class, given in person or by proxy, either in writing or at a meeting:

- amend or alter the provisions of the Company's restated certificate of incorporation so as to authorize or create, or increase the authorized amount of, any class or series of stock ranking senior to the Series C Preferred Stock with respect to payment of dividends or the distribution of assets upon liquidation, dissolution or winding up of the Company;
- amend, alter or repeal the provisions of the Company's restated certificate of incorporation so as to materially and adversely affect the special rights, preferences, privileges and voting powers of the Series C Preferred Stock, taken as a whole; or
- consummate a binding share exchange or reclassification involving the Series C Preferred Stock or a merger or consolidation of the Company with another entity, unless in each case (i) the shares of Series C Preferred Stock remain outstanding or, in the case of any such merger or consolidation with respect to which the Company is not the surviving or resulting entity, are converted into or exchanged for preference securities of the surviving or resulting entity or its ultimate parent, and (ii) such shares remaining outstanding or such preference securities, as the case may be, have such rights, preferences, privileges and voting powers, taken as a whole, as are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Series C Preferred Stock, taken as a whole;

provided, however, that any increase in the amount of the authorized or issued Series C Preferred Stock or authorized preferred stock or the creation and issuance, or an increase in the authorized or issued amount, of other series of preferred stock ranking equally with and/or junior to the Series C Preferred Stock with respect to the payment of dividends (whether such dividends are cumulative or non-cumulative) and/or the distribution of assets upon liquidation, dissolution or winding up of the Company will not be deemed to adversely affect the rights, preferences, privileges or voting powers of the Series C Preferred Stock. In addition, any conversion of the Series C Preferred Stock upon the occurrence of certain regulatory events, as discussed above under "— Regulatory Changes Relating to Capital Adequacy," will not be deemed to adversely affect the rights, preferences, privileges or voting powers of the Series C Preferred Stock.

If an amendment, alteration, repeal, share exchange, reclassification, merger or consolidation described above would adversely affect one or more but not all series of voting preferred stock (including the Series C Preferred Stock for this purpose), then only the series affected and entitled to vote shall vote as a class in lieu of all such series of preferred stock.

Without the consent of the holders of the Series C Preferred Stock, so long as such action does not adversely affect the rights, preferences, privileges and voting powers of the Series C Preferred Stock, the Company may amend, alter, supplement or repeal any terms of the Series C Preferred Stock:

- to cure any ambiguity, or to cure, correct or supplement any provision contained in the certificate of designation for the Series C Preferred Stock that may be defective or inconsistent; or
- to make any provision with respect to matters or questions arising with respect to the Series C Preferred Stock that is not inconsistent with the provisions of the certificate of designations.

The foregoing voting provisions will not apply if, at or prior to the time when the act with respect to which such vote would otherwise be required shall be effected, all outstanding shares of Series C Preferred Stock shall have been redeemed or called for redemption upon proper notice and sufficient funds shall have been set aside by the Company for the benefit of the holders of the Series C Preferred Stock to effect such redemption.

DESCRIPTION OF DEPOSITARY SHARES

Please note that as used in this section, references to “holders” of depositary shares mean those who own depositary shares registered in their own names, on the books that the Company or the depositary maintain for this purpose, and not indirect holders who own beneficial interests in depositary shares registered in street name or issued in book-entry form through The Depository Trust Company.

General

The Company has issued fractional interests in shares of preferred stock in the form of depositary shares, each representing a 1/1,000th ownership interest in a share of Series C Preferred Stock and evidenced by a depositary receipt. The shares of Series C Preferred Stock represented by depositary shares are deposited under a deposit agreement among the Company, the depositary and the holders from time to time of the depositary receipts evidencing the depositary shares. Subject to the terms of the deposit agreement, each holder of a depositary share is entitled, through the depositary, in proportion to the applicable fraction of a share of Series C Preferred Stock represented by such depositary share, to all the rights and preferences of the Series C Preferred Stock represented thereby (including dividend, voting, redemption and liquidation rights).

Dividends and Other Distributions

The depositary will distribute any cash dividends or other cash distributions received in respect of the deposited Series C Preferred Stock to the record holders of depositary shares relating to the underlying Series C Preferred Stock in proportion to the number of depositary shares held by the holders. The depositary will distribute any property received by it other than cash to the record holders of depositary shares entitled to those distributions, unless it determines that the distribution cannot be made proportionally among those holders or that it is not feasible to make a distribution. In that event, the depositary may, with the Company’s approval, sell the property and distribute the net proceeds from the sale to the holders of the depositary shares in proportion to the number of depositary shares they hold.

Record dates for the payment of dividends and other matters relating to the depositary shares will be the same as the corresponding record dates for the Series C Preferred Stock.

The amounts distributed to holders of depositary shares will be reduced by any amounts required to be withheld by the depositary or by the Company on account of taxes or other governmental charges.

Redemption of Depositary Shares

If the Company redeems the Series C Preferred Stock represented by the depositary shares, the depositary shares will be redeemed from the proceeds received by the depositary resulting from the redemption of the

Series C Preferred Stock held by the depositary. The redemption price per depositary share will be equal to 1/1,000th of the redemption price per share payable with respect to the Series C Preferred Stock (or \$25 per depositary share). Whenever the Company redeems shares of Series C Preferred Stock held by the depositary, the depositary will redeem, as of the same redemption date, the number of depositary shares representing shares of Series C Preferred Stock so redeemed.

In case of any redemption of less than all of the outstanding depositary shares, the depositary shares to be redeemed will be selected by the depositary *pro rata* or in such other manner determined by the depositary to be equitable. In any such case, the Company will redeem depositary shares only in increments of 1,000 shares and any multiple thereof.

Voting the Series C Preferred Stock

When the depositary receives notice of any meeting at which the holders of the Series C Preferred Stock are entitled to vote, the depositary will mail the information contained in the notice to the record holders of the depositary shares relating to the Series C Preferred Stock. Each record holder of the depositary shares on the record date, which will be the same date as the record date for the Series C Preferred Stock, may instruct the depositary to vote the amount of the Series C Preferred Stock represented by the holder's depositary shares. To the extent possible, the depositary will vote the amount of the Series C Preferred Stock represented by depositary shares in accordance with the instructions it receives. The Company will agree to take all reasonable actions that the depositary determines are necessary to enable the depositary to vote as instructed. If the depositary does not receive specific instructions from the holders of any depositary shares representing the Series C Preferred Stock, it will vote all depositary shares of that series held by it proportionately with instructions received.

Listing

The depositary shares are listed on the New York Stock Exchange under the ticker symbol "GS PrC."

Form of Preferred Stock and Depositary Shares

The depositary shares are issued in book-entry form through The Depository Trust Company. The Series C Preferred Stock is issued in registered form to the depositary.

DESCRIPTION OF THE DEPOSITARY SHARES, EACH REPRESENTING 1/1,000TH INTEREST IN A SHARE OF FLOATING RATE NON-CUMULATIVE PREFERRED STOCK, SERIES D

DESCRIPTION OF SERIES D PREFERRED STOCK

The depositary is the sole holder of the Company's Floating Rate Non-Cumulative Preferred Stock, Series D (the "Series D Preferred Stock"), and all references herein to the holders of the Series D Preferred Stock shall mean the depositary. However, the holders of depositary shares are entitled, through the depositary, to exercise the rights and preferences of the holders of the Series D Preferred Stock, as described below under "Description of Depositary Shares."

The following is a brief description of the material terms of the Series D Preferred Stock. The following summary of the terms and provisions of the Series D Preferred Stock does not purport to be complete and is qualified in its entirety by reference to the pertinent sections of the Company's restated certificate of incorporation, which is an exhibit to the Annual Report of which this exhibit is a part. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

General

The Company's authorized capital stock includes 150,000,000 shares of preferred stock, par value \$0.01 per share. The Series D Preferred Stock is part of a single series of authorized preferred stock. The Company may from time to time, without notice to or the consent of holders of the Series D Preferred Stock, issue additional shares of the Series D Preferred Stock, up to the maximum number of authorized but unissued shares.

Shares of the Series D Preferred Stock rank senior to the Company's common stock, equally with each other series of the Company's preferred stock outstanding as of December 31, 2020 and at least equally with each other series of preferred stock that the Company may issue (except for any senior series that may be issued with the requisite consent of the holders of the Series D Preferred Stock), with respect to the payment of dividends and distributions of assets upon liquidation, dissolution or winding up. In addition, the Company will generally be able to pay dividends and distributions upon liquidation, dissolution or winding up only out of lawfully available funds for such payment (*i.e.*, after taking account of all indebtedness and other non-equity claims). The Series D Preferred Stock is fully paid and nonassessable, which means that its holders have paid their purchase price in full and that the Company may not ask them to surrender additional funds. Holders of Series D Preferred Stock do not have preemptive or subscription rights to acquire more stock of the Company.

The Series D Preferred Stock is not convertible into, or exchangeable for, shares of any other class or series of stock or other securities of the Company, except under certain limited circumstances described below under "— Regulatory Changes Relating to Capital Adequacy." The Series D Preferred Stock has no stated maturity and is not subject to any sinking fund or other obligation of the Company to redeem or repurchase the Series D Preferred Stock. The Series D Preferred Stock represents non-withdrawable capital, is not a bank deposit and is not insured by the FDIC or any other governmental agency, nor is it the obligation of, or guaranteed by, a bank.

Dividends

Dividends on shares of the Series D Preferred Stock are not mandatory. Holders of Series D Preferred Stock are entitled to receive, when, as and if declared by the Company's board of directors (or a duly authorized committee of the board), out of funds legally available for the payment of dividends under Delaware law, non-cumulative cash dividends from the original issue date, quarterly in arrears on the 10th day of February, May, August, and November of each year (each, a "dividend payment date"). These dividends accrue, with respect to each dividend period, on the liquidation preference amount of \$25,000 per share (equivalent to \$25 per depositary share) at a rate per annum equal to the greater of (1) 0.67% above LIBOR (as described below) on the related LIBOR determination date (as described below) or (2) 4.00%. In the event that the Company issues

additional shares of Series D Preferred Stock after the original issue date, dividends on such shares may accrue from the original issue date or any other date the Company specifies at the time such additional shares are issued.

Dividends will be payable to holders of record of Series D Preferred Stock as they appear on the Company's books on the applicable record date, which shall be the 15th calendar day before that dividend payment date or such other record date fixed by the Company's board of directors (or a duly authorized committee of the board) that is not more than 60 nor less than 10 days prior to such dividend payment date (each, a "dividend record date"). These dividend record dates will apply regardless of whether a particular dividend record date is a business day. The corresponding record dates for the depository shares are the same as the record dates for the Series D Preferred Stock.

A dividend period is the period from and including a dividend payment date to but excluding the next dividend payment date. Dividends payable on the Series D Preferred Stock are computed on the basis of a 360-day year and the actual number of days elapsed in the dividend period. If any date on which dividends would otherwise be payable is not a business day, then the dividend payment date will be the next succeeding business day unless such day falls in the next calendar month, in which case the dividend payment date will be the immediately preceding day that is a business day.

For any dividend period, LIBOR shall be determined by the calculation agent on the second London business day immediately preceding the first day of such dividend period in the following manner:

- LIBOR will be the offered rate per annum for three-month deposits in U.S. dollars, beginning on the first day of such period, as that rate appears on Moneyline Telerate Page 3750 (or any successor or replacement page) as of 11:00 A.M., London time, on the second London business day immediately preceding the first day of such dividend period.
- If the rate described above does not appear on Moneyline Telerate page 3750 (or any successor or replacement page), LIBOR will be determined on the basis of the rates, at approximately 11:00 A.M., London time, on the second London business day immediately preceding the first day of such dividend period, at which deposits of the following kind are offered to prime banks in the London interbank market by four major banks in that market selected by the calculation agent: three-month deposits in U.S. dollars, beginning on the first day of such dividend period, and in a Representative Amount. The calculation agent will request the principal London office of each of these banks to provide a quotation of its rate. If at least two quotations are provided, LIBOR for the second London business day immediately preceding the first day of such dividend period will be the arithmetic mean of the quotations.
- If fewer than two quotations are provided as described above, LIBOR for the second London business day immediately preceding the first day of such dividend period will be the arithmetic mean of the rates for loans of the following kind to leading European banks quoted, at approximately 11:00 A.M. New York City time on the second London business day immediately preceding the first day of such dividend period, by three major banks in New York City selected by the calculation agent: three-month loans of U.S. dollars, beginning on the first day of such dividend period, and in a Representative Amount.
- If fewer than three banks selected by the calculation agent are quoting as described above, LIBOR for the new dividend period will be LIBOR in effect for the prior dividend period.

Under the FRB's final rule and the LIBOR Act, on the first London business day following June 30, 2023, LIBOR was replaced with three-month term SOFR plus the statutorily prescribed tenor spread, with three-month term SOFR determined by the calculation agent on the second U.S. government securities business day immediately preceding the first day of the applicable dividend period.

The calculation agent's determination of any dividend rate, and its calculation of the amount of dividends for any dividend period, will be on file at the Company's principal offices, will be made available to any stockholder upon request and will be final and binding in the absence of manifest error.

This subscription uses several terms that have special meanings relevant to calculating LIBOR. Those terms have the following meanings:

The term “Representative Amount” means an amount that, in the calculation agent’s judgment, is representative of a single transaction in the relevant market at the relevant time.

The term “Moneyline Telerate Page” means the display on Moneyline Telerate, Inc., or any successor service, on the page or pages specified herein or any replacement page or pages on that service.

The term “business day” means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is not a day on which banking institutions in New York City generally are authorized or obligated by law or executive order to close.

The term “London business day” means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is a day on which dealings in U.S. dollars are transacted in the London interbank market.

Dividends on shares of Series D Preferred Stock are not cumulative. Accordingly, if the board of directors of the Company (or a duly authorized committee of the board) does not declare a dividend on the Series D Preferred Stock payable in respect of any dividend period before the related dividend payment date, such dividend will not accrue and the Company will have no obligation to pay a dividend for that dividend period on the dividend payment date or at any future time, whether or not dividends on the Series D Preferred Stock are declared for any future dividend period.

So long as any share of Series D Preferred Stock remains outstanding, no dividend shall be paid or declared on the Company’s common stock or any other shares of the Company’s junior stock (as defined below) (other than a dividend payable solely in junior stock), and no common stock or other junior stock shall be purchased, redeemed or otherwise acquired for consideration by the Company, directly or indirectly (other than as a result of a reclassification of junior stock for or into other junior stock, or the exchange or conversion of one share of junior stock for or into another share of junior stock and other than through the use of the proceeds of a substantially contemporaneous sale of junior stock), during a dividend period, unless the full dividends for the latest completed dividend period on all outstanding shares of Series D Preferred Stock have been declared and paid (or declared and a sum sufficient for the payment thereof has been set aside). However, the foregoing provision shall not restrict the ability of Goldman Sachs & Co. LLC, or any of the Company’s other affiliates, to engage in any market-making transactions in the Company’s junior stock in the ordinary course of business.

As used in this description of the Series D Preferred Stock, “junior stock” means any class or series of stock of the Company that ranks junior to the Series D Preferred Stock either as to the payment of dividends or as to the distribution of assets upon any liquidation, dissolution or winding up of the Company. Junior stock includes the Company’s common stock.

When dividends are not paid (or duly provided for) on any dividend payment date (or, in the case of parity stock, as defined below, having dividend payment dates different from the dividend payment dates pertaining to the Series D Preferred Stock, on a dividend payment date falling within the related dividend period for the Series D Preferred Stock) in full upon the Series D Preferred Stock and any shares of parity stock, all dividends declared upon the Series D Preferred Stock and all such equally ranking securities payable on such dividend payment date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Series D Preferred Stock, on a dividend payment date falling within the related dividend period for the Series D Preferred Stock) shall be declared *pro rata* so that the respective amounts of such dividends shall bear the same ratio to each other as all accrued but unpaid dividends per share on the Series D Preferred Stock and all parity stock payable on such dividend payment date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Series D Preferred Stock, on a dividend payment date falling within the related dividend period for the Series D Preferred Stock) bear to each other.

As used in this description of the Series D Preferred Stock, “parity stock” means any other class or series of stock of the Company that ranks equally with the Series D Preferred Stock in the payment of dividends and in the distribution of assets on any liquidation, dissolution or winding up of the Company.

Subject to the foregoing, such dividends (payable in cash, stock or otherwise) as may be determined by the Company’s board of directors (or a duly authorized committee of the board) may be declared and paid on the Company’s common stock and any other stock ranking equally with or junior to the Series D Preferred Stock from time to time out of any funds legally available for such payment, and the shares of the Series D Preferred Stock shall not be entitled to participate in any such dividend.

Liquidation Rights

Upon any voluntary or involuntary liquidation, dissolution or winding up of the Company, holders of the Series D Preferred Stock are entitled to receive out of assets of the Company available for distribution to stockholders, after satisfaction of liabilities to creditors, if any, before any distribution of assets is made to holders of common stock or of any of the Company’s other shares of stock ranking junior as to such a distribution to the shares of Series D Preferred Stock, a liquidating distribution in the amount of \$25,000 per share (equivalent to \$25 per depositary share) plus declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of the Series D Preferred Stock will not be entitled to any other amounts from the Company after they have received their full liquidation preference.

In any such distribution, if the assets of the Company are not sufficient to pay the liquidation preferences in full to all holders of the Series D Preferred Stock and all holders of any other shares of the Company’s stock ranking equally as to such distribution with the Series D Preferred Stock, the amounts paid to the holders of Series D Preferred Stock and to the holders of all such other stock will be paid *pro rata* in accordance with the respective aggregate liquidation preferences of those holders. In any such distribution, the “liquidation preference” of any holder of preferred stock means the amount payable to such holder in such distribution, including any declared but unpaid dividends (and any unpaid, accrued cumulative dividends in the case of any holder of stock on which dividends accrue on a cumulative basis). If the liquidation preference has been paid in full to all holders of the Company’s Series D Preferred Stock and any other shares of the Company’s stock ranking equally as to the liquidation distribution, the holders of the Company’s other stock shall be entitled to receive all remaining assets of the Company according to their respective rights and preferences.

For purposes of this description of the Series D Preferred Stock, the merger or consolidation of the Company with any other entity, including a merger or consolidation in which the holders of Series D Preferred Stock receive cash, securities or property for their shares, or the sale, lease or exchange of all or substantially all of the assets of the Company for cash, securities or other property shall not constitute a liquidation, dissolution or winding up of the Company.

Redemption

The Series D Preferred Stock is not subject to any mandatory redemption, sinking fund or other similar provisions. The Series D Preferred Stock is currently redeemable at the Company’s option, in whole or in part, upon not less than 30 nor more than 60 days’ notice, at a redemption price equal to \$25,000 per share (equivalent to \$25 per depositary share), plus any declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of Series D Preferred Stock have no right to require the redemption or repurchase of the Series D Preferred Stock.

If shares of the Series D Preferred Stock are to be redeemed, the notice of redemption shall be given by first class mail to the holders of record of the Series D Preferred Stock to be redeemed, mailed not less than 30 days nor more than 60 days prior to the date fixed for redemption thereof (*provided that*, if the depositary shares representing the Series D Preferred Stock are held in book-entry form through The Depository Trust Company, or “DTC,” the Company may give such notice in any manner permitted by the DTC). Each notice of redemption will include a statement setting forth: (i) the redemption date, (ii) the number of shares of the Series

D Preferred Stock to be redeemed and, if less than all the shares held by such holder are to be redeemed, the number of such shares to be redeemed from such holder, (iii) the redemption price and (iv) the place or places where holders may surrender certificates evidencing shares of Series D Preferred Stock for payment of the redemption price. If notice of redemption of any shares of Series D Preferred Stock has been given and if the funds necessary for such redemption have been set aside by the Company for the benefit of the holders of any shares of Series D Preferred Stock so called for redemption, then, from and after the redemption date, dividends will cease to accrue on such shares of Series D Preferred Stock, such shares of Series D Preferred Stock shall no longer be deemed outstanding and all rights of the holders of such shares will terminate, except the right to receive the redemption price.

In case of any redemption of only part of the shares of the Series D Preferred Stock at the time outstanding, the shares to be redeemed shall be selected either *pro rata* or in such other manner as the Company may determine to be fair and equitable.

See “Description of Depositary Shares” below for information about redemption of the depositary shares relating to the Company’s Series D Preferred Stock.

Regulatory Changes Relating to Capital Adequacy

The Company was previously regulated by the SEC as a consolidated supervised entity (“CSE”) pursuant to the SEC’s rules relating to CSEs (referred to as the “CSE Rules”). The Company treated the Series D Preferred Stock as allowable capital in accordance with the CSE Rules (such capital is referred to below as “Allowable Capital”).

If the regulatory capital requirements that apply to the Company change in the future, the Series D Preferred Stock may be converted, at the Company’s option and without consent of the holders, into a new series of preferred stock, subject to the limitations described below. The Company will be entitled to exercise this conversion right as follows.

If both of the following occur:

- the Company (by election or otherwise) becomes subject to any law, rule, regulation or guidance (together, “regulations”) relating to the Company’s capital adequacy, which regulation (i) modifies the existing requirements for treatment as Allowable Capital, (ii) provides for a type or level of capital characterized as “Tier 1” or its equivalent pursuant to regulations of any governmental body having jurisdiction over the Company (or any of the Company’s subsidiaries or consolidated affiliates) and implementing capital standards published by the Basel Committee on Banking Supervision, the SEC, the Federal Reserve Board or any other United States national governmental body, or any other applicable regime based on capital standards published by the Basel Committee on Banking Supervision or its successor, or (iii) provides for a type of capital that in the Company’s judgment (after consultation with counsel of recognized standing) is substantially equivalent to such “Tier 1” capital (such capital described in either (ii) or (iii) is referred to below as “Tier 1 Capital Equivalent”), and
- the Company affirmatively elects to qualify the Series D Preferred Stock for such Allowable Capital or Tier 1 Capital Equivalent treatment without any sublimit or other quantitative restriction on the inclusion of the Series D Preferred Stock in Allowable Capital or Tier 1 Capital Equivalent (other than any limitation the Company elects to accept and any limitation requiring that common equity or a specified form of common equity constitute the dominant form of Allowable Capital or Tier 1 Capital Equivalent) under such regulations,

then, upon such affirmative election, the Series D Preferred Stock shall be convertible at the Company’s option into a new series of preferred stock having terms and provisions substantially identical to those of the Series D Preferred Stock, *except* that such new series may have such additional or modified rights, preferences, privileges and voting powers, and such limitations and restrictions thereof, as are necessary, in the Company’s judgment (after consultation with counsel of recognized standing), to comply with the Required Unrestricted Capital Provisions (defined below), *provided* that the Company will not cause any such conversion unless the

Company determines that the rights, preferences, privileges and voting powers of such new series of preferred stock, taken as a whole, are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Series D Preferred Stock, taken as a whole. For example, the Company could agree to restrict its ability to pay dividends on or redeem the new series of preferred stock for a specified period or indefinitely, to the extent permitted by the terms and provisions of the new series of preferred stock, since such a restriction would be permitted in the Company's discretion under the terms and provisions of the Series D Preferred Stock.

The Company will provide notice to holders of the Series D Preferred Stock of any election to qualify the Series D Preferred Stock for Allowable Capital or Tier 1 Capital Equivalent treatment and of any determination to convert the Series D Preferred Stock into a new series of preferred stock, promptly upon the effectiveness of any such election or determination. A copy of any such notice and of the relevant regulations will be on file at the Company's principal offices and, upon request, will be made available to any stockholder.

As used above, the term "Required Unrestricted Capital Provisions" means the terms that are, in the Company's judgment (after consultation with counsel of recognized standing), required for preferred stock to be treated as Allowable Capital or Tier 1 Capital Equivalent, as applicable, without any sublimit or other quantitative restriction on the inclusion of such preferred stock in Allowable Capital or Tier 1 Capital Equivalent (other than any limitation the Company elects to accept and any limitation requiring that common equity or a specified form of common equity constitute the dominant form of Allowable Capital or Tier 1 Capital Equivalent) pursuant to applicable regulations.

Voting Rights

Except as provided below, the holders of the Series D Preferred Stock have no voting rights.

Whenever dividends on any shares of the Series D Preferred Stock shall have not been declared and paid for the equivalent of six or more dividend payments, whether or not for consecutive dividend periods (as used in this section, a "Nonpayment"), the holders of such shares, voting together as a class with holders of any and all other series of voting preferred stock (as defined below) then outstanding, will be entitled to vote for the election of a total of two additional members of the Company's board of directors (as used in this section, the "Preferred Stock Directors"), *provided* that the election of any such directors shall not cause the Company to violate the corporate governance requirement of the New York Stock Exchange (or any other exchange on which the Company's securities may be listed) that listed companies must have a majority of independent directors and *provided further* that the Company's board of directors shall at no time include more than two Preferred Stock Directors. In that event, the number of directors on the Company's board of directors shall automatically increase by two, and the new directors shall be elected at a special meeting called at the request of the holders of record of at least 20% of the Series D Preferred Stock or of any other series of voting preferred stock (unless such request is received less than 90 days before the date fixed for the next annual or special meeting of the stockholders, in which event such election shall be held at such next annual or special meeting of stockholders), and at each subsequent annual meeting. These voting rights will continue until dividends on the shares of the Series D Preferred Stock and any such series of voting preferred stock for at least four dividend periods, whether or not consecutive, following the Nonpayment shall have been fully paid (or declared and a sum sufficient for the payment of such dividends shall have been set aside for payment).

As used in this description of the Series D Preferred Stock, "voting preferred stock" means any other class or series of preferred stock of the Company ranking equally with the Series D Preferred Stock either as to dividends or the distribution of assets upon liquidation, dissolution or winding up and upon which like voting rights have been conferred and are exercisable. Whether a plurality, majority or other portion of the shares of Series D Preferred Stock and any other voting preferred stock have been voted in favor of any matter shall be determined by reference to the liquidation amounts of the shares voted.

If and when dividends for at least four dividend periods, whether or not consecutive, following a Nonpayment have been paid in full (or declared and a sum sufficient for such payment shall have been set aside), the

holders of the Series D Preferred Stock shall be divested of the foregoing voting rights (subject to revesting in the event of each subsequent Nonpayment) and, if such voting rights for all other holders of voting preferred stock have terminated, the term of office of each Preferred Stock Director so elected shall terminate and the number of directors on the board of directors shall automatically decrease by two. In determining whether dividends have been paid for four dividend periods following a Nonpayment, the Company may take account of any dividend the Company elects to pay for such a dividend period after the regular dividend date for that period has passed. Any Preferred Stock Director may be removed at any time without cause by the holders of record of a majority of the outstanding shares of the Series D Preferred Stock when they have the voting rights described above (voting together as a class with all series of voting preferred stock then outstanding). So long as a Nonpayment shall continue, any vacancy in the office of a Preferred Stock Director (other than prior to the initial election after a Nonpayment) may be filled by the written consent of the Preferred Stock Director remaining in office, or if none remains in office, by a vote of the holders of record of a majority of the outstanding shares of Series D Preferred Stock and all voting preferred stock when they have the voting rights described above (voting together as a class). The Preferred Stock Directors shall each be entitled to one vote per director on any matter.

So long as any shares of Series D Preferred Stock remain outstanding, the Company will not, without the affirmative vote or consent of the holders of at least two-thirds of the outstanding shares of the Series D Preferred Stock and all other series of voting preferred stock entitled to vote thereon, voting together as a single class, given in person or by proxy, either in writing or at a meeting:

- amend or alter the provisions of the Company's restated certificate of incorporation so as to authorize or create, or increase the authorized amount of, any class or series of stock ranking senior to the Series D Preferred Stock with respect to payment of dividends or the distribution of assets upon liquidation, dissolution or winding up of the Company;
- amend, alter or repeal the provisions of the Company's restated certificate of incorporation so as to materially and adversely affect the special rights, preferences, privileges and voting powers of the Series D Preferred Stock, taken as a whole; or
- consummate a binding share exchange or reclassification involving the Series D Preferred Stock or a merger or consolidation of the Company with another entity, unless in each case (i) the shares of Series D Preferred Stock remain outstanding or, in the case of any such merger or consolidation with respect to which the Company is not the surviving or resulting entity, are converted into or exchanged for preference securities of the surviving or resulting entity or its ultimate parent, and (ii) such shares remaining outstanding or such preference securities, as the case may be, have such rights, preferences, privileges and voting powers, taken as a whole, as are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Series D Preferred Stock, taken as a whole;

provided, however, that any increase in the amount of the authorized or issued Series D Preferred Stock or authorized preferred stock or the creation and issuance, or an increase in the authorized or issued amount, of other series of preferred stock ranking equally with and/or junior to the Series D Preferred Stock with respect to the payment of dividends (whether such dividends are cumulative or non-cumulative) and/or the distribution of assets upon liquidation, dissolution or winding up of the Company will not be deemed to adversely affect the rights, preferences, privileges or voting powers of the Series D Preferred Stock. In addition, any conversion of the Series D Preferred Stock upon the occurrence of certain regulatory events, as discussed above under "— Regulatory Changes Relating to Capital Adequacy," will not be deemed to adversely affect the rights, preferences, privileges or voting powers of the Series D Preferred Stock.

If an amendment, alteration, repeal, share exchange, reclassification, merger or consolidation described above would adversely affect one or more but not all series of voting preferred stock (including the Series D Preferred Stock for this purpose), then only the series affected and entitled to vote shall vote as a class in lieu of all such series of preferred stock.

Without the consent of the holders of the Series D Preferred Stock, so long as such action does not adversely affect the rights, preferences, privileges and voting powers of the Series D Preferred Stock, the Company may amend, alter, supplement or repeal any terms of the Series D Preferred Stock:

- to cure any ambiguity, or to cure, correct or supplement any provision contained in the certificate of designation for the Series D Preferred Stock that may be defective or inconsistent; or
- to make any provision with respect to matters or questions arising with respect to the Series D Preferred Stock that is not inconsistent with the provisions of the certificate of designations.

The foregoing voting provisions will not apply if, at or prior to the time when the act with respect to which such vote would otherwise be required shall be effected, all outstanding shares of Series D Preferred Stock shall have been redeemed or called for redemption upon proper notice and sufficient funds shall have been set aside by the Company for the benefit of the holders of the Series D Preferred Stock to effect such redemption.

DESCRIPTION OF DEPOSITARY SHARES

Please note that as used in this section, references to “holders” of depositary shares mean those who own depositary shares registered in their own names, on the books that the Company or the depositary maintain for this purpose, and not indirect holders who own beneficial interests in depositary shares registered in street name or issued in book-entry form through The Depository Trust Company.

General

The Company has issued fractional interests in shares of preferred stock in the form of depositary shares, each representing a 1/1,000th ownership interest in a share of Series D Preferred Stock and evidenced by a depositary receipt. The shares of Series D Preferred Stock represented by depositary shares are deposited under a deposit agreement among the Company, the depositary and the holders from time to time of the depositary receipts evidencing the depositary shares. Subject to the terms of the deposit agreement, each holder of a depositary share is entitled, through the depositary, in proportion to the applicable fraction of a share of Series D Preferred Stock represented by such depositary share, to all the rights and preferences of the Series D Preferred Stock represented thereby (including dividend, voting, redemption and liquidation rights).

Dividends and Other Distributions

The depositary will distribute any cash dividends or other cash distributions received in respect of the deposited Series D Preferred Stock to the record holders of depositary shares relating to the underlying Series D Preferred Stock in proportion to the number of depositary shares held by the holders. The depositary will distribute any property received by it other than cash to the record holders of depositary shares entitled to those distributions, unless it determines that the distribution cannot be made proportionally among those holders or that it is not feasible to make a distribution. In that event, the depositary may, with the Company’s approval, sell the property and distribute the net proceeds from the sale to the holders of the depositary shares in proportion to the number of depositary shares they hold.

Record dates for the payment of dividends and other matters relating to the depositary shares will be the same as the corresponding record dates for the Series D Preferred Stock.

The amounts distributed to holders of depositary shares will be reduced by any amounts required to be withheld by the depositary or by the Company on account of taxes or other governmental charges.

Redemption of Depositary Shares

If the Company redeems the Series D Preferred Stock represented by the depositary shares, the depositary shares will be redeemed from the proceeds received by the depositary resulting from the redemption of the

Series D Preferred Stock held by the depositary. The redemption price per depositary share will be equal to 1/1,000th of the redemption price per share payable with respect to the Series D Preferred Stock (or \$25 per depositary share). Whenever the Company redeems shares of Series D Preferred Stock held by the depositary, the depositary will redeem, as of the same redemption date, the number of depositary shares representing shares of Series D Preferred Stock so redeemed.

In case of any redemption of less than all of the outstanding depositary shares, the depositary shares to be redeemed will be selected by the depositary *pro rata* or in such other manner determined by the depositary to be equitable. In any such case, the Company will redeem depositary shares only in increments of 1,000 shares and any multiple thereof.

Voting the Series D Preferred Stock

When the depositary receives notice of any meeting at which the holders of the Series D Preferred Stock are entitled to vote, the depositary will mail the information contained in the notice to the record holders of the depositary shares relating to the Series D Preferred Stock. Each record holder of the depositary shares on the record date, which will be the same date as the record date for the Series D Preferred Stock, may instruct the depositary to vote the amount of the Series D Preferred Stock represented by the holder's depositary shares. To the extent possible, the depositary will vote the amount of the Series D Preferred Stock represented by depositary shares in accordance with the instructions it receives. The Company will agree to take all reasonable actions that the depositary determines are necessary to enable the depositary to vote as instructed. If the depositary does not receive specific instructions from the holders of any depositary shares representing the Series D Preferred Stock, it will vote all depositary shares of that series held by it proportionately with instructions received.

Listing

The depositary shares are listed on the New York Stock Exchange under the ticker symbol "GS PrD."

Form of Preferred Stock and Depositary Shares

The depositary shares are issued in book-entry form through The Depository Trust Company. The Series D Preferred Stock is issued in registered form to the depositary.

DESCRIPTION OF (i) 5.793% FIXED-TO-FLOATING RATE NORMAL AUTOMATIC PREFERRED ENHANCED CAPITAL SECURITIES OF GOLDMAN SACHS CAPITAL II (FULLY AND UNCONDITIONALLY GUARANTEED BY THE GOLDMAN SACHS GROUP, INC.) AND (ii) FLOATING RATE NORMAL AUTOMATIC PREFERRED ENHANCED CAPITAL SECURITIES OF GOLDMAN SACHS CAPITAL III (FULLY AND UNCONDITIONALLY GUARANTEED BY THE GOLDMAN SACHS GROUP, INC.)

The following is a brief description of the terms of the 5.793% Fixed-to-Floating Rate Normal Automatic Preferred Enhanced Capital Securities of Goldman Sachs Capital II and the Floating Rate Normal Automatic Preferred Enhanced Capital Securities of Goldman Sachs Capital III (the "APEX") and of the Trust Agreements (both as defined below) under which they are issued. It does not purport to be complete. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

The APEX and the Common Securities of Goldman Sachs Capital II ("Capital II") and Goldman Sachs Capital III ("Capital III"), each a Delaware statutory trust (a "Trust"), represent beneficial interests in the relevant Trust. The Trust in respect of Capital II holds the Company's Perpetual Non-Cumulative Preferred Stock, Series E (the "Series E Preferred Stock"), and the Trust in respect of Capital III holds the Company's Perpetual Non-Cumulative Preferred Stock, Series F (the "Series F Preferred Stock," and collectively with the Series E Preferred Stock, the "Preferred").

Each holder of APEX has a beneficial interest in the relevant Trust but does not own any specific shares of the Preferred held by that Trust. However, the applicable trust agreement among the Company, The Bank of New York Mellon, BNY Mellon Trust of Delaware, the administrative trustees and the several holders of the relevant Trust securities (each, a "Trust Agreement") under which each Trust operates defines the financial entitlements of its APEX in a manner that causes those financial entitlements to correspond to the financial entitlements of that Trust in the Preferred it holds. Accordingly, each APEX of Capital II corresponds to 1/100th of a share of Series E Preferred Stock held by Capital II and each APEX of Capital III corresponds to 1/100th of a share of Series F Preferred Stock held by Capital III.

The Trusts

Each Trust is a statutory trust organized under Delaware law pursuant to a Trust Agreement and the filing of a certificate of trust with the Delaware Secretary of State.

The Trusts are used solely for the following purposes:

- issuing the APEX and the Common Securities;
- holding shares of the Preferred; and
- engaging in other activities that are directly related to the activities described above.

The Company owns all of the Common Securities, either directly or indirectly. The Common Securities rank equally with the APEX and the Trusts make payment on their Trust securities *pro rata*, except that if the Company pays less than the full dividend on or redemption price of the Preferred, the rights of the holders of the Common Securities to payment in respect of distributions and payments upon liquidation, redemption and otherwise are subordinated to the rights of the holders of the APEX.

Each Trust is perpetual, but may be dissolved earlier as provided in its Trust Agreement.

The Company pays all fees and expenses related to the Trusts.

DESCRIPTION OF THE APEX

General

The APEX are securities of each Trust and are issued pursuant to the applicable Trust Agreement. The Property Trustee, The Bank of New York Mellon, acts as indenture trustee for the APEX under the Trust Agreement for purposes of compliance with the provisions of the Trust Indenture Act. Each APEX has a liquidation amount of \$1,000.

The terms of the APEX of each Trust include those stated in the Trust Agreement for such Trust, including any amendments thereto and those made part of the Trust Agreement by the Trust Indenture Act and the Delaware Statutory Trust Act.

In addition to the APEX, each Trust Agreement authorizes the administrative trustees of the Trust to issue the Common Securities on behalf of the Trust. The Company owns, directly or indirectly, all of the Common Securities. The Common Securities rank on a parity, and payments upon redemption, liquidation or otherwise are made on a proportionate basis, with the APEX except as set forth below under “— Ranking of Common Securities.” The Trust Agreements do not permit the Trust to issue any securities other than the Common Securities and the APEX or to incur any indebtedness.

Under the Trust Agreement, the Property Trustee on behalf of the relevant Trust holds the Preferred for the benefit of the holders of its APEX and Common Securities.

The payment of distributions out of money held by a Trust, and payments upon redemption of the APEX or liquidation of the Trust, are guaranteed by the Company to the extent described under “Description of the Guarantees.” Each Guarantee, when taken together with the Company’s obligations under the applicable Trust Agreement, including its obligations to pay costs, expenses, debts and liabilities of the Trust, other than with respect to its Common Securities and APEX, has the effect of providing a full and unconditional guarantee of amounts due on the APEX. The Bank of New York Mellon, as the Guarantee Trustee, holds each Guarantee for the benefit of the holders of the APEX. The Guarantees do not cover payment of distributions when the Trusts do not have sufficient available funds to pay those distributions.

When the term “holder” is used in this description of the APEX with respect to a registered APEX, it means the person in whose name such APEX is registered in the security register. The APEX are currently held in book-entry form only, and are held in the name of DTC or its nominee.

Capital II’s APEX are listed on the New York Stock Exchange under the symbol “GS/PE” and Capital III’s APEX are listed on the New York Stock Exchange under the symbol “GS/PF.”

The financial entitlements as a holder of APEX generally correspond to the applicable Trust’s financial entitlements as a holder of the Preferred. The corresponding asset for each APEX is a 1/100th, or \$1,000, interest in one share of Preferred held by the Trust. Each Trust will pass through to the holder amounts that it receives on the corresponding assets for the APEX as distributions on, or the liquidation preference of, APEX. Holders of a Trust’s APEX are entitled to receive distributions corresponding to non-cumulative dividends on the Preferred held by the Trust. These cash dividends are payable if, as and when declared by the Company’s board of directors, on the Dividend Payment Dates (as defined below), which are: quarterly in arrears on each March 1, June 1, September 1 and December 1 (or if such day is not a business day, the next business day).

Assuming that the Company does not elect to pay partial dividends or to skip dividends on the Preferred, holders of APEX will receive distributions on the \$1,000 liquidation amount per APEX at a rate per annum equal to the greater of (x) three-month LIBOR for the related distribution period plus 0.765% (in the case of Capital II’s APEX) or 0.77% (in the case of Capital III’s APEX) and (y) 4.000%, payable quarterly on each March 1, June 1, September 1 and December 1 (or if any such date is not a business day, on the next business day).

Dividends are calculated on the basis of a 360-day year and the number of days actually elapsed in the dividend period. Distributions on the APEX and dividends on the Preferred are non-cumulative.

The Bank of New York Mellon acts as registrar and transfer agent, or “Transfer Agent,” for the APEX. If The Bank of New York Mellon should resign or be removed, the Company or the Trust will designate a successor and the term “Transfer Agent” as used herein will refer to that successor. A “business day” as used in this section means any day other than a Saturday, Sunday or any other day on which banking institutions and trust companies in New York, New York or Wilmington, Delaware are permitted or required by any applicable law to close.

Each Trust must make distributions on its APEX on each distribution date to the extent that it has funds available therefor. A Trust’s funds available for distribution to a holder of its APEX will be limited to payments received from the Company on the Preferred held by the Trust. The Company guarantees the payment of distributions on the APEX out of moneys held by each Trust to the extent of available Trust funds, as described under “Description of the Guarantees” below.

Distributions on the APEX are payable to holders as they appear in the security register of the Trust on the relevant record dates. The record dates are the fifteenth calendar day immediately preceding the next succeeding distribution date. Distributions are paid through the Property Trustee or paying agent, who hold amounts received in respect of the Preferred for the benefit of the holders of the APEX.

For more information about dividends on the Preferred, see “— Dividends” below.

Agreed Tax Treatment of the APEX

As a beneficial owner of APEX, by acceptance of the beneficial interest therein, the holder will be deemed to have agreed, for all U.S. federal income tax purposes:

- to treat the holder as the owner of a 1/100th interest in a share of the relevant Preferred; and
- to treat the Trust as one or more grantor trusts or agency arrangements.

Mandatory Redemption of APEX upon Redemption of the Preferred

The APEX have no stated maturity but must be redeemed on the date the Company redeems the Preferred, and the Property Trustee or paying agent will apply the proceeds from such repayment or redemption to redeem a like amount, as defined below, of the APEX. The Preferred is perpetual but the Company may redeem it on any Dividend Payment Date, subject to certain limitations. See “— Redemption” below. The redemption price per APEX will equal the redemption price of the Preferred. See “— Redemption” below. If notice of redemption of any Preferred has been given and if the funds necessary for the redemption have been set aside by the Company for the benefit of the holders of any shares of the Preferred so called for redemption, then, from and after the redemption date, those shares shall no longer be deemed outstanding and all rights of the holders of those shares (including the right to receive any dividends) will terminate, except the right to receive the redemption price.

If less than all of the shares of the Preferred held by the Trust are to be redeemed on a redemption date, then the proceeds from such redemption will be allocated *pro rata* to the redemption of the APEX and the Common Securities, except as set forth below under “— Ranking of Common Securities.”

The term “*like amount*” as used above means APEX having a liquidation amount equal to that portion of the liquidation amount of the Preferred to be contemporaneously redeemed, the proceeds of which will be used to pay the redemption price of such APEX.

Distributions to be paid on or before the redemption date for any APEX called for redemption will be payable to the holders as of the record dates for the related dates of distribution. If the APEX called for redemption are no

longer in book-entry form, the Property Trustee, to the extent funds are available, will irrevocably deposit with the paying agent for the APEX funds sufficient to pay the applicable redemption price and will give such paying agent irrevocable instructions and authority to pay the redemption price to the holders thereof upon surrender of their certificates evidencing the APEX.

If notice of redemption shall have been given and funds deposited as required, then upon the date of such deposit:

- all rights of the holders of such APEX called for redemption will cease, except the right of the holders of such APEX to receive the redemption price and any distribution payable in respect of the APEX on or prior to the redemption date, but without interest on such redemption price; and
- the APEX called for redemption will cease to be outstanding. If any redemption date is not a business day, then the redemption amount will be payable on the next business day (and without any interest or other payment in respect of any such delay). However, if payment on the next business day causes payment of the redemption amount to be in the next calendar month, then payment will be on the preceding business day.

If payment of the redemption amount for the Preferred held by a Trust called for redemption is improperly withheld or refused and accordingly the redemption amount of the Trust's APEX is not paid either by the Trust or by the Company under the applicable Guarantee, then dividends on the Preferred called for redemption will continue to accrue and distributions on such series of APEX called for redemption will continue to accumulate at the applicable rate then borne by such APEX from the original redemption date scheduled to the actual date of payment. In this case, the actual payment date will be considered the redemption date for purposes of calculating the redemption amount.

Redemptions of the APEX will require prior approval of the Federal Reserve Board.

The Company will not exercise its option to redeem any shares of the Preferred without obtaining the approval of the Federal Reserve Board (or any successor appropriate federal banking agency) as required by applicable law. Unless the Federal Reserve Board (or any successor appropriate federal banking agency) authorizes the Company to do otherwise in writing, the Company will redeem the Preferred only if it is replaced with other tier 1 capital that is not a restricted core capital element (e.g., common stock or another series of noncumulative perpetual preferred stock).

If less than all of the outstanding shares of the Preferred held by a Trust are to be redeemed on a redemption date, then the aggregate liquidation amount of APEX and Common Securities of that Trust to be redeemed shall be allocated *pro rata* to the APEX and Common Securities based upon the relative liquidation amounts of such series, except as set forth below under "— Ranking of Common Securities." The Property Trustee will select the particular APEX to be redeemed on a *pro rata* basis not more than 60 days before the redemption date from the outstanding APEX not previously called for redemption by any method the Property Trustee deems fair and appropriate, or if the APEX are in book-entry only form, in accordance with the procedures of DTC. The Property Trustee shall promptly notify the Transfer Agent in writing of the APEX selected for redemption and, in the case of any APEX selected for redemption in part, the liquidation amount to be redeemed.

For all purposes of the Trust Agreement, unless the context otherwise requires, all provisions relating to the redemption of APEX shall relate, in the case of any APEX redeemed or to be redeemed only in part, to the portion of the aggregate liquidation amount of APEX that has been or is to be redeemed. If less than all of the APEX, the APEX held through the facilities of DTC will be redeemed *pro rata* in accordance with DTC's internal procedures.

Liquidation Distribution upon Dissolution

Pursuant to each Trust Agreement, the applicable Trust shall dissolve on the first to occur of:

- certain events of bankruptcy, dissolution or liquidation of the Company;
- redemption of all of its APEX as described above; and
- the entry of an order for the dissolution of the Trust by a court of competent jurisdiction.

Except as set forth in the next paragraph, if an early dissolution occurs as a result of certain events of bankruptcy, dissolution or liquidation of the Company, the Property Trustee and the administrative trustees will liquidate the Trust as expeditiously as they determine possible by distributing, after satisfaction of liabilities to creditors of the Trust as provided by applicable law, to each holder of its APEX a like amount of the Preferred held by the Trust as of the date of such distribution. Except as set forth in the next paragraph, if an early dissolution occurs as a result of the entry of an order for the dissolution of the Trust by a court of competent jurisdiction, the Property Trustee will liquidate the Trust as expeditiously as it determines to be possible by distributing, after satisfaction of liabilities to creditors of the Trust as provided by applicable law, to each holder of its APEX a like amount of the Preferred held by the Trust as of the date of such distribution. The Property Trustee shall give notice of liquidation to each holder of APEX at least 30 days and not more than 60 days before the date of liquidation.

If, whether because of an order for dissolution entered by a court of competent jurisdiction or otherwise, the Property Trustee determines that distribution of the Preferred in the manner provided above is not possible, or if the early dissolution occurs as a result of the redemption of all the APEX, the Property Trustee shall liquidate the property of the Trust and wind up its affairs. In that case, upon the winding up of the Trust, except with respect to an early dissolution that occurs as a result of the redemption of all the APEX, the holders will be entitled to receive out of the assets of the Trust available for distribution to holders and after satisfaction of liabilities to creditors of the Trust as provided by applicable law, an amount equal to the aggregate liquidation amount per Trust security plus accrued and unpaid distributions to the date of payment. If, upon any such winding up, the Trust has insufficient assets available to pay in full such aggregate liquidation distribution, then the amounts payable directly by the Trust on its Trust securities shall be paid on a *pro rata* basis, except as set forth below under “— Ranking of Common Securities.”

The term “like amount” as used above means Preferred having a liquidation preference equal to the liquidation amount of the APEX of the holder to whom such Preferred would be distributed.

Distribution of Trust Assets

Upon liquidation of a Trust other than as a result of an early dissolution upon the redemption of all the APEX and after satisfaction of the liabilities of creditors of the Trust as provided by applicable law, the assets of the Trust will be distributed to the holders of such Trust securities in exchange therefor.

After the liquidation date fixed for any distribution of assets of the Trust:

- the APEX will no longer be deemed to be outstanding;
- DTC or its nominee, as the record holder of the APEX, will receive a registered global certificate or certificates representing the Preferred to be delivered upon such distribution;
- any certificates representing the APEX not held by DTC or its nominee will be deemed to represent shares of the Preferred having a Liquidation Preference equal to the APEX until such certificates are so surrendered for transfer and reissuance; and
- all rights of the holders of the APEX will cease, except the right to receive Preferred upon such surrender.

Since each APEX corresponds to 1/100th of a share of Preferred, holders of APEX may receive fractional shares of the Preferred or depositary shares representing the Preferred upon this distribution.

Ranking of Common Securities

If on any distribution date a Trust does not have funds available from dividends on the Preferred it holds to make full distributions on its APEX and Common Securities, then the available funds from dividends on the Preferred it holds shall be applied first to make distributions then due on its APEX on a *pro rata* basis on such distribution date up to the amount of such distributions corresponding to dividends on the Preferred (or if less, the amount of the corresponding distributions that would have been made on the APEX had the Company paid a full dividend on the Preferred) before any such amount is applied to make a distribution on the Trust's Common Securities on such distribution date.

If on any date where APEX and Common Securities must be redeemed because the Company is redeeming Preferred and a Trust does not have funds available from the Company's redemption of the Preferred it holds to pay the full redemption price then due on all of its outstanding APEX and Common Securities to be redeemed, then (i) the available funds shall be applied first to pay the redemption price on the APEX to be redeemed on such redemption date and (ii) Common Securities shall be redeemed only to the extent funds are available for such purpose after the payment of the full redemption price on the APEX to be redeemed.

If an early dissolution event occurs in respect of a Trust, no liquidation distributions shall be made on its Common Securities until full liquidation distributions have been made on its APEX.

In the case of any event of default under the Trust Agreement of a Trust resulting from the Company's failure to comply in any material respect with any of its obligations as issuer of the Preferred held by the Trust, including obligations set forth in its restated certificate of incorporation, as amended, or "restated certificate of incorporation," or arising under applicable law, the Company, as holder of its Common Securities, will be deemed to have waived any right to act with respect to any such event of default under the Trust Agreement until the effect of all such events of default with respect to its APEX have been cured, waived or otherwise eliminated. Until all events of default under the Trust Agreement have been so cured, waived or otherwise eliminated, the Property Trustee shall act solely on behalf of the holders of its APEX and not on the Company's behalf, and only the holders of its APEX will have the right to direct the Property Trustee to act on their behalf.

Events of Default; Notice

Any one of the following events constitutes an event of default under a Trust Agreement, or a "Trust Event of Default," regardless of the reason for such event of default and whether it shall be voluntary or involuntary or be effected by operation of law or pursuant to any judgment, decree or order of any court or any order, rule or regulation of any administrative or governmental body:

- the failure to comply in any material respect with the Company's obligations as issuer of the Preferred, under its restated certificate of incorporation, or those of the Trust, or arising under applicable law;
- the default by the Trust in the payment of any distribution on any Trust security of the Trust when such becomes due and payable, and continuation of such default for a period of 30 days;
- the default by the Trust in the payment of any redemption price of any Trust security of the Trust when such becomes due and payable;
- the failure to perform or the breach, in any material respect, of any other covenant or warranty of the trustees in the Trust Agreement for 90 days after the defaulting trustee or trustees have received written notice of the failure to perform or breach in the manner specified in such Trust Agreement; or
- the occurrence of certain events of bankruptcy or insolvency with respect to the Property Trustee and the Company's failure to appoint a successor Property Trustee within 90 days.

Within 30 days after any Trust Event of Default with respect to a Trust actually known to the Property Trustee occurs, the Property Trustee will transmit notice of such Trust Event of Default to the holders of its APEX and to the administrative trustees, unless such Trust Event of Default shall have been cured or waived. The Company, as sponsor, and the administrative trustees are required to file annually with the Property Trustee a certificate as to whether or not the Company or the administrative trustees are in compliance with all the conditions and covenants applicable to the Company and to the administrative trustees under the Trust Agreement.

Mergers, Consolidations, Amalgamations or Replacements of a Trust

A Trust may not merge with or into, consolidate, amalgamate, or be replaced by, or convey, transfer or lease its properties and assets substantially as an entirety to the Company or any other person, except as described below or as otherwise described in its Trust Agreement. A Trust may, at the Company's request, with the consent of the administrative trustees but without the consent of the holders of its APEX, the Property Trustee or the Delaware Trustee, merge with or into, consolidate, amalgamate, or be replaced by, or convey, transfer or lease its properties and assets substantially as an entirety to, a trust organized as such under the laws of any state if:

- such successor entity either:
 - expressly assumes all of the obligations of the Trust with respect to its APEX, or
 - substitutes for its APEX other securities having substantially the same terms as its APEX, or the "Successor Securities," so long as the Successor Securities rank the same as its APEX in priority with respect to distributions and payments upon liquidation, redemption and otherwise;
- a trustee of such successor entity possessing the same powers and duties as the Property Trustee is appointed to hold the Preferred then held by or on behalf of the Property Trustee;
- such merger, consolidation, amalgamation, replacement, conveyance, transfer or lease does not cause its APEX, including any Successor Securities, to be downgraded by any nationally recognized statistical rating organization;
- such merger, consolidation, amalgamation, replacement, conveyance, transfer or lease does not adversely affect the rights, preferences and privileges of the holders of its APEX, including any Successor Securities, in any material respect;
- such successor entity has purposes substantially identical to those of the Trust;
- prior to such merger, consolidation, amalgamation, replacement, conveyance, transfer or lease, the Property Trustee has received an opinion from counsel to the Trust experienced in such matters to the effect that:
 - such merger, consolidation, amalgamation, replacement, conveyance, transfer or lease does not adversely affect the rights, preferences and privileges of the holders of its APEX, including any Successor Securities, in any material respect, and
 - following such merger, consolidation, amalgamation, replacement, conveyance, transfer or lease, neither the Trust nor such successor entity will be required to register as an investment company under the Investment Company Act of 1940, or "Investment Company Act";
- the Trust has received an opinion of counsel experienced in such matters that such merger, consolidation, amalgamation, conveyance, transfer or lease will not cause the Trust or the successor entity to be classified as an association or a publicly traded partnership taxable as a corporation for U.S. federal income tax purposes; and

- the Company or any permitted successor or assignee owns all of the common securities of such successor entity and guarantees the obligations of such successor entity under the Successor Securities at least to the extent provided by the Guarantee.

Notwithstanding the foregoing, a Trust may not, except with the consent of holders of 100% in liquidation amount of its APEX, consolidate, amalgamate, merge with or into, or be replaced by or convey, transfer or lease its properties and assets substantially as an entirety to any other entity or permit any other entity to consolidate, amalgamate, merge with or into, or replace it if such consolidation, amalgamation, merger, replacement, conveyance, transfer or lease would cause the Trust or the successor entity to be classified as other than one or more grantor trusts or agency arrangements or to be classified as an association or a publicly traded partnership taxable as a corporation for U.S. federal income tax purposes.

Voting Rights; Amendment of a Trust Agreement

Except as provided herein and under “— Amendments and Assignment” below and as otherwise required by law and the Trust Agreement, the holders of a Trust’s APEX have no voting rights or control over the administration, operation or management of the Trust or the obligations of the parties to its Trust Agreement, including in respect of the Preferred held by the Trust. Under the Trust Agreement, however, the Property Trustee is required to obtain their consent before exercising some of its rights in respect of these securities.

Trust Agreement. The Company and the administrative trustees may amend a Trust’s Trust Agreement without the consent of the holders of its APEX, the Property Trustee or the Delaware Trustee, unless in the case of the first two bullets below such amendment will materially and adversely affect the interests of any holder of APEX or the Property Trustee or the Delaware Trustee or impose any additional duty or obligation on the Property Trustee or the Delaware Trustee, to:

- cure any ambiguity, correct or supplement any provisions in the Trust Agreement that may be inconsistent with any other provision, or to make any other provisions with respect to matters or questions arising under the Trust Agreement, which may not be inconsistent with the other provisions of the Trust Agreement;
- modify, eliminate or add to any provisions of the Trust Agreement to such extent as shall be necessary to ensure that the Trust will be classified for U.S. federal income tax purposes as one or more grantor trusts or agency arrangements and not as an association or a publicly traded partnership taxable as a corporation at all times that any Trust securities are outstanding, or to ensure that the Trust will not be required to register as an “investment company” under the Investment Company Act;
- provide that certificates for the APEX may be executed by an administrative trustee by facsimile signature instead of manual signature, in which case such amendment(s) shall also provide for the appointment by the Company of an authentication agent and certain related provisions;
- require that holders that are not U.S. persons for U.S. federal income tax purposes irrevocably appoint a U.S. person to exercise any voting rights to ensure that the Trust will not be treated as a foreign trust for U.S. federal income tax purposes; or
- conform the terms of the Trust Agreement to the description of the Trust Agreement, the APEX and the Common Securities in the prospectus dated December 5, 2006, of the Company and the Trusts, as supplemented by the prospectus supplement, dated May 8, 2007, in the manner provided in the Trust Agreement.

Any such amendment shall become effective when notice thereof is given to the Property Trustee, the Delaware Trustee and the holders of the APEX.

The Company and the administrative trustees may generally amend a Trust’s Trust Agreement with:

- the consent of holders representing not less than a majority, based upon liquidation amounts, of its APEX; and
- receipt by the administrative trustees of the Trust of an opinion of counsel to the effect that such amendment or the exercise of any power granted to the administrative trustees of the Trust or the administrative trustees in accordance with such amendment will not affect the Trust's status as one or more grantor trusts or agency arrangements for U.S. federal income tax purposes or affect the Trust's exemption from status as an "investment company" under the Investment Company Act.

However, without the consent of each affected holder of Trust securities, a Trust Agreement may not be amended to:

- change the amount or timing, or otherwise adversely affect the amount, of any distribution required to be made in respect of Trust securities as of a specified date; or
- restrict the right of a holder of Trust securities to institute a suit for the enforcement of any such payment on or after such date.

Preferred Stock. So long as Preferred is held by the Property Trustee on behalf of a Trust, the trustees of the Trust will not waive any rights in respect of the Preferred without obtaining the prior approval of the holders of at least a majority in liquidation amount of its APEX then outstanding. The trustees of the Trust shall also not consent to any amendment to the Trust's or the Company's governing documents that would change the dates on which dividends are payable or the amount of such dividends, without the prior written consent of each holder of APEX. In addition to obtaining the foregoing approvals from holders, the administrative trustees shall obtain, at the Company's expense, an opinion of counsel to the effect that such action shall not cause the Trust to be taxable as a corporation or classified as a partnership for U.S. federal income tax purposes.

General. Any required approval of holders of APEX may be given at a meeting of holders convened for such purpose or pursuant to written consent. The Property Trustee will cause a notice of any meeting at which holders are entitled to vote, or of any matter upon which action by written consent of such holders is to be taken, to be given to each record holder in the manner set forth in the Trust Agreement.

No vote or consent of the holders of APEX will be required for a Trust to redeem and cancel the APEX in accordance with a Trust Agreement.

Notwithstanding that holders of the APEX are entitled to vote or consent under any of the circumstances described above, any of the APEX that are owned by the Company or its affiliates or the trustees shall be deemed not to be outstanding.

Payment and Paying Agent

Payments on the APEX shall be made to DTC, which shall credit the relevant accounts on the applicable distribution dates. If any APEX are not held by DTC, such payments shall be made by check mailed to the address of the holder as such address shall appear on the register.

The paying agent is The Bank of New York Mellon and any co-paying agent chosen by the Property Trustee and acceptable to the Company and to the administrative trustees.

Registrar and Transfer Agent

The Bank of New York Mellon acts as registrar and transfer agent, or "Transfer Agent," for the APEX.

Information Concerning the Property Trustee

Other than during the occurrence and continuance of a Trust Event of Default, the Property Trustee undertakes to perform only the duties that are specifically set forth in the Trust Agreement. After a Trust Event of Default, the Property Trustee must exercise the same degree of care and skill as a prudent individual would exercise or use in the conduct of his or her own affairs. Subject to this provision, the Property Trustee is under no obligation to exercise any of the powers vested in it by the Trust Agreement at the request of any holder of APEX unless it is offered indemnity satisfactory to it by such holder against the costs, expenses and liabilities that might be incurred. If no Trust Event of Default has occurred and is continuing and the Property Trustee is required to decide between alternative courses of action, construe ambiguous provisions in the Trust Agreement or is unsure of the application of any provision of the Trust Agreement, and the matter is not one upon which holders of APEX are entitled under the Trust Agreement to vote, then the Property Trustee will take any action that the Company directs. If the Company does not provide direction, the Property Trustee may take any action that it deems advisable and in the interests of the holders of the Trust securities and will have no liability except for its own bad faith, negligence or willful misconduct.

The Company and its affiliates may maintain certain accounts and other banking relationships with the Property Trustee and its affiliates in the ordinary course of business.

Trust Expenses

Pursuant to each Trust Agreement, the Company, as sponsor, agrees to pay:

- all debts and other obligations of the Trust (other than with respect to its APEX);
- all costs and expenses of the Trust, including costs and expenses relating to the organization of the Trust, the fees, expenses and indemnities of the trustees and the cost and expenses relating to the operation of the Trust; and
- any and all taxes and costs and expenses with respect thereto, other than U.S. withholding taxes, to which the Trust might become subject.

Governing Law

Each Trust Agreement is governed by and construed in accordance with the laws of the State of Delaware.

Miscellaneous

The administrative trustees are authorized and directed to conduct the affairs of and to operate each Trust in such a way that it will not be required to register as an “investment company” under the Investment Company Act or characterized as other than one or more grantor trusts or agency arrangements for U.S. federal income tax purposes.

In this regard, the Company and the administrative trustees are authorized to take any action, not inconsistent with applicable law, the certificate of trust of a Trust or its Trust Agreement, that the Company and the administrative trustees determine to be necessary or desirable to achieve such end, as long as such action does not materially and adversely affect the interests of the holders of the APEX.

Holders of the APEX have no preemptive or similar rights. The APEX are not convertible into or exchangeable for the Company’s common stock or preferred stock.

Subject to any applicable rules of the Federal Reserve Board (or any successor appropriate federal banking agency), the Company or its affiliates may from time to time purchase any of the APEX that are then outstanding by tender, in the open market or by private agreement.

The Trust may not borrow money or issue debt or mortgage or pledge any of its assets.

DESCRIPTION OF THE GUARANTEES

The following is a brief description of the terms of the Guarantee (as defined below) pursuant to the Guarantee Agreement for Goldman Sachs Capital II (formerly known as Goldman Sachs Capital IV), dated as of March 23, 2016, and the Guarantee Agreement for Goldman Sachs Capital III (formerly known as Goldman Sachs Capital V), dated as of March 23, 2016 (collectively, the "Guarantee Agreements"). The description does not purport to be complete.

General

The following payments on each Trust's APEX, also referred to as the "guarantee payments," if not fully paid by the Trust, will be paid by the Company under a guarantee, or "Guarantee," that the Company has executed and delivered for the benefit of the holders of such APEX. Pursuant to each Guarantee, the Company irrevocably and unconditionally agrees to pay in full the guarantee payments, without duplication:

- any accumulated and unpaid distributions required to be paid on the APEX, to the extent the Trust has funds available to make the payment;
- the redemption price for any APEX called for redemption, to the extent the Trust has funds available to make the payment; and
- upon a voluntary or involuntary dissolution, winding up or liquidation of the Trust, other than in connection with a distribution of a like amount of corresponding assets to the holders of the APEX, the lesser of:
 - the aggregate of the liquidation amount and all accumulated and unpaid distributions on the APEX to the date of payment, to the extent the Trust has funds available to make the payment; and
 - the amount of assets of the Trust remaining available for distribution to holders of the APEX upon liquidation of the Trust.

The Company's obligation to make a guarantee payment may be satisfied by direct payment of the required amounts by the Company to the holders of the APEX or by causing the Trust to pay the amounts to the holders.

If the Company does not make a regular dividend payment on the Preferred held by a Trust, the Trust will not have sufficient funds to make the related payments on the relevant series of APEX. The Guarantee does not cover payments on the APEX when the Trust does not have sufficient funds to make these payments. Because the Company is a holding company, its rights to participate in the assets of any of its subsidiaries upon the subsidiary's liquidation or reorganization will be subject to the prior claims of the subsidiary's creditors except to the extent that the Company may itself be a creditor with recognized claims against the subsidiary. The Guarantee does not limit the incurrence or issuance by the Company of other secured or unsecured indebtedness.

Each Guarantee is qualified as an indenture under the Trust Indenture Act. The Bank of New York Mellon acts as "Guarantee Trustee" for each Guarantee for purposes of compliance with the provisions of the Trust Indenture Act. The Guarantee Trustee holds each Guarantee for the benefit of the holders of APEX.

Effect of the Guarantees

Each Guarantee, when taken together with the Company's obligations and the Trust's obligations under the Trust Agreement, including the obligations to pay costs, expenses, debts and liabilities of the applicable Trust, other than with respect to its Trust securities, has the effect of providing a full and unconditional guarantee, on a subordinated basis, of payments due on its APEX.

The Company has also agreed separately to irrevocably and unconditionally guarantee the obligations of each Trust with respect to its Common Securities to the same extent as the Guarantee.

Status of the Guarantees

Each Guarantee is unsecured and ranks:

- subordinate and junior in right of payment to all of the Company's senior and subordinated debt; and
- equally with any of the Company's other present or future obligations that by their terms rank *pari passu* with such Guarantee.

Each Guarantee constitutes a guarantee of payment and not of collection, which means that the guaranteed party may sue the guarantor to enforce its rights under the Guarantee without suing any other person or entity. Each Guarantee is held for the benefit of the holders of APEX. Each Guarantee will be discharged only by payment of the guarantee payments in full to the extent not paid by the applicable Trust.

Amendments and Assignment

A Guarantee may be amended only with the prior approval of the holders of not less than a majority in aggregate liquidation amount of the applicable outstanding APEX. No vote is required, however, for any changes that do not adversely affect the rights of holders of APEX in any material respect. All guarantees and agreements contained in the Guarantee bind the Company's successors, assignees, receivers, trustees and representatives and are for the benefit of the holders of the applicable APEX then outstanding.

Termination of the Guarantees

A Guarantee will terminate:

- upon full payment of the redemption price of all applicable APEX; or
- upon full payment of the amounts payable in accordance with the Trust Agreement upon liquidation of the Trust.

A Guarantee will continue to be effective or will be reinstated, as the case may be, if at any time any holder of APEX must restore payment of any sums paid under the APEX or the Guarantee.

Events of Default

An event of default under a Guarantee will occur if the Company fails to perform any payment obligation or if the Company fails to perform any other obligation under the Guarantee and such default remains unremedied for 30 days.

The holders of a majority in liquidation amount of the applicable APEX have the right to direct the time, method and place of conducting any proceeding for any remedy available to the Guarantee Trustee in respect of a Guarantee or to direct the exercise of any trust or power conferred upon the Guarantee Trustee under the Guarantee. Any holder of APEX may institute a legal proceeding directly against the Company to enforce the Guarantee Trustee's rights and the Company's obligations under a Guarantee, without first instituting a legal proceeding against the Trust, the Guarantee Trustee or any other person or entity.

As guarantor, the Company is required to file annually with the Guarantee Trustee a certificate as to whether or not the Company is in compliance with all applicable conditions and covenants under the Guarantee.

Information Concerning the Guarantee Trustee

Prior to the occurrence of an event of default relating to a Guarantee, the Guarantee Trustee is required to perform only the duties that are specifically set forth in the Guarantees. Following the occurrence of an event of default, the Guarantee Trustee will exercise the same degree of care as a prudent individual would exercise in the conduct of his or her own affairs. Provided that the foregoing requirements have been met, the Guarantee Trustee is under no obligation to exercise any of the powers vested in it by the Guarantees at the request of any holder of APEX, unless offered indemnity satisfactory to it against the costs, expenses and liabilities which might be incurred thereby.

The Company and its affiliates may maintain certain accounts and other banking relationships with the Guarantee Trustee and its affiliates in the ordinary course of business.

Governing Law

The Guarantees are governed by, and construed in accordance with, the laws of the State of New York.

Limited Purpose of Trust

The Trust securities evidence beneficial interests in the Trust. A principal difference between the rights of a holder of a Trust security and a holder of Preferred is that a holder of Preferred would be entitled to receive from the issuer the dividends, redemption payments and payment upon liquidation in respect of Preferred while a holder of Trust securities is entitled to receive distributions from a Trust, or from the Company under a Guarantee, if and to the extent the Trust has funds available for the payment of such distributions.

Rights upon Dissolution

Upon any voluntary or involuntary dissolution of the Trust, holders of each series of APEX will receive the distributions described under “—Liquidation Distribution upon Dissolution” above. Upon any voluntary or involuntary liquidation or bankruptcy of the Company, the holders of the Preferred would be preferred shareholders of the Company, entitled to the preferences upon liquidation described under “Description of the Preferred” below. Since the Company is the guarantor under the Guarantee and has agreed to pay for all costs, expenses and liabilities of the Trust, other than the Trust’s obligations to the holders of the Trust securities, the positions of a holder of APEX relative to other creditors and to the Company’s shareholders in the event of liquidation or bankruptcy are expected to be substantially the same as if that holder held the corresponding assets of the Trust directly.

DESCRIPTION OF THE PREFERRED

The following is a brief description of the terms of the Preferred held by the relevant Trust. This summary does not purport to be complete and is subject to and qualified in its entirety by reference to the Company’s restated certificate of incorporation, which is an exhibit to the Annual Report of which this exhibit is a part.

General

The Company’s authorized capital stock includes 150,000,000 shares of preferred stock, par value \$0.01 per share (including the Preferred).

Shares of the Preferred rank senior to the Company’s common stock, equally with each other series of the Company’s preferred stock outstanding as of December 31, 2020, and at least equally with each other series of preferred stock that the Company may issue (except for any senior series that may be issued with the requisite consent of the holders of the Preferred), with respect to the payment of dividends and distributions of assets upon liquidation, dissolution or winding up. The Preferred is fully paid and nonassessable, which means that its holders have paid their purchase price in full and that the Company may not ask them to surrender additional

funds. Holders of the Preferred do not have preemptive or subscription rights to acquire more preferred stock of the Company. The Preferred is not convertible into, or exchangeable for, shares of any other class or series of stock or other securities of the Company. The Preferred has no stated maturity and is not subject to any sinking fund or other obligation of the Company to redeem or repurchase the Preferred.

The Preferred has a fixed liquidation preference of \$100,000 per share. If the Company liquidates, dissolves or winds up its affairs, holders of the Preferred will be entitled to receive, out of the Company's assets that are available for distribution to shareholders, an amount per share equal to the liquidation preference per share, plus any declared and unpaid dividends, without regard to any undeclared dividends.

Unless the Trust is dissolved prior to the redemption of the Preferred, holders of APEX will not receive shares of the Preferred, and their interest in the Preferred will be represented by their APEX. If the Trust is dissolved, the Company may elect to distribute depository shares representing the Preferred instead of fractional shares. Since the Preferred is held by the Property Trustee, holders of APEX are only able to exercise voting or other rights with respect to the Preferred through the Property Trustee.

Dividends

Dividends on shares of the Preferred are not mandatory. Holders of the Preferred are entitled to receive, when, as and if declared by the Company's board of directors (or a duly authorized committee of the board), out of funds legally available for the payment of dividends under Delaware law, non-cumulative cash dividends from the date of their issuance. These dividends are payable on March 1, June 1, September 1 and December 1 of each year (each, for purposes of this section, a "Dividend Payment Date"), with respect to the Dividend Period, or portion thereof, ending on the day preceding the respective Dividend Payment Date, at a rate per annum equal to the greater of (x) three-month LIBOR for the related distribution period plus 0.765% (in the case of the Series E Preferred Stock) or 0.77% (in the case of the Series F Preferred Stock) and (y) 4.000%.

Dividends are payable to holders of record of the Preferred as they appear on the Company's books on the applicable record date, which shall be the 15th calendar day before that Dividend Payment Date or such other record date fixed by the Company's board of directors (or a duly authorized committee of the board) that is not more than 60 nor less than 10 days prior to such Dividend Payment Date (each, for purposes of this section, a "Dividend Record Date"). These Dividend Record Dates apply regardless of whether a particular Dividend Record Date is a business day.

A "Dividend Period" is the period from and including a Dividend Payment Date to but excluding the next Dividend Payment Date. If any that would otherwise be a Dividend Payment Date is not a business day, then the next business day will be the applicable Dividend Payment Date.

The amount of dividends payable per share of the Preferred on each Dividend Payment Date is calculated by multiplying the per annum Dividend Rate in effect for that Dividend Period by a fraction, the numerator of which is the actual number of days in that Dividend Period and the denominator of which is 360, and multiplying the rate obtained by \$100,000.

For any Dividend Period, LIBOR shall be determined by Goldman Sachs & Co. LLC, as calculation agent for the Preferred, on the second London business day immediately preceding the first day of such Dividend Period, as the case may be, in the following manner:

- LIBOR will be the offered rate per annum for three-month deposits in U.S. dollars, beginning on the first day of such period, as that rate appears on Reuters Screen LIBOR01 (or any successor or replacement page) as of 11:00 A.M., London time, on the second London business day immediately preceding the first day of such Dividend Period or Interest Period, as the case may be.
- If the rate described above does not appear on Reuters Screen LIBOR01 (or any successor or replacement page), LIBOR will be determined on the basis of the rates, at approximately

11:00 A.M., London time, on the second London business day immediately preceding the first day of such Dividend Period or Interest Period, as the case may be, at which deposits of the following kind are offered to prime banks in the London interbank market by four major banks in that market selected by the calculation agent: three-month deposits in U.S. dollars, beginning on the first day of such Dividend Period or Interest Period, as the case may be, and in a Representative Amount. The calculation agent will request the principal London office of each of these banks to provide a quotation of its rate. If at least two quotations are provided, LIBOR for the second London business day immediately preceding the first day of such Dividend Period or Interest Period, as the case may be, will be the arithmetic mean of the quotations.

- If fewer than two quotations are provided as described above, LIBOR for the second London business day immediately preceding the first day of such Dividend Period or Interest Period, as the case may be, will be the arithmetic mean of the rates for loans of the following kind to leading European banks quoted, at approximately 11:00 A.M. New York City time on the second London business day immediately preceding the first day of such Dividend Period or Interest Period, as the case may be, by three major banks in New York City selected by the calculation agent: three-month loans of U.S. dollars, beginning on the first day of such Dividend Period, and in a Representative Amount.
- If fewer than three banks selected by the calculation agent are quoting as described above, LIBOR for the new Dividend Period will be LIBOR in effect for the prior Dividend Period or Interest Period, as the case may be.

The calculation agent's determination of any dividend rate, and its calculation of the amount of dividends for any Dividend Period, will be on file at the Company's principal offices, will be made available to any stockholder upon request and will be final and binding in the absence of manifest error.

This subsection uses several terms that have special meanings relevant to calculating LIBOR. Those terms have the following meanings:

The term "Representative Amount" means an amount that, in the calculation agent's judgment, is representative of a single transaction in the relevant market at the relevant time.

"Reuters Screen LIBOR01 Page" means the display designated on the Reuters 3000 Xtra (or such other page as may replace that page on that service or such other service as may be nominated by the British Bankers' Association for the purpose of displaying London interbank offered rates for U.S. Dollar deposits).

The term "business day" means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is not a day on which banking institutions in New York City generally are authorized or obligated by law or executive order to close.

The term "London business day" means a day that is a Monday, Tuesday, Wednesday, Thursday or Friday and is a day on which dealings in U.S. dollars are transacted in the London interbank market. If the Company determines not to pay any dividend or a full dividend, the Company will provide prior written notice to the Property Trustee, who will notify holders of APEX, and the administrative trustees.

Dividends on the Preferred are not cumulative. Accordingly, if the board of directors of the Company (or a duly authorized committee of the board) does not declare a dividend on the Preferred payable in respect of any Dividend Period before the related Dividend Payment Date, such dividend will not accrue and the Company will have no obligation to pay a dividend for that Dividend Period on the Dividend Payment Date or at any future time, whether or not dividends on the Preferred are declared for any future Dividend Period.

So long as any share of Preferred remains outstanding, no dividend shall be paid or declared on the Company's common stock or any other shares of the Company's junior stock (as defined below) (other than a dividend payable solely in junior stock), and no common stock or other junior stock shall be purchased, redeemed or otherwise acquired for consideration by the Company, directly or indirectly (other than as a result of a reclassification of junior stock for or into other Junior Stock, or the exchange or conversion of one share of

junior stock for or into another share of junior stock and other than through the use of the proceeds of a substantially contemporaneous sale of junior stock), during a Dividend Period, unless the full dividends for the latest completed Dividend Period on all outstanding shares of Preferred have been declared and paid (or declared and a sum sufficient for the payment thereof has been set aside). However, the foregoing provision shall not restrict the ability of Goldman Sachs & Co. LLC, or any of the Company's other affiliates, to engage in any market-making transactions in the Company's junior stock in the ordinary course of business.

As used in this description of the Preferred, "junior stock" means any class or series of stock of the Company that ranks junior to the Preferred either as to the payment of dividends or as to the distribution of assets upon any liquidation, dissolution or winding up of the Company's junior stock includes the Company's common stock.

When dividends are not paid (or declared and a sum sufficient for payment thereof set aside) on any Dividend Payment Date (or, in the case of parity stock, as defined below, having Dividend Payment Dates different from the Dividend Payment Dates pertaining to the Preferred, on a Dividend Payment Date falling within the related Dividend Period for the Preferred) in full upon the Preferred and any shares of parity stock, all dividends declared upon the Preferred and all such equally ranking securities payable on such Dividend Payment Date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Preferred, on a dividend payment date falling within the related Dividend Period for the Preferred) shall be declared *pro rata* so that the respective amounts of such dividends shall bear the same ratio to each other as all accrued but unpaid dividends per share on the Preferred and all parity stock payable on such Dividend Payment Date (or, in the case of parity stock having dividend payment dates different from the dividend payment dates pertaining to the Preferred, on a dividend payment date falling within the related Dividend Period for the Preferred) bear to each other.

As used in this description of the Preferred, "parity stock" means any other class or series of stock of the Company that ranks equally with the Preferred in the payment of dividends and in the distribution of assets on any liquidation, dissolution or winding up of the Company.

Subject to the foregoing, such dividends (payable in cash, stock or otherwise) as may be determined by the Company's board of directors (or a duly authorized committee of the board) may be declared and paid on the Company's common stock and any other stock ranking equally with or junior to the Preferred from time to time out of any funds legally available for such payment, and the shares of the Preferred shall not be entitled to participate in any such dividend.

Redemption

The Preferred may be redeemed (but subject to applicable regulatory limits) in whole or in part, at the Company's option. Any such redemption will be at a cash redemption price of \$100,000 per share, plus any declared and unpaid dividends including, without regard to any undeclared dividends. Holders of Preferred have no right to require the redemption or repurchase of the Preferred. If notice of redemption of any Preferred has been given and if the funds necessary for the redemption have been set aside by the Company for the benefit of the holders of any shares of the Preferred so called for redemption, then, from and after the redemption date, those shares shall no longer be deemed outstanding and all rights of the holders of those shares (including the right to receive any dividends) will terminate, except the right to receive the redemption price.

If fewer than all of the outstanding shares of the Preferred are to be redeemed, the shares to be redeemed will be selected either *pro rata* from the holders of record of shares of the Preferred in proportion to the number of shares held by those holders or by lot or in such other manner as the Company's board of directors or a committee thereof may determine to be fair and equitable.

The Company will mail notice of every redemption of Preferred by first class mail, postage prepaid, addressed to the holders of record of the Preferred to be redeemed at their respective last addresses appearing on the Company's books. This mailing will be at least 30 days and not more than 60 days before the date fixed for

redemption (*provided* that if the Preferred is held in book-entry form through DTC, the Company may give this notice in any manner permitted by DTC). Any notice mailed or otherwise given as provided in this paragraph will be conclusively presumed to have been duly given, whether or not the holder receives this notice, and failure duly to give this notice by mail or otherwise, or any defect in this notice or in the mailing or provision of this notice, to any holder of Preferred designated for redemption will not affect the redemption of any other Preferred. If the Company redeems the Preferred, the Trust, as holder of the Preferred, will redeem the corresponding APEX as described under “— Mandatory Redemption of APEX upon Redemption of Preferred.”

Each notice shall state:

- the redemption date;
- the number of shares of the Preferred to be redeemed and, if less than all shares of the Preferred held by the holder are to be redeemed, the number of shares to be redeemed from the holder;
- the redemption price; and
- the place or places where the Preferred is to be redeemed.

The Company’s right to redeem the Preferred once issued is subject to prior approval of the Federal Reserve Board (or any successor banking agency).

Liquidation Rights

Upon any voluntary or involuntary liquidation, dissolution or winding up of the Company, holders of the Preferred are entitled to receive out of assets of the Company available for distribution to stockholders, after satisfaction of liabilities to creditors, if any, before any distribution of assets is made to holders of common stock or of any of the Company’s other shares of stock ranking junior as to such a distribution to the shares of the Preferred, a liquidating distribution in the amount of \$100,000 per share, plus declared and unpaid dividends, without accumulation of any undeclared dividends. Holders of the Preferred are not entitled to any other amounts from the Company after they have received their full liquidation preference.

In any such distribution, if the assets of the Company are not sufficient to pay the liquidation preferences in full to all holders of the Preferred and all holders of any other shares of the Company’s stock ranking equally as to such distribution with the Preferred, the amounts paid to the holders of the Preferred and to the holders of all such other stock will be paid *pro rata* in accordance with the respective aggregate liquidation preferences of those holders. In any such distribution, the “liquidation preference” of any holder of preferred stock means the amount payable to such holder in such distribution, including any declared but unpaid dividends (and any unpaid, accrued cumulative dividends in the case of any holder of stock on which dividends accrue on a cumulative basis). If the liquidation preference has been paid in full to all holders of Preferred and any other shares of the Company’s stock ranking equally as to the liquidation distribution, the holders of the Company’s other stock shall be entitled to receive all remaining assets of the Company according to their respective rights and preferences.

For purposes of this description of the Preferred, the merger or consolidation of the Company with any other entity, including a merger or consolidation in which the holders of Preferred receive cash, securities or property for their shares, or the sale, lease or exchange of all or substantially all of the assets of the Company for cash, securities or other property shall not constitute a liquidation, dissolution or winding up of the Company.

Voting Rights

Except as provided below, the holders of the Preferred have no voting rights.

Whenever dividends on any shares of the Preferred shall have not been declared and paid for a period the equivalent of six or more dividend payments, whether or not consecutive, equivalent to at least 18 months Dividend Periods (as used in this section, a “Nonpayment”), the holders of such shares, voting together as a class with holders of any and all other series of voting preferred stock (as defined below) then outstanding, will be entitled to vote for the election of a total of two additional members of the Company’s board of directors (as used in this section, the “Preferred Stock Directors”), *provided* that the election of any such director shall not cause the Company to violate the corporate governance requirement of the New York Stock Exchange (or any other exchange on which the Company’s securities may be listed) that listed companies must have a majority of independent directors and provided further that the Company’s board of directors shall at no time include more than two Preferred Stock Directors. In that event, the number of directors on the Company’s board of directors shall automatically increase by two, and the new directors shall be elected at a special meeting called at the request of the holders of record of at least 20% of the Preferred or of any other series of voting preferred stock (unless such request is received less than 90 days before the date fixed for the next annual or special meeting of the stockholders, in which event such election shall be held at such next annual or special meeting of stockholders), and at each subsequent annual meeting. These voting rights will continue until dividends on the shares of the Preferred and any such series of voting preferred stock for at least one year four Dividend Periods, whether or not consecutive, following the Nonpayment shall have been fully paid (or declared and a sum sufficient for the payment of such dividends shall have been set aside for payment).

As used in this description of the Preferred, “voting preferred stock” means any other class or series of preferred stock of the Company ranking equally with the Preferred either as to dividends or the distribution of assets upon liquidation, dissolution or winding up and upon which like voting rights have been conferred and are exercisable. Whether a plurality, majority or other portion of the shares of the Preferred and any other voting preferred stock have been voted in favor of any matter shall be determined by reference to the liquidation amounts of the shares voted.

If and when dividends for at least four Dividend Periods, whether or not consecutive, following a Nonpayment have been paid in full (or declared and a sum sufficient for such payment shall have been set aside), the holders of the Preferred shall be divested of the foregoing voting rights (subject to revesting in the event of each subsequent Nonpayment) and, if such voting rights for all other holders of voting preferred stock have terminated, the term of office of each Preferred Stock Director so elected shall terminate and the number of directors on the board of directors shall automatically decrease by two. In determining whether dividends have been paid for at least four Dividend Periods, whether or not consecutive, the Company may take account of any dividend the Company elects to pay for a Dividend Period after the regular dividend date for that period has passed. Any Preferred Stock Director may be removed at any time without cause by the holders of record of a majority of the outstanding shares of the Preferred when they have the voting rights described above (voting together as a class with all series of voting preferred stock then outstanding). So long as a Nonpayment shall continue, any vacancy in the office of a Preferred Stock Director (other than prior to the initial election after a Nonpayment) may be filled by the written consent of the Preferred Stock Director remaining in office, or if none remains in office, by a vote of the holders of record of a majority of the outstanding shares of the Preferred and all voting preferred stock when they have the voting rights described above (voting together as a class). The Preferred Stock Directors shall each be entitled to one vote per director on any matter.

So long as any shares of the Preferred remain outstanding, the Company will not, without the affirmative vote or consent of the holders of at least two-thirds of the outstanding shares of the Preferred and all other series of voting preferred stock entitled to vote thereon, voting together as a single class, given in person or by proxy, either in writing or at a meeting:

- authorized amount of, any class or series of stock ranking senior to the Preferred with respect to payment of dividends or the distribution of assets upon liquidation, dissolution or winding up of the Company;
- amend, alter or repeal the provisions of the Company’s restated certificate of incorporation so as to materially and adversely affect the special rights, preferences, privileges and voting powers of the Preferred, taken as a whole; or

- consummate a binding share exchange or reclassification involving the Preferred or a merger or consolidation of the Company with another entity, unless in each case (i) the shares of the Preferred remain outstanding or, in the case of any such merger or consolidation with respect to which the Company is not the surviving or resulting entity, are converted into or exchanged for preference securities of the surviving or resulting entity or its ultimate parent, and (ii) such shares remaining outstanding or such preference securities, as the case may be, have such rights, preferences, privileges and voting powers, taken as a whole, as are not materially less favorable to the holders thereof than the rights, preferences, privileges and voting powers of the Preferred, taken as a whole;

provided, however, that any increase in the amount of the authorized or issued Preferred or other authorized preferred stock or the creation and issuance, or an increase in the authorized or issued amount, of other series of preferred stock ranking equally with and/or junior to the Preferred with respect to the payment of dividends (whether such dividends are cumulative or non-cumulative) and/or the distribution of assets upon liquidation, dissolution or winding up of the Company will not be deemed to adversely affect the rights, preferences, privileges or voting powers of the Preferred.

If an amendment, alteration, repeal, share exchange, reclassification, merger or consolidation described above would adversely affect one or more but not all series of voting preferred stock (including the Preferred for this purpose), then only the series affected and entitled to vote shall vote as a class in lieu of all such series of preferred stock.

Without the consent of the holders of the Preferred, so long as such action does not adversely affect the rights, preferences, privileges and voting powers of the Preferred, the Company may amend, alter, supplement or repeal any terms of the Preferred:

- to cure any ambiguity, or to cure, correct or supplement any provision contained in the certificate of designations for the Preferred that may be defective or inconsistent; or
- to make any provision with respect to matters or questions arising with respect to the Preferred that is not inconsistent with the provisions of the certificate of designations.

The foregoing voting provisions do not apply if, at or prior to the time when the act with respect to which such vote would otherwise be required shall be effected, all outstanding shares of the Preferred shall have been redeemed or called for redemption upon proper notice and sufficient funds shall have been set aside by the Company for the benefit of the holders of the Preferred to effect such redemption.

Form

The Preferred are issued only in fully registered form. Other than the fractional shares currently held by each Trust, no fractional shares will be issued unless a Trust is dissolved and the Company delivers the shares, rather than depositary receipts representing the shares, to the registered holders of its APEX. If a Trust is dissolved and depositary receipts or shares of the Preferred held by the Trust are distributed to holders of its APEX, the Company would intend to distribute them in book-entry form only and the procedures governing holding and transferring beneficial interests in the Preferred, and the circumstances in which holders of beneficial interests will be entitled to receive certificates evidencing their shares or depositary receipts. If the Company determines to issue depositary shares representing fractional interests in the Preferred, each depositary share will be represented by a depositary receipt. In such an event, the Preferred represented by the depositary shares will be deposited under a deposit agreement among the Company, a depositary and the holders from time to time of the depositary receipts representing depositary shares. Subject to the terms and conditions of any deposit agreement, each holder of a depositary share will be entitled, through the depositary, in proportion to the applicable fraction of a share of the Preferred represented by such depositary share, to all the rights applicable fraction of a share of the Preferred represented by such depositary share, to all the rights and preferences of the Preferred represented thereby (including dividends, voting, redemption and liquidation rights).

Title

The Company, the transfer agent and registrar for the Preferred held by a Trust, and any of their agents may treat the registered owner of the Preferred, which shall be the Property Trustee unless and until the Trust is dissolved, as the absolute owner of that stock, whether or not any payment for the Preferred shall be overdue and despite any notice to the contrary, for any purpose.

Transfer Agent and Registrar

If a Trust is dissolved and shares of the Preferred held by the Trust or depositary receipts representing the Preferred are distributed to holders of APEX, the Company may appoint a transfer agent, registrar, calculation agent, redemption agent and dividend disbursement agent for the Preferred. The registrar for the Preferred will send notices to shareholders of any meetings at which holders of the Preferred have the right to vote on any matter.

DESCRIPTION OF MEDIUM-TERM NOTES, SERIES F, CALLABLE FIXED AND FLOATING RATE NOTES DUE MARCH 2031 OF GS FINANCE CORP. (FULLY AND UNCONDITIONALLY GUARANTEED BY THE GOLDMAN SACHS GROUP, INC.)

The following is a brief description of the terms of the Callable Fixed and Floating Rate Notes, an issuance of Medium-Term Notes, Series F of GS Finance Corp. ("GSFC") (the "Callable Notes"), which are fully and unconditionally guaranteed by the Company. It does not purport to be complete. This description is subject to and qualified in its entirety by reference to the Senior Debt Indenture, dated as of October 10, 2008, among GSFC, as issuer, the Company, as guarantor, and The Bank of New York Mellon, as trustee, as supplemented by the First Supplemental Indenture, dated as of February 20, 2015, the Fourth Supplemental Indenture, dated as of August 21, 2018 and the Seventh Supplemental Indenture, dated as of July 1, 2020 (collectively, the "GSFC 2008 Indenture"), which are exhibits to the Annual Report of which this exhibit is a part. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

The GSFC 2008 indenture permits GSFC to issue, from time to time, different series of debt securities and, within each different series of debt securities, different debt securities. The Medium-Term Notes, Series F are a single, distinct series of debt securities. GSFC may, however, issue notes in such amounts, at such times and on such terms as GSFC wishes. The notes of the Medium-Term Notes, Series F may differ from one another, and from other series, in their terms.

In this description, references to a series of debt securities mean a series issued under the GSFC 2008 Indenture, such as the notes issued under GSFC's Medium-Term Notes, Series F program.

Terms of the Callable Notes

The Callable Notes were originally issued on March 11, 2021 and have a stated maturity date of March 11, 2031 (the "stated maturity date"). As noted above, the Callable Notes are part of a series of debt securities, entitled "Medium-Term Notes, Series F," that GSFC may issue under the GSFC 2008 Indenture from time to time. The Callable Notes are listed on the New York Stock Exchange Bonds Market under the ticker symbol "GSJ31B."

The payment of principal of, and any interest and premium on, the Callable Notes is fully and unconditionally guaranteed by the Company. The guarantee will remain in effect until the entire principal of, and interest and premium, if any, on, the Callable Notes has been paid in full or discharged in accordance with the provisions of the GSFC 2008 Indenture, or otherwise fully defeased by GSFC or by the Company. The guarantee of senior debt securities of GSFC, such as the Callable Notes, will rank equally in right of payment to all senior indebtedness of the Company.

Payment of Principal on Stated Maturity Date

GSFC will pay holders of Callable Notes that have not been redeemed by the stated maturity date an amount in cash equal to the outstanding face amount of such holders' Callable Notes. The stated maturity date of the Callable Notes is March 11, 2031, subject to GSFC's early redemption right. If the stated maturity date falls on a day that is not a business day, payment of principal otherwise due on such day will be made on the next succeeding business day, and no interest on such payment shall accrue for the period from and after the stated maturity date.

Interest Payments

For each fixed rate interest period, the fixed interest rate on the Callable Notes will equal 5% per annum. For each floating rate interest period, the floating interest rate on the Callable Notes will be based upon the CMS spread on the relevant interest determination date for such floating rate interest period and will be a rate per annum equal to:

- if (i) the CMS spread minus 0.25% times (ii) 6.5 is greater than or equal to the maximum interest rate, the maximum interest rate;
- if (i) the CMS spread minus 0.25% times (ii) 6.5 is less than the maximum interest rate but greater than the minimum interest rate, (i) the CMS spread minus 0.25% times (ii) 6.5; or
- if (i) the CMS spread minus 0.25% times (ii) 6.5 is equal to or less than the minimum interest rate, the minimum interest rate.

The maximum interest rate on the Callable Notes is 10% per annum. Based on the formula used to calculate the floating interest rate on the Callable Notes, holders will not benefit from any increases in the CMS spread minus 0.25% above approximately 1.54%. The minimum interest rate on the Callable Notes is 0% per annum.

The term “fixed rate interest periods” means the periods from and including a fixed rate interest payment date (or the original issue date, in the case of the first fixed rate interest period) to but excluding the next succeeding fixed rate interest payment date, where the term “fixed rate interest payment dates” means March 11, June 11, September 11 and December 11 of each year, commencing on June 11, 2021 and ending on March 11, 2022, subject to adjustments as described below.

The term “floating rate interest periods” means the periods from and including a floating rate interest payment date (or the final fixed rate interest payment date, in the case of the first floating rate interest period) to but excluding the next succeeding floating rate interest payment date (or the stated maturity date, in the case of the final floating rate interest period), where the term “floating rate interest payment dates” means March 11, June 11, September 11 and December 11 of each year, beginning on June 11, 2022 and ending on the stated maturity date, subject to adjustments as described below.

The term “interest determination dates” means, for each floating rate interest period, the second U.S. government securities business day preceding such floating rate interest period.

The term “CMS spread” means, on any interest determination date, the 30-year CMS rate minus the 5-year CMS rate, as described under “—CMS Rate” below.

The calculation agent will calculate the amount of interest payable on each fixed rate interest payment date and floating rate interest payment date (each, an “interest payment date”) for the applicable fixed rate interest period and floating rate interest period (each, an “interest period”) in the following manner. For each \$1,000 face amount of the Callable Notes and for each interest period, the calculation agent will calculate the amount of interest to be paid by calculating the product of (i) the \$1,000 face amount times (ii) the applicable fixed interest rate or floating interest rate times (iii) the applicable day count convention on a 30/360 (ISDA) basis.

Interest will be paid on the Callable Notes on each quarterly interest payment date. If an interest payment date (other than the interest payment date that falls on the stated maturity date) falls on a day that is not a business day, the payment due on such interest payment date will be postponed to the next day that is a business day; *provided* that interest due with respect to such interest payment date shall not accrue from and including such interest payment date to and including the date of payment of such interest as so postponed. If the stated maturity date falls on a day that is not a business day, payment of interest otherwise due on such day will be made on the next succeeding business day, and no interest on such payment shall accrue for the period from and after the stated maturity date.

CMS Rate

References to the 30-year CMS rate or the 5-year CMS rate on an interest determination date mean the rate appearing on the Refinitiv page ICESWAP1 for 30-year or 5-year index maturity, as the case may be, as of approximately 11:00 A.M., New York City time, on such interest determination date. If a CMS rate cannot be determined in this manner on the relevant interest determination date, the following procedures will apply to the Callable Notes.

If the calculation agent determines on an interest determination date that a CMS rate has been discontinued, then the calculation agent will use a substitute or successor rate that it has determined in its sole discretion is most comparable to the applicable CMS rate, provided that if the calculation agent determines there is an industry-accepted successor rate, then the calculation agent shall use such successor rate. If the calculation agent has determined a substitute or successor rate in accordance with the foregoing, the calculation agent in its sole discretion may determine the business day convention, the applicable business days and the interest determination dates to be used, and any other relevant methodology for calculating such substitute or successor rate, including any adjustment factor needed to make such substitute or successor rate comparable to the applicable CMS rate, in a manner that is consistent with any industry-accepted practices for such substitute or successor rate.

Unless the calculation agent uses a substitute or successor rate as so provided, if the CMS rate cannot be determined in the manner described above, the applicable CMS rate for that interest determination date will be determined by the calculation agent, after consulting such sources as it deems comparable to the foregoing display page, or any other source it deems reasonable, in its sole discretion.

The applicable CMS rate will be subject to the corrections, if any, published on the Refinitiv page ICESWAP1 within one hour of the time that rate was first displayed on such source.

The term “Refinitiv page ICESWAP1” means the display on the Refinitiv Eikon service, or any successor or replacement service, on the page ICESWAP1, or any successor or replacement page on that service.

Manner of Payment

Any payment on the Callable Notes at maturity or upon redemption will be made to an account designated by the holder of such Callable Notes and approved by GSFC, or at the office of the trustee in New York City, but only when such Callable Notes are surrendered to the trustee at that office. GSFC may pay interest on any interest payment date by check mailed to the person who is the holder on the regular record date. GSFC also may make any payment in accordance with the applicable procedures of the depository.

Modified Business Day

Any payment on the Callable Notes that would otherwise be due on a day that is not a business day may instead be paid on the next day that is a business day, with the same effect as if paid on the original due date. For the Callable Notes, however, the term business day may have a different meaning than it does for other Series F medium-term notes, as described under “— Special Calculation Provisions” below.

Role of Calculation Agent

The calculation agent in its sole discretion will make all determinations regarding the CMS spread, the 30-year CMS rate, the 5-year CMS rate, the interest determination dates, the regular record dates, the interest payable on each interest payment date, U.S. government securities business days, business days, postponement of the stated maturity date and the amount payable on the Callable Notes at maturity or redemption, as applicable. Absent manifest error, all determinations of the calculation agent will be final and binding on holders of the Callable Notes and GSFC, without any liability on the part of the calculation agent.

Goldman Sachs & Co. LLC (“GS&Co.”), an affiliate of GSFC, is currently serving as the calculation agent as of the date of this description. GSFC may change the calculation agent for the Callable Notes at any time after the date of this prospectus supplement without notice and GS&Co. may resign as calculation agent at any time upon 60 days’ written notice to GSFC.

Early Redemption Right

GSFC may redeem the Callable Notes, at GSFC's option, in whole but not in part, on the interest payment date falling on March 11, 2022 and on each interest payment date occurring thereafter, for an amount equal to 100% of the face amount plus any accrued and unpaid interest to, but excluding, the redemption date.

If GSFC chooses to exercise its early redemption right, it will notify the holder of the Callable Notes and the trustee by giving at least five business days' prior notice. The day GSFC gives the notice, which will be a business day, will be the redemption notice date and the immediately following interest payment date, which GSFC will state in the redemption notice, will be the redemption date. GSFC will not give a redemption notice that results in a redemption date later than the stated maturity date.

If GSFC gives the holder a redemption notice, GSFC will redeem the entire outstanding face amount of such holder's Callable Notes. On the redemption date, GSFC will pay to the holder of record on the business day immediately preceding the redemption date, the redemption price in cash, together with any accrued and unpaid interest to, but excluding, the redemption date, in the manner described under "—Manner of Payment" above.

Special Calculation Provisions

The term "business day" with respect to the Callable Notes means each Monday, Tuesday, Wednesday, Thursday and Friday that is not a day on which banking institutions in New York City generally are authorized or obligated by law, regulation or executive order to close.

References to a U.S. government securities business day with respect to the Callable Notes mean any day except for a Saturday, Sunday or a day on which the Securities Industry and Financial Markets Association recommends that the fixed income department of its members be closed for the entire day for purposes of trading in U.S. government securities.

Defeasance and Covenant Defeasance

The provisions for full defeasance and covenant defeasance in the GSFC 2008 Indenture do not apply to the Callable Notes.

Default, Remedies and Waiver of Default

A holder will have special rights if an event of default with respect to his or her series of debt securities occurs and is continuing, as described in this subsection.

Events of Default

References to an event of default with respect to any series of debt securities mean any of the following:

- GSFC or the Company does not pay the principal or any premium on any debt security of that series within 30 days after the due date;
- GSFC or the Company does not pay interest on any debt security of that series within 30 days after the due date; or
- GSFC files for bankruptcy or other events of bankruptcy, insolvency or reorganization relating to GSFC occur. Those events must arise under U.S. federal or state law, unless GSFC merges, consolidates or sells its assets as described above and the successor firm is a non-U.S. entity. If that happens, then those events must arise under U.S. federal or state law or the law of the jurisdiction in which the successor firm is legally organized.

As described below under “Remedies If an Event of Default Occurs,” under the GSFC 2008 Indenture, events of bankruptcy, insolvency or reorganization relating to the Company will not cause any of GSFC’s debt securities issued under such indenture to be automatically accelerated. In the event that the Company becomes subject to certain events of bankruptcy, insolvency or reorganization (but GSFC does not), any series of debt securities issued under the GSFC 2008 Indenture will not be immediately due and repayable. In addition, under the GSFC 2008 Indenture, a breach of a covenant or warranty by the Company (including, for example, a breach of the Company’s covenants and warranties with respect to mergers and similar transactions or restrictions on liens) will not have the potential to cause any of GSFC’s debt securities issued under the GSFC 2008 Indenture to be declared due and payable immediately. Instead, under the GSFC 2008 Indenture, the trustee or holder will need to wait until the earlier of the time that (i) GSFC itself becomes subject to certain events of bankruptcy, insolvency or reorganization or otherwise defaults on the terms of the debt securities, (ii) the Company otherwise defaults on the terms of the debt securities and (iii) the final maturity of the debt securities. The return the holder receives on any series of debt securities issued under the GSFC 2008 Indenture may be significantly less than what a holder would have otherwise received had the debt securities been automatically accelerated upon certain events of bankruptcy, insolvency or reorganization relating to the Company or declared due and payable immediately following the breach of a covenant or warranty by the Company.

Covenant Breaches

References to a covenant breach with respect to any series of debt securities mean any of the following:

- GSFC or the Company does not deposit a required sinking fund payment with regard to any debt security of that series on the due date;
- GSFC remains in breach of any other covenant it makes in the GSFC 2008 Indenture for the benefit of the relevant series for 60 days after GSFC and the Company receive a notice of default stating that GSFC is in breach and requiring GSFC to remedy the breach. The notice must be sent by the trustee or the holders of at least 10% in principal amount of the relevant series of debt securities then outstanding;
- Except as provided by the GSFC 2008 Indenture, the debt security of that series and the related guarantee, the guarantee ceases to be effective, or a court finds the guarantee to be unenforceable or invalid, or the Company denies its obligations as the guarantor.

A covenant breach shall not be an event of default with respect to any security.

Remedies If an Event of Default or Covenant Breach Occurs

If an event of default has occurred with respect to any series of debt securities and has not been cured or waived, the trustee or the holders of not less than 25% in principal amount of all debt securities of that series then outstanding may declare the entire principal amount of the debt securities of that series to be due immediately. If the event of default occurs because of events in bankruptcy, insolvency or reorganization relating to GSFC, the entire principal amount of the debt securities of that series will be automatically accelerated, without any action by the trustee or any holder.

Each of the situations described above is called an acceleration of the stated maturity of the affected series of debt securities. If the stated maturity of any series is accelerated and a judgment for payment has not yet been obtained, the holders of a majority in principal amount of the debt securities of that series may cancel the acceleration for the entire series.

If an event of default or a covenant breach occurs, the trustee will have special duties. In that situation, the trustee will be obligated to use those of its rights and powers under the GSFC 2008 Indenture, and to use the same degree of care and skill in doing so, that a prudent person would use in that situation in conducting his or her own affairs.

Except as described in the prior paragraph, the trustee is not required to take any action under the GSFC 2008 Indenture at the request of any holders unless the holders offer the trustee reasonable protection from expenses and liability (*i.e.*, an indemnity). If the trustee is provided with an indemnity reasonably satisfactory to it, the holders of a majority in principal amount of all debt securities of the relevant series may direct the time, method and place of conducting any lawsuit or other formal legal action seeking any remedy available to the trustee with respect to that series. These majority holders may also direct the trustee in performing any other action under the GSFC 2008 Indenture with respect to the debt securities of that series.

Before a holder bypasses the trustee and bring its own lawsuit or other formal legal action or take other steps to enforce its rights or protect its interests relating to any debt security, all of the following must occur:

- The holder must give the trustee written notice that an event of default or a covenant breach has occurred, and the event of default or covenant breach must not have been cured or waived;
- The holders of not less than 25% in principal amount of all debt securities of a holder's series must make a written request that the trustee take action because of the default, and they or other holders must offer to the trustee indemnity reasonably satisfactory to the trustee against the cost and other liabilities of taking that action;
- The trustee must not have taken action for 60 days after the above steps have been taken; and
- During those 60 days, the holders of a majority in principal amount of the debt securities of a holder's series must not have given the trustee directions that are inconsistent with the written request of the holders of not less than 25% in principal amount of the debt securities of a holder's series.

A holder is entitled at any time, however, to bring a lawsuit for the payment of money due on his or her debt security on or after its stated maturity (or, if the debt security is redeemable, on or after its redemption date).

Waiver of Default

The holders of not less than a majority in principal amount of the debt securities of any series may waive a default for all debt securities of that series. If this happens, the default will be treated as if it has not occurred. No one can waive a payment default on a holder's debt security, however, without the approval of the particular holder of that debt security.

GSFC and the Company Will Give the Trustee Information About Defaults Annually

GSFC and the Company will furnish to the trustee every year a written statement, respectively, of two of their officers certifying that to their knowledge GSFC or the Company, as the case may be, is in compliance with the GSFC 2008 Indenture and the debt securities issued under it, or else specifying any default under the relevant debt indenture. For the purpose of this paragraph, the term "default" means any event which is, or after notice or lapse of time or both would become, an event of default or covenant breach.

Default Amount on Acceleration

If an event of default occurs and the maturity of the Callable Notes is accelerated, GSFC will pay the default amount in respect of the principal of the Callable Notes at the maturity, instead of the amount payable on the Callable Notes as described earlier.

For the purpose of determining whether the holders of GSFC's Medium-Term Notes Series F, which include the Callable Notes, are entitled to take any action under the GSFC 2008 Indenture, GSFC will treat the outstanding face amount of each Callable Note as the outstanding principal amount of that note. Although the terms of the Callable Notes differ from those of the other Medium-Term Notes Series F, holders of specified percentages in

principal amount of all Medium-Term Notes Series F, together in some cases with other series of GSFC's debt securities, will be able to take action affecting all the Medium-Term Notes Series F, including the Callable Notes, except with respect to certain Medium-Term Notes Series F, if the terms of such notes specify that the holders of specified percentages in the principal amount of all such notes must also consent to such action. This action may involve changing some of the terms that apply to the Medium-Term Notes Series F, accelerating the maturity of the Medium-Term Notes Series F, after a default or waiving some of GSFC's obligations under the GSFC 2008 Indenture. In addition, certain changes to the GSFC 2008 Indenture and the Callable Notes that only affect certain debt securities may be made with the approval of holders of a majority of the principal amount of such affected debt securities.

Guarantee by the Company

The Company has fully and unconditionally guaranteed the payment of principal of, and any interest and premium on, the Medium-Term Notes, Series F, which include the Callable Notes, when due and payable, whether at the stated maturity, by declaration of acceleration, call for redemption or otherwise, in accordance with the terms of the security and the GSFC 2008 Indenture. The guarantee will remain in effect until the entire principal of, and interest and premium, if any, on, the debt securities has been paid in full or discharged in accordance with the provisions of the GSFC 2008 Indenture, or otherwise fully defeased by the Company.

The guarantee by the Company of its debt securities issued under the GSFC 2008 Indenture will rank equally in right of payment with all senior indebtedness of the Company.

Mergers and Similar Transactions

GSFC and the Company are generally permitted to merge or consolidate with another corporation or other entity. GSFC and the Company also permitted to sell their assets substantially as an entirety to another corporation or other entity. With regard to any series of debt securities, however, GSFC or the Company may not take any of these actions unless all the following conditions are met:

- If the successor entity in the transaction is not GSFC or the Company, as the case may be, the successor entity must be organized as a corporation, partnership or trust and must expressly assume the obligations of GSFC or the Company under the debt securities of that series and the GSFC 2008 Indenture with respect to that series. The successor entity may be organized under the laws of any jurisdiction, whether in the United States or elsewhere.
- Immediately after the transaction, no default under the debt securities of that series or the related guarantees has occurred and is continuing. For this purpose, "default under the debt securities of that series or the related guarantees" means an event of default or a covenant breach with respect to that series or the related guarantees or any event that would be an event of default with respect to that series or the related guarantees if the requirements for giving GSFC or the Company default notice and for GSFC's or the Company's default having to continue for a specific period of time were disregarded.

If the conditions described above are satisfied with respect to debt securities of any series, neither GSFC nor the Company will need to obtain the approval of the holders of those debt securities in order to merge or consolidate or to sell assets of GSFC or the Company. Also, these conditions will apply only if GSFC or the Company wishes to merge or consolidate with another entity or sell assets of GSFC or the Company substantially as an entirety to another entity. Neither GSFC nor the Company will need to satisfy these conditions if GSFC or the Company enters into other types of transactions, including any transaction in which GSFC or the Company acquire the stock or assets of another entity, any transaction that involves a change of control of GSFC or the Company but in which GSFC or the Company does not merge or consolidate and any transaction in which GSFC or the Company sells less than substantially all assets of GSFC or the Company. While GSFC is currently a wholly owned subsidiary of the Company, there is no requirement that it remain a subsidiary.

Also, if GSFC or the Company merges, consolidates or sells assets of GSFC or the Company substantially as an entirety and the successor is a non-U.S. entity, neither GSFC nor any successor would have any obligation to compensate a holder for any resulting adverse tax consequences relating to his or her debt securities.

Notwithstanding the foregoing and for the avoidance of doubt, GSFC may sell or transfer its assets substantially as an entirety, in one or more transactions, to one or more entities, provided that GSFC's assets and the assets of its direct or indirect subsidiaries in which it owns a majority of the combined voting power, taken together, are not sold or transferred substantially as an entirety to one or more entities that are not majority-owned subsidiaries of the Company, and the Company may sell or transfer its assets substantially as an entirety in one or more transactions, to one or more entities, provided that the assets of the Company and its direct or indirect subsidiaries in which it owns a majority of the combined voting power, taken together, are not sold or transferred substantially as an entirety to one or more entities that are not such subsidiaries.

Restriction on Liens

In the GSFC 2008 Indenture, the Company promises, with respect to each series of senior debt securities, not to create, assume, incur or guarantee any debt for borrowed money that is secured by a lien on the voting or profit participating equity ownership interests that the Company or any of its subsidiaries own in Goldman Sachs & Co. LLC, or in any subsidiary of the Company that beneficially owns or holds, directly or indirectly, those interests in Goldman Sachs & Co. LLC, unless the Company also secures the senior debt securities of that series on an equal or priority basis with the other secured debt. The promise of the Company, however, is subject to an important exception: it may secure debt for borrowed money with liens on those interests without securing the senior debt securities of any series if its board of directors determines that the liens do not materially detract from or interfere with the value or control of those interests, as of the date of the determination.

Except as noted above, the GSFC 2008 Indenture does not restrict the Company's ability to put liens on its interests in its subsidiaries other than Goldman Sachs & Co. LLC, nor does the indenture restrict the Company's ability to sell or otherwise dispose of its interests in any of its subsidiaries, including Goldman Sachs & Co. LLC. In addition, the restriction on liens in the GSFC 2008 Indenture applies only to liens that secure debt for borrowed money. For example, liens imposed by operation of law, such as liens to secure statutory obligations for taxes or workers' compensation benefits, or liens the Company creates to secure obligations to pay legal judgments or surety bonds, would not be covered by this restriction.

Modification of the Debt Indenture and Waiver of Covenants

There are four types of changes GSFC and the Company can make to the GSFC 2008 Indenture and the debt securities or series of debt securities and related guarantees issued under the GSFC 2008 Indenture.

Changes Requiring Each Holder's Approval

First, there are changes that cannot be made without the approval of the holder of each debt security affected by the change under the GSFC 2008 Indenture. Here is a list of those types of changes:

- change the stated maturity for any principal or interest payment on a debt security;
- reduce the principal amount, the amount payable on acceleration of the stated maturity after a default, the interest rate or the redemption price for a debt security;
- permit redemption of a debt security if not previously permitted;
- impair any right a holder may have to require repayment of its debt security;
- change the currency of any payment on a debt security;
- change the place of payment on a debt security;
- impair a holder's right to sue for payment of any amount due on its debt security;
- reduce the percentage in principal amount of the debt securities of any one or more affected series, taken

- separately or together, as applicable, and whether comprising the same or different series or less than all of the debt securities of a series, the approval of whose holders is needed to change the applicable debt indenture or those debt securities;
- reduce the percentage in principal amount of the debt securities of any one or more affected series, taken separately or together, as applicable, and whether comprising the same or different series or less than all of the debt securities of a series, the consent of whose holders is needed to waive GSFC's compliance with the applicable debt indenture or to waive defaults; and
- change the provisions of the applicable debt indenture dealing with modification and waiver in any other respect, except to increase any required percentage referred to above or to add to the provisions that cannot be changed or waived without approval of the holder of each affected debt security.

Changes Not Requiring Approval

The second type of change does not require any approval by holders of the debt securities affected. These changes are limited to clarifications and changes that would not adversely affect any debt securities of any series in any material respect. Neither GSFC nor the Company needs any approval to make changes that affect only debt securities to be issued under the applicable indenture after the changes take effect.

GSFC and the Company may also make changes or obtain waivers that do not adversely affect a particular debt security, even if they affect other debt securities. In those cases, neither GSFC nor the Company needs to obtain the approval of the holder of the unaffected debt security; GSFC and the Company need only obtain any required approvals from the holders of the affected debt securities.

Changes Requiring Majority Approval

Any other change to the GSFC 2008 Indenture and the debt securities issued under such debt indenture would require the following approval:

- If the change affects only particular debt securities within a series, it must be approved by the holders of a majority in principal amount of such particular debt securities.
- If the change affects multiple debt securities of one or more series, it must be approved by the holders of a majority in principal amount of all debt securities affected by the change, with all such affected debt securities voting together as one class for this purpose (and by the holders of a majority in principal amount of any affected debt securities that by their terms are entitled to vote separately).

In each case, the required approval must be given by written consent.

This would mean that modification of terms with respect to certain debt securities of a series could be effectuated under the GSFC 2008 Indenture without obtaining the consent of the holders of a majority in principal amount of other securities of such series that are not affected by such modification.

The same majority approval would be required for GSFC to obtain a waiver of any of its covenants in the GSFC 2008 Indenture. GSFC's covenants include the promises GSFC and the Company make about merging and, with respect to the Company, putting liens on GSFC's interests in Goldman Sachs & Co. LLC. If the holders approve a waiver of a covenant, neither GSFC nor the Company will have to comply with it. The holders, however, cannot approve a waiver of any provision in a particular debt security, or in the GSFC 2008 Indenture as it affects that debt security, that neither GSFC nor the Company can change without the approval of the holder of that debt security as described above in "— Changes Requiring Each Holder's Approval," unless that holder approves the waiver.

Special Rules for Action by Holders

When holders take any action under the GSFC 2008 Indenture, such as giving a notice of default, notice of covenant breach, declaring an acceleration, approving any change or waiver or giving the trustee an instruction, GSFC will apply the following rules.

Only Outstanding Debt Securities Are Eligible

Only holders of outstanding debt securities or the outstanding debt securities of the applicable series, as applicable, will be eligible to participate in any action by holders of such debt securities or the debt securities of that series. Also, GSFC will count only outstanding debt securities in determining whether the various percentage requirements for taking action have been met. For these purposes, a debt security will not be “outstanding” if:

- it has been surrendered for cancellation;
- GSFC has deposited or set aside, in trust for its holder, money for its payment or redemption;
- GSFC has fully defeased it; or
- GSFC or one of its affiliates, such as Goldman Sachs & Co. LLC, is the owner.

Determining Record Dates for Action by Holders

GSFC and the Company will generally be entitled to set any day as a record date for the purpose of determining the holders that are entitled to take action under a particular debt indenture. In certain limited circumstances, only the trustee will be entitled to set a record date for action by holders. If GSFC, the Company or the trustee set a record date for an approval or other action to be taken by holders, that vote or action may be taken only by persons or entities who are holders on the record date and must be taken during the period that GSFC specifies for this purpose, or that the trustee specifies if it sets the record date. GSFC, the Company or the trustee, as applicable, may shorten or lengthen this period from time to time.

This period, however, may not extend beyond the 180th day after the record date for the action. In addition, record dates for any global debt security may be set in accordance with procedures established by the depository from time to time. Accordingly, record dates for global debt securities may differ from those for other debt securities.

Form of Callable Notes

The Callable Notes are issued in book-entry form through The Depository Trust Company and represented by a global note. GSFC will not issue definitive notes in exchange for the global note except in limited circumstances.

DESCRIPTION OF MEDIUM-TERM NOTES, SERIES F, CALLABLE FIXED AND FLOATING RATE NOTES DUE MAY 2031 OF GS FINANCE CORP. (FULLY AND UNCONDITIONALLY GUARANTEED BY THE GOLDMAN SACHS GROUP, INC.)

The following is a brief description of the terms of the Callable Fixed and Floating Rate Notes, an issuance of Medium-Term Notes, Series F of GS Finance Corp. ("GSFC") (the "Callable Notes"), which are fully and unconditionally guaranteed by the Company. It does not purport to be complete. This description is subject to and qualified in its entirety by reference to the Senior Debt Indenture, dated as of October 10, 2008, among GSFC, as issuer, the Company, as guarantor, and The Bank of New York Mellon, as trustee, as supplemented by the First Supplemental Indenture, dated as of February 20, 2015, the Fourth Supplemental Indenture, dated as of August 21, 2018 and the Seventh Supplemental Indenture, dated as of July 1, 2020 (collectively, the "GSFC 2008 Indenture"), which are exhibits to the Annual Report of which this exhibit is a part. Unless the context otherwise provides, all references to the Company in this description refer only to The Goldman Sachs Group, Inc. and does not include its consolidated subsidiaries.

The GSFC 2008 indenture permits GSFC to issue, from time to time, different series of debt securities and, within each different series of debt securities, different debt securities. The Medium-Term Notes, Series F are a single, distinct series of debt securities. GSFC may, however, issue notes in such amounts, at such times and on such terms as GSFC wishes. The notes of the Medium-Term Notes, Series F may differ from one another, and from other series, in their terms.

In this description, references to a series of debt securities mean a series issued under the GSFC 2008 Indenture, such as the notes issued under GSFC's Medium-Term Notes, Series F program.

Terms of the Callable Notes

The Callable Notes were originally issued on May 28, 2021 and have a stated maturity date of May 28, 2031 (the "stated maturity date"). As noted above, the Callable Notes are part of a series of debt securities, entitled "Medium-Term Notes, Series F," that GSFC may issue under the GSFC 2008 Indenture from time to time. The Callable Notes are listed on the New York Stock Exchange Bonds Market under the ticker symbol "GS/31X."

The payment of principal of, and any interest and premium on, the Callable Notes is fully and unconditionally guaranteed by the Company. The guarantee will remain in effect until the entire principal of, and interest and premium, if any, on, the Callable Notes has been paid in full or discharged in accordance with the provisions of the GSFC 2008 Indenture, or otherwise fully defeased by GSFC or by the Company. The guarantee of senior debt securities of GSFC, such as the Callable Notes, will rank equally in right of payment to all senior indebtedness of the Company.

Payment of Principal on Stated Maturity Date

GSFC will pay holders of Callable Notes that have not been redeemed by the stated maturity date an amount in cash equal to the outstanding face amount of such holders' Callable Notes. The stated maturity date of the Callable Notes is May 28, 2031, subject to GSFC's early redemption right. If the stated maturity date falls on a day that is not a business day, payment of principal otherwise due on such day will be made on the next succeeding business day, and no interest on such payment shall accrue for the period from and after the stated maturity date.

Interest Payments

For each fixed rate interest period, the fixed interest rate on the Callable Notes will equal 4.25% per annum. For each floating rate interest period, the floating interest rate on the Callable Notes will be based upon the CMS

spread on the relevant interest determination date for such floating rate interest period and will be a rate per annum equal to:

- if (i) the CMS spread times (ii) 3.25 is greater than or equal to the maximum interest rate, the maximum interest rate;
- if (i) the CMS spread times (ii) 3.25 is less than the maximum interest rate but greater than the minimum interest rate, (i) the CMS spread times (ii) 3.25; or
- if (i) the CMS spread times (ii) 3.25 is equal to or less than the minimum interest rate, the minimum interest rate.

The maximum interest rate on the Callable Notes is 8% per annum. Based on the formula used to calculate the floating interest rate on the Callable Notes, holders will not benefit from any increases in the CMS spread above approximately 2.46%. The minimum interest rate on the Callable Notes is 1% per annum.

The term “fixed rate interest periods” means the periods from and including a fixed rate interest payment date (or the original issue date, in the case of the first fixed rate interest period) to but excluding the next succeeding fixed rate interest payment date, where the term “fixed rate interest payment dates” means February 28, May 28, August 28 and November 28 of each year, commencing on August 28, 2021 and ending on May 28, 2023, subject to adjustments as described below.

The term “floating rate interest periods” means the periods from and including a floating rate interest payment date (or the final fixed rate interest payment date, in the case of the first floating rate interest period) to but excluding the next succeeding floating rate interest payment date (or the stated maturity date, in the case of the final floating rate interest period), where the term “floating rate interest payment dates” means February 28, May 28, August 28, and November 28 of each year, beginning on May 28, 2023 and ending on the stated maturity date, subject to adjustments as described below.

The term “interest determination dates” means, for each floating rate interest period, the second U.S. government securities business day preceding such floating rate interest period.

The term “CMS spread” means, on any interest determination date, the 30-year CMS rate minus the 5-year CMS rate, as described under “— CMS Rate” below.

The calculation agent will calculate the amount of interest payable on each fixed rate interest payment date and floating rate interest payment date (each, an “interest payment date”) for the applicable fixed rate interest period and floating rate interest period (each, an “interest period”) in the following manner. For each \$1,000 face amount of the Callable Notes and for each interest period, the calculation agent will calculate the amount of interest to be paid by calculating the product of (i) the \$1,000 face amount times (ii) the applicable fixed interest rate or floating interest rate times (iii) the applicable day count convention on a 30/360 (ISDA) basis.

Interest will be paid on the Callable Notes on each quarterly interest payment date. If an interest payment date (other than the interest payment date that falls on the stated maturity date) falls on a day that is not a business day, the payment due on such interest payment date will be postponed to the next day that is a business day; *provided* that interest due with respect to such interest payment date shall not accrue from and including such interest payment date to and including the date of payment of such interest as so postponed. If the stated maturity date falls on a day that is not a business day, payment of interest otherwise due on such day will be made on the next succeeding business day, and no interest on such payment shall accrue for the period from and after the stated maturity date.

CMS Rate

References to the 30-year CMS rate or the 5-year CMS rate on an interest determination date mean the rate appearing on the Refinitiv page ICESWAP1 for 30-year or 5-year index maturity, as the case may be, as of approximately 11:00 A.M., New York City time, on such interest determination date. If a CMS rate cannot be determined in this manner on the relevant interest determination date, the following procedures will apply to the Callable Notes.

If the calculation agent determines on an interest determination date that a CMS rate has been discontinued, then the calculation agent will use a substitute or successor rate that it has determined in its sole discretion is most comparable to the applicable CMS rate, provided that if the calculation agent determines there is an industry-accepted successor rate, then the calculation agent shall use such successor rate. If the calculation agent has determined a substitute or successor rate in accordance with the foregoing, the calculation agent in its sole discretion may determine the business day convention, the applicable business days and the interest determination dates to be used, and any other relevant methodology for calculating such substitute or successor rate, including any adjustment factor needed to make such substitute or successor rate comparable to the applicable CMS rate, in a manner that is consistent with any industry-accepted practices for such substitute or successor rate.

Unless the calculation agent uses a substitute or successor rate as so provided, if the CMS rate cannot be determined in the manner described above, the applicable CMS rate for that interest determination date will be determined by the calculation agent, after consulting such sources as it deems comparable to the foregoing display page, or any other source it deems reasonable, in its sole discretion.

The applicable CMS rate will be subject to the corrections, if any, published on the Refinitiv page ICESWAP1 within one hour of the time that rate was first displayed on such source.

The term “Refinitiv page ICESWAP1” means the display on the Refinitiv Eikon service, or any successor or replacement service, on the page ICESWAP1, or any successor or replacement page on that service.

Manner of Payment

Any payment on the Callable Notes at maturity or upon redemption will be made to an account designated by the holder of such Callable Notes and approved by GSFC, or at the office of the trustee in New York City, but only when such Callable Notes are surrendered to the trustee at that office. GSFC may pay interest on any interest payment date by check mailed to the person who is the holder on the regular record date. GSFC also may make any payment in accordance with the applicable procedures of the depository.

Modified Business Day

Any payment on the Callable Notes that would otherwise be due on a day that is not a business day may instead be paid on the next day that is a business day, with the same effect as if paid on the original due date. For the Callable Notes, however, the term business day may have a different meaning than it does for other Series F medium-term notes, as described under “— Special Calculation Provisions” below.

Role of Calculation Agent

The calculation agent in its sole discretion will make all determinations regarding the CMS spread, the 30-year CMS rate, the 5-year CMS rate, the interest determination dates, the regular record dates, the interest payable on each interest payment date, U.S. government securities business days, business days, postponement of the stated maturity date and the amount payable on the Callable Notes at maturity or redemption, as applicable. Absent manifest error, all determinations of the calculation agent will be final and binding on holders of the Callable Notes and GSFC, without any liability on the part of the calculation agent.

Goldman Sachs & Co. LLC ("GS&Co."), an affiliate of GSFC, is currently serving as the calculation agent as of the date of this description. GSFC may change the calculation agent for the Callable Notes at any time after the date of this prospectus supplement without notice and GS&Co. may resign as calculation agent at any time upon 60 days' written notice to GSFC.

Early Redemption Right

GSFC may redeem the Callable Notes, at GSFC's option, in whole but not in part, on the interest payment date falling on May 28, 2022 and on each interest payment date occurring thereafter, for an amount equal to 100% of the face amount plus any accrued and unpaid interest to, but excluding, the redemption date.

If GSFC chooses to exercise its early redemption right, it will notify the holder of the Callable Notes and the trustee by giving at least five business days' prior notice. The day GSFC gives the notice, which will be a business day, will be the redemption notice date and the immediately following interest payment date, which GSFC will state in the redemption notice, will be the redemption date. GSFC will not give a redemption notice that results in a redemption date later than the stated maturity date.

If GSFC gives the holder a redemption notice, GSFC will redeem the entire outstanding face amount of such holder's Callable Notes. On the redemption date, GSFC will pay to the holder of record on the business day immediately preceding the redemption date, the redemption price in cash, together with any accrued and unpaid interest to, but excluding, the redemption date, in the manner described under "—Manner of Payment" above.

Special Calculation Provisions

The term "business day" with respect to the Callable Notes means each Monday, Tuesday, Wednesday, Thursday and Friday that is not a day on which banking institutions in New York City generally are authorized or obligated by law, regulation or executive order to close.

References to a U.S. government securities business day with respect to the Callable Notes mean any day except for a Saturday, Sunday or a day on which the Securities Industry and Financial Markets Association recommends that the fixed income department of its members be closed for the entire day for purposes of trading in U.S. government securities.

Defeasance and Covenant Defeasance

The provisions for full defeasance and covenant defeasance in the GSFC 2008 Indenture do not apply to the Callable Notes.

Default, Remedies and Waiver of Default

A holder will have special rights if an event of default with respect to his or her series of debt securities occurs and is continuing, as described in this subsection.

Events of Default

References to an event of default with respect to any series of debt securities mean any of the following:

- GSFC or the Company does not pay the principal or any premium on any debt security of that series within 30 days after the due date;
- GSFC or the Company does not pay interest on any debt security of that series within 30 days after the due date; or

- GSFC files for bankruptcy or other events of bankruptcy, insolvency or reorganization relating to GSFC occur. Those events must arise under U.S. federal or state law, unless GSFC merges, consolidates or sells its assets as described above and the successor firm is a non-U.S. entity. If that happens, then those events must arise under U.S. federal or state law or the law of the jurisdiction in which the successor firm is legally organized.

As described below under “Remedies If an Event of Default Occurs,” under the GSFC 2008 Indenture, events of bankruptcy, insolvency or reorganization relating to the Company will not cause any of GSFC’s debt securities issued under such indenture to be automatically accelerated. In the event that the Company becomes subject to certain events of bankruptcy, insolvency or reorganization (but GSFC does not), any series of debt securities issued under the GSFC 2008 Indenture will not be immediately due and repayable. In addition, under the GSFC 2008 Indenture, a breach of a covenant or warranty by the Company (including, for example, a breach of the Company’s covenants and warranties with respect to mergers and similar transactions or restrictions on liens) will not have the potential to cause any of GSFC’s debt securities issued under the GSFC 2008 Indenture to be declared due and payable immediately. Instead, under the GSFC 2008 Indenture, the trustee or holder will need to wait until the earlier of the time that (i) GSFC itself becomes subject to certain events of bankruptcy, insolvency or reorganization or otherwise defaults on the terms of the debt securities, (ii) the Company otherwise defaults on the terms of the debt securities and (iii) the final maturity of the debt securities. The return the holder receives on any series of debt securities issued under the GSFC 2008 Indenture may be significantly less than what a holder would have otherwise received had the debt securities been automatically accelerated upon certain events of bankruptcy, insolvency or reorganization relating to the Company or declared due and payable immediately following the breach of a covenant or warranty by the Company.

Covenant Breaches

References to a covenant breach with respect to any series of debt securities mean any of the following:

- GSFC or the Company does not deposit a required sinking fund payment with regard to any debt security of that series on the due date;
- GSFC remains in breach of any other covenant it makes in the GSFC 2008 Indenture for the benefit of the relevant series for 60 days after GSFC and the Company receive a notice of default stating that GSFC is in breach and requiring GSFC to remedy the breach. The notice must be sent by the trustee or the holders of at least 10% in principal amount of the relevant series of debt securities then outstanding;
- Except as provided by the GSFC 2008 Indenture, the debt security of that series and the related guarantee, the guarantee ceases to be effective, or a court finds the guarantee to be unenforceable or invalid, or the Company denies its obligations as the guarantor.

A covenant breach shall not be an event of default with respect to any security.

Remedies If an Event of Default or Covenant Breach Occurs

If an event of default has occurred with respect to any series of debt securities and has not been cured or waived, the trustee or the holders of not less than 25% in principal amount of all debt securities of that series then outstanding may declare the entire principal amount of the debt securities of that series to be due immediately. If the event of default occurs because of events in bankruptcy, insolvency or reorganization relating to GSFC, the entire principal amount of the debt securities of that series will be automatically accelerated, without any action by the trustee or any holder.

Each of the situations described above is called an acceleration of the stated maturity of the affected series of debt securities. If the stated maturity of any series is accelerated and a judgment for payment has not yet been obtained, the holders of a majority in principal amount of the debt securities of that series may cancel the acceleration for the entire series.

If an event of default or a covenant breach occurs, the trustee will have special duties. In that situation, the trustee will be obligated to use those of its rights and powers under the GSFC 2008 Indenture, and to use the same degree of care and skill in doing so, that a prudent person would use in that situation in conducting his or her own affairs.

Except as described in the prior paragraph, the trustee is not required to take any action under the GSFC 2008 Indenture at the request of any holders unless the holders offer the trustee reasonable protection from expenses and liability (*i.e.*, an indemnity). If the trustee is provided with an indemnity reasonably satisfactory to it, the holders of a majority in principal amount of all debt securities of the relevant series may direct the time, method and place of conducting any lawsuit or other formal legal action seeking any remedy available to the trustee with respect to that series. These majority holders may also direct the trustee in performing any other action under the GSFC 2008 Indenture with respect to the debt securities of that series.

Before a holder bypasses the trustee and bring its own lawsuit or other formal legal action or take other steps to enforce its rights or protect its interests relating to any debt security, all of the following must occur:

- The holder must give the trustee written notice that an event of default or a covenant breach has occurred, and the event of default or covenant breach must not have been cured or waived;
- The holders of not less than 25% in principal amount of all debt securities of a holder's series must make a written request that the trustee take action because of the default, and they or other holders must offer to the trustee indemnity reasonably satisfactory to the trustee against the cost and other liabilities of taking that action;
- The trustee must not have taken action for 60 days after the above steps have been taken; and
- During those 60 days, the holders of a majority in principal amount of the debt securities of a holder's series must not have given the trustee directions that are inconsistent with the written request of the holders of not less than 25% in principal amount of the debt securities of a holder's series.

A holder is entitled at any time, however, to bring a lawsuit for the payment of money due on his or her debt security on or after its stated maturity (or, if the debt security is redeemable, on or after its redemption date).

Waiver of Default

The holders of not less than a majority in principal amount of the debt securities of any series may waive a default for all debt securities of that series. If this happens, the default will be treated as if it has not occurred. No one can waive a payment default on a holder's debt security, however, without the approval of the particular holder of that debt security.

GSFC and the Company Will Give the Trustee Information About Defaults Annually

GSFC and the Company will furnish to the trustee every year a written statement, respectively, of two of their officers certifying that to their knowledge GSFC or the Company, as the case may be, is in compliance with the GSFC 2008 Indenture and the debt securities issued under it, or else specifying any default under the relevant debt indenture. For the purpose of this paragraph, the term "default" means any event which is, or after notice or lapse of time or both would become, an event of default or covenant breach.

Default Amount on Acceleration

If an event of default occurs and the maturity of the Callable Notes is accelerated, GSFC will pay the default amount in respect of the principal of the Callable Notes at the maturity, instead of the amount payable on the Callable Notes as described earlier.

For the purpose of determining whether the holders of GSFC's Medium-Term Notes Series F, which include the Callable Notes, are entitled to take any action under the GSFC 2008 Indenture, GSFC will treat the outstanding face amount of each Callable Note as the outstanding principal amount of that note. Although the terms of the Callable Notes differ from those of the other Medium-Term Notes Series F, holders of specified percentages in principal amount of all Medium-Term Notes Series F, together in some cases with other series of GSFC's debt securities, will be able to take action affecting all the Medium-Term Notes Series F, including the Callable Notes, except with respect to certain Medium-Term Notes Series F, if the terms of such notes specify that the holders of specified percentages in the principal amount of all such notes must also consent to such action. This action may involve changing some of the terms that apply to the Medium-Term Notes Series F, accelerating the maturity of the Medium-Term Notes Series F, after a default or waiving some of GSFC's obligations under the GSFC 2008 Indenture. In addition, certain changes to the GSFC 2008 Indenture and the Callable Notes that only affect certain debt securities may be made with the approval of holders of a majority of the principal amount of such affected debt securities.

Guarantee by the Company

The Company has fully and unconditionally guaranteed the payment of principal of, and any interest and premium on, the Medium-Term Notes, Series F, which include the Callable Notes, when due and payable, whether at the stated maturity, by declaration of acceleration, call for redemption or otherwise, in accordance with the terms of the security and the GSFC 2008 Indenture. The guarantee will remain in effect until the entire principal of, and interest and premium, if any, on, the debt securities has been paid in full or discharged in accordance with the provisions of the GSFC 2008 Indenture, or otherwise fully defeased by the Company.

The guarantee by the Company of its debt securities issued under the GSFC 2008 Indenture will rank equally in right of payment with all senior indebtedness of the Company.

Mergers and Similar Transactions

GSFC and the Company are generally permitted to merge or consolidate with another corporation or other entity. GSFC and the Company also permitted to sell their assets substantially as an entirety to another corporation or other entity. With regard to any series of debt securities, however, GSFC or the Company may not take any of these actions unless all the following conditions are met:

- If the successor entity in the transaction is not GSFC or the Company, as the case may be, the successor entity must be organized as a corporation, partnership or trust and must expressly assume the obligations of GSFC or the Company under the debt securities of that series and the GSFC 2008 Indenture with respect to that series. The successor entity may be organized under the laws of any jurisdiction, whether in the United States or elsewhere.
- Immediately after the transaction, no default under the debt securities of that series or the related guarantees has occurred and is continuing. For this purpose, "default under the debt securities of that series or the related guarantees" means an event of default or a covenant breach with respect to that series or the related guarantees or any event that would be an event of default with respect to that series or the related guarantees if the requirements for giving GSFC or the Company default notice and for GSFC's or the Company's default having to continue for a specific period of time were disregarded.

If the conditions described above are satisfied with respect to debt securities of any series, neither GSFC nor the Company will need to obtain the approval of the holders of those debt securities in order to merge or consolidate or to sell assets of GSFC or the Company. Also, these conditions will apply only if GSFC or the Company wishes to merge or consolidate with another entity or sell assets of GSFC or the Company

substantially as an entirety to another entity. Neither GSFC nor the Company will need to satisfy these conditions if GSFC or the Company enters into other types of transactions, including any transaction in which GSFC or the Company acquire the stock or assets of another entity, any transaction that involves a change of control of GSFC or the Company but in which GSFC or the Company does not merge or consolidate and any transaction in which GSFC or the Company sells less than substantially all assets of GSFC or the Company. While GSFC is currently a wholly owned subsidiary of the Company, there is no requirement that it remain a subsidiary.

Also, if GSFC or the Company merges, consolidates or sells assets of GSFC or the Company substantially as an entirety and the successor is a non-U.S. entity, neither GSFC nor any successor would have any obligation to compensate a holder for any resulting adverse tax consequences relating to his or her debt securities.

Notwithstanding the foregoing and for the avoidance of doubt, GSFC may sell or transfer its assets substantially as an entirety, in one or more transactions, to one or more entities, provided that GSFC's assets and the assets of its direct or indirect subsidiaries in which it owns a majority of the combined voting power, taken together, are not sold or transferred substantially as an entirety to one or more entities that are not majority-owned subsidiaries of the Company, and the Company may sell or transfer its assets substantially as an entirety in one or more transactions, to one or more entities, provided that the assets of the Company and its direct or indirect subsidiaries in which it owns a majority of the combined voting power, taken together, are not sold or transferred substantially as an entirety to one or more entities that are not such subsidiaries.

Restriction on Liens

In the GSFC 2008 Indenture, the Company promises, with respect to each series of senior debt securities, not to create, assume, incur or guarantee any debt for borrowed money that is secured by a lien on the voting or profit participating equity ownership interests that the Company or any of its subsidiaries own in Goldman Sachs & Co. LLC, or in any subsidiary of the Company that beneficially owns or holds, directly or indirectly, those interests in Goldman Sachs & Co. LLC, unless the Company also secures the senior debt securities of that series on an equal or priority basis with the other secured debt. The promise of the Company, however, is subject to an important exception: it may secure debt for borrowed money with liens on those interests without securing the senior debt securities of any series if its board of directors determines that the liens do not materially detract from or interfere with the value or control of those interests, as of the date of the determination.

Except as noted above, the GSFC 2008 Indenture does not restrict the Company's ability to put liens on its interests in its subsidiaries other than Goldman Sachs & Co. LLC, nor does the indenture restrict the Company's ability to sell or otherwise dispose of its interests in any of its subsidiaries, including Goldman Sachs & Co. LLC. In addition, the restriction on liens in the GSFC 2008 Indenture applies only to liens that secure debt for borrowed money. For example, liens imposed by operation of law, such as liens to secure statutory obligations for taxes or workers' compensation benefits, or liens the Company creates to secure obligations to pay legal judgments or surety bonds, would not be covered by this restriction.

Modification of the Debt Indenture and Waiver of Covenants

There are four types of changes GSFC and the Company can make to the GSFC 2008 Indenture and the debt securities or series of debt securities and related guarantees issued under the GSFC 2008 Indenture.

Changes Requiring Each Holder's Approval

First, there are changes that cannot be made without the approval of the holder of each debt security affected by the change under the GSFC 2008 Indenture. Here is a list of those types of changes:

- change the stated maturity for any principal or interest payment on a debt security;

- reduce the principal amount, the amount payable on acceleration of the stated maturity after a default, the interest rate or the redemption price for a debt security;
- permit redemption of a debt security if not previously permitted;
- impair any right a holder may have to require repayment of its debt security;
- change the currency of any payment on a debt security;
- change the place of payment on a debt security;
- impair a holder's right to sue for payment of any amount due on its debt security;
- reduce the percentage in principal amount of the debt securities of any one or more affected series, taken separately or together, as applicable, and whether comprising the same or different series or less than all of the debt securities of a series, the approval of whose holders is needed to change the applicable debt indenture or those debt securities;
- reduce the percentage in principal amount of the debt securities of any one or more affected series, taken separately or together, as applicable, and whether comprising the same or different series or less than all of the debt securities of a series, the consent of whose holders is needed to waive GSFC's compliance with the applicable debt indenture or to waive defaults; and
- change the provisions of the applicable debt indenture dealing with modification and waiver in any other respect, except to increase any required percentage referred to above or to add to the provisions that cannot be changed or waived without approval of the holder of each affected debt security.

Changes Not Requiring Approval

The second type of change does not require any approval by holders of the debt securities affected. These changes are limited to clarifications and changes that would not adversely affect any debt securities of any series in any material respect. Neither GSFC nor the Company needs any approval to make changes that affect only debt securities to be issued under the applicable indenture after the changes take effect.

GSFC and the Company may also make changes or obtain waivers that do not adversely affect a particular debt security, even if they affect other debt securities. In those cases, neither GSFC nor the Company needs to obtain the approval of the holder of the unaffected debt security; GSFC and the Company need only obtain any required approvals from the holders of the affected debt securities.

Changes Requiring Majority Approval

Any other change to the GSFC 2008 Indenture and the debt securities issued under such debt indenture would require the following approval:

- If the change affects only particular debt securities within a series, it must be approved by the holders of a majority in principal amount of such particular debt securities.
- If the change affects multiple debt securities of one or more series, it must be approved by the holders of a majority in principal amount of all debt securities affected by the change, with all such affected debt securities voting together as one class for this purpose (and by the holders of a majority in principal amount of any affected debt securities that by their terms are entitled to vote separately).

In each case, the required approval must be given by written consent.

This would mean that modification of terms with respect to certain debt securities of a series could be effectuated under the GSFC 2008 Indenture without obtaining the consent of the holders of a majority in principal amount of other securities of such series that are not affected by such modification.

The same majority approval would be required for GSFC to obtain a waiver of any of its covenants in the GSFC 2008 Indenture. GSFC's covenants include the promises GSFC and the Company make about merging and, with respect to the Company, putting liens on GSFC's interests in Goldman Sachs & Co. LLC. If the holders

approve a waiver of a covenant, neither GSFC nor the Company will have to comply with it. The holders, however, cannot approve a waiver of any provision in a particular debt security, or in the GSFC 2008 Indenture as it affects that debt security, that neither GSFC nor the Company can change without the approval of the holder of that debt security as described above in “— Changes Requiring Each Holder’s Approval,” unless that holder approves the waiver.

Special Rules for Action by Holders

When holders take any action under the GSFC 2008 Indenture, such as giving a notice of default, notice of covenant breach, declaring an acceleration, approving any change or waiver or giving the trustee an instruction, GSFC will apply the following rules.

Only Outstanding Debt Securities Are Eligible

Only holders of outstanding debt securities or the outstanding debt securities of the applicable series, as applicable, will be eligible to participate in any action by holders of such debt securities or the debt securities of that series. Also, GSFC will count only outstanding debt securities in determining whether the various percentage requirements for taking action have been met. For these purposes, a debt security will not be “outstanding” if:

- it has been surrendered for cancellation;
- GSFC has deposited or set aside, in trust for its holder, money for its payment or redemption;
- GSFC has fully defeased it; or
- GSFC or one of its affiliates, such as Goldman Sachs & Co. LLC, is the owner.

Determining Record Dates for Action by Holders

GSFC and the Company will generally be entitled to set any day as a record date for the purpose of determining the holders that are entitled to take action under a particular debt indenture. In certain limited circumstances, only the trustee will be entitled to set a record date for action by holders. If GSFC, the Company or the trustee set a record date for an approval or other action to be taken by holders, that vote or action may be taken only by persons or entities who are holders on the record date and must be taken during the period that GSFC specifies for this purpose, or that the trustee specifies if it sets the record date. GSFC, the Company or the trustee, as applicable, may shorten or lengthen this period from time to time.

This period, however, may not extend beyond the 180th day after the record date for the action. In addition, record dates for any global debt security may be set in accordance with procedures established by the depository from time to time. Accordingly, record dates for global debt securities may differ from those for other debt securities.

Form of Callable Notes

The Callable Notes are issued in book-entry form through The Depository Trust Company and represented by a global note. GSFC will not issue definitive notes in exchange for the global note except in limited circumstances.

The Goldman Sachs Group, Inc.
Outside Director _____ RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of RSUs (your “Award”) [that will be granted to you as set forth on your Award Statement]. You should read carefully this entire Award Agreement, which includes the Award Statement and any attached Appendix.

Documents that Govern Your Award; Definitions

1. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement.
2. **Your Award Statement.** The Award Statement delivered to you contains some of your Award’s specific terms. For example, it contains the [applicable] Date[s] of Grant, the [calculation that will be used to determine the] number of RSUs [that will be] awarded to you [on any such Date of Grant] and the Delivery Date.
3. **Definitions.** Unless otherwise defined herein, including in the Definitions Appendix or any other Appendix, capitalized terms have the meanings provided in the Plan.

Delivery of Your RSU Shares

4. **Delivery.** RSU Shares (less applicable withholding) will be delivered in respect of your Outstanding RSUs reasonably promptly (but no more than 30 Business Days) after the Delivery Date listed on your Award Statement. Unless otherwise determined by the Committee, delivery of the RSU Shares will be effected by book-entry credit to your Account and no delivery of RSU Shares will be made unless you have timely established your Account. Until such delivery, you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc.

Dividend Equivalent Rights

5. **Dividend Equivalent Rights.** Each RSU includes a Dividend Equivalent Right, which entitles you to receive an amount (less applicable withholding), at or after the time of distribution of any regular cash dividend paid by GS Inc. in respect of a share of Common Stock, equal to any regular cash dividend payment that would have been made in respect of an RSU Share underlying your Outstanding RSUs for any record date that occurs on or after the [applicable] Date of Grant.

(a) **Accelerated Delivery**

6. **Accelerated Delivery in the Event of Conflicted Employment or Death.** In the event of your Conflicted Employment or death, your Outstanding Award will be treated as described in this Paragraph 6, and all other terms of this Award Agreement continue to apply.

(a) You Are Determined to Have Accepted Conflicted Employment.

(i) Generally. Unless prohibited by applicable law or regulation, if you accept Conflicted Employment, as soon as practicable after the Committee has received satisfactory documentation relating to your Conflicted Employment, RSU Shares will be delivered in respect of your Outstanding RSUs (including in the form of cash as described in Paragraph 7(b)).

(ii) You May Have to Take Other Steps to Address Conflicts of Interest. The Committee retains the authority to exercise its rights under the Award Agreement or the Plan (including Section 1.3.2 of the Plan) to take or require you to take other steps it determines in its

sole discretion to be necessary or appropriate to cure an actual or perceived conflict of interest (which may include a determination that the accelerated delivery described in Paragraph 6(a)(i) will not apply because such actions are not necessary or appropriate to cure an actual or perceived conflict of interest).

(b) **Death.** If you die, the RSU Shares underlying your Outstanding RSUs [(and, notwithstanding anything to the contrary in any applicable award agreement, any previously granted Outstanding RSUs)] will be delivered to your Account as soon as practicable after the date of death and after such documentation as may be requested by the Committee is provided to the Committee.

Other Terms, Conditions and Agreements

7. Additional Terms, Conditions and Agreements.

(a) **You Must Satisfy Applicable Tax Withholding Requirements.** Delivery of RSU Shares is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, provided that the Committee may determine not to apply the withholding rate described in Section 3.2.2 of the Plan.

(b) **Firm May Deliver Cash or Other Property Instead of RSU Shares.** In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the RSU Shares, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to deliveries of RSU Shares will include such deliveries of cash, other securities, other awards under the Plan or other property.

(c) **Firm May Affix Legends and Place Stop Orders on RSU Shares.** GS Inc. may affix to Certificates representing RSU Shares any legend that the Committee determines to be necessary or advisable. GS Inc. may advise the transfer agent to place a stop order against any legended RSU Shares.

(d) **You Agree to Certain Consents.** By accepting this Award, you have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

Non-Transferability

8. **Non-transferability.** Except as otherwise may be provided in this Paragraph 8 or as otherwise may be provided by the Committee, the limitations on transferability set forth in Section 3.5 of the Plan will apply to this Award. Any purported transfer or assignment in violation of the provisions of this Paragraph 8 or Section 3.5 of the Plan will be void. The Committee may adopt procedures pursuant to which you may transfer some or all of your RSUs through a gift for no consideration to any child, stepchild, grandchild, parent, stepparent, grandparent, spouse, sibling, niece, nephew, mother-in-law, father-in-law, son-in-law, daughter-in-law, brother-in-law or sister-in-law, including adoptive relationships, any person sharing the recipient's household (other than a tenant or employee), a trust in which these persons have more than 50% of the beneficial interest, and any other entity in which these persons (or the recipient) own more than 50% of the voting interests.

Governing Law

9. **Governing Law.** This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

10. **Compliance of Award Agreement and Plan with Section 409A.** The provisions of this Paragraph 10 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 10 and the other provisions of this Award Agreement, this Paragraph 10 will govern.

(b) Delivery of RSU Shares will not be delayed beyond the date on which all applicable conditions or restrictions on delivery of RSU Shares required by this Agreement (including those specified in Paragraphs 4, 6(b) and 7 and the consents and other items specified in Section 3.3 of the Plan) are satisfied, and will occur by December 31 of the calendar year in which the Delivery Date occurs unless, in order to permit such conditions or restrictions to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay delivery of RSU Shares to a later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 7(b) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any securities, other Awards or other property that the Firm may deliver in respect of your RSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the RSU Shares that would otherwise have been deliverable (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Notwithstanding the timing provisions of Paragraph 6(b), the delivery of RSU Shares referred to therein will be made after the date of death and during the calendar year that includes the date of death (or on such later date as may be permitted under Section 409A).

(e) Notwithstanding any provision of Paragraph 5 or Section 2.8.2 of the Plan to the contrary, the Dividend Equivalent Rights with respect to each of your Outstanding RSUs will be paid to you within the calendar year that includes the date of distribution of any corresponding regular cash dividends paid by GS Inc. in respect of a share of Common Stock the record date for which occurs on or after the [applicable] Date of Grant. The payment will be in an amount (less applicable withholding) equal to such regular dividend payment as would have been made in respect of the RSU Shares underlying such Outstanding RSUs.

(f) The timing of delivery or payment referred to in Paragraph 6(a)(i) will be the earlier of (i) the Delivery Date or (ii) within the calendar year in which the Committee receives satisfactory documentation relating to your Conflicted Employment, provided that such delivery or payment will be made, and any Committee action referred to in Paragraph 6(a)(i) will be taken, only at such time as, and if and to the extent that it, as reasonably determined by the Firm, would not result in the imposition of any additional tax to you under Section 409A.

(g) Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(h) Delivery of RSU Shares in respect of this Award may be made, if and to the extent elected by the Committee, later than the Delivery Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later delivery is permitted under Section 409A).

(i) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Amendment, Construction and Regulatory Reporting

11. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided*, that, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further*, that the Committee expressly reserves the right to accelerate the delivery of the RSU Shares and in its discretion to provide that such Shares may not be transferable until the Delivery Date. A modification that impacts the tax consequences of this Award or the timing of delivery of RSU Shares will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

12. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words "include," "includes" and "including" will be deemed to be followed by the words "without limitation." The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

13. **Providing Information to the Appropriate Authorities.** In accordance with applicable law, nothing in this Award Agreement or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the [applicable] Date of Grant [for each Award granted hereunder].

THE GOLDMAN SACHS GROUP, INC.

Accepted and Agreed:

By: _____
Print Name:

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings:

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Agreement” means the written document or documents by which each Award is evidenced, including any related Award Statement and signature card.

(c) “Award Statement” means a written statement that reflects certain Award terms.

(d) “Board” means the Board of Directors of GS Inc.

(e) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Committee” means the committee appointed by the Board to administer the Plan pursuant to Section 1.3, and, to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, shall be a committee or subcommittee of the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m), and which, to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3. Unless otherwise determined by the Board, the Committee shall be the Compensation Committee of the Board.

(h) “Common Stock” means common stock of GS Inc., par value \$0.01 per share.

(i) “Conflicted Employment” means the Grantee’s employment at any U.S. Federal, state or local government, any non-U.S. government, any supranational or international organization, any self-regulatory organization, or any agency or instrumentality of any such government or organization, or any other employer determined by the Committee, if, as a result of such employment, the Grantee’s continued holding of any Outstanding Award or Shares at Risk would result in an actual or perceived conflict of interest.

(j) “Date of Grant” means the date specified in the Grantee’s Award Agreement as the date of grant of the Award.

(k) "Delivery Date" means each date specified in the Grantee's Award Agreement as a delivery date, provided, unless the Committee determines otherwise, such date is during a Window Period or, if such date is not during a Window Period, the first trading day of the first Window Period beginning after such date.

(l) "Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.

(m) "Exchange Act" means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(n) "Firm" means GS Inc. and its subsidiaries and affiliates.

(o) "GS Inc." means The Goldman Sachs Group, Inc., and any successor thereto.

(p) "Outstanding" means any Award to the extent it has not been forfeited, canceled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(q) "RSU" means a restricted stock unit Award granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(r) "RSU Shares" means shares of Common Stock that underlie an RSU.

(s) "Section 409A" means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(t) "Window Period" means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (provided that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
Outside Director _____ RSU Award (Cash-Settled)

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of RSUs (your “Award”). You should read carefully this entire Award Agreement, which includes the Award Statement and any attached Appendix.

Documents that Govern Your Award; Definitions

1. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement.
2. **Your Award Statement.** The Award Statement delivered to you contains some of your Award’s specific terms. For example, it contains the Date of Grant, the number of RSUs awarded to you, the Settlement Date and the Payment Date.
3. **Definitions.** Unless otherwise defined herein, including in the Definitions Appendix or any other Appendix, capitalized terms have the meanings provided in the Plan.

Payment of Your RSU Payment Amount

4. **Payment.** The RSU Payment Amount (less applicable withholding) will be paid in respect of your Outstanding RSUs on the Payment Date listed on your Award Statement. Until such payment, you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc.

Dividend Equivalent Rights

5. **Dividend Equivalent Rights.** Each RSU includes a Dividend Equivalent Right, which entitles you to receive an amount (less applicable withholding), at or after the time of distribution of any regular cash dividend paid by GS Inc. in respect of a share of Common Stock, equal to any regular cash dividend payment that would have been made in respect of an RSU Share underlying your Outstanding RSUs for any record date that occurs on or after the Date of Grant.

(a) **Accelerated Payment**

6. **Accelerated Payment in the Event of Conflicted Employment or Death.** In the event of your Conflicted Employment or death, your Outstanding Award will be treated as described in this Paragraph 6, and all other terms of this Award Agreement continue to apply.

(a) **You Are Determined to Have Accepted Conflicted Employment.**

(i) **Generally.** Unless prohibited by applicable law or regulation, if you accept Conflicted Employment, as soon as practicable after the Committee has received satisfactory documentation relating to your Conflicted Employment, the RSU Payment Amount will be paid in respect of your Outstanding RSUs.

(ii) **You May Have to Take Other Steps to Address Conflicts of Interest.** The Committee retains the authority to exercise its rights under the Award Agreement or the Plan (including Section 1.3.2 of the Plan) to take or require you to take other steps it determines in its sole discretion to be necessary or appropriate to cure an actual or perceived conflict of interest (which may include a determination that the accelerated payment described in Paragraph 6(a)(i) will not apply because such actions are not necessary or appropriate to cure an actual or perceived conflict of interest).

(b) **Death.** If you die, the payment of the RSU Payment Amount with respect to your Outstanding RSUs [(and, notwithstanding anything to the contrary in any applicable award agreement, any previously granted Outstanding RSUs)] will be delivered to your Account as soon as practicable after the date of death and after such documentation as may be requested by the Committee is provided to the Committee.

Other Terms, Conditions and Agreements

7. Additional Terms, Conditions and Agreements.

(a) **You Must Satisfy Applicable Tax Withholding Requirements.** Payment of the RSU Payment Amount is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, provided that the Committee may determine not to apply the withholding rate described in Section 3.2.2 of the Plan.

(b) **You Agree to Certain Consents.** By accepting this Award, you have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

Non-Transferability

8. **Non-transferability.** Except as otherwise may be provided in this Paragraph 8 or as otherwise may be provided by the Committee, the limitations on transferability set forth in Section 3.5 of the Plan will apply to this Award. Any purported transfer or assignment in violation of the provisions of this Paragraph 8 or Section 3.5 of the Plan will be void. The Committee may adopt procedures pursuant to which you may transfer some or all of your RSUs through a gift for no consideration to any child, stepchild, grandchild, parent, stepparent, grandparent, spouse, sibling, niece, nephew, mother-in-law, father-in-law, son-in-law, daughter-in-law, brother-in-law or sister-in-law, including adoptive relationships, any person sharing the recipient's household (other than a tenant or employee), a trust in which these persons have more than 50% of the beneficial interest, and any other entity in which these persons (or the recipient) own more than 50% of the voting interests.

Governing Law

9. **Governing Law.** This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

10. **Compliance of Award Agreement and Plan with Section 409A.** The provisions of this Paragraph 10 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 10 and the other provisions of this Award Agreement, this Paragraph 10 will govern.

(b) Payment of the RSU Payment Amount in respect of your RSUs will not be delayed beyond the date on which all applicable conditions or restrictions on payment required by

this Agreement (including those specified in Paragraphs 4, 6(b) and 7 and the consents and other items specified in Section 3.3 of the Plan) are satisfied, and will occur by December 31 of the calendar year in which the Payment Date occurs unless, in order to permit such conditions or restrictions to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay payment of the RSU Payment Amount to a later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any payments that the Firm may make in respect of your RSUs will not have the effect of deferring payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the RSU Payment Amount that would otherwise have been payable (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Notwithstanding the timing provisions of Paragraph 6(b), the Payment of the RSU Payment Amount referred to therein will be made after the date of death and during the calendar year that includes the date of death (or on such later date as may be permitted under Section 409A).

(e) Notwithstanding any provision of Paragraph 5 or Section 2.8.2 of the Plan to the contrary, the Dividend Equivalent Rights with respect to each of your Outstanding RSUs will be paid to you within the calendar year that includes the date of distribution of any corresponding regular cash dividends paid by GS Inc. in respect of a share of Common Stock the record date for which occurs on or after the Date of Grant. The payment will be in an amount (less applicable withholding) equal to such regular dividend payment as would have been made in respect of the RSU Shares underlying such Outstanding RSUs.

(f) The timing of payment referred to in Paragraph 6(a)(i) will be the earlier of (i) the Payment Date or (ii) within the calendar year in which the Committee receives satisfactory documentation relating to your Conflicted Employment, provided that such payment will be made, and any Committee action referred to in Paragraph 6(a)(i) will be taken, only at such time as, and if and to the extent that it, as reasonably determined by the Firm, would not result in the imposition of any additional tax to you under Section 409A.

(g) Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(h) Payment of the RSU Payment Amount in respect of any RSUs may be made, if and to the extent elected by the Committee, later than the Payment Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later payment is permitted under Section 409A).

(i) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the payment.

Amendment, Construction and Regulatory Reporting

11. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided*, that, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially

adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further*, that the Committee expressly reserves the right to accelerate the payment of the RSU Payment Amount and in its discretion to provide that such payment may not be transferable until the Payment Date. A modification that impacts the tax consequences of this Award or the timing of payment of the RSU Payment Amount will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

12. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words “include,” “includes” and “including” will be deemed to be followed by the words “without limitation.” The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

13. **Providing Information to the Appropriate Authorities.** In accordance with applicable law, nothing in this Award Agreement or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

Accepted and Agreed:

By: _____
Print Name:

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings:

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Agreement” means the written document or documents by which each Award is evidenced, including any related Award Statement and signature card.

(c) “Award Statement” means a written statement that reflects certain Award terms.

(d) “Board” means the Board of Directors of GS Inc.

(e) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Committee” means the committee appointed by the Board to administer the Plan pursuant to Section 1.3, and, to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, shall be a committee or subcommittee of the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m), and which, to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3. Unless otherwise determined by the Board, the Committee shall be the Compensation Committee of the Board.

(h) “Common Stock” means common stock of GS Inc., par value \$0.01 per share.

(i) “Conflicted Employment” means the Grantee’s employment at any U.S. Federal, state or local government, any non-U.S. government, any supranational or international organization, any self-regulatory organization, or any agency or instrumentality of any such government or organization, or any other employer determined by the Committee, if, as a result of such employment, the Grantee’s continued holding of any Outstanding Award or Shares at Risk would result in an actual or perceived conflict of interest.

(j) “Date of Grant” means the date specified in the Grantee’s Award Agreement as the date of grant of the Award.

(k) “Dividend Equivalent Right” means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.

(l) “Exchange Act” means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(m) “Fair Market Value” means, with respect to a share of Common Stock on any day, the fair market value as determined in accordance with a valuation methodology approved by the Committee.

(n) “Firm” means GS Inc. and its subsidiaries and affiliates.

(o) “GS Inc.” means The Goldman Sachs Group, Inc., and any successor thereto.

(p) “Outstanding” means any Award to the extent it has not been forfeited, canceled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(q) “RSU” means a restricted stock unit Award granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(r) “RSU Payment Amount” means the Fair Market Value of a share of Common Stock at the close of trading on the Settlement Date.

(s) “RSU Shares” means shares of Common Stock that underlie an RSU.

(t) “Section 409A” means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(u) “Settlement Date” means the last trading day prior to the Payment Date; provided that, if the Committee determines that such date is impracticable under the circumstances, the Committee may designate a trading day prior to the Payment Date that is practicable.

(v) “Window Period” means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (provided that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
[One-Time][Year-End] RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of RSUs (your “Award”). You should read carefully this entire document (including its [Appendix][Appendices]), the Award Statement and the Signature Card (together, the “Award Agreement”), as well as the other documentation presented to you in connection with acceptance of your Award (collectively, “Award Documentation”).

Acceptance

1. **You Must Decide Whether to Accept Your Award.** To be eligible for your Award, you must by the Acceptance Deadline: (a) open and activate an Account; (b) accept any Additional Terms presented to you as part of the Award Documentation; and (c) execute (including by electronic means) the accompanying Signature Card in accordance with its instructions. By executing the Signature Card, you confirm your agreement to all of the terms of the Award Agreement (including the Award Statement and the Signature Card), as well as any Additional Terms.

Documents that Govern Your Award; Definitions

2. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement. The Plan and its Prospectus are included with the Award Documentation.

3. **Your Award Statement.** The accompanying Award Statement contains the number of RSUs awarded to you and some of your Award’s specific terms. For example, it specifies any applicable Vesting Dates, Delivery Dates and Transferability Dates.

4. **Definitions.** Unless otherwise defined herein, including in the **Definitions Appendix** or any other Appendix, capitalized terms have the meanings provided in the Plan.

Vesting of Your RSUs

5. **Vesting.** On each Vesting Date listed on your Award Statement, you will become Vested in the amount of Outstanding RSUs listed next to that date. When an RSU becomes Vested, it means **only** that your continued active Employment is not required for delivery of that portion of RSU Shares. **Vesting does not mean you have a non-forfeitable right to the Vested portion of your Award. The terms of this Award Agreement (including conditions to delivery and any applicable Transfer Restrictions) continue to apply to Vested RSUs, and you can still forfeit Vested RSUs and any RSU Shares.**

Delivery of Your RSU Shares

6. **Delivery.** Reasonably promptly (but no more than 30 Business Days) after each Delivery Date identified on your Award Statement, RSU Shares (less applicable withholding as described in Paragraph 13(b)) will be delivered (by book entry credit to your Account) in respect of the amount of Outstanding RSUs listed next to that date. The Committee may select multiple dates within the 30-Business-Day period following the Delivery Date to deliver RSU Shares in respect of all or a portion of the RSUs with the same Delivery Date listed on the Award Statement, and all such dates will be treated as a single Delivery Date for purposes of this Award. Until such delivery, you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc. Without limiting the Committee’s authority under Section 1.3.2(h) of the Plan, the Firm may accelerate any Delivery Date by up to 30 days.

Transfer Restrictions Following Delivery

7. **Transfer Restrictions and Shares at Risk.** Within 30 Business Days after any applicable Transferability Date identified on your Award Statement, GS Inc. will remove the Transfer Restrictions on any Shares at Risk in respect of the amount of Outstanding RSUs listed next to that date. If your Award Statement reflects a Transferability Date, [% of the RSU Shares delivered on any Delivery Date **before** tax withholding (or, if the tax withholding rate is greater than 50%, all RSU Shares)][all of the RSU Shares delivered on any Delivery Date **after** tax withholding] will be Shares at Risk. The Committee may select multiple dates within the 30-Business-Day period on which to remove Transfer Restrictions for all or a portion of the Shares at Risk with the same Transferability Date, and all such dates will be treated as a single Transferability Date for purposes of this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition in violation of the Transfer Restrictions on Shares at Risk will be void. **[If your Award Statement does not reflect any Transferability Date, then references to Transfer Restrictions and Shares at Risk in this Award Agreement are inapplicable to your Award and this Paragraph 7 does not apply.]**

Dividends

8. **[Dividend Equivalent Rights and] Dividends.** [Each RSU includes a Dividend Equivalent Right, which entitles you to receive an amount (less applicable withholding), at or after the time of distribution of any regular cash dividend paid by GS Inc. in respect of a share of Common Stock, equal to any regular cash dividend payment that would have been made in respect of an RSU Share underlying your Outstanding RSUs for any record date that occurs on or after the Date of Grant. In addition, you][You] will be entitled to receive on a current basis any regular cash dividend paid in respect of any Shares at Risk. [The RSUs do **not** include Dividend Equivalent Rights.]

Forfeiture of Your Award

9. **How You May Forfeit Your Award.** This Paragraph 9 sets forth the events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. More than one event may apply, and in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event. Paragraph 11 (relating to certain circumstances under which you will not forfeit your unvested RSUs upon Employment termination) and Paragraph 12 (relating to certain circumstances under which vesting, delivery and/or release of Transfer Restrictions may be accelerated) provide for exceptions to one or more provisions of this Paragraph 9.

(a) Unvested RSUs Forfeited if Your Employment Terminates. If your Employment terminates for any reason or you are otherwise no longer actively Employed with the Firm (which includes off-premises notice periods, "garden leaves," pay in lieu of notice or any other similar status), your rights to your Outstanding RSUs that are not Vested will terminate, and no RSU Shares will be delivered in respect of such RSUs.

(b) Vested and Unvested RSUs Forfeited [if You Solicit Clients or Employees, Interfere with Client or Employee Relationships or Participate in the Hiring of Employees][Upon Certain Events]. If any of the following occurs before the applicable Delivery Date, [except as prohibited by applicable law] your rights to all of your Outstanding RSUs (whether or not Vested) will terminate, and no RSU Shares will be delivered in respect of such RSUs:

(i) [You Solicit Clients or Employees, Interfere with Client or Employee Relationships or Participate in the Hiring of Employees. Except as prohibited by applicable law, either:]

(A) you, in any manner, directly or indirectly, [(A)][(1)] Solicit any Client to transact business with a Covered Enterprise or to reduce or refrain from doing any business with the Firm, [(B)][(2)] interfere with or damage (or attempt to interfere with or

damage) any relationship between the Firm and any Client, (C)[(3)] Solicit any person who is an employee of the Firm to resign from the Firm, [(D)][(4)] Solicit any Selected Firm Personnel to apply for or accept employment (or other association) with any person or entity other than the Firm or (E) participate in the hiring of any Selected Firm Personnel by any person or entity other than the Firm (including, without limitation, participating in the identification of individuals for potential hire, and participating in any hiring decision), whether as an employee or consultant or otherwise, or

(B) Selected Firm Personnel are Solicited, hired or accepted into partnership, membership or similar status by any entity where you have, or will have, direct or indirect managerial responsibility for such Selected Firm Personnel, unless the Committee determines that you were not involved in such Solicitation, hiring or acceptance.

(ii) [GS Inc. Fails to Maintain the Minimum Tier 1 Capital Ratio]. GS Inc. fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days.]

(iii) [GS Inc. Is Determined to Be in Default]. The Board of Governors of the Federal Reserve or the Federal Deposit Insurance Corporation (the "FDIC") makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Wall Street Reform and Consumer Protection Act for the appointment of the FDIC as a receiver of GS Inc. based on a determination that GS Inc. is "in default" or "in danger of default."]

(c) Vested and Unvested RSUs and Shares at Risk Forfeited upon Certain Events. If any of the following occurs (i) your rights to all of your Outstanding RSUs (whether or not Vested) will terminate, and no RSU Shares will be delivered in respect of such RSUs and (ii) your rights to any Shares at Risk will terminate and all such Shares at Risk will be cancelled, in each case, as may be further described below:

(i) You Failed to Consider Risk. You Failed to Consider Risk during ____.

(ii) Your Conduct Constitutes Cause. Any event that constitutes Cause has occurred before the applicable Delivery Date for RSUs or the applicable Transferability Date for any Shares at Risk.

(iii) You Breach an Obligation to the Firm. The Committee determines that, before the applicable Delivery Date for RSUs or the applicable Transferability Date for any Shares at Risk, you Breached an Obligation to the Firm.

(iv) You Do Not Provide Timely Certifications or Comply with Your Certifications. You fail to certify to GS Inc. that you have complied with all of the terms of the Plan and this Award Agreement as required by Paragraph 13(d), or the Committee determines that you have failed to comply with a term of the Plan or this Award Agreement to which you have certified compliance.

(v) You Do Not Follow Dispute Resolution Procedures. You attempt to have any dispute, controversy or claim under the Plan or this Award Agreement resolved in any manner that is not provided for by Paragraph 16 or Section 3.17 of the Plan, or you attempt to arbitrate a dispute, controversy or claim without first having exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(vi) You Bring an Action that Results in a Determination that Any Award Agreement Term Is Invalid. As a result of any action brought by you, it is determined that any term of this Award Agreement is invalid.

(vii) You Receive Compensation in Respect of Your Award from Another Employer. Your Employment terminates for any reason or you otherwise are no longer actively Employed with the Firm and another entity grants you cash, equity or other property (whether vested or unvested) to replace, substitute for or otherwise in respect of any Outstanding RSUs or Shares at Risk; *provided, however*, that your rights will only be terminated in respect of the RSUs and Shares at Risk that are replaced, substituted for or otherwise considered by such other entity in making its grant.

(d) [Additional Forfeiture and Repayment Conditions for Material Risk Takers]. [If at any point in ____][Since] the Firm identified you as a Material Risk Taker (“MRT”) [in ____] then the **EU and UK Material Risk Taker Appendix** and/or the **GSBE Material Risk Taker Appendix** (as applicable to you, the “MRT Appendix”) supplements this Paragraph 9. The MRT Appendix sets forth additional events that (i) result in forfeiture of up to all of your RSUs and Shares at Risk and (ii) may require repayment to the Firm of up to all amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. Your MRT classification, as defined in the MRT Appendix, was communicated to you separately. [If you are unsure of whether the Firm identified you as an MRT, or which classification(s) apply, you must contact HCM prior to accepting this Award.] The Firm’s records with respect to such classification(s) will be controlling as to whether and which MRT Appendix applies to your Award. The MRT classifications with their corresponding MRT Appendix are as follows:

MRT Classification	MRT Appendix
EU MRT, GSAMI MRT, GSAMI Senior Management and/or UK MRT	EU and UK Material Risk Taker Appendix
GSBE MRT and GSBE Senior Management	GSBE Material Risk Taker Appendix
GSBE MRT or GSBE Senior Management <i>and</i> any combination of EU MRT, GSAMI MRT, GSAMI Senior Management or UK MRT	GSBE Material Risk Taker Appendix <i>and</i> EU and UK Material Risk Taker Appendix]

(e) Actions Pending Forfeiture. In connection with any investigation as to whether any of the events that would result in forfeiture under the Plan [or][,] this Paragraph 9 [or the MRT Appendix] have occurred, the Firm reserves the right to (i) suspend vesting of Outstanding RSUs, [payments under Dividend Equivalent Rights,] delivery of RSU Shares or release of any Transfer Restrictions; (ii) deliver any RSU Shares[, payments under Dividend Equivalent Rights] or dividends into an escrow account in accordance with Paragraph 13(e)(iii); or (iii) apply Transfer Restrictions to any RSU Shares.

Repayment of Your Award

10. **When You May Be Required to Repay Your Award**. If the Committee determines that any term of this Award was not satisfied, you will be required, immediately upon demand therefor, to repay to the Firm the following:

(a) Any RSU Shares (which, for the avoidance of doubt, includes any Shares at Risk) for which the terms (including the terms for delivery) of the related RSUs were not satisfied, in accordance with Section 2.6.3 of the Plan.

(b) Any Shares at Risk for which the terms (including the terms for the release of Transfer Restrictions) were not satisfied, in accordance with Section 2.5.3 of the Plan.

(c) Any RSU Shares that were delivered (but not subject to Transfer Restrictions) at the same time any Shares at Risk that are cancelled or required to be repaid were delivered.

(d) [Any payments under Dividend Equivalent Rights for which the terms were not satisfied (including any such payments made in respect of RSUs that are forfeited or RSU Shares that are cancelled or required to be repaid), in accordance with Section 2.8.3 of the Plan.]

(e) Any dividends paid in respect of any RSU Shares that are cancelled or required to be repaid.

(f) Any amount applied to satisfy tax withholding or other obligations with respect to any RSUs, RSU Shares[,] [and] dividend payments [and payments under Dividend Equivalent Rights] that are forfeited or required to be repaid.

Exceptions to the Vesting, Delivery and/or Transferability Dates

11. **Circumstances Under Which You Will Not Forfeit Your Unvested RSUs on Employment Termination (but the Original Delivery Dates and Transferability Dates Continue to Apply)**. If your Employment terminates at a time when you meet the requirements for [Retirement,] Extended Absence[,] [“downsizing”] [or] [Approved Termination], [each] as described below, then Paragraph 9(a) will not apply, and your Outstanding RSUs will be treated as described in this Paragraph 11. All other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

(a) [Retirement or] Extended Absence and No Association With a Covered Enterprise.

(i) **Generally.** If your Employment terminates by [Retirement or] Extended Absence, your Outstanding RSUs that are not Vested will be treated as Vested. **However, your rights to any Outstanding RSU that are treated as Vested by this Paragraph 11(a)(i) will terminate and no RSU Share will be delivered in respect of that RSU if you Associate With a Covered Enterprise on or before the originally scheduled Vesting Date for that RSU.** You acknowledge and agree that if any portion of the preceding sentence is unenforceable under applicable law, or is found to be unenforceable, then the entirety of this Paragraph 11(a)(i) is inapplicable to you and your rights to any Outstanding RSUs that were treated as Vested, or otherwise would have been treated as Vested, will be cancelled in accordance with Paragraph 9(a).

(ii) Special Treatment for Involuntary or Mutual Agreement Termination. The second sentence of Paragraph 11(a)(i) (relating to forfeiture if you Associate With a Covered Enterprise) will not apply if (A) the Firm characterizes your Employment termination as “involuntary” or by “mutual agreement” (and, in each case, you have not engaged in conduct constituting Cause) and (B) you execute a general waiver and release of claims and an agreement to pay any associated tax liability, in each case, in the form the Firm prescribes. No Employment termination that you initiate, including any purported “constructive termination,” a “termination for good reason” or similar concepts, can be “involuntary” or by “mutual agreement.”

(b) [Downsizing. If (i) the Firm terminates your Employment solely by reason of a “downsizing” (and you have not engaged in conduct constituting Cause) and (ii) you execute a general waiver and release of claims and an agreement to pay any associated tax liability, in each case, in the form the Firm prescribes, your Outstanding RSUs that are not yet Vested will become Vested. Whether or not your Employment is terminated solely by reason of a “downsizing” will be determined by the Firm in its sole discretion.]

(c) [Approved Terminations of Fixed-Term Employees. If the Firm classifies you as a “fixed-term” employee and your Employment terminates solely by reason of an Approved Termination (and you have not engaged in conduct constituting Cause), your Outstanding RSUs that are not yet Vested will become Vested.]

12. **Accelerated Vesting, Delivery and/or Release of Transfer Restrictions in the Event of [a Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] Death.** In the event of your [Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] death, [each] as described below, then Paragraph 9(a) will not apply, your Outstanding RSUs and any Shares at Risk will be treated as described in this Paragraph 12, and, except as set forth in Paragraph

12(a), all other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

(a) [You Have a Qualifying Termination After a Change in Control]. If your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, the RSU Shares underlying your Outstanding RSUs (whether or not Vested) will be delivered, and any Transfer Restrictions will cease to apply. In addition, the forfeiture events in Paragraph 9 [(except the MRT Appendix)] will not apply to your Award.]

(b) [You Are Determined to Have Accepted Conflicted Employment].

(i) Generally. To the extent consistent with applicable law, regulation and regulatory guidance, the following will apply as soon as practicable after the Committee has received satisfactory documentation relating to your Conflicted Employment.

(A) Vesting. If your Employment terminates solely because you resign to accept Conflicted Employment and, according to the Firm's records, you have completed at least three years of continuous service with the Firm as of the date your Employment terminates, your Outstanding RSUs will Vest; otherwise, you will forfeit any Outstanding RSUs that are not Vested in accordance with Paragraph 9(a).

(B) Delivery and Release of Transfer Restrictions. If your Employment terminates solely because you resign to accept Conflicted Employment or if, following your termination of Employment, you notify the Firm that you are accepting Conflicted Employment, RSU Shares will be delivered in respect of your Outstanding Vested RSUs (including in the form of cash as described in Paragraph 13(e)(ii)) and any Transfer Restrictions will cease to apply.

(ii) You May Have to Take Other Steps to Address Conflicts of Interest. The Committee retains the authority to exercise its rights under the Award Agreement or the Plan (including Section 1.3.2 of the Plan) to take or require you to take other steps it determines in its sole discretion to be necessary or appropriate to cure an actual or perceived conflict of interest (which may include a determination that the accelerated vesting, delivery and/or release of Transfer Restrictions described in Paragraph 12(b)(i) will not apply because such actions are not necessary or appropriate to cure an actual or perceived conflict of interest).]

(c) Death. If you die, the RSU Shares underlying your Outstanding RSUs (whether or not Vested) will be delivered to your Account and any Transfer Restrictions will cease to apply as soon as practicable after the date of death and receipt of any documentation as may be requested by the Committee or a SIP Administrator.

Other Terms and Conditions of Your Award

13. Additional Terms, Conditions and Consents.

(a) [You Will Be Required to Accept Additional Terms]. In addition to the terms of the Award Agreement, the Award Documentation includes other contractual terms that you must accept as a condition of the Award (the "Additional Terms"). By accepting the Award and executing the Signature Card, you agree to any such Additional Terms applicable to you as set forth in the Award Documentation, which will include but are not limited to the following:

(i) Notice Period Policy. You must agree to abide by the terms, as applicable to you, of the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy").

(ii) Agreement Regarding Arbitration of Employment-Related Matters. You must agree to the Agreement Regarding Arbitration of Employment-Related Matters, which imposes obligations in addition to those contained in Paragraph 16.

(iii) APAC Restrictive Covenants. If you are employed by the Firm in the Asia-Pacific Region (e.g., Australia, the People's Republic of China, the Hong Kong Special Administrative Region, India, Indonesia, Japan, the Republic of Korea, New Zealand, Singapore, Taiwan), you must agree to the APAC Restrictive Covenants Agreement.

(iv) Role-, Business- or Division-Specific Agreements. If applicable to you, you must accept Additional Terms related to your specific role, business unit or division as set forth in the Award Documentation.]

(b) You Must Satisfy Applicable Tax Withholding Requirements. Delivery of RSU Shares is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, which includes the Firm deducting or withholding amounts from any payment or distribution to you. In addition, to the extent permitted by applicable law, the Firm, in its sole discretion, may require you to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations imposed on you or the Firm in connection with the grant, Vesting or delivery of this Award by requiring you to choose between remitting the amount (i) in cash (or through payroll deduction or otherwise) or (ii) in the form of proceeds from the Firm's executing a sale of RSU Shares delivered to you under this Award. In no event, however, does this Paragraph 13(b) give you any discretion to determine or affect the timing of the delivery of RSU Shares or the timing of payment of tax obligations.

(c) You Must Comply with Applicable Deadlines and Procedures to Appeal. If you disagree with a determination made by the Committee, the SIP Administrators or any of their delegates or designees relating to the Plan or the Award Agreement and you wish to appeal such determination, you must submit a written request to the Committee for review within 180 days after the determination at issue. You must exhaust your internal administrative remedies (i.e., submit your appeal and wait for resolution of that appeal) before seeking to resolve a dispute, controversy or claim through arbitration pursuant to Paragraph 16 and Section 3.17 of the Plan.

(d) You May Be Required to Certify Compliance with Award Terms; You Are Responsible for Providing Updated Address and Contact Information After Your Departure from the Firm. If your Employment terminates while you continue to hold RSUs or Shares at Risk, the Firm may require you to provide certifications of your compliance with all of the terms of the Plan and this Award Agreement. You understand and agree that (i) your contact information as reflected in the Firm's personnel records at the time any certification is requested will be deemed current; (ii) it is your responsibility to inform HCM of any changes to your contact information to ensure timely receipt of the certification materials, regardless of whether your contact information is provided to another part of the Firm; and (iii) you are responsible for contacting the Firm to obtain such certification materials if not received. Your failure to return properly completed certification materials by the specified deadline (including because you did not provide HCM with updated contact information) will result in the forfeiture of all of your RSUs and any Shares at Risk and subject previously delivered amounts to repayment under Paragraph 9(c)(iv).

(e) You Agree to Other Terms, Conditions and Consents. By accepting this Award you understand and agree that:

(i) Amounts May Be Rounded to Avoid Fractional RSU Shares. RSUs that become Vested, RSU Shares that become deliverable and RSU Shares subject to Transfer Restrictions may, in each case, be rounded to avoid fractional RSU Shares.

(ii) Firm May Deliver Cash or Other Property Instead of RSU Shares. In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the RSU Shares, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to deliveries of RSU Shares will include such deliveries of cash, other securities, other awards under the Plan or other property.

(iii) Firm May Deliver Your Award into an Escrow Account. The Firm may establish and maintain an escrow account on such terms (which may include your executing any documents related to, and your paying for any costs associated with, such account) as it may deem necessary or appropriate, and the delivery of RSU Shares (including any Shares at Risk), and any payment of cash (including dividends [and payments under Dividend Equivalent Rights]) or other property, may initially be made into and held in that escrow account until such time as the Committee has received such documentation as it may have requested or until the Committee has determined that any other conditions or restrictions on delivery of RSU Shares, cash or other property required by this Award Agreement have been satisfied.

(iv) Firm May Affix Legends and Place Stop Orders on RSU Shares. GS Inc. may affix to Certificates representing RSU Shares any legend that the Committee determines to be necessary or advisable (including to reflect any restrictions to which you may be subject under the Additional Terms, a separate agreement or any Firm policy applicable to you). GS Inc. may advise the transfer agent to place a stop order against any legended RSU Shares. For the avoidance of doubt, RSU Shares as used herein includes, without limitation, Restricted Shares.

(v) You Authorize the Firm to Register, in Its or Its Designee's Name, Any Restricted Shares and Sell, Assign or Transfer Any Forfeited Restricted Shares. You are granting to the Firm the full power and authority to register any Restricted Shares in its or its designee's name and authorizing the Firm or its designee to sell, assign or transfer any Restricted Shares if you forfeit your Restricted Shares.

(vi) You Agree to Certain Consents as a Condition to the Award. You have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

(vii) You Are Subject to the Firm's Policies, Rules and Procedures. You are subject to the Firm's policies in effect from time to time concerning trading in RSU Shares and hedging or pledging RSU Shares and equity-based compensation or other awards (including, without limitation, the "Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards" or any successor policies), and confidential or proprietary information, and you will effect sales of RSU Shares in accordance with such rules and procedures as may be adopted from time to time (which may include, without limitation, restrictions relating to the timing of sale requests, the manner in which sales are executed, pricing method, consolidation or aggregation of orders and volume limits determined by the Firm).

(viii) You Are Responsible for Costs Associated with Your Award. You will be responsible for all brokerage costs and other fees or expenses associated with your RSUs, including those related to the sale of RSU Shares.

(ix) You Will Be Deemed to Represent Your Compliance with All the Terms of Your Award if You Accept Delivery of, or Sell, RSU Shares. You will be deemed to have represented and certified that you have complied with all of the terms of the Plan and this Award Agreement when RSU Shares are delivered to you[, you receive payment in respect of Dividend Equivalent Rights] and you request the sale of RSU Shares following the release of any Transfer Restrictions.

(x) You Agree that Covered Persons Will Not Have Liability. In addition to and without limiting the generality of the provisions of Section 1.3.5 of the Plan, neither the Firm nor any Covered Person will have any liability to you or any other person for any action taken or omitted in respect of this or any other Award.

14. **Non-transferability.** Except as otherwise may be provided by the Committee, the limitations set forth in Section 3.5 of the Plan will apply to this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition (including

through the use of any cash-settled instrument), whether voluntary or involuntary, in violation of the provisions of this Paragraph 14 or Section 3.5 of the Plan will be void.

15. **Right of Offset.** Except as provided in Paragraph 18(h), the obligation to deliver RSU Shares, to pay dividends [or payments under Dividend Equivalent Rights] or to remove any Transfer Restrictions under this Award Agreement is subject to Section 3.4 of the Plan, which provides for the Firm's right to offset against such obligation any outstanding amounts you owe to the Firm and any amounts the Committee deems appropriate pursuant to any tax equalization policy or agreement.

Arbitration, Choice of Forum and Governing Law

16. **Arbitration; Choice of Forum.**

(a) **By accepting this award, you are indicating that you understand and agree that the arbitration and choice of forum provisions set forth in Section 3.17 of the Plan will apply to this Award to the fullest extent permitted by applicable law. These provisions, which are expressly incorporated herein by reference, provide among other things that any dispute, controversy or claim between the Firm and you arising out of or relating to or concerning the Plan or this Award Agreement will be finally settled by arbitration, pursuant to the terms more fully set forth in Section 3.17 of the Plan; provided that nothing herein shall preclude you from filing a charge with or participating in any investigation or proceeding conducted by any governmental authority, including but not limited to the SEC, the Equal Employment Opportunity Commission and a state or local human rights agency, as well as law enforcement.**

(b) Nothing in the Plan or this Award Agreement includes an agreement to arbitrate claims on a collective, class or representative basis. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider class, collective or representative claims, to order consolidation or to join different claimants or grant relief other than on an individual basis to the individual claimant involved.

(c) Notwithstanding any applicable forum rules to the contrary, to the extent there is a question of enforceability of the Plan or this Award Agreement arising from a challenge to the arbitrator's jurisdiction or to the arbitrability of a claim, it will be decided by a court and not an arbitrator.

(d) Prior to filing any arbitration in accordance with Paragraph 16(a), you must first exhaust your internal administrative remedies in accordance with Paragraph 13(c), which includes awaiting the Committee's final determination of any such appeal. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider any dispute, controversy or claim as to which you have not first exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(e) The Federal Arbitration Act governs interpretation and enforcement of all arbitration provisions under the Plan and this Award Agreement, and all arbitration proceedings thereunder.

(f) Nothing in the Plan or this Award Agreement creates a substantive right to bring a claim under U.S. Federal, state, or local employment laws.

(g) By accepting your Award, you irrevocably appoint each General Counsel of GS Inc., or any person whom the General Counsel of GS Inc. designates, as your agent for service of process in connection with any suit, action or proceeding arising out of or relating to or concerning the Plan or any Award which is not arbitrated pursuant to the provisions of Section 3.17.1 of the Plan, who shall promptly advise you of any such service of process.

(h) **As a condition of your Award, you also must accept the terms of the Agreement Regarding Arbitration of Employment-Related Matters, which is included among the Additional Terms described in Paragraph 13(a) and sets forth obligations independent of this Paragraph 16. For the avoidance of doubt, the obligation to exhaust your internal remedies described in**

Paragraph 13(c) and Paragraph 16(d) only applies to disputes, controversies or claims relating to the Plan or the Award Agreement.

17. Governing Law. This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

18. Compliance of Award Agreement and Plan with Section 409A. The provisions of this Paragraph 18 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 18 and the other provisions of this Award Agreement, this Paragraph 18 will govern.

(b) Delivery of RSU Shares will not be delayed beyond the date on which all applicable conditions or restrictions on delivery of RSU Shares required by this Award Agreement (including those specified in Paragraphs 6, 7, 11(a)(ii), 11(b), 12(c) and 13 and the consents and other items specified in Section 3.3 of the Plan) are satisfied. To the extent that any portion of this Award is intended to satisfy the requirements for short-term deferral treatment under Section 409A, delivery for such portion will occur by the March 15 coinciding with the last day of the applicable "short-term deferral" period described in Reg. 1.409A-1(b)(4) in order for the delivery of RSU Shares to be within the short-term deferral exception unless, in order to permit all applicable conditions or restrictions on delivery to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay delivery of RSU Shares to a later date within the same calendar year or to such later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 13(e) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any securities, other Awards or other property that the Firm may deliver in respect of your RSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the RSU Shares that would otherwise have been deliverable (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Notwithstanding the timing provisions of Paragraph 12(c), the delivery of RSU Shares referred to therein will be made after the date of death and not later than the end of the calendar year following the calendar year that includes the date of death (or on such later date as may be permitted under Section 409A).

(e) The timing of delivery or payment pursuant to Paragraph 12(a) will occur on the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the termination of Employment occurs; *provided, however*, that if you are a "specified employee" (as defined by the Firm in accordance with Section 409A(a)(2)(i)(B) of the Code), delivery will occur on the earlier of the Delivery Date or (to the extent required to avoid the imposition of additional tax under Section 409A) the date that is six months after your termination of Employment (or, if the latter date is not during a Window Period, the first trading day of the next Window Period). For purposes of Paragraph 12(a), references in this

Award Agreement to termination of Employment mean a termination of Employment from the Firm (as defined by the Firm) which is also a separation from service (as defined by the Firm in accordance with Section 409A).

(f) [Notwithstanding any provision of Paragraph 8 or Section 2.8.2 of the Plan to the contrary, the Dividend Equivalent Rights with respect to each of your Outstanding RSUs will be paid to you within the calendar year that includes the date of distribution of any corresponding regular cash dividends paid by GS Inc. in respect of a share of Common Stock the record date for which occurs on or after the Date of Grant. The payment will be in an amount (less applicable withholding) equal to such regular dividend payment as would have been made in respect of the RSU Shares underlying such Outstanding RSUs.]

(g) The timing of delivery or payment referred to in Paragraph 12(b)(i) will be the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the Committee receives satisfactory documentation relating to your Conflicted Employment, *provided* that such delivery or payment will be made, and any Committee action referred to in Paragraph 12(b)(ii) will be taken, only at such time as, and if and to the extent that it, as reasonably determined by the Firm, would not result in the imposition of any additional tax to you under Section 409A.

(h) Paragraph 15 and Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(i) Delivery of RSU Shares in respect of any Award may be made, if and to the extent elected by the Committee, later than the Delivery Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later delivery is permitted under Section 409A).

(j) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Committee Authority, Amendment, Construction and Regulatory Reporting

19. **Committee Authority.** The Committee has the authority to determine, in its sole discretion, that any event triggering forfeiture or repayment of your Award will not apply, to limit the forfeitures and repayments that result under Paragraphs 9 and 10 and to remove any Transfer Restrictions before the applicable Transferability Date. In addition, the Committee, in its sole discretion, may determine whether Paragraphs 11(a)(ii) and 11(b) will apply upon a termination of Employment and whether a termination of Employment constitutes an Approved Termination under Paragraph 11(c).

20. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided* that, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further* that the Committee expressly reserves its rights to amend the Award Agreement and the Plan as described in Sections 1.3.2(h)(1), (2) and (4) of the Plan. A modification that impacts the tax consequences of this Award or the timing of delivery of RSU Shares will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

21. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words "include," "includes" and "including" will be deemed to be followed by the words "without limitation." The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

22. **Providing Information to the Appropriate Authorities; Protected Communications.** In accordance with applicable law, nothing in this Award Agreement (including the forfeiture and repayment provisions in Paragraphs 9 and 10) or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

[EU and UK Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider certain factors to determine whether and what portion of your Award will terminate, including the reason for the Loss Event or Risk Event, as well as the extent to which you participated in the Loss Event or Risk Event and your compensation for ____ may or may not have been adjusted to take into account the risk associated with the Loss Event, Risk Event, your Serious Misconduct or the Serious Misconduct of a Supervised Employee. Your compensation may be adjusted for the year in which the Loss Event, Risk Event, your Serious Misconduct or a Supervised Employee's Serious Misconduct is discovered.

(a) A Loss Event Occurs Prior to Delivery. If a Loss Event occurs prior to the delivery of RSU Shares, your rights in respect of all or a portion of your RSUs (whether or not Vested) which are scheduled to deliver on the next Delivery Date immediately following the date that the Loss Event is identified (or, if not practicable, then the next following Delivery Date) will terminate, and no RSU Shares will be delivered in respect of such RSUs.

(i) **"Loss Event"** means (A) an annual pre-tax loss at GS Inc. or (B) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm), or annual negative revenues in the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm) of \$5 billion or more, *provided* in either case that you are employed in a business within such reporting segment.

(b) A Risk Event Occurs On or Before the Applicable End Date. If a Risk Event occurs on or before the Applicable End Date, (i) your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Risk Event occurred and (B) the date that the repayment request is made.

(i) **"Risk Event"** means there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting and Operational Risk Events Policy (or any successor policy).

(c) You Engage in Serious Misconduct On or Before the Applicable End Date. If you engage in Serious Misconduct during the period beginning on the applicable Transferability Date through the Applicable End Date, you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (i) the date the Serious Misconduct occurred and (ii) the date that the repayment request is made.

(i) **“Serious Misconduct”** means that you engage in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under applicable employment law.

(d) **A Supervised Employee Engages in Serious Misconduct.** If the Committee determines that it is appropriate to hold you accountable in whole or in part for Serious Misconduct related to compliance, control or risk that occurred during ____ by a Supervised Employee, your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs and your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled.

(i) **“Supervised Employee”** means an individual with respect to whom the Committee determines you had supervisory responsibility as a result of direct or indirect reporting lines or your management responsibility for an office, division or business.

(e) Certain Definitions.

(i) **“Applicable End Date”** means[_____] , with respect to each classification in the table below, the following:

Classification	Applicable End Date
EU MRT	_____
GSAMI MRT	_____
GSAMI Senior Management	_____
UK MRT	_____]

(ii) **“EU MRT”** means an individual who, at any point in ____, was classified by the Firm as a (A) Material Risk Taker in accordance with the applicable remuneration rules of the Alternative Investment Fund Managers Directive or Undertakings for Collective Investment in Transferable Securities V (“AIFMD-UCITS”), including “Identified Staff” under AIFMD-UCITS (“AIFMD-UCITS Identified Staff”), as applicable to the business operations of Goldman Sachs Asset Management, B.V. (“GSAMBV”), Goldman Sachs Asset Management Belgium, S.A. (“GSAM Belgium”) or Goldman Sachs Towarzystwo Funduszy Inwestycyjnych S.A. (“GS TFI”), including any branch of subsidiary thereof (collectively, “GSAM Europe”), whether or not GSAM Europe has employed such individual; or (B) an individual who, at any point in ____, was classified by the Firm with respect to Goldman Sachs Paris Inc. et Cie in accordance with the applicable remuneration rules of either the Capital Requirements Directive V or the Investment Firms Regulation/Investment Firms Directive, as a Material Risk Taker (“GSPIC MRT”) or Senior Management (“GSPIC Senior Management”). EU MRT does not include AIFMD-UCITS Identified Staff of Goldman Sachs Asset Management Fund Services Ltd. (“GSAMFSL”).

(iii) **“GSAMI MRT”** means an individual who, at any point in ____, was classified by the Firm as a GSAMI Material Risk Taker in accordance with the remuneration code under the UK Investment Firms Prudential Regime (“IFPR”) as applicable to the business operations of Goldman Sachs Asset Management International Ltd. (“GSAMI”).

(iv) **“GSAMI Senior Management”** means an individual who, at any point in ____, was classified by the Firm as a GSAMI Senior Management in accordance with the remuneration code under the IFPR as applicable to the business operations of GSAMI.

(v) **“UK MRT”** means an individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (including UK Risk Managers and UK PRA Senior Managers) in accordance with the remuneration codes applicable to dual-regulated UK firms.

(vi) For purposes of these definitions, if during ____ you were classified by the Firm as any combination of EU MRT and GSAMI MRT, GSAMI Senior Management or UK MRT

then your classification in the table above is EU MRT; any combination of GSAMI MRT or GSAMI Senior Management and UK MRT then your classification in the table above is UK MRT; or as a combination of GSAMI MRT and GSAMI Senior Management then your classification in the table above is GSAMI Senior Management.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by Paragraphs (b) and (c) of this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

[GSBE Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider all relevant factors in making the decision, in its sole discretion as to whether an Adjustment Event or Actionable Misconduct have occurred.

This Appendix applies to all RSUs in your Award if you are a GSBE MRT or GSBE Senior Management.

(a) An Adjustment Event Occurs with Respect to ____. If an Adjustment Event occurs with respect to ____, (i) your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled, and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Adjustment Event occurred and (B) the date that the repayment request is made.

(i) **“Adjustment Event”** means that one of the following has occurred:

(A) you significantly contributed to, or were responsible for, any conduct that resulted in a loss of 0.75% or more of the total capital of GS Inc.;

(B) a material regulatory sanction for the Firm comprising one or more of the following:

1. a moratorium pursuant to sec. 46g of the German Banking Act,
 2. a measure in case of danger pursuant to sec. 46 of the German Banking Act,
 3. the revocation of appointment of a manager pursuant to sec. 36 German Banking Act,
 4. a fine pursuant to sec. 56 of the German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of GS Inc.,
 5. the cancellation of the banking permit pursuant to sec. 35 of the German Banking Act,
 6. an order to increase the capital requirements Goldman Sachs Bank Europe SE (“GSBE”) by at least 0.5% pursuant to sec. 10 of the German Banking Act,
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7. a measure in case of organizational deficiencies,
8. a comparable regulatory order, or
9. a material supervisory measure; or

(C) you acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that a violation is considered serious if it suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct pursuant to sec. 1 German Termination Protection Act.

(b) You Engage in Actionable Misconduct. If the Committee determines that you engaged in Actionable Misconduct, which if known (or if known, adequately considered) would have resulted in a reduction to your annual variable compensation when your variable compensation for that year was determined, your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs.

(i) **“Actionable Misconduct”** means either or both of:

- (A) “Unethical behavior” within the meaning of the IVV;
- (B) “Breach of duty” within the meaning of the IVV.

(ii) **“IVV”** means the German Remuneration Ordinance for Institutions (*Institutsvergütungsverordnung*) and applicable guidance from the German Federal Financial Supervisory Authority (BaFin) or the European Central Bank.

(c) Scope of Appendix. This Appendix applies to any individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (“GSBE MRT”) or Senior Management (“GSBE Senior Management”) in connection with the business operations of GSBE or any branch or subsidiary thereof, whether or not GSBE or any branch or subsidiary thereof employed such individual.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings:

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

(b) “Acceptance Deadline” means the date communicated to you by the Firm as the deadline to accept your Award. The Acceptance Deadline will be no less than 14 days after the Award Documentation is first made available to you in full. To the extent you are entitled to a translation in accordance with the Language Notice for Equity Awards provided to you as part of the Award Documentation, the Acceptance Deadline will be extended by the time required to furnish you with any such translation.

(c) “Approved Termination” means that you are classified by the Firm as a “fixed-term employee” and you (i) successfully complete the fixed-term engagement, as determined by the Firm in its sole discretion, including remaining Employed through the completion date specified by the Firm, and (ii) terminate Employment immediately after the completion date without any “stay-on” or other agreement or understanding to continue Employment with the Firm. If you agree to stay with the Firm as an employee after your fixed-term engagement ends and then later terminate Employment, you will not have an Approved Termination.

(d) “Associate With a Covered Enterprise” means that you (i) form, or acquire a 5% or greater equity ownership, voting or profit participation interest in, any Covered Enterprise or (ii) associate in any capacity (including association as an officer, employee, partner, director, consultant, agent or advisor) with any Covered Enterprise. Associate With a Covered Enterprise may include, as determined in the discretion of the Committee, (i) becoming the subject of any publicly available announcement or report of a pending or future association with a Covered Enterprise and (ii) unpaid associations, including an association in contemplation of future employment. “Association With a Covered Enterprise” will have its correlative meaning.

(e) “Breached an Obligation to the Firm” means that you failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with your Employment or this Award (including any applicable Additional Terms); including, without limitation, any offer letter, employment agreement or any shareholders’ agreement relating to the Firm. Your failure to pay or reimburse the Firm, on demand, for any amount you owe to the Firm will constitute (i) failure to meet an obligation you have under an agreement, regardless of whether such obligation arises under a written agreement, and/or (ii) a material violation of Firm policy constituting Cause.

(f) “Committee” means the Compensation Committee unless otherwise determined by the Board; *provided, however*, that: (i) the authority of the Compensation Committee to administer the Plan and the Award may be granted or delegated to the SIP Committee, in which case “Committee” means the SIP Committee when acting pursuant to such a grant or delegation; (ii) to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3; and (iii) to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, “Committee” shall be a committee or subcommittee of

the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m).

(g) “Compensation Committee” means the Compensation Committee of the Board.

(h) “Conflicted Employment” means your employment at any U.S. Federal, state or local government, any non-U.S. government, any supranational or international organization, any self-regulatory organization, or any agency or instrumentality of any such government or organization, or any other employer (other than an “Accounting Firm” within the meaning of SEC Rule 2-01(f)(2) of Regulation S-X or any successor thereto) determined by the Committee, if, as a result of such employment, your continued holding of any Outstanding RSUs or Shares at Risk would result in an actual or perceived conflict of interest.

(i) “Covered Enterprise” means a Competitive Enterprise and any other existing or planned business enterprise that: (i) offers, holds itself out as offering or reasonably may be expected to offer products or services that are the same as or similar to those offered by the Firm or that the Firm reasonably expects to offer (“Firm Products or Services”) or (ii) engages in, holds itself out as engaging in or reasonably may be expected to engage in any other activity that is the same as or similar to any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage (“Firm Activities”). For the avoidance of doubt, Firm Activities include any activity that requires the same or similar skills as any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage, irrespective of whether any such financial activity is in furtherance of an advisory, agency, proprietary or fiduciary undertaking.

The enterprises covered by this definition include enterprises that offer, hold themselves out as offering or reasonably may be expected to offer Firm Products or Services, or engage in, hold themselves out as engaging in or reasonably may be expected to engage in Firm Activities directly, as well as those that do so indirectly by ownership or control (e.g., by owning, being owned by or by being under common ownership with an enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services or that engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities). The definition of Covered Enterprise includes, solely by way of example, any enterprise that offers, holds itself out as offering or reasonably may be expected to offer any product or service, or engages in, holds itself out as engaging in or reasonably may be expected to engage in any activity, in any case, associated with investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than you or your family members (including, for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading. An enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services, or engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities is a Covered Enterprise, **irrespective of whether the enterprise is a customer, client or counterparty of the Firm or is otherwise associated with the Firm and, because the Firm is a global enterprise, irrespective of where the Covered Enterprise is physically located.**

(j) “Delivery Date” means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a delivery date, so long as any such date occurs before the 29th day of the month specified on the Award Statement; if there is no such date, the Delivery Date will

be the earlier of another date selected by the Committee in the month and year specified on the Award Statement or the first trading day in the subsequent Window Period.

(k) “Failed to Consider Risk” means that you participated (or otherwise oversaw or were responsible for, depending on the circumstances, another individual’s participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where you have improperly analyzed such risk or where you have failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, the Committee determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, your business unit or the broader financial system. This definition is not limited to financial risks and is designed to encourage the consideration of the full range of risks associated with activities (e.g., legal, compliance or reputational). This definition also does not require that a material adverse impact actually occur, but, rather, may be triggered if it is determined that there is a reasonable expectation of such an impact.

(l) “HCM” means the Firm’s Human Capital Management Division or its successor.

(m) “Qualifying Termination After a Change in Control” means that the Firm terminates your Employment other than for Cause or you terminate your Employment for Good Reason, in each case, within 18 months following a Change in Control.

(n) “SEC” means the U.S. Securities and Exchange Commission.

(o) “Selected Firm Personnel” means any individual who is or in the three months preceding the conduct prohibited by Paragraph 9(b)(i) was (i) a Firm employee or consultant with whom you personally worked while employed by the Firm, (ii) a Firm employee or consultant who, at any time during the year preceding the date of the termination of your Employment, worked in the same division in which you worked or (iii) an Advisory Director, a Managing Director (Participating or Extended) or a Senior Advisor of the Firm.

(p) “Shares at Risk” means RSU Shares subject to Transfer Restrictions.

(q) “Signature Card” means the document and/or website presented to you with your Award that you are required to execute (including electronically) to indicate your acceptance of the Award, the Award Agreement (including the Award Statement and the terms contained in the Signature Card) and Additional Terms applicable to you, as well your receipt of the Award Documentation.

(r) “Transferability Date” means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a transferability date; *provided, however*, that if there is no trading day in a Window Period that occurs in the specified month and year, the Transferability Date will be the first trading day in the subsequent Window Period.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Statement” means a written statement that reflects certain Award terms.

(c) “Board” means the Board of Directors of GS Inc.

(d) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(e) “Cause” means (i) the Grantee’s conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion, or (B) on a felony charge, or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations, (ii) the Grantee’s engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the Exchange Act), (iii) the Grantee’s willful failure to perform the Grantee’s duties to the Firm, (iv) the Grantee’s violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member, (v) the Grantee’s violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or the Grantee’s material violation of any other Firm policy as in effect from time to time, (vi) the Grantee’s engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm or (vii) the Grantee’s engaging in any conduct detrimental to the Firm. The determination as to whether Cause has occurred shall be made by the Committee in its sole discretion and, in such case, the Committee also may, but shall not be required to, specify the date such Cause occurred (including by determining that a prior termination of Employment was for Cause). Any rights the Firm may have hereunder and in any Award Agreement in respect of the events giving rise to Cause shall be in addition to the rights the Firm may have under any other agreement with a Grantee or at law or in equity.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Change in Control” means the consummation of a merger, consolidation, statutory share exchange or similar form of corporate transaction involving GS Inc. (a “Reorganization”) or sale or other disposition of all or substantially all of GS Inc.’s assets to an entity that is not an affiliate of GS Inc. (a “Sale”), that in each case requires the approval of GS Inc.’s shareholders under the law of GS Inc.’s jurisdiction of organization, whether for such Reorganization or Sale (or the issuance of securities of GS Inc. in such Reorganization or Sale), unless immediately following such Reorganization or Sale, either: (i) at least 50% of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of (A) the entity resulting from such Reorganization, or the entity which has acquired all or substantially all of the assets of GS Inc. in a Sale (in either case, the “Surviving Entity”), or (B) if applicable, the ultimate parent entity that directly or indirectly has beneficial

ownership (within the meaning of Rule 13d-3 under the Exchange Act, as such Rule is in effect on the date of the adoption of the 1999 SIP) of 50% or more of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of the Surviving Entity (the "Parent Entity") is represented by GS Inc.'s securities (the "GS Inc. Securities") that were outstanding immediately prior to such Reorganization or Sale (or, if applicable, is represented by shares into which such GS Inc. Securities were converted pursuant to such Reorganization or Sale) or (ii) at least 50% of the members of the board of directors (or similar officials in the case of an entity other than a corporation) of the Parent Entity (or, if there is no Parent Entity, the Surviving Entity) following the consummation of the Reorganization or Sale were, at the time of the Board's approval of the execution of the initial agreement providing for such Reorganization or Sale, individuals (the "Incumbent Directors") who either (A) were members of the Board on the Effective Date or (B) became directors subsequent to the Effective Date and whose election or nomination for election was approved by a vote of at least two-thirds of the Incumbent Directors then on the Board (either by a specific vote or by approval of GS Inc.'s proxy statement in which such persons are named as nominees for director).

(h) "Client" means any client or prospective client of the Firm to whom the Grantee provided services, or for whom the Grantee transacted business, or whose identity became known to the Grantee in connection with the Grantee's relationship with or employment by the Firm.

(i) "Code" means the Internal Revenue Code of 1986, as amended from time to time, and the applicable rulings and regulations thereunder.

(j) "Common Stock" means common stock of GS Inc., par value \$0.01 per share.

(k) "Competitive Enterprise" means an existing or planned business enterprise that (i) engages, or may reasonably be expected to engage, in any activity; (ii) owns or controls, or may reasonably be expected to own or control, a significant interest in any entity that engages in any activity or (iii) is, or may reasonably be expected to be, owned by, or a significant interest in which is, or may reasonably be expected to be, owned or controlled by, any entity that engages in any activity that, in any case, competes or will compete anywhere with any activity in which the Firm is engaged. The activities covered by this definition include, without limitation: financial services such as investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the Grantee and members of the Grantee's family (including for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading.

(l) "Covered Person" means a member of the Board or the Committee or any employee of the Firm.

(m) "Date of Grant" means the date specified in the Grantee's Award Agreement as the date of grant of the Award.

(n) ["Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.]

(o) "Effective Date" means the date this Plan is approved by the shareholders of GS Inc. pursuant to Section 3.15 of the Plan.

(p) "Employment" means the Grantee's performance of services for the Firm, as determined by the Committee. The terms "employ" and "employed" shall have their correlative meanings. The Committee in its sole discretion may determine (i) whether and when a Grantee's leave of absence results in a termination of Employment (for this purpose, unless the Committee determines otherwise, a Grantee shall be treated as terminating Employment with the Firm upon the occurrence of an Extended Absence), (ii) whether and when a change in a Grantee's association with the Firm results in a termination of Employment and (iii) the impact, if any, of any such leave of absence or change in association on Awards theretofore made. Unless expressly provided otherwise, any references in the Plan or any Award Agreement to a Grantee's Employment being terminated shall include both voluntary and involuntary terminations.

(q) "Exchange Act" means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(r) "Extended Absence" means the Grantee's inability to perform for six (6) continuous months, due to illness, injury or pregnancy-related complications, substantially all the essential duties of the Grantee's occupation, as determined by the Committee.

(s) "Firm" means GS Inc. and its subsidiaries and affiliates.

(t) "Good Reason" means, in connection with a termination of employment by a Grantee following a Change in Control, (i) as determined by the Committee, a materially adverse alteration in the Grantee's position or in the nature or status of the Grantee's responsibilities from those in effect immediately prior to the Change in Control or (ii) the Firm's requiring the Grantee's principal place of Employment to be located more than seventy-five (75) miles from the location where the Grantee is principally Employed at the time of the Change in Control (except for required travel on the Firm's business to an extent substantially consistent with the Grantee's customary business travel obligations in the ordinary course of business prior to the Change in Control).

(u) "Grantee" means a person who receives an Award.

(v) "GS Inc." means The Goldman Sachs Group, Inc., and any successor thereto.

(w) "1999 SIP" means The Goldman Sachs 1999 Stock Incentive Plan, as in effect prior to the effective date of the 2003 SIP.

(x) "Outstanding" means any Award to the extent it has not been forfeited, cancelled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(y) "Restricted Share" means a share of Common Stock delivered under the Plan that is subject to Transfer Restrictions, forfeiture provisions and/or other terms and conditions specified herein and in the Restricted Share Award Agreement or other applicable Award Agreement. All references to Restricted Shares include "Shares at Risk."

(z) "Retirement" means termination of the Grantee's Employment (other than for Cause) on or after the Date of Grant at a time when (i) (A) the sum of the Grantee's age plus years of service with

the Firm (as determined by the Committee in its sole discretion) equals or exceeds 60 and (B) the Grantee has completed at least 10 years of service with the Firm (as determined by the Committee in its sole discretion) or, if earlier, (ii) (A) the Grantee has attained age 50 and (B) the Grantee has completed at least five years of service with the Firm (as determined by the Committee in its sole discretion).

(aa) "RSU" means a restricted stock unit granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(ab) "RSU Shares" means shares of Common Stock that underlie an RSU.

(ac) "Section 409A" means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(ad) "SIP Administrator" means each person designated by the Committee as a "SIP Administrator" with the authority to perform day-to-day administrative functions for the Plan.

(ae) "SIP Committee" means the persons who have been delegated certain authority under the Plan by the Committee.

(af) "Solicit" means any direct or indirect communication of any kind whatsoever, regardless of by whom initiated, inviting, advising, suggesting, encouraging or requesting any person or entity, in any manner, to take or refrain from taking any action. The terms "Solicited," "Soliciting" and "Solicitation" will have their correlative meanings.

(ag) "Transfer Restrictions" means restrictions that prohibit the sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposal (including through the use of any cash-settled instrument), whether voluntarily or involuntarily by the Grantee, of an Award or any shares of Common Stock, cash or other property delivered in respect of an Award.

(ah) "Vested" means, with respect to an Award, the portion of the Award that is not subject to a condition that the Grantee remain actively employed by the Firm in order for the Award to remain Outstanding. The fact that an Award becomes Vested shall not mean or otherwise indicate that the Grantee has an unconditional or nonforfeitable right to such Award, and such Award shall remain subject to such terms, conditions and forfeiture provisions as may be provided for in the Plan or in the Award Agreement.

(ai) "Vesting Date" means each date specified in the Grantee's Award Agreement as a date on which part or all of an Award becomes Vested.

(aj) "Window Period" means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (*provided* that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
[One-Time][Year-End] RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of RSUs (your “Award”). You should read carefully this entire document (including its [Appendix][Appendices]), the Award Statement and the Signature Card (together, the “Award Agreement”), as well as the other documentation presented to you in connection with acceptance of your Award (collectively, “Award Documentation”).

Acceptance

1. **You Must Decide Whether to Accept Your Award.** To be eligible for your Award, you must by the Acceptance Deadline: (a) open and activate an Account; (b) accept any Additional Terms presented to you as part of the Award Documentation; and (c) execute (including by electronic means) the accompanying Signature Card in accordance with its instructions. By executing the Signature Card, you confirm your agreement to all of the terms of the Award Agreement (including the Award Statement and the Signature Card), as well as any Additional Terms.

Documents that Govern Your Award; Definitions

2. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement. The Plan and its Prospectus are included with the Award Documentation.

3. **Your Award Statement.** The accompanying Award Statement contains the number of RSUs awarded to you and some of your Award’s specific terms. For example, it specifies any applicable Delivery Dates and Transferability Dates.

4. **Definitions.** Unless otherwise defined herein, including in the Definitions Appendix or any other Appendix, capitalized terms have the meanings provided in the Plan.

Vesting of Your RSUs

5. **Vesting.** All of your RSUs are Vested. When an RSU is Vested, it means **only** that your continued active Employment is not required for delivery of that portion of RSU Shares. **Vesting does not mean you have a non-forfeitable right to the Vested portion of your Award. The terms of this Award Agreement (including conditions to delivery and any applicable Transfer Restrictions) continue to apply to Vested RSUs, and you can still forfeit Vested RSUs and any RSU Shares.**

Delivery of Your RSU Shares

6. **Delivery.** Reasonably promptly (but no more than 30 Business Days) after each Delivery Date identified on your Award Statement, RSU Shares (less applicable withholding as described in Paragraph 12(b)) will be delivered (by book entry credit to your Account) in respect of the amount of Outstanding RSUs listed next to that date. The Committee may select multiple dates within the 30-Business-Day period following the Delivery Date to deliver RSU Shares in respect of all or a portion of the RSUs with the same Delivery Date listed on the Award Statement, and all such dates will be treated as a single Delivery Date for purposes of this Award. Until such delivery, you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc. Without limiting the Committee’s authority under Section 1.3.2(h) of the Plan, the Firm may accelerate any Delivery Date by up to 30 days.

Transfer Restrictions Following Delivery

7. **Transfer Restrictions and Shares at Risk.** Within 30 Business Days after any applicable Transferability Date identified on your Award Statement, GS Inc. will remove the Transfer Restrictions on any Shares at Risk in respect of the amount of Outstanding RSUs listed next to that date. If your Award Statement reflects a Transferability Date, [] % of the RSU Shares delivered on any Delivery Date **before** tax withholding (or, if the tax withholding rate is greater than 50%, all RSU Shares)[all of the RSU Shares delivered on any Delivery Date **after** tax withholding] will be Shares at Risk. The Committee may select multiple dates within the 30-Business-Day period on which to remove Transfer Restrictions for all or a portion of the Shares at Risk with the same Transferability Date, and all such dates will be treated as a single Transferability Date for purposes of this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition in violation of the Transfer Restrictions on Shares at Risk will be void. **[If your Award Statement does not reflect any Transferability Date, then references to Transfer Restrictions and Shares at Risk in this Award Agreement are inapplicable to your Award and this Paragraph 7 does not apply.]**

Dividends

8. **[Dividend Equivalent Rights and] Dividends.** [Each RSU includes a Dividend Equivalent Right, which entitles you to receive an amount (less applicable withholding), at or after the time of distribution of any regular cash dividend paid by GS Inc. in respect of a share of Common Stock, equal to any regular cash dividend payment that would have been made in respect of an RSU Share underlying your Outstanding RSUs for any record date that occurs on or after the Date of Grant. In addition, you][You] will be entitled to receive on a current basis any regular cash dividend paid in respect of any Shares at Risk. [The RSUs do **not** include Dividend Equivalent Rights.]

Forfeiture of Your Award

9. **How You May Forfeit Your Award.** This Paragraph 9 sets forth the events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. More than one event may apply, and in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event. Paragraph 11 (relating to certain circumstances under which delivery and/or release of Transfer Restrictions may be accelerated) provides for exceptions to one or more provisions of this Paragraph 9.

(a) **RSUs Forfeited Upon Certain Events.** If any of the following occurs, your rights to your Outstanding RSUs will terminate, and no RSU Shares will be delivered in respect of such RSUs, as may be further described below:

(i) **You Associate With a Covered Enterprise.** Except as prohibited by applicable law, you Associate With a Covered Enterprise before the applicable Restriction Date. The "Restriction Date" is the first day of the month and year specified as the Delivery Date on your Award Statement; e.g., if the specified Delivery Date is _____, then the applicable Restriction Date is _____.

(ii) **You Solicit Clients or Employees, Interfere with Client or Employee Relationships or Participate in the Hiring of Employees.** Before the applicable Delivery Date and except as prohibited by applicable law, either:

A. you, in any manner, directly or indirectly, (1) Solicit any Client to transact business with a Covered Enterprise or to reduce or refrain from doing any business with the Firm, (2) interfere with or damage (or attempt to interfere with or damage) any relationship between the Firm and any Client, (3) Solicit any person who is an employee of the Firm to resign from the Firm, (4) Solicit any Selected Firm Personnel to apply for or accept employment (or other association) with any person or entity other than the Firm or (5) participate

in the hiring of any Selected Firm Personnel by any person or entity other than the Firm (including, without limitation, participating in the identification of individuals for potential hire, and participating in any hiring decision), whether as an employee or consultant or otherwise, or

B. Selected Firm Personnel are Solicited, hired or accepted into partnership, membership or similar status by any entity where you have, or will have, direct or indirect managerial responsibility for such Selected Firm Personnel, unless the Committee determines that you were not involved in such Solicitation, hiring or acceptance.

(iii) [GS Inc. Fails to Maintain the Minimum Tier 1 Capital Ratio]. Before the applicable Delivery Date, GS Inc. fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days.]

(iv) [GS Inc. Is Determined to Be in Default]. Before the applicable Delivery Date, the Board of Governors of the Federal Reserve or the Federal Deposit Insurance Corporation (the "FDIC") makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Wall Street Reform and Consumer Protection Act for the appointment of the FDIC as a receiver of GS Inc. based on a determination that GS Inc. is "in default" or "in danger of default."

(b) RSUs and Shares at Risk Forfeited upon Certain Events. If any of the following occurs (i) your rights to all of your Outstanding RSUs will terminate, and no RSU Shares will be delivered in respect of such RSUs and (ii) your rights to any Shares at Risk will terminate and all such Shares at Risk will be cancelled, in each case, as may be further described below:

(i) You Failed to Consider Risk. You Failed to Consider Risk during ____.

(ii) Your Conduct Constitutes Cause. Any event that constitutes Cause has occurred before the applicable Delivery Date for RSUs or the applicable Transferability Date for any Shares at Risk.

(iii) You Breach an Obligation to the Firm. The Committee determines that, before the applicable Delivery Date for RSUs or the applicable Transferability Date for any Shares at Risk, you Breached an Obligation to the Firm.

(iv) You Do Not Provide Timely Certifications or Comply with Your Certifications. You fail to certify to GS Inc. that you have complied with all of the terms of the Plan and this Award Agreement as required by Paragraph 12(d), or the Committee determines that you have failed to comply with a term of the Plan or this Award Agreement to which you have certified compliance.

(v) You Do Not Follow Dispute Resolution Procedures. You attempt to have any dispute, controversy or claim under the Plan or this Award Agreement resolved in any manner that is not provided for by Paragraph 15 or Section 3.17 of the Plan, or you attempt to arbitrate a dispute, controversy or claim without first having exhausted your internal administrative remedies in accordance with Paragraph 12(c).

(vi) You Bring an Action that Results in a Determination that Any Award Agreement Term Is Invalid. As a result of any action brought by you, it is determined that any term of this Award Agreement is invalid.

(vii) You Receive Compensation in Respect of Your Award from Another Employer. Your Employment terminates for any reason or you otherwise are no longer actively Employed with the Firm and another entity grants you cash, equity or other property (whether vested or unvested) to replace, substitute for or otherwise in respect of any Outstanding RSUs or Shares at

Risk; *provided, however*, that your rights will only be terminated in respect of the RSUs and Shares at Risk that are replaced, substituted for or otherwise considered by such other entity in making its grant.

(c) [Additional Forfeiture and Repayment Conditions for Material Risk Takers. If at any point in ____][Since] the Firm identified you as a Material Risk Taker (“MRT”) [in ____] then the **EU and UK Material Risk Taker Appendix** and/or the **GSBE Material Risk Taker Appendix** (as applicable to you, the “MRT Appendix”) supplements this Paragraph 9. The MRT Appendix sets forth additional events that (i) result in forfeiture of up to all of your RSUs and Shares at Risk and (ii) may require repayment to the Firm of up to all amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. Your MRT classification, as defined in the MRT Appendix, was communicated to you separately. [If you are unsure of whether the Firm identified you as an MRT, or which classification(s) apply, you must contact HCM prior to accepting this Award.] The Firm’s records with respect to such classification(s) will be controlling as to whether and which MRT Appendix applies to your Award. The MRT classifications with their corresponding MRT Appendix are as follows:

MRT Classification	MRT Appendix
EU MRT, GSAMI MRT, GSAMI Senior Management and/or UK MRT	EU and UK Material Risk Taker Appendix
GSBE MRT and GSBE Senior Management	GSBE Material Risk Taker Appendix
GSBE MRT or GSBE Senior Management <i>and</i> any combination of EU MRT, GSAMI MRT, GSAMI Senior Management or UK MRT	GSBE Material Risk Taker Appendix <i>and</i> EU and UK Material Risk Taker Appendix]

(d) **Actions Pending Forfeiture.** In connection with any investigation as to whether any of the events that would result in forfeiture under the Plan [or][,] this Paragraph 9 [or the MRT Appendix] have occurred, the Firm reserves the right to (i) suspend [payments under Dividend Equivalent Rights,] delivery of RSU Shares or release of any Transfer Restrictions; (ii) deliver any RSU Shares[, payments under Dividend Equivalent Rights] or dividends into an escrow account in accordance with Paragraph 12(e)(iii); or (iii) apply Transfer Restrictions to any RSU Shares.

Repayment of Your Award

10. **When You May Be Required to Repay Your Award.** If the Committee determines that any term of this Award was not satisfied, you will be required, immediately upon demand therefor, to repay to the Firm the following:

(a) Any RSU Shares (which, for the avoidance of doubt, includes any Shares at Risk) for which the terms (including the terms for delivery) of the related RSUs were not satisfied, in accordance with Section 2.6.3 of the Plan.

(b) Any Shares at Risk for which the terms (including the terms for the release of Transfer Restrictions) were not satisfied, in accordance with Section 2.5.3 of the Plan.

(c) Any RSU Shares that were delivered (but not subject to Transfer Restrictions) at the same time any Shares at Risk that are cancelled or required to be repaid were delivered.

(d) [Any payments under Dividend Equivalent Rights for which the terms were not satisfied (including any such payments made in respect of RSUs that are forfeited or RSU Shares that are cancelled or required to be repaid), in accordance with Section 2.8.3 of the Plan.]

(e) Any dividends paid in respect of any RSU Shares that are cancelled or required to be repaid.

(f) Any amount applied to satisfy tax withholding or other obligations with respect to any RSUs, RSU Shares[,] [and] dividend payments [and payments under Dividend Equivalent Rights] that are forfeited or required to be repaid.

Exceptions to Delivery and/or Transferability Dates

11. **Accelerated Delivery and/or Release of Transfer Restrictions in the Event of [a Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] Death.** In the event of your [Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] death, [each] as described below, your Outstanding RSUs and any Shares at Risk will be treated as described in this Paragraph 11, and, except as set forth in Paragraph 11(a), all other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

(a) [You Have a Qualifying Termination After a Change in Control]. If your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, the RSU Shares underlying your Outstanding RSUs will be delivered, and any Transfer Restrictions will cease to apply. In addition, the forfeiture events in Paragraph 9 [(except the MRT Appendix)] will not apply to your Award.]

(b) [You Are Determined to Have Accepted Conflicted Employment].

(i) Generally. To the extent consistent with applicable law, regulation and regulatory guidance, if your Employment terminates solely because you resign to accept Conflicted Employment or if, following your termination of Employment, you notify the Firm that you are accepting Conflicted Employment, RSU Shares will be delivered in respect of your Outstanding RSUs (including in the form of cash as described in Paragraph 12(e)(ii)) and any Transfer Restrictions will cease to apply as soon as practicable after the Committee has received satisfactory documentation relating to your Conflicted Employment.

(ii) You May Have to Take Other Steps to Address Conflicts of Interest. The Committee retains the authority to exercise its rights under the Award Agreement or the Plan (including Section 1.3.2 of the Plan) to take or require you to take other steps it determines in its sole discretion to be necessary or appropriate to cure an actual or perceived conflict of interest (which may include a determination that the accelerated delivery and/or release of Transfer Restrictions described in Paragraph 11(b)(i) will not apply because such actions are not necessary or appropriate to cure an actual or perceived conflict of interest).]

(c) Death. If you die, the RSU Shares underlying your Outstanding RSUs will be delivered to your Account and any Transfer Restrictions will cease to apply as soon as practicable after the date of death and receipt of any documentation as may be requested by the Committee or a SIP Administrator.

Other Terms and Conditions of Your Award

12. Additional Terms, Conditions and Consents.

(a) You Will Be Required to Accept Additional Terms. In addition to the terms of the Award Agreement, the Award Documentation includes other contractual terms that you must accept as a condition of the Award (the "Additional Terms"). By accepting the Award and executing the Signature Card, you agree to any such Additional Terms applicable to you as set forth in the Award Documentation, which will include but are not limited to the following:

(i) Notice Period Policy. You must agree to abide by the terms, as applicable to you, of the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy").

(ii) Agreement Regarding Arbitration of Employment-Related Matters. You must agree to the Agreement Regarding Arbitration of Employment-Related Matters, which imposes obligations in addition to those contained in Paragraph 15.

(iii) APAC Restrictive Covenants. If you are employed by the Firm in the Asia-Pacific Region (e.g., Australia, the People's Republic of China, the Hong Kong Special Administrative Region, India, Indonesia, Japan, the Republic of Korea, New Zealand, Singapore, Taiwan), you must agree to the APAC Restrictive Covenants Agreement.

(iv) Role-, Business- or Division-Specific Agreements. If applicable to you, you must accept Additional Terms related to your specific role, business unit or division as set forth in the Award Documentation.

(b) You Must Satisfy Applicable Tax Withholding Requirements. Delivery of RSU Shares is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, which includes the Firm deducting or withholding amounts from any payment or distribution to you. In addition, to the extent permitted by applicable law, the Firm, in its sole discretion, may require you to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations imposed on you or the Firm in connection with the grant or delivery of this Award by requiring you to choose between remitting the amount (i) in cash (or through payroll deduction or otherwise) or (ii) in the form of proceeds from the Firm's executing a sale of RSU Shares delivered to you under this Award. In no event, however, does this Paragraph 12(b) give you any discretion to determine or affect the timing of the delivery of RSU Shares or the timing of payment of tax obligations.

(c) You Must Comply with Applicable Deadlines and Procedures to Appeal. If you disagree with a determination made by the Committee, the SIP Administrators or any of their delegates or designees relating to the Plan or the Award Agreement and you wish to appeal such determination, you must submit a written request to the Committee for review within 180 days after the determination at issue. You must exhaust your internal administrative remedies (i.e., submit your appeal and wait for resolution of that appeal) before seeking to resolve a dispute, controversy or claim through arbitration pursuant to Paragraph 15 and Section 3.17 of the Plan.

(d) You May Be Required to Certify Compliance with Award Terms; You Are Responsible for Providing Updated Address and Contact Information After Your Departure from the Firm. If your Employment terminates while you continue to hold RSUs or Shares at Risk, the Firm may require you to provide certifications of your compliance with all of the terms of the Plan and this Award Agreement. You understand and agree that (i) your contact information as reflected in the Firm's personnel records at the time any certification is requested will be deemed current; (ii) it is your responsibility to inform HCM of any changes to your contact information to ensure timely receipt of the certification materials, regardless of whether your contact information is provided to another part of the Firm; and (iii) you are responsible for contacting the Firm to obtain such certification materials if not received. Your failure to return properly completed certification materials by the specified deadline (including because you did not provide HCM with updated contact information) will result in the forfeiture of all of your RSUs and any Shares at Risk and subject previously delivered amounts to repayment under Paragraph 9(b)(iv).

(e) You Agree to Other Terms, Conditions and Consents. By accepting this Award you understand and agree that:

(i) Amounts May Be Rounded to Avoid Fractional RSU Shares. RSUs that become Vested, RSU Shares that become deliverable and RSU Shares subject to Transfer Restrictions may, in each case, be rounded to avoid fractional RSU Shares.

(ii) Firm May Deliver Cash or Other Property Instead of RSU Shares. In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the RSU Shares, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to deliveries of RSU Shares will include such deliveries of cash, other securities, other awards under the Plan or other property.

(iii) Firm May Deliver Your Award into an Escrow Account. The Firm may establish and maintain an escrow account on such terms (which may include your executing any documents related to, and your paying for any costs associated with, such account) as it may deem necessary or appropriate, and the delivery of RSU Shares (including any Shares at Risk), and any payment of cash (including dividends [and payments under Dividend Equivalent Rights]) or other property, may initially be made into and held in that escrow account until such time as the Committee has received such documentation as it may have requested or until the Committee has determined that any other conditions or restrictions on delivery of RSU Shares, cash or other property required by this Award Agreement have been satisfied.

(iv) Firm May Affix Legends and Place Stop Orders on RSU Shares. GS Inc. may affix to Certificates representing RSU Shares any legend that the Committee determines to be necessary or advisable (including to reflect any restrictions to which you may be subject under the Additional Terms, a separate agreement or any Firm policy applicable to you). GS Inc. may advise the transfer agent to place a stop order against any legended RSU Shares. For the avoidance of doubt, RSU Shares as used herein includes, without limitation, Restricted Shares.

(v) You Authorize the Firm to Register, in Its or Its Designee's Name, Any Restricted Shares and Sell, Assign or Transfer Any Forfeited Restricted Shares. You are granting to the Firm the full power and authority to register any Restricted Shares in its or its designee's name and authorizing the Firm or its designee to sell, assign or transfer any Restricted Shares if you forfeit your Restricted Shares.

(vi) You Agree to Certain Consents as a Condition to the Award. You have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

(vii) You Are Subject to the Firm's Policies, Rules and Procedures. You are subject to the Firm's policies in effect from time to time concerning trading in RSU Shares and hedging or pledging RSU Shares and equity-based compensation or other awards (including, without limitation, the "Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards" or any successor policies), and confidential or proprietary information, and you will effect sales of RSU Shares in accordance with such rules and procedures as may be adopted from time to time (which may include, without limitation, restrictions relating to the timing of sale requests, the manner in which sales are executed, pricing method, consolidation or aggregation of orders and volume limits determined by the Firm).

(viii) You Are Responsible for Costs Associated with Your Award. You will be responsible for all brokerage costs and other fees or expenses associated with your RSUs, including those related to the sale of RSU Shares.

(ix) You Will Be Deemed to Represent Your Compliance with All the Terms of Your Award if You Accept Delivery of, or Sell, RSU Shares. You will be deemed to have represented and certified that you have complied with all of the terms of the Plan and this Award Agreement when RSU Shares are delivered to you[, you receive payment in respect of Dividend Equivalent Rights] and you request the sale of RSU Shares following the release of any Transfer Restrictions.

(x) You Agree that Covered Persons Will Not Have Liability. In addition to and without limiting the generality of the provisions of Section 1.3.5 of the Plan, neither the Firm nor any Covered Person will have any liability to you or any other person for any action taken or omitted in respect of this or any other Award.

13. **Non-transferability.** Except as otherwise may be provided by the Committee, the limitations set forth in Section 3.5 of the Plan will apply to this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition (including

through the use of any cash-settled instrument), whether voluntary or involuntary, in violation of the provisions of this Paragraph 13 or Section 3.5 of the Plan will be void.

14. **Right of Offset.** Except as provided in Paragraph 17(h), the obligation to deliver RSU Shares, to pay dividends [or payments under Dividend Equivalent Rights] or to remove any Transfer Restrictions under this Award Agreement is subject to Section 3.4 of the Plan, which provides for the Firm's right to offset against such obligation any outstanding amounts you owe to the Firm and any amounts the Committee deems appropriate pursuant to any tax equalization policy or agreement.

Arbitration, Choice of Forum and Governing Law

15. **Arbitration; Choice of Forum.**

(a) **By accepting this award, you are indicating that you understand and agree that the arbitration and choice of forum provisions set forth in Section 3.17 of the Plan will apply to this Award to the fullest extent permitted by applicable law. These provisions, which are expressly incorporated herein by reference, provide among other things that any dispute, controversy or claim between the Firm and you arising out of or relating to or concerning the Plan or this Award Agreement will be finally settled by arbitration, pursuant to the terms more fully set forth in Section 3.17 of the Plan; provided that nothing herein shall preclude you from filing a charge with or participating in any investigation or proceeding conducted by any governmental authority, including but not limited to the SEC, the Equal Employment Opportunity Commission and a state or local human rights agency, as well as law enforcement.**

(b) Nothing in the Plan or this Award Agreement includes an agreement to arbitrate claims on a collective, class or representative basis. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider class, collective or representative claims, to order consolidation or to join different claimants or grant relief other than on an individual basis to the individual claimant involved.

(c) Notwithstanding any applicable forum rules to the contrary, to the extent there is a question of enforceability of the Plan or this Award Agreement arising from a challenge to the arbitrator's jurisdiction or to the arbitrability of a claim, it will be decided by a court and not an arbitrator.

(d) Prior to filing any arbitration in accordance with Paragraph 15(a), you must first exhaust your internal administrative remedies in accordance with Paragraph 12(c), which includes awaiting the Committee's final determination of any such appeal. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider any dispute, controversy or claim as to which you have not first exhausted your internal administrative remedies in accordance with Paragraph 12(c).

(e) The Federal Arbitration Act governs interpretation and enforcement of all arbitration provisions under the Plan and this Award Agreement, and all arbitration proceedings thereunder.

(f) Nothing in the Plan or this Award Agreement creates a substantive right to bring a claim under U.S. Federal, state, or local employment laws.

(g) By accepting your Award, you irrevocably appoint each General Counsel of GS Inc., or any person whom the General Counsel of GS Inc. designates, as your agent for service of process in connection with any suit, action or proceeding arising out of or relating to or concerning the Plan or any Award which is not arbitrated pursuant to the provisions of Section 3.17.1 of the Plan, who shall promptly advise you of any such service of process.

(h) **As a condition of your Award, you also must accept the terms of the Agreement Regarding Arbitration of Employment-Related Matters, which is included among the Additional Terms described in Paragraph 12(a) and sets forth obligations independent of this Paragraph 15. For the avoidance of doubt, the obligation to exhaust your internal remedies described in**

Paragraph 12(c) and Paragraph 15(d) only applies to disputes, controversies or claims relating to the Plan or the Award Agreement.

16. Governing Law. This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

17. Compliance of Award Agreement and Plan with Section 409A. The provisions of this Paragraph 17 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 17 and the other provisions of this Award Agreement, this Paragraph 17 will govern.

(b) Delivery of RSU Shares will not be delayed beyond the date on which all applicable conditions or restrictions on delivery of RSU Shares required by this Award Agreement (including those specified in Paragraphs 6, 7, 11(c) and 12 and the consents and other items specified in Section 3.3 of the Plan) are satisfied. To the extent that any portion of this Award is intended to satisfy the requirements for short-term deferral treatment under Section 409A, delivery for such portion will occur by the March 15 coinciding with the last day of the applicable "short-term deferral" period described in Reg. 1.409A-1(b)(4) in order for the delivery of RSU Shares to be within the short-term deferral exception unless, in order to permit all applicable conditions or restrictions on delivery to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay delivery of RSU Shares to a later date within the same calendar year or to such later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 12(e) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any securities, other Awards or other property that the Firm may deliver in respect of your RSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the RSU Shares that would otherwise have been deliverable (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Notwithstanding the timing provisions of Paragraph 11(c), the delivery of RSU Shares referred to therein will be made after the date of death and not later than the end of the calendar year following the calendar year that includes the date of death (or on such later date as may be permitted under Section 409A).

(e) The timing of delivery or payment pursuant to Paragraph 11(a) will occur on the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the termination of Employment occurs; *provided, however*, that if you are a "specified employee" (as defined by the Firm in accordance with Section 409A(a)(2)(i)(B) of the Code), delivery will occur on the earlier of the Delivery Date or (to the extent required to avoid the imposition of additional tax under Section 409A) the date that is six months after your termination of Employment (or, if the latter date is not during a Window Period, the first trading day of the next Window Period). For purposes of Paragraph 11(a), references in this

Award Agreement to termination of Employment mean a termination of Employment from the Firm (as defined by the Firm) which is also a separation from service (as defined by the Firm in accordance with Section 409A).

(f) [Notwithstanding any provision of Paragraph 8 or Section 2.8.2 of the Plan to the contrary, the Dividend Equivalent Rights with respect to each of your Outstanding RSUs will be paid to you within the calendar year that includes the date of distribution of any corresponding regular cash dividends paid by GS Inc. in respect of a share of Common Stock the record date for which occurs on or after the Date of Grant. The payment will be in an amount (less applicable withholding) equal to such regular dividend payment as would have been made in respect of the RSU Shares underlying such Outstanding RSUs.]

(g) The timing of delivery or payment referred to in Paragraph 11(b)(i) will be the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the Committee receives satisfactory documentation relating to your Conflicted Employment, *provided* that such delivery or payment will be made, and any Committee action referred to in Paragraph 11(b)(ii) will be taken, only at such time as, and if and to the extent that it, as reasonably determined by the Firm, would not result in the imposition of any additional tax to you under Section 409A.

(h) Paragraph 14 and Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(i) Delivery of RSU Shares in respect of any Award may be made, if and to the extent elected by the Committee, later than the Delivery Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later delivery is permitted under Section 409A).

(j) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Committee Authority, Amendment, Construction and Regulatory Reporting

18. **Committee Authority.** The Committee has the authority to determine, in its sole discretion, that any event triggering forfeiture or repayment of your Award will not apply, to limit the forfeitures and repayments that result under Paragraphs 9 and 10 and to remove any Transfer Restrictions before the applicable Transferability Date.

19. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided* that, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further* that the Committee expressly reserves its rights to amend the Award Agreement and the Plan as described in Sections 1.3.2(h)(1), (2) and (4) of the Plan. A modification that impacts the tax consequences of this Award or the timing of delivery of RSU Shares will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

20. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words "include," "includes" and "including" will be deemed to be followed by the words "without limitation." The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

21. **Providing Information to the Appropriate Authorities; Protected Communications.** In accordance with applicable law, nothing in this Award Agreement (including the forfeiture and repayment provisions in Paragraphs 9 and 10) or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

[EU and UK Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider certain factors to determine whether and what portion of your Award will terminate, including the reason for the Loss Event or Risk Event, as well as the extent to which you participated in the Loss Event or Risk Event and your compensation for ____ may or may not have been adjusted to take into account the risk associated with the Loss Event, Risk Event, your Serious Misconduct or the Serious Misconduct of a Supervised Employee. Your compensation may be adjusted for the year in which the Loss Event, Risk Event, your Serious Misconduct or a Supervised Employee's Serious Misconduct is discovered.

If you are an EU MRT, Paragraphs (a), (b), (c) and (d) apply to all RSUs in your Award. If you are a GSAMI MRT, GSAMI Senior Management or UK MRT, Paragraphs (a), (b) and (c) of this Appendix apply to all RSUs in your Award, but Paragraph (d) applies only to your Base RSUs and Additional RSUs, as applicable, but not any Supplemental RSUs. Paragraph (e) of this Appendix contains certain definitions not otherwise contained in the Award Agreement.

(a) **A Loss Event Occurs Prior to Delivery.** If a Loss Event occurs prior to the delivery of RSU Shares, your rights in respect of all or a portion of your RSUs which are scheduled to deliver on the next Delivery Date immediately following the date that the Loss Event is identified (or, if not practicable, then the next following Delivery Date) will terminate, and no RSU Shares will be delivered in respect of such RSUs.

(i) **"Loss Event"** means (A) an annual pre-tax loss at GS Inc. or (B) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm), or annual negative revenues in the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm) of \$5 billion or more, *provided* in either case that you are employed in a business within such reporting segment.

(b) **A Risk Event Occurs On or Before the Applicable End Date.** If a Risk Event occurs on or before the Applicable End Date, (i) your rights in respect of all or a portion of your RSUs will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Risk Event occurred and (B) the date that the repayment request is made.

(i) **"Risk Event"** means there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting and Operational Risk Events Policy (or any successor policy).

(c) You Engage in Serious Misconduct On or Before the Applicable End Date. If you engage in Serious Misconduct during the period beginning on the applicable Transferability Date through the Applicable End Date, you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (i) the date the Serious Misconduct occurred and (ii) the date that the repayment request is made.

(i) **“Serious Misconduct”** means that you engage in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under applicable employment law.

(d) A Supervised Employee Engages in Serious Misconduct. If the Committee determines that it is appropriate to hold you accountable in whole or in part for Serious Misconduct related to compliance, control or risk that occurred during ____ by a Supervised Employee, your rights in respect of all or a portion of your RSUs will terminate and no RSU Shares will be delivered in respect of such RSUs and your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled.

(i) **“Supervised Employee”** means an individual with respect to whom the Committee determines you had supervisory responsibility as a result of direct or indirect reporting lines or your management responsibility for an office, division or business.

(e) **Certain Definitions.**

(i) **"Applicable End Date"** means [_____] , with respect to each classification in the table below, the following:

Classification	Applicable End Date
EU MRT	
GSAMI MRT	
GSAMI Senior Management	
UK MRT	

(ii) **“EU MRT”** means an individual who, at any point in _____, was classified by the Firm as a (A) Material Risk Taker in accordance with the applicable remuneration rules of the Alternative Investment Fund Managers Directive or Undertakings for Collective Investment in Transferable Securities V (“AIFMD-UCITS”), including “Identified Staff” under AIFMD-UCITS (“AIFMD-UCITS Identified Staff”), as applicable to the business operations of Goldman Sachs Asset Management, B.V. (“GSAMBV”), Goldman Sachs Asset Management Belgium, S.A. (“GSAM Belgium”) or Goldman Sachs Towarzystwo Funduszy Inwestycyjnych S.A. (“GS TFI”), including any branch of subsidiary thereof (collectively, “GSAM Europe”), whether or not GSAM Europe has employed such individual; or (B) an individual who, at any point in _____, was classified by the Firm with respect to Goldman Sachs Paris Inc. et Cie in accordance with the applicable remuneration rules of either the Capital Requirements Directive V or the Investment Firms Regulation/Investment Firms Directive, as a Material Risk Taker (“GSPIC MRT”) or Senior Management (“GSPIC Senior Management”). EU MRT does not include AIFMD-UCITS Identified Staff of Goldman Sachs Asset Management Fund Services Ltd. (“GSAMFSL”).

(iii) **“GSAMI MRT”** means an individual who, at any point in _____, was classified by the Firm as a GSAMI Material Risk Taker in accordance with the remuneration code under the UK Investment Firms Prudential Regime (“IFPR”) as applicable to the business operations of Goldman Sachs Asset Management International Ltd. (“GSAMI”).

(iv) **“GSAMI Senior Management”** means an individual who, at any point in ____, was classified by the Firm as a GSAMI Senior Management in accordance with the remuneration code under the IFPR as applicable to the business operations of GSAMI.

(v) **“UK MRT”** means an individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (including UK Risk Managers and UK PRA Senior Managers) in accordance with the remuneration codes applicable to dual-regulated UK firms.

(vi) For purposes of these definitions, if during ____ you were classified by the Firm as any combination of EU MRT and GSAMI MRT, GSAMI Senior Management or UK MRT then your classification in the table above is EU MRT; any combination of GSAMI MRT or GSAMI Senior Management and UK MRT then your classification in the table above is UK MRT; or as a combination of GSAMI MRT and GSAMI Senior Management then your classification in the table above is GSAMI Senior Management.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by Paragraphs (b) and (c) of this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

[GSBE Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider all relevant factors in making the decision, in its sole discretion as to whether an Adjustment Event or Actionable Misconduct have occurred.

This Appendix applies to all RSUs in your Award if you are a GSBE MRT or GSBE Senior Management.

(a) An Adjustment Event Occurs with Respect to ____. If an Adjustment Event occurs with respect to ____, (i) your rights in respect of all or a portion of your RSUs will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled, and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Adjustment Event occurred and (B) the date that the repayment request is made.

(i) **“Adjustment Event”** means that one of the following has occurred:

- A. you significantly contributed to, or were responsible for, any conduct that resulted in a loss of 0.75% or more of the total capital of GS Inc.;
- B. a material regulatory sanction for the Firm comprising one or more of the following:
 - 1. a moratorium pursuant to sec. 46g of the German Banking Act,
 - 2. a measure in case of danger pursuant to sec. 46 of the German Banking Act,
 - 3. the revocation of appointment of a manager pursuant to sec. 36 German Banking Act,
 - 4. a fine pursuant to sec. 56 of the German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of GS Inc.,
 - 5. the cancellation of the banking permit pursuant to sec. 35 of the German Banking Act,
 - 6. an order to increase the capital requirements Goldman Sachs Bank Europe SE (“GSBE”) by at least 0.5% pursuant to sec. 10 of the German Banking Act,

7. a measure in case of organizational deficiencies,
8. a comparable regulatory order, or
9. a material supervisory measure; or

C. you acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that a violation is considered serious if it suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct pursuant to sec. 1 German Termination Protection Act.

(b) You Engage in Actionable Misconduct. If the Committee determines that you engaged in Actionable Misconduct, which if known (or if known, adequately considered) would have resulted in a reduction to your annual variable compensation when your variable compensation for that year was determined, your rights in respect of all or a portion of your RSUs will terminate and no RSU Shares will be delivered in respect of such RSUs.

(i) **“Actionable Misconduct”** means either or both of:

- A. “Unethical behavior” within the meaning of the IVV;
- B. “Breach of duty” within the meaning of the IVV.

(ii) **“IVV”** means the German Remuneration Ordinance for Institutions (*Institutsvergütungsverordnung*) and applicable guidance from the German Federal Financial Supervisory Authority (BaFin) or the European Central Bank.

(c) Scope of Appendix. This Appendix applies to any individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (“GSBE MRT”) or Senior Management (“GSBE Senior Management”) in connection with the business operations of GSBE or any branch or subsidiary thereof, whether or not GSBE or any branch or subsidiary thereof employed such individual.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings:

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

(b) “Acceptance Deadline” means the date communicated to you by the Firm as the deadline to accept your Award. The Acceptance Deadline will be no less than 14 days after the Award Documentation is first made available to you in full. To the extent you are entitled to a translation in accordance with the Language Notice for Equity Awards provided to you as part of the Award Documentation, the Acceptance Deadline will be extended by the time required to furnish you with any such translation.

(c) “Associate With a Covered Enterprise” means that you (i) form, or acquire a 5% or greater equity ownership, voting or profit participation interest in, any Covered Enterprise or (ii) associate in any capacity (including association as an officer, employee, partner, director, consultant, agent or advisor) with any Covered Enterprise. Associate With a Covered Enterprise may include, as determined in the discretion of the Committee, (i) becoming the subject of any publicly available announcement or report of a pending or future association with a Covered Enterprise and (ii) unpaid associations, including an association in contemplation of future employment. “Association With a Covered Enterprise” will have its correlative meaning.

(d) “Breached an Obligation to the Firm” means that you failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with your Employment or this Award (including any applicable Additional Terms); including, without limitation, any offer letter, employment agreement or any shareholders’ agreement relating to the Firm. Your failure to pay or reimburse the Firm, on demand, for any amount you owe to the Firm will constitute (i) failure to meet an obligation you have under an agreement, regardless of whether such obligation arises under a written agreement, and/or (ii) a material violation of Firm policy constituting Cause.

(e) “Committee” means the Compensation Committee unless otherwise determined by the Board; *provided, however*, that: (i) the authority of the Compensation Committee to administer the Plan and the Award may be granted or delegated to the SIP Committee, in which case “Committee” means the SIP Committee when acting pursuant to such a grant or delegation; (ii) to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3; and (iii) to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m).

(f) “Compensation Committee” means the Compensation Committee of the Board.

(g) “Conflicted Employment” means your employment at any U.S. Federal, state or local government, any non-U.S. government, any supranational or international organization, any self-regulatory organization, or any agency or instrumentality of any such government or organization, or any

other employer (other than an “Accounting Firm” within the meaning of SEC Rule 2-01(f)(2) of Regulation S-X or any successor thereto) determined by the Committee, if, as a result of such employment, your continued holding of any Outstanding RSUs or Shares at Risk would result in an actual or perceived conflict of interest.

(h) “Covered Enterprise” means a Competitive Enterprise and any other existing or planned business enterprise that: (i) offers, holds itself out as offering or reasonably may be expected to offer products or services that are the same as or similar to those offered by the Firm or that the Firm reasonably expects to offer (“Firm Products or Services”) or (ii) engages in, holds itself out as engaging in or reasonably may be expected to engage in any other activity that is the same as or similar to any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage (“Firm Activities”). For the avoidance of doubt, Firm Activities include any activity that requires the same or similar skills as any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage, irrespective of whether any such financial activity is in furtherance of an advisory, agency, proprietary or fiduciary undertaking.

The enterprises covered by this definition include enterprises that offer, hold themselves out as offering or reasonably may be expected to offer Firm Products or Services, or engage in, hold themselves out as engaging in or reasonably may be expected to engage in Firm Activities directly, as well as those that do so indirectly by ownership or control (e.g., by owning, being owned by or by being under common ownership with an enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services or that engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities). The definition of Covered Enterprise includes, solely by way of example, any enterprise that offers, holds itself out as offering or reasonably may be expected to offer any product or service, or engages in, holds itself out as engaging in or reasonably may be expected to engage in any activity, in any case, associated with investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than you or your family members (including, for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading. An enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services, or engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities is a Covered Enterprise, **irrespective of whether the enterprise is a customer, client or counterparty of the Firm or is otherwise associated with the Firm and, because the Firm is a global enterprise, irrespective of where the Covered Enterprise is physically located.**

(i) “Failed to Consider Risk” means that you participated (or otherwise oversaw or were responsible for, depending on the circumstances, another individual’s participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where you have improperly analyzed such risk or where you have failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, the Committee determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, your business unit or the broader financial system. This definition is not limited to financial risks and is designed to encourage the consideration of the full range of risks associated with activities (e.g., legal, compliance or reputational). This definition also does not

require that a material adverse impact actually occur, but, rather, may be triggered if it is determined that there is a reasonable expectation of such an impact.

(j) "HCM" means the Firm's Human Capital Management Division or its successor.

(k) "Qualifying Termination After a Change in Control" means that the Firm terminates your Employment other than for Cause or you terminate your Employment for Good Reason, in each case, within 18 months following a Change in Control.

(l) "SEC" means the U.S. Securities and Exchange Commission.

(m) "Selected Firm Personnel" means any individual who is or in the three months preceding the conduct prohibited by Paragraph 9(a)(ii) was (i) a Firm employee or consultant with whom you personally worked while employed by the Firm, (ii) a Firm employee or consultant who, at any time during the year preceding the date of the termination of your Employment, worked in the same division in which you worked or (iii) an Advisory Director, a Managing Director (Participating or Extended) or a Senior Advisor of the Firm.

(n) "Shares at Risk" means RSU Shares subject to Transfer Restrictions.

(o) "Signature Card" means the document and/or website presented to you with your Award that you are required to execute (including electronically) to indicate your acceptance of the Award, the Award Agreement (including the Award Statement and the terms contained in the Signature Card) and Additional Terms applicable to you, as well your receipt of the Award Documentation.

(p) "Transferability Date" means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a transferability date; *provided, however*, that if there is no trading day in a Window Period that occurs in the specified month and year, the Transferability Date will be the first trading day in the subsequent Window Period.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Statement” means a written statement that reflects certain Award terms.

(c) “Board” means the Board of Directors of GS Inc.

(d) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(e) “Cause” means (i) the Grantee’s conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion, or (B) on a felony charge, or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations, (ii) the Grantee’s engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the Exchange Act), (iii) the Grantee’s willful failure to perform the Grantee’s duties to the Firm, (iv) the Grantee’s violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member, (v) the Grantee’s violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or the Grantee’s material violation of any other Firm policy as in effect from time to time, (vi) the Grantee’s engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm or (vii) the Grantee’s engaging in any conduct detrimental to the Firm. The determination as to whether Cause has occurred shall be made by the Committee in its sole discretion and, in such case, the Committee also may, but shall not be required to, specify the date such Cause occurred (including by determining that a prior termination of Employment was for Cause). Any rights the Firm may have hereunder and in any Award Agreement in respect of the events giving rise to Cause shall be in addition to the rights the Firm may have under any other agreement with a Grantee or at law or in equity.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Change in Control” means the consummation of a merger, consolidation, statutory share exchange or similar form of corporate transaction involving GS Inc. (a “Reorganization”) or sale or other disposition of all or substantially all of GS Inc.’s assets to an entity that is not an affiliate of GS Inc. (a “Sale”), that in each case requires the approval of GS Inc.’s shareholders under the law of GS Inc.’s jurisdiction of organization, whether for such Reorganization or Sale (or the issuance of securities of GS Inc. in such Reorganization or Sale), unless immediately following such Reorganization or Sale, either: (i) at least 50% of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of (A) the entity resulting from such Reorganization, or the entity which has acquired all or substantially all of the assets of GS Inc. in a Sale (in either case, the “Surviving Entity”), or (B) if applicable, the ultimate parent entity that directly or indirectly has beneficial

ownership (within the meaning of Rule 13d-3 under the Exchange Act, as such Rule is in effect on the date of the adoption of the 1999 SIP) of 50% or more of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of the Surviving Entity (the "Parent Entity") is represented by GS Inc.'s securities (the "GS Inc. Securities") that were outstanding immediately prior to such Reorganization or Sale (or, if applicable, is represented by shares into which such GS Inc. Securities were converted pursuant to such Reorganization or Sale) or (ii) at least 50% of the members of the board of directors (or similar officials in the case of an entity other than a corporation) of the Parent Entity (or, if there is no Parent Entity, the Surviving Entity) following the consummation of the Reorganization or Sale were, at the time of the Board's approval of the execution of the initial agreement providing for such Reorganization or Sale, individuals (the "Incumbent Directors") who either (A) were members of the Board on the Effective Date or (B) became directors subsequent to the Effective Date and whose election or nomination for election was approved by a vote of at least two-thirds of the Incumbent Directors then on the Board (either by a specific vote or by approval of GS Inc.'s proxy statement in which such persons are named as nominees for director).

(h) "Client" means any client or prospective client of the Firm to whom the Grantee provided services, or for whom the Grantee transacted business, or whose identity became known to the Grantee in connection with the Grantee's relationship with or employment by the Firm.

(i) "Code" means the Internal Revenue Code of 1986, as amended from time to time, and the applicable rulings and regulations thereunder.

(j) "Common Stock" means common stock of GS Inc., par value \$0.01 per share.

(k) "Competitive Enterprise" means an existing or planned business enterprise that (i) engages, or may reasonably be expected to engage, in any activity; (ii) owns or controls, or may reasonably be expected to own or control, a significant interest in any entity that engages in any activity or (iii) is, or may reasonably be expected to be, owned by, or a significant interest in which is, or may reasonably be expected to be, owned or controlled by, any entity that engages in any activity that, in any case, competes or will compete anywhere with any activity in which the Firm is engaged. The activities covered by this definition include, without limitation: financial services such as investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the Grantee and members of the Grantee's family (including for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading.

(l) "Covered Person" means a member of the Board or the Committee or any employee of the Firm.

(m) "Date of Grant" means the date specified in the Grantee's Award Agreement as the date of grant of the Award.

(n) ["Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.]

(o) "Effective Date" means the date this Plan is approved by the shareholders of GS Inc. pursuant to Section 3.15 of the Plan.

(p) "Employment" means the Grantee's performance of services for the Firm, as determined by the Committee. The terms "employ" and "employed" shall have their correlative meanings. The Committee in its sole discretion may determine (i) whether and when a Grantee's leave of absence results in a termination of Employment (for this purpose, unless the Committee determines otherwise, a Grantee shall be treated as terminating Employment with the Firm upon the occurrence of an Extended Absence), (ii) whether and when a change in a Grantee's association with the Firm results in a termination of Employment and (iii) the impact, if any, of any such leave of absence or change in association on Awards theretofore made. Unless expressly provided otherwise, any references in the Plan or any Award Agreement to a Grantee's Employment being terminated shall include both voluntary and involuntary terminations.

(q) "Exchange Act" means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(r) "Extended Absence" means the Grantee's inability to perform for six (6) continuous months, due to illness, injury or pregnancy-related complications, substantially all the essential duties of the Grantee's occupation, as determined by the Committee.

(s) "Firm" means GS Inc. and its subsidiaries and affiliates.

(t) "Good Reason" means, in connection with a termination of employment by a Grantee following a Change in Control, (i) as determined by the Committee, a materially adverse alteration in the Grantee's position or in the nature or status of the Grantee's responsibilities from those in effect immediately prior to the Change in Control or (ii) the Firm's requiring the Grantee's principal place of Employment to be located more than seventy-five (75) miles from the location where the Grantee is principally Employed at the time of the Change in Control (except for required travel on the Firm's business to an extent substantially consistent with the Grantee's customary business travel obligations in the ordinary course of business prior to the Change in Control).

(u) "Grantee" means a person who receives an Award.

(v) "GS Inc." means The Goldman Sachs Group, Inc., and any successor thereto.

(w) "1999 SIP" means The Goldman Sachs 1999 Stock Incentive Plan, as in effect prior to the effective date of the 2003 SIP.

(x) "Outstanding" means any Award to the extent it has not been forfeited, cancelled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(y) "Restricted Share" means a share of Common Stock delivered under the Plan that is subject to Transfer Restrictions, forfeiture provisions and/or other terms and conditions specified herein and in the Restricted Share Award Agreement or other applicable Award Agreement. All references to Restricted Shares include "Shares at Risk."

(z) “RSU” means a restricted stock unit granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(aa) “RSU Shares” means shares of Common Stock that underlie an RSU.

(ab) “Section 409A” means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(ac) “SIP Administrator” means each person designated by the Committee as a “SIP Administrator” with the authority to perform day-to-day administrative functions for the Plan.

(ad) “SIP Committee” means the persons who have been delegated certain authority under the Plan by the Committee.

(ae) “Solicit” means any direct or indirect communication of any kind whatsoever, regardless of by whom initiated, inviting, advising, suggesting, encouraging or requesting any person or entity, in any manner, to take or refrain from taking any action. The terms “Solicited,” “Soliciting” and “Solicitation” will have their correlative meanings.

(af) “Transfer Restrictions” means restrictions that prohibit the sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposal (including through the use of any cash-settled instrument), whether voluntarily or involuntarily by the Grantee, of an Award or any shares of Common Stock, cash or other property delivered in respect of an Award.

(ag) “Vested” means, with respect to an Award, the portion of the Award that is not subject to a condition that the Grantee remain actively employed by the Firm in order for the Award to remain Outstanding. The fact that an Award becomes Vested shall not mean or otherwise indicate that the Grantee has an unconditional or nonforfeitable right to such Award, and such Award shall remain subject to such terms, conditions and forfeiture provisions as may be provided for in the Plan or in the Award Agreement.

(ah) “Window Period” means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (*provided* that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
[One-Time][Year-End] RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of RSUs (your “Award”). You should read carefully this entire document (including its [Appendix][Appendices]), the Award Statement and the Signature Card (together, the “Award Agreement”), as well as the other documentation presented to you in connection with acceptance of your Award (collectively, “Award Documentation”).

Acceptance

1. **You Must Decide Whether to Accept Your Award.** To be eligible for your Award, you must by the Acceptance Deadline: (a) open and activate an Account; (b) accept any Additional Terms presented to you as part of the Award Documentation; and (c) execute (including by electronic means) the accompanying Signature Card in accordance with its instructions. By executing the Signature Card, you confirm your agreement to all of the terms of the Award Agreement (including the Award Statement and the Signature Card), as well as any Additional Terms.

Documents that Govern Your Award; Definitions

2. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement. The Plan and its Prospectus are included with the Award Documentation.

3. **Your Award Statement.** The accompanying Award Statement contains the type and number of RSUs awarded to you and some of your Award’s specific terms. For example, it specifies any applicable Vesting Dates, Delivery Dates and Transferability Dates. Your Award consists of up to [two][three] types of RSUs that are designated in the Award Statement and referred to in this Award Agreement as:

Award Statement Designation	Award Agreement Reference
Base RSUs	Base RSUs
Supplemental RSUs	Supplemental RSUs
[Additional Base RSUs]	[Additional RSUs]

4. All references to RSUs in this Award Agreement include Outstanding RSUs of all types, whether Base RSUs[,][or] Supplemental RSUs [or Additional RSUs] (as applicable to you). This Award Agreement does not govern the terms and conditions of any RSUs designated on your Award Statement as “Short-Term RSUs,” which (if applicable) are addressed in a separate Award Agreement.

5. **Definitions.** Unless otherwise defined herein, including in the **Definitions Appendix** or any other Appendix, capitalized terms have the meanings provided in the Plan.

Vesting of Your RSUs

6. **Vesting.** On each Vesting Date listed on your Award Statement, you will become Vested in the amount of any Base RSUs [and Additional RSUs] listed next to that date. All of your Supplemental RSUs, if any, are Vested. When an RSU becomes Vested, it means **only** that your continued active Employment is not required for delivery of that portion of RSU Shares. **Vesting does not mean you have a non-forfeitable right to the Vested portion of your Award. The terms of this Award Agreement (including conditions to delivery and any applicable Transfer Restrictions) continue to apply to Vested RSUs, and you can still forfeit Vested RSUs and any RSU Shares.**

Delivery of Your RSU Shares

7. **Delivery.** Reasonably promptly (but no more than 30 Business Days) after each Delivery Date identified on your Award Statement, RSU Shares (less applicable withholding as described in Paragraph 13(b)) will be delivered (by book entry credit to your Account) in respect of the type and amount of Outstanding RSUs listed next to that date. The Committee may select multiple dates within the 30-Business-Day period following the Delivery Date to deliver RSU Shares in respect of all or a portion of the RSUs with the same Delivery Date listed on the Award Statement, and all such dates will be treated as a single Delivery Date for purposes of this Award. Until such delivery, you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc. Without limiting the Committee's authority under Section 1.3.2(h) of the Plan, the Firm may accelerate any Delivery Date by up to 30 days.

Transfer Restrictions Following Delivery

8. **Transfer Restrictions and Shares at Risk.** Within 30 Business Days after any applicable Transferability Date identified on your Award Statement, GS Inc. will remove the Transfer Restrictions on any Shares at Risk in respect of the type and amount of Outstanding RSUs listed next to that date. If your Award Statement reflects a Transferability Date, [% of the RSU Shares delivered on any Delivery Date **before** tax withholding (or, if the tax withholding rate is greater than 50%, all RSU Shares)][all of the RSU Shares delivered on any Delivery Date **after** tax withholding] will be Shares at Risk. The Committee may select multiple dates within the 30-Business-Day period on which to remove Transfer Restrictions for all or a portion of the Shares at Risk with the same Transferability Date, and all such dates will be treated as a single Transferability Date for purposes of this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition in violation of the Transfer Restrictions on Shares at Risk will be void. **[If your Award Statement does not reflect any Transferability Date [as applying to a type of Outstanding RSUs], then references to Transfer Restrictions and Shares at Risk in this Award Agreement are inapplicable to your [Award][RSU Shares delivered in respect of such RSUs] and this Paragraph 7 does not apply [to RSU Shares delivered in respect of such RSUs].**

Dividends

9. **[Dividend Equivalent Rights and] Dividends.** [Each RSU includes a Dividend Equivalent Right, which entitles you to receive an amount (less applicable withholding), at or after the time of distribution of any regular cash dividend paid by GS Inc. in respect of a share of Common Stock, equal to any regular cash dividend payment that would have been made in respect of an RSU Share underlying your Outstanding RSUs for any record date that occurs on or after the Date of Grant. In addition, you][You] will be entitled to receive on a current basis any regular cash dividend paid in respect of any Shares at Risk. [The RSUs do **not** include Dividend Equivalent Rights.]

Forfeiture of Your Award

10. **How You May Forfeit Your Award.** This Paragraph 9 sets forth the events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. More than one event may apply, and in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event. Paragraph 11 (relating to certain circumstances under which you will not forfeit your unvested RSUs upon Employment termination) and Paragraph 12 (relating to certain circumstances under which vesting, delivery and/or release of Transfer Restrictions may be accelerated) provide for exceptions to one or more provisions of this Paragraph 9.

(a) **Unvested RSUs Forfeited if Your Employment Terminates.** If your Employment terminates for any reason or you are otherwise no longer actively Employed with the Firm (which includes off-premises notice periods, "garden leaves," pay in lieu of notice or any other similar status), your rights to your Outstanding Base RSUs [and Additional RSUs] that are not Vested will terminate, and no RSU Shares will be delivered in respect of such RSUs.

(b) Vested and Unvested Base RSUs [and Additional RSUs] Forfeited Upon Certain Events. If any of the following occurs before the applicable Delivery Date, your rights to all of your Outstanding Base RSUs [and Additional RSUs] (whether or not Vested) will terminate, and no RSU Shares will be delivered in respect of such RSUs:

(i) You Solicit Clients or Employees, Interfere with Client or Employee Relationships or Participate in the Hiring of Employees. Except as prohibited by applicable law, either:

A. you, in any manner, directly or indirectly, (1) Solicit any Client to transact business with a Covered Enterprise or to reduce or refrain from doing any business with the Firm, (2) interfere with or damage (or attempt to interfere with or damage) any relationship between the Firm and any Client, (3) Solicit any person who is an employee of the Firm to resign from the Firm, (4) Solicit any Selected Firm Personnel to apply for or accept employment (or other association) with any person or entity other than the Firm or (5) participate in the hiring of any Selected Firm Personnel by any person or entity other than the Firm (including, without limitation, participating in the identification of individuals for potential hire, and participating in any hiring decision), whether as an employee or consultant or otherwise, or

B. Selected Firm Personnel are Solicited, hired or accepted into partnership, membership or similar status by any entity where you have, or will have, direct or indirect managerial responsibility for such Selected Firm Personnel, unless the Committee determines that you were not involved in such Solicitation, hiring or acceptance.

(ii) [GS Inc. Fails to Maintain the Minimum Tier 1 Capital Ratio. GS Inc. fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days.]

(iii) [GS Inc. Is Determined to Be in Default. The Board of Governors of the Federal Reserve or the Federal Deposit Insurance Corporation (the "FDIC") makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Wall Street Reform and Consumer Protection Act for the appointment of the FDIC as a receiver of GS Inc. based on a determination that GS Inc. is "in default" or "in danger of default."]

(c) Vested and Unvested RSUs and Shares at Risk Forfeited upon Certain Events. If any of the following occurs (i) your rights to all of your Outstanding RSUs (whether or not Vested) will terminate, and no RSU Shares will be delivered in respect of such RSUs and (ii) your rights to any Shares at Risk will terminate and all such Shares at Risk will be cancelled, in each case, as may be further described below:

(i) You Failed to Consider Risk. You Failed to Consider Risk during ____.

(ii) Your Conduct Constitutes Cause. Any event that constitutes Cause has occurred before the applicable Delivery Date for RSUs or the applicable Transferability Date for any Shares at Risk.

(iii) You Breach an Obligation to the Firm. The Committee determines that, before the applicable Delivery Date for RSUs or the applicable Transferability Date for any Shares at Risk, you Breached an Obligation to the Firm.

(iv) You Do Not Provide Timely Certifications or Comply with Your Certifications. You fail to certify to GS Inc. that you have complied with all of the terms of the Plan and this Award Agreement as required by Paragraph 13(d), or the Committee determines that you have failed to comply with a term of the Plan or this Award Agreement to which you have certified compliance.

(v) You Do Not Follow Dispute Resolution Procedures. You attempt to have any dispute, controversy or claim under the Plan or this Award Agreement resolved in any manner that is not provided for by Paragraph 16 or Section 3.17 of the Plan, or you attempt to arbitrate a dispute, controversy or claim without first having exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(vi) You Bring an Action that Results in a Determination that Any Award Agreement Term Is Invalid. As a result of any action brought by you, it is determined that any term of this Award Agreement is invalid.

(vii) You Receive Compensation in Respect of Your Award from Another Employer. Your Employment terminates for any reason or you otherwise are no longer actively Employed with the Firm and another entity grants you cash, equity or other property (whether vested or unvested) to replace, substitute for or otherwise in respect of any Outstanding RSUs or Shares at Risk; *provided, however*, that your rights will only be terminated in respect of the RSUs and Shares at Risk that are replaced, substituted for or otherwise considered by such other entity in making its grant.

(d) [Additional Forfeiture and Repayment Conditions for Material Risk Takers]. [If at any point in ____][Since] the Firm identified you as a Material Risk Taker (“MRT”) [in ____] then the **EU and UK Material Risk Taker Appendix** and/or the **GSBE Material Risk Taker Appendix** (as applicable to you, the “MRT Appendix”) supplements this Paragraph 9. The MRT Appendix sets forth additional events that (i) result in forfeiture of up to all of your RSUs and Shares at Risk and (ii) may require repayment to the Firm of up to all amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. Your MRT classification, as defined in the MRT Appendix, was communicated to you separately. [If you are unsure of whether the Firm identified you as an MRT, or which classification(s) apply, you must contact HCM prior to accepting this Award.] The Firm’s records with respect to such classification(s) will be controlling as to whether and which MRT Appendix applies to your Award. The MRT classifications with their corresponding MRT Appendix are as follows:

MRT Classification	MRT Appendix
EU MRT, GSAMI MRT, GSAMI Senior Management and/or UK MRT	EU and UK Material Risk Taker Appendix
GSBE MRT and GSBE Senior Management	GSBE Material Risk Taker Appendix
GSBE MRT or GSBE Senior Management <i>and any</i> combination of EU MRT, GSAMI MRT, GSAMI Senior Management or UK MRT	GSBE Material Risk Taker Appendix <i>and</i> EU and UK Material Risk Taker Appendix]

(e) Actions Pending Forfeiture. In connection with any investigation as to whether any of the events that would result in forfeiture under the Plan [or][,] this Paragraph 9 [or the MRT Appendix] have occurred, the Firm reserves the right to (i) suspend vesting of Outstanding RSUs, [payments under Dividend Equivalent Rights,] delivery of RSU Shares or release of any Transfer Restrictions; (ii) deliver any RSU Shares[, payments under Dividend Equivalent Rights] or dividends into an escrow account in accordance with Paragraph 13(e)(iii); or (iii) apply Transfer Restrictions to any RSU Shares.

Repayment of Your Award

11. **When You May Be Required to Repay Your Award**. If the Committee determines that any term of this Award was not satisfied, you will be required, immediately upon demand therefor, to repay to the Firm the following:

(a) Any RSU Shares (which, for the avoidance of doubt, includes any Shares at Risk) for which the terms (including the terms for delivery) of the related RSUs were not satisfied, in accordance with Section 2.6.3 of the Plan.

(b) Any Shares at Risk for which the terms (including the terms for the release of Transfer Restrictions) were not satisfied, in accordance with Section 2.5.3 of the Plan.

(c) Any RSU Shares that were delivered (but not subject to Transfer Restrictions) at the same time any Shares at Risk that are cancelled or required to be repaid were delivered.

(d) [Any payments under Dividend Equivalent Rights for which the terms were not satisfied (including any such payments made in respect of RSUs that are forfeited or RSU Shares that are cancelled or required to be repaid), in accordance with Section 2.8.3 of the Plan.]

(e) Any dividends paid in respect of any RSU Shares that are cancelled or required to be repaid.

(f) Any amount applied to satisfy tax withholding or other obligations with respect to any RSUs, RSU Shares[,] [and] dividend payments [and payments under Dividend Equivalent Rights] that are forfeited or required to be repaid.

Exceptions to the Vesting, Delivery and/or Transferability Dates

12. **Circumstances Under Which You Will Not Forfeit Your Unvested RSUs on Employment Termination (but the Original Delivery Dates and Transferability Dates Continue to Apply)**. If your Employment terminates at a time when you meet the requirements for [Retirement,] Extended Absence[,] ["downsizing"] [or] [Approved Termination], [each] as described below, then Paragraph 9(a) will not apply, and your Outstanding RSUs will be treated as described in this Paragraph 11. All other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

(a) [Retirement or] Extended Absence and No Association With a Covered Enterprise.

(i) Generally. If your Employment terminates by [Retirement or] Extended Absence, your Outstanding RSUs that are not Vested will be treated as Vested. **However, your rights to any Outstanding RSU that are treated as Vested by this Paragraph 11(a)(i) will terminate and no RSU Share will be delivered in respect of that RSU if you Associate With a Covered Enterprise on or before the originally scheduled Vesting Date for that RSU.** You acknowledge and agree that if any portion of the preceding sentence is unenforceable under applicable law, or is found to be unenforceable, then the entirety of this Paragraph 11(a)(i) is inapplicable to you and your rights to any Outstanding RSUs that were treated as Vested, or otherwise would have been treated as Vested, will be cancelled in accordance with Paragraph 9(a).

(ii) Special Treatment for Involuntary or Mutual Agreement Termination. The second sentence of Paragraph 11(a)(i) (relating to forfeiture if you Associate With a Covered Enterprise) will not apply if (A) the Firm characterizes your Employment termination as "involuntary" or by "mutual agreement" (and, in each case, you have not engaged in conduct constituting Cause) and (B) you execute a general waiver and release of claims and an agreement to pay any associated tax liability, in each case, in the form the Firm prescribes. No Employment termination that you initiate, including any purported "constructive termination," a "termination for good reason" or similar concepts, can be "involuntary" or by "mutual agreement."

(b) [Downsizing. If (i) the Firm terminates your Employment solely by reason of a "downsizing" (and you have not engaged in conduct constituting Cause) and (ii) you execute a general waiver and release of claims and an agreement to pay any associated tax liability, in each case, in the form the Firm prescribes, your Outstanding RSUs that are not yet Vested will become Vested. Whether or not your Employment is terminated solely by reason of a "downsizing" will be determined by the Firm in its sole discretion.]

(c) [Approved Terminations of Fixed-Term Employees]. If the Firm classifies you as a “fixed-term” employee and your Employment terminates solely by reason of an Approved Termination (and you have not engaged in conduct constituting Cause), your Outstanding RSUs that are not yet Vested will become Vested.]

13. **Accelerated Vesting, Delivery and/or Release of Transfer Restrictions in the Event of [a Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] Death.** In the event of your [Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] death, [each] as described below, then Paragraph 9(a) will not apply, your Outstanding RSUs and any Shares at Risk will be treated as described in this Paragraph 12, and, except as set forth in Paragraph 12(a), all other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

(a) [You Have a Qualifying Termination After a Change in Control]. If your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, the RSU Shares underlying your Outstanding RSUs (whether or not Vested) will be delivered, and any Transfer Restrictions will cease to apply. In addition, the forfeiture events in Paragraph 9 [(except the MRT Appendix)] will not apply to your Award.]

(b) [You Are Determined to Have Accepted Conflicted Employment].

(i) Generally. To the extent consistent with applicable law, regulation and regulatory guidance, the following will apply as soon as practicable after the Committee has received satisfactory documentation relating to your Conflicted Employment.

A. Vesting. If your Employment terminates solely because you resign to accept Conflicted Employment and, according to the Firm’s records, you have completed at least three years of continuous service with the Firm as of the date your Employment terminates, your Outstanding RSUs will Vest; otherwise, you will forfeit any Outstanding RSUs that are not Vested in accordance with Paragraph 9(a).

B. Delivery and Release of Transfer Restrictions. If your Employment terminates solely because you resign to accept Conflicted Employment or if, following your termination of Employment, you notify the Firm that you are accepting Conflicted Employment, RSU Shares will be delivered in respect of your Outstanding Vested RSUs (including in the form of cash as described in Paragraph 13(e)(ii)) and any Transfer Restrictions will cease to apply.

(ii) You May Have to Take Other Steps to Address Conflicts of Interest. The Committee retains the authority to exercise its rights under the Award Agreement or the Plan (including Section 1.3.2 of the Plan) to take or require you to take other steps it determines in its sole discretion to be necessary or appropriate to cure an actual or perceived conflict of interest (which may include a determination that the accelerated vesting, delivery and/or release of Transfer Restrictions described in Paragraph 12(b)(i) will not apply because such actions are not necessary or appropriate to cure an actual or perceived conflict of interest).]

(c) Death. If you die, the RSU Shares underlying your Outstanding RSUs (whether or not Vested) will be delivered to your Account and any Transfer Restrictions will cease to apply as soon as practicable after the date of death and receipt of any documentation as may be requested by the Committee or a SIP Administrator.

Other Terms and Conditions of Your Award

14. **Additional Terms, Conditions and Consents.**

(a) You Will Be Required to Accept Additional Terms. In addition to the terms of the Award Agreement, the Award Documentation includes other contractual terms that you must accept as a condition of the Award (the "Additional Terms"). By accepting the Award and executing the Signature Card, you agree to any such Additional Terms applicable to you as set forth in the Award Documentation, which will include but are not limited to the following:

(i) Notice Period Policy. You must agree to abide by the terms, as applicable to you, of the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy").

(ii) Agreement Regarding Arbitration of Employment-Related Matters. You must agree to the Agreement Regarding Arbitration of Employment-Related Matters, which imposes obligations in addition to those contained in Paragraph 16.

(iii) APAC Restrictive Covenants. If you are employed by the Firm in the Asia-Pacific Region (e.g., Australia, the People's Republic of China, the Hong Kong Special Administrative Region, India, Indonesia, Japan, the Republic of Korea, New Zealand, Singapore, Taiwan), you must agree to the APAC Restrictive Covenants Agreement.

(iv) Role-, Business- or Division-Specific Agreements. If applicable to you, you must accept Additional Terms related to your specific role, business unit or division as set forth in the Award Documentation.

(b) You Must Satisfy Applicable Tax Withholding Requirements. Delivery of RSU Shares is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, which includes the Firm deducting or withholding amounts from any payment or distribution to you. In addition, to the extent permitted by applicable law, the Firm, in its sole discretion, may require you to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations imposed on you or the Firm in connection with the grant, Vesting or delivery of this Award by requiring you to choose between remitting the amount (i) in cash (or through payroll deduction or otherwise) or (ii) in the form of proceeds from the Firm's executing a sale of RSU Shares delivered to you under this Award. In no event, however, does this Paragraph 13(b) give you any discretion to determine or affect the timing of the delivery of RSU Shares or the timing of payment of tax obligations.

(c) You Must Comply with Applicable Deadlines and Procedures to Appeal. If you disagree with a determination made by the Committee, the SIP Administrators or any of their delegates or designees relating to the Plan or the Award Agreement and you wish to appeal such determination, you must submit a written request to the Committee for review within 180 days after the determination at issue. You must exhaust your internal administrative remedies (*i.e.*, submit your appeal and wait for resolution of that appeal) before seeking to resolve a dispute, controversy or claim through arbitration pursuant to Paragraph 16 and Section 3.17 of the Plan.

(d) You May Be Required to Certify Compliance with Award Terms; You Are Responsible for Providing Updated Address and Contact Information After Your Departure from the Firm. If your Employment terminates while you continue to hold RSUs or Shares at Risk, the Firm may require you to provide certifications of your compliance with all of the terms of the Plan and this Award Agreement. You understand and agree that (i) your contact information as reflected in the Firm's personnel records at the time any certification is requested will be deemed current; (ii) it is your responsibility to inform HCM of any changes to your contact information to ensure timely receipt of the certification materials, regardless of whether your contact information is provided to another part of the Firm; and (iii) you are responsible for contacting the Firm to obtain such certification materials if not received. Your failure to return properly completed certification materials by the specified deadline (including because you did not provide HCM with updated contact information) will result in the forfeiture of all of your RSUs and any Shares at Risk and subject previously delivered amounts to repayment under Paragraph 9(c)(iv).

(e) You Agree to Other Terms, Conditions and Consents. By accepting this Award you understand and agree that:

(i) Amounts May Be Rounded to Avoid Fractional RSU Shares. RSUs that become Vested, RSU Shares that become deliverable and RSU Shares subject to Transfer Restrictions may, in each case, be rounded to avoid fractional RSU Shares.

(ii) Firm May Deliver Cash or Other Property Instead of RSU Shares. In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the RSU Shares, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to deliveries of RSU Shares will include such deliveries of cash, other securities, other awards under the Plan or other property.

(iii) Firm May Deliver Your Award into an Escrow Account. The Firm may establish and maintain an escrow account on such terms (which may include your executing any documents related to, and your paying for any costs associated with, such account) as it may deem necessary or appropriate, and the delivery of RSU Shares (including any Shares at Risk), and any payment of cash (including dividends [and payments under Dividend Equivalent Rights]) or other property, may initially be made into and held in that escrow account until such time as the Committee has received such documentation as it may have requested or until the Committee has determined that any other conditions or restrictions on delivery of RSU Shares, cash or other property required by this Award Agreement have been satisfied.

(iv) Firm May Affix Legends and Place Stop Orders on RSU Shares. GS Inc. may affix to Certificates representing RSU Shares any legend that the Committee determines to be necessary or advisable (including to reflect any restrictions to which you may be subject under the Additional Terms, a separate agreement or any Firm policy applicable to you). GS Inc. may advise the transfer agent to place a stop order against any legended RSU Shares. For the avoidance of doubt, RSU Shares as used herein includes, without limitation, Restricted Shares.

(v) You Authorize the Firm to Register, in Its or Its Designee's Name, Any Restricted Shares and Sell, Assign or Transfer Any Forfeited Restricted Shares. You are granting to the Firm the full power and authority to register any Restricted Shares in its or its designee's name and authorizing the Firm or its designee to sell, assign or transfer any Restricted Shares if you forfeit your Restricted Shares.

(vi) You Agree to Certain Consents as a Condition to the Award. You have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

(vii) You Are Subject to the Firm's Policies, Rules and Procedures. You are subject to the Firm's policies in effect from time to time concerning trading in RSU Shares and hedging or pledging RSU Shares and equity-based compensation or other awards (including, without limitation, the "Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards" or any successor policies), and confidential or proprietary information, and you will effect sales of RSU Shares in accordance with such rules and procedures as may be adopted from time to time (which may include, without limitation, restrictions relating to the timing of sale requests, the manner in which sales are executed, pricing method, consolidation or aggregation of orders and volume limits determined by the Firm).

(viii) You Are Responsible for Costs Associated with Your Award. You will be responsible for all brokerage costs and other fees or expenses associated with your RSUs, including those related to the sale of RSU Shares.

(ix) You Will Be Deemed to Represent Your Compliance with All the Terms of Your Award if You Accept Delivery of, or Sell, RSU Shares. You will be deemed to have represented and certified that you have complied with all of the terms of the Plan and this Award Agreement when RSU Shares are delivered to you[, you receive payment in respect of Dividend Equivalent Rights] and you request the sale of RSU Shares following the release of any Transfer Restrictions.

(x) You Agree that Covered Persons Will Not Have Liability. In addition to and without limiting the generality of the provisions of Section 1.3.5 of the Plan, neither the Firm nor any Covered Person will have any liability to you or any other person for any action taken or omitted in respect of this or any other Award.

15. **Non-transferability.** Except as otherwise may be provided by the Committee, the limitations set forth in Section 3.5 of the Plan will apply to this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition (including through the use of any cash-settled instrument), whether voluntary or involuntary, in violation of the provisions of this Paragraph 14 or Section 3.5 of the Plan will be void.

16. **Right of Offset.** Except as provided in Paragraph 18(h), the obligation to deliver RSU Shares, to pay dividends [or payments under Dividend Equivalent Rights] or to remove any Transfer Restrictions under this Award Agreement is subject to Section 3.4 of the Plan, which provides for the Firm's right to offset against such obligation any outstanding amounts you owe to the Firm and any amounts the Committee deems appropriate pursuant to any tax equalization policy or agreement.

Arbitration, Choice of Forum and Governing Law

17. Arbitration; Choice of Forum.

(a) **By accepting this award, you are indicating that you understand and agree that the arbitration and choice of forum provisions set forth in Section 3.17 of the Plan will apply to this Award to the fullest extent permitted by applicable law. These provisions, which are expressly incorporated herein by reference, provide among other things that any dispute, controversy or claim between the Firm and you arising out of or relating to or concerning the Plan or this Award Agreement will be finally settled by arbitration, pursuant to the terms more fully set forth in Section 3.17 of the Plan; provided that nothing herein shall preclude you from filing a charge with or participating in any investigation or proceeding conducted by any governmental authority, including but not limited to the SEC, the Equal Employment Opportunity Commission and a state or local human rights agency, as well as law enforcement.**

(b) Nothing in the Plan or this Award Agreement includes an agreement to arbitrate claims on a collective, class or representative basis. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider class, collective or representative claims, to order consolidation or to join different claimants or grant relief other than on an individual basis to the individual claimant involved.

(c) Notwithstanding any applicable forum rules to the contrary, to the extent there is a question of enforceability of the Plan or this Award Agreement arising from a challenge to the arbitrator's jurisdiction or to the arbitrability of a claim, it will be decided by a court and not an arbitrator.

(d) Prior to filing any arbitration in accordance with Paragraph 16(a), you must first exhaust your internal administrative remedies in accordance with Paragraph 13(c), which includes awaiting the Committee's final determination of any such appeal. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider any dispute, controversy or claim as to which you have not first exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(e) The Federal Arbitration Act governs interpretation and enforcement of all arbitration provisions under the Plan and this Award Agreement, and all arbitration proceedings thereunder.

(f) Nothing in the Plan or this Award Agreement creates a substantive right to bring a claim under U.S. Federal, state, or local employment laws.

(g) By accepting your Award, you irrevocably appoint each General Counsel of GS Inc., or any person whom the General Counsel of GS Inc. designates, as your agent for service of process in connection with any suit, action or proceeding arising out of or relating to or concerning the Plan or any Award which is not arbitrated pursuant to the provisions of Section 3.17.1 of the Plan, who shall promptly advise you of any such service of process.

(h) **As a condition of your Award, you also must accept the terms of the Agreement Regarding Arbitration of Employment-Related Matters, which is included among the Additional Terms described in Paragraph 13(a) and sets forth obligations independent of this Paragraph 16. For the avoidance of doubt, the obligation to exhaust your internal remedies described in Paragraph 13(c) and Paragraph 16(d) only applies to disputes, controversies or claims relating to the Plan or the Award Agreement.**

18. Governing Law. This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

19. Compliance of Award Agreement and Plan with Section 409A. The provisions of this Paragraph 18 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 18 and the other provisions of this Award Agreement, this Paragraph 18 will govern.

(b) Delivery of RSU Shares will not be delayed beyond the date on which all applicable conditions or restrictions on delivery of RSU Shares required by this Award Agreement (including those specified in Paragraphs 6, 7, 11(a)(ii), 11(b), 12(c) and 13 and the consents and other items specified in Section 3.3 of the Plan) are satisfied. To the extent that any portion of this Award is intended to satisfy the requirements for short-term deferral treatment under Section 409A, delivery for such portion will occur by the March 15 coinciding with the last day of the applicable "short-term deferral" period described in Reg. 1.409A-1(b)(4) in order for the delivery of RSU Shares to be within the short-term deferral exception unless, in order to permit all applicable conditions or restrictions on delivery to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay delivery of RSU Shares to a later date within the same calendar year or to such later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 13(e) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any securities, other Awards or other property that the Firm may deliver in respect of your RSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the RSU Shares that would otherwise have been deliverable (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Notwithstanding the timing provisions of Paragraph 12(c), the delivery of RSU Shares referred to therein will be made after the date of death and not later than the end of the calendar year following the calendar year that includes the date of death (or on such later date as may be permitted under Section 409A).

(e) The timing of delivery or payment pursuant to Paragraph 12(a) will occur on the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the termination of Employment occurs; *provided, however*, that if you are a “specified employee” (as defined by the Firm in accordance with Section 409A(a)(2)(i)(B) of the Code), delivery will occur on the earlier of the Delivery Date or (to the extent required to avoid the imposition of additional tax under Section 409A) the date that is six months after your termination of Employment (or, if the latter date is not during a Window Period, the first trading day of the next Window Period). For purposes of Paragraph 12(a), references in this Award Agreement to termination of Employment mean a termination of Employment from the Firm (as defined by the Firm) which is also a separation from service (as defined by the Firm in accordance with Section 409A).

(f) [Notwithstanding any provision of Paragraph 8 or Section 2.8.2 of the Plan to the contrary, the Dividend Equivalent Rights with respect to each of your Outstanding RSUs will be paid to you within the calendar year that includes the date of distribution of any corresponding regular cash dividends paid by GS Inc. in respect of a share of Common Stock the record date for which occurs on or after the Date of Grant. The payment will be in an amount (less applicable withholding) equal to such regular dividend payment as would have been made in respect of the RSU Shares underlying such Outstanding RSUs.]

(g) The timing of delivery or payment referred to in Paragraph 12(b)(i) will be the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the Committee receives satisfactory documentation relating to your Conflicted Employment, *provided* that such delivery or payment will be made, and any Committee action referred to in Paragraph 12(b)(ii) will be taken, only at such time as, and if and to the extent that it, as reasonably determined by the Firm, would not result in the imposition of any additional tax to you under Section 409A.

(h) Paragraph 15 and Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(i) Delivery of RSU Shares in respect of any Award may be made, if and to the extent elected by the Committee, later than the Delivery Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later delivery is permitted under Section 409A).

(j) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Committee Authority, Amendment, Construction and Regulatory Reporting

20. **Committee Authority.** The Committee has the authority to determine, in its sole discretion, that any event triggering forfeiture or repayment of your Award will not apply, to limit the forfeitures and repayments that result under Paragraphs 9 and 10 and to remove any Transfer Restrictions before the applicable Transferability Date. In addition, the Committee, in its sole discretion, may determine whether Paragraphs 11(a)(ii) and 11(b) will apply upon a termination of Employment and whether a termination of Employment constitutes an Approved Termination under Paragraph 11(c).

21. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided* that, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further* that the Committee expressly reserves its rights to amend the Award Agreement and the

Plan as described in Sections 1.3.2(h)(1), (2) and (4) of the Plan. A modification that impacts the tax consequences of this Award or the timing of delivery of RSU Shares will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

22. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words “include,” “includes” and “including” will be deemed to be followed by the words “without limitation.” The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

23. **Providing Information to the Appropriate Authorities; Protected Communications.** In accordance with applicable law, nothing in this Award Agreement (including the forfeiture and repayment provisions in Paragraphs 9 and 10) or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

[EU and UK Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider certain factors to determine whether and what portion of your Award will terminate, including the reason for the Loss Event or Risk Event, as well as the extent to which you participated in the Loss Event or Risk Event and your compensation for the ____ may or may not have been adjusted to take into account the risk associated with the Loss Event, Risk Event, your Serious Misconduct or the Serious Misconduct of a Supervised Employee. Your compensation may be adjusted for the year in which the Loss Event, Risk Event, your Serious Misconduct or a Supervised Employee's Serious Misconduct is discovered.

If you are an EU MRT, Paragraphs (a), (b), (c) and (d) apply to all RSUs in your Award. If you are a GSAMI MRT, GSAMI Senior Management or UK MRT, Paragraphs (a), (b) and (c) of this Appendix apply to all RSUs in your Award, but Paragraph (d) applies only to your Base RSUs [and Additional RSUs, as applicable], but not any Supplemental RSUs. Paragraph (e) of this Appendix contains certain definitions not otherwise contained in the Award Agreement.

(a) A Loss Event Occurs Prior to Delivery. If a Loss Event occurs prior to the delivery of RSU Shares, your rights in respect of all or a portion of your RSUs (whether or not Vested) which are scheduled to deliver on the next Delivery Date immediately following the date that the Loss Event is identified (or, if not practicable, then the next following Delivery Date) will terminate, and no RSU Shares will be delivered in respect of such RSUs.

(i) **"Loss Event"** means (A) an annual pre-tax loss at GS Inc. or (B) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm), or annual negative revenues in the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm) of \$5 billion or more, *provided* in either case that you are employed in a business within such reporting segment.

(b) A Risk Event Occurs On or Before the Applicable End Date. If a Risk Event occurs on or before the Applicable End Date, (i) your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Risk Event occurred and (B) the date that the repayment request is made.

(i) **"Risk Event"** means there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting and Operational Risk Events Policy (or any successor policy).

(i) **“Serious Misconduct”** means that you engage in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under applicable employment law.

(i) **“Supervised Employee”** means an individual with respect to whom the Committee determines you had supervisory responsibility as a result of direct or indirect reporting lines or your management responsibility for an office, division or business.

(i) **“Applicable End Date”** means [_____], with respect to each classification in the table below, the following:

(ii) **“EU MRT”** means an individual who, at any point in _____, was classified by the Firm as a (A) Material Risk Taker in accordance with the applicable remuneration rules of the Alternative Investment Fund Managers Directive or Undertakings for Collective Investment in Transferable Securities V (“AIFMD-UCITS”), including “Identified Staff” under AIFMD-UCITS (“AIFMD-UCITS Identified Staff”), as applicable to the business operations of Goldman Sachs Asset Management, B.V. (“GSAMBV”), Goldman Sachs Asset Management Belgium, S.A. (“GSAM Belgium”) or Goldman Sachs Towarzystwo Funduszy Inwestycyjnych S.A. (“GS TFI”), including any branch of subsidiary thereof (collectively, “GSAM Europe”), whether or not GSAM Europe has employed such individual; or (B) an individual who, at any point in _____, was classified by the Firm with respect to Goldman Sachs Paris Inc. et Cie in accordance with the applicable remuneration rules of either the Capital Requirements Directive V or the Investment Firms Regulation/Investment Firms Directive, as a Material Risk Taker (“GSPIC MRT”) or Senior Management (“GSPIC Senior Management”). EU MRT does not include AIFMD-UCITS Identified Staff of Goldman Sachs Asset Management Fund Services Ltd. (“GSAMFSL”).

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(iv) **“GSAMI Senior Management”** means an individual who, at any point in ____, was classified by the Firm as a GSAMI Senior Management in accordance with the remuneration code under the IFPR as applicable to the business operations of GSAMI.

(v) **“UK MRT”** means an individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (including UK Risk Managers and UK PRA Senior Managers) in accordance with the remuneration codes applicable to dual-regulated UK firms.

(vi) For purposes of these definitions, if during ____ you were classified by the Firm as any combination of EU MRT and GSAMI MRT, GSAMI Senior Management or UK MRT then your classification in the table above is EU MRT; any combination of GSAMI MRT or GSAMI Senior Management and UK MRT then your classification in the table above is UK MRT; or as a combination of GSAMI MRT and GSAMI Senior Management then your classification in the table above is GSAMI Senior Management.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by Paragraphs (b) and (c) of this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

[GSBE Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider all relevant factors in making the decision, in its sole discretion as to whether an Adjustment Event or Actionable Misconduct have occurred.

This Appendix applies to all RSUs in your Award if you are a GSBE MRT or GSBE Senior Management.

(a) An Adjustment Event Occurs with Respect to ____. If an Adjustment Event occurs with respect to ____, (i) your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled, and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Adjustment Event occurred and (B) the date that the repayment request is made.

(i) **“Adjustment Event”** means that one of the following has occurred:

- A. you significantly contributed to, or were responsible for, any conduct that resulted in a loss of 0.75% or more of the total capital of GS Inc.;
- B. a material regulatory sanction for the Firm comprising one or more of the following:
 - 1. a moratorium pursuant to sec. 46g of the German Banking Act,
 - 2. a measure in case of danger pursuant to sec. 46 of the German Banking Act,
 - 3. the revocation of appointment of a manager pursuant to sec. 36 German Banking Act,
 - 4. a fine pursuant to sec. 56 of the German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of GS Inc.,
 - 5. the cancellation of the banking permit pursuant to sec. 35 of the German Banking Act,
 - 6. an order to increase the capital requirements Goldman Sachs Bank Europe SE (“GSBE”) by at least 0.5% pursuant to sec. 10 of the German Banking Act,

7. a measure in case of organizational deficiencies,
8. a comparable regulatory order, or
9. a material supervisory measure; or

C. you acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that a violation is considered serious if it is suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct pursuant to sec. 1 German Termination Protection Act.

(b) You Engage in Actionable Misconduct. If the Committee determines that you engaged in Actionable Misconduct, which if known (or if known, adequately considered) would have resulted in a reduction to your annual variable compensation when your variable compensation for that year was determined, your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs.

(i) **"Actionable Misconduct"** means either or both of:

- A. "Unethical behavior" within the meaning of the IVV;
- B. "Breach of duty" within the meaning of the IVV.

(ii) **"IVV"** means the German Remuneration Ordinance for Institutions (*Institutsvergütungsverordnung*) and applicable guidance from the German Federal Financial Supervisory Authority (BaFin) or the European Central Bank.

(c) Scope of Appendix. This Appendix applies to any individual who, at any point in ____, was classified by the Firm as a Material Risk Taker ("GSBE MRT") or Senior Management ("GSBE Senior Management") in connection with the business operations of GSBE or any branch or subsidiary thereof, whether or not GSBE or any branch or subsidiary thereof employed such individual.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings:

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

(b) “Acceptance Deadline” means the date communicated to you by the Firm as the deadline to accept your Award. The Acceptance Deadline will be no less than 14 days after the Award Documentation is first made available to you in full. To the extent you are entitled to a translation in accordance with the Language Notice for Equity Awards provided to you as part of the Award Documentation, the Acceptance Deadline will be extended by the time required to furnish you with any such translation.

(c) “Approved Termination” means that you are classified by the Firm as a “fixed-term employee” and you (i) successfully complete the fixed-term engagement, as determined by the Firm in its sole discretion, including remaining Employed through the completion date specified by the Firm, and (ii) terminate Employment immediately after the completion date without any “stay-on” or other agreement or understanding to continue Employment with the Firm. If you agree to stay with the Firm as an employee after your fixed-term engagement ends and then later terminate Employment, you will not have an Approved Termination.

(d) “Associate With a Covered Enterprise” means that you (i) form, or acquire a 5% or greater equity ownership, voting or profit participation interest in, any Covered Enterprise or (ii) associate in any capacity (including association as an officer, employee, partner, director, consultant, agent or advisor) with any Covered Enterprise. Associate With a Covered Enterprise may include, as determined in the discretion of the Committee, (i) becoming the subject of any publicly available announcement or report of a pending or future association with a Covered Enterprise and (ii) unpaid associations, including an association in contemplation of future employment. “Association With a Covered Enterprise” will have its correlative meaning.

(e) “Breached an Obligation to the Firm” means that you failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with your Employment or this Award (including any applicable Additional Terms); including, without limitation, any offer letter, employment agreement or any shareholders’ agreement relating to the Firm. Your failure to pay or reimburse the Firm, on demand, for any amount you owe to the Firm will constitute (i) failure to meet an obligation you have under an agreement, regardless of whether such obligation arises under a written agreement, and/or (ii) a material violation of Firm policy constituting Cause.

(f) “Committee” means the Compensation Committee unless otherwise determined by the Board; *provided, however*, that: (i) the authority of the Compensation Committee to administer the Plan and the Award may be granted or delegated to the SIP Committee, in which case “Committee” means the SIP Committee when acting pursuant to such a grant or delegation; (ii) to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3; and (iii) to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, “Committee” shall be a committee or subcommittee of

the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m).

(g) “Compensation Committee” means the Compensation Committee of the Board.

(h) “Conflicted Employment” means your employment at any U.S. Federal, state or local government, any non-U.S. government, any supranational or international organization, any self-regulatory organization, or any agency or instrumentality of any such government or organization, or any other employer (other than an “Accounting Firm” within the meaning of SEC Rule 2-01(f)(2) of Regulation S-X or any successor thereto) determined by the Committee, if, as a result of such employment, your continued holding of any Outstanding RSUs or Shares at Risk would result in an actual or perceived conflict of interest.

(i) “Covered Enterprise” means a Competitive Enterprise and any other existing or planned business enterprise that: (i) offers, holds itself out as offering or reasonably may be expected to offer products or services that are the same as or similar to those offered by the Firm or that the Firm reasonably expects to offer (“Firm Products or Services”) or (ii) engages in, holds itself out as engaging in or reasonably may be expected to engage in any other activity that is the same as or similar to any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage (“Firm Activities”). For the avoidance of doubt, Firm Activities include any activity that requires the same or similar skills as any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage, irrespective of whether any such financial activity is in furtherance of an advisory, agency, proprietary or fiduciary undertaking.

The enterprises covered by this definition include enterprises that offer, hold themselves out as offering or reasonably may be expected to offer Firm Products or Services, or engage in, hold themselves out as engaging in or reasonably may be expected to engage in Firm Activities directly, as well as those that do so indirectly by ownership or control (e.g., by owning, being owned by or by being under common ownership with an enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services or that engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities). The definition of Covered Enterprise includes, solely by way of example, any enterprise that offers, holds itself out as offering or reasonably may be expected to offer any product or service, or engages in, holds itself out as engaging in or reasonably may be expected to engage in any activity, in any case, associated with investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than you or your family members (including, for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading. An enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services, or engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities is a Covered Enterprise, **irrespective of whether the enterprise is a customer, client or counterparty of the Firm or is otherwise associated with the Firm and, because the Firm is a global enterprise, irrespective of where the Covered Enterprise is physically located.**

(j) “Delivery Date” means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a delivery date, so long as any such date occurs before the 29th day of the month specified on the Award Statement; if there is no such date, the Delivery Date will

be the earlier of another date selected by the Committee in the month and year specified on the Award Statement or the first trading day in the subsequent Window Period.

(k) “Failed to Consider Risk” means that you participated (or otherwise oversaw or were responsible for, depending on the circumstances, another individual’s participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where you have improperly analyzed such risk or where you have failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, the Committee determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, your business unit or the broader financial system. This definition is not limited to financial risks and is designed to encourage the consideration of the full range of risks associated with activities (e.g., legal, compliance or reputational). This definition also does not require that a material adverse impact actually occur, but, rather, may be triggered if it is determined that there is a reasonable expectation of such an impact.

(l) “HCM” means the Firm’s Human Capital Management Division or its successor.

(m) “Qualifying Termination After a Change in Control” means that the Firm terminates your Employment other than for Cause or you terminate your Employment for Good Reason, in each case, within 18 months following a Change in Control.

(n) “SEC” means the U.S. Securities and Exchange Commission.

(o) “Selected Firm Personnel” means any individual who is or in the three months preceding the conduct prohibited by Paragraph 9(b)(i) was (i) a Firm employee or consultant with whom you personally worked while employed by the Firm, (ii) a Firm employee or consultant who, at any time during the year preceding the date of the termination of your Employment, worked in the same division in which you worked or (iii) an Advisory Director, a Managing Director (Participating or Extended) or a Senior Advisor of the Firm.

(p) “Shares at Risk” means RSU Shares subject to Transfer Restrictions.

(q) “Signature Card” means the document and/or website presented to you with your Award that you are required to execute (including electronically) to indicate your acceptance of the Award, the Award Agreement (including the Award Statement and the terms contained in the Signature Card) and Additional Terms applicable to you, as well your receipt of the Award Documentation.

(r) “Transferability Date” means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a transferability date; *provided, however*, that if there is no trading day in a Window Period that occurs in the specified month and year, the Transferability Date will be the first trading day in the subsequent Window Period.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Statement” means a written statement that reflects certain Award terms.

(c) “Board” means the Board of Directors of GS Inc.

(d) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(e) “Cause” means (i) the Grantee’s conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion, or (B) on a felony charge, or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations, (ii) the Grantee’s engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the Exchange Act), (iii) the Grantee’s willful failure to perform the Grantee’s duties to the Firm, (iv) the Grantee’s violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member, (v) the Grantee’s violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or the Grantee’s material violation of any other Firm policy as in effect from time to time, (vi) the Grantee’s engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm or (vii) the Grantee’s engaging in any conduct detrimental to the Firm. The determination as to whether Cause has occurred shall be made by the Committee in its sole discretion and, in such case, the Committee also may, but shall not be required to, specify the date such Cause occurred (including by determining that a prior termination of Employment was for Cause). Any rights the Firm may have hereunder and in any Award Agreement in respect of the events giving rise to Cause shall be in addition to the rights the Firm may have under any other agreement with a Grantee or at law or in equity.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Change in Control” means the consummation of a merger, consolidation, statutory share exchange or similar form of corporate transaction involving GS Inc. (a “Reorganization”) or sale or other disposition of all or substantially all of GS Inc.’s assets to an entity that is not an affiliate of GS Inc. (a “Sale”), that in each case requires the approval of GS Inc.’s shareholders under the law of GS Inc.’s jurisdiction of organization, whether for such Reorganization or Sale (or the issuance of securities of GS Inc. in such Reorganization or Sale), unless immediately following such Reorganization or Sale, either: (i) at least 50% of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of (A) the entity resulting from such Reorganization, or the entity which has acquired all or substantially all of the assets of GS Inc. in a Sale (in either case, the “Surviving Entity”), or (B) if applicable, the ultimate parent entity that directly or indirectly has beneficial

ownership (within the meaning of Rule 13d-3 under the Exchange Act, as such Rule is in effect on the date of the adoption of the 1999 SIP) of 50% or more of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of the Surviving Entity (the "Parent Entity") is represented by GS Inc.'s securities (the "GS Inc. Securities") that were outstanding immediately prior to such Reorganization or Sale (or, if applicable, is represented by shares into which such GS Inc. Securities were converted pursuant to such Reorganization or Sale) or (ii) at least 50% of the members of the board of directors (or similar officials in the case of an entity other than a corporation) of the Parent Entity (or, if there is no Parent Entity, the Surviving Entity) following the consummation of the Reorganization or Sale were, at the time of the Board's approval of the execution of the initial agreement providing for such Reorganization or Sale, individuals (the "Incumbent Directors") who either (A) were members of the Board on the Effective Date or (B) became directors subsequent to the Effective Date and whose election or nomination for election was approved by a vote of at least two-thirds of the Incumbent Directors then on the Board (either by a specific vote or by approval of GS Inc.'s proxy statement in which such persons are named as nominees for director).

(h) "Client" means any client or prospective client of the Firm to whom the Grantee provided services, or for whom the Grantee transacted business, or whose identity became known to the Grantee in connection with the Grantee's relationship with or employment by the Firm.

(i) "Code" means the Internal Revenue Code of 1986, as amended from time to time, and the applicable rulings and regulations thereunder.

(j) "Common Stock" means common stock of GS Inc., par value \$0.01 per share.

(k) "Competitive Enterprise" means an existing or planned business enterprise that (i) engages, or may reasonably be expected to engage, in any activity; (ii) owns or controls, or may reasonably be expected to own or control, a significant interest in any entity that engages in any activity or (iii) is, or may reasonably be expected to be, owned by, or a significant interest in which is, or may reasonably be expected to be, owned or controlled by, any entity that engages in any activity that, in any case, competes or will compete anywhere with any activity in which the Firm is engaged. The activities covered by this definition include, without limitation: financial services such as investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the Grantee and members of the Grantee's family (including for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading.

(l) "Covered Person" means a member of the Board or the Committee or any employee of the Firm.

(m) "Date of Grant" means the date specified in the Grantee's Award Agreement as the date of grant of the Award.

(n) ["Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.]

(o) "Effective Date" means the date this Plan is approved by the shareholders of GS Inc. pursuant to Section 3.15 of the Plan.

(p) "Employment" means the Grantee's performance of services for the Firm, as determined by the Committee. The terms "employ" and "employed" shall have their correlative meanings. The Committee in its sole discretion may determine (i) whether and when a Grantee's leave of absence results in a termination of Employment (for this purpose, unless the Committee determines otherwise, a Grantee shall be treated as terminating Employment with the Firm upon the occurrence of an Extended Absence), (ii) whether and when a change in a Grantee's association with the Firm results in a termination of Employment and (iii) the impact, if any, of any such leave of absence or change in association on Awards theretofore made. Unless expressly provided otherwise, any references in the Plan or any Award Agreement to a Grantee's Employment being terminated shall include both voluntary and involuntary terminations.

(q) "Exchange Act" means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(r) "Extended Absence" means the Grantee's inability to perform for six (6) continuous months, due to illness, injury or pregnancy-related complications, substantially all the essential duties of the Grantee's occupation, as determined by the Committee.

(s) "Firm" means GS Inc. and its subsidiaries and affiliates.

(t) "Good Reason" means, in connection with a termination of employment by a Grantee following a Change in Control, (i) as determined by the Committee, a materially adverse alteration in the Grantee's position or in the nature or status of the Grantee's responsibilities from those in effect immediately prior to the Change in Control or (ii) the Firm's requiring the Grantee's principal place of Employment to be located more than seventy-five (75) miles from the location where the Grantee is principally Employed at the time of the Change in Control (except for required travel on the Firm's business to an extent substantially consistent with the Grantee's customary business travel obligations in the ordinary course of business prior to the Change in Control).

(u) "Grantee" means a person who receives an Award.

(v) "GS Inc." means The Goldman Sachs Group, Inc., and any successor thereto.

(w) "1999 SIP" means The Goldman Sachs 1999 Stock Incentive Plan, as in effect prior to the effective date of the 2003 SIP.

(x) "Outstanding" means any Award to the extent it has not been forfeited, cancelled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(y) "Restricted Share" means a share of Common Stock delivered under the Plan that is subject to Transfer Restrictions, forfeiture provisions and/or other terms and conditions specified herein and in the Restricted Share Award Agreement or other applicable Award Agreement. All references to Restricted Shares include "Shares at Risk."

(z) "Retirement" means termination of the Grantee's Employment (other than for Cause) on or after the Date of Grant at a time when (i) (A) the sum of the Grantee's age plus years of service with

the Firm (as determined by the Committee in its sole discretion) equals or exceeds 60 and (B) the Grantee has completed at least 10 years of service with the Firm (as determined by the Committee in its sole discretion) or, if earlier, (ii) (A) the Grantee has attained age 50 and (B) the Grantee has completed at least five years of service with the Firm (as determined by the Committee in its sole discretion).

(aa) "RSU" means a restricted stock unit granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(ab) "RSU Shares" means shares of Common Stock that underlie an RSU.

(ac) "Section 409A" means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(ad) "SIP Administrator" means each person designated by the Committee as a "SIP Administrator" with the authority to perform day-to-day administrative functions for the Plan.

(ae) "SIP Committee" means the persons who have been delegated certain authority under the Plan by the Committee.

(af) "Solicit" means any direct or indirect communication of any kind whatsoever, regardless of by whom initiated, inviting, advising, suggesting, encouraging or requesting any person or entity, in any manner, to take or refrain from taking any action. The terms "Solicited," "Soliciting" and "Solicitation" will have their correlative meanings.

(ag) "Transfer Restrictions" means restrictions that prohibit the sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposal (including through the use of any cash-settled instrument), whether voluntarily or involuntarily by the Grantee, of an Award or any shares of Common Stock, cash or other property delivered in respect of an Award.

(ah) "Vested" means, with respect to an Award, the portion of the Award that is not subject to a condition that the Grantee remain actively employed by the Firm in order for the Award to remain Outstanding. The fact that an Award becomes Vested shall not mean or otherwise indicate that the Grantee has an unconditional or nonforfeitable right to such Award, and such Award shall remain subject to such terms, conditions and forfeiture provisions as may be provided for in the Plan or in the Award Agreement.

(ai) "Vesting Date" means each date specified in the Grantee's Award Agreement as a date on which part or all of an Award becomes Vested.

(aj) "Window Period" means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (*provided* that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
[One-Time][Year-End] Short-Term RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of Short-Term RSUs (your “Award”). You should read carefully this entire document (including its [Appendix][Appendices]), the Award Statement and the Signature Card (together, the “Award Agreement”), as well as the other documentation presented to you in connection with acceptance of your Award (collectively, “Award Documentation”).

Acceptance

1. **You Must Decide Whether to Accept Your Award.** To be eligible for your Award, you must by the Acceptance Deadline: (a) open and activate an Account; (b) accept any Additional Terms presented to you as part of the Award Documentation; and (c) execute (including by electronic means) the accompanying Signature Card in accordance with its instructions. By executing the Signature Card, you confirm your agreement to all of the terms of the Award Agreement (including the Award Statement and the Signature Card), as well as any Additional Terms.

Documents that Govern Your Award; Definitions

2. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement. The Plan and its Prospectus are included with the Award Documentation.

3. **Your Award Statement.** The accompanying Award Statement contains the number of Short-Term RSUs awarded to you and some of your Award’s specific terms. For example, it specifies the Delivery Date.

4. **Definitions.** Unless otherwise defined herein, including in the Definitions Appendix or any other Appendix, capitalized terms have the meanings provided in the Plan.

Vesting of Your RSUs

5. **Vesting.** All of your Short-Term RSUs are Vested. When an RSU is Vested, it means **only** that your continued active Employment is not required for delivery of that portion of RSU Shares. **Vesting does not mean you have a non-forfeitable right to the Vested portion of your Award. The terms of this Award Agreement (including conditions to delivery) continue to apply to Vested Short-Term RSUs, and you can still forfeit Vested Short-Term RSUs and any RSU Shares.**

Delivery of Your RSU Shares

6. **Delivery.** Reasonably promptly (but no more than 30 Business Days) after each Delivery Date identified on your Award Statement, RSU Shares (less applicable withholding as described in Paragraph 11(b)) will be delivered (by book entry credit to your Account) in respect of the amount of Outstanding Short-Term RSUs listed next to that date. The Committee may select multiple dates within the 30-Business-Day period following the Delivery Date to deliver RSU Shares in respect of all or a portion of the Short-Term RSUs with the same Delivery Date listed on the Award Statement, and all such dates will be treated as a single Delivery Date for purposes of this Award. Until such delivery, you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc. Without limiting the Committee’s authority under Section 1.3.2(h) of the Plan, the Firm may accelerate any Delivery Date by up to 30 days.

Dividends

7. **Dividend Equivalent Rights and Dividends.** Each Short-Term RSU includes a Dividend Equivalent Right, which entitles you to receive an amount (less applicable withholding), at or after the time of distribution of any regular cash dividend paid by GS Inc. in respect of a share of Common Stock, equal to any regular cash dividend payment that would have been made in respect of an RSU Share underlying your Outstanding Short-Term RSUs for any record date that occurs on or after the Date of Grant.

Forfeiture of Your Award

8. **How You May Forfeit Your Award.** This Paragraph 8 sets forth the events that result in forfeiture of up to all of your Short-Term RSUs and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 9. More than one event may apply, and in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event. Paragraph 10 (relating to certain circumstances under which delivery may be accelerated) provides for exceptions to one or more provisions of this Paragraph 8.

(a) Short-Term RSUs Forfeited Upon Certain Events. If any of the following occurs, your rights to all of your Outstanding Short-Term RSUs will terminate, and no RSU Shares will be delivered in respect of such Short-Term RSUs, as may be further described below:

(i) You Failed to Consider Risk. You Failed to Consider Risk during ____.

(ii) Your Conduct Constitutes Cause. Any event that constitutes Cause has occurred before the Delivery Date.

(iii) You Breach an Obligation to the Firm. The Committee determines that, before the Delivery Date, you Breached an Obligation to the Firm.

(iv) You Do Not Provide Timely Certifications or Comply with Your Certifications. You fail to certify to GS Inc. that you have complied with all of the terms of the Plan and this Award Agreement as required by Paragraph 11(d), or the Committee determines that you have failed to comply with a term of the Plan or this Award Agreement to which you have certified compliance.

(v) You Do Not Follow Dispute Resolution Procedures. You attempt to have any dispute, controversy or claim under the Plan or this Award Agreement resolved in any manner that is not provided for by Paragraph 14 or Section 3.17 of the Plan, or you attempt to arbitrate a dispute, controversy or claim without first having exhausted your internal administrative remedies in accordance with Paragraph 11(c).

(vi) You Bring an Action that Results in a Determination that Any Award Agreement Term Is Invalid. As a result of any action brought by you, it is determined that any term of this Award Agreement is invalid.

(vii) You Receive Compensation in Respect of Your Award from Another Employer. Your Employment terminates for any reason or you otherwise are no longer actively Employed with the Firm and another entity grants you cash, equity or other property (whether vested or unvested) to replace, substitute for or otherwise in respect of any Outstanding Short-Term RSUs; *provided, however*, that your rights will only be terminated in respect of the Short-Term RSUs that are replaced, substituted for or otherwise considered by such other entity in making its grant.

(b) [Additional Forfeiture and Repayment Conditions for Material Risk Takers. [If at any point in ____][Since] the Firm identified you as a Material Risk Taker ("MRT") [in ____] then the **EU and UK Material Risk Taker Appendix** and/or the **GSBE Material Risk Taker Appendix** (as

applicable to you, the “MRT Appendix”) supplements this Paragraph 8. The MRT Appendix sets forth additional events that (i) result in forfeiture of up to all of your Short-Term RSUs and (ii) may require repayment to the Firm of up to all amounts previously delivered or paid to you under your Award in accordance with Paragraph 9. Your MRT classification, as defined in the MRT Appendix, was communicated to you separately. [If you are unsure of whether the Firm identified you as an MRT, or which classification(s) apply, you must contact HCM prior to accepting this Award.] The Firm’s records with respect to such classification(s) will be controlling as to whether and which MRT Appendix applies to your Award. The MRT classifications with their corresponding MRT Appendix are as follows:

MRT Classification	MRT Appendix
EU MRT, GSAMI MRT, GSAMI Senior Management and/or UK MRT	EU and UK Material Risk Taker Appendix
GSBE MRT and GSBE Senior Management	GSBE Material Risk Taker Appendix
GSBE MRT or GSBE Senior Management <i>and</i> any combination of EU MRT, GSAMI MRT, GSAMI Senior Management or UK MRT	GSBE Material Risk Taker Appendix <i>and</i> EU and UK Material Risk Taker Appendix]

(c) Actions Pending Forfeiture. In connection with any investigation as to whether any of the events that would result in forfeiture under the Plan [or][,] this Paragraph 8 [or the MRT Appendix] have occurred, the Firm reserves the right to (i) suspend payments under Dividend Equivalent Rights or delivery of RSU Shares; (ii) deliver any RSU Shares, payments under Dividend Equivalent Rights or dividends into an escrow account in accordance with Paragraph 11(e)(iii); or (iii) apply Transfer Restrictions to any RSU Shares.

Repayment of Your Award

9. **When You May Be Required to Repay Your Award**. If the Committee determines that any term of this Award was not satisfied, you will be required, immediately upon demand therefor, to repay to the Firm the following:

- (a) Any RSU Shares for which the terms (including the terms for delivery) of the related Short-Term RSUs were not satisfied, in accordance with Section 2.6.3 of the Plan.
- (b) Any payments under Dividend Equivalent Rights for which the terms were not satisfied (including any such payments made in respect of Short-Term RSUs that are forfeited or RSU Shares that are cancelled or required to be repaid), in accordance with Section 2.8.3 of the Plan.
- (c) Any dividends paid in respect of any RSU Shares that are cancelled or required to be repaid.
- (d) Any amount applied to satisfy tax withholding or other obligations with respect to any Short-Term RSUs, RSU Shares, dividend payments and payments under Dividend Equivalent Rights that are forfeited or required to be repaid.

Exceptions to the Delivery Date

10. **Accelerated Delivery in the Event of [a Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] Death**. In the event of your [Qualifying Termination After a Change in Control][,] [Conflicted Employment] [or] death, [each] as described below, your Outstanding Short-Term RSUs will be treated as described in this Paragraph 10, and[, except as set forth in Paragraph 10(a),] all other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 8 and 9 [and the MRT Appendix], continue to apply.

- (a) [You Have a Qualifying Termination After a Change in Control]. If your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, the

RSU Shares underlying your Outstanding Short-Term RSUs will be delivered. In addition, the forfeiture events in Paragraph 8 [(except the MRT Appendix)] will not apply to your Award.]

(b) [You Are Determined to Have Accepted Conflicted Employment.

(i) Generally. To the extent consistent with applicable law, regulation and regulatory guidance, if your Employment terminates solely because you resign to accept Conflicted Employment or if, following your termination of Employment, you notify the Firm that you are accepting Conflicted Employment, RSU Shares will be delivered in respect of your Outstanding Short-Term RSUs (including in the form of cash as described in Paragraph 11(e)(ii)) as soon as practicable after the Committee has received satisfactory documentation relating to your Conflicted Employment.

(ii) You May Have to Take Other Steps to Address Conflicts of Interest. The Committee retains the authority to exercise its rights under the Award Agreement or the Plan (including Section 1.3.2 of the Plan) to take or require you to take other steps it determines in its sole discretion to be necessary or appropriate to cure an actual or perceived conflict of interest (which may include a determination that the accelerated delivery described in Paragraph 10(b)(i) will not apply because such actions are not necessary or appropriate to cure an actual or perceived conflict of interest).]

(c) Death. If you die, the RSU Shares underlying your Outstanding Short-Term RSUs will be delivered to your Account as soon as practicable after the date of death and receipt of any documentation as may be requested by the Committee or a SIP Administrator.

Other Terms And Conditions of Your Award

11. Additional Terms, Conditions and Consents.

(a) You Will Be Required to Accept Additional Terms. In addition to the terms of the Award Agreement, the Award Documentation includes other contractual terms that you must accept as a condition of the Award (the "Additional Terms"). By accepting the Award and executing the Signature Card, you agree to any such Additional Terms applicable to you as set forth in the Award Documentation, which will include but are not limited to the following.

(i) Notice Period Policy. You must agree to abide by the terms, as applicable to you, of the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy").

(ii) Agreement Regarding Arbitration of Employment-Related Matters. You must agree to the Agreement Regarding Arbitration of Employment-Related Matters, which imposes obligations in addition to those contained in Paragraph 14.

(iii) APAC Restrictive Covenants. If you are employed by the Firm in the Asia-Pacific Region (e.g., Australia, the People's Republic of China, the Hong Kong Special Administrative Region, India, Indonesia, Japan, the Republic of Korea, New Zealand, Singapore, Taiwan), you must agree to the APAC Restrictive Covenants Agreement.

(iv) Role-, Business- or Division-Specific Agreements. If applicable to you, you must accept Additional Terms related to your specific role, business unit or division as set forth in the Award Documentation.

(b) You Must Satisfy Applicable Tax Withholding Requirements. Delivery of RSU Shares is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, which includes the Firm deducting or withholding amounts from any payment or distribution to you. In addition, to the extent permitted by applicable law, the Firm, in its sole discretion, may require you to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations

imposed on you or the Firm in connection with the grant or delivery of this Award by requiring you to choose between remitting the amount (i) in cash (or through payroll deduction or otherwise) or (ii) in the form of proceeds from the Firm's executing a sale of RSU Shares delivered to you under this Award. In no event, however, does this Paragraph 11(b) give you any discretion to determine or affect the timing of the delivery of RSU Shares or the timing of payment of tax obligations.

(c) You Must Comply with Applicable Deadlines and Procedures to Appeal. If you disagree with a determination made by the Committee, the SIP Administrators or any of their delegates or designees relating to the Plan or the Award Agreement and you wish to appeal such determination, you must submit a written request to the Committee for review within 180 days after the determination at issue. You must exhaust your internal administrative remedies (*i.e.*, submit your appeal and wait for resolution of that appeal) before seeking to resolve a dispute, controversy or claim through arbitration pursuant to Paragraph 14 and Section 3.17 of the Plan.

(d) You May Be Required to Certify Compliance with Award Terms; You Are Responsible for Providing Updated Address and Contact Information After Your Departure from the Firm. If your Employment terminates while you continue to hold Short-Term RSUs, the Firm may require you to provide certifications of your compliance with all of the terms of the Plan and this Award Agreement. You understand and agree that (i) your contact information as reflected in the Firm's personnel records at the time any certification is requested will be deemed current; (ii) it is your responsibility to inform HCM of any changes to your contact information to ensure timely receipt of the certification materials, regardless of whether your contact information is provided to another part of the Firm; and (iii) you are responsible for contacting the Firm to obtain such certification materials if not received. Your failure to return properly completed certification materials by the specified deadline (including because you did not provide HCM with updated contact information) will result in the forfeiture of all of your Short-Term RSUs and subject previously delivered amounts to repayment under Paragraph 8(a)(iv).

(e) You Agree to Other Terms, Conditions and Consents. By accepting this Award you understand and agree that:

(i) Amounts May Be Rounded to Avoid Fractional RSU Shares. RSU Shares that become deliverable may be rounded to avoid fractional RSU Shares.

(ii) Firm May Deliver Cash or Other Property Instead of RSU Shares. In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the RSU Shares, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to deliveries of RSU Shares will include such deliveries of cash, other securities, other awards under the Plan or other property.

(iii) Firm May Deliver Your Award into an Escrow Account. The Firm may establish and maintain an escrow account on such terms (which may include your executing any documents related to, and your paying for any costs associated with, such account) as it may deem necessary or appropriate, and the delivery of RSU Shares, and any payment of cash (including dividends and payments under Dividend Equivalent Rights) or other property, may initially be made into and held in that escrow account until such time as the Committee has received such documentation as it may have requested or until the Committee has determined that any other conditions or restrictions on delivery of RSU Shares, cash or other property required by this Award Agreement have been satisfied.

(iv) Firm May Affix Legends and Place Stop Orders on RSU Shares. GS Inc. may affix to Certificates representing RSU Shares any legend that the Committee determines to be necessary or advisable (including to reflect any restrictions to which you may be subject under the Additional Terms, a separate agreement or any Firm policy applicable to you). GS Inc. may advise the transfer agent to place a stop order against any legended RSU Shares.

(v) You Agree to Certain Consents as a Condition to the Award. You have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying

to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

(vi) You Are Subject to the Firm's Policies, Rules and Procedures. You are subject to the Firm's policies in effect from time to time concerning trading in RSU Shares and hedging or pledging RSU Shares and equity-based compensation or other awards (including, without limitation, the "Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards" or any successor policies), and confidential or proprietary information, and you will effect sales of RSU Shares in accordance with such rules and procedures as may be adopted from time to time (which may include, without limitation, restrictions relating to the timing of sale requests, the manner in which sales are executed, pricing method, consolidation or aggregation of orders and volume limits determined by the Firm).

(vii) You Are Responsible for Costs Associated with Your Award. You will be responsible for all brokerage costs and other fees or expenses associated with your Short-Term RSUs, including those related to the sale of RSU Shares.

(viii) You Will Be Deemed to Represent Your Compliance with All the Terms of Your Award if You Accept Delivery of, or Sell, RSU Shares. You will be deemed to have represented and certified that you have complied with all of the terms of the Plan and this Award Agreement when RSU Shares are delivered to you and you receive payment in respect of Dividend Equivalent Rights.

(ix) You Agree that Covered Persons Will Not Have Liability. In addition to and without limiting the generality of the provisions of Section 1.3.5 of the Plan, neither the Firm nor any Covered Person will have any liability to you or any other person for any action taken or omitted in respect of this or any other Award.

12. **Non-transferability.** Except as otherwise may be provided by the Committee, the limitations set forth in Section 3.5 of the Plan will apply to this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition (including through the use of any cash-settled instrument), whether voluntary or involuntary, in violation of the provisions of this Paragraph 12 or Section 3.5 of the Plan will be void.

13. **Right of Offset.** Except as provided in Paragraph 16(h), the obligation to deliver RSU Shares or to pay dividends or payments under Dividend Equivalent Rights under this Award Agreement is subject to Section 3.4 of the Plan, which provides for the Firm's right to offset against such obligation any outstanding amounts you owe to the Firm and any amounts the Committee deems appropriate pursuant to any tax equalization policy or agreement.

Arbitration, Choice of Forum and Governing Law

14. Arbitration; Choice of Forum.

(a) **By accepting this award, you are indicating that you understand and agree that the arbitration and choice of forum provisions set forth in Section 3.17 of the Plan will apply to this Award to the fullest extent permitted by applicable law. These provisions, which are expressly incorporated herein by reference, provide among other things that any dispute, controversy or claim between the Firm and you arising out of or relating to or concerning the Plan or this Award Agreement will be finally settled by arbitration, pursuant to the terms more fully set forth in Section 3.17 of the Plan; provided that nothing herein shall preclude you from filing a charge with or participating in any investigation or proceeding conducted by any governmental authority, including but not limited to the SEC, the Equal Employment Opportunity Commission and a state or local human rights agency, as well as law enforcement.**

(b) Nothing in the Plan or this Award Agreement includes an agreement to arbitrate claims on a collective, class or representative basis. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider class, collective or representative claims, to order consolidation or to join different claimants or grant relief other than on an individual basis to the individual claimant involved.

(c) Notwithstanding any applicable forum rules to the contrary, to the extent there is a question of enforceability of the Plan or this Award Agreement arising from a challenge to the arbitrator's jurisdiction or to the arbitrability of a claim, it will be decided by a court and not an arbitrator.

(d) Prior to filing any arbitration in accordance with Paragraph 14(a), you must first exhaust your internal administrative remedies in accordance with Paragraph 11(c), which includes awaiting the Committee's final determination of any such appeal. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider any dispute, controversy or claim as to which you have not first exhausted your internal administrative remedies in accordance with Paragraph 11(c).

(e) The Federal Arbitration Act governs interpretation and enforcement of all arbitration provisions under the Plan and this Award Agreement, and all arbitration proceedings thereunder.

(f) Nothing in the Plan or this Award Agreement creates a substantive right to bring a claim under U.S. Federal, state, or local employment laws.

(g) By accepting your Award, you irrevocably appoint each General Counsel of GS Inc., or any person whom the General Counsel of GS Inc. designates, as your agent for service of process in connection with any suit, action or proceeding arising out of or relating to or concerning the Plan or any Award which is not arbitrated pursuant to the provisions of Section 3.17.1 of the Plan, who shall promptly advise you of any such service of process.

(h) **As a condition of your Award, you also must accept the terms of the Agreement Regarding Arbitration of Employment-Related Matters, which is included among the Additional Terms described in Paragraph 11(a) and sets forth obligations independent of this Paragraph 14. For the avoidance of doubt, the obligation to exhaust your internal remedies described in Paragraph 11(c) and Paragraph 14(d) only applies to disputes, controversies or claims relating to the Plan or the Award Agreement.**

15. Governing Law. This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

16. Compliance of Award Agreement and Plan with Section 409A. The provisions of this Paragraph 16 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 16 and the other provisions of this Award Agreement, this Paragraph 16 will govern.

(b) Delivery of RSU Shares will not be delayed beyond the date on which all applicable conditions or restrictions on delivery of RSU Shares required by this Award Agreement (including those specified in Paragraphs 6, 10(c) and 11 and the consents and other items specified in Section 3.3 of the Plan) are satisfied. To the extent that any portion of this Award is intended to satisfy the requirements for

short-term deferral treatment under Section 409A, delivery for such portion will occur by the March 15 coinciding with the last day of the applicable “short-term deferral” period described in Reg. 1.409A-1(b)(4) in order for the delivery of RSU Shares to be within the short-term deferral exception unless, in order to permit all applicable conditions or restrictions on delivery to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay delivery of RSU Shares to a later date within the same calendar year or to such later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a “series of installment payments” as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 11(e) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any securities, other Awards or other property that the Firm may deliver in respect of your Short-Term RSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the RSU Shares that would otherwise have been deliverable (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Notwithstanding the timing provisions of Paragraph 10(c), the delivery of RSU Shares referred to therein will be made after the date of death and not later than the end of the calendar year following the calendar year that includes the date of death (or on such later date as may be permitted under Section 409A).

(e) The timing of delivery or payment pursuant to Paragraph 10(a) will occur on the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the termination of Employment occurs; *provided, however*, that if you are a “specified employee” (as defined by the Firm in accordance with Section 409A(a)(2)(i)(B) of the Code), delivery will occur on the earlier of the Delivery Date or (to the extent required to avoid the imposition of additional tax under Section 409A) the date that is six months after your termination of Employment (or, if the latter date is not during a Window Period, the first trading day of the next Window Period). For purposes of Paragraph 10(a), references in this Award Agreement to termination of Employment mean a termination of Employment from the Firm (as defined by the Firm) which is also a separation from service (as defined by the Firm in accordance with Section 409A).

(f) Notwithstanding any provision of Paragraph 7 or Section 2.8.2 of the Plan to the contrary, the Dividend Equivalent Rights with respect to each of your Outstanding Short-Term RSUs will be paid to you within the calendar year that includes the date of distribution of any corresponding regular cash dividends paid by GS Inc. in respect of a share of Common Stock the record date for which occurs on or after the Date of Grant. The payment will be in an amount (less applicable withholding) equal to such regular dividend payment as would have been made in respect of the RSU Shares underlying such Outstanding Short-Term RSUs.

(g) The timing of delivery or payment referred to in Paragraph 10(b)(i) will be the earlier of (i) the Delivery Date or (ii) a date that is within the calendar year in which the Committee receives satisfactory documentation relating to your Conflicted Employment, *provided* that such delivery or payment will be made, and any Committee action referred to in Paragraph 10(b)(ii) will be taken, only at such time as, and if and to the extent that it, as reasonably determined by the Firm, would not result in the imposition of any additional tax to you under Section 409A.

(h) Paragraph 13 and Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(i) Delivery of RSU Shares in respect of any Award may be made, if and to the extent elected by the Committee, later than the Delivery Date or other date or period specified hereinabove (but,

in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later delivery is permitted under Section 409A).

(j) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Committee Authority, Amendment, Construction and Regulatory Reporting

17. **Committee Authority.** The Committee has the authority to determine, in its sole discretion, that any event triggering forfeiture or repayment of your Award will not apply and to limit the forfeitures and repayments that result under Paragraphs 8 and 9.

18. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided* that, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further* that the Committee expressly reserves its rights to amend the Award Agreement and the Plan as described in Sections 1.3.2(h)(1), (2) and (4) of the Plan. A modification that impacts the tax consequences of this Award or the timing of delivery of RSU Shares will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

19. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words "include," "includes" and "including" will be deemed to be followed by the words "without limitation." The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

20. **Providing Information to the Appropriate Authorities; Protected Communications.** In accordance with applicable law, nothing in this Award Agreement (including the forfeiture and repayment provisions in Paragraphs 8 and 9) or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

[EU and UK Material Risk Taker Appendix

This Appendix supplements Paragraph 8 and sets forth additional events that result in forfeiture of up to all of your Short-Term RSUs and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 9. As with the events described in Paragraph 8, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 8(c).

With respect to the events described in this Appendix, the Committee will consider certain factors to determine whether and what portion of your Award will terminate, including the reason for the Risk Event, as well as the extent to which you participated in the Risk Event and your compensation for ____ may or may not have been adjusted to take into account the risk associated with the Risk Event or your Serious Misconduct. Your compensation may be adjusted for the year in which the Risk Event or your Serious Misconduct is discovered.

(a) A Risk Event Occurs On or Before the Applicable End Date. If a Risk Event occurs on or before the Applicable End Date, (i) your rights in respect of all or a portion of your Short-Term RSUs will terminate and no RSU Shares will be delivered in respect of such Short-Term RSUs and (ii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments and payments under Dividend Equivalent Rights) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Risk Event occurred and (B) the date that the repayment request is made.

(i) **“Risk Event”** means there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting and Operational Risk Events Policy (or any successor policy).

(b) You Engage in Serious Misconduct On or Before the Applicable End Date. If you engage in Serious Misconduct during the period beginning on the Delivery Date through the Applicable End Date, you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments and payments under Dividend Equivalent Rights) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (i) the date the Serious Misconduct occurred and (ii) the date that the repayment request is made.

(i) **“Serious Misconduct”** means that you engage in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under applicable employment law.

(c) Certain Definitions.

(i) **“Applicable End Date”** means[_____], with respect to each classification in the table below, the following:

Classification	Applicable End Date
EU MRT	
GSAMI MRT	
GSAMI Senior Management	
UK MRT	

(ii) **"EU MRT"** means an individual who, at any point in [____], was classified by the Firm as a (A) Material Risk Taker in accordance with the applicable remuneration rules of the Alternative Investment Fund Managers Directive or Undertakings for Collective Investment in Transferable Securities V ("AIFMD-UCITS"), including "Identified Staff" under AIFMD-UCITS ("AIFMD-UCITS Identified Staff"), as applicable to the business operations of Goldman Sachs Asset Management, B.V. ("GSAMBV"), Goldman Sachs Asset Management Belgium, S.A. ("GSAM Belgium") or Goldman Sachs Towarzystwo Funduszy Inwestycyjnych S.A. ("GS TFI"), including any branch or subsidiary thereof (collectively, "GSAM Europe"), whether or not GSAM Europe has employed such individual; or (B) an individual who, at any point in [____], was classified by the Firm with respect to Goldman Sachs Paris Inc. et Cie in accordance with the applicable remuneration rules of either the Capital Requirements Directive V or the Investment Firms Regulation/Investment Firms Directive, as a Material Risk Taker ("GSPIC MRT") or Senior Management ("GSPIC Senior Management"). EU MRT does not include AIFMD-UCITS Identified Staff of Goldman Sachs Asset Management Fund Services Ltd. ("GSAMFSL").

(iii) **"GSAMI MRT"** means an individual who, at any point in [____], was classified by the Firm as a GSAMI Material Risk Taker in accordance with the remuneration code under the UK Investment Firms Prudential Regime ("IFPR") as applicable to the business operations of Goldman Sachs Asset Management International Ltd. ("GSAMI").

(iv) **"GSAMI Senior Management"** means an individual who, at any point in [____], was classified by the Firm as a GSAMI Senior Management in accordance with the remuneration code under the IFPR, as applicable to the business operations of GSAMI.

(v) **"UK MRT"** means an individual who, at any point in [____], was classified by the Firm as a Material Risk Taker (including UK Risk Managers and UK PRA Senior Managers) in accordance with the remuneration codes applicable to dual-regulated UK firms.

(vi) For purposes of these definitions, if during [____] you were classified by the Firm as any combination of EU MRT and GSAMI MRT, GSAMI Senior Management or UK MRT then your classification in the table above is EU MRT; any combination of GSAMI MRT or GSAMI Senior Management and UK MRT then your classification in the table above is UK MRT; or as a combination of GSAMI MRT and GSAMI Senior Management then your classification in the table above is GSAMI Senior Management.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by Paragraphs (a) and (b) of this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 14 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

[GSBE Material Risk Taker Appendix

This Appendix supplements Paragraph 8 and sets forth additional events that result in forfeiture of up to all of your Short-Term RSUs and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 9. As with the events described in Paragraph 8, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 8(c).

With respect to the events described in this Appendix, the Committee will consider all relevant factors in making the decision, in its sole discretion as to whether an Adjustment Event or Actionable Misconduct have occurred.

This Appendix applies to all Short-Term RSUs in your Award if you are a GSBE MRT or GSBE Senior Management.

(a) An Adjustment Event Occurs with Respect to ____. If an Adjustment Event occurs with respect to ____, (i) your rights in respect of all or a portion of your Short-Term RSUs will terminate and no RSU Shares will be delivered in respect of such Short-Term RSUs, and (ii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Adjustment Event occurred and (B) the date that the repayment request is made.

(i) **“Adjustment Event”** means that one of the following has occurred:

- A. you significantly contributed to, or were responsible for, any conduct that resulted in a loss of 0.75% or more of the total capital of GS Inc.;
- B. a material regulatory sanction for the Firm comprising one or more of the following:
 - 1. a moratorium pursuant to sec. 46g of the German Banking Act,
 - 2. a measure in case of danger pursuant to sec. 46 of the German Banking Act,
 - 3. the revocation of appointment of a manager pursuant to sec. 36 German Banking Act,
 - 4. a fine pursuant to sec. 56 of the German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of GS Inc.,
 - 5. the cancellation of the banking permit pursuant to sec. 35 of the German Banking Act,
 - 6. an order to increase the capital requirements Goldman Sachs Bank Europe SE (“GSBE”) by at least 0.5% pursuant to sec. 10 of the German Banking Act,

7. a measure in case of organizational deficiencies,
8. a comparable regulatory order, or
9. a material supervisory measure; or

C. you acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that a violation is considered serious if it is suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct pursuant to sec. 1 German Termination Protection Act.

(b) You Engage in Actionable Misconduct. If the Committee determines that you engaged in Actionable Misconduct, which if known (or if known, adequately considered) would have resulted in a reduction to your annual variable compensation when your variable compensation for that year was determined, your rights in respect of all or a portion of your Short-Term RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such Short-Term RSUs.

(i) **“Actionable Misconduct”** means either or both of:

- A. “Unethical behavior” within the meaning of the IVV;
- B. “Breach of duty” within the meaning of the IVV.

(ii) **“IVV”** means the German Remuneration Ordinance for Institutions (*Institutsvergütungsverordnung*) and applicable guidance from the German Federal Financial Supervisory Authority (BaFin) or the European Central Bank.

(c) Scope of Appendix. This Appendix applies to any individual who, at any point in [____], was classified by the Firm as a Material Risk Taker (“GSBE MRT”) or Senior Management (“GSBE Senior Management”) in connection with the business operations of GSBE or any branch or subsidiary thereof whether or not GSBE or any branch or subsidiary thereof employed such individual.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 14 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings:

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

(b) “Acceptance Deadline” means the date communicated to you by the Firm as the deadline to accept your Award. The Acceptance Deadline will be no less than 14 days after the Award Documentation is first made available to you in full. To the extent you are entitled to a translation in accordance with the Language Notice for Equity Awards provided to you as part of the Award Documentation, the Acceptance Deadline will be extended by the time required to furnish you with any such translation.

(c) “Breached an Obligation to the Firm” means that you failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with your Employment or this Award (including any applicable Additional Terms); including, without limitation, any offer letter, employment agreement or any shareholders’ agreement relating to the Firm. Your failure to pay or reimburse the Firm, on demand, for any amount you owe to the Firm will constitute (i) failure to meet an obligation you have under an agreement, regardless of whether such obligation arises under a written agreement, and/or (ii) a material violation of Firm policy constituting Cause.

(d) “Committee” means the Compensation Committee unless otherwise determined by the Board; *provided, however*, that: (i) the authority of the Compensation Committee to administer the Plan and the Award may be granted or delegated to the SIP Committee, in which case “Committee” means the SIP Committee when acting pursuant to such a grant or delegation; (ii) to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3; and (iii) to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m).

(e) “Compensation Committee” means the Compensation Committee of the Board.

(f) “Conflicted Employment” means your employment at any U.S. Federal, state or local government, any non-U.S. government, any supranational or international organization, any self-regulatory organization, or any agency or instrumentality of any such government or organization, or any other employer (other than an “Accounting Firm” within the meaning of SEC Rule 2-01(f)(2) of Regulation S-X or any successor thereto) determined by the Committee, if, as a result of such employment, your continued holding of any Outstanding Short-Term RSUs would result in an actual or perceived conflict of interest.

(g) “Delivery Date” means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a delivery date, so long as any such date occurs before the 29th day of the month specified on the Award Statement; if there is no such date, the Delivery Date will

be the earlier of another date selected by the Committee in the month and year specified on the Award Statement or the first trading day in the subsequent Window Period.

(h) "Failed to Consider Risk" means that you participated (or otherwise oversaw or were responsible for, depending on the circumstances, another individual's participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where you have improperly analyzed such risk or where you have failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, the Committee determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, your business unit or the broader financial system. This definition is not limited to financial risks and is designed to encourage the consideration of the full range of risks associated with activities (e.g., legal, compliance or reputational). This definition also does not require that a material adverse impact actually occur, but, rather, may be triggered if it is determined that there is a reasonable expectation of such an impact.

(i) "HCM" means the Firm's Human Capital Management Division or its successor.

(j) "Qualifying Termination After a Change in Control" means that the Firm terminates your Employment other than for Cause or you terminate your Employment for Good Reason, in each case, within 18 months following a Change in Control.

(k) "SEC" means the U.S. Securities and Exchange Commission.

(l) "Signature Card" means the document and/or website presented to you with your Award that you are required to execute (including electronically) to indicate your acceptance of the Award, the Award Agreement (including the Award Statement and the terms contained in the Signature Card) and Additional Terms applicable to you, as well your receipt of the Award Documentation.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Statement” means a written statement that reflects certain Award terms.

(c) “Board” means the Board of Directors of GS Inc.

(d) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(e) “Cause” means (i) the Grantee’s conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion, or (B) on a felony charge, or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations, (ii) the Grantee’s engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the Exchange Act), (iii) the Grantee’s willful failure to perform the Grantee’s duties to the Firm, (iv) the Grantee’s violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member, (v) the Grantee’s violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or the Grantee’s material violation of any other Firm policy as in effect from time to time, (vi) the Grantee’s engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm or (vii) the Grantee’s engaging in any conduct detrimental to the Firm. The determination as to whether Cause has occurred shall be made by the Committee in its sole discretion and, in such case, the Committee also may, but shall not be required to, specify the date such Cause occurred (including by determining that a prior termination of Employment was for Cause). Any rights the Firm may have hereunder and in any Award Agreement in respect of the events giving rise to Cause shall be in addition to the rights the Firm may have under any other agreement with a Grantee or at law or in equity.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Change in Control” means the consummation of a merger, consolidation, statutory share exchange or similar form of corporate transaction involving GS Inc. (a “Reorganization”) or sale or other disposition of all or substantially all of GS Inc.’s assets to an entity that is not an affiliate of GS Inc. (a “Sale”), that in each case requires the approval of GS Inc.’s shareholders under the law of GS Inc.’s jurisdiction of organization, whether for such Reorganization or Sale (or the issuance of securities of GS Inc. in such Reorganization or Sale), unless immediately following such Reorganization or Sale, either: (i) at least 50% of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of (A) the entity resulting from such Reorganization, or the entity which has acquired all or substantially all of the assets of GS Inc. in a Sale (in either case, the “Surviving Entity”), or (B) if applicable, the ultimate parent entity that directly or indirectly has beneficial

ownership (within the meaning of Rule 13d-3 under the Exchange Act, as such Rule is in effect on the date of the adoption of the 1999 SIP) of 50% or more of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of the Surviving Entity (the "Parent Entity") is represented by GS Inc.'s securities (the "GS Inc. Securities") that were outstanding immediately prior to such Reorganization or Sale (or, if applicable, is represented by shares into which such GS Inc. Securities were converted pursuant to such Reorganization or Sale) or (ii) at least 50% of the members of the board of directors (or similar officials in the case of an entity other than a corporation) of the Parent Entity (or, if there is no Parent Entity, the Surviving Entity) following the consummation of the Reorganization or Sale were, at the time of the Board's approval of the execution of the initial agreement providing for such Reorganization or Sale, individuals (the "Incumbent Directors") who either (A) were members of the Board on the Effective Date or (B) became directors subsequent to the Effective Date and whose election or nomination for election was approved by a vote of at least two-thirds of the Incumbent Directors then on the Board (either by a specific vote or by approval of GS Inc.'s proxy statement in which such persons are named as nominees for director).

(h) "Client" means any client or prospective client of the Firm to whom the Grantee provided services, or for whom the Grantee transacted business, or whose identity became known to the Grantee in connection with the Grantee's relationship with or employment by the Firm.

(i) "Code" means the Internal Revenue Code of 1986, as amended from time to time, and the applicable rulings and regulations thereunder.

(j) "Common Stock" means common stock of GS Inc., par value \$0.01 per share.

(k) "Covered Person" means a member of the Board or the Committee or any employee of the Firm.

(l) "Date of Grant" means the date specified in the Grantee's Award Agreement as the date of grant of the Award.

(m) "Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.

(n) "Effective Date" means the date this Plan is approved by the shareholders of GS Inc. pursuant to Section 3.15 of the Plan.

(o) "Employment" means the Grantee's performance of services for the Firm, as determined by the Committee. The terms "employ" and "employed" shall have their correlative meanings. The Committee in its sole discretion may determine (i) whether and when a Grantee's leave of absence results in a termination of Employment (for this purpose, unless the Committee determines otherwise, a Grantee shall be treated as terminating Employment with the Firm upon the occurrence of an Extended Absence), (ii) whether and when a change in a Grantee's association with the Firm results in a termination of Employment and (iii) the impact, if any, of any such leave of absence or change in association on Awards theretofore made. Unless expressly provided otherwise, any references in the Plan or any Award Agreement to a Grantee's Employment being terminated shall include both voluntary and involuntary terminations.

(p) “Exchange Act” means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(q) “Firm” means GS Inc. and its subsidiaries and affiliates.

(r) “Good Reason” means, in connection with a termination of employment by a Grantee following a Change in Control, (i) as determined by the Committee, a materially adverse alteration in the Grantee’s position or in the nature or status of the Grantee’s responsibilities from those in effect immediately prior to the Change in Control or (ii) the Firm’s requiring the Grantee’s principal place of Employment to be located more than seventy-five (75) miles from the location where the Grantee is principally Employed at the time of the Change in Control (except for required travel on the Firm’s business to an extent substantially consistent with the Grantee’s customary business travel obligations in the ordinary course of business prior to the Change in Control).

(s) “Grantee” means a person who receives an Award.

(t) “GS Inc.” means The Goldman Sachs Group, Inc., and any successor thereto.

(u) “1999 SIP” means The Goldman Sachs 1999 Stock Incentive Plan, as in effect prior to the effective date of the 2003 SIP.

(v) “Outstanding” means any Award to the extent it has not been forfeited, cancelled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(w) “RSU” means a restricted stock unit granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(x) “RSU Shares” means shares of Common Stock that underlie an RSU.

(y) “Section 409A” means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(z) “SIP Administrator” means each person designated by the Committee as a “SIP Administrator” with the authority to perform day-to-day administrative functions for the Plan.

(aa) “SIP Committee” means the persons who have been delegated certain authority under the Plan by the Committee.

(ab) “Transfer Restrictions” means restrictions that prohibit the sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposal (including through the use of any cash-settled instrument), whether voluntarily or involuntarily by the Grantee, of an Award or any shares of Common Stock, cash or other property delivered in respect of an Award.

(ac) “Vested” means, with respect to an Award, the portion of the Award that is not subject to a condition that the Grantee remain actively employed by the Firm in order for the Award to remain Outstanding. The fact that an Award becomes Vested shall not mean or otherwise indicate that the

Grantee has an unconditional or nonforfeitable right to such Award, and such Award shall remain subject to such terms, conditions and forfeiture provisions as may be provided for in the Plan or in the Award Agreement.

(ad)“Window Period” means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (*provided* that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
[One-Time][Year-End] Performance-Based RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your _____ award of performance-based RSUs (your “Award” or “PSUs”). You should read carefully this entire document (including its [Appendix][Appendices]), the Award Statement and the Signature Card (together, the “Award Agreement”), as well as the other documentation presented to you in connection with acceptance of your Award (collectively, “Award Documentation”).

Acceptance

1. **You Must Decide Whether to Accept Your Award.** To be eligible for your Award, you must by the **Acceptance Deadline**: (a) open and activate an Account; (b) accept any Additional Terms presented to you as part of the Award Documentation; and (c) execute (including by electronic means) the accompanying Signature Card in accordance with its instructions. By executing the Signature Card, you confirm your agreement to all of the terms of the Award Agreement (including the Award Statement and the Signature Card), as well as any Additional Terms.

Documents that Govern Your Award; Definitions

2. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement. The Plan and its Prospectus are included with the Award Documentation.

3. **Your Award Statement.** The accompanying Award Statement contains the number of PSUs awarded to you and some of your Award’s specific terms. For example, it specifies the Performance Period and the Performance Goal[s] applicable to your Award. It also contains the Determination Date and the Settlement Date[s] for your Award and [the] [any applicable] Transferability Date[s] for any Shares at Risk that may be delivered to you in respect of any Settlement Amount that you may earn. The number of PSUs on your Award Statement is not necessarily the number of PSUs in respect of which the Settlement Amount will be earned, but is merely the basis for determining the amount (if any) that will be delivered to you.

4. **Definitions.** Unless otherwise defined herein, including in the Definitions Appendix or any other Appendix, capitalized terms have the meanings provided in the Plan.

Vesting of Your PSUs

5. **Vesting.** All of your PSUs are Vested. When a PSU is Vested, it means **only** that your continued active Employment is not required to earn delivery in respect of that PSU. **Vesting does not mean you have a non-forfeitable right to the Vested portion of your Award. The terms of this Award Agreement (including conditions to delivery, satisfaction of the Performance Goal[s] and any applicable Transfer Restrictions) continue to apply to Vested PSUs, and you can still forfeit Vested PSUs and any Shares at Risk.**

Performance Goal[s]

6. **Performance.** The Settlement Amount is dependent, and may vary based, on achievement of the Performance Goal[s] over the Performance Period. On the Determination Date, the Firm will determine whether or not, and to what extent, the Performance Goal[s] for that Performance Period [has][have] been satisfied. All your rights with respect to the Settlement Amount [(and any Dividend Equivalent Payments)] are dependent on the extent to which the Performance Goal[s] [is][are] achieved, and any rights to delivery in respect of your Outstanding PSUs immediately will terminate and

no Settlement Amount will be delivered in respect of such PSUs upon the Committee's determination, in its sole discretion, that the Performance Goal[s] [has][have] not been satisfied to the extent necessary to result in delivery in respect of the PSUs.

Settlement Amount

7. Settlement.

(a) In General. Subject to satisfaction of the terms of your Award, including satisfaction of the Performance Goal[s], on [the] [each] Settlement Date [listed on your Award Statement], you will receive delivery (less applicable withholding as described in Paragraph 13(b)) of the [applicable portion of the] Settlement Amount [and payment of any Dividend Equivalent Payments] as further described in this Award Agreement and in your Award Statement. Until such delivery [and payment], you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc. with respect to either the PSUs or the Settlement Amount. Without limiting the Committee's authority under Section 1.3.2(h) of the Plan, the Firm may accelerate any Settlement Date by up to 30 days.

(b) Form of Delivery. The Settlement Amount will be delivered [in the form of Shares [at Risk] (by book entry credit to your Account)] [as follows:

(i) ___% in the form of cash.

(ii) ___% in the form of Shares at Risk (by book entry credit to your Account)].

(c) Shares at Risk. [All] [___ percent] of the Shares delivered to you in respect of the Settlement Amount will be subject to Transfer Restrictions until the [applicable] Transferability Date. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition in violation of the Transfer Restrictions on Shares at Risk will be void.

(d) Release of Transfer Restrictions. Within 30 Business Days after the [applicable] Transferability Date identified on your Award Statement, GS Inc. will remove the Transfer Restrictions on any Shares at Risk [in respect of the amount of Outstanding PSUs listed next to that date]. The Committee may select multiple dates within the 30-Business-Day period on which to remove Transfer Restrictions for all or a portion of the Shares at Risk with the same Transferability Date, and all such dates will be treated as a single Transferability Date for purposes of this Award.

Dividends

8. [Dividend Equivalent Rights and] Dividends. [To the extent described in your Award Statement, each PSU will include a Dividend Equivalent Right, which will be subject to the provisions of Section 2.8 of the Plan. Accordingly, for each of your Outstanding PSUs with respect to which delivery is made under the Settlement Amount, you will be entitled to payments under Dividend Equivalent Rights equal to any regular cash dividend paid by GS Inc. in respect of a Share for which the record date occurs on or after the Date of Grant. The payment to you of amounts under Dividend Equivalent Rights (less applicable withholding as described in Paragraph 13(b)) is conditioned upon the delivery under the Settlement Amount in respect of the PSUs to which such Dividend Equivalent Rights relate, and you will have no right to receive any Dividend Equivalent Payments relating to PSUs for which you do not receive delivery under the Settlement Amount (including, without limitation, due to a failure to satisfy the Performance Goal[s]). Dividend Equivalent Payments will be paid on the [applicable] Settlement Date.] [In addition, you][You] will be entitled to receive on a current basis any regular cash dividend paid in respect of any Shares at Risk. [The PSUs do not include Dividend Equivalent Rights.]

Forfeiture of Your Award

9. How You May Forfeit Your Award. This Paragraph 9 sets forth the events that result in forfeiture of up to all of your PSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with

Paragraph 10. More than one event may apply, and in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event. Paragraph 12 (relating to certain circumstances under which release of Transfer Restrictions may be accelerated) provides for exceptions to one or more provisions of this Paragraph 9.

(a) PSUs Forfeited Upon Certain Events. If any of the following occurs, your rights to all of your Outstanding PSUs will terminate, and no Settlement Amount will be delivered in respect of such PSUs, as may be further described below:

(i) You Associate With a Covered Enterprise. Except as prohibited by applicable law, you Associate With a Covered Enterprise during the Performance Period.

(ii) You Solicit Clients or Employees, Interfere with Client or Employee Relationships or Participate in the Hiring of Employees. Before the [applicable] Settlement Date and except as prohibited by applicable law, either:

A. (A) you, in any manner, directly or indirectly, (1) Solicit any Client to transact business with a Covered Enterprise or to reduce or refrain from doing any business with the Firm, (2) interfere with or damage (or attempt to interfere with or damage) any relationship between the Firm and any Client, (3) Solicit any person who is an employee of the Firm to resign from the Firm, (4) Solicit any Selected Firm Personnel to apply for or accept employment (or other association) with any person or entity other than the Firm or (5) participate in the hiring of any Selected Firm Personnel by any person or entity other than the Firm (including, without limitation, participating in the identification of individuals for potential hire, and participating in any hiring decision), whether as an employee or consultant or otherwise, or

B. (B) Selected Firm Personnel are Solicited, hired or accepted into partnership, membership or similar status by any entity where you have, or will have, direct or indirect managerial responsibility for such Selected Firm Personnel, unless the Committee determines that you were not involved in such Solicitation, hiring or acceptance.

(iii) [GS Inc. Fails to Maintain the Minimum Tier 1 Capital Ratio]. Before the [applicable] Settlement Date, GS Inc. fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days.]

(iv) [GS Inc. Is Determined to Be in Default]. Before the [applicable] Settlement Date, the Board of Governors of the Federal Reserve or the FDIC makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Wall Street Reform and Consumer Protection Act for the appointment of the FDIC as a receiver of GS Inc. based on a determination that GS Inc. is "in default" or "in danger of default."]

(b) PSUs and Shares at Risk Forfeited upon Certain Events. If any of the following occurs (i) your rights to all of your Outstanding PSUs will terminate, and no Settlement Amount will be delivered in respect of such PSUs and (ii) your rights to any Shares at Risk will terminate and all such Shares at Risk will be cancelled, in each case, as may be further described below:

(i) You Failed to Consider Risk. You Failed to Consider Risk during ____.

(ii) Your Conduct Constitutes Cause. Any event that constitutes Cause has occurred before the [applicable] Settlement Date for PSUs or the [applicable] Transferability Date for any Shares at Risk.

(iii) You Breach an Obligation to the Firm. The Committee determines that, before the [applicable] Settlement Date for PSUs or the [applicable] Transferability Date for any Shares at Risk, you Breached an Obligation to the Firm.

(iv) You Do Not Provide Timely Certifications or Comply with Your Certifications. You fail to certify to GS Inc. that you have complied with all of the terms of the Plan and this Award Agreement as required by Paragraph 13(d), or the Committee determines that you have failed to comply with a term of the Plan or this Award Agreement to which you have certified compliance.

(v) You Do Not Follow Dispute Resolution Procedures. You attempt to have any dispute, controversy or claim under the Plan or this Award Agreement resolved in any manner that is not provided for by Paragraph 16 or Section 3.17 of the Plan, or you attempt to arbitrate a dispute, controversy or claim without first having exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(vi) You Bring an Action that Results in a Determination that Any Award Agreement Term Is Invalid. As a result of any action brought by you, it is determined that any term of this Award Agreement is invalid.

(vii) You Receive Compensation in Respect of Your Award from Another Employer. Your Employment terminates for any reason or you otherwise are no longer actively Employed with the Firm and another entity grants you cash, equity or other property (whether vested or unvested) to replace, substitute for or otherwise in respect of any Outstanding PSUs or Shares at Risk; *provided, however*, that your rights will only be terminated in respect of the PSUs or Shares at Risk that are replaced, substituted for or otherwise considered by such other entity in making its grant.

(viii) [Accounting Restatement Required Under Sarbanes-Oxley]. GS Inc. is required to prepare an accounting restatement due to GS Inc.'s material noncompliance, as a result of misconduct, with any financial reporting requirement under the securities laws as described in Section 304(a) of Sarbanes-Oxley; *provided, however*, that your rights will only be terminated in respect of PSUs or Shares at Risk to the same extent that would be required under Section 304(a) of Sarbanes-Oxley had you been a "chief executive officer" or "chief financial officer" of GS Inc. (regardless of whether you actually hold such position at the relevant time).]

(ix) Accounting Restatement Required and Dodd-Frank Clawback Policy Applies. GS Inc. is required to prepare an "Accounting Restatement" as described in GS Inc.'s Policy for Recovery of Erroneously Awarded Compensation (the "Dodd-Frank Clawback Policy"); in each case if, and to the same extent as, required under the Dodd-Frank Clawback Policy. For the avoidance of doubt, PSUs previously granted to you and Shares at Risk in respect of PSUs previously granted to you are also subject to forfeiture and clawback under this Paragraph 9(b)[(viii)/(ix)].

(c) [Additional Forfeiture and Repayment Conditions for Material Risk Takers]. Since the Firm identified you as a Material Risk Taker ("MRT") in ____, the **EU and UK Material Risk Taker Appendix** and/or the **GSBE Material Risk Taker Appendix** (as applicable to you, the "MRT Appendix") supplements this Paragraph 9. The MRT Appendix sets forth additional events that (i) result in forfeiture of up to all of your PSUs and Shares at Risk and (ii) may require repayment to the Firm of up to all amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. Your MRT classification, as defined in the MRT Appendix, was communicated to you separately. If you are unsure of which classification(s) apply, you must contact HCM prior to accepting this Award. The Firm's records with respect to such classification(s) will be controlling as to whether the MRT Appendix applies to your Award. The MRT classifications with their corresponding MRT Appendix are as follows:

MRT Classification	MRT Appendix
EU MRT, GSAMI MRT, GSAMI Senior Management and/or UK MRT	EU and UK Material Risk Taker Appendix

GSBE MRT and GSBE Senior Management	GSBE Material Risk Taker Appendix
GSBE MRT or GSBE Senior Management <i>and</i> any combination of EU MRT, GSAMI MRT, GSAMI Senior Management or UK MRT	GSBE Material Risk Taker Appendix <i>and</i> EU and UK Material Risk Taker Appendix]

(d) Actions Pending Forfeiture. In connection with any investigation as to whether any of the events that would result in forfeiture under the Plan[,] [or] this Paragraph 9 [or the MRT Appendix] have occurred, the Firm reserves the right to (i) suspend [any Dividend Equivalent Payments,] delivery of the Settlement Amount or release of any Transfer Restrictions; (ii) deliver the Settlement Amount[,] [any Dividend Equivalent Payments] [or] dividends into an escrow account in accordance with Paragraph 13(e)(iii); or (iii) apply Transfer Restrictions to any Shares.

Repayment of Your Award

10. When You May Be Required to Repay Your Award.

(a) Repayment Generally. If the Committee determines that any term of this Award was not satisfied, you will be required, immediately upon demand therefor, to repay to the Firm the following:

(i) Any Settlement Amount (including any Shares at Risk) for which the terms (including the terms for delivery) of the related PSUs were not satisfied, in accordance with Section 2.6.3 of the Plan.

(ii) Any Shares at Risk for which the terms (including the terms for the release of Transfer Restrictions) were not satisfied, in accordance with Section 2.5.3 of the Plan.

(iii) [Any Shares that were delivered (but not subject to Transfer Restrictions) at the same time any Shares at Risk that are cancelled or required to be repaid were delivered.]

(iv) [Any Dividend Equivalent Payments for which the terms were not satisfied (including any such payments made in respect of PSUs that are forfeited or any Settlement Amount that is required to be repaid), in accordance with Section 2.8.3 of the Plan.]

(v) Any dividends paid in respect of any [delivered Shares (including] Shares at Risk[] that are cancelled or required to be repaid.

(vi) Any amount applied to satisfy tax withholding or other obligations with respect to any PSUs, Settlement Amount (including Shares at Risk)[,] [and] dividend payments [and Dividend Equivalent Payments] that are forfeited or required to be repaid.

(b) Repayment Upon Materially Inaccurate Financial Statements. If any delivery is made under this Award Agreement based on materially inaccurate financial statements (which includes, but is not limited to, statements of earnings, revenues or gains) or other materially inaccurate performance criteria, you will be obligated to repay to the Firm, immediately upon demand therefor, any excess amount delivered, as determined by the Committee in its sole discretion. For the avoidance of doubt, this Paragraph 10(b) includes any repayment required following an "Accounting Restatement" under the Dodd-Frank Clawback Policy.

(c) [Repayment Upon Accounting Restatement Required Under Sarbanes-Oxley. If an event described in Paragraph 9(b)(viii) (relating to a requirement under Sarbanes-Oxley that GS Inc. prepare an accounting restatement) occurs, any Settlement Amount (including Shares at Risk), dividend payments, [Dividend Equivalent Payments,] cash or other property delivered, paid or withheld in respect of this Award will be subject to repayment as described in Paragraph 10(a) to the same extent that would be required under Section 304(a) of Sarbanes-Oxley had you been a "chief executive officer" or "chief financial officer" of GS Inc. (regardless of whether you actually hold such position at the relevant time).]

Terminations of Employment

11. **Treatment of PSUs.** Unless the Committee determines otherwise, if your Employment terminates for any reason or you are otherwise no longer actively Employed with the Firm (which includes off-premises notice periods, "garden leaves," pay in lieu of notice or any other similar status), the Performance Goal[s] applicable to your Outstanding PSUs will continue to apply and the determination of the Settlement Amount will continue to be subject to whether, and to what extent, the Performance Goal[s] [has][have] been achieved, in each case, as provided in Paragraph 6. All other terms of this Award Agreement, including the forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

12. **Accelerated Release of Transfer Restrictions on Shares at Risk in the Event of [a Qualifying Termination After a Change in Control or] Death.** In the event of your [Qualifying Termination After a Change in Control or] death, [each] as described below, any Shares at Risk [(and delivery of your Settlement Amount in the case of death in certain circumstances)] will be treated as described in this Paragraph 12, and[, except as set forth in Paragraph 12(a),] all other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply. [In each case, the Performance Goal[s] applicable to your Outstanding PSUs will continue to apply and the determination of the Settlement Amount will continue to be subject to whether or not, and to what extent, the Performance Goal[s] [has][have] been achieved, in each case, as provided in Paragraph 6.]

(a) [You Have a Qualifying Termination After a Change in Control]. If your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, any Transfer Restrictions will cease to apply to your Shares at Risk. In addition, the forfeiture events in Paragraph 9 [(except the MRT Appendix)] will not apply to your Shares at Risk.]

(b) Death. If you die, [as soon as practicable] after receipt of any [such] documentation as may be requested by the Committee or a SIP Administrator, [any Transfer Restrictions will cease to apply to your Shares at Risk] [unless prohibited by applicable law or regulation:

(i) If you die prior to the Determination Date or an applicable Settlement Date, the representative of your estate will receive delivery of any undelivered portion of the Settlement Amount that would have otherwise been made pursuant to Paragraph 6 as soon as practicable after the later of the date of death and the Determination Date.

(ii) Any Transfer Restrictions will cease to apply to your Shares at Risk.]

Other Terms And Conditions of Your Award

13. Additional Terms, Conditions and Consents.

(a) You Will Be Required to Accept Additional Terms. In addition to the terms of the Award Agreement, the Award Documentation includes other contractual terms that you must accept as a condition of the Award (the "Additional Terms"). By accepting the Award and executing the Signature Card, you agree to any such Additional Terms applicable to you as set forth in the Award Documentation, which will include but are not limited to the following:

(i) Notice Period Policy. You must agree to abide by the terms, as applicable to you, of the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy").

(ii) Agreement Regarding Arbitration of Employment-Related Matters. You must agree to the Agreement Regarding Arbitration of Employment-Related Matters, which imposes obligations in addition to those contained in Paragraph 16.

(iii) APAC Restrictive Covenants. If you are employed by the Firm in the Asia-Pacific Region (e.g., Australia, the People's Republic of China, the Hong Kong Special Administrative Region, India, Indonesia, Japan, the Republic of Korea, New Zealand, Singapore, Taiwan), you must agree to the APAC Restrictive Covenants Agreement.

(iv) Role-, Business- or Division-Specific Agreements. If applicable to you, you must accept Additional Terms related to your specific role, business unit or division as set forth in the Award Documentation.

(b) You Must Satisfy Applicable Tax Withholding Requirements. Delivery of the Settlement Amount is conditioned on your satisfaction of any applicable withholding taxes in accordance with Section 3.2 of the Plan, which includes the Firm deducting or withholding amounts from any payment or distribution to you. In addition, to the extent permitted by applicable law, the Firm, in its sole discretion, may require you to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations imposed on you or the Firm in connection with the grant or delivery of this Award by requiring you to choose between remitting the amount (i) in cash (or through payroll deduction or otherwise) or (ii) in the form of proceeds from the Firm's executing a sale of Shares delivered to you under this Award. In no event, however, does this Paragraph 13(b) give you any discretion to determine or affect the timing of the delivery of the Settlement Amount or the timing of payment of tax obligations.

(c) You Must Comply with Applicable Deadlines and Procedures to Appeal. If you disagree with a determination made by the Committee, the SIP Administrators or any of their delegates or designees relating to the Plan or the Award Agreement and you wish to appeal such determination, you must submit a written request to the Committee for review within 180 days after the determination at issue. You must exhaust your internal administrative remedies (i.e., submit your appeal and wait for resolution of that appeal) before seeking to resolve a dispute, controversy or claim through arbitration pursuant to Paragraph 16 and Section 3.17 of the Plan.

(d) You May Be Required to Certify Compliance with Award Terms; You Are Responsible for Providing Updated Address and Contact Information After Your Departure from the Firm. If your Employment terminates while you continue to hold PSUs or Shares at Risk, the Firm may require you to provide certifications of your compliance with all of the terms of the Plan and this Award Agreement. You understand and agree that (i) your contact information as reflected in the Firm's personnel records at the time any certification is requested will be deemed current; (ii) it is your responsibility to inform HCM of any changes to your contact information to ensure timely receipt of the certification materials, regardless of whether your contact information is provided to another part of the Firm; and (iii) you are responsible for contacting the Firm to obtain such certification materials if not received. Your failure to return properly completed certification materials by the specified deadline (including because you did not provide HCM with updated contact information) will result in the forfeiture of all of your PSUs and any Shares at Risk and subject previously delivered amounts to repayment under Paragraph 10.

(e) You Agree to Other Terms, Conditions and Consents. By accepting this Award you understand and agree that:

(i) Amounts May Be Rounded to Avoid Fractional Shares. Any amounts delivered in respect of the Settlement Amount, including Shares at Risk, may be rounded to avoid fractional Shares.

(ii) Firm May Deliver Cash or Other Property in Respect of the Settlement Amount. In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the Settlement Amount, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to delivery of the Settlement Amount will include such deliveries of cash, other securities, other awards under the Plan or other property.

(iii) Firm May Deliver Your Award into an Escrow Account. The Firm may establish and maintain an escrow account on such terms (which may include your executing any documents related to, and your paying for any costs associated with, such account) as it may deem necessary or appropriate, and the Settlement Amount may initially be delivered, and any [Dividend Equivalent Payments and] dividends may initially be paid, into and held in that escrow account until such time as the Committee has

received such documentation as it may have requested or until the Committee has determined that any other conditions or restrictions on deliveries required by this Award Agreement have been satisfied.

(iv) Firm May Affix Legends and Place Stop Orders on Shares. GS Inc. may affix to Certificates representing Shares any legend that the Committee determines to be necessary or advisable (including to reflect any restrictions to which you may be subject under the Additional Terms, a separate agreement or any Firm policy applicable to you). GS Inc. may advise the transfer agent to place a stop order against any legended Shares. For the avoidance of doubt, Shares as used herein includes, without limitation, Restricted Shares.

(v) You Authorize the Firm to Register in Its or Its Designee's Name Any Restricted Shares and Sell, Assign or Transfer Any Forfeited Restricted Shares. You are granting to the Firm the full power and authority to register any Restricted Shares in its or its designee's name and authorizing the Firm or its designee to sell, assign or transfer any Restricted Shares if you forfeit your Restricted Shares.

(vi) You Agree to Certain Consents as a Condition to the Award. You have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

(vii) You Are Subject to the Firm's Policies, Rules and Procedures. You are subject to the Firm's policies in effect from time to time concerning trading in Shares and hedging or pledging Shares and equity-based compensation or other awards (including, without limitation, the "Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards" or any successor policies), and confidential or proprietary information, and you will effect sales of Shares in accordance with such rules and procedures as may be adopted from time to time (which may include, without limitation, restrictions relating to the timing of sale requests, the manner in which sales are executed, pricing method, consolidation or aggregation of orders and volume limits determined by the Firm).

(viii) You Are Responsible for Costs Associated with Your Award. You will be responsible for all brokerage costs and other fees or expenses associated with your Award, including those related to the sale of Shares.

(ix) You Will Be Deemed to Represent Your Compliance with All the Terms of Your Award if You Accept Delivery of, or Sell, Shares. You will be deemed to have represented and certified that you have complied with all of the terms of the Plan and this Award Agreement when any Settlement Amount [and Dividend Equivalent Payments] [is][are] delivered to you and you request the sale of Shares following the release of Transfer Restrictions on any Shares at Risk.

(x) You Agree that Covered Persons Will Not Have Liability. In addition to and without limiting the generality of the provisions of Section 1.3.5 of the Plan, neither the Firm nor any Covered Person will have any liability to you or any other person for any action taken or omitted in respect of this or any other Award.

14. **Non-transferability.** Except as otherwise may be provided by the Committee, the limitations set forth in Section 3.5 of the Plan will apply to this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition (including through the use of any cash-settled instrument), whether voluntary or involuntary, in violation of the provisions of this Paragraph 14 or Section 3.5 of the Plan will be void.

15. **Right of Offset.** Except as provided in Paragraph 18(d), the obligation to deliver the Settlement Amount, pay dividends [or Dividend Equivalent Payments] or release any Transfer Restrictions under this Award Agreement is subject to Section 3.4 of the Plan, which provides for the Firm's right to offset against such obligation any outstanding amounts you owe to the Firm and any amounts the Committee deems appropriate pursuant to any tax equalization policy or agreement.

Arbitration, Choice of Forum and Governing Law

16. Arbitration; Choice of Forum.

(a) By accepting this award, you are indicating that you understand and agree that the arbitration and choice of forum provisions set forth in Section 3.17 of the Plan will apply to this Award to the fullest extent permitted by applicable law. These provisions, which are expressly incorporated herein by reference, provide among other things that any dispute, controversy or claim between the Firm and you arising out of or relating to or concerning the Plan or this Award Agreement will be finally settled by arbitration, pursuant to the terms more fully set forth in Section 3.17 of the Plan; provided that nothing herein shall preclude you from filing a charge with or participating in any investigation or proceeding conducted by any governmental authority, including but not limited to the SEC, the Equal Employment Opportunity Commission and a state or local human rights agency, as well as law enforcement.

(b) Nothing in the Plan or this Award Agreement includes an agreement to arbitrate claims on a collective, class or representative basis. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider class, collective or representative claims, to order consolidation or to join different claimants or grant relief other than on an individual basis to the individual claimant involved.

(c) Notwithstanding any applicable forum rules to the contrary, to the extent there is a question of enforceability of the Plan or this Award Agreement arising from a challenge to the arbitrator's jurisdiction or to the arbitrability of a claim, it will be decided by a court and not an arbitrator.

(d) Prior to filing any arbitration in accordance with Paragraph 16(a), you must first exhaust your internal administrative remedies in accordance with Paragraph 13(c), which includes awaiting the Committee's final determination of any such appeal. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider any dispute, controversy or claim as to which you have not first exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(e) The Federal Arbitration Act governs interpretation and enforcement of all arbitration provisions under the Plan and this Award Agreement, and all arbitration proceedings thereunder.

(f) Nothing in the Plan or this Award Agreement creates a substantive right to bring a claim under U.S. Federal, state, or local employment laws.

(g) By accepting your Award, you irrevocably appoint each General Counsel of GS Inc., or any person whom the General Counsel of GS Inc. designates, as your agent for service of process in connection with any suit, action or proceeding arising out of or relating to or concerning the Plan or any Award which is not arbitrated pursuant to the provisions of Section 3.17.1 of the Plan, who shall promptly advise you of any such service of process.

(h) As a condition of your Award, you also must accept the terms of the Agreement Regarding Arbitration of Employment-Related Matters, which is included among the Additional Terms described in Paragraph 13(a) and sets forth obligations independent of this Paragraph 16. For the avoidance of doubt, the obligation to exhaust your internal remedies described in Paragraph 13(c) and Paragraph 16(d) only applies to disputes, controversies or claims relating to the Plan or the Award Agreement.

17. Governing Law. This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

18. Compliance of Award Agreement and Plan with Section 409A. The provisions of this Paragraph 18 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any conflict or potential inconsistency between this Paragraph 18 and the other provisions of this Award Agreement, this Paragraph 18 will govern.

(b) Settlement will not be delayed beyond the date on which all applicable conditions or restrictions on settlement in respect of your PSUs required by this Award Agreement (including the consents and other items specified in Section 3.3 of the Plan) are satisfied. To the extent that any portion of this Award is intended to satisfy the requirements for short-term deferral treatment under Section 409A, settlement in respect of such portion will occur by the March 15 coinciding with the last day of the applicable "short-term deferral" period described in Reg. 1.409A-1(b)(4) in order for settlement to be within the short-term deferral exception unless, in order to permit all applicable conditions or restrictions on settlement to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay settlement to a later date within the same calendar year or to such later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 13(e) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any delivery or payment [(including in the form of Shares at Risk or other property)] that the Firm may make in respect of your PSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the payment or delivery that would otherwise have been made (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Paragraph 15 and Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(e) Settlement in respect of any portion of the Award may be made, if and to the extent elected by the Committee, later than the [applicable] Settlement Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later payment or delivery, as applicable, is permitted under Section 409A).

(f) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Committee Authority, Amendment, Construction and Regulatory Reporting

19. **Committee Authority.** The Committee has the authority to determine, in its sole discretion, that any event triggering forfeiture or repayment of your Award will not apply, to limit the forfeitures and repayments that result under Paragraphs 9 and 10 and to remove any Transfer Restrictions before the [applicable] Transferability Date.

20. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided that*, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further* that the Committee expressly reserves its rights to amend the Award Agreement and the

Plan as described in Sections 1.3.2(h)(1), (2) and (4) of the Plan. A modification that impacts the tax consequences of this Award or the timing of delivery of the Settlement Amount will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

21. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words “include,” “includes” and “including” will be deemed to be followed by the words “without limitation.” The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

22. **Providing Information to the Appropriate Authorities; Protected Communications.** In accordance with applicable law, nothing in this Award Agreement (including the forfeiture and repayment provisions in Paragraphs 9 and 10) or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

[EU and UK Material Risk Taker Appendix

Qualitative Overlay Reduction

(g) In addition to and without limiting the Firm's rights under the forfeiture and repayment provisions set forth in Paragraphs 9 and 10 or in the "Forfeiture and Repayment" section below, the Committee may determine to reduce the Settlement Amount as described in the section entitled "Qualitative Overlay Reduction" in the Award Statement.

Forfeiture and Repayment

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your PSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply, in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(d).

With respect to the events described in this Appendix, the Committee will consider certain factors to determine whether and what portion of your Award will terminate, including the reason for the Loss Event or Risk Event, as well as the extent to which you participated in the Loss Event or Risk Event and your compensation for the Firm's ____ fiscal year may or may not have been adjusted to take into account the risk associated with the Loss Event, Risk Event, your Serious Misconduct or the Serious Misconduct of a Supervised Employee. Your compensation may be adjusted for the year in which the Loss Event, Risk Event, your Serious Misconduct or a Supervised Employee's Serious Misconduct is discovered.

(a) **A Loss Event Occurs Prior to Delivery.** If a Loss Event occurs prior to the delivery of any portion of the Settlement Amount, your rights in respect of all or a portion of your PSUs which are scheduled to deliver on the next Settlement Date immediately following the date that the Loss Event is identified (or, if not practicable, then the next following Settlement Date) will terminate, and no Shares will be delivered in respect of such PSUs.

(i) **"Loss Event"** means (A) an annual pre-tax loss at GS Inc. or (B) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm), or annual negative revenues in the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm) of \$5 billion or more, *provided* in either case that you are employed in a business within such reporting segment.

(b) **A Risk Event Occurs On or Before the Applicable End Date.** If a Risk Event occurs on or before the Applicable End Date, (i) your rights in respect of all or a portion of your PSUs will terminate and no Settlement Amount will be delivered in respect of such PSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Risk Event occurred and (B) the date that the repayment request is made.

(i) **"Risk Event"** means there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting and Operational Risk Events Policy (or any successor policy).

(iv) **“GSAMI Senior Management”** means an individual who, at any point in ____, was classified by the Firm as a GSAMI Senior Management in accordance with the remuneration code under the IFPR as applicable to the business operations of GSAMI.

(v) **“UK MRT”** means an individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (including UK Risk Managers and UK PRA Senior Managers) in accordance with the remuneration codes applicable to dual-regulated UK firms.

(vi) For purposes of these definitions, if during ____ you were classified by the Firm as any combination of EU MRT and GSAMI MRT, GSAMI Senior Management or UK MRT then your classification in the table above is EU MRT; any combination of GSAMI MRT or GSAMI Senior Management and UK MRT then your classification in the table above is UK MRT; or as a combination of GSAMI MRT and GSAMI Senior Management then your classification in the table above is GSAMI Senior Management.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by Paragraphs (b) and (c) of this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

[GSBE Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider all relevant factors in making the decision, in its sole discretion as to whether an Adjustment Event or Actionable Misconduct have occurred.

This Appendix applies to all RSUs in your Award if you are a GSBE MRT or GSBE Senior Management.

(a) An Adjustment Event Occurs with Respect to ____. If an Adjustment Event occurs with respect to ____, (i) your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled, and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Adjustment Event occurred and (B) the date that the repayment request is made.

(i) **“Adjustment Event”** means that one of the following has occurred:

- A. you significantly contributed to, or were responsible for, any conduct that resulted in a loss of 0.75% or more of the total capital of GS Inc.;
- B. a material regulatory sanction for the Firm comprising one or more of the following:
 - 1. a moratorium pursuant to sec. 46g of the German Banking Act,
 - 2. a measure in case of danger pursuant to sec. 46 of the German Banking Act,
 - 3. the revocation of appointment of a manager pursuant to sec. 36 German Banking Act,
 - 4. a fine pursuant to sec. 56 of the German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of GS Inc.,
 - 5. the cancellation of the banking permit pursuant to sec. 35 of the German Banking Act,
 - 6. an order to increase the capital requirements Goldman Sachs Bank Europe SE (“GSBE”) by at least 0.5% pursuant to sec. 10 of the German Banking Act,

7. a measure in case of organizational deficiencies,
8. a comparable regulatory order, or
9. a material supervisory measure; or

C. you acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that a violation is considered serious if it is suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct pursuant to sec. 1 German Termination Protection Act.

(b) You Engage in Actionable Misconduct. If the Committee determines that you engaged in Actionable Misconduct, which if known (or if known, adequately considered) would have resulted in a reduction to your annual variable compensation when your variable compensation for that year was determined, your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs.

(i) **"Actionable Misconduct"** means either or both of:

- A. "Unethical behavior" within the meaning of the IVV;
- B. "Breach of duty" within the meaning of the IVV.

(ii) **"IVV"** means the German Remuneration Ordinance for Institutions (*Institutsvergütungsverordnung*) and applicable guidance from the German Federal Financial Supervisory Authority (BaFin) or the European Central Bank.

(c) Scope of Appendix. This Appendix applies to any individual who, at any point in _____, was classified by the Firm as a Material Risk Taker ("GSBE MRT") or Senior Management ("GSBE Senior Management") in connection with the business operations of GSBE or any branch or subsidiary thereof, whether or not GSBE or any branch or subsidiary thereof employed such individual.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings.

(a) "409A Deferred Compensation" means a "deferral of compensation" or "deferred compensation" as those terms are defined in the regulations under Section 409A.

(b) "Acceptance Deadline" means the date communicated to you by the Firm as the deadline to accept your Award. The Acceptance Deadline will be no less than 14 days after the Award Documentation is first made available to you in full. To the extent you are entitled to a translation in accordance with the Language Notice for Equity Awards provided to you as part of the Award Documentation, the Acceptance Deadline will be extended by the time required to furnish you with any such translation.

(c) "Associate With a Covered Enterprise" means that you (i) form, or acquire a 5% or greater equity ownership, voting or profit participation interest in, any Covered Enterprise or (ii) associate in any capacity (including association as an officer, employee, partner, director, consultant, agent or advisor) with any Covered Enterprise. Associate With a Covered Enterprise may include, as determined in the discretion of the Committee, (i) becoming the subject of any publicly available announcement or report of a pending or future association with a Covered Enterprise and (ii) unpaid associations, including an association in contemplation of future employment. "Association With a Covered Enterprise" will have its correlative meaning.

(d) "Breached an Obligation to the Firm" means that you failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with your Employment or this Award (including any applicable Additional Terms); including, without limitation, any offer letter, employment agreement or any shareholders' agreement relating to the Firm. Your failure to pay or reimburse the Firm, on demand, for any amount you owe to the Firm will constitute (i) failure to meet an obligation you have under an agreement, regardless of whether such obligation arises under a written agreement, and/or (ii) a material violation of Firm policy constituting Cause.

(e) "Committee" means the Compensation Committee unless otherwise determined by the Board; *provided, however*, that: (i) the authority of the Compensation Committee to administer the Plan and the Award may be granted or delegated to the SIP Committee, in which case "Committee" means the SIP Committee when acting pursuant to such a grant or delegation; (ii) to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, "Committee" shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a "non-employee director" within the meaning of Rule 16b-3; and (iii) to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered "performance based" compensation under Section 162(m) of the Code, "Committee" shall be a committee or subcommittee of the Board composed of two or more members, each of whom is an "outside director" within the meaning of Code Section 162(m).

(f) "Compensation Committee" means the Compensation Committee of the Board.

(g) "Covered Enterprise" means a Competitive Enterprise and any other existing or planned business enterprise that: (i) offers, holds itself out as offering or reasonably may be expected to offer products or services that are the same as or similar to those offered by the Firm or that the Firm

reasonably expects to offer ("Firm Products or Services") or (ii) engages in, holds itself out as engaging in or reasonably may be expected to engage in any other activity that is the same as or similar to any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage ("Firm Activities"). For the avoidance of doubt, Firm Activities include any activity that requires the same or similar skills as any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage, irrespective of whether any such financial activity is in furtherance of an advisory, agency, proprietary or fiduciary undertaking.

The enterprises covered by this definition include enterprises that offer, hold themselves out as offering or reasonably may be expected to offer Firm Products or Services, or engage in, hold themselves out as engaging in or reasonably may be expected to engage in Firm Activities directly, as well as those that do so indirectly by ownership or control (e.g., by owning, being owned by or by being under common ownership with an enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services or that engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities). The definition of Covered Enterprise includes, solely by way of example, any enterprise that offers, holds itself out as offering or reasonably may be expected to offer any product or service, or engages in, holds itself out as engaging in or reasonably may be expected to engage in any activity, in any case, associated with investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than you or your family members (including, for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading. An enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services, or engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities is a Covered Enterprise, **irrespective of whether the enterprise is a customer, client or counterparty of the Firm or is otherwise associated with the Firm and, because the Firm is a global enterprise, irrespective of where the Covered Enterprise is physically located.**

(h) "Determination Date" means the date specified on your Award Statement as the date on which the Committee will determine whether or not, and to what extent, the Performance Goal[s] [was][were] achieved for the Performance Period.

(i) ["Dividend Equivalent Payments"] means any payments made in respect of Dividend Equivalent Rights.].

(j) "FDIC" means the Federal Deposit Insurance Corporation or any successor thereto.

(k) "Failed to Consider Risk" means that you participated (or otherwise oversaw or were responsible for, depending on the circumstances, another individual's participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where you have improperly analyzed such risk or where you have failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, the Committee determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, your business unit or the broader financial system. This definition is not limited to financial risks and is designed to encourage the consideration of the full range of risks associated with activities (e.g., legal, compliance or reputational). This definition also does not

require that a material adverse impact actually occur, but, rather, may be triggered if it is determined that there is a reasonable expectation of such an impact.

(l) "HCM" means the Firm's Human Capital Management Division or its successor.

(m) "Performance Goal[s]" means the performance goal[s] determined by the Committee that [is][are] specified on your Award Statement.

(n) "Performance Period" means the performance period determined by the Committee that is specified on your Award Statement.

(o) "Qualifying Termination After a Change in Control" means that the Firm terminates your Employment other than for Cause or you terminate your Employment for Good Reason, in each case, within 18 months following a Change in Control.

(p) ["Sarbanes-Oxley" means the Sarbanes-Oxley Act of 2002, as amended.]

(q) "SEC" means the U.S. Securities and Exchange Commission.

(r) "Selected Firm Personnel" means any individual who is or in the three months preceding the conduct prohibited by Paragraph 9(a)(ii) was (i) a Firm employee or consultant with whom you personally worked while employed by the Firm, (ii) a Firm employee or consultant who, at any time during the year preceding the date of the termination of your Employment, worked in the same division in which you worked or (iii) an Advisory Director, a Managing Director (Participating or Extended) or a Senior Advisor of the Firm.

(s) "Settlement Amount" means the amount deliverable to you in respect of your PSUs, which will be determined as described in the Award Statement.

(t) "Settlement Date" means each date specified on your Award Statement as the date on which the Settlement Amount will be delivered, *provided*, unless the Committee determines otherwise, such date is during a Window Period or, if such date is not during a Window Period, the first trading day of the first Window Period beginning after such date.

(u) "Share" means a share of Common Stock.

(v) "Shares at Risk" means Shares subject to Transfer Restrictions.

(w) "Signature Card" means the document and/or website presented to you with your Award that you are required to execute (including electronically) to indicate your acceptance of the Award, the Award Agreement (including the Award Statement and the terms contained in the Signature Card) and Additional Terms applicable to you, as well your receipt of the Award Documentation.

(x) "Transferability Date" means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a transferability date; *provided, however*, that if there is no trading day in a Window Period that occurs in the specified month and year, the Transferability Date will be the first trading day in the subsequent Window Period.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) "Account" means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) "Award Statement" means a written statement that reflects certain Award terms.

(c) "Board" means the Board of Directors of GS Inc.

(d) "Business Day" means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(e) "Cause" means (i) the Grantee's conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion, or (B) on a felony charge, or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations, (ii) the Grantee's engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the Exchange Act), (iii) the Grantee's willful failure to perform the Grantee's duties to the Firm, (iv) the Grantee's violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member, (v) the Grantee's violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or the Grantee's material violation of any other Firm policy as in effect from time to time, (vi) the Grantee's engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm or (vii) the Grantee's engaging in any conduct detrimental to the Firm. The determination as to whether Cause has occurred shall be made by the Committee in its sole discretion and, in such case, the Committee also may, but shall not be required to, specify the date such Cause occurred (including by determining that a prior termination of Employment was for Cause). Any rights the Firm may have hereunder and in any Award Agreement in respect of the events giving rise to Cause shall be in addition to the rights the Firm may have under any other agreement with a Grantee or at law or in equity.

(f) "Certificate" means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) "Change in Control" means the consummation of a merger, consolidation, statutory share exchange or similar form of corporate transaction involving GS Inc. (a "Reorganization") or sale or other disposition of all or substantially all of GS Inc.'s assets to an entity that is not an affiliate of GS Inc. (a "Sale"), that in each case requires the approval of GS Inc.'s shareholders under the law of GS Inc.'s jurisdiction of organization, whether for such Reorganization or Sale (or the issuance of securities of GS Inc. in such Reorganization or Sale), unless immediately following such Reorganization or Sale, either: (i) at least 50% of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of (A) the entity resulting from such Reorganization, or the entity which has acquired all or substantially all of the assets of GS Inc. in a Sale (in either case, the "Surviving Entity"), or (B) if applicable, the ultimate parent entity that directly or indirectly has beneficial

ownership (within the meaning of Rule 13d-3 under the Exchange Act, as such Rule is in effect on the date of the adoption of the 1999 SIP) of 50% or more of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of the Surviving Entity (the "Parent Entity") is represented by GS Inc.'s securities (the "GS Inc. Securities") that were outstanding immediately prior to such Reorganization or Sale (or, if applicable, is represented by shares into which such GS Inc. Securities were converted pursuant to such Reorganization or Sale) or (ii) at least 50% of the members of the board of directors (or similar officials in the case of an entity other than a corporation) of the Parent Entity (or, if there is no Parent Entity, the Surviving Entity) following the consummation of the Reorganization or Sale were, at the time of the Board's approval of the execution of the initial agreement providing for such Reorganization or Sale, individuals (the "Incumbent Directors") who either (A) were members of the Board on the Effective Date or (B) became directors subsequent to the Effective Date and whose election or nomination for election was approved by a vote of at least two-thirds of the Incumbent Directors then on the Board (either by a specific vote or by approval of GS Inc.'s proxy statement in which such persons are named as nominees for director).

(h) "Client" means any client or prospective client of the Firm to whom the Grantee provided services, or for whom the Grantee transacted business, or whose identity became known to the Grantee in connection with the Grantee's relationship with or employment by the Firm.

(i) "Code" means the Internal Revenue Code of 1986, as amended from time to time, and the applicable rulings and regulations thereunder.

(j) "Common Stock" means common stock of GS Inc., par value \$0.01 per share.

(k) "Competitive Enterprise" means an existing or planned business enterprise that (i) engages, or may reasonably be expected to engage, in any activity; (ii) owns or controls, or may reasonably be expected to own or control, a significant interest in any entity that engages in any activity or (iii) is, or may reasonably be expected to be, owned by, or a significant interest in which is, or may reasonably be expected to be, owned or controlled by, any entity that engages in any activity that, in any case, competes or will compete anywhere with any activity in which the Firm is engaged. The activities covered by this definition include, without limitation: financial services such as investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the Grantee and members of the Grantee's family (including for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading.

(l) "Covered Person" means a member of the Board or the Committee or any employee of the Firm.

(m) "Date of Grant" means the date specified in the Grantee's Award Agreement as the date of grant of the Award.

(n) ["Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.]

(o) "Effective Date" means the date this Plan is approved by the shareholders of GS Inc. pursuant to Section 3.15 of the Plan.

(p) "Employment" means the Grantee's performance of services for the Firm, as determined by the Committee. The terms "employ" and "employed" shall have their correlative meanings. The Committee in its sole discretion may determine (i) whether and when a Grantee's leave of absence results in a termination of Employment (for this purpose, unless the Committee determines otherwise, a Grantee shall be treated as terminating Employment with the Firm upon the occurrence of an Extended Absence), (ii) whether and when a change in a Grantee's association with the Firm results in a termination of Employment and (iii) the impact, if any, of any such leave of absence or change in association on Awards theretofore made. Unless expressly provided otherwise, any references in the Plan or any Award Agreement to a Grantee's Employment being terminated shall include both voluntary and involuntary terminations.

(q) "Exchange Act" means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(r) "Extended Absence" means the Grantee's inability to perform for six (6) continuous months, due to illness, injury or pregnancy-related complications, substantially all the essential duties of the Grantee's occupation, as determined by the Committee.

(s) "Firm" means GS Inc. and its subsidiaries and affiliates.

(t) "Good Reason" means, in connection with a termination of employment by a Grantee following a Change in Control, (i) as determined by the Committee, a materially adverse alteration in the Grantee's position or in the nature or status of the Grantee's responsibilities from those in effect immediately prior to the Change in Control or (ii) the Firm's requiring the Grantee's principal place of Employment to be located more than seventy-five (75) miles from the location where the Grantee is principally Employed at the time of the Change in Control (except for required travel on the Firm's business to an extent substantially consistent with the Grantee's customary business travel obligations in the ordinary course of business prior to the Change in Control).

(u) "Grantee" means a person who receives an Award.

(v) "GS Inc." means The Goldman Sachs Group, Inc., and any successor thereto.

(w) "1999 SIP" means The Goldman Sachs 1999 Stock Incentive Plan, as in effect prior to the effective date of the 2003 SIP.

(x) "Outstanding" means any Award to the extent it has not been forfeited, cancelled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(y) "Restricted Share" means a share of Common Stock delivered under the Plan that is subject to Transfer Restrictions, forfeiture provisions and/or other terms and conditions specified herein and in the Restricted Share Award Agreement or other applicable Award Agreement. All references to Restricted Shares include "Shares at Risk."

(z) “RSU” means a restricted stock unit granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(aa) “Section 409A” means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(ab) “SIP Administrator” means each person designated by the Committee as a “SIP Administrator” with the authority to perform day-to-day administrative functions for the Plan.

(ac) “SIP Committee” means the persons who have been delegated certain authority under the Plan by the Committee.

(ad) “Solicit” means any direct or indirect communication of any kind whatsoever, regardless of by whom initiated, inviting, advising, suggesting, encouraging or requesting any person or entity, in any manner, to take or refrain from taking any action. The terms “Solicited,” “Soliciting” and “Solicitation” will have their correlative meanings.

(ae) “Transfer Restrictions” means restrictions that prohibit the sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposal (including through the use of any cash-settled instrument), whether voluntarily or involuntarily by the Grantee, of an Award or any shares of Common Stock, cash or other property delivered in respect of an Award.

(af) “Vested” means, with respect to an Award, the portion of the Award that is not subject to a condition that the Grantee remain actively employed by the Firm in order for the Award to remain Outstanding. The fact that an Award becomes Vested shall not mean or otherwise indicate that the Grantee has an unconditional or nonforfeitable right to such Award, and such Award shall remain subject to such terms, conditions and forfeiture provisions as may be provided for in the Plan or in the Award Agreement.

(ag) “Window Period” means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (*provided* that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.
[One-Time][Year-End] Performance-Based RSU Award

This Award Agreement, together with The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the “Plan”), governs your ____ award of performance-based RSUs (your “Award” or “PSUs”). You should read carefully this entire document (including its [Appendix][Appendices]), the Award Statement and the Signature Card (together, the “Award Agreement”), as well as the other documentation presented to you in connection with acceptance of your Award (collectively, “Award Documentation”).

Acceptance

1. **You Must Decide Whether to Accept Your Award.** To be eligible for your Award, you must by the Acceptance Deadline: (a) open and activate an Account; (b) accept any Additional Terms presented to you as part of the Award Documentation; and (c) execute (including by electronic means) the accompanying Signature Card in accordance with its instructions. By executing the Signature Card, you confirm your agreement to all of the terms of the Award Agreement (including the Award Statement and the Signature Card), as well as any Additional Terms.

Documents that Govern Your Award; Definitions

2. **The Plan.** Your Award is granted under the Plan, and the Plan’s terms apply to, and are a part of, this Award Agreement. The Plan and its Prospectus are included with the Award Documentation.

3. **Your Award Statement.** The accompanying Award Statement contains the number of PSUs awarded to you and some of your Award’s specific terms. For example, it specifies the Performance Period and the Performance Goal[s] applicable to your Award. It also contains the Vesting Date[s], the Determination Date and the Settlement Date[s] for your Award and [the] [any applicable] Transferability Date[s] for any Shares at Risk that may be delivered to you in respect of any Settlement Amount that you may earn. The number of PSUs on your Award Statement is not necessarily the number of PSUs in respect of which the Settlement Amount will be earned, but is merely the basis for determining the amount (if any) that will be delivered to you.

4. **Definitions.** Unless otherwise defined herein, including in the Definitions Appendix or any other Appendix, capitalized terms have the meanings provided in the Plan.

Vesting of Your PSUs

5. **Vesting.** On each Vesting Date listed on your Award Statement, you will become Vested in the amount of Outstanding PSUs listed next to that date. When a PSU becomes Vested, it means **only** that your continued active Employment is not required to earn delivery in respect of that PSU. **Vesting does not mean you have a non-forfeitable right to the Vested portion of your Award. The terms of this Award Agreement (including conditions to delivery, satisfaction of the Performance Goal[s] and any applicable Transfer Restrictions) continue to apply to Vested PSUs, and you can still forfeit Vested PSUs and any Shares at Risk.**

Performance Goal[s]

6. **Performance.** The Settlement Amount is dependent, and may vary based, on achievement of the Performance Goal[s] over the Performance Period. On the Determination Date, the Firm will determine whether or not, and to what extent, the Performance Goal[s] for that Performance Period [has][have] been satisfied. All your rights with respect to the Settlement Amount [(and any Dividend Equivalent Payments)] are dependent on the extent to which the Performance Goal[s] [is][are] achieved, and any rights to delivery in respect of your Outstanding PSUs immediately will terminate and

no Settlement Amount will be delivered in respect of such PSUs upon the Committee's determination, in its sole discretion, that the Performance Goal[s] [has][have] not been satisfied to the extent necessary to result in delivery in respect of the PSUs.

Settlement Amount

7. Settlement.

(a) In General. Subject to satisfaction of the terms of your Award, including satisfaction of the Performance Goal[s], on [the] [each] Settlement Date [listed on your Award Statement], you will receive delivery (less applicable withholding as described in Paragraph 13(b)) of the [applicable portion of the] Settlement Amount [and payment of any Dividend Equivalent Payments] as further described in this Award Agreement and in your Award Statement. Until such delivery [and payment], you have only the rights of a general unsecured creditor, and no rights as a shareholder of GS Inc. with respect to either the PSUs or the Settlement Amount. Without limiting the Committee's authority under Section 1.3.2(h) of the Plan, the Firm may accelerate any Settlement Date by up to 30 days.

(b) Form of Delivery. The Settlement Amount will be delivered in the form of Shares (by book entry credit to your Account).

(c) Shares at Risk. ___ percent of the Shares delivered to you in respect of the Settlement Amount will be subject to Transfer Restrictions until the [applicable] Transferability Date. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition in violation of the Transfer Restrictions on Shares at Risk will be void.

(d) Release of Transfer Restrictions. Within 30 Business Days after the [applicable] Transferability Date identified on your Award Statement, GS Inc. will remove the Transfer Restrictions on any Shares at Risk [in respect of the amount of Outstanding PSUs listed next to that date]. The Committee may select multiple dates within the 30-Business-Day period on which to remove Transfer Restrictions for all or a portion of the Shares at Risk with the same Transferability Date, and all such dates will be treated as a single Transferability Date for purposes of this Award.

(e) [Voluntary Deferral of Settlement Date[s]]. Subject to any procedures and agreement terms adopted by the Committee to govern any such election, at least 12 months prior to the last day of the original Performance Period set forth on the Award Statement (or such other date as may be permitted under Section 409A), you may make an irrevocable election to defer the Settlement Date[s] (and the delivery of the Settlement Amount [and payment of any Dividend Equivalent Payments, each] determined as of the Determination Date) until the fifth anniversary of the originally scheduled Settlement Date[s] set forth on the Award Statement (or such other date as may be permitted by Section 409A). Any such election will be in accordance with the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b) or otherwise as may be permitted under Section 409A.]

Dividends

8. **[Dividend Equivalent Rights and] Dividends.** [To the extent described in your Award Statement, each PSU will include a Dividend Equivalent Right, which will be subject to the provisions of Section 2.8 of the Plan. Accordingly, for each of your Outstanding PSUs with respect to which delivery is made under the Settlement Amount, you will be entitled to payments under Dividend Equivalent Rights equal to any regular cash dividend paid by GS Inc. in respect of a Share for which the record date occurs on or after the Date of Grant. The payment to you of amounts under Dividend Equivalent Rights (less applicable withholding as described in Paragraph 13(b)) is conditioned upon the delivery under the Settlement Amount in respect of the PSUs to which such Dividend Equivalent Rights relate, and you will have no right to receive any Dividend Equivalent Payments relating to PSUs for which you do not receive delivery under the Settlement Amount (including, without limitation, due to a failure to satisfy the Performance Goal[s]). Dividend Equivalent Payments will be paid on the [applicable] Settlement Date.] [In addition, you][You] will be entitled to receive on a current basis any regular cash dividend paid in respect of any Shares at Risk. [The PSUs do not include Dividend Equivalent Rights.]

Forfeiture of Your Award

9. **How You May Forfeit Your Award.** This Paragraph 9 sets forth the events that result in forfeiture of up to all of your PSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. More than one event may apply, and in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event. Paragraph 11 (relating to certain circumstances under which you will not forfeit your unvested PSUs upon Employment termination) and Paragraph 12 (relating to certain circumstances under which vesting and/or release of Transfer Restrictions may be accelerated) provide for exceptions to one or more provisions of this Paragraph 9.

(a) **Unvested PSUs Forfeited if Your Employment Terminates.** If your Employment terminates for any reason or you are otherwise no longer actively Employed with the Firm (which includes off-premises notice periods, "garden leaves," pay in lieu of notice or any other similar status), your rights to your Outstanding PSUs that are not Vested will terminate, and no Settlement Amount will be delivered in respect of such PSUs.

(b) **Vested and Unvested PSUs Forfeited Upon Certain Events.** If any of the following occurs, your rights to all of your Outstanding PSUs (whether or not Vested) will terminate, and no Settlement Amount will be delivered in respect of such PSUs, as may be further described below:

(i) **You Associate With a Covered Enterprise.** Except as prohibited by applicable law, you Associate With a Covered Enterprise during the Performance Period.

(ii) **You Solicit Clients or Employees, Interfere with Client or Employee Relationships or Participate in the Hiring of Employees.** Before the [applicable] Settlement Date and except as prohibited by applicable law, either:

A. (A) you, in any manner, directly or indirectly, (1) Solicit any Client to transact business with a Covered Enterprise or to reduce or refrain from doing any business with the Firm, (2) interfere with or damage (or attempt to interfere with or damage) any relationship between the Firm and any Client, (3) Solicit any person who is an employee of the Firm to resign from the Firm, (4) Solicit any Selected Firm Personnel to apply for or accept employment (or other association) with any person or entity other than the Firm or (5) participate in the hiring of any Selected Firm Personnel by any person or entity other than the Firm (including, without limitation, participating in the identification of individuals for potential hire, and participating in any hiring decision), whether as an employee or consultant or otherwise, or

B. (B) Selected Firm Personnel are Solicited, hired or accepted into partnership, membership or similar status by any entity where you have, or will have, direct or indirect managerial responsibility for such Selected Firm Personnel, unless the Committee determines that you were not involved in such Solicitation, hiring or acceptance.

(iii) **[GS Inc. Fails to Maintain the Minimum Tier 1 Capital Ratio.** Before the [applicable] Settlement Date, GS Inc. fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days.]

(iv) **[GS Inc. Is Determined to Be in Default.** Before the [applicable] Settlement Date, the Board of Governors of the Federal Reserve or the FDIC makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Wall Street Reform and Consumer Protection Act for the appointment of the FDIC as a receiver of GS Inc. based on a determination that GS Inc. is "in default" or "in danger of default."]

(c) PSUs and Shares at Risk Forfeited upon Certain Events. If any of the following occurs (i) your rights to all of your Outstanding PSUs (whether or not Vested) will terminate, and no Settlement Amount will be delivered in respect of such PSUs and (ii) your rights to any Shares at Risk will terminate and all such Shares at Risk will be cancelled, in each case, as may be further described below:

(i) You Failed to Consider Risk. You Failed to Consider Risk during ____.

(ii) Your Conduct Constitutes Cause. Any event that constitutes Cause has occurred before the [applicable] Settlement Date for PSUs or the [applicable] Transferability Date for any Shares at Risk.

(iii) You Breach an Obligation to the Firm. The Committee determines that, before the [applicable] Settlement Date for PSUs or the [applicable] Transferability Date for any Shares at Risk, you Breached an Obligation to the Firm.

(iv) You Do Not Provide Timely Certifications or Comply with Your Certifications. You fail to certify to GS Inc. that you have complied with all of the terms of the Plan and this Award Agreement as required by Paragraph 13(d), or the Committee determines that you have failed to comply with a term of the Plan or this Award Agreement to which you have certified compliance.

(v) You Do Not Follow Dispute Resolution Procedures. You attempt to have any dispute, controversy or claim under the Plan or this Award Agreement resolved in any manner that is not provided for by Paragraph 16 or Section 3.17 of the Plan, or you attempt to arbitrate a dispute, controversy or claim without first having exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(vi) You Bring an Action that Results in a Determination that Any Award Agreement Term Is Invalid. As a result of any action brought by you, it is determined that any term of this Award Agreement is invalid.

(vii) You Receive Compensation in Respect of Your Award from Another Employer. Your Employment terminates for any reason or you otherwise are no longer actively Employed with the Firm and another entity grants you cash, equity or other property (whether vested or unvested) to replace, substitute for or otherwise in respect of any Outstanding PSUs or Shares at Risk; *provided, however*, that your rights will only be terminated in respect of the PSUs or Shares at Risk that are replaced, substituted for or otherwise considered by such other entity in making its grant.

(viii) [Accounting Restatement Required Under Sarbanes-Oxley]. GS Inc. is required to prepare an accounting restatement due to GS Inc.'s material noncompliance, as a result of misconduct, with any financial reporting requirement under the securities laws as described in Section 304(a) of Sarbanes-Oxley; *provided, however*, that your rights will only be terminated in respect of PSUs or Shares at Risk to the same extent that would be required under Section 304(a) of Sarbanes-Oxley had you been a "chief executive officer" or "chief financial officer" of GS Inc. (regardless of whether you actually hold such position at the relevant time).]

(ix) Accounting Restatement Required and Dodd-Frank Clawback Policy Applies. GS Inc. is required to prepare an "Accounting Restatement" as described in GS Inc.'s Policy for Recovery of Erroneously Awarded Compensation (the "Dodd-Frank Clawback Policy"); in each case if, and to the same extent as, required under the Dodd-Frank Clawback Policy. For the avoidance of doubt, PSUs previously granted to you and Shares at Risk in respect of PSUs previously granted to you are also subject to forfeiture and clawback under this Paragraph 9(c)[(viii)/(ix)].

(d) [Additional Forfeiture and Repayment Conditions for Material Risk Takers]. Since the Firm identified you as a Material Risk Taker ("MRT") in ____, the **EU and UK Material Risk Taker Appendix** and/or the **GSBE Material Risk Taker Appendix** (as applicable to you, the "MRT Appendix") supplements this Paragraph 9. The MRT Appendix sets forth additional events that (i) result in forfeiture of up to all of your PSUs and Shares at Risk and (ii) may require repayment to the Firm of up to all amounts previously delivered or paid to you under your Award in accordance with Paragraph 10.

Your MRT classification, as defined in the MRT Appendix, was communicated to you separately. If you are unsure of which classification(s) apply, you must contact HCM prior to accepting this Award. The Firm's records with respect to such classification(s) will be controlling as to whether the MRT Appendix applies to your Award. The MRT classifications with their corresponding MRT Appendix are as follows:

MRT Classification	MRT Appendix
EU MRT, GSAMI MRT, GSAMI Senior Management and/or UK MRT	EU and UK Material Risk Taker Appendix
GSBE MRT and GSBE Senior Management	GSBE Material Risk Taker Appendix
GSBE MRT or GSBE Senior Management <i>and</i> any combination of EU MRT, GSAMI MRT, GSAMI Senior Management or UK MRT	GSBE Material Risk Taker Appendix <i>and</i> EU and UK Material Risk Taker Appendix]

(e) Actions Pending Forfeiture. In connection with any investigation as to whether any of the events that would result in forfeiture under the Plan[,] [or] this Paragraph 9 [or the MRT Appendix] have occurred, the Firm reserves the right to (i) suspend vesting of Outstanding PSUs[, any Dividend Equivalent Payments], delivery of the Settlement Amount or release of any Transfer Restrictions; (ii) deliver the Settlement Amount[,] [any Dividend Equivalent Payments] [or] dividends into an escrow account in accordance with Paragraph 13(e)(iii); or (iii) apply Transfer Restrictions to any Shares.

Repayment of Your Award

10. When You May Be Required to Repay Your Award.

(a) Repayment Generally. If the Committee determines that any term of this Award was not satisfied, you will be required, immediately upon demand therefor, to repay to the Firm the following:

(i) Any Settlement Amount (including any Shares at Risk) for which the terms (including the terms for delivery) of the related PSUs were not satisfied, in accordance with Section 2.6.3 of the Plan.

(ii) Any Shares at Risk for which the terms (including the terms for the release of Transfer Restrictions) were not satisfied, in accordance with Section 2.5.3 of the Plan.

(iii) [Any Shares that were delivered (but not subject to Transfer Restrictions) at the same time any Shares at Risk that are cancelled or required to be repaid were delivered.]

(iv) [Any Dividend Equivalent Payments for which the terms were not satisfied (including any such payments made in respect of PSUs that are forfeited or any Settlement Amount that is required to be repaid), in accordance with Section 2.8.3 of the Plan.]

(v) Any dividends paid in respect of any [delivered Shares (including] Shares at Risk[]]) that are cancelled or required to be repaid.

(vi) Any amount applied to satisfy tax withholding or other obligations with respect to any PSUs, Settlement Amount (including Shares at Risk)[,] [and] dividend payments [and Dividend Equivalent Payments] that are forfeited or required to be repaid.

(b) Repayment Upon Materially Inaccurate Financial Statements. If any delivery is made under this Award Agreement based on materially inaccurate financial statements (which includes, but is not limited to, statements of earnings, revenues or gains) or other materially inaccurate performance criteria, you will be obligated to repay to the Firm, immediately upon demand therefor, any excess amount delivered, as determined by the Committee in its sole discretion. For the avoidance of doubt, this Paragraph 10(b) includes any repayment required following an "Accounting Restatement" under the Dodd-Frank Clawback Policy.

(c) [Repayment Upon Accounting Restatement Required Under Sarbanes-Oxley]. If an event described in Paragraph 9(c)(viii) (relating to a requirement under Sarbanes-Oxley that GS Inc. prepare an accounting restatement) occurs, any Settlement Amount (including Shares at Risk), dividend payments, [Dividend Equivalent Payments,] cash or other property delivered, paid or withheld in respect of this Award will be subject to repayment as described in Paragraph 10(a) to the same extent that would be required under Section 304(a) of Sarbanes-Oxley had you been a “chief executive officer” or “chief financial officer” of GS Inc. (regardless of whether you actually hold such position at the relevant time).]

Terminations of Employment

11. Treatment of Vested PSUs.

(a) Generally. Unless the Committee determines otherwise, if your Employment terminates for any reason or you are otherwise no longer actively Employed with the Firm (which includes off-premises notice periods, “garden leaves,” pay in lieu of notice or any other similar status), the Performance Goal[s] applicable to your Outstanding PSUs will continue to apply with respect to your Vested PSUs (and PSUs that are treated as Vested) and the determination of the Settlement Amount will continue to be subject to whether, and to what extent, the Performance Goal[s] [has][have] been achieved with respect to such PSUs, in each case, as provided in Paragraph 6. All other terms of this Award Agreement, including the forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply.

(b) [Retirement or] Extended Absence. If your Employment terminates by [Retirement or] Extended Absence, then Paragraph 9(a) will not apply, and your Outstanding PSUs that are not Vested will be treated as Vested and, on that basis, be subject to Paragraph 11(a). However, your rights to any Outstanding PSUs that are treated as Vested by this Paragraph 11(b) will terminate and no Settlement Amount will be delivered in respect of such PSUs if you Associate With a Covered Enterprise during the Performance Period, as described in Paragraph 9(b)(i). You acknowledge and agree that if any portion of the preceding sentence is unenforceable under applicable law, or is found to be unenforceable, then the entirety of this Paragraph 11(b) is inapplicable to you and your rights to any Outstanding PSUs that were treated as Vested, or otherwise would have been treated as Vested, will be cancelled in accordance with Paragraph 9(a).

(c) [Qualified Termination]. If the Firm terminates your Employment solely by reason of a Qualified Termination and you execute a general waiver and release of claims and an agreement to pay any associated tax liability in the form the Firm prescribes, then Paragraph 9(a) will not apply, and a pro rata portion of your Outstanding PSUs (based on the portion of the Performance Period during which you were Employed) will become Vested and, on that basis, be subject to Paragraph 11(a). For the avoidance of doubt, your rights to any Outstanding PSUs will be terminated and no Settlement Amount will be delivered in respect thereof if you Associate With a Covered Enterprise during the Performance Period, as described in Paragraph 9(b)(i).]

12. Accelerated Vesting and/or Release of Transfer Restrictions in the Event of [a Change in Control,] [a Qualifying Termination After a Change in Control] [or] Death. In the event of [a Change in Control][,] your [Qualifying Termination After a Change in Control] [or] death, [each] as described below, your Outstanding PSUs will become Vested and Paragraph 9(a) will not apply; any delivery of your Settlement Amount and Shares at Risk will be treated as described in this Paragraph 12, and[, except as set forth in Paragraph 12(a),] all other terms of this Award Agreement, including the other forfeiture and repayment events in Paragraphs 9 and 10 [and the MRT Appendix], continue to apply. [In each case, the Performance Goal[s] applicable to your Outstanding PSUs will continue to apply and the determination of the Settlement Amount will continue to be subject to whether, and to what extent, the Performance Goal[s] [has][have] been achieved, in each case, as provided in Paragraph 6.]

(a) [Change in Control].

(i) In the event of a Change in Control which would result in none of GS Inc., its successors (including any surviving or acquiring entity or its affiliates) or affiliates being listed or publicly traded on any national securities exchange (“Delisting Change in Control”): (A) the last day of

the Performance Period will be deemed to be the effective date of the Change in Control, (B) the Determination Date will be as soon as practicable after the effective date of the Change in Control and (C) the Settlement Date[s] will occur as soon as practicable following the Determination Date, but no later than March 15 coinciding with the last day of the applicable "short-term deferral" period described in Reg. 1.409A-1(b)(4). Any Transfer Restrictions will cease to apply to your Shares at Risk, and the forfeiture events in Paragraph 9 will not apply to your Shares at Risk.

(ii) If the Change in Control is not a Delisting Change in Control described in Paragraph 12(a)(i) and if your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, then (i) Paragraph 9(a) will not apply and your PSUs that are not Vested will become Vested, and (ii) you will receive delivery of the Settlement Amount [and payment of the Dividend Equivalent Payments] on the [applicable] Settlement Date that would have otherwise been made pursuant to Paragraph 6, and (iii) any Transfer Restrictions will cease to apply to your Shares at Risk, and the forfeiture events in Paragraph 9 will not apply to your Shares at Risk.]

(b) You Have a Qualifying Termination After a Change in Control. If your Employment terminates when you meet the requirements of a Qualifying Termination After a Change in Control, (i) you will, on the [applicable] Settlement Date, receive delivery of the Settlement Amount [and payment of the Dividend Equivalent Payments] that[, in each case,] would have otherwise been made pursuant to Paragraph 6, and (ii) any Transfer Restrictions will cease to apply to your Shares at Risk. In addition, the forfeiture events in Paragraph 9 [(except the MRT Appendix)] will not apply to your Shares at Risk.]

(c) Death. If you die, [as soon as practicable] after receipt of any documentation as may be requested by the Committee or a SIP Administrator, (i) your Account will receive delivery of the Settlement Amount [and payment of the Dividend Equivalent Payments] on the [applicable] Settlement Date that[, in each case,] would have otherwise been made pursuant to Paragraph 6, and (ii) any Transfer Restrictions will cease to apply to your Shares at Risk.]

Other Terms And Conditions of Your Award

13. Additional Terms, Conditions and Consents.

(a) You Will Be Required to Accept Additional Terms. In addition to the terms of the Award Agreement, the Award Documentation includes other contractual terms that you must accept as a condition of the Award (the "Additional Terms"). By accepting the Award and executing the Signature Card, you agree to any such Additional Terms applicable to you as set forth in the Award Documentation, which will include but are not limited to the following:

(i) Notice Period Policy. You must agree to abide by the terms, as applicable to you, of the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy").

(ii) Agreement Regarding Arbitration of Employment-Related Matters. You must agree to the Agreement Regarding Arbitration of Employment-Related Matters, which imposes obligations in addition to those contained in Paragraph 16.

(iii) APAC Restrictive Covenants. If you are employed by the Firm in the Asia-Pacific Region (e.g., Australia, the People's Republic of China, the Hong Kong Special Administrative Region, India, Indonesia, Japan, the Republic of Korea, New Zealand, Singapore, Taiwan), you must agree to the APAC Restrictive Covenants Agreement.

(iv) Role-, Business- or Division-Specific Agreements. If applicable to you, you must accept Additional Terms related to your specific role, business unit or division as set forth in the Award Documentation.

(b) You Must Satisfy Applicable Tax Withholding Requirements. Delivery of the Settlement Amount is conditioned on your satisfaction of any applicable withholding taxes in accordance with

Section 3.2 of the Plan, which includes the Firm deducting or withholding amounts from any payment or distribution to you. In addition, to the extent permitted by applicable law, the Firm, in its sole discretion, may require you to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations imposed on you or the Firm in connection with the grant, Vesting or delivery of this Award by requiring you to choose between remitting the amount (i) in cash (or through payroll deduction or otherwise) or (ii) in the form of proceeds from the Firm's executing a sale of Shares delivered to you under this Award. In no event, however, does this Paragraph 13(b) give you any discretion to determine or affect the timing of the delivery of the Settlement Amount or the timing of payment of tax obligations.

(c) You Must Comply with Applicable Deadlines and Procedures to Appeal. If you disagree with a determination made by the Committee, the SIP Administrators or any of their delegates or designees relating to the Plan or the Award Agreement and you wish to appeal such determination, you must submit a written request to the Committee for review within 180 days after the determination at issue. You must exhaust your internal administrative remedies (*i.e.*, submit your appeal and wait for resolution of that appeal) before seeking to resolve a dispute, controversy or claim through arbitration pursuant to Paragraph 16 and Section 3.17 of the Plan.

(d) You May Be Required to Certify Compliance with Award Terms; You Are Responsible for Providing Updated Address and Contact Information After Your Departure from the Firm. If your Employment terminates while you continue to hold PSUs or Shares at Risk, the Firm may require you to provide certifications of your compliance with all of the terms of the Plan and this Award Agreement. You understand and agree that (i) your contact information as reflected in the Firm's personnel records at the time any certification is requested will be deemed current; (ii) it is your responsibility to inform HCM of any changes to your contact information to ensure timely receipt of the certification materials, regardless of whether your contact information is provided to another part of the Firm; and (iii) you are responsible for contacting the Firm to obtain such certification materials if not received. Your failure to return properly completed certification materials by the specified deadline (including because you did not provide HCM with updated contact information) will result in the forfeiture of all of your PSUs and any Shares at Risk and subject previously delivered amounts to repayment under Paragraph 10.

(e) You Agree to Other Terms, Conditions and Consents. By accepting this Award you understand and agree that:

(i) Amounts May Be Rounded to Avoid Fractional Shares. PSUs that become Vested and any amounts delivered in respect of the Settlement Amount, including Shares at Risk, may be rounded to avoid fractional Shares.

(ii) Firm May Deliver Cash or Other Property in Respect of the Settlement Amount. In accordance with Section 1.3.2(i) of the Plan, in the sole discretion of the Committee, in lieu of all or any portion of the Settlement Amount, the Firm may deliver cash, other securities, other awards under the Plan or other property, and all references in this Award Agreement to delivery of the Settlement Amount will include such deliveries of cash, other securities, other awards under the Plan or other property.

(iii) Firm May Deliver Your Award into an Escrow Account. The Firm may establish and maintain an escrow account on such terms (which may include your executing any documents related to, and your paying for any costs associated with, such account) as it may deem necessary or appropriate, and the Settlement Amount may initially be delivered, and any [Dividend Equivalent Payments and] dividends may initially be paid, into and held in that escrow account until such time as the Committee has received such documentation as it may have requested or until the Committee has determined that any other conditions or restrictions on deliveries required by this Award Agreement have been satisfied.

(iv) Firm May Affix Legends and Place Stop Orders on Shares. GS Inc. may affix to Certificates representing Shares any legend that the Committee determines to be necessary or advisable (including to reflect any restrictions to which you may be subject under the Additional Terms, a separate agreement or any Firm policy applicable to you). GS Inc. may advise the transfer agent to place a stop order against any legended Shares. For the avoidance of doubt, Shares as used herein includes, without limitation, Restricted Shares.

(v) You Authorize the Firm to Register, in Its or Its Designee's Name, Any Restricted Shares and Sell, Assign or Transfer Any Forfeited Restricted Shares. You are granting to the Firm the full power and authority to register any Restricted Shares in its or its designee's name and authorizing the Firm or its designee to sell, assign or transfer any Restricted Shares if you forfeit your Restricted Shares.

(vi) You Agree to Certain Consents as a Condition to the Award. You have expressly consented to all of the items listed in Section 3.3.3(d) of the Plan, including the Firm's supplying to any third-party recordkeeper of the Plan or other person such personal information of yours as the Committee deems advisable to administer the Plan, and you agree to provide any additional consents that the Committee determines to be necessary or advisable.

(vii) You Are Subject to the Firm's Policies, Rules and Procedures. You are subject to the Firm's policies in effect from time to time concerning trading in Shares and hedging or pledging Shares and equity-based compensation or other awards (including, without limitation, the "Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards" or any successor policies), and confidential or proprietary information, and you will effect sales of Shares in accordance with such rules and procedures as may be adopted from time to time (which may include, without limitation, restrictions relating to the timing of sale requests, the manner in which sales are executed, pricing method, consolidation or aggregation of orders and volume limits determined by the Firm).

(viii) You Are Responsible for Costs Associated with Your Award. You will be responsible for all brokerage costs and other fees or expenses associated with your Award, including those related to the sale of Shares.

(ix) You Will Be Deemed to Represent Your Compliance with All the Terms of Your Award if You Accept Delivery of, or Sell, Shares. You will be deemed to have represented and certified that you have complied with all of the terms of the Plan and this Award Agreement when any Settlement Amount [and Dividend Equivalent Payments] [is][are] delivered to you and you request the sale of Shares following the release of Transfer Restrictions on any Shares at Risk.

(x) You Agree that Covered Persons Will Not Have Liability. In addition to and without limiting the generality of the provisions of Section 1.3.5 of the Plan, neither the Firm nor any Covered Person will have any liability to you or any other person for any action taken or omitted in respect of this or any other Award.

14. **Non-transferability.** Except as otherwise may be provided by the Committee, the limitations set forth in Section 3.5 of the Plan will apply to this Award. Any purported sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposition (including through the use of any cash-settled instrument), whether voluntary or involuntary, in violation of the provisions of this Paragraph 14 or Section 3.5 of the Plan will be void.

15. **Right of Offset.** Except as provided in Paragraph 18(d), the obligation to deliver the Settlement Amount, pay dividends [or Dividend Equivalent Payments] or release any Transfer Restrictions under this Award Agreement is subject to Section 3.4 of the Plan, which provides for the Firm's right to offset against such obligation any outstanding amounts you owe to the Firm and any amounts the Committee deems appropriate pursuant to any tax equalization policy or agreement.

Arbitration, Choice of Forum and Governing Law

16. **Arbitration; Choice of Forum.**

(a) **By accepting this award, you are indicating that you understand and agree that the arbitration and choice of forum provisions set forth in Section 3.17 of the Plan will apply to this Award to the fullest extent permitted by applicable law. These provisions, which are expressly incorporated herein by reference, provide among other things that any dispute, controversy or**

claim between the Firm and you arising out of or relating to or concerning the Plan or this Award Agreement will be finally settled by arbitration, pursuant to the terms more fully set forth in Section 3.17 of the Plan; provided that nothing herein shall preclude you from filing a charge with or participating in any investigation or proceeding conducted by any governmental authority, including but not limited to the SEC, the Equal Employment Opportunity Commission and a state or local human rights agency, as well as law enforcement.

(b) Nothing in the Plan or this Award Agreement includes an agreement to arbitrate claims on a collective, class or representative basis. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider class, collective or representative claims, to order consolidation or to join different claimants or grant relief other than on an individual basis to the individual claimant involved.

(c) Notwithstanding any applicable forum rules to the contrary, to the extent there is a question of enforceability of the Plan or this Award Agreement arising from a challenge to the arbitrator's jurisdiction or to the arbitrability of a claim, it will be decided by a court and not an arbitrator.

(d) Prior to filing any arbitration in accordance with Paragraph 16(a), you must first exhaust your internal administrative remedies in accordance with Paragraph 13(c), which includes awaiting the Committee's final determination of any such appeal. To the fullest extent permitted by applicable law, no arbitrator will have the authority to consider any dispute, controversy or claim as to which you have not first exhausted your internal administrative remedies in accordance with Paragraph 13(c).

(e) The Federal Arbitration Act governs interpretation and enforcement of all arbitration provisions under the Plan and this Award Agreement, and all arbitration proceedings thereunder.

(f) Nothing in the Plan or this Award Agreement creates a substantive right to bring a claim under U.S. Federal, state, or local employment laws.

(g) By accepting your Award, you irrevocably appoint each General Counsel of GS Inc., or any person whom the General Counsel of GS Inc. designates, as your agent for service of process in connection with any suit, action or proceeding arising out of or relating to or concerning the Plan or any Award which is not arbitrated pursuant to the provisions of Section 3.17.1 of the Plan, who shall promptly advise you of any such service of process.

(h) **As a condition of your Award, you also must accept the terms of the Agreement Regarding Arbitration of Employment-Related Matters, which is included among the Additional Terms described in Paragraph 13(a) and sets forth obligations independent of this Paragraph 16. For the avoidance of doubt, the obligation to exhaust your internal remedies described in Paragraph 13(c) and Paragraph 16(d) only applies to disputes, controversies or claims relating to the Plan or the Award Agreement.**

17. Governing Law. This Award will be governed by and construed in accordance with the laws of the State of New York, without regard to principles of conflict of laws.

Certain Tax Provisions

18. Compliance of Award Agreement and Plan with Section 409A. The provisions of this Paragraph 18 apply to you only if you are a U.S. taxpayer.

(a) This Award Agreement and the Plan provisions that apply to this Award are intended and will be construed to comply with Section 409A (including the requirements applicable to, or the conditions for exemption from treatment as, 409A Deferred Compensation), whether by reason of short-term deferral treatment or other exceptions or provisions. The Committee will have full authority to give effect to this intent. To the extent necessary to give effect to this intent, in the case of any conflict or potential inconsistency between the provisions of the Plan (including Sections 1.3.2 and 2.1 thereof) and this Award Agreement, the provisions of this Award Agreement will govern, and in the case of any

conflict or potential inconsistency between this Paragraph 18 and the other provisions of this Award Agreement, this Paragraph 18 will govern.

(b) Settlement will not be delayed beyond the date on which all applicable conditions or restrictions on settlement in respect of your PSUs required by this Award Agreement (including [those specified in Paragraph 11 (execution of waiver and release of claims agreement to pay associated tax liability) and] the consents and other items specified in Section 3.3 of the Plan) are satisfied. To the extent that any portion of this Award is intended to satisfy the requirements for short-term deferral treatment under Section 409A, settlement in respect of such portion will occur by the March 15 coinciding with the last day of the applicable "short-term deferral" period described in Reg. 1.409A-1(b)(4) in order for settlement to be within the short-term deferral exception unless, in order to permit all applicable conditions or restrictions on settlement to be satisfied, the Committee elects, pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted in accordance with Section 409A, to delay settlement to a later date within the same calendar year or to such later date as may be permitted under Section 409A, including Reg. 1.409A-3(d). For the avoidance of doubt, if the Award includes a "series of installment payments" as described in Reg. 1.409A-2(b)(2)(iii), your right to the series of installment payments will be treated as a right to a series of separate payments and not as a right to a single payment.

(c) Notwithstanding the provisions of Paragraph 13(e) and Section 1.3.2(i) of the Plan, to the extent necessary to comply with Section 409A, any delivery or payment [(including in the form of Shares at Risk or other property)] that the Firm may make in respect of your PSUs will not have the effect of deferring delivery or payment, income inclusion, or a substantial risk of forfeiture, beyond the date on which such delivery, payment or inclusion would occur or such risk of forfeiture would lapse, with respect to the payment or delivery that would otherwise have been made (unless the Committee elects a later date for this purpose pursuant to Reg. 1.409A-1(b)(4)(i)(D) or otherwise as may be permitted under Section 409A, including and to the extent applicable, the subsequent election provisions of Section 409A(a)(4)(C) of the Code and Reg. 1.409A-2(b)).

(d) Paragraph 15 and Section 3.4 of the Plan will not apply to Awards that are 409A Deferred Compensation except to the extent permitted under Section 409A.

(e) Settlement in respect of any portion of the Award may be made, if and to the extent elected by the Committee, later than the [applicable] Settlement Date or other date or period specified hereinabove (but, in the case of any Award that constitutes 409A Deferred Compensation, only to the extent that the later payment or delivery, as applicable, is permitted under Section 409A).

(f) You understand and agree that you are solely responsible for the payment of any taxes and penalties due pursuant to Section 409A, but in no event will you be permitted to designate, directly or indirectly, the taxable year of the delivery.

Committee Authority, Amendment, Construction and Regulatory Reporting

19. **Committee Authority.** The Committee has the authority to determine, in its sole discretion, that any event triggering forfeiture or repayment of your Award will not apply, to limit the forfeitures and repayments that result under Paragraphs 9 and 10 and to remove any Transfer Restrictions before the [applicable] Transferability Date. [In addition, the Committee, in its sole discretion, may determine whether Paragraph 11(b)[or (c)] will apply upon a termination of Employment.]

20. **Amendment.** The Committee reserves the right at any time to amend the terms of this Award Agreement, and the Board may amend the Plan in any respect; *provided that*, notwithstanding the foregoing and Sections 1.3.2(f), 1.3.2(h) and 3.1 of the Plan, no such amendment will materially adversely affect your rights and obligations under this Award Agreement without your consent; and *provided further* that the Committee expressly reserves its rights to amend the Award Agreement and the Plan as described in Sections 1.3.2(h)(1), (2) and (4) of the Plan. A modification that impacts the tax consequences of this Award or the timing of delivery of the Settlement Amount will not be an amendment that materially adversely affects your rights and obligations under this Award Agreement. Any amendment of this Award Agreement will be in writing.

21. **Construction, Headings.** Unless the context requires otherwise, (a) words describing the singular number include the plural and vice versa, (b) words denoting any gender include all genders and (c) the words “include,” “includes” and “including” will be deemed to be followed by the words “without limitation.” The headings in this Award Agreement are for the purpose of convenience only and are not intended to define or limit the construction of the provisions hereof. References in this Award Agreement to any specific Plan provision will not be construed as limiting the applicability of any other Plan provision.

22. **Providing Information to the Appropriate Authorities; Protected Communications.** In accordance with applicable law, nothing in this Award Agreement (including the forfeiture and repayment provisions in Paragraphs 9 and 10) or the Plan prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the SEC, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in this Award Agreement or the Plan prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in this Award Agreement or the Plan limits your ability to use the internal and external reporting channels that are available to you, as described in the Firmwide Policy on Escalation.

IN WITNESS WHEREOF, GS Inc. has caused this Award Agreement to be duly executed and delivered as of the Date of Grant.

THE GOLDMAN SACHS GROUP, INC.

[EU and UK Material Risk Taker Appendix

Qualitative Overlay Reduction

(g) In addition to and without limiting the Firm's rights under the forfeiture and repayment provisions set forth in Paragraphs 9 and 10 or in the "Forfeiture and Repayment" section below, the Committee may determine to reduce the Settlement Amount as described in the section entitled "Qualitative Overlay Reduction" in the Award Statement.

Forfeiture and Repayment

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your PSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply, in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider certain factors to determine whether and what portion of your Award will terminate, including the reason for the Loss Event or Risk Event, as well as the extent to which you participated in the Loss Event or Risk Event and your compensation for the Firm's ____ fiscal year may or may not have been adjusted to take into account the risk associated with the Loss Event, Risk Event, your Serious Misconduct or the Serious Misconduct of a Supervised Employee. Your compensation may be adjusted for the year in which the Loss Event, Risk Event, your Serious Misconduct or a Supervised Employee's Serious Misconduct is discovered.

(a) A Loss Event Occurs Prior to Delivery. If a Loss Event occurs prior to the delivery of any portion of the Settlement Amount, your rights in respect of all or a portion of your PSUs which are scheduled to deliver on the next Settlement Date immediately following the date that the Loss Event is identified (or, if not practicable, then the next following Settlement Date) will terminate, and no Shares will be delivered in respect of such PSUs.

(i) **"Loss Event"** means (A) an annual pre-tax loss at GS Inc. or (B) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm), or annual negative revenues in the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm) of \$5 billion or more, *provided* in either case that you are employed in a business within such reporting segment.

(b) A Risk Event Occurs On or Before the Applicable End Date. If a Risk Event occurs on or before the Applicable End Date, (i) your rights in respect of all or a portion of your PSUs will terminate and no Settlement Amount will be delivered in respect of such PSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Risk Event occurred and (B) the date that the repayment request is made.

(i) **"Risk Event"** means there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting and Operational Risk Events Policy (or any successor policy).

(c) You Engage in Serious Misconduct On or Before the Applicable Date. If you engage in Serious Misconduct during the period beginning on the applicable Transferability Date through the Applicable End Date, you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (i) the date the Serious Misconduct occurred and (ii) the date that the repayment request is made.

(i) **“Serious Misconduct”** means that you engage in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under applicable employment law.

(d) A Supervised Employee Engages in Serious Misconduct. If the Committee determines that it is appropriate to hold you accountable in whole or in part for Serious Misconduct related to compliance, control or risk that occurred during the Firm's ____ fiscal year by a Supervised Employee, your rights in respect of all or a portion of your PSUs will terminate and no Settlement Amount will be delivered in respect of such PSUs and your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled.

(i) **“Supervised Employee”** means an individual with respect to whom the Committee determines you had supervisory responsibility as a result of direct or indirect reporting lines or your management responsibility for an office, division or business.

(e) **Certain Definitions.**

(i) “**Applicable End Date**” means [_____] [, with respect to each classification in the table below, the following:

Classification	Applicable End Date
EU MRT	
GSAMI MRT	
GSAMI Senior Management	
UK MRT	

(ii) **“EU MRT”** means an individual who, at any point in _____, was classified by the Firm as a (A) Material Risk Taker in accordance with the applicable remuneration rules of the Alternative Investment Fund Managers Directive or Undertakings for Collective Investment in Transferable Securities V (“AIFMD-UCITS”), including “Identified Staff” under AIFMD-UCITS (“AIFMD-UCITS Identified Staff”), as applicable to the business operations of Goldman Sachs Asset Management, B.V. (“GSAMBV”), Goldman Sachs Asset Management Belgium, S.A. (“GSAM Belgium”) or Goldman Sachs Towarzystwo Funduszy Inwestycyjnych S.A. (“GS TFI”), including any branch of subsidiary thereof (collectively, “GSAM Europe”), whether or not GSAM Europe has employed such individual; or (B) an individual who, at any point in _____, was classified by the Firm with respect to Goldman Sachs Paris Inc. et Cie in accordance with the applicable remuneration rules of either the Capital Requirements Directive V or the Investment Firms Regulation/Investment Firms Directive, as a Material Risk Taker (“GSPIC MRT”) or Senior Management (“GSPIC Senior Management”). EU MRT does not include AIFMD-UCITS Identified Staff of Goldman Sachs Asset Management Fund Services Ltd. (“GSAMFSL”).

(iii) **"GSAMI MRT"** means an individual who, at any point in _____, was classified by the Firm as a GSAMI Material Risk Taker in accordance with the remuneration code under the UK Investment Firms Prudential Regime ("IFPR") as applicable to the business operations of Goldman Sachs Asset Management International Ltd. ("GSAMI").

(iv) **“GSAMI Senior Management”** means an individual who, at any point in ____, was classified by the Firm as a GSAMI Senior Management in accordance with the remuneration code under the IFPR as applicable to the business operations of GSAMI.

(v) **“UK MRT”** means an individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (including UK Risk Managers and UK PRA Senior Managers) in accordance with the remuneration codes applicable to dual-regulated UK firms.

(vi) For purposes of these definitions, if during ____ you were classified by the Firm as any combination of EU MRT and GSAMI MRT, GSAMI Senior Management or UK MRT then your classification in the table above is EU MRT; any combination of GSAMI MRT or GSAMI Senior Management and UK MRT then your classification in the table above is UK MRT; or as a combination of GSAMI MRT and GSAMI Senior Management then your classification in the table above is GSAMI Senior Management.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by Paragraphs (b) and (c) of this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

[GSBE Material Risk Taker Appendix

This Appendix supplements Paragraph 9 and sets forth additional events that result in forfeiture of up to all of your RSUs and any Shares at Risk and may require repayment to the Firm of up to all other amounts previously delivered or paid to you under your Award in accordance with Paragraph 10. As with the events described in Paragraph 9, more than one event may apply; in no case will the occurrence of one event limit the forfeiture and repayment obligations as a result of the occurrence of any other event; and the Firm reserves the same rights in connection with any investigation of whether any of the events that result in forfeiture under this Appendix have occurred as set forth in Paragraph 9(e).

With respect to the events described in this Appendix, the Committee will consider all relevant factors in making the decision, in its sole discretion as to whether an Adjustment Event or Actionable Misconduct have occurred.

This Appendix applies to all RSUs in your Award if you are a GSBE MRT or GSBE Senior Management.

(a) An Adjustment Event Occurs with Respect to ____. If an Adjustment Event occurs with respect to ____, (i) your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs, (ii) your rights to all or a portion of any Shares at Risk will terminate and such Shares at Risk will be cancelled, and (iii) you will be obligated immediately upon demand therefor to pay the Firm an amount not in excess of the greater of the Fair Market Value of the RSU Shares (plus any dividend payments) delivered in respect of the Award (without reduction for any amount applied to satisfy tax withholding or other obligations) determined as of (A) the date the Adjustment Event occurred and (B) the date that the repayment request is made.

(i) **“Adjustment Event”** means that one of the following has occurred:

- A. you significantly contributed to, or were responsible for, any conduct that resulted in a loss of 0.75% or more of the total capital of GS Inc.;
- B. a material regulatory sanction for the Firm comprising one or more of the following:
 - 1. a moratorium pursuant to sec. 46g of the German Banking Act,
 - 2. a measure in case of danger pursuant to sec. 46 of the German Banking Act,
 - 3. the revocation of appointment of a manager pursuant to sec. 36 German Banking Act,
 - 4. a fine pursuant to sec. 56 of the German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of GS Inc.,
 - 5. the cancellation of the banking permit pursuant to sec. 35 of the German Banking Act,
 - 6. an order to increase the capital requirements Goldman Sachs Bank Europe SE (“GSBE”) by at least 0.5% pursuant to sec. 10 of the German Banking Act,

7. a measure in case of organizational deficiencies,
8. a comparable regulatory order, or
9. a material supervisory measure; or

C. you acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that a violation is considered serious if it suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct pursuant to sec. 1 German Termination Protection Act.

(b) You Engage in Actionable Misconduct. If the Committee determines that you engaged in Actionable Misconduct, which if known (or if known, adequately considered) would have resulted in a reduction to your annual variable compensation when your variable compensation for that year was determined, your rights in respect of all or a portion of your RSUs (whether or not Vested) will terminate and no RSU Shares will be delivered in respect of such RSUs.

(i) **“Actionable Misconduct”** means either or both of:

- A. “Unethical behavior” within the meaning of the IVV;
- B. “Breach of duty” within the meaning of the IVV.

(ii) **“IVV”** means the German Remuneration Ordinance for Institutions (*Institutsvergütungsverordnung*) and applicable guidance from the German Federal Financial Supervisory Authority (BaFin) or the European Central Bank.

(c) Scope of Appendix. This Appendix applies to any individual who, at any point in ____, was classified by the Firm as a Material Risk Taker (“GSBE MRT”) or Senior Management (“GSBE Senior Management”) in connection with the business operations of GSBE or any branch or subsidiary thereof, whether or not GSBE or any branch or subsidiary thereof employed such individual.

Notwithstanding any provision in the Plan, this Award Agreement or any other agreement or arrangement you may have with the Firm, the parties agree that to the extent that there is any dispute, controversy or claim arising out of or relating to the payment required by this Appendix (including your refusal to remit payment) the parties will submit to arbitration in accordance with Paragraph 16 of this Award Agreement and Section 3.17 of the Plan as the sole means of resolution of such dispute, controversy or claim (including the recovery by the Firm of the payment amount).]

Definitions Appendix

The following capitalized terms are used in this Award Agreement with the following meanings.

(a) “409A Deferred Compensation” means a “deferral of compensation” or “deferred compensation” as those terms are defined in the regulations under Section 409A.

(b) “Acceptance Deadline” means the date communicated to you by the Firm as the deadline to accept your Award. The Acceptance Deadline will be no less than 14 days after the Award Documentation is first made available to you in full. To the extent you are entitled to a translation in accordance with the Language Notice for Equity Awards provided to you as part of the Award Documentation, the Acceptance Deadline will be extended by the time required to furnish you with any such translation.

(c) “Associate With a Covered Enterprise” means that you (i) form, or acquire a 5% or greater equity ownership, voting or profit participation interest in, any Covered Enterprise or (ii) associate in any capacity (including association as an officer, employee, partner, director, consultant, agent or advisor) with any Covered Enterprise. Associate With a Covered Enterprise may include, as determined in the discretion of the Committee, (i) becoming the subject of any publicly available announcement or report of a pending or future association with a Covered Enterprise and (ii) unpaid associations, including an association in contemplation of future employment. “Association With a Covered Enterprise” will have its correlative meaning.

(d) “Breached an Obligation to the Firm” means that you failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with your Employment or this Award (including any applicable Additional Terms); including, without limitation, any offer letter, employment agreement or any shareholders’ agreement relating to the Firm. Your failure to pay or reimburse the Firm, on demand, for any amount you owe to the Firm will constitute (i) failure to meet an obligation you have under an agreement, regardless of whether such obligation arises under a written agreement, and/or (ii) a material violation of Firm policy constituting Cause.

(e) “Committee” means the Compensation Committee unless otherwise determined by the Board; *provided, however*, that: (i) the authority of the Compensation Committee to administer the Plan and the Award may be granted or delegated to the SIP Committee, in which case “Committee” means the SIP Committee when acting pursuant to such a grant or delegation; (ii) to the extent the Board determines it is appropriate for Awards under the Plan to qualify for the exemption available under Rule 16b-3(d)(1) or Rule 16b-3(e) promulgated under the Exchange Act, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is a “non-employee director” within the meaning of Rule 16b-3; and (iii) to the extent the Board determines it is appropriate for the compensation realized from Awards under the Plan to be considered “performance based” compensation under Section 162(m) of the Code, “Committee” shall be a committee or subcommittee of the Board composed of two or more members, each of whom is an “outside director” within the meaning of Code Section 162(m).

(f) “Compensation Committee” means the Compensation Committee of the Board.

(g) “Covered Enterprise” means a Competitive Enterprise and any other existing or planned business enterprise that: (i) offers, holds itself out as offering or reasonably may be expected to offer products or services that are the same as or similar to those offered by the Firm or that the Firm

reasonably expects to offer ("Firm Products or Services") or (ii) engages in, holds itself out as engaging in or reasonably may be expected to engage in any other activity that is the same as or similar to any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage ("Firm Activities"). For the avoidance of doubt, Firm Activities include any activity that requires the same or similar skills as any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage, irrespective of whether any such financial activity is in furtherance of an advisory, agency, proprietary or fiduciary undertaking.

The enterprises covered by this definition include enterprises that offer, hold themselves out as offering or reasonably may be expected to offer Firm Products or Services, or engage in, hold themselves out as engaging in or reasonably may be expected to engage in Firm Activities directly, as well as those that do so indirectly by ownership or control (e.g., by owning, being owned by or by being under common ownership with an enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services or that engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities). The definition of Covered Enterprise includes, solely by way of example, any enterprise that offers, holds itself out as offering or reasonably may be expected to offer any product or service, or engages in, holds itself out as engaging in or reasonably may be expected to engage in any activity, in any case, associated with investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than you or your family members (including, for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading. An enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services, or engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities is a Covered Enterprise, **irrespective of whether the enterprise is a customer, client or counterparty of the Firm or is otherwise associated with the Firm and, because the Firm is a global enterprise, irrespective of where the Covered Enterprise is physically located.**

(h) "Determination Date" means the date specified on your Award Statement as the date on which the Committee will determine whether or not, and to what extent, the Performance Goal[s] [was][were] achieved for the Performance Period.

(i) ["Dividend Equivalent Payments"] means any payments made in respect of Dividend Equivalent Rights.]

(j) "FDIC" means the Federal Deposit Insurance Corporation or any successor thereto.

(k) "Failed to Consider Risk" means that you participated (or otherwise oversaw or were responsible for, depending on the circumstances, another individual's participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where you have improperly analyzed such risk or where you have failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, the Committee determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, your business unit or the broader financial system. This definition is not limited to financial risks and is designed to encourage the consideration of the full range of risks associated with activities (e.g., legal, compliance or reputational). This definition also does not

require that a material adverse impact actually occur, but, rather, may be triggered if it is determined that there is a reasonable expectation of such an impact.

(l) "HCM" means the Firm's Human Capital Management Division or its successor.

(m) "Performance Goal[s]" means the performance goal[s] determined by the Committee that [is][are] specified on your Award Statement.

(n) "Performance Period" means the performance period determined by the Committee that is specified on your Award Statement.

(o) ["Qualified Termination"] means the termination of your Employment by the Firm where none of the forfeiture and repayment events described in Paragraphs 9 and 10 has occurred. No Employment termination that you initiate, including any purported "constructive termination," a "termination for good reason" or similar concepts, can be a Qualified Termination.]

(p) "Qualifying Termination After a Change in Control" means that the Firm terminates your Employment other than for Cause or you terminate your Employment for Good Reason, in each case, within 18 months following a Change in Control.

(q) ["Sarbanes-Oxley"] means the Sarbanes-Oxley Act of 2002, as amended.]

(r) "SEC" means the U.S. Securities and Exchange Commission.

(s) "Selected Firm Personnel" means any individual who is or in the three months preceding the conduct prohibited by Paragraph 9(b)(ii) was (i) a Firm employee or consultant with whom you personally worked while employed by the Firm, (ii) a Firm employee or consultant who, at any time during the year preceding the date of the termination of your Employment, worked in the same division in which you worked or (iii) an Advisory Director, a Managing Director (Participating or Extended) or a Senior Advisor of the Firm.

(t) "Settlement Amount" means the amount deliverable to you in respect of your PSUs, which will be determined as described in the Award Statement.

(u) "Settlement Date" means each date specified on your Award Statement as the date on which the Settlement Amount will be delivered, *provided*, unless the Committee determines otherwise, such date is during a Window Period or, if such date is not during a Window Period, the first trading day of the first Window Period beginning after such date.

(v) "Share" means a share of Common Stock.

(w) "Shares at Risk" means Shares subject to Transfer Restrictions.

(x) "Signature Card" means the document and/or website presented to you with your Award that you are required to execute (including electronically) to indicate your acceptance of the Award, the Award Agreement (including the Award Statement and the terms contained in the Signature Card) and Additional Terms applicable to you, as well your receipt of the Award Documentation.

(y) "Transferability Date" means the first trading day in a Window Period that occurs in the month and year specified on your Award Statement as a transferability date; *provided, however*, that if

there is no trading day in a Window Period that occurs in the specified month and year, the Transferability Date will be the first trading day in the subsequent Window Period.

The following capitalized terms are used in this Award Agreement with the meanings that are assigned to them in the Plan.

(a) “Account” means any brokerage account, custody account or similar account, as approved or required by GS Inc. from time to time, into which shares of Common Stock, cash or other property in respect of an Award are delivered.

(b) “Award Statement” means a written statement that reflects certain Award terms.

(c) “Board” means the Board of Directors of GS Inc.

(d) “Business Day” means any day other than a Saturday, a Sunday or a day on which banking institutions in New York City are authorized or obligated by Federal law or executive order to be closed.

(e) “Cause” means (i) the Grantee’s conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion, or (B) on a felony charge, or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations, (ii) the Grantee’s engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the Exchange Act), (iii) the Grantee’s willful failure to perform the Grantee’s duties to the Firm, (iv) the Grantee’s violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member, (v) the Grantee’s violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or the Grantee’s material violation of any other Firm policy as in effect from time to time, (vi) the Grantee’s engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm or (vii) the Grantee’s engaging in any conduct detrimental to the Firm. The determination as to whether Cause has occurred shall be made by the Committee in its sole discretion and, in such case, the Committee also may, but shall not be required to, specify the date such Cause occurred (including by determining that a prior termination of Employment was for Cause). Any rights the Firm may have hereunder and in any Award Agreement in respect of the events giving rise to Cause shall be in addition to the rights the Firm may have under any other agreement with a Grantee or at law or in equity.

(f) “Certificate” means a stock certificate (or other appropriate document or evidence of ownership) representing shares of Common Stock.

(g) “Change in Control” means the consummation of a merger, consolidation, statutory share exchange or similar form of corporate transaction involving GS Inc. (a “Reorganization”) or sale or other disposition of all or substantially all of GS Inc.’s assets to an entity that is not an affiliate of GS Inc. (a “Sale”), that in each case requires the approval of GS Inc.’s shareholders under the law of GS Inc.’s jurisdiction of organization, whether for such Reorganization or Sale (or the issuance of securities of GS Inc. in such Reorganization or Sale), unless immediately following such Reorganization or Sale, either: (i) at least 50% of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of (A) the entity resulting from such Reorganization, or the entity which has acquired all or substantially all of the assets of GS Inc. in a Sale (in either case, the “Surviving Entity”), or (B) if applicable, the ultimate parent entity that directly or indirectly has beneficial

ownership (within the meaning of Rule 13d-3 under the Exchange Act, as such Rule is in effect on the date of the adoption of the 1999 SIP) of 50% or more of the total voting power (in respect of the election of directors, or similar officials in the case of an entity other than a corporation) of the Surviving Entity (the "Parent Entity") is represented by GS Inc.'s securities (the "GS Inc. Securities") that were outstanding immediately prior to such Reorganization or Sale (or, if applicable, is represented by shares into which such GS Inc. Securities were converted pursuant to such Reorganization or Sale) or (ii) at least 50% of the members of the board of directors (or similar officials in the case of an entity other than a corporation) of the Parent Entity (or, if there is no Parent Entity, the Surviving Entity) following the consummation of the Reorganization or Sale were, at the time of the Board's approval of the execution of the initial agreement providing for such Reorganization or Sale, individuals (the "Incumbent Directors") who either (A) were members of the Board on the Effective Date or (B) became directors subsequent to the Effective Date and whose election or nomination for election was approved by a vote of at least two-thirds of the Incumbent Directors then on the Board (either by a specific vote or by approval of GS Inc.'s proxy statement in which such persons are named as nominees for director).

(h) "Client" means any client or prospective client of the Firm to whom the Grantee provided services, or for whom the Grantee transacted business, or whose identity became known to the Grantee in connection with the Grantee's relationship with or employment by the Firm.

(i) "Code" means the Internal Revenue Code of 1986, as amended from time to time, and the applicable rulings and regulations thereunder.

(j) "Common Stock" means common stock of GS Inc., par value \$0.01 per share.

(k) "Competitive Enterprise" means an existing or planned business enterprise that (i) engages, or may reasonably be expected to engage, in any activity; (ii) owns or controls, or may reasonably be expected to own or control, a significant interest in any entity that engages in any activity or (iii) is, or may reasonably be expected to be, owned by, or a significant interest in which is, or may reasonably be expected to be, owned or controlled by, any entity that engages in any activity that, in any case, competes or will compete anywhere with any activity in which the Firm is engaged. The activities covered by this definition include, without limitation: financial services such as investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the Grantee and members of the Grantee's family (including for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading.

(l) "Covered Person" means a member of the Board or the Committee or any employee of the Firm.

(m) "Date of Grant" means the date specified in the Grantee's Award Agreement as the date of grant of the Award.

(n) ["Dividend Equivalent Right" means a dividend equivalent right granted under the Plan, which represents an unfunded and unsecured promise to pay to the Grantee amounts equal to all or any portion of the regular cash dividends that would be paid on shares of Common Stock covered by an Award if such shares had been delivered pursuant to an Award.]

(o) "Effective Date" means the date this Plan is approved by the shareholders of GS Inc. pursuant to Section 3.15 of the Plan.

(p) "Employment" means the Grantee's performance of services for the Firm, as determined by the Committee. The terms "employ" and "employed" shall have their correlative meanings. The Committee in its sole discretion may determine (i) whether and when a Grantee's leave of absence results in a termination of Employment (for this purpose, unless the Committee determines otherwise, a Grantee shall be treated as terminating Employment with the Firm upon the occurrence of an Extended Absence), (ii) whether and when a change in a Grantee's association with the Firm results in a termination of Employment and (iii) the impact, if any, of any such leave of absence or change in association on Awards theretofore made. Unless expressly provided otherwise, any references in the Plan or any Award Agreement to a Grantee's Employment being terminated shall include both voluntary and involuntary terminations.

(q) "Exchange Act" means the Securities Exchange Act of 1934, as amended from time to time, and the applicable rules and regulations thereunder.

(r) "Extended Absence" means the Grantee's inability to perform for six (6) continuous months, due to illness, injury or pregnancy-related complications, substantially all the essential duties of the Grantee's occupation, as determined by the Committee.

(s) "Firm" means GS Inc. and its subsidiaries and affiliates.

(t) "Good Reason" means, in connection with a termination of employment by a Grantee following a Change in Control, (i) as determined by the Committee, a materially adverse alteration in the Grantee's position or in the nature or status of the Grantee's responsibilities from those in effect immediately prior to the Change in Control or (ii) the Firm's requiring the Grantee's principal place of Employment to be located more than seventy-five (75) miles from the location where the Grantee is principally Employed at the time of the Change in Control (except for required travel on the Firm's business to an extent substantially consistent with the Grantee's customary business travel obligations in the ordinary course of business prior to the Change in Control).

(u) "Grantee" means a person who receives an Award.

(v) "GS Inc." means The Goldman Sachs Group, Inc., and any successor thereto.

(w) "1999 SIP" means The Goldman Sachs 1999 Stock Incentive Plan, as in effect prior to the effective date of the 2003 SIP.

(x) "Outstanding" means any Award to the extent it has not been forfeited, cancelled, terminated, exercised or with respect to which the shares of Common Stock underlying the Award have not been previously delivered or other payments made.

(y) "Restricted Share" means a share of Common Stock delivered under the Plan that is subject to Transfer Restrictions, forfeiture provisions and/or other terms and conditions specified herein and in the Restricted Share Award Agreement or other applicable Award Agreement. All references to Restricted Shares include "Shares at Risk."

(z) "Retirement" means termination of the Grantee's Employment (other than for Cause) on or after the Date of Grant at a time when (i) (A) the sum of the Grantee's age plus years of service with

the Firm (as determined by the Committee in its sole discretion) equals or exceeds 60 and (B) the Grantee has completed at least 10 years of service with the Firm (as determined by the Committee in its sole discretion) or, if earlier, (ii) (A) the Grantee has attained age 50 and (B) the Grantee has completed at least five years of service with the Firm (as determined by the Committee in its sole discretion).

(aa)“RSU” means a restricted stock unit granted under the Plan, which represents an unfunded and unsecured promise to deliver shares of Common Stock in accordance with the terms of the RSU Award Agreement.

(ab)“Section 409A” means Section 409A of the Code, including any amendments or successor provisions to that Section and any regulations and other administrative guidance thereunder, in each case as they, from time to time, may be amended or interpreted through further administrative guidance.

(ac)“SIP Administrator” means each person designated by the Committee as a “SIP Administrator” with the authority to perform day-to-day administrative functions for the Plan.

(ad)“SIP Committee” means the persons who have been delegated certain authority under the Plan by the Committee.

(ae)“Solicit” means any direct or indirect communication of any kind whatsoever, regardless of by whom initiated, inviting, advising, suggesting, encouraging or requesting any person or entity, in any manner, to take or refrain from taking any action. The terms “Solicited,” “Soliciting” and “Solicitation” will have their correlative meanings.

(af)“Transfer Restrictions” means restrictions that prohibit the sale, exchange, transfer, assignment, pledge, hypothecation, fractionalization, hedge or other disposal (including through the use of any cash-settled instrument), whether voluntarily or involuntarily by the Grantee, of an Award or any shares of Common Stock, cash or other property delivered in respect of an Award.

(ag)“Vested” means, with respect to an Award, the portion of the Award that is not subject to a condition that the Grantee remain actively employed by the Firm in order for the Award to remain Outstanding. The fact that an Award becomes Vested shall not mean or otherwise indicate that the Grantee has an unconditional or nonforfeitable right to such Award, and such Award shall remain subject to such terms, conditions and forfeiture provisions as may be provided for in the Plan or in the Award Agreement.

(ah)“Vesting Date” means each date specified in the Grantee’s Award Agreement as a date on which part or all of an Award becomes Vested.

(ai)“Window Period” means a period designated by the Firm during which all employees of the Firm are permitted to purchase or sell shares of Common Stock (*provided* that, if the Grantee is a member of a designated group of employees who are subject to different restrictions, the Window Period may be a period designated by the Firm during which an employee of the Firm in such designated group is permitted to purchase or sell shares of Common Stock).

The Goldman Sachs Group, Inc.Signature Card for _____ Awards
and Consent to Receive Electronic Delivery**[IMPORTANT: PLEASE REVIEW, EXECUTE AND RETURN THIS FORM TO: _____.]****YOU MUST PROPERLY EXECUTE THIS FORM TO ACKNOWLEDGE ACCEPTANCE OF THE TERMS AND CONDITIONS OF YOUR AWARD(S) AND RELATED MATTERS.]**

1. I have received and agree to be bound by The Goldman Sachs Amended and Restated Stock Incentive Plan (2021) (the "Plan"), the Award Agreement(s) and the Award Statement applicable to me in connection with each Award (collectively, the "Award") that I have been granted by the Firm (as defined in the Plan). I confirm that I am accepting the Award subject to the terms and conditions contained in the Plan, the applicable Award Agreement and Award Statement, the Language Notice for Equity Awards, this Signature Card and the other documentation presented to me in connection with acceptance of the Award (collectively, the "Award Documentation"). I understand that I will have no less than 14 days after the Award Documentation is first made available to me in full to accept my Award by executing this Signature Card (including by electronic means). To the extent that I am entitled to request a translation in accordance with the Language Notice for Equity Awards, and such a request is made and accepted, I further understand that the Acceptance Deadline (as defined in the applicable Award Agreement) will be extended by a reasonable time period (of at least 14 days) after receipt of the translation, as communicated to me together with the translation.

For the avoidance of doubt, I understand and agree that to be eligible to receive any award under the Plan, I must not have engaged in any conduct constituting "Cause" (as defined in the Plan) prior to the grant of the Award, and by accepting this Award, I represent and warrant that I have not engaged in any conduct constituting Cause.

As a condition of this grant, I understand that the Award (and any other award that the Firm may grant to me under the Plan) is subject to governing law provisions in the Plan or applicable Award Agreement; and, as a condition to receiving such awards, I agree to be bound thereby. I also agree, as a condition of this grant, to the electronic consent in Paragraph 6 below.

2. By accepting my Award and executing this Signature Card, I confirm that I am agreeing to the additional contractual terms presented to me with the Award Documentation (the "Additional Terms") and acknowledge that my agreement to the Additional Terms is a condition of my Award. I acknowledge that such Additional Terms include the Policy on Notice Periods for Recipients of Year-End Equity-Based Awards and Other Deferred Compensation (the "Notice Period Policy") and the Agreement Regarding Arbitration of Employment-Related Matters. I further acknowledge that such Additional Terms may also include the APAC Restrictive Covenants and any agreement(s) specific to my role, business unit or division.

I acknowledge and understand that, to the extent the Award Documentation subjects me to any post-employment restriction, I have the right to consult with an attorney prior to accepting my Award. I further acknowledge and understand that my eligibility for the Award constitutes good and valuable consideration for any post-employment restriction contained in the Award Documentation.

3. As a condition of this grant, I agree to provide upon request an appropriate certification regarding my United States ("U.S.") tax status on Form W-8BEN, Form W-9, or other appropriate form, and I understand that failure to supply a required form may result in the imposition of backup withholding on certain payments I receive pursuant to this grant.

Further, as a condition of this grant, if I am a person who has worked in the United Kingdom at any time during the earnings period relating to any award under the Plan, as determined by the Firm, when requested and as directed by the Firm, I will agree to a Joint Election under s431 ITEPA 2003 of the laws of the United Kingdom for full or partial disapplication of Chapter 2 Income Tax (Earnings and Pensions) Act 2003 under the laws of the United Kingdom and will sign and return such election in respect of all future deliveries of shares of GS Inc. common stock ("Shares") underlying the Award and any previous grants made to me under the Plan and understand that the Firm intends to meet its delivery obligations in Shares with respect to my Award, except as may be prohibited by law or described in the accompanying Award Agreement(s) or supplementary materials.

If I have worked in Switzerland at any time during the earnings period relating to the Award granted to me as determined by the Firm, (a) I acknowledge that my Award is subject to tax in accordance with the rulings and method of calculation of taxable values to be agreed by the Firm with the Federal and/or Zurich/Geneva cantonal/communal tax authorities or as otherwise directed by the Firm, and (b) I hereby agree to be bound by any rulings agreed by the Firm in respect of any Award, which is expected to result in taxation at the time of delivery of Shares, and (c) I undertake to declare and make a full and accurate income tax declaration in respect of my Award in accordance with the above ruling or as directed by the Firm.

4. I agree to open and activate any brokerage, trust, sub-trust, custody or similar account (an "Account"), as required or approved by the Firm in its sole discretion. I agree to access, review, execute and be bound by any agreements that govern any such Account, including any provisions that provide for the applicable restrictions on transfers, pledges and withdrawals of Shares; permitting the Firm to monitor any such Account; and the limitations on the liability of the party (which may not be affiliated with the Firm) providing the Account and the Firm. I understand and agree that the Firm may direct the transfer of securities, cash or other assets in my Account to the Firm in connection with any indebtedness or any other obligation that I have to the Firm, as determined by the Firm in its sole discretion, however such obligation may have arisen. I also agree to open an Account with any other custodian, broker, trustee, transfer agent or similar party selected by the Firm, if the Firm, in its sole discretion, requires me to open an Account with such custodian, broker, trustee, transfer agent or similar party as a condition to delivery of Shares underlying the Award.
5. If the Firm advanced or loaned me funds to pay certain taxes (including income taxes and Social Security, or similar contributions) in connection with the Award (or does so in the future), and if I have not signed a separate loan agreement governing repayment, I authorize the Firm to withhold from my compensation any amounts required to reimburse it for any such advance or loan to the extent permitted by applicable law.

I understand and agree that, if I leave the Firm, I am required immediately to repay any outstanding amount. I further understand and agree that the Firm has the right to offset, to the extent permitted by the Award Agreement(s) and applicable law (including Section 409A (as defined in the Award Agreement), which limits the Firm's ability to offset in the case of U.S. taxpayers under certain circumstances), any outstanding amounts that I then owe the Firm against its delivery obligations under the Award, against any obligations to remove restrictions and/or other terms and conditions in respect of any Restricted Shares or Shares at Risk (each as defined in the applicable Award Agreement) or against any other amounts the Firm then owes me, including payments of dividends or dividend equivalent payments. I understand that the delivery of Shares pursuant to the Award is conditioned on my satisfaction of any applicable taxes or Social Security contributions (collectively referred to as "tax" or "taxes" for purposes of the Award Documentation) in accordance with the Plan. To the extent permitted by applicable law, the Firm, in its sole discretion, may require me to provide amounts equal to all or a portion of any Federal, state, local, foreign or other tax obligations imposed on me or the Firm in connection with the grant, vesting or delivery of the Award by requiring me to choose between remitting such amount (a) in cash (or through payroll deduction or otherwise), (b) in the form of proceeds from the Firm's executing a sale of Shares delivered to me pursuant to the Award or (c) as otherwise permitted in the applicable Award Agreement; *provided, however*, that in no event shall any such choice determine, or give me any discretion to affect, the timing of the delivery of Shares or payment of tax obligations.

6. In connection with any Award Agreement or other interest I may receive in the Plan or any Shares that I may receive in connection with the Award or any award I have previously received or may receive, or in connection with any amendment or variation thereof or any documents listed in Paragraph 7, I hereby consent to (a) the acceptance by me of the Award electronically, (b) the giving of instructions in electronic form whether by me or the Firm, and (c) the receipt in electronic form at my email address maintained at Goldman Sachs or via Goldman Sachs' intranet site (or, if I am no longer employed by the Firm, at such other email address as I may specify, or via such other electronic means as the Firm and I may agree) all notices and information that the Firm is required by law to send to me in connection therewith including, without limitation, any document (or part thereof) constituting part of a prospectus covering securities that have been registered under the U.S. Securities Act of 1933, the information contained in any such document and any information required to be delivered to me under Rule 428 of the U.S. Securities Act of 1933, including, for example, the annual report to security holders or the annual report on Form 10-K of GS Inc. for its latest fiscal year, and that all prior elections that I may have made relating to the delivery of any such document in physical form are hereby revoked and superseded. I agree to check Goldman Sachs' intranet site (or, if I am no longer employed by the Firm, such other electronic site as notified to me by the Firm) periodically as I deem appropriate for any new notices or information concerning the Plan. I understand that I am not required to consent to the receipt of such documents in electronic form in order to receive the Award and that I may decline to receive such documents in electronic form by contacting _____, which will provide me with hard copies of such documents upon request. I also understand that this consent is voluntary and may be revoked at any time on three business days' written notice.
7. I hereby acknowledge that I have received in electronic form in accordance with my consent in Paragraph 6 the following documents:
- The Goldman Sachs Amended and Restated Stock Incentive Plan (2021);
 - Summary of The Goldman Sachs Amended and Restated Stock Incentive Plan (2021);
 - The most recent annual report on Form 10-K for The Goldman Sachs Group, Inc.;
 - The Award Agreement(s);
 - The Award Statement;
 - The Language Notice for Equity Awards; and
 - Summaries of the Award ("Award Summary").
8. I expressly authorize any appropriate representative of the Firm to make any notifications, filings or remittances of funds that may be required in connection with the Plan or otherwise on my behalf. I acknowledge that any such authorization is effective from the date of acceptance of my Award until such time as I expressly revoke the authorization by written notice to any appropriate representative of the Firm. I understand that this authorization does

not create any obligation on the Firm to deal with any such notifications, filings or remittances of funds that I may be required to make in connection with the Plan and I accept full responsibility in this regard.

9. The granting of the Award, the delivery of the underlying Shares and any subsequent dividends or dividend equivalent payments, and the receipt of any proceeds in connection with the Award may result in legal or regulatory requirements in some jurisdictions. I understand and agree that it is my responsibility to ensure that I comply with any legal or regulatory requirements in respect of the Award.
10. I confirm that I have filed all tax returns that I am required to file and paid all taxes I am required to pay with respect to awards previously granted to me by the Firm, and I agree, with respect to both the Award as well as awards previously granted to me by the Firm, to file all tax returns I am required to file in connection with the Award and any sales of any Shares or other property delivered pursuant to the Award and to pay all taxes I am required to pay.
11. The goodwill associated with the relationships between the Firm and its clients and prospective clients is a valuable asset of the Firm that is built and preserved through the combined services and efforts of the Firm and all of its personnel. The Firm provides its employees with a unique platform of financial products and services, confidential and proprietary information, professional training, access to specialized expertise, research, analytical, operational, and business development support, travel and entertainment expenses and other valuable resources to build and enhance the goodwill associated with the relationships between the Firm and its clients, as well as to foster and establish such relationships with prospective clients. Accordingly, I acknowledge and agree that (a) because the Firm contributes valuable resources to build and enhance client relationships, including those for which I provide services, it has a legitimate and essential business interest in protecting the goodwill associated with those relationships; (b) by my continued employment, I confirm that I have assigned and will assign to the Firm all goodwill I have developed or will develop with persons or entities with whom I interact while at the Firm and/or who are or will become clients or prospective clients of the Firm in connection with my employment with the Firm, even if I did business with such persons or entities prior to joining the Firm; and (c) while at the Firm I do not have and will not acquire any property, proprietary, contractual or other legal right or interest whatsoever in or to any client or prospective client with whom I interact or conduct business while employed by the Firm or (except to the extent otherwise provided in a written agreement between the Firm and me that governs my compensation) to any current or prospective revenues associated with such client or prospective client (all such interests being referred to herein as "Intangible Interests"). For the avoidance of doubt, I am hereby assigning all Intangible Interests to the Firm. I acknowledge and agree that my compensation during the term of my employment with the Firm is adequate financial consideration in this regard, and that no further consideration is necessary (including in respect of obligations applicable to me after my employment with the Firm has ended).

Data Collection, Processing and Transfers

Grantees residing outside of the European Economic Area ("EEA"), the United Kingdom ("UK") and the PRC (defined below): If I am located outside of the EEA, the UK and the PRC, I consent to the processing of my personal data in accordance with the information set out below.

Grantees residing in the EEA or the UK: If I am located in the EEA or the UK, my personal data will be processed in accordance with the information set out below and in the GS HCM - Fair Processing Notice which can be found at <http://www.goldmansachs.com/notices/hcm-fpn.html>. In the event of any inconsistency, the GS HCM - Fair Processing Notice shall prevail over this section.

Grantees residing in the People's Republic of China ("PRC") (which, for the purpose of this section, excludes Hong Kong Special Administrative Region, Macau Special Administrative Region and Taiwan): If I am located in the PRC, I give my consent to the processing of my personal data as follows by signing this Signature Card:

- **The Firm will process my personal data as set out in this section. I consent to such use.**
- **The Firm will process and retain my sensitive personal data as set out in this section and underlined. I consent to such use.**
- **The Firm will transfer my personal data to third parties within the PRC, as described by this section and the hyperlinks within it. I consent to such transfers.**
- **The Firm will transfer my personal data to third parties outside the PRC, as described by this section and the hyperlinks within it. I consent to such transfers.**

For All Grantees:

In connection with the Plan and any equity-based award under the Plan, including my Award (collectively, the "Program"), to the extent permitted under the laws of the applicable jurisdiction, the Firm may process (including, where applicable, collecting, transferring/transmitting internationally and/or domestically, disclosing, using and storing) various data that is personal to me, and my data might be deemed sensitive personal data in certain jurisdictions, including but not limited to my name, address, work location, hire date, Social Security or Social Insurance or taxpayer identification number (as required for tax purposes), type and amount of Plan or other benefit plan award, other compensation-related data, citizenship or residency (required for tax purposes) and other similar information reasonably necessary for the administration of the Program (collectively referred to as "Information") and provide such Information to its affiliates, Computershare Limited and its affiliates (collectively "Computershare") and Fidelity Stock Plan Services, LLC, Fidelity Personal Trust Company, FSB and any of their affiliates (collectively "Fidelity") or any other service provider that assists in the administration of the Program (e.g., brokerage, custody and nominee services), whether in the U.S. or elsewhere, as is reasonably necessary for the administration of the Program and under the laws of these jurisdictions.

In certain circumstances and subject to any applicable restrictions regarding cross-border data transfers in the jurisdiction where I reside, where required by law, foreign courts, law enforcement agencies or regulatory agencies may be entitled to access the Information. Unless I explicitly authorize otherwise, the Firm, its affiliates and its service providers (through their respective employees in charge of the relevant electronic and manual processing) will process this Information only for purposes of administering the Program. In the U.S. and in other countries to which such Information may be transferred for the administration of the Program, the level of data protection may not be equivalent to data protection standards in the member states of the EEA, Switzerland, New Zealand, Canada or certain Canadian provinces or my home country and U.S. public authorities may potentially access such Information.

If I am employed in Argentina, the PRC, Peru, South Korea or Turkey, I have also read the text in bold in the respective Argentina, the PRC, Peru and Turkey legal notices below (the "Argentina Clause", the "PRC Clause", the "Peru Clause" or the "Turkey Clause") in conjunction with this Data Collection, Processing and Transfers section, and I acknowledge that such text forms part of this section and that in the event of any inconsistency the Argentina Clause, PRC Clause, Peru Clause or Turkey Clause, as applicable, shall prevail over this section.

Upon request to Equity Compensation (division of HCM), 200 West Street, 19th Floor, New York, NY 10282, telephone (212) 357-1444, email EquityCompensation@ny.email.gs.com, to the extent required under the laws of the applicable jurisdiction, I may have access to and obtain communication of the Information and may exercise any of my rights in respect of such Information, in each case free of charge, including objecting to any type of processing of the Information and requesting that the Information be updated or corrected (if wrong), completed or clarified (if incomplete or equivocal), or erased (if it cannot legally be collected or kept). Upon request, to the extent required under the laws of the applicable jurisdiction, a team member or representative of Equity Compensation (division of HCM) will also provide me, free of charge, with: (i) written information about the Firm's policies and practices with respect to service providers outside of my jurisdiction, including a list of all the service providers used in connection with the Program at the time of request; and/or (ii) answers to my questions about the processing of my Information by service providers or affiliates outside of my jurisdiction.

There is no legal obligation for me to provide the Firm with the Information and any Information is provided at my own will and consent. If I refuse to authorize the processing of the Information consistent with the above, I may not benefit from the Program. The processing of the Information will be consistent with the above for the period of administration of the Program. In particular (within the limits described above): (i) the Firm will process data (Firm means GS Inc. and any of its subsidiaries and affiliates); (ii) Fidelity or Computershare will process data; (iii) the Firm's other service providers will process data; and (iv) data will be transferred to the U.S. and other countries, as described above for the purposes set forth herein. A list of the Firm's international offices and countries to which data that is personal to me can be transferred is set forth at <https://www.goldmansachs.com/our-firm/locations>. The Information may be retained by the aforementioned persons beyond the period of administration of the Program to the extent permitted under the laws of the applicable jurisdiction.

Other Legal Notices

12. By accepting (whether expressly or by implication) any benefit granted to me by the Firm, including, without limitation, my Award, I acknowledge and agree to each of the following:

- **No Public Offer:** Awards under the Plan and the Firm's other compensation and benefit programs are strictly limited to eligible participants and are not intended to constitute a public offer in any jurisdiction, nor intended for registration in any jurisdiction outside of the U.S. I must keep all Award Documentation confidential and I may not reproduce, distribute or otherwise make public any part of such documents without the Firm's express written consent. If I have received any such documents and I am not the intended recipient, I will disregard and destroy them.
- **Transferability:** Any provisions permitting transfers to a third party in the Award Documentation will not apply to me (i) to the extent that the applicability of those provisions would affect the availability of relevant exemptions or tax favorable treatment, or (ii) otherwise in circumstances determined by the Firm in its sole discretion from time to time.
- **Adequate Information:** I acknowledge that (i) I have been provided with all relevant information and materials with respect to the Firm's operations and financial conditions and the terms and conditions of my Award, (ii) I have read and understood such information and materials, (iii) I am fully aware and knowledgeable of the terms and conditions of my Award, and (iv) I completely and voluntarily agree to the terms and conditions of my Award.
- **Independent Advice Recommended:** The information provided by the Firm or its service providers in respect of an Award does not take into account the individual circumstances of recipients and does not constitute legal, investment or tax advice. Awards under the Plan involve certain risks and I should exercise caution. The Firm recommends that I consult my own independent legal, investment and tax advisors in all cases, and I acknowledge that I am provided with adequate opportunity to do so.
- **Share Price and Currency Risks:** There is a risk that Shares may fall as well as rise in value. Market forces will impact the price of Shares and, in the worst case, the market value of the Shares may become zero. If the Shares are traded in a currency which is not the currency in my jurisdiction, the value of the Shares to me may also be affected by movements in the exchange rate. The Firm is not liable for any loss due to movements in the price of Shares or any applicable exchange rate.
- **No Employer Involvement:** Except to the extent required by applicable law, all Awards are offered, issued and administered by GS Inc., a Delaware corporation, and my employer (if it is not GS Inc.) is not involved in the grant of my Award or any other GS Inc. equity compensation. All Award Documentation and the link by which I access these documents are originated and maintained in the U.S.
- **No Effect on Employment-Related Rights:** Any compensation I receive (even on a regular and repeated basis) in connection with the Plan is discretionary and does not constitute part of my base or normal salary or wages. It does not affect my rights and obligations under the terms of my employment and it will not be taken into account (except to the extent otherwise required by applicable law) in determining any other employment-related rights I may have, including, without limitation, rights in relation to severance, redundancy or end-of-service payments, bonuses, long-service awards, pension or retirement benefits. In particular, I waive any and all rights to compensation or damages in consequence of the termination of my employment for any reason whatsoever insofar as those rights arise or may arise from me ceasing to have rights under, or be entitled to receive payment in respect of, the Plan, or from the loss or diminution in value of such rights, as a result of such termination. This waiver applies whether or not such termination amounts to wrongful or unfair dismissal.
- **No Additional Entitlements:** The grant of an Award is strictly discretionary and voluntary. Neither this Award (even if grants are made to me under the Plan on a regular and repeated basis) nor my employment contract implies any expectation or right in relation to (i) the grant of any award under the Plan or similar compensation in the future, (ii) the terms, conditions and amount of any award under the Plan or similar compensation that the Firm may decide to grant in the future, or (iii) continued employment or engagement.
- **Severability:** Unless another result is specifically provided for in an agreement to the contrary, if any provision (in whole or in part) of the Award Documentation (including but not limited to the Agreement Regarding Arbitration of Employment-Related Matters) is to any extent illegal, otherwise invalid, or incapable of being enforced, that provision will be excluded to the extent (only) of such invalidity or unenforceability. All other provisions will remain in full effect and, to the extent possible, the invalid or unenforceable provision will be deemed replaced by a provision that is valid and enforceable and that comes closest to expressing the intention of such invalid or unenforceable provision.
- **Country-Specific Legal Notices:** I have read the country-specific legal notices below that pertain to my place of employment and/or residence (and also the location of my employer, if different), if any, and understand that they apply throughout the term of my Award.

Country-Specific Legal Notices

FOR EMPLOYEES IN CERTAIN EUROPEAN UNION JURISDICTIONS (BELGIUM, CZECH REPUBLIC, DENMARK, FRANCE, GERMANY, GREECE, IRELAND, ITALY, LUXEMBOURG, SPAIN AND SWEDEN)

You are being offered an Award under the Plan in order to provide an additional incentive and to encourage employee share ownership and to increase your interest in GS Inc.'s success. The Award is offered to you by GS Inc. in accordance with the terms of the Plan which are summarized in the Award Summary. Further details on the rights attaching to your Award can be found in the Award Summary. More information about GS Inc. is available at www.gs.com.

The Shares subject to the Award are new or existing ordinary shares in GS Inc. and information on the total maximum number of Shares which can be offered under the Plan rules can be found in the section entitled *Shares Available for Awards* in the Plan. The obligation to publish a prospectus does not apply because of Article 1(4)(i) of the EU Prospectus Regulation 2017/1129.

If applicable, you should also refer to any additional specific notices below in relation to these jurisdictions.

FOR ARGENTINA EMPLOYEES ONLY

Your Award is being offered to you in connection with your employment or appointment by the Firm, as applicable. The Plan is not offered to the general public. By receiving and accepting your Award, you are deemed to: (i) acknowledge that the Firm has not made, and will not make, any application to obtain an authorization from the National Securities Commission (*Comisión Nacional de Valores*) for the public offering of the underlying Shares in Argentina, or otherwise taken any action that would permit a public offering of the underlying Shares in Argentina within the meaning of Argentine Capital Markets Law No. 26,831, as amended, supplemented or otherwise modified from time to time (the "CML") and of the Argentine Securities Exchange Commission General Resolution No. 622/2013, as amended, supplemented or otherwise modified from time to time, and ancillary regulations (the "CNV Rules"); (ii) acknowledge that the Argentine Securities Exchange Commission (the "CNV") has not previously reviewed nor approved the offering of the underlying Shares nor any document relating to their offering and therefore that this offer is being delivered to you on a private basis in accordance with Sub-title I, Chapter I, Title XX of the CNV Rules or Sub-title II, Chapter I, Title XX of the CNV Rules, as applicable; (iii) acknowledge that the CNV has not rendered and will not render any opinion in respect of the documents governing the Plan, and all information contained herein is the responsibility of the Firm and additional intervening parties; (iv) acknowledge that the Plan is exempted from complying with the general rules applicable to public offerings; and (v) agree that you will not sell or offer to sell any Shares acquired upon settlement of your Award in Argentina within the following 6 months after acquisition if it is not a primary placement pursuant to section 17 of Sub-title II, Chapter I, Title XXX or to Qualified Investors within 3 months, or to Non-Qualified Investors within 6 months from the date of its acquisition pursuant to section 10 of Sub-title I, Chapter I, Title XX, as applicable, of the CNV Rules.

Receipt and acceptance of the Award Documentation constitutes your agreement that the information contained in the Award Documentation may not be: (i) reproduced or used, in whole or in part, for any purpose whatsoever other than as a representation of your holding of your Award, or (ii) furnished to or discussed with any person (other than your personal advisors on a confidential basis) without the express written permission of GS Inc.

You acknowledge that the Access to Public Information Agency, as the enforcing authority of Act 25.326, has the power to attend the reports and claims from those who are affected in their rights as a consequence of non-fulfilment of data protection provisions. (*La Agencia de Acceso a la Información Pública, en su carácter de Órgano de Control de la Ley Nº 25.326, tiene la atribución de atender las denuncias y reclamos que interpongan quienes resulten afectados en sus derechos por incumplimiento de las normas vigentes en materia de protección de datos personales.*)

Additional data protection information for Argentina employees (which should be read in conjunction with, and forms part of, the Data Collection, Processing and Transfers section above):

You understand that your data may be stored in a database duly registered with the Argentine National Data Protection Agency, under the name and responsibility of GS Inc. or one of its subsidiaries or affiliates.

FOR AUSTRALIA EMPLOYEES ONLY

Offers under the Plan are made under Division 1A of Part 7.12 of the Corporations Act.

GS Inc. undertakes that it will, within a reasonable period of you so requesting and at no charge, provide you with a copy of the rules of the Plan. The market price of a Share can be accessed at the following link: <https://www.nyse.com/index>. The Australian dollar equivalent of that market price can be ascertained by applying the prevailing U.S. dollar ("USD") / Australia dollar ("AUD") exchange rate published by the Reserve Bank of Australia, which can be accessed at the following link: <http://www.rba.gov.au/statistics/frequency/exchange-rates.html>.

Any advice given by GS Inc. in connection with the Plan is general advice only. The Award Documentation does not take into account the objectives, financial situation or needs of any particular person. Before acting on the information contained in the Award Documentation, or making a decision to participate, you should consider obtaining your own

financial product advice from a person who is licensed by the Australian Securities and Investments Commission to give such advice.

Throughout the period in which you hold a dividend equivalent right you may obtain copies of all information filed by GS Inc. with the U.S. Securities and Exchange Commission ("SEC") which is accessible by GS Inc.'s shareholders and the general public ("shareholder information") by going to the SEC's website (<https://www.sec.gov>) or to the GS Inc. website (www.gs.com), and at <http://www.goldmansachs.com/investor-relations/financials/>. You should be aware that shareholder information can affect the value of your dividend equivalent rights from time to time.

The actual value you receive in respect of the Shares acquired by you will depend on the number of Shares you receive, the market value of a Share, the value of any dividend and dividend equivalent payments made in respect of a Share, and the USD/AUD exchange rate.

There are risks associated with an investment in Shares and the value of any Shares you receive may be less than the value of those Shares today. Some of those risks are specific to GS Inc.'s business activities while others are of a more general nature. For more detail on those risks, please refer to GS Inc.'s most recent annual report. Individually or in combination, those risks may affect the value of Shares.

FOR BRAZIL EMPLOYEES ONLY

The Award referred to in this Signature Card and the underlying Shares have not been and will not be publicly issued, placed, distributed, offered or negotiated in the Brazilian capital markets and, as a result, will not be registered with the Brazilian Securities Commission (*Comissão de Valores Mobiliários*). The Award and the underlying Shares will not be offered or sold in Brazil under any circumstances that constitute a public offering, placement, distribution or negotiation under the Brazilian capital markets regulation.

By accepting the Award, you agree and acknowledge that (i) neither your employer or contracting company, as applicable, nor any person or entity acting on their behalf has provided you with financial advice with respect to your Award or the Shares acquired upon settlement of your Award; and (ii) your employer or contracting company, as applicable, does not guarantee a specified level of return on your Award or the Shares.

FOR CHILE EMPLOYEES ONLY

By accepting the Signature Card connected to your Award, you acknowledge that this legend applies to all Award Documentation and communications.

General Warning

Neither GS Inc., the Plan nor the Shares have been registered in the *Registro de Valores* ("Securities Registry") or in the *Registro de Valores Extranjeros* ("Foreign Securities Registry") of the *Comisión para el Mercado Financiero* (Chilean Commission for the Financial Market or "CMF") and they are not subject to the control of the CMF. If such securities are offered within Chile, they will be offered and sold only pursuant to *Norma de Carácter General 336* (General Regulation 336) of the CMF, an exemption to the registration requirements, or in circumstances which do not constitute a public offer of securities in Chile within the meaning of Article 4 of the Chilean Securities Market Law 18,045. As the Shares are not registered, the issuer has no obligation under Chilean law to deliver public information regarding the Shares in Chile. The Shares shall not be subject to public offering in Chile unless they are registered in the Foreign Securities Registry of the CMF. The commencement date of the offer is the date of the email in which the Award Documentation (including this Signature Card) was first communicated to you.

Statement by Investors (Grantees)

Pursuant to General Regulation 336 (as amended), you are required to inform GS Inc. whether or not you are a "Qualified Investor" as defined in *Norma de Carácter General 216* (General Regulation 216) of the CMF. By accepting the Signature Card connected to your Award, you state to GS Inc. that you are not a Qualified Investor, or if you are a Qualified Investor, that you have provided a separate statement acknowledging this to GS Inc. (please note that any such statement may be emailed to GS Inc. at EquityCompensation@ny.email.gs.com). By accepting the Signature Card connected to your Award, you further state to GS Inc. that you acknowledge:

(i) that the securities acquired in connection with the Plan will not be registered in the CMF Securities Registry or Foreign Securities Registry and, therefore, such securities may not be publicly offered in Chile; and

(ii) that since GS Inc. is not registered in the registries kept by the CMF, GS Inc. will not be subject to the CMF's oversight nor to the ongoing disclosure obligations imposed by the law and regulation on registered issuers.

Accessibility of Award Documentation

The Firm undertakes to provide you with a copy of the Award Documentation in a form that remains accessible even if your employment is terminated.

SÓLO PARA EMPLEADOS DE CHILE

Al aceptar la "Signature Card" relacionada con su plan, usted reconoce que esta leyenda se aplica a todos los documentos y comunicaciones relacionados con el Plan.

Advertencia General

Ni GS Inc., ni el Plan, ni las Acciones, han sido registradas en el Registro de Valores o Registro de Valores Extranjeros que lleva la Comisión para el Mercado Financiero ("CMF") y ninguno de ellos está sujeto a la fiscalización de la CMF. Si dichos valores son ofrecidos dentro de Chile, serán ofrecidos y colocados sólo de acuerdo a la Norma de Carácter General 336 de la CMF, una excepción a la obligación de registro, o en circunstancias que no constituyan una oferta pública de valores en Chile según lo definido por el Artículo 4 de la Ley 18.045 de Mercado de Valores de Chile. Por tratarse de valores no inscritos, el emisor de las Acciones no tiene obligación bajo la ley chilena de entregar en Chile información pública acerca de las Acciones. Las Acciones no pueden ser ofrecidas públicamente en Chile en tanto éstas no se registren en el Registro de Valores Extranjeros de la CMF. Se informa que la fecha de inicio de la presente oferta es la fecha del correo electrónico en el que se le comunicaron por primera vez los documentos del Premio (incluida esta Signature Card).

Declaración de los Inversionistas (Destinatarios)

De conformidad con la Norma de Carácter General N°336 (modificada), se requiere que usted informe a GS Inc. si califica como "Inversionista Calificado" en los términos de la Norma de Carácter General N°216 de la CMF. Por medio de la aceptación de la Signature Card referida a su Plan, usted declara a GS Inc., que no califica como "Inversionista Calificado", o que si califica como "Inversionista Calificado", usted ha enviado por separado una declaración reconociendo dicha circunstancia a GS Inc. (favor tenga en cuenta que cualquier declaración puede ser enviada a GS Inc., al correo EquityCompensation@ny.email.gs.com). Por medio de la aceptación de la Signature Card referida a su plan, usted declara a GS Inc., que está en conocimiento de que:

(i) los valores adquiridos en conexión con el Plan no estarán inscritos en el Registro de Valores o en el Registro de Valores Extranjeros de la CMF y, por lo tanto, dichos valores no podrán ser públicamente ofrecidos en Chile; y

(ii) dado que GS Inc. no está inscrita en los registros mantenidos por la CMF, GS Inc. no estará sujeta a la fiscalización de la CMF ni a las obligaciones de información continua impuestas por la ley y la regulación a los emisores inscritos.

Accesibilidad a los Documentos del Plan

La empresa se obliga a entregarle una copia de los documentos clave relativos al plan de forma tal que permanezcan accesibles a usted incluso luego de terminada su relación laboral con ésta.

FOR THE PEOPLE'S REPUBLIC OF CHINA EMPLOYEES ONLY

All Award Documentation is intended for your personal use and in your capacity as an employee of the Firm (and/or its affiliate) and is being given to you solely for the purpose of providing you with information concerning the Award which the Firm may grant to you as an employee of the Firm (and/or its affiliate) in accordance with the terms of the Award Documentation. The grant of the Award has not been and will not be registered with the China Securities Regulatory Commission of the People's Republic of China pursuant to relevant securities laws and regulations, and the Award may not be offered or sold within the mainland of the People's Republic of China by means of any of the Award Documentation through a public offering or in circumstances which require a registration or approval of the China Securities Regulatory Commission of the People's Republic of China in accordance with the relevant securities laws and regulations.

You agree that notwithstanding anything to the contrary under the Plan or the Award Agreement(s), the Award may be settled in cash in Renminbi or such other currency, payable by your employing entity in the mainland of the People's Republic of China or such other entity, in each case, as may be determined by the Firm in its sole discretion.

Additional data protection information for the People's Republic of China employees (which should be read in conjunction with, and forms part of, the Data Collection, Processing and Transfers section above):

You have the ability to request the name, contact information, processing purpose and processing methods for any onshore data controller to which the Firm discloses your personal data in connection with your Award. You also have the ability to request the information listed in the preceding sentence, as well as information on the procedures that you may follow to exercise your data rights, in relation to any foreign party to which the Firm

discloses your personal data in connection with your Award. The Firm will respond to any such request by you as soon as is reasonably practicable for the Firm.

FOR FRANCE EMPLOYEES ONLY

The provisions of the Award Agreement will apply only in respect of the year to which the Award Agreement relates and will not in any circumstances create any right or entitlement to you for any future fiscal years.

POUR LES SALARIES EN FRANCE

Les dispositions relatives à la Convention d'Attribution s'appliquent uniquement à l'année concernée par la Convention d'Attribution et ne créent en aucun cas un droit s'agissant des années fiscales à venir.

FOR HONG KONG EMPLOYEES ONLY

WARNING:

The contents of this document have not been reviewed by any regulatory authority in Hong Kong. You are advised to exercise caution in relation to the Award. If you are in doubt about any of the contents of this document, you should obtain independent professional advice.

This document has not been, and will not be, registered as a "prospectus" in Hong Kong under the Companies (Winding Up and Miscellaneous Provisions) Ordinance (Cap 32) nor has it been authorised by the Securities and Futures Commission in Hong Kong pursuant to the Securities and Futures Ordinance (Cap 571) of the Laws of Hong Kong. This document does not constitute an offer or invitation to the public in Hong Kong to acquire any securities nor an advertisement of securities in Hong Kong. This document is distributed on a confidential basis.

By accepting the Award, you acknowledge and agree that you will not be permitted to transfer the Award to persons who fall outside the definition of "qualifying persons" in the Companies (Winding Up and Miscellaneous Provisions) Ordinance (Cap 32) (*i.e.*, a person who is neither a current or former director, employee, officer or consultant of the Firm, nor your wife, husband, widow, widower, child or step-child under the age of 18 years, or as otherwise defined), even if otherwise permitted under the Plan or any of the related documents.

FOR INDIA EMPLOYEES ONLY

The Award Documentation (including any related website) does not invite offers from the public for subscription or purchase of the securities of any body corporate under any law for the time being in force in India. The Award Documentation is not a prospectus under the applicable laws for the time being in force in India. GS Inc. does not intend to market, promote, invite offers for subscription or purchase of the securities of any body corporate by the Award Documentation. The information provided in the Award Documentation is for the record only. Any person who subscribes or purchases securities of any body corporate should consult their own investment advisors before making any investments. GS Inc. shall not be liable or responsible for any such investment decision made by any person.

FOR INDONESIA EMPLOYEES ONLY

The Award does not constitute an offer or sale of securities in Indonesia and are being granted to fewer than 51 individuals in Indonesia.

FOR NEW ZEALAND EMPLOYEES ONLY

GS Inc. is offering you an Award under the Plan in reliance upon clause 8 of Schedule 1 of the Financial Markets Conduct Act 2013 (offers under employee share purchase schemes) ("Exclusion"). In accordance with the requirements of the Exclusion, the following information has been made available to you:

1. GS Inc.'s most recent annual report via <https://www.goldmansachs.com/investor-relations/index.html>.
2. The Plan documentation (which constitutes the current rules of the employee share purchase scheme for the purposes of the Exclusion) on <https://hcm.web.gs.com/newaward>.
3. A copy of the Award Agreement on <https://hcm.web.gs.com/newaward>.
4. GS Inc.'s most recent published audited and unaudited financial statements on <https://www.goldmansachs.com/investor-relations/financials/index.html>.
5. A copy of the auditor's report on the above financial statements (if any) at <https://www.goldmansachs.com/investor-relations/financials/index.html>.

You may obtain a copy of the documents listed above by electronic means from the internet site addresses given in each case. You may request hard copies of the documents listed above free of charge from Head of Securities Compliance – Goldman Sachs Australia Pty Ltd.

For further information, including the form, dividend payments, vesting, delivery, and transfer restrictions of the Awards, please refer to the documents listed above.

Warning

The Award is an offer of Shares (although the Award may in limited circumstances be settled in cash or other property). The Shares give you a stake in the ownership of GS Inc. You may receive a return if dividends are paid.

If GS Inc. runs into financial difficulties and is wound up, you will be paid only after all creditors and holders of any preference shares have been paid. You may lose some or all of your investment.

New Zealand law normally requires people who offer financial products to give information to investors before they invest. This information is designed to help investors to make an informed decision.

The usual rules do not apply to this offer because it is made under an employee share purchase scheme. As a result, you may not be given all the information usually required. You will also have fewer other legal protections for this investment.

Ask questions, read all documents carefully, and seek independent financial advice before committing yourself.

The Shares are quoted on a stock exchange. GS Inc. intends to quote these Shares on the New York Stock Exchange (“NYSE”). This means you may be able to sell them on the NYSE if there are interested buyers. You may get less than you invested. The price will depend on the demand for the Shares.

FOR PERU EMPLOYEES ONLY

If you are employed in Peru, the following statement is hereby made part of the Award Documentation: the Shares which may be issued upon settlement of your Award have not been registered with the Public Registry of the Securities Market administered by the Peruvian Securities Market Superintendence (*Superintendencia del Mercado de Valores* – SMV) and may not be offered or sold publicly in Peru. In addition, the contents of the Plan documents have not been reviewed by any Peruvian regulatory authority.

Additional data protection information for Peru employees (which should be read in conjunction with, and forms part of, the Data Collection, Processing and Transfers section above):

GS Inc., identified by Tax ID Number (EIN) No. 134019460, with registered office at 200 West Street, New York 10282 - United States, will use, collect, transfer and store your following personal data for the purposes of administering the Plan: name, address, work location, hire date, Social Security or Social Insurance or taxpayer identification number (required for tax purposes), type and amount of Plan or other benefit plan award, citizenship or residency (required for tax purposes) and other similar information. Your personal data is necessary for the administration of the Plan and so, if you express your refusal to provide it, GS Inc. will not be able to administer the Plan in respect of your Award. Your personal data will be stored in a data bank owned by GS Inc., the identity and physical address of which can be obtained upon your request, for the period of administration of the Plan and potentially beyond that to the extent permitted under the laws of the applicable jurisdiction(s). Your personal data may be sent to GS Inc. affiliates (detailed at <https://www.goldmansachs.com/our-firm/locations>) and service providers (including Computershare Limited and its affiliates; and Fidelity Stock Plan Services, LLC, Fidelity Personal Trust Company, FSB and any of their affiliates) whether in the U.S. or elsewhere, as is reasonably necessary for the administration of the Plan and under the laws of these jurisdictions. You may exercise your rights of access, rectification and cancellation by contacting Equity Compensation (division of HCM) at _____.

Información adicional sobre protección de datos para los empleados de Perú (que debe leerse junto con la sección Recolección, el Tratamiento y la Transferencia de Datos anterior y forma parte de ella):

GS, Inc., identificado con el número de identificación fiscal (Employer Identification Number, EIN) No. 134019460, con domicilio legal en 200 West Street, New York 10282 - Estados Unidos (en adelante, “GS Incs.”), utilizará, recopilará, transferirá y almacenará sus siguientes datos personales con el fin de administrar el Plan de Incentivos en Acciones Modificado y Reiterado de Goldman Sachs (en adelante, el “Programa”): nombre, dirección, lugar de trabajo, fecha de contratación, número de identificación de Seguridad Social, Seguro Social o de contribuyente (requeridos para efectos fiscales), tipo y cantidad del Plan u otras concesiones de planes de beneficios, ciudadanía o residencia (requeridos para efectos fiscales) y otros datos similares. Sus datos personales son necesarios para la ejecución del Programa; por lo tanto, si usted manifiesta su negativa a

proporcionarlos, Goldman Sachs no podrá administrar dicho Programa con respecto a su Premio. Sus datos personales serán almacenados en un banco de datos personales de propiedad de GS Inc., cuya identidad y dirección física del mismo se le proporcionará a su requerimiento, durante el tiempo necesario para la administración del Programa y potencialmente durante más tiempo para el cumplimiento de las obligaciones legales que establezca la regulación aplicable. Sus datos personales podrían ser enviados a las afiliadas de GS Inc., detalladas en <https://www.goldmansachs.com/our-firm/locations>; y a proveedores de servicios o encargados de tratamiento (incluyendo a Computershare Limited y sus filiales; y Fidelity Stock Plan Services, LLC, Fidelity Personal Trust Company, FSB y cualquiera de sus filiales) ubicados en Estados Unidos de América y en cualquier otro país; ello en tanto sea razonablemente necesario para la administración del Programa y bajo las leyes de tales jurisdicciones. Podrá ejercer sus derechos de acceso, rectificación y cancelación contactando a Equity Compensation (división de HCM) a través del correo _____.

FOR RUSSIA EMPLOYEES ONLY

None of the information contained in the Award Documentation constitutes an advertisement of the Award in Russia and must not be passed on to third parties or otherwise be made publicly available in Russia. The Award has not been and will not be registered in Russia and are not intended for "placement" or "public circulation" in Russia.

You understand and agree that the Firm may, in its sole discretion or where required by legal or regulatory requirements, grant Awards as cash-settled instruments or settle Awards in cash and that you will not have any rights to the Shares underlying your Award (if any). Your Award will remain subject to the terms and conditions contained in the Award Documentation.

FOR SAUDI ARABIA EMPLOYEES ONLY

The Award Documentation may not be distributed in the Kingdom except to such persons as are permitted under the Rules on the Offer of Securities and Continuing Obligations issued by the Capital Market Authority. The Capital Market Authority has not reviewed the Award Documentation, nor does it make any representation as to the accuracy or completeness of the Award Documentation. You acknowledge the risks associated with any investment made pursuant to the Award Documentation, including what may result in loss of the full amount of the investment. You further acknowledge that the Firm does not have to notify the Capital Market Authority of the suitability of any investment you make pursuant to the Award Documentation. The Capital Market Authority expressly disclaims any liability whatsoever for any loss arising from, or incurred in reliance upon, any part of the Award Documentation. Prospective purchasers of the securities offered hereby should conduct their own due diligence on the accuracy of the information relating to the securities. The informational materials prepared in connection with the Plan have been produced for information purposes only and are not intended to induce investment. If you do not understand the contents of the Award Documentation you should consult an authorized financial adviser.

FOR SINGAPORE EMPLOYEES ONLY

The Award is a prescribed capital markets product (as defined in the Securities and Futures (Capital Markets Products) Regulations 2018) and Excluded Investment Product (as defined in MAS Notice SFA 04-N12: Notice on the Sale of Investment Products and MAS Notice FAA-N16: Notice on Recommendations on Investment Products).

The Shares or the Award may not be offered or sold, or be made the subject of an invitation for subscription or purchase, whether directly or indirectly, to persons in Singapore other than pursuant to, and in accordance with the conditions of, an exemption under any provision of Subdivision (4) of Division 1 of Part 13 of the Securities and Futures Act 2001.

FOR SOUTH KOREA EMPLOYEES ONLY

If you are located in South Korea, by signing this Signature Card and accepting your Award, you consent to the processing of your personal data in accordance with the information set out in the Data Collection, Processing and Transfers section above. Specifically:

- **The Firm will collect and use your personal data as set out in the Data Collection, Processing and Transfers section. You consent to such use.**
- **The Firm will collect and use your unique identification information (foreigner registration number, passport number, driver's licence number) as set out and underlined in the Data Collection, Processing and Transfers section. You consent to such use.**
- **The Firm will provide your personal data to third parties, as described in the Data Collection, Processing and Transfers section and the hyperlinks within it. You consent to such transfers.**

- **The Firm will provide your unique identification information to third parties, as described by the Data Collection, Processing and Transfers section and the hyperlinks within it. You consent to such transfers.**
- **The Firm will transfer your personal data and unique identification number (including foreigner registration number, passport number and driver's license number) overseas to the United States and other countries, as described by the Data Collection, Processing and Transfers section and the hyperlinks within it. You consent to such overseas transfers.**

FOR SPAIN EMPLOYEES ONLY

You may be subject to certain reporting obligations for the acquisition or disposal of Shares under the Plan, the opening of cash or brokerage bank accounts abroad and the transfer or receipt of funds. Please consult your own advisors regarding these and other legal or tax obligations that may be applicable.

FOR TURKEY EMPLOYEES ONLY

This offer is not a public offering in terms of the Turkish Capital Markets legislation and the information provided herein cannot be construed as a public offering. The grant of the Award should not be construed as a public offering or a private placement and is made to you as an employee of the Firm. You are not obligated to accept your Award. Your decision to accept or reject the Award is entirely up to you and will have no impact on your employment or your career, either positive or negative. The grant of your Award does not change or supplement the terms of your employment in any way. The Award Documentation does not constitute an employee handbook or an employment contract between you and the Firm.

The information set forth in the Award Documentation is solely for informative reasons and the Firm is not hereby giving you nor purports to be giving you investment or other financial advice. The Firm reserves the right to suspend, change, amend or supplement the terms of the Award Documentation, in whole or in part, for any reason at any time. If you are in doubt about the merits of the Award Documentation, you should contact your financial advisor.

Additional data protection information for Turkey employees (which should be read in conjunction with, and forms part of, the Data Collection, Processing and Transfers section above):

In terms of the Law on the Protection of Personal Data No. 6698 and its relevant legislation, GOLDMAN SACHS TK DANIŞMANLIK HİZMETLERİ ANONİM ŞİRKETİ acts as the data controller regarding your personal data in Turkey.

FOR UK EMPLOYEES ONLY

This document does not have regard to the specific investment objectives, financial situation and particular needs of any specific person who may receive it. Recipients should seek their own financial advice.

The Award is subject to the terms and conditions set forth in the Documentation. The price of Shares and the income from such Shares (if any) can fluctuate and may be affected by changes in the exchange rate for U.S. dollars. Past performance will not necessarily be repeated. Levels and bases of taxation may change from time to time. Investors should consult their own tax advisors in order to understand tax consequences. GS Inc. has (and its associates may have) a material interest in the Shares and the investments that are the subject of this document.

CERTAIN INFORMATION, IDENTIFIED BY AND REPLACED WITH A MARK OF “[],” HAS BEEN EXCLUDED FROM THIS DOCUMENT BECAUSE IT IS BOTH (I) NOT MATERIAL AND (II) THE TYPE THAT THE REGISTRANT TREATS AS PRIVATE OR CONFIDENTIAL.

**Form of
Amended and Restated
Agreement of Limited Partnership**

for

Participants in the Long Term Executive Carried Interest Incentive Program

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**AMENDED AND RESTATED
AGREEMENT OF LIMITED PARTNERSHIP**

AMENDED AND RESTATED AGREEMENT OF LIMITED PARTNERSHIP, dated as of [], by and among [], as general partner (the “General Partner”), the Persons who have signed this Amended and Restated Agreement of Limited Partnership (together with the subscription agreement or other related materials entered into by any Limited Partner, this “Agreement”), as limited partners (together with the General Partner, the “Partners”), the Initial Limited Partner and [] (the “Original General Partner”).

WHEREAS, the General Partner is a wholly owned subsidiary of The Goldman Sachs Group, Inc., a Delaware corporation (“GS Group” and, together with Goldman Sachs & Co. LLC and GS Group’s other subsidiaries and affiliates, including the General Partner, “Goldman Sachs”) or the “Firm”;

WHEREAS, the limited partners of the Fund primarily consist of (A) “Eligible Investors” who are (i) participants in the Goldman Sachs Partner Compensation Plan for fiscal year [], Participating Managing Directors, Senior Advisors, Extended Managing Directors of Goldman Sachs and certain other eligible investors; and/or (ii) the affiliated entities (“Affiliated Investor Entities”) of certain of the investors described in clause (i); (B) Goldman Sachs, upon the acquisition of interests in the Fund pursuant to Article VI and/or Article VIII; (C) the “Institutional Limited Partners,” which are one or more entities organized by Goldman Sachs primarily for investment by Eligible Investors in the Fund; and (D) SOX Insider Preferred Limited Partners, who are the holders from time to time of SOX Insider Preferred Interests;

WHEREAS, Goldman Sachs caused the formation of the Fund for the benefit of the Limited Partners and the Limited Partners seek to pool investment resources under the sponsorship of Goldman Sachs;

WHEREAS, the General Partner and the Initial Limited Partner formed a series limited partnership (the “Fund”) pursuant to the terms and provisions of an agreement of limited partnership (the “Original Agreement”), dated as of [], and the filing of the certificate of limited partnership of the Fund (the “Certificate of Limited Partnership”) in accordance with the statutes and laws of the State of Delaware, relating to limited partnerships, including the Limited Partnership Act of the State of Delaware (the “Delaware Act”); and

WHEREAS, the Partners and the Initial Limited Partner desire to amend and restate the terms and provisions of the Original Agreement to *inter alia*, admit [] as the General Partner and evidence the withdrawal of the Original General Partner.

NOW, THEREFORE, in consideration of the mutual agreements and covenants set forth herein and for good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, and intending to be legally bound hereby, the parties agree to amend and restate the Original Agreement in its entirety to read as follows:

Article I

Name and Place of Business; Powers;
Purpose; Term; Continuance and Qualification

1.1. Name. The name of the Fund shall be [] or such other name as the General Partner may from time to time designate. The business of the Fund may be conducted under any other name deemed necessary or desirable by the General Partner in order to comply with local law.

1.2. Place of Business. The principal place of business of the Fund shall be at [], and/or at such other place within or outside [] as the General Partner may hereafter designate in writing to the Limited Partners from time to time.

1.3. Registered Office. The initial address of the Fund's registered office is [].

1.4. Purpose

(a) The purposes of the Fund are to engage in any lawful act or activity for which limited partnerships may be formed under the Delaware Act, including investing all or substantially all of its committed capital in a diversified set of equity-oriented funds sponsored by Goldman Sachs Asset & Wealth Management, as is more fully described in the Offering Memorandum (any such investment of any type held, directly or indirectly, by the Fund, any wholly or partly owned subsidiary (a "Subsidiary"), any Investment Vehicle or Alternative Investment Vehicle from time to time, a "Fund Investment"), and engaging in any other activities which may be directly or indirectly related or incidental to any of the foregoing. In particular, the Fund expects to invest in: [] (collectively, the "Underlying Funds" and each, separately, an a "Underlying Fund"). In the sole discretion of the General Partner, the Fund may determine not to invest in one or more of the Underlying Funds, or may invest in other underlying funds. The Underlying Funds are expected to invest alongside funds formed for clients of Goldman Sachs (the "External Funds").

(b) The Fund, together with one or more of [], the "Funds"), may not invest in an Underlying Fund (or its offshore analogue), or will invest in an Underlying Fund using different structures. The Underlying Funds may make their investments through one or more U.S. or non-U.S. investment or finance vehicles organized by or for the applicable Underlying Fund, that may be wholly owned by such Underlying Fund or owned by the applicable Underlying Fund together with any wholly or partially owned Subsidiary, Alternative Investment Vehicles, other investment funds or vehicles managed by Goldman Sachs (the "Other Funds"), and/or other investors (each, together with their respective permitted assignees or designees, an "Investment Vehicle").

(c) It is understood by all the parties hereto that the interests of the clients of Goldman Sachs and the legal and ethical obligations of Goldman Sachs to its clients are paramount and prior to the interests of the Fund or any Partners thereof. Consequently, the Fund shall not make any investment which Goldman Sachs determines to be inappropriate in light of such obligations. The Fund shall promptly dispose of any investment which Goldman Sachs deems to be inappropriate in light of such obligations. The Fund shall not enter into any transaction with any Person, nor accept any investment opportunity from any Person, if deemed by Goldman Sachs to be inappropriate in light of such obligations.

1.5. Status for U.S. Federal Income Tax Purposes. The Fund and, if established, each Series intends, and shall take all appropriate steps, to be treated as a partnership for U.S. federal income tax purposes and, in the discretion of the General Partner, for purposes of the tax laws of any other relevant jurisdiction for which such treatment is available.

1.6. Powers and Authority. The Fund and each Series shall possess and may exercise all powers, authority and privileges granted by the Delaware Act, any other law or this Agreement, including the power and authority to perform any act or carry out any activity or transaction contemplated by or in furtherance of this Agreement or as is in the General Partner's determination necessary, appropriate, desirable, incidental or convenient to or for the furtherance or accomplishment of the foregoing objectives and purposes.

1.7. Term. Subject to the provisions of Section 9.1, the Fund will dissolve and terminate one (1) year after the date by which all of the Fund's investments have been liquidated and the Fund's obligations (including, without limitation, contingent obligations) have been terminated (or reasonable provision has been otherwise made for such obligations), provided that the General Partner may amend this Agreement without the consent of the Limited Partners: (a) in accordance with Section 11.2(d)(vii), to set the date on which the Fund shall dissolve and terminate on any date not later than the fifteenth anniversary of the formation of the Fund and (b) to limit the duration of the Fund's investments in any Fund Assets to a period of 15 years to satisfy certain requirements under the U.S. Bank Holding Company Act of 1956, as amended (the "BHCA"), subject in all cases to the General Partner's right, pursuant to Section 9.1, to liquidate the Fund at any time.

1.8. Continuance/Qualification. In the event that it shall be necessary for the Fund to exist or qualify to do business under the laws of any jurisdiction other than or in addition to the State of Delaware, the parties hereby

agree that the Fund shall take such action as may be necessary to exist or qualify to do business in any state in which such existence or qualification shall be required; provided, that, in any such event the Fund shall at all times continue to be a limited partnership formed under and governed by the provisions of the Delaware Act.

1.9. Establishment of Series.

(a) Each Limited Partner understands and acknowledges that this Agreement grants the General Partner the right to establish one or more separate Series of partners or partnership interests pursuant to Section 17-218 of the Delaware Act, with such terms as may be determined by the General Partner in its sole discretion. Each Limited Partner understands and acknowledges that such Limited Partner may not hold an interest in each Series. In addition, each Limited Partner understands and acknowledges that, except as expressly provided in this Agreement, such Limited Partner has no right to participate, and the General Partner has no duty or obligation to offer to such Limited Partner the right to participate, in any Series.

(b) The terms of each Series shall be as set forth in this Agreement, and any supplement to this Agreement relating to a particular Series (each, a "Series Supplement"). For all purposes of the Delaware Act, this Agreement together with each Series Supplement, if any, constitute the "partnership agreement" of the Fund within the meaning of the Delaware Act. Notwithstanding any provision of this Agreement to the contrary, the terms and provisions of a Series Supplement may have the effect of altering, supplementing or amending the terms and provisions of this Agreement, and to the extent that any of the terms or provisions of a Series Supplement conflict with any of the terms or provisions of this Agreement, the terms or provisions of such Series Supplement shall control with respect to such Series. As established from time to time in accordance with this Agreement, a Series may be designated as having separate rights, powers or duties with respect to specified assets, property, obligations or liabilities or profits and losses associated with specified assets, property, obligations or liabilities and, to the extent provided in this Agreement or the Series Supplement that corresponds to such Series, as having a separate business purpose or investment objective than the Fund or any other Series. A Person may be admitted as a limited partner of the Fund associated with more than one Series. The General Partner shall be the sole general partner of the Fund associated with each Series.

(c) All capital contributions, assets, income, fees, earnings, profits and proceeds held or received by the Fund in respect of a Series shall be deemed to be and may be referred to herein as "assets belonging to" that Series (and to no other Series) for all purposes hereof.

(d) The assets belonging to a particular Series shall be so recorded upon the books of the Fund, and shall be held in trust for the benefit of the Limited Partners of such Series. The assets belonging to each particular Series shall be charged with the liabilities of that Series, and with all expenses, costs, charges and reserves attributable to that Series. The debts, liabilities, obligations and expenses incurred, contracted for or otherwise existing with respect to a particular Series shall be enforceable against the assets belonging to such Series only, and not against the assets of the Fund generally or of any other Series. No Limited Partner or former Limited Partner of any Series shall have any claim on or right to any assets allocated or belonging to any other Series.

(e) Separate and distinct records shall be maintained by the Fund for each Series and the assets associated with or belonging to any such Series shall be held and accounted for separately from the assets of the Fund or of any other Series. To the fullest extent permitted under the Delaware Act, the Limited Partners of a Series shall only have access to the records of such Series, including the Series Supplement that corresponds to such Series.

(f) "Series" shall mean any separate series of partners or partnership interests established by the General Partner pursuant to Section 17-218 of the Delaware Act and this Section 1.9. References in this Agreement to the "Fund" shall, to the extent the context requires, be deemed to be references to one or more Series.

Article II

The General Partner: Relationship with Goldman Sachs

2.1. The General Partner. The name, registered office, business address and capital commitment of the General Partner are set forth on Exhibit A hereto, as amended from time to time, and are incorporated herein by reference. The General Partner shall acquire a Class O Interest as a Partner in exchange for its capital commitment. Goldman Sachs is authorized to admit any Person to the Fund as a general partner, to cause the withdrawal of any general partner of the Fund and to effect the transfer of a general partner interest from any general partner to any successor general partner, subject to the restrictions set forth herein. All references in this Agreement to the General Partner shall be deemed to include [] and any successor general partner.

2.2. Management. The General Partner shall have the exclusive right to manage, conduct and control the affairs of the Fund. To the extent permitted by the General Partner's organizational documents, the General Partner is authorized at any time to designate one or more managers or successor managers, to remove any manager or successor manager and to designate a replacement manager or managers.

2.3. Delegation of Managerial Authority. It is understood that the General Partner may delegate any or all of its duties, rights and responsibilities in respect of the Fund (other than investment decisions) to its managers, officers, affiliates and/or to third parties selected by the General Partner. In addition, subject to guidelines in keeping with the investment objectives and policies of the Fund as set forth in the confidential offering memorandum relating to the Fund, dated [] (as supplemented from time to time, the "Offering Memorandum"), the General Partner may delegate final investment decisions to committees of one or more officers, directors or employees of Goldman Sachs. The General Partner initially delegates managerial authority to [] (the "Investment Manager"), and shall, on behalf of and in the name of the Fund, enter into an investment management agreement with the Investment Manager, all without the further approval of any Limited Partner. It is understood that the Investment Manager may delegate its managerial authority to other or additional officers, directors or employees of Goldman Sachs. In addition, the General Partner hereby admits [] as a Limited Partner in the Fund to serve as the "qualified institutional investor" in respect of the Fund for purposes of non-U.S. securities laws. The Limited Partners acknowledge and agree that, so long as such delegation remains in effect, (i) the General Partner shall have no responsibility for making any investment, disposition or management decisions on behalf of the Fund, and (ii) the Investment Manager shall have sole authority to make such investment, disposition and management decisions, including the authority to approve all investments in Underlying Funds.

2.4. No Compensation of Members and Officers. Neither the members of any investment committee that evaluates potential investments in Underlying Funds nor the officers of the General Partner shall be entitled to receive any compensation from the Fund for their services.

2.5. Reliance by Third Parties. Persons dealing with the Fund are entitled to rely conclusively upon a certificate of the General Partner to the effect that it is then acting as the General Partner and upon the power and authority of the General Partner and of any Person to whom the General Partner may have delegated all or part of such power and authority as herein set forth.

2.6. Services by Goldman Sachs; Expenses.

(a) The Fund (and each of the Other Funds) shall be responsible for its *pro rata* share (based on its aggregate capital commitments as a proportion of the aggregate capital commitments to the Fund and the Other Funds) of all the expenses and fees incurred in connection with (1) the organization of, and the offering of limited partnership interests in the Fund and the Other Funds and (2) any Fund Expenses incurred by the Fund; provided that, in circumstances where the General Partner reasonably believes that such expenses and fees allocated to the Fund and/or the Other Funds pursuant to the foregoing procedures would produce an inequitable result, or are directly attributable or allocable to the Fund and/or an Other Fund, the General Partner may allocate such expenses and fees to the Fund or such Other Fund, as it determines in its sole discretion, in a fair and equitable manner, which may not be on a *pro rata* basis. All Fund Expenses shall be paid out of capital contributions, borrowings or proceeds or other cash funds of the Fund that are determined by the General Partner to be available for such purpose; provided that the General Partner and/or the Investment Manager may advance funds or arrange for one of its affiliates to advance funds to the Fund for the payment of Fund Expenses and the General Partner or such affiliate shall be entitled to the reimbursement, without interest, of any funds so advanced. As used herein, "Fund

Expenses" shall include, whether incurred directly or indirectly by or on behalf of the Fund, including by the Investment Manager, or any Investment Vehicle, but not be limited to, the following (to the extent not reimbursed by the Portfolio Companies or any other Person):¹

(i) all costs and expenses² necessary to register or qualify the Fund, and, if established, any Series, under any applicable laws, rule or directive or any other regulatory requirement, including U.S. federal or state or foreign laws, to maintain such registrations or qualifications or to obtain or maintain exemptions under such laws, or compliance with the foregoing requirements by Goldman Sachs or its affiliates to the extent such compliance relates to the Fund's activities;

(ii) all costs and expenses charged to the Fund, including all fees or expenses that the General Partner determines to be related to the acquisition or potential acquisition, holding or disposing of investments and potential investments in the Underlying Funds as well as any other taxes, governmental charges and other costs and expenses associated with Fund Investments and obtaining leverage;

(iii) administration of the Fund, including (x) all external legal, tax and accounting costs and expenses, including costs and expenses for preparation of annual audited financial statements, tax return preparation (or preparation of other tax filings or elections), routine tax and legal advice, other tax-related expenses, including any expenses incurred by the Partnership Representative, external insurance premiums charged to the Fund (which insurance premiums may cover numerous Goldman Sachs advisory accounts, in which case the Goldman Sachs advisory accounts shall be allocated a share of such premiums), wire transfer fees, mailing costs, amendments to this Agreement and any costs and expenses related to winding-up, dissolution, liquidation or termination of the Fund and (y) fees and expenses charged to the Fund in connection with services performed by internal staff of Goldman Sachs and/or entities affiliated with Goldman Sachs for internal accounting services in an amount equal to \$[] per each Fund per quarter;

(iv) all costs and expenses incurred in relation to the registration of any Fund Investments in the name of the Fund or its nominee or the custody of the documents of title thereto (including bank charges, insurance of documents of title against loss in shipment, transit or otherwise and charges made by agents of the Fund for retaining documents in safe custody);

(v) indemnification and contribution obligations of the Fund, including legal advice and other external costs and expenses with respect to any existing or potential claim, dispute or litigation and the amount of any judgment or settlement paid in connection therewith; and

(vi) any other costs and expenses, as reasonably determined by the Investment Manager in good faith, that may be authorized by this Agreement or as may be necessary or appropriate in managing the Fund's business.

(b) [] In addition, the Limited Partners will indirectly bear a proportionate share of the organizational and operating fees and expenses of the Underlying Funds. Where appropriate, the General Partner may specifically allocate a portion of Fund Expenses to one or more particular Limited Partners.

¹ In all relevant cases, including any travel and entertainment expenses, including private air travel when deemed appropriate by the General Partner in its reasonable discretion (after taking into account the risks associated with public health crises such as the COVID-19 pandemic).

² In all relevant cases, costs shall include fees for purpose of the definition of "Fund Expenses."

(c) It is understood that stock brokerage and investment advisory services for the Fund generally will be performed, and other administrative and similar services for the Fund may be performed, by Goldman Sachs. In connection with investments in money market funds advised or administered by Goldman Sachs, the assets will be subject to the customary advising and/or administrative fee charged to and paid by third party customers participating in the same fund. In addition, Goldman Sachs may, to the extent permitted by law, require the Fund to bear customary commissions and fees for brokerage and other services performed by Goldman Sachs [].

2.7. Investment Banking and Other Activities. The Limited Partners acknowledge and agree that, subject to applicable law:

(a) the Investment Manager, the General Partner or any of their agents or affiliates (on behalf of the Fund) and any entity in which an Underlying Fund invests, directly or indirectly (each, a “Portfolio Company”) may engage Goldman Sachs (and Goldman Sachs may act in its individual capacity and for its own account) as investment bankers, underwriters, financial advisors, asset managers, placement agents, selling agents, or brokers, dealers or traders in securities (including warrants and options), real estate, hedging and derivative instruments, structured financial products, foreign exchange and commodities, including in connection with the acquisition, holding, disposition, liquidation or bankruptcy of any investment;

(b) Goldman Sachs may make interest-bearing loans to Portfolio Companies and the Fund, and may act as agent in connection with the placement or syndication of indebtedness or other instruments or securities of Portfolio Companies and the Fund, and may acquire, hold or dispose of any notes, debt securities, other evidences of indebtedness or other instruments or securities of Portfolio Companies and the Fund;

(c) Goldman Sachs has established and may establish entities that may provide investment, fund accounting, finance, loan servicing, asset management, portfolio management, risk management, cash management, due diligence, tax coordination, information technology and/or administrative or other services to one or more other Goldman Sachs entities;

(d) the Fund, any Portfolio Companies and any competitor or counterparty of any of the foregoing, may pay, and Goldman Sachs may receive and retain from the Fund, Portfolio Companies and any competitor or counterparty of any of the foregoing, fees (including sponsor fees), commissions, discounts, interest and other sums, and Goldman Sachs may earn profits in connection with any of the foregoing;

(e) Goldman Sachs may sponsor, manage or advise investment funds or vehicles that may seek to make private investments in securities or other instruments, sectors or strategies in which the Fund may invest, and may create subsequent funds (including any successor fund), and Goldman Sachs may earn profits in connection therewith;

(f) Goldman Sachs may invest in investments of the type sought by the Fund or any Portfolio Company, and Goldman Sachs may earn profits in connection therewith; and

(g) neither the Fund nor any Limited Partner shall have any interest in any such profits, fees, commissions, discounts, interest and other sums by virtue of this Agreement or the partnership relation created hereby and any such amounts shall not reduce the [] or otherwise be shared with the Fund or the Limited Partners; provided, that in the case of clauses (a), (b), (c) and (d) above, with respect to services rendered to the Fund, the services rendered are determined by the General Partner in good faith to be appropriate and useful; the Persons or entities rendering such services are qualified to do so and the fees or other amounts charged in respect of such services are determined by the General Partner to be commercially reasonable.

Article III

Limited Partners

3.1. Initial Limited Partner. The initial limited partner (the "Initial Limited Partner") has become such only for the purpose of organizing the Fund and has made a capital commitment of \$[] to the Fund. On the first day of admission of additional Limited Partners, the Initial Limited Partner shall withdraw from the Fund, the Initial Limited Partner's funded capital (if any) shall be returned, together with a return thereon (if any), in accordance with the Original Agreement, and the Initial Limited Partner shall have no other rights, obligations or liabilities whatsoever with respect to the Fund in the Initial Limited Partner's capacity as Initial Limited Partner.

Unless the context otherwise specifically requires, references in this Agreement to the Limited Partners, their capital and their rights and obligations shall not be references to the Initial Limited Partner.

3.2. Additional Limited Partners; Classes of Interests.

(a) The General Partner is authorized to admit additional limited partners, including Limited Partners and SOX Insider Preferred Limited Partners, to the Fund pursuant to the terms contained in the Offering Memorandum and this Agreement. A Person shall be admitted as a Limited Partner of the Fund at the time that: (i) this Agreement, a counterpart hereof or an instrument incorporating by reference the terms of this Agreement is executed by or on behalf of such Person, the General Partner and each other Limited Partner and (ii) the General Partner consents to the admission of such Person as a Limited Partner.

The Fund generally shall issue Class A limited partnership interests ("Class A Interests") [] and/or Class O limited partnership interests ("Class O Interests") to each subscribing Eligible Investor and Institutional Limited Partner. Such Eligible Investors and Institutional Limited Partners holding Class A Interests and any other holder of Class A Interests shall be referred to as "Class A Limited Partners," [] and such Eligible Investors and Institutional Limited Partners holding Class O Interests and any other holder of Class O Interests shall be referred to as "Class O Limited Partners." The General Partner, Goldman Sachs or their affiliates are also authorized to issue or to convert the interest, or a portion thereof, of any Class A Limited Partner [] into separate classes of limited partnership interests, to be referred to as [] and collectively, with the Class A Interests [], together with the Class O Interests, are the "Participating Interests," and, together with SOX Insider Preferred Interests, the "Interests," and the holders thereof, other than the General Partner, the "Limited Partners").

With respect to the Interests:

(i) []

(ii) []

(iii) []

(iv) Class O Interests. The Class O Interests shall, subject to the final sentence of this Section 3.2(a)(iv), share in a portion (such portion, the "Sharing Percentage") of any carried interest (the "Carried Interest") that is earned in respect of one or more (but not necessarily all) of the External Funds (as determined, and subject to change, in the General Partner's sole discretion) as well as other external funds, entities, separate accounts and/or single investor vehicles managed or advised by [], in each case, as determined by Goldman Sachs in its sole discretion (all such funds, and (as the context requires) any subsidiary investment vehicles and/or alternative investment vehicles of such external funds, entities, separate accounts and/or single investor vehicles in which the Fund will acquire, directly or indirectly, a special limited partnership or equivalent interest, the "Carried Interest Paying Funds"). The Class O Interests shall not otherwise share in the profits and losses from the Underlying Funds' investment

activities and operations and will not be subject to a management fee or carried interest. The Sharing Percentage may be adjusted from time to time, including in connection with adjustments as described in Section 6.6(c)(vi) below. The Carried Interest will not include any amounts received in respect of distributions from the Carried Interest Paying Funds of Goldman Sachs' direct or indirect capital investment in the Carried Interest Paying Funds, any returns on Goldman Sachs' investment in the Carried Interest Paying Funds (including carried interest with respect thereto) or in respect of any other amounts received from the Carried Interest Paying Funds other than the Carried Interest.

(b) ☐

(c) ☐

(d) To the extent permitted by law, including the Dodd-Frank Wall Street Reform and Consumer Protection Act (including its Section 619 known as the "Volcker Rule," (the "Volcker Rule")) as amended, together with the regulations promulgated thereunder (the "Dodd-Frank Act"), from time to time Goldman Sachs may (but is not obligated to) acquire Interests from Limited Partners who are no longer employees and in certain other circumstances. Goldman Sachs may, but shall not be obligated to, offer these Interests on a discretionary basis to certain investors, including to Persons who are currently not Eligible Investors, but are Eligible Investors at such time. To the extent that the General Partner, Goldman Sachs or any of their affiliates purchase or acquire an Interest, they shall be treated, for all purposes, as being the owner of that Interest, and entitled to the economics associated therewith.

(e) Following the initial closing date and through such date as the General Partner shall determine, the General Partner is authorized to admit additional Eligible Investors and Institutional Limited Partners, including SOX Insider Preferred Limited Partners, to the Fund and accept additional commitments from existing Limited Partners. ☐ At the time that any additional Limited Partners are admitted to the Fund, the Capital Accounts of the General Partner and/or Limited Partners other than such additional Limited Partners shall be fairly adjusted to give effect to the intent of the provisions of this Section 3.2(e). ☐

(f) The manner of the offering of the Interests, the terms and conditions under which subscriptions for such Interests will be accepted, the manner of and conditions to the sale of Interests to subscribers therefor and the admission of such subscribers as additional Limited Partners will be as provided in the Offering Memorandum, their respective subscription agreements and this Agreement and subject to any provisions thereof and hereof. Consistent with the foregoing and Articles VII and VIII, the Fund may also admit additional Limited Partners or SOX Insider Preferred Limited Partners.

(g) The General Partner may, without the consent of any Limited Partner, amend this Agreement in order to create and offer one or more new Series or classes of interests, convert existing classes of interests or make any other changes necessary or advisable, in the determination of the General Partner, for purposes of allocating special opportunities, such as carry distributions, or otherwise, as determined by the General Partner in its sole discretion, in which case any such Series or classes of interests shall also be deemed an Interest. In connection with the issuing of any additional Series or class of interests in the Fund in accordance with this Section 3.2 (or the conversion of interests in accordance with this Section 3.2), the General Partner may amend this Agreement, without the consent of or notice to the Limited Partners, as necessary to provide for the establishment of the additional class or Series of interests in the Fund (including a class or Series issued upon conversion) and to reflect the economics of such additional class or Series of Interests and the impact on the existing Series, class or classes of interests in the Fund.

(h) The General Partner may offer additional Series or classes of interests in the Fund to the Partners, in proportion to their Interests in the Fund, or otherwise. In addition, subject to the terms and conditions of the application filed on ☐ with the Securities and Exchange Commission (the "SEC") by Goldman Sachs seeking an order exempting the Fund from most, but not all, of the provisions of the Investment Company Act, as amended and/or supplemented (the "Application," a copy of which will be furnished to Limited Partners upon written request), and any order approving the Application, and applicable law, including the Dodd-Frank Act (including the

Volcker Rule), the General Partner may offer additional Series or classes of interests in the Fund, for subscription by other Eligible Investors and third parties, for any purpose, including to raise sufficient capital to fund expenses or meet the other obligations of the Fund if the Fund does not have sufficient funds to pay such expenses or meet such other obligations.

3.3. List of Limited Partners; Investment Records. The General Partner shall maintain a list, as amended from time to time, of Limited Partners setting forth the name, residence or business address, capital commitment and class of each Limited Partner.

3.4. No Management by Limited Partners. No Limited Partner, in its capacity as a limited partner of the Fund, shall participate in the management, control and conduct of the business of the Fund, transact any business in the Fund's name, have the power to sign documents for or otherwise bind the Fund, or deal with any Persons who are not Partners on behalf of the Fund.

3.5. []

Article IV

Liability of Partners

4.1. General Partner. Except as provided in Section 6.1(a), the General Partner shall not be required to contribute to the capital of, or loan to, the Fund any funds. Neither the General Partner nor any of its affiliates shall have any personal liability for the return or repayment of the capital contribution of any Limited Partner. Except as provided herein or by the Delaware Act, the General Partner shall have the liabilities of a partner in a partnership without limited partners.

4.2. Limited Partners. In no event shall any Limited Partner (a) be obligated to make any contribution or payment to the Fund or Goldman Sachs, except as expressly provided under his, her or its subscription agreement or this Agreement or (b) have any personal liability for the repayment or discharge of debts and obligations of the Fund, except in each case of clause (a) or (b), as provided under the Delaware Act.

[]

Article V

Powers of General Partner; Prohibited Transactions and Restrictions; Duties of General Partner; Indemnification and Contribution

5.1. Powers of the General Partner. In addition to and not in limitation of any rights and powers conferred by law or other provisions of this Agreement, and except as limited, restricted or prohibited by the express provisions of this Agreement, including Section 5.5, the General Partner (and any of its assignees or designees) shall have and may exercise on behalf of itself, the Fund or any entity controlled or owned by the Fund all powers and rights necessary, proper, convenient or advisable (as determined by the General Partner) to effectuate and carry out the purpose, business and objectives of the Fund or any entity controlled by the Fund. Such powers shall include the following powers, which may be carried out by the General Partner or directly or indirectly through the Fund or one or more Investment Vehicles or other wholly or partly owned subsidiaries by the General Partner, on behalf of the Fund or any entities controlled or owned by the Fund, in each case, to the extent permitted by applicable law including the Dodd-Frank Act (including the Volcker Rule):

(a) to invest the capital contributions of the Partners (after setting aside suitable reserves) and reinvest proceeds of the Fund in accordance with the provisions of this Agreement;

(b) to purchase or otherwise acquire, to invest in, to hold for investment, to sell, exchange or convert or otherwise dispose of, to mortgage, to grant security interests over, to pledge or otherwise deal in, (i) capital stock, pre-organization certificates and subscriptions; (ii) warrants, partnership interests, equity interests in trusts and limited liability companies, convertible debt securities and other equity and equity-related securities and similar securities or instruments; (iii) mezzanine debt, subordinated debt, notes (including notes issued pursuant to indentures, note purchase agreements and other credit facilities), loans, and other debt instruments and securities, and other evidences of indebtedness (whether secured or unsecured and including loans acquired with or without

equity or equity related securities); (iv) preferred stock; (v) convertible securities and forward, swap and option contracts, short sales, or any other financial instruments with similar characteristics (including interest rate swaps, exchanges and caps, collars, floors, currency options, futures and options on futures, commodity swaps and commodity options, total return swaps, back to back agreements and/or similar agreements) or participation interests in respect thereof; (vi) trust receipts; and (vii) assets and other property, including real property, intellectual property, and property leased or operating under a concession from a regulatory body or a sovereign government; and in the case of securities, regardless of whether such securities are readily marketable or not and whether of a speculative nature or not, and in rights and options relating thereto and/or to exercise all rights, powers, privileges, and other incidents of ownership or possession with respect to securities held or owned by the Fund (including (i) any voting rights and other rights of the Fund as a security holder or as party to a shareholder or similar agreement and (ii) rights to elect to adjust the tax basis of Fund assets);

(c) to enter into, make and perform all such contracts and other undertakings, and engage in all such activities and transactions (including financings), as the General Partner may deem necessary or advisable for or incidental to the carrying out of the foregoing objects and purposes, including:

(i) to pursue, negotiate and enter into transactions with respect to Fund Assets and, in connection therewith, to facilitate investments in Portfolio Companies, including providing or arranging for financing for investments in Portfolio Companies;

(ii) either by itself or by contract with others, including a corporation, partnership or limited liability company whose shareholders, partners, members, directors, officers or employees are shareholders, partners, members, directors, officers or employees of the General Partner or any affiliate of the General Partner, to maintain for the conduct of the Fund's affairs one or more offices and in connection therewith to rent or acquire office space, engage personnel, whether part-time or full-time, and to do, or cause to be done, such other acts as the General Partner may deem necessary or desirable in connection with the maintenance and administration of the affairs of the Fund;

(iii) to form, or cause the Fund to join with one or more of the Other Funds to form, one or more corporations, partnerships, limited liability companies, or other appropriate entities (including Investment Vehicles) for the purpose of financing, making, holding or disposing of any investment or investments (including one or more such entities formed for the purposes contemplated by Section 5.2), holding commitments of the Partners and receiving the proceeds of draws thereon, or incurring indebtedness for borrowed money, to cause any such entities to issue common or preferred equity interests or evidences of indebtedness or other debt interests;

(iv) to engage independent attorneys, accountants, consultants, appraisers and such other Persons as the General Partner may deem necessary or desirable; and

(v) to execute and deliver the investment management agreement (the "Investment Management Agreement") on behalf of the Fund;

(d) to cause the Fund to make any investment, or engage in any transaction, that is permitted by the Application or any order approving the Application, including:

(i) permitting Goldman Sachs, acting as principal, to engage in any transaction directly or indirectly with the Fund or any entity controlled by the Fund;

(ii) causing the Fund to invest in any entity or engage in any transaction with any entity, acting as principal, (A) in which the Fund, any entity controlled by the Fund or Goldman Sachs has invested or will invest, or (B) with which the Fund, any entity controlled by the Fund or Goldman Sachs is or will become otherwise affiliated;

(iii) causing the Fund or any entity controlled by the Fund to engage in any transaction directly or indirectly with any partner or other investor in an Investment Vehicle or account in which Persons who are not affiliated with Goldman Sachs may invest and over which a Goldman Sachs entity exercises investment discretion (a "Third Party Fund") acting as principal;

(iv) causing the Fund to engage in transactions in which affiliated Persons of the Fund or affiliated Persons of any of those Persons (including any partner or investor in a Third Party Fund) may also be participants; and

(v) permitting Goldman Sachs, acting as an agent or broker, to receive placement fees, advisory fees, commissions or other compensation from the Fund or a Portfolio Company or any entity controlled by the Fund or a Portfolio Company;

(e) to make temporary investments of Fund capital in all types of securities, including short-term U.S. government and government agency securities, certificates of deposit, interest-bearing deposits in United States banks, securities issued by or on behalf of states, municipalities and their instrumentalities, the interest from which is exempt from U.S. federal income tax, prime-grade commercial paper and securities issued by other investment companies (including unit investment trusts and taxable and tax-exempt money market funds sponsored and/or advised by affiliates of the General Partner), prior to long-term investment or pending cash distributions to the Partners;

(f) to perform all acts and file all documents, including tax returns and registration statements, necessary or advisable to effect compliance with applicable laws, rules and regulations applicable to the Fund or the conduct of the Fund's business and to conduct all other tax affairs of the Fund (including elections and audits);

(g) to organize, invest in and conduct business through one or more Investment Vehicles or other wholly or partly owned subsidiaries;

(h) to open, maintain and close bank, money market, brokerage and other accounts (and make deposits therein, withdrawals therefrom and draw checks or other orders thereon) whether located inside or outside the United States, in the Fund's name;

(i) to cause securities owned by the Fund to be registered in the Fund's name or in the name of a bank's nominee or to be held in street name, as the General Partner shall elect;

(j) to incur and pay on behalf of the Fund all expenses necessary for or incidental to the Fund's business, and, to the extent that funds of the Fund are available or callable, pay and establish reserves in respect of, all expenses, debts and obligations of the Fund;

(k) to execute, acknowledge, deliver, seal, file and record all instruments, exercise all voting or other rights and collect dividends and other income on the assets of the Fund;

(l) to employ and dismiss from employment any and all financial advisors, underwriters, investment bankers, attorneys, accountants, consultants, appraisers, asset managers or custodians of the assets of the Fund or other agents (who may be officers of the General Partner or the Fund), to represent, audit the books of, and provide other appropriate services to the Fund, on such terms and for such compensation as the General Partner may determine, whether or not any such Person is an affiliate of the General Partner or otherwise employed by any affiliate of the General Partner, the General Partner being expressly authorized to engage any affiliate of the General Partner to act in any such capacity;

(m) to submit any Fund claim or liability to arbitration or reference, or agree to submit any future Fund claims or liabilities to arbitration or reference;

(n) to bring and defend actions at law or in equity;

(o) to enter into, execute, supplement, acknowledge and deliver any and all contracts, agreements, guarantees, indemnities or other instruments (including insurance policies and contracts, of any type and coverage) and make commitments (including guarantees and indemnities) on behalf of the Fund with and to any party, including Goldman Sachs (except that the Fund shall not lend the Fund's money to any individual Partner) and, in general to do and perform all acts which may be necessary, advisable, suitable or proper for the conduct of the Fund's business and for the carrying out of the purposes and objectives hereinbefore enumerated and to give effect to any of the provisions of this Agreement, including the delegation to any Person or Persons of such functions and authority as the General Partner may determine;

(p) to utilize financial instruments, including forward contracts, options and swaps, caps, collars and floors and other derivative instruments;

(q) to lend, either with or without security, any securities, funds or other properties of the Fund and borrow or raise funds and secure the payment of obligations of the Fund by pledges, charges or hypothecation or granting of security interests (including, without limitation, the General Partner granting security interest over and/or assigning its right to call unfunded capital commitments and accept capital contributions) of or over all or any part of the property or rights of the Fund, and to exercise any and all rights of the Fund as a creditor or as a party to a credit or similar agreement;

(r) to effect conversions, issuances and purchases of Interests under certain circumstances;

(s) to make distributions to Partners in cash or (to the extent permitted hereunder) otherwise;

(t) to determine the accounting methods and conventions to be used in the preparation of any accounting or financial records of the Fund;

(u) to manage the Fund's investments, including the administration of Fund investments and the realization of those Fund Investments; and

(v) to take all actions necessary to, in connection with, or incidental to any of the foregoing.

Each of the Partners agrees that subject to applicable law, all determinations, decisions, and actions made or taken by or on behalf of the General Partner in good faith and in accordance with this Agreement shall be conclusive and absolutely binding upon the Fund, the Partners, and their respective successors, assigns, and personal representatives.

5.2. Alternative Investment Vehicles.

(a) Based on legal, tax, regulatory and other structuring considerations, in connection with particular Fund Investments, the General Partner, in coordination with the Investment Manager, may create one or more "Alternative Investment Vehicles") for purposes of making, holding or disposing of one or more Fund Investments. An Alternative Investment Vehicle may be a parallel partnership, corporation or other entity or such an entity that invests in the Fund. [] The economic terms of any Alternative Investment Vehicle shall be substantially similar in all material respects to those of the Fund; provided that the General Partner or the Investment Manager may vary such terms based on the structure of legal, tax and regulatory considerations; and provided, further, that such Alternative Investment Vehicle's economic provisions may not be integrated with those of the Fund. Any such Alternative Investment Vehicle will be structured in a manner whereby the Limited Partners [] participating in such Alternative Investment Vehicle shall bear the incremental costs of the alternative

arrangement (including taxes) and such costs will be taken into consideration in the distribution waterfall for such Alternative Investment Vehicle.

(b) The economic and other terms of any Alternative Investment Vehicle may vary from the terms of this Agreement based in part on the structure of the relevant transaction, legal requirements, tax, regulatory or other considerations, as determined by the General Partner. In the event that the General Partner forms one or more Alternative Investment Vehicles, the provisions of this Agreement (including as to allocations, distributions, costs and expenses), as well as the operative agreements of any such Alternative Investment Vehicles, shall be interpreted as determined by the General Partner as necessary and appropriate to give effect to the intended economic relationship set out in this Agreement and other agreements between the applicable parties [] among the Limited Partners [] with respect to the activities and investments of the Alternative Investment Vehicle(s) and the Fund. []

5.3. Fund Investment Holding Entities.

(a) The Fund may invest directly or indirectly in one or more corporations or other appropriate entities that will elect to be treated as corporations for U.S. federal income tax purposes (each, a "Fund Investment Holding Corporation"), for the purpose of holding the pro rata share of any Indirect Partner of any Fund Investment and/or its entitlement to Carried Interest with respect to any Fund Investment (any such capital interest and/or entitlement to Carried Interest, as applicable, a "Fund Asset").

(b) The General Partner may, but shall not be required to, cause one or more Partner's *pro rata* share of any Fund Asset to be held, directly or indirectly, through a Fund Investment Holding Corporation, and in any such case, the General Partner shall provide for the acquisition by, or transfer to, such Fund Investment Holding Corporation of a portion of such Fund Asset corresponding to the aggregate percentage of all such Partners (the "Indirect Partners"). Partners that are not Indirect Partners shall be defined, for the purposes of this Section 5.3, as "Direct Partners." For the avoidance of doubt, the determination by the General Partner of which Partners (including some or all of the Limited Partners) shall be Indirect Partners with respect to applying any Fund Asset, and which Partners shall be Direct Partners with respect to such Fund Asset, shall be made in the General Partner's discretion.

(c) Any Fund Asset in which some but less than all of the Fund's interest is held by a Fund Investment Holding Corporation shall be treated, for the purposes of this Section 5.3, as two "Special Fund Investments." One Special Fund Investment shall be that portion of the Fund Asset that is the direct or indirect investment in the equity and/or debt instruments, if any, of such Fund Investment Holding Corporation (the "Corporate Portion"), and the other Special Fund Investment shall be the remaining portion of the direct or indirect investment in such Fund Asset (the "Remaining Portion"). The Indirect Partners shall be treated as participating in the Special Fund Investment ("Participating Partners") only with respect to the Special Fund Investment that is the Corporate Portion, and the Direct Partners shall be treated as Participating Partners only with respect to the Special Fund Investment that is the Remaining Portion.

(d) The (i) allocation of income, gain, loss and expense, (ii) the distribution of material net proceeds from the sale, refinancing, redemption or other disposition of all, or any portion, of an investment and amounts received as extraordinary dividends or distributions in recapitalizations with regard to a Fund Asset ("Net Proceeds from Investment Dispositions") and (iii) distributions in respect of interest and other dividends ("Current Cash Flow") received by the Fund from Fund Assets in respect of the Special Fund Investment shall be made separately in respect of the Corporate Portion and the Remaining Portion in accordance with the provisions of this Section 5.3. An amount equal to the corporate-level tax paid or accrued by the Fund Investment Holding Corporation, and any expenses incurred by the Fund Investment Holding Corporation, will be deemed to have been allocated and distributed to the Participating Partners in such Fund Investment Holding Corporation for purposes of determining allocations and distributions under this Section 5.3.

(e) Distributions made to the Indirect Partners in respect of the Special Fund Investment that is the Corporate Portion shall be made solely out of amounts received by the Fund in respect of dividends,

distributions, interest or principal paid by, and redemptions or retirements with respect to the equity or debt instruments of, the Fund Investment Holding Corporation.

(f) The Fund Investment Holding Corporation may enter into an agreement with the General Partner or an affiliate of Goldman Sachs providing for the management of such Fund Investment Holding Corporation.

(g) The Fund may invest directly or indirectly in one or more partnerships, limited liability companies or other appropriate entities that will elect to be treated as partnerships for U.S. federal income tax purposes (each, a “Fund Investment Holding Partnership”), for the purpose of holding all or any portion of any Fund Asset, including the portion attributable to a Fund Investment Holding Corporation. The General Partner may, in its discretion cause the Remaining Portion of any Fund Asset (as described in Section 5.3(c)) to be owned by the Fund Investment Holding Partnership, in which case the partnership interests of the Fund Investment Holding Partnership (other than those held by the Fund Investment Holding Corporation) will be considered to be a Special Fund Investment in which some or all of [] the Direct Partners will be the Participating Partners, as determined by the General Partner.

(h) The General Partner may form or cause the Fund to form, or cause the Fund to join with one or more of the Other Funds in forming, in each case, directly or indirectly, one or more corporations, partnerships, limited liability companies, or other appropriate entities to make or hold Fund Investments (each, an “Other Vehicle”).

(i) In any case where a Fund Asset is made through an Other Vehicle, the Other Vehicle will be directly or indirectly owned by the Fund for the benefit of all of the Partners; provided, however, that in any case in which the General Partner determines it would be advantageous from a legal, tax, regulatory or other perspective, the Other Vehicle may be utilized on behalf of less than all of the Partners (which may include or exclude [] some or all of the Class O Limited Partners) and the investment in such Fund Investment on behalf of the remaining Partners (which may include or exclude [] some or all of the Class O Limited Partners) may be made directly by the Fund, or through an Other Vehicle, in which case, each such portion shall be treated, for purposes of this Section 5.3, as a Special Fund Investment made for the benefit of the Partners participating therein, respectively, and shall be attributed as the General Partner shall deem appropriate, in its sole discretion. In any case where the Fund receives or is allocated disproportionate distributions and allocations from an Other Vehicle, such disproportionate distributions or allocations shall be distributed and allocated to the [] relevant Class O Limited Partners (or to the Fund for the benefit of the [] relevant Class O Limited Partners) as appropriate.

(j) Subject to paragraphs (a) – (i) of this Section 5.3, the allocation of income, gain, loss and expense, and the distribution of Net Proceeds from Investment Dispositions and Current Cash Flow from Fund Assets, in respect of a Special Fund Investment shall be made in accordance with this Article V as if such Special Fund Investment were a Fund Asset; *provided*, that any allocations and distributions in respect of a Special Fund Investment shall be made solely to the Participating Partners []; and *provided further*, that in applying those Sections, each such Participating Partner’s Special Percentage shall be substituted for such Partner’s [] Class O Percentage, as applicable.

(k) The term “Special Percentage” shall mean, with respect to each Participating Partner in any Special Fund Investment, the result, expressed as a percentage, of dividing such Participating Partner’s contributions with respect to such Special Fund Investment by the sum of the contributions of all the Participating Partners with respect to such Special Fund Investment (adjusted as necessary, in the reasonable discretion of the General Partner, to reflect (i) any funded contributions which are funded disproportionately to relative capital commitments and/or (ii) the appropriate economic terms with respect to Class O Limited Partners).

5.4. **Prohibited Transactions.** Notwithstanding anything to the contrary contained herein and except to the extent permitted under the terms and conditions of the Application or any order approving the Application, and except for any joint trading account through which the Fund and affiliates of the General Partner may engage in the transactions contemplated by Section 5.1, no funds of the Fund shall be kept in any account other than a Fund

account, and funds shall not be commingled with the funds of any other Person, and the General Partner shall not employ, or permit any other Person to employ, such funds in any manner except for the benefit of the Fund (it being understood that this provision does not prohibit any investments otherwise permitted to be made by the Fund and does not apply to funds owned by Portfolio Companies).

5.5. Restrictions on the Authority of the General Partner.

(a) The General Partner shall not have the authority to:

- (i) do any act in contravention of this Agreement or the Delaware Act or any other applicable law;
- (ii) do any act that would make it impossible to carry on the ordinary business of the Fund, which, for the avoidance of doubt shall not prohibit the General Partner from selling any or all of the assets of the Fund at any time;
- (iii) possess Fund property or assign, pledge, grant a security interest in or over, charge or hypothecate its rights in specific Fund property, for other than a Fund purpose;
- (iv) except as provided in Section 8.10, admit another Person as a general partner, unless such Person is an entity wholly owned by Goldman Sachs; or
- (v) admit a Person as a Limited Partner or SOX Insider Preferred Limited Partner, except as provided in this Agreement and in the Offering Memorandum or as otherwise permitted by the terms and conditions of the Application or any order approving the Application.

(b) The General Partner shall not perform any act that would subject any Limited Partner or SOX Insider Preferred Limited Partner to liability as a general partner in any jurisdiction.

5.6. Indemnification and Contribution.

(a) To the fullest extent permitted by law, none of the General Partner, the Investment Manager, Goldman Sachs, the Partnership Representative, the Fund persons controlling, controlled by or under common control with any of the foregoing or any of their respective officers, directors, managers, partners, managing directors, stockholders, members, other equity holders, employees and controlling Persons (if any) (each, a “Management Indemnified Person” and collectively, the “Management Indemnified Persons”), shall be liable to the Fund or to any Limited Partners or SOX Insider Preferred Limited Partners for: (i) any act or omission performed or omitted by any such Management Indemnified Person (or by another Management Indemnified Person) or for any Losses arising therefrom, in the absence of gross negligence, willful misfeasance, bad faith or any material violation of any applicable securities law or any other intentional or criminal wrongdoing on the part of such Management Indemnified Person or any willful and material breach by such Management Indemnified Person of this Agreement or any side letter that is reasonably expected to have a material adverse effect on the Fund; (ii) any tax liability (including any liability for withholding, transfer, or other taxes (including (x) any interest and penalties thereon, (y) withholding tax liabilities pursuant to Section 1446(f) of the Code, and (z) any “imputed underpayment” attributable to any such Limited Partner, as determined by the Partnership Representative in its sole discretion, under Section 6225 of the Code)) imposed on the Fund, any Subsidiary or other entity in which the Fund invests, directly or indirectly, or any Limited Partner or SOX Insider Preferred Limited Partner (in excess of such Management Indemnified Person’s proportionate share of any such tax liability to the extent such Management Indemnified Person is a Partner); or (iii) any Losses due to any act or omission performed or omitted by brokers or other agents of the Fund or their respective employees (whether or not such Persons are directly employed by any Management Indemnified Person), as long as such brokers or other agents were selected with reasonable care. Subject to applicable law, no member of the Advisory Committee (each, an “Advisory Committee Indemnified Person” and, collectively, the “Advisory Committee Indemnified Persons”), shall be liable to the Fund or to any Limited Partner for (i) any act or omission performed or omitted by any such Advisory Committee Indemnified Person in connection with serving on the Advisory Committee (including any acts or omissions of or by an

Management Indemnified Person or another Advisory Committee Indemnified Person), or for any costs, damages or liabilities arising therefrom in the absence of willful misfeasance or bad faith by such Advisory Committee Indemnified Person; or (ii) any losses due to any act or omission performed or omitted by agents or advisors of the Advisory Committee or the Investor Advisory Committee (or their respective employees) in the absence of willful misfeasance or bad faith by such Advisory Committee Indemnified Person (whether or not such Persons are directly employed by the Advisory Committee).

(b) To the fullest extent permitted by law and subject to certain limits herein, the Fund will indemnify (and advance funds pursuant to clause (d) below to cover) any Management Indemnified Person, jointly and severally, for any expenses, losses, costs, damages, liabilities, demands, charges and claims (including without limitation any legal expenses and costs and expenses relating to investigating or defending any demands, charges or claims) (any such expense, loss, cost, damage, liability, demand, charge or claim, a “Loss” and collectively, “Losses”) to which such Management Indemnified Person may become subject in connection with: (i) any matter arising out of or in connection with the Fund’s business or affairs (including any Losses arising out of or in connection with the Fund’s indemnification, contribution, reimbursement or similar obligations to any of its investments or to any director, manager, officer, employee, partner, agent, or any other similar Person or entity of any such investment), except, with respect to any Management Indemnified Person, to the extent that any such Loss results solely from the gross negligence, willful misfeasance, bad faith, fraud or any material violation of any applicable securities law or any other intentional or criminal wrongdoing on the part of such Management Indemnified Person or any willful and material breach by such Management Indemnified Person of this Agreement or any side letter that is reasonably expected to have a material adverse effect on the Fund; (ii) any tax liability (including any liability for withholding, transfer, or other taxes (including (x) any interest and penalties thereon, (y) withholding tax liabilities pursuant to Section 1446(f) of the Code, and (z) any “imputed underpayment” attributable to any such Limited Partner, as determined by the Partnership Representative in its sole discretion, under Section 6225 of the Code)) imposed on the Fund, any Subsidiary of the Fund or other entity in which the Fund invests, directly or indirectly, or any Limited Partner or SOX Insider Preferred Limited Partner (in excess of such Management Indemnified Person’s proportionate share of any such tax liability as a Partner); or (iii) any act or omission performed by or omitted by brokers or other agents of the Fund or their respective employees (unless such employee, broker or agent is a Management Indemnified Person, in which case clause (i) of this sentence would apply, as applicable) so long as such Persons were selected with reasonable care. Notwithstanding the foregoing, the Fund will not indemnify any Management Indemnified Person for Losses incurred in connection with proceedings where one or more Management Indemnified Persons are suing one or more other Management Indemnified Persons unless one of the Funds is a plaintiff, defendant or other participant in such proceedings or will (or could reasonably be expected to) receive any monetary benefit from the outcome of such proceeding. Additionally, each Limited Partner will be required to indemnify the Fund and each Management Indemnified Person against taxes attributable to such Limited Partner.

(c) To the fullest extent permitted by law, the Fund will indemnify (and advance funds pursuant to clause (d) below to cover) any Advisory Committee Indemnified Person, jointly and severally, for any Losses to which such Advisory Committee Indemnified Person may become subject in connection with serving on the Advisory Committee or the Investor Advisory Committee, except to the extent that any such Loss results solely from the willful misfeasance or bad faith of such Advisory Committee Indemnified Person.

(d) In the event that any Management Indemnified Person or Advisory Committee Indemnified Person (each, an “Indemnified Person” and, collectively, the “Indemnified Persons”) becomes involved in any capacity in any action, proceeding or investigation brought by or against any Person (including any Limited Partner or SOX Insider Preferred Limited Partner) in connection with any matter arising out of or in connection with the Fund’s business or affairs (including a breach by any Limited Partner or SOX Insider Preferred Limited Partner (other than any Limited Partner or SOX Insider Preferred Limited Partner affiliated with Goldman Sachs) of this Agreement or the Limited Partner’s subscription agreement), the Fund will periodically reimburse such Management Indemnified Person or Advisory Committee Indemnified Person for its legal and other expenses (including the cost of any investigation and preparation) incurred in connection therewith; provided, that such Management Indemnified Person or Advisory Committee Indemnified Person shall promptly repay to the Fund the amount of any such reimbursed expenses paid to it if it shall ultimately be determined, by a court having appropriate jurisdiction in a

decision that is not subject to appeal, that such Management Indemnified Person or Advisory Committee Indemnified Person is not entitled to be indemnified by the Fund in connection with such action, proceeding or investigation as provided in clause (b), clause (c) or this clause (d), as applicable; *provided, further*, that the Fund shall not pursuant to this Section 5.6(d) advance any legal or other expenses incurred in connection with the defense of any action or proceeding brought by a majority in interest of the investors in the Underlying Funds against a Management Indemnified Person.

(e) If for any reason (other than, in the case of any Management Indemnified Person, the gross negligence, willful misfeasance, bad faith or fraud of such Management Indemnified Person or any material violation of any applicable securities law or any other intentional criminal wrongdoing on the part of such Management Indemnified Person and in the case of any Advisory Committee Indemnified Person, the willful misfeasance or bad faith of such Advisory Committee Indemnified Person) the foregoing indemnification, reimbursement or advance is unavailable to such Management Indemnified Person or Advisory Committee Indemnified Person, or is insufficient to hold either harmless, then the Fund shall contribute to the amount paid or payable by such Management Indemnified Person or Advisory Committee Indemnified Person as a result of such Loss in such proportion as is appropriate to reflect not only the relative benefits received by the Fund on the one hand and such Management Indemnified Person or Advisory Committee Indemnified Person on the other hand but also the relative fault of the Fund and such Management Indemnified Person or Advisory Committee Indemnified Person, as well as any relevant equitable considerations.

(f) The reimbursement, indemnity and contribution obligations of the Fund (or the General Partner in the event the Fund has been dissolved) under this Section 5.6 shall be in addition to any liability which the Fund may otherwise have, shall extend upon the same terms and conditions to the officers, directors, managers, partners, managing directors, stockholders, members, other equity holders, employees and controlling Persons (if any) of each Management Indemnified Person and shall be binding upon and inure to the benefit of any successors, assigns, heirs and personal representatives of any Management Indemnified Persons.

(g) The reimbursement, indemnity and contribution obligations provided by this Section 5.6 shall not be deemed to be exclusive of any other rights to which any Management Indemnified Person may be entitled under any agreement, as a matter of law or otherwise, both as to action in a Management Indemnified Person's official capacity and to action in another capacity, and shall continue as to a Management Indemnified Person who shall have ceased to have an official capacity for acts or omissions during such official capacity (or otherwise when acting at the request of the Fund, the General Partner or the Investment Manager) and shall be binding upon and inure to the benefit of any successors, assigns, heirs and personal representatives of any Management Indemnified Persons.

(h) The General Partner shall have power to purchase and maintain reasonable insurance on behalf of the General Partner and the other Management Indemnified Persons at the expense of the Fund, against any liability that may be asserted against or incurred by them in any such capacity or arising out of the General Partner's or such Management Indemnified Person's status as such, whether or not the Fund would have the power to indemnify the Management Indemnified Persons against such liability under the provisions of this Agreement.

(i) Nothing in this Agreement, the subscription agreements or any related document or agreement shall constitute a waiver or limitation of the General Partner's, Investment Manager's, and/or Goldman Sachs's potential liability, or of any rights which a Partner or the Fund may have, under applicable securities laws or other laws which may not be waived, including any fiduciary duty arising under the Investment Advisers Act of 1940 (the "Advisers Act").

(j) Notwithstanding any of the foregoing to the contrary, the provisions of this Section 5.6 shall not be construed so as to provide for the indemnification of any Management Indemnified Person for any liability (including liability under U.S. federal securities laws which, under certain circumstances, impose liability on Persons that act in good faith) to the extent (but only to the extent) that such indemnification would be in violation of applicable law, but shall be construed so as to effectuate the provisions of this Section 5.6 to the fullest extent permitted by law.

(k) The assets of the Fund and the amount of the Limited Partners' and the SOX Insider Preferred Limited Partners' aggregate capital commitments shall be available to satisfy the indemnity and contribution obligations of the Fund under this Section 5.6; in addition, the General Partner may require the other Partners (or Persons who were formerly Partners) to contribute again to the Fund any such obligations an amount equal to amounts previously distributed to Limited Partners or the SOX Insider Preferred Limited Partners. The General Partner may provide each affected Partner with an invoice setting forth the amount owed by the Limited Partner or the SOX Insider Preferred Limited Partner for amounts due in respect of any indemnity and contribution obligations. Each Partner shall be obligated to pay the invoiced amount promptly upon request by the General Partner. Any unpaid amount shall constitute an obligation of the Partner and shall accrue interest at the Interest Rate or a higher commercially reasonable rate as determined by the General Partner, and that amount, together with accrued interest, shall be debited from any distributions otherwise to be made to the Partner.

(l) Each Management Indemnified Person may rely upon and shall be protected in acting or refraining from acting upon any resolution, certificate, statement, instrument, opinion, report, notice, request, consent, order, bond, debenture or other paper or document believed by it to be genuine and to have been signed or presented by the proper party or parties.

(m) Each Management Indemnified Person shall be entitled, to the fullest extent of the law, to rely in good faith upon any and all sources enumerated in Section 17-407(c) of the Delaware Act, and any act or omission taken or suffered by such Management Indemnified Person in reasonable reliance on such source or sources shall in no event subject the Management Indemnified Person to liability to the Fund or to any Partner or to any other Person. All Partners hereby acknowledge and agree that each Management Indemnified Person is entitled to the same right of reliance and protection from liability as the General Partner.

(n) The General Partner may execute any of the trusts or powers hereunder or perform any duties hereunder either directly or by or through agents or attorneys, and the General Partner shall not be responsible for any misconduct or negligence on the part of any agent or attorney appointed with reasonable care by it hereunder.

(o) The General Partner is specifically authorized and empowered, for and on behalf of the Fund, to enter into any agreement or undertaking with any Management Indemnified Person or Advisory Committee Indemnified Person not itself a party to this Agreement that the General Partner considers to be necessary or advisable to give full effect to the foregoing indemnification provisions of this Agreement.

(p) The foregoing provisions of this Section 5.6 shall survive any termination of this Agreement and any Limited Partner ceasing to be a limited partner of the Fund.

Article VI

Capital Commitments, Defaults and Accounts; Aggregate Capital Contributions; Accounting and Valuation; Allocations; Reinvestment and Distributions

6.1. Capital Commitments, Defaults and Accounts.

(a) []

The General Partner at any time may cause the Fund to retain proceeds to pay (or establish reserves for) any of the Fund's current or anticipated direct or indirect obligations (including indebtedness, management fees and other Fund Expenses, as well as obligations relating to current or additional investments) and may also offset an amount that would otherwise be distributed to Limited Partners against amounts due pursuant to capital calls or otherwise. One or more of the Underlying Funds may have the ability to retain and reinvest proceeds or distribute proceeds to investors (including the Fund) and subsequently recall those proceeds for re-investment, as described in the governing documents of the Underlying Funds, (any such proceeds, "Underlying Fund Recyclable

Proceeds). Accordingly, the Fund may retain distributions that would otherwise be made to, or recall distributions previously made to, its Limited Partners. A Limited Partner's available commitment will not be impacted if the Fund retains, or distributes and subsequently recalls, any Underlying Fund Recyclable Proceeds.

(b) If any Limited Partner fails to pay any installment of such Limited Partner's capital commitment or any other payment required by this Agreement or his, her or its subscription agreement when due, the General Partner will have the right (in addition to any available recourse), but not the obligation, to charge such Limited Partner interest on the late payment from the due date until the day it is paid. Interest shall be charged at the Interest Rate or such higher commercially reasonable rate as determined by the General Partner from and including the due date of the installment until, but not including, the date it is paid.

(c) If any Limited Partner (a "Defaulting Limited Partner") other than an Institutional Limited Partner, fails to pay any installment of such Limited Partner's capital commitment or fails to make any other payment required by this Agreement or his, her or its subscription agreement within thirty (30) calendar days (unless such period is otherwise extended by the General Partner) (a "Default") from the date on which such installment is due, (the due date of any such delinquent capital contribution or other payment, the "Default Date") the General Partner will have the right, but not the obligation, to redeem, purchase or cause the Transfer of the Defaulting Limited Partner's entire interest in the Fund to one or more Persons, including Goldman Sachs, another Limited Partner or other Eligible Investors, on the first Purchase Date following Goldman Sachs' notification of its election to exercise this right at a price (the "Default Price") equal to seventy-five percent (75%) of the Minimum Purchase Price [] payable in cash, in kind or by delivery of a note (or any combination of the foregoing) and on such other terms and conditions as the General Partner may determine [].

In addition, the General Partner may undertake any one or more of the following remedies: (i) [], (ii) allowing the Defaulting Limited Partner to withdraw from the Fund at the Default Price and in accordance with the terms of Section 8.9, (iii) causing the Defaulting Limited Partner to forfeit its Participating Interest for no consideration (whether or not distributed or vested), and (iv) taking any other actions available to the General Partner under law and equity.

(d) If an Institutional Limited Partner fails to pay the entire amount of any installment or other required payment, the foregoing purchase rights shall apply only to the portion of such Institutional Limited Partner's Interest that is attributable to the Indirect Investor in respect of the amount that was not paid. The failure to pay any installment or other required payment when due shall constitute a breach of this Agreement, and such Defaulting Limited Partner shall remain liable for such breach regardless of any purchase by Goldman Sachs of the Defaulting Limited Partner's Participating Interest in the Fund (or portion thereof, in the case of an Institutional Limited Partner). Goldman Sachs may, but is not required to, offer any Participating Interest so acquired to other Limited Partners or any other Persons, who meet, in the judgment of the General Partner, the requirements to become a Limited Partner. Any such sale by Goldman Sachs shall be treated as a Transfer for purposes of this Agreement and shall be subject to and effected only in accordance with Article VIII.

(e) []

(f) By subscribing to the Fund, each Limited Partner authorizes Goldman Sachs to deduct without prior notice any Default amount(s) due by such Limited Partner from (i) any other Goldman Sachs employee special investments in which such Limited Partner is invested (now or at any time in the future), including by withholding from distributions, (ii) any cash, any proceeds from the liquidation of any marketable securities, including Goldman Sachs stock, and/or any other amounts held in the Limited Partner's Goldman Sachs brokerage account(s), (iii) any amounts paid in connection with the Limited Partner's departure from Goldman Sachs, and (iv) any other amounts owed to such Limited Partner by Goldman Sachs (including income, compensation, or equity-based awards), to the extent permitted by applicable law and any applicable plans, programs or agreements. In addition, the Fund (1) may withdraw funds credited to a Limited Partner's Capital Account in the Fund if

necessary to satisfy any obligation owed by such Limited Partner to Goldman Sachs and (2) to the extent such funds are insufficient, may cause such Limited Partner's Interest in the Fund to be forfeited or reallocated.

6.2. Additional Capital Contributions. Except for [] the indemnity obligations set forth in Section 8.5 arising in connection with certain Transfers, [], the obligation to return distributions in connection with the indemnification and contribution obligations set forth in Section 5.6 and the obligation to contribute capital up to the amount of Underlying Fund Recyclable Proceeds in accordance with Section 6.1(a) and the clawback set forth in Section 6.9, no Limited Partner shall be required or, except with the express written consent of the General Partner, permitted to make aggregate payments or contributions of capital to the Fund in excess of such Limited Partner's capital commitment, which shall be contributed as and when required by Section 6.1 and Article VII. The General Partner may make, but will not be obligated to make, voluntary additional capital contributions in such amounts and at such times as it may choose, upon notification thereof to all Limited Partners and the failure of three-quarters in interest of the Limited Partners to object thereto in writing within thirty (30) calendar days of notice thereof.

6.3. Fiscal and Taxable Year. Unless the General Partner at any time shall otherwise determine the fiscal year and taxable year of the Fund shall each end on December 31 of each year.

6.4. Partners' Capital Accounts.

(a) A capital account ("Capital Account") shall be established and maintained in the records of the Fund for each Partner and in respect of each class of Interests held by such Partner. A Partner's Capital Account in respect of a class of Interests will, pursuant to the terms of this Agreement, be:

(i) increased by [] allocations to such Partner in respect of that class of Interests of Fund income and gain []; and

(ii) decreased by (A) the amount of money or the net fair value of property distributed to such Partner by the Fund in respect of that class of Interests, (B) allocations to such Partner in respect of that class of Interests of Fund loss and deduction [].

If any Limited Partner Transfers all or any portion of his, her or its Interests (including as provided in Section 6.1), the Capital Account of such transferor Limited Partner that is attributable to such portion will carry over to the transferee Limited Partner. A Partner's aggregate Capital Account shall equal the sum of his, her or its Capital Accounts maintained in respect of each class of Interests held by such Partner.

As used herein, a Partner Capital Account means a Partner's aggregate Capital Account or a Partner's Capital Account in respect of a particular class of Interests as the context so requires.

(b) The Investment Manager shall determine the value or oversee the valuation of each Limited Partner's Capital Account as of the close of (i) the annual accounting period and (ii) the period ending with the distribution pursuant to Section 9.2 by treating each Fund Investment as having been disposed of for its fair value on the earlier of the last day of such period or the day such Fund Investment was actually disposed of or an In-kind Distribution was made, as the case may be, during such period, and by applying the allocation principles set forth in Section 6.5. Unrealized gains and losses shall be determined with respect to each Fund Investment owned by the Fund by reference to the value at which such Fund Investment is carried on the balance sheet of the Fund as determined by the Investment Manager in accordance with the valuation principles used in the quarterly financial statements of the Fund prepared in accordance with U.S. generally accepted accounting principles ("GAAP") (the "Financial Statement Value"). The reasonableness and appropriateness of the procedure utilized to determine such Financial Statement Values shall be reviewed annually by the Fund's independent auditors in connection with the examination of the Fund's annual financial statements. The determination of capital gains and losses with respect to any Fund Investment shall take into account any related fees and expenses incurred in connection with the acquisition and disposition of such Fund Investment, including Fund Expenses which are capitalized into the basis of a Fund Investment.

6.5. Allocation of Profits and Losses.

(a) The Fund's profits and losses for any year shall be determined by the General Partner in accordance with GAAP and procedures using the accrual method (with investments carried at their fair value as determined by the General Partner or Persons selected by it). The General Partner shall determine the allocation periods under this Section 6.5, and any period in respect of which an allocation is made is referred to herein as an "Allocation Period."

(b) Profits and losses of the Fund for each Allocation Period shall be allocated as follows, subject, in each case, to Section 6.5(c) and any adjustments the General Partner determines appropriate so as to take into account (i) any Default by a Limited Partner, (ii) ☐ (iii) any deemed distribution of withholding and/or other taxes that are allocable or attributable to a particular Limited Partner and (iv) the forfeiture of any Class O Interests held by each Limited Partner, (v) ☐, and (vi) any clawback of Carried Interest:

(i) ☐;

(ii) ☐;

(iii) ☐;

(iv) ☐

(v) an amount equal to the Carried Interest received directly from the Carried Interest Paying Funds (including, for purposes hereof, any investment vehicle directly or indirectly held by one or more of the Carried Interest Paying Funds by which Carried Interest is paid) shall be allocated among the Class O Limited Partners in proportion to their respective Class O Percentages.

Notwithstanding the foregoing, the General Partner may determine not to allocate Borrowing Expenses to any, all or certain Participating Interests held by SOX Insiders, including Institutional Limited Partners, if any (to the extent of the portion of any Institutional Limited Partner's Participating Interests held directly or indirectly by SOX Insiders), in the event SOX Insiders have pre-funded their capital calls or purchased SOX Insider Preferred Interests as contemplated by Section 6.1. Allocations with respect to SOX Insider Preferred Interests shall be made in accordance with Section 7.2.

(c) Allocations for U.S. federal income tax purposes will generally follow the allocations described above (except that investments will not generally be marked to market).

(d) It is the intention of the parties that, to the extent possible and consistent with the economics of this Agreement, the allocations made by the General Partner be respected for U.S. federal income tax purposes and comply with Section 704 of the Internal Revenue Code of 1986, as amended (the "Code"), and Treasury Regulations promulgated thereunder (the "Treasury Regulations"), and shall be interpreted and applied in a manner consistent with Section 704 and such Treasury Regulations. In furtherance of this intention a "qualified income offset provision" and any such other provision described in applicable regulations and deemed desirable by the General Partner shall be incorporated by reference into this Agreement. Notwithstanding any implication to the contrary contained herein, the General Partner shall have authority to make or refrain from making available tax elections and to choose from all available tax accounting methodologies in implementing the foregoing. To the extent consistent with applicable law, the General Partner may specially allocate income to any Partner, the status of which resulted in recognition of such income or otherwise alters the distribution or allocation provisions herein so that such Partner bears the consequences of such recognition. The General Partner's determination of allocations shall be binding upon all parties.

(e) Allocation of Certain Taxes and Income. Taxes (including any U.S. or non-U.S. income, withholding and similar taxes paid by Subsidiaries) will be allocated annually to the Partners in the same proportion

that the income in that year in respect of which such taxes are paid is allocated; provided, however, that where any such tax payable by the Fund or a Subsidiary is calculated, under applicable law, with respect to income allocable to some but not all of the Partners, or, to the extent income allocable to some of the Partners is exempt from tax in the hands of the Fund (or such taxes are reduced in respect of income allocated to certain Partners), or such tax arises as a result of the status of one or more Partners (or class of Partners), the failure by one or more Partners to comply with their obligations under Section 10.5, or the treatment or characterization for applicable tax purposes of any entity, payment or instrument by one or more Partners (or class of Partners), such taxes shall be allocated, as reasonably determined by the General Partner, among such Partners to whom allocations of income are subject to tax in the hands of the Fund (or to reflect such reductions) or to whom such taxes are otherwise properly allocable. To the extent consistent with applicable law, the General Partner may specially allocate income to any Limited Partner the status of which resulted in recognition of such income or otherwise alter the distribution or allocation provisions herein so that such Limited Partner bears the consequences of such recognition.

(f) Certain Definitions.

(i) []

(ii) "Borrowing Expenses" means all amounts in respect of premium, discount and interest on, and all costs, expenses, commissions and fees relating to, any incurrence, modification, amendment, restructuring, refinancing, cancellation or repayment of the direct or indirect indebtedness of the Fund and relating to notes and guarantees and providing any credit support.

(iii) []

(iv) []

(v) []

(vi) []

(vii) []

(viii) []

(ix) "Class O Percentage" means, with respect to any Class O Limited Partner and any date of determination, a fraction, the numerator of which is the Class O Interest of such Limited Partner, and the denominator of which is the aggregate Class O Interests. Each Class O Limited Partner's initial Class O Percentage will be determined with respect to a Class O Limited Partner based upon the Class O Interests offered to a Class O Limited Partner (or in the case of an investor who is an Affiliated Investor Entity, its related Offeree) as described in the Offering Memorandum. A Class O Limited Partner's Class O Percentage may be adjusted from time to time, including in connection with any whole or partial withdrawal, Transfer or forfeiture of a Class O Limited Partner's Class O Interest, or any grant of additional Class O Interests to a Class O Limited Partner in accordance with this Agreement, as determined by the General Partner in its discretion. The General Partner's determination of the adjusted Class O Percentage of a Class O Limited Partner for purposes of any distribution shall be conclusive and binding on all Limited Partners.

(x) "FATCA" means, Sections 1471 to 1474 of the Code (or any amended or successor provisions), any current or future regulations thereunder or official interpretations thereof, any agreement entered into pursuant to Section 1471(b)(1) of the Code, any fiscal or regulatory legislation, rules or practices adopted pursuant to any applicable intergovernmental agreement entered into in connection with the implementation of such Sections of the Code, and any other similar legislation, regulations or guidance enacted in any other jurisdiction which seeks to implement similar financial account information reporting and/or withholding tax regimes.

(xi) []

(xii) []

(xiii) “Person” means, any individual, partnership, limited liability company, corporation, trust, estate, association or other entity, or any other custodian or nominee or any individual or entity in its own or any representative capacity.

6.6. Distributions to Limited Partners.

(a) General. The amount and timing of distributions to each Limited Partner shall be at the sole discretion of the General Partner. The General Partner will have the right to withhold distributions if a Limited Partner fails to comply with any Goldman Sachs policy or request. In addition, (i) the timing of distributions may vary among the classes of Limited Partners, in the sole discretion of the General Partner and (ii) the General Partner may take into account legal, tax, regulatory or other considerations, including to comply with regulatory or legislative rules, when determining the timing of any payments (including, but not limited to, distributions) with respect to one or more Limited Partners (including considerations applicable to a Limited Partner’s status as an employee of Goldman Sachs) and/or classes, as it or Goldman Sachs determines is necessary or advisable, it being understood that varying regional considerations may result in one or more Limited Partners receiving such payments earlier than one or more other Limited Partners. Underlying Fund Recyclable Proceeds and other amounts held by the Fund may be retained in the Fund for reinvestment or distributed to the Limited Partners and subsequently recalled for reinvestment by the Fund, as the case may be, in accordance with Section 6.1. Any Underlying Fund Recyclable Proceeds so retained or recalled shall not reduce the available capital commitments of the Limited Partners. The General Partner at any time may cause the Fund to retain any proceeds from any investment (including the amount of the capital contribution relating to that investment) it receives to pay (or establish reserves for) any of the Fund’s current or anticipated direct or indirect obligations (including indebtedness, management fees and other Fund Expenses, as well as obligations relating to current or additional investments) and may also offset an amount that would otherwise be distributed to Limited Partners (including amounts payable in respect of a different class of Interests owned by the Limited Partner) against amounts due pursuant to capital calls or otherwise. [] The General Partner does not intend to make any distributions in respect of Carried Interest (excluding the distribution of Carried Interest Tax Distribution Amounts in respect of Carried Interest) until at least []. Moreover, the General Partner does not intend to make any distributions of Carried Interest until the likelihood of clawback is remote.

(b) Notwithstanding anything else in this Agreement to the contrary, to the extent that distributions from, or proceeds from the disposition of, Fund Investments are retained (rather than distributed, subject to recall), the General Partner will have the authority to adjust distributions and allocations to cause each Partner to receive, to the extent possible, the same distributions and allocations (as determined without giving effect to the individual tax treatment of any Partner) that each such Partner would have received had the amounts been distributed and the Partners made capital contributions in accordance with Section 6.1 (so that each Limited Partner bears his, her or its share of Fund Expenses that are not borne by the Partners in proportion to their capital commitments, bears amounts attributable to such Limited Partner pursuant to Section 6.5(e), and bears any other Fund Expenses according to such Partner’s capital commitment, and so that each Limited Partner receives an appropriate amount of distributions and allocations as contemplated herein, in each case as determined by the General Partner).

Any amounts that become available for distribution but are not distributed, including the Carried Interest, may be invested in money market instruments and/or other marketable securities, either directly or through funds that are managed by Goldman Sachs. The General Partner may determine to make distributions to the Institutional Limited Partners prior to making distributions to the other Limited Partners, in which case the General Partner shall make subsequent distributions disproportionately to the other Limited Partners until the amounts distributed to each other Limited Partner hereunder would equal the amount each such Limited Partner would have received had no such special distributions been made to the Institutional Limited Partners. All distributions will be

made net of any taxes payable by the Fund or by entities in which the Fund invests (including, for example, any local, non-U.S. withholding or other taxes).

The General Partner in its discretion may withhold from amounts otherwise distributable to a Limited Partner pursuant to this Section 6.6 [] amounts withheld pursuant to Section 6.6(e) in respect of taxes allocable to such Limited Partner.

Notwithstanding anything in this Agreement to the contrary, no provision shall limit in any way the General Partner's discretion to maintain reserves to meet potential investment needs or other obligations of the Fund.

(c) Distributions will generally be made as follows, subject, in each case, to any adjustments the General Partner determines appropriate so as to take into account (i) any Default by a Limited Partner, (ii) [], (iii) any deemed distribution of withholding and/or other taxes that are allocable or attributable to a particular Limited Partner and (iv) the forfeiture of any Class O Interests held by each Limited Partner, (v) [], and (vi) any clawback of Carried Interest.

(i) []

(ii) []

(iii) []

(iv) []

(v) Distributions to Holders of Class O Interests. Subject to Section 6.6(a) and Section 6.6(c), the available cash to be distributed, with respect to the Carried Interest, including any tax distribution amount (each, a "Carried Interest Tax Distribution Amount") received by the Fund in its capacity as a direct or indirect recipient of Carried Interest for purposes of the respective operating agreements of the Carried Interest Paying Funds (including, for purposes hereof, any investment vehicle directly or indirectly held by one or more of Carried Interest Paying Funds by which Carried Interest is paid), will be distributed among the Class O Limited Partners in proportion to their Class O Percentages, subject to any "Incremental Hurdle" or "Notch Amount" as described in the Offering Memorandum.

(vi) Adjustments to Class O and Sharing Percentages. Each Class O Limited Partner acknowledges that the Sharing Percentage and/or the Class O Limited Partners' Class O Percentages may be adjusted if there are aggregate commitments to the Carried Interest Paying Funds in excess of the aggregate amount of commitments that were projected in respect of the Carried Interest Paying Funds in connection with the determination of the total number of Carried Interest points offered to Class O Limited Partners (or in the case of investors who are Affiliated Investor Entities, their related Offerees), in order to give effect to changes in the points as described in the Offering Memorandum. The General Partner may also determine that any commitments in excess of the initial projection will not be considered part of the Carried Interest Paying Funds for purposes of this Agreement. Any determination described in this Section 6.6(c)(vi) will be made in Goldman Sachs' sole discretion and without prior notice to the holders of Class O Interests. Goldman Sachs may also determine whether or not any additional capital commitments subsequent to the final closing of the Carried Interest Paying Funds are included in the Carried Interest Paying Funds.

(vii) Distributions to Holders of SOX Insider Preferred Interests. Distributions with respect to SOX Insider Preferred Interests shall be made in accordance with Section 7.2.

(d) In-kind Distributions. The General Partner may make in kind distributions of property ("In-kind Distributions") (including interests denominated in currencies other than the U.S. dollar) which generally shall be distributed among the Partners in the same proportions as cash of equivalent value would be distributed if In-kind Distributions had been sold and the proceeds distributed pursuant to this Section 6.6, provided that the Fund may make a non-*pro rata* distribution of property (including distributing certain types of property to certain Partners and cash or other types of property to other Partners), pursuant to the redemption or purchase of the Interests by the Fund of one or more Partners (including pursuant to Section 8.7 or 8.9) or otherwise where the General Partner determines a non-*pro rata* distribution is appropriate or necessary. In-kind Distributions shall be valued at their fair value as determined by the investment managers of the Underlying Funds; provided that the investment managers of the Underlying Funds may hire third parties to make valuation determination or estimation, or may delegate valuation determination or estimation to third parties, and the investment managers of the Underlying Funds (or the Investment Manager) shall be entitled to rely upon such determinations or estimates. Where securities are to be distributed in kind among all or a group of Partners, securities will be distributed to all such Partners who would receive a specified minimum number of securities, and, as an administrative convenience, the securities distributable to Partners who would receive fewer securities may, instead, be distributed to a selling pool for sale on behalf of these Partners. In this way, all Partners entitled to such distribution will be treated as receiving a distribution at the same time. Notwithstanding anything in this Agreement to the contrary, money or other property wrongfully distributed to any Partners in violation of this Agreement may be recovered by the Fund.

(e) Withholding and Other Taxes.

(i) The General Partner may withhold and/or pay any taxes with respect to any Limited Partner's share of Fund taxable income (or which relate to a predecessor Interest or to a transfer (or deemed transfer) of such Limited Partner's Interest) (including any taxes paid by the Fund as a result of an audit adjustment or a change in law), and any such taxes may be withheld from any distribution otherwise payable to such Limited Partner. Taxes so withheld (or paid), or taxes withheld on amounts directly or indirectly payable to the Fund, a financing vehicle, a Fund Investment Holding Corporation, a Fund Investment Holding Partnership, or a pass-through vehicle in which the Fund has an investment and taxes otherwise directly or indirectly paid by the Fund, a financing vehicle or such pass-through vehicle may, in the General Partner's discretion, be treated for purposes of this Agreement (including Section 6.6) as distributed to the Partners to which such taxes are properly allocable (including as a result of the status of a Partner (or class of Partners) or the treatment or characterization for applicable tax purposes of any entity, payment or instrument by any Partner (or class of Partners)) and deemed paid by such Partners to the relevant taxing jurisdiction [].

(ii) Each Limited Partner (or former Limited Partner) hereby agrees to indemnify and hold harmless the Fund, any Subsidiary, the Management Indemnified Persons and the Advisory Committee Indemnified Persons and the other Limited Partners from and against any withholding or other tax-related liability (including any liability for taxes, penalties, additions to tax, interest or failure to withhold taxes) directly or indirectly payable (or otherwise borne) by the Fund, including through Investment Vehicles, that arise with respect to, or as a result of, income attributable to such Limited Partner, distributions or other payments to such Limited Partner, the status of such Limited Partner (or class of Limited Partners including such Limited Partner), the failure by such Limited Partner to comply with its obligations under Section 10.5, or the treatment or characterization for applicable tax purposes of any entity, payment or instrument by such Limited Partner (or class of Limited Partners including such Limited Partner) (including taxes indemnified under Section 5.6). For the avoidance of doubt, any such indemnity may be structured as a contribution to the Fund and will not be considered a capital contribution for purposes of this Agreement, nor shall any requirement that any such contribution be paid be considered a capital call. The amount of any taxes in respect of which a Limited Partner makes such a contribution shall not be treated as deemed distributed as described above. The foregoing provision of this Section 6.6(e) shall survive any termination of this Agreement.

Recognizing the complexity of the distributions pursuant to this Section 6.6, the General Partner is authorized to modify these distributions to ensure that they achieve the intended results without the consent of a Limited Partner or the Limited Partners, notwithstanding Section 11.2.

6.7. Allocation in Case of Transfers. In the event of a Transfer of an Interest during a taxable year of the Fund, allocations of income, gain, loss, deductions and credits and items of tax preference and other tax items of the Fund between the transferor and the transferee will be based on the portions of such taxable year during which each owned the Interest or as the transferor and the transferee may agree, provided that in either case the General Partner shall have consented to such allocation and the transferor and the transferee shall have agreed to reimburse the Fund for any incremental accounting fees and other expenses incurred by the Fund in making such allocation.
[]

6.8. []

6.9. Clawback

. Any distributions of Carried Interest to a Limited Partner (including the General Partner) in excess of the Carried Interest allocable to such Limited Partner as described in Section 6.5 will be subject to clawback by the General Partner. In addition, the Fund may become obligated to recontribute all or a portion of the Carried Interest to one or more of the Carried Interest Paying Funds (including, for purposes hereof, any investment vehicle directly or indirectly held by one or more of the Carried Interest Paying Funds by which Carried Interest is paid) if the investors in such Carried Interest Paying Fund have not received the preferred return provided for in the applicable operating agreement of such Carried Interest Paying Fund or are otherwise entitled to clawback payments pursuant to the operating agreement of such Carried Interest Paying Fund. Moreover, the Fund may become obligated to recontribute all or a portion of the Carried Interest received from one or more of the Carried Interest Paying Funds if the distributions made by such Carried Interest Paying Fund in respect of such amounts exceed the respective Carried Interest accrued as of certain measurement dates. Each of the General Partner and each Class O Limited Partner agrees that, in any of the foregoing cases, such Class O Limited Partner will recontribute to the Fund such Limited Partner's appropriate share of the Carried Interest previously distributed to such Limited Partner that must be recontributed to such Carried Interest Paying Fund (including, for purposes hereof, any investment vehicle directly or indirectly held by one or more of the Carried Interest Paying Funds by which Carried Interest is paid) (net of any Carried Interest Tax Distribution Amount distributed in respect of such recontributed Carried Interest), promptly upon request by the General Partner. The General Partner may provide each Class O Limited Partner with an invoice setting forth the amount owing by the Class O Limited Partner for amounts due in respect of any clawback obligations, and each Class O Limited Partner shall be obligated to pay the invoiced amount promptly upon request by the General Partner. Any unpaid amount shall constitute an obligation of the Class O Limited Partner and shall accrue interest at the Interest Rate, and that amount, together with accrued interest, shall be debited from any distributions otherwise to be made to the Class O Limited Partner or from other sources (including but not limited to any source listed in Section 6.1(f)).

6.10. Modifications to Allocations and/or Distributions. The General Partner may make allocations and/or distributions in a manner other than as described in this Article VI, including disproportionate allocations of gain or loss and/or disproportionate distributions to the extent required under applicable laws, rules and regulations, or otherwise deemed advisable or necessary by the General Partner in its discretion in order to give economic effect to the provisions of this Agreement; *provided*, however, that all such allocations are intended to be made in a manner such that all Class O Interests will be treated as "profits interests" for U.S. federal income tax purposes.

Article VII

Sarbanes-Oxley Act; SOX Insider Preferred Interests; Modification of SOX Insiders' Interests; Limitation on Pledges

7.1. Purchase of SOX Insider Preferred Interests; Issuance of SOX Insider Preferred Interests.

(a) For purposes of complying with the Sarbanes-Oxley Act of 2002 ("SOX"), including Section 402 thereof and any rules, regulations or interpretations of the U.S. Securities and Exchange Commission or

its staff promulgated thereunder ("SOX § 402"), the Fund is authorized to issue, from time to time, one or more classes of preferred limited partnership interests ("SOX Insider Preferred Interests," and the holder or holders of such SOX Insider Preferred Interests, "SOX Insider Preferred Limited Partners"), in such amounts as the General Partner determines, and the General Partner may require the purchase thereof by such of the Limited Partners as could, at any time and from time to time, in the judgment of the General Partner, be deemed to be (or to be Affiliated Investor Entities affiliated with) directors or executive officers (or equivalent thereof) of GS Group within the meaning of SOX § 402 ("SOX Insiders"). Any such SOX Insider will be required to purchase the SOX Insider Preferred Interest no later than a date and time set by the General Partner. The General Partner shall use commercially reasonable efforts to provide no less than five (5) days' notice to any Limited Partner required to purchase a SOX Insider Preferred Interest.

(b) The General Partner may require any subsequently admitted Institutional Limited Partner to (i) purchase a SOX Insider Preferred Interest if and to the extent that the General Partner determines, or the manager or general partner of such Institutional Limited Partner notifies the General Partner, that one or more Indirect Investors in such Institutional Limited Partner are SOX Insiders and (ii) allocate such SOX Insider Preferred Interest to such Indirect Investors under such methods as the General Partner or that the manager or general partner of the Institutional Limited Partner deems appropriate.

(c) The General Partner will use commercially reasonable efforts to avoid requiring a Limited Partner to contribute, in the aggregate, more than such Limited Partner's capital commitment towards such Limited Partner's Participating Interest and any SOX Insider Preferred Interest.

7.2. Terms. SOX Insider Preferred Interests shall be entitled to the following rights and subject to the following terms:

(a) Distributions. Each SOX Insider Preferred Interest shall entitle the holder thereof to receive a preferred distribution in respect of the capital contributed, as an expense of the Fund, at a rate and at times to be determined by the General Partner and notified to the Limited Partner required to purchase such SOX Insider Preferred Interest. The General Partner expects that the distribution rate will equal the rate payable from time to time, directly or indirectly, by the Fund, or any Investment Vehicle in which it has invested, on borrowings generally secured by unfunded capital commitments of the direct or indirect investors in the applicable borrower.

(b) Preference. Each SOX Insider Preferred Interest shall have a preference over all Interests (other than other SOX Insider Preferred Interests) as to the payment of distributions and as to the payment of the liquidation preference (which shall be equal to the amount invested in such SOX Insider Preferred Interest plus any accrued but unpaid distributions) upon the redemption of any SOX Insider Preferred Interest or the winding up of the Fund. SOX Insider Preferred Interests shall be *pari passu* with respect to each other.

(c) Redemption; Application to Satisfy Capital Calls. Notwithstanding anything to the contrary herein, the General Partner may from time to time redeem any outstanding SOX Insider Preferred Interests in whole or in part. The General Partner may, but shall not be obligated to, apply the proceeds from the redemption of a SOX Insider Preferred Interest to satisfy amounts required to be contributed by applicable SOX Insider Preferred Limited Partners in respect of called capital.

(d) Transfer Restricted. SOX Insider Preferred Interests shall be subject to the transfer restrictions set forth in Article VIII to the same extent as the other Interests.

(e) Allocation of Profits and Losses. Profits or losses of the Fund (if any) attributable to a SOX Insider Preferred Interest shall, unless otherwise determined by the General Partner, be allocated solely to the holder of such SOX Insider Preferred Interest.

(f) No Other Rights. Except as expressly provided in this Agreement, a SOX Insider Preferred Interest will not entitle the holder thereof to any rights to participate in the profits or assets of the Fund or any other rights against the Fund or the Limited Partners with respect to the SOX Insider Preferred Interest.

7.3. Modifications of SOX Insiders' Interests; Voting Rights of SOX Insider Preferred Interests.

(a) The Limited Partners acknowledge that the provisions of SOX § 402 may impose constraints on the ability of the General Partner, exercising prudence, to permit Limited Partners that are SOX Insiders or Affiliated Investor Entities affiliated with SOX Insiders to hold Interests on the same terms as other Limited Partners. In light of these potential constraints, it is understood and agreed that the General Partner is authorized, at any time and without notice to or the consent of any Limited Partner, to modify or redeem, in whole or in part, Interests held from time to time by SOX Insiders or Affiliated Investor Entities affiliated with SOX Insiders, as the General Partner deems necessary or appropriate to ensure compliance by Goldman Sachs with SOX § 402; provided, however, that the General Partner shall not effect any modification to an Interest that would materially adversely affect the Limited Partners who are not SOX Insiders or Affiliated Investor Entities affiliated with SOX Insiders without first complying with the procedure set forth in Section 11.2(a).

(b) Notwithstanding any provision of this Agreement to the contrary, the General Partner may at its election (but may determine not to) pledge, assign or otherwise make available as credit support the unfunded capital commitments of any Limited Partner who is a SOX Insider or the Affiliated Investor Entity of a SOX Insider, unless the General Partner determines that such pledge, assignment or availability may not comply with SOX § 402.

Article VIII

Transferability of Limited Partners' Interests; Consequences of Departure, Breach of GS Agreements, etc.

8.1. Restrictions on Transfer.

(a) No Limited Partner may, directly or indirectly, sell, exchange, transfer, assign, make a gift of, donate, bequeath, devise, pledge, hypothecate, or otherwise dispose, directly or indirectly (including effecting any direct or indirect transfer or other disposition of interests in, or any direct or indirect change in control of, any entity directly or indirectly investing in the Fund), of all or any portion of such Limited Partner's Interest (each, a "Transfer") except in accordance with this Article VIII. The General Partner may waive any requirement of this Article VIII with respect to any particular Transfer, except the provisions of Section 8.2.

(b) Subject to the other requirements of this Article VIII, with the prior written consent of the General Partner (which consent may be withheld by the General Partner in its discretion), a Limited Partner may Transfer all or any portion of such Limited Partner's Interest to any Person approved by the General Partner (a "Transferee").

(c) If any purported Transfer of an Interest by a Limited Partner is made other than in accordance with the terms of this Agreement (i) it shall be void, and neither the Fund nor the General Partner shall be required to recognize any equitable or other claims to such Interest on the part of the purported Transferee thereof and (ii) the transferring Limited Partner shall be deemed to be a Defaulting Limited Partner. Such default shall be governed by Section 6.1, and the General Partner may undertake any one or more of the remedies described in Section 6.1 with respect to such Defaulting Limited Partner.

(d) Notwithstanding the foregoing provisions of this Section 8.1, no Transfer of any kind may be made unless it also complies with the requirements of Section 8.2.

(e) []

8.2. Certain Conditions. Any Transfer with the consent of the General Partner may be made only if such Transfer:

(a) would not violate the terms and conditions of the Application or any order approving the Application;

(b) would not violate the Delaware Act, any United States securities laws, or any state securities or “blue sky” laws (including any investor suitability standards), or any other jurisdiction applicable to the Fund or the Interest to be sold, exchanged, transferred or assigned;

(c) would not create a material risk of the Fund being treated as a publicly traded partnership or other adverse tax consequences;

(d) would not require the Fund to register as an “investment company” under the Investment Company Act of 1940, as amended from time to time (the “Investment Company Act”) or to register under any other United States securities laws;

(e) would not violate any applicable law or regulation, including the BHCA and the Dodd-Frank Act (including the Volcker Rule); and

(f) is made to an “accredited investor” under Regulation D under the Securities Act of 1933, as amended from time to time (the “Securities Act”).

8.3. Transferee’s Agreement to be Bound; Effective Date of Transfer.

(a) Any Transfer with the consent of the General Partner shall not be effective until the Transferee shall have:

- (i) accepted and assumed, in form reasonably satisfactory to the General Partner, all terms and provisions of this Agreement,
- (ii) executed and delivered to the General Partner a power of attorney as provided in Section 11.1,
- (iii) executed transfer documents satisfactory to the General Partner, and
- (iv) executed such other documents or instruments as the General Partner may reasonably require to effect the admission of such Transferee as a substituted Limited Partner and such Transferee is admitted as a substituted Limited Partner.

(b) In addition, both the transferor Limited Partner and the Transferee must notify the General Partner in writing of the Transfer and must represent and warrant to the Fund and the General Partner that the Transfer was made in compliance with applicable law. The transferor Limited Partner and the Transferee further agree to provide any information in connection with a Transfer that the General Partner requires in order to comply with mandatory basis adjustment rules and/or withholding or other tax requirements.

(c) Any Transfer shall be recognized as effective as of the last day of the semi-annual period during which the notice required by Section 8.3(b) is delivered or such other date as determined by the General Partner.

(d) Any Person who acquires all or any portion of the Interest of a Limited Partner shall be obligated to contribute to the Fund the appropriate portion of any amounts thereafter becoming due in respect of the capital commitment made by his, her or its predecessor in respect of such Interest.

8.4. Substituted Partners.

(a) The Limited Partners hereby consent to the admission of a substituted Limited Partner whose admission has been consented to by the General Partner. Any such consent by the General Partner and the Limited Partners may be evidenced by the execution by the General Partner of an amendment to this Agreement (or

an instrument incorporating by reference the basis of this Agreement) on its behalf and on behalf of all Limited Partners evidencing the admission of such Person as a Limited Partner and the making of any filing required by law.

(b) The General Partner shall make such filings with the appropriate authorities of each state in which the Fund transacts business as may be required to reflect changes in its Partners.

8.5. Indemnity; Expenses of Transfer.

(a) Each transferor Limited Partner shall indemnify and hold harmless the Fund, the General Partner and every Limited Partner who was or is a party or is threatened to be made a party to any threatened, pending or completed action, suit or proceeding, whether civil, criminal, administrative or investigative, by reason of or arising from any actual or alleged misrepresentation or misstatement of facts (or omission to state facts) made (or omitted to be made) by such transferor Limited Partner in connection with any Transfer of all or any part of an Interest, or in connection with the admission of a substituted Limited Partner to the Fund, against expenses for which the Fund or such other Person has not otherwise been reimbursed (including attorneys' fees, judgments, fines and amounts paid in settlement) actually and reasonably incurred in connection with such action, suit or proceeding.

(b) Each Limited Partner agrees to be responsible for and shall indemnify and hold the Fund and the General Partner harmless from and against all costs, including any transfer or other taxes due as a result of a Limited Partner's Transfer and the cost of accounting for Transfers for U.S. income tax purposes, associated with an attempted or realized Transfer of any portion of its Interest regardless of whether or not the General Partner consents to the Transfer.

8.6. Death or Incapacity of Limited Partner, etc.; Acquisition of Such Partner's Interest by the General Partner.

(a) If a Limited Partner dies, his or her executor, administrator or trustee, or if he or she becomes an adjudicated incompetent, his or her committee, guardian or conservator, or if he, she or it becomes bankrupt, the trustee or receiver of his or her estate, shall have all the rights of a Limited Partner for the purpose of settling or managing the Interest of such incapacitated Limited Partner, and such power as the incapacitated Limited Partner possessed to Transfer all or any part of the incapacitated Limited Partner's Interest in accordance with this Article VIII.

(b) In the event of (i) the death or adjudication of incompetence of a Limited Partner or (ii) except as otherwise consented to by the General Partner, including any direct or indirect Transfer of interests in, or any direct or indirect change in control of, any entity directly or indirectly investing in the Fund, Goldman Sachs (or one of its affiliates, including the General Partner) may, but will have no obligation to, cause the Fund to purchase, redeem or cause the Transfer of the Limited Partner's vested Interest [] and/or forfeiture of any unvested Interests. If the General Partner elects to exercise its right to cause the Fund to purchase, redeem or cause the Transfer of the Limited Partner's vested Interest and/or forfeiture of any unvested Interests, then such Limited Partner or such Limited Partner's executor, administrator, committee, guardian or conservator, as the case may be, shall be obligated to sell such vested Interests and/or forfeit such unvested Interests to the General Partner or such affiliate or designee. Any such purchase will be at the then Fair Value of such vested Interests, and any such forfeiture will be for no consideration.

(c) The Fund may, but is not obligated to, purchase from the Fund's funds for the Fund's account any Interest tendered to the General Partner pursuant to this Section 8.6.

8.7. Consequences of Departure, Breach of GS Agreements, etc. Except as set forth in this Section 8.7, a Limited Partner's Equity Interest [] shall vest on December 31, []. Except as set forth in this Section 8.7, a Class O Limited Partner's Class O Interest will vest on the following schedule: (i) one-third of a Class O Limited Partner's Class O Interest will vest on December 31, [], (ii) an additional one-third of such Class O Limited Partner's Class O Interest will vest on December 31, [] (for a total amount of two-thirds of such Class O Limited Partner's Class O Interest having vested as of such date, including the amount vested on December 31, []), and (iii) the final one-third of such Class O Limited Partner's Class O Interest will vest on December 31, [] (for a total

amount of 100% of such Class O Limited Partner's Class O Interest having vested as of such date, including the amounts vested on December 31, [] and December 31, [] (the foregoing vesting schedule set forth in this sentence, the "Class O Vesting Schedule").

(a) If (i) an Unlevered Limited Partner leaves Goldman Sachs, for any reason (whether due to voluntary or involuntary termination of employment) or (ii) an event described in clause (i) occurs with respect to any employee of Goldman Sachs whose Affiliated Investor Entity or Eligible Family Member is an Unlevered Limited Partner (an "Unlevered Limited Partner Related Employee"), in each case, before December 31, [●], then the General Partner will have the right, but not the obligation, to (a) convert such Unlevered Limited Partner's Class A Interest to a Class C Interest (as described in Section 8.7(g)), or (b) cause the redemption, purchase or Transfer of the Unlevered Limited Partner's Class A Interest or Class C Interest on the next Purchase Date, upon the terms set forth in this Section 8.7. A Transfer of a Class A Interest or Class C Interest as described above may be made to Goldman Sachs, one or more Limited Partners or Eligible Investors, to the extent permitted by law, as determined by the General Partner in its sole discretion. "Eligible Family Member" has the meaning ascribed to it in the Application.

(b) []

(c) Each Class O Limited Partner's Class O Interests are subject to the Class O Vesting Schedule. Upon a Class O Limited Partner's or Related Employee's departure from Goldman Sachs for any reason (whether due to the voluntary or involuntary termination of employment) prior to December 31, [], the General Partner will have the right, but not the obligation, to cause the forfeiture, for no consideration, of the unvested portion of the Limited Partner's Class O Interest. This Section 8.7(c) shall apply irrespective of retirement, involuntary termination (including due to downsizing), death or adjudication of incompetence prior to December 31, [] (i.e., vesting will not be accelerated for any reason).

(d)

(i) If Goldman Sachs determines that a Limited Partner or Related Employee has failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with the Limited Partner's employment, including the Firm's notice period requirements, applicable restrictive covenants (such as restrictions on association with a Competitive Enterprise, Association with a Covered Enterprise, Engaging in Solicitation, and similar terms or concepts), any offer letter, employment agreement or any shareholders' agreement relating to Goldman Sachs (the "GS Agreements"), any vesting achieved within the twelve-month period prior to the termination of such Limited Partner's employment will be reversed for any Goldman Sachs employee special investments in which such Limited Partner is invested. If Goldman Sachs determines that a Limited Partner or Related Employee has breached a GS Agreement applicable to such Limited Partner or Related Employee, the General Partner will have the right, but not the obligation, to [] cause the redemption, purchase or transfer of any Interest held by such Limited Partner or Related Employee on the next Purchase Date, upon the terms set forth in this Section 8.7[].

(ii) The General Partner will have the right, but not the obligation, to cause the forfeiture for no consideration of some or all of the Class O Interests of a Class O Limited Partner whether or not vested, if (A) such Class O Limited Partner is a UK investor that is considered a "Material Risk Taker" and any of the following events occurs prior to the seventh anniversary of the issuance of the applicable Class O Interest(s) (which period may be extended by the General Partner in the event of any action, proceeding or investigation in connection with any of the following events is initiated by Goldman Sachs or a third party prior to the end of such seven year period): (i) the GS Group fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to the GS Group for a period of 90 consecutive business days; (ii) the Board of Governors of the Federal Reserve or the Federal Deposit Insurance Corporation (the "FDIC") makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Act for the appointment of the FDIC as a receiver of the GS Group based on a determination that the GS Group is "in default" or "in danger of

default"; (iii) (x) an annual pre-tax loss at the GS Group or (y) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset Management segment, or annual negative revenues in the Asset Management segment of \$5 billion or more, provided in either case that such Class O Limited Partner is employed in a business within such reporting segment; (iv) there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting Operational Risk Events Policy; (v) such Class O Limited Partner engages in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under English law; or (vi) an individual with respect to whom the committee appointed by the Board of Directors of the GS Group determines such Class O Limited Partner had supervisory responsibility as a result of direct or indirect reporting lines or such Class O Limited Partner's management responsibility for an office, division or business, engages during the calendar year in which such Class O Limited Partner's Class O Interest was issued, in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under English law, (B) such Class O Limited Partner is a German Class O Limited Partner who is a material risk taker designated by the Firm in accordance with applicable law and regulation (including a German Class O Limited Partner that is a management board member) (a "German MRT") and prior to the seventh anniversary of the issuance of the relevant Class O Interests (which period may be extended by the General Partner in the event of any action, proceeding or investigation in connection with any of the following events is initiated by Goldman Sachs or a third party prior to the end of such seven year period): (i) the applicable German MRT significantly contributed to, or was responsible for, any conduct that resulted in (a) a considerable loss (in particular, but not limited to, an unexpected loss of 0.75% (which may also be reached in sum by several individual losses) or more of the total capital of the GS Group), (b) a material regulatory sanction for the Firm comprising one or more of the following (1) a moratorium pursuant to sec. 46g German Banking Act ("Kreditwesengesetz" / "KWG"), (2) a measure in case of danger pursuant to sec. 46 German Banking Act, (3) the revocation of appointment of a manager pursuant to sec. 36 German Banking Act, (4) a fine pursuant to sec. 56 German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of the GS Group, (5) the cancellation of the banking permit pursuant to sec. 35 German Banking Act, (6) an order to increase the capital requirements by at least 0.5% pursuant to sec. 10 German Banking Act, (7) a measure in case of organizational deficiencies pursuant to sec. 45b German Banking Act, or (8) a comparable regulatory order or (c) a material supervisory measure; or (ii) the Applicable German MRT acted in serious violation of relevant external or internal rules with respect to suitability and conduct, *provided* that relevant regulations relating to the suitability and conduct of the members of the Board of Management include all regulations relating to conduct and professional suitability, compliance with which is necessary for the maintenance of a proper business organization within the meaning of Section 25a (1) sentence 1 German Banking Act, and provided that a violation is always considered serious if it is due to grossly negligent or intentional behavior or if it is suitable to justify a termination of employment for cause pursuant to sec. 626 German Civil Code or a termination of employment for misconduct ("verhaltensbedingte Kündigung") pursuant to sec. 1 German Termination Protection Act (Kündigungsschutzgesetz), (C) such Class O Limited Partner is a Dutch investor who works under the responsibility of GSAM BV (a "Dutch Class O Limited Partner"), and any of the following events occurs prior to the fifth anniversary of the issuance of the applicable Class O Interest(s): (i) the Dutch Class O Limited Partner has not met appropriate standards of competence and proper conduct; or (ii) the Dutch Class O Limited Partner was responsible for behaviors that have significantly deteriorated the financial position of GSAM BV, or (D) such Class O Limited Partner is a Dutch Class O Limited Partner and any of the following occurs prior to the fifth anniversary of the issuance of the applicable Class O Interest(s): (i) the award of the Class O Interest(s) to the Class O Limited Partner would be unacceptable according to standards of reasonableness and fairness; or (ii) the Class O Interest(s) have been awarded on the basis of incorrect information about the achievement of the goals underlying the award or about the circumstances on which the award was dependent. For the avoidance of doubt, an Affiliated Investor Entity will be considered a Material Risk Taker, German MRT

or Dutch Class O Limited Partner, if the Related Employee of such Affiliated Investor Entity is a Material Risk Taker, German MRT or Dutch Class O Limited Partner, as applicable.

(iii) Furthermore, if Goldman Sachs determines that (A) any employee of Goldman Sachs who is a Limited Partner or (B) any Related Employee (i) Fails to Consider Risk at any time prior to the liquidation of the Fund, (ii) at any time engages in any conduct that constitutes Cause, or (iii) at any time fails to certify to Goldman Sachs, in accordance with procedures established by the General Partner, that such Limited Partner or such Related Employee is in compliance with, or the General Partner determines that such Limited Partner or Related Employee in fact has failed to comply with all the terms and conditions of this Agreement and his, her or its subscription agreement, then the General Partner will have the right, but not the obligation, to [] cause the forfeiture for no consideration of any Class O Interests of such Limited Partner []; in each case, whether such Limited Partner's Interests have vested or not. []

(e) Notwithstanding any other provision of this Section 8.7, at any time Goldman Sachs may, in its sole discretion, determine to suspend a Limited Partner's vesting period (for instance, if such Limited Partner is placed on administrative leave), and any such suspension of a Limited Partner's vesting period (the "Suspension Period") shall automatically extend the time period required to vest by the number of days the Suspension Period continues. Any termination of the Suspension Period shall be at the sole discretion of Goldman Sachs. In addition to the other remedies set forth above, at any time on or after December 31, [] (or, with respect to Class O Interests at any time on or after December 31, [], December 31, [], and/or December 31, [], as applicable), the General Partner will have the right, but not the obligation, to cause the redemption, purchase or transfer of a Limited Partner's vested interest at a price equal to the then-current fair market value of such Limited Partner's vested Interest, as determined by the General Partner in its sole discretion, payable in cash, in kind and / or by delivery of a note (or any combination of the foregoing) as described in Section 8.7(i).

(f) Upon any forfeiture of Class O Interests, a Class O Limited Partner may be required, in Goldman Sachs' sole discretion, to return any Carried Interest previously distributed to such Class O Limited Partner (even though such Class O Limited Partner may have been taxed on income allocated in respect of such Carried Interest and/or paid expenses incurred in connection with the formation and implementation of any such Carried Interest arrangement). Amounts required to be returned in connection with a forfeiture will not include amounts corresponding to tax distributions, tax advances or similar amounts from Carried Interest Paying Funds. Subject to applicable law, forfeited Carried Interest shall be reallocated to Goldman Sachs (and therefore may be reallocated at a later date) and/or to new or existing Limited Partners. Any forfeited Carried Interest that is reallocated to new and/or existing Limited Partners may be subject to a different vesting schedule (including a shorter or longer vesting schedule). Upon the forfeiture of all or a portion of a Limited Partner's Class O Interest, such Limited Partner may be required, in the Investment Manager's sole discretion, to return any Carried Interest Tax Distribution Amount previously distributed in respect of such forfeited Class O Interest. A conversion, purchase or Transfer of Interests or forfeiture of Class O Interests as described above may have adverse tax consequences.

(g) []

(h) []

(i) The purchase price for any Interest acquired pursuant to Section 8.7(a), (b), (d), (e) or (l) shall be an amount equal to the lesser of (i) such Limited Partner's capital contributions plus interest at the Interest Rate (net of any distributions) (the "Net Contributed Capital"), and (ii) an amount equal to the then Fair Value of such Limited Partner's Interest (such lesser amount, the "Minimum Purchase Price"), payable in cash, in kind or by delivery of a note (or any combination of the foregoing). [] For purposes hereof, the "Fair Value" of an Interest in the Fund shall be determined by the General Partner in its discretion at the time any such redemption, purchase, Transfer or withdrawal takes place (as determined without regard to the tax consequences of the receipt of income or gain therefrom) as an amount equal to the fair market value of such Interest at such point in time[]. A Transfer of an Interest as described herein may be made to Goldman Sachs, its affiliates, one or

more Limited Partners or Eligible Investors, to the extent permitted by law, as determined by the General Partner in its sole discretion. For the avoidance of doubt, such purchase price will take into account any distributions previously made, and, in the case of Interests that are subject to retroactive conversion, any future distributions in respect of these Interests may be adjusted by taking account of the distributions made before the date of conversion. The purchase price may reflect an illiquidity discount. The Limited Partners hereby acknowledge that (x) if an Interest is sold to a third party buyer, the purchase price will reflect only what such third party buyer would be willing to pay for the Interests, (y) in each case, the price of such Interests may represent a substantial discount to such Limited Partner's Net Contributed Capital or Capital Account value, and (z) any redemption, purchase or Transfer of Interests may be payable in cash, in kind or by delivery of a note (or combination of any of the foregoing).

(j) The "Interest Rate" will be six-month SOFR plus 115 basis points, subject to change from time to time, and at any time, in the discretion of the General Partner (including, for the avoidance of doubt, that the General Partner may at any time decide in its sole discretion to use a different benchmark).

(k) Payment of the purchase price shall be made on May 30 of the year immediately following the year during which Goldman Sachs or the Fund exercises its election to purchase or cause the Transfer of any Limited Partner's Interest pursuant to this Agreement, or any other date as the General Partner may determine from time to time (any such date, a "Purchase Date"). For this purpose, a payment will be deemed to have been paid (i) if payment is made by check, when it has been deposited by Goldman Sachs in the U.S. mail or delivered to a reputable courier service or (ii) if payment is made by deposit into a brokerage account, when the payment is deposited into the Limited Partner's or Related Employee's (as the case may be) brokerage account. Goldman Sachs will have no obligation to provide any notice of purchase other than the notice accompanying payment. The effective date of the purchase and the date of payment may be different days. Notwithstanding anything to the contrary in this Agreement, any Transfer upon a purchase pursuant to this Section 8.7 shall, for all purposes of this Agreement, be effective as of the close of business on the date determined by the General Partner. If the effective date of the Transfer is any date other than the date on which the price was calculated, then for purposes of determining the purchase price, the Limited Partner's Capital Account balance shall be appropriately adjusted to reflect any contributions and distributions made in respect of such Limited Partner's Interest after the date on which the price was calculated.

(l) In addition to the other rights that Goldman Sachs has under this Section 8.7, if Goldman Sachs determines that a Limited Partner or a Related Employee may have breached any of his, her or its representations, warranties or covenants made in the subscription agreement or otherwise made in connection with the acquisition of an Interest, then the General Partner shall have the right to cause the redemption, purchase, Transfer or forfeiture, or require the withdrawal, of the Limited Partner's Interest on the Purchase Date, using the purchase price calculation set forth in Section 8.7(h), payable in cash, in kind or by delivery of a note (or any combination of the foregoing), or, in the case of forfeiture, for no consideration. Additionally, the General Partner, in its sole discretion and without the consent of any Limited Partner, may take any action, including (i) causing the redemption, purchase or Transfer of any Limited Partner's Interests in the Fund to Goldman Sachs, its affiliates or one or more other Limited Partners or Eligible Investors, to the extent permitted by law, (ii) causing the forfeiture of any Limited Partner's unvested Class O Interest or (iii) requiring the withdrawal of any Limited Partner, at a price equal to the then Fair Value of such Limited Partner's Interest for payment in cash, in kind, by delivery of a note (or any combination of the foregoing), or, in the case of forfeiture, for no consideration and/or in exchange for a participation interest (or any combination of the foregoing) and on such other terms and conditions as the General Partner may determine, in each case, without regard to any applicable tax consequences (including consequences of Transfer) and (x) as the General Partner or Goldman Sachs in its discretion determines is necessary or advisable for Goldman Sachs or any of its affiliates to comply with applicable laws, rules or regulations (including the BHCA or the Dodd-Frank Act, including the Volcker Rule) affecting the Fund, the Limited Partners, Goldman Sachs or any of its affiliates or (y) if the General Partner determines that the continued participation of the Limited Partner in the Fund may cause the Fund, the General Partner or any of their respective affiliates to be (or remain) in violation of the terms and conditions of the Application or any order approving the Application, or any U.S. federal or state law, rule or regulation, including SOX or the Dodd-Frank Act, including the Volcker Rule, or have an adverse effect on the business or operations of Goldman Sachs, or otherwise raise tax, legal (including contractual restrictions) or

regulatory issues. [] By exercising any of its rights, or taking any of the actions described above, the General Partner may cause distributions in kind of non-marketable securities, forced sales of Interests by some or all Limited Partners (for notes or other consideration), or resignation of Goldman Sachs as the Investment Manager and/or the General Partner. Furthermore, the General Partner may exercise any such rights or authorized actions in its sole discretion and with respect to certain or all Limited Partners in a manner that may disproportionately affect some Limited Partners relative to other Limited Partners (even similarly situated Limited Partners) and the General Partner will have no duty (implied or otherwise) to mitigate the effect on any Limited Partners or to treat any similarly situated Limited Partners similarly.

(m) The General Partner may waive, on a case-by-case basis, its rights under this Section 8.7 to cause the Transfer or forfeiture of, convert and/or purchase Interests, [] and/or the retroactivity of any conversion or forfeiture. In addition to any other rights described herein, the General Partner shall have the right to cause the redemption, purchase, Transfer or forfeiture of any Interests in any manner that does not adversely affect a Limited Partner.

(n) Nothing in this Agreement shall be construed as giving any Limited Partner or any Related Employee any right to continued employment with Goldman Sachs or as affecting any right that Goldman Sachs may have to terminate or alter the terms and conditions of any such Limited Partner's or Related Employee's employment.

(o) []

(p) All references to the termination of employment with or departure from Goldman Sachs are deemed to occur immediately following any "garden leave" or similar notice period.

(q) "Association with a Covered Enterprise" means that a Limited Partner or Related Employee (i) forms, or acquires a 5% or greater equity ownership, voting or profit participation interest in, any Covered Enterprise or (ii) associates in any capacity (including association as an officer, employee, partner, director, consultant, agent or advisor) with any Covered Enterprise. "Association with a Covered Enterprise" may include, as determined in the discretion of Goldman Sachs, (i) becoming the subject of any publicly available announcement or report of a pending or future association with a Covered Enterprise and (ii) unpaid associations, including an association in contemplation of future employment.

(r) "Cause" means the Limited Partner's (i) conviction, whether following trial or by plea of guilty or *nolo contendere* (or similar plea), in a criminal proceeding (A) on a misdemeanor charge involving fraud, false statements or misleading omissions, wrongful taking, embezzlement, bribery, forgery, counterfeiting or extortion or (B) on a felony charge or (C) on an equivalent charge to those in clauses (A) and (B) in jurisdictions which do not use those designations; (ii) engaging in any conduct which constitutes an employment disqualification under applicable law (including statutory disqualification as defined under the U.S. Securities Exchange Act of 1934, as amended); (iii) willful failure to perform the Limited Partner's duties to the Firm; (iv) violation of any securities or commodities laws, any rules or regulations issued pursuant to such laws, or the rules and regulations of any securities or commodities exchange or association of which the Firm is a member; (v) violation of any Firm policy concerning hedging or pledging or confidential or proprietary information, or material violation of any other Firm policy as in effect from time to time; (vi) engaging in any act or making any statement which impairs, impugns, denigrates, disparages or negatively reflects upon the name, reputation or business interests of the Firm, including any of its portfolio companies; or (vii) engaging in any conduct detrimental to the Firm, including any of its portfolio companies. All determinations of whether any act or omission constitutes "Cause" in any particular case will be made by Goldman Sachs and will be final and binding on all parties.

(s) "Client" means any client or prospective client of the Firm to whom the applicable Limited Partner or Related Employee provided services, or for whom the applicable Limited Partner or Related Employee transacted business, or whose identity became known to the applicable Limited Partner or Related Employee in connection with such Limited Partner's or Related Employee's relationship with or employment by the Firm.

(t) “Commencement of the Fund’s Operations” shall mean the first date on which subscriptions of Limited Partners (other than the Initial Limited Partner) for investment in the Fund are accepted by the General Partner.

(u) “Competitive Enterprise” means an existing or planned business enterprise that (i) engages, or may reasonably be expected to engage, in any activity, (ii) owns or controls, or may reasonably be expected to own or control, a significant interest in any entity that engages in any activity or (iii) is, or may reasonably be expected to be, owned by, or a significant interest in which is, or may reasonably be expected to be, owned or controlled by, any entity that engages in any activity, that, in any case, competes or will compete anywhere with any activity in which the Firm is engaged. The activities covered by this definition include: financial services such as investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the applicable Limited Partner or Related Employee and members of the applicable Limited Partner’s or Related Employee’s family (including, for the avoidance of doubt, any proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading.

(v) “Covered Enterprise” means a Competitive Enterprise and any other existing or planned business enterprise that: (i) offers, holds itself out as offering or reasonably may be expected to offer products or services that are the same as or similar to those offered by the Firm or that the Firm reasonably expects to offer (“ Firm Products or Services.”) or (ii) engages in, holds itself out as engaging in or reasonably may be expected to engage in any other activity that is the same as or similar to any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage (“ Firm Activities.”). For the avoidance of doubt, Firm Activities include any activity that requires the same or similar skills as any financial activity engaged in by the Firm or in which the Firm reasonably expects to engage, irrespective of whether any such financial activity is in furtherance of an advisory, agency, proprietary or fiduciary undertaking.

The enterprises covered by this definition include enterprises that offer, hold themselves out as offering or reasonably may be expected to offer Firm Products or Services, or engage in, hold themselves out as engaging in or reasonably may be expected to engage in Firm Activities directly, as well as those that do so indirectly by ownership or control (e.g., by owning, being owned by or being under common ownership with an enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services or that engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities). The definition of Covered Enterprise includes, solely by way of example, any enterprise that offers, holds itself out as offering or reasonably may be expected to offer any product or service, or engages in, holds itself out as engaging in or reasonably may be expected to engage in any activity, in any case, associated with investment banking; public or private finance; lending; financial advisory services; private investing for anyone other than the applicable Limited Partner or Related Employee or family members of the applicable Limited Partner or Related Employee (including, for the avoidance of doubt, any type of proprietary investing or trading); private wealth management; private banking; consumer or commercial cash management; consumer, digital or commercial banking; merchant banking; asset, portfolio or hedge fund management; insurance or reinsurance underwriting or brokerage; property management; or securities, futures, commodities, energy, derivatives, currency or digital asset brokerage, sales, lending, custody, clearance, settlement or trading. An enterprise that offers, holds itself out as offering or reasonably may be expected to offer Firm Products or Services, or engages in, holds itself out as engaging in or reasonably may be expected to engage in Firm Activities is a Covered Enterprise, **irrespective of whether the enterprise is a customer, client or counterparty of the Firm or is otherwise associated with the Firm and, because the Firm is a global enterprise, irrespective of where the Covered Enterprise is physically located.**

(w) “Engages in Solicitation” means (i) a Limited Partner, in any manner, directly or indirectly, (A) Solicits any Client to transact business with a Covered Enterprise or to reduce or refrain from doing any business with the Firm, (B) interferes with or damages (or attempt to interfere with or damage) any relationship between the Firm and any Client, (C) Solicits any person who is an employee of the Firm to resign from the Firm, (D) Solicits any Selected Firm Personnel to apply for or accept employment (or other association) with any person or entity other than the Firm, or (E) hires or participates in the hiring of any Selected Firm Personnel by any person or entity other than the Firm (including participating in the identification of individuals for potential hire, and participating in any hiring decision), whether as an employee or consultant or otherwise, or (ii) Selected Firm Personnel are solicited, hired or accepted into partnership, membership or similar status by (A) any entity that a Limited Partner forms, that bears a Limited Partner’s name, or in which a Limited Partner possesses or controls greater than a *de minimis* equity ownership, voting or profit participation, or (B) any entity where a Limited Partner has, or will have, direct or indirect managerial responsibility for such Selected Firm Personnel. Alternative formulations of this term (e.g., “Engaging in Solicitation”) will have their correlative meanings.

(x) “Fails to Consider Risk” means that the Limited Partner participated (or otherwise oversaw or was responsible for, depending on the circumstances, another individual’s participation) in the structuring or marketing of any product or service, or participated on behalf of the Firm or any of its clients in the purchase or sale of any security or other property, in any case without appropriate consideration of the risk to the Firm or the broader financial system as a whole (for example, where the Limited Partner has improperly analyzed such risk or where the Limited Partner has failed sufficiently to raise concerns about such risk) and, as a result of such action or omission, Goldman Sachs determines there has been, or reasonably could be expected to be, a material adverse impact on the Firm, the Limited Partner’s business unit or the broader financial system.

(y) “Firm” means The Goldman Sachs Group, Inc. and its subsidiaries and affiliates.

(z) “Selected Firm Personnel” means any individual who is or in the three months preceding the prohibited conduct was (i) a Firm employee or consultant with whom the Limited Partner personally worked while employed by the Firm, (ii) a Firm employee or consultant who, at any time during the year preceding the date of the termination of the Limited Partner’s employment, worked in the same division in which the Limited Partner worked or (iii) an Advisory Director, a Managing Director or a Senior Advisor of the Firm.

(aa) “Solicit” means any direct or indirect communication of any kind whatsoever, regardless of by whom initiated, inviting, advising, suggesting, encouraging or requesting any Person or entity, in any manner, to take or refrain from taking any action.

8.8. General Partner Purchase. If the General Partner or Goldman Sachs purchases any Interest pursuant to Sections 6.1(c), 8.6, 8.7 or 8.9 for its own account and not for the account of the Fund, the General Partner or Goldman Sachs, as the case may be, shall be entitled to the rights of a Transferee of such Interest and shall be entitled to vote (except as provided herein) such Interest as if it were a substituted Limited Partner or to be admitted as a substituted Limited Partner. Subject to this Article VIII, the General Partner or Goldman Sachs may Transfer any Interest acquired by it under the provisions of such Sections on such terms as are acceptable to it, and if the transferee of such Interest is not a Limited Partner of the Fund, such transferee will be entitled to be admitted to the Fund as a substituted Limited Partner with respect to such Interest. The effective date of any such Transfer shall be the last day of the semi-annual period during which payment is made by the transferee of such Interest or such other day as the General Partner, the transferor and the transferee may agree.

8.9. Withdrawal of Limited Partners. No Limited Partner may withdraw from the Fund or withdraw or require redemption of some or all of its Capital Account without the written consent of the General Partner, which consent may be granted or withheld by the General Partner in its sole discretion. In addition, the General Partner may require the partial or total withdrawal of a Limited Partner from the Fund (a “Required Withdrawal”) upon at least five (5) days’ prior written notice if the General Partner determines, that the continued participation of such Limited Partner may adversely affect any of the Funds (including for any tax or regulatory purposes), any Limited Partner, the General Partner (or other parties with authority to act) or any of their respective affiliates, or otherwise give rise to any business, tax, legal or regulatory issues. Without limiting the generality of the foregoing, the General Partner may require the partial or total withdrawal of any Limited Partner from the Fund if:

(a) the General Partner determines that the continued participation of that Limited Partner with respect to all or a portion of its Interest may:

(i) require registration of the interests of any of the Funds (under the Securities Act, or any other securities laws applicable to any of the Funds);

(ii) cause any of the Funds to fail to qualify for an exemption from registration under the Investment Company Act (including causing one or more of the Funds to violate the terms and conditions set forth in the Application or any order approving the Application), or any other law or regulation requiring the regulation of collective investment schemes that the General Partner believes to be otherwise than in the best interests of any of the Funds;

(iii) result in any direct or indirect material adverse tax consequences to any of the Funds (or any of the Underlying Funds) (or any of the Funds' (or any of the Underlying Funds') investors generally), including any of the Funds' (or any of the Underlying Funds') failing to comply with FATCA or being subject to withholding tax under FATCA;

(iv) result in any violation of applicable laws or regulations, including the BHCA, SOX or the Dodd-Frank Act (including the Volcker Rule); or

(v) otherwise cause the Fund or any of its affiliates, including Goldman Sachs, to be in violation of any applicable law, rule or regulation or have an adverse business, legal or regulatory consequence to the Fund or any of its affiliates, including Goldman Sachs;

(b) such Limited Partner (i) engages in illegal conduct or gross misconduct which the General Partner determines could result in reputational harm to the Fund, Goldman Sachs or its affiliates or (ii) illegally or fraudulently obtained the funds which such Limited Partner utilized to make its contributions;

(c) there is any breach of such Limited Partner's representations, warranties or covenants in the subscription agreement, this Agreement or related documents executed by such Limited Partner; or

(d) such Limited Partner is convicted of, or pleads *nolo contendere* to, a felony or serious misdemeanor.

If a Limited Partner is subject to a Required Withdrawal described in (i) Section 8.9(a) above, the General Partner will have the right, but not the obligation, to cause (1) the partial or total redemption, purchase or Transfer of any [] vested Class O Interests of such Limited Partner (including to Goldman Sachs, its affiliates, one or more other employees or other purchasers) on the next Purchase Date, at a price equal to the then Fair Value of such Limited Partner's Interest, payable in cash, in kind or by delivery of a note (or any combination of the foregoing) and/or (2) the forfeiture, for no consideration, of [] unvested Class O Interests of such Limited Partner, or, (ii) Section 8.9(b), 8.9(c) or 8.9(d) above, the General Partner will have the right, but not the obligation, to cause [] the forfeiture, for no consideration, of [] any Class O Interests of such Limited Partner (whether or not vested). []

Upon any Required Withdrawal of a Limited Partner described above, Goldman Sachs, the Fund and their respective affiliates may be restricted under certain circumstances from purchasing all or a portion of such Limited Partner's Interest due to legal and other regulatory restrictions applicable to Goldman Sachs and its affiliates. A Transfer of an Interest upon withdrawal as described above may be made, in whole or in part, to Goldman Sachs, its affiliates, one or more Limited Partners or Eligible Investors, to the extent permitted by law.

Each Limited Partner acknowledges and agrees that, because the investment will be illiquid and is subject to legal and contractual transfer restrictions and because the Fund's ability to permit transfers of interests in the Fund is limited, actual payment in respect of a Limited Partner's Interest in the Fund may take a substantial amount of time (potentially years) from the date a Limited Partner's Interest in the Fund is terminated.

8.10. Withdrawal or Substitution of the General Partner. Subject to the requirements of the Fund and under the Delaware Act, the General Partner may, at any time, in its sole discretion, (a) assign all or a portion of its interest to any affiliate of Goldman Sachs or any other Person, as deemed necessary or advisable by the General Partner or Goldman Sachs in order to comply with applicable laws, rules, regulations or other legal requirements (including the Dodd-Frank Act (including the Volcker Rule)), (b) in the General Partner's discretion, admit an affiliate or such other Person or Persons as an additional or substitute General Partner, or (c) resign or withdraw from the Fund.

Article IX

Winding-up, Dissolution and Liquidation of the Fund

9.1. Events Causing Winding-up and Dissolution.

(a) The Fund shall be wound-up and dissolved upon the happening of any of the following events:

(i) the retirement, bankruptcy, commencement of liquidation proceedings, resignation or withdrawal, insolvency or dissolution of the General Partner shall cause the immediate dissolution of the Fund, unless (1) the General Partner has admitted an additional or substitute General Partner pursuant to Section 8.10 or (2) a majority in interest of the Limited Partners, within ninety (90) days of such date of dissolution, elect one or more new general partners, in which case, the business of the Fund shall be continued; provided that if the Limited Partners elect to continue the Fund's business, an appropriate amendment to the Fund's Certificate of Limited Partnership shall be filed immediately after such election is made, but if no such election is made by a majority in interest of the Limited Partners then liquidation shall proceed in accordance with the terms of Section 9.2;

(ii) the insolvency or bankruptcy of the Fund;

(iii) the General Partner's election at any time to wind-up and dissolve the Fund, including in the event that the general partners of the Underlying Funds, liquidate and dissolve the Underlying Funds prior to the end of the commitment period of the Underlying Funds in accordance with the governing documents of the Underlying Funds; or

(iv) at any time that there are no Limited Partners of the Fund, unless the Fund is continued in accordance with the Delaware Act, or upon entry of a decree of judicial dissolution under Section 17-802 of the Delaware Act.

(b) Subject to Section 9.1(a)(iii), winding-up and dissolution of the Fund shall commence on the date on which the event occurs giving rise to the winding-up and dissolution, but the Fund shall not terminate until the Fund's Certificate of Limited Partnership has been canceled and the assets of the Fund have been distributed as provided in Section 9.2.

(c) None of the death, incapacity, bankruptcy, insolvency, termination or dissolution of any Limited Partner (or Related Employee), the retirement from Goldman Sachs of any Limited Partner (or Related Employee), or the termination of any Limited Partner's (or Related Employee's) employment or consulting arrangement with Goldman Sachs, as applicable, shall result in the winding-up and dissolution of the Fund.

(d) The General Partner may elect to terminate a Series at any time upon the disposition by that Series of all or substantially all of the assets relating to that Series. The Fund shall not dissolve solely upon the

termination of a Series under this Agreement; however, upon the dissolution of the Fund, each Series shall terminate without any further action by the General Partner or any other Person.

9.2. Liquidation. The winding-up and dissolution of the Fund's business shall be liquidated in an orderly manner. The General Partner, or its designee, shall be the liquidator to wind up the affairs of the Fund pursuant to this Agreement, except that if there shall be no General Partner or, in the event that the General Partner has resigned without replacement, withdrawn without replacement, dissolved or is bankrupt, a majority-in-interest of the Limited Partners (including any Institutional Limited Partners) shall elect a Person to perform the functions of the General Partner in liquidating the assets of the Fund and in winding up its affairs. At the termination of the Fund, without limiting the provisions of Section 1.7, the General Partner (or other liquidator appointed by the Limited Partners pursuant to this Section 9.2) shall determine which assets of the Fund shall be sold and which assets of the Fund shall be retained for distribution in kind to the Partners. After all fees, expenses, liabilities and other obligations of the Fund have been paid or duly provided for, the remaining assets and cash (if any) of the Fund shall be distributed to the Limited Partners in accordance with the positive balance in their Capital Accounts determined as of the date of distribution in accordance with Articles VI and VII. In performing its duties, the liquidator is authorized to sell, distribute, exchange or otherwise dispose of the assets of the Fund in any reasonable manner that the liquidator shall determine to be in the best interest of the Partners. During the liquidation of the Fund, the liquidator shall furnish to the Partners the financial statements and other information specified in Article X. The expenses incurred by the liquidator in connection with winding up the Fund, all other losses or liabilities of the Fund incurred in accordance with the terms of this Agreement, and reasonable compensation for the services of the liquidator shall be borne by the Fund. The liquidator shall dispose of or distribute all Fund assets (other than those being held to make reasonable provision for the payment of liabilities of the Fund) within a reasonable period of time, taking into account the nature of the Fund's remaining assets, the need to comply with any contractual restrictions, including the terms of indebtedness to which the Fund or any financing vehicle is subject, the degree of participation of the liquidator in the management of entities whose securities are owned by the Fund, the financial and other effects of a distribution to the Partners of assets in kind and the objective of maximizing the value of the Fund's assets. The liquidator shall not be liable to any Partner for any loss attributable to any act or omission of the liquidator taken in good faith on its part in connection with the liquidation of the Fund and the distribution of its assets. Notwithstanding the foregoing, the General Partner has the right to liquidate the Fund, in the manner described in this Section 9.2, at any time or as otherwise provided by operation of law.

9.3. Potential Early Liquidation of Investments. In order to provide for an orderly and timely liquidation of the Fund, any investments of the Fund may be, subject to applicable law, purchased by Goldman Sachs, its affiliates, one or more other employees or other purchasers, at any time on or after the tenth (10th) anniversary of the final closing of the Carried Interest Paying Funds. Any such purchase would be made using a purchase price equal to the then net asset value of such interests or other investments (as determined by Goldman Sachs in its sole discretion), payable in cash, in kind and/or by delivery of a note (or any combination of the foregoing). Any proposed transactions pursuant to this Section 9.3 will be effected only in accordance with the terms and conditions of the Application or any order approving the Application. The Application permits such transactions upon a determination by the General Partner that the terms of the transactions are fair and reasonable to the investors in the Fund and do not involve overreaching of the Fund or its investors on the part of any person concerned, and that the transactions are consistent with the interests of the investors in the Fund. In making this determination, the General Partner may consider (1) the terms upon which Goldman Sachs or any Third Party Fund are participating in the transaction, (2) the consideration being paid or received by the Fund, (3) the methodology for allocating the investment / disposition opportunity among the Fund and the other participants in the transaction and (4) other factors that it deems relevant. Each Limited Partner acknowledges that by entering into this Agreement, such Limited Partner consents to all such transactions in which Goldman Sachs or an affiliate acts as a principal on the other side of a transaction with the Fund, provided that such transactions meet the requirements set forth in the Application or any order approving the Application.

Article X

Books and Records; Accounting; Valuation; Tax Matters and Elections

10.1. Books and Records. The General Partner shall provide the Limited Partners with such access to the books and records of the Fund and the General Partner as they are entitled under the Delaware Act, provided, however, that the General Partner may in its discretion, to the fullest extent permitted by law, withhold from such Limited Partners certain information to which they are otherwise entitled to receive. Such information may be used for Fund purposes only. The Fund shall maintain books and records as required by the Delaware Act and may maintain such other books and records and may provide such financial or other statements as the General Partner deems advisable.

10.2. Accounting Basis. The books and records, and the financial statements and reports, of the Fund shall be kept in accordance with GAAP.

10.3. Bank Accounts. The General Partner shall maintain the Fund's bank account, and withdrawals shall be made only in the ordinary course of the Fund business on such signature or signatures as the General Partner may determine. Temporary investments are deemed activities in the ordinary course of Fund business.

10.4. Reports.

(a) After the close of the Fund's taxable year, the General Partner will send or cause to be sent to each Person who was a Limited Partner at any time during the taxable year then ended tax information concerning the Fund for use in the preparation by the Limited Partner of the Limited Partner's U.S. federal and state income tax returns. The foregoing tax information may be prepared by internal staff of Goldman Sachs or by a nationally recognized accounting firm. Whether internal staff of Goldman Sachs or an accounting firm prepares the tax information, the costs of preparation and delivery will be borne by the Fund. The Fund does not expect that it will provide a Schedule K-1 to the Limited Partners in advance of April 15th; therefore, each Limited Partner should be prepared to file for an extension of the filing date for his, her or its U.S. federal income tax returns (and any required U.S. state, local, or non-U.S., tax returns). Each Limited Partner will be responsible for the preparation and filing of such Limited Partner's own income tax returns.

(b) The General Partner will send to each Person who was a Limited Partner at the end of the fiscal year then ended financial statements of the Fund audited by the accountants, including the portfolio valuation as of that date. The Fund will use commercially reasonable efforts to send such annual audited financial statements within 180 calendar days after the end of the Fund's fiscal year, or as soon thereafter as practicable. As soon as practicable after the end of the Fund's fiscal year (commencing with the first fiscal year during which the Fund makes an investment), the General Partner will also send to each Person who was a Limited Partner at the end of such fiscal year a report of the investment activities of the Fund during that year. Each Limited Partner acknowledges that the General Partner's ability to provide Limited Partners with financial reports on the Fund depends, in part, on the information the General Partner and the Investment Manager receive (directly or indirectly) from the Underlying Funds, when the General Partner or the Investment Manager receives such information and whether the Underlying Funds agree to permit the Fund to provide such information to Limited Partners.

(c) Notwithstanding clause (b) and anything else herein to the contrary, if a Limited Partner leaves Goldman Sachs (or, in the case of an Affiliated Investor Entity, the related Eligible Investor leaves Goldman Sachs), then the General Partner may, in its sole discretion, withhold from that Limited Partner some of the information Limited Partners are otherwise entitled to receive, to the fullest extent permitted by applicable law.

10.5. Tax Matters and Elections.

(a) The General Partner, or such eligible person as the General Partner may designate, shall be designated as the "partnership representative" of the Fund within the meaning of Section 6223 of the Code and under any similar provision of state, local or non-U.S. tax law. The Fund shall be entitled to appoint a "designated individual" to the extent required or allowed by Treasury Regulations or other official guidance promulgated under or with respect to Subchapter C of Chapter 63 of the Code and under any similar provision of state, local or non-U.S. tax law (such partnership representative or designated individual, as applicable, the "Partnership Representative"). The Partnership Representative is specifically directed and authorized to take whatever steps the Partnership Representative deems necessary or desirable to perfect any such designation, including filing any forms or documents with the Internal Revenue Service (the "IRS") and taking such other action as may from time to time be required under Treasury Regulations. Each Limited Partner does hereby agree that, to the extent permitted by law, (i) the Partnership Representative will have the right to control all administrative and judicial proceedings in respect of tax matters of the Fund, and (ii) the Limited Partners will be bound by the outcome of final administrative adjustments resulting from an audit, as well as by the outcome of judicial review of adjustments. Each Limited Partner further agrees, to the extent permitted by law, to file all tax returns that the Limited Partner is required to file in a manner consistent with the Fund's tax returns and to waive his, her or its rights to participate in any administrative or judicial proceedings with respect to the determination or allocation of the Fund's taxable income.

Each Limited Partner does hereby agree that any action taken by the General Partner in connection with audits of the Fund under applicable tax law will be binding upon such Limited Partner.

(b) The General Partner or the Partnership Representative may, in its discretion, cause the Fund to make or refrain from making all elections required or permitted to be made by the Fund under applicable tax law, including but not limited to an election to treat the Fund as an Electing Investment Partnership as defined in Section 743(e)(5) of the Code and an election under Section 754 of the Code. The General Partner does not expect that the Fund will make an election under Section 754 of the Code. If the General Partner makes an election under Section 754 of the Code and it is required to adjust the tax basis of the Fund's assets as a result thereof, or is otherwise required to adjust the tax basis of the Fund's assets under Section 734 or 743 of the Code, the General Partner shall endeavor to make such adjustments in good faith.

(c) The General Partner is hereby authorized and empowered to prepare or have prepared, to execute or have executed and to file, on behalf of and in the name of the Fund, any returns, applications, agreements, elections and other instruments or documents, under applicable tax law, which it deems desirable or advisable.

(d) If the Fund receives a notice of final partnership adjustment from the IRS, the Partnership Representative may, as determined in its good faith discretion and with respect to any applicable year, cause the Fund to (x) elect the application of Section 6226 of the Code with respect to any imputed underpayment arising from such adjustment, and (y) furnish to each Limited Partner (or former Limited Partner, as applicable) a statement of such Limited Partner's (or former Limited Partner's) share of any adjustment to income, gain, loss, deduction or credit (as determined in the notice of final partnership adjustment).

(e) Each Limited Partner further agrees that such Limited Partner will, upon request by the General Partner or the Partnership Representative, execute any forms or documents (including a power of attorney or settlement or closing agreement), provide any information and take any further action requested by the General Partner or the Partnership Representative, and that the General Partner or the Partnership Representative may execute any forms or documents or obtain any information on such Limited Partner's behalf that relate to such Limited Partner's investment in the Fund, in connection with any tax matter (including in connection with a tax audit or proceeding) affecting the Fund, including as reasonably necessary to effectuate any of the foregoing provisions of this Section 10.5, including with respect to any forms, documents or information reasonably necessary for the Fund to comply with FATCA or avoid being subject to withholding tax under FATCA.

(f) If a Limited Partner fails to comply with its obligations under this Section 10.5 or as required under the withholding rules related to transfers of Interests by non-U.S. persons and such failure results in any taxes, penalties, interest and/or any related costs or expenses (a "Tax Cost"), the General Partner shall, to the extent commercially practicable, cause such Limited Partner to bear the economic burden of such Tax Cost by specially allocating the Tax Cost to such Limited Partner and/or withholding the Tax Cost from proceeds otherwise distributable to such Limited Partner. In the event that the General Partner does not withhold such amounts, the General Partner may require the Limited Partner to reimburse the Fund or the General Partner, as applicable, for any such Tax Costs. In addition, the General Partner shall have full authority to take any steps that the General Partner reasonably determines are necessary or appropriate to mitigate the consequences to the Fund, any entity in which the Fund holds an equity or debt interest and/or any other Limited Partner of such Limited Partner's failure to comply with its obligations under this Section 10.5. Moreover, any Limited Partner that fails to comply with this Section 10.5 shall, to the fullest extent permitted by law, exculpate the General Partner, the Fund and its affiliates, including any member of the Fund's "expanded affiliated group" within the meaning of Section 1471(e)(2) of the Code, for any liabilities related to such failure and indemnify the General Partner, the Fund, and any such affiliates.

(g) Each Limited Partner acknowledges and agrees that:

(i) the Fund is required to comply with the provisions of FATCA;

(ii) it will provide, in a timely manner, such information regarding the Limited Partner and its beneficial owners and such forms or documentation as may be requested from time to time by the Fund (whether by its General Partner or other agents to enable the Fund to comply with the requirements and obligations imposed on it pursuant to FATCA);

(iii) any such forms or documentation requested by the Fund or its agents pursuant to paragraph (ii), or any financial or account information with respect to the Limited Partner's investment in the Fund, may be disclosed to the applicable tax authorities and to any withholding agent where the provision of that information is required by such agent to avoid the application of any withholding tax on any payments to the Fund;

(iv) it waives, and/or shall cooperate with the Fund to obtain a waiver of, the provisions of any law which;

(A) prohibit the disclosure by the Fund, or by any of its agents, of the information or documentation requested from the Limited Partner pursuant to paragraph (ii); or

(B) prohibit the reporting of financial or account information by the Fund or its agents required pursuant to FATCA; or

(C) otherwise prevent compliance by the Fund with its obligations under FATCA;

(v) if it provides information and documentation that is in anyway misleading, or it fails to provide the Fund or its agents with the requested information and documentation necessary in either case to satisfy the Fund's obligations under FATCA, the General Partner reserves the right (whether or not such action or inaction leads to compliance failures by the Fund, or a risk of the Fund or its investors being subject to withholding tax or other costs, debts, expenses, obligations or liabilities (whether external, or internal, to the Fund) (together, "costs") under FATCA), in its sole discretion, to take any action and/or pursue all remedies at its disposal including:

(A) to establish separate sub-accounts within a Limited Partner's capital account for the purpose of calculating FATCA-related costs; and/or

(B) to allocate any or all FATCA-related costs among capital accounts (or sub-accounts within a Limited Partner's capital account) on a basis determined solely by the General Partner; and/or

(C) to compulsorily withdraw such Limited Partner from the Fund; and/or

(D) to hold back or deduct from any withdrawal proceeds or from any other payments or distributions due to such Limited Partner any costs caused (directly or indirectly) by the Limited Partner's action or inaction;

(vi) it shall have no claim against the Fund, the General Partner or any of its or their agents, for any form of damages or liability as a result of actions taken or remedies pursued by or on behalf of the Fund in order to comply with FATCA; and

(vii) it hereby indemnifies the Fund, the General Partner, the Partnership Representative and each of their respective principals, members, partners, managers, officers, directors, stockholders, employees and agents and holds them harmless from and against any FATCA-related liability, action, proceeding, claim, demand, costs, damages, expenses (including legal expenses), penalties or taxes whatsoever which such parties may incur as a result of any action or inaction (directly or indirectly) of such Limited Partner (or any related person) described in the preceding paragraphs. This indemnification shall survive such Limited Partner's ceasing to be a Limited Partner of the Fund and/or the termination, dissolution, liquidation and winding up of the Fund.

(h) By executing this Agreement, each Limited Partner authorizes and directs the Fund to elect to have the "Safe Harbor" described in the proposed Revenue Procedure set forth in IRS Notice 2005-43 (the "Notice") apply to any interest in the Fund transferred to a service provider by the Fund on or after the effective date of such Revenue Procedure in connection with services provided to or for the benefit of the Fund. For purposes of making such Safe Harbor election, the General Partner is hereby designated as the "partner who has responsibility for federal income tax reporting" by the Fund and, accordingly, execution of such Safe Harbor election by the General Partner constitutes execution of a "Safe Harbor Election" in accordance with Section 3.03(1) of the Notice. The Fund and each Limited Partner hereby agree (i) to use commercially reasonable efforts to comply with all requirements of the Safe Harbor described in the Notice, and (ii) that each Limited Partner shall prepare and file all U.S. federal income tax returns reporting the income tax effects of each Interest issued by the Fund that is subject to the Safe Harbor in a manner consistent with the requirements of the Notice. Each Limited Partner authorizes the General Partner to amend this subsection to the extent reasonably necessary to achieve substantially the same tax treatment with respect to any Interest transferred to a service provider by the Fund in connection with services provided to the Fund as set forth in Section 4 of the Notice (e.g., to reflect changes from the rules set forth in the Notice in subsequent IRS guidance); provided that such amendment is not materially adverse to such Limited Partner (as compared with the after-tax consequences that would result if the provisions of the Notice applied to all interests in the Fund transferred to a service provider by the Fund in connection with services provided to the Fund).

(i) A Limited Partner's obligation to comply with the requirements of this Section 10.5 shall survive such Limited Partner's ceasing to be a Limited Partner of the Fund and/or the termination, dissolution, liquidation and winding up of the Fund, and, for purposes of this Section 10.5, the Fund shall be treated as continuing in existence.

Article XI

Miscellaneous Provisions

11.1. Appointment of the General Partner as Attorney-in-Fact.

(a) Each Limited Partner, by such Limited Partner's execution hereof (which execution may be on such Limited Partner's behalf pursuant to a power of attorney or other authority contained in a subscription agreement for Interests), hereby irrevocably makes, constitutes and appoints the General Partner (including any successor General Partner), acting by any of its officers or designees, such Limited Partner's true and lawful agent and attorney-in-fact, with full power of substitution and full power and authority in such Limited Partner's name, place and stead to make, complete, change, correct, execute, sign, acknowledge, swear to, deliver, record and file, on behalf of such Limited Partner and on behalf of the Fund, such documents, instruments and conveyances that may be necessary or appropriate to carry out the provisions or purposes of this Agreement, including:

(i) instruments or agreements required or permitted by law or the provisions of this Agreement including those to effect the admission to or withdrawal from the Fund of Partners and substituted Partners, and all instruments that the General Partner deems necessary or appropriate to reflect a change, modification or novation of this Agreement, in accordance with this Agreement;

(ii) all certificates and other instruments deemed necessary or advisable by the General Partner to carry out the provisions of this Agreement including such certificates, agreements and other documents as may be necessary (A) to organize, invest in and conduct business through one or more Investment Vehicles, (B) to permit the Fund to become or to continue as a limited partnership or a company wherein the limited partners have limited liability in the jurisdictions where the Fund may be doing business, or (C) in the event that Goldman Sachs makes a loan to or guarantees any debt of, the Fund, or in the event that Goldman Sachs or any third party makes a loan to, or guarantees any debt of, any financing vehicle, to permit the Fund and/or the General Partner to pledge, charge or assign, or otherwise make available as credit support for any such loan or guarantee, assets of the Fund, the capital commitments of the Partners, or both;

(iii) all conveyances and other instruments or papers deemed advisable by the General Partner to effect the winding-up and dissolution of the Fund;

(iv) all fictitious or assumed name certificates required or permitted to be filed on behalf of the Fund;

(v) any and all documentation necessary to pledge, assign or otherwise transfer or encumber Fund assets, including the right to make and collect capital contributions;

(vi) any and all documentation necessary to permit the Fund or the General Partner or its respective designee to redeem, purchase, sell, Transfer, convey and assign, including as contemplated by Sections 6.1, 8.7 or 8.9, the Interest (or any portion thereof) of any Limited Partner to any Transferee, including the Fund, the General Partner or Goldman Sachs; and

(vii) all other instruments or papers which may be required or permitted by law to be filed on behalf of the Fund, or which the General Partner considers necessary or desirable to carry out the business of the Fund.

(b) The foregoing power of attorney:

(i) is coupled with an interest, shall be irrevocable, shall not be affected by and shall survive the death, incapacity, dissolution, termination or bankruptcy of the Limited Partner;

(ii) may be exercised by the General Partner without notice to or any additional action on the part of any Limited Partner, either by signing separately as attorney-in-fact for each Limited Partner or, after listing all of the Limited Partners executing an instrument, by a single signature of the General Partner acting as attorney-in-fact for all of them; and

(iii) shall survive the Transfer by a Limited Partner of the whole or any portion of such Limited Partner's Interest; except that, where the Transferee of the whole of such Limited Partner's Interest has been approved by the General Partner for admission to the Fund as a substituted Limited Partner, the power of attorney of the transferor Limited Partner shall survive the delivery of such Transfer for the sole purpose of enabling the General Partner to execute, swear to, acknowledge and file any instrument necessary or appropriate to effect such substitution.

11.2. Amendments to this Agreement.

(a) Except as expressly provided herein, in order to effect an amendment to this Agreement, the General Partner must propose such amendment by submitting (i) the text of such amendment and (ii) a statement of the purpose of such amendment to all Limited Partners. The proposed amendment shall be deemed adopted thirty (30) calendar days after the General Partner submits such notice, unless either the majority of materially adversely affected SOX Insider Preferred Limited Partners or two-thirds in interest of the Limited Partners (including any Institutional Limited Partners but excluding any Participating Interests beneficially owned by Goldman Sachs) have,

by the end of such notice period, delivered its or their written objection to the proposed amendment to the General Partner.

(b) No amendment may:

(i) disproportionately increase a Limited Partner's liability or obligations or disproportionately reduce its rights to distributions without the consent of the disproportionately affected Limited Partner;

(ii) amend Section 5.5 or this Section 11.2 without the consent of all the Partners (except that Section 5.5 may be amended as provided in Section 11.2(d)(v)-(vi)); or

(iii) modify the method provided in Article VI of determining and allocating or distributing, as the case may be, items of income, gain, loss, and deduction and other tax items without the consent of the General Partner and three-quarters in interest of the Limited Partners adversely affected by such modification.

(c) Upon the adoption of any amendment to this Agreement, the amended Agreement or instrument incorporating by reference the terms thereof shall be executed by the General Partner for itself and on behalf of the Limited Partners and, if such amendment requires any filing with respect to the Fund under the Delaware Act or any other filing made in any other state, the General Partner, pursuant to the power of attorney granted in Section 11.1 shall execute and file proper amendments and filings in each jurisdiction in which such action is necessary for the Fund to conduct business or to preserve the limited liability of the Limited Partners.

(d) Notwithstanding Section 11.2(a) or anything in this Agreement to the contrary, the General Partner may amend this Agreement without the consent of the Limited Partners for any of the following purposes:

(i) amendments reflecting admissions or withdrawals of Limited Partners, substitution of Limited Partners or conversions, purchases or forfeitures of Participating Interests;

(ii) amendments with regard to any additional series or classes of Interests issued to Limited Partners, to the extent such amendment does not materially affect the economic rights of the other Limited Partners;

(iii) technical and clarifying amendments that are not materially adverse to the Limited Partners, *provided* that the General Partner provides prompt notice of any such amendment to the Limited Partners;

(iv) amendments necessary to comply with SOX as it may be interpreted from time to time, including amendments to modify the Interest held by any Limited Partner who becomes a SOX Insider during the term of the Fund, *provided* that the amendments are not adverse to the interests of the Limited Partners who are not SOX Insiders;

(v) amendments that make any change (including the removal of any or all restrictions imposed by the terms of this Agreement that were designed to ensure compliance with the Application or any order approving the Application) that the General Partner in its discretion determines is necessary or advisable, to comply with (or reduce the burdens of complying with) the BHCA, the Dodd-Frank Act, including the Volcker Rule, or any other current or future laws, rules, regulations or legal requirements applicable to Goldman Sachs or one or more of the Funds or to reduce, eliminate or otherwise modify the impact, or applicability to, Goldman Sachs or any of its affiliates, or any vehicle organized, offered and/or managed by Goldman Sachs (including the Funds) of any bank regulatory or other regulatory restrictions that might otherwise be imposed upon or affect any such Person or any Limited Partner as a result of Goldman Sachs' status as a bank holding company or a financial holding company

under the BHCA, as an entity otherwise subject to the Dodd-Frank Act (including the Volcker Rule), in each case, including amendments in connection with (A) changes to the structure, ownership or management of the General Partner, or the reorganization or restructuring of the Fund; (B) the liquidation, reorganization, or other modifications to, or restructuring of any of the Funds; (C) distributions in kind of non-marketable securities and/or forced sales of interests by some or all of the Limited Partners (for notes or other consideration); and (D) the resignation of Goldman Sachs as the Investment Manager and/or a change in the General Partner effected in accordance with this Agreement;

(vi) amendments that make any necessary or advisable change, that the General Partner in its discretion determines is necessary or advisable, in connection with any replacement of the Investment Manager, or any assignment or Transfer by the General Partner or the Investment Manager (as permitted by applicable law and/or the Application (or any order approving the Application)) of any of its rights or obligations (or by the General Partner of its interests in the Fund) or any redemption, assignment or Transfer of interests Goldman Sachs may hold in one or more of the Funds, in each case to the extent such redemption, assignment or Transfer is deemed necessary or advisable to comply with (or reduce the burdens of complying with) the BHCA, the Dodd-Frank Act (including the Volcker Rule) or any other applicable statute;

(vii) amendments to cause the Fund to automatically wind-up and dissolve no later than the date that is 15 years from the date of formation, unless the Fund is terminated earlier pursuant to this Agreement;

(viii) amendments to correct any inconsistency with the Offering Memorandum;

(ix) amendments to otherwise facilitate the General Partner's ability to carry out its powers;

(x) amendments the General Partner deems necessary or advisable to comply with the Application (or any order approving the Application);

(xi) amendments to make any change in any provision of this Agreement that requires any action to be taken by or on behalf of the General Partner or the Fund pursuant to the requirements of applicable Delaware law if the provisions of applicable Delaware law are amended, modified or revoked so that the taking of such action is no longer required;

(xii) amendments to address the application to the Fund of the European Union Alternative Investment Fund Managers Directive (the "AIFM Directive"), the UK Alternative Investment Fund Managers Regulations (the "AIFM Regulations") and any laws, regulations and administrative provisions that are adopted by member states of the EEA and/or the UK in compliance with the AIFM Directive, the AIFM Regulations or any similar laws, regulations or administrative provisions relating to the regulation of alternative investment funds or their managers;

(xiii) amendments that are necessary or advisable, in the opinion of the General Partner (including upon the advice or recommendation of the Investment Manager), to qualify the Fund or, if established, any Series, as an entity in which the Limited Partners have limited liability under the laws of any state or other jurisdiction, or to ensure that the Fund will not be treated for U.S. federal income tax purposes as an association taxable as a corporation;

(xiv) if during the term of the Fund (including any extensions thereof), in the discretion of the General Partner, there is a material change in law that affects the U.S. federal income tax treatment of [] the Carried Interest, amendments to restructure the Fund or the Offshore Funds and/or restructure the method by which [] the Carried Interest is allocated, distributed or otherwise paid, or make any other changes to this Agreement or any other relevant agreements in connection therewith, in a

manner intended to reduce or eliminate any adverse impact of any such change in law while preserving the intended economic arrangement [], as reasonably determined by the General Partner [];

(xv) amendments the General Partner determines in its sole discretion to be necessary or advisable in order to address any changes in tax law or interpretations thereof applicable to the Funds or Goldman Sachs, including a change in law or interpretations that affects the U.S. federal income tax treatment of some or all of the income allocations or distributions by the Funds, including any necessary or advisable changes, in the opinion of the General Partner, in order to restructure the Fund or the Offshore Funds and/or the way in which amounts are allocated, distributed or otherwise paid, changes to the relevant agreements in connection therewith, subdividing or combining one or more classes of interests in the Fund or converting one or more classes of interests in the Fund to one or more Series of Fund interests (including any segregation of assets and liabilities in connection therewith), each of which will be treated as a separate entity for U.S. federal income tax purposes;

(xvi) amendments the General Partner determines in its sole discretion to be necessary or advisable to change the domicile of, or otherwise continue, the Fund, the General Partner or any other Fund entity to another jurisdiction to address legal, regulatory or tax concerns applicable to the Fund, its investments or Limited Partners;

(xvii) amendments that do not, as reasonably determined by the General Partner, adversely affect the Limited Partners considered as a whole in any material respect;

(xviii) amendments to prevent the assets of the Fund and/or a Subsidiary from being treated as "plan assets" that are subject to the Employee Retirement Income Security Act of 1974 (as the same can be amended from time to time) ("ERISA") or Section 4975 of the Code (or a comparable law or regulation);

(xix) if the Application is withdrawn without the SEC issuing an order approving the Application, amendments to ensure that the Fund complies with the order of the SEC dated August 14, 1998 (Release No. IC 23390);

(xx) amendments that are substantially similar to any amendments made to the governing documents of one or more of the Underlying Funds (whether or not approved by the investors in such funds); and

(xxi) amendments necessary to make any other modification or amendments similar to the foregoing

provided that, in the event that this Agreement is amended pursuant to clause (iii), the General Partner shall promptly provide notice of such amendment to the Limited Partners.

(e) In the event that an Institutional Limited Partner, in connection with a vote or consent solicitation with respect to its Interest, informs the General Partner of the results of a poll of the Indirect Investors with respect to such Interest, the General Partner shall permit such Institutional Limited Partner to vote or consent with respect to its Interest in accordance with results of such poll.

11.3. **Arbitration.** Any dispute, controversy or claim arising out of or in connection with or relating to this Agreement or any breach or alleged breach hereof shall (to the extent not prohibited by governing law) be determined and settled by arbitration in the City of New York pursuant to the rules then in effect of the American Arbitration Association. Any award rendered shall be final and conclusive upon the parties and a judgment thereon may be entered in the highest court of the forum, state or federal, having jurisdiction.

11.4. **Notices.** Any notice to the Fund or the General Partner shall be sent to the principal office of the Fund to the attention of the General Counsel, and any notice to a Limited Partner shall be sent to such Limited

Partner's business or residence address, as set forth on the list, as amended from time to time, maintained by the General Partner as provided in Section 3.3.

11.5. Binding Provisions. The covenants and agreements contained herein shall be binding upon and inure to the benefit of the heirs, executors, administrators, successors and assigns of the respective parties hereto.

11.6. Governing Law. This Agreement shall be construed and enforced in accordance with the laws of the State of Delaware. If it is determined by a court of competent jurisdiction that any provision of this Agreement is invalid under applicable law, such provision will be ineffective only to the extent of such prohibition or invalidity, without invalidating the remainder of this Agreement.

11.7. Counterparts. This Agreement may be executed in several counterparts, all of which together shall constitute one agreement binding on all parties hereto, notwithstanding that not all the parties have signed the same counterpart, except that no counterpart shall be binding unless signed by the General Partner. The General Partner may execute any document by facsimile signature of a duly authorized officer.

11.8. Severability of Provisions. If for any reason any provision or provisions hereof that are not material to the purposes or business of the Fund or the Limited Partners' Interests are determined to be invalid and contrary to any existing or future law, such invalidity shall not impair the operation of or affect those portions of this Agreement that are valid.

11.9. Interpretation. For all purposes of this Agreement, except as otherwise expressly provided herein:

(a) the words "including", "includes", "include", and other words of similar import shall be deemed to be followed by the phrase "without limitation";

(b) references to a Limited Partner's subscription agreement shall be deemed to include all of a Limited Partner's subscription materials, including his, her or its subscriber signature page, consent and power of attorney into which the Limited Partner's subscription agreement is incorporated; and

(c) in any case where Goldman Sachs, the Investment Manager or the General Partner is authorized or required to take or omit an action, make any decision, determination, valuation, or grant or withhold any approval, consent or waiver,

(i) it may do so (or not do so) in its sole and absolute discretion or judgment;

(ii) it shall be entitled to (but not required to) take into account any considerations, interests or factors it deems appropriate or desirable, including its own interests and interests of affiliates;

(iii) such action (or inaction), decision, determination, approval, valuation, or grant or withholding of consent shall be final, binding and conclusive as to the Fund, the Limited Partners and their respective successors, assigns or personal representatives; and

(iv) when so doing, it shall not be subject to any duties or standards (including fiduciary or similar duties or standards) existing under the Delaware Act or other Delaware law (or in equity).

It is intended that the terms of this Agreement be construed in accordance with their fair meanings and not against any particular Person, including the General Partner.

11.10. Entire Agreement. This Agreement and the subscription agreements for the Interests constitute the entire agreement among the parties relating to the subject matter hereof. Except for the subscription agreements, this Agreement supersedes any prior agreement or understanding among the parties and may not be modified or amended in any manner other than as set forth herein.

11.11. Headings. The headings in this Agreement are for descriptive purposes only and shall not control or alter the meaning of this Agreement as set forth in the text.

11.12. **Further Assurances.** Each Limited Partner (including each SOX Insider Preferred Limited Partner) shall do and perform or cause to be done and performed all further acts and things and shall execute and deliver all other agreements, certificates, instruments and documents as the General Partner reasonably may request in order to carry out the intent and accomplish the purposes of this Agreement and the consummation of the transactions contemplated hereby.

11.13. **Effectiveness.** This Agreement shall become effective as of the day and year first above written upon execution hereof by the General Partner and the Initial Limited Partner and, as to each additional Limited Partner, when the subscription hereto by such party has been accepted by the General Partner.

11.14. **FCC Representations and Covenants.** Each Limited Partner hereby acknowledges, covenants and agrees that it will not be materially involved, directly or indirectly, in the management or operation of the media- or carrier-related activity of the Fund or any Portfolio Company that, directly or indirectly, owns, controls or operates a broadcast station, cable television system, a wireless or wireline telecommunications business, daily newspaper, any other communications facility or any other activity regulated by, or otherwise subject to ownership restrictions imposed by, the U.S. Federal Communications Commission (the “FCC”) (including restrictions imposed by Section 310(b) of the Communications Act of 1934, as amended (the “Communications Act”)) (a “Media Company”) and that neither it nor any of its directors, officers, partners, members or greater-than-five-percent (5%) voting or equity holders will:

(a) act as an employee of the Fund or any Media Company (directly or through the officers, directors, partners, members or affiliates of such Limited Partner) if such Limited Partner’s functions (or those of its officers, directors, partners, members or affiliates), directly or indirectly, relate to the media or carrier enterprises of the Fund or any Media Company;

(b) serve, in any material capacity, as an independent contractor or agent with respect to the media or carrier enterprises of the Fund or any Media Company;

(c) communicate with the Fund, the management of any Media Company or with the General Partner on matters pertaining to the day-to-day operations of any Media Company or the Fund;

(d) vote to admit any additional or replacement General Partner to the Fund unless such additional or replacement General Partner has been approved by the General Partner(s) then existing;

(e) perform any services for the Fund or any Media Company materially relating to its media or carrier activities; or

(f) become actively involved in the management or operation of any Media Company or the media or carrier businesses of the Fund.

Notwithstanding the foregoing limitations, a Limited Partner may with the consent of the General Partner, have or establish a direct relationship with, or obtain an interest in, a Media Company other than through, by or on behalf of the Fund which would otherwise be proscribed by Sections 11.14(b), 11.14(c) and 11.14(f); provided that such Limited Partner acknowledges that it may be deemed thereby to have an attributable interest in such Media Company as provided in the rules, regulations and policies of the FCC as a result of such interest or relationship; and provided further that such Limited Partner agrees that it shall (i) be responsible for ensuring its own compliance with the Communications Act, and all applicable rules, regulations and policies of the FCC (the “FCC Rules”); (ii) provide the General Partner with such information reasonably requested by the General Partner to ensure that the Fund and any Media Company remain in compliance with the Communications Act and the FCC Rules; and (iii) take any and all action necessary or reasonably requested by the General Partner, including divesting or transferring its interest in the Fund to ensure that the Fund and any Media Company (x) remain in compliance with the Communications Act and the FCC Rules, or (y) are not precluded from holding or acquiring an interest in any other business.

The foregoing provisions of this Section 11.14 are designed to ensure, to the extent possible, that the Limited Partners will not be deemed to hold an attributable interest in a Media Company. Accordingly, such provisions shall be read to preclude the Limited Partners from engaging in any activity which is inconsistent with the insulation criteria of the FCC, as such may be amended, modified or clarified from time to time and the General Partner is authorized to amend the provisions of this Section 11.14 as may be deemed necessary to ensure that this Agreement insulates the Limited Partners from attribution pursuant to the FCC Rules and for purposes of the foreign ownership restrictions set forth in Section 310(b) of the Communications Act. Limited Partners, however, are not precluded by this Section 11.14 from engaging in any activity which the FCC would deem consistent with its insulation criteria and the non-attributable status of such Limited Partner.

11.15. Compliance with Applicable Law. Without limitation of any of the General Partner's rights and/or authorities herein, the General Partner, in its sole discretion and without the consent of any Limited Partner, may take any action, including (a) liquidating or reorganizing the Fund or taking other conforming steps or (b) otherwise modifying, amending or restructuring the Fund, in each case, without regard to any applicable tax consequences, and as the General Partner or Goldman Sachs in its discretion determines is necessary or advisable for Goldman Sachs or any of its affiliates to comply with the Application, any order approving the Application, applicable laws, rules or regulations (including the BHCA and/or the Dodd-Frank Act, including the Volcker Rule) affecting the Fund, the Limited Partners, Goldman Sachs or any of its affiliates. By exercising any of its rights, or taking any of the actions described in this Section 11.15, the General Partner may cause distributions in kind of non-marketable securities, forced sales of Interests by some or all Limited Partners (for notes or other consideration), or resignation of Goldman Sachs as the Investment Manager and/or the General Partner. Furthermore, the General Partner may exercise any such rights or authorized actions in its sole discretion and with respect to certain or all Limited Partners in a manner that may disproportionately affect some Limited Partners relative to other Limited Partners (even similarly situated Limited Partners) and the General Partner will have no duty (implied or otherwise) to mitigate the effect on any Limited Partners or to treat any similarly situated Limited Partners similarly.

11.16. Confidentiality.

(a) Subject to the exception in the fourth sentence of this Section 11.16(a), each Limited Partner agrees to keep confidential all communications between the General Partner and/or the Investment Manager and/or any of their respective affiliates, on the one hand, and such Limited Partner, on the other hand, and any other information regarding the other Limited Partners, the Fund, any existing, past or prospective Fund Investment (including any Subsidiary, Investment Vehicle or Alternative Investment Vehicle), Underlying Fund or Portfolio Company or the affairs of the Fund generally which such Limited Partner may obtain or have access to by reason of being a Limited Partner, including (i) any information regarding any other Limited Partner (including the identity of such Limited Partner); (ii) any information, financial or otherwise, regarding the Fund, including quarterly and annual reports, financial results, market projections and other information provided to the Limited Partners as part of the Fund's ongoing reporting obligations and investor communications; and (iii) any information, financial or otherwise, regarding any existing, past or prospective Fund Investment or Portfolio Company (all such communications and any such information whether obtained from the Fund, the General Partner or any other source, collectively, the "Confidential Information"). Each Limited Partner acknowledges and agrees that the Confidential Information shall be deemed non-public, confidential and proprietary in nature and shall constitute trade secrets under applicable law with respect to the Fund and its investments, the disclosure of which could have adverse effects on the Fund. No Limited Partner will seek to obtain, either lawfully or unlawfully, the identity of any other Limited Partner or any information regarding any other Limited Partner, whether or not such information is available generally to Persons who are Limited Partners. Without the prior written consent of the General Partner (which may be withheld in the discretion of the General Partner), each Limited Partner agrees that he, she or it shall (x) maintain in strict confidence and not disclose any Confidential Information to any Person who is not an officer, employee, accountant, consultant, attorney, portfolio administrator, tax advisor or any other advisor who is involved in such Limited Partner's investments, except to the extent such information is required by applicable law to be reflected in such Limited Partner's tax filings, and (y) not use Confidential Information for any purpose, including contacting other Limited Partners, other than the preparation of such Limited Partner's tax returns and evaluation of the performance of such Limited Partner's investment in the Fund. Each Limited Partner shall first advise any such officer, employee, accountant, consultant, attorney, portfolio administrator, tax advisor or other advisor involved in such Limited Partner's investments of the confidential nature of such information and such Limited Partner's

obligations with respect to the Confidential Information prior to the disclosure of any such information. Without limiting the generality of the foregoing, each Limited Partner agrees that it shall not use Confidential Information in contravention of applicable securities laws, including in respect of any direct or indirect purchase or sale of any publicly-traded securities (or derivatives thereof) while in possession of material non-public information. Each Limited Partner further agrees that the General Partner may refuse a Limited Partner's request to furnish any correspondence, documents or other Confidential Information to any Person who is not an officer, employee, accountant, consultant, attorney, portfolio administrator, tax advisor or any other advisor who is involved in such Limited Partner's investments. In the event disclosure of any such information is permitted by the General Partner or the exceptions set forth above, such Limited Partner is responsible for the compliance by any such recipient with the foregoing restrictions. Each Limited Partner acknowledges and agrees that monetary damages would not be a sufficient remedy for any breach of this Section 11.16 by a Limited Partner or its representatives and that in addition to any other remedies available to the Fund in respect of any such breach, the Fund shall be entitled to specific performance and injunctive or other equitable relief as a remedy for any such breach, without the obligation of posting a bond or other security.

(b) Each Limited Partner acknowledges that the other Limited Partners are relying on such Limited Partner to maintain the confidentiality of any information relating to such other Limited Partners, the Fund, any existing, past or prospective Fund Investment or Portfolio Company, the affairs of the Fund generally and any other Confidential Information. Accordingly, the Limited Partners hereby acknowledge and agree that to the fullest extent permitted by Section 17-305(a) of the Delaware Act, the rights of a Limited Partner to obtain information from the General Partner and the Fund shall be restricted to only those rights provided for in this Agreement, and that any other rights provided under Section 17-305(f) of the Delaware Act shall not be available to the Limited Partners or applicable to the Fund, except as otherwise provided by the General Partner. The General Partner and the Fund may also keep confidential and not disclose to any or all Limited Partners any Confidential Information. Notwithstanding anything in this Agreement to the contrary, any information to be provided or disclosed to a Limited Partner may be adjusted, such that the data that identifies any other Limited Partner and any existing, past or prospective Fund Investment or Portfolio Company is not disclosed to the Limited Partner. Each Limited Partner acknowledges and agrees that the Confidential Information shall be deemed non-public, confidential and proprietary in nature and shall constitute trade secrets under applicable law with respect to the Fund and its Fund Investments, the disclosure of which could have adverse effects on the Fund, the General Partner, the Investment Manager, Fund Investments and Portfolio Companies and that the General Partner may withhold some or all information which would otherwise be provided to such Limited Partner under the terms of this Agreement and shall be entitled to terminate the Interest of any Limited Partner that discloses Confidential Information in a manner not expressly permitted in this Section 11.16.

(c) []

(d) Each Limited Partner (x) acknowledges that the General Partner may release confidential information about such Limited Partner and, if applicable, any underlying beneficial owners of such Limited Partner (i) if the General Partner reasonably determines that such release is in the best interest of the Fund, including in light of applicable laws and regulations concerning money laundering and similar activities, (ii) if required by any law or regulation of any jurisdiction applicable to the Fund, (iii) in connection with any incurrence of any Financing, or (iv) in connection with preparing for or the making of any Fund Investment, and (y) agrees to provide the General Partner with any additional information that the General Partner deems necessary to ensure compliance with any laws or regulations applicable to the Fund or its business.

(e) In the event a Limited Partner (or his, her or its representatives) is requested to disclose any Confidential Information (i) to any governmental regulatory body having jurisdiction over the Limited Partner, (ii) in response to any court order, subpoena, civil investigative demand or similar process or (iii) in connection with any disclosure obligation under any law or regulation of any jurisdiction applicable to the Limited Partner (including FOIA or any comparable law or regulation of any other jurisdiction (including any non-U.S. jurisdiction)), such Limited Partner shall provide written notice to the General Partner promptly after such request and prior to responding, unless such notice is prohibited by applicable law, so that the General Partner may seek a protective order or other appropriate remedy (and such Limited Partner agrees to cooperate with the General Partner in

connection with seeking such order or other remedy). In the event that such protective order or other remedy is not obtained, such Limited Partner agrees to furnish only that portion of the information that it determines, after consultation with counsel, is legally required, and to exercise best efforts to obtain assurance that confidential treatment will be accorded such information. No such notice shall be required with respect to disclosure to a governmental regulatory body pursuant to periodic regular regulatory examinations. In addition, upon receipt by the General Partner of written notice from any such Limited Partner of such public disclosure request, the General Partner (x) may facilitate the sale or Transfer or may purchase the interest of such Limited Partner or (y) may withhold some or all information which would otherwise be provided to such Limited Partner under the terms of this Agreement if, in the case of either clause (x) or clause (y), the General Partner reasonably determines that the disclosure of such information could adversely affect the Fund.

(f) Each Limited Partner understands and acknowledges that the Fund, the General Partner and Goldman Sachs make no representation or warranty as to the accuracy or completeness of that portion of the Confidential Information provided to the Limited Partner which is provided to the Fund by any unaffiliated third party, including any Fund Investment or Portfolio Company ("Third Party Information"), and that to the extent the Fund provides any such Third Party Information to any Limited Partner, each Limited Partner acknowledges and agrees that such information is provided for informational purposes only. The Fund, the General Partner and Goldman Sachs shall have no liability to any Limited Partner or any other Person resulting from reliance on or use of the Third Party Information.

(g) Notwithstanding anything contained in this Agreement to the contrary, except as reasonably necessary to comply with applicable securities laws, each Limited Partner (and any employee, representative or other agent thereof) may disclose to any and all Persons the tax treatment and tax structure of, and all tax strategies relating to, the Fund, the Limited Partner's ownership of an Interest in the Fund, and any Fund transaction and all materials of any kind (including opinions and other tax analyses) that are provided to the Limited Partner relating to such tax treatment, tax structure and tax strategies. For this purpose, "tax structure" means any facts relevant to the U.S. federal, state or local income tax treatment of the Fund, the Limited Partner's ownership of an Interest in the Fund, and any Fund transaction, and does not include information relating to the identity of the Limited Partners, any Fund Investment, any Underlying Fund, any Portfolio Company or any of their respective affiliates. Nothing in this Section 11.16 shall be deemed to require the General Partner to disclose to any Limited Partner any information that the General Partner is permitted or is required to keep confidential in accordance with this Agreement or otherwise.

11.17. Advisory Committee Authorization.

(a) Each Limited Partner hereby authorizes the General Partner, on behalf of such Limited Partner, to cause the formation of a committee (the "Advisory Committee"), the purpose of which will be to consider and, on behalf of the Limited Partners, approve or disapprove, to the extent determined by the General Partner to be required by applicable law (including the Advisers Act) or otherwise appropriate, principal transactions, certain other related party transactions, transactions involving a potential conflict of interest and such other matters as the General Partner shall determine appropriate and to select one or more Persons, who shall not be affiliated with the General Partner, to serve on the committee. In no event shall any such transaction be entered into unless it complies with applicable law. A consent by the Advisory Committee acting on behalf of the Limited Partners and/or the Fund with respect to any matter shall be deemed to constitute the consent of the Limited Partners and the Fund. There may be a single Advisory Committee for all of the Funds, notwithstanding that such committee may not include (or may underrepresent) investors in any particular Fund.

(b) Each Limited Partner further acknowledges that any investor advisory committee of the Underlying Funds (each, an "Investor Advisory Committee") will consider matters similar to those described in clause (a) above as well as other matters in connection with such Underlying Funds. The General Partner may delegate to an Investor Advisory Committee the authority to approve, on behalf of the Fund, any such matter that the General Partner determines in its discretion affects the applicable Underlying Fund and the Funds as a whole. In such a case the consent of the Investor Advisory Committee with respect to such matter shall be deemed to

constitute the consent of the Limited Partners and the Fund with respect to such matter (without, for the avoidance of doubt, any separate consent of the Advisory Committee).

(c) No member of an Investor Advisory Committee or the Advisory Committee, if formed, shall be liable to any Partner or the Fund for any act or omission performed or omitted by him, her or it, as the case may be, arising out of such Person's activities in connection with serving on an Investor Advisory Committee or the Advisory Committee (other than willful misfeasance, or bad faith on the part of such member) including for any mistake in judgment, any action or inaction taken or omitted to be taken, or for any loss due to any mistake, action or inaction arising out of such Person's activities in connection with serving on an Investor Advisory Committee or the Advisory Committee. The General Partner is authorized and empowered to cause the Fund to provide members of the Advisory Committee similar rights of reimbursement, advance, indemnity, contribution and other similar rights as are provided to Indemnified Persons above.

(d) To the extent that an Investor Advisory Committee approves any matter (including any approval required by or appropriate (as determined by Goldman Sachs) in light of applicable law, including the Advisers Act, principal transactions, certain other related party transactions, transactions involving a potential conflict of interest and such other matters, in each case, that are presented to the Investor Advisory Committee) such matter will also be deemed approved by the Partners and the Fund.

11.18. Interests Constitute Article 8 Securities. The Fund and each Limited Partner expressly acknowledge and agree that (a) each Interest in the Fund is a security governed by Article 8 of the Uniform Commercial Code in effect in the state of Delaware (the "DEUCC") and the Uniform Commercial Code of any other relevant jurisdiction and (b) this Agreement establishes the terms of the Interests in the Fund. The "issuer's jurisdiction" (within the meaning of Section 8-110 of the DEUCC) of the Fund shall be the State of Delaware.

[The remainder of this page intentionally left blank.]

IN WITNESS WHEREOF, the parties hereto have executed and delivered this Agreement as of the date first above written.

[],
as General Partner

By: ____
Name:
Title:

[],
as Original General Partner

By: ____
Name:
Title:

[],
as Initial Limited Partner

By: ____
Name:
Title:

LIMITED PARTNERS

Executed and delivered on behalf of all those Limited Partners whose names, residence or business address appear on the list of Limited Partners which is maintained by the General Partner as provided in Section 3.3, by [] pursuant to a duly granted power of attorney or other authority.

[], as Attorney-in-Fact for each Limited Partner

By: ____
Name:
Title:

AMENDED AND RESTATED AGREEMENT OF LIMITED PARTNERSHIP

EXHIBIT A

General Partner

Name: []

Registered Office: []

Business Address: []

Capital Commitment []

CERTAIN INFORMATION, IDENTIFIED BY AND REPLACED WITH A MARK OF “[],” HAS BEEN EXCLUDED FROM THIS DOCUMENT BECAUSE IT IS BOTH (I) NOT MATERIAL AND (II) THE TYPE THAT THE REGISTRANT TREATS AS PRIVATE OR CONFIDENTIAL.

**FORM OF SUBSCRIPTION AGREEMENT AND MATERIALS FOR
PARTICIPANTS IN THE LONG TERM EXECUTIVE CARRIED INTEREST INCENTIVE PROGRAM**

Individual and Entity

**FORM OF SUBSCRIPTION MATERIALS FOR
PARTICIPANTS IN THE LONG TERM EXECUTIVE CARRIED INTEREST INCENTIVE PROGRAM**

[] (the "Fund")
Class A Limited Partnership Interests¹
Class O Limited Partnership Interests

[]

SUBSCRIBER SIGNATURE PAGE, CONSENT AND POWER OF ATTORNEY

I. PERSONAL INFORMATION

GS Offeree's Name _____

Employee ID # _____

Office Location _____

These Subscription Agreement and Materials, including the supplement provided to you as Annex B (the "Supplement"), relate to the issuance of Class O Interests (as defined below) in [] (the "Fund"). All references herein to the "Partnership Agreement" shall refer to the agreement of limited partnership of the Fund, as amended or restated through the date hereof and (iii) the "Offering Memorandum" shall refer to the Offering Memorandum of the Fund, as supplemented or amended through the date hereof (other than by the Supplement) and (iv) the "Subscription Agreement" shall refer to the Subscription Agreement of the Fund provided to you as Annex A. Capitalized terms not otherwise defined in these Subscription Materials have the same meaning set forth in the Partnership Agreement or the Offering Memorandum.

You may either invest directly in the Fund (in which case you are both the "Subscriber" and the "GS Offeree" for purposes of this agreement) or offer your investment opportunity to an entity that you control (in which case you are the "GS Offeree" and such entity is the "Subscriber" for purposes of this agreement). Please enter the Subscriber name and check the appropriate box indicating the Subscriber for your investment opportunity and, if the Subscriber is an entity, complete the "Investor Certification - Entity" form:

Subscriber Name: _____

☐ Self	☐ Corporation	☐ Partnership	☐ LLC	☐ Trust
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¹ Class A Limited Partnership Interests represent the required minimum equity capital commitment with respect to the Class O Limited Partnership Interests.

I. ACCEPTANCE OF CLASS O INTERESTS AND SUBSCRIPTION REQUEST

The minimum allowable equity capital commitment is detailed in the table below:

Fund	Minimum Equity Capital Commitment
[]	[]

ACCEPTANCE OF CLASS O INTERESTS

I irrevocably **ACCEPT** the opportunity to participate in the Fund with respect to the Class O Interests ("Class O Interests") being offered to the GS Offeree, that will be granted as of []. The Class O Interests offered to the GS Offeree have been communicated separately.

I acknowledge and affirm that my participation in this opportunity is subject to the acceptance of my request, in whole or in part, by the Fund.

Class O Interests are entitled to share in a portion of any Carried Interest that may be earned from the applicable external funds as described in the Offering Memorandum (including, as applicable, one or more of the External Funds, and any other funds, entities, separate accounts and/or single investor vehicles managed or advised by Goldman Sachs Asset & Wealth Management, in each case, as described in the Offering Memorandum) (each, a "Carried Interest Paying Fund," and collectively, the "Carried Interest Paying Funds") subject to the return hurdles, notch amounts and other limits described in the offering memorandum of each Carried Interest Paying Fund and/or the Supplement.

I understand (i) that the vesting dates for the Class O Interests are set forth in the Supplement and are one year later than the vesting dates set forth in the Offering Memorandum, (ii) my obligations under the GS Agreements (including the terms of the non-compete provisions), (iii) the illiquid nature of the Class O Interests and (iv) the tax consequences associated with the Carried Interest (as defined in the Offering Memorandum). I further understand that (a) I should not expect to receive any distributions (other than potentially tax distributions) in respect of the Carried Interest before [] (if ever), (b) I may recognize substantial amounts of taxable income in each year, the taxes on which may exceed the amount of any distribution from the Fund in that year, (c) by accepting this opportunity, I am granting Goldman Sachs the right to net any amounts owed under the Employee Special Investments Program, or from other sources, as described in the Offering Memorandum and Subscription Agreement, (d) notwithstanding anything to the contrary in the Partnership Agreement or the Offering Memorandum, distributions in respect of the Carried Interests will be subject to the Incremental Hurdle and "catch-up" mechanics described in the Supplement and (e) the Fund reserves the right, in its sole discretion, to establish and offer to investors one or more additional series or classes of interests with terms that may be different from those described herein and in the Supplement, at any time (any such offering, a "Re-Offer"), including interests that are entitled to share in a portion of any Carried Interest that may be earned from the applicable Carried Interest Paying Funds and, if any Re-Offers are conducted, the amount of Carried Interest allocated to me may be delayed or reduced (as compared to the amounts that would have been allocated if no Re-Offers had occurred).

By executing the Subscriber Signature Page, I am confirming my understanding of, and agreement to, the terms as described in the provisions set forth herein and in the Offering Memorandum and the Supplement with respect to the Carried Interest and its potential consequences, notwithstanding anything to the contrary in any

documents I have received or may receive in connection with any other Goldman Sachs employee funds in which I may own an interest.

PLEASE ENTER SUBSCRIPTION REQUEST AMOUNT ON THE NEXT PAGE.

SUBSCRIPTION REQUEST

To the extent different from the GS Offeree, the Subscriber identified in Section I irrevocably **REQUESTS** the opportunity to participate in the Fund with respect to the Class O Interests being offered to the GS Offeree. Both the Subscriber and the undersigned GS Offeree (a) re-affirm the Subscription Agreement, including the representations, consents, agreements and power of attorney contained therein, which Subscription Agreement the Subscriber and GS Offeree had acknowledged and affirmed in connection with the most recent prior offering by the Fund, as of the date hereof and (b) acknowledge and affirm that the Subscriber's participation in the Fund is subject to acceptance of the Subscriber's subscription request, in whole or in part, by the Fund. Notwithstanding the provisions of the Subscription Agreement of the Fund, these Subscription Materials are governed by and shall be construed in accordance with the internal laws of the State of Delaware applicable to a contract made and performed wholly within the State of Delaware.

Interests	Amount
Class A Unlevered Equity Capital Commitment	US\$ _____
Class O Carried Interest	Election to Receive Class O Interests (PLEASE INITIAL AS APPROPRIATE) I ACCEPT _____ I DECLINE _____

Please note that in order to participate in the Fund, the GS Offeree (or its controlled entity that is the Subscriber) must qualify as an "accredited investor" under the income test set forth in Rule 501(a)(6) of Regulation D under the U.S. Securities Act of 1933, as detailed on pages [] and [] of these Subscription Materials, or in the case of a controlled entity, under another provision of Rule 501(a).

Goldman Sachs reserves the right, in its sole discretion, to reduce in whole or in part your subscription amount proportionally or otherwise, including, without limitation, the proportion of Carried Interest to which any Class O Interest(s) issued to you relate, with or without adjusting any related capital commitments.

PLEASE CONTINUE ON THE NEXT PAGE.

II. PAYMENT INSTRUCTIONS

[]

III. AUTHORIZATION TO DEBIT BROKERAGE ACCOUNT AND TO DEPOSIT DISTRIBUTION INTO BROKERAGE ACCOUNT

[]

PLEASE COMPLETE THE APPLICABLE INVESTOR CERTIFICATIONS (PAGE [] IF THE SUBSCRIBER IS AN INDIVIDUAL AND PAGES [] IF THE SUBSCRIBER IS AN ENTITY).

INVESTOR CERTIFICATION - INDIVIDUAL

[]

INVESTOR CERTIFICATION - ENTITY

[]

[CERTAIN TAX FORMS]

[]

CERTIFICATION

In connection with the Fund's obligations relating to "Disqualifying Events" pursuant to Rule 506(d) of the U.S. Securities Act of 1933, as amended (the "1933 Act"), please indicate below whether any of the following (which are "Disqualifying Events" as provided in Rule 506(d) of the 1933 Act) have occurred with respect to you (or any other person with whom you are subscribing, including your spouse, if applicable).

Check all applicable boxes and provide the details of any such Disqualifying Events below. "Disqualifying Event" includes any of (i) - (viii) below. You must check at least one box below.

- | | |
|--------------------------|--|
| ☐ | None of the below have occurred with respect to you (or any other person with whom you are subscribing, including your legal spouse, if applicable). |
| ☐ | (i) a conviction, within the past ten years, of any felony or misdemeanor: (A) in connection with the purchase or sale of any security; (B) involving the making of any false filing with the U.S. Securities and Exchange Commission (the "Commission"); or (C) arising out of the conduct of the business of an underwriter, broker, dealer, municipal securities dealer, investment adviser or paid solicitor of purchasers of securities; |
| ☐ | (ii) being subject to any order, judgment or decree of any court of competent jurisdiction, entered within the past five years, that, as of the date hereof, restrains or enjoins such person from engaging or continuing to engage in any conduct or practice: (A) in connection with the purchase or sale of any security; (B) involving the making of any false filing with the Commission; or (C) arising out of the conduct of the business of an underwriter, broker, dealer, municipal securities dealer, investment adviser or paid solicitor of purchasers of securities; |
| ☐ | (iii) being subject to a final order of a state securities commission (or an agency or officer of a state performing like functions); a state authority that supervises or examines banks, savings associations, or credit unions; a state insurance commission (or an agency or officer of a state performing like functions); an appropriate federal banking agency; the U.S. Commodity Futures Trading Commission (the "CFTC"); or the National Credit Union Administration that: (A) as of the date hereof, bars the person from: (1) association with an entity regulated by such commission, authority, agency, or officer; (2) engaging in the business of securities, insurance or banking; or (3) engaging in savings association or credit union activities; or (B) constitutes a final order based on a violation of any law or regulation that prohibits fraudulent, manipulative, or deceptive conduct entered within the past ten years; |
| ☐ | (iv) being subject to an order of the Commission entered pursuant to section 15(b) or 15B(c) of the U.S. Commodity Exchange Act, as amended, and its rules and regulations (the "Exchange Act") (15 U.S.C. 78o(b) or 78o-4(c)) or section 203(e) or (f) of the Investment Advisers Act (15 U.S.C. 80b-3(e) or (f)) that, as of the date hereof: (A) suspends or revokes such person's registration as a broker, dealer, municipal securities dealer or investment adviser; (B) places limitations on the activities, functions or operations of such person; or (C) bars such person from being associated with any entity or from participating in the offering of any penny stock; |

- ☐ (v) being subject to any order of the Commission entered within the past five years that, as of the date hereof, orders the person to cease and desist from committing or causing a violation or future violation of: (A) any scienter-based anti-fraud provision of the federal securities laws, including without limitation section 17(a)(1) of the 1933 Act (15 U.S.C. 77q(a)(1)), section 10(b) of the Exchange Act (15 U.S.C. 78j(b)) and 17 CFR 240.10b-5, section 15(c)(1) of the Exchange Act (15 U.S.C. 78o(c)(1)) and section 206(1) of the Investment Advisers Act (15 U.S.C. 80b-6(1)), or any other rule or regulation thereunder; or (B) section 5 of the 1933 Act (15 U.S.C. 77e);
- ☐ (vi) being suspended or expelled from membership in, or suspended or barred from association with a member of, a registered national securities exchange or a registered national or affiliated securities association for any act or omission to act constituting conduct inconsistent with just and equitable principles of trade;
- ☐ (vii) having filed (as a registrant or issuer), or was named as an underwriter in, any registration statement or Regulation A offering statement filed with the Commission that, within the past five years, was the subject of a refusal order, stop order, or order suspending the Regulation A exemption, or is, as of the date hereof, the subject of an investigation or proceeding to determine whether a stop order or suspension order should be issued; or
- ☐ (viii) being subject to a United States Postal Service false representation order entered within the past five years, or is, as of the date hereof, subject to a temporary restraining order or preliminary injunction with respect to conduct alleged by the United States Postal Service to constitute a scheme or device for obtaining money or property through the mail by means of false representations.

If you check any of (i) - (viii) above, please contact Employee Special Investments and provide the dates and the summary of each Disqualifying Event in the space below. You may be required to provide additional information.

Disqualified Person. A person who has committed a Disqualifying Event.

Disqualified Person Disclosure. Any information provided to the Subscriber disclosing whether certain persons are Disqualified Persons, including the Fund, any affiliated issuer, any director, executive officer, other officer participating in the offering, general partner or managing member of the Fund; any beneficial owner of 20% or more of the Fund's outstanding voting equity securities, calculated on the basis of voting power; any promoter connected with the Fund in any capacity; any investment manager of the Fund; any person that has been or will be paid (directly or indirectly) remuneration for solicitation of purchasers in connection with the sale of Interests; any general partner or managing member of any such investment manager or solicitor; or any

director, executive officer or other officer participating in the offering of any such investment manager or solicitor or general partner or managing member of such investment manager or solicitor.

Disqualified Person Disqualification. Except as disclosed in your response to the question above and any additional information requested by Goldman Sachs (as defined in the Offering Memorandum), no Disqualifying Event exists with respect to you. You agree to provide the Fund to which you are subscribing pursuant to this Subscription Agreement any information that the Fund may reasonably request in order to determine whether you are a Disqualified Person, including, without limitation, filings with, and records of, courts and regulators. You agree to provide the Fund any information that the Fund may reasonably request in connection with the Fund's compliance with section (e) of Rule 506 of the 1933 Act. You further agree that the Fund may disclose to investors and prospective investors in the Fund (i) information provided by you in your response to the questions above and any other information that you provide in connection therewith and (ii) any other information that the Fund determines is necessary to disclose in connection with its obligations under section (e) of Rule 506 of the 1933 Act, including without limitation, your identity. You agree to promptly notify the Fund if a Disqualifying Event occurs with respect to you. You further agree that if any Disqualifying Event occurs with respect to you on or after September 23, 2013, if you would otherwise have the right to vote more than 20% of the Fund's outstanding voting equity securities (calculated on the basis of voting power), notwithstanding anything to the contrary in the Partnership Agreement (as defined in the Offering Memorandum), your voting rights with respect to the Fund will be limited to 19.9% of the Fund's outstanding voting equity securities (calculated on the basis of voting power) unless and until the Fund determines otherwise in its sole discretion. Furthermore, upon the occurrence of a Disqualifying Event with respect to you, the General Partner and the Fund may, in their sole discretion, take any action they determine necessary or advisable in connection with compliance with applicable regulations, including, without limitation, redeeming all or a portion of your Interest in accordance with the Partnership Agreement, or treating you as a defaulting Limited Partner (as defined in the Offering Memorandum) of the Fund and imposing any or all of the penalties applicable to a defaulting Limited Partner, in each case, in accordance with, and to the extent set forth in the Partnership Agreement.

**PLEASE SIGN AND COMPLETE THE SUBSCRIBER
AND GS OFFEREE SIGNATURE BLOCKS ON PAGE []**

SIGNATURE PAGE

Subscriber Name	_____	
Signature	_____	Date _____
Title of Authorized Signatory (if applicable)	_____	

(If the GS Offeree is different from the Subscriber, also complete the signature block below.)

GS Offeree's Name	_____	
GS Offeree's Signature	_____	Date _____

**FORM OF SUBSCRIPTION AGREEMENT FOR
PARTICIPANTS IN THE LONG TERM EXECUTIVE
CARRIED INTEREST INCENTIVE PROGRAM**

**Class A Limited Partnership Interests
Class O Limited Partnership Interests**

By signing the Subscriber Signature Page, Consent and Power of Attorney (the "Subscriber Signature Page") into which this subscription agreement (collectively with the Subscriber Signature Page, this "Subscription Agreement") is incorporated by reference, the subscriber (the "Subscriber") hereby tenders a subscription and applies for the purchase or acquisition of Class A limited partnership interests ("Class A Interests") and/or Class O limited partnership interests ("Class O Interests" and together with Class A Interests, the "Interests") in [] (the "Fund"), in the aggregate amount set forth in the Subscriber Signature Page. The Subscriber understands that the Fund reserves the right to reduce in whole or in part, in any order (relative to other Subscribers), the amount subscribed for in the Subscriber Signature Page, including, without limitation, the proportion of Carried Interest to which any Class O Interest(s) issued to you relates, such reduction to be in the Fund's sole discretion. The amount, if any, of the Subscriber's subscription that the Fund accepts and approves is referred to herein as the "Subscription Amount" or "capital commitment." If the Subscriber is an entity, each of the Subscriber and the individual named as the GS Offeree in the Subscriber Signature Page (the "GS Offeree") represents and warrants to the Fund and the Authorized Person (as defined below) that the entity is controlled by the GS Offeree. Capitalized terms not otherwise defined herein have the meaning set forth in the Partnership Agreement (as defined below) or the offering memorandum for the Fund, dated [], as supplemented through the date hereof (the "Offering Memorandum").

[]

Further, the Subscriber hereby acknowledges and agrees as follows:

- 1. Receipt of the Offering Memorandum.** The Subscriber acknowledges receipt of the Offering Memorandum for the Fund, which includes the private placement memorandum, dated [] (as amended and supplemented from time to time), for the Fund. The Offering Memorandum describes the terms and conditions of the offering of the Interests and the special risks of an investment in the Fund. The Access Funds are being established to invest all or substantially all of their committed capital in a diversified set of equity-oriented funds sponsored by Goldman Sachs Asset & Wealth Management, including []. The [] Funds are collectively referred to as the "Underlying Funds" and each, separately, an "Underlying Fund." The Access Funds are expected to invest directly or indirectly in the Underlying Funds, and the Underlying Funds are, in turn, expected to invest alongside the [] funds formed for clients of Goldman Sachs (such funds, the "External Funds").
- 2. Adoption of the Partnership Agreement, Subscription Agreement and Other Terms and Conditions.**
 - (a) The Subscriber has carefully read and fully understands, or will carefully read and fully understand, the form of the Amended and Restated Agreement of Limited Partnership Agreement of the [], a copy of which has been, or will be, made available to the Subscriber (as the same may be amended from time to time, the "Partnership Agreement"). The Subscriber hereby accepts, adopts and agrees to be bound by each and every provision contained in the Partnership Agreement, hereby agrees to be a limited partner under the Partnership Agreement and hereby acknowledges and agrees that if the Subscriber's subscription is accepted in whole or in part, the Subscriber shall,

with no further action on the Subscriber's part, become a limited partner in the Fund. By signing the Subscriber Signature Page, the Subscriber further agrees that, effective on the date the Subscriber becomes a limited partner in the Fund, the Subscriber will become a party to the Partnership Agreement and the Subscriber Signature Page will also serve as a counterpart signature page to the Partnership Agreement of the Fund. The Subscriber agrees to fund the total Subscription Amount, including, without limitation, funds necessary to satisfy any of the Subscriber's indemnification and Carried Interest clawback obligations (and any applicable fees and expenses and any interest due on late payments), in accordance with the terms of the Partnership Agreement and this Subscription Agreement.

- (b) The Subscriber has carefully read and fully understands this Subscription Agreement and the Disqualified Person Disclosure included in the Subscription Agreement and Materials into which this Subscription Agreement is incorporated by reference (the "Disqualified Person Disclosure"). The Subscriber hereby accepts, adopts and agrees to be bound by this Subscription Agreement.
- (c) The Subscriber has carefully read and fully understands the Offering Memorandum. The Subscriber hereby accepts, adopts and agrees to be bound by the various provisions, terms and obligations of an investment and ongoing participation in the Fund as described in the Offering Memorandum including, without limitation, the provisions, terms and obligations described in the following sections of the Offering Memorandum: [].

3. Provision of Credit Support.

- (a) **The following is applicable only to Subscribers that are GS Offerees who are executive officers or directors of The Goldman Sachs Group, Inc. or its successor or affiliated companies ("SOX Insiders") and to Subscribers that are controlled by SOX Insiders.** The Subscriber understands that as a result of Section 402 of the United States Sarbanes-Oxley Act of 2002, as amended (the "Sarbanes-Oxley Act"), Goldman Sachs & Co. LLC (together with its affiliates, "Goldman Sachs") is prohibited from extending, maintaining or arranging for the extension of credit in the form of a personal loan to or for the Subscriber. The Subscriber hereby acknowledges that The Goldman Sachs Group, Inc. or any of its affiliates, including, without limitation, Goldman Sachs, or any third party may, subject to applicable law, make or arrange a loan to or guarantee debt of the Fund, any Underlying Fund, [] (the "Main Offshore Fund"), [] (the "UK Offshore Fund" and, together with the Fund and the Main Offshore Fund, the "Access Funds") or any investment or financing vehicle organized by or for the Fund (an "Investment Vehicle"), which will be supported by a pledge by the Fund of its rights under the Partnership Agreement and its subscription agreements with limited partners in the Fund, including, without limitation, this Subscription Agreement, to require the limited partners in the Fund who are not SOX Insiders to fund their commitments under their subscription agreements and/or the Partnership Agreement. The Subscriber further acknowledges that Goldman Sachs may lend money to, or on behalf of, any Underlying Fund or any Investment Vehicle to finance an investment prior to the funding of a commitment under this Subscription Agreement. The Subscriber understands that for the purposes of compliance with the Sarbanes-Oxley Act, the Fund may require the Subscriber to fund its commitments under this Subscription Agreement in advance of limited partners in the Fund who are not SOX Insiders upon not less than five calendar days' notice and/or in disproportionately larger amounts, or in lieu of pre-funding commitments under this Subscription Agreement, the Fund may require SOX Insiders to

purchase Preferred Interests (as defined below) or invest through Alternative Investment Vehicles which do not incur leverage in connection with a particular investment. The Subscriber understands that the Subscriber's unfunded capital commitments under this Subscription Agreement may, but need not, be pledged to support any such loan or guarantee. The Subscriber understands that other circumstances may arise in which considerations under the Sarbanes-Oxley Act will make it necessary or advisable for the Fund, in its sole discretion, to require pre-funding of commitments by the Subscriber under this Subscription Agreement in advance of other limited partners of the Fund who are not SOX Insiders and upon not less than five calendars days' notice.

- (b) **[The following is applicable only to Subscribers that are neither a SOX Insider nor controlled by a SOX Insider.** The Subscriber hereby acknowledges that, in the event Goldman Sachs or any third party makes a loan to or guarantees any debt of any Underlying Fund or any Investment Vehicle, the Fund may pledge or assign, or otherwise make available as credit support for any loan or guarantee (collectively, "Credit Support"), assets of the Fund. Without limiting the foregoing, the Fund may contribute or pledge the rights and interests of the Fund under this Subscription Agreement and/or the Partnership Agreement in respect of the Subscriber's uncalled capital commitment evidenced by the Interest acquired by the Subscriber and any other payment obligations of the Subscriber to the Fund, and the Fund may assign or pledge the rights and interests of the Fund in the capital commitment or other payment obligations to the Fund from the Subscriber, including, without limitation, the right to call and receive capital pursuant to the capital commitment, as collateral for the obligations of any Underlying Fund or any Investment Vehicle in respect of any loan or guarantee. In the event of a default on any loan or performance under any guarantee, the Subscriber understands that the lender and/or guarantor may have recourse to the Subscriber's obligations to make additional contributions or other payments to the Fund, including, without limitation, any remainder of the Subscriber's Subscription Amount to be paid hereunder, which obligations the Subscriber will perform in full and without condition regardless of any defense to, or right of offset against, or counterclaim with respect to, the obligation to fund the Subscriber's contributions.]

4. Consent to Principal and Agency Cross Transactions with Goldman Sachs. Subject to applicable law, the Subscriber hereby consents to any Underlying Fund, or any Investment Vehicle engaging in securities transactions with Goldman Sachs in which Goldman Sachs will act as agent or principal, including, without limitation, the purchase of securities from or sale of securities to Goldman Sachs as principal and the entering into of derivative transactions such as interest rate or currency swaps, forwards or other hedging devices with Goldman Sachs or its affiliates. Without limiting the foregoing, the Subscriber also consents to:

- (a) any Underlying Fund or any Investment Vehicle selling securities in a public offering in which Goldman Sachs acts as managing underwriter or a member of an underwriting syndicate;
- (b) Goldman Sachs effecting, on behalf of any Underlying Fund or any Investment Vehicle, transactions in securities where Goldman Sachs is also acting as broker for, receives commissions from, and may have a potentially conflicting division of loyalty and responsibility regarding, both parties to the transaction, known as agency cross transactions;

- (c) the aggregation of orders for securities for the account of any Underlying Fund or any Investment Vehicle with orders for other accounts of Goldman Sachs, which may be disadvantageous to the Fund;
- (d) all transactions in which Goldman Sachs acts as a principal, counterparty, underwriter or broker on the other side of a transaction with any Underlying Fund, or any Investment Vehicle, and the Subscriber acknowledges that, in the event that Goldman Sachs purchases investments on behalf of and for sale to any Underlying Fund or any Investment Vehicle, the Underlying Fund or the Investment Vehicle, as the case may be, will pay to Goldman Sachs, as the purchase price for these investments, the amount paid by Goldman Sachs for such investments together with interest at commercially reasonable arm's length rates;
- (e) all investments or co-investments in entities in which Goldman Sachs has or is making a principal investment or for which Goldman Sachs is providing or arranging financing at the time of investment by any Underlying Fund or any Investment Vehicle;
- (f) the appointment of an Investment Committee of Goldman Sachs Asset & Wealth Management or an advisory committee, if formed, to approve (i) any transaction for which consent is required under the United States Investment Advisers Act of 1940, as amended, or the Application (as defined below) (or any order of the SEC (as defined below) approving the Application) and (ii) all the transactions in which Goldman Sachs acts as a principal (including, without limitation, in connection with a "riskless principal" trade or "block trade") or underwriter, as broker for any Underlying Fund or any Investment Vehicle, or as broker on the other side of a transaction with the Fund or bunches or aggregates transactions with others;
- (g) the payment of commissions to Goldman Sachs and retention by Goldman Sachs of any commissions, remuneration or other profits which Goldman Sachs may receive or realize in the aforementioned transactions; and
- (h) any Underlying Fund or any Investment Vehicle retaining Goldman Sachs to perform investment banking and other services, including, without limitation, underwriting, merger advisory, placement agency, selling agency, stock brokerage, real estate brokerage and other advisory services.

The Subscriber acknowledges and agrees that in connection with the transactions described above, Goldman Sachs will receive customary fees, plus expenses and indemnities.

5. **Preferred Interests. The following is applicable only to Subscribers that are either a SOX Insider or controlled by a SOX Insider.** The Subscriber hereby agrees that the Fund in its sole discretion may issue to the Subscriber preferred limited partnership interests ("Preferred Interests"), representing all or a portion of the proceeds of any funding of commitments under the Subscription Agreement and paying a fixed or floating rate of return, but not otherwise participating in the performance of the Fund. The Subscriber further understands that the return on such Preferred Interests may be less than if the same portion of the proceeds of the funding were invested in the Subscriber's Interests and that the Fund in its sole discretion may redeem the Subscriber's Preferred Interests in whole or in part, whereupon the funds contributed in respect of the Preferred Interests will participate in the performance of the Fund.
6. **Carried Interest.** The Subscriber understands that the Subscriber's Class O Interests (if any) will share in a portion of any Carried Interest that is earned in respect of the External Funds of

the Underlying Funds as well as other external funds, entities, separate accounts and/or single investor vehicles managed or advised by Goldman Sachs Asset Management, in each case, as determined by Goldman Sachs in its sole discretion (all such funds, the "Carried Interest Paying Funds"). The Subscriber further understands that the Subscriber may be required, under the circumstances described in the Offering Memorandum, to recontribute to the Fund all or a portion of any Carried Interest the Subscriber may have indirectly received from the Carried Interest Paying Funds. The Subscriber agrees that the Subscriber will recontribute these amounts as and when required by the Fund.

- 7. Services Required of Class O Interest Holders.** In connection with the Subscriber's acquisition of Class O Interests (if any), the Subscriber agrees (or if the Subscriber is an entity, the Subscriber agrees to cause its related GS Offeree, and the related GS Offeree agrees) to consult or provide other services, consistent with his or her role as a partner of Goldman Sachs, as reasonably requested by Goldman Sachs Asset & Wealth Management or Goldman Sachs, regarding matters relevant to the Carried Interest Paying Funds, including, without limitation, regarding potential or current investments. The Subscriber acknowledges that any such consulting or other services, would only be required to be provided upon reasonable notice and subject to (i) such services not conflicting with the Subscriber's (or its related GS Offeree's) other duties and responsibilities to Goldman Sachs and (ii) applicable legal, tax, regulatory and other internal policy considerations (including information barriers).
- 8. Power of Attorney.** The Subscriber hereby irrevocably makes, constitutes and appoints the General Partner (including, without limitation, any successor General Partner) (the "Authorized Person") of the Fund, acting by any of its officers or designees, the true and lawful agent and attorney-in-fact of the Subscriber, with full power of substitution and full power and authority in the Subscriber's name, place and stead to (i) change or correct on the Subscriber's behalf all documents (including this Subscription Agreement and the Subscriber Signature Page) executed by the Subscriber in connection with his, her or its subscription to the Fund, including, without limitation, filling in, changing or amending amounts, dates, name of the Fund, or other pertinent information or changing or providing answers to the questions contained in this Subscription Agreement or related documents, in each case, based upon written or verbal instructions from the Subscriber, and the Subscriber hereby agrees that he, she or it will be bound by the terms of any such document as so modified, and (ii) make, complete, change, correct, execute, sign, acknowledge, swear to, deliver, record and file, on the Subscriber's behalf and on behalf of the Fund, such documents, instruments and conveyances that may be necessary or appropriate to carry out the provisions or purposes of this Subscription Agreement and the Partnership Agreement, including, without limitation:
- (a) the Partnership Agreement and any instruments or agreements required or permitted by law or the provisions of the Partnership Agreement, including, without limitation, those to effect the admission to or withdrawal from the Fund of Limited Partners (as defined in the Partnership Agreement) and substituted Limited Partners, and all instruments that the Authorized Person deems necessary or appropriate to reflect a change, modification or novation of the Partnership Agreement, in accordance therewith;
 - (b) all certificates and other instruments deemed necessary or advisable by the Authorized Person to carry out the provisions of the Partnership Agreement or the Subscriber's obligations hereunder or thereunder, including, without limitation, such certificates, agreements and other documents as may be necessary (i) to organize, invest in and conduct business through one or more investment or finance vehicles, including, without limitation, any Investment Vehicles ("Fund Entities"), (ii) to permit the Fund to become or

to continue as a limited partnership or a company wherein the limited partners have limited liability in the jurisdictions where the Fund may be doing business, or (iii) in the event that Goldman Sachs or any third party makes a loan to or guarantees any debt of the Fund or any Fund Entities, to permit the Fund to pledge or assign, or otherwise make available as Credit Support for any such loan or guarantee, assets of the Fund, the capital commitments of the Limited Partners, or both;

- (c) all conveyances and other instruments or papers deemed advisable by the Authorized Person to effect the winding-up and dissolution of the Fund;
- (d) all fictitious or assumed name certificates required or permitted to be filed on behalf of the Fund;
- (e) any and all documentation necessary to pledge, assign or otherwise transfer or encumber Fund assets, including, without limitation, the right to make and collect capital contributions;
- (f) any and all documentation necessary to permit the Fund or the Authorized Person or its respective designee to redeem, purchase, sell, transfer, convey and assign, including, without limitation, as contemplated by the Partnership Agreement, the Interest (or any portion thereof) of any Limited Partner (including, without limitation, all or any portion of the Subscriber's own Interest) to any transferee, including, without limitation, the Authorized Person, the Fund or Goldman Sachs;
- (g) *U.K. Investors Only*: the elections under Section 431 of the Income Tax (Earnings and Pensions) Act 2003, included in the Subscription Agreement and Materials into which this Subscription Agreement is incorporated by reference;
- (h) the election under Section 83(b) of the United States Internal Revenue Code of 1986, as amended (the "Code"), included in the Subscription Agreement and Materials into which this Subscription Agreement is incorporated by reference; and
- (i) all other instruments or papers which may be required or permitted by law to be filed on behalf of the Fund, or which the Authorized Person considers necessary or desirable to carry out the business of the Fund.

The foregoing power of attorney:

- (a) is coupled with an interest, shall be irrevocable, shall not be affected by and shall survive the death, incapacity, dissolution, termination or bankruptcy of the Subscriber;
- (b) may be exercised by the Authorized Person without notice to or any additional action on the part of any Subscriber, either by signing separately as attorney-in-fact for each Subscriber or, after listing all of the Subscribers executing an instrument, by a single signature of the Authorized Person acting as attorney-in-fact for all of them; and
- (c) shall survive the transfer by the Subscriber of the whole or any portion of the Subscriber's Interest; except that, where the transferee of the whole of the Interest has been approved by the Authorized Person for admission to the Fund as a substituted Limited Partner, the power of attorney of the transferor shall survive the delivery of the transfer for the sole purpose of enabling the Authorized Person to execute, swear to, acknowledge and file any instrument necessary or appropriate to effect the substitution.

9. Representations and Warranties. The Subscriber represents and warrants to the Fund and the Authorized Person that:

- (a) The Subscriber is purchasing the Interests for the Subscriber's account for investment purposes only and not for distribution or resale to others. The Subscriber agrees that the Subscriber will not sell or otherwise transfer these securities unless they are registered under the United States Securities Act of 1933, as amended (the "1933 Act"), or unless an exemption from registration is available therefrom.
- (b) The Subscriber understands that the Interests are speculative investments that involve a high degree of risk of loss, has adequate means of providing for the Subscriber's current needs and possible personal contingencies, has no need for liquidity of this investment, and is financially able to withstand the loss of the Subscriber's entire investment in the Fund.
- (c) The Subscriber will pay the Subscription Amount only from the Subscriber's own funds (including, without limitation, in the case of an entity controlled by a GS Offeree, funds gifted or contributed to the Subscriber by the GS Offeree). The Subscriber understands that investing funds provided by any person other than the GS Offeree is strictly prohibited.
- (d) The Subscriber understands that the Subscriber has no right to require the purchase of the Subscriber's Interests; that the transferability of the Interests is severely limited and that any transfer requires the consent of the Authorized Person, which may be withheld in its sole discretion; that there will be no public market for Interests and it will therefore not be readily possible to sell or dispose of the Interests being acquired; and that the Subscriber will not be readily able to liquidate this investment.
- (e) The Subscriber understands that the Interests have not been registered under the 1933 Act in reliance on the private placement exemption thereunder, and thus the Subscriber will not be afforded the protections of registration under that act. Further, the Subscriber understands that Goldman Sachs has filed an application (together with any amendments thereto, the "Application") with the United States Securities and Exchange Commission (the "SEC") seeking an order exempting the Fund, as an "employees' securities company," from most, but not all, of the provisions of the U.S. Investment Company Act of 1940 (as amended, the "1940 Act"), including registration thereunder and certain prohibitions on transactions with Goldman Sachs. The Fund will rely on Rule 6b-1 under the 1940 Act, pursuant to which the Fund will be exempt from the provisions of the 1940 Act, subject to the terms and conditions set forth in the Application (or any order of the SEC approving the Application). Thus, Subscribers of the Fund thus will not be afforded most of the protections afforded by the 1940 Act. If an order approving the Application is not obtained, the Fund will rely on the order dated August 14, 1998 (Release No. IC 23390), which was previously obtained from the SEC, to be exempt from the provisions of the 1940 Act, subject to the terms of that order.
- (f) The Subscriber has carefully read and fully understands the Offering Memorandum, this Subscription Agreement and the Disqualified Person Disclosure, including, without limitation, the risks of an investment set forth in the Offering Memorandum, and has been given the opportunity to ask questions and receive answers concerning the terms and conditions of the offering and other matters pertaining to an investment in the Fund.

- (g) The Subscriber has carefully read and fully understands the discussion of consequences of departure in the Offering Memorandum, including, without limitation, the impact of a breach by the GS Offeree or Subscriber of any provision of any agreement, arrangement or understanding with Goldman Sachs, and the impact of engaging in certain competitive activities, or failing to provide certain certifications to Goldman Sachs, and understands that, upon the occurrence of certain termination events described in the Offering Memorandum under the caption [] and in the Partnership Agreement, when available, as well as certain other events relating to the GS Offeree or Subscriber, (1) some or all of the Subscriber's applicable Class A Interests will be subject to (i) conversion (including, without limitation, to a Class C Interest) and/or (ii) purchase or transfer, and/or (2) some or all of the Subscriber's Class O Interests will be subject to forfeiture for no consideration. If the Subscriber is an entity, the events of conversion, withdrawal, redemption, purchase or transfer described therein will be triggered by the associated employee engaging in the conduct described therein.

The Subscriber understands that any such purchase, conversion, transfer or forfeiture of Class O Interests may have adverse economic and tax consequences. The method for determining the purchase price is more completely set forth in the Partnership Agreement.

- (h) If the Subscriber is an entity, it understands that (i) it may be subject to special tax considerations not discussed in the Offering Memorandum, (ii) it has consulted with its own tax advisor as to all of the U.S. federal, state, local and non-U.S. tax consequences of an investment in the Fund and (iii) except as otherwise consented to by the General Partner, in the event of any direct or indirect Transfer of interests in, or any direct or indirect change in control of, any entity directly or indirectly investing in the Fund, the General Partner may, but will be under no obligation to, cause the Fund to purchase or cause the Transfer or forfeiture of the Subscriber's Interest.
- (i) If the Subscriber is an entity, it is not, nor has it elected, nor will it elect under Treas. Reg. § 301.7701-3 to be (i) treated as a disregarded entity for U.S. federal income tax purposes or (ii) treated as a "grantor trust" for U.S. federal income tax purposes under Sections 671-674 of the Code. The Subscriber has full power and authority to execute and deliver the Subscription Agreement and to subscribe for and purchase the Interests in the manner provided herein. The execution and delivery of the Subscription Agreement have been duly authorized by all necessary action on the part of the Subscriber and will constitute the legal, valid and binding obligations of the Subscriber, enforceable in accordance with their respective terms.
- (j) The Subscriber, by signing the Subscriber Signature Page, hereby agrees that if the Subscriber ceases or expects to cease to be a "United States person" as defined in Section 7701(a)(30) of the Code, the Subscriber shall first provide prior written notice to the General Partner of the change in the Subscriber's status as a United States person at least 60 days prior to such change in status. The Subscriber agrees that, subject to the conditions set forth in the Partnership Agreement, the General Partner in its discretion may cause the Fund (i) to transfer the Subscriber's interest in the Fund to another United States person or (ii) to redeem the Subscriber's interest in the Fund, in each case prior to such change in status.
- (k) The Subscriber, by signing the Subscriber Signature Page, hereby agrees that in the event that the Subscriber (i) does not fund any funding notices or capital calls duly made or make other contributions duly called pursuant to the partnership agreement (or other

governing documents) of any employee special investments in which the Subscriber owns an interest, including, without limitation, the Partnership Agreement and the Fund, or (ii) does not recontribute any amounts previously distributed as permitted by the partnership agreement (or other governing documents) of such investments, including, without limitation, the Partnership Agreement and the Fund, the Subscriber authorizes Goldman Sachs or any of its affiliates without further notice or presentment, to the extent permitted under applicable law, including, without limitation, Section 409A of the Code, to deduct any or all amount(s) due from any or all of the following (each of which shall include any such amounts due, delivered or paid to the GS Offeree, if the Subscriber is an entity):

- (1) any amount due or to become due to the Subscriber in distributions from investments in which the Subscriber is invested under the Employee Special Investments Program now or at any time in the future;
 - (2) any stock deliveries made under the GS Stock Incentive Plan if otherwise permitted under the terms of the applicable award agreement;
 - (3) any cash and/or any proceeds from the liquidation of any marketable securities, including, without limitation, Goldman Sachs stock, held in the Subscriber's Goldman Sachs brokerage account(s) (including, without limitation, joint accounts);
 - (4) any amounts paid to the Subscriber in connection with the Subscriber's departure from Goldman Sachs, including, without limitation, severance payments; and
 - (5) any amounts due to the Subscriber as T&E reimbursements.
- (l) The Subscriber, by signing the Subscriber Signature Page, hereby authorizes the Fund to (i) withdraw funds allocated to the Subscriber's capital account if necessary to satisfy any obligation owed by the Subscriber to Goldman Sachs, and (ii) to the extent such funds are insufficient, cause the Subscriber's Interest to be forfeited or reallocated.
- (m) The Subscriber, by signing the Subscriber Signature Page, hereby agrees to provide to the Fund any information, documentation, waiver, or certification that the Fund or the General Partner may reasonably request or require in order to comply with applicable United States and non-United States laws, including tax laws. In addition, the Subscriber agrees to update such information if and when any such information is no longer true or correct and to provide any additional true and correct information required pursuant to any change in law, or the application or interpretation thereof.
- (n) Without limiting the foregoing, the Subscriber acknowledges that certain terms described herein (including, without limitation, the terms described in clauses (g), (k) and (l) above) materially affect Interests the Subscriber acquires in this offer, as well as interests previously acquired in other offers. By signing the Subscriber Signature Page, the Subscriber is confirming the Subscriber's understanding of, and agreement to, the terms as described in the provisions set forth herein and in the Offering Memorandum and the Partnership Agreement and their potential consequences, notwithstanding anything to the contrary, if applicable, in the constituent documents of other employee special investments in which the Subscriber may own an interest.

- (o) The Subscriber represents that they are not named on a list of prohibited entities and individuals maintained by the U.S. Treasury Department's Office of Foreign Assets Control ("OFAC") or under the European Union ("EU") and United Kingdom ("U.K.") Regulations (as extended to the Cayman Islands by statutory instrument), and is not operationally based or domiciled in a country or territory in relation to which current sanctions have been issued by the United Nations, EU or U.K. (collectively, "Sanctions Lists").

- 10. Indemnification.** The Subscriber agrees that the amount of the Subscriber's capital commitment plus distributions received from the Fund will be available to satisfy indemnity and contribution obligations on the terms described in the Offering Memorandum. The Fund may provide each affected Subscriber with an invoice setting forth the amount owed by the Subscriber in respect of any such indemnity and contribution obligations. The affected Subscriber agrees that it will be obligated to pay the invoiced amount promptly upon request by the Fund. Any amount not so paid will constitute an obligation of the Subscriber and will accrue interest at an annual rate of 8% (or a higher commercially reasonable rate determined by the General Partner in its sole discretion), and that amount, together with accrued interest, will be debited from any distributions otherwise to be made to the Subscriber.
- 11. Access to Information.** The Subscriber acknowledges that he, she or it has had the opportunity, a reasonable period of time before the due date of this Subscription Agreement, to ask questions and receive answers concerning the terms and conditions of this offering of Interests and to obtain any additional information necessary to verify the accuracy of information furnished to the Subscriber. The Subscriber acknowledges that all documents, records, and books pertaining to this investment have been made available for inspection by the Subscriber and by the Subscriber's attorney, accountant or other representative, and that the books and records of the Fund will be available for inspection by its limited partners as required by applicable law. The Subscriber further understands that the Fund will provide annual audited financial statements to its limited partners. The Subscriber acknowledges and understands that if Goldman Sachs determines a Subscriber has failed to comply with GS Agreements, then the Fund may withhold from such Subscriber information the Subscriber would otherwise be entitled to receive. Notwithstanding the foregoing, the Subscriber acknowledges and understands that if a Limited Partner leaves Goldman Sachs (or in the case of an Affiliated Investor Entity the related GS Offeree leaves Goldman Sachs), then the General Partner may, in its sole discretion, withhold from that Limited Partner some of the information Limited Partners are otherwise entitled to receive, to the fullest extent permitted by applicable law.
- 12. Guarantee and Consent to Pledge by the GS Offeree.** The GS Offeree, by signing the Subscriber Signature Page, hereby guarantees the complete and timely fulfillment of (a) the Subscriber's obligations to pay the Subscription Amount, as set forth in the second paragraph and in Section 2 of this Subscription Agreement, (b) the Subscriber's obligation to recontribute certain amounts to the Fund as and when required by the General Partner, including, without limitation, to satisfy any of the Subscriber's indemnification obligations, as described under the caption [] in the Offering Memorandum or to satisfy any clawback of Carried Interest and (c) any other amounts that the Subscriber is obligated to pay to the Fund pursuant to the Partnership Agreement or this Subscription Agreement. [In addition, Subscribers that are neither a SOX Insider nor controlled by a SOX Insider hereby consent to the pledging by the General Partner of the guarantee given in the preceding sentence for the same purposes as set forth in Section 3(b) of this Subscription Agreement.]

13. **Sarbanes-Oxley Act.** The following is applicable only to Subscribers that are either a SOX Insider or controlled by a SOX Insider. The Subscriber understands that the Authorized Person has the authority to modify the Subscriber's Interests or, in some cases, to redeem the Subscriber's Interests in whole or in part as the Authorized Person deems necessary or appropriate to comply with the Sarbanes-Oxley Act. Any such modification or redemption may have adverse financial and tax consequences for the Subscriber.
14. **Governing Law Clause.** The Subscriber Signature Page (including this Subscription Agreement which is incorporated by reference therein) is governed by and shall be construed in accordance with the internal laws of the State of Delaware applicable to a contract made and performed wholly within the State of Delaware.
15. **Arbitration.** The arbitration provisions in the Partnership Agreement shall apply to this Subscription Agreement.

**FORM OF SUPPLEMENT TO THE SUBSCRIPTION AGREEMENT FOR
PARTICIPANTS IN THE LONG TERM EXECUTIVE
CARRIED INTEREST INCENTIVE PROGRAM**

**Class A Limited Partnership Interests
Class O Limited Partnership Interests**

This Supplement does not contain a complete description of the Fund or other matters appearing in the Offering Memorandum, and should be read in conjunction with the Partnership Agreement and the Offering Memorandum. If you have any questions about the Fund or this offering, please contact the designated contact person in the launch email.

The Offering

Goldman Sachs is granting participants in the Long Term Executive Carried Interest Program (“Executive Participants”) and/or certain controlled entities of such Executive Participants Class O limited partnership interests (“Class O Interests”) and, in connection with that grant, Executive Participants will make an unlevered equity capital commitment (“Class A Interests,” and together with the Class O Interests, the “Interests”) in the [] (the “Fund”).

Subscription Deadline

To subscribe, you must complete, date, execute and deliver an executed Subscription Agreement and Materials on or before []. The grant will be effective [].

Future Capital Calls

As of the date of this Supplement, the Fund has called [●%] of equity capital commitments. By agreeing to acquire Interests pursuant to this offering, you are agreeing to fund all future capital calls as and when instructed via capital call notice from the Fund.

Class A Interests

Class A Interests will share in the profits and losses from the Fund’s investment activities and operations without being subject to a management fee or carried interest payable to Goldman Sachs at the level of the Fund or the Underlying Funds, subject to the Class A Vesting Date described below. Class A Interests will, however, indirectly bear management fees, carried interest and/or similar fees and expenses charged by Non-GS External Fund managers at the level of the Non-GS External Funds. The Class A Interests will not be entitled to the benefits, nor subject to the risks and detriments, of leverage. The holders of Class A Interests will be required to make capital contributions in respect of the Class A Interests for which they subscribe.

Class A Interests are governed by the terms of the Partnership Agreement and the Offering Memorandum; provided, however, that contrary to the vesting dates set forth in the Partnership Agreement and the Offering Memorandum (as it relates to Class A Interests), any Class A Interests acquired pursuant to this offering will vest on December 31, [] (the “Class A Vesting Date”). **If an Executive Participant’s Employment terminates, for any reason (whether due to voluntary or involuntary termination of employment), before the Class A Vesting Date, the General Partner will have the right, but not the obligation, to (a) convert the Executive Participant’s Class A Interest to a Class C Interest, which, subject to applicable law, will be subject to the Class C**

Management Fee and Class C Carried Interest, or (b) cause the redemption, purchase or transfer of any Class A Interest or Class C Interest of such Executive Participant at a price equal to the Minimum Purchase Price.

[]

Class O Interests

Class O Interests are entitled to share in a portion of any Carried Interest that may be earned in respect of the applicable Carried Interest Paying Funds subject to the return hurdles, notch amounts and other limits described in the offering memorandum of each Carried Interest Paying Fund and/or the Supplement. If the return hurdles and notch amounts are achieved or exceeded as of [], the Class O Interests will only be entitled to share in a portion of any Carried Interest that may be earned in respect of the applicable Carried Interest Paying Fund to the extent that (i) the applicable Carried Interest Paying Fund's limited partners have achieved or exceeded the return hurdle or notch amount described in the offering memorandum of each applicable Carried Interest Paying Fund (the "Original Hurdle") and (ii) the aggregate realized and estimated value of the Carried Interest Paying Funds' investments (calculated on the assumption that all unrealized investments were disposed of for their fair values immediately prior to a distribution or measurement date for such Class O Interests) as a percentage of the Carried Interest Paying Funds' aggregate investment costs represents at least the applicable percentage as of [] (as finally determined by Goldman Sachs in accordance with its valuation procedures) plus []% (the "Class O Incremental Hurdle"). For the avoidance of doubt, because the Class O Interests may be subject to the Class O Incremental Hurdle, prior Class O Interests that are not subject to an additional hurdle, or that are subject to a different additional hurdle, may have value while Class O Interests acquired pursuant to this offering have no value.

To the extent the Class O Incremental Hurdle is required, until the Class O Incremental Hurdle has been achieved, any Carried Interest associated with the Class O Interests shall be allocated to Goldman Sachs. Once the Class O Incremental Hurdle has been achieved, each holder of Class O Interests will be allocated, in respect of its Class O Interest, *pro rata* based on all Class O Interests, (i) the Carried Interest associated with the applicable Class O Interest earned after that point and (ii) a portion of the Carried Interest held by Goldman Sachs, until the holders of Class O Interests have been allocated an amount of Carried Interest that they would have been allocated in the absence of the Original Hurdle, Class O Incremental Hurdle or any notch amount set forth in the Partnership Agreement (the amount described in clause (ii), the "Catch-up Amount"). Each holder of a Class O Interest will receive the Carried Interest associated with the applicable Class O Interest earned after holders of Class O Interests have received the Catch-up Amount in respect of their Class O Interests. The foregoing may be applied at the level of a subsidiary of the Fund.

As a result of the mechanic described in the previous paragraph, in order for holders of Class O Interests to be allocated the entire Catch-up Amount, there must be sufficient gains (as determined for U.S. Federal income tax purposes) with respect to the Carried Interest allocated to the holders of Class O Interests and to Goldman Sachs after the Class O Incremental Hurdle has been achieved. If any Re-Offers are conducted, such gains would be allocated among additional holders of Carried Interest, and the amounts allocated to holders of Class O Interests in respect of the Catch-up Amount may be delayed or reduced (as compared to the amounts that would have been allocated if no Re-Offers had occurred). Moreover, if as of the termination of the applicable Carried Interest Paying Fund (or at such earlier time in the sole discretion of Goldman Sachs) both the Original Hurdle (if any) and the Class O Incremental Hurdle have not been satisfied, any Carried Interest that otherwise would have been distributed to the holders of Class O Interests shall instead be distributed to Goldman Sachs.

Under the circumstances described in [] of the Offering Memorandum, you may be required to re-contribute to the Fund all or a portion of the Carried Interest that has been distributed to you (e.g., if the Fund becomes obligated to re-contribute to the Carried Interest Paying Funds all or a portion of any Carried Interest previously received from the Carried Interest Paying Funds because the investors in the Carried Interest Paying Funds have not received their priority return amount). By agreeing to acquire Interests pursuant to this offering, you are agreeing that you will re-contribute these amounts as and when required. Class O Interests are being granted for no consideration as the Class O Interests are currently estimated to have no value. []

Holders of Class O Interests should be aware that regulations under Section 1061 of the Internal Revenue Code that were finalized in 2021 could cause a recharacterization of Carried Interest, with the effect that a substantial portion of your income from the Funds may be treated as ordinary income for U.S. tax purposes.

Vesting

Contrary to the vesting dates set forth in the Offering Memorandum and the Partnership Agreement (as it relates to Class O Interests), any Class O Interests acquired pursuant to this offering will vest one-third on December 31, [], an additional one-third on December 31, [] (for a total of two-thirds having vested as of such date, including the amount vested on December 31, []) and the final one-third on December 31, [] (for a total of 100% having vested as of such date, including the amounts vested on December 31, [] and December 31, []) (each date, a "Class O Vesting Date" and together, the "Class O Vesting Dates"). **Except in the case of Retirement (as described below), a departure from Goldman Sachs (or any successors thereto) for any reason (whether due to voluntary or involuntary termination of employment) prior to the applicable Class O Vesting Date will result in the automatic forfeiture, for no consideration, of the unvested portion of your Class O Interests. These vesting provisions will apply irrespective of involuntary termination (including due to downsizing) or death or adjudication of incompetence (i.e., vesting will not be accelerated for any reason).**

Notwithstanding the foregoing, if the Employment of an Executive Participant is terminated by Retirement, such Executive Participant's Class O Interests will not be subject to forfeiture. Instead, such Interests that have not yet vested will vest according to the Class O Vesting Dates set forth above; provided, however, that **such Executive Participant's rights to any Class O Interest that become vested by virtue of Retirement will terminate and such Executive Participant will have no further rights in respect of that Class O Interest following such Executive Participant's Association with a Covered Enterprise on or before the applicable Class O Vesting Date.** Class O Interests will not be eligible for continued vesting as described in the preceding sentence if: (i) such Executive Participant or Goldman Sachs gives notice of termination, or such Executive Participant's Employment terminates for any reason, prior to the date such Executive Participant would otherwise qualify for Retirement; (ii) such Executive Participant's Employment terminates while Goldman Sachs is considering whether such Executive Participant engaged in conduct constituting Cause or after Goldman Sachs determines that such Executive Participant has engaged in conduct constituting Cause; or (iii) such Executive Participant has Breached an Obligation to the Firm. Goldman Sachs will determine, in its sole discretion, whether the Executive Participant meets the eligibility criteria for continued vesting described in this paragraph, and the Interests to which continued vesting may apply. The terms and conditions of an investment in the Fund applicable to an Executive Participant's Interests will remain in full force and effect; including, without limitation, provisions that provide for termination of some or all of an Executive Participant's rights such as redemption, transfer and/or forfeiture of vested Class O Interests, as well as the Additional Conditions Applicable to Class O Interests described below.

[]

Additional Conditions Applicable to Class O Interests

Notwithstanding the terms of the Partnership Agreement and the Offering Memorandum, the conditions set forth below apply to an Executive Participant's Class O Interests, in addition to any other terms or conditions that permit forfeiture, cancellation, conversion or liquidation of the Executive Participant's Interests.

Certain Forfeiture Events

If Goldman Sachs determines that an Executive Participant has engaged in one or more of the following events, the General Partner will cause the forfeiture for no consideration of the Executive Participant's Class O Interests (whether or not vested):

- Fails to Consider Risk,
- conduct that constitutes Cause,
- Breached an Obligation to the Firm,
- Engages in Solicitation prior to the applicable Class O Vesting Date, or
- Associates with a Covered Enterprise prior to the applicable Class O Vesting Date.

Deferral of Payment upon Occurrence of Certain Events

If Goldman Sachs determines that one or both of the following events occurs, the General Partner will defer any distribution with respect to Class O Interests (other than tax distributions), without interest owing on any deferred amount, until the time as Goldman Sachs determines such event is no longer occurring:

- GS Group's ROE is anticipated or reported to be less than 5%; or
- GS Group fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days.

[Forfeiture and Repayment upon Accounting Restatement]

[Class O Interests (whether or not vested) will be forfeited, and any amounts previously paid in respect of Class O Interests will be subject to repayment, if GS Group is required to prepare an account restatement due to GS Group's material noncompliance, as a result of misconduct, with any financial reporting requirement under the securities laws as described in Section 304(a) of SOX; provided, however, that an Executive Participant's Class O Interests will only be subject to forfeiture or repayment to the same extent that would be required under Section 304(a) of SOX had the Executive Participant been a "chief executive officer" or "chief financial officer" of GS Group (regardless of whether the Executive Participant actually holds such position at the relevant time).]

Additional Consequences of Forfeiture

Upon any forfeiture, you may be required, in Goldman Sachs' sole discretion, to return any Class O Interest distributions previously received (even though you may have been taxed on income allocated in respect of such Class O Interest and/or paid expenses incurred in connection with the formation and implementation of any such carried interest arrangement). Subject to applicable law, forfeited Class O Interest shall be reallocated to Goldman Sachs (and therefore may be reallocated at a later date) and/or to new or existing Limited Partners. Any forfeited Class O Interest that is reallocated to new and/or existing Limited Partners may be subject to a different vesting schedule (including a shorter or longer vesting schedule). Upon the forfeiture of all or a portion of your Class O Interests, you may be required,

in the Investment Manager's sole discretion, to return any tax distribution amount previously distributed in respect of such forfeited Class O Interest.

Definitions

As used in this Supplement:

"Association with a Covered Enterprise," "Cause," "Covered Enterprise," "Engages in Solicitation," Firm," "Goldman Sachs," "GS Group," and "SOX" have the same meaning as set forth in the Partnership Agreement.

"Breached an Obligation to the Firm" means that Goldman Sachs has determined, in its sole discretion, that an Executive Participant has failed to meet, in any respect, any obligation under any agreement with the Firm, or any agreement entered into in connection with the Executive Participant's Employment, including the Firm's notice period requirements, applicable restrictive covenants (such as restrictions on association with a Competitive Enterprise or Covered Enterprise, Engaging in Solicitation, and similar terms or concepts), any offer letter, employment agreement (e.g., Managing Director Agreement) or any shareholders' agreement relating to Goldman Sachs.

"Employment" means an Executive Participant's employment by the Firm.

"ROE" means the return on average common shareholders' equity as calculated by the Firm.

"Retirement" means termination of an Executive Participant's Employment at a time when (i) (A) the sum of the Executive Participant's age plus years of service with the Firm equals or exceeds 60 and (B) the Executive Participant has completed at least 10 years of service with the Firm or, if earlier, (ii) (A) the Executive Participant has attained age 50 and (B) the Executive Participant has completed at least five years of service with the Firm; in each case, as determined by the Firm in its sole discretion.

Certain U.K. Investors Only

The General Partner will have the right, but not the obligation, to cause the forfeiture for no consideration of some or all of the Class O Interest(s) of a U.K. investor who is considered a "Material Risk Taker", if any of the following events occurs prior to the seventh anniversary of the issuance of the applicable Class O Interest(s) (which period may be extended by the General Partner in the event of any action, proceeding or investigation in connection with any of the following events is initiated by Goldman Sachs or a third party prior to the end of such seven-year period):

- (i) The Goldman Sachs Group, Inc. (and any successor thereto, "GS Inc.") fails to maintain the required "Minimum Tier 1 Capital Ratio" as defined under Federal Reserve Board Regulations applicable to GS Inc. for a period of 90 consecutive business days;
- (ii) the Board of Governors of the Federal Reserve or the Federal Deposit Insurance Corporation (the "FDIC") makes a written recommendation under Title II (Orderly Liquidation Authority) of the Dodd-Frank Act for the appointment of the FDIC as a receiver of GS Inc. based on a determination that GS Inc. is "in default" or "in danger of default;"
- (iii) (A) an annual pre-tax loss at GS Inc. or (B) annual negative revenues in one or more reporting segments as disclosed in the Firm's Form 10-K other than the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm),

or annual negative revenues in the Asset & Wealth Management segment (or any successor or equivalent segment or sub-segment as determined by the Firm) of \$5 billion or more, provided in either case that such Class O Limited Partner is employed in a business within such reporting segment;

- (iv) there occurs a loss of 5% or more of firmwide total capital from a reportable operational risk event determined in accordance with the firmwide Reporting Operational Risk Events Policy;
- (v) such Class O Limited Partner engages in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under English law; or
- (vi) an individual with respect to whom the committee appointed by the Board of Directors of GS Inc. determines such Class O Limited Partner had supervisory responsibility as a result of direct or indirect reporting lines or such Class O Limited Partner's management responsibility for an office, division or business, engages during the calendar year in which such Class O Limited Partner's Class O Interest was issued, in conduct that the Firm reasonably considers, in its sole discretion, to be misconduct sufficient to justify summary termination of employment under English law.

Certain German Investors Only

The General Partner will have the right, but not the obligation, to cause the forfeiture for no consideration of some or all of the Class O Interest(s) of a German investor who is considered a "Material Risk Taker" (a "German MRT"), if any of the following events occurs prior to the seventh anniversary of the issuance of the applicable Class O Interest(s) (which period may be extended by the General Partner in the event of any action, proceeding or investigation in connection with any of the following events is initiated by Goldman Sachs or a third party prior to the end of such seven-year period):

- (i) the applicable German MRT significantly contributed to, or was responsible for, any conduct that resulted in (a) a considerable loss (in particular, but not limited to, an unexpected loss of 0.75% (which may also be reached in sum by several individual losses) or more of the total capital of the GS Group), (b) a material regulatory sanction for the GS Group and its subsidiaries and affiliates comprising one or more of the following (i) a moratorium pursuant to sec. 46g German Banking Act ("*Kreditwesengesetz*" / "*KWG*"), (ii) a measure in case of danger pursuant to sec. 46 German Banking Act, (iii) the revocation of appointment of a manager pursuant to sec. 36 German Banking Act, (iv) a fine pursuant to sec. 56 German Banking Act or a threatened penalty payment, if the fine or penalty payment amounts to 0.75% or more of the total capital of the GS Group, (v) the cancellation of the banking permit pursuant to sec. 35 German Banking Act, (vi) an order to increase the capital requirements by at least 0.5% pursuant to sec. 10 German Banking Act, (vii) a measure in case of organizational deficiencies pursuant to sec. 45b German Banking Act, or (viii) a comparable regulatory order or (c) a material supervisory measure; or
- (ii) The applicable German MRT acted in serious violation of relevant external or internal rules with respect to suitability and conduct, provided that relevant regulations relating to the suitability and conduct of the members of the Board of Management include all regulations relating to conduct and professional suitability, compliance with which is necessary for the maintenance of a proper business organization within the meaning of Section 25a (1) sentence 1 German Banking Act, and provided that a violation is always considered serious if it is due to grossly negligent or intentional behaviour or if it is suitable to justify a termination of employment for cause pursuant

to sec. 626 German Civil Code or a termination of employment for misconduct ("*verhaltensbedingte Kündigung*") pursuant to sec. 1 German Termination Protection Act (*Kündigungsschutzgesetz*).

Certain GSAM B.V. Investors Only

In accordance with Articles 1:126 and 1:127 of the Dutch Financial Supervision Act (*Wet op het financieel toezicht*) in conjunction with Article 2:135 of the Dutch Civil Code (*Burgerlijk Wetboek*), the board of GSAM BV will have the obligation to cause the forfeiture of or to claw back, for no consideration, some or all of the Class O Interest(s) of a Dutch investor who works under the responsibility of GSAM BV (a "Dutch O Investor"), if any of the following events occurs prior to the fifth anniversary of the board of GSAM BV becoming aware of those events:

- (i) the Dutch O Investor has not met appropriate standards of competence and proper conduct; or
- (ii) the person was responsible for behaviours that have significantly deteriorated the financial position of GSAM BV.

In addition, the board of GSAM BV will have the right, but not the obligation, to cause the forfeiture for no consideration of some or all of the Class O Interest(s) of a Dutch O Investor, if any of the following events occurs prior to the fifth anniversary of the board of GSAM BV becoming aware of the event:

- (i) the award of the Class O Interest(s) to the Dutch O Investor would be unacceptable according to standards of reasonableness and fairness.

Finally, the board of GSAM BV will have the right, but not the obligation, to claw back for no consideration some or all of the Class O Interest(s) of a Dutch O Investor, if the following event occurs prior to the fifth anniversary of the board of GSAM BV becoming aware of the event:

- (i) the Class O Interest(s) have been awarded on the basis of incorrect information about the achievement of the goals underlying the award or about the circumstances on which the award was dependent.

Providing Information to the Appropriate Authorities and Protected Communications

In accordance with applicable law, nothing in the Fund Documentation (including the forfeiture, clawback and repayment provisions) prevents you from providing information you reasonably believe to be true to the appropriate governmental authority, including a regulatory, judicial, administrative, or other governmental entity; reporting possible violations of law or regulation; making other disclosures that are protected under any applicable law or regulation; or filing a charge or participating in any investigation or proceeding conducted by a governmental authority. For the avoidance of doubt, governmental authority includes Federal, state and local government agencies such as the U.S. Securities and Exchange Commission, the Equal Employment Opportunity Commission and any state or local human rights agency (e.g., the New York State Division of Human Rights, the New York City Commission on Human Rights, the California Civil Rights Department), as well as law enforcement (e.g., the state Attorney General and the U.S. Department of Justice). Similarly, nothing in the Fund Documentation prohibits you from engaging in protected concerted activity pursuant to applicable law protecting such activity, including discussing wages, hours, or other terms and conditions of your employment; or speaking with your own attorney regarding your own legal rights or obligations. In addition, nothing in the Fund Documentation limits your ability to use the internal and external reporting

channels that are available to you, as described in the Firmwide Policy on Escalation. “Fund Documentation” means this Supplement, the Subscription Materials, the Subscription Agreement, the Partnership Agreement and the Offering Memorandum, as well as any other documentation presented to you in connection with the Fund.

Tax Considerations

For a general description of certain of the U.S. federal income tax consequences and certain other tax considerations applicable to you of an investment in the Fund, including in respect of the Class O Interests [].

Certain Tax Implications - Funds

The Class O Interests are being granted for no consideration as such Class O Interests are currently estimated to have no value. [] As a result, you are not expected to be subject to U.S. federal income taxes or to have a tax payment obligation in connection with acquiring interests pursuant to this offering. However, as valuation principles are inherently uncertain, the assets are illiquid and difficult to value, and timely information may not be readily available, there can be no assurance that the valuations of the Interests assumed herein will not be challenged by the Internal Revenue Service (“IRS”), or that any such challenge would not be upheld. If the IRS determines that the fair market value of the Interests you are acquiring exceeds the amount you include in income, you will be required to include in income the excess. Accordingly, in agreeing to acquire Interests pursuant to this offering, you agree to accept the risk of incurring tax liability. Goldman Sachs cannot quantify the size of any such potential tax exposure. If a tax liability occurs, you may be required to make a tax payment to Goldman Sachs for any applicable income and employment taxes required to be withheld by, or otherwise paid by, Goldman Sachs in connection with the acquisition of the Interests.

As discussed above, in the event that the Class O Incremental Hurdle is established, until the Class O Incremental Hurdle has been achieved, any Carried Interest associated with the Class O Interests shall be allocated to Goldman Sachs. Once the Class O Incremental Hurdle has been achieved, each holder of the Class O Interests will be allocated (i) allocations in respect of the Carried Interest earned after that point and (ii) a portion of allocations in respect of the Carried Interest held by Goldman Sachs, until the holders of Class O Interests have been allocated income equal to the amount of Carried Interest that they would have been allocated in the absence of the Original Hurdle and the Class O Incremental Hurdle (the amount described in clause (ii), the “Catch-up Amount”). Adverse consequences, including with respect to character of income for tax purposes, may apply to the extent that distributions of Carried Interest to a holder of Class O Interests exceed the allocations described in the previous sentence. Each Class O Limited Partner should consult his or her own tax advisors with respect to the tax consequences of a Class O Incremental Hurdle and related mechanics.

If you are a non-U.S. investor, you should be prepared to be subject to similar rules and similar risks. These taxes will generally depend upon the specific laws of your jurisdiction of tax residence, as well as the values ascribed to the Interests by your applicable taxing authority. Accordingly, non-U.S. investors should also carefully consider these consequences and risks.

If you are a U.K. investor and you would like to hold your Interests through a brokerage account separate from the brokerage account through which you hold other employee fund interests, if applicable, please contact [].

Potential Changes to Tax Laws and Regulations

Each Limited Partner should be aware that developments in the tax laws of the United States, or other jurisdictions (including as a result of BEPS, ATAD, ATAD 2 and ATAD 3, and DAC6), could, directly or indirectly, have a material effect on the tax consequences to the Limited Partners and/or the Fund and that a Limited Partner may be required to provide certain additional information to the Fund (which may be provided to the IRS or other taxing authorities) or may be subject to other adverse consequences as a result of such developments.

Goldman Sachs is both the owner of the Interests and the Investment Manager of the Funds. As a result, Goldman Sachs may be in possession of certain non-public information regarding the Funds, assets owned by the Funds, and/or the fair market value of the Interests. Goldman Sachs is under no obligation to and may not affirmatively disclose such information. However, you may discuss such information with the General Partner. Please contact the designated contact person in the launch email with any questions.

An investment in the Fund will involve substantial risks and there are conflicts of interest.

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The Goldman Sachs Group, Inc. Firmwide Insider Trading Policy Governing Transactions in GS Securities by Covered Persons

Applicability: All Goldman Sachs

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A. Scope and Summary

This Firmwide Insider Trading Policy Governing Transactions in GS Securities by Covered Persons (this “policy”) applies to transactions entered into by Firm Individuals, including Covered Persons, involving Goldman Sachs (“Goldman Sachs” or “GS”) Securities and other companies’ securities. The transactions covered by this policy are also subject to the Personal Trading Policy, applicable divisional policies and the Firmwide Policy With Respect to Personal Transactions Involving GS Securities and GS Equity Awards (the “Firmwide Trading Policy”), as well as local regulatory requirements.¹

Terms used but not defined in this policy have the definitions set forth in Appendix A.

1. Framework Linkages

This Policy does not have linkages to any Frameworks.

2. Regulatory Linkages

This Policy has linkages to the following LRR obligations:

- 17 C.F.R. § 240.10b-5.

3. Risk Taxonomy Linkages

Applicable risks for this document include:

- Unauthorized Personal Investments or Trading and
- Misuse or Improper Sharing of Information for Trading, Transactions or Other Firm Activities.

B. Governance and Oversight

This policy may be amended or modified in any respect at any time.

The Deputy General Counsel of the Legal Division is responsible for approving this Policy. The Legal Division owns the Policy and is responsible for maintaining and overseeing the Policy, reviewing conformance with the Policy requirements, and providing guidance to divisions on consistency of the associated divisional Standards / Procedures created in support of this Policy. The Goldman Sachs Group, Inc. Firmwide Insider Trading Policy Governing Transactions in GS Securities by Covered Persons is a Tier I document. As such, this document is required to be reviewed at least annually by the Legal Division.

Risk is owned by all firm personnel and responsibility for oversight lies with the Legal Division and Compliance Division.

C. Policy Requirements

1. Prohibitions Against Insider Trading

U.S. federal and state securities laws impose important restrictions on trading by individuals who are in possession of material nonpublic information regarding a company. Under the federal and

¹ See the Firmwide Trading Policy for more information on transactions involving GS Equity Awards and for specific restrictions, limitations and procedures, as well as reporting requirements, relating to the trading of GS Securities. If any provisions of this policy are inconsistent with the provisions of the Personal Trading Policy or the Firmwide Trading Policy, this policy shall govern.

state securities laws and this policy, no Firm Individual who is in possession of material nonpublic information related to the firm, may (i) trade in any GS Securities, or securities issued by another company, on the basis of material nonpublic information or (ii) “tip” any material nonpublic information to another person who may use such information to trade in GS Securities or such other company’s securities.

2. Material Nonpublic Information

i Determining Materiality

Whether a piece of information is material depends on the facts and circumstances in which it is created and distributed.

Information is typically deemed “material” if:

- There is substantial likelihood that it would be viewed by a reasonable investor as important in making an investment decision;
- Its disclosure would be viewed by a reasonable investor as having significantly altered the total mix of information otherwise available; or
- It is information that could reasonably be expected to affect the price of a security.

Both favorable and unfavorable information can be material. However, there is no precise definition of “materiality,” and the question of whether information is material is subjective and may be questioned in hindsight. Accordingly, Firm Individuals are advised to take a cautious view when evaluating whether information is “material.” Note that information may be considered material even if it relates to contingent, uncertain, or rumored events. Some examples of “material information” may include, depending on the particular circumstances:

- Potential merger and acquisition activities;
- Potential public offerings or other financings;
- Financial forecasts or projections;
- Significant changes in management; and

Significant product developments. This list is intended to be illustrative only and is not exhaustive. Many other types of information may be material at any particular time depending upon the circumstances.

ii. Determining Whether Information is Nonpublic

“Nonpublic information” is information that is not available to the public about an entity or its securities, loans, certificates of deposit, partnership interests, certificates of participation, investment contracts and trade claims, as well as any securities- or loan-based swaps or derivatives or other instruments that are convertible into or exchangeable or exercisable for, or the value of which is affected or dependent upon the value of, such instruments (whether physically or cash settled).

Information is typically considered nonpublic until it has been disseminated to the financial marketplace and sufficient time has elapsed to allow the markets to absorb and evaluate the information. Examples of public dissemination include: public filing of a press release or disclosure

to a regulatory authority, and official statement widely distributed to the media (e.g., a statement to the media by an authorized senior executive).

Press speculation and/or market rumors about nonpublic information do not constitute adequate dissemination of such information.

3. Consequences of Violations

Violations of the insider trading laws can carry both criminal and civil penalties. Potential legal penalties include liability for the firm and for Firm Individuals. For an individual, insider trading may result in criminal penalties, including fines, jail time or both, and the Securities and Exchange Commission has the authority to seek civil monetary penalties. Private parties may also bring civil actions seeking damages against any person purchasing or selling a security while in the possession of material, nonpublic information. In addition, violations of this policy may lead to closure of the applicable account, the voiding or reversal of the relevant transaction (and the violator is responsible for any losses, costs or expenses arising in connection with such voiding or reversal), as well as disciplinary actions, up to and including termination.

D. General Rules Applicable to Covered Persons

E.

The following rules (the “General Rules”) are applicable to all transactions involving GS Securities by all Covered Persons, unless a specific exception is applicable.

1. Window Periods/Blackout Periods

Covered Persons (excluding Senior Directors of GS Inc. (“Senior Directors”) who are Non-Covered Senior Directors) may only enter into a transaction involving GS Securities during an applicable Window Period (and not during an applicable Blackout Period, except as set forth below).

The following transactions are permitted outside of the applicable Window Period:

- The automatic exercise or assignment of listed options on GS Shares that are U.S. \$0.01 or more in-the-money on the expiration date
- Transactions involving GS Structured Debt Securities, GS Structured Warrants and GS Deposit Obligations; provided, that Executive Officers of GS Inc. (as defined for purposes of Section 16 of the U.S. Securities Exchange Act of 1934, as amended from time to time, the “Executive Officers”) and GS Inc. Directors may only enter into such transactions during an applicable Window Period
- Transfers of GS Securities between certain accounts with the same tax identification, so long as such transfer does not create a taxable event
- Transfers of GS Securities from a trust account to certain brokerage accounts with the same tax identification, so long as such transfer does not create a taxable event
- Executive Officers, members of the Management Committee of GS Inc., Non-Employee Directors and, in certain circumstances, PMDs specified by an Authorized Person (“Specified PMDs”), but not other Covered Persons, are permitted to sell GS Shares pursuant to a firm-approved Rule 10b5-1 Plan with respect to the sale of their GS Shares
 - o Entry into, amendment of or termination of any firm-approved Rule 10b5-1 Plan must be pre-approved by the General Counsel of GS Inc. or her or his designee

Regardless, the firm may impose Special Blackout Periods at any time and for any period relating to any or all types of GS Securities or any or all categories of Covered Persons.

2. Market, Limit, Good-Til-Date and Good-Til-Cancelled Orders

Covered Persons are permitted to place the following types of orders for purchases and sales of freely transferable GS Securities:

- Market orders
- Limit orders
- Good-Til-Date orders (so long as the expiration date is within the applicable Window Period)

Covered Persons are not permitted to place Good-Til-Cancelled orders.

3. Certification of No Material Nonpublic Information and Compliance with This Strategy

Prior to entering into any transaction involving GS Securities (other than GS Deposit Obligations), a Covered Person must certify, among other things, that they are not aware of material, nonpublic information with respect to GS Inc., subject to certain conditions set forth in the Firmwide Trading Policy.

Additionally, with respect to transactions in GS Structured Debt Securities and GS Structured Warrants, Executive Officers and GS Inc. Directors must certify that they are not aware of material, nonpublic information with respect to any underlying securities referenced by the GS Structured Debt Securities or GS Structured Warrants.

4. Pre-Clearance/Notification Process for Transactions Involving GS Securities

Executive Officers and members of the Board of Directors of GS Inc. (each, a "GS Inc. Director") are not permitted to enter into any transactions involving GS Securities (excluding transactions involving GS Deposit Obligations) without first obtaining approval from the General Counsel of GS Inc. or her or his designee (or, in the case of the General Counsel, approval from the Chief Compliance Officer of GS Inc. or her or his designee) and providing an attestation that they are not aware of material nonpublic information pursuant to Section 3 above. Pre-clearance is effective until the end of the next business day, except for any individuals who are held to the Goldman Sachs Asset Management Code of Ethics, who must trade within the same day of approval.

Transactions involving Non-GS Instruments Linked to GS Securities must go through all required firm, divisional, departmental and desk pre-clearances required.

F. Specific Transaction Rules Involving GS Securities by Covered Persons

The following rules (the "Specific Transaction Rules") apply to specific types of transactions involving GS Securities by Covered Persons.

1. Selling GS Securities

Covered Persons are permitted to sell their freely transferable GS Securities. Short sales of GS Securities are not permitted.

2. Hedging GS Shares

Subject to certain conditions set forth in the Firmwide Trading Policy, Covered Persons may:

- Hedge their GS Shares, but only by writing (selling) listed covered calls with a maturity of at least 30 days or buying listed protective puts with a remaining maturity of at least 30 days.
- Buy listed cash-secured calls on GS Shares with a remaining maturity of at least 30 days and write (sell) listed cash-secured puts on GS Shares with a remaining maturity of at least 30 days.

Covered Persons are not permitted to establish over-the-counter ("OTC") option or other derivative positions to hedge their GS Shares.

Covered Persons are generally not permitted to enter into other short derivative or hedging transactions involving GS Securities or other transactions involving options on GS Securities (including entering into forward contracts, equity swaps or any similar agreements referencing GS Securities). Exceptions include permitted purchases of listed covered puts and writing of listed covered calls on GS Shares.

Covered Persons may not transfer GS Shares that provide cover for any call options written by or protective puts purchased by a Covered Person (except for delivery of such GS Shares upon exercise of such call or put).

Covered Persons are not permitted to enter into other transactions involving the purchase or sale of options on GS Shares other than those described above.

3. Pledging GS Securities

Covered Persons are generally permitted to pledge, subject to certain exceptions, their GS Securities as security in connection with an extension of credit from a third party.

4. Transactions Involving Non-GS Instruments Linked to GS Securities

Covered Persons are not permitted to enter into transactions involving any Non-GS Instruments linked to GS Securities (listed or OTC), other than those which are linked to an Approved Index, Approved ETF or Approved Basket (each as defined in the Firmwide Trading Policy).

When entering into permitted transactions involving Non-GS Instruments Linked to GS Securities, Covered Persons may not hedge out all or a portion of the risk not involving GS Securities and thereby create exposure to GS Securities in excess of the permitted percentage of their value in the index, basket or exchange-traded fund ("ETF").

5. Transactions Involving GS Structured Debt Securities and GS Structured Warrants

Covered Persons are permitted to enter into transactions involving GS Structured Debt Securities and GS Structured Warrants.

These transactions:

- Are not subject to the Window Periods/Blackout Periods (other than Executive Officers and GS Inc. Directors);
- Are subject to the 30-Day Rule (as described in the Personal Trading Policy);
- If entered into by an Executive Officer or GS Inc. Director, must be pre-approved by the General Counsel of GS Inc. or her or his designee (or, in the case of the General Counsel, pre-approved by the Chief Compliance Officer of GS Inc. or her or his designee); and

- Are subject to an attestation that the Covered Person is not in possession of material nonpublic information pursuant to Section B.3.

When entering into permitted transactions involving GS Structured Debt or GS Structured Warrants, Covered Persons may not hedge out all or a portion of the risk not involving GS Securities, which would create exposure to GS Securities in excess of the permitted percentage of the value of GS Securities in an Approved Index, Approved ETF or Approved Basket.

6. Donating, Gifting or Bequeathing GS Securities

Subject to the following conditions, Covered Persons are permitted to donate, gift or bequeath GS Securities:

- Except for GS Equity Awards, which are subject to special policies, Covered Persons are permitted to make donations under their will of any GS Securities that they own at the time of their death, or during their life to a trust that is for their sole benefit and revocable by them acting alone until their death or earlier incapacity.
- A donation or gift of GS Securities must be made during a Window Period.

7. Prohibition Against Lending GS Securities

Covered Persons are not permitted to lend their GS Securities.

8. Prohibition Against Margining GS Securities

Covered Persons are not permitted to margin their GS Securities, except as set forth under the headings “PMDs” and “Non-Employee Directors” and as permitted by an Authorized Person and the Goldman Sachs Private Wealth Management Department.

9. Prohibition Against Transactions Involving GS Credit Derivatives

Covered Persons are not permitted to engage in any transactions involving GS Credit Derivatives.

G. Additional Covered Person Rules

The following rules (the “Additional Covered Person Rules”) are applicable to all transactions involving GS Securities by certain categories of Covered Persons in Covered Accounts or otherwise and must be read in conjunction with the General Rules and the Specific Transaction Rules.

1. Executive Officers

Transactions involving GS Securities entered into by an Executive Officer are subject to the General Rules and the Specific Transaction Rules, but are also subject to the additional rules set forth below.

Executive Officers are not permitted to:

- Enter into any transaction involving GS Securities (other than GS Deposit Obligations) without first obtaining approval from the General Counsel of GS Inc. or her or his designee (or, in the case of the General Counsel, approval from the Chief Compliance Officer of GS Inc. or her or his designee) and providing an attestation that they are not aware of material nonpublic

information pursuant to Section B.3, even if the Access Person Window Period is open. No such transactions are permitted during a closed window.

- Hedge, buy or sell options on GS Shares, including GS Shares in which they have a pecuniary interest.
- Pledge their freely transferable GS Securities as security in connection with an extension of credit from a third party without first obtaining approval from the General Counsel of GS Inc. or her or his designee (or, in the case of the General Counsel, approval from the Chief Compliance Officer of GS Inc. or her or his designee), but neither GS Inc. and its affiliates nor any of their people can arrange, directly or indirectly, such extension of credit or otherwise assist an Executive Officer with obtaining an extension of credit.

Executive Officers are permitted to sell GS Shares pursuant to a firm-approved Rule 10b5-1 Plan with respect to the sale of their GS Shares. Entry into, amendment of or termination of any firm-approved Rule 10b5-1 Plan must be pre-approved by the General Counsel of GS Inc. or her or his designee.

2. Members of the Management Committee of GS Inc.

Transactions involving GS Securities entered into by a member of the Management Committee who is not an Executive Officer are subject to the General Rules and the Specific Transaction Rules, but are also subject to the additional rules set forth below.

If the Access Person Window Period is open, notification to Global Compliance Employee Services of a transaction involving GS Securities (other than GS Deposit Obligations) is required, but pre-clearance is not required.

If the Access Person Window Period is closed or there is another special circumstance, no transaction involving GS Securities (excluding transactions involving GS Deposit Obligations, GS Structured Debt Securities (except with respect to Executive Officers) and GS Structured Warrants (except with respect to Executive Officers)) can be made until senior compliance / business manager approval is received.

Members of the Management Committee are permitted to sell GS Shares pursuant to a firm-approved Rule 10b5-1 Plan with respect to the sale of their GS Shares. Entry into, amendment of or termination of any firm-approved Rule 10b5-1 Plan must be pre-approved by the General Counsel of GS Inc. or her or his designee.

3. PMDs and Senior Advisors

Transactions involving GS Securities entered into by a PMD or a Senior Advisor are subject to the General Rules and the Specific Transaction Rules. Senior Advisors and PMDs who are not Executive Officers are permitted to obtain margin with respect to their freely transferable GS Securities during the Access Person Window Period, subject to prior approval.

4. Specified PMDs

Transactions involving GS Securities entered into by a Specified PMD are subject to the General Rules and the Specific Transaction Rules (as well as the Additional Covered Person Rules with respect to PMDs), as well as additional rules for such period of time as any Authorized Person may deem appropriate. Specified PMDs may be permitted to sell GS Shares pursuant to a firm-

approved Rule 10b5-1 Plan with respect to the sale of their GS Shares. Entry into, amendment of or termination of any firm-approved Rule 10b5-1 Plan must be pre-approved by the General Counsel of GS Inc. or her or his designee.

5. Senior Directors and Covered Retired Limited Partners

Transactions involving GS Securities entered into by Senior Directors and Covered Retired Limited Partners are subject to the General Rules and the Specific Transaction Rules, except as set forth below.

Senior Directors (including Covered Senior Directors) and Covered Retired Limited Partners are not subject to the 30-Day Rule.

Non-Covered Senior Directors are not subject to trading only in the Window Periods.

Covered Senior Directors and Covered Retired Limited Partners are either Access Persons or Non-Access Persons depending on their access to information as determined by an Authorized Person and are subject to the applicable Window Periods.

6. Non-Employee Directors

Transactions involving GS Securities entered into by Non-Employee Directors are subject to the General Rules and the Specific Transaction Rules, except as set forth below.

- Except as otherwise provided below, Non-Employee Directors are not permitted to:
 - o Enter into any transaction involving GS Securities (other than a transaction involving GS Deposit Obligations) without first obtaining approval from the General Counsel or her or his designee of the applicable GS Entity of which they are a director.
 - o Hedge, buy or sell GS Shares, including GS Shares in which they have a pecuniary interest, except Non-Employee Directors that are Subsidiary Directors may hedge, buy or sell freely transferable GS Shares in an Access Person Window Period, but pre-clearance is required from the General Counsel or her or his designee of the GS Entity of which they are a director.
 - o Pledge their GS Shares as security in connection with an extension of credit from a third party without first obtaining approval from the General Counsel or her or his designee of the GS Entity of which they are a director, but neither GS Inc. and its affiliates nor any of their people can arrange, directly or indirectly, such extension of credit for a GS Inc. Director or otherwise assist a GS Inc. Director with obtaining an extension of credit.
- Even though margin is generally prohibited, Non-Employee Directors that are Subsidiary Directors are permitted to obtain margin with respect to their freely transferable GS Securities during the Access Person Window Period, but pre-clearance is required from the General Counsel or her or his designee of the GS Entity of which they are a director.
- Non-Employee Directors are permitted to sell GS Shares pursuant to a firm-approved Rule 10b5-1 Plan with respect to the sale of their GS Shares.
 - o Entry into, amendment of or termination of any firm-approved Rule 10b5-1 Plan must be pre-approved by the General Counsel of GS Inc. or her or his designee.

- This policy does not apply once a person ceases to be a Non-Employee Director and is not otherwise a Covered Person.
- Non-Employee Directors will be asked to represent that they are not aware of material, non-public information concerning GS Inc. and the GS Entity of which they are a director and, with respect to transactions in GS Structured Debt Securities and GS Structured Warrants, GS Inc. Directors must certify that they are not aware of material, nonpublic information concerning any underlying securities referenced by the GS Structured Debt Securities or GS Structured Warrant, as applicable.

7. Contingent Workers, Advisory Directors and Regional Advisors

Transactions involving GS Securities entered into by Contingent Workers, Advisory Directors and Regional Advisors are subject to the General Rules and the Specific Transaction Rules.

Contingent Workers, Advisory Directors and Regional Advisors are either Access Persons or Non-Access Persons depending on their access to information (as determined by an Authorized Person) and are subject to the applicable Window Periods.

H. Rules Applicable to Transactions by the Firm

Transactions by the firm in GS Securities are subject to applicable securities laws and must be done in accordance with the resolutions and policies adopted by the GS Inc. Board of Directors or a committee thereof. Issuances by the firm of GS Securities other than GS Structured Debt Securities and GS Structured Warrants are also generally subject to blackout periods, established by Corporate Treasury, during which pricing, sales and settlements are generally not permitted.

I. Contacts; Additional Information

If you have any questions regarding this policy or would like to request an exception from this policy, contact one of the people listed for “General Questions” in the Firmwide Trading Policy.

J. Roles and Responsibilities

- Who owns the risk: Firm Personnel
- Who manages the risk: Legal and Compliance
- Who reviews the risk: Legal and Compliance

K. Exceptions

L.

All exceptions to this policy must be escalated to the Policy Owner for approval.

M. Reporting and Escalations

N.

All transactions involving GS securities by Covered Persons may be subject to monitoring by the firm.

O. Related Documents

- Firmwide Policy with Respect to Personal Transactions Involving GS Securities and GS Equity Awards

APPENDIX A: DEFINITIONS

General

As used in this Policy, Advisory Directors, Contingent Workers, MDs, PMDs, Regional Advisors and Senior Advisors are individuals designated as such in the personnel systems of the firm.

Specific Defined Terms

1. Access/Non-Access Persons

Covered Persons are designated as either "Access Persons" or "Non-Access Persons" and departments, businesses or desks of the firm can be designated as "Access" or "Non-Access," in each case based generally on their potential access to material, non-public information about the firm. Access Persons include the following Covered Persons:

- All PMDs and MDs of the firm, as well as Senior Advisors
- All Non-Employee Directors
- All Firm Individuals in departments, businesses or desks of the firm designated as Access by an Authorized Person
- A Firm Individual or any other person designated as an Access Person by an Authorized Person

Non-Access Persons are all Covered Persons who are not designated as Access Persons.

2. Authorized Persons are the General Counsel of GS Inc., Chief Compliance Officer of GS Inc. or such other individuals to whom any of the foregoing shall delegate authority to make exceptions to all or certain rules under this policy.

3. Blackout Periods, including Access Person Blackout Periods and Non-Access Person Blackout Periods, are any periods of time that are not part of the applicable Window Periods. A Special Blackout Period is any period of time determined by an Authorized Person when trading by Access Persons and/or Non-Access Persons or any other group of Firm Individuals is not permitted for certain or all GS Securities, even if that period of time falls within a previously announced Access Person Window Period or Non-Access Person Window Period.

4. Covered Accounts include accounts in which a financial instrument can be purchased, sold or held that are controlled or directed by a Covered Person or in which a Covered Person has a beneficial interest.

5. Covered Persons are:

- Firm Individuals
- Non-Employee Directors
- Related Persons of a Covered Person
- Other persons who, or entities which, by the terms of any other agreement with the firm, are made subject to this policy (e.g., Covered Retired Limited Partners and Covered Senior Directors)

- 6. Covered Retired Limited Partners** are those Retired Limited Partners who have access to the firm's e-mail, data or computer systems (except to the extent clients of the firm have the same access) or unescorted access to the firm's premises (not including unescorted access which is limited to the 200 West Street ground floor lobby, the 11th floor Sky Lobby and the elevators). An Authorized Person may designate such person to be "Non-Covered."
- 7. Covered Senior Directors** are those Senior Directors who have access to the firm's e-mail, data or computer systems (except to the extent clients of the firm have the same access) or unescorted access to the firm's premises (not including unescorted access which is limited to the 200 West Street ground floor lobby, the 11th floor Sky Lobby and the elevators). An Authorized Person may designate such person to be "Non-Covered."
- 8. Firm** means GS Inc. and its consolidated subsidiaries, other than those that are treated as consolidated investments for financial statement purposes.
- 9. Firm Individuals** include:
- Employees (also "people") of the firm²
 - Contingent Workers
 - Regional Advisors
 - Senior Directors
 - Covered Retired Limited Partners
 - Other persons that any Authorized Person may designate from time to time
- 10. GS Credit Derivatives** are any obligation issued by any entity other than a GS Entity that by its terms pays principal or interest based in whole or in part on the credit of a GS Entity (e.g., based on the bankruptcy, default on obligations or the credit spreads of a GS Entity), as well as any similar obligation or any other obligation designated by an Authorized Person.
- 11. GS Deposit Obligations** are any obligation of a GS Entity that is regulated as a bank and that under the applicable law governing such GS Entity is deemed to be a "deposit," whether or not insured by the Federal Deposit Insurance Corporation or any similar organization. Exclusions include any deposit that is a GS Structured Debt Security, as well as any similar obligation or any other obligation designated by an Authorized Person.
- 12. GS Entity** means GS Inc. or any of its consolidated subsidiaries.
- 13. GS Equity Awards** include all equity-based compensation awards or other equity-based awards or any other award, arrangement or agreement the value of which is based in whole or in part on the value of GS Shares, other than interests in the Goldman Sachs Stock Fund in the

² A person ceases to be an employee of the firm for purposes of this policy once their "active employment" has been terminated and such person ceases to receive a salary from the firm. To the extent that any "notice period" is applicable to the termination of employment, active employment will for the purposes of this policy be deemed to continue through the end of such notice period. If the firm and, to the extent required, the person agree to waive all or a portion of such notice period, then active employment for purposes of this policy will be deemed to terminate on the date that the person ceases to receive a salary from the firm.

"Severance," "salary continuation periods," and any other periods of time with a similar status (as determined by the firm) are not deemed to be periods of active employment.

Goldman Sachs 401(k) Plan, as well as any similar obligation or any other obligation designated by an Authorized Person.

14. GS Inc. means The Goldman Sachs Group, Inc.

15. GS Securities are securities issued or guaranteed by GS Inc. or any other GS Entity, as well as Non-GS Instruments Linked to GS Securities.

16. GS Shares are shares of the common stock of GS Inc. and, if issued, voting shares of preferred stock of GS Inc.

17. GS Structured Debt Securities are any debt obligations (including notes, certificates, certificates of deposit or deposit accounts) issued or guaranteed by a GS Entity where the obligation is structured so that the amount received as principal by the holder of such obligation at maturity or the amount received as periodic interest is determined by reference to one or more equity securities, currencies, commodities or the credit of one or more entities (including indexes, ETFs and baskets) and include any similar obligation or any other obligation designated by an Authorized Person.

18. GS Structured Warrants are any warrant or similar instrument issued or guaranteed by a GS Entity that is exercisable into anything other than GS Securities, as well as any similar obligation or any other obligation designated by an Authorized Person.

19. Non-Employee Directors are GS Inc. Directors and/or Subsidiary Directors who are not employed by the firm.

20. Non-GS Instruments Linked to GS Securities are financial instruments issued by an entity that is not a GS Entity where more than 10% (5% in the case of instruments that are linked to baskets or sector/industry indexes and ETFs) of the payout and/or market value at the time they are issued is linked to one or more GS Securities, as well as any similar obligation or any other obligation designated by an Authorized Person.

21. Related Persons are individuals with the following relationships to a Firm Individual:

- Spouses or domestic partners
- Minor children of the Firm Individual or such Firm Individual's domestic partner
- Any other family members residing in the same household

Family members and any other relatives not living with a Firm Individual may be deemed Related Persons if they are financially dependent on that individual. Their status as Related Persons will be determined on a case-by-case basis.

22. Retired Limited Partner is a retired Partner of The Goldman Sachs Group, LP or its subsidiaries or a retired PMD.

23. Rule 10b5-1 Plan is a trading plan designed to comply with the requirements of Rule 10b5-1(c) under the U.S. Securities Exchange Act of 1934, as amended.

24. Subsidiary Directors are individuals (other than a Firm Individual) who serve as members of the board of directors of a GS Entity other than GS Inc.

25. Window Periods, including Access Person Window Periods and Non-Access Person Window Periods, are such periods as are determined by an Authorized Person.

EXHIBIT 21.1

Significant Subsidiaries of the Registrant

The following are significant subsidiaries of The Goldman Sachs Group, Inc. as of December 31, 2024 and the states or jurisdictions in which they are organized. Each subsidiary is indented beneath its principal parent. The Goldman Sachs Group, Inc. owns, directly or indirectly, at least 99% of the voting securities of substantially all of the subsidiaries included below. The names of particular subsidiaries have been omitted because, considered in the aggregate as a single subsidiary, they would not constitute, as of the end of the year covered by this report, a “significant subsidiary” as that term is defined in Rule 1-02(w) of Regulation S-X under the Securities Exchange Act of 1934.

Name	State or Jurisdiction of Organization of Entity
The Goldman Sachs Group, Inc.	Delaware
Goldman Sachs & Co. LLC	New York
Goldman Sachs Funding LLC	Delaware
GS European Funding I S.A R.L. LLC	Delaware
Murray Street Corporation	Delaware
Sphere Fundo De Investimento Multimercado - Investimento No Exterior Credito Privado	Brazil
Sphere Fund	Cayman Islands
GS Financial Services II, LLC	Delaware
Goldman Sachs Global Funding I, CO.	Delaware
Goldman Sachs Global Funding II LTD	United Kingdom
Goldman Sachs (UK) L.L.C.	Delaware
Goldman Sachs UK Funding Limited	United Kingdom
Goldman Sachs Group UK Limited	United Kingdom
Goldman Sachs International Bank	United Kingdom
Goldman Sachs International	United Kingdom
J. Aron & Company LLC	New York
GSAM Holdings LLC	Delaware
GSAMI Holdings I LLC	Delaware
GSAMI Holdings II Ltd	United Kingdom
Goldman Sachs Asset Management International Holdings Ltd	United Kingdom
Goldman Sachs Asset Management International	United Kingdom
Goldman Sachs Asset Management, L.P.	Delaware
Goldman Sachs Asset Management Holdings LLC	Delaware
Goldman Sachs Asset Management UK Holdings I Ltd	United Kingdom
Goldman Sachs Asset Management UK Holdings II Ltd	United Kingdom
Goldman Sachs Asset Management Holdings I B.V.	Netherlands
Goldman Sachs Asset Management Holdings II B.V.	Netherlands
Goldman Sachs Asset Management Holdings B.V.	Netherlands
Goldman Sachs Asset Management International Holdings B.V.	Netherlands
Goldman Sachs Asset Management B.V.	Netherlands
Goldman Sachs (Asia) Corporate Holdings L.L.C.	Delaware
Goldman Sachs Holdings (Asia Pacific) Limited	Hong Kong
Goldman Sachs (Japan) Ltd.	British Virgin Islands
Goldman Sachs Japan Co., Ltd.	Japan
GS Lending Partners Holdings LLC	Delaware
Goldman Sachs Lending Partners LLC	Delaware
Goldman Sachs Bank USA	New York
Goldman Sachs Bank Europe SE	Germany
Goldman Sachs Mortgage Company	New York
GSSG Holdings LLC	Delaware
ALQ Holdings (Del) LLC	Delaware
GLQ International Partners LP	Jersey
GLQ International Holdings Ltd	Jersey
GLQ Holdings (UK) Ltd	United Kingdom
GLQ Holdings (UK) II LTD	United Kingdom
GLQL S.A R.L.	Luxembourg
GLQC Holdings S.A R.L.	Luxembourg
GLQC II Designated Activity Company	Ireland
Goldman Sachs Non-US Americas Holdings LLC	Delaware
Goldman Sachs Non-US Americas Holdings II LLC	Delaware
Goldman Sachs Canada Holdings LLC	Delaware
Goldman Sachs Canada Inc.	Canada
Broad Street Principal Investments Superholdco LLC	Delaware
GS Fund Holdings, L.L.C.	Delaware
Broad Street Principal Investments, L.L.C.	Delaware

CONSENT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

We hereby consent to the incorporation by reference in the Registration Statements on Form S-3 (No. 333-284538) and on Form S-8 (Nos. 333-80839, 333-42068, 333-106430, 333-120802, 333-235973 and 333-261673) of The Goldman Sachs Group, Inc. of our report dated February 26, 2025 relating to the consolidated financial statements and the effectiveness of internal control over financial reporting, which appears in this Form 10-K.

/s/ PricewaterhouseCoopers LLP
New York, New York
February 26, 2025

CERTIFICATION

I, David Solomon, certify that:

- 1. I have reviewed this Annual Report on Form 10-K for the year ended December 31, 2024 of The Goldman Sachs Group, Inc.;
- 2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
- 3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
- 4. The registrant’s other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) Evaluated the effectiveness of the registrant’s disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) Disclosed in this report any change in the registrant’s internal control over financial reporting that occurred during the registrant’s most recent fiscal quarter (the registrant’s fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant’s internal control over financial reporting; and
- 5. The registrant’s other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant’s auditors and the audit committee of the registrant’s board of directors (or persons performing the equivalent functions):
 - a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant’s ability to record, process, summarize and report financial information; and
 - b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant’s internal control over financial reporting.

Date: February 26, 2025

/s/	David Solomon
Name:	David Solomon
Title:	Chief Executive Officer

CERTIFICATION

I, Denis P. Coleman III, certify that:

1. I have reviewed this Annual Report on Form 10-K for the year ended December 31, 2024 of The Goldman Sachs Group, Inc.;
2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;
3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;
4. The registrant's other certifying officer(s) and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:
 - a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;
 - b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;
 - c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and
 - d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and
5. The registrant's other certifying officer(s) and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):
 - a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and
 - b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: February 26, 2025

/s/

Denis P. Coleman III

Name:

Denis P. Coleman III

Title:

Chief Financial Officer

CERTIFICATION

Pursuant to 18 U.S.C. § 1350, the undersigned officer of The Goldman Sachs Group, Inc. (the Company) hereby certifies that the Company’s Annual Report on Form 10-K for the year ended December 31, 2024 (the Report) fully complies with the requirements of Section 13(a) or 15(d), as applicable, of the Securities Exchange Act of 1934 and that the information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: February 26, 2025

/s/	David Solomon
Name:	David Solomon
Title:	Chief Executive Officer

The foregoing certification is being furnished solely pursuant to 18 U.S.C. § 1350 and is not being filed as part of the Report or as a separate disclosure document.

CERTIFICATION

Pursuant to 18 U.S.C. § 1350, the undersigned officer of The Goldman Sachs Group, Inc. (the Company) hereby certifies that the Company's Annual Report on Form 10-K for the year ended December 31, 2024 (the Report) fully complies with the requirements of Section 13(a) or 15(d), as applicable, of the Securities Exchange Act of 1934 and that the information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: February 26, 2025

/s/ Denis P. Coleman III

Name: Denis P. Coleman III

Title: Chief Financial Officer

The foregoing certification is being furnished solely pursuant to 18 U.S.C. § 1350 and is not being filed as part of the Report or as a separate disclosure document.